
Towards a Dynamics of Prosodic Change: Corpus-Based and Computational Studies in the Synchronic and Diachronic Prosodic Phonology of Indic, Greek, and Germanic

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This work concerns itself with language change, but strives to obtain not only descriptive adequacy, but also greater explanatory adequacy for the phenomena of change under discussion. The phenomena treated all fall within the domain of PROSODIC PHONOLOGY (see generally Nespor and Vogel 1986), and more specifically, phenomena of prosodic phonology at the level of the word, especially concerning the calculation of word-level prominence relations, falling under the label of “stress” or “accent” (see generally Hayes 1995, Hyman 2006). The empirical basis for study will consist of Indo-European languages, both ancient (e.g., Vedic Sanskrit) and modern (e.g., Italian), as well as their reconstructed ancestor (i.e., Proto-Indo-European). The ultimate goal is to successfully model attested changes in the domain of word-level prosody across the histories of these languages, and thereby shed light on *why* as well as *how* the attested changes (could have) occurred. My basic contention is that, at least as far as the question of word-level prosodic prominence is concerned, too little is understood about the conditions that drive changes in phonological grammars, and that existing explanations that are invoked in the literature suffer from not being grounded in explicit, predictive models that permit sufficiently fine-grained examination of causes and consequences. A toolkit containing techniques from constraint-based phonological analysis, learnability theory, agent-based modeling, and old-fashioned philology supports this endeavor.

In contradistinction to much scholarship on the historical linguistics of prosody and “accentuation” in the oldest attested Indo-European languages (e.g., Sanskrit, Ancient Greek, Latin), the primary concern in this work does not lie with the detailed reconstruction of the prosodic phonology or stress grammar of their last common ancestor. Rather, I am concerned with working out the diachronic developments that have led to clear changes in the prosodic systems of the attested stages of various Indo-European languages. This research objective means that focus will first be applied to the analysis of synchronic prosodic grammars, and then to the description and attempted modeling of pathways of change between the given grammars. Particular languages, aspects of whose synchronic prosody is considered in greater and lesser detail include Vedic Sanskrit (Chapter 4), Classical Sanskrit (Chapter 5), Greek dialects of the 1st Millennium BCE (especially Attic-Ionic and Doric), Proto-Germanic and its earliest-attested daughters (Chapter 7), Old and Classical Latin (Sections 2.2.7.1 and 3.3.1.1), and Tocharian B (2.2.7.2).

The basic question ultimately to be confronted is: to what extent are stress (accent) systems diachronically stable, and what factors (especially factors internal to the phonological grammar) can condition diachronic change? To answer this question, however, a much more detailed and precise conception of the workings of the relevant synchronic grammars is needed.

Part I: Theoretical Bases for the Synchronic and Diachronic Study of Prosody

Part I concerns itself with the development of the phonological analysis of prosodic structure and stress assignment, as well as further techniques from the domains of learnability theory and relevant results from research on psycholinguistics and language acquisition that will be brought to bear in Part II.

Chapter 1 opens with a brief introduction to the phonological phenomenon perhaps most central to the work as a whole: lexically contrastive stress of the sort found represented in minimal pairs of the English type *cónvert* (noun) vs. *convért* (verb) or Vedic Sanskrit (*viśás*

‘settlement.GEN/ABL.SG’ vs. *vísas* ‘settlement.NOM/ACC.PL. After describing the range of linguistic phenomena covered by the term *prosody*, the core of the chapter is devoted to discussing evidence for the existence of prosodic constituents in the phonological component of the grammar, how those constituents are composed, and how they interact with one another. These constituents (prosodic words, feet, syllables) will be familiar to most of the readership, but how they are situated within the PROSODIC HIERARCHY and what status they have as units of phonological analysis requires some justification. Although the psychological reality of these constituents has been called into question, and alternatives to the by-now standard array of constituents found in, e.g., Féry 2017, their utility in the analysis of myriad phenomena makes their continued use attractive. The latter part of the chapter presents an introduction to prosodic typology with regard to intonation, tone, and stress. Of particular relevance here is the typological status of so-called “pitch-accent” languages, among which some of the older Indo-European languages dealt with here are often classed.

After restricting the scope of investigation in this work to phenomena at the level of the prosodic word and below, and especially to the principles by which word-level prominence is assigned, the matter of how to analyze a wide variety of stress systems is taken up in Chapter 2. At 2.1, I advance a provisional typology of the factors relevant to stress assignment and how they may combine with one another to produce an array of typologically distinct systems. How to analyze stress assignment in “fixed-stress” systems, in which the position of primary and secondary stress depends essentially on syllable count, relation to word edges, and potentially syllable weight, is treated from two perspectives: the parametric system developed in Hayes 1995 and the constraint system for the analysis of stress in Optimality Theory proposed in Kager 1999 (with some modifications and extensions). An introduction to stress systems controlled by reference to syllable weight, tone, and lexically specified prosody (i.e., lexical stress/accent) completes the typological portrait of stress systems. With respect to lexical accent systems in particular, I distinguish primarily, on the one hand, “simple” systems in which only lexically prespecified properties and edge orientation play a role, and, following Revithiadou (1999), head-controlled lexical accent systems, in

which the internal morphological structure of a word motivates the usage of special prosodic markedness (HEADSTRESS) and faithfulness (HEADFAITH) constraints. Particularly the latter, head faithfulness, proves to be of considerable utility in the description of word prosody for older Indo-European languages.

Linking Chapters 2 and 3 is discussion of the computational description of stress systems, and discussion of research results concerning the computational complexity of certain terms of systems. Taking center stage in Chapter 3 is then an overview of the approaches to the learnability of constraint-based phonological grammars developed since Tesar 1995. There, the central concern is: how are constraint interactions (in terms of their *ranking* or *weighting*) learned through exposure to (complete and error-free) data? Log-linear models (so-called “Maximum Entropy Grammars”), their properties and behaviors in the context of phonological systems, occupy much of the discussion. Of most direct relevance to the larger questions of the book is how stress systems are acquired, despite the fact the prosodic constituents frequently assumed to be responsible for determining stress placement have no direct perceptual correlates themselves (so-called “hidden structure”).

How these approaches to the learnability of phonological grammars and stress grammars in particular relate to change in prosody (stress systems in particular) is treated under 3.3. I argue there for a distinction between two types of changes in stress grammars. On the one hand, the partial dependency of stress rules on the segmental structure or intonational patterns of prosodic words may be affected if one of these aspects undergoes some sound change. In such cases, stress rules may be opacified or the phonetic percepts of stress may be subject to confusion. In such cases, a change in the structure of the learning data is apparently a cause. On the other hand, there may be cases of reanalysis motivated by the frequency distribution and learnability of stress patterns. In such instances, the change in grammars and/or lexica might be said to rest purely in the phonology. Two small case studies from the histories of Latin and Ancient Greek are presented as plausible examples of the latter type of change. The question of how lexically contrastive stress may undergo attrition and loss diachronically is also treated through a toy simulation, where I contend that Max-

Ent grammars are able to adequately capture such a development, provided that at least a small amount of variation exists in the learning data. A closer look at the status of word prosody in two contemporary cases, one from English and one from South Kyungsang Korean, indicates that variation in stress and pitch accent patterns can exist synchronically not only at the population level, but at the level of individual speakers. Such data lends support to the view that variation in lexical prosody – however introduced – constitutes a plausible precursor to prosodic change.

The latter portions of Chapter 3 then survey the literature on the acquisition of stress, as well as the psycholinguistics of stress perception. Studies on L1 prosodic acquisition suggest that all types of word-prosodic systems (excluding canonical tone systems, not discussed here) are acquired so early and so accurately that change in this domain of the grammar might come as something of a surprise. Conversely, available research on L2 prosodic acquisition and the psycholinguistics of stress perception indicates that L1 speakers whose prosodic systems are relatively simple (“surface observable”, in the terminology of Peperkamp 2004) may have relatively high error rates when attempting to produce novel word-level stress distinctions. This literature thus provides grounds to believe that L2 speakers can constitute a source of either patterned (if biased towards outputs matching their L1 grammars) or unpatterned variation in word prosody. This chapter, and Part I, concludes with a brief introduction to agent-based learning as a potential tool for the study of language change.

Part II: Cases Studies in the Analysis of Prosodic Systems of Older Indo-European Languages

The second part of this work constitutes the empirical and analytical core of the study. In this part, essentially synchronic issues in the analysis and modeling of prosody for a range of Indo-European languages are addressed. In some instances, the analysis presented rests on new insights or new pieces of data; in others, the existing competing analyses are compared, and recast in the light of recent theoretical advances. In Chapters 5 and 7, in particular,

iterative agent-based models are employed in order to gain insight into developments of the stress systems of Classical Sanskrit and Proto-Germanic, out of the earlier Vedic and Proto-Indo-European systems, respectively.

Chapter 4 aims at a comprehensive analysis of the synchronic word-level prosody of Vedic Sanskrit, primarily as known from the *R̥gveda* and *Atharvaveda*. Three major contentions are advanced in the course of the chapter. First, apparent exceptions to the OBLIGATORINESS of word-level stress are argued to largely result from intonational factors. Second, the range of stress-attracting and stress-rejecting properties exhibited by morphemes in Vedic is shown to fit well with a representation of lexical accent as consisting in underlying foot structure. By exploiting the combinatorial possibilities of left and right foot edges, a finer distinction in degrees of stress-sensitivity is obtained, which empirically compares favorably with the predictions of earlier analyses (in particular Kiparsky 2010a). Third, Vedic Sanskrit must be sensitive to a distinction between category-changing derivational affixes, as opposed to non-category-changing and inflectional affixes. The strong tendency of category-changing derivational affixes to impose their prosodic preferences on the prosodic word fits well with the theoretical model of lexical accent advanced in Revithiadou 1999. In particular, the failure of many derivational suffixes in the verb to exhibit such “accentual dominance” is proposed to result from the fact that they are non-category determining, instead inheriting the lexical category of VERB from roots.

The subsequent Chapter 5 looks into the later history of prosody in Indic, specifically by attempting to reconstruct a metrical stress rule for Classical Sanskrit. This task requires exploiting and examining a range of diverse data: variant, non-standard forms in Epic and Classical Sanskrit, phonological developments in Middle Indic languages, phonological developments in Modern Indic languages, and the testimony of 19th-century fieldwork in India. The sum of this evidence is argued to point towards a rather unique stress system, which however exhibits some parallels with the stress systems of Old Latin and some varieties of Arabic. In short, primary stress was typically assigned to the rightmost non-final heavy syllable up to the preantepenult; in disyllables, stress was permitted on a final syllable with a long

vowel or diphthong, if the initial syllable was light. Computational simulation demonstrates that this stress system, despite its typological rarity, is learnable, assuming a standard array of prosodic markedness constraints (and variation-free learning data).

The second half of the chapter considers how such a fixed-stress pattern might have arisen out of the lexically and morphologically sensitive stress pattern of Vedic Sanskrit. I consider the frequency distributions of stress in the *R̥gveda*, examine the over- and under-representedness of stress in particular positions, and compare the Vedic frequency distributions to the predictions made by a variety of fixed-stress grammars (none of which correctly predict the position of stress in Vedic for more than $\sim 45\%$ of tokens). A further attempt to determine whether the Vedic stress distributions could have directly led to the acquisition of the Classical Sanskrit pattern is made by undertaking a series of iterated agent-based learning simulations. Learners equipped with Maximum Entropy grammars outfitted with standard prosodic markedness constraints are forced to learn through Robust Interpretative Parsing of overt forms, and propagate stress data to subsequent generations. Results suggest that a solution rooted simply in the frequencies of stress in overt forms of Vedic is insufficient as a historical explanation, but do indicate that some specific aspects of the Classical Sanskrit pattern (in particular final syllable extrametricality) are perhaps attributable to the original Vedic situation. Furthermore, these simulations again confirm the clear simplicity bias of Maximum Entropy grammars: simpler patterns are more likely to emerge and stabilize out of a pool of variation.

Chapter 6 investigates the word-level prosody of Ancient Greek in the 1st Millennium BCE. With regard to the well-known Attic-Ionic system, I advance a revised phonological analysis for the placement of stress and association of tone that takes inspiration from an array of existing analyses (Golston 1990, Steriade 2014, Revithiadou 2018). The principal factors governing this system are argued to consist in a strict association between primary stress and starred tones, within a two-syllable window at the right edge of the word, where a weight hierarchy $VV > VC > V$ holds. A new treatment of accentuation in the Doric dialect argues that the word prosody of Doric differs from that of Attic-Ionic with respect to just a single pa-

parameter: word-final consonant extrametricality (active in Attic-Ionic, not in Doric). A brief comparison of Attic-Ionic with Modern Greek prosody reveals striking overlaps between the position of High tone in Ancient Greek and primary stress in Modern Greek. Greek thus appears to reflect an instance of “persistence” of surface patterns: different grammars and lexica are constructed in order to faithfully learn the position of word-level prominence in the face of other phonological changes.

The final case study in Chapter 7 is devoted to the earliest attested Germanic languages and their last common ancestor, Proto-Germanic. Primarily on the basis of data from Gothic and Old English, I contend, contrary to some approaches (e.g., Hutton 1995), that these languages possessed no lexically specified prosodic features. Instead, primary stress was determined by the construction of moraic trochees at the left edge of minimal prosodic words. Apparent exceptions, in particular seeming non-initial stress in prefixed verbs, are proposed to result from cliticization. For Proto-Germanic, and its development from Proto-Indo-European, this means that a Vedic-like system with lexically contrastive stress has been lost in favor of a comparatively simple fixed-stress grammar. Explaining this loss is not straightforward, given the complications in attempting to reconstruct the stress patterns of Early Proto-Germanic, prior to the development of an initial stress rule. Agent-based simulation suggests, however, that language contact with “stress-deaf” speakers of a language with just such a rule, could plausibly have both introduced inexplicable variation at the level of individual lexemes and contributed to an overall greater proportion of initial stress in the learning data.

Finally, a concluding chapter then summarizes the principal results of each chapter in greater detail, and lays out directions for future research in light of the results obtained here. In particular, more quantitatively oriented philological work could help to clarify some persistent uncertainties about the Vedic and Greek data. The development and refinement of agent-based simulations (along the lines of, e.g., Ito and Feldman 2022) should also continue to play a role in all future approaches to prosodic change, both within and outside the Indo-European language family.

For Leone,
who has been with this project from beginning to end,
but knows almost nothing of it.

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SYMBOLS

/ markedness constraint	V any vowel or syllable peak
* [preceding] reconstructed form	Ǟ short vowel
* [following] assured but unattested form	Ǡ long vowel
* [following] marker of pitch accents or site for association of pitch accents	ǡ, ǣ vowel or syllable with high tone
^x hypothetical / incorrect / ungrammatical form	Ǥ, ǥ vowel or syllable with low tone
* markedness constraint	Ǧ, ǧ vowel or syllable with falling tone
# word boundary	X any segment
- morpheme boundary	$P(A)$ probability of event A
= clitic boundary	$P(A B)$ conditional probability of event A given event B
. syllable boundary	η learning rate
(left boundary of $\mathfrak{f}$	υ utterance
) right foot boundary of $\mathfrak{f}$	ı intonational phrase
[left boundary of a constituent	ϕ phonological/prosodic phrase
] right boundary of a constituent	ω prosodic word
/X/ underlying representation	$\bar{\omega}$ recursive prosodic word
[X] surface representation	$\mathfrak{f}$ foot
$\langle X \rangle$ orthographic representation	σ syllable
$\langle X \rangle$ extrametrical /	$\bar{\sigma}$ heavy syllable
prosodically unparsed constituent	$\check{\sigma}$ light syllable
< develops by sound change from	– heavy syllable
> develops by sound change to	~ light syllable
≫ outranks, crucially dominates	μ mora
→ becomes in a synchronic derivation	
← becomes in a synchronic derivation	
⇔ mutual implication	
∅ empty set; nothing	
Δ change in	
C any consonant	
F falling tone (= HL)	
G Glide	
H heavy syllable; high tone	
L light syllable; low tone	
HL falling tone	
LH rising tone	
M mid tone	
N Nasal	
R Sonorant	

ABBREVIATIONS

1 first person	M. Māhārāṣṭrī
2 second person	MaxEnt Maximum Entropy Grammar
3 third person	MBh Mahābhārata
AB Aitareya Brāhmaṇa	Mg. Māgadhi
ABL ablative	MS Maitrāyaṇī Saṃhitā
ACC accusative	MID middle
ACT active	NOM nominative
AMg. Ardhamāgadhi	NPST non-past
AOR aorist	OCP Obligatory Contour Principle
(Y/O)Av. (Younger/Old) Avestan	Od. <i>Odyssey</i>
AV(Ś/P) <i>Atharvaveda</i> (Śaunaka/Paippalada)	OE Old English
Att.-Ion. Attic-Ionic	OHG Old High German
C consonant	OPT optative
C. Classical	OT Optimality Theory
cmpv comparative	PDE present-day English
DAT dative	PERF perfect
Dor. Doric	(P)IE (Proto)-Indo-European
DTS Default-to-Same	(P)Iir. (Proto)-Indo-Iranian
DTO Default-to-Opposite	PART participle
DU dual	PL plural
E Epic	PPP past passive participle
F feminine	PRES present
Fo fundamental frequency	QI quantity insensitive
FTBIN FOOTBINARITY	QS quantity sensitive
G. Gujarātī	RV Ṛgveda
GEN genitive	Skt. Sanskrit
G(r)k. Greek	SPE <i>The Sound Pattern of English</i> (Chomsky and Halle 1968)
Germ. Germanic	SUBJ subjunctive
(P)Gmc. (Proto-)Germanic	SG singular
Goth. Gothic	SV Sāmaveda
Hom. Homer(ic)	Ś Śaurasenī
Hz Hertz (periods per second)	ŚB Śatapatha Brāhmaṇa
(P)IE (Proto-)Indo-European	SR surface representation
Il. <i>Iliad</i>	TLG <i>Thesaurus Linguae Graecae</i> (Pantelia 2021)
IMPV imperative	TB Tocharian B
INJ injunctive	TS Taittirīya Saṃhitā
INST instrumental	V vowel; verb
JB Jaiminīya Brāhmaṇa	Ved. Vedic
JM Jainamāhārāṣṭrī	UR underlying representation
Lat. Latin	WSP WEIGHT-TO-STRESS Principle
loc. locative	YV Yajurveda
log (natural) logarithm	
M masculine	

TRANSCRIPTION AND TRANSLITERATION

Sanskrit

For convenience and easier legibility throughout the dissertation, I will largely cite Sanskrit forms using the standard International Alphabet of Sanskrit Transliteration (IAST) of the Devanāgarī script; symbols largely have IPA values, but please note the following conventions:

- Consonants with an underdot indicate retroflex place of articulation, but with features otherwise identical to corresponding dental consonants (e.g., $\dot{t}$ = [t̪]). One exception is $\dot{h}$, which represents [h].
- Consonants with an underring are syllabic (e.g., $r̥$ = [r̥]).
- c and j are voiceless and voiced palatal stops (i.e., [c] and [j]); $\tilde{n}$ is a palatal nasal (i.e., [ɲ]); y is a palatal glide (i.e., [j]); $\acute{s}$ is a voiceless palatal fricative (i.e., [ç]).
- Stops followed by h are aspirated in the case of voiceless stops (e.g., th = [tʰ]) or breathy voiced in the case of voiced stops (e.g., dh = [d̪ʱ]); h alone is a voiced glottal fricative (i.e., [ɦ]).
- A macron over a vowel indicates a long vowel (e.g., $\bar{i}$ = [iː]).
- a is [ə], while $\bar{a}$ is [aː].
- The vowels e and o are long (= [eː] and [oː]).

The reader is further referred to the useful presentation of the Sanskrit sound inventory in Ryan 2017.

(Ancient) Greek

(Ancient) Greek forms throughout will usually be given both in Greek script and an accompanying phonemic transcription in IPA, though I resort to transcription alone in some places, and in others provide only the Greek. Romanization of Greek script is never used. The approximate phonetic values assumed are for 8th c. BCE Ionic Greek, unless otherwise indicated. The table below provides the correspondences between the Greek and the usual IPA transcription employed.

Greek Alphabet	Approximate Phonemic Transcription (8th c. Ionic)
α	[a] or [a:]
β	[b]
γ	[g]
δ	[d]
ε	[e]
ει	[ei] or [e:]
ζ	[zɗ]
η	[ɛ:]
θ	[tʰ]
ι	[i] or [i:]
κ	[k]
λ	[l]
μ	[m]
ν	[n]
ξ	[ks]
ο	[o]
ου	[ou] or [o:]
π	[p]
ρ	[r]
σ / ς	[s]
τ	[t]
υ	[u]
φ	[pʰ]
χ	[kʰ]
ψ	[ps]
ω	[ɔ:]

English and Other Languages

Other languages will usually be cited in their standard orthographies, or standard romanized transcriptions (e.g., for Gothic), with an accompanying broad transcription where relevant. Present-Day English transcriptions are largely based on the author's pronunciations as a native speaker of North American English (Western United States of America). Transcriptions of other modern languages are largely based on available phonological descriptions of "standard" dialects (e.g., for German or Italian).

FOREWORD

The first seeds for this project were planted in 2016 and 2017, after having made a first attempt at the formal analysis of Vedic Sanskrit accentuation (Sandell 2015a: Ch. 7) and having applied iterated agent-based learning to the problem of the development of Germanic strong verbs out of Proto-Indo-European (Sandell and Zukoff 2017). I was consistently struck and puzzled by one simple fact about the prosody of Proto-Germanic in relation to Vedic Sanskrit (and Ancient Greek), which to me seemed to lack a well-founded explanation (in handbook discussions, at least): how and why could primary stress in Proto-Germanic have come to consistently rest on the initial syllable (or root syllable) of every word? Such a relatively simple-looking stress system (which is indeed simple to acquire, following results in Staubs 2014, substantially replicated under 5.3) stands in stark contrast to the intricate and unpredictable-looking accentual systems of Vedic Sanskrit and Ancient Greek. Reflecting on the fact that the Vedic Sanskrit and Ancient Greek prosodic grammars, in virtue of their very existence, must constitute learnable phonological systems, and considering the fact that most work on the learnability of phonological grammars (see the survey under 3.2) takes the faithful acquisition of a synchronic system as its objective, my sense of perplexity only grew. What historical chain of events – sudden or gradual, catastrophic or subtle – could have resulted in such strikingly different grammars for stress assignment, given that all these languages (Vedic, Ancient Greek, Proto-Germanic) share a common ancestor?

Precisely why this question does not allow itself to be answered easily is summarized in the “core conundrum” stated below. With respect to the grammars and lexica constructed for the purposes of determining word-level prosodic prominence (“stress”), this conundrum becomes all the more vexing in light of research on phonological learnability and acquisition (Chapter 3): stress grammars can mostly be acquired quickly and accurately, and variation in the acoustic correlates of stress does not readily entail misinterpretation concerning the position of stress.

(1) **The Core Conundrum of Diachronic Phonological Learnability**

- a. Any synchronically attested phonological grammar (plus lexicon) must be acquirable and diachronically faithfully transmissible: what is learnable is transmissible.
- b. But phonological grammars (and lexica) are subject to change diachronically (i.e., are presumably not always transmitted faithfully): transmission is not always faithful.

Although the present work has concerns of language change at heart, a major methodological principle adopted is the following: explanation of change requires knowledge of synchrony. In dealing with grammars of stress and prosodic structure, this principle in practice means that the inner workings of synchronic grammars must be well understood, in order to be able to adequately compare systems and determine where, exactly similarities and differences lie. Consequently, more space in the case studies of Part II is devoted to the analysis of synchronic systems than to descriptive or explanatory accounts of changes

identified between two synchronic states. Hence the title: “Towards a Dynamics of Prosodic Change”. Much more remains to be said about such dynamics, in both Indo-European and non-Indo-European languages, than is addressed here.

The audience for this book is meant to be wide-ranging, and for that reason, Part I aims to provide the general background necessary for successful engagement with the case studies undertaken in Part II. The only background assumed is a basic knowledge of methods of phonological analysis and of Optimality Theory (Prince and Smolensky 1993 [2004]) in particular. A graduate student of linguists with an interest prosodic phonology and language change would likely benefit from reading the book from cover to cover. Phonologists already possessed of a good working knowledge of prosodic phonology, metrical phonology, Optimality Theory, and learnability might wish to begin with the second half of Chapter 3 for an orientation on the central questions surrounding prosodic change before delving into the case studies. Historical linguists and philologists of the older Indo-European languages looking for accounts of specific developments in the prosody of the languages treated here might prefer to simply read the cases studies of interest, but may often find it necessary to return to relevant portions of Part I for theoretical background. A reader approaching this work principally from the perspective of language acquisition and/or learnability theory, with sufficient background in phonological analysis might find it most fruitful to work through Chapters 2 and 3, then skip to the latter portions and conclusions from each case study to pick out points of greatest interest.

Fundamentally, Part I reflects a thorough synthesis of literature relevant to the problems investigated in Part II, as well as an attempt to motivate many, if not all, of the theoretical assumptions brought to bear. Part II then applies the tools from Part I to studies on the synchrony, diachrony, and learnability of word prosody in Vedic Sanskrit (Chapter 4), Classical Sanskrit (Chapter 5), Ancient Greek (Chapter 6), and Chapter 7 (Germanic). The overarching goal is to, at a minimum, provide more precise, formal descriptions of *what* has undergone a change in phonological terms in the specific cases considered.

Since many of the studies undertaken here, both larger and smaller, are supported in one way or another through software, electronic data, or code, some of these materials are made available through a GitHub repository at:

<https://github.com/rpsandell/Sandell-LMU-Habilitation>.

Materials there are divided by chapter. Notes at relevant points throughout will usually alert the reader to the availability of these digital supporting resources.

Part I

**Theoretical Bases for the Synchronic and
Diachronic Study of Prosody**

CHAPTER 1

The Object of Study, Historical Background, and Theoretical Preliminaries

In the earliest Vedic Sanskrit (spoken ~ 1200–1000 BCE in the Punjab), one finds genuine minimal pairs distinguished not by any segmental differences, but merely a differences in the position in a word of the *prosodic prominence* (here, marked with an acute), traditionally referred to as “accent”, as shown in (1)a. Beyond the not-infrequent minimal pairs, words of identical prosodic shapes (same number of syllables, same sequence of heavy and light syllables) may also show prosodic prominence on different syllables (as in (1)b.), and prominence may be found at opposite edges of the word (as in (1)c.). In these respects, the situation concerning the placement of prosodic prominence in Vedic Sanskrit is not unlike the situation of stress in Present-Day English, in which true minimal pairs (e.g., *import* ['impɔ:t] vs. *impórt* [ɪm'pɔ:t]) and prosodic contrast pairs (e.g., *misapprehénsion* [misæpi'hɛnʃɪn] vs. *unattáinable* [ʌnə'tʰɛɪnəbl̩]) in the position of stress are all visible.

(1) Prosodic Prominence and Contrast in Vedic Sanskrit

- a. *bráhmanasya* [brá:fɪmənəsɟə] ‘pious:M.GEN.SG’ vs.
brāhmanáasya [bra:fɪmənəsɟə] ‘Brahman:M.GEN.SG’
- b. *puróhitam* [puró:ɦitəm] ‘domestic priest:M.ACC.SG’
vs. *purūtámam* [puru:təməm] ‘much:MPV.M.ACC.SG’
- c. *ápratidhr̥ṣṭaśavasam* [ə.prə.ti.dʰr̥.ʃtə.ʃə.wə.səm]
‘having irresistible power:MASC.ACC.SG’
vs. *abhibhañj-at-ī-nám* [əbʰibʰəɲɟ-ət-i:-ná:m] ‘smashing against-PART.ACT-FEM.-GEN.PL’

The situation of prosodic prominence is substantially different in the Classical Sanskrit of some centuries later (from ~ 500 BCE): the first word of each pair shown in (1)a. and (1)b. would carry the word-level prosodic prominence on the same syllable, while the forms in (1)c. would both show a change in the position of prominence, moving closer to or farther from the right edge (cf. Bühler 1883: Schrifttafel; see further chapter 5), as illustrated in (2); forms in which the position of the prominence would have changed are given in bold. Similarly, the rules of stress assignment in the Old English dialects spoken a millennium ago would have generated patterns only faintly resembling the present-day stress distributions (cf. Hogg 2011: §2.84–2.90 and further Chapter 7).

(2) Prosodic Prominence in Classical Sanskrit

- a. *bráhmanasya* [brá:ɦmənəsɟə] ‘pious:M.GEN.SG’ vs.
bráhmanasya [brá:ɦmənəsɟə] ‘Brahman:M.GEN.SG’
- b. *puróhitam* [puró:ɦitəm] ‘domestic priest:M.ACC.SG’
vs. *purútamam* [purú:təməm] ‘much:CMPV.M.ACC.SG’
- c. *apratidhṛṣṭásavasam* [ə.prə.ti.d^hṛ.ʃtə.çə.wə.səm]
‘having irresistible power:MASC.ACC.SG’
vs. *abhibhañj-at-í-nām* [əb^hib^həɲɟ-ət-í:-nɑ:m] ‘smashing against-PART.ACT-FEM.-
GEN.PL’

Why should it be the case that the word-level prominence (“stress” or “accent”) does not remain diachronically stable between Vedic Sanskrit and Classical Sanskrit, or Old English and Present-Day English, and more importantly *how could such changes come to pass?* This question is the central one to be investigated in the present work. The primary aims consist in adequately characterizing the prosodic systems of a number of (older) Indo-European languages, and then attempting to account for changes in the prosodic system that have taken place diachronically. As the term “prosodic system” should suggest, the focus in this work will lie not with changes impacting individual lexical items – though such changes may provide crucial evidence about prosodic grammars or serve as harbingers of larger systemic change – but rather on the components of phonological grammars responsible for generating the attested patterns.

The case of Sanskrit mentioned here, to be pursued in detail in Chapter 5, further exemplifies a particular species of prosodic change, which lacks a straightforward explanation. Specifically, Vedic Sanskrit possesses a lexical accent system (cf. Kiparsky 1984, Kiparsky 2010a; on the analysis of such systems generally, see 2.4 in Chapter 2), in which the assignment of word-level prosodic prominence is determined through the interaction of lexical properties of the constituent morphemes in a morphologically complex word, or simply arbitrary lexical distinctions of some kind in the case of morphologically simplex words. Classical Sanskrit, meanwhile, assigns stress predictably based on the construction of metrical structure alone. The perhaps puzzling fact in this and similar cases is that, despite extensive evidence in the Vedic data for contrastive lexical accent, Classical Sanskrit has constructed a phonological grammar and lexicon in which lexical accent plays no role. Both intuitively and under some models of phonological learnability (e.g., Tesar 2014 for Optimality Theory, Dresher 2016 for a parameter-setting model), the presence of genuine minimal stress pairs or even other sorts of phonological contrast pairs in the primary linguistic data ought to lead a learner to conclude that lexical accent is a necessary feature of the phonological lexicon under construction. Thus, that such a change is possible at all, whether occurring gradually over many generations or relatively suddenly, raises a host of difficult and interesting questions about how prosodic grammars and lexica, in particular surrounding stress assignment, are learned. The nature of this issue will be discussed more extensively from the perspective of learnability theory and language change in Chapter 3 (especially 3.3)

Before proceeding with a general introduction to the history and study of prosodic phenomena, I would like to encourage the reader to reflect for a moment on the particular em-

pirical difficulties that the present undertaking faces in centering its inquiries around languages such as Sanskrit, Ancient Greek, and the oldest attested Germanic languages. That a linguist operating in the present day knows anything at all, let alone finds themselves in the position to comment in detail on and analyze the prosody as part of the larger phonological system – i.e., aspects of pitch, duration, rhythm, and “accent” or “stress” – for languages that were spoken and acquired as first languages thousands of years ago, is an enormous stroke of luck. More specifically, as this work and some of its predecessors show (among others, Devine and Stephens 1994, Allen 1973, Kuryłowicz 1968, Hirt 1895), that we should be able to so deeply investigate and argue about prosodic features of languages like Hittite, Vedic Sanskrit, and Ancient Greek cannot be taken for granted, given that our records of these languages are exclusively written.

Consider, for example, the representation of suprasegmentals (in particular, stress) in the orthographies of modern languages such as Russian, Italian, or English: even though stress in all three of these languages is potentially contrastive (i.e., differences in the position are sufficient for building minimal pairs, and in case other segmental differences occur, these are attributable to stress-sensitive phonological processes), none of their standard orthographies consistently represents the position of primary stress on words; see the examples in (3); only Italian regularly marks final stress in contrast to penultimate, antepenultimate, or (very rare) pre-antepenultimate stress.¹

(3) Minimal Stress Pairs in Russian, Italian, and English²

- a. Russian (cf. further examples in Melvold 1989: 12)
 - i. *pláču* [ˈplatʃʊ] ‘weep:PRES.1SG’; *mýka* [ˈmukə] ‘torture, pain’
 - ii. *plačú* [pləˈtʃu] ‘pay:PRES.1SG’; *myká* [mɔˈka] ‘flour’
- b. Italian
 - i. *príncipi* [ˈprintʃipi] ‘princes:PL’
 - ii. *princìpi* [prinˈtʃi:pi] ‘principles:PL’
- c. English
 - i. *ímport* [ˈɪmpɔɪt]
 - ii. *impórt* [ɪmˈpɔɪt]

¹Allen (1973: 88) nicely observes that, in these and languages with similarly potentially contrastive stress, a functional account for such potentially contrastive stress is not to be sought: the number of minimal pairs distinguished exclusively by the position of the stress is typically not large (Allen speaks of “at most a few dozen pairs”), for which reason the functional load of contrastive stress in a given language is comparatively low. The English noun–verb pairs of the *ímport*~*impórt* type perhaps constitute an exception, quantitatively speaking: Sondregger and Niyogi (2013) indicate that 1143 pairs of orthographically identical noun–verb forms occur in the CELEX corpus (Baayen et al. 1995), though here, only some proportion of the total exhibit distinct stress patterns, while others (e.g., *fracture* [ˈfɹæktʃɹ]; *review* [ˈiːvju]) are identical in the noun and verb.

²Note that the use of acute and grave in these examples is not employed in the standard orthographies of these languages (either at all, or for the relevant lexemes, in the case of Italian). I have added these diacritics for the reader to signal the position of the primary stress.

Were a linguist interested in describing the stress systems specifically and prosody more generally of these languages based exclusively on written documents employing the standard orthography, they would find themselves in a difficult situation. To a some extent, processes of vowel alternation and deletion in English or Russian, insofar as orthographically represented, could be used to diagnose the position of primary stress in some lexical items, but because stress in these languages is inherently unpredictable on the basis of segmental structure, syllable count, and syllable weight alone, any description of prosodic phenomena in these languages on the basis of written records alone would at best be approximative, and surely be beset with uncertainties.

In the case of a language possessing a predictable, “fixed” metrical stress, such as Latin spoken in the 1st Millennium BCE (cf. discussion in 2.2.7.1 and 3.3.1.1), that the written representation of the language offers little direct information on the position of word-level prosodic prominence is not perhaps terribly surprising: insofar as the position of stress is predictable, a speaker of the language might find the explicit indication of stress unnecessary, perhaps even wholly redundant.³ At the same, for the precise reason that orthographies often act more like an *aide-mémoire* to competent speakers, information on phonological contrasts is simply not always integrated; speakers of Latin did not, for instance, regularly distinguish long and short vowels in writing, and diacritic marks for position of “accents” in Ancient Greek texts only appear as knowledge of the accent system of the classical (and earlier) language declined among native speakers, and even then were first used systematically in cases of minimal pairs, for the purpose of disambiguation (cf. Allen 1973: 4–5, Allen 1968 [1987]: 125).

The important point at hand is that grammarians and orthoepists of some of the earliest and most well-attested Indo-European languages saw fit to describe prosodic properties of their languages and even to systematically outfit texts with diacritic marks. From the composers of the *prāṭisākhya*s (manuals of phonetics to accompany the Vedas; see Whitney 1869 for summary and synthesis of their descriptions of Vedic prosody), to the Greek grammarians (working at primarily at Alexandria; cf. Allen 1968 [1987]: 124–6, Probert 2006b: 21–45), whose work in turn influenced Latin grammarians such as Quintilian (see now Probert 2019 on doctrines relating to accent), their work forms the basis and provides the crucial data for the analysis of prosodic phonology in these languages. Precisely because these languages happen to be relatively closely related and at the same time gave rise to a numerous modern languages (from Marathi to Portuguese), they enable the examination of prosodic systems and prosodic change from on a vast diachronic scale: the prosody of a language like Ancient Attic-Ionic Greek may be analyzed synchronically on its own terms, feed the reconstruction of prosodic phonology in the last common ancestor of the Indo-European languages, and be contrasted with the prosody of Modern Greek. One can thus observe a series of of prosodic changes spanning millennia, from (reconstructable) Proto-Indo-European and (reconstructable) Proto-Greek, to the attested Attic-Ionic and the related Greek dialects of the

³Already worth noting here is that the precise rule for the assignment of primary stress in Latin almost certainly underwent changes during the course of the 1st Millennium BCE, becoming the well-known “penultimate rule” probably only in the last two centuries BCE or so. See further Allen 1978: 83; 91 fn. 1 and the recent treatment of stress assignment and foot construction during different periods of Latin in Sen 2012.

1st Millennium BCE, through to the present-day Greek language.

That languages undergo change over time at all levels and in all domains of the grammar is a truism. In the domain of prosody, however, and in the domain of stress assignment in particular, the motivations underlying change are perhaps less obvious, especially perhaps in comparison to phonological change at the segmental level, even if accentual “laws” are not novel in historical–comparative Indo-European linguistics (see, e.g., the entries on Wheeler’s Law, Hirt’s Law I, or Dybo’s Law in Collinge 1985). While the impact of hypo- and hyperarticulation as phonetic bases for segmental sound changes are reasonably well understood (cf. Ohala 1981, Ohala 1993b, Ohala 2003), and likewise have evident effects on tone and intonation diachronically (see, e.g., Kingston 2005 on the diachrony of tone in Athabaskan languages), a phonetically grounded pathway from a tangible acoustic phenomenon to potential reanalysis and novel phonological interpretation is less clear, or perhaps altogether absent, when dealing with “abstract” phonological units, i.e., syllables, feet, and “accent” or “stress”. This point will be discussed at greater length in Chapter 3.3. Precisely because of their relative intangibility – syllables, feet, and “accent” all famously lack clear phonetic correlates or may have quite different correlates in different languages – the very existence of such “things” as genuine categories that shape the phonological knowledge of speakers has been doubted (see further discussion in this chapter). Insofar as one accepts the psychological reality of such units, or admits to their necessity in phonological analysis, one confronts particular difficulties in attempting to understand changes affecting prosody: what leads a learner to make a different analytical choice about the rules of their phonological grammar? Or to put it more concretely: how and why did speakers of Classical Sanskrit so profoundly alter the patterns of stress assignment inherited from Vedic Sanskrit (cf. examples (1) and (2) above).

This introductory chapter now devotes itself to an overview of the study of prosodic phenomena in language, beginning with a treatment of the history of the term “prosody” in 1.1, before introducing the notions of prosodic constituents, prosodic structure, and controversies surrounding the elements that comprise the prosodic hierarchy in 1.2. Once the object of investigation is narrowed more specifically to “word-level prosodic prominence”, I address continuing difficulties concerning the categorization of phenomena and usage of terms applied word-level prosody, seeking to establish a uniform set of terms to be used throughout this work, while simultaneously presenting a short survey of prosodic typology in Section 1.3. I then in Chapter 2 introduce the primary theoretical frameworks that will undergird the research presented here (namely, generative approaches to the construction of metrical structure, implemented in Optimality Theory and related constraint-based formalisms).

1.1 Historical and Philological Background: προσωδία

Since a not insubstantial portion of the present work dedicates itself to the treatment of prosody in Ancient Greek, it seems appropriate to briefly sketch the history of the term προσωδία [prosoːdíːa], its possible interpretations, and current usages in linguistic research. I will then proceed to further delimit the territory for this object of study in this work, so as

to make clear precisely the domain of application generally intended by the term *προσῳδίᾱ* [prosɔːdíɑː] within these pages. For more on the historical background outlined here, see in particular Allen 1973: 3–16.

Many of the earliest attested usages of the word *προσῳδίᾱ*, related terms, and discussion of their meaning and application, are to be found in the Aristotelian corpus (11×). An explicit early definition of the lexeme is found in a fragment attributed to the Pre-Socratic philosopher Critias (Diels and Kranz 1952: Fragment 57:375–99): τὰς πρὸς κιθάραν ᾠδὰς προσῳδίας ἀρέσκει καλεῖν ‘it is suitable to call the *προσῳδίας* songs [ᾠδίας] proceeding from the *κιθάρα* (a kind of lyre).’⁴ Other later definitions (e.g., in Hesychius 3946; Latte 1953–1966) agree in considering “prosodies” to be songs with instrumental accompaniment. The form (acc.pl.) *προσῳδίας* itself occurs once in the Platonic corpus (*Republic* 399a8), but appears extensively in the Aristotelian corpus and thereafter appears as a common term in works of or related to the Alexandrian grammarians. Indeed, the attested usage in Greek suggests a term which had early become a grammatical term, since after Aristotle it is employed almost exclusively in grammatical treatises. As Allen (1973: 3–4) indicates, the original denotation of the term was restricted to the “accent” of Greek words, and later extended to refer to vowel length and aspiration in the language.

Etymologically, the connection of *προσῳδίᾱ* with ᾠδή [ɔːdéː] ‘song’ (borrowed into English as *ode*), in turn a derivative of the verb αἰδῶ [aeíðɔː] ‘sing’ (earlier form αἰοδῆ [aoidéː], without vowel contraction, mostly attested in Homer), is transparent, and indeed then *προσῳδίᾱ* is correctly to be regarded as a type of song or singing. Literally, it is a ‘singing’ (ᾠδία [ɔːdíɑː]) ‘against’ (πρὸς [prós] < PIE *[próti]; cf. Skt. *práti* ‘to, against’). This etymological transparency also permitted a literal translation of the term into Latin as *accentus* (< *ad* ‘to, towards’ + *canere* ‘to sing’, in the form of a past passive participle); cf. Probert 2003: 3 and Allen 1968 [1987]: 116.

Given that *προσῳδίᾱ* in origin concerns singing (along to something), *prosody* can intuitively be understood as comprising a component of both duration (including organized rhythm) and pitch changes. In its application to the study of language, what the *prosody* is “sung against” are the phonological segments (whether viewed as decomposable into features or not). To draw a very gross division, the study of phonological components and processes can be divided into *segmental phonology* and *prosodic phonology*, where the prosodic phonology may be viewed as encompassing all phonology that operates “above and beyond” the mere segmental level.⁵ If the parallel to music is strictly construed, then prosody in language should likewise concern itself with three aspects: *pitch* (intonation, change), *intensity* (volume), and the (relative) *duration* of units (which in absolute terms will be affected by *tempo*, and which may be organized into repeating units of *rhythm*). In the most naive sense, human speech takes place in a linear time course, and segments and groups or sequences of

⁴One attested occurrence which is likely earlier (probably early 5. c. BCE) is in a fragment of tragedy attributed to Aeschylus; see Radt 1985: Fragment 299.

⁵See, however, the anecdotal accounts summarized in Beckman 1986: 28–9 of the perceptions of native speakers of Burmese and Mandarin that the tone borne by a syllabic segment is inextricable from the syllabic segment itself. In this sense, the strict separation between “segmental” and “suprasegmental” phonology may not be (psycholinguistically, at least) valid for all languages.

segments may exhibit and avail themselves of differences in duration, pitch (frequency) and intensity (amplitude), as even the most superficial observation of a phonetic waveform will show.

From the Classical period onward, knowledge concerning accentuation, vowel length, syllable weight, and “breathings” (i.e., the occurrence of word-initial [h] or delayed VOT in vowel-initial words) in the Greek language must have been gradually accumulated and worked out by grammarians, especially as the Greek language began to undergo changes in prosody (see further 6.4) that led to differences with respect to the original prosody known for certain works (especially Homer). Much of this knowledge was gathered together in the treatise of Aelius Herodian (2nd c. CE), *Περὶ καθολικῆς προσωδίας* ‘On prosody in general’, which only survives to the present day in fragments and summaries of later grammarians (Dickey 2007: 75). For later investigations into questions of prosody, both general and for particularities of Classical and Pre-Classical Greek, Herodian’s work “was extremely influential, to the extent that all later grammarians who treated of accentuation depended on Herodian, whether directly or indirectly (so Vendryes 1904:11)” (Probert 2006b: 22).

Via the Greek grammarians, the terms *naturā* ‘by nature’ and *positu* (or *positione*) ‘by position’ for describing vowel length and syllable weight entered the Latin language alongside *accentus*, calquing the Greek φύσει [p^húsei] and θέσει [t^hései] (where the latter may be understood as ‘by convention’, at least from the work Τέχνη γραμματική ‘Grammatical Technique’ of Dionysius Thrax); see further Allen 1978: 91–2. From the Latin tradition into the early modern period, scholarship could at least distinguish between some type of “prominence” or “pitch change” as indicated by *accentus*, and between syllable and/or vowel weight or length with the terms *naturā* and *positu*.⁶ Acquaintance with the Sanskrit grammatical tradition in the area of prosody (*chandas*,⁷ more specifically focused on the form of poetic meter) in Europe in the 19th Century allowed for an understanding of a prosodic distinction between short and long (Skt. *hrasva*- and *dīrgha*-) vowels versus heavy and light (*laghu*- and *guru*-) syllables.⁸ The basic categories of accented/unaccented, long/short, and heavy/light, plus intuitive notions of the SYLLABLE and the WORD, thus formed the groundwork for treatments of prosody in the 19th Century and later theoretical investigations into the area in the 20th Century. Representative investigations into aspects of prosodic structure and suprasegmental phonology of the Indo-European languages in the period ~ 1880–1940 include Hirt 1895 (on “accent”) Hermann (1923) (on syllabification).

Likewise inherited from the Greek and Sanskrit grammatical traditions is terminology used in the analysis of rhythmic structure as applied to poetic meter. Greek and Sanskrit

⁶As Allen (1978: 91) further notes, φύσει/*naturā* and θέσει/*positu* were insufficient for the purposes of distinguishing length as applied to vowels and weight as applied to syllables, sometimes leading to confused notions about length along these two different dimensions. Scholarship on Greek and Latin lacking familiarity with better linguistic terminology still sometimes applies the terms “long” and “short” to syllables.

⁷Note that the term *chandas* in the Indic grammatical tradition resting on Pāṇini can also denote the language of Vedic Sanskrit as opposed to “standard” or “classical” Sanskrit (*bhāṣā*). See Scharfe 1977: 106 on the distinction between *bhāṣā* and *chandas*.

⁸See now also Ryan 2019: 1–2 on the weight categories distinguished by Sanskrit grammarians and their conception of weight as interval-based rather than syllable-based (on which see further 1.2.4 and 1.2.5 below).

employ the etymologically related terms ποῦς [pó:s] and *páda*- both ‘foot’, designating, however, different units: the Greek ποῦς refers to a small rhythmic unit, such as a trochee, iamb, or dactyl, whereas Sanskrit *páda*- refers to an entire line.⁹ The functional equivalent of the Greek ποῦς is instead Sanskrit *gāṇa*- ‘group’. As with the term *accentus*, Greek ποῦς was literally taken over into Latin as *pēs* ‘foot’, whence then further literal translation into English *foot* to designate small repeating units internal to a metrical line. The extension of *foot* to refer to a constituent in a prosodic hierarchy above the level of the syllable and below the level of the prosodic word seems already established in Liberman 1975, who employs the term without much further comment.¹⁰

Much of the terminology relating to “prosody” and, in large part, conceptions of its scope thus reflect a relatively direct line of inheritance from Greek grammarians, with some admixture from the Sanskrit grammatical tradition. A crucial turn in the domain of “prosody” must, however, have occurred in the early modern period, as observed by Allen (1973: 5): at least in English, ‘prosody’ from the 15th century onward can denote phenomena above the level of the word, incorporating thereby notions such as phrasal stress and intonation. As presently understood, the study of prosody or prosodic phenomena may include any and all phonological constituents and phenomena operating above the level of the segment. The range of phenomena discussed in Nespor and Vogel 1986, Fox 2000 (the title of which explicitly equates “prosody” and “suprasegmental phonology”), and Féry 2017 may serve as a relevant guide to the research questions now conventionally placed within the scope of prosody:

- the duration of phonological segments, whether inherent or derived.
- the construction of hierarchical constituent structures above the level of the segment, including (but not necessarily limited to) the syllable, prosodic word, phonological phrase, and utterance.
- the assignment of relations of relative strength or prominence between smaller prosodic constituents falling within the domain of a common larger constituent (“stress” or “accent”).
- the usage of pitch in the phonology of a language, whether as a pre-associated lexical property (“tone”) or a melodic contour to be associated and aligned with elements of the prosodic structure (“intonation”).

⁹The Greek ποῦς [pó:s] is a root noun, PIE */pod-/ , while the Sanskrit *páda*- is a thematic derivative to this root, a virtual */pod-o-/. The root noun *pád-* is, of course, present in Sanskrit, but generally refers only to the body part itself, with few attested metaphorical extensions. In the *Rgveda*, the instances of *páda*- attested are equivalent to *pad-* and simply mean ‘foot’ (cf. *dvi-pád-* ‘two-footed’ vs. *pāñca-pāda-* ‘five-footed’).

¹⁰Liberman (1975: 211) writes: “We will describe these conditions [viz. for marking a “strong” element in a prosodic constituent within a word] as rules inserting a “foot boundary”, symbolized |, into the string of segments. The meaning of “foot boundaries” is that they divide the string into groups which the metrical bracketing cannot violate.” Since Liberman does not explicitly cite a source for this terminological usage, I can only infer that “foot” in the sense of “sub-word level prosodic constituent” had some currency in the already early 1970’s. It may be the case that the method of designating a hierarchy of strong and weak elements in lines of poetic meter in correspondence to tradition metrical feet was simply taken over by Liberman and others at the time without much further thought; cf. Liberman 1975: 46, where “feet” in the traditional sense are mentioned.

With a general conception of prosody now in hand, I turn in the following two sections first to the matter of abstract prosodic structure in the phonology (Section 1.2) and then to the perceptible usage of suprasegmental phenomena – i.e., intonation, tone, and stress – and how those phenomena interact in the development of a general prosodic typology (Section 1.3). While the latter suprasegmental phenomena often provide crucial evidence concerning the more abstract phonological entities to be discussed in the following section, those abstract entities also in part govern the occurrence of intonation and stress in particular, while at the same time finding support in segmental phenomena and independently generated syntactic structure. For this reason, I begin with a treatment of the prosodic hierarchy, stopping to then consider what entities and phenomena this book principally aims to investigate along the way (Section 1.4).

1.2 Prosodic Structure and Prosodic Hierarchies

Investigations into prosodic structure are principally concerned with providing answers and insights into the following questions:

- what types of units and domains that are relevant to the phonology exist above the level of the segment (e.g., word, phrase)?
- on what basis are such units and domains constructed (e.g., by direct reference to syntactic structure)?
- what effects can the units of prosodic structure exercise on the phonology, e.g., what processes may they condition?

The fundamental empirical difficulty posed by the investigation into prosodic structure, i.e., the suprasegmental phonological organization of speech, is that its units are, for the most part, in themselves imperceptible and must therefore be examined indirectly through the effects that they exercise. Such effects may potentially be visible in the intonational phonology (alignment of pitch tunes), metrical phonology (stress assignment), or the application or non-application, respectively, of some segmental process. Historically, prosodic structure and its relation to morphological and syntactic structure has been approached through reference to “edges” or “boundaries” in the Prague School (*Grenzen* in Trubetskoy 1939 [1989]), “juncture” in Structuralism and Descriptivism, boundary symbols in the Generative phonology of the 1960’s and 1970’s, the Prosodic Hierarchy and its domains as codified by Nespor and Vogel (1986), and most recently, Match Theory (Selkirk 2009, Ito and Mester 2012, Ito and Mester 2020). I will briefly survey the distinctions between the theory of boundary symbols, the Prosodic Hierarchy, and Match Theory as concerns especially units above the level of the prosodic word before giving more detailed attention to constituents below the level of the word, namely, the mora, syllable, and foot.

The species of phonological analysis exemplified by Chomsky and Halle (1968) (*The Sound Pattern of English* = SPE) takes an essentially horizontal or linear view of phonological constituent structure, and, insofar as constituents are present in or relevant to the phonology at

all, they are assumed to derive directly from the morphology or syntax. Namely, constituents of morphology or syntax produce intangible boundary or juncture that are legible by the phonological component, as illustrated in the rules in (4). These junctures take the form of boundary symbols in the phonology: + designates a morpheme boundary, while # marks a word boundary, with the latter being potentially iterable in representation of “stronger” boundaries, where different syntactic relations hold between the words in question. In this regard, the use of boundary symbols in SPE, its forerunners, and subsequent research in the same framework strongly parallels conditions on (non)-application of sandhi rules in Pāṇini (cf. generally Allen 1962 [1972], or Kiparsky Forthcoming: §2.5 for reinterpretation in terms of prosodic domains). See Aronoff 1980 for a survey of juncture in American Descriptivism as a precedent for boundaries in SPE, and Selkirk 1980a: 107–8 or Nespor and Vogel 1986: 1–5 for a brief overview of the underlying theory and application of boundaries in SPE.

(4) Morphological and Syntactic Juncture in SPE

a. HIATUS RESOLUTION in Japanese (+)

[+syllabic] → ∅ / [+syllabic]+__

/tabe-anai/ ‘eat-NEG’ → [tabenai]

(cf. /kae-ru/ ‘change-PRES’ → [kaeru], not ^x[karu])

b. OBSTRUENT VOICING in Sanskrit (#; cf. Allen 1962 [1972]: 91, Selkirk 1980a: 115)

[–sonorant] → [+voice] / __#[+voice]

/gamat vājebhis/ → *gamad vājebhis* ‘may go with the prizes’

(cf. /gamat sas/ → *gamat sas*)

c. *visarga*-FORMATION in Sanskrit (/ / = utterance boundary, before a pause; cf. Macdonell 1916 [1993]: 34)¹¹

$\left\{ \begin{matrix} s \\ r \end{matrix} \right\}^{12} \rightarrow [h] / _ //$

/ déva-s/ / → *dévaḥ* ‘god-NOM.SG’

¹¹In SPE, only the boundary symbols “+” and “#” are actually introduced, where the latter is taken to be inserted “at the beginning and end of every string dominated by a major category, i.e., by one of the lexical categories...or by a category...which dominates a lexical category” (SPE: 366), with the result that an utterance should always terminate in a sequence of at least three “#” symbols, since, under the model of syntax presumed by SPE, every utterance will minimally have the structure [[[#]_X #]_{XP} #]_S. The symbol “//” is employed by Selkirk (1980a), and could be understood in the context of SPE as the boundary symbol inserted at the beginning and end of the category “sentence”.

¹²Only the dental sibilant and the (likely retroflex) rhotic of Sanskrit are subject to *visarga*-FORMATION.

Compare Ryan 2017: 302 on the consonantal inventory of Sanskrit. The natural class $\left[\begin{matrix} +\text{continuant} \\ +\text{consonantal} \\ +\text{coronal} \\ -\text{dorsal} \\ -\text{lateral} \end{matrix} \right]$ would

be appealing, but also captures the retroflex sibilant /ʂ/, which is realized as a voiceless retroflex stop [ɭ] before a pause (e.g., *ṣát* ‘six’ for /ṣáʂ/, not ^xṣáh).

Selkirk 1980a is of note for advancing the explicit claim that “the theory of prosodic domains is preferable to a theory of boundary-defined domains” (pg. 110), i.e., a model under which processes may occur within a particular domain, across the juncture between two particular domains, or at the edge of some specified domain. These conditions correspond to Selkirk’s *domain span*, *domain juncture*, and *domain limit* rules, respectively (cf. also Nespor and Vogel 1986: 15). Moreover, the domains need not be isomorphic to morphological or syntactic constituents; Nespor and Vogel (1986: 5) offer the empirical objection to the SPE theory of boundary symbols that it cannot correctly account for the application of LINKING-*r* in Received Pronunciation British English (see (5)); instead, they contend, the phonological component of the grammar must construct domains consisting of hierarchically layered constituents independently of the syntax.¹³

(5) LINKING-*r* in Received Pronunciation British English

a. Non-application at an utterance boundary:

I won’t bother. [aɪwʊntbʊðə]_{Utterance}

b. Application across large syntactic constituents within the same utterance:

I won’t bother; I’m too busy. [aɪwʊntbʊðə.ɪaɪmt^hubɪzɪ]_{Utterance}

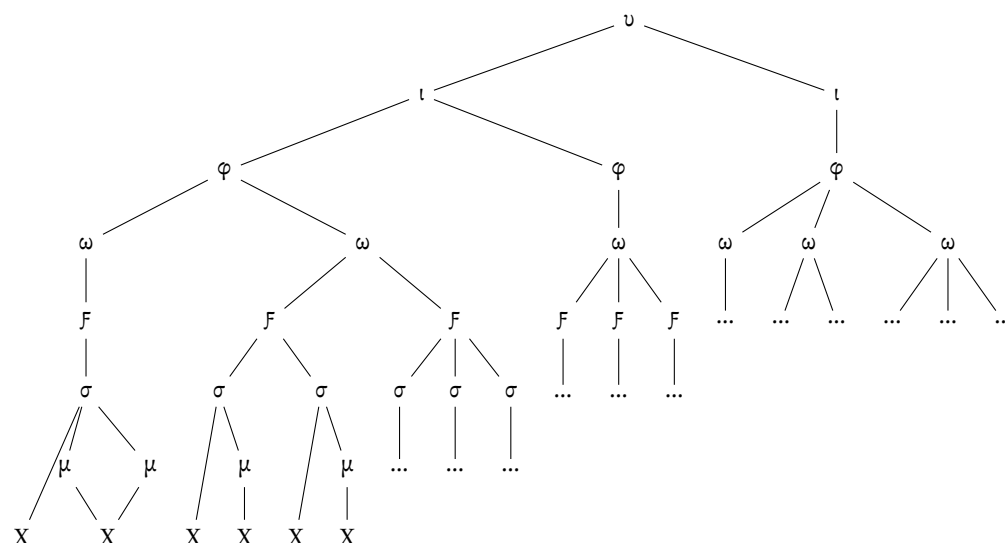
Under the model proposed in Nespor and Vogel 1986, a PROSODIC HIERARCHY (PH) built up of seven units between the syllable (σ) and utterance (υ) was proposed, to which the mora (μ) is frequently added between the syllable and segment terminals (see further Section 1.2.5) below).¹⁴ This extended prosodic hierarchy is illustrated with unary-, binary-, and ternary-branching constituents in 6 to indicate how and where a higher level constituents may incorporate multiple lower level constituents; the labels are defined in 8.¹⁵ On the prosodic hierarchy in this or a very similar form, see Nespor and Vogel 1986: 16, Fox 2000: 338–9, Gussenhoven 2004: 123–6, Selkirk 2011: 437, Byrd 2015: 51, Goldstein 2015: 44–6, or Féry 2017: 36.

¹³Notice in the prosodic hierarchy discussed below that the morpheme boundary “+” of SPE finds no approximate correspondent. Processes supposedly sensitive to a morpheme boundary might, at least in some case, be reframed by reference to the distinction between stem- and word-level processes in Lexical Phonology or Stratal Optimality Theory (Kiparsky 1982b, Kiparsky 2010b).

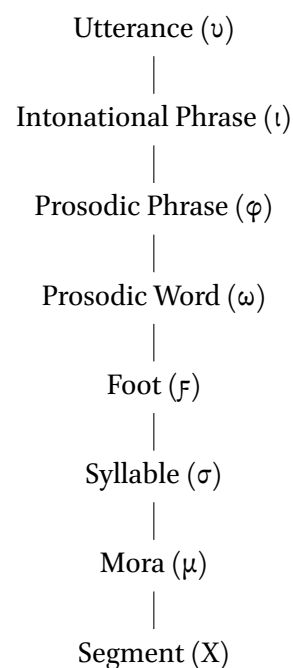
¹⁴Nespor and Vogel (1986: 25) state that morae were left out of consideration for the reason that morae do not ever seem to function as a domain for or boundary on the application of phonological processes.

¹⁵Nespor and Vogel (1986: 7–9) hold that prosodic constituents are *n*-ary branching, i.e., they may dominate theoretically infinitely many constituents of the immediately lower level. This view contrasts with an assumption of strictly unary- or binary branching constituents, commonly assumed especially for the domain of the foot (cf. the universal foot inventory proposed in Hayes 1995).

(6) Constituents and Construction of the PROSODIC HIERARCHY^{16, 17}



(7) Schematic PROSODIC HIERARCHY



¹⁶For more detail on the formal properties of tree structures employed in prosodic representations, see Ito and Mester 2009: 136–8, with reference to Wall 1972: 144–52 concerning tree structures generally.

¹⁷The major prosodic constituent not included here, but present in the version of the hierarchy given in Nespor and Vogel 1986: 16, is the so-called *clitic group* (see the treatment in Nespor and Vogel 1986: 145–64), which was posited as an intermediate constituent between the phonological word and phonological phrase. As the name suggests, the clitic group was originally posited to provide a prosodic constituent able to combine a host capable of forming an independent prosodic word with a prosodically weak clitic below the level of the phonological phrase. The admission of recursion of prosodic words into well-formed prosodic structures renders the clitic group superfluous, and most current representations of the prosodic hierarchy thus exclude it. See Anderson 2005: Ch. 3 or Anderson 2011 on clitics and the prosodic hierarchy.

(8) Definitions of Constituents in the PROSODIC HIERARCHY (cf. Féry 2017: 36)¹⁸

Abbreviation	Prosodic Constituent	Definition (~ Syntactic Equivalent)
υ (U)	Utterance	Complete expression without pause, including multiple clauses
ι (I)	Intonational Phrase	Clause (multiple XPs)
φ (Φ)	Phonological/Prosodic Phrase	Phrase (XP)
ω	Prosodic Word	Grammatical Word (X)
ƒ (φ, F, f, Σ)	Foot	Metrical Unit (e.g., trochee)
σ (Σ)	Syllable	Segmental string (e.g., CVC)
μ	Mora	Timing unit / syllable weight
X/C/V	Segment	Concrete syllable nucleus (V) or margin (C)

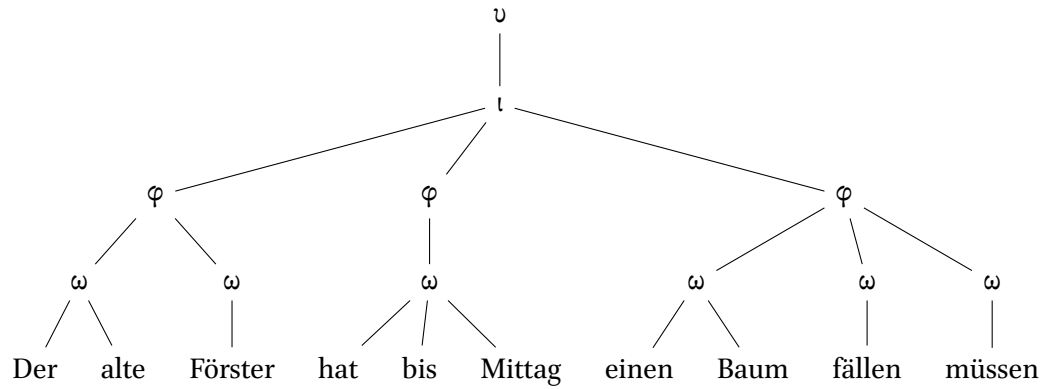
As a concrete example, consider the following sentence of German. Glossing is provided in 9a., while 9b. shows the prosodic constituency above the level of the word, and finally 9b. provides the structures at the word level and below. For the sake of illustration, I assume that function words in this example, including disyllabic ones and sequences of multiple function words, are clitics parsed into prosodic words with neighboring content words, though variation may occur based on e.g., speech style. See further Ito and Mester 2009 for specific arguments in favor of prosodic-word adjunction in German.

(9) Prosodic Structure in German

- a. Der alte Förster hat bis Mittag einen Baum fällen
the.NOM.M old forester.M have.PRES.3SG until noon a.ACC.M tree.M fell.INF
müssen.
must.PTCP.PST
‘The old forester had to (= *hat müssen*) fell a tree by noon.’

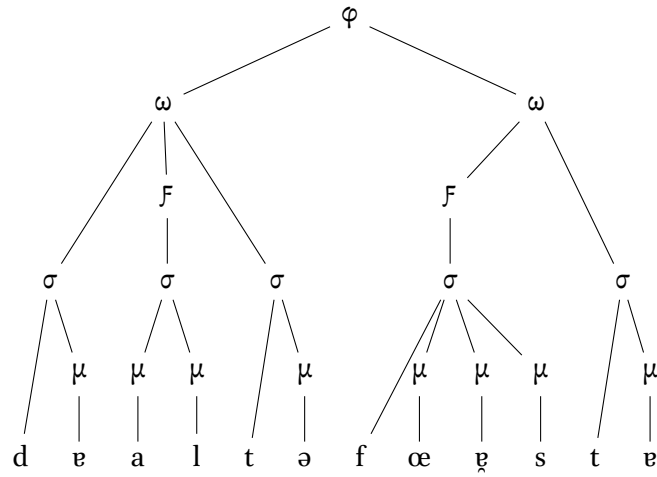
¹⁸Practice in the abbreviation conventions for the prosodic hierarchy varies; almost universally employed are σ (syllable) and ω (prosodic word), though for the form one sometimes encounters the uppercase sigma, Σ. The primary symbol shown here will be used throughout this work; the alternative in parentheses is also encountered reasonably often. The labels here follow Féry 2017, with the exception of the foot, for which I introduce the sign lowercase digamma (ƒ), in order to consistently apply lowercase Greek letters for all constituents of the prosodic hierarchy (note that the Greek letter digamma, while not part of the familiar Ionic alphabet, is the source for Latin ⟨F⟩). Compare further Selkirk 2011 or Ito and Mester 2013: 26 on the correspondences between prosodic categories and syntactic categories.

b. Prosodic Structure at ω and above

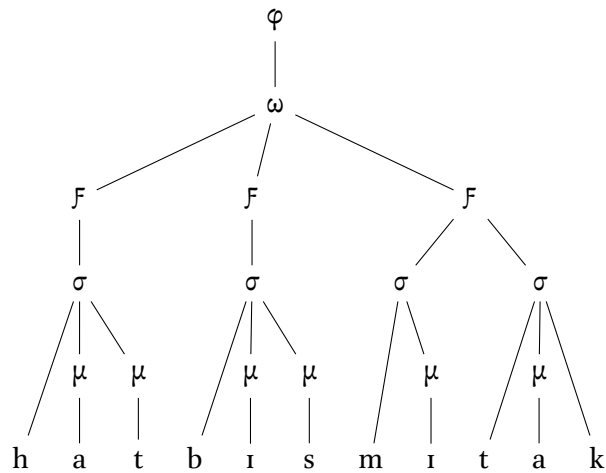


c. Prosodic Structure from ω and below, by φ ¹⁹

i. [Der alte Förster] _{φ}

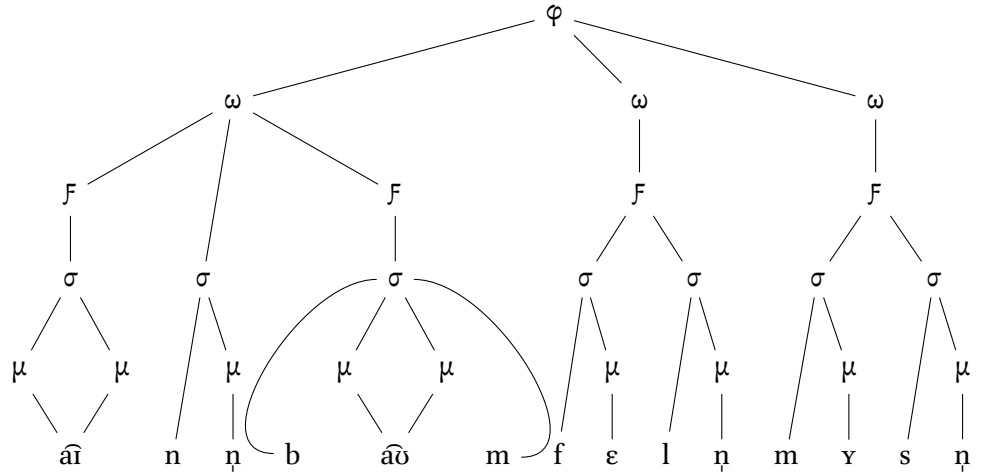


ii. [hat bis Mittag] _{φ}



¹⁹I assume non-exhaustive parsing of prosodic words into feet, constructing strictly bimoraic feet, and assume the extrametricality of word-final consonants. See further Alber 1998 for an analysis of foot construction in German.

iii. [einen Baum fällen müssen]_φ



Notice, in example (9), that the boundaries of prosodic (phonological) phrases do not align perfectly with the boundaries of syntactic phrases: not every ϕ constituent perfectly coextensive with a syntactic XP. In particular, the second prosodic phrase, [hat bis Mittag]_φ is made up of both material from a complete prepositional phrase (*bis Mittag*) and an auxiliary verb (*hat*), which does not build a syntactic constituent with the PP to the exclusion of the other material to its right (*einen Baum fällen müssen*).²⁰ Compare similarly the discussion of *Gorgia Toscana* in Nespor and Vogel 1986: 42–4: the domain of application for the lenition of stops /p, t, k/ to corresponding fricatives in Tuscan Italian is not isomorphic with the domain syntactic XPs. The Prosodic Hierarchy, in comparison with the theory of boundary symbols in SPE, posits a less direct relationship between phonological and syntactic structure (though note that boundary symbols involve a mediated relationship between syntax and phonology – the latter cannot directly see the structure of the former), which in turn contrast with literal direct reference to labeled syntactic constituents (as assumed in, e.g., Liberman and Prince 1977 or argued for in Kaisse 1985).²¹

According to the conception of the Prosodic Hierarchy advanced by Nespor & Vogel, well-formed prosodic structures are subject to a set of four specific design principles (laid out in Nespor and Vogel 1986: 7), the first two of which are frequently summarized under the label of the STRICT LAYER HYPOTHESIS (though this term originates with Selkirk 1984: 26; cf. also Inkelas 1990: §2.2, Ito and Mester 2009: 146, or Selkirk 2011: 437). Nespor & Vogel's Principle 1 and 2 are reproduced here, followed by the version of the Strict Layer Hypothesis

²⁰Specifically, the auxiliary *hat* can be analyzed as having undergone head adjunction with T, which in turn has undergone head adjunction with C, thereby producing the V2 phenomenon in a main clause. See generally Vikner 1995 or Sternefeld 2006 on syntactic structures in German.

²¹The contrast between hypotheses of “direct reference” and “indirect reference” of the syntax–phonology interface drawn by, e.g., Selkirk (2011: 435–6), is somewhat different from the distinction made in Féry 2017: 35, where Féry associates “direct reference” instead with a strictly functionalist conception of intonation (e.g., Xu and Xu 2005) that “assumes a direct relation between acoustic events on the one hand and morphosyntax and semantics on the other, without being mediated by phonology and prosodic domains.” I will not consider further conceptions of prosody of this latter type, and instead use the term “direct reference” in Selkirk's sense.

stated in Selkirk 2011.

- (10) Principles 1 and 2 of Prosodic Structure Construction (Nespor and Vogel 1986: 7)
Principle 1. A given nonterminal unit of the prosodic hierarchy, X^p , is composed of one or more units of the immediately lower category, X^{p-1} .
Principle 2. A unit of a given level of the hierarchy is exhaustively contained in the superordinate unit of which it is a part.
- (11) The STRICT LAYER HYPOTHESIS (Selkirk 2011: 437)
 A constituent of category-level n in the prosodic hierarchy immediately dominates only a (sequence of) constituents at category-level $n-1$ in the hierarchy.

The abstract presentation of the Prosodic Hierarchy shown in (6) adheres to strict layering: every constituent shown there has as its daughter at least one constituent of the next lower category on the hierarchy. The example given in (9), on the other hand, does not, at least below the level of the prosodic phrase (ϕ): (9) shows instances of syllables dominated by prosodic words without inclusion in a foot, and segments directly associated with syllables without an intervening mora. Whether the Strict Layer Hypothesis represents an inviolable set of principles, establishes a set of violable guidelines for ideal prosodic structure, or does not adequately capture conditions on the construction of prosodic structures at all, is discussed further below.

For the moment, worth observing is that the constituents posited for the PH can usefully be invoked to determine the domain of application for a wide array of phonological processes. The two rules given under (4)b. and c. above may be readily reformulated in terms of the PH: Sanskrit OBSTRUENT VOICING applies at word boundaries within the domain of the same utterance, while *visarga*-FORMATION applies at the right edge of an utterance.²² These two rules are restated in (12)b. and c., with a. provided as an example of a domain span rule.

- (12) Prosodic Domain Rules
- a. INTERVOCALIC SPIRANTIZATION in Tuscan Italian (cf. Nespor and Vogel 1986: 209)
- $$\begin{bmatrix} +\text{delayed release} \\ -\text{continuant} \end{bmatrix} \rightarrow [+continuant] / [\dots [-\text{consonantal}] ___ [-\text{consonantal}] \dots]_i$$
- Affricates are realized as corresponding fricatives between $[-\text{consonantal}]$ segments contained within the same intonational phrase.
- /[il mio kritʃeto tʃɛrka il suo tʃibo neʎi angoli del:a gab:ja]_i/
 → [il mio krifeto ʃɛrka il suo ʃibo neʎi aŋgoli del:a gab:ja]_i
- ‘my hamster looks for its food in the corners of the cage’

²²See further discussion in (4.1.3) for reasons to believe that *visarga*-formation (more properly, the creation of [h], *visarjanīya*) in Vedic Sanskrit perhaps applied at the edge of a domain smaller than the utterance, the intonational phrase, depending upon how the mapping between metrical and prosodic constituents is interpreted.

- $$[-\text{sonorant}] \rightarrow [+voice] / [...[... ______]_{\omega} [[+voice]...]_{\omega} ...]_{\nu}$$

c. *visarga*-FORMATION in Sanskrit (cf. 4c. above)

$$\begin{Bmatrix} \mathbf{s} \\ \mathbf{r} \end{Bmatrix} \rightarrow [\mathbf{h}] / ______]_v$$

20

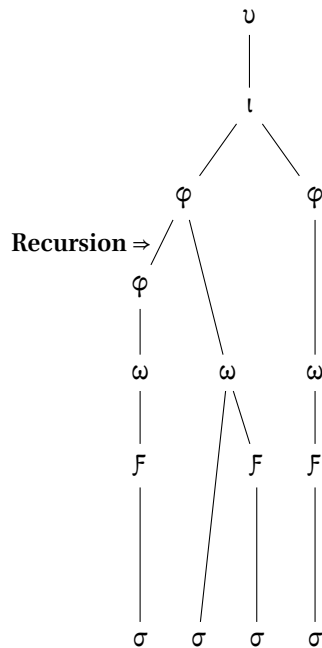
3. For constituents below the word level: Is the assumption of such constituents adequately motivated?

The first two of these questions are considered in the immediately following section and carried on into discussion surrounding the prosodic word under 1.2.2. Debates surrounding the nature and theoretical necessity of sub-word constituents (foot, syllable, and mora) are then each reviewed in turn. This survey then concludes with a summary of theoretical, empirical, and practical motivations behind the adoption of the specific prosodic constituents shown in (6).

1.2.1 Weak Layering and Match Theory

A prosodic structure can potentially deviate from the Strict Layer Hypothesis in one of two ways: it may exhibit recursivity, that is, structures in which a constituent of a given category dominates another instance of the same category (e.g., a φ dominates a φ), or it may exhibit level-skipping, that is, structures in which a constituent of a given category does not dominate an instance of the next immediately lower category in the hierarchy (e.g., a ω dominates a σ); cf. Selkirk 2011: 438. Example (13) illustrates precisely these sorts of structures that are in principle forbidden by the Strict Layer Hypothesis.

(13) Recursion and Level Skipping in a Prosodic Tree



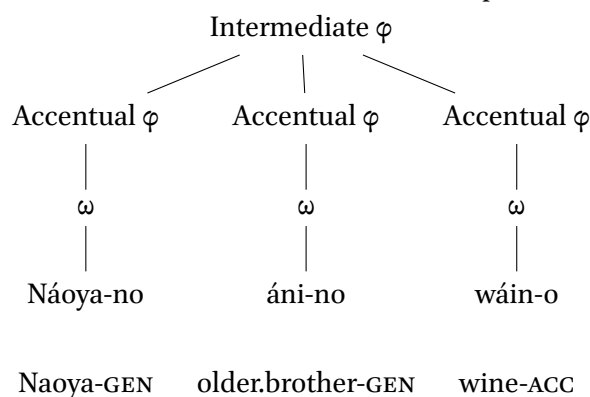
If structures like those in (13) are to be categorically avoided, then the strict layering of prosodic constituents poses certain problems. First, strict layering necessitates the introduction of prosodic constituents for which much empirical support is wanting (e.g, the clitic group, not included in (6) above) or for which the evidence is cross-linguistically inconsistent (e.g, the distinction between a major phrase and minor phrase between ω and

i).²⁴ Furthermore, assumptions of exhaustive parsing of all segments into the constituents below the level of the prosodic word may force the construction of structures that seem at best seem unintuitive (e.g., the domination of weightless onsets by the mora projected from the syllable nucleus) and at worst might lead to incorrect predictions (e.g., the incorporation of all segments into syllables, despite evidence for extrasyllabic segments; cf. Kiparsky 2002 or Vaux and Wolfe 2009).

Concerning the first issue, Ito and Mester (2007: 97) object that prosodic hierarchies become both “overarticulated” and lose cross-linguistic generality when further categories must be posited to fit the data of one language, when empirical evidence for those categories appears, in other languages, to be totally lacking. For instance, Beckman and Pierrehumbert (1986) and Pierrehumbert and Beckman (1988) distinguish an “accentual phrase” (= “minor phrase”) and an “intermediate phrase” (= “major phrase”) between the levels of prosodic word and intonational phrase on the basis of intonational and accentual data in Japanese data (cf. Ito and Mester 2013: 21). Examples for Japanese are given in 14, where “accented” NPs build an accentual phrase, resulting in pitch reset at each accentual phrase boundary contained within the larger intermediate phrase. However, independent substantiation for each further category added to the prosodic hierarchy in any particular language is difficult to establish; indeed, Beckman and Pierrehumbert (1986: 298–305) were, for English, compelled to reanalyze intonational properties originally attributed to the intonational phrase by Pierrehumbert (1980) as belonging to an “intermediate phrase”, in order to salvage the generality of the proposal grounded primarily in the facts of Japanese.

- (14) Intonational Evidence for the “Accentual” and “Intermediate” Phrase in Japanese (data from Ishihara 2015: 572; cf. discussion in Féry 2017: 241–2)²⁵

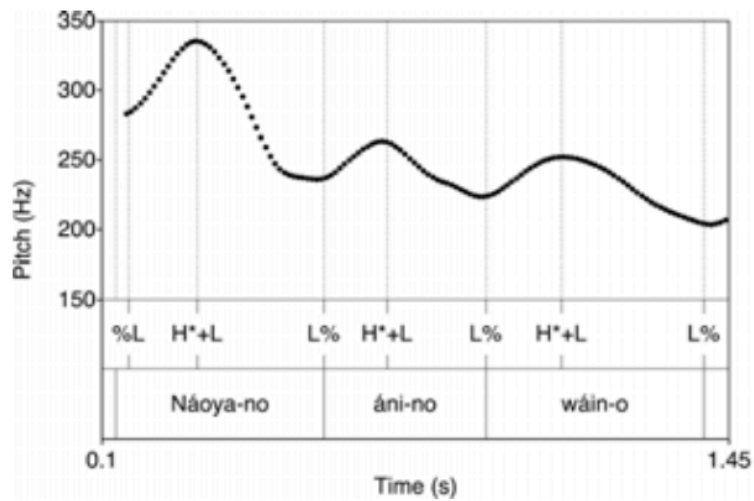
- a. Three accented NPs $\Rightarrow$ three accentual phrases within the intermediate phrase



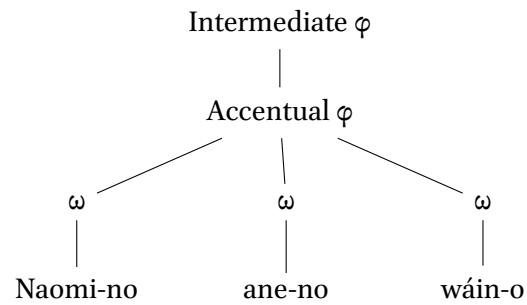
²⁴A proliferation of prosodic categories also renders any possible mapping between syntactic structure and prosodic structure more opaque. Insofar as prosodic constituents are presumed to be truly wholly independent of syntactic constituents, category proliferation does not necessarily represent a concern. Conversely, if syntactic constituency is thought to inform prosodic structure to at least some extent, prosodic categories without any clear syntactic analogue become troubling.

²⁵Intonationally, the accentual phrase is characterized by one lexically specified H*L “pitch accent” and an initial boundary tone %L. The crucial point here is that the lexical accents of /náoya/ and /áni/ supposedly induce the formation of accentual phrases, which leads to the insertion of initial %L boundary tones and the drops in Fo in a. here. See Section 1.3.1 for more details on the grammar of intonational tones.

'Naoya's older brother's wine'

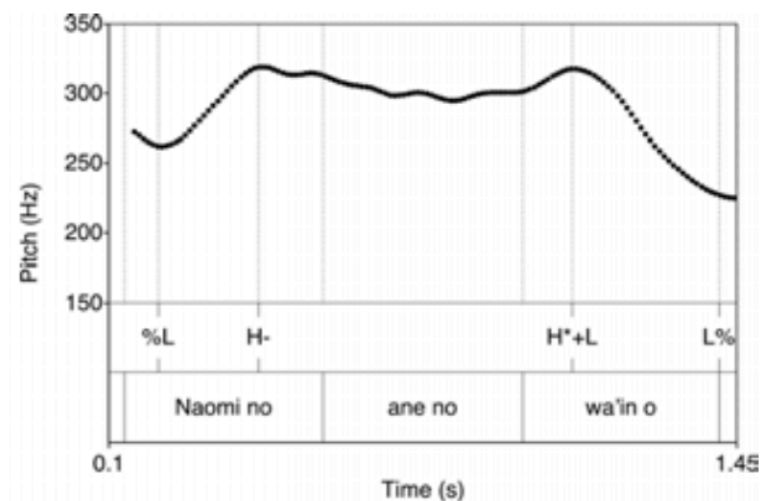


- b. Two unaccented NPs preceding an accented NP $\Rightarrow$ accentual and intermediate phrase are coextensive



Naomi-GEN older.sister-GEN wine-ACC

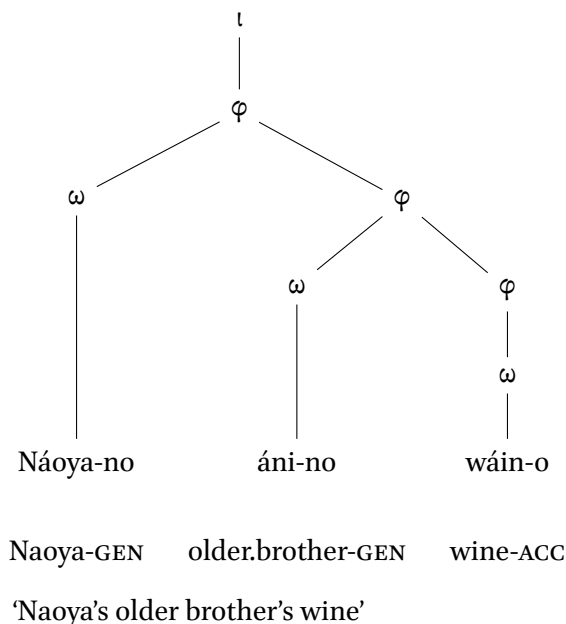
'Naomi's older sister's wine'



If a distinction between, e.g., accentual/minor and intermediate/major phrase cannot be

consistently justified across the prosodic phonology of wide variety of languages, then the hierarchy appears to include too many different categories. Conversely, if those categories were to be excluded, then the hierarchy would contain an insufficiently articulated structure in the cases of those languages where such categories are empirically motivated. The alternative in the Japanese case under (14) would be to allow for recursion of a single prosodic category, here, the phonological phrase, where, in a right-branching syntactic structure like in (14), a minimal phonological phrase projected from the then builds an intermediate and finally maximal phonological phrase dominated by an intonational phrase; precisely this analysis of the Japanese facts is pursued in Ito and Mester 2013 (cf. similarly Kabak and Revithiadou 2009b or Féry 2010 for proposals of prosodic recursion along these lines).

(15) Phonological Phrase Recursion in Japanese (cf. (14)a. above)²⁶



At the same time, even among the generally accepted set of prosodic categories, the exhaustive parsing of certain constituents into all higher-level prosodic categories has the potential to create issues. At the level of the foot, for instance, Ito and Mester (2009: 148) point out that exhaustive parsing in English should be expected to create secondary stress and inhibit vowel reduction in cases where vowel reduction does in fact occur (i.e., *banana* is [ω bə (φ 'nænə)] not χ [ω (φ ,beɪ) (φ 'nænə)]); the alternative would be to assume that degenerate feet in such cases are exempt from stress assignment, which then effectively vitiates a major analytical purpose of feet or introduces a new category distinct from the normal foot (cf. also Pierrehumbert and Beckman 1988: 147–9 concerning *banana*-shaped words of

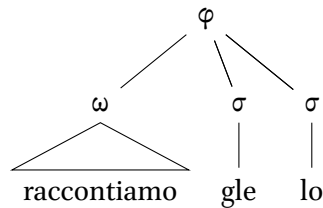
²⁶Here, only the φ directly dominating a ω alone, [wain-o], is minimal, whereas the subsequent two, which dominate another φ , are non-minimal, and the final φ , dominated by the higher-level category ι , is a maximal φ . A recursive structure of this type admits of exactly one minimal and one maximal projection, but theoretically infinitely many intermediate projections dominating the same category but not dominated by the next higher level category.

English).²⁷ In such cases, adjunction of a syllable directly to the prosodic word seems more plausible. At a higher level of prosodic constituency, the interaction of stress assignment with dative and object clitics in Standard Italian, as exemplified in (16), shows that such clitics stand outside the domain of primary stress assignment, which must be taken to be the prosodic word, and which is normally restricted to a three-syllable window at the right edge of the word (cf. 2.4.2.1 on stress assignment in Italian). By extension, the likely cite of incorporation of these clitics into the prosodic hierarchy is the prosodic phrase, which implies level-skipping, insofar as the clitics do not build their own prosodic words (i.e., they are *free clitics* in the sense of Anderson 2005: Ch. 3; see 1.2.2 below for more concerning clitics).

(16) Clitics in Standard Italian (cf. Peperkamp 1997: 177; 193–9)

- a. *raccontiamo* [rak:ɔn'tjamɔ]
tell:1.PL.IMPV
'let's tell'
- b. *raccontiamo=gli* [rak:ɔn'tjamɔli]
tell:1.PL.IMPV=DAT.3SG
'let's tell him'
- c. *raccontiamo=gle=lo* [rak:ɔn'tjamɔləlo]
tell:1.PL.IMPV=DAT.3SG=ACC.3SG
'let's tell it to him'

(17) Prosodic Structure of Standard Italian Clitics



This brief discussion should convey that the Strict Layer Hypothesis often creates serious problems in the analysis of prosodic structure, which called ever more pressingly for satisfying resolution, as the issues of “too much” and “too little” prosodic structure continued to rear their heads. Indeed, not long after the explicit appearance of the Strict Layer Hypothesis, Ladd (1986) argued in favor of the admission of recursive structures on the basis of intonational evidence in English. Likewise, Ito and Mester (2003a) (originally circulated as a draft in 1992) adduced arguments in favor of structures containing level-skipping below the prosodic word. Thus Selkirk (1995: 6–8) observed that the Strict Layer Hypothesis lends

²⁷ Hayes (1995: 86–105) provides thorough discussion on the matter of degenerate feet in relation to stress assignment. Hayes finds that languages may either categorically ban degenerate feet (i.e., fail to parse a [light] syllable into a foot) or allow them in a metrically strong position (i.e., when a monosyllabic/monomoraic word must receive stress). Weaker evidence suggests that a degenerate foot might occur in some languages at a word-edge at the end of a parse (98–101), but in such cases might be repaired by vowel lengthening or any in case, host a secondary stress. That the first syllable of *banána* self-evidently bears no stress excludes the option of treating it as a “normal” degenerate foot.

itself to decomposition into a set of four “constraints on prosodic domination”, reproduced in (18) (cf. similarly Anderson 2005: 47–8).

- (18) Constraints on Prosodic Domination (Selkirk 1995: 6)
- a. LAYEREDNESS
No C^i dominates a C^j , where $j > i$
(= no category dominates a category of a higher level)
 - b. HEADEDNESS
Any C^i must dominate a C^{i-1}
(= every category dominates at least one instance of a category no more than one level below it)²⁸
 - c. EXHAUSTIVITY (cf. *Principle 2* under 10)
No C^i immediately dominates a constituent C^j , where $j < i-1$
(= every element of a constituent C_i consists exhaustively of elements of the next lower category, C_{i-1})
 - d. NON-RECURSIVITY (cf. *Principle 1* under 10)
No C^i dominates C^j , where $j = i$
(= no element of a constituent C_i directly dominates another instances of the same category as C_i)

The first two constraints, LAYEREDNESS and HEADEDNESS, would appear to be universally inviolable, whereas the latter two, EXHAUSTIVITY and NON-RECURSIVITY, defined over specific prosodic categories, could admit of specific types of level-skipping and recursivity, if dominated by other constraints in a constraint ranking that would be violated in case EXHAUSTIVITY or NON-RECURSIVITY were to be satisfied. Allowing recursivity allows resolves the “too little structure” problem that leads to category proliferation; allowing non-exhaustivity in the parsing of constituents resolves the “too much structure” problem that compels the inclusion of unmotivated prosodic structure. EXHAUSTIVITY and NON-RECURSIVITY both admit of formulation as optimality-theoretic constraints (with a version of the former present already in Prince and Smolensky 1993 [2004]), which can be further broken out and specified with reference to particular categories (cf. Ito and Mester 2009: 139).

- (19) PARSE-INTO-X (e.g., PARSE-INTO- σ , PARSE-INTO- ω , etc.)
Assign one violation * for each element of the terminal string that is not parsed at the X-level.²⁹

²⁸This constraint does not apply to the lowest level constituent. Selkirk, for instance, sets the lower boundary at the syllable.

²⁹More commonly seen in the phonological literature after its introduction in Prince and Smolensky 1993 [2004]: 62 is the constraint family PARSE-X.

- (1) PARSE-X (e.g., PARSE- σ , PARSE- ω , etc.)
Assign one violation mark * for each instance of a category X^i that is not parsed into the next immediately higher category X^{i+1} .

Formulated in these terms, a syllable that is directly parsed into a phonological phrase will violate PARSE- σ , but

(20) NORECURSION(-X)

Assign one violation mark * for each instance of a category X that is dominated by an instance that same category X.³⁰

Worth mentioning at this juncture is the observation that violations of exhaustivity and recursion largely occur in separate domains: non-exhaustivity (violations of PARSE-INTO-X) are more common, perhaps even systematic at and *below* the level of the prosodic word, while recursion is encountered much more frequently at and *above* the level of the prosodic word, and perhaps entirely absent for constituents below the word level. This distinction in behavior is perhaps motivated precisely if one accepts that the prosodic constituents at and above the word level are not defined exclusively with reference to the phonology, while the constituents below the word are indeed largely the province of the phonological component. The underlying motivation for recursion in prosodic structure would then lie with the presence of recursion in the syntax.

Placing the responsibility for recursion on the syntax then would strongly suggest that the construction of phonological constituents at and above the word level may be directly due to constituent structure inherited from the syntax. At the same time, the obvious non-isomorphisms visible throughout the survey of Nespor and Vogel (1986) means that phonological influences must be able to disrupt a faithfulness relationship between syntactic constituent structure and prosodic constituent structure; in other words, syntax-prosody isomorphism represents the default case, but can be disrupted by phonological requirements, e.g., that a phonological phrase dominate minimally two prosodic words.

MATCH THEORY advanced in Selkirk 2011 (cf. also, e.g., Selkirk 2009, Ito and Mester 2013, Elordieta 2015, or Ito and Mester 2020 for further discussion) captures the assumption of default isomorphism through violable faithfulness constraints on the correspondence of syntactic constituents to prosodic constituents (and vice-versa). Selkirk (2011: 451) formulates

would not violate PARSE- $\mathcal{F}$, because there is no $\mathcal{F}$ that fails to be parsed. One could reformulate the constraint to call for further violation marks for each successive higher level category, such that a σ parsed directly into a φ would incur two violation marks (one for the foot-level, one for the word-level), but this solution could potentially result in tableaux in which the failure to parse two syllables into a foot would be evaluated identically with the failure to parse one syllable into both a foot and a word. The shift proposed by Ito and Mester (2009: 139) to a family of constraints PARSE-INTO-X thus helps to distinguish what would appear to be two different sorts of violations: a syllable parsed directly into a phonological phrase would incur one violation of PARSE-INTO- $\mathcal{F}$ and PARSE-INTO- ω , while two syllables parsed directly into a word would result in two violations of PARSE-INTO- $\mathcal{F}$. Since the two PARSE-INTO-X constraints can be ranked differently, the distinction between the two candidates exhibiting non-exhaustivity of parsing with respect to different prosodic categories is maintained.

³⁰Work by Ito & Mester does not distinguish a family of NORECURSION constraints, but rather applies a single general constraint. Selkirk's (2011) family of MATCH-Prosody-to-Syntax constraints, insofar as it penalizes prosodic constituents without a matching syntactic constituent, will effectively penalize recursive prosodic constituents in most cases (cf. Ito and Mester 2013: 29, fn. 8). The common instances in which prosodic category recursion is sensibly invoked are cases in which a syntactic element is prosodically too light to build its own prosodic word or phrase. Because such elements are syntactically typically function words and particles, they are unlikely to instantiate their own syntactic XPs. I leave open the question of how exactly a (family of) markedness constraint(s) should be formulated to penalize prosodic recursion. Practically speaking, in this book, no configurations involving the recursion of multiple different prosodic constituents in the same output will be encountered, for which reason a simple NORECURSION will be sufficient.

the syntax-to-prosody correspondence as in (21); the reverse prosody-to-syntax mapping is equivalently achieved by NORECURSION in (20), which will be employed here.

(21) MATCH (α, π)

The left and right edge of a constituent of type α in the input syntactic representation must correspond to the left and right edge of a constituent of type π in the output phonological representation.

The exact formulation of MATCH constraints, their mappings to different types of syntactic constituents, and possible distinctions for minimal, intermediate, and maximal projections of prosodic constituents, are discussed in Elfner 2012, Ito and Mester 2013: 26–8, and Ito and Mester 2020: 265–8. I now briefly illustrate some of the possibilities afforded by Match Theory and the MinMax Model through pitch downstep and pitch reset in Lekeitio Basque as analyzed in Elordieta 2015.

The word-level prosody of Lekeitio Basque is characterized principally by a lexical distinction between “accented” and “unaccented” lexemes where the former receive a pitch accent H*L on the penultimate syllable; certain inflectional suffixes introduce accentedness to otherwise unaccented stems (cf. Gussenhoven 2004: 170–2, with comparison to the dialect of Gernika). The domain of accentuation may be taken to be the minimal phonological phrase ($\phi_{\min}$), where the pitch accent marks its right edge. According to Elordieta, downstep may be seen applying to any further accented word beyond the first, but pitch reset is observable instead under certain configurations; specifically, downstep occurs within the same DP, but pitch reset may be seen if the head noun is followed by a modifier or when a DP precedes the V head inside *vP* after another constituent. The four configurations of pitch patterns and their corresponding syntactic structures investigated by Elordieta are given schematically and with an example in (22) ($\downarrow$ indicates downstep, $\uparrow$ pitch reset; cf. Elordieta 2015: 53 and also 1.3.1 below) A strong pitch reset is thus found in Type D as a consequence of the fact that it contains the left boundary of two non-minimal phonological phrases between the second and third noun.

(22) Downstep and Pitch Reset in Lekeitio Basque by Syntactic Constituent Structure

- a. Type A: [[[N $\downarrow$ N]_{DP} [$\downarrow$ N]_{DP} $\downarrow$ VP]]_{TP}: second DP consisting of a single noun grouped with a first DP.

Mirén-en lagún-en liburú-ak gustaten dxáraz.
Miren-GEN.SG friend-GEN.PL book-ABS.PL like AUX
'I like Miren's friends' books'

- b. Type B: [[[N $\downarrow$ N]_{DP} [$\uparrow$ N A]]_{DP} $\downarrow$ VP]_{TP}: second DP with an adjectival modifier grouped with a first DP.

Mirén-en lagún-en libúru lodídx-ak gustaten dxáraz.
Miren-GEN.SG friend-GEN.PL book thick-ABS.PL like AUX
'I like Miren's friends' thick books'

- c. Type C: [[N ↓N]_{DP} [[↑N]_{DP} ↓VP]_{VP}]_{TP}: second DP consisting of a single noun grouped with the VP.

Mirén-en amumá-ri liburú-ak emon dótzez
 Miren-GEN.SG grandmother-DAT.SG book-ABS.PL give AUX
 ‘They have given the books to Miren’s grandmother’

- d. Type D: [[N ↓N]_{DP} [[↑↑N A]_{DP} VP]_{VP}]_{TP}: second DP with an adjectival modifier grouped with the VP.

Mirén-en amumá-ri liburú lodídx-ak emon dótzez
 Miren-GEN.SG grandmother-DAT.SG book thick-ABS.PL give AUX
 ‘They have given the thick books to Miren’s grandmother’

Undominated in Lekeitio Basque prosody is ACCENTAsHEAD, which requires each accent to be the head of a minimal phrase $\varphi_{\min}$; any candidate in which a $\varphi_{\min}$ dominates multiple accented words will be excluded. Beyond this driving feature, the prosodic phrases are constructed so as to minimize recursion and minimize the number of unary prosodic constituents at the level of the φ , at the cost of non-isomorphic mappings between XPs and phonological phrases. Complete definitions of the constraints required for the analysis of the sentences in (22) are given in (23)–(26), following the analysis of Japanese prosodic phrasing in Ito and Mester 2013.

- (23) ACCENTAsHEAD
 Every accent is the head of a $\varphi_{[\text{+min}]}$. Assign one violation * for each accent that is not the head of a minimal phonological phrase φ .
- (24) NORECURSION (cf. (20) above)
 No recursive structures. Assign one violation * for each node of category α immediately dominated by another node of category α .
- (25) MINBIN- φ
 φ is minimally binary. Assign one violation * for each φ that does not dominate two ω .³¹
- (26) MATCH-XP-TO- φ
 Suppose there is a syntactic phrase XP in the syntactic representation that exhaustively dominates a set of one or more terminal nodes α . Assign one violation mark * if there is no phonological phrase φ in the phonological representation that exhaustively dominates all and only the phonological exponents of the terminal nodes in α .

³¹This constraint will only be violated in cases where a minimal phonological phrase is projected from a single prosodic word, or where phonological phrase directly dominating one ω and one or more σ is built (cf. the structure of Standard Italian clitics under 17).

In all of the sentence types shown in (22), only the lexical verb is unaccented, and consequently, only it may become part of a binary minimal phonological phrase; unary minimal phonological phrases must be constructed around each accented word. In the case of Type A, candidate b. in (27) violates ACCENTAsHEAD by building a minimal φ around two accented words. Candidate b., meanwhile, achieves a better mapping between XPs and phonological phrases than candidate a., in virtue of constructing a recursive φ corresponding to vP , but the consequence is then more prosodic recursion.³² Candidate d., finally, performs worse on MATCH-XP-TO- φ because it does not construct a φ matching the DP consisting of the leftmost two NPs.

(27) Basque Type A

		ACCENTAsHEAD	NO RECURSION	MINBIN- φ	MATCH-XP-TO- φ
a.	$\mathbb{E} \quad (((\acute{N})_{\varphi} (\acute{N})_{\varphi})_{\bar{\varphi}} (\acute{N})_{\bar{\varphi}})_{\bar{\varphi}} (V \acute{Aux})_{\varphi})_{\varphi[\max]}$		***	***	**
b.	$((((\acute{N})_{\varphi} (\acute{N})_{\varphi})_{\bar{\varphi}} (\acute{N})_{\bar{\varphi}} V)_{\bar{\varphi}} (\acute{Aux})_{\varphi})_{\varphi[\max]}$		****!	****	*
c.	$((((\acute{N} N)_{\varphi} (\acute{N})_{\varphi})_{\bar{\varphi}} (V \acute{Aux})_{\varphi})_{\varphi[\max]}$	*!	**	*	****
d.	$((\acute{N})_{\varphi} ((\acute{N})_{\varphi} (\acute{N})_{\varphi}))_{\bar{\varphi}} (V \acute{Aux})_{\varphi})_{\varphi[\max]}$		***	***	***!

The situation is much the same for all other types: once each accented word projects the head of a minimal phonological phrase, the minimal phrases are combined so as to minimize recursion, and then to maximize syntax-prosody isomorphism. In Type D, for instance,

³²Candidate b. may be independently excluded if the lexical item projecting the T head is to be taken as a clitic and it forms a prosodic word with the lexical verb. The φ dominating the lexical verb and auxiliary in that case would also be unary, rather than binary. Although Elordieta (2015: 61) then represents a φ consisting of *gustaten dxáraz* as mapping to the VP, the effect of cliticization does not, I think, change the fact that no prosodic constituent corresponds to the right edge of VP and vP , resulting in two violations of MATCH-XP-TO- φ , as in candidate a. in 27.

evaluated in 28, once ACCENTAsHEAD is satisfied (candidates a., b., and d.), then candidate A simultaneously minimizes recursion and creates more binary structure (in contrast to candidate b.), and also better aligns the phonological phrases with the syntactic phrases (in contrast to candidate d.).³³

(28) Basque Type D

		AccentAsHead	NoRecursion	MinBin- ϕ	Match-XP-to- ϕ
a.	$((\acute{N}P)_{\phi} (\acute{N}P)_{\phi})_{\bar{\phi}} ((\acute{N}P)_{\phi} (\acute{F})_{\phi})_{\bar{\phi}} (V \acute{A}ux)_{\phi})_{\bar{\phi}})_{\phi[\max]}$		****	****	***
b.	$((\acute{N}P)_{\phi} (\acute{N}P)_{\phi})_{\bar{\phi}} (((\acute{N}P)_{\phi} (\acute{F})_{\phi})_{\bar{\phi}} V)_{\bar{\phi}} (\acute{A}ux)_{\phi})_{\bar{\phi}})_{\phi[\max]}$		*****!	*****!	*
c.	$((\acute{N}P)_{\phi} NP)_{\bar{\phi}} ((\acute{N}P)_{\phi} (\acute{F})_{\phi})_{\bar{\phi}} (V \acute{A}ux)_{\phi})_{\bar{\phi}})_{\phi[\max]}$	*!	****	***	****
d.	$((\acute{N}P)_{\phi} ((\acute{N}P)_{\phi} (\acute{N}P)_{\phi})_{\bar{\phi}})_{\bar{\phi}} ((\acute{F})_{\phi} (V \acute{A}ux)_{\phi})_{\bar{\phi}})_{\phi[\max]}$		****	****	*****!

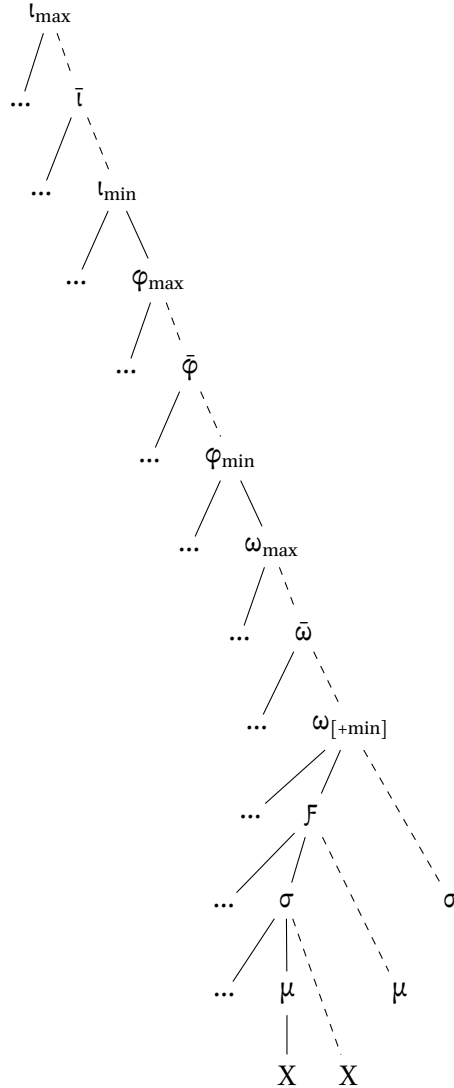
Thus, under the conception of the prosodic hierarchy as regulated by MATCH constraints above the word while also allowing recursion, there is no utterance as an independent prosodic domain: the utterance is equivalent to the maximal projection of the intonational phrase, in much the same way that a syntactic sentence could be taken as the maximal projection of a complementizer phrase (CP), rather than grouping all of the constituents of the sentence under an arbitrary label S (cf. Ito and Mester 2012: 287–90). The admission of weak layering with recursive prosodic constituents has the effect of creating prosodic constituents that, in abstract terms, much more closely resemble the structure of syntactic constituents under

³³Empirically, distinguishing between candidates a. and b. is difficult, since the perceptible evidence in these cases consists of the strong pitch reset between the second and third noun, which would be expected in both cases.

$\bar{X}$ -Theory, which admits of a theoretically infinite number of adjuncts between the minimal and maximal projection of a phrase. With violations of EXHAUSTIVITY readily permitted below the word, but uncommon or perhaps even disallowed above the word, the structure of the prosodic hierarchy then assumes the shape given in 29.³⁴ This prosodic hierarchy represents strict headedness via solid lines: every constituent from the word level and above must have a minimal and maximal projection (which of course may be identical), and every sub-word category must dominate at least one instance of the lower category. Optional instances of recursion and level-skipping adjunction are signaled by a dashed line.

³⁴The question of sub-word constituent recursion will be treated under the following subsections, but generally found to be unlikely.

(29) PROSODIC HIERARCHY under Match Theory and MinMax Model³⁵



With a well-delineated portrait of the general prosodic structures to be adopted in hand, I will now, in the succeeding subsections, more closely examine the constituents at the word level and below, since precisely these constituents will be of greatest relevance throughout.

³⁵A fundamentally different alternative to the model developed by Selkirk on the one hand and Ito and Mester on the other is presented in Vogel 2020. The primary innovation proposed in Vogel's COMPOSITE PROSODY MODEL to account for the excessive restrictiveness of the Strict Layer Hypothesis is to introduce a new prosodic constituent, the Composite Group, between the prosodic word and phonological phrase. Vogel argues that this constituent eliminates the need for recursion (at least of the prosodic word), serving as the specific site for the incorporation of various stray elements (certain affixes, clitics, function words), thereby also avoiding some amount of level-skipping. The advantages of this alternative, at least without further detailed case studies, are not apparent, insofar as it cannot immediately exclude the recursion of phonological or intonational phrases, nor level-skipping that is exclusively phonologically motivated (e.g., extrasyllabicity due to sonority sequencing).

1.2.2 The Prosodic Word and its Recursion

On the one hand, the prosodic word, at least in guises where it is fully isomorphic to a morphosyntactic (grammatical) word, is wholly unproblematic, and no conception of prosodic structure, to my knowledge, has attempted to do without a category of WORD that is relevant to the phonology, where the left and right boundaries will be coextensive with the left and right boundaries of one (or perhaps more) grammatical words. On the other hand, the exact size and composition of well-formed prosodic words, and the extent to which they can encompass multiple grammatical words is often less self-evident, and in a constituent consisting of multiple grammatical words, distinguishing between a relatively larger ω and a relatively smaller φ (or other logical alternatives) becomes necessary. The two primary questions to be addressed in this section are:

- What exactly can constitute a ω , and how is a ω to be diagnosed on a language-specific basis?
- Allowing for its recursion, under what conditions does recursion of the ω occur, and how is such prosodic word recursion to be identified?

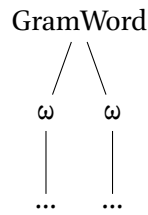
Building upon and modifying Nespor & Vogel's (1986: Ch. 4) work, the most systematic treatment of issues surrounding the construction and identification of prosodic words remains Peperkamp 1997; briefer surveys are available in Hall 1999, and Féry 2017: 51–7. The set of studies in Hall and Kleinhenz 1999 complements Peperkamp 1997 and Nespor and Vogel 1986: Ch. 4 to give a sense of the full range of phonological phenomena potentially attributable to the ω as a prosodic constituent. Under any and all approaches, for any language, the null hypothesis is that a grammatical word maps directly onto (i.e., is coextensive with or “matches”) a prosodic word.³⁶ Nevertheless, grammatical words may often simultaneously project higher-level prosodic categories (as in the analysis of Lekeitio Basque above), and certain common or familiar phenomena might seem to point in the direction of mismatches, e.g., non-cohering affixes in Germanic (are these their own prosodic words, despite being morphologically subordinate?) or clitics (are these not prosodic words, despite being syntactically independent?). The potential non-isomorphism of prosodic word and grammatical word thus runs in both directions: one the one hand, a single grammatical word might be treated as consisting of multiple prosodic words (e.g., a compound that contains a single site for inflectional morphology), and on the other hand, a single prosodic word might be analyzed as containing multiple grammatical words (e.g., a function word plus a content/lexical word). Both options are represented schematically in 30; the precise prosodic configurations that might arise in case of such non-isomorphisms are various.³⁷

³⁶Compare the constraint $Lx \approx PR$ (*MCat* in Prince and Smolensky 1993 [2004]: 51–61, which penalizes cases in which a given morphological category *MCat* does not perfectly correspond to a prosodic word, and earlier proposals by Selkirk (1986) and Chen (1987) for the matching of left or right edges of grammatical and prosodic words.

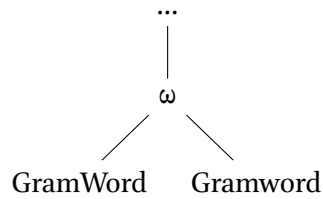
³⁷Grammatical words in polysynthetic languages offer particularly tricky and potentially extreme cases of non-isomorphism: in a grammatical word consisting of a dozen syllables, perhaps half a dozen or more prosodic words might be present. The study of Russell (1999) on Dakota (Siouan) and Cree (Algonquian) repre-

(30) Non-isomorphism between Grammatical and Prosodic Words

a. One Grammatical Word $\Leftrightarrow$ Multiple Prosodic Words



b. One Prosodic Word $\Leftrightarrow$ Multiple Grammatical Words



For the purposes of diagnosing prosodic words, Hall (1999: 3) lists several distinct types of evidence that may be employed for identifying a type prosodic constituent that frequently appears to map onto a grammatical word: it can act as a bounding or triggering domain for certain segmental processes; it frequently delimits the domain within which syllables and feet are constructed, and within which stress or similar (obligatory and culminative) prominence is assigned; it circumscribes certain phonotactic restrictions; and it may be subject to minimal size requirements (in terms of syllables or morae). Raffelsiefen (1999) contends, however, that, at least in the case of English, the only truly reliable diagnostics of a prosodic word are the construction of lower-level prosodic constituents; hence stress can be a diagnostic, as well as stress-dependent segmental processes, whereas other purely segmental processes would not be considered reliable. Concretely, Raffelsiefen observes that trisyllabic morphologically simplex words of English, when produced in isolation, exhibit primary stress on the initial syllable and secondary stress on the final syllable (e.g., *difficùlt* or *táci-tùrn*); conversely, trisyllabic words built with the prefixes *un-* or *in-* to a monomorphemic, disyllabic base exhibit primary stress on the final syllable and secondary stress on the initial (e.g., *ùncorrúpt*, *ìmpolíte*). Although a supposed distinction in application of nasal assimilation to *un-* vs. *in-* has often been taken as indicative of a difference affix class (i.e., “Level 1” vs. “Level 2”; cf. Kiparsky 1982a), which may be attributed to a distinction in prosodic composition (i.e., separate ω or single ω Aronoff and Sridhar 1983), the stress patterns of trisyllabic words with either suffix are identical.³⁸ Raffelsiefen’s proposal might be generalized and

sents perhaps one of the few attempts to address the status of prosodic words in languages in which especially “long” grammatical words do not arise primarily through composition. At least in the case of Cree, Russell finds that both prosodic and segmental processes point towards generally “short” prosodic words, and concurs with the observation of Rice (1994) concerning Slave (Athabaskan), that the “long” verbal complexes should probably be analyzed as phonological phrases.

³⁸For an approach to English stress centered around faithfulness to the stress of the base in morphologically

moderated to state that other prosodic evidence (stress assignment, foot and syllable structure) should constitute the primary evidence for prosodic words, which may be supported by segmental evidence; in cases where the segmental and prosodic evidence conflict, the prosodic evidence should be given more weight.

Hall (1999: 14) also points to syllable structure as a potential diagnostic of prosodic word boundaries, stating a strict generalization on the relationship between prosodic words and syllabification: “If C_a [some consonant] is non-ambisyllabic and syllable-initial, then C_a is not in pword-final position.” This statement essentially means that any consonant that can be identified as an onset must be internal to the same prosodic word as the other segments belonging to the same syllable, and that the onset C_a cannot represent the final segment in a ω , i.e., onsets may only be initial or medial in a ω . Languages in which resyllabification of consonants that are final in a grammatical word into an onset position would then have the effect of potentially creating mismatches between grammatical and prosodic words as well (cf. e.g. Nespor and Vogel 1986: 47–52 on resyllabification in various Romance languages): if two prosodic words are present, then a consonant crosses a prosodic word boundary, and disrupts perfect edge alignment of grammatical and prosodic words; if the two grammatical words form parts of one prosodic word, there can similarly be no perfect match between the segmental constituency of the grammatical and prosodic words involved.

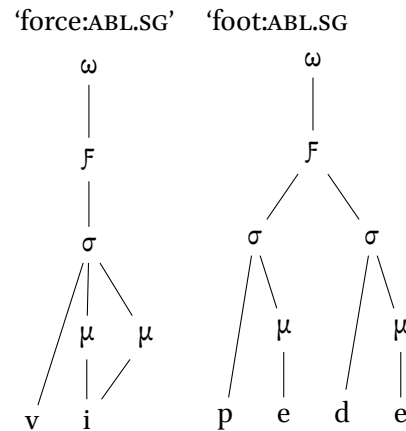
One further frequent phonological diagnostic of wordhood is a prosodic minimality requirement, whereby a well-formed prosodic word must dominate at least one heavy (bimoraic) syllable or two light syllables. The motivation behind such minimality requirements may be HEADEDNESS (cf. 18b.), which would require a ω to dominate at least one $\mathfrak{f}$; if a well-formed $\mathfrak{f}$ is in turn restricted to a disyllabic or bimoraic configuration (depending upon licit foot structures in the language in question), then the interaction of HEADEDNESS and constraints on licit feet may conspire to create a minimal word effect (but see Garrett 1999 or Pigott 2010 for arguments in favor of the independence of word minimality from foot well-formedness). Thus, in order for isomorphism between a grammatical word (syntactically, the terminal node X of a phrase) and prosodic word to hold, a grammatical word that would be subminimal must be prosodically “filled out”; alternatively, if a match between the grammatical word and prosodic word may is not required (i.e., MATCH-X-TO- ω may be violated), then the prosodically sub-minimal grammatical word may be effectively cliticized in some form (see further below). The evident ban on monosyllabic $/\langle C \rangle \check{V}/$ words in Latin (discussed *inter alia* by Kuryłowicz 1968: 191, Allen 1972: 51, Golston 1991: 101–23, Mester 1994: 20–23, Prince and Smolensky 1993 [2004]: 49–50, and Byrd 2015: 113–7), clearly holding for at least lexical words, exemplifies just such a typical minimality requirement.³⁹

complex words, see Stanton and Steriade 2014. Under such an account, forms like *ūncorrūpt* and *īmpolīte* would exhibit the same stress pattern primarily in virtue of the fact that their bases exhibit the same stress pattern and both are morphologically parsable.

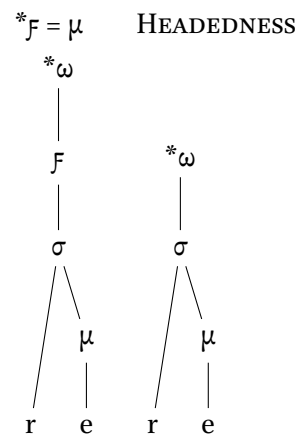
³⁹ Golston offers the most detailed available discussion of word minimality in Latin, and argues in favor of extrametricality of word-final obstruents. Under this account, Latin has function words of the shape $/C\check{V}C/$ that are not obliged to satisfy a minimal word requirement, while all instances of lexical words that surface as $[(C)\check{V}P]$ (e.g., *lac* ‘milk:NOM/ACC.SG’, *os* ‘bone:NOM/ACC.SG’) are argued to be underlyingly $/CVCC/$ and undergo a post-lexical cluster simplification, after the prosodic word has already been constructed.

(31) Minimal Prosodic Words in Latin

a. Well-Formed Bimoraic Feet ($/\langle C \rangle V:/$, $/\langle C \rangle \check{V}C\check{V}/$)



b. No Degenerate Monomoraic Foot or Violation of HEADEDNESS



Assuming that segmental processes, prosody, and minimality requirements align for a given language in the identification of prosodic words, how does the interface between phonology, morphology, and syntax function to create a ω ? For Nespor and Vogel (1986: 141–3), a syntactic terminal node (= grammatical word) is taken to constitute the structural upper bound on a prosodic word: prosodic words may be smaller than X (e.g., a prefix or suffix may constitute an independent ω), but not span multiple X . This restriction automatically requires all sequences of multiple grammatical words to be grouped into a higher-level prosodic constituent (for Nespor and Vogel, the clitic group or phonological phrase, assuming the Strict Layer Hypothesis), and would categorically exclude the incorporation of grammatical words projecting their own X into the ω of another X (i.e., internal clitics). Given that a ω must minimally be isomorphic to a morpheme, in the case of languages for which a ω is not determinable solely on the basis of a consistent set of syntactic ($X = \omega$), morphological (e.g., stem or prefix = ω), or phonological criteria (minimal foot = ω), diacritic marking for the capacity to build a prosodic word is permitted. The possible types of prosodic word

construction under Nespor & Vogel's account are then listed and exemplified in (32).

(32) Options for ω -Construction after Nespor and Vogel 1986

- a. Languages in which ω is exactly equivalent to a syntactic terminal node (Modern Greek, Latin).
- b. Languages in which a ω is built around any stem, with any affixes intervening between two stems assigned to the same morphological word as the stem or, if the morphological word contains multiple prosodic words, to the nearest prosodic word (Sanskrit, Turkish).
- c. Languages in which a ω is built around any stem, but in which certain classes of affixes build their own ω , and minimality requirements may compel an affix to be incorporated into a prosodic word with a stem (Hungarian, Italian, Yidip).
- d. Languages in which a ω is built around any stem, as well as any morphemes diacritically marked for prosodic-wordhood (Dutch, Germanic more generally).

Much subsequent thinking on the construction of prosodic word domains follows Nespor & Vogel in admitting morphological units below the level of the full morphological word (X) to build independent prosodic words, but the restriction to X as the maximal morphologically defined domain for a prosodic word poses some potential issues. Consider the well-known distinction in stress pattern of noun phrases with adjectival modifiers versus adjective + noun compounds in Germanic languages, illustrated with examples from English and German in (33) (cf. Booij 2012: 177–9, Féry 2017: 72–8 on the West Germanic COMPOUND STRESS RULE). The phenomenon of stress subordination (and deletion to avoid stress clash) in compounds, which does not apply in phrases, indicates that such compounds and phrases in Germanic languages have distinct prosodic structures. If the Strict Layer Hypothesis is assumed, and the compounds cannot constitute a single prosodic word, what prosodic structures and/or constituents can distinguish (33)a. from (33)b.?

(33) Compound Stress vs. Phrasal Stress Germanic

a. Adjective + Noun Compound

English	German
<i>bláckbird</i>	<i>Rótwein</i>
['blækbɪd]	['ʁo:tvain]
' <i>turdus merula</i> '	'red wine (e.g., merlot)'

b. Adjective Modifier of NP

English	German
<i>bláck bírd</i>	<i>róter Weín</i>
['blæk 'bɪd]	['ʁo:tə 'vaɪn]
'a bird which is black'	'red wine, i.e., a wine which is red'

Given that a unit like *black bird* or *roter Wein* is too small to constitute an intonational phrase, the two prosodic words of the compounds could be dominated by perhaps the same clitic group or the same minor phrase; if stress subordination is taken to be a property of the clitic group or minor phrase, then the forms in a. would fall under just one clitic group or minor phrase, whereas the forms under b. would contain two independent clitic groups or minor phrases. The fundamental problem with this analytical move is that the assumption of a distinction intermediate phrasal prosodic constituent between the ω and ϕ is difficult to consistently motivate cross-linguistically; cf. the recursion of phonological phrases discussed at (14) and (15) above.

At the same time, the evidence amassed by Peperkamp (1997: 176–99) (cf. also Anderson 2005: 52–55) on the behavior of clitics in three varieties of Italian (Standard, Lucanian, and Neapolitan) indicates that no clitic group can be the common site of prosodic incorporation for clitics across all three languages; differences in stress phenomena indicate that the prosodic structures must be distinct, as illustrated in 34 (cf. also 17 above).⁴⁰ Under the assumption that a minimal prosodic word is a domain for obligatory and culminative stress in each language (see further 1.3.3 on OBLIGATORINESS and CULMINATIVITY), the position of primary stress(es) serves as a telling diagnostic as to prosodic structure. In Lucanian, only a single primary stress every appears in a sequences of host plus clitics, but progressively shifts rightwards as more syllables are added, thereby indicating that all elements constitute part of the same minimal prosodic word. Neapolitan, meanwhile, presents multiple (primary) stresses, provided that the clitic sequence can build a minimal disyllabic foot; this phenomenon could be due either the inactivity of CULMINATIVITY above the domain of $\omega_{[+min]}$, or to a requirement for stress within the last three syllables of the outermost prosodic word (see also Bafle 1994).⁴¹ Finally, in Standard Italian, primary stress remains on the same position of the host and no further stresses appear, thereby indicating that clitics are incorporated at the level of the ϕ , not ω . These distinctions are only possible to clearly encode by either admitting an infelicitous multitude of language-specific prosodic constituents, or

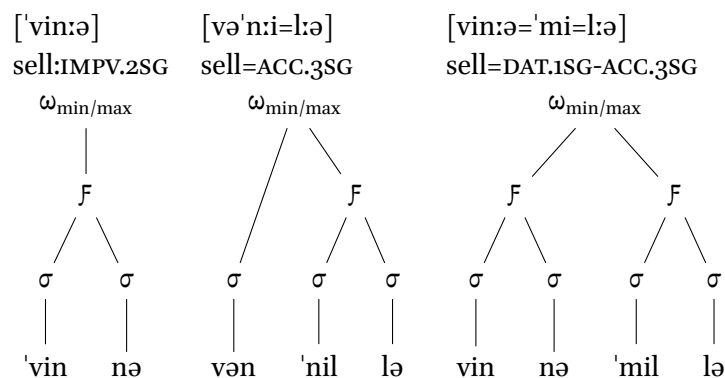
⁴⁰For an overview of explicit arguments against the clitic group as a prosodic constituent, see Peperkamp 1997: 167–76 or Anderson 2005: 42–44.

⁴¹Bafle (1994: 16) in fact marks the rightmost stress as primary, e.g., [pɔ:ta'til.lə] 'bring:IMPV.2SG-DAT.2SG-ACC.3SG'. Loporcaro (2000: 137) marks Neapolitan stress similarly, indicating forms with two enclitics as having secondary stress on the stem and primary stress on the penult, e.g., [pi'la:a'til:ə] 'take:IMPV.2SG-DAT.2SG-ACC.3SG'. Peperkamp (1997: 190) reports that the rightmost stress "is perceived as strongest...due to phrasal stress, which is indeed rightmost as well in Neapolitan." An instrumental analysis of one token of *accattate=vi=llo* [ak:a'ta:tə'vil:ə] 'get:IMPV.2PL=DAT.2PL=ACC.3SG' (https://www.youtube.com/watch?v=-e4AgY3_UP8, Timestamp: 26 seconds) suggests that the two stresses are distinguished only by Fo, with higher Fo on the rightmost, which supports Peperkamp's claim. Note further that stress on a monosyllabic stem may be suppressed in order to avoid a violation of *CLASH: compare [fa] 'do:IMPV.2SG' and [fa.'til.lə] 'do:IMPV.2SG-DAT.2SG-ACC.3SG'.

by admitting recursivity of the prosodic word, as well as the non-exhaustivity of parsing of lower level prosodic constituents into higher levels.⁴²

(34) Prosodic Structure in Three Italian Varieties

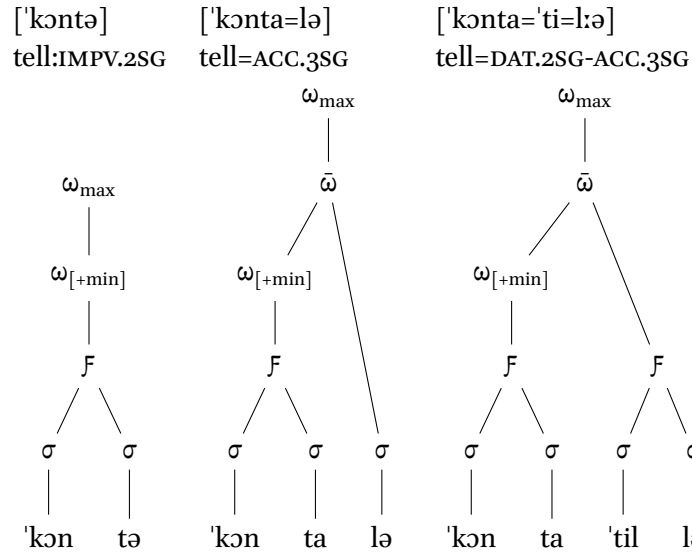
a. Lucanian (Prosodic Word Incorporation = Internal Clitic)⁴³



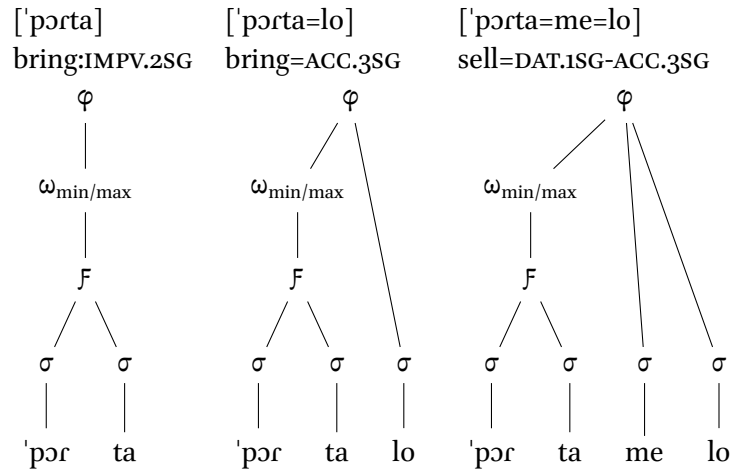
⁴²Loporcaro (2000) advances an alternative analysis in terms of Lexical Phonology, under which enclitics in these and all similar varieties of Italian are “adjoined to the (lexical) P(rosodic)W(ord) and form a postlexical PW with their host” and “whenever varieties other than S(tandard)I(talian) display genuinely PW-bound segmental rules, best described if clitics are analyzed as PW-internal at the postlexical level” (164). Loporcaro’s arguments are, however, essentially negative: he argues that differences in stress assignment simply follow a distinction between lexical and post-lexical processes, and that segmental processes potentially indicative of prosodic word boundaries are either non-probative or similarly reducible to a lexical vs. post-lexical distinction.

⁴³Peperkamp (1997: 193) represents the form with a single clitic, $[və'n:i=l:ə]$, as having an unfooted first syllable, with a foot built around the penultimate and ultimate syllables. Depending upon other factors, such as quantity sensitivity or final syllable extrametricality, the correct foot structures may be different, but do not materially affect the larger point concerning the domain of attachment for these clitics, for which reason I take over Peperkamp’s structures below the word level.

b. Neapolitan (Prosodic Word Adjunction = Affixal Clitic)⁴⁴



c. Standard Italian (Phonological Phrase Incorporation = Free Clitic)



Selkirk (1995), Truckenbrodt (1999), and Ito and Mester (2009) all share the assumption that function words and clitics (more specifically, words projecting functional categories like Determiner or Preposition, or projecting inflectional categories) are not subject to lexical mapping, which instead only applies to lexical words (i.e., words projecting functional categories like Noun or Verb). Lexical mapping may be formulated in terms of generalized alignment constraints (cf. McCarthy and Prince 1993 [2001]), which are weaker than MATCH constraints: the latter require simultaneous isomorphism of the left and right edges of syntactic and prosodic constituents, while the former refer only to either the left and/or right

⁴⁴Lai (2017) observes empirically similar stress behavior in Nuorese Sardinian, but where proclitics are implicated as well. Lai therefore argues that single monosyllabic clitics in Nuorese behave as affixal clitics, but disyllabic clitics or clitic sequences build a prosodic word which is nevertheless part of a recursive prosodic word as a sister to the host, i.e., $[[\text{HOST}]_{\omega} [\text{CLITIC}]_{\omega}]_{\bar{\omega}}$.

edges of a prosodic and a morphological or syntactic constituent. Ito and Mester (2009: 142) propose two particular LEX-MAPPING constraints as relevant to the interface between lexical words, prosodic words, and phonological phrases, reproduced in 35.⁴⁵

(35) LEX-MAPPING

- a. LEX-TO- ω (L, R): ALIGN(Lex, Left/Right, ω , Left/Right)
Every lexical word is left/right aligned with a prosodic word.
- b. LEX-TO- ϕ (L, R): ALIGN(LexP, Left/Right, ϕ , Left/Right)
Every lexical maximal projection is left/right aligned with a phonological phrase.

So how exactly is the incorporation or adjunction of clitics regulated in the three Italian varieties above? The relevant constraints have for the most part already been introduced in the preceding subsection. In these examples, all segments are parsed into syllables, all syllables are ultimately parsed into a phonological phrase, and the phonological phrase dominates at least one prosodic word, which in turn dominates at least one foot. Leaving aside an analysis of precisely how the stresses in these languages are assigned, the construction of prosodic structures is decided by the ranking of FOOTBINARITY, PARSE-INTO- ϕ , PARSE-INTO- ω , NORECUSION, and LEX-to- ω (R).⁴⁶

The defining distinctions in prosodic structure are principally regulated by the relative rankings of NORECUSION, PARSE-INTO- ω , and LEX-TO- ω (R). In all three languages, the ranking FOOTBINARITY $\gg$ PARSE-INTO- ϕ holds, thereby excluding subminimal feet. In cases of prosodic word incorporation (Lucanian; see 36), the crucial constraint relationship is that NORECUSION and PARSE-INTO- ω dominate LEX-TO- ω (R); the result is that the clitic(s) are included inside the same prosodic word as the lexical verb.⁴⁷ When instead the clitic(s) are adjoined to the prosodic word (Neapolitan; see 37), instead NORECUSION must fall in the stratum below LEX-TO- ω (R) and PARSE-INTO- ω . Finally, in cases of prosodic phrase incorporation (Standard Italian; see 38), PARSE-INTO- ω represents the lower ranked constraint, with both NORECUSION and LEX-to- ω (R) necessarily being satisfied.⁴⁸ For further computational

⁴⁵Such lexical mapping constraints amount to an observation that, cross-linguistically, lexical items projecting lexical categories are substantially more likely to “match” a prosodic word, though function words and particles projecting inflectional categories are by no means excluded from forming prosodic independent prosodic words. See, for instance, Goldstein 2015: Ch. 3 (and 6.2.1.5 in 6) for arguments that the class of “postpositive” clitics in Ancient Attic-Ionic Greek represent prosodic word clitics: such clitics are able to form independent, non-recursive prosodic words, but are unable to ever form independent phonological phrases, and thus must fall within a phonological phrase headed by a prosodic word projected by a host.

⁴⁶For Lucanian, the constraint ALL-FEET-RIGHT and/or RIGHTMOST further ensures that a foot is built close to the right edge and that it receives primary stress, while for Neapolitan, OBLIGATORINESS requires that one primary stress occurs among the feet immediately dominated by a prosodic word. In the following, constraints regulating stress assignment are ignored for the purposes of simplification of presentation; see 2.3 in Chapter 2 on relevant constraints.

⁴⁷I indicate further the violation of PARSE-INTO- ϕ based on the foot structure proposed by Peperkamp (1997: 193), following the earlier analysis of Kenstowicz (1991).

⁴⁸The further distinction of prosodic word clitics, of structures like [[HOST] $_{\omega}$ [CLITIC] $_{\omega}$] $_{\phi}$, requires that NORECUSION, LEX-to- ω (L, R), and PARSE-INTO- ω all be satisfied, but some further constraint must also be at work. In languages with no prosodic word clitics, it may be assumed that minimality requirements on prosodic

perspectives on structure building operations in the terms of finite-state grammars employing precisely these same examples, see now Yu 2021.

(36) Lucanian Enclitics

a. Single Enclitics

vinni lə		FOOTBINARITY	NORECURSION	PARSE-INTO- ω	LEX-TO- ω (R)	PARSE-INTO-F
a.	$[vən.('nil.lə)]_{\omega}$			*	*	
b.	$[(['vin.nə])_{\omega-min} lə]_{\bar{\omega}}$		*!			*
c.	$[(['vin.nə])_{\omega-min} lə]_{\varphi}$			*!		*
d.	$[('vin.nə).(lə)]_{\omega}$	*!		*		

b. Two Enclitics

vinni mi lə		FOOTBINARITY	NORECURSION	PARSE-INTO- ω	LEX-TO- ω (R)	PARSE-INTO-F
a.	$[(vin.nə).('mil.lə)]_{\omega}$			*		
b.	$[(['vin.nə])_{\omega-min} ('mil.lə)]_{\bar{\omega}}$		*!			
c.	$[(['vin.nə])_{\omega-min} mil.lə]_{\varphi}$			*!		**
d.	$[('vin.nə).(mil).(lə)]_{\omega}$	*!		*		

words ($\sim$ well-formed feet combined with HEADEDNESS), outranking the other three constraints, excludes them; in languages showing only prosodic word clitics, regardless of phonological size of the clitic, presumably the three constraints and PARSE-INTO-F dominate FOOTBINARITY to allow for degenerate feet. In languages with both prosodic word clitics and clitics of other kinds, if the certain clitics are all able to build well-formed feet, then a clear phonological distinction may be drawn; if one finds clitics of the prosodic word type and another type in the same language, then it would seem that the prosodic word clitics must simply be lexically marked to enforce the formation of a prosodic word. Ito and Mester (2009: 175–88) consider yet another possibility in their treatment of disyllabic function words in English and German: grammatical words projecting a functional category may build an independent minimal prosodic word, but nevertheless be adjoined to the prosodic word of a lexical category, producing a recursive prosodic word. A further requirement on the relationship between heads in prosodic structure and morphosyntactic categories, HEAD-TO-LEX (“Prosodic heads must be contained in a lexical category”) can regulate whether the phonological material projecting the head of a foot or prosodic word is allowed to come from a functional category or not. See further Vigário 2010 or Bennett 2018 for discussion as to whether “balanced recursion”, where two instances of a lower-level category are dominated by a higher level category, e.g., $[[X]_{\omega-min} [Y]_{\omega-min}]_{\omega-max}$, is at all possible.

(37) Neapolitan Enclitics

a. Single Enclitics

kɔnta lə		FOOTBINARITY	LEX-TO- ω (R)	PARSE-INTO- ω	NORECUSION	PARSE-INTO-f
a.	$\llbracket [(['kɔn.ta])_{\omega-min} lə]_{\bar{\omega}} \rrbracket$			*	*	
b.	$\llbracket [kɔn.('ta.lə)]_{\omega} \rrbracket$		*!			*
c.	$\llbracket [(['kɔn.ta)]_{\omega} lə]_{\varphi} \rrbracket$			*!		
d.	$\llbracket [(['kɔn.ta)]_{\omega-min} ('lə)]_{\bar{\omega}} \rrbracket$	*!				

b. Two Enclitics

kɔnta ti lə		FOOTBINARITY	LEX-TO- ω (R)	PARSE-INTO- ω	NORECUSION	PARSE-INTO-f
a.	$\llbracket [(['kɔn.ta)]_{\omega-min} ('til.lə)]_{\bar{\omega}} \rrbracket$			*		
b.	$\llbracket [(kɔn.ta).('til.lə)]_{\omega} \rrbracket$		*!			
c.	$\llbracket [(['kɔn.ta)]_{\omega} til.lə]_{\varphi} \rrbracket$			*!		
d.	$\llbracket [(['kɔn.ta)]_{\omega-min} ('til).(lə)]_{\bar{\omega}} \rrbracket$	*!				

(38) Standard Italian Enclitics

a. Single Enclitics

pɔrta lo		FOOTBINARITY	LEX-TO- ω (R)	NORECUSION	PARSE-INTO- ω	PARSE-INTO-f
a.	$\llbracket [(['pɔr.ta)]_{\omega} lo]_{\varphi} \rrbracket$			*	*	
b.	$\llbracket [(['pɔr.ta)]_{\omega} lo]_{\bar{\omega}} \rrbracket$		*!		*	
c.	$\llbracket [pɔr.('ta.lo)]_{\omega} \rrbracket$		*!		*	
d.	$\llbracket [(['pɔr.ta)]_{\omega} (lo)]_{\varphi} \rrbracket$	*!				

b. Two Enclitics

pɔrta me lo		FOOTBINARITY	LEX-TO- ω (R)	NORECUSION	PARSE-INTO- ω	PARSE-INTO-f
a.	$\llbracket [(['pɔr.ta)]_{\omega} me.lo]_{\varphi} \rrbracket$			*	**	
b.	$\llbracket [(['pɔr.ta)]_{\omega} ('me.lo)]_{\bar{\omega}} \rrbracket$		*!			
c.	$\llbracket [(pɔr.ta).('me.lo)]_{\omega} \rrbracket$		*!			
d.	$\llbracket [(['pɔr.ta)]_{\omega} (me).(lo)]_{\varphi} \rrbracket$	*!*				

As the evidence from the closely related Italian varieties discussed here shows, recursive structures with only weak layering of prosodic constituents, whose construction is regulated by differences in the ranking of universally available constraints, enables an account

of differences in both overt prosody (in this case, stress), and can similarly be leveraged to account for differences in the domain of application of segmental processes. Bennett (2018), for instance, employs recursive adjunction under the prosodic word (i.e., the type of structure seen in Neapolitan under (34)b.) to account for a variety of segmental phenomena in Kaqchikel (Mayan). The insertion of a glottal stop at the left edge of any otherwise onsetless prosodic word at any level of recursion, and failure of consonant degemination to apply across a the boundary of a (recursive) prosodic word, indicates that the prefixes implicated in these phenomena belong to the same type of constituent, rather than constructing some new type of prosodic constituent.

- (39) Multiple high-attaching prefixes in Kaqchikel (cf. Bennett 2018: 10)
 /at=at̃=aχ=mak^h/
- [ʔat=ʔat̃=ʔaχ=mak^h]
- 2SG.ABS=COM=AGT=sin
- ‘You are an accomplice’

The prosodic structure implied by (39) is thus [ABS [COM [AGT [ROOT]_{ω-min}]_ω]_{ω-max}: the ω recurs for as many high-attaching prefixes as occur in a string, resulting in glottal stop insertion, since resyllabification does not apply across ω-boundaries (^X[ʔa.ta.t̃aχ.mak^h]). An analysis whereby these simply build independent syllables (whether via lexical specification of a syllable boundary, morpheme-specific alignment constraints, or cyclic syllabification) is ruled out, since the same prefixes fail to trigger consonant degemination, as opposed to low-attaching prefixes, which do (and which also fail to trigger glottal stop insertion).⁴⁹

- (40) Low- vs. high-attaching prefixes: contrast in consonant degemination
 (Bennett 2018: 7)
- a. /f-oχ-χot-eʔ/ → [foχoteʔ] (^X[foχ:oteʔ])
- CPL-1PL.ABS-elevate-INTR.POS
- ‘we climbed’
- b. /aχ=χukuʔ/ → [ʔaχ:ukuʔ] (^X[ʔaχukuʔ])
- AGT=boat
- ‘boatman’

From the morphosyntactic point of view, cases like this one represent the mirror image of many clitics: while clitics often correspond to independent grammatical words, their prosodic deficiency is typically responsible for some type of adjunction or incorporation; in the present case, that affixes expressing inflectional categories are not incorporated into the ω built around a stem is atypical. In general, adjunction to rather than incorporation into

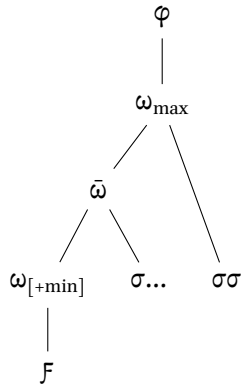
⁴⁹In the data discussed by Bennett, there is, however, no evidence for the non-application of degemination across the boundaries of the high prefixes, which may be accidental: the set is limited to the four prefixes /aχ/ ‘AGT’, /if/ ‘FEM’, /at̃/ ‘COM’ and /jaχ/ ‘related by marriage’. The environment for degemination thus cannot be created between any high prefixes. A left prosodic word boundary is thus truly in evidence only at the boundary between the innermost high prefix and the root. In principle, all high attaching prefixes could then be treated as independent syllables outside the minimal prosodic word.

a ω may be regulated by LEX-TO- ω constraints; contrast (36) with (37) above, noting that, in the former, the lower ranking of LEX-TO- ω (R) admits of incorporation. In the Kaqchikel case, the situation is made more complex by the fact that roots do allow for affix incorporation (in the case of low-attaching prefixes; cf. 40a.), and that every high-attaching prefix apparently builds the left edge of another prosodic word, and treats all existing material as contained in a separate prosodic word; for this reason, Bennett opts for the following prosodic subcategorization frame (cf. generally Inkelas 1990, Kabak and Revithiadou 2009b) borne lexically by the high-attaching prefixes.⁵⁰

- (41) Prosodic subcategorization frame for high-attaching prefixes in Kaqchikel:
 $[\text{HIGHPREFIX } [X]_{\omega}]_{\omega}$

All of the foregoing discussion in this section serves to establish that the prosodic word, as a phonological entity, represents a site of considerable potential complexity. Both cross-linguistically and language-internally, multiple different prosodic word structures can be found, though diagnosing and distinguishing them often remains empirically fraught. Nevertheless, the current state of research generally speaks in favor of permitting structures like the one seen in 42 as a means of handling non-isomorphisms between grammatical words and the phonological diagnostics of wordhood.

- (42) Schematic Prosodic word (ω) Recursion with Enclitics



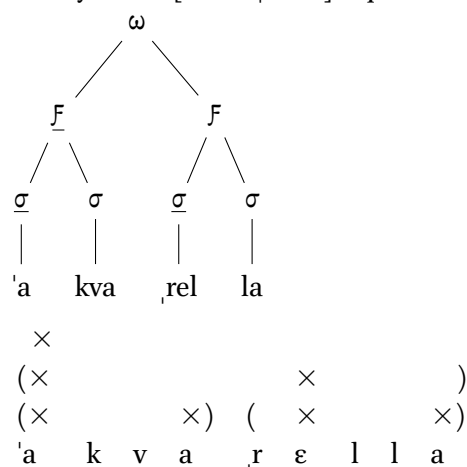
1.2.3 Feet and Footing

Although the exact prosodic constituents to be found between the prosodic word and the utterance remains debated (i.e., whether clitic groups, accentual phrases, etc., are available universally or on a language-specific basis), and, as discussed above, some hypothesize relatively strict isomorphism between grammatical and prosodic words, the FOOT arguably con-

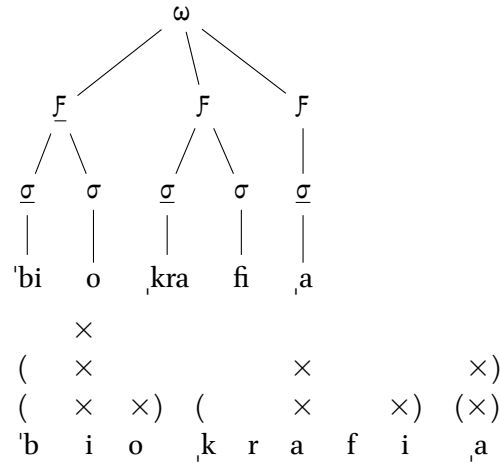
⁵⁰A third possibility would be to operate with faithfulness to lexically specified left or right prosodic word boundaries. Such a solution will not help in the case at hand, since fully specified left and right edges would result in independent minimal prosodic words, specified right edges would predict that the outermost prefix forms its own minimal prosodic word, and specified left edges would result in the incorporation of the innermost prefix.

stitutes the most-disputed prosodic constituent of all, in large part due to its status as a “hidden structure” in the phonology: feet are neither heard in the phonetics, nor do they directly interface with syntactic structures, and unlike the syllable, seem less available than syllables to naive intuition. Fundamentally, metrical feet represent a unit of (rhythmic) organization that captures relations of prominence between syllables contained within the same prosodic word; in the phonological systems of many languages, the primary and most visible role accorded to the foot is thus the calculation and assignment of stress (see further 1.3.3 below and Chapter 2). In relatively simple cases, in which a language exhibits an alternating pattern of primary and secondary stress beginning from some word edge, e.g., the first syllable in Icelandic (Árnason 1985, Hayes 1995: 188–98), Mansi (Hayes 1995: 100, 200, Bennett 2012: 15), or Pintupi (Hansen and Hansen 1969: 163, Hayes 1995: 62–3), a sequence of two syllables may be grouped into a foot, with the leftmost syllable of each foot constituting the head, and the leftmost foot in each word constituting the head of the ω ; these stress systems exhibit syllabic trochees, and differ principally in terms of whether stress can occur on the final syllable of a word with an odd number of syllables (perhaps so in Icelandic, almost certainly not in Pintupi). Example (43) illustrates the foot structures assumed for Icelandic as both a prosodic tree and metrical grid with foot constituents, containing a final degenerate foot built around a single syllable in b.; note on the grid that the single mark on the highest line corresponds to primary stress, and each mark on the next-highest line corresponds to secondary stress.

- (43) Foot Structure and Stress in Icelandic
 a. Four-syllable: ['a.kva.rɛl.la] 'aquarelle'



b. Five-syllable: ['bi.o,kra.fi,a] 'biography'



Doubts as to the necessity of feet in prosodic phonology generally or for the purposes of calculating stress assignment are nearly as old as proposals for such hierarchical constituents: not longer after their debut in Liberman 1975, Liberman and Prince 1977, and Selkirk 1980b, it was quickly observed that processes of stress assignment need not refer to prosodic constituents whatsoever – the metrical grid may be sufficient (so, e.g., Prince 1983, Selkirk 1984). While a foot- or constituency-based approach to stress assignment has remained predominant (Halle and Vergnaud 1987, Hayes 1995, Halle and Isardi 1995), foot-free models of stress assignment remain current as well (e.g., Gordon 2002, van der Hulst 2009, Vaxman 2016; see further discussion in Chapter 2). Precisely the availability of this latter alternative undercuts a substantial amount of the empirical grounding in favor of feet as a prosodic constituent: if all phenomena directly pertaining to stress and rhythm are explicable without reference to feet, what purpose do they really serve? Gordon (2016: 11) further observes that a theory of prominence relations without feet (i.e., the grid alone) is more parsimonious.

Beyond views of prosody that deny the foot altogether, another recent strain of thought has argued that the employ of foot-structures is language-specific, i.e., that feet do not belong to a universal repertoire of prosodic constituents (so Özçelik 2017 and Garcia and Goad 2021), whose presence or absence on a language-particular basis may be regulated by HEAD-EDNESS (cf. 18b.), which is often assumed to be universally inviolable; see further discussion below. Independent of the foot's universality in the Prosodic Hierarchy is then the further question of whether feet are *n*-ary branching (as argued in e.g., Nespor and Vogel 1986: 84–90), as appears to be the case for most prosodic constituents, or whether they are strictly binary branching (so Hayes 1995: 55–6, Bennett 2012), and if so, why feet should generally exhibit a more restricted range of licit structures. In the following, I will focus primarily on the first of these questions, i.e., evidence for feet as prosodic constituents, before finally considering whether a category of languages with stress but without feet constitutes a viable type; throughout, I will assume that feet are maximally binary branching.⁵¹

⁵¹From an optimality-theoretic perspective, the construction of feet with different sorts of branching structures may be regulated through constraint rankings. Thus, different languages can prefer the construction of

If the assignment of stress cannot provide evidence of foot structure in a given language, segmental processes, and in particular segmental processes that cannot be formulated in terms of reference to stress, must instead be relied upon. Kiparsky (1979) offers an early example for analyses of this sort, attributing the tapping of coronals in North American English to their presence in the weak branch of a foot. Bennett (2012: 20) formulates a useful heuristic for the identification of segmental processes that may be driven by foot structure: “whenever the phonological properties of particular syllables depend on the properties of an adjacent syllable, we may be dealing with a foot-based phonological pattern. This is especially true when the phenomenon in question is sensitive to stress, syllable weight, or syllable count.” Certainly the number of such processes to be found cross-linguistically is not insignificant.

Bennett 2012 represents an attempt to amass data on such phenomena, as, for instance, the lenition of non-geminate stops /p t k/ in Ibibio to [β t γ],⁵² apparently limited to the onset of the weak branch of a trochaic foot, whose left edge is aligned with the left edge of the root (see Bennett 2012: 21–4, Harris and Urua 2001, Akinlabi and Urua 2002, Harris 2004). Much of the data would be compatible with a general process of intervocalic lenition from which root-initial stops are exempt, but at least the data in 44c. and d. does not admit of such an interpretation: in c., the negative suffix /ké/ undergoes lenition to [yé] when preceded by a monosyllabic root, but not when preceded by a disyllabic root, while in d., lenition applies in the medial syllable of a phrase built around two grammatical words. The examples in 44c. thus exclude the interpretation that stop lenition is restricted to root segments, while those in d. exclude the interpretation that the process applies only in the final syllable of disyllabic forms.

(44) Ibibio Stop Lenition

- a. Suffix /-V/ induces lenition
 - i. [(bèt)] ‘push’
 - ii. [(bèr-é)] ‘be pushed’ (^x[bèté])
- b. Prefix /V-/ does not induce lenition of a root initial vowel
 - i. [í-(bàt-tá)] ‘s/he is not counting’ (^x[íbàttá])
 - ii. [á-[tòòr-ó)] ‘s/he is praising’ (^x[áròòró])
- c. Negative /ké/ undergoes lenition after monosyllabic roots
 - i. [(séé-yé)] ‘not look’ (^x[sééké])
 - ii. [(sà.ŋá)-ké] ‘not walk’ (^x[sà.ŋà.yé])

feet of different sizes, and within the same language, different-sized feet may be permitted, depending upon the exact conditions. Foot binarity seems quite strongly enforced in many, if not all languages, perhaps especially on the upper end; this feature contrasts with most other constituents of the prosodic hierarchy (see the preceding sections on phonological phrases and prosodic words), which otherwise appear to be *n*-ary branching. However, the free deployment of recursive structures at and above the level of the prosodic word may starkly reduce the number of ternary or larger constituents necessary in prosodic structures.

⁵²Harris (2004: 16) indicates that the lenited segments transcribed as [β] and [γ] are to be understood as bilabial and velar tapped approximants, rather than proper voiced fricatives.

- d. Lenition applies at the boundary to a vowel-initial word
- i. /bèt ówó/ → [(bèr.ó).wó] ‘push someone’ (^x[bè.tó.wó])
 - ii. /kλk úsάη/ → [(kλ.ýú).sάη] ‘shut the door’ (^x[kλ.kú.sάη])

Although one might conceivably claim that stop lenition applies following the first syllable of a root, analyzing this phenomenon without a phonological constituent regularly built out of two adjacent syllables would seemingly result in a substantial loss of generality. Further striking in this case is the fact that Ibibio apparently lacks any sort of stress prominence: the strong member of the hypothesized foot is not obviously marked via any phonetic cues typical of stress (Harris and Urua 2001); the language exhibits only contrastive lexical tone (cf. 1.3.2 below). See also the study of Penner 2019 for extensive evidence in favor of foot-based generalizations to capture a variety of phonological and morphophonological phenomena in Ixtayutla Mixtec, likewise usually built around a root.

Similarly attributable to considerations of foot structure is an instance of apparent trochaic shortening (cf. generally Hayes 1995: 142–9) involving word-formation in Ancient Attic-Ionic Greek, described in Gunkel 2010: Ch. 1 and Gunkel 2011. In this case, neuter nouns built with a suffix -μα(τ)- /-ma(t)-/ (typically with resultative semantics) such as ῥήγ-μα [rê:g-ma] ‘break, tear’ frequently present a diachronically older form with a long vowel or diphthong (e.g., χεῦμα [k^heûma] ‘that which is poured’), but a diachronically younger form with a short monophthong (e.g., χύμα [k^húma] ‘id.’); see examples listed in (45). This change in the phonological structure of -μα(τ)-stems is subject to two conditions: the root must not end in a consonant (e.g., always ῥήγμα [rê:g-ma], no innovative ^x[régma]), and the “short” allomorph of the root must exist independently in the language.

(45) Trochaic Shortening in Ancient Attic-Ionic Greek

Other Nominal Form	Gloss	Older μα-stem	Younger μα-stem	Gloss
k ^h ú-sis	‘(act of) pouring’	x ^h eû-ma	k ^h ú-ma	‘that which is poured’
heure-sis	‘discovery’	heure:-ma	heure-ma	‘invention’
klí-sis	‘inclination’	klí:-ma	klí-ma	‘direction, region’
rú-sis	‘flow’	reû-ma	rú-ma	‘stream, current’
ap ^h -aíre-sis	‘taking away’	par-aíre:-ma	ap ^h -aíre-ma	‘tribute’
pó-sis	‘drink(ing)’	pô:-ma	pó-ma	‘drink’
súm-bá-sis	‘going together’	be:-ma	para-súm-ba-ma	‘secondary circumstance’
p ^h óre-sis	‘wearing (of clothes)’	p ^h óre:-ma	p ^h óre-ma	‘that which is carried/worn’
aná-t ^h e-sis	‘dedication’	aná-t ^h e:-ma	aná-t ^h e-ma	‘that which is dedicated’
dó-sis	‘(act of) giving’	*dô:-ma	dó-ma	‘gift’
én-e-sis	‘injection’	hê:-ma	én-e-ma	‘injection’
dé-sis	‘binding’	hupó-de:-ma	dé-ma	‘band, rope’
kρί-sis	‘decision’	krí:-ma	krí-ma	‘(matter for) decision’
*épse-sis	‘(act of) boiling’	épse:-ma	épse-ma	‘that which is boiled’
sk ^h é-sis	‘condition’	sk ^h ê:-ma	sk ^h é-ma	‘form, figure’
el-lip-é:s	‘omitting’	leîm-ma	leîm-ma	‘remains’
haimo-rag-ía	‘hemorrhage’	re:g-ma	re:g-ma	‘break, tear’
lab-é:	‘grip, hold’	lê:m-ma	lê:m-ma	‘that which is received’
dák-os	‘bite, sting’	dê:g-ma	dê:g-ma	‘bite, sting’
sú-zdug-os	‘paired’	zdeûg-ma	zdeûg-ma	‘something used for joining’
pág-ε:	‘thing that fastens’	pê:g-ma	pê:g-ma	‘thing that is fastened’

Under the assumption that moraic trochees are the preferred foot form in Ancient Attic-Ionic, selection of the allomorph with a short vowel serves to maximize syllable parsing and allow the right edge of the foot to be nearer to the right edge of the word in the nominative-accusative singular of the $-\mu\alpha(\tau)$ -stems: $[(k^heû).ma]$ is of the shape heavy–light and thus leaves the final syllable unparsed (or builds an overlong trimoraic foot $[(k^heû.ma)]$), while $[(kú.ma)]$ is of the shape light–light. The emergence of innovative forms like $[k^húma]$ is lexically conservative (in the sense of Steriade 2001), in that the preference for an optimal moraic trochee does not induce the creation of novel allomorphs, and when a root-final consonant excludes the possibility of a light first syllable, no short-voweled allomorph is introduced (cf. $[(dê:g).ma]$ ‘bite, sting’, never $^x[(dág).ma]$, alongside $[dákos]$).⁵³ While the innovative $-\mu\alpha(\tau)$ -stems alone might be explained via a prosodic preference for disyllables consisting of two light syllables, without reference to foot structure, such an interpretation becomes less appealing in the light of accentual analyses that similarly rely on the moraic trochee (e.g., Golston 1990; see further Chapter 6) and phenomena of vowel lengthening that may (originally have) serve(d) to avoid unparsed light syllables (e.g., the so-called “tribrach law”, on which see most recently Steriade 2018).

Certain phenomena of prosodic morphology, in particular from the domain of reduplication, much like minimal word requirements (cf. under 1.2.2 above), also allow for attractive interpretations in terms in foot structure when a prosodic template may be satisfied in multiple ways, e.g., through two light syllables or one heavy syllable in prosodic systems in which the preferred foot shape is a moraic trochee. (Vedic) Sanskrit presents a wealth of productive and semi-productive verbal formations built through reduplication (see Kulikov 2005 for an overview), most of which generally exhibit prefixing partial reduplicants consisting of a single syllable.⁵⁴ The reduplicant of the so-called “intensive” exhibits perhaps the greatest diversity of reduplicants, appearing as CV, CVC, and CVCV, illustrated in 46 (all data from Schaefer 1994); this diversity of shapes may be unified as a moraic trochee.⁵⁵ In other words, RED = $\mathcal{F}$ for this morphological category.

⁵³Outside of the nominative-accusative singular, other case forms of $-\mu\alpha(\tau)$ -stems become trisyllabic, e.g., GEN.SG $[-mat-os]$, NOM.PL $[-mat-a]$, where the short root allomorphy would not be optimizing in terms of foot structure. Presumably the substantially higher token frequency of the nominative-accusative singular means that it is treated as a paradigmatic base and precisely its phonological structure is subject to optimization.

⁵⁴The literature on the reduplicated verbal categories of (Vedic) Sanskrit is extensive. Three of the categories (reduplicated present, desiderative, and perfect) generally build CV-reduplicants that may become heavy by position; useful formal analyses of the perfect especially are given in Steriade 1988, Sandell 2015a: Ch. 8, Zukoff 2017b: Ch. 5, and Cooper 2015: Ch. 3 on the explicit avoidance of superheavy reduplicants.

⁵⁵At first glance, the Vedic reduplicated causative aorist (e.g., $(a)jījanat$ ‘caused to be born, generated’; $śísraṭhat$ ‘made pierce’) similarly might appear to require a bimoraic reduplicant, but the phonological restriction appears to hold over the stem, requiring that its first syllable be heavy: the reduplicant of the causative aorist allows only the vowels $[i:]$ and $[u:]$, which are subject to lengthening just in case the root does not have at least two consonants at its left edge (contrast $(a)jījanat$ and $śísraṭhat$). The requirements on the shape of the intensive reduplicant apply independently of the shape of the following syllable, that is, the intensive’s reduplicant must be bimoraic as a free-standing unit, whereas the reduplicated aorist’s reduplicant may be bimoraic by position.

(46) Sanskrit Intensive Reduplication

- a. Reduplicant = CV:
 - i. *cā-kaśī-mi* ($\sqrt{kāśī}$) ‘RED-consider-NPST.ACT.1SG’
 - ii. *ce-kit-e* ($\sqrt{cīt}$) ‘RED-recognize-NPST.MID.3SG’
- b. Reduplicant = CVC
 - i. *car-kr̥ṣ-a-t* ($\sqrt{kr̥ṣ}$) ‘RED-plow-SUBJ.-ACT.3SG’
 - ii. *cañ-kram-ata* ($\sqrt{kram}$) ‘RED-stride-MID.3PL’
- c. Reduplicant = CVCV
 - i. *kari-kr-at* ($\sqrt{kṛ}$) ‘RED-make-PART.ACT.NOM.SG.M’
 - ii. *kani-kran-ti* ($\sqrt{kran}$) ‘RED-shout-NPST.ACT.3SG’

Beyond evidence from segmental alternations, optimal allomorph selection, and prosodic morphology, some amount of psycholinguistic and neurolinguistic evidence in favor of the presence of foot structures may be available. Knaus and Domahs (2008) and Domahs et al. (2008), for instance, interpret ERP experiments involving certain incorrectly stressed lexemes of German as most readily explained by the need to reparse the incorrect stresses with different metrical structures. The results of experiment 1 in Domahs et al. 2008: 18–20, for instance, found enhanced late positivity effects (P300) in the lexical identification of lexemes heard with an incorrect stress pattern, depending upon the position of the correct lexical stress and the position of the incorrect stress in the test item.⁵⁶ Among trisyllabic words with correct antepenultimate and final stress (e.g., *Ánanas* ‘pineapple’ or *Vitamín*, both presumed to contain a foot structure of the form $[(\sigma\sigma)(\sigma)]$), only the incorrect penultimate stress produced this P300 effect, while among trisyllabic words with correct penultimate stress (e.g., *Bikíni*, presumed to contain a foot structure of the form $[\sigma(\sigma\sigma)]$), both an incorrect antepenultimate or final stress resulted in the P300 effect. The interpretation provided by the authors is that, since trisyllabic forms with initial or final stress share the same foot structure, no reparsing of prosodic constituents is needed to identify the form with the opposite primary stress (i.e., final or initial, respectively); conversely, identifying a trisyllabic lexeme with correct initial or final stress when presented with penultimate stress, or one with correct penultimate stress in any other position, does necessitate a parsing of prosodic constituency contrary to the expected constituency, thereby yielding the demonstrated P300 effect.

Artificial grammar learning experiments conducted by Bennett (2012: Ch. 4) with native speakers of English and Japanese (see also Bennett 2014 on the latter), meanwhile, may be interpreted to suggest a bias for learners to understand segmental alternations in terms of foot structure when such an interpretation is available, even when a stress-based phonotactic explanation of the learning data would be equally possible. In these experiments, training data, consisting of trisyllabic CV sequences with either initial or peninitial stress, excluded the vowel [u] from post-tonic syllables; the training data would thus be compatible with either the foot-based generalization that [u] cannot occur in the weak branch of a trochaic

⁵⁶P300 is an event-related potential that shows a positive movement in voltage as measured by an electroencephalogram. Delays in the peak of P300 are often reported in psychological and neurolinguistic literature where the stimulus is unexpected or of lower probability.

foot (e.g., *[(CV.Cu).CV]), or with a purely stress-based generalization that disallows [u] in the syllable following the stressed syllable (e.g., *[CV.'CV.Cu]). In two testing phases, participants were first presented again with trisyllabic stimuli, then with five-syllable stimuli, and asked to decide whether a given word could belong to the “same language” as heard in the training phase, or not. The real interest is to determine whether participants regard [u] as ungrammatical in the weak branch of a covert foot not signaled phonetically by stress, i.e., in the penultimate syllable of words with initial stress (*['(CV.CV).(CV.Cu).CV]) or the final syllable of words with peninitial stress (*[CV.(CV.CV).(CV.Cu)]). For both native speakers of English and native speakers of Japanese, ungrammaticality of [u] was extended to the covert foot to statistically significant degrees when stress was peninitial, and at (Japanese-speakers) or approaching significance (English-speakers) when stress was initial.⁵⁷

Such experimental results help to make sense of cases such as the Ibibio one seen in 44, which raise the question of how foot structure is to be acquired without overt phonetic cues to the constituent. As it stands, the synchronic process of stop lenition is evidently acquisible and productively applicable so long as speakers have the option to refer to feet as a universal prosodic category. In origin, I suspect that restriction of stop lenition to a weak foot branch reflects a form of domain generalization, whereby originally unbounded phonological processes are attributed to progressively smaller phonological domains of the word, foot, or syllable (Padgett 2015, Kiparsky 2015), under the assumption that these categories are available for learners to exploit in organizing learning data.

The combined force of the phenomena and experimental evidence reviewed here makes the usage of feet as an analytical device in prosodic structures at minimum very attractive; to account for the same range of facts without reference foot structure appears to be a difficult, if not impossible task. Having seen that foot structure can be profitably employed free of overt stress indicative of rhythm, the further question is whether stress may occur as a parameter independent of feet (contrary to the STRESS-HEADEDNESS HOMOMORPHISM of Bennett (2012: 34): “all stressed syllables are foot heads, though not all foot heads are stressed”). Özçelik (2017) advances the argument that the availability of the Foot as a constituent of the prosodic hierarchy involves a parameter setting, and that certain languages may then indeed be entirely “footless”. Such a move requires, however, that HEADEDNESS (cf. 18 above) be treated as a violable constraint on prosodic structures, along with EXHAUSTIVITY and RECURSION. In the typology developed by Garcia and Goad (2021), four language types emerge, depending upon the violability of HEADEDNESS and OBLIGATORINESS of word-level stress (see 1.3.3.2 below), instantiated by the languages listed in 47.

⁵⁷Bennett (2012: 288–93) attributes the failure to clearly extend the phonotactic restriction to initially-stressed 5σ forms among English speakers primarily the possibility that initially-stressed forms could be understood as containing non-adjacent feet (i.e., [(CV.CV).CV.(CV.CV)]), which would lead to the expectation that [u] would not occur post-tonically or in final syllables.

(47) Foot and Word Stress Typology (after Garcia and Goad 2021: 26)

Foot	OBLIGATORINESS	Language
yes	yes	English
no	yes	Portuguese
yes	no	(Québec) French (cf. Guzzo et al. 2018)
no	no	Turkish

Granting that stress is not a phonological universal (in accord with standard assumptions) and that not all foot heads need to be instantiated through stress, the existence of languages of the (Québec) French type is unsurprising, and insofar as prosodic HEADEDNESS is indeed violable, the Turkish-type without feet or obligatory word-level stress is similarly possible.⁵⁸ The postulated Portuguese type – obligatory word-level stress without feet – raises the question of how stress should be assigned when feet are theoretically available but supposedly not employed (in contrast to foot-free models of stress assignment).⁵⁹ Garcia and Goad (2021) do not explicitly address this question, and the evidence mustered in favor of the footlessness of Portuguese (inconsistent stress assignment, attraction of stress to heavy antepenultimate syllables, absence of bimoraic word minimum) strikes me as insufficient. Similar issues concerning exceptional stress, WEIGHT-TO-STRESS phenomena, and unclarity of foot shape may be observed in Standard Italian, for which a credible and systematic analysis in terms of right-aligned moraic trochees with additional WEIGHT-TO-STRESS effects and faithfulness to exceptional lexical stress is available (see Krämer 2009: Ch. 6). Insofar as entirely different foot shapes (iambs and trochees) are necessary to capture the range of Portuguese data, this in itself may not pose a problem, given the increasing number of recent analyses argue for the coexistence of different foot shapes within a single language (e.g., the analysis of accentuation in Uspanteko developed in Köhnlein 2019). I therefore adopt the position that stress signals foot structure, while the absence of word-level stress is compatible with either the presence or absence of foot structure.

Under 1.2.2, a possible distinction between intrinsic and extrinsic motivations for prosodic structure was mentioned, whereby inheritance from the syntax would account for prosodic recursion. Conversely, recursive constituents below the word are not expected. Nevertheless, recursive feet have been proposed to account for supposed cases of ternary rhythm in a handful of languages (e.g., Chugach Alutiiq Yupik, Tripura Bangla, Cayuva; cf. Rice 2011: 1230–1, Paricio and Kager 2015). Golston (2021), however, argues against the need for recursive feet in general, and against their necessity in the well-known cases of apparent ternary rhythm specifically. The recursivity of feet is thus evidently rare, if it occurs whatsoever.

In summary, I will apply the following assumptions concerning feet and foot structure

⁵⁸As Özçelik (2017: 161–75) discusses, Turkish does possess “exceptional” stress on non-final syllables introduced by underlying foot boundaries (on Özçelik’s account; cf. also Özçelik 2014); feet may then be present in outputs, but are not present in the output just in case no foot boundaries are present in the input. Crucial for the entire approach is the claim that the “regular” stress of Turkish on the final syllable is in fact not stress at all, but rather a high boundary tone (see Özçelik 2014: 234–8 and further references there)

⁵⁹For foot-free analyses of stress assignment in Optimality Theory see Kager 2001, Gordon 2002, or Stanton 2015.

throughout this work: 1) in accord with HEADEDNESS, all prosodic words (ω) must dominate at least one foot ($\mathcal{F}$); 2) feet are maximally disyllabic (i.e., no unbounded feet encompassing all syllables in a prosodic word), but may be defective (monosyllabic/monomoraic) or over-long (having more than two morae); 3) stress is to be interpreted as an indication of a foot head.⁶⁰

1.2.4 Syllables and Syllabification

Since the view promulgated in Stetson (1951), namely, that syllables find a phonetic correlate in the form of “chest pulses”, has long been abandoned (see the counterevidence of Ladefoged 1967), the syllable constitutes a similarly abstract unit embedded in the phonology, motivated essentially on the grounds that reference to it enables simplification and generalization in the statement of phonological processes. While early generative phonology denied the reality or necessity of syllables in phonological analysis (syllables play no role in *SPE*), investigations by Vennemann (1974), Hooper (1976), and especially Kahn (1976) (cf. Féry and van de Vijver 2003: 3) largely rehabilitated reference to syllables as a succinct means of capturing conditions on process application.⁶¹ Kahn (1980) systematically demonstrates the greater ease with which several processes relating to /t/ in North American English (aspiration, glottalization, and tapping) can be stated in contrast with the notional devices made available by *SPE*. Similarly, without the syllable as a phonological domain, the at least six syllable-sensitive rules of Mexican Spanish originally posited by Harris 1969 (cf. Nespor and Vogel 1986: 75) would require reformulation with the environment $\left\{ \begin{smallmatrix} C \\ \# \end{smallmatrix} \right\}$ as opposed to $]_{\sigma}$.

As a basic example, consider the fate of voiced obstruents in German: these are regularly devoiced when preceding another voiceless obstruent, many morpheme boundaries, or a word boundary; without reference to the syllable, the devoicing process might appear to apply in several disparate contexts.⁶² Following the general treatment of Hall 1992 (cf. also Kleinhenz 1998: 6;), given that syllabification is word-bounded in Standard German, obstruent devoicing can then apply to any obstruent in a syllable coda, even when the following segment may (underlyingly) be a vowel (as in 49b.), sonorant consonant (49c.), or voiced obstruent (49d.), even when morphologically simplex (49e., though in such cases the UR is

⁶⁰On cases of apparent conflict between segmental diagnostics of foot structure and stress diagnostics, see the detailed study of Nganasan in Vaysman 2009, as well as Hermans 2011, and for a reinterpretation of some of such cases in terms of intonational pitch accent, Gordon 2016.

⁶¹Goldsmith (2011: 172–3) observes, however, that investigations into the syllable were alive and well outside of the generative phonology of the 1960's; he notes that Pulgram 1970, for instance, contains many observations encoded into later approaches to the analysis of syllables, e.g., onset maximization, coda avoidance, and ambisyllabicity of segments.

⁶²Arguably, the morpheme boundaries at which obstruent devoicing applies could be subsumed under the word boundary: obstruent devoicing occurs the boundary between prefix and root (e.g., *ab-artig* [ʔap,ʔaɐ̯.tɪç] ‘abnormal’), as well as preceding many derivational suffixes, e.g., *-lich* ‘-like’, which might all be treated as a containing prosodic word boundary, with a recursive ω then incorporating these further affixes. The *SPE* boundary symbol # could be used to capture both contexts, but an analysis in terms of prosodic constituency would find itself referring to a left and right word boundary, and thus be unable to nicely generalize over the two contexts without the syllable.

perhaps likely restructured to contain a voiceless obstruent).⁶³ While the environment of application for obstruent devoicing could, in most cases, be rewritten with curly brackets to $\left\{ \begin{smallmatrix} C \\ \# \end{smallmatrix} \right\}$, cases like 49b. are arguably problematic, since the glottal stop [ʔ] preceding the devoiced /b/ is derived; the consequence is that glottal stop epenthesis would necessarily feed obstruent devoicing, but glottal stop epenthesis itself is apparently motivated by the need to provide a syllable onset, which the preceding /b/ in 49b. could, in principle, provide. Most instances of glottal stop epenthesis can be captured through phonotactic constraints (*#___V and *V___V_[+stress]; cf. Hall 1992: 58–9), but cases like /ab-art-ɪg/ do not fulfill those conditions (unless, again, an *SPE* #-boundary is placed both before and after the prefix /ab-/). The sum of these facts allows the (non)-interaction of obstruent devoicing and glottal stop epenthesis to be most efficiently captured by reference to syllables (and feet).

(48) German Syllable-Final Obstruent Devoicing: Rule

$[-\text{sonorant}] \rightarrow [-\text{voice}] / ___\sigma$

(49) German Syllable-Final Obstruent Devoicing: Examples

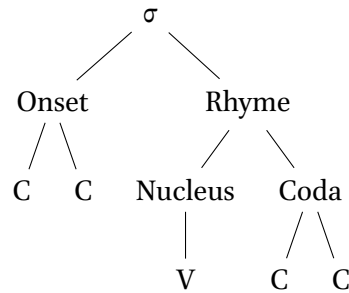
- a. /haʊz/ → ['haʊs] *Haus* 'house'
- b. /ab-art-ɪg/ → [ʔap.ʔaɐ̯.tɪç] *abartig* 'abnormal'
- c. /bynd-nɪs/ → ['bynt.nɪs] *Bündnis* 'alliance'
- d. /leːz-bar/ → ['leːs.baɐ̯] *lesbar* 'readable'
- e. /ʁaɡbi/ → ['ʁak.bi] *Rugby* 'rugby' (cf. English ['ɹʌɡ.bi])

For the purposes of the present work, the exact nature of syllabic representations and their precise relation to syllable weight (on which see further 1.2.5 below) is not crucial. Whether syllables contain internal constituents of onset, rhyme, nucleus and coda (as represented in 50a.), directly dominate moras or other timing units (as in 50b.), or exhibit a flat structure directly dominating segments and are themselves bearers of distinctions in duration, will not have crucial implications in most cases. As long as syllables, as potential bearers of stress or similar prosodic prominence, are available to be organized into feet, the analyses of the phenomena in and Part II will not founder on a faulty choice of syllabic representation. I will generally operate with the type of representation shown in 50b., and, given the violability of EXHAUSTIVITY (cf. 18c.), accept that “extrasyllabic” segments may adjoin to higher-level prosodic constituents above the syllable (at least the ω); see Vaux and Wolfe 2009 for extensive arguments and evidence in favor of extrasyllabic segments, which in their force serve as evidence for syllables as prosodic constituents generally, and Byrd 2015 on their role in older Indo-European languages in particular.

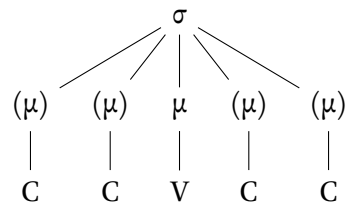
⁶³In cases of clusters of obstruents in a syllable coda in which an obstruent not at the right edge of the syllable undergoes devoicing, e.g., [lo:pst] ‘praise:PRS.2SG’ from /lo:b-st/, one could reckon with a version of 48 that also applies when further consonants follow (i.e., /___C_o]σ), or with an independent process of regressive voicing assimilation.

(50) Three Models of Syllable Structure and Constituency (cf. Goldsmith 2011: 168–71)

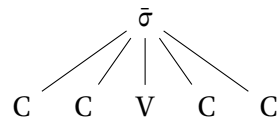
a. Internal Hierarchical Constituency



b. Flat Structure Dominating Timing Units



c. Flat Structure Dominating Segments



Nevertheless, worth addressing at this juncture are two general alternative approaches syllabic constituency (or lack thereof): on the one hand, there are essentially phonetically grounded approaches to phenomena that appear to be conditioned by syllable structure, which would deny the psychological reality of any syllable-like prosodic constituent for the organization of segments (e.g., Dressler and Dziubalska-Kolaczyk 1992, Ohala and Kawasaki-Fukumori 1997: 355–60, Steriade 1999, Samuels 2009: 99–130); on the other, work by Steriade has proposed to revive interval theory, where the “intervals” as alternatives to syllables may or may not be interpreted as constituents. First, one must admit that segmental processes attributed to the effects of syllable structure can generally be recast in terms of linear phonotactic restrictions (so Steriade 1999, Côté 2000, and Blevins 2003), even if doing so may appear necessitate greater complexity in the grammar (i.e., more constraints) or result in a loss of generality.

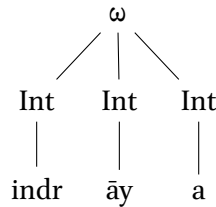
One commonly adduced argument against syllabic organization is that the number of syllables in a word is not distinguishable from the number of sonority peaks (i.e., one could simply count peaks), but that participants in experimental settings cannot reliably identify the edges of syllables (Steriade 1999: 228–32).⁶⁴ Nevertheless, some studies (e.g., Treiman

⁶⁴The argument made by Samuels (2009: 103–4) concerning English *fucking*-infixation at [s]+stop clusters, namely, that, because speakers tendentially produce forms like *des-fucking-spotic* or *dis-fucking-structable*, with “copy” of the [s], they do not know how to parse the string syllabically, does not really hold water. As Samuel’s own discussion indicates, all other alternatives – with the exception of vowel lengthening of the first

and Danis 1988) indicate “substantial agreement” among speakers concerning the placement of syllable boundaries. Furthermore, interspeaker differences in syllabification in some cases could be interpreted as reflecting slightly different phonological grammars, or, in some well-known instances (e.g., English *lemon* /ləmən/), complicating factors could be responsible for the differences in response: the task of producing word forms with indications of syllable boundaries requires treating each syllable as its own prosodic word, for which reason, in a case like /ləmən/, the ban on [ɛ] as a word-final segment in English may be responsible for the reluctance of speakers to produce the phrase [[lə]_ω [mən]_ω]_φ to indicate the syllabification of *lemon* (cf. Steriade 2002: 10).

Interval Theory, as an alternative view of segmental organization, potentially holds more promise. Work in this direction has been carried out in largely unpublished form by Steriade (Steriade 2008, Steriade 2011, Steriade 2012, Steriade 2019: 174–6), but the major insights are available in summary form in Ryan 2016 and Ryan 2019: 153–8, to which I refer the reader for arguments in favor of operating with intervals.⁶⁵ As Ryan (*op. cit.*) observes, an intervallic conception of segmental organization with implications for syllable weight has a basis in thought surrounding prosody much older than 20th-century conceptions of the syllable: both Pāṇini (*Āṣṭdhyāyī* 1.4.11) and some philologists in the 19th and early 20th centuries (e.g., Pipping 1903: 1, cited by Ryan 2016: 725) treat words as divisible into units that span the interval from the beginning of one vowel to the beginning of the next vowel. A Sanskrit form such as *indrāya* ‘Indra:DAT.SG’ would thus be parsed into three intervals: *indr-āy-a*. Insofar as intervals are taken to be constituents, the prosodic organization of *indrāya* would be as in 51, and, the process of constructing intervals would, furthermore, be simpler and cross-linguistically invariable: every consonant belongs to the interval constructed on the preceding vowel (or sonority peak).

(51) Prosodic Structure of Sanskrit *indrāya* under Interval Theory (Int = Interval)



As I understand it, the abandonment of the syllable in favor of an interval-based analysis of prosodic weight would not, in turn, necessarily entail the abandonment of the foot, as usually understood, as a further prosodic unit below the level of the word: a foot could be reconceived as a prosodic constituent encompassing intervals of specified sizes. While intervals themselves, as illustrated above, would be structurally different from syllables, they could well play a role in the prosodic hierarchy essentially equivalent to the role

syllable, evident in forms like *de-fucking-pressing* [(di:).(fakɪŋ).(‘pɪɛsɪŋ)] – are ill-formed. Instead, the phenomenon of [s]-splitting in such cases may be interpreted as a strategy for creating a heavy syllable (= foot) preceding the infix, when the [s] properly belongs to the onset of the following syllable.

⁶⁵Interval Theory may also be seen as influencing Yun 2016 and Zukoff 2017b, in which syllables tend to be tacitly avoided as a device for phonological generalization.

usually afforded to the syllable. At the same time, available experimental evidence that has explicitly contrasted syllables and intervals (Garcia 2017b) does not offer decisive psycholinguistic support for one or the other. A major practical issue, however, with attempting to implement interval theory systematically in the context of the present work is simply that no comprehensive theoretical treatment is available – this book would then find itself in the position of simultaneously developing a coherent generative (optimality-theoretic) implementation of interval theory, while simultaneously investigating the diachrony of prosodic systems. Given, however, that Ryan’s (2019) comparative survey of predictions made by interval theory and syllable theory indicates considerable overlap in their predictions, I am cautiously optimistic that the syllable-dependent aspects of the present work could largely be reformulated with intervals without much loss of insight.

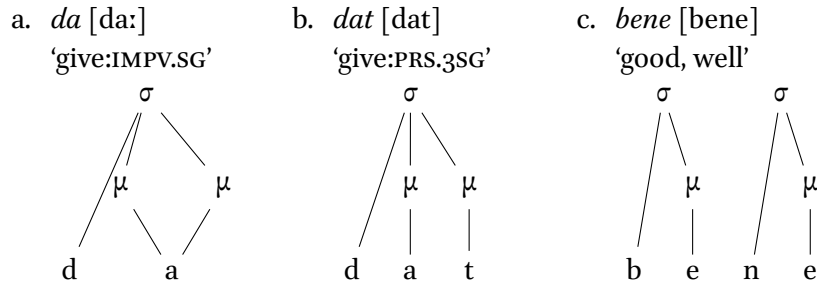
To conclude this section, I wish to consider whether syllables (or something similar like intervals) constitute a consistently obligatory category of prosodic organization cross-linguistically or whether, as has been argued with respect to feet, some languages (e.g., Turkish, on the account of Özçelik) may, by default, skip the construction of such constituents. Ohala and Kawasaki-Fukumori (1997: 355) rightly emphasize that syllables are “not logically necessary” for the transmission of a stream of segments, a view which accords with a bottom-up view of the construction of prosodic constituents; that syllables might be exploited by most (or all?) languages, however, could well be attributed to UG, if prosodic categories are indeed universally provided through an inborn language faculty. Syllables could reflect the phonologization of the identification of local maxima in degree of opening of the vocal tract (*op cit.*: 356), which would make them an epiphenomenon, whether originating in the process of language evolution acquired by each human learner anew.

Some of the most salient empirical evidence bearing on the general question is furnished by Gokana (Niger-Congo, Ogoni subgroup; Nigeria). Hyman (1983, 1985: Ch. 3) originally argued that the phonology of Gokana offers no evidence for the syllable as a prosodic constituent: any and all segmental, morphological, or prosodic phenomena of the language were argued to be insightfully analyzable purely by reference to morae. A more recent reconsideration by Hyman (2011) of the original and further Gokana data led him, however, to the conclusion that some marginal evidence for syllabic constituency in Gokana may exist: the Gokana “prosodic stem” has a maximal size restriction of CVV(C)VV (where each V = μ), which may be interpreted as a maximum of two heavy syllables; only this characterization among conceivable alternatives (e.g., that the prosodic stem contains maximally four morae) accounts for the absence of prosodic stems containing more than two C (e.g., *CVCVCV) or having more than two V separating the two C (e.g., *CVVVCV). The larger point, with respect to possible prosodic universals, is that it is probably correct to say that “all languages exhibit them [viz. syllables] somewhere in their phonology, even if marginally”, though the syllable may not in all languages — as Gokana shows — always constitute the “main organizational principle of phonological constituency”, as Hyman (2011: 81) puts it.

1.2.5 Moras and Other Timing Units: Where is Duration?

The smallest or lowest unit often included in the prosodic hierarchy is the mora (μ), conceived of as an explicit prosodic representation of duration accessible to higher-level prosodic categories. Moraic representations of the two Latin monosyllables shown in (52), in contrast to the disyllabic word of the shape [CVCV], are intended to capture the traditional binary distinction between heavy and light syllables, alongside the seeming equivalence of [CV:] with [CVC] with [CVCV] in meter or the description of certain phonological processes.⁶⁶ In (52), each form contains precisely two units of weight (= two morae). Approaches to the representation of segmental duration and syllable weight are diverse, but may be subdivided primarily into a distinction between moraic theory and skeletal slot theories (CV and X-Slot theory).⁶⁷ Beyond the choice of representation for syllable weight is the question of whether the mora is, in fact, a proper constituent element of the prosodic hierarchy, or whether the syllable in fact constitutes the smallest proper prosodic constituent. In other words, the question is whether prosodic constituents larger than the syllable have access to the moraic or skeletal and segmental information contained within the syllable, or whether they only see different types of syllables (e.g., light and heavy) constructed on the basis of the sub-syllabic constituents that do not belong to the prosodic hierarchy proper. Here, I will concentrate here on a description of the moraic representation that I will subsequently employ throughout, and then briefly discuss some further empirical issues concerning syllable weight, based largely on the treatments of weight phenomena in Gordon 2006 and Ryan 2019 (see also the briefer surveys in Ryan 2016 and Gordon 2017).

(52) Weight Equivalence of Bimoraic Sequences in Classical Latin under Moraic Theory



In the classical version of moraic theory, as expounded by Hayes 1989 (building on earlier similar proposals in Hyman 1985 and McCarthy and Prince 1986 [1996]), the relationship between segment, timing, and syllable is encoded as seen in (52): segmental feature matrices (segments) are linked to prosodic units of timing, morae; both morae and the segments

⁶⁶At least within the domain of weight-sensitive stress phenomena, a categorically binary distinction is adequate for the overwhelming majority of cases: according to Gordon (2006: 21), 86.7% (118/136) of languages in which stress assignment depends on syllable weight have a simple binary criterion (but cf. Ryan 2017: 23 for indications that some of those supposedly binary systems may, in fact, exhibit subtler and richer weight phenomena).

⁶⁷X theory, as first fleshed out in Levin 1985, constitutes a development of the former, rendering CV models seen in, e.g., Steriade 1982 or Clements and Keyser 1983, largely deprecated.

themselves may be dominated by a syllable, though in case a segment projects no unit of timing, violations of EXHAUSTIVITY are in principle incurred, since a syllable directly dominates a segment, in that case.⁶⁸ Moras may be both underlying (e.g., short vowels with one μ , geminate consonants with μ , long vowels with $\mu\mu$) or derived (through Weight-By-Position, a mora is assigned to segment without an underlying mora, when located in a given configuration). Although moraic theory sometimes assumed a limit of maximally two μ slots per syllable — thus fitting the typologically common binary distinction between heavy and light σ — the formal representation itself does not exclude syllables with a theoretically limitless number of morae; see discussion in Steriade 1991: 255–6 on theoretical advantages of a two-mora maximum, and already Hayes 1989: 291–3 on evidence for trimoraic syllables.⁶⁹

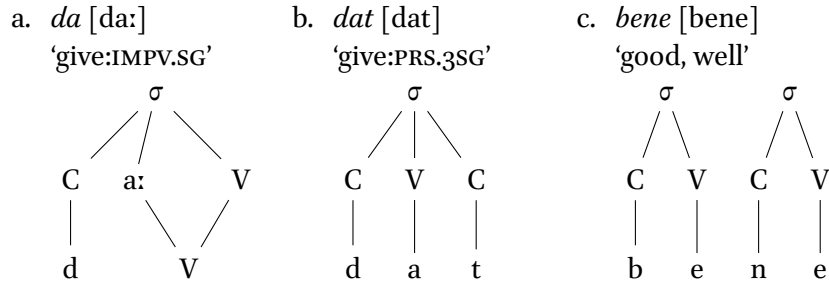
In contradistinction to CV (see Clements and Keyser 1983) and X (see Levin 1985) theory, moraic theory does not operate with any segmental tier, that is, feature matrices of the melodic tier need not be linked to an autosegmental segmental tier, which could be assumed to provide the relevant timing structure as well. Contrast the representations of the same Classical Latin forms from (52) with representations in terms of CV and X theory in (53) and 54; what primarily distinguishes the CV and X representation from moraic representations is that each set of segmental features is necessarily associated with an element (C, V, or X) on a segmental tier. Much of the empirical testing ground for the adequacy of the three models has taken place in the domain of compensatory lengthening (CL) phenomena (e.g., Clements 1986, Hayes 1989, Yun 2013), though at least in the most typologically common cases (e.g., CL from loss of coda segments), the three models handle the facts equally well.⁷⁰ Combining moraic theory and OT presents, however, a special challenge in formalizing CL synchronically where the weight of a segment is derived (e.g., by Weight by Position), at least as long as no additional derivational architecture is admitted (e.g., Stratal OT, Harmonic Serialism); see, however, Topintzi 2010: Ch. 3 and Topintzi 2012 for some possible solutions.

⁶⁸Such violations of EXHAUSTIVITY, frequent where onsets do not contribute to syllable weight, can be avoided if the weightless segment is linked to an independently justified mora; Hyman (1985) in fact regularly opts for such a representation, where his “x” (effectively equivalent to μ) is allowed to dominate all onsets that do not contribute to syllable weight. Arguments for preferring one treatment of weightless segments in moraic theory to the other are not easily forthcoming. However, since the research of Topintzi (2010) indicates that onsets may indeed contribute to syllable weight in some languages (e.g., Trukese, Arrernte, Bella Coola), a distinction between moraic and non-moraic onsets seems clearest if the latter associate directly to syllables. One further possible theoretical argument in favor of the direct attachment of weightless onsets to syllables is developed by Ryan (2019: 81–3).

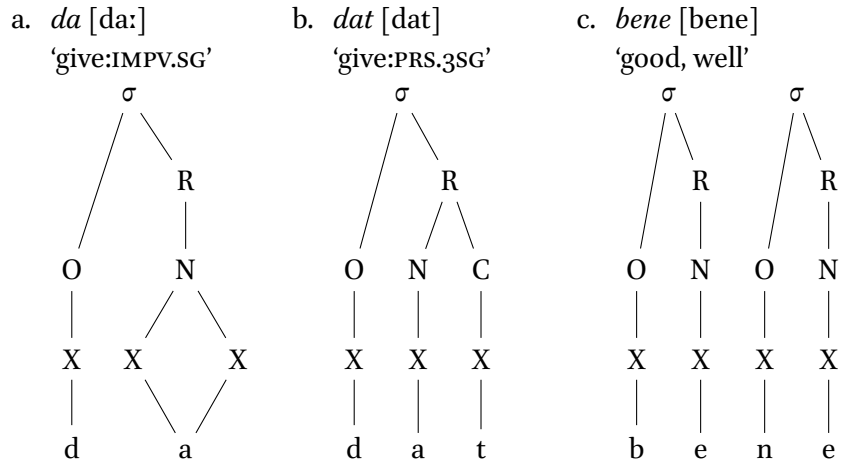
⁶⁹One motivation behind a bimoraic maximum was that it simplified the formulation of the rule of Weight by Position (see Hayes 1989: 258–9). Under an Optimality Theoretic interpretation of Weight by Position, a penalty can simply be assigned to any consonant that does not head a mora, and the weightfulness or weightlessness of consonants in a coda or onset can be regulated by markedness constraints (*MORAIC-ONSET, *MORAIC-CODA) and their relative ranking, also with respect to constraints like $*_3\mu$, which penalize syllables with three or more morae.

⁷⁰Ito (1989) has also argued that some phenomena of vowel epenthesis crucially depend upon a moraic, rather than CV or X representation of subsyllabic timing, though approaches to the treatment of epenthesis in constraint-based formalisms may render Ito’s arguments on this point moot.

(53) Heavy Sequences in Classical Latin under CV Theory



(54) Heavy Sequences in Classical Latin under X Theory⁷¹



In principle, a wholly non-prosodic representation of segmental duration remains, at least in some cases: the distinction between a long and a short consonant or in parallel, a geminate and simple consonant, can be represented with a binary feature such as [+/-long]. Segmental duration would then be treated no from a place feature like [coronal], manner feature like [nasal], or glottal state feature like [voice] (insofar as all of these sorts of features are regarded as strictly non-prosodic). As far as the assignment of segments to syllables is concerned, no obvious problems emerge: a [+long] vowel may be assigned to a syllable nucleus just like a short vowel; a [+long] consonant can be assigned to an onset or coda, or parsed ambisyllabically between a coda and onset. Where, however, an interpretation of segmental duration in terms of a non-prosodic feature potentially breaks down is in its relation to syllable weight. In languages of the Latin or Sanskrit type, a general phonological equivalence between syllables containing a long vowel and syllables with at least one segment in the coda exists: both constitute “heavy” syllables (but see now Ryan 2010 or Ryan 2019 for a more fine-grained approach), though no single segmental feature can be inherited by the syllable in the case of syllables containing a rhyme of the shape [-VC]. Nor can a syllable merely containing a long vowel in its rhyme be readily identified as having a segmentally complex rhyme. Thus, while this segmental approach to weight cannot generalize well to all systems, in some limited cases (e.g., where only long vowels build heavy syllables, as in

⁷¹Here: O = Onset, R = Rhyme, N = Nucleus, C = Coda.

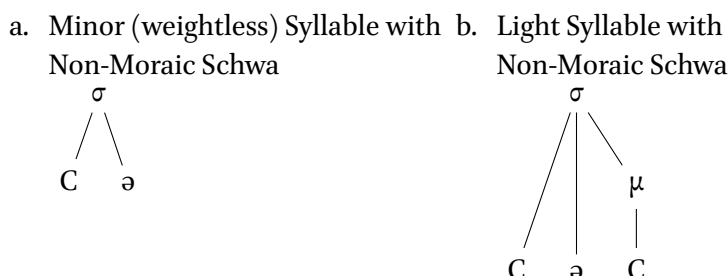
Khalka Mongolian; cf. Zec 1994), it may be descriptively adequate. Compare further Hyman 1985: 5–9 for an overview of different possible configurations and ways of distinguishing light and heavy syllables across a range of languages.

Ito and Mester (2003b: 7–17), meanwhile, explicitly exclude moras or other timing units from the prosodic hierarchy; they adopt rather the position that inherent quantity distinctions and derived syllable weight are precisely properties of segments and syllables, which effectively means that no prosodic constituent exists between the segment and the syllable (cf. also Ito and Mester 2009: 136–7). Assuming a binary distinction between light and heavy syllables, an σ would be designated as either $\check{\sigma}$ or $\bar{\sigma}$, depending upon what sort of and how many segments it dominates. Given the restriction on the occurrence of morae, MORA CONFINEMENT as stated in 55, one peculiarity of morae in contrast to all other elements of the prosodic hierarchy, is that they are evidently *always* dominated by syllables — violations of EXHAUSTIVITY through the association of a μ directly to a $\mathfrak{f}$ seem not to exist whatsoever. Furthermore, one could also do away with the frequent violations of EXHAUSTIVITY entailed by the Hayesian conception of moraic theory, where onset consonants that do not contribute to syllable weight are directly dominated by the syllable, if morae or similar units of duration stand outside the prosodic hierarchy proper.

(55) MORA CONFINEMENT (cf. Ito and Mester 2003b: 11): μ is licensed only by σ .

Violations of EXHAUSTIVITY potentially resulting from prosodic structures created between segment, mora, and syllable may indeed be even more extensive, given the reasonable evidence for so-called “minor syllables” or “semi-syllables” that do not dominate a mora. Such minor syllables are most clearly in evidence in the cases of syllables that evidently possess a vocalic nucleus, but behave prosodically as though the syllable does not contribute to the weight or mora count of a $\mathfrak{f}$ or ω at all, or, in languages in which coda consonants usually contribute to syllable weight, CVC syllables with certain nuclei behave as though they are monomoraic. Evidence for non-moraic schwa in particular, shown in the representations in (56), has now been thoroughly substantiated by Shih (2018a) (building on proposals to this effect in Kager 1989, Féry 1995, and Crosswhite 1999).

(56) Syllable Configurations with Non-Moraic Schwa



Ryan (2019)’s combination of mora-based and sonority-based approaches to syllable weight has the interesting effect that syllables containing different segments can sometimes appear

to behave identically with respect to syllable weight, but at other times may act differently. This approach would allow, as Ryan (2019: 20) discusses, rime sequences of V + obstruent and V + sonorant to both be treated as bimoraic, but nevertheless allow the latter (V + sonorant) to sometimes act as heavier or stronger than the former (e.g., with respect to stress through a mapping constraint that refers explicitly to [+sonorant]: $V_{\mu}[+son]_{\mu} \rightarrow \text{STRESS}$). While Ryan primarily adduces evidence from stress phenomena to support the need for more subtle sonority effects on syllable weight, segmental phenomena may offer support as well. The synchronic activity of Osthoff's Law in Classical Latin (see Yates 2015b), which shortens long vowels in syllables closed by sonorants, suggests that sonorants behave as phonologically longer or heavier than obstruents (in accord with their phonetics; compare, for instance, the data on duration of different syllable types in Gordon 2006: 173–9), even though syllables closed by obstruents and sonorants (as well as syllables containing long vowels) generally behave in a phonologically undifferentiated fashion (i.e., as bimoraic) for the purposes of stress assignment in the language.

Given the different sorts of subsyllabic configurations that may exist, and to which stress is evidently sensitive (to say nothing of tone), the relevant question for present purposes is whether to position of Ito and Mester 2003b mentioned above should be justified. If indeed subsyllabic elements (segments, moras) are wholly invisible to higher-level prosodic structures, and presumably invisible to phenomena dependent upon prosodic constituency and properties (stress assignment, tone linking), at least two potential problems become evident. First, in the domain of stress assignment, insofar as stress assignment may depend upon more subtle weight distinctions that in turn depend upon that properties of the segments involved, or depend upon genuine weight distinctions between not only heavy and light, but also tri- or quadrimoraic syllables, the typology of syllable labelling begins to explode. Theoretically, a language might need to distinguish between σ_{μ} , $\sigma_{\mu\mu}$, and $\sigma_{\mu\mu\mu}$, as well as $\sigma_{V[+son]}$ and $\sigma_{V[-son]}$, creating a six-way distinction in syllable types. From this point of view, allowing higher-level prosodic constituents and prosodic phenomena to read such information directly from an arboreal representation of subsyllabic elements seems more efficient. Second, as far as non-finality effects are concerned (see especially the constraint NONFINALITY-Foot under 2.3.4 in Chapter 2), if a foot to be constructed cannot necessarily distinguish between a syllable with and without a coda consonant, how is the foot to know whether it stands at the right edge of the word or not? I will thus tentatively treat moras throughout as part of the prosodic hierarchy, while acknowledging the somewhat peculiar status that (55) imposes upon them.

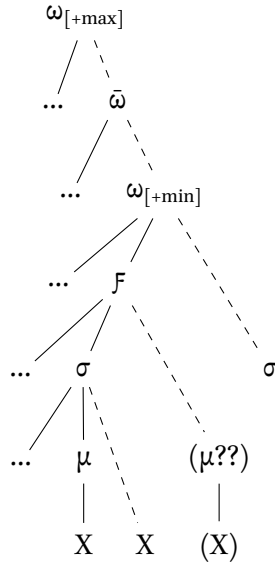
1.2.6 Summary: Adopting a Prosodic Hierarchy

Although the treatment of prosodic constituency and structure in the preceding sections has been fairly extensive, it is hardly innovative, being in its essence and in most details very similar to the view of prosodic constituency presented in e.g., Féry 2017 or assumed without extensive argumentation in e.g., Ryan 2019: 17. Indeed, the version of the prosodic hierarchy shown in (29), with its attendant constituents and admission of violations of EXHAUSTIVITY, NON-RECURSIVITY, and perhaps even HEADEDNESS, might well be said to constitute the pre-

dominant *communis opinio* among phonologists principally concerned with prosodic phenomena at present — though many differing opinions exist and open questions remain, to which the discussion in the preceding sections thoroughly attests.

In the present work, which will primarily investigate word-level prosodic prominence, the prosodic hierarchy above the level of $\omega_{\max}$ will therefore play but a marginal role; discussion of phonological (φ) and intonational phrases (ι) in the languages dealt with most extensively in Part II ([Vedic] Sanskrit, Ancient Greek, Old Germanic), will be comparatively infrequent. The abbreviated prosodic hierarchy shown in (57) may serve the reader as a quick reference of the types of structures relevant at the level of the prosodic word and below that I take to be theoretically licit. Note that, although a “skipped level” showing the association of a mora (and by implication the segment below it) directly to a foot is shown, whether this type of structure exists at all (in contrast to the good evidence for extrasyllabic segments or syllables not parsed into feet) is questionable.

(57) Abbreviated PROSODIC HIERACHY below the $\omega_{\max}$



1.3 Stress, (Stress-)Accent, Pitch(-Accent), Tone, and Intonation

Three seeming universals of (spoken) human language, alluded to under 1.1 above, are the presence of duration, pitch, and volume: spoken language, as an acoustic phenomenon, is perforce measurable along the dimensions of time, frequency, and amplitude of the sound generated. Regardless of whether and to what extent these acoustic phenomena, which can be variously controlled by the human vocal apparatus, play any role in the phonological system of any given language, utterances of spoken language can more or less readily be converted to the form of a musical score, explicitly marking the absolute speed, relative duration, volume, and absolute or relative pitch of “groupings” of segments in an utterance (cf. more or less explicit implementations and similar observations on music–language parallels

in Jackendoff and Lehrdahl 1983, Lehrdahl and Jackendoff 1983, Gilbers 1987, and Schreuder 2006: Ch. 1).⁷² The crucial and interesting questions in the description and analysis of phonological systems concern the *sources* in the phonological component of the grammar and lexicon for distinct patterns in the organization of duration, pitch, and amplitude.

Attempts to adequately describe and characterize the different roles that duration, pitch (= acoustic frequency), and amplitude may play in a given language's phonological system is both an old and ever-present problem: see, for instance, Beckman 1986: 11–26 for an overview and summary of views on notions of “stress”, “(pitch-)accent”, and “tone” in linguistic literature from the 19th and first half of the 20th century; compare in recent literature the contrasting views of e.g., van der Hulst 2012 and Hyman 2014 on the universality of “accent”. Substantial confusion also potentially arises insofar as the same terms are employed to refer to distinct concepts or phenomena, e.g., “pitch accent” to label a (supposed) type of prosodic system or to designate an intonational pattern attracted to a “prominent” element. This overview of questions, problems, and terminology surrounding the phonetic and phonological analysis of stress(-accent), (pitch-)accent, tone, and intonation does not aspire to cut any Gordian knots, but merely to alert the reader to these issues and establish the terms and definitions that I will employ in this work.

This relatively extensive excursus is necessary precisely because some of the languages central to this study, in particularly Ancient Greek (cf. Probert 2003: 1) and Vedic Sanskrit (cf. Lubotsky 1995: 515, Beguš 2016: 3), have regularly been labeled “pitch-accent languages”. Precisely this category label and its appropriateness (i.e., whether it should ever be employed at all, and if so, under which conditions) constitute a source of particularly deep confusion and debate (see, e.g., Hyman 2009 on this matter). The treatment of the issues given here thus proceeds from the comparatively clear and uncontroversial notion of “intonation” and its relation to the somewhat more disputed status of “tone”, before delving into the more fraught issues surrounding “stress”, “accent”, and above all, “pitch accent”. The results of the survey here then have implications for the investigations in Part II, insofar as they provide guidelines and guardrails for the description and analysis of prosody for the languages investigated there.

1.3.1 Intonation

The term INTONATION refers to the exploitation of pitch excursions – phonetically, changes in fundamental frequency (Fo) – in speech. By Féry (2017: 6)’s definition, intonation is “the tonal structure of speech expressed by the melody produced by our larynx. It has a phonetic aspect, the fundamental frequency (Fo), and a grammatical (phonological) aspect.” In its broadest sense, as the phonologically regulated use of pitch, intonation could be understood to encompass tone (see the following subsection) as a special case of intonation (i.e., where pitch patterns are associated with specific lexical items having segmental content). Typically, however, intonation is understood in a somewhat more restricted sense as

⁷²For a brief review of prosodic components of non-spoken (signed) human language, there taken to consist of the temporal properties of and non-manual gestures accompanying signing, see Bentari et al. 2018: 2–4. A more extensive treatment of the matter is available in Sandler et al. 2020.

the set of melodic patterns employed, on the one hand, to grammatically express information about discourse and information structure (e.g., question, focus), and on the other, to mark prosodic constituents above the level of the prosodic word; intonational “tunes” may be mapped onto or spread across phrases or entire utterances (cf. Gussenhoven 2004: 22). The use of pitch in this fashion may well constitute a universal of spoken language, i.e., essentially every spoken language exhibits phrase- and utterance-level changes in pitch that may be analyzed as mapping onto phonological constituent structures (usually interfacing with the syntax) or expressing aspects of information structure. The grammatical usage of pitch for encoding information structure may be found regardless of other usages of pitch in the language (i.e., as tone); see, e.g., Jannedy 2008 and Wang and Féry 2015 on focus intonation in Vietnamese and Mandarin, respectively.⁷³ See further Gussenhoven 2004: Ch. 4 on possible biological codes in intonation and Brunelle et al. 2012: 22–5 on their role in languages with lexical tone.

Acoustically, intonation has one clear and readily measurable acoustic correlate: fundamental frequency (F₀). Although imposing a phonological interpretation on pitch excursions requires care and consideration – it is self-evidently not the case that a given absolute pitch, e.g., 300 Hz can be associated with a phonological pitch entity like a high tone – determining the phonetic realization of a phonological pitch entity is substantially less fraught than in the case of a phonological stress entity. Furthermore, since fundamental frequency is generated essentially by the rate of vibration of the vocal cords, an articulatory correlate for intonation is likewise readily available. A given pitch is also necessarily produced with a particular intensity (it is louder or quieter) and a particular duration (it is shorter or longer in time), and so intensity and duration constitute further acoustic correlates of intonation (cf. Féry 2017: 28–30), though the phonological analysis of intonation does not typically specify intensity and duration as aspects of an intonational tune. Figure 1.1 (from Jannedy 2008: 216) shows a waveform, spectrogram, and pitch track of the Vietnamese phrase *Phuông đi xe đạp* ‘Phuong is riding a bicycle’, with the verb *đi* under focus, illustrating the changes in F₀ over the time-course of an utterance (note the rise in F₀ coinciding with the focused constituent, and fall in F₀ as the phrase continues).

The effects and role of intonation are most evident in languages in which intonation is the primary source of excursions in F₀ during an utterance, i.e., in so-called “intonation languages”. Most West Germanic languages (Standard Dutch, Standard German, and most varieties of English) constitute such intonation languages, in which intonational tunes are associated to a text, where pitches are aligned with and associated to elements of the prosodic structure (cf. 1.2 above), in particular, metrically prominent elements and the edges of prosodic constituents. The situation is somewhat more complicated in the case of languages that exhibit a minimal amount of lexical tone or in which pitch is standardly considered to constitute a primary phonetic realization of abstract “accent” (see the substantial literature on the intonational phonology of Japanese, e.g., Ito and Mester 2013, Ishihara 2015) and most com-

⁷³Brunelle et al. (2012: 25) suggest, however, that, in Vietnamese, “intonation patterns do not appear to be fully grammaticalized”: the authors observe tendencies for questions, for instance, to exhibit rising intonation, but not among all speakers. The functional load of intonation in Vietnamese might therefore be more limited in comparison to clear intonation languages.

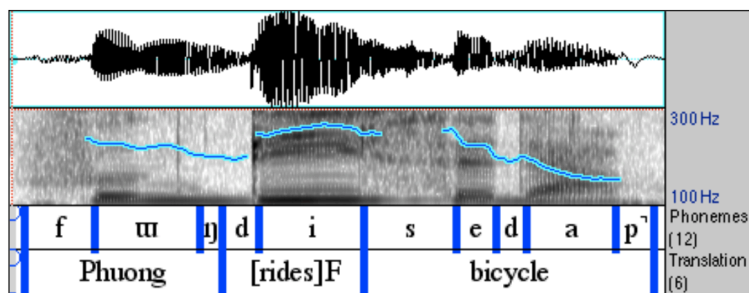


Figure 1.1: Waveform, Spectrogram, and Pitch Track of *Phuồng đi xe đạp* in Vietnamese (from Jannedy 2008: 216)

plicated for languages with relatively dense lexical tone (cf. Féry 2017: 257–69 for a short overview of sentence intonation in Mandarin, Cantonese, and Chichêwa).

Because the intonation will play a comparatively minor role in this book as compared to the analysis of stress and metrical structure, I will forego a detailed introduction to the phonological analysis of intonation, and instead refer the reader generally to Gussenhoven 2004. A brief introduction to the concepts and principles of the Autosegmental-Metrical (AM) model of intonational phonology is nevertheless in order here; see further Pierrehumbert 1980 (in large part inspired by the earlier work in Liberman 1975 and Bruce 1977), Ladd 1996 for a detailed introduction to the topic, or Arvaniti and Fletcher 2020 or Féry 2017: Ch. 5 for a briefer overviews.

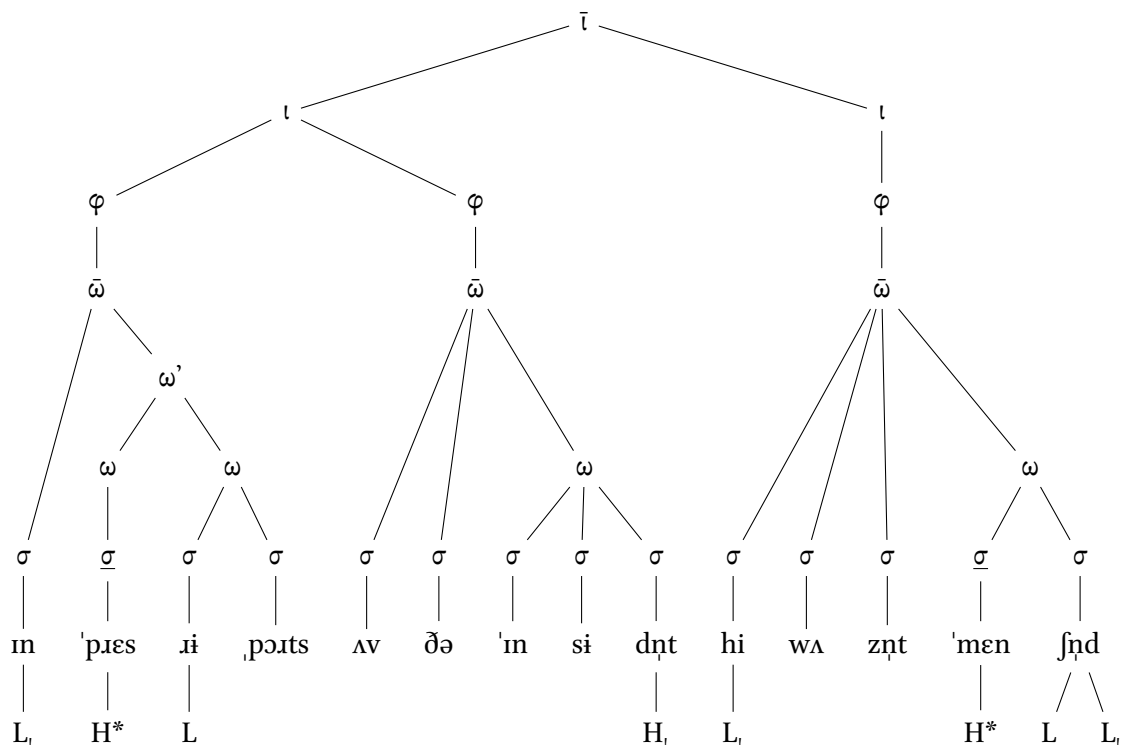
First, as the name implies, AM treats intonation through the interaction of an autosegmental tier for melodies with metrical structure and prosodic constituents. The AM model assumes that languages possess a lexicon of tonal morphemes (“tunes”) that contribute to the semantic and/or pragmatic interpretation of an utterance. Phonologically, the tunes consist of (sequences of) tones specified for pitch height and alignment with segmental material. These tunes generally consist of Low (L) and High (H) autosegments (tonemes), though some analysts also include Mid (M) tones, which may further be specified to associate at or near “accented” tone-bearing units or at the edges of prosodic constituents. The former are standardly indicated with * (e.g., H*); these are intonational pitch accents. The latter are standardly marked with are preceding or trailing % (e.g., L%); these are boundary tones, which may be more precisely indicated as aligning with the boundary of particular types of prosodic constituents (e.g., L_φ).⁷⁴ I will follow the latter practice (with Gussenhoven 2004), explicitly signaling the level of prosodic constituency to which a boundary tone belongs, where possible. Bitonal tonemes (variously notated as, e.g., H*+L or H* L) are also commonly employed. These phonological representational devices are “underspecified in the sense that they do not account (and are not meant to account) for all pitch movements” (Arvaniti and Fletcher 2020: 80).⁷⁵

⁷⁴Pierrehumbert’s grammar of English intonation distinguished between phrasal accents H⁺ and L⁺ versus the boundary tones of intonational phrases, marked with %. In practice, the distinction is often either ignored, % being used for all boundary tones, or, the specific constituent is signaled, as in Gussenhoven 2004.

⁷⁵At present, most work on intonational phonology assumes no unitary tones with multiple targets (e.g.,

Example (58), borrowed from Gussenhoven 2004: 23, is an utterance comprised of two intonational phrases, each of which carries a declarative intonation (cf. Gussenhoven 2004: 296–8). Both intonational phrases are defined by a low boundary tone at their left edges, L_i and a $H^* L$ pitch accent; a non-final ι receives a high boundary tone at its right edge, H_i , while the final ι receives a low boundary tone, L_i . The starred portion of the $H^* L$ pitch accent is associated with the metrically most prominent element in the intonational phrase, the focused constituent *press* and the only content word *mentioned*, respectively.

(58) Declarative Intonation: *In press reports of the incident, he wasn't mentioned.*



This simple example abstractly illustrates the central aspects of the phonetic implementation of intonational tunes. First, the boundaries of the prosodic constituents and the metrically most prominent syllables in the respective intonational phrases serve as anchors for the pitch accents (starred tones) and boundary tones, respectively. Since the syllable serves as the tone-bearing unit (TBU) in English, the phonological tonal targets aim to align themselves with syllables, though, as Silverman and Pierrehumbert 1990 (American English) Arvaniti et al. 1998 (Greek) and D’Imperio 2001 (Neapolitan Italian) have shown, there is a tendency in many languages for the true peak of F_0 to come slightly after the TBU with which a tone is associated; this phenomenon is referred to as “peak delay”. Second is the general downward scaling of pitch in the course of the utterance, i.e., downstep: the same type of

a contour tone L^*H that would attempt to associate to one intonation-bearing unit); in other words, pitch levels constitute the primitives of intonation. See Arvaniti et al. 1998 and Dilley and Brown 2007 for empirical investigations supporting this view.

tone occurring later in the utterance is realized as lower (cf. the discussion of downstep and pitch reset in Lekeitio Basque at (22) above, based on Elordieta 2015, or the decline in *F₀* seen in 1.1) above. Third, where no tone is associated to a potential TBU, the *F₀* is determined through interpolation: typically, the pitch level set by a tone remains relatively constant until the next pitch target occurs. Thus, in (58), *F₀* remains relatively low from the L portion of the H* L pitch accent associated with the syllable [ɫi] up until the H_i associated with [dɲt]. In contrast to stretches of sparse tone, cases of so-called “tonal crowding” may occur when there are fewer TBUs than tones between two given anchoring TBUs. If multiple tones would need to be associated to the same TBU, several repair strategies may be implemented: a tone is not realized (truncation or elision), the tones may fail reach their phonologically specified targets (undershoot), or the tones may be temporally realigned, moved away from their preferred hosts. See Gussenhoven 2004: Ch. 8 for on the optimality-theoretic implementation of the alignment of intonational tones and the application of repair strategies in cases of conflict.

1.3.2 Tone

In comparison to the phenomena of stress and “pitch accent” to be treated below, TONE as both a phonetic and phonological object is generally less complicated to describe. Under one common understanding, languages with tone possess word-level lexically specified pitch features, i.e., the phonological element of a lexical entry bears information on specific *F₀* excursions in combination with segments, and the *F₀* excursions realized on the surface are not attributable primarily to pure intonation. Phonetically, tone in such languages is not different from intonation, in that the phonetic correlate of both intonation and lexical tone is simply fundamental frequency. Languages with relatively dense lexical tone, in which contrastive lexical specifications are possible, minimal pairs are abundant, and the realization of tone is largely orthogonal to other aspects of prosody, include Iau (Lakes Plain, West Papua; Bateman 1990), Thai, Mandarin Chinese, and Vietnamese. Although *F₀* is the primary or only acoustic correlate of tone in many languages, tone may also be implicated in differences of phonation type (as in Hanoi Vietnamese; Kirby 2011: 386) or duration (as in Mandarin Chinese; Xu 1997: 67); cf. generally Hyman and Leben 2020: 47 on the interaction of pitch and other prosodic properties in languages with tone.

Example (59) reproduces a typical set of minimal pairs for pitch in Ibibio, which underlyingly distinguishes a High (H) and a Low (L) tone, and which may combine into a surface falling contour tone; any TBU (a vocalic mora) may bear either tone, unpredictably. See similarly McCawley 1978 or Yip 2002 for this phonological view of tone as based on lexical specification of pitch in the phonological entry of morphemes.⁷⁶

⁷⁶Nevertheless, in intonation-only languages a small portion of the lexicon, consisting essentially of discourse particles, might be specified for tone, given that they invariably occur with the same intonation (so Gussenhoven 2004: 45–6).

(59) Contrastive Lexical Tone in Ibibio (Niger-Congo, Nigeria)⁷⁷

- a. H–H: /ák pá/ → [ʔák pá] ‘expanse of ocean’
- b. L–L: /àk pà/ → [ʔàk pà] ‘small ant species’
- c. H–L: /ákù/ → [ʔákù] ‘priest’
- d. L–H: /àkú/ → [ʔàkú] ‘first’
- e. H–HL: /ák pân/ → [ʔák pân] ‘square woven basket’
- f. L–HL: /àk p̂/ → [ʔàk p̂] ‘rubber tree’

Complications arise where surface word-level pitch features are sparse, predictable (i.e., essentially syntagmatic), or crucially dependent upon other aspects of prosody (e.g., stress). In some such cases, the necessity of pitch specification in the lexicon remains phonologically necessary, in which case the criterion for TONE is satisfied, while in others, the proper phonological representation is less evident. Sparsity of surface pitch distinctions, where phonetically only High and Low are observable, and phonologically at most a privative distinction between a lexically specified tone and a default tone would be required, are hardly uncommon. Most Athabaskan languages, for instance, exhibit only a surface H vs. L distinction (Krauss 2005), as do many varieties of North Germanic languages (Oslo Norwegian, Stockholm Swedish), Seneca (Iroquoian; Chafe 1977, Melinger 2002), Chimwiini (Bantu; cf. Kisseberth and Abasheikh 2011) varieties of Basque (Hualde 2000, Hualde 2012), and most varieties of Japanese (including the most well-described dialect, that of Tōkyō). The fundamental question, to be discussed further under 1.3.3.3 below, is whether the lexical representation of tone in such cases is truly necessary, whether the occurrence of the marked tone is predictable (as in Seneca) or not (as in Basque or Japanese).

To briefly illustrate the type of issues involved, consider the analysis of the distribution of H in Seneca offered by Melinger 2002 (cf. also Hyman 2009: 227–8): high tone occurs on the first syllable of every syllabic trochee (built starting from the peninitial syllable, left-to-right), provided that at least one of the syllables in the foot is closed. Insofar as pitch peaks in prosodic words Seneca are fully predictable based on metrical structure, the lexical specification requirement for tone is not satisfied. The fact, however, that more than one H is allowed, or no H is required, contrasts with the canonical enforcement of CUMULATIVITY and OBLIGATORINESS in stress systems (see further below)

(60) Predictability of the Distribution of H in Seneca

- a. One H (leftmost foot has a closed syllable):
[oʔ.(dís.wa).(de.njê:).doh] ‘you (PL) waded’
- b. One H (rightmost foot has a closed syllable):
[a.(gě.ni).(yás.da).yeʔ] ‘I’m willing’
- c. Multiple H (first and third feet have closed syllables):
[de.(wá.geʔ).(nyo.da).(gě.ʔõh)] ‘I’m busy’

⁷⁷Data and recordings available from the UCLA Phonetics Lab at <http://www.phonetics.ucla.edu/appendix/languages/ibibio/ibibio.html>.

- d. No H (all feet contain only open syllables):
[de.(ga.de).(nye.o).dẽ?] ‘I’ll put a necktie on’

The prosody of Lithuanian (Kenstowicz 1972, Young 1991, Blevins 1993, Kushnir 2018) illustrates a radically different case, one in which the position of stress is unpredictable, and the realization of tone is dependent upon stress: in heavy syllables of the shape containing [V:] or [VR], if under the stress, one of two distinct pitch contours may be realized. See 61, where preceding ' marks stress and a circumflex or acute represent distinct tones; the traditional “circumflex” represents a sustained or rising tone, while the traditional “acute” represents a falling tone. Note that, when stress moves away from a syllable specified for tone, the tone does not surface; as the near-minimal pair between 61c. and d. indicates, the distinction between rising and falling tone is best treated as lexically specified, though insofar as all stressed heavy syllables exhibit one or the other tone, the surface distinction may be treated as phonologically privative, i.e. /Marked Tone/ vs. /Ø/.

(61) Lituanian Stress–Tone Interaction

- a. *va'dõvas* [va'dõ:vas] ‘leader:M.NOM.SG’
- b. *vado'vu* [vado:'vu] ‘leader:M.INST.SG’
- c. *'kálti* ['kálti] ‘to forge’
- d. *'káltas* ['káltas] ‘guilty:NOM.SG.M’

The fact that these pitch contours are unpredictable necessitates their (partial) lexical specification, and thus principal criterion for tone (lexically specified pitch features) is fulfilled. The stress dependence of tone requires that stress be assigned, such that then the underlying specifications for tone may be accessed and realized. The closely related Latvian (Karinš 1996) requires a similar lexical specification of tone, but differs in that stress assignment is fully predictable (leftmost syllable of the prosodic word), with the result that non-intonational pitch contrasts occur only on initial syllables. These such cases, combined with the evidence for stress and metrical structure in languages like Mandarin, where stress and tone largely operate orthogonally to one another (see Duanmu 2004 and citations there), indicate that stress and tone are independent prosodic variables, which may or may not play a substantial role in the phonology of a language, and may or may not interact with one another (see also 2.4.1.2 on further types of tone–stress interaction).

Finally, tone behaves differently from stress in that the units prosodic assignment may vary from language to language: a tone may associate with either a mora or a syllable, and, in languages with quantity distinctions, heavy syllables may host multiple tones if the mora is the tone bearing unit. Stress, on the other hand, is typically assumed to be a property of syllables (cf. Hayes 1995: 48–9, Hyman 2009: 217, and discussion in the following section); if stress plays an active role in the phonology of a language, it will be assigned at the level of the syllable.⁷⁸ Lexical (or grammatical) tone, as well as intonation, may associate with

⁷⁸Halle and Vergnaud (1987: 190–203) in fact treat the surface tonal contrast of Lithuanian as regulated by a distinction in stress at the level of the mora, but such a maneuver would violate the apparent universality of the syllable as a stress bearing unit, and has not won acceptance in the face of more straightforward analyses involving lexical tone.

sub-syllabic constituents or to syllables. Interestingly, I am not aware of clear evidence from an intonation-only language in which the site of association would appear to be the mora, rather than the syllable.

1.3.3 Stress, Accent, and Pitch Accent

This section provides an introduction to the phonetic properties and phonological definition of the suprasegmental phenomenon that will occupy that most substantial role in this work: stress. After developing a description of stress from all possible phonetic perspectives (physiological, acoustic, and perceptual), I attempt to work out a phonological definition of stress, and to clarify the relation of that term to the term “accent”. The third part of this section then discusses the term “pitch accent”, and largely argues in favor of the perspective taken by Hyman (Hyman 2006, Hyman 2009), that the term is unhelpful insofar as it covers a variety of obviously distinct prosodic systems.

1.3.3.1 The Physiological, Acoustic, and Perceptual Correlates of Stress

Phonetic research into the acoustic properties associated with segments or syllables intuitively classified by native speakers as bearing stress in languages assumed to be possessed of stress, as well as the particular physiology involved in producing a stress-bearing go back well over a century. The work of Henry Sweet (Sweet 1877 [1970]: 59–60, Sweet 1889, Sweet 1906: 47–50) was path-breaking in this regard; Sweet’s view was that stress is manifested acoustically mainly through intensity (as opposed to frequency in the case of pitch) and physiologically with force or effort in expiration. Detailed instrumental examinations of the acoustics and physiology related to stress begin in earnest in the 1930’s, while Fry 1958 pioneered the first psycholinguistic experiments on stress perception. See Fox 2000: 117–24 for a short summary of the history of research. The presentation of stress correlates given here is based primarily on the recent surveys of Erickson and Kawahara 2016, Gordon and Roettger 2017 (a database of 110 acoustic studies encompassing 75 different languages or language varieties), Roettger and Gordon 2017, van Heuven 2019, and van Heuven and Turk 2020. These surveys also examine the possibility of different phonetic correlates in languages with fixed metrical vs. lexical stress, as well as differences in stress correlates due potentially to the presence of other sorts of phonological contrasts (e.g., contrastive vowel length) in the language (i.e., the Functional Load Hypothesis; cf. Martinet 1955, Bernstein 1979). Beckman (1986: Ch. 3–5) and Sluijter (1995) present somewhat older comprehensive treatments of stress correlates or correlates of “accentual prominence”.

Crucial on the side of acoustics and perception, at least, is the fact that more recent investigations into stress strive to clearly distinguish between *word* stress, on the one hand, and *sentence* stress on the other. At least in the case of intonation languages such as Dutch, English, Italian, or Modern Greek, the heads of focused constituents in particular are known to be subject to additional prosodic enhancement, for which reason instrumental measurements or intuitions regarding words spoken in isolation is not reliable: possible properties of stressed elements particular to focus or when out of focus must be distinguished from

one another. The “special intonation” attributed by Jones (1950: 147), for instance, to the properties of stress in intonation languages, it turns out, should best be regarded as an intonational property attracted to the stress under the right conditions, rather than a property of (word) stress in itself. As van Heuven (2019: 19–21) emphasizes, disentangling word stress in particular and isolating its acoustic correlates from the acoustic properties of the segments themselves or the correlates of other boundary effects can be challenging. Precisely because they possess at least some minimal stress pairs and stressed syllables readily occur away from the initial or final syllable of a word, languages with lexical stress of the English, Russian, or Italian sort are probably most suitable for investigations into stress correlates generally.

For the purposes of prosodic acquisition and, presumably, prosodic change, the perception of stress presumably is of greatest relevance, since the extent to which a syllable can be (correctly or not) classified by a learner as “under stress” directly influences how the primary linguistic data will be interpreted. Thus, in addition to knowing where prosodic prominence was assigned in a language on what phonological basis, recognizing what phonetic properties that realization of prominence may have had, and what phonetic properties of prosodic relevance elsewhere in a word or utterance were present, is of utmost importance in order to potentially obtain a picture of what phonological interpretation a learner might have applied to a phonetic percept (see further discussion along these lines in 3.3).

Stress and Physiology Insofar as stress is taken to be a marking of prominence, a syllable standing under stress may be generated with a greater degree of physiological effort (Sweet’s “force”), which can potentially occur in any and all of the physiological mechanisms involved in speech production, from the lungs to the oral articulators. The limitations on [ə] and similar centralized vowels in English, for instance, points towards increased articulatory effort being exercised in stressed syllables, given that the production of a schwa necessitates less movement or displacement of oral articulators. From some acoustic properties of stressed syllables (see immediately below), reasonable inferences about the physiology involved may also be made, e.g., if stressed units show greater acoustic intensity, this feature may be attributed to increased effort in the subglottal mechanism, expelling more air in a shorter amount of time from the lungs.

Ladefoged (1967: 46) in fact goes so far as to assert that “stress is best described in physiological, rather than acoustic terms”. In this electro-myographic study, Ladefoged found that an increase in activity of the intercostal muscles immediately preceding a stress syllable could be registered, indicative of greater subglottal pressure in the production of the stressed syllable. Ladefoged’s claim might be valid, and is interesting for the pursuit of an absolute common factor behind “stress” in all of its distinct acoustic manifestations, but because facts about stress are learned and transmitted intergenerationally via the sound stream, one must nevertheless devote greater attention to the divergences in the acoustic properties of stress and how they are perceived.

Beyond evidence for increased muscular activity that could translate to increased subglottal pressure and effects of increased airflow on the laryngeal level, a several studies on English have found an increase in jaw displacement (i.e., a greater opening of the jaw) dur-

ing the production of syllables standing under contrastive focus. Similar patterns of jaw displacement have been reported for Japanese and French, while somewhat different language-specific patterns have been found among speakers of Mandarin and Spanish. See the literature cited in Erickson and Kawahara 2016: 2; 10. In their own study on English, Kawahara & Erickson specifically found greatest jaw displacement on the focused word with utterance-level stress in English, while Kawahara’s work on Japanese indicates greater displacement in utterance-initial and utterance-final syllables. Kawahara & Erickson’s interpretation is that jaw displacement is associated with greater metrical prominence, falling on the head of a focused constituent in English, but on utterance edges in Japanese (cf. Jun 2014 for other evidence of initial and final syllables in Japanese as metrically strong).

Stress and Acoustics The acoustic correlates of stress may be considered from two perspectives: on the one hand, whether a given correlate distinguishes a stressed syllable from unstressed syllables in the same word whatsoever, and on the other, how effectively a correlate can be used to correctly classify a syllable as stressed or unstressed. The reader should bear in mind that acoustic correlates do not automatically translate into *cues* of stress for a listener in perception (see immediately below). Figure 1.2 below reproduces Figure 2 from Gordon and Roettger 2017, depicting the number of studies in their database of phonetic studies on stress in which a possible stress correlate was or was not investigated, and whether or not that correlate was found to successfully differentiate stress levels (e.g., syllables with primary stress from unstressed syllables) in a given study on a given language.

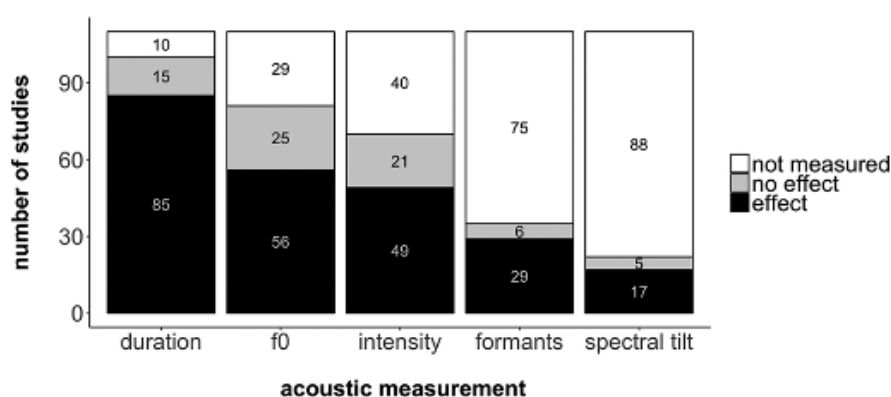


Figure 1.2: From Gordon and Roettger 2017: 4: “Number of (sub-)studies for which various acoustic correlates of stress were successful (black bars), unsuccessful (grey bars), and not measured (white bars) in differentiating stress level.”

Perhaps the clearest and most consistent correlate of word stress, going back to investigation of Fry 1955 using disyllabic minimal stress pairs in English, is duration: a stressed syllable is typically longer than unstressed syllable. According to the database compiled by Gordon and Roettger (2017), duration was found to successfully distinguish at least two degrees of stress in 85% (85/100) studies in which duration was measured, amounting to 90%

(62/72) languages.⁷⁹ Most work has focused on the duration of the syllable nuclei alone, though some experiments have found support for longer duration of consonants belonging to stressed syllables as well (Nooteboom 1972 on Dutch), especially in syllable codas (Sluijter and van Heuven 1995). Gordon and Roettger (2017) report that some few studies have found lengthening of an onset (e.g., Estonian; Lehiste 1966) or coda (e.g., Welsh; Williams 1999) without concomitant lengthening of the syllable nucleus. In Dutch and English at least, the effects of duration are substantial: vowels in stressed syllables are typically 40–50% longer than their unstressed counterparts (van Heuven and Turk 2020: 153), and with regard to stress classification, duration alone could correctly classify 98% of tokens in the study of Fry 1955.

Likewise often reported as a correlate with word stress is an increase in intensity of a syllable nucleus, measured as an increase in the root-mean-square average of amplitude within a small time span encompassing minimally two periods of glottal vibration. Intensity is reported in decibels (dB), being a measure of relative difference in sound pressure. In human speech at a normal volume, vowel intensities typically fall in the range of 55 to 75 dB, where a stressed syllable often exhibits a greater amplitude of approximately 5 dB. In comparisons of minimal pairs, the study of Fry 1955 on English found that stressed vowel (e.g., the first syllable of Eng. *pérmit*) had greater intensity in dB than its unstressed counterpart (i.e., the first syllable of Eng. *permít*) in 89% of cases (107/120). Among the studies in the database of Gordon and Roettger (2017), 70% (56/81) report an effect of greater intensity associated with stress, in 39 different languages (whereas 13 languages studies showed no difference in intensity). At the same time, Gordon and Roettger (2017: 6) caution that “few of the studies that controlled for phrase-level prominence found intensity to be a robust exponent of stress”, and observe a perhaps greater tendency for intensity to register as significant in languages with lexical tone, though not exclusively (e.g., intensity in non-focused words registered as significant for Hungarian in Vogel et al. 2016).

Three further possible correlates have also not infrequently been examined in connection with word stress: fundamental frequency (F₀), the formant frequencies (F₁ and F₂) of stressed vowels (also labeled “spectral expansion”), and the spectral tilt (also “spectral balance”) of stressed vowels. A change in fundamental frequency, however, as recognized at least since Bolinger 1958, occurs so often as a phrasal or utterance-level “pitch accent” (in the sense discussed under 1.3.1 above) that it cannot be reliably interpreted as a correlate of word stress without careful attempts to disentangle the correlates of different stress levels. For instance, Gordon (2014: 83) worries that many analyses or typological surveys of stress systems may be compromised by a failure to properly distinguish word-level stress

⁷⁹Hungarian represents a language whose stress correlates have been investigated in multiple studies, and for which duration is claimed to *not* or only weakly predict stress. Both White and Mády (2008) and Vogel et al. 2016, as reported in Lunden et al. 2017, supposedly found that Hungarian does not employ duration as a stress cue. Indeed, the data in Vogel et al. 2016 indicate that duration in Hungarian did not show significant increases in syllables under word or sentence stress – but the same is also holds true for Vogel et al.’s Spanish and Turkish data. van Heuven (2019: 51–2) and van Heuven and Turk (2020: 161), therefore, assert that the investigation in Vogel et al. 2016 precisely does not establish duration as a weaker stress cue in Hungarian than in the languages all languages with which it was compared (Spanish, Turkish); data from Greek in Vogel et al.’s study stood out, as word-stressed items were somewhat and sentence-stressed items were indeed much longer.

and prominence proper to higher-level prosodic constituents. van Heuven (2019) and van Heuven and Turk (2020) in fact do not even include fundamental frequency among the correlates of word-stress surveyed, and at least for Dutch and English, and attribute changes in fundamental frequency found on certain stressed syllables in those languages entirely to sentence stress (see further below). Compare similarly Cutler (2005: 283): “effects due to intrinsic characteristics of stressed versus unstressed syllables must be distinguished from effects which arise from differences in the applicability of intonational prominence.”

Gordon and Roettger (2017: 5) nevertheless note that some studies that have attempted to control for the possible effects of sentence stress have found F_0 effects associated with word stress, namely for Finnish (Suomi et al. 2001), and Greek, Iberian Spanish, and Hungarian (Vogel et al. 2016; the latter authors caution, however, that an effect of phrasal pitch accent cannot be completely excluded). The role of fundamental frequency as a cue to “accent” in languages of the Japanese or Basque type to which the label “pitch-accent language” is often applied constitutes a partly separate question (see 1.3.3.3 below). Worth noting is also that, although higher F_0 is typically found in association with stress, a handful of studies have instead found *lower* F_0 to correlate with stress (Hussain 1997 on Urdu; Eriksson et al. 2016 on words in isolation in Italian). For the moment, it suffices to recognize that differences in F_0 (typically higher pitch) frequently co-occur with stress, even if those changes in F_0 are not ultimately to be attributed to *word*-level stress in many (or most) languages.

Possible relationships between stress and spectral expansion or spectral tilt have been investigated in comparatively few studies (cf. Figure 1.2 above), though both correlates seem potentially robust. Although the tendency for stressed vowels to be realized with a less peripheral quality is well-known, complicating the potential for cross-linguistic generalizations is the fact that vowel reduction in unstressed syllables often appears phonologized (as in English or Russian), and that many studies of spectral expansion have examined only a limited number of vowels or only one of F_1 and F_2 (Gordon and Roettger 2017: 6). Some relatively detailed and systematic examinations, e.g., van Bergem 1993 on the Dutch long vowels /e:/, o:/, a:/, have found, as might be expected, greatest centralization (i.e., less distance from /ə/) in unaccented words when not under sentence stress (cf. van Heuven 2019: 29–30). Spectral tilt, expressed as the ratio of decibels to Hertz (db/Hz) for the slope of a regression line fit to the long-term average spectrum of a vowel in the range of 0–4000 Hz, has proven to be generally flatter when stressed, at least in English and Dutch.

Looking to the domain of sentence stress, namely, the stress on the heads of focused constituents that receive nuclear pitch accents, the correlates in West Germanic languages (English and Dutch) are much as for word stress, with the major difference that changes in F_0 (of at least four semitones) constitute extremely robust correlates that allow for near-perfect discrimination of stressed and unstressed syllables (Sluijter et al. 1995). Further durational effects have been measured in the form of lengthening applied to the entire word and specific components of the word under sentence stress: both stressed syllables and word edges obtain greater duration when the word bears sentence stress.

Which of these acoustic properties constitutes the strongest and most reliable correlate of stress generally? In their survey of acoustic examinations of Dutch and English, van

Heuven and Turk (2020: 155) conclude that duration is the most reliable correlate, followed by intensity, with spectral tilt and spectral expansion correlating with stress less robustly; recall that *F₀* was excluded from consideration Gordon and Roettger (2017: 6–7). Meanwhile, considering eight studies with data on eleven languages in which either logistic regression or linear discriminant analysis was applied to distinguish the strength of correlates, Gordon and Roettger (2017) found that either mean *F₀* or a change in *F₀* constituted the most reliable predictor in six cases (K'ekchi, Tongan, Greek, Hungarian, Spanish, Turkish), while duration was the best predictor for five of the languages (Thai, Dutch, Ma'ya, Papiamentu, and Hebrew). In broad typological terms, increased duration of segments surrounding the location of the stress may be the most regularly occurring stress correlate with some predictive power, but insofar as changes in *F₀* are not ascribed purely to sentence stress, *F₀* may often be more robust.⁸⁰

A final issue to be mentioned here, which will return in 3.3, concerns possible differences in the acoustics of stress between languages with phonologically predictable (“fixed”) versus languages with (partly) lexically specified (“free”) stress. The general hypotheses has been forwarded that the marking of stressed elements will be comparatively “weaker” (i.e., exhibit less differentiation between stressed and unstressed syllables) in languages belonging to the first category (so, e.g., Rigault 1970, Dogil 1999). The summary of the literature by Cutler (2005: 273–4) suggests that the role of stress in languages with entirely or highly predictable stress may be limited to providing a site for the association of intonational pitch accents. The acoustic investigation of Polish stress (largely fixed penultimate stress) in Dogil 1999: 284–91 thus concludes that “lexical stress in Polish seems to have no context invariant correlates”.⁸¹ Nonetheless, other studies of languages with delimitative stress have indeed found systematic distinctions between stressed and unstressed syllables, e.g., higher *F₀* in Finnish (Suomi et al. 2001) and increased spectral tilt in both Polish and Macedonian (Crosswhite 2003). The balance of evidence at present suggests that languages exhibiting regularly predictable stress patterns may indeed employ an acoustically weaker complex of cues to stress, but whether any languages with word-level prominence at all (as opposed to French-like languages exhibiting only phrasal prominence) altogether lack some consistent acoustic manifestation of stress remains to be established.

Stress Perception The potential acoustic manifestations of stress surveyed immediately above need not necessarily align with the cues actually employed by speakers for distinguishing between stressed and unstressed syllables: the acoustically most robust cue to the presence of phonological stress may not be the cue that speakers choose to employ. Indeed,

⁸⁰From a phonological perspective, that duration and stress should exhibit a close relationship in acoustic terms is perhaps not surprising, and indeed desirable: the typological frequency with which stress is phonologically attracted to heavy syllables (WEIGHT-TO-STRESS) or in which stressed syllables are obligatorily made phonologically heavy (STRESS-TO-WEIGHT) becomes entirely comprehensible if stressed syllables cross-linguistically exhibit a tendency to be durationally longer.

⁸¹Dogil (1980) also reports a phenomenon whereby secondary stress is promoted to primary stress in Polish when the word occurs under narrow focus; in effect, the secondary stress on the initial (or second?) syllable becomes the site for the association of a pitch accent, and, according to the data in Dogil 1999: 290, the syllable with secondary stress becomes durationally longest in the word.

comparison of the results of acoustic studies on stress with psycholinguistic studies on the perception of stress reveal that there is not a clear and direct correlation between *acoustic* correlates of stress, on the one hand, and the strength of *perceptual* cues to stress on the other. For surveys of perceptual cues to stress, see Cutler 2005: 264–70, van Heuven 2019: 36–47, or van Heuven and Turk 2020: 158–160.

A series of studies conducted by Fry (Fry 1955, Fry 1958, and Fry 1965) examined the utility of various factors in the perception of stress (peak intensity, *F₀*, and vowel quality) with respect to a baseline condition of vowel duration; these studies thus mainly compare the extent to which duration on the one hand and one of intensity, *F₀*, or vowel quality, on the other, serve to help speakers pick out the position of stress. All of Fry’s studies employed English disyllabic noun/verb minimal pairs of the *pérmit*~*permít* type, synthesized to present forms with minor gradations of difference in amplitude, *F₀*, or further vowel formant ratios.

Concretely, Fry (1955) found that duration outweighed intensity in the discrimination of stress by a factor of 15; a similar result was obtained by Turk and Sawush 1997 and replicated for Dutch by Sluijter et al. 1997. Fry’s finding, namely, that amplitude serves as a comparatively weak perceptual cue to stress in English, was paralleled by the study of Mol and Uhlenbeck (1956), in which the reversal of amplitude on stressed and unstressed syllables was found not to impact stress judgments. Fry (1958), meanwhile, found that English speakers were more sensitive to changes in *F₀* than to changes in duration (see also the detailed analysis of Fry’s results in van Heuven 2019: 43–7); similar results were obtained by Bolinger (1958), who found that variation of amplitude parameters on syllables with intonational pitch accents did not impact stress judgments, and van Katwijk (1974: 76–88) found the *F₀* differences could robustly trigger identification of stress position in trisyllabic Dutch forms. Finally, Fry’s (1965) study found vowel reduction, as synthesized by reducing vowel quality along *F₁* and *F₂*, to constitute only a weak perceptual cue to stress, while even relatively small differences in duration allowed stress to be convincingly distinguished (cf. also van Heuven and de Jonge 2011 for similar results with Dutch speakers).

The results of Fry’s experiments, and parallel experiments conducted with speakers of Dutch, indicate that, in such languages, excursions in *F₀* may constitute the strongest perceptual cue, followed by duration, while intensity plays a less important role (except perhaps as a component of spectral balance, as suggested in Sluijter et al. 1997). The exact timing of the *F₀* movement within a syllable is, however, crucial for triggering the perceptual effect (cf. Thorsen 1982 on Danish, van Heuven and Turk 2020: 159–60). Since changes in *F₀* in English and Dutch usually arise when a lexical item stands under sentence stress, it is perhaps unsurprising that changes in *F₀* would seem to be able to override any other single cue to stress, since those other cues or combinations thereof may be implicated in word-level stress. Further complicating the relationship between *F₀* and stress, however, are possible influences of *F₀* on both perceived duration and perceived intensity (see Niebuhr et al. 2020). All things being equal, duration is likely the safest cue to stress, since, with van Heuven (2019: 47), “differences in vowel duration are both perceptually strong and acoustically highly reliable, for both word stress and sentence stress”.

From these results one might infer that, in learning stress in an English-like language, a

learner can probably pick up on reliable cues in the form of both a rise in *F₀* and increased vowel duration, when a word stands under sentence stress. If, however, a given lexical item rarely or never happens to occur under sentence stress, then the learner will be compelled to rely first and foremost on duration in order to discriminate between stressed and unstressed syllables.

1.3.3.2 Stress (and Accent) as Phonological Phenomenon

In comparison to the fraught question of stress identification at the phonetic level, some phonologists apply a perhaps refreshingly simple definition of stress, which, however, requires considerable unpacking: “Stress characterizes the abstract property of being prominent,” writes Féry (2017: 178), referring to Liberman and Prince 1977, Halle and Vergnaud 1987, and Halle and Isardi 1995 for the representation of relative prominence and rhythm on a metrical grid. Such an abstract property of prominence identifiable through relations between prosodic units (in particular, syllables) is equivalent to what many others label “accent”, e.g., Garde 1976 or van der Hulst 2012; compare further Gordon and van der Hulst 2020: fn. 1, who explicitly note the terminological overlap between “stress” and “accent” when speaking of the phenomenon at the phonological level. One might thus essentially equate the conception embodied in the statement of Ambergcrombie (1976 [1991]) that “accent is ineffable” with Féry’s “abstract property”.

Usage of the term “stress” to denote an abstract relation of prominence is distinct from the usage of this term applied at the phonetic level, from Sweet 1877 [1970] onward, where it designates the confluence of (potential) acoustic properties discussed under 1.3.3.1 above. There exists a long tradition of separating phonological “accent” or “stress” from any particular set of articulatory or acoustic properties involved in the phonetic realization of this abstract property. Hirt (1895: 10), for instance, draws the following distinction: “Es kann die eine Silbe vor der andern durch schärfere resp. stärkere Expiration oder durch grössere Höhe oder Tiefe hervorgehoben sein. Je nachdem eines oder das andere überwiegt, sprechen wir vom expiratorischem oder Nachdrucksakzent und musikalischem oder chromatischem Akzent.”⁸² Taken forward nearly a century later, Hirt’s “Nachdrucksakzent” might be mapped onto Beckman (1986)’s “stress-accent” (exemplified by English) and his “musikalischer Akzent” onto Beckman’s “non-stress-accent” (exemplified by Japanese), with the latter also susceptible to the label “pitch-accent”.

The question to be addressed in the following pages is whether a consistent and coherent set of phonetic or phonological properties can be identified for which the label “stress” is unambiguously appropriate. In a series of papers (Hyman 2006, Hyman 2009, Hyman 2014, Hyman 2019), Larry Hyman has attempted to more precisely delimit the boundaries of stress systems by identifying possible common phonological properties, while at the same time programmatically emphasizing a need to examine prosody from the perspective of in-

⁸²“A given syllable can be more prominent in relation to another on account of a sharper or stronger expiration or on account of higher or lower pitch [grössere Höhe oder Tiefe]. Depending upon whether the one [expiration] or the other [pitch] predominates, we speak of an expiratory or emphatic accent and of a musical or chromatic accent.” Hirt’s “Nachdrucksakzent” could equally be rendered as “dynamic accent”.

dividual phonological properties (e.g., CULMINATIVITY, METRICITY) rather than categories (e.g., “tone language”, “pitch-accent language”, “stress language”). Hyman’s line of reasoning will form the basis for much of the succeeding discussion, and will be explicitly contrasted with the perspective on accent represented by van der Hulst 2009, van der Hulst 2011, van der Hulst 2012, though Hyman and van der Hulst agree on one important point: “language typology [viz., prosodic typology] should refer to properties of languages rather than languages as a whole (van der Hulst 2012: 1516). In other words, categorical labels such as “stress-accent language” should generally be eschewed in favor of a finer-grained set of properties or parameters.

In a real sense, approaches to investigating acoustic and perceptual correlates of stress, as surveyed under 1.3.3.1, have proceeded from the assumption that a given language exhibits stress. Given the evident diversity of stress correlates, adopting an essentially phonetically grounded definition of stress does not appear promising. If, for instance, the perspective on Polish stress presented in Dogil 1999 is correct, namely, that stress in Polish has no consistent acoustic correlates, a strictly phonetic definition of stress might then exclude Polish from languages that possess stress. One might attempt to defend such a position, but I think that most researchers into stress from a phonological perspective would find it odd to claim “Polish has no stress”, even if stress in Polish is of a phonetically qualitatively different nature as compared to stress in, e.g., Dutch. In view of this line of argumentation, a purely phonetic definition of stress will not yield particularly helpful or meaningful linguistic classifications. For this reason, a fundamentally phonological definition of stress is likely to be more successful.

In proceeding with an attempt to phonologically identify stress, a first desideratum to fulfill is to cleanly distinguish “stress” from “accent”, if possible, since the two terms are often employed interchangeably (or at least indistinguishably; cf. above). Hyman (2014: 56) offers a nicely operationalizable pair of definitions: he describes *word accent* as “the phonological marking of one most prominent position in a word”, and he explicitly distinguishes this notion from a concept of *word stress*, to which he does not, however, ascribe any particular properties in its phonetic realization, but rather a specific phonological property, namely, the determination of prominence by reference to metrical structure. If accepted as valid definitions, such an opposition between “stress” and “accent” would imply that there exist languages that determine word-level prominence without reference to metrical structure whatsoever. Granting that one accepts reference to metrical structure as the best fashion in which to analyze languages such as Pintupi (cf. 2.2.2), exhibiting a clear rhythmic distribution of stress and distinguishing between multiple degrees of stress, languages like Tōkyō Japanese could reasonably be treated as expressing a single most prominent position on the basis of lexical diacritic marking or lexically specified tone alone. Nonetheless, the latter might also conceivably be treated by reference to lexically specified metrical structure, in which case prominence could nevertheless be seen as due to metrical structure, and thereby satisfy the metricity condition for stress (as argued for lexical stress in, e.g., Turkish by Özçelik 2014). Metricity should be seen as a necessary condition for stress, but is perhaps not fully sufficient for delimiting stress from accent.

Hyman himself (Hyman 2006: 231, Hyman 2009: 217, Hyman 2014: 60–1), in accord with

most other phonologically oriented approaches to stress (e.g., Hayes 1995: 24–6) additionally recognizes three other properties as constituting “a minimum that any system must meet” in order to be classified as exhibiting stress:

- (62) Minimum Properties of Languages with Stress (after Hyman 2014: 60–1)
- a. SYLLABLE-DESIGNATION: the stress-bearing unit is the syllable (cf. Hayes 1995: 49–50)
 - b. OBLIGATORINESS: every lexical word has at least one syllable marked for the highest degree of metrical prominence (= primary stress).
 - c. CULMINATIVITY: every lexical word has at most one syllable marked for the highest degree of metrical prominence.

These properties of canonical phonological stress also imply fourth property, PRIVATIVITY (Hyman 2009: 215; 224): a given syllable is either (relatively) more prominent or not. Expressed in terms of metrical structure, privacy can be understood to mean that a syllable is either the head of a foot or it is not, and only foot heads are licit bearers of stress. The typologically diverse array of languages surveyed in Hayes 1995: Ch. 6, from varieties of Arabic, to Icelandic, to various Iroquoian languages (e.g., Potawatomi) show that this canonical package of characteristics is far from uncommon.

Whether word-level prominence is identified impressionistically (with good confidence), identified instrumentally through some acoustic property, or reported by native speakers according to their intuitions, if the distribution of that “prominence” exhibits OBLIGATORINESS and CULMINATIVITY, is recognizable as a property of syllables, and is open to analysis as arising from metrical structure (whether determined by rule or prespecified in a phonological representation), that “prominence” may be identified as stress. Less clear, however, is the role of rhythm: do languages in which only a single foot per word may be constructed (i.e., in which the parsing of syllables into feet is less important than avoidance of foot-edges at too great a distance from the word edge; cf. 2.3.3) properly exhibit stress? I would tentatively answer this question in the affirmative: a language may still be seen as a canonical instantiation of stress, even if no rhythmic alternation between primary and secondary stress occurs, and even if the primary (perhaps only) acoustic or perceptual cue to stress is an intonational “pitch accent” associated to the metrically most prominent syllable.

In contrast to Hyman’s view of stress as the co-occurrence of several properties that together constitute a phonological prototype for stress, van der Hulst (2012) completely subsumes stress under the label of “word accent”.⁸³ Departing from the view that word accent is a universal or near-universal property (so also van der Hulst 2009: 1), van der Hulst takes stress to be a particular manifestation – a phonetic realization – of either a lexical property (“stress^{Accent}”) or some other phonological property (edge prominence, rhythm, or weight); word accent, whether determined lexically or through some other property, may have other manifestations, e.g., pitch. In this sense, van der Hulst basically allows for the distinction

⁸³Somewhat confusingly, van der Hulst distinguishes “word accent” and “accent”, where the former is a superordinate label, encompassing word-level prominence, however assigned, while the latter specifically means an underlying lexical property marking the position of prominence.

between pitch-accent and stress-accent as expressed by Hirt (1895: 10) cited above. In van der Hulst's conception, primary word prominence is determined by *accent*, where heavy syllables may have an influence on the placement accent.

Analyses presented in van der Hulst 2009: 230–7 and van der Hulst 2012: 1498–1507 make clear that accent may either be assigned to (a) heavy syllable(s) within a specified domain, or be borne as lexical properties; in case multiple accents exist within the domain, a single one is *SELECTED* and all others are deleted, and in case no accent exists within the domain, a *DEFAULT* accent is assigned. Ultimately, the single abstract accent, whether deriving from a lexical feature, syllable weight, or default preference, may phonetically manifest itself in various ways (as higher pitch, as increased duration, etc.). This primary prominence is determined independently of metrical structure, and serves as a pre-specified word head in a case the (independent) rhythmic component builds up metrical structures (and so creates secondary stress), leading to possible cases of “top-down” prominence (cf. 2.2.4); this primary prominence may also show pitch as a direct exponent. Note that the role of the parameters *SELECT* and *DEFAULT* in van der Hulst's system are essentially equivalent to the properties of *CULMINATIVITY* and *OBLIGATORINESS*: if *SELECT* is inactive, multiple accents may remain and may be phonetically manifested equivalently; if *DEFAULT* is inactive, no accent is assigned, and words without any more prominent syllables may occur. In short, van der Hulst's notion of “accent” completely subsumes stress on the one hand, but encompasses an array of other possible distributions of prosodic prominence that do not fit the profile of canonical stress systems.

Fundamentally, the debate outlined above can be summarized into 1) a view that clearly delimitates languages with stress from languages without it on the basis of phonological properties, and 2) a view that takes stress as a specific subtype of a more general concept of prosodic prominence (“accent”) or as an outgrowth of rhythm or edge prominence. In other words, view 1) makes “stress languages” into a potentially coherent phonological category, while view 2) would only potentially unify “stress languages” in a negative sense, i.e., languages in which pitch is not the exclusive or primary realization of “accent” combined with non-accentual languages. If “stress” is to be taken seriously as a phonologically coherent category, view 2) is unsatisfactory; conversely, if “accent” is to be taken seriously as a phonologically coherent category, then view 1) is unsatisfactory.

(63) STRESS versus ACCENT

- a. Phonological STRESS (Hyman): languages possessed of *STRESS* share an obligatory and culminative expression of prominence at the level of the syllable, whether that prominence is determined through the construction of prosodic constituents (feet) or as a lexical property, and distinguishes syllables “with prominence” and “without prominence” privatively.
- b. Phonological ACCENT (van der Hulst): languages possessed of *ACCENT* possess a lexical specification of prominence or an interpretation of syllable weight as prominence (though lexical prominence may be restricted to a bounded domain near a word edge).

In phonological terms, the second view treats prosodic prominence in e.g., Russian as more akin to prosodic prominence in Tōkyō Japanese (since both languages possess lexical “accent”) than to prosodic prominence in e.g., Hungarian (where “stress” might merely be a manifestation of an edge prominence). The set of parameters available under the second view, however, lead to a potentially peculiar prediction: languages without accent but possessing rhythmic stress might phonetically manifest stress (i.e., some syllables would exhibit acoustically greater duration and amplitude) without any syllable being more prominent (i.e., no CULMINATIVITY). As the immediately following section will argue, and studies in subsequent chapters will strive to demonstrate, the view of STRESS as a phonologically coherent category, when combined with principles of intonational phonology, is more attractive overall.⁸⁴

1.3.3.3 Pitch Accent and Restricted Tone: Phonological Categorization

There is a certain attraction to both the views expressed by van der Hulst, on the one hand, and by Hyman, on the other as concerns the phonological analysis of languages whose word-level prosody fits neither the portrait of a canonical “tone language” like Vietnamese or a canonical “stress language”, whether with lexical stress like English or with edge-prominence stress like Hungarian. The point emphasized in Hyman 2009, namely, that languages to which the label “pitch-accent language” is often applied in fact differ in terms of a number of phonological properties, is self-evidently correct: for instance, there are word-prosodic systems like that of Chuave (Swick 1966), in which H tone is obligatory, but not culminative, at the word level, which may be contrasted with word-prosodic systems like those of Tinputz (Hostetler and Hostetler 1975), in which H tone is culminative, but not obligatory, at the word-level. Thus, while the designations “stress accent” or “stress language” are coherent insofar as they indicate whether a language exhibits a particular set of phonological properties, the same cannot be said for “pitch accent” and “pitch-accent language”, except perhaps, unless one wishes to use the term in a negative sense to mean a language that fits neither the canonical profile of a stress language nor the canonical profile of a tone language.

To make matters yet more confusing, the label “pitch-accent language” has been applied to many prosodic systems – including the prosodic systems of numerous Indo-European languages to be treated in detail here – in which the phonological definition of STRESS discussed above appears to hold. Languages including Serbian dialects (Inkelas and Zec 1988, Zec and Zsiga 2022), Croatian dialects (Langston 2006), Lithuanian (Kenstowicz 1972, Young 1991, Blevins 1993, Kushnir 2018 and examples in 61 above), and Scandinavian languages (Swedish: Riad 2014; Norwegian: Kristoffersen 2000) all possess culminative, obligatory, privative, and syllable-designated prominence, which may, however, be realized with different pitch patterns. Such prosodic systems combine aspects of metrical and/or lexical stress with lexical tone.⁸⁵ Although essentially tonal analyses of many of these prosodic systems have been

⁸⁴This approach does not, however, preclude the possibility of analyzing prosodic prominence of a Russian sort and prosodic prominence of a Japanese sort in similar fashions, as in Alderete 2001b.

⁸⁵In the case of certain Serbian dialects, namely those designated as “Old Štokavian” stress and high tone always coincide, with no difference in pitch patterns occurring within stressed syllables (Zec and Zsiga 2022:

proposed, the generally culminative and privative nature of the site of pitch contrasts, plus independent phonetic or phonological evidence for prominence marking in some cases (e.g., increased duration on stressed syllables in Swedish), means that such prosodic systems are most felicitously to be regarded as combining canonical stress with (comparatively limited) lexical tone.

Outside of the Indo-European language family, many further cases in which stress and tone coexist in a single prosodic system have been reported and analyzed (cf. Gordon and van der Hulst 2020: 76). In some languages, e.g., Thai (Potisuk et al. 1996) or Papiamentu (Remijnsen and van Heuven 2005), stress and lexical tone appear to operate largely independently of one another, but are both demonstrably present in word-level prosody. In other cases, such as Ayutla Mixtec, the position of lexical tone predicts the position of stress; this type of tone-stress relationship has been analyzed most extensively in de Lacy 2002 (see further 2.4.1.2). Finally, DiCanio (2008: 20) has argued that Itunyoso Trique exhibits tonal distributions that are dependent upon stress. Indeed, the Trique case is most reminiscent of the Indo-European cases mentioned above, where only the stressed syllable (in Itunyoso Trique, always the final syllable of the word) licenses the realization of the full inventory of nine lexical tones.

Besides languages exhibiting canonical stress, whether metrically or lexically determined, and languages fulfilling the criteria for stress but also possessing lexical tone, to at least some degree, come languages in which the surface inventory of tones is very restricted, but not all major conditions for canonical stress (obligatoriness, culminativity, privativity, metricity; cf. Hyman 2009: 220) are satisfied. In much the same way as the prosodic systems with “stress plus tone” discussed above have traditionally (e.g., by Hirt (1895)) and by others at present (e.g., by van der Hulst (2011)) been treated as realizing an abstract prominence as pitch, so too have many of these prosodic systems that supposedly possess a “non-stress accent” (in the terms of Beckman 1986). As Hyman (2009) has argued with considerable systematicity, however, many such cases are usually best to be analyzed as possessing lexical tone, but with a surface distribution that is strongly restricted by one or more of the constraints typical of canonical stress. For instance, Puinave (Guirón Higueta and Wetzels 2007) should be treated as possessing a tonal inventory of /H/, /HL/ and /LH/ for the reason that all three can contrast on monosyllabic words (in addition to surface L, which is analyzable as /Ø/) and the surface pattern [H-L] contrasts with [HL-L] on disyllables. Culminativity applicable to high tone, however, limits the occurrence of [H] to exactly one per prosodic word. Such systems in general are most appropriately to be labeled as “restricted tone systems”.

Making a decision about the appropriate phonological perspective is most difficult in cases (including some of the most well-known examples of supposed “pitch-accent” systems) where the surface distribution of tone is both highly restricted and independent phonological evidence for lexical tone is wanting. The word prosody of Seneca (exemplified under 60 above) instantiates a case in which neither lexical tone is present nor the conditions for canonical stress are satisfied, yet metrical structure can induce the assignment of high tone.

69–70). This type of prosodic system is potentially open to analysis as possessing stress alone, plus an word-level intonational pitch accent /H*/. Compare the treatment of Egyptian Arabic prosody in Hellmuth 2006 and Hellmuth 2007 (cf. also Gordon 2014: 88).

Insofar as metrical structure governs the distribution of a syllable-designated prominence, Seneca possesses features that bring it closer to stress systems, but is lacking in culminativity or obligatoriness. A further phonological process, whereby vowels in penultimate syllables undergo lengthening if they constitute the strong portion of a trochee, further resembles the portrait of stress – penultimate lengthening may be the phonologization of enhanced duration in a stressed syllable. The fact that foot heads (of feet containing heavy syllables) in Seneca manifest apparently phonetically manifest themselves as high pitch may be seen either as “promotion” of stressed syllables to pitch-accent status (cf. Gordon’s (2014: 88) reading of Hellmuth’s (2007) Egyptian Arabic data) or as the association of a word-level pitch accent /H*/ with a stressed syllable, and even with multiple stressed syllables, given that no stress is promoted to primary status.⁸⁶

In the case of certain almost “canonical” examples of languages supposedly possessing word-level pitch accent, such as Tōkyō Japanese or varieties of Basque (see Hualde 2022 for a recent survey), the very restrictedness of surface tones and the fact that at least CULMINATIVITY is satisfied lead to the intuition of abstract accentedness. Nevertheless, treating diacritic “accent” in the phonological analysis of Japanese or Basque prosody as lexically specified tone, subject to culminativity restrictions, is equally possible. As concerns Tōkyō more specifically, given both established and growing evidence for metrical structures in at least some subset of lexical items (Poser 1990, Kubozono 2008, Poppe 2015, Kubozono 2020), one might wonder whether all “accented” words are words containing a foot, creating a metrically prominent syllable that can act as a host for a word-level pitch accent; “unaccented” words would then be interpreted as prosodic words in which the ω dominates no $\mathfrak{f}$, in violation of HEADEDNESS, where the absence of a metrically prominent syllable leaves the word-level pitch accent without an appropriate host. Parallel considerations apply to the word prosody of Nubi (Gussenhoven 2006) where the choice of analysis between a single metrical prominence that provides the site of association for a H* (whatever its provenance) or lexical representations containing H tones, which are then subject to obligatoriness and culminativity on the surface, is virtually impossible to make on the basis of available evidence.

The ultimate fundamental questions to be answered in the analysis of a language’s word-level prosody, as correctly emphasized by Hyman (2009: 234), are the extent to which it employs metrical structure, the extent to which it employs lexical tone, and how metrical structure and lexical tone further interact with elements of the intonational system. Given as the term “pitch accent” has, historically, been applied a prosodically quite diverse array of languages, ones that exhibit the combination of “stress plus lexical tone” (like Lithuanian), languages demonstrably possessing lexical tone that is relatively restricted (like Puinave), languages with probably having metrical stress not restricted by culminativity (like Seneca),

⁸⁶Treating the word-level prosody of Seneca in the latter fashion assumes that CULMINATIVITY and OBLIGATORINESS are low-ranked in any case. When multiple feet containing heavy syllables arise in a word, effectively multiple primary stresses are present, and ‘ $\sigma \leftarrow T^*$ ’ (“all primary stressed syllables must be associated with a starred tone”) dominates *NOSPREAD (“a tone is associated with at most one TBU”). If one does not worry about the source of surface H in Seneca, simply allows for stress to be “converted” into H tone, or for stress to be primarily manifested as raised F_0 (as reported for Turkish; cf. Levi 2005), then CULMINATIVITY and OBLIGATORINESS can be assumed to hold over H tone rather than stress (cf. Hyman 2009: 233).

and languages with highly restricted tone or possibly lexical stress (like Nubi), it is of little value. In this point, I will follow Hyman (2009), and eschew use of the term “pitch accent” as a label for some class of languages; rather, “pitch accent” will always mean a “starred tone” that preferentially associates with metrically prominent elements (cf. 1.3.1 above). In discussing the prosodic systems of older Indo-European languages, in particular the systems of Vedic Sanskrit (Chapter 4) and Ancient Greek (Chapter 6), more work will be necessary to precisely determine the interaction of metrical/lexical stress, possible lexical tone, and word- and phrase-level intonational tones.

1.4 Defining the Object of Investigation

The preceding sections have aimed to describe in some detail the aspects of phonology pertaining to the domain of prosody, from the categories of the prosodic hierarchy and their construction to the distinctions between intonation, tone, and stress, considering their potential phonetic and characteristic phonological properties. These descriptions now facilitate a clearer definition of the notion “prosodic change” first mentioned in the introduction this this chapter; more detailed examinations of consequences relating to the object of investigation defined here will be the topic of 3.3. Here, I answer the following questions: what is then, exactly, “prosodic change”, and what aspects will be studied in this work?

Definitionally, a prosodic change in the broadest sense can be a phonological change that results in changes to either phonological processes or phonological representations involving the categories treated in 1.2 and 1.3. A prosodic change may thus consist in changes to the constraint rankings governing the construction of prosodic structure (e.g., whether a prosodic word may be built recursively or not), a change in a process assigning tone, stress, or intonation (e.g., whether all heavy syllables must receive stress), a change in the lexicon of intonational tunes (e.g., questions marked as HL* instead of LH*), or a change in prosodic properties associated with the phonological representation of some element in the lexicon (e.g., whether a certain prefix behaves as “stress attracting” or not). Even a change in the bounding domain of a phonological process (e.g., whether the application of a process is conditioned by a syllable- rather than word-edge) might arguably constitute a prosodic change as well; however, if one accepts that all phonological processes are defined by prosodic domains, then expanding the field of “prosodic change” to encompass changes of this type arguably risks trivializing a distinction between “prosodic” and “non-prosodic” changes. Evidently “non-prosodic” in nature would be, e.g., a change in a phonological feature of a certain class of segments, the merger of two previously contrastive phonemes, or the emergence of a phonological process, applying to some segment(s) under segmental conditions (though such a process may well be circumscribed by a prosodic domain).

To take a famous and well-studied case of an evident “prosodic change”, the surface distinction between two different word-level pitch patterns found in many Scandinavian languages (see details for Central Swedish under 6.2.4), if assumed to involve a distinction in the lexical representation involving tone, historically requires changes in the specific representation of particular lexemes, as well as more generally in the sort of prosodically relevant

information to be found in the language. That is, prior to the development of the distinction between “Accent 1” and “Accent 2”, no lexical entries were specified for tone; afterwards, granting the privative analysis proposed by Riad (2012) (for Swedish), a subset of the lexicon came to bear a tone specification (H^* or L^* , depending on the language or dialect). Understanding factors that conditioned the emergence of lexical tone in Scandinavian languages would be desirable for grasping the nature of a particular type of prosodic change (see Riad 2003 and Kiparsky 2013 for a proposal centered around foot structure and stress), but at least recognizing this historical development as a case of “prosodic change” helps to define its domain.

The extent to which all language change may be profitably investigated, including prosodic change, naturally depends on the type and extent of testimony available. In instances where changes are reflected in relatively shallow divergences among closely related languages or dialects, and the languages themselves are still actively spoken, as is the case for the Scandinavian languages or varieties of Basque, one can trace prosodic changes via reconstructed (or even partly attested) language states into languages for which rich phonological description aided by instrumental phonetic analysis may be undertaken. Even in cases where the nature of the data facilitates the study of word-level prosody, reconstructing prosodic characteristics above the level of the prosodic word is challenging for much the same reason that the reconstruction of syntax poses special challenges (see Walkden 2013): syntax and the “putting together” of lexical / prosodic words into larger units depends principally on synchronic rules that are highly *productive*. These synchronic rules themselves are often not stored as lexical units, and hence not readily transmitted diachronically as a whole cloth.

The other obvious problem for the study of prosody above the word level where the data consists largely of linguistic representations originally recorded on vellum, papyrus, or stone is that the phonetic correlates and/or phonological processes that betray the presence and features of phonological and intonational phrases are transmitted much less systematically and reliably. Indeed, the vast majority of the orthographies applied to Indo-European languages, ancient and modern alike, are decidedly deficient in their representation of prosodic features. Nevertheless, a fair number of observations about prosodic phonology at and above the word level can be drawn in the case of Sanskrit and Ancient Greek; for most other earlier-attested Indo-European languages, the traces of phonological (ϕ) and intonational phrases (ι) are yet more sporadic and of uncertain testimony. Even if the domains of segmental processes may aid in diagnosing larger prosodic units, the details of intonational melodies will remain difficult to access, though the testimony of Ancient Greek music (see generally West 1992 and Pöhlmann and West 2001 and some applications in 6.2.1.6) potentially allowing for insights.

For the methodological and empirical reasons just enumerated, the treatment of prosody in the oldest-attested Indo-European languages and their reconstructed ancestor(s) will be largely limited to phonology that takes place at the level of the prosodic word and below. Since phenomena at the level of the syllable still benefit from insightful recent treatments (especially Byrd 2015 and Cooper 2015), the interest of this work will lie principally with the patterns of pitch and stress that occur within the domain of the (maximal) prosodic word. These patterns, depending upon the language, may refer to syllables, syllable weight, and

feet, or simply to segments with some given featural composition and word edges; appropriate and elegant description will be language dependent.

The study of change involving these systems will, in turn, concentrate in large part on changes in the determination of word-level prosodic prominence, or simply stress. Approaching this aspect of prosody in particular is particularly feasible because both the theoretical apparatus for the study of stress in phonological terms is so extensive (see Chapter 2) and research on the learnability of stress systems (including lexical stress) is also relatively well developed (see Chapter 3). The overarching objective for this book, particularly in Part II, will be to provide up-to-date phonological analyses of word-level prosody in several older Indo-European languages, to describe notable diachronic and diatopic developments in word prosody affecting those languages, and finally, to attempt to model, and thereby provide a partial account, from a strictly phonological perspective, of how some of those changes might have come about.

1.5 Summary and Prospectus

This chapter has offered a short introduction to the study of linguistic prosody, with particular emphasis on the phonological analysis of prosody at the level of the prosodic word. The theoretical positions adopted fit within a relatively “mainstream” or “standard” view of prosodic structure. The prosodic hierarchy as presented in (6) is accepted, but with the adjustments introduced under Match Theory that admit of recursive constituents (at least at the level of the prosodic word and above). The foot and syllable are adopted as prosodic constituents in preference to the major theoretical alternatives, and morae are generally employed for the representation of weight contrasts, at least where a generally binary contrast between heavy and light syllables is adequate.⁸⁷ This perspective largely matches the assumptions made in, e.g., Féry 2017 or Ryan 2019. Throughout this work, morae, syllables and foot structure will be consistently adopted in the analysis of phenomena of stress, tone, and intonation; the possibility for recursion of the prosodic word will come into play where the prosody of cliticization or compounding is concerned.

Section 1.3 has also provided a brief overview of the concepts of intonation, stress, tone, and “pitch accent”. In general, I adopt the assumption that intonational use of pitch constitutes a phonological universal, whether a language possesses a lexicon of tunes associated with particular semantics, or whether intonation merely serves to mark prosodic structure and constituency in some fashion. Stress, meanwhile, I understand in the most abstract sense possible, namely, the presence of metrical constituency and/or privative prominence marking internal to the prosodic word. In accord with the view of Hyman (2009), I avoid the use of “pitch-accent language” as a label for a typology of prosodic system, and follow Hyman’s injunction to more carefully distinguish between phonological properties that can occur independently of one another. Since most Indo-European languages, ancient and mod-

⁸⁷See the treatment of “recessive” accentuation in Ancient Attic-Ionic Greek in 6.2.3 for a concrete case that may require a finer distinction between relatively heavier syllables with long vowels or diphthongs versus relatively lighter syllables with short vowels and coda consonants.

ern alike, as will be seen throughout this book, tend to exhibit obligatory, culminative, and privative word-level prominence as a property designated on syllables, and often transparently or potentially attributable to metrical structure, I consider it appropriate to apply the term “stress” in discussing the word-level prosody of these languages. More detailed treatments at 4.1.3 and 6.2.1.7 will justify this choice with respect to details of Vedic and Ancient Greek word prosody.

These two major components – a theoretical framework for prosodic structures and a grounded classification of suprasegmental categories – undergird the rest of the present work. They will now enable a detailed exposition of stress systems and some models for the assignment of word stress in the next chapter, which will in turn feed the overview of perspectives on learnability and change in prosodic phonology. With this complete set of tools in hand, detailed investigations into the prosodic systems of older Indo-European languages can be undertaken in Part II.

CHAPTER 2

The Phonological Analysis of Stress Assignment: Metrical Structure and Lexical Stress

This chapter presents an introduction to the phonological analysis of stress systems, as a necessary prerequisite to the study of diachronic change in prosodic systems. A detailed presentation of approaches to the analysis of stress, their underlying principles, and the tools involved is indispensable for ultimately discussing prosodic change in an insightful way. For example, as concerns the assignment of primary stress, if one observes that some language at a given stage regularly assigned primary stress to the penultimate syllable of a prosodic word, but later assigned primary stress to a heavy penult or else the antepenult, the first descriptive task is state which rules, parameters, or constraint rankings have undergone some change. Since stress will stand at the center of the studies in Part II, an adequate means of characterizing *what* has changed, precisely, must precede any attempts to understand how and why those changes might have come about.

Before delving into descriptive formalisms, I first present a brief overview of the typology of stress systems in Section 2.1. The core of the chapter is then developed in sections devoted to the parametric system of foot structures and stress rules employed in Hayes 1995 (Section 2.2) and further extensions of those insights implemented in Optimality Theory (Section 2.3). The range of systems treated there is initially deliberately restricted to systems of the rhythmic and bounded type (see further below). Neither the alternative parametric model (“simplified bracketed grid”; SBG) of stress assignment worked out in Isardi 1992, Halle and Isardi 1995, and Isardi 2009, nor the optimality-theoretic implementation of a foot-free approach to stress assignment seen in Gordon 2000, will be systematically explicated, though scholarship relying on SBG or Gordon’s foot-free constraints will occasionally play a role. The view on stress and accent taken by van der Hulst (see discussion under 1.3.3.2) will be mentioned only in passing. Finally, the model from Section 2.3 is extended to stress systems exhibiting less regularity or predictability, i.e., languages with unbounded stress, or stress which is sensitive to morphological structure or lexical marking (2.4). I conclude the chapter with brief discussion of some computational perspectives on the learnability of stress systems (2.5), which will serve as a bridge to Chapter 3.

Within generative approaches to analysis of stress and accent, a series of historical developments and major approaches may be distinguished. Prior to the work of Liberman and Prince, stress was essentially treated as phonological feature assigned by (iterative) rule, where, however, degrees of stress could be distinguished (e.g., [1stress], [2stress], etc.). This approach is found in SPE; its conventions and legacy are discussed in Anderson 1974: Ch. 9. The works of Liberman (1975) and Liberman and Prince (1977), with the introduction of

prosodic tree structures to represent strong and weak constituents, as well as the explicit notion of the foot as a prosodic constituent, represent a turning point. Essentially all subsequent work on the analysis of stress is indebted to this work, whether through direct inheritance or by way of contrast. During the 1980's, work on two closely related but distinct approaches to stress assignment, culminating in two canonical publications on stress, Hayes 1995, on the one hand, and Halle and Isardi 1995, on the other. These two approaches share the assumption that stress assignment is determined fundamentally by parameters that decide how prosodic constituents are to be constructed, and that prosodic structure relating to stress is (normally) built from the bottom up. They differ from one another essentially in their view on feet: Hayes 1995 attempts to restrict the range of typologically occurring feet to a minimum; Halle & Isardi do not assume a strict inventory foot structures, but instead admit of a relatively freer combination of elements for the construction and projection of constituents. Parallel to these models is the “primary first” approach developed by van der Hulst (1984, 2009, 2012). van der Hulst's model shares with Halle and Isardi (1995) a rejection of a limited foot inventory feet – though differs from the latter in determining stress from the top down, assuming that prosodic constituency is irrelevant to the assignment of primary stress; thus, the assignment of primary and secondary stress is strictly divorced. These three analytical models may fairly be said to represent the theoretical underpinnings for the majority of research into stress systems on the market.

2.1 Where Stress Comes From: Basics of Stress Typology

In accord with the view of prosodic structure developed in Chapter 1, I depart from the basic assumption that stress phonologically derives, on the one hand, from rhythmic organization internal to the word (Lieberman and Prince 1977), and, on the other, from the marking of a most prominent constituent dominated by the prosodic word. In languages in which the parsing of syllables into feet is generally preferred (i.e., the direct domination of σ by ω should be generally avoided), primary stress may be seen as the promotion of one constituent within the larger rhythmic structure; in this sense, stress relations are constructed in a bottom-up fashion. The selection of a most prominent constituent – a head foot, and a head syllable within that foot – may depend on numerous factors, from position of a syllable in the prosodic word to morphological structure, including scopal relations.

The classic principal division in stress typology is the distinction between **fixed** and **free** stress, where “fixed stress is predictable in location, and may be derived by rule, while free stress is unpredictable and must be lexically listed” (Hayes 1995: 31; cf. Revithiadou 1999: 11). While intuitively appealing, the definition given by Hayes is perhaps not fully cogent. First, stress assignment in principle always requires the operation of a rule ($\sim$ constraint interaction) of some sort, whether or not that rule may make reference to some arbitrary lexical properties. Second, non-derived instances of weight and tone are themselves lexical properties (if perhaps not arbitrary the same way as a stress diacritic), and may determine the position of primary stress; in unbounded stress systems (cf. Hayes 1995: 296–99) the position of primary stress in an arbitrarily long word is unpredictable if one does not

know the relevant properties (e.g., length, height, tone) of every potential syllable nucleus in the input.¹ Third, even in languages with diacritic or morphologically determined stress, its position may nonetheless be at least partly predictable, if confined to some window at the edge of the word.

The other conceivable primary division would be the distinction between bounded and unbounded systems, i.e., whether or not “stress can fall an unlimited distance from a [word] boundary or another stress, provided the appropriate conditions are met” (Hayes 1995: 32). Among both bounded and unbounded systems, primary stress might be assigned on the basis of some preferred property (syllable weight, lexical diacritic, etc.), the distinction lying with the domain in which the language may search to find a syllable with the relevant property (i.e., the entire word vs. some smaller window).

I will take the distinction of “fixed” vs. “free” stress systems to be the most basic typological distinction, but provide an unambiguously explicit definition of the former: a fixed stress system is a system of stress assignment whose rules need refer only to phonological properties and prosodic constituents that do not directly refer to either abstract, lexical stress or lexical foot structure directly (see 2.4.2.2 on the representation of lexically specified stress properties). Falling within the scope of this definition are then both properly “metrical” stress systems and “prominence-driven” stress systems (in the sense of de Lacy 2002). That is, if the position of stress may be predicted by reference to a word edge and derived foot constituents (“metrical” stress), or by reference to a word edge and another contrastive and non-hidden phonological property (e.g., weight or tone, thus “prominence-driven” stress), then stress may be regarded as “fixed”, or perhaps better “phonologically predictable”.² In contrast, systems with “free” stress possess either lexical diacritic marking of some sort that is primarily relevant to stress assignment, and/or refer to the morphological constituency of the word.³ The term “interface system” preferred by Revithiadou (1999), seems appropriate to systems in which stress assignment refers to morphological structure, but perhaps not for systems that merely possess lexical accents, since both prominence-driven and simple lexical accent systems may exhibit very similar stress assignment algorithms, as shown in 1; note further that both stress assignment algorithms possess a purely metrical stress component (the *else* clause).

¹ Compare the observation of Dresher and Kaye 1990: 183: “Accented syllables in languages with lexical accent play much the same role as heavy syllables in Q[uan]tity S[ensitive] languages.”

² Apoussidou (2007: 11) also employs the term “grammatical stress”, which is certainly meant to include metrical stress systems, but it is not clear to me that Apoussidou means to include prominence-driven systems there as well. Staubs (2014: 19–20), meanwhile, seems to apply the term “fixed stress” to mean “metrical” stress, where primary stress always falls at some predictable distance from the edge of the word.

³ One could though arguably class the prominence-driven systems together with lexical systems against the metrical systems, since the prominence-driven systems may depend upon lexical properties like vowel length or lexical tone, or distinguish three major categories (“metrical”, “prominence-driven”, “lexical”).

- (1) Comparison: Prominence-Driven vs. Simple Lexical Stress
 - a. PROMINENCE-DRIVEN STRESS: place primary stress on the rightmost heavy syllable, *else* on the leftmost syllable.
 - b. SIMPLE LEXICAL STRESS: place primary stress on the rightmost lexically accented syllable, *else* on the leftmost syllable.

Indeed, even within languages that impose strong morphological restrictions on the domain of primary stress assignment, a purely metrical component nevertheless remains necessary to decide the position of primary stress in cases where the requirements deriving from interface with the morphology could be satisfied by more than one syllable. In Norton Sound Yupik (data from Jacobson 1985: 29–41; cf. discussion in van de Vijver 1998: 131 and Revithiadou 1999: 23), for instance, stress must always occur within the stem, determined by an iambic foot built within the stem (cf. 2.2 below on foot patterns). If the stem is monosyllabic and heavy (2a.), stress falls on this syllable; if it is disyllabic, stress falls on the second syllable (2b.); if the only stem syllable would be light, then it is made heavy by consonant gemination to provide a well-formed iamb within the stem.⁴

- (2) Stress in Norton Sound Yupik: iamb built within the stem
 - a. /ang-yamini/ → [(áŋ).(ya.mí:).ni] ‘boat-his own’
 - b. /qaya-ni/ → [(qayá:).ni] ‘kayak-his own’
 - c. /kuvə-uq/ → [(kúv).vuq] ‘it spills’ (not ^x[(ku.vúq)])

Finally, recall from discussion under 1.3.3.2 that stress, however, assigned, is usually taken to be a property of syllables or syllable peaks. Hayes (1995), for instance, although he adopts the grid representation of prosodic relationships of Halle and Vergnaud 1987, takes the terminal nodes of a grid to be syllables, which are grouped together with brackets to indicate their organization. In this sense, Hayes is more explicit, and leaves less room for language-particular parameters, than Halle & Vergnaud, for whom the terminal grid marks could theoretically refer to any type of phonological unit and differ from language to language.⁵

A provisional typology of stress-assignment systems is presented in 2.1. Following the discussion above, the basic division falls between systems that are wholly phonologically

⁴In most cases in Jacobson’s data, stress could be analyzed as purely metrical, with iambs being constructed left to right. Precisely the instance of gemination to ensure that stress falls within the domain of the stem demonstrates the necessity of a component of morphological interface. Note further that Jacobson marks all stressed syllables equally with an acute, and does not comment on the relative degree of prominence between stresses. Compare also Hayes 1995: 242–7 for metrical analysis and treatment of segmental lengthening processes in the same data.

⁵The assumption that the syllable constitutes the stress-bearing unit (Hayes 1995: 49–50, following Jakobson 1931 [1962]) has important implications for the analysis of prosodic prominence in many Indo-European languages (Lithuanian, Latvian, Serbo-Croatian, Ancient Greek) in which pitch contours may potentially be interpreted as referring to some kind of unit smaller than the syllable (e.g., moras). Insofar as such languages are taken to have stress in the same phonological sense that, e.g., Latin or Russian do, an additional set of rules and/or representations that can derive those pitch contours is necessary; see particularly Chapter 6 on Ancient Greek.

predictable, and systems that are not, or, between systems with lexically specified stress (or foot structure) and systems in which preferences for prosodic constituency are not prespecified. Among the phonologically unpredictable systems, the fundamental division assumed there is between systems that genuinely require reference to morphological structures and systems that do not. Among the morphologically sensitive systems, I follow the distinction of drawn by Revithiadou (1999) between head-stress and head-dependent systems, though these two properties, as well as the presence or absence of lexical accent, and presence or absence of prominence attraction. I assume that all properties may, in principle, be freely combined, with further requirement that all head-dependent systems must have lexical accents; other gaps in the typology of stress calculation might exist as well, e.g., a system combining attracting of stress to heavy syllables *and/or* syllables with lexical high tone.⁶ Systems for which good evidence is unknown to me are placed in square brackets.

(3) Typology of Stress Calculation

I. Phonologically Predictable

- a. Pure Metrical Stress (e.g., Pintupi, Classical Latin)
- b. Prominence-Driven Stress
 - i. Weight-to-Stress (e.g., Selkup)
 - ii. Tone-to-Stress (e.g., Ayutla Mixtec)
 - iii. [Weight-to-Stress and Tone-to-Stress]

II. Phonologically Unpredictable

- a. Morphologically Insensitive
 - i. Simple Lexical Accent (e.g., Cupeño)
 - ii. Lexical Accent and Prominence (Weight-to-Stress or Tone-to-Stress; e.g., Stockholm Swedish)
- b. Morphologically Sensitive
 - i. [Head-dependence, head-stress, lexical accent, and prominence]
 - ii. Head-dependence, lexical accent, and prominence (perhaps Ancient Attic-Ionic Greek? See Chapter 6)
 - iii. [Head-stress, lexical accent, and prominence]
 - iv. [Head-stress and prominence]
 - v. Head-dependence, head-stress, and lexical accent (e.g., Thompson Salish)⁷

⁶I have not attempted to document the actual attestation of all logically possible systems suggested in this typology, especially as concerns the role of prominence-driven stress. Systems I.a., I.b., and II.a.i. are amply documented; morphologically sensitive systems without further prominence conditions (II.b.v–viii.) are all claimed to exist by Revithiadou (1999). The major gap in empirical description seems to be the possible interaction of weight-to-stress or tone-to-stress with lexical accent and types of head-controlled stress.

⁷Thompson Salish seems, however, to possess separate stress subsystems: a stem-level system in which lexical accent and head-dependence is active, as in Russian, and a word-level system in which head-stress is the overriding principle. Thus, lexical accents and preference for stress on morphological heads do not truly interact. See Revithiadou 1999: Ch. 5.

- vi. [Head stress and lexical accent (perhaps Tahltan?)]
- vii. Head-stress (e.g., Norton Sound Yupik)
- viii. Head-dependence and lexical accent (e.g., Russian)

In the following sections, I present the approaches to the analysis of word stress that underlie and are indispensable for an understanding of all further chapters of this book. This introduction, though especially 2.3, is recommended to all readers without much familiarity with the phonological literature on stress patterns or stress assignment.

2.2 Parameters in *Metrical Stress Theory*

The most typologically thorough and perhaps theoretically most restrictive treatment of word stress in a pre-OT framework is Hayes 1995. In a real sense, the model of stress analysis developed and presented there represents the culmination of a “standard theory” of word stress, departing from the organization phrases and words into strong and weak prosodic constituents in Liberman 1975 and Liberman and Prince 1977, and successively further developed in Hayes 1980, Hammond 1984, Halle and Vergnaud 1987, and Kager 1989, among many others. Perhaps the fundamental feature that characterizes this strain of research is the assumption that word stress may be calculated by reference constituent structure below the word level, i.e., metrical feet (cf. 1.2.3 in Ch. 1); in this respect, the “standard theory” of word stress may be contrasted with proposals to derive stress exclusively by reference to a grid (Prince 1983, Selkirk 1984; cf. Hayes 1995: 37–9), as well as with the simplified bracketed grid theory (Isardi 1992, Halle and Isardi 1995). Like most work on stress within the strain of research belonging to the “standard theory”, Hayes (1995) captures the construction of prosodic constituents and assignment of prominence relations among those constituents through the setting of various prosodic **parameters**. Such a parametric approach to prosodic systems was and remains influential in thinking concerning the learnability of stress systems, as well as how stress systems might undergo change diachronically (e.g., Dresher and Kaye 1990, Lahiri 2015, Nazarov and Jarosz 2017; see further discussion in 3.3). Furthermore, most subsequent optimality-theoretic treatments of stress may be rightly regarded as an adaptation and extension of this “standard theory”; as discussion in this and the following section will make clear, many of the parameter settings of the standard theory have more or less direct analogues in terms of constraints, or correspond to the interaction between certain constraints.

2.2.1 Canonical Feet

Here and in the following, I outline the major parameters involved in the calculation of stress employed by Hayes, and illustrate throughout the type of metrical stress systems that emerge from different parameter values. First, one truly restrictive aspect of Hayes’ overall model derives from the limited inventory of foot types that he argues to be typologically supported. Namely, Hayes claims that all (bounded and rhythmic) metrical stress systems can be analyzed as possessing one of three types of feet: syllabic trochees, moraic trochees, or

iamb, shown in (4). If broken down into smaller parameters, one could contrast quantity-sensitive feet (moraic trochees, iambs) vs. quantity-insensitive feet (syllabic trochees), and left-headed (trochees) vs. right-headed feet (iambs). Precisely because Hayes holds the typologically available inventory of feet to be unbalanced (contrast the inventory proposed in Hayes 1980; cf. Hayes 1995: 75–77), the parameter FOOT FORM, with three possible values, is employed instead of binary parameters referencing quantity-sensitivity and the direction of headedness.

(4) Typology of Canonical Feet (cf. Hayes 1995: 71, Apoussidou 2007: 12)

- a. SYLLABIC TROCHEE: a foot of exactly two syllables with left-syllable prominence.

$$\begin{array}{cc} (\times & .) \\ \sigma & \sigma \end{array}$$

- b. MORAIC TROCHEE: a foot of two morae, distributed over one or two syllables, with left-syllable prominence.

$$\begin{array}{ccc} (\times & .) & (\times) \\ \smile & \smile & \text{or} \quad \text{—} \end{array}$$

- c. IAMB: a foot of two syllables, with right-syllable prominence where the first is light, or a single heavy syllable.

$$\begin{array}{ccc} (. & \times) & (\times) \\ \smile & \sigma & \text{or} \quad \text{—} \end{array}$$

(5) FOOT FORM Parameter: syllabic trochee; moraic trochee; iamb.

A fundamental rule for any language with rhythmic stress will be to iteratively construct constituents of one or more syllables according to the specified foot shape employed by the language.⁸ From a bottom-up perspective, once segments are organized into syllables, the syllables are in turn to be grouped into iterating patterns of strong and weak elements within a foot. Moraic trochees and iambs both share the feature that a single heavy syllable may constitute a foot; this feature requires that a language be quantity sensitive in order to be able to use moraic trochees or iambs. Languages without distinctions in syllable weight that nevertheless exhibit bounded rhythmic stress are limited to the syllabic trochee, and indeed, languages that exhibit syllabic trochees tend to lack syllable quantity contrasts. Indeed, the only typological fact that would seem to necessitate the distinction between syllabic and moraic trochees is the fact that some languages with quantity contrasts nevertheless preferentially build consistently disyllabic feet (but cf. below on degenerate feet); Kager (1992) terms such languages “truly quantity insensitive”.⁹

⁸See Section 2.6 for some discussion as to whether a language might employ different foot shapes. As will become evident already in Section 2.3, a grammar consisting of ranking violable constraints rather than fixed parameters readily admits of the possibility that different foot structures might be created under different conditions in one and the same language.

⁹Finno-Ugric languages are particularly well represented among the “truly quantity insensitive languages”; cf. the list in Hayes 1995: 102.

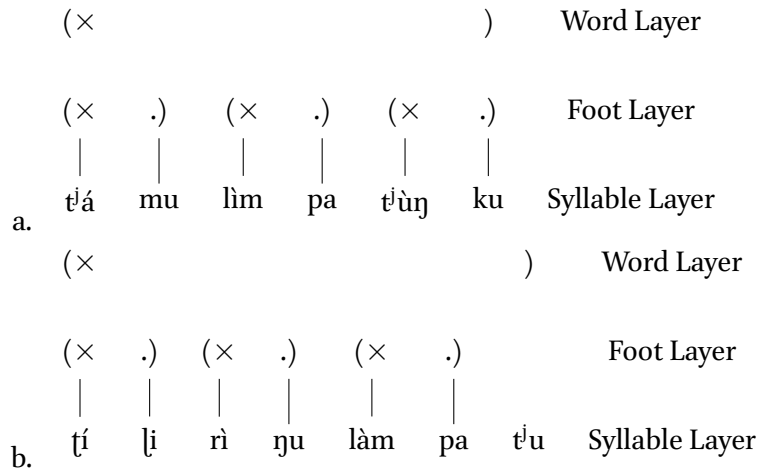
2.2.2 Direction of Parsing and End Rule

The two further most basic parameters required for the description of any metrical stress system are the DIRECTION OF PARSING and the END RULE. Both parameters, shown below, refer to left or right edges. The former decides from which edge of the word foot parsing begins, while the latter determines which foot (leftmost or rightmost) will be promoted to the primary prominence of the word (i.e., which foot head will be promoted to the head of the word in the construction of the word layer).

- (6) **DIRECTION OF PARSING Parameter:** left-to-right; right-to-left.
- (7) **END RULE Parameter:** left; right.

These three parameters are sufficient for describing a number of basic and common stress patterns, representing a total typology of 12 possible different languages ($3 \times 2 \times 2$).¹⁰ Three languages that can arise and need refer only to these parameters (though further parameters discussed below must, in principle, be set as well), assuming that a minimal and maximal foot must fit one of the templates in (4), include the following illustrated in (8)–(10). As the examples there show, the quantity-sensitive iambs and moraic trochees admit of more diversity in stress patterns, since the exact sequence of heavy and light syllables impacts foot construction. The data from these languages will recur in Section 2.3.

- (8) Pintupi (Pama-Nyungan; cf. Hayes 1995: 62–4): syllabic trochees; left-to-right-parsing; end rule left = primary stress on initial syllable, alternating secondary stress on every odd-numbered syllable from the left.



¹⁰Whether in fact all possible systems potentially arising from this small set of parameters actually occur is unclear. Some strong typological tendencies certainly exist, e.g., among syllabic trochee systems, the direction of parsing strongly correlates with the end rule, i.e., left-to-right systems have initial primary stress, while right-to-left systems have final primary stress. Among the languages sketched in Hayes 1995: 198–203, only Piro (Arawakan) and Warao (isolate) exhibit an end rule of a direction opposite their direction of parsing, but at the same time both obligatorily build a binary foot aligned with the word edge that corresponds to the direction of the end rule. In effect, there is little to no evidence for syllabic trochee systems that vary between either initial and peninitial stress or between antepenultimate and penultimate stress because of opposite directional settings for foot parsing and the end rule.

- (9) Seminole/Creek (Muskogean; cf. Hayes 1995: 64–7): iambs; left-to-right parsing; end rule right = primary stress on final or penultimate syllable.
- a. All light syllables
 - i. /˘˘˘˘/ (a.pa).(ta.ká) ‘pancake’
 - ii. /˘˘˘˘˘/ (a.ma).(pa.tá).ka ‘my pancake’
 - b. Final heavy syllable: always final primary stress, e.g., /—˘ —/ (ak).ha.(sí:) ‘lake’.
 - c. Penultimate heavy followed by final light: always penultimate primary stress, e.g., /— —˘/ (ak).(cáwh).ka ‘stork’
 - d. Pre-penultimate heavy followed by light.
 - i. /—˘˘/ (ta:s).(ki.tá) ‘jump:SUBJ.SG’
 - ii. /—˘˘˘/ (ta:s).(ho.kí).ta ‘jump:SUBJ.DU’
- (10) Fijian (Austronesian; cf. Hayes 1995: 142–9): moraic trochees; right-to-left parsing; end rule right = primary stress on the final syllable (if heavy) or penultimate syllable (if final is light), alternating secondary stress on every second light syllable and every heavy syllable counting from the right edge.
- a. Final heavy syllable: always final primary stress
 - i. /˘˘ —/ (n̄rè:).(n̄ré:) ‘difficult’
 - ii. /˘˘˘˘ —/ pa.(là.si).(tá:) ‘plaster’
 - b. Light final syllable: always penultimate primary stress¹¹
 - i. /˘˘˘˘/ βi.(ná.ka) ‘good’
 - ii. /˘˘˘˘˘/ ta.(ràu).(sé.se) ‘trousers’

2.2.3 Degenerate Feet

Examples like (8)b., (9)a.ii, or (10)b.i. all contain forms where a syllable at the left or right edge of the word fails to be parsed into a foot. Had the parse begun at the opposite edge, the unparsed syllables would in turn have fallen at the opposite edge, besides generating a distinct alternating stress pattern. (8)b., if instead generated using left-to-right parsing, would be *[t̪i(l̪i.ri).(ŋù.lam).(pà.t̪u)]. In all of these cases, the absence of a primary or secondary stress on a syllable at the edge of the word may be attributed to a strict prohibition on degenerate feet. That is, Pintupi, Seminole/Creek, and Fijian all apparently do not permit for a foot to be subminimal, i.e., consist of only a single syllable (in the case of syllabic trochees) or a single light syllable (in the case of iambs or moraic trochees). However, a ban on subminimal feet is evidently not a prosodic universal, and instead constitutes another parameter. Hayes (1995: 89–105) makes the case that the parameter in has only two values, strong (no degenerate feet) and weak (degenerate foot allowed when used to project primary stress),

¹¹There are no cases of a stressed heavy penultimate syllable in Fijian, due to the process of TROCHAIC SHORTENING, which shortens long vowels or diphthongs in penultimate syllables preceding a final light syllable. In effect, trochaic shortening conspires to ensure that a well-formed moraic trochee is always aligned with the right edge of the word.

but allows for the possibility that some languages might freely degenerate feet at word edges (e.g., perhaps Icelandic; see (43) in Section 1.2.3).

- (11) **DEGENERATE FOOT Parameter:** strong; weak; (free?)
- a. Strong: Feet of the shape ($\sim$) are absolutely banned.
 - b. Weak: Feet of the shape ($\sim$) are permitted when dominated by another grid mark (i.e., the word layer = primary stress).

Auca (language isolate of Ecuador), for instance, would appear to select for the “weak” option of the degenerate foot parameter: primary stress always falls on the final (in words with an odd number of syllables) or penultimate (in words with an even number of syllables) syllable, and a secondary stress in words of three or more syllables always falls on the antepenult (in odd-numbered words) or fourth-to-last syllable (in even-numbered words). Such a pattern emerges through left-to-right parsing of syllabic trochees, where the crucial diagnostic for the degenerate foot emerges in unsuffixed words with an odd number of syllables (cf. Hayes 1995: 90; 182–8).¹² In (12), example b. consists of two well-formed syllabic trochees, while example a. contains a degenerate foot to which primary stress comes to be assigned.

- (12) Stress Assignment in Auca (Unsuffixed): syllabic trochees, left-to-right parsing, end rule right, weak prohibition of degenerate feet.
- a. /mɔ̃iko/ → [(mɔ̃.i).(kó)] ‘blanket’
 - b. /bodæpoka/ → [(bò.dæ).(pó.ka)] ‘anthill’

The relation of the degenerate foot parameter to the inventory of canonical feet has interesting implications for languages that have syllable quantity contrasts, but preferentially construct syllabic trochees. Since a foot built around a heavy syllable /–/ is not degenerate, such languages are permitted to construct feet of the form (–), even when the foot form parameter is set to syllabic trochee. Estonian presents just such a case, where final heavy syllables may receive secondary stress, but final light syllables may not. Thus, (13)a. and (13)c. do not have secondary stress on the final syllable, whereas (13)b. does; (13)c. further shows that the canonical foot is the syllabic trochee, rather than moraic trochee, since a binary foot of the shape ($\sim$ –) is built there.¹³

¹²The analysis of stress in Auca is complicated by the fact that two rounds of stress assignment appear to occur: left-to-right syllabic trochees in the stem, then right-to-left syllabic trochees in the suffix complex. Hence, only unsuffixed words like (12)a. are truly diagnostic of degenerate feet.

¹³The prosody of Estonian presents a host of interesting challenges, most famously the contrast of long (bi-moraic) and overlong (trimoraic) syllables, as well as the possibility of ternary (as opposed to strictly binary) alternations in stress. Besides the discussion in Hayes 1995: 316–29, see the data in Hint 1973 and extensive analysis of Prince 1980. For more recent phonological analyses, see Anttila and Cho 1998, Prillop 2013, and Golston 2021: 13; for phonetic descriptions of Estonian stress, which in part cast doubt on the patterns of secondary stress, see Gordon 1997 and Asu and Lippus 2018.

- (13) Stress Assignment in Estonian: syllabic trochees, left-to-right parsing, end rule left, strong prohibition of degenerate feet.
- a. /osava/ → [(ó.sa).va] = [(˘ ˘) ˘]
 - b. /osavatt/ → [(ó.sa).(vätt)] = [(˘ ˘) (—)]
 - c. /pimestavasse/ → [(pí.mes).(tà.vas).se] = [(˘ —) (˘ —) ˘]; OR
 /pimestavasse/ → [(pí.mes).ta.(väs.se)] = [(˘ —) ˘ (— ˘)]

2.2.4 Location of Parsing

Exactly how degenerate feet are to be treated in combination with particular parameters for foot structure will occupy us again in Chapter 7, where the question of how to properly generate initial stress in words beginning with the sequence /˘ —/ in systems where the canonical foot is the moraic trochee requires special consideration. More generally, in quantity-sensitive systems in which stress persistently occurs at a word edge, regardless of the weight of the first two or last two syllables, such a result can only be obtained if the word edge is obligatorily aligned with the edge of a foot. In the context of the Hayesian theory, this can obtain from a parametric difference in the LOCATION OF PARSING: top down (the word layer is constructed first) vs. bottom up (the foot layer is constructed first). All of the systems discussed thus far apply bottom-up parsing, constructing the foot layer first; in contrast, if the word layer is constructed first, then a word layer grid mark will be placed over the first or last syllable (based on the END RULE parameter), with the consequence that the first/last syllable cannot be skipped in construction of the foot layer.

- (14) LOCATION OF PARSING Parameter: bottom up; top down

In Gothic (East Germanic), the generally persistent occurrence of primary stress on the initial syllable, despite evidence of moraic trochees as the preferred foot shape (cf. Kiparsky 1998b), can be obtained by applying top-down parsing, with the end rule parameter set to left; see the much more extensive discussion in Chapter 7. If construction of the foot layer were to apply first, then 15b. would be expected to exhibit primary stress on the second syllable, ^x[si.(˘ po:).(˘ ni:)(˘ s)], skipping the initial syllable, since (˘ —) is not a canonical moraic trochee and (˘ ˘) would constitute a degenerate foot.

- (15) Gothic: moraic trochees, top-down parsing, end rule left, left-to-right parsing, strong prohibition of degenerate feet.

a. Heavy–Light: *airkniþa* ('ɛrk).(ni.θa) 'sincerity:F.NOM.SG'

i. Word Layer Construction

(×) Word Layer

ɛrk ni θa Syllable Layer

ii. Foot Layer Construction

(×) Word Layer

(×) (× .) Foot Layer

ɛrk ni θa Syllable Layer

b. Light–Heavy: *siponeis* ('si.po:).(ni:)<s>. 'disciple:M.NOM.SG'

i. Word Layer Construction

(×) Word Layer

si po: ni:<s> Syllable Layer

ii. Foot Layer Construction

(×) Word Layer

(× .) (×) Foot Layer

si po: ni:<s> Syllable Layer

Meanwhile, the variation in the position of secondary stress seen in 13c. concerns a further, comparatively minor parameter, namely, LOCAL PARSING, with two possible settings, weak and strong, stated in (16) (cf. Hayes 1995: 308). This parameter is employed to generate languages that appear to exhibit patterns of ternary alternation, where stress may appear to (sometimes) occur on every third syllable, rather than in the (much more common) binary alternating pattern.¹⁴

¹⁴On apparent ternary stress patterns, see most recently Golston 2021, where an optimality-theoretic im-

- (16) **LOCAL PARSING Parameter:** strong; weak
- a. Strong: align the window for foot parsing on the next unfooted syllable immediately following a foot.
 - b. Weak: align the window for foot parsing by skipping over a light syllable /~/ immediately following a foot.

In analyzing Estonian, Hayes accounts for the variation seen in cases like (13)c. by allowing the weak setting of the local parsing parameter to be optionally invoked. Thus, the first parse shown there, [(pí.mes).(tà.vas).se], obtains when strong local parsing is employed (after creating the first foot at the left edge, parsing continues on the syllable /ta/); the second parse, [(pí.mes).ta.(väs.se)], on the other hand, results from weak local parsing, where the medial light syllable /ta/ is skipped.

2.2.5 Iterativity and Persistency

Finally, two further parameters concerning foot parsing govern whether all potentially well-formed feet are, in fact, constructed. These are **ITERATIVITY** and **PERSISTENCY**: the first determines whether a string is exhaustively parsed into feet, or whether foot construction ceases once one foot has been built; the second determines what occurs with material that is defooted through the application of segmental processes or destressing to avoid stress clash (i.e., the occurrence of stress on two adjacent syllables). All examples shown in the preceding subsections reflect systems with iterative stress, and persistent stress (as far as can be determined).¹⁵

- (17) **ITERATIVITY Parameter:** iterative; non-iterative.¹⁶

- (18) **PERSISTENCY Parameter:** persistent; non-persistent.

Non-iterativity may be observed in languages that exhibit an obligatory primary stress, but present no evidence for secondary stress.¹⁷ Macedonian (cf. Halle 1990: 153–6, Kenstowicz 1991: 175–9) would appear to be such a language: primary stress regularly falls on the antepenultimate syllable (though also exceptionally on the penultimate or final syllable),

plementation of weak local parsing (i.e., skipping some light syllables) is argued to account for three of the empirically best supported instances of ternary patterns (Chugach Alutiiq Yupik, Tripura Bangla, and Cayuvava), as opposed to the recursive feet posited by Paricio and Kager (2015).

¹⁵The Gothic form in (15)a. contains an instance of stress clash that might, in fact, result in destressing. See 7.2.2.2 in Chapter 7 for discussion of stress in Gothic.

¹⁶This parameter is discussed only fleetingly in Hayes 1995: 113, though included in a summary list of parameters (400); see rather Hayes 1980 for more discussion. Halle and Vergnaud (1987) and Halle (1990) instead assume that an Exhaustivity Condition always results in exhaustive parsing of a string into constituents according to the rules of the layer or line at which they apply, but Conflation of heads on one or more layers/lines of the stress grid will eliminate all stresses except for the primary word stress.

¹⁷See, however, McCarthy 2003: 111–4 for discussion as to whether the phonetic absence of clear secondary stress necessarily implies that only a single foot per word is constructed. Exhaustive foot parsing, even in the absence of secondary stress (or indeed any stress) would be in accord with segmental evidence for foot structure discussed at 1.2.3 in Chapter 1.

and no secondary stress occurs elsewhere to the left of the single primary stress. After skipping the rightmost syllable (see immediately below on extrametricality), a syllabic trochee is built around the second- and third-to-last syllables, at which point the application of foot parsing ceases. In both (19)a. and b., the availability of further disyllabic sequences to the left of the primary stress would in principle allow for the construction of additional feet where a secondary stress could occur.¹⁸

(19) Non-Iterative Footing in Macedonian:

- a. /vodeničar-ot/ ‘miller-DEF’ → [vo.de.(‘ni.ča.).(rot)]
 b. /momitš-inja-ta/ ‘girl-PL-DEF’ → [mo.mit.(‘šin.ja).(ta)]

Non-persistent footing is systematically evident in languages that avoid stress clash. Quite naturally, if a previously footed constituent is defooted in order to remove a clash, the same previously stressed syllable should not be parsed again as a foot head. Such a situation can most readily obtain in moraic trochee or iambic systems, where a heavy syllable that could build a foot stands adjacent to a stressed syllable. The optional destressing in clash applied in Cahuilla (Uto-Aztecan), which builds moraic trochees from left-to-right, with a degenerate foot allowed at the left edge, illustrates such non-persistent footing (cf. Hayes 1995: 136).¹⁹

(20) Cahuilla: Optional Destressing in Clash

/kavi:čiwen/	/kakawla:qa/	UR
(‘ka).(vi:).(či.wen)	(‘ka).(kaw).(la:).qa	Foot Construction / End Rule Left
(ka).vi:.(či.wen)	(‘ka).kaw.(la:).qa	Destressing in Clash
[(ka).vi:).(či.wen)]	[(‘ka).kaw).(la:).qa]	SR

A destressed foot head might, however, become a foot tail, if footing is persistent and an adjacent syllable can allow for refooting in such a fashion that no clash will result (cf. Harris 1989 [*non vidi*] and Hayes 1995: 96–7 on the initial dactyl effect in Spanish; see also Hualde 2010 and Buckley 2016 for more recent treatments of secondary stress in Spanish).

More interesting is the contrast between persistent and non-persisting footing due to the application of segmental processes that alter the number of syllables or the weight of syllables. Contrast here the persistent footing applied after the shortening of long vowels in Maithili (Indo-Aryan; 21) with the non-persistent footing of outputs resulting from vowel deletion in Cyrenaican Bedouin Arabic (22). In the former case, any underlying long vowel to the left of the main stress is shortened, with the consequence that new adjacent sequences of two light syllables can build a moraic trochee. In the latter case, processes of syncope and subsequent epenthesis substantially disrupt the footing applied to the underlying sequence

¹⁸Questions surrounding the iterativity parameter are further complicated by cases in which foot-dependent segmental processes provide evidence exhaustive parsing, but no secondary stress is phonetically evident. Kashaya represents a language of this sort: the process of iambic lengthening requires for the word to be exhaustively parsed into iambic feet, but phonetically, only a primary stress on the leftmost foot is evident. See further Buckley 1994, Buckley 1998, and Bennett 2012: 24–6.

¹⁹For a recent partial reanalysis of the Cahuilla stress system, see Pruitt 2019: 54–63, where, however, the cases of stress clash are not explicitly treated.

of syllables, but the same foot heads are preserved throughout, with the result that degenerate feet may appear on the surface (as in [(sɪ).⟨(jɪr.tih)⟩] below), although they are never created when footing first applies.

(21) Persistent Footing after Pretonic Shortening in Maithili

'muffler'	(no gloss)	
/gala:bɒnna/	/jami:nada:ra/	UR
(,ga).(,la).('bɒn).na	(,ja).(,mi:).na.('da:).ra	Foot Construction / End Rule Right
(,ga).(,la).('bɒn).na	(,ja).(mi).na.('da:).ra	Pretonic Shortening
(,ga.la).('bɒn).na	(,ja).(mi.na).('da:).ra	Persistent Footing
(,ga.lə).('bɒn).nə	(,ja).(mi.nə).('da:).rə	Vowel Reduction
[(,ga.lə).('bɒn).nə]	[(,ja).(mi.nə).('da:).rə]	SR

Note in the following examples from Cyrenaican Bedouin Arabic that foot boundaries, once created, never shift, though syllabification is evidently persistent, and a segment may be resyllabified as an onset to a new syllable created within another foot by Epenthesis.

(22) Non-Persistent Footing in Cyrenaican Bedouin Arabic

'his tree'	'his fruit'	
/ʃaʔar-it-ih/	/fa:kiḥ-it-ih/	UR
(ʃa.ʔa).⟨(ri.tih)⟩	(fa:).(ki.'hi).⟨(tih)⟩	Iambs (L-to-R), End Rule Right
(ʃ.j).⟨(ri.tih)⟩	—	Trisyllabic Syncope
(ʃ.j).⟨(r.tih)⟩	(fa:).(k.'h).⟨(tih)⟩	High Vowel Syncope
('ʃi).⟨(r.tih)⟩	(fa:).('kiḥ).⟨(tih)⟩	Epenthesis (1)
('ši).⟨(jɪr.tih)⟩	—	Epenthesis (2)
[('ši).⟨(jɪr.tih)⟩]	[(fa:).('kiḥ).⟨(tih)⟩]	SR

2.2.6 Extrametricality

The final major set of parameters governing the Hayesian system concern the deliberate exclusion of particular prosodic elements from the construction of higher-level prosodic constituents. These constituents are then rendered effectively invisible in the construction of higher level prosodic constituents. Extrametricality parameters serve essentially one primary objective: to account for consistent stress patterns or some particular cases within a stress system where the primary stress occurs farther from the edge of the word than would be predicted by the foot construction and end rule parameters alone. Without extrametricality conditions, given the inventory of feet in (4), primary stress could never fall outside a two-syllable window from the left or right edge of the word. The principal types of extrametricality invoked by Hayes apply to the segment (consonant), the syllable, and the foot, though morphologically sensitive extrametricality of suffixes also appears in the system.²⁰ Since languages are permitted to treat multiple constituents as extrametrical, there is not

²⁰The list of parameters in Hayes 1995: 401 also provides for word extrametricality, which might conceivably come into play in the assignment of compound or phrasal stress, but seems nowhere to be explicitly employed (despite a reference to pg. 369 on the topic in the index).

merely a single multi-valued parameter, but rather a set of several extrametricality parameters, each of which may potentially be active or not.²¹ Whether a language must further select for the edge at which constituents are extrametrical remains debated: the left edge is at best highly marked.²²

(23) **EXTRAMETRICALITY Parameters:**

- a. Consonant Extrametricality: yes; no
- b. Syllable Extrametricality: yes; no
- c. Foot Extrametricality: yes; no
- d. Affix Extrametricality: yes; no
- e. (Edge of Extrametricality: right; left)

Restricting extrametricality to the right edge, for the moment, the different varieties produce the following effects:

- Final consonant extrametricality renders final CVC syllables light in languages in which (some) such syllables count as heavy. The instances of final stress in moraic trochee systems with rightmost primary stress are thereby reduced.
- Final syllable extrametricality categorically excludes stress from final syllables. Regular antepenultimate primary stress is then the rule in syllabic trochee systems with rightmost primary stress (cf. Macedonian in example (19) above), and is permitted in moraic trochee systems with rightmost primary stress. Iambic systems with rightmost primary stress will exhibit primary stress on the penultimate rather than final syllable.
- Final foot extrametricality renders final heavy syllables or sequences of two light syllables at the right edge unable to bear stress. Preantepenultimate stress is thereby permitted in trochaic systems, and antepenultimate stress in iambic systems, if primary stress is rightmost.

The effects of final consonant and final foot extrametricality may be observed by parallel comparison of Hayes' analyses of Palestinian, Egyptian Radio Arabic (Hayes 1995: 125–32), further contrasted with stress in Diegueño (Yuman; Langdon 1966: 19–23): the first has final consonant extrametricality and final foot extrametricality, whereas the second has no final consonant extrametricality and final foot extrametricality only optionally, while the third has neither type

²¹In fact, only final consonant and final foot extrametricality appear together as active, for the reason that final syllable extrametricality would render final foot extrametricality unable to apply, since the final foot could not be at the edge of the word.

²²Indeed, precisely none of the analyses adopted by Hayes (1995) in fact applies extrametricality to a constituent at the left edge of the word. Hyde (2011: 1047) argues that essentially that all of the best-known possible cases involving initial extrametricality (Western Arrente, Kashaya, Winnebago) can be accounted for without recourse to left-edge extrametricality. Weber (2020), meanwhile, has argued, primarily on the basis of data from Blackfoot (Algonquian), that a constraint against the alignment of the left edge of a foot with the left edge of the word (NONINITIALITY-Foot) is available; see further discussion under 2.3.4 below.

of extrametricality. The table in (24) illustrates how these different extrametricality settings generate different stress patterns in systems where the foot shape (moraic trochee), direction of parsing (left-to-right), and end rule (right) are otherwise identical.²³

(24) Contrast in Extrametricality Settings between Palestinian Arabic, Egyptian Radio Arabic, and Diegueño

Prosodic Shape	Palestinian Arabic	Egyptian Radio Arabic	Diegueño
/σ ₁ VCC/	[da.('ras)<t>]	[di.('maš)<q>]	—
/σ ₁ V:(C)/	[(duk).('ka:<n>)]	[sa.('la:<m>)]	[(ʔəx).('ma:ʔ)]
/~ VC/	[('ka.ta)]	[('ma.li)<k>]	[ʔə.('mat)]
/σ ₁ – V/	[ka.('tab).na]	[(qad).('dam).na]	[(mi:).('ca:).a]
/– ~ V(C)/	[('ba:).<(ra.ko)>]	[(ha:).('ði.hi)]	[(mi:).('ca:).a]
/~ ~ ~ /	[('ka.ta).bu]	[('ʔa.ba).da(n)]	[ʔə.('mə.ti)]
/~ ~ ~ ~ /	[('ša.ʃa).<(ra.tu)><n>]	[(ka.ta).('ba.hu)]	—
/≡ ~ ~ ~ ~ /	[('ša.ʃa).('ra.tu).hu]	[(mux).('ta.li).fa]	—

The absence of final consonant extrametricality in Diegueño results in the restriction of stress to the final two syllables, occurring on the final syllable whenever it contains a long vowel or coda consonant; in contrast, stress in both varieties of Arabic cannot fall on the final syllable if it contains a short vowel followed by a single consonant. The (optional) absence of final foot extrametricality in Egyptian Radio Arabic essentially accounts for why, in words containing four morae in total ending in two light syllables, stress may fall on the penult; in contrast, the extrametricality of the final foot in Palestinian results in stress on the antepenultimate or fourth-to-last syllable in such cases.

A potentially worrisome gap, however, is the apparent absence of languages with consistent preantepenultimate stress in all words of four or more syllables, which could result in a language with syllabic trochees, final foot extrametricality, and primary stress on the rightmost foot. Hayes (1995: 78) admits: “The absence of [foot] extrametricality for syllabic trochees is unexplained.” As will become clear below, final foot extrametricality is transparently superfluous in systems with left-to-right parsing; the same effect is captured with final syllable extrametricality, leaving a penultimate light syllable trapped and unparsed; this includes all iambic systems analyzed in detail by Hayes.²⁴ In moraic trochee systems with right-to-left parsing, the motivation for final foot extrametricality is overall thin: the Paamese data are open to analysis without final foot extrametricality (see Lee 1999), while Maithili (pg. 153) and Manam (pg. 182) avoid feet at the right edge only under stress clash conditions.

This leaves Hindi, where the contrast between initial stress in four-syllable words of the shape /~ ~ ~ ~ / ([‘anumati] ‘approval’) as opposed to peninitial stress in four syllable words of the shape /~ ~ ~ ~ / (i.e., with final heavy syllable; *titāliya*: ‘butterfly’) seems to require foot

²³The Diegueño data that I have examined are not sufficient to state whether foot parsing occurs left-to-right or right-to-left, given that [ə] may often be epenthetic and appear after foot construction.

²⁴Likewise, the distinction between Palestinian and Egyptian Radio Arabic illustrated above can be captured by variable application of final syllable extrametricality in the latter.

extrametricality.²⁵ Depending on what the actual facts of Hindi stress, and what variety is under investigation, a constraint-based analysis may be able to capture this pattern, if feet are preferentially aligned rightward, but primary stress is assigned to the leftmost foot. More likely, though, are problems with the data. Note that Hindi pattern analyzed by Hayes, based largely on a combination of data Grierson 1895b, Fairbanks 1987, and Pandey 1989, seems rather different from the variety described by Kelkar 1968 (analyzed in Prince and Smolensky 1993 [2004]: 40–2, Walker 1996: 14–5), which admits of a comparatively simple analysis: primary stress falls on the heaviest (given a three-way weight distinction), rightmost non-final syllable; in words consisting only of light syllables, stress falls on the penult. A most recent treatment of Hindi stress, Pandey 2021, attempts to clarify some of the variation in the data reported in and across other treatments. For more discussion of stress in Hindi in historical relation to stress in Sanskrit, see Section 5.2.2.2 of Chapter 5.

The parameters of extrametricality may then potentially be restricted to final consonant and final syllable extrametricality alone. With this discussion of a parametric theory of stress complete, I can now demonstrate its application in detail, with reference to the usage of such parameters for describing changes in stress systems.

2.2.7 Applications: Stress Assignment in Early Latin and Tocharian B

In this section, I provide illustrative analyses in the Hayesian parametric framework for two languages with metrical stress systems of particular interest to specialists of older Indo-European languages: Early Latin and Tocharian B. Treatment of the Early Latin data allows for parametric comparison with the entirely familiar and uncontroversial stress grammar of Classical Latin; thereby, the effects of a diachronic change in a single parameter setting are made evident. Presentation of the stress system of Tocharian B is worthwhile, insofar as its metrical stress grammar constitutes a typologically less-common system (defective iambs), and which in its preferred foot form is seemingly without parallel among other older Indo-European languages (iambic as opposed to trochaic).

2.2.7.1 Old Latin Stress

The observation that stress assignment in Old Latin (*Altlatein*, or also Preclassical Latin), distinguishing this phase of the language from the yet older [before the 3rd C. BCE] Archaic Latin (*Frühlatein*),²⁶ must have functioned somewhat differently from stress assignment in

²⁵In fact, the form *títálya*, which crucially motivates final foot extrametricality for Hindi, is reported by Grierson (1895b: 140) to vary between initial and antepenultimate stress. Grierson states: “Sometimes, however, even an I[ndo-]A[ryan-]V[ernacular] word, which has the accent on the antepenultimate, may retain the accent on the same syllable, when it becomes the last syllable but three by the addition of a secondary suffix. This is by no means universal, pronunciation varying with people. Thus. II. *títali* ‘a butterfly’, long form, properly, *títáliyā*, but also sometimes *títáliyā*.”

²⁶Complete terminological consistency in English for the periodization of Latin does not exist. 75 BCE is generally set as the boundary between Classical Latin and earlier phases of the language; in English, the terms “old”, “early”, and “archaic” are all found applied to Latin materials older than this date. Weiss (2009: 23–4) helpfully applies renderings of the German *Frühlatein* and *Altlatein* as “Very Old Latin” (6th to 3rd c. BCE) and

Classical Latin goes back at least to Corssen 1870: 892–906, who noticed that the position of the so-called *ictus* of some tetrasyllabic words in the meter of the early dramatists Plautus and Terence was indicative of primary stress on the initial syllable (cf. Sturtevant 1919 on the usual correspondence between stress and ictus in the early Latin drama of Plautus and Terence).²⁷ Most of the relevant words are all of the shape / ~ ~ ~ σ/, where the parameter settings of Classical Latin, given in (25), would have produced antepenultimate stress. Some of the relevant forms are listed in (26), and contrasted with the stress known for Classical Latin for the same word forms; cf. also Lindsay 1894: 173–4 or Allen 1973: 188–90 on this data. Note further that many of these forms attest by-forms with syncope of the vowel that would be expected to bear stress under the Classical Latin stress rule, further supporting the notion that stress fell on the preantepenultimate syllable in such cases. Mercado (2012) also presumes that forms of the prosodic shape in (26) bore primary stress in the initial syllable in Old Latin as part of his analysis of the Saturnian metrical scheme.

(25) Metrical Stress Parameters of Classical Latin

- a. Foot Form: Moraic Trochee
- b. Direction of Parsing: Right-to-Left
- c. End Rule: Right
- d. Degenerate Feet: Banned
- e. Syllable Extrametricality: Yes

(26) Stress of / ~ ~ ~ σ/ Words in Old Latin

- a. Old Lat. ['ba.li.ne.um] 'bath' vs. C. Lat. [ba.'li.ne.um] (cf. syncopated *balneum*)
- b. Old Lat. ['o.pi.tu.mus] 'best' vs. C. Lat. ['op.ti.mus] with syncope and weakening of [u] to [i] (cf. inscriptional OPTUMO)
- c. Old Lat. ['fa.ki.li.us] 'easy' vs. C. Lat. [fa.'ki.li.us]
- d. Old Lat. ['ba.si.li.kus] 'royal' vs C. Lat. [ba.'si.li.kus]
- e. Old Lat. ['mu.li.e.rem] 'woman' vs. C. Lat. [mu.'li.e.rem]

Given that four-syllable words in Old Latin with a heavy penult or heavy antepenult and light penult are bear primary stress on those heavy syllables just as in Classical Latin, primary stress on the initial syllable of the forms in (26) can be generated via a single parametric difference: the direction of parsing is instead left-to-right, rather than right to left. For this conclusion, compare similarly McCarthy 1979: 462, Hayes 1995: 180, and Jacobs 2003a: 232–3. The parametric stress grammar of Old Latin would then look as given in (27). Parallel stress

“Old Latin” (3rd to end of 2nd c. BCE). Alternatively, the term “Preclassical Latin” is sometimes encountered. Regardless, intended here is the stress system that seems to have held in the period roughly 300 – 100 BCE. Prior to ~300 BCE, most scholarship assumes that Latin consistently assigned primary stress to the initial syllable of a prosodic word (see generally Nishimura 2014).

²⁷ Although the notion of ictus as an organizing principle of Latin verse is not universally accepted (Fortson 2008: 30–3, for instance, expresses doubts), the demonstrably strong statistical cooccurrence between where an ictus would be presumed to fall and the expected word stress would suggest that the rhythmic organization of the line was not altogether insensitive to stress.

derivations applied to the UR /fakilius/ ‘easy’ using the respective Old and Classical Latin grammars shows how the difference results: after the final syllable is marked as extrametrical, depending upon the edge at which parsing begins, an unparsed light syllable remains either before the final syllable (in Early Latin) or at the left edge (in Classical Latin). Compare also Nishimura 2011: 8–14 and Nishimura 2014: 181–3 for an analysis in nearly identical terms.

(27) Metrical Stress Parameters of Classical Latin

- a. Foot Form: Moraic Trochee
- b. Direction of Parsing: **Left-to-Right**
- c. End Rule: Right
- d. Degenerate Feet: Banned
- e. Syllable Extrametricality: Yes

(28) Parallel Derivation of /fakilius/ ‘easy’ in Early and Classical Latin

O. Lat. /fakilius/	C. Lat. /fakilius/	UR
fa.ki.li.⟨us⟩ (fa.ki).li.⟨us⟩	fa.ki.li.⟨us⟩ fa.(ki.li).⟨us⟩	Syllable Extrametricality
(‘fa.ki).li.⟨us⟩	fa.(‘ki.li).⟨us⟩	Foot Construction: L-to-R vs. R-to-L
[(‘fa.ki).li.⟨us⟩]	[fa.(‘ki.li).⟨us⟩]	End Rule: Right
		SR

Besides the evidence for preantepenultimate stress in four-syllable words without heavy syllables, Sturtevant (1919: 242–3) further observes that data from the metrical ictus in five-syllable words consistently supports stress on the antepenultimate syllable (given a short penult), regardless of the weight of that syllable or the preceding syllables, except in the configuration /– ∪ ∪ ∪ σ/. In a word with an odd number of syllables, where the antepenult is heavy and the penult is light, the antepenult will be footed and constitute the rightmost foot in the word, regardless of the footing of the initial and second syllable; where both the antepenult and penult are light, they will build the rightmost foot together, provided that the fourth-to-last syllable is heavy, or both initial and second syllable are light; only in words of the shape /– ∪ ∪ ∪ σ/ will the footing result in stress in the preantepenult, since a parse [(-)(∪ ∪) ∪ ⟨σ⟩] can be generated. Thus, in words with an odd-number of syllables, the Old Latin and C. Latin footing algorithms will produce largely identical parses, as illustrated for the forms /domikilium/ ‘house’, /ami:kitia/ ‘friendship’, and /adsimiliter/ ‘similarly’ in 29; cf. once again Allen 1973: 188–91 for further discussion.²⁸

²⁸Concerning /adsimiliter/, an ictus *adsímiliter* at Plautus, *Bacchus* 951 supports primary stress on the second syllable; cf. Sturtevant 1919: 241.

(29) Derivation of Stress in Five-Syllable Words in Early Latin

/domikilium/	/ami:kitia/	/adsimiliter/	UR
do.mi.ki.li.⟨um⟩	a.mi:ki.ti.⟨a⟩	ad.si.mi.li.⟨ter⟩	Syllable Extrametricality
(do.mi).(ki.li).⟨um⟩	a.(mi:).(ki.ti).⟨a⟩	(ad).(si.mi).li.⟨ter⟩	Foot Construction: L-to-R
(do.mi).('ki.li).⟨um⟩	a.(mi:).('ki.ti).⟨a⟩	(ad).('si.mi).li.⟨ter⟩	End Rule: Right
[(do.mi).('ki.li).⟨um⟩]	[a.(mi:).('ki.ti).⟨a⟩]	[(ad).('si.mi).li.⟨ter⟩]	SR

The analysis developed by Sen (2012: 221–5) of stress in Old Latin, however, instead proceeds on the presumption that five-syllable words of the shape $\sim \sim \sim \sim \sigma$ received primary stress on the preantepenultimate syllable. Theoretically, Sen’s analysis is comparatively unattractive, in that it must assume that ternary feet of the shape $(\sim \sim \sim)$ could be built, producing parses like $^X[\text{do}.('mi.ki.li).\langle um \rangle]$.²⁹ Empirically, the evidential basis for such an alternative stress pattern is marginal.³⁰ I therefore hold to the view that the grammar in (27) is essentially correct, and primary stress fell on the antepenult in words of the shape $\sim \sim \sim \sim \sigma$.

For words of six or more syllables, evidence of the stress pattern in words with light penult and antepenult is not readily forthcoming.³¹ Under the grammar developed thus far, six-syllable words and all other words with an even number of syllables would be predicted to behave just like four-syllable words (i.e., preantepenultimate primary stress in $\sigma \sigma \sim \sim \sigma$ words). Indeed, with the theoretical apparatus employed in this section, no other behavior would be possible, except by invoking (optional) weak local parsing, as in Estonian (cf. (13) and (16) above).³²

²⁹I have not considered whether recursive foot construction following Paricio and Kager (2015) might make such a ternary foot more attractive.

³⁰In essence, the evidence rests upon a single occurrence of the word *pueritia* ‘boyhood:NOM.SG’ in Terence (*Heauton Timorumenos*, 183) where the metrical ictus would presumably fall on the second syllable (thus *puérítia*), combined with an instance of the same lexeme in Horace, *puertiae* (*Carmina* 1.36.8), with apparent syncope of the antepenultimate /i/, which would normally have been expected to receive the primary stress, i.e., [pu.e.ri.ti.a]. Sen (2012: 221) follows Lindsay (1894: 174) in interpreting the coincidence of these two facts as evidence for a stress [pu.e.ri.ti.a] in Early Latin. I am inclined to see the matter as a peculiar coincidence. First, as concerns the Terentian *pueritia*, although tendency for cooccurrence of word stress and ictus offers support for Early Latin stress patterns, it is far from absolute, and must be weighed against the evidence in Sturtevant 1919 pointing precisely to usual avoidance of the ictus on the second syllable of words of the shape $\sim \sim \sim \sim \sigma$. Second, the syncopated form *puertiae* in Horace need not necessarily be indicative of an earlier preantepenultimate stress, for several reasons: 1) the poetry of Horace attests unusual syncopes in a number of lexemes that are not attested with syncope elsewhere in Classical Latin (cf. Maclean 1853: 80 and Shorey and Laing 1910); 2) some manuscripts of the poem in question instead have the normal *pueritiae* (though *puertiae* is obviously the *lectio difficilior*; cf. Garrod 1901); 3) an alternative scansion of an original *pueritia* as *puerítia* (i.e., with synthesis of the first two vowels and *brevis in longo* of the third under stress) is not impossible (in which case the medieval manuscripts with *puertia* might reflect an invention to repair a perceived metrical fault).

³¹Jacobs (2003a: 233) offers [si.mi.li.'tu.di.nem], with primary stress on the antepenult, as such an instance, but I cannot find discussion of stress in longer words of such a shape in the sources cited by Jacobs (Allen 1973, Fraenkel 1928, and Lindsay 1894). Words longer than five syllables are not investigated in Sturtevant 1919. The parametric grammar of 27 would predict primary stress on the preantepenult, just as in four-syllable $\sim \sim \sim \sim \sigma$, thus [(si.mi).('li.tu).di.(nem)]. If indeed evidence for a primary stress on the pre-antepenult in words of this shape in Old Latin exists, the stress grammar employed here would be insufficient.

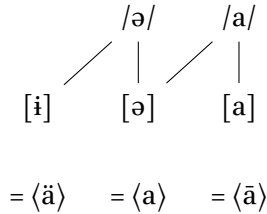
³²Jacobs (2003a: 233) thus generates [(si.mi).li.('tu.di).⟨nem⟩], leaving an unparsed syllable word-medially,

2.2.7.2 Tocharian B Stress

Tocharian B (also known as West Tocharian) is, alongside Tocharian A (or East Tocharian) one of the representatives of the Tocharian branch of the Indo-European family. The Tocharian languages are attested from the period approximately 500–1000 CE in the Tarim Basin (Xinjiang, China) primarily in the form of Buddhist literature translated from Sanskrit. The fundamental generalizations concerning non-exceptional stress assignment in Tocharian B (TB) have been known since Krause 1952: 10 (cf. also Krause and Thomas 1960: 42–7, Hackstein 1995: 15–6, Pinault 2008, Malzahn 2010: 5, Hackstein 2017: 1304–8). Burrioni (2017) has provided an analysis of just aspects of phonologically predictable stress following the framework of Gordon 2002; Koller (2021) has recently subjected stress assignment in Tocharian B, including its morphologically conditioned aspects and interaction with other phonological processes, to a rigorous analysis in the parametric framework of Hayes 1995. The presentation here draws essentially on the data and analysis provided in the unpublished work of Burrioni and Koller. See also Kim 2011 for discussion on the prehistory and diachrony of stress in Tocharian.

The position of stress in a given word can often be established on the basis of stress-conditioned allophony of the central vowel phonemes /ə/ and /a/, represented in (30); since [i] ⟨ä⟩ never occurs under the primary stress, while [a] ⟨ā⟩ is found exclusively when bearing primary stress, syllables with one of these vowels as its nucleus must be unstressed or stressed, respectively. Depending upon the pattern of alternation in which a given vowel in a given root or affix is implicated, its underlying quality can be established, as well as the position of the stress. Some examples of stress-conditioned vowel alternation are given in 31.

(30) Allophony of Central Vowel Phonemes in Tocharian B



(31) Stress-conditioned Vowel Alternations (Unstressed Vowel Raising) in Tocharian B

- a. Stressed [ʼə] alternates with unstressed [i] = /ə/
 /tərkər/ → *tarkär* [ʼtərkir] ‘cloud:NOM.SG’
 vs. /tərkərwa/ → *tärkarwa* [tirʼkərwə] ‘cloud:NOM.PL’

due to the effect of preferences for a foot edge to stand both at the left edge of the word and at the right edge of the word; antepenultimate stress is generated in preference to preantepenultimate stress because, in a [(si.mi).(ʼli.tu).di.(nem)], the rightmost foot stands one syllable farther from the right edge. The crucial theoretical question is whether the relevant ALIGN constraint should be evaluated gradiently or not; see further 2.3.3 below. Regardless, a constraint-based analysis is certainly able to accommodate primary stress on the preantepenult just in /˘˘˘σ/ words, but within a three-syllable window otherwise.

- b. Stressed [a] alternates with unstressed [ə] = /a/
 /çarsa/ → *śarsa* [ˈçarsə] ‘know:PST.3SG’
 /çarsare/ → *śärsäre* [çirˈsare] ‘know:PST.3PL’

Given that the vowel [a], which only surfaces under the stress, does not occur in final syllables (except when an underlying word-final /ə/ is deleted), and that an [a] or [ə] may be found in the first syllable of a disyllable, which then becomes [ə] or [i], respectively, when a further syllable is added, final syllables must be treated as extrametrical. Compare *lakte* [ˈləkle] ‘pain’ versus *läklenta* [likˈlentə] ‘pains’. In general, barring cases where stress assignment is rendered opaque by processes of syncope, as well as some classes of morphologically motivated exceptional stress, primary stress falls on the second syllable of all words with three or more syllables. Compare the 5- and 6-syllable forms *parākäṣṣeñca* [pəˈrakiṣːeɲçɛə] ‘prosper:CAUS.PART’ and *parākäskemane* [pəˈrakiskeməne] ‘prosper:CAUS.PART’; see Burroni 2017 for a collection of data organized according to length in syllables. (32) provides data on words between one and six syllables in length, also illustrating vowel alternations in morphologically related forms.

(32) Stress Patterns of Tocharian B

- a. /σ/: /ak/ → *āk* [ˈak] ‘zeal:NOM.SG’
 b. /σ σ/: /ləkle/ → *lakte* [ˈləkle] ‘pain:NOM.SG’
 c. /σ σ σ/: /ləkle-ntə/ → *läklenta* [likˈlentə] ‘pain:NOM.PL’
 d. /σ σ σ σ/: /enkək-e-mtər/ → *enkaskemtär* [eɲˈkəskemtär] ‘take:PRES.1PL’
 e. /σ σ σ σ σ/: /parak-əṣːençə/ → *parākäṣṣeñca* [pəˈrakiṣːeɲçɛə] ‘prosper:CAUS.PART’
 f. /σ σ σ σ σ σ/: /parak-əskemane/ → *parākäskemane* [pəˈrakiskeməne] ‘prosper:CAUS.PART’;

Tocharian B thus appears to construct left-to-right iambs, with primary stress assigned to the leftmost foot. A difference in any of the parameters foot form, direction of parsing, or end rule would result in a different overall pattern, rather than consistent peninitial stress. Example (34) provides illustrative derivations for two-, three-, and six-syllable forms.

(33) Metrical Stress Parameters of Tocharian B

- a. Foot Form: Iamb
 b. Direction of Parsing: Left-to-Right
 c. End Rule: Left
 d. Degenerate Feet: Banned
 e. Syllable Extrametricality: Yes

(34) Derivation of Primary Stress in Tocharian B

/ləkle/	/ləklentə/	/parakəskemane/	UR
lək.⟨le⟩	lək.len.⟨te⟩	pa.ra.kəs.ke.ma.⟨ne⟩	Final Syllable Extrametricality
(lək).⟨le⟩	(lək.len).⟨te⟩	(pa.ra).(kəs.ke).ma.⟨ne⟩	Foot Construction: Iamb, L-to-R
(ˈlək).⟨le⟩	(lək.ˈlen).⟨te⟩	(pa.ˈra).(kəs.ke).ma.⟨ne⟩	End Rule: Left
(ˈlək).⟨le⟩	(lik.ˈlen).⟨te⟩	(pə.ˈra).(kis.ke).mə.⟨ne⟩	Unstressed Vowel Raising
[(ˈlək).⟨le⟩]	[(lik.ˈlen).⟨te⟩]	[(pə.ˈra).(kis.ke).mə.⟨ne⟩]	SR

The absence of contrastive vowel length, as well as the fact that Tocharian B meters evidently do not exhibit any distinction in syllable quantity, together indicate that the Tocharian B prosody generally lacked distinctions in syllable weight. Typologically, the absence of weight distinctions in iambic systems is comparatively rare, since the canonical iamb of the shape ($\sim \acute{\text{~}}$) cannot be constructed, nor the alternative iamb of the shape ($\acute{\text{~}}$); all feet will have the form ($\sim \sim$). As observed by Koller (2021: 6), this feature places TB among the languages labeled by Hayes (1995: 267–8) as “defective iambic languages”; as observed by Burroni 2017, TB may be unique in constituting a quantity-insensitive system with regular peninitial stress but also final syllable extrametricality (cf. Gordon 2002: 512–4; 540). The fact that the orthographic equivalent of Sanskrit long [a:] is used to represent TB [‘a] (= /‘a/), and that occasionally an orthographic (ī) is found for stressed [‘i] might be an indication that stressed vowels were subject to phonetic lengthening, or that length was a primary phonetic correlate of stress (cf. 1.3.3 in Chapter 1), which would allow for TB to conform to the Iambic/Trochaic Law in a fashion similar to Onondaga and Seneca, which exhibit vowel lengthening in / $\sim \sim$ / feet (Hayes 1995: 225–6).

Footing in TB furthermore appears to be non-persistent: a process of schwa deletion (PRE-CORONAL DELETION, stated in (35); see Winter 1993 and Kim 2011) may delete a schwa that would have been expected to bear primary stress, with the result that stress then instead falls on the initial syllable in affected forms, on a now monosyllabic foot. The derivations of inflected forms of the root /nek-/ ‘disappear’ illustrate apparent intraparadigmatic stress mobility between peninitial and initial stress due to PRE-CORONAL DELETION.

- (35) PRE-CORONAL DELETION (cf. Koller 2021: 6)

$$\text{ə} \rightarrow \emptyset / V(C_i)C_j \left[\begin{array}{c} +\text{coronal} \\ -\text{sonorant} \end{array} \right] V$$

where C_i , if present, is more sonorous than C_j

A vowel preceding a coronal obstruent immediately before another vowel is deleted if it is the second in a sequence of three vowels, and there is not a cluster of rising or flat sonority preceding the target vowel.

- (36) Non-Persistent Footing in Cases of Schwa-Deletion³³

/nek-ə-sta/	/nek-əsa-te/	UR
ne.kə.⟨sta⟩	ne.kə.sa.⟨te⟩	Final Syllable Extrametricality
(ne.kə).⟨sta⟩	(ne.kə).sa.⟨te⟩	Foot Construction: Iamb, L-to-R
(ne.‘kə).⟨sta⟩	(ne.‘kə).sa.⟨te⟩	End Rule: Left
(ne.‘kə).⟨sta⟩	(‘nek).sa.⟨te⟩	Pre-Coronal Deletion
(ne.‘kə).⟨sta⟩	(‘nek).sə.⟨te⟩	Unstressed Vowel Raising
[(ne.‘kə).⟨sta⟩]	[(‘nek).sə.⟨te⟩]	SR

³³Koller interprets the presence of primary stress on the initial syllable in cases like [(‘nek).sə.⟨te⟩] where PRE-CORONAL DELETION has applied as evidence for both a need for iambic foot structure (as opposed to the foot-free analysis of Burroni 2017) and need for a derivational architecture, given that PRE-CORONAL DELETION opacifies the expected stress assignment ($^X[(\text{nek.‘sə}).\langle\text{te}\rangle]$). Although coda consonants generally do not contribute to syllable weight in TB, it is conceivable that the deletion of /ə/ creates a heavy syllable with moraic coda consonant in order to preserve the mora associated with /ə/ in the input (MAX- $\mu \gg$ *MORAIC CODA).

From the point of view of the comparative prosody of the (older) Indo-European languages, Tocharian B is somewhat remarkable in exhibiting iambic feet, and when the parameter settings are directly compared with the settings of Classical Latin (cf. (25) above), one notes that the major parameters of foot shape, parsing direction, and end rule all have different values. Regardless of precisely how these distinct stress systems are ultimately (best) formalized, the very fact of their divergence, despite descent from a common ancestor, is telling of the extent to which related languages may, at least within the domain of stress assignment, go down rather different prosodic paths.

2.2.8 Summary on Parametric Stress

The preceding subsections have systematically illustrated the analytical possibilities permitted under a parametric model of stress assignment that assumes foot structure, a stress grid, and the serial application of rules that construct prosodic structure alongside other phonological processes. The five major parameters of foot form, direction of parsing, end rule, degenerate feet, and extrametricality, are themselves able to provide extensive empirical coverage of attested stress systems, and when supplemented with additional parameters (e.g., location of parsing) or rules (e.g., destressing in clash), yet more patterns become readily generable. Examining differences in parameter settings also offers a convenient means of describing diachronic changes in rules of stress assignment (cf. Lahiri 2015) and can be leveraged towards modeling the learnability of stress grammars (see Dresher and Kaye 1990, Nazarov and Jarosz 2017). The specific approach to the analysis of surveyed in this section will not, however, play a major role in the development of novel analyses in 3.4; this survey was nevertheless necessary in order to provide the reader with sufficient conceptual background for understanding treatments of stress couched in a parametric framework. Furthermore, knowledge of the prosodic parameters just treated will facilitate the understanding of stress analyses in a constraint-based framework, beginning with the following section.

2.3 Stress Assignment in Optimality Theory

Work on stress in Optimality Theory (beginning with already with Prince and Smolensky 1993 [2004]: 33–72) differs from major approaches discussed in the preceding sections in that constraint rankings, rather than fixed parameters, govern stress assignment. Much, probably most, of the work on stress in a constraint-based framework operates with constraint implementations that approximately reproduce the major parameters seen in Hayes 1995. However, the possible permutations of constraint rankings, and the focus on output well-formedness, both allow for the generation of more distinct stress patterns and for the introduction of constraints that have no clear antecedent in the parametric frameworks (e.g., *CLASH; compare the rules designed to eliminate stress clash used throughout Hayes 1995). The presentation found in Kager 1999: Ch. 4 may be fairly taken as representing a fairly standard approach to constraint-based analyses of stress; compare also Apoussidou 2007: 14–21, Gordon 2011b, and especially Kager 2005 for overviews of commonly used constraints on metrical stress, as well as Tesar and Smolensky 2000: Ch. 4 for such constraints in the

context of the learnability of stress grammars (on which see further Chapter 4). The assumptions of a foot-free metrical grid can also be readily implemented as constraints, as Gordon (2001) has shown for quantity-insensitive stress systems; similar work on a constraint-based implementation of quantity-sensitive systems in a footless grid remains a desideratum.

This section will survey that analysis of stress using constraint-based, parallel derivations, based largely upon the standard repertoire of constraints found in the works just cited. Effort will be made to explicitly bring out the relationship between the constraints introduced here and parameters introduced above. The constraints introduced here will also be of substantial further import in the rest of this work, recurring in much of the treatment of learnability in Chapter 3, as well as many of the analyses developed in Part II.

2.3.1 Foot Shape, Quantity Sensitivity, and Rhythmic Contours

While the FOOT FORM parameter given at (5) above provided an inventory of three prespecified foot shapes (syllabic trochee, moraic trochee, and iamb), differing from one another in terms of location of the foot head and whether the feet are quantity sensitive, these properties are usually permitted to be governed by independent constraints in OT grammars.³⁴ First are constraints on the rhythmic structure of feet, which govern whether the foot head should be leftmost (trochaic) or rightmost (iambic) within a foot (cf. Kager 1999: 172).³⁵ Note that these constraints in no way restrict the size of the feet involved: any foot consisting of two or more syllables will violate one or the other constraint, regardless of how many syllables are parsed into a single foot; conversely, both constraints are satisfied by any monosyllabic foot.

- (37) TROCHEE: Feet should be trochaic (= feet have initial prominence). Assign one violation * for each polysyllabic foot in the output which the strongest member is **not** leftmost in the foot.
- (38) IAMB: Feet should be iambic (= feet have final prominence). Assign one violation * for each polysyllabic foot in the output in which the strongest member is not rightmost the foot.

For languages such as Pintupi or Icelandic, which exhibit syllabic trochees throughout, the ranking TROCHEE $\gg$ IAMB will result, all other things being equal, in the consistent construction of feet with initial prominence, or monosyllabic feet (which will however, be dispreferred in many cases). The tableau in 39 shows that, as long as the general preference

³⁴In principle, nothing *a priori* excludes the existence or usage of a constraint such as FOOTFORM = MORAIC TROCHEE, which penalizes the construction of any foot that does not conform to the canonical shape of the moraic trochee (as done, for instance in Kiparsky 1998b). The general intuition of most analysts seems, however, to favor the generation of patterns through the interaction of constraints that cannot be decomposed. This theoretical predisposition comes at the cost of larger factorial typologies that might permit the existence of grammars that can generate rare or unattested (if not indeed perhaps impossible) patterns.

³⁵The exact labels given to these constraints vary slightly throughout the literature (e.g., FOOTFORM = ..., RHYTHMTYPE = ...), but that constraints intended to be evaluated in an identical fashion is usually self-evident.

for trochees over iambs holds, and no other higher-ranked constraints militate against the construction of trochees (under particular conditions), trochaic feet will always result.

- (39) TROCHEE $\gg$ IAMB in Icelandic: [(‘a.kva).(rɛl.la)] ‘aquarelle’

/akvarella/		TROCHEE	IAMB
a.	☞ (‘a.kva).(rɛl.la)		**
b.	(‘a.kva).(rɛl.la)	*!	*
c.	(‘a.kva).(rɛl.la)	*!	*
d.	(‘a.kva).(rɛl.la)	*!*	

The simple ranking seen in (39), when combined with the appropriate constraints on the formation of prosodic constituents (see 2.3.2 below), can result in exhaustive parsing of trochaic feet, and, where no syllable weight distinctions are available or are ignored, syllabic trochees of the Icelandic or Pintupi sort will emerge. If, however, a distinction between light and heavy syllables is available, activity of the constraint WEIGHT-TO-STRESS PRINCIPLE (SWP) can provide for quantity sensitivity, and thereby provide for the formation of alternative heavy monosyllabic feet in trochaic systems, as well as canonical iambs of the form (˘˘).

- (40) WEIGHT-TO-STRESS PRINCIPLE (WSP): Heavy syllables should be stressed. Assign one violation * for each heavy syllable in the output that is unstressed (i.e., parsed into the weak part of a foot or not parsed into a foot at all).

Because the WSP militates in favor of treating every heavy syllable as a foot head, sequences of multiple heavy syllables may then each build a foot, as well. In systems that appear to build moraic trochees, such as Fijian (cf. 10 above), feet of the shapes (˘˘), (˘˘), and potentially (˘˘) as well will all satisfy TROCHEE and WSP, as shown in (41). Further constraints determine exactly how the trochaic feet with consistently stressed heavy syllables should be built across the string.

- (41) Quantity Sensitivity in Fijian due to WSP

- a. [pa.(la.si).('ta:)] ‘plaster’

/palasita:/		WSP	TROCHEE	IAMB
a.	☞ pa.(la.si).('ta:)			*
b.	(pa.la).('si.ta:)	*!		
c.	pa.('la.si).ta:	*!		
d.	(pa.la).(si.'ta:)		*!*	

b. [ta.(,rau).('se.se)] 'trousers'

/tarausese/		WSP	TROCHEE	IAMB
a.	ta.(,rau).('se.se)			*
b.	(,ta.rau).('se.se)	*!		
c.	ta.rau.('se.se)	*!		
d.	(ta,(rau).(se.'se)		*!*	

What, however, might speak against the construction of a candidate with a foot of the shape (ˈ˘) in 41b., e.g., [ta.(ˈrau.se).se], where both WSP and TROCHEE would be satisfied? If such feet were freely allowed or even desirable, the success with which most moraic trochee systems can be parsed into feet exclusively of the shapes (˘˘) and (ˈ˘) alone. To the end of working against feet of the shape (ˈ˘), as well as the defective iamb (˘˘), Kager (1999: 174) proposes the constraint RHYTHM-CONTOUR (RH-CONTOUR), which prefers for a strong–weak sequence to obtain at the *moraic* level. The canonical trochee (˘˘) and iamb (˘ˈ) both satisfy this constraint, as do alternative monosyllabic heavy feet, (ˈˈ).

- (42) RH-CONTOUR: A foot must end in a strong–weak contour at the moraic level. Assign one violation * for each trochaic foot of the shape (ˈ˘) or iambic foot of the shape (˘ˈ).

In this fashion, candidates containing such anti-rhythmic feet, as in candidate e. of the tableau immediately below, may be ruled out. Conversely, in languages in which anti-rhythmic feet appear to be freely permitted, such as the defective iambic languages like Tocharian B discussed above, or languages with truly quantity-insensitive stress, such as Estonian, RH-CONTOUR (as well as WSP) must be comparatively unimportant (i.e., low-ranked). Further constraints that can lead to the generation of such systems will be introduced in the following subsection.

To summarize, the four constraints just introduced, TROCHEE, IAMB, WSP, and RH-CONTOUR, can together work to reproduce the inventory of common foot patterns employed in Hayes 1995:

- In systems in which TROCHEE, WSP, and RH-CONTOUR are inviolable, only moraic trochees (4b.) may be generated.
- In systems in which IAMB, WSP, and RH-CONTOUR are inviolable, only iambs (4c.) may be generated.
- In systems where TROCHEE dominates IAMB, and no weight contrast is observed, or WSP and RH-CONTOUR are instead relatively low-ranked, syllabic trochees (4a.) may be generated.

2.3.2 The Parsing of Prosodic Constituents

Consider again the simple parse of Icelandic /akvarella/ under 39 into two syllabic trochees. Given that both TROCHEE and IAMB are satisfied in the case of a monosyllabic foot, or where no foot is constructed at all, what factors operate to obtain the apparently exhaustive parsing of syllables into disyllabic feet seen there (cf. the ITERATIVITY parameter under 2.2.5)? In principle, candidates containing only monosyllabic feet, fewer feet, or no feet at all could be more harmonic, avoiding violations of TROCHEE and/or IAMB altogether. Similarly, how can the quantity-insensitive feet of Estonian as seen in 13 be generated, if WSP and RH-CONTOUR conspire to allow for only canonical trochees and iambs? These questions may be answered by considering the overall construction of prosodic structure at and below the level of the ω , and interface with the constraints on prosodic domination seen at 18 in Chapter 1.

First, the possibility of a null parse, in which no foot structure is created and thus no relations of stress prominence may be excluded from multiple perspectives. The presumably inviolable principle of prosodic HEADEDNESS, namely, that each category of the prosodic hierarchy must dominated at least one instance of the immediately lower-level category, will require that a prosodic word, ω , if constructed out of morphological material, must dominate at least one foot, $\mathfrak{f}$. Then, given the preference for a grammatical word to correspond to a prosodic word, such a preference can only be satisfied where a ω exists, which can only be well-formed if it dominates at least one $\mathfrak{f}$. Similarly, OBLIGATORINESS (cf. 62 in Chapter 1) requires a primary stress, which similarly necessitates the presence of at least one foot, insofar as stress is treated as dependent on foot structure. Kager (1999: 152) proposes the constraint given in 43 below; Prince and Smolensky (1993 [2004]: 45) instead employ the constraint $LX \approx PR$ (MCAT), which may be precisely specified for the syntactic category of the word in question, while Ito and Mester (2009: 142) employ LEX-MAPPING constraints to a similar effect. See discussion on these constraints in 1.2.2 in Chapter 1.

- (43) GRAMMATICALWORD = PROSODICWORD (GRWD = PRWD): A grammatical word must be a prosodic word. Assign one violation * for each grammatical word (= terminal syntactic category node) in the input that does not have a matching prosodic word in the output.

Second, the exhaustive parsing of syllables into feet, beyond the requirement for at least a single foot, may be regulated either by PARSE-Syllable (PARSE- σ ; cf. Prince and Smolensky 1993 [2004]: 62, Kager 1992: 162) or the approximately equivalent PARSE-INTO- $\mathfrak{f}$ (Ito and Mester 2009: 139).³⁶ For present purposes, and wherever the distinction is immaterial, I will employ the largely more familiar constraint of the PARSE-X type. Violations of PARSE- σ result in weakly (as opposed to strictly) layered prosodic structures below the ω .

- (44) PARSE-Syllable: All syllables should be parsed into feet. Assign one violation * for each syllable in the output that is not parsed into a foot.

³⁶See the discussion of the distinction between constraints of the PARSE-X versus the PARSE-INTO-X families at footnote 28 in Chapter 1.

Precisely the preference to incorporate as many syllables as possible into a feet can drive the creation of structures contrary to the preferences on foot structure governed by WSP and RH-CONTOUR. If the ranking $\text{PARSE-}\sigma \gg \text{WSP}$ and RH-CONTOUR obtains, then the creation of feet in which heavy syllables do not receive stress, or where anti-rhythmic structures result, will be permitted, if the result parses a larger number of syllables into feet. The tableau in (45) shows that Estonian has just such a constraint ranking.³⁷ Both candidates a. and b. appear as the winner there, because candidates a. and b. parse the greatest number of trochaic feet, and b. incurs one fewer violation of WSP compared to a., while b. contains more violations of RH-CONTOUR compared to a. Candidate c. interestingly satisfies lower-ranking WSP and RH-CONTOUR perfectly, but at the cost of building iambic feet; candidate d. similarly avoids violation of those lower ranked constraints, and by building feet only around the heavy syllables, incurs no violations of either TROCHEE or IAMB, but consequently contains more unparsed syllables.

(45) Maximization of Syllable Parsing in Estonian

/pimestavasse/		GRWD=PRWD	TROCHEE	PARSE- σ	WSP	RH-CONTOUR	IAMB
a.	 ('pi.mes).(ta.vas).se			*	**	*	**
b.	 ('pi.mes).ta.(vas.se)			*	*	**	**
c.	(pi.'mes).(ta.vas).se		*!*	*			
d.	pi.(mes).ta.(vas).se			**!*			
e.	pi.mes.ta.vas.se	*!		*****	**		

Nonetheless, $\text{PARSE-}\sigma$ in the above tableau might yet be better satisfied, if a monosyllabic foot were, for instance, created around the medial syllable (ta), thus [(pi.mes).(ta).(vas.se)]. That such a candidate is not preferred may be attributed to the effects of a constraint against degenerate feet, FOOTBINARITY (compare the DEGENERATE FOOT Parameter at 11 above).

- (46) FOOTBINARITY (FTBIN): Feet should be binary at either the moraic or syllabic level. Assign one violation * for each foot in the output that does not dominate any branching constituent (i.e., all feet consisting of a single light syllable).

In the Estonian case, the ranking $\text{FTBIN} \gg \text{PARSE-}\sigma$ must hold, in order to prevent the unrestricted formation of degenerate feet, as shown in 47. Even in languages in which the reverse ranking holds, such as Icelandic (shown in 48), FTBIN will ensure that binary feet are constructed wherever possible, rather than a sequence of monomoraic feet.

³⁷ Recall from (13) that words like /pimestavasse/ exhibit variation in the output.

(47) No Degenerate Feet in Estonian: FTBIN $\gg$ PARSE- σ

pimestavasse		TROCHEE	FTBIN	PARSE- σ	WSP	RH-CONTOUR	LAMB
a.	☞ ('pi.mes).(ta.vas).se			*	**!	*	**
b.	☞ ('pi.mes).ta.('vas.se)			*	*	**	**
c.	(pi.'mes).(ta.vas).se	*!*		*			
d.	pi.('mes).ta.(vas).se			**!*			
e.	('pi.mes).(ta).(vas.se)		*!		**		**

(48) Degenerate Feet Permitted in Icelandic: PARSE- σ $\gg$ FTBIN

/biokrafia/		TROCHEE	PARSE- σ	FTBIN
a.	☞ ('bi.o).(kra.fi).(a)			*
b.	('bi.o).(kra.fi).a		*!	

2.3.3 The Direction of Parsing and Position of Primary Stress: Foot Edge, Head Foot, and Word Edge Alignment

The exposition of constraints concerning metrical stress up to this point has taken the directionality of foot construction (e.g., L-to-R in Icelandic, R-to-L in Fijian) and placement of primary stress (e.g., leftmost in Icelandic, rightmost in Fijian) for granted. Optimality-theoretic pendants to the parameters of directionality and end rule (cf. 2.2.2 above) take the form of constraints on generalized alignment (cf. McCarthy and Prince 1993), which strive to enforce the edge alignment of one given prosodic or morphological category with another prosodic or morphological category.

The effect of directionality in parsing in a system that constructs iterative feet (i.e., strives to exhaustively parse syllables into feet, as constraint by WSP and RH-CONTOUR) can be reproduced, then, by requirements to align the left or right edge of every foot with the left or right edge of every prosodic word, shown in 49.

(49) ALIGN-($\mathcal{F}$, Left/Right, ω , Left/Right) = ALL-FEET-LEFT/RIGHT (ALL-Ft-L/R): Every foot should stand at the left/right edge of the prosodic word. Assign one violation * for each syllable that intervenes between the left/right edge of a foot and the left/right edge of the prosodic word.

In systems where the ranking FTBIN $\gg$ PARSE- σ holds, the relative ranking of ALL-Ft-L and ALL-Ft-R directly corresponds to the setting left or right of the parameter on the direction of parsing. Thus, in 50, candidate a. is selected as the winner because the well-formed

binary foot constructed out the sequence of three light syllables stands closer to the right edge of the word (incurring only one violation of ALL-Ft-R) than the left edge. Violations of all the constraints on foot edge alignment could be minimized by reducing the total number of feet constructed, but at the cost of violations of higher-ranking PARSE- σ , thereby eliminating candidate e. Notice crucially that the evaluation of ALL-Ft-L/R is gradient: more feet necessarily result in more violations of these constraints.

(50) Right-to-Left Parsing of Moraic Trochees in Fijian: ALL-Ft-R $\gg$ ALL-Ft-L

/palasita:/		TROCHEE	FtBIN	PARSE- σ	ALL-Ft-R	ALL-Ft-L
a.	pa.(la.si).('ta:)			*	*	*****
b.	(,pa.la).si.('ta:)			*	**!	***
c.	(,pa).(la.si).(ta:)		*!		*****	*****
d.	(,pa.la).(si).('ta:)		*!		***	*****
e.	pa.la.si.('ta:)			**!		***

The gradient evaluation of ALL-Ft-L/R also readily accounts for systems in which foot construction is apparently non-iterative, that is, only a single foot supporting the primary stress is built. Stress systems like that found in Macedonian (cf. 19 above) result from the ranking of PARSE- σ below the decisive constraint on word edge alignment, as illustrated in 51 (final syllable extrametricality is tacitly applied).

(51) Non-iterative Footing in Macedonian: ALL-Ft-R $\gg$ PARSE- σ

/vodeničar-ot/		TROCHEE	FtBIN	ALL-Ft-R	PARSE- σ	ALL-Ft-L
a.	vo.de.('ni.ča).(rot)			*	***	**
b.	(,vo.de).('ni.ča).(rot)			**!	*	**
c.	('vo.de).ni.ča.(rot)			**!*	***	

Where iterative foot construction does occur and multiple feet may appear in a prosodic word, the head foot must also be determined from the available foot, thereby establishing the position of the primary stress. Again, an alignment constraint, this time concerned with the alignment of some edge of the head foot alone relative to an edge of the prosodic word, can be employed.

(52) ALIGN-(Head- $\mathcal{F}$, Left/Right, ω , Left/Right) = LEFTMOST/RIGHTMOST:³⁸ The head foot should be leftmost/rightmost in the prosodic word. Assign one violation * for each

³⁸Compare constraints are also labeled EDGEMOST in Prince and Smolensky 1993 [2004] or MAIN-LEFT/RIGHT in Tesar and Smolensky 2000: 54.

syllable in the prosodic word that intervenes between the foot with primary stress and leftmost/rightmost syllable of the prosodic word.

In the both the Fijian and Macedonian cases seen immediately above, RIGHTMOST must dominate LEFTMOST, ensuring that the head foot stands as close to the right edge of the ω as possible (modulo final syllable extrametricality in Macedonian). Like ALL-Ft-L/R, LEFTMOST/RIGHTMOST is evaluated gradiently.³⁹ The tableau in 53 reproduces the Fijian example, illustrating the selection of the rightmost foot as the head foot. The failure of candidates b. and c. there can essentially be ascribed to the fact that a foot is available the right edge of the word, which is not the head foot.

(53) Primary Stress is Rightmost in Fijian: RIGHTMOST $\gg$ LEFTMOST

/palasita:/		TROCHEE	FtBIN	RIGHTMOST	PARSE- σ	ALL-Ft-R	ALL-Ft-L	LEFTMOST
a.	pa.(la.si).(ta:)				*	*	*****	***
b.	pa.(la.si).(ta:)			*!	*	*	*****	*
c.	(pa.la).si.(ta:)			*!*	*	**	***	

Finally, the equivalent of languages exhibiting “top down” parsing may be generated by alignment constraints that invert the relationship between foot edge and prosodic word edge, effectively requiring that a prosodic word begin or end with a left or right foot boundary.

(54) ALIGN-(ω , Left/Right, f, Left/Right) = ALIGN-WD-L/R: Every prosodic word ω should begin/end with a foot. Assign one violation * the left/right edge of a prosodic word is not aligned with the left/right edge of a prosodic word.

ALIGN-WD-L/R plays a crucial role in two varieties of stress systems. First are complex bidirectional systems of the Indonesian type (Cohn 1989, Kager 1999: 169–70), in which primary stress and (at least one) secondary stress appear to be aligned with different word

³⁹As a constraint of the generalized alignment type, gradient evaluation for LEFTMOST/RIGHTMOST is to be expected. However, whether a constraint that governs the position of the primary stress in a prosodic word with multiple feet must be gradiently evaluated is less clear. One could instead imagine a non-gradient constraint that requires the head foot to be the leftmost or rightmost from among the available feet. Since systems with iterative feet always select either the leftmost or the rightmost foot from among the available feet as the head foot, whether the relevant constraint is evaluated gradiently or absolutely has no effect on the outcome. See McCarthy 2003: 110–1 for a proposal to instead employ categorical constraints, END-RULE-LEFT/RIGHT. In the case of non-iterative systems, ALL-Ft-L/R is sufficient to ensure that the single foot sits as close to a word boundary as possible. Note further that, in constraint-based grammars employing weighted constraints (e.g., Maximum Entropy Grammar; see Chapter 3), the gradient evaluation of LEFTMOST/RIGHTMOST will cause most outputs to generally be more disharmonic, which might impact the calculation of weights or constraint ganging in undesirable ways. I find McCarthy’s proposal of a non-gradient constraint for establishing the head foot attractive, though empirically whether any real difference results from the usage of a gradient versus a categorical constraint has not yet, to my knowledge, been investigated in detail.

edges. As the data in 55 show, primary stress exceptionlessly falls close to the right edge of the word, on the penult, but in word with an odd number of syllables and multiple feet, a secondary stress always falls on the initial syllable. Such behavior falls out if the left edge of the word should be aligned with the left edge of a foot, when the preferred directional parsing given by ALL-Ft-L/R would not consistently generate a foot boundary at the left edge of the word.

- (55) Indonesian Stress: FTBIN, RIGHTMOST $\gg$ ALIGN-WD-L $\gg$ PARSE- σ $\gg$ ALL-Ft-R $\gg$ ALL-Ft-L
- a. Three Syllables: [bi.(ˈca.ra)] ‘speak’ – RIGHTMOST forces the head foot closer to the right edge.
 - b. Five Syllables: [(ˌkon.ti).nu.(ˈa.si)] ‘continuation’ – ALIGN-WD-L compels construction of a foot at the left edge.
 - c. Six Syllables: [(ˌe.ro).(ˌdi.na).(ˈmi.ka)] ‘aerodynamics’ – ALL-Ft-R and ALL-Ft-L equally satisfied.
 - d. Seven Syllables: [(ˌa.me).ri.(ˌka.ni).(ˈsa.si)] ‘americanization’ – secondary stress on the fourth-to-last syllable shows that directional parsing is right-to-left (ALL-Ft-R $\gg$ ALL-Ft-L).

The second type of system in which ALIGN-WD-L/R may be decisive are systems in which a foot is forcibly constructed at a word edge, even when the resulting would violate constraints on the well-formedness of feet (FTBIN, WSP, and/or RH-CONTOUR) that are otherwise generally respected in the language. The earliest attested Germanic languages and their last common ancestor probably instantiate a system of this sort; see further Chapter 7.

Before continuing with the exposition of constraints on metrical structure, brief mention of a known issue with generalized alignment constraints of the sort introduced above is warranted. Since at least Eisner 1997, worked out more explicitly in Hyde 2008 with respect to stress patterns, generalized alignment constraints have been known to potentially generate patterns whereby some particular element (e.g., stress, tone) comes to consistently be placed at or near the midpoint of a string of substring; this behavior is termed the Midpoint Pathology in Hyde 2008.⁴⁰ Example 56 illustrates a pathological pattern in Hyde’s sense for a string of syllables: the stress given there always falls on the middle syllable (in odd-numbered strings) or one of the two middle syllables (in even-numbered strings).

- (56) A Stress Pattern with a Midpoint Pathology (cf. Hyde 2015: 2)
- a. 1 syllable $\acute{\sigma}$
 - b. 2 syllables $\acute{\sigma}\sigma$ or $\sigma\acute{\sigma}$
 - c. 3 syllables $\sigma\acute{\sigma}\sigma$

⁴⁰Kager (2012) employs the same label, but to describe a different sort of behavior of stress patterns that can be generated by anti-lapse constraints. From the point of view of stress typology, the pattern described by Kager appears unusual (stress restricted to a medial syllable, though still within a window), but may not necessarily be pathological (cf. Hyde 2015: 3).

d. 4 syllables	σóσσ or σσóσ
e. 5 syllables	σσóσσ
f. 6 syllables	σσóσσσ or σσσóσσ
g. 7 syllables	σσσóσσσ
h. 8 syllables	σσσóσσσσ or σσσσóσσσ
i. 9 syllables	σσσσóσσσσ

Without going into detail here, because generalized alignment constraints are in theory capable of generating Midpoint Pathologies, Hyde (2015) (see also Hyde 2016) proposes instead to operate with “relation-specific alignment constraints”. The corresponding version of ALL-Ft-L as a relation-specific alignment constraint is shown in 57 (Hyde 2015: 22).

- (57) Relation-specific ALL-FEET-LEFT: $\langle \omega, f, \sigma \rangle / [\dots \sigma \dots f \dots]_{\omega}$
Assign one violation * for every set $\langle \omega, f, \sigma \rangle$ such that σ precedes f within ω .

Empirically and practically speaking, how great a problem theoretically possible midpoint pathologies are is not totally clear. As the discussion in Hyde 2015: 13–17 indicates, a generalized alignment constraint like ALL-Ft-L/R will not exhibit pathological behavior when the input is restricted to a single prosodic word; only in strings consisting of multiple prosodic words might the foot edges within the bounds of one ω be drawn to the edges of a neighboring ω . Insofar as the analysis of word-level stress almost exclusively considers single prosodic words (including clitics), the possibility for generalized alignment constraints to behave pathologically practically never arises. While Hyde’s proposed relation-specific alignment constraints may ultimately prove preferable to generalized alignment constraints, I will continue to employ the generalized alignment constraints in this work on account of their familiarity.

2.3.4 Extrametricality through NONFINALITY and NONINITIALITY

In both Macedonian, where primary stress regularly falls on the antepenultimate syllable, and Early as well as Classical Latin (see 2.2.7.1 above), where primary stress is banned from final syllables, the parameter of syllable extrametricality was invoked above in order to exclude the final syllable of the word from the rules of foot parsing. Similarly, the contrast in stress assignment between Diegueño, on the one hand, and Egyptian Radio Arabic, on the other, seen in 24, could be attributed to a difference in the setting of final consonant extrametricality, with the result that final CVC syllables behave as light in Egyptian Radio Arabic. Since particular prosodic constituents cannot be rendered invisible to parsing into higher-level prosodic constituents by rule application in a constraint based grammar, the prosodic structural configurations present when, e.g., stress falls on a final syllable or a final consonant is treated as weight-bearing, must be penalized in the output.

The effects of final syllable and final consonant extrametricality, respectively, may be reproduced in constraint grammars by head-based NONFINALITY constraints, given below;

cf. Prince and Smolensky 1993 [2004]: 41–62.⁴¹ Basic NONFINALITY, or syllable NONFINALITY, bans primary stress (= the prosodic head of ω) from the final syllable, though it does not necessarily exclude a final syllable from being parsed into a foot. NONFINALITY-Foot, on the other hand, is violated when a foot is final in the prosodic word, coextensive with the right edge.⁴²

- (58) NONFINALITY (NONFIN; also more specifically NONFINALITY-Syllable): The prosodic head of the word (i.e., the syllable with primary stress) may not be final in the prosodic word. Assign one violation * if the constituent bearing primary stress in the output (i.e., the syllable with primary stress) is aligned with the right edge of the prosodic word.
- (59) NONFINALITY-Foot (NONFIN-F): No foot is final in the prosodic word. Assign one violation * if the right edge of the prosodic word in the output is aligned with the right edge of a foot.⁴³

Violations of both species of head-based NONFINALITY constraints can be satisfied by either failing to parse the final consonant of the ω into a foot, or failing to parse a word-final consonant into a syllable. Thus, the relative rankings of the two constraints in 58–59 to PARSE- σ and the parallel PARSE-Segment determine what sort of output configurations are licit, and descriptively, what extrametricality appears to obtain. The crucial typology of generable patterns is listed here:

- PARSE- σ , PARSE-Seg $\gg$ NONFIN, NONFIN-F: no extrametricality effects; feet at the right edge of the word and primary stress on final syllables permitted.⁴⁴
- PARSE-Seg $\gg$ NONFIN, NONFIN-F $\gg$ PARSE- σ : essentially equivalent to final syllable extrametricality; word-final syllables are neither included in feet nor given primary stress.
- PARSE- σ $\gg$ NONFIN, NONFIN-F $\gg$ PARSE-Seg: essentially equivalent to final consonant extrametricality; the rightmost syllable may be parsed into a foot and/or bear primary stress, at least as long as a consonant follows it; how exactly wording ending in /-V/ will be treated depends in part on other constraints.

⁴¹On alternative grid-based NONFINALITY constraints, see Hyde 2007 and Hyde 2011.

⁴²The definition of NONFINALITY provided by Prince and Smolensky (1993 [2004]: 56) allows for a semi-gradient interpretation: one violation is assessed if the head foot is final in the prosodic word, and two violations are assessed if the final syllable in addition bears stress.

⁴³This constraint should not be confused with the constraint FOOTNONFINAL introduced by Tesar and Smolensky (2000: 54), which penalizes polysyllabic feet whose heads are final in the foot; this constraint is thus similar to TROCHEE, in that all disyllabic iambs will violate it, but unlike TROCHEE and IAMB, also assigns a violation to monosyllabic feet, since such feet also have heads that are final in the foot (cf. Apoussidou 2007: 20–1).

⁴⁴NONFINALITY can also, potentially interact with IAMB and TROCHEE in a crucial way: if the ranking PARSE- σ , textscParse-Seg, NONFIN $\gg$ RIGHTMOST, IAMB $\gg$ TROCHEE obtains, a language might build generally iambic feet, but invert the foot shape to a trochee for the final foot, in order to satisfy NONFIN and the parsing requirements.

- NONFIN, NONFIN-f $\gg$ PARSE- σ $\gg$ PARSE-Seg: words ending in /-V/ will not be parsed into a syllable, resulting in the rightmost foot ending with the penult in such words; words ending in /-C/ may treat it as extrametrical and include the rightmost syllable in a foot and let it bear primary stress.
- NONFIN, PARSE- σ , PARSE-Seg $\gg$ NONFIN-f: word-final syllables may freely be included in a foot, but primary stress may not fall on the final syllable.
- NONFIN, PARSE- σ $\gg$ NONFIN-f $\gg$ PARSE-Seg: primary stress on the final syllable is strictly avoided; words ending in /-V/ may have a foot boundary aligned with the right edge of the word, while words ending in /-C/ will avoid this.

Typical patterns of final syllable and final consonant extrametricality using NONFINALITY may be exemplified based on data from Classical Latin and Egyptian Radio Arabic, respectively. In the former, PARSE- σ is lowest ranked, with the result that final syllables are weakly layered except in the case of monosyllabic and disyllabic words (where they are parsed in order to ensure the satisfaction of GRWD = PRWD). Primary stress falls on the rightmost trochaic foot, parsed from right-to-left, ignoring the final syllable. Since PARSE-Seg $\gg$ NONFIN, NONFIN-f are all undominated, both light and heavy final syllables will remain unparsed; RIGHTMOST and ALL-Ft-R then determine the placement of primary stress, given that RH-CONTOUR excludes ill-formed trochees of the shape (ˈ ˘) and (˘ ˘) is penalized by WSP. Crucial to observe throughout is that the undominated status of PARSE-Segment precisely prevents primary stress from occurring closer to the right edge than it actually does.

(60) Stress Assignment with Final Syllable Extrametricality in Classical Latin

a. /sapie:ns/ $\rightarrow$ [(ˈsa.pi).⟨e:ns⟩] ‘know:PTCP.NOM.SG’

/sapie:ns/	NONFIN	NONFIN-f	PARSE-Seg	RH-CONTOUR	PARSE- σ	RIGHTMOST	ALL-Ft-R	WSP
a.  (ˈsa.pi).⟨e:ns⟩					*	**	*	*
b. (ˌsa.pi).⟨ˈe:ns⟩	*!	*!					*	
c. (ˌsa.pi).⟨ˈe:n⟩⟨s⟩			*!				*	

b. /sapiente:s/ $\rightarrow$ [(ˌsa.pi).⟨ˈen⟩.⟨te:s⟩] ‘know:PTCP.NOM.PL’

/sapiente:s/	NONFIN	NONFIN-f	PARSE-Seg	RH-CONTOUR	PARSE- σ	RIGHTMOST	ALL-Ft-R	WSP
a.  (ˌsa.pi).⟨ˈen⟩.⟨te:s⟩					*	*	***	*
b. (ˌsa.pi).⟨ˌen⟩.⟨ˈte:s⟩	*!	*!					***	
c. (ˌsa.pi).⟨ˌen⟩.⟨ˈte:⟩⟨s⟩			*!				***	
d. (ˈsa.pi).⟨ˌen⟩.⟨te:s⟩					*	**!	***	*

c. /sapientior/ → [(,sa.pi).(‘en).ti.(or)] ‘know:PTCP.CMPV.NOM.SG’

/sapientior/		NONFIN	NONFIN-f	PARSE-Seg	RH-CONTOUR	PARSE-σ	RIGHTMOST	ALL-Ft-R	WSP
a.	ⲉⲛⲧⲓⲟⲣ (,sa.pi).(‘en).ti.(or)					**	**	*****	*
b.	(,sa.pi).(,en).ti.(‘or)	*!	*!			*		*****	
c.	(,sa.pi).(,en).(‘ti.o)(r)			*!			*	*****	
d.	(,sa.pi).(‘en.ti).(or)				*!	*	**	****	*

As seen in the Classical Latin examples just above, failure to parse a word-final consonant into a syllable can, however, prevent violations of NONFINALITY constraints. A final [CV(:)C(C)] or [CV(:)C(C)] syllable, when parsed into a foot, whether as a foot head or tail, will not violate any sort of non-finality, because the unparsed consonant necessarily intervenes between the foot boundary and the right edge of the prosodic word. In Early and Classical Latin, evidently, PARSE-Segment outranks NONFINALITY generally, which means that final syllables cannot be rescued through final consonant extrametricality. In Egyptian Radio Arabic, in contrast, final consonant extrametricality rescues words ending in [VCC] and [V:C] from violations of NONFINALITY, resulting in stress on those final heavy syllables. In cases of words ending in [VC] sequences, stress moves onto the penult or antepenult instead, in order to avoid NONFINALITY violations that would result from treating the final syllable as heavy. Interestingly, the combination of left-to-right foot parsing and licit exclusion of final consonants from a syllable produces the an effect of rendering final /CVC/ sequences excluded from syllable parsing as well.

(61) Stress Assignment with Final Consonant Extrametricality in Egyptian Radio Arabic

a. /sala:m/ → [sa.(‘la:)<m>]

/sala:m/		NONFIN	WSP	RH-CONTOUR	PARSE-σ	PARSE-Seg	NONFIN-f	ALL-Ft-L	RIGHTMOST
a.	ⲉⲗⲁⲙ sa.(‘la:)<m>				*	*		*	
b.	sa.(‘la:m)	*!			*		*	*	
c.	(‘sa.la:)<m>		*!	*!		*			

b. /malik/ → [(‘ma.li)<k>]

/malik/		NONFIN	WSP	RH-CONTOUR	PARSE-σ	PARSE-Seg	NONFIN-f	ALL-Ft-L	RIGHTMOST
a.	ⲙⲁⲗⲓⲕ (‘ma.li)<k>					*		*	*
b.	ma.(‘lik)	*!			*		*	*	
c.	(‘ma.lik)	*!		*!			*		*

c. /ha:ðihi/ → [(ha:).(‘ði.hi)]

/ha:ðihi/	NONFIN	WSP	RH-CONTOUR	PARSE-σ	PARSE-Seg	NONFIN-f	ALL-Ft-L	RIGHTMOST
a.  (ha:).(‘ði.hi)						*	*	*
b. (‘ha:).(ði.hi)						*	*	**!
c. (‘ha:).ði.hi				*!*				**
d. (‘ha:ði).hi			*!	*				**

d. /ʔabadan/ → [(‘ʔa.ba).da⟨n⟩]

/ʔabadan/	NONFIN	WSP	RH-CONTOUR	PARSE-σ	PARSE-Seg	NONFIN-f	ALL-Ft-L	RIGHTMOST
a.  (‘ʔa.ba).da⟨n⟩				*	*			**
b. ʔa.(‘ba.da)⟨n⟩				*	*		*!	*
c. (ʔa.ba).(‘dan)	*!					*	**	

e. /katabahu/ → [(ka.ta).(‘ba.hu)]

/katabahu/	NONFIN	WSP	RH-CONTOUR	PARSE-σ	PARSE-Seg	NONFIN-f	ALL-Ft-L	RIGHTMOST
a.  (ka.ta).(‘ba.hu)						*	**	*
b. ka.(‘ta.ba).⟨hu⟩				*!			*	**
c. (‘ka.ta).ba.⟨hu⟩				*!*				***

While the interaction of PARSE-X constraints with NONFINALITY constraints can generate classic extrametricality effects, the existence of parallel NONINITIALITY constraints is much less clear. As discussed already at 2.2.6 above, the empirical evidence for extrametricality effects at the left edge of the word is slim, if not non-existent. The recent survey of Weber (2020) indicates that perhaps only a parallel NONFINALITY-f might be necessary to capture rare cases of persistent third syllable stress; that is, if feet are allowed to be aligned with the left edge of the word at all, primary stress on the initial syllable will not be particularly penalized (general NONINITIALITY is not necessary). Weber’s observation accords with results obtained by Staubs (2014: 119–21) through computational experimentation, which suggest that NONFINALITY-σ would be superfluous.

2.3.5 *CLASH

Already in early work on the prosodic analysis of stress (e.g., Liberman 1975, Liberman and Prince 1977), rules for the movement of strong word-level stress positions in phrasal contexts were observed, while Prince (1983) and Hammond (1984) observed the apparent need for

rules to “weaken” stress in contexts where an expected secondary stress seemed to be absent. Such rules of DESTRESSING IN CLASH are employed extensively in Hayes 1995 (cf. pg. 35–6) to remove cases of adjacent stress that would be generated by the exhaustive parsing of syllables into feet. Stress clash is in general liable to emerge in quantity-sensitive systems where either a sequence of more than one heavy syllable occurs, since each syllable could build a foot and bear stress on its own, or when a heavy syllable immediately precedes (in trochaic systems) or follows (in iambic systems) a disyllabic foot. The optional destressing in clash observed in Cahuilla (cf. Hayes 1995: 136 and 2.2.5 above) regularly removes secondary stress from a heavy penult after an initial light syllable bearing primary stress (i.e., stress assignment appears to be “top-down”); conversely, secondary stress after an initial heavy syllable is generally permitted, though stress on a final light syllable is not.⁴⁵ The effects of such destressing in clash can be essentially reproduced through a markedness constraint that penalizes the presence of stress on adjacent syllables.

- (62) *CLASH: No stressed syllables are adjacent. Assign one violation * for each pairwise contiguous sequence of two stressed syllables.

For cases in Cahuilla where peninitial heavy syllables lack secondary stress, the ranking *CLASH $\gg$ WSP crucially enables the “skipping” of those heavy syllables, seen in 63a. and b.⁴⁶ When no stress clash at results, because the word begins with a / $\sim$ $\sim$ / sequence, as in 63c., all heavy syllables are footed.

⁴⁵The actual facts of secondary stress are not entirely clear. The treatment of Pruitt 2019: 54–63 follows closely the reports of Seiler (1965, 1967, 1977), which imply exhaustive parsing of syllables (PARSE- σ $\gg$ FTBIN) and many stress clashes (e.g., [(‘su).(kaʔ).(ti)] ‘deer:OBJ’). In the interest of exposition of constraint interactions, I will follow Hayes’ reinterpretation of Seiler’s data (Hayes 1995: 136–7), which assumes that final light syllables do not actually bear stress, contrary to Seiler’s transcriptions (a discrepancy acknowledged by Hayes); the reader interested in the Cahuilla facts should carefully reevaluate the data themselves.

⁴⁶I take the optionality of destressing in clash in forms like [‘ka.vi.:.či.wen] to mean that the grammar allows for the choice between a violation of *CLASH or not, and that violations of RH-CONTOUR are strictly disallowed, while violations of FTBIN are permitted. An alternative grammar could interpret (optional) absence of secondary stress on heavy peninitial syllables as allowing for the formation of ($\sim$ –) feet with the ranking FTBIN $\gg$ RH-CONTOUR.

Note further that the grammar employed in 63 predicts formation of a single degenerate foot at the left edge in words of the shape / $\sim$ – $\sim$ /, which Seiler apparently regularly transcribes with secondary stress on the heavy second syllable, (e.g., [(‘su).(kaʔ).(ti)] ‘deer:OBJ’). An analysis with weighted constraints (see further Chapter 4) could generate presence of secondary stress in / $\sim$ – $\sim$ / words as [$\sim$ – $\sim$], with only a single violation of *CLASH, but absence of secondary stress in longer words like [‘ka.vi.:.či.wen], where two violations of *CLASH could outweigh one violation each of FTBIN and WSP.

(63) (Optional) Avoidance of Stress Clash in /~ – σ σ/ words in Cahuilla⁴⁷

(63) /kavi:čiwen/ → [('ka).vi:(či.wen)] 'I was surprised'

		LEFTMOST	RH-CONTOUR	FTBIN	*CLASH	WSP	PARSE-σ	ALL-Ft-L
	kavi:čiwen							
a.	 ('ka).vi:(či.wen)			*		*	*	**
b.	('ka).(vi:).(či.wen)			*	*!*			***
c.	('ka.vi:).(či.wen)		*!			*		**

(64) /kakawla:qa/ → [('ka).kaw.(law).qa] 'his legs are bow-shaped'

		LEFTMOST	RH-CONTOUR	FTBIN	*CLASH	WSP	PARSE-σ	ALL-Ft-L
	kakawla:qa							
a.	 ('ka).kaw.(la:).qa			*		*	**	**
b.	('ka).(kaw).(la:).qa			*	*!*		*	***
c.	('ka.kaw).(la:).qa		*!			*	*	**
d.	('ka).(kaw).(la:).(qa)			**!	***			*****

(65) /taxmuʔaʔti/ → [('tax.mu).(ʔaʔ).ti] 'the song:OBJ'

		LEFTMOST	RH-CONTOUR	FTBIN	*CLASH	WSP	PARSE-σ	ALL-Ft-L
	taxmuʔaʔti							
a.	 ('tax.mu).(ʔaʔ).ti						*	**
b.	('tax.mu).(ʔaʔ).(ti)			*!	*			*****
c.	('tax.mu).(ʔaʔ.ti)		*!					

*CLASH can also play a further interesting role in regulating the directionality of parsing in systems in which PARSE-σ outranks FTBIN and drives the exhaustive parsing of all syllables into feet. Granting that Icelandic constitutes such a language, words with an odd number of syllables build degenerate monosyllabic feet at the right edge of the word. As seen under 2.3.3, where syllable parsing is non-exhaustive, the relative ranking of ALL-Ft-L and ALL-Ft-R matches the iterative directionality of parsing in parametric terms. Where, however, an additional monosyllabic foot is created at a word edge, the relative ranking of ALL-Ft-L and ALL-Ft-R does not directly reflect the parametric direction of parsing: when feet are trochaic, aligning the right edge of feet closest to the right edge of the word produces the expected alternating stress pattern, as seen in (66). Alternatively, alignment of feet with

⁴⁷Recall that only syllables containing long vowels or closed by [ʔ] count as heavy in Cahuilla.

word edges can be ignored entirely, and *CLASH can regulate the foot parses; incidentally, an exhaustive parsing that minimizes violations of ALL-Ft-L would produce a stress clash between the initial and peninitial syllables in words with an odd number of syllables. Thus, simply ranking *CLASH above the constraints of the alignment of feet with word edges can equally generate the desired parse, as shown in (67).

(66) ALL-Ft-R $\gg$ ALL-Ft-L in Icelandic

biokrafia		LEFTMOST	PARSE- σ	ALL-Ft-R	ALL-Ft-L	Ft-BIN
a.	☞ ('bi.o).(kra.fi).(a)			****	*****	*
b.	('bi).(o.kra).(fi.a)			*****!	****	*
c.	('bi.o).(kra.fi).a		*!	****	**	

(67) Icelandic Alternating Stress as *CLASH Avoidance

biokrafia		LEFTMOST	PARSE- σ	*CLASH	ALL-Ft-R	ALL-Ft-L	Ft-BIN
a.	☞ ('bi.o).(kra.fi).(a)				****	*****	*
b.	('bi).(o.kra).(fi.a)			*!	*****	****	*
c.	('bi.o).(kra.fi).a		*!		****	**	

2.3.6 Summary: Metrical Stress in OT and its Factorial Typology

The constraints and types of miniature sample analyses of stress provided in the foregoing add up to an overall theory of stress assignment with the following major features: prosodic constituents of FEET are built below the word, with the principles of HEADEDNESS, OBLIGATORINESS, and CULMINATIVITY generally ensuring at least one and at most one primary stress; weak layering allows for lower-level prosodic constituents (syllables, segments) to escape parsing into higher level constituents in service of constraint satisfaction; gradient generalized alignment constraints organize the position of feet and the head foot relative to word edges; constraints on foot structure govern the well-formedness of foot shapes; and a dispreference for stress clash may ensure rhythmic alternation at the syllable level. Table 2.1 summarizes the constraint types and total number of constraints introduced in the preceding sections.

The total number of constraints assumed to be minimally necessary for generating the known major patterns of metrical stress is not insubstantial: even ignoring GRWD = PRWD for the moment, the remaining 15 constraints admit of a factorial typology of more than one trillion ($1307674368000 = 15!$) distinct grammars (i.e., possible total rankings), though

	Foot Structure	Foot/Word Edge Alignment	Prosodic Parsing	Extrametricality	Syllable Rhythm
	IAMB TROCHEE FTBIN WSP RH-CONTOUR	ALL-Ft-L ALL-Ft-R LEFTMOST RIGHTMOST ALIGN-Wd-L ALIGN-Wd-R	GRWD = PRWD PARSE- σ PARSE-C	NONFINALITY- σ NONFINALITY-f NONINITIALITY-f	*CLASH
Total #	5	6	3	3	1

Table 2.1: Metrical Stress Constraints in Treated in Section 2.2

the actual number of distinct output patterns must be considerably smaller.⁴⁸ Even though this constraint set can generate weight-governed unbounded systems, including default-to-opposite patterns (cf. generally Baković 1998), the total number of distinct patterns will not be enormous, though the particular typological predictions of this constraint set have been nowhere systematically investigated. In the main, exploration of the possibilities predicted by the constraints on foot structure, in terms of different quantity-sensitive systems, represents a gap in the literature; only Baković (1998) considers a set of patterns that can result from undominated WSP.

By way of summary, let us consider how the stress patterns of Old Latin and Tocharian B, treated under 2.2.7.1 and 2.2.7.2, are generated by the interaction of the constraints listed in Table 2.1. First, to recall the basic facts concerning primary stress in the languages, Tocharian B places primary stress on the peninitial syllable in words of three or more syllables, but on the initial syllable in disyllabic forms; Old Latin places primary stress on a heavy penult or heavy antepenult, but in longer sequences of all light syllables in four- or five-syllable words, on either the pre-antepenult (/LLL σ /) or antepenult (/LLLL σ /).⁴⁹ Although direct evidence of secondary stress is wanting for both languages, evidence from patterns of syncope in Latin points towards exhaustive foot parsing without permitting degenerate feet (see generally Mester 1994), and exhaustively parsed but well-formed feet may be assumed for TB as well without encountering any obvious problems.⁵⁰

The stress grammar of Tocharian B is comparatively simple, since the system is not quantity sensitive, meaning that constraints relating to quantity and foot form (WSP and RH-

⁴⁸For comparison, the factorial typologies worked out by Kager (2005: 17–27) for a set of 8 metrical stress constraints using gradient alignment, versus 11 metrical stress constraints using rhythmic licensing, produce factorial typologies of 40320 and 39916800 total rankings, respectively, but with only 35 and 24 distinct output patterns. Similarly, the 12 constraints employed in Gordon 2002 for quantity-insensitive systems generated a set of 479001600 total rankings, but merely 152 distinct output patterns of primary and secondary stress.

⁴⁹As noted above the position of primary stress in words of six or more syllables is not entirely certain, and primary stress on either the pre-antepenult or antepenult given sequence /LLLLL σ / can be generated.

⁵⁰If a form like TB *alāṣmontaṃtsa* [ə.ˈlaʃ.mon.tən.sə] ‘sick:PERL.PL’ indeed reflects underlying /alaṣmontasa/, the application of unstressed vowel raising in the penultimate syllable, [tən], which might be predicted to bear secondary stress if footing is exhaustive, could be taken as an indication that only a single foot is built at the left edge of the word. Alternatively, unstressed vowel raising may be taken to apply to any /a/ or /ə/ not under the primary stress. Since, to my knowledge, the vowels suffixes like [-ontə] never exhibit any alternations on the surface, regardless of whether the word contains an even or odd number of syllables, any possible secondary stress seems to play no role in governing unstressed vowel raising.

CONTOUR) may be ignored and placed in the lowest stratum. Likewise, constraints that would prefer feet or stress closer to the right edge of the word (ALL-Ft-R, ALIGN-WD-R, RIGHTMOST) are irrelevant and thus belong to the lowest stratum of the grammar. Since NONINITIALITY- $\mathcal{F}$ is consistently violated by all winning candidates, it belongs to the lowest stratum as well. Insofar as words of five or more syllables build multiple feet, e.g., [(pə.'ra).(kis.ke).mə.(ne)] 'prosper:CAUS.PART', ALL-Ft-L, too, can be placed in the lowest stratum, below PARSE- σ . This constraint and FTBIN belong in a stratum together below NONFINALITY- σ and NONFINALITY- $\mathcal{F}$, PARSE- σ is consistently violated in order to avoid violation of these two constraints, and, as stress on the initial syllable in disyllabic forms like [(.'lak).(le)] shows, degenerate monosyllabic feet are permitted to satisfy NONFINALITY. Ranking above the NONFINALITY constraints, on the other hand, are the constraints satisfied by every winner, which always has a foot with primary stress at the left edge of the word (ALIGN-WD-L, LEFTMOST), is always iambic when disyllabic (IAMB), always has a primary stress, even in monosyllabic forms (GRWD = PRWD); *CLASH and PARSE-C are always untouched as well. The total ranking is given in (68).

(68) Total Constraint Ranking for the Stress Grammar of Tocharian B:

GRWD = PRWD, IAMB, ALIGN-WD-L, LEFTMOST, *CLASH, PARSE-Seg $\gg$ NONFINALITY- σ , NONFINALITY- $\mathcal{F}$ $\gg$ FTBIN, PARSE- σ $\gg$ ALL-Ft-L, ALL-Ft-R, ALIGN-WD-R, RIGHTMOST WSP, RH-CONTOUR, NONINITIALITY- $\mathcal{F}$, TROCHEE

The stress grammar of the quantity-sensitive Old Latin is rather more complex, and depends on a more articulated constraint ranking in order to generate the desired patterns. Furthermore, some facts from beyond the domain of stress assignment alone, in particular, alternations in vowel length or variable treatment of coda consonants as weight-bearing (*brevis brevians* or “iambic shortening”; see Mester 1994: 11–20 or Jacobs 2003a: 234–42 for formal treatments, with further references) is crucial for determining certain ranking relationships. First, two major constraints on foot form are inviolable and must belong to the highest stratum: FTBIN and TROCHEE — all feet are disyllabic or bimoraic, and all disyllabic feet must be trochaic. Since there is no evidence that feet of the shape ('H L) were ever constructed, RH-CONTOUR may belong to the highest stratum as well; PARSE-SEG trivially belongs here as well, as does GRWD=PRWD. Stress is also categorically excluded from final heavy syllables, which indicates that NONFINALITY- σ was undominated as well. Meanwhile, the fact that words of the (seeming) shape /L H H σ /, such as *voluptātem* 'desire:ACC.SG', are subject to *brevis brevians*, being parsed as [(vo.lup).(ta:).(tem)] (= [(L L) ('H) (σ)], with the /p/ of the second syllable not making position), rather than [vo.(lup).(ta:).(tem)], constitutes an indication that *CLASH is inviolable as well; contrast [do.(mes).ti.(ka:).tim] (= [L (H) L ('H) (σ)]), where the light syllable [ti] intervening between two heavy syllables excludes the possibility of clash, and allows [mes] to be treated as heavy, as preferred. Since heavy syllables nearly always receive stress, WSP belongs to the second-highest stratum; NON-FINALITY- $\mathcal{F}$ proves violable in just one special circumstance, namely, in disyllabic words of the shape /L L/ (or /L H/, with *brevis brevians* to [L L]), where it must be violated in order to satisfy FTBIN.

The third stratum is populated by ALIGN-WD-L alone, which is essential for generating

initial stress in words of the *facilius* type (/L L L σ /), thus [(‘L L) L (σ)]. ALIGN-WD-L crucially outranks RIGHTMOST and ALIGN-WD-R, which conspire for a foot to stand as close to the right edge of the word as possible, and for the rightmost foot to bear primary stress. Under the assumption that all words of five or more syllables containing no heavy non-final syllables place primary stress on the antepenult (e.g., [(‘si.mi).li.(‘tu.di).(<nem>)] instead of ^x[(‘si.mi).(‘li.tu).di.<nem>)], the ranking of RIGHTMOST $\gg$ ALL-Ft-L is crucial: the head foot should be closer to the right edge of the word, as long as ALIGN-WD-L can be satisfied. Since multiple feet are permitted per word, PARSE- σ belongs in the stratum above ALL-Ft-L and ALL-Ft-R as well. Finally, IAMB and LEFTMOST are also placed in the lowest stratum, since no candidate with an iambic foot ever wins, and no candidate with multiple feet ever has fewer violations of LEFTMOST than the winner.⁵¹ As already correctly observed by Jacobs (2003a: 236), if this grammar is altered to place ALIGN-WD-L in the lowest stratum (i.e., crucially below ALIGN-WD-R and RIGHTMOST), the grammar will then generate the familiar pattern of Classical Latin.

(69) Total Constraint Ranking for the Stress Grammar of Old Latin:

TROCHEE, FTBIN, RH-CONTOUR, PARSE-SEG, GrWD=PrWD, *CLASH $\gg$ WSP, NON-FINALITY-f $\gg$ ALIGN-WD-L $\gg$ RIGHTMOST, ALIGN-WD-R, PARSE- σ , NONINITIALITY-f $\gg$ IAMB, ALL-Ft-L, ALL-Ft-R, LEFTMOST

2.4 Extensions to Unbounded, Morphological, and Lexical Stress Systems

Discussion in Section 2.1 has already introduced a distinction between prosodic systems in which stress is calculated on the basis of metrical structure alone, versus systems with unbounded attraction of stress to a phonologically prominent element, as well as systems that refer to lexical properties and morphological structure. This section will extend the optimality-theoretic model of Section 2.3 with additional families of constraints designed to account for these further possibilities.

The stress systems reviewed from parametric and optimality-theoretic perspectives in Sections 2.2 and 2.3, respectively, all instantiate pure metrical stress systems: these are systems in which the position of primary and secondary stress (if present) is strictly phonologically predictable, and in which the dependence of stress on the construction of feet and marking of relative prominence among the feet by an end rule (or functionally similar alignment constraints) tends to restrict primary stress to a window at the left or right edge of the word. Primary stress in such systems is essentially limited to two- or three-syllable windows at the left and right edges, and typically follows from a canonical foot form, thus resulting in the regular occurrence of the primary stress at some fixed distance from an edge, countable in syllables, morae, or feet.

⁵¹See the file `Old_Lat_Stress.txt` at <https://github.com/rpsandell/Sandell-LMU-Habilitation> (Chapter 2), which includes the further constraints WEIGHT-BY-POSITION (WBP) and MAX- μ for capturing the effects of *brevis brevis*.

This section now introduces families of constraints that serve primarily to analyze the position of primary stress in languages in which its position appears not to fall out directly from parameters or constraints on metrical stress alone. The position of primary stress may nevertheless be phonologically predictable, if dependent upon a properly phonological property (such as syllable weight or tone), but exceeds the capacity of purely metrical stress if primary stress breaks the boundaries set by canonical foot shapes and principles word edge alignment or foot parsing. If stress appears, in some or many instances, to be divorced from properly phonological properties entirely, then its analysis must depend on reference to morphological structure and/or arbitrary lexical properties (however represented). Striking in both types of systems, in contrast to the rhythmic metrical systems surveyed above, is that reference to foot structure is, at least for the calculation of primary stress, less transparently motivated.

With respect to the prosody of early-attested Indo-European languages and diachronic changes in prosody more broadly, the type of stress systems treated in this section are of particular relevance. Many of the languages treated in detail in Part II evince interactions between stress and tone to some extent, and many of those same languages exhibit a complex interface between the lexicon, morphological structure, and phonological calculation of stress. Precisely these prosodic characteristics will then stand in the foreground in the discussion of diatopic variation and diachronic change relating to those languages. For this reason, an understanding of the mechanisms involved in prominence-driven stress and lexical accent will serve the reader well in subsequent chapters.

2.4.1 Prominence-Driven Stress

The term “prominence-driven stress” is due primarily to Walker (1996: 1), who characterizes them as “not rhythmic...where prominence depends on factors contributing to increased salience, such as syllabic weight, peripherality, and nonfinality.”; compare also the definition of de Lacy (2002: 1), who states that “prominence-driven systems allow syllables with certain properties to override edge-attraction, with stress attracted to syllables with high-sonority nuclei, long vowels, onsets, or any of a number of other properties”. I will engage here, however, with prominence attraction of two just two sorts, which represent the typologically best-established varieties: attraction to heavy syllables and attraction to syllables with high tone. The former sort is treated extensively already by Walker (1996), while the latter sort is examined in detail by de Lacy (2002). Since systems in which primary stress strictly aligns with a prosodic word edge (modulo nonfinality) are readily generated as a species of metrical stress, I will regard them as such, despite Walker (1996: 7–8)’s arguments to regard systems with peripheral primary stress and no secondary stress as instances of prominence-driven stress.⁵²

⁵²In this connection, see also Baković 1998: 3–7 on the distinction between genuine unbounded feet (i.e., a foot of indeterminate size encompassing all syllables in a ω) with edge prominence versus a single foot at an edge, potentially leaving many syllables unparsed. I adopt the view that genuine unbounded feet probably do not exist: in the absence of evidence secondary stress or segmental processes potentially indicating foot structure outside the site of primary stress, I assume that only a single foot per word constructed.

Whether yet other types of prominence-driven stress exist at all is, at this juncture, somewhat doubtful; at minimum, cases of prominence attraction dependent upon vowel sonority or syllable structure, independent of syllable weight considerations, are typologically exceptional. As concerns sonority-driven stress in particular, earlier work by Kenstowicz (1997) and de Lacy (2006: 232–43) had also argued, primarily on the basis of Gujarati data, that stress might be attracted to more sonorous vowels as well; subsequent closer phonetic investigation of Gujarati by Shih (2016, 2018) and Bowers (2019) has however, indicated that no attraction of stress to more sonorous vowels is really in evidence. For discussion of another possible case of (partly) sonority-driven stress in Nanti (Kampa, Peru), see Crowhurst and Michael 2005: 52–5. Yet others (Shih and de Lacy 2019, McCollum 2020) have proposed that apparent cases of sonority-driven stress should be handled by encoding sonority distinctions as weight distinctions; under such an approach, all cases of sonority-driven stress are then effectively reducible to weight-driven prominence, and can be analyzed identically to the cases seen in 2.4.1.1.

Formally, the constraint mechanisms supposedly needed for treating cases of prominence driven stress were introduced already by Prince and Smolensky (1993 [2004]: 38–44) (cf. also Walker 1996: 9), taking the form of the constraint **PEAK-PROMINENCE**, which essentially demands that stress peaks be preferentially associated with syllable nuclei of greater “intrinsic prominence” (however determined).

- (70) **PEAK-PROMINENCE (PK-PROM):** $\text{Peak}(x) \succ \text{Peak}(y)$ if $|x| > |y|$.
Assign one violation * for each stress peak associated with syllable nucleus of lower prominence when a syllable nucleus of greater prominence is available.

In general, PK-PROM should probably be regarded as a cover constraint for a stringency hierarchy of distinct binary constraint rankings. More precise constraint hierarchies relating to syllable weight and pitch will be introduced below.

2.4.1.1 Weight-Driven Prominence

In languages with a binary distinction between light and heavy syllables, the set of constraints introduced in the preceding section 2.3 are potentially sufficient for capturing the typology of “unbounded” stress systems with attraction of primary stress to a heavy syllable described in Hayes 1995: 296–99 and Walker 1996: 8–40; see also further Baković 1998 on “unbounded” stress systems, in which stress is definitionally non-iterative. Following Walker’s exposition, the existing systems may be classified into four varieties, depending upon the position of primary stress when no heavy syllables are present and whether or not **NONFINALITY** is relevant:

- default-to-same (DTS) edge without nonfinality (e.g., rightmost heavy syllable, else rightmost syllable in Aguacatec [Mayan, Guatemala])
- default-to-opposite (DTO) edge without nonfinality (e.g., rightmost heavy syllable, else leftmost syllable in Selkup [Ostyak-Smoyed, Siberia])

- default-to-same edge (DTS) with nonfinality (e.g., rightmost non-final syllable heavy, else the only heavy, else rightmost non-final syllable in Sindhi [Indo-Aryan])
- default-to-opposite edge (DTO) with nonfinality (e.g., rightmost non-final heavy syllable, else the only heavy, else leftmost syllable in Khalkha Mongolian)

To test concretely how restrictively weight-driven prominence systems allow themselves to be analyzed, I limit the constraints employed for the moment to the following: WSP, LEFTMOST/RIGHTMOST, ALL-Ft-X, PARSE- σ and NONFINALITY. The position of primary stress in default-to-same-edge systems derives straightforwardly from the ranking of WSP over LEFTMOST and ALL-Ft-L or RIGHTMOST and ALL-Ft-R, which in turn outrank other metrical constraints. The right-edge-oriented system of Aguatic produces final stress in sequences of exclusively light syllables, and stress on the only available heavy syllable in strings with only one heavy syllable, as shown in (71); in these cases, just a single foot is constructed.⁵³

- (71) Default-to-same weight-driven prominence (right side, as in Aguatic or Kolya Yukaghir):

WSP $\gg$ ALL-Ft-R $\gg$ RIGHTMOST $\gg$ ALL-Ft-L, LEFTMOST, PARSE- σ , NONFINALITY

- a. Only Light Syllables (else rightmost)

		WSP	ALL-Ft-R	RIGHTMOST	ALL-Ft-L	LEFTMOST	PARSE- σ	NONFINALITY
	L L L L L							
a.	 L L L (L 'L)				***	*****	***	*
b.	('L L) L L L		*!***	*!***			***	
c.	L L L ('L L)			*!	***	***	***	

- b. Rightmost single heavy syllable

		WSP	ALL-Ft-R	RIGHTMOST	ALL-Ft-L	LEFTMOST	PARSE- σ	NONFINALITY
	L L L H L							
a.	 L L L ('H) L		*	*	***	***	*****	
b.	L L L (H 'L)	*!			***	*****	***	*
c.	('L L) L (H) L		*!***	*!***	***		**	
d.	('L L) L H L	*!	***	****			***	

⁵³In Aguatic, only syllables containing long vowels count as heavy. Furthermore, although McArthur and McArthur (1956: 76) explicitly state that “within the word, stress occurs on the final long vowel or on the final short vowel if no long vowels are present”, the absence of longer words in their data (used also by Walker 1996: 10) makes it in fact impossible to distinguish between the unbounded stress analysis presented here and right-aligned iambs. The available data indicate stress only on a final syllable or penultimate heavy syllable; no word with a heavy antepenult or multiple heavy syllables are offered. For this reason, the tableaux in 71 show rather abstract sequences of heavy and light syllables. In fact, a clear case of right-edge weight-driven prominence (as opposed to tone, as in Golin) without nonfinality effects (as in Sindhi or Western Cheremis) is difficult to find. The stress of monomorphemic forms in Kolya Yukaghir constitutes perhaps the best case (Maslova 2003: 58–9) known to me of a right-aligned DTS system.

As recognized already by Baković (1998: 10–12), if WSP is undominated, a foot will necessarily be constructed around every heavy syllable, which would potentially predict the occurrence of secondary stress on every heavy syllable to the left of the rightmost heavy syllable (at least when separated by at least one light syllable, where *CLASH would be irrelevant). Unless foot structure is presumed to not necessarily project a secondary stress on the stress grid (or be removed via conflation, as in Halle and Vergnaud 1987: 50–5), a constraint like Walker’s PK-PROM, specified for syllable weight, might be desirable in order to derive systems in which only a single primary prominence, at an edge or on a heavy syllable, actually occurs. The predictions made by undominated WSP versus undominated PK-PROM in cases of forms with multiple heavy syllables are represented in (72). In (72)a., candidate a. wins because the alternatives violate WSP at least once; in (72)b., on the other hand, only the rightmost heavy syllable need be footed in order to satisfy PK-PROM, at which point ALL-Ft-R restricts the number of feet actually constructed.⁵⁴

(72) Default-to-same weight-driven prominence: multiple heavy syllables

a. Undominated WSP

		WSP	ALL-Ft-R	RIGHTMOST	ALL-Ft-L	LEFTMOST	PARSE-σ	NONFINALITY
	L H L H L							
a.	☞ L (,H) L ('H) L		*****		*****	***	***	
b.	L H L ('H) L	*!	*	*	***	***	*****	
c.	L (,H) L (H 'L)	*!	*****		*****	*****	**	*

b. Undominated PK-PROM

		PK-PROM	ALL-Ft-R	RIGHTMOST	ALL-Ft-L	LEFTMOST	PARSE-σ	NONFINALITY
	L H L H L							
a.	L (,H) L ('H) L		**!*		*****	***	***	
b.	☞ L H L ('H) L		*	*	***	***	*****	
c.	L (,H) L (H 'L)	*!	*****		*****	*****	**	*
d.	L ('H) L H L		**!*	***	*	*	*****	

Now, consider simple ranking for deriving default-to-opposite prominence, in which primary stress lands at the opposite edge when the sequence consists solely of light syllables. In comparison to default-to-same-side systems, default-to-opposite systems are slightly more interesting, and can be generated by ranking one of ALL-Ft-L/R above one of

⁵⁴Perhaps unintuitively, the constraints on foot form, TROCHEE, IAMB, and FtBIN can also be employed to restrict the number of feet: if these three constraints dominate other metrical constraints, feet will preferentially be just of the shape (–), since feet consisting of a single heavy syllable satisfy all three constraints. However, the derivation of true default-to-opposite systems then becomes impossible: if words are restricted to containing a single foot, the single foot must always be oriented toward the same edge in all or a subset of cases containing multiple heavy syllables as well as in words consisting exclusively of light syllables.

LEFTMOST/RIGHTMOST and the other of ALL-Ft-L/R, in turn dominating the other of LEFTMOST/RIGHTMOST, with, again WSP dominating all of these constraints. Concretely, in the case of Selkup (cf. Kuznetsova et al. 1980, Isardi 1992: 10, Walker 1996: 21–3), primary stress falls on the rightmost heavy syllable, else on the leftmost syllable, and can result from the ranking given in (73), with examples. Note that only syllables with long vowels constitute heavy syllables. In cases involving only light syllables, reference to a constraint on foot form like TROCHEE is apparently necessary, in order to rule out an alternative position of the primary stress within the foot (contrast candidates a. and d. in (73)c.).

(73) Default-to-opposite weight-driven prominence in Selkup:

WSP $\gg$ ALL-Ft-L $\gg$ RIGHTMOST, ALL-Ft-R $\gg$ LEFTMOST

a. Only heavy syllable: /pynakisə:/ $\rightarrow$ [py.na.ki.(‘sə:)] ‘giant’

/pynakisə:/		WSP	ALL-Ft-L	RIGHTMOST	ALL-Ft-R	LEFTMOST
a.	py.na.ki.(‘sə:)		***			***
b.	(‘py.na).ki.sə:	*!		***	**	

b. Rightmost heavy syllable: /qumo:qli:lr:/ $\rightarrow$ [qu.(mo:).qli.(‘lr:)] ‘your two friends’

		WSP	ALL-Ft-L	RIGHTMOST	ALL-Ft-R	LEFTMOST
a.	qu.(mo:).qli.(‘lr:)		****		**	***
b.	qu.(‘mo:).qli.(lr:)		****	*!*	**	*
c.	qu.mo:qli.(‘lr:)	*!	***			***

c. Else Leftmost: /qol^hcimpat/ $\rightarrow$ [(‘qol^h.cim).pa.ti] ‘found’

qol ^h cimpat		WSP	TROCHEE	ALL-Ft-L	RIGHTMOST	ALL-Ft-R	LEFTMOST
a.	(‘qol ^h .cim).pa.ti				***	**	
b.	qol ^h .cim.pa.(‘ti)			*!*			***
c.	qol ^h .cim.(‘pa.ti)			*!*	*		**
d.	(qol ^h .‘cim).pa.ti		*!		***	**	

In at least one respect is an extension of the basic set of stress constraints potentially necessary. First, in cases where more than two degrees of syllable weight appear to be distinguished, with primary stress being attracted to superheavy (trimoraic or heavier) syllables in preference to bimoraic syllables (as in, e.g., the variety of Hindi described in Kelkar 1968;

cf. Prince and Smolensky 1993 [2004]: 40–2, Hayes 1995: 276–8, Walker 1996: 14), a fixed ranking $WSP_{\mu\mu\mu} \gg WSP_{\mu\mu}$ will suffice; compare also the analysis using weight-to-stress mapping constraints $\sigma_{\mu\mu} \rightarrow \text{STRESS}$ and $\sigma_{\mu\mu\mu} \rightarrow \text{STRESS}$ in Ryan 2019: 35–6.⁵⁵

The ultimate conclusions to take away from this discussion are the following: first, there descriptively exist systems in which attraction of stress to heavy syllables and a word edge, rather than clear alternating rhythmic stress, governs primary stress assignment; second, the set of constraints outlined in Section 2.3, with perhaps minor gradient extensions to WSP, are sufficient for generating the attested stress pattern, independent of any special assumptions about relationships between foot structure and the realization of stress. Nonetheless, the same set of constraints likely may typologically overgenerate, predicting, e.g., a pattern in which primary stress falls on the rightmost heavy syllable, else on the peninitial syllable; just such a pattern, similar to that of Selkup, would be predicted to exist if TROCHEE were substituted with IAMB in (73)c. In the absence of a consensus on more minimalistic constraint set that successfully generates both bounded and unbounded stress patterns, I prefer to adhere to the constraint set from Section 2.3, rather than extending it with other constraints of more dubious effect and utility (e.g., *GRIDSTRUC in Walker 1996: 41, or separation of foot heads from stress proposed in Hyde 2001).

2.4.1.2 Pitch-Driven Prominence

Much like weight-driven prominence systems treated in the immediately preceding section, in which stress is preferentially attracted to (superheavy syllables in preference to) heavy syllables in preference to light syllables, a number of stress systems are attested in which the position of stress is regulated by lexical tone. Just as WSP expresses a stringency hierarchy, whereby stress always preferentially maps onto heavier syllables, de Lacy (2002) argues that the relationship between stress and tone in systems with pitch-driven prominence is regulated by two sets of constraints that govern the relationship between tones and foot heads, which similarly constitute a stringency hierarchy. These constraint families, and their presumably fixed rankings, are given in (74); the first set, shown in a., penalizes the association of relatively lower-pitched tones with foot heads more strongly, while the second set, given in b., penalizes the association of non-heads with relatively higher pitched tones more strongly.⁵⁶

(74) Constraint Families Regulating Pitch Prominence (de Lacy 2002: 2)

- a. *HD/L $\gg$ *HD/M $\gg$ *HD/H
- b. *NON-HD/H $\gg$ NON-HD/M $\gg$ *NON-HD/L

⁵⁵The particular ternary weight distinction described by Kelkar, $V < \{VC, VV\} < VVC$ is apparently unique (cf. Gordon 2006: 126–7 and Ryan 2019: 36 on Eipomek). When combined with a lack of any edge restriction, the Kelkar’s Hindi exhibits a degree of computational complexity that is unusual for stress systems (see Heinz 2009: 322–4).

⁵⁶Note that H and L in this context stand for “high tone” and “low tone”, respectively, rather than “heavy” and “light”.

The parallel with weight-driven prominence is most clearly in evidence in cases in which a language definitely possesses lexically specified tone. There, syllables lexically specified for /H/ and /M/ play the role of heavy or superheavy syllables, consistently attracting stress in preference to low-toned syllables; stress occurs on a syllable specified for /L/ tone only when no syllables bearing higher-pitched tones are available. The stress system of Ayutla Mixtec (Pankratz and Pike 1967: 289–94, de Lacy 2002: 5–14), which distinguishes /H/, /M/, and /L/ tones, appears to depend upon lexical tone in just such a fashion: stress is assigned to the /H/ of the leftmost /HL/ sequence, else on the leftmost /H/, else on the leftmost /ML/ sequence, and finally on the leftmost syllable.⁵⁷ Given a preference for binary trochaic feet, (active TROCHEE FTBIN), sequences in which a /L/ follows an /H/ or /M/ will allow for the satisfaction of FTBIN (Ayutla has no syllable weight contrast), without incurring serious violations of the constraints on tone/foot-relations in (74). In terms of regulation the edge orientation of the stress, ALL-Ft-L suffices for expressing the preference for stress to occur closer to the left edge, all else being equal. The tableaux in (75) illustrate the derivation of Ayutla stress, which may be characterized essentially as a default-to-same system, with a further interesting wrinkle caused by the absence of heavy syllables, such that FTBIN can only be satisfied by building disyllabic feet.

(75) Default-to-same pitch-driven prominence in Ayutla Mixtec:

*HD/L, *NONHD-H >> *HD/M, *NON-HD/M >> FTBIN >> ALL-Ft-L

- a. Stress the leftmost /HL/ sequence/: /kátʃiràá/ → [ká.(ʔʃi.rà).á] ‘there is none of his cotton’

/kátʃiràá/	*HD/L	*NONHD-H	*HD/M	*NON-HD/M	FTBIN	ALL-Ft-L
a. (ʔká.ʔʃi).rà.á		*!				
b. ká.ʔʃi.(ʔrà.á)	*!					**
c. ☞ ká.(ʔʃi.rà).á						*
d. (ʔká).ʔʃi.rà.á					*!	

- b. Else stress the leftmost /H/: /ʃitórirá/ → [(ʔʃi).tó.rí.rá] ‘he (an animal) is watching him (a man)’

/ʃitórirá/	*HD/L	*NONHD-H	*HD/M	*NON-HD/M	FTBIN	ALL-Ft-L
a. (ʔʃi.tó).rí.rá/		*!				
b. ʃi.(ʔtó).rí.rá					*	*!
c. ☞ (ʔʃi).tó.rí.rá/					*	

⁵⁷Since /M/ tone exhibits a restricted distribution in the language, there are no words containing only /M/ and /L/ in which a /ML/ sequence does not arise.

- c. Else stress the leftmost /ML/ sequence: /lāfārà/ → [lā.(‘fā.rà)] ‘his orange’

/lāfārà/	*HD/L	*NONHD-H	*HD/M	*NONHD/M	FtBIN	ALL-Ft-L
a.  lā.(‘fā.rà)			*			*
b. (‘lā.fā).rà			*	*!		
c. (‘lā).fā.rà			*		*!	

- d. Else stress the leftmost syllable in a sequence consisting exclusively of /L/ or /M/: /fātùì/ → [(‘fā.tù).ì] ‘my trousers’

/fātùì/	*HD/L	*NONHD-H	*HD/M	*NONHD/M	FtBIN	ALL-Ft-L
a.  (‘fā.tù).ì	*					
b. fā.(‘tù.ì)	*					*!

Ayutla thus constitutes a case in which tones are always faithfully realized on their hosts (i.e., IDENT(Tone) is undominated), and metrical structure is constructed so as to best satisfy the constraints on relations between tone and foot heads. Lamba (Bantu; Bickmore 1995), on the other hand, constitutes a case in which the construction of metrical structure does not need to refer to lexical tone, but any lexical tones in the input may be freely reassociated to as to satisfy the constraints in (74) (see the treatment in de Lacy 2002: 14–9). Thus, in the Ayutla case, the position of metrical structures depends on lexical tone (“tone-driven stress”), whereas in Lamba, the position of lexical tone depends upon metrical structure (“stress-driven tone”). In cases in which high pitch and stress seem to regularly or frequently co-occur, as in Vedic Sanskrit (see Chapter 4), Lithuanian (Blevins 1993, Kushnir 2018), or some varieties of Serbo-Croatian (see Zec 1999, Langston 2006, Zec and Zsiga 2022), to speak of a clear case of “tone-driven stress”, as de Lacy (2002: 23), does, is difficult. In each of those languages, the constraints *HD/L and *NON-HD/H may often be satisfied, but it is far from obvious that the placement of stress is primarily affected by lexical tone (or that lexical tone in the sense discussed in 1.3.2 is present whatsoever, in the Vedic case).⁵⁸

2.4.2 Phonologically Unpredictable Stress: Lexical Accent and Headedness

This chapter has thus far examined systems of metrical structure construction and stress assignment in which other independent prosodic properties are responsible for determining the position of primary (and where relevant, secondary) stress. These prosodic factors consist principally of the boundaries of the ω, the number of syllables, the weight of syllables, and lexical tone. Such systems exhibit stress that is fundamentally predictable

⁵⁸To be fair, Zsiga and Zec (2012) advance the argument that stress in Standard Serbian indeed has stress distributions that are determined by contrastive tonal specifications (though the language possesses only a single privative lexical tone, /H/).

based on phonological factors alone. In contrast, the nature of stress assignment to be surveyed in this system is fundamentally *morphological* (taking entries in the lexicon to fall under the purview of the morphology); with Revithiadou (1999), the prosodic systems of such languages constitute “interface systems” in which the phonological component evaluates prosodic information received from the morphology or aspects of morphological structure itself in order to determine the position of stress. What stands out in languages possessing such interface systems is that the position of stress in a word is fundamentally unpredictable on the basis of independent phonological properties of a word.

Perhaps typologically most common among languages with phonologically unpredictable stress is to find apparent “marking” of some kind for the position of the word-level prosodic prominence in the lexical entries of morphemes. In the case of morphologically simplex lexemes, a surface contrast in the position of prominence may be equated with an underlying contrast, with vocalic peaks likely to become syllable heads marked with a “lexical accent”. An underlying distinction in such marking, e.g., Russian /múka/ ‘pain’ versus /muká/ ‘flour’, is held responsible for the surface contrast in the position of stress: [‘muka] versus [mo’ka]. If a language consistently satisfies CULMINATIVITY, in the event that multiple morphemes within a prosodic word are “marked” in some fashion (as will be seen below, possible “markings” go beyond a preference to bear primary stress), markedness constraints on prosodic structure or special prosodic faithfulness constraints (see further 2.4.2.3 below) adjudicate the outcome. If a language consistently satisfies OBLIGATORINESS, in the event that no morpheme in an input is “marked”, prosodic markedness constraints alone will calculate the position of a “default” prominence. Historically, insights into the analysis of lexical accent – i.e., deriving some surface prominence from competition between underlying lexical properties – goes back to Garde 1965 and exponents of the Moscow School of Accentology (e.g., Illich-Svitych 1963).

Seemingly less common – or at least less investigated – are cases in which morphological structure directly influences the position of prosodic prominence. Revithiadou (1999) has argued for the special role of morphological heads, either as subject to special faithfulness (HEADFAITH) or through the direct attraction of prominence to a syllable comprising the morphological head of the word (HEADSTRESS). As discussion below and the studies in Part II will show, special faithfulness to the lexical accents of morphological heads is not uncommon across Indo-European languages, both ancient and modern. Cases of HEADSTRESS are harder to come by, and new cases beyond those adduced by Revithiadou (1999: 29) have not been readily forthcoming, with perhaps the exception of the further Salishan language, Squamish, argued to exhibit stress attraction to morphological heads in Dyck 2004: Ch. 6.⁵⁹

Besides the already cited work of Revithiadou, lexical accent or “morphological interface” systems have received comprehensive treatments in Alderete 2001a and, with a focus on polysynthetic languages, in Bogomolets 2020. In this section, I first illustrate some sim-

⁵⁹ Furthermore, the existence of systems with both active head-attraction effects and lexical accents is questionable. Revithiadou (1999: 29) claims that Tahltan (Athabaskan, Alaska and British Columbia) and Hua (Gorokan, East New Guinea) both instantiate prosodic systems of this sort, but closer examination of Revithiadou’s sources on Tahltan (Cook 1972, Nater 1989) suggests that the facts of Tahltan prosody cannot be interpreted quite so straightforwardly; see further below.

ple lexical accent systems, in which morphological structure *per se* plays no direct role, in order to provide a sense of the typology of possible prosodic marking in underlying representations (2.4.2.1). Exactly how lexical accents are then to be represented is then the topic of the subsequent section, 2.4.2.2, and this topic will return in Chapters 4 and 6. To complete this comprehensive overview on sources of stress, further effects deriving from the status of morphemes as roots, heads, or inflection are treated in 2.4.2.3.

2.4.2.1 Simple Lexical Accent Systems

As indicated above, languages that exhibit genuine minimal pairs distinguished only through the position of stress must, by definition, possess some lexically specified property that determines the position of stress. When morphological structure itself, beyond the fact of being polymorphemic, plays no role in the calculation of stress, the language may be said to possess a simple lexical accent system. In the classification applied by Bogomolets (2020: 12), such systems resolve conflict between lexical accents directionally or post-cyclically, using only proper prosodic markedness constraints to resolve conflicts. Arbitrary lexical accent and its interaction with considerations of metrical stress is both robustly attested and reasonably well understood. The fundamental distinction between a simple lexical accent system and a pure metrical stress system lies in the relative ranking of faithfulness constraints referring to the lexical accents and prosodic markedness constraints: in the former, faithfulness outranks (some subset of) prosodic markedness constraints, allowing lexical accents to determine the position of stress in a manner contrary to what the ranking of the prosodic markedness constraints might prefer.

Cupeño The Uto-Aztecan language Cupeño serves as a paradigm case for a simple lexical accent system (at least under the analysis of Yates 2017: Ch. 3, followed by Bogomolets (2020: 137–44); see Alderete 2001a: 43–65 for a different view). Under Yates’ analysis, morphemes of Cupeño belong to one of three prosodic classes: **ACCENTED** (stress attracting), **PREACCENTING** (stress repelling, but locally, onto the immediately preceding syllable), and **UNACCENTED** (stress neutral).⁶⁰ When a phonological string contains no accented or preaccenting morphemes, stress surfaces on the leftmost syllable of the word (76a.) ; when a single accented or preaccenting morpheme occurs, stress is determined by the preference of that morpheme (76b.); when multiple accented or preaccenting morphemes occur in a string, only a single stress occurs on the surface, such that this primary stress falls as close to the left edge as possible (76c-e.)⁶¹ Take particular note of the stress on the antepenult in (76)c., which results from the preaccent projected by the morpheme /-’ŋa/, whose segmental content makes up the penultimate syllable.

⁶⁰In this initial presentation, I remain agnostic as to the nature of the lexical accents. Notationally, an acute is used to mark an **ACCENTED** morpheme, a preceding acute marks a **PREACCENTING** morpheme, and a following grave marks a **POSTACCENTING** morpheme.

⁶¹Yates (2017: 43), following Hill and Hill (1968: 236) and Alderete (2001b), assumes that Cupeño has no secondary stress; hence, the option of faithfully realizing several or even all lexical accents via secondary stress is excluded. A larger question is whether cases of competing lexical accents realized as different degrees of stress are attested whatsoever.

- (76) Cupeño (Yates 2017: Ch. 3; examples pp. 64–66): leftmost underlying lexical stress wins, else stress on the leftmost syllable.
- a. /pəm - təw - wəni/ → [ˈpəm.tə.wə.ni] ‘we see’ (1PL-see-DS.PL)
 - b. /ʔə - yax - qalí/ → [ʔə.ya.qa.liʔ] ‘you say’ (2SG-say-DS.SG)
 - c. /nə - təwi - ʔa - ʔaw/ → [nə.tə.wi.ʔa.ʔaw] ‘on my chest’ (1SG-chest-INL-LOC)
 - d. /nə - míx - á:n - βə/ → [nə.mi.xan.βə] ‘(where) I did (something)’ (3SG-be-AAN-SUB.RL)
 - e. /taβ - á:n - pə - qál/ → [ta.βa:n.pə.qal] he placed (it) in’ (put.in-AAN-3SG-PST.IPFV.SG)

Since the position of the primary stress is completely unrestricted with respect to either word edge, a faithfulness constraint ensuring the realization of the lexical accent can be assumed to outrank almost all prosodic markedness constraints, except for CULMINATIVITY, which ensures only a single primary stress per word. For the moment, the relatively neutral constraint MAX-PROMINENCE (defined in 77) will be employed to this end. Among the prosodic markedness constraints, ALL-Ft-L, LEFTMOST, TROCHEE, and FOOTBINARITY strive to ensure that a single minimally bimoraic trochee emerges; the epenthesis of a glottal stop when stress falls on a final otherwise light syllable (as in 76b.) avoids violations of FtBIN. ALL-Ft-L or LEFTMOST can be held responsible for the selection of the leftmost lexical accent, on the one hand, and, in cooperation with TROCHEE, for the appearance of primary stress on the leftmost syllable in cases of inputs without any lexical accents.⁶² Throughout, whether a given morpheme constitutes a root, derivational morpheme, or inflectional morpheme plays no role whatsoever.

- (77) MAX-PROMINENCE:
Lexical accents should be preserved as stress in the output. Assign one violation * for each lexical accent $\mathcal{A}_i$ in the input that does not correspond to a stress $\mathcal{S}_i$ in the output.

The tableaux in 78 illustrate the three major cases of stress assignment in Cupeño-like simple lexical accent systems. In a., the need to satisfy OBLIGATORINESS (i.e., that each ω possess a primary stress) or EXHAUSTIVITY (here, that each ω dominate at least one $\mathfrak{f}$) excludes stressless/footless candidates; since no lexical accents are present in the input, the lower-ranking markedness constraints result in the construction of a trochaic foot at the left edge of the word. In b., meanwhile, the single lexical accent, in virtue of dominating the markedness constraints, decides the position of the stress/foot, and brings with it epenthesis of [ʔ] to satisfy FtBIN. Finally, the tableau in c. shows that MAX-PROM can be violated to satisfy CULMINATIVITY, but the choice of which correspondence relation to satisfy is determined by the lower-ranking markedness constraints.

⁶²Given that secondary stress does not occur, the ranking ALL-Ft-L $\gg$ MAX-PROM $\gg$ LEFTMOST is probably necessary: high-ranking ALL-Ft-L excludes the construction of multiple feet, while LEFTMOST is then responsible for positioning the single foot as close to the left edge as possible. In the tableau below, I simply ignore candidates containing secondary stress, and show all markedness constraints below MAX-PROM. Candidates showing migration of a lexical accent to another vocalic peak are likewise ignored below.

(78) Derivations of Stress in Cupeño

a. /pəm - təw - wəni/ → [(pəm.tə).wə.ni] ‘we see’ (1PL-see-DS.PL)

/pəm - təw - wəni/	OBLIG	CULM	MAX-PROM	ALL-Ft-L	TROCHEE	FTBIN	PARSE-σ
a.  (pəm.tə).wə.ni							**
b. pəm.tə.wə.ni	*!						
c. pəm.(tə.wə).ni				*!			**
d. (pəm.tə).wə.ni					*!		

b. /ʔə - yax - qalí/ → [ʔə.ya.qa.(liʔ)] ‘you say’ (2SG-say-DS.SG)

/ʔə - yax - qalí/	OBLIG	CULM	MAX-PROM	ALL-Ft-L	TROCHEE	FTBIN	PARSE-σ	DEP-[ʔ]
a.  ʔə.ya.qa.(liʔ)				***		*	***	*
b. (ʔə.ya).qa.(liʔ)		*!		***		*		*
c. (ʔə.ya).qa.li			*!				**	

c. /taβ - á:n - pə - qál/ → [ta.(βa:n.pə).qal] ‘he placed (it) in’ (put.in-AAN-3SG-PST.IPFV.SG)

taβ - á:n - pə - qál	OBLIG	CULM	MAX-PROM	ALL-Ft-L	TROCHEE	FTBIN	PARSE-σ
a.  ta.(βa:n.pə).qal			*	*			**
b. ta.(βa:n.pə).(qal)		*!		****			*
c. ta.βa:n.pə.(qal)			*	***!			***
d. (ta.βa:n).pə.qal			**!				**

Cupeño thus constitutes a paradigm case in which the position of the word-level prosodic prominence is unrestricted. In this regard, stress in languages having lexically prespecified preferences can appear to behave in a fashion similar to stress in languages with prominence-driven stress (see 2.4.1); in the Cupeño case, lexical accent plays a role similar to syllable weight in Kolyma Yukaghir or high tone in Ayutla Mixtec.

Standard Italian The prosody of Standard Italian can similarly be analyzed as possessing a simple lexical accent system, but is subject to the activity of more prosodic markedness

constraints that more strongly constrain where primary stress may occur (see the extensive treatment of Italian word stress in Krämer 2009: Ch. 6; 187–94). Specifically, primary stress nearly always occurs within a window of three syllables at the right edge of the word, and tends strongly to occur on the penultimate syllable if it is closed by a consonant (i.e., heavy).⁶³ These restrictions are a residue of inheritance from the metrical stress system of Latin. Exceptions to these restrictions serve to further prove, however, that Italian possesses lexically specified prosody, as opposed to, e.g., long vowels that could be used to largely predict the distribution of stress within the three-syllable window. The forms given in 79 shows that stress may, in fact, occur on up to the preantepenultimate syllable (if rarely, and then only in verbs), and that stress may fall on the antepenult regardless of the weight of the penult (again, if rarely).

(79) Basic Italian Stress Patterns

- a. Final Stress: *università* [u.ni.ver.si.'ta] 'university'
- b. Penultimate Stress (open penult): *lumaca* [lu.'ma:..ka] 'snail'
- c. Penultimate Stress (closed penult): [in.gom.'bran.te] 'bulky:SG'
- d. Antepenultimate Stress (open penult): ['bru.go.la] 'hexagon wrench, Allen key'
- e. Antepenultimate Stress (closed penult): *Taranto* ['ta.ran.to] '(name of a city)'
- f. Preantepenultimate Stress: *telefonano* [te.'le.fo.na.no] 'call on the telephone:PRES.3PL'

Without delving too deeply into the details of a phonological analysis for stress assignment in Standard Italian, I will illustrate here on the basis of stress in verbal inflection that simple lexical accents and a preference for stress to occur towards the right edge of the prosodic word are largely sufficient to generate the facts. Whether the default stress pattern assigned through the activity of markedness constraints alone always selects the penultimate syllable, or in some configurations prefers the antepenult, is disputed; D'Imperio and Rosenthal (1999) make the claim that the penult is the default throughout the lexicon, and Krämer (2009: 187–8) holds that the penult constitutes the default stress at least in verbs, where most instances of interaction between lexical accents may be observed. I hold that the virtual four-syllable window – somewhat comparable to stress systems claimed to exist in Old Latin (cf. 2.2.7.1) and Classical Sanskrit (cf. Chapter 5) – follows from (violable) preferences for final syllable extrametricality (NONFIN- $\mathcal{F}$), binary feet (FTBIN, selection of the rightmost foot in the word as the locus for primary stress (RIGHTMOST) and preservation of lexical accents (MAX-Prom).

Verbal inflectional paradigms given in (80) and (81) show surface stress patterns for two verbs of the 1st conjugation (theme vowel *-a-*) and two of the second conjugation (theme vowel in *-e-*). Given the assumption of default penultimate stress, a contrast between lexically accented (stress-attracting) and lexically unaccented (stress-neutral) stems, combined

⁶³Borrelli (2002) offers the following frequencies for the position of primary stress based on 120 consecutive entries in a dictionary: 76.67% penultimate, 20.83% antepenultimate, 2.5% final; these percentages add up to 100%. Since a small residue of forms on the surface do exhibit preantepenultimate stress, Borrelli's figures do not capture the full picture of where stress is permitted to occur.

with a three-way distinction between unaccented, accented (stress-attracting), and preaccenting (stress-rejecting) inflectional suffixes, can generate the attested patterns. For the cases at hand, one need only specify that the rightmost lexical accent determines the position of primary stress. Compare the data and analysis in Krämer 2009: 187–90.

(80) Italian Stress Patterns in Verbal Inflection (Present Indicative) 1: Unaccented Stems

INF	[dis.lo.'ka:.re] 'dislocate, displace'	[ka.'de:.re] 'fall'
1SG	[dis.'lɔ.ko]	['ka:.do]
2SG	[dis.'lɔ.ki]	['ka:.di]
3SG	[dis.'lɔ.ka]	['ka.de]
1PL	[dis.lo.'kja:.mo]	[ka.'dja:.mo]
2PL	[dis.lo.'ka:.te]	[ka.'de:.te]
3PL	[dis.'lɔ.ka.no]	['ka.do.no]

(81) Italian Stress Patterns in Verbal Inflection (Present Indicative) 2: Accented Stems

INF	[te.le.fo.'na:.re] 'call on the telephone'	['ri.de.re] 'laugh'
1SG	[te.'le.fo.no]	['ri:.do]
2SG	[te.'le.fo.ni]	['ri:.di]
3SG	[te.'le.fo.na]	['ri:.de]
1PL	[te.le.fo.'nja:.mo]	[ri.'dja:.mo]
2PL	[te.le.fo.'na:.te]	[ri.'de:.te]
3PL	[te.'le.fo.na.no]	['ri.do.no]

Specifically, the inflectional endings for the infinitive of the 1st class, /-áre/, as well as the endings for 1PL (/ -jámo/) and 2PL (/ -áte/, / -éte/), may be treated as accented: primary stress always falls on the ending and penultimate syllable, regardless of the properties of the stem. The inflectional endings of the 1SG (/ -o/) and 2SG /-i/), on the other hand, may be treated as unaccented, as well as the infinitive of the 2nd class (/ -ere/): stress falls on the stem, which may be either the penultimate syllable (if the stem is unaccented, or accented on its only syllable) or antepenultimate syllable (if the stem is polysyllabic and lexically accented on a vowel other than its rightmost vowel). Finally, the endings for the 3PL, -ano and -ono cannot be either stress attracting or stress neutral, since penultimate stress would be predicted in such cases, at least on unaccented stems (e.g., ^x[dis.lo.'ka:.no]). If these suffixes instead reject stress or place stress to its right, then antepenultimate stress (e.g., [dis.'lɔ.ka.no]) or the lexical accent of the stem (e.g., [te.'le.fo.na.no]) may result.⁶⁴

In strings of derivational suffixes, stress often appears to shift rightwards within the window. This rightward stress shift again falls out directly from a combination of simple lexical accents and a preference for stress to fall closer to the right edge of the word. Consider the derivational chain illustrated in 82: the nouns ['bru.go.la] 'hexagon wrench' and [lu.'ma:.ka] 'snail' must contrast in terms of lexically specified stress under any analysis of the default

⁶⁴Exactly how such a property of stress-rejection or preaccentedness should be represented will be discussed below. For the moment, I assume that this property and the ranking of relevant faithfulness constraints can both induce stress on the syllable preceding these endings or allow the stress-attracting syllable of a stem to determine primary stress.

stress; one or the other must be marked for special stress attraction. However, an adjective built with the adjective-forming suffix /-óso/, from which in turn a noun with the suffix /-itá/ can be derived, attest to rightward stress shift, regardless of the stress of the base. Granting that both suffixes are specified as stress-attracting, their lexical accents will then be the rightmost in the word.

(82) Lexical Stress in Derivation in Italian

- a. ['bru.go.la] 'hexagon wrench' ⇒ [bru.go.'lo:.zo] 'hexagon-wrench-like' ⇒
[bru.go.lo.zi.'ta] 'having the properties of being hexagon-wrench-like'
- b. [lu.'ma:ka] 'snail' ⇒ [lu.ma.'ko:.zo] 'snail-like' ⇒
[lu.ma.ko:zi.'ta] 'having the properties of being snail-like'

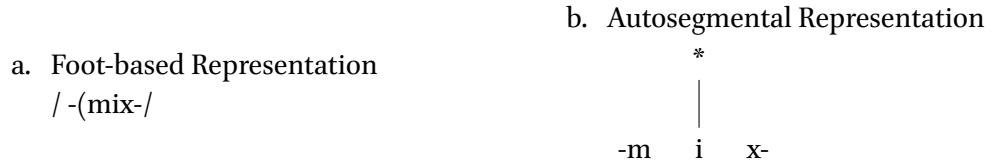
2.4.2.2 The Representation of Lexical Accent

Having now illustrated the fundamental property of competition between lexical entries with distinct prosodic properties on the basis of the “simple” systems of Cupeño and Standard Italian, I turn to the question of the phonological representation of such preferences. The fundamental question is: what does it mean to be “stress attracting”, “preaccenting”, etc.? Two major approaches to the representation of lexical accent have been developed in the literature. One characterizes lexical accents in terms of feet, the other in terms of autosegments. I will refer to these two representational proposals as the FOOT REPRESENTATION and AUTOSEGMENTAL REPRESENTATION of lexical accent, respectively.⁶⁵ A third option, to treat lexical stress as arising through lexically specific constraint rankings, has also been considered, though employed less often in the literature. All three approaches will be described below.

Two possible representations for an “accented” or “stress-attracting” morpheme in Cupeño, such as /-míx-/ ‘be’ (cf. 76 above) are shown in 83. Given a preference for constructing trochaic feet in the language, a left foot edge specifies where a foot should begin, leading to prominence on its first syllable in a trochaic system, while a vowel or morpheme linked to a strong prominence autosegment will be eligible for stress, and, insofar as feet are constructed, lead to the placement of a foot boundary to its left. In the case of a foot boundary, the foot should be seen as a hidden structure (see further 3.2.2 on this concept) that is part of a lexical representation, not dependent upon the segmental content of that lexical entry; in the case of an autosegment, the marking may be seen as a property of some segment within the lexical entry or as an element simply carried within the lexical entry.

⁶⁵In her survey of theoretical approaches to the representation of lexical accent, Revithiadou (1999: 56–7) distinguishes rather between what she terms LEXICAL CONSTITUENCY and LEXICAL REPRESENTATION. Under approaches corresponding to the former type, “lexical stress is an autonomous prosodic element that is affiliated to morphological or segmental material”, while under the latter view, “[lexical stress] marking is the projection of intrinsic metrical structure from vocalic peaks”. Revithiadou’s distinction is essentially tied to the use of a bracketed grid model of stress (Halle and Vergnaud 1987, Halle and Isardi 1995) as opposed to the use of bounded foot types, where bracketed grids fall into the LEXICAL REPRESENTATION group.

(83) Foot-based vs. Autosegmental Representation of “Accentedness”



Fundamentally, the former proposal tends to make all lexical accent “stress-like”: underlying foot edges compete for preservation and, in case multiple feet are permitted in the output, promotion to the head foot of the prosodic word; the only foot or head foot then manifests prominence in some fashion, but is then effectively tied to the claim that this prominence is a kind of stress, or that stress determines the position of some other prominence (such as a starred tone). The autosegmental representation allows for accent to be somewhat more flexible: it can be taken to stand for an abstract prominence, ultimately implemented more concretely as a grid mark, a tone, or perhaps even intensity or duration independent of stress or tone.

The choice of foot-based or autosegmental representations is orthogonal to a constraint-based or derivational model of stress assignment. Revithiadou (1999) and Alderete (2001a), for instance, both treat lexical accent as an autosegment in the context of constraint grammars, while Halle and Vergnaud 1987: Ch. 3 represent lexical stress similarly, namely, as a head on line 1 of the stress grid, then deriving stress placement through the application of ordered rules for the construction of metrical structure. Conversely, Özçelik (2014) represents lexical accent as underlying foot edges, then deriving stress in OT grammars, while Drescher (2016) similarly takes lexical accent to consist in lexical representations of certain syllables enriched with left or right brackets (foot edges), used to derive stress in a version of Simplified Bracketed Grid theory (Halle and Isardi 1995).

In the following, I review how the three different representational approaches introduced above (foot, autosegment, and lexical constraints) may, in practice be implemented in conjunction with constraint-based grammars of stress assignment. The objective here is merely to familiarize the reader with the existence of these options, rather than to systematically muster empirical or theoretical arguments for or against a particular representational choice. Indeed, in many cases, the choice between an autosegmental representation or prespecified foot makes no practical difference. Nevertheless, some stress phenomena in Vedic Sanskrit appear more readily accommodated under the assumption of prespecified foot structure (see 4.3.2), for which reason I will generally favor the representation of lexical accent as prespecified prosodic structure throughout the present work. For further literature on the representation of lexical accent including and beyond discussions mentioned below, see Bogomolets 2020: 10-12.

Lexical Accent as Diacritic or Autosegment One of the most extensive discussions concerning the representation of is to be found in Revithiadou 1999: Ch. 2. Revithiadou takes lexical accents to be autosegmental features, in line with the view of lexical accent as pure prominence promoted in Alderete 2001a. By and large, Revithiadou (1999) and Alderete

(2001b) agree on certain fundamental properties of lexical accents: they are abstract, autosegmental, and contain no cues as to phonetic realization. The representations for lexical accent proposed by Revithiadou and Alderete seem to generally proceed from the assumption that there exists a unified class of “lexical accent systems”, which share the property of having some sort of ‘prominence’, whose position is unpredictable. Such prominence is permitted to take the form of phonetic stress, tone, or duration (supposedly independent of stress).⁶⁶

Alderete (2001a: 23) is explicit about the relationship between lexical accent and stress as represented on the stress grid: “accent is encoded as a lexical prominence, i.e., a grid mark over an accented vowel in the underlying representation”.⁶⁷ For Alderete, these grid marks are autosegments similar to tonal representations. The relationship between lexical accents in the input and output is then governed by the set of prosodic faithfulness constraints defined in 84; see the brief analysis of Cupeño data above for the use of MAX-PROM, for instance. Alderete does not adopt representations with foot structure for the reason that cases pre- and post-accentuation can be generated using anti-faithfulness constraints, without the need to refer to foot structure whatsoever.⁶⁸

- (84) Prosodic Faithfulness Constraints Governing the Realization of Lexical Accent (cf. Alderete 2001a: 24)⁶⁹
- a. MAX-PROM: Assign one violation * for each prominence in the input that lacks a correspondent prominence in the output.
 - b. DEP-PROM: Assign one violation * for each prominence in the output that lacks a correspondent prominence in the input.
 - c. NO-FLOP-PROM: Assign one violation * for each prominence in the input that is autosegmentally associated to a given host in the input, H_1 , if that prominence is associated to some different host, H_2 , in the output.⁷⁰

In simple lexical accent systems, like the cases of Cupeño and Italian reviewed above, ranking all three of these prosodic faithfulness constraints below OBLIGATORINESS and CULMINATIVITY, but above other prosodic markedness constraints that govern the position of a default prominence, produces exactly the effects seen there, namely:

⁶⁶Revithiadou, following Thompson and Thompson 1996, assumes that Thompson Salish possesses metrically predictable secondary stress, but unpredictable primary stress (marked by strong lexical accents) as well as vowels that are unpredictably protected from reduction to $\emptyset$ (marked by weak lexical accents).

⁶⁷Compare also Kabak and Revithiadou 2009a: 2, fn. 4, where the autosegmental representation is defined in a similar fashion: “A lexical accent is treated as an autosegmental unit, a grid mark, which is projected onto the stress plane as an idiosyncratic property of a vocalic peak of a particular morpheme (Alderete 1999, Alderete 2001b, Alderete 2001a; Revithiadou 1999, Revithiadou 2007).”

⁶⁸This choice, however, constitutes a considerable theoretical problem: see, for instance, Apoussidou 2003 or Trommer 2006 for compelling arguments against the employ of anti-faithfulness constraints.

⁶⁹These constraints are functionally equivalent to MAX(L[exical]A[ccent]), DEP(LA) and *FLOP(LA) used by Revithiadou (1999: 52–3).

⁷⁰The exact interpretation of NO-FLOP-PROM in cases of floating prominence autosegments is not clear, and in fact Alderete and Revithiadou apply different interpretations. See 4.3.2 in Chapter 4 on the effects of these different interpretations on the analysis of certain facts in Vedic Sanskrit.

- inputs containing multiple lexical accents satisfy CULMINATIVITY by violating MAX-PROM, in a fashion that minimizes violations of other prosodic markedness constraints.
- inputs lacking any lexical accents satisfy OBLIGATORINESS by violating DEP-PROM, in a fashion that minimizes violations of other prosodic markedness constraints.
- inputs do not reassociate underlying (non-floating) prominences to new hosts in order to better satisfy the lower-ranked prosodic markedness constraints.

Revithiadou (1999: Ch. 2), meanwhile permits lexical accents to be equipped with multiple valences, i.e., to be “strong” or weak. For Revithiadou, a “strong” lexical accent will specify the position of a foot head, provided that feet are relevant to prominence in the language, or that the language builds feet at all, while a “weak” accent effectively marks a foot tail.⁷¹ In a language assumed to construct metrical feet (e.g., Modern Greek), strong lexical accents will be parsed as foot heads and thus receive stress, whereas weak lexical accents will be parsed as foot tails. In pitch-accent systems, on the other hand, strong lexical accents are generally realized as high tone, weak accents as low tone (cf. 2.4.1.2 above). In yet other languages, weak lexical accents are permitted to have other phonological effects (e.g., production from vowel reduction, independent of the position of primary or secondary stress).

These valences are marked by Revithiadou by two distinct autosegments, “*” for strong lexical accents and “.” for weak lexical accents. These autosegments may be either pre-associated with a vocalic peak, or permitted to enter the derivation as a floating accent, carried as part of a lexical entry. These three autosegmental structures are shown in 85, and all contrast with prosodically unmarked lexical entries; dashed lines indicate that the floating lexical accent would be expected to associated somewhere outside of its sponsor. Interestingly, both weak and floating lexical accents typically characterize morphemes that appear “unaccentable” on the surface: the former because attract the absence of prominence (foot tail, low tone), the latter because the floating accent is taken to prefer to associate outside the domain of the morpheme that sponsors it. In order to account for the behavior of posited “unaccentable” morphemes, Revithiadou borrows the constraint *DOMAIN from Myers and Carleton 1996. This constraint penalizes the association of a lexical accent α to a syllable within the same morphological domain α that sponsors it; the consequence is then that LA_α must link a syllable in some other morpheme.

⁷¹As discussed under 1.2.3 in Chapter 1, I adopt the strong position that foot structure is generally universally available, and that a form of HEADEDNESS will usually enforce the construction of at least one foot in each prosodic word. Minimal word requirements (word binarity) may not, however, necessarily depend on HEADEDNESS.

(85) Typology of Lexical Accents in Revithiadou 1999: 43–51⁷²

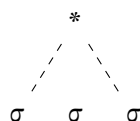
a. Strong Lexical Accent



b. Weak Lexical Accent



c. Floating (Strong) Lexical Accent



The availability of floating autosegments constitutes a clear difference between the foot representation (see immediately below) and autosegmental representation of lexical accent. An underlying foot boundary – even one potentially inserted without any other associated segmental content – can, at most, condition a strictly local effect (without violation of any faithfulness constraints on anchoring; see below); a floating accent can, at least in principle, reassociate at some unbounded distance from its sponsor. Consider, for instance, the case of Ancient Greek derivational suffixes like /-ia-/ that do not attract high tone, but rather condition the default word-prosodic pattern (so-called “recessive accent”), even on bases where another derivational suffix must be presumed to bear a lexical accent that can defeat the “recessive accent”, at least when its sponsor is the outermost or rightmost in a derivation (see Chapter 6 on all of these facts, and 6.2.1.4 and 6.2.3.4 in particular on Ancient Greek prosody in derivation). Relevant data is shown in 86. One potential analysis of this phenomenon could be to assign a floating accent to the suffix /-ia-/, which then associates outside its domain, to the syllable that would by default be preferred by lower-ranking prosodic markedness constraints; see Sandell 2015a: 190–2 for a sketch of precisely such an analysis.

(86) Ancient Greek NML.FEM /-ia-/ induces “recessive accent”

- a. βασιλεια [ba.sí.lei.a] ‘queen’ (cf. βασιλεύς [basileús] ‘king’)
- b. ἀλήθεια [a.lé.t^hei.a] ‘truth’ (cf. ἀληθής [ale:t^hés:s] ‘true’)
- c. ἀσέβεια [a.sé.bei.a] ‘impiety’ (cf. ἀσεβής [asebé:s] ‘impious’)

The autosegmental approach to lexical accent overall enables good empirical coverage, and, at least in the version implemented by Revithiadou, is without serious theoretical problems. One potentially attractive feature of the autosegmental representation of lexical ac-

⁷²Revithiadou provides only for strong floating accents, both for the reason that the cases surveyed by her do not call for the usage, and that, more generally, a floating weak accent might condition unattested effects, such as inducing the appearance of prominence two syllables away from the floating weak accent’s host.

cent is that it enables the cross-linguistic presumption of a unified class of “lexical accent systems”, all of which would share the property of exhibiting some sort of prominence whose surface position is unpredictable. Both restricted tone systems, lacking lexical contrast in tone, but not satisfying one or both of OBLIGATORINESS or CUMULATIVITY, as well as languages with unpredictable stress, can be readily bundled together, being equipped with abstract autosegments in the underlying forms of phonological representations. In the case of languages such as Japanese in which the position of a pitch prominence often seems to have little to do with conceivable foot structure, an autosegment – whether abstract or tonally specified – seems more readily implemented, as opposed to deriving surface contrasts between accented and unaccented words through foot structure.⁷³

Nonetheless, there may exist some patterns of prominence alternation that pose problems for control of accent assignment by the constraints in 84 and *DOMAIN alone. One issue of this type is found in Modern Greek (cf. Revithiadou 2007), where high vowels that must bear a “strong” lexical accent exhibit reassociation of the accent to a vowel to their right when the high vowel undergoes glide formation (see Yates 2020 and 4.3.2 on a fundamentally similar case in Vedic Sanskrit). An example of this type is seen in 87, and applies generally to neuter nouns built with the suffix /-í-/: rightward accent shift is expected in the GEN.SG/PL, since the endings /-ú/ and /-ón/ must have strong accents; the NOM.PL, however, is problematic, because its inflectional ending /-a/, never bears stress outside of this class. Hence, the lexical accent of /-í-/ appears to associate to [-a] on the surface, despite the fact that such a reassociation violates both *DOMAIN and the default prosodic markedness constraints, which would prefer stress on the antepenult (or penult in disyllables). Thus, a NOM.PL ^X[péd̥ja] might have been expected.

(87) NML.N /-í-/ in Modern Greek: /peð-í-/ ‘child’

	SG	PL
NOM	peðí	peðjá
GEN	peðjú	peðjón

This particular case can receive a relatively straightforward account under the assumption of prespecified foot structure as a representational device for lexical accent, to which we now turn.⁷⁴

Lexical Accent as Prespecified Foot Structure The view of exceptional lexical stress as resulting from underlying, prespecified feet has numerous defenders in the literature, e.g.,

⁷³Accounting for such cases through interaction of tone and foot structure might be somewhat more complex, but is nevertheless feasible. Assuming that all prosodic words of Tōkyō Japanese for instance receive an underlying prominence tone /H*L/, and that only stressed syllables may host the starred tone, the distinction between “accented” and “unaccented” words of that language might reduce to the presence versus absence of underlying foot structure (much like Özçelik (2014) posits for Turkish), where, in the latter, /H*L/ is deleted rather than associated due to lack of an appropriate host.

⁷⁴Revithiadou (2007) operates with the Turbidity Theory of Goldrick (2000) combined with the Colored Containment model of van Oostendorp 2007 to account for these data. Under this analysis, NML.N /-i-/ is not associated with a strong lexical accent, but rather introduces a floating accent that associates within the same syllable as its host.

Inkelas 1990, Hayes 1995: 115, Ito et al. 1996, Benua 1997, McCarthy 2000, Özçelik 2014, Yawney 2018, or Yates 2020. Note also that the approach to the treatment of lexical accent in Halle 1997 also derives exceptional stress from prespecified constituent marking on Line 0 of the stress grid (employing the model of Halle and Isardi 1995; see also the discussion of Drescher 2016 at 3.2.2.3.) Under this approach, the phonological component of a lexical entry is assumed to be able to carry information on the relation of certain segments to the edges of (partly) preformed prosodic constituents. As with autosegmental lexical accents, the interaction of faithfulness constraints referring to the prespecified constituent boundaries with prosodic markedness constraints determines winning outputs. Prespecified constituent marking fits more easily with a typically “bottom-up” approach to stress assignment (as in Hayes 1995), where the construction of lower-level constituent structure terminates in choosing prominent elements; the autosegmental approach reviewed above, on the other hand, is more straightforwardly compatible with a “top-down” model (e.g., van der Hulst 1996), where lower-level prosodic constituents are constructed in accord with the position of an already-specified prominence relation.

As a small example of how prespecified foot structure may be analytically implemented, consider again the Cupeño data analyzed under 2.4.2.1. To the data from (76), repeated in (88), I add further examples of an “accented” morpheme combined with “preaccenting” morphemes in (88)f., g., and h.⁷⁵ Given a preference for binary, trochaic feet in Cupeño, morphemes marked in (76) with an acute as “accented” may be marked with a left foot edge to the left of the morpheme’s vowel, while the “preaccenting” morphemes may be marked with a right foot edge to the right of the morpheme’s vowel. The relevant faithfulness constraint, borrowed from Özçelik 2014: 238 and Yates 2020: 258, is assumed to be $\text{ANCHOR}-(L/R)_f$, defined in (89), which assesses the relationship between feet and (potential) syllable peaks in the input and output.⁷⁶

⁷⁵I have inferred 88h. from other examples in Yates 2017.

⁷⁶In the definition originally employed by Özçelik (“the left/right edge of every foot in the input corresponds to the left/right edge of some foot in the output”), $\text{ANCHOR}-L/R_f$ behaves more like a MAX constraint. Yates’ redefinition of the ANCHOR constraints on feet seems better to fit the spirit of the ANCHOR constraints introduced in McCarthy and Prince 1995. I nevertheless harbor some doubts as to whether ANCHOR is really the appropriate constraint to invoke at all. Consider the respective definitions of McCarthy and Prince 1995: 123 and Kager 1999: 213 (the latter referring explicitly to Base-Reduplicant relations):

- “{RIGHT, LEFT}-ANCHOR(S_1, S_2): Any element at the designated periphery of S_1 has a correspondent at the designated periphery of S_2 .”
- “ANCHORING-BR: Correspondence preserves alignment in the following sense: the left (right) peripheral element of R[eduplicant] corresponds to the left (right) peripheral element of B[ase], if R is to the left (right) of B.”

Perhaps the technically correct way to understand the use of ANCHOR made by Özçelik, Yawney, and Yates is that a given (potential) syllable peak V_1 found at the left/right periphery of a foot f_1 in the input must be found the left/right edge of a foot f_1 in the output. Thus, if an underlyingly prespecified foot edge is either absent in the output or shifted so that some other syllable peak V_2 intervenes between V_1 and the left/right edge of f_1 , $\text{ANCHOR}-L/R_f$ is violated. See Yates 2020: 258–9 for further discussion on assessing violations of this constraint and consideration of the question of why consonants seem to be irrelevant for foot anchoring.

- (88) Cupeño Lexical Accent as Underlying Foot Structure (cf. Yates 2017: Ch. 3; examples pp. 64–66)
- a. /pəm - təw - wəni/ → [(‘pəm.tə).wə.ni] ‘we see’ (1PL-see-DS.PL)
 - b. /ʔə - yax - qa(li/ → [ʔə.ya.qa.(‘liʔ)] ‘you say’ (2SG-say-DS.SG)
 - c. /nə - təwi - ŋa) - ʔaw/ → [nə.tə.(‘wi.ŋa).ʔaw] ‘on my chest’ (1SG-chest-INL-LOC)
 - d. /nə - (mix - (a:n - βə/ → [nə.(‘mi.xan).βə] ‘(where) I did (something)’ (3SG-be-AAN-SUB.RL)
 - e. /taβ - (a:n - pə - (qal/ → [ta.(‘βa:n.pə).qal] he placed (it) in’ (put.in-AAN-3SG-PST.IPFV.SG)
 - f. /(səwə-t-ŋa) - ʔaw/ → [(‘sə.wət).ŋa.ʔaw] ‘on the rattlesnake’ (rattlesnake-ABSL-INL-LOC)
 - g. /pə-wən-(i-βə-ŋa)/ → [pə.wə.(‘ni.βə).ŋa] ‘in which it lay’ (3SG-lie-NML-SUB.RL-INL)
 - h. /pə-(ʔəw-ŋa)/ [pə.(‘ʔəw.ŋa)] ‘in his blood’ (3SG-blood-LOC)
- (89) ANCHOR-L/R_f-IO: The left/right edge of every foot *i* in the input corresponds to the left/right edge of a foot *i* in the output. Assign one violation (*) if any syllable peak intervenes between *f*_i in the input and *f*_i in the output.

In practical terms, left foot edges and root foot edges here are equivalent to strong or weak lexical accents in Revithiadou’s (1999) treatment of Modern Greek stress, which likewise is assumed to prefer syllabic trochees. Since primary stress in Cupeño always occurs on the leftmost possible syllable without incurring any violations of prosodic faithfulness, the constraint MAX-PROM from (78) may simply be substituted by ANCHOR-(L/R)_f, and the correct results will emerge. Note, in a case like, 88h., an underlying left and an underlying right foot edge agree on the position and structure of a foot in the output, and thus no faithfulness violations are incurred. In principle, ANCHOR-L_f and ANCHOR-R_f may be ranked differently, as in Özçelik’s (2014) analysis of Turkish stress. While not seen in the Cupeño data, a morpheme containing multiple (underlying) syllable peaks might be specified with a foot built around both peaks, e.g., /(CVCV)/ would constitute a prespecified disyllabic foot (compare the stress-attracting suffix [-‘indʒe] ‘when’ of Turkish, analyzed by Özçelik 2014: 238 as /-(indʒe)/).

The use of prespecified foot edges also allows for a flexible treatment of many different “flavors” of lexical accent. As an example, consider a language with stress, which regularly builds trochaic feet. Given a morpheme with a single syllable peak, the underlying foot edges will generate the effects in terms of stress placement laid out in Table 2.2.

Invoking multiple foot edges in an input can thus be used to create genuinely unaccentable vocalic peaks because of the high cost to ANCHOR-L/R_f would be incurred by placing that vocalic peak in the strong part of a foot: a representation /σ/(/ in a trochaic system would effectively mark a syllable peak as not only as a foot tail, but also the following peak as a foot head, so that in a string /σ-σ/(-σ)/, the output [(‘σσ).(‘σ)] would minimize faithfulness violations. Furthermore, multiple foot edges in an input can simulate an effect of strength of

Underlying Foot Structure	Stress Position Relative to σ
σ	Unaccented
$(\sigma \text{ or } (\sigma)$	Accented
$\sigma)$	Pre-accenting
$\sigma($	Post-accenting
$)\sigma$	Pre-pre-accenting
$\sigma)(\text{ or })\sigma($	Unaccentable

Table 2.2: Stress behaviors generated by underlying lexical foot edges in a trochaic system

a lexical accent, since placing stress in a way that creates an entirely different foot structure would become more costly in terms of faithfulness violations. For instance, given an underlying string $/(\sigma-\sigma-(\sigma)/$, containing two species of “accented” morpheme, the output $[\sigma\sigma(')\sigma]$ better incurs fewer violations of ANCHOR-L/R_F than an alternative $[(')\sigma\sigma)\sigma]$.

Three possible objections against the employ of lexically prespecified foot structure are worth bearing in mind. First is the general concern that prosodic structure does not belong in underlying representations, instead being predictable from markedness constraints alone; however, this usual premise applies just in case the prosodic structures concerned (syllables, feet) are indeed non-contrastive, i.e., predictable based on segmental conditions. Second is the fact that the use of underlying foot edges essentially presupposes a particular foot shape: a left foot bracket has a different effect on the output, depending on whether the foot being constructed is a trochee or an iamb. Hence, if a learner is faced with a percept like $[\text{miti}'\text{ko}]$, and knows the she needs a lexical representation of stress (e.g., because it builds a minimal pair with $[\text{'mitiko}]$), she must also have already decided or be able to simultaneously decide whether an iambic parse violating TROCHEE ($[\text{mi}.\text{(ti.'ko)}]$) or non-iambic parse violating FTBIN ($[\text{mi.ti.('ko)}]$) is preferable. In principle, some existing approaches to learning URs and the ranking of prosodic markedness constraints should be able to handle this problem (see 3.2.2.2).

Finally, Kabak and Revithiadou (2009a: 26–8) make the specific empirical objection that underlying trochaic feet in lexical entries of Modern Greek would make incorrect predictions for words with consistently antepenultimate stress, such as the stem *furnarið-* ‘baker’, with forms shown in 90. Because primary stress never shifts onto a the syllable of pre-accenting inflectional endings like M.GEN.SG $/\text{-on}/$, the stem must be assumed to possess its own stress-determining properties. Kabak and Revithiadou correctly observe that, were the representation of the stem $/(\text{funa})\text{rið-}/$, when combined with pre-accenting $/(*\text{-on})/$ (see the representations in 91), since primary stress would be assigned to the rightmost foot, stress should fall on the penult, thus GEN.PL $^x\text{furnariðon}$. However, this objection depends on assuming the the representations contain fully specified feet. If instead only right foot edges are specified, given that NONFINALITY-F dominates FTBIN and PARSE- σ (compare the rankings of prosodic markedness constraints for Modern Greek in Revithiadou 1999: 116-8; 209), a representation of ‘baker’ as $/\text{fur})\text{narið-}/$ will be able to generate expected genitive plural $[\text{fur'nariðon}]$ in a way that is with other patterns of stress in Modern Greek nouns.⁷⁷

⁷⁷The other major patterns (cf. Revithiadou 1999: 193 have either: 1) mobile stress conditioned by prosod-

In essence, antepenultimate stress can be generated in this case because it avoids stress on the final syllable and better satisfies other lower-ranking markedness constraints.

(90) Partial Paradigm of Greek *furnarið*- ‘baker’

	SG	PL
NOM	fúnaris	furnárides
GEN	furnáriðu	furnáriðon

(91) Foot Representations Assumed by Kabak and Revithiadou 2009a: 26–8

a. ‘baker’

(* .)

f u r n a r i ð-

b. M.GEN.PL

(* .)

o n

(92) Derivation of GEN.PL /fur)narið-on)/ → [fur.(‘na.rið).on]
(cf. NOM.SG /fur)narið-s/ → [(‘fur).na.ris])

/fur) _i narið) _j -on) _k /		*EXTENDED LARSE-R	NONFIN-σ	ANCHOR-R _f	NONFIN-f	FtBIN	ALL-Ft-R	PARSE-σ
a.	(_i fur.na) _{i,j} .(‘ri.ðon) _k			**	*!		**	
b.	fur.(‘na) _{i,j,k} .ri.ðon			***!		*	**	***
c.	(‘fur) _{i,j,k} .na.ri.ðon	*!		**		*	**	**
d. 	fur.(‘na.ri) _{i,j,k} .ðon			**			*	**
e.	fur.(_i na.ri) _{i,j} .(‘ðon) _k		*!	*		*	*	*

One final issue of relevance concerns cases of possible misalignment between foot structure and prosodic prominence, as described by Vaysman (2009) for e.g., Ngansan. If the entire approach advocated by Vaysman is correct, it may be more sensible, as a first pass in each case, to approach lexical accent as grid prominence. In case a language aligns prominence and foot construction, it may prefer to create lexical representations enriched with feet, rather than autosegmental prominence marks; in case prominence has nothing to do with feet, or indeed is not aligned with feet, the lexical representations might instead be assumed to contain autosegments. An empirical question is whether, in fact, any languages

ically unmarked stems interacting with accented or preaccenting endings (e.g., /anθrop-/ ‘man’; 2) fixed stem stress, due to both both a left and right foot edge (e.g., /sta(fið)-/ ‘raisin’; 3) or an unstressable stem, due to a left foot edge at the right edge of the stem (e.g., /uran(-/ ‘sky’). Stress on final syllables, otherwise ruled out by high-ranking NONFIN-σ in g2, can be generated by ranking the anchoring of left foot edges (ANCHOR-L_f) above NONFIN-f. See the file `Mod_Gk_Stress.txt` in the accompanying GitHub repository for this chapter.

with lexical accent systems that clearly do not align prominence with foot structure actually exist: Gordon (2016) suggests that many cases of seeming incoherence between foot structure and prominence may involve pitch accents pertaining to prosodic constituents larger than the ω , not a primary stress that is “out of phase” with word’s the foot parse.

Overall, both the autosegmental and foot representation of lexical accent prove to be sufficiently powerful and flexible to be able to successfully describe most usual patterns of (simple) lexical accent to have thus far received any extensive treatment in the literature. Whether any genuine empirical test cases exist that cannot be covered by one representation or the other whatsoever is, at present, unclear. Cases like the Modern Greek type illustrated in (87) or so-called “secondary mobility” in Vedic Sanskrit (see 4.3.2) seem to constitute serious issues for the autosegmental representation, which are nevertheless generable through adopting (considerable) additional machinery. Similarly, cases like the Modern Greek type seen under (90) pose some complications for the foot representation.

Lexical Accent as Lexically Specific Constraint Ranking Although most theoretical approaches to lexical accentuation or analyses of languages exhibiting extensive unpredictable stress presume some type of prominence specification in the lexicon, whether through autosegments or underlying constituent structure, an entirely different approach is in principle possible. Specifically, the phonological representations of lexical items might be associated with lexically specified “clones” of particular markedness constraints. While lexically specific versions of (markedness) constraints have been explored generally in the literature (see especially Pater 2010, or Zukoff 2017b for an application to reduplication patterns), their application to the domain of stress assignment has been comparatively limited: Kraska-Szlenk (1995) (also Kraska-Szlenk 2003) proposes an analysis of exceptional stress in Polish along such lines, Pater (2000: 258–66) handles some patterns of secondary stress in English through a combination of lexically specified faithfulness and markedness constraints, and Moore-Cantwell and Pater (2016: 54–9) have briefly explored the learnability of lexical stress patterns using lexically specified rankings; some further attempts to deal with exceptional stress in terms of morpheme-specific constraints may be found in Hammond 1995. As long as the lexically specified constraints are indeed limited to markedness constraints, the problem of representation for lexical stress is largely resolved: URs need not carry prosodic representations relating to stress, but instead be indexed to (some) particular constraint(s).

Whether operating with lexically specific constraints beyond otherwise intractable cases – or at all – is open to question. Özçelik (2014: 240–2), for instance, argues that lexically specified constraints (or cophonologies) are in principle undesirable: by stipulating “that certain morphemes violate or do not violate a given constraint” we are prevented “from being able to benefit from constraint violability or from the universal nature of OT constraints.” At least from the perspective of learnability, creating lexically specific constraints is no great obstacle; constraints may be “cloned” under conditions of inconsistency employed by Tesar (2004, 2014) for helping to set features in underlying representations. For the moment, it suffices illustrate their behavior with a toy example based on Polish, which exhibits regular quantity-insensitive penultimate stress, but (rare) exceptional antepenultimate stress. To generate regular penultimate stress, a disyllabic trochaic foot may be constructed at the right edge

of the word, in violation of RIGHTMOST and NONFINALITY. Given a lexical item associated with the constraint diacritic j , the higher-ranked version of NONFINALITY, NONFINALITY _{j} can then be activated, and will result in the production of candidates with antepenultimate stress, in violation of ALL-FT-R.

- (93) Exceptional Stress in Polish through Lexically Indexed NONFINALITY _{j}
 a. Regular Penultimate Stress $/\sigma \sigma \sigma \sigma / \rightarrow [\sigma \sigma \sigma (' \sigma \sigma)]$

$/\sigma \sigma \sigma \sigma /$	NONFINALITY _{j}	ALL-FT-R	TROCHEE	RIGHTMOST	NONFINALITY	LAMB
a.  $\sigma \sigma \sigma (' \sigma \sigma)$				*	*	*
b. $\sigma \sigma \sigma (\sigma ' \sigma)$			*!		*	
c. $\sigma \sigma (' \sigma \sigma) \sigma$		*!		**		*

- b. Exceptional Antepenultimate Stress: $/\sigma \sigma \sigma \sigma /_j \rightarrow [\sigma \sigma (' \sigma \sigma) \sigma]$

$\sigma \sigma \sigma \sigma$	NONFINALITY _{j}	ALL-FT-R	TROCHEE	RIGHTMOST	NONFINALITY	LAMB
a. $\sigma \sigma \sigma (' \sigma \sigma)$	*!			*	*	*
b. $\sigma \sigma \sigma (\sigma ' \sigma)$	*!		*		*	
c.  $\sigma \sigma (' \sigma \sigma) \sigma$		*		**		*
d. $\sigma \sigma (\sigma ' \sigma) \sigma$		*	*!	*		

An approach to lexical stress in these terms appears attractive precisely in cases of the Polish sort, in which the analytical and statistical choice between a “regular” and an “exceptional” pattern is clear, and the total number of possible patterns is limited, and both patterns are unproblematically generable with the standard repertoire of metrical stress constraints. In terms of formal representation, the choice between an analysis employing a prosodic diacritic in the UR and a faithfulness constraint, on the one hand, or an abstract lexical class diacritic and a duplicate markedness constraint, on the other, is effectively equivalent in such cases. In cases in which exceptionality is rampant, and a regular pattern does not readily emerge (even if a thorough analysis may allow for the choice of some default stress), such as in Russian or Vedic Sanskrit, the number of lexically specific constraints needed might multiply considerably. Furthermore, as ensuing discussion will make clear, reference to underlying lexical properties or lexical constraints alone is insufficient in some prosodic systems with unpredictable prominence.

2.4.2.3 Head Dependency and Head Attraction

A number of languages evidently possessing lexical accent in some form exhibit patterns of prominence alternation that cannot be reduced to the selection of a leftmost or rightmost underlying prominence, as illustrated in 2.4.2.1, or to a lexical accent associated with any other phonological property (e.g., syllable weight). Instead, stress assignment across derivationally related forms appears to depend on the construction of morphological structure. An example of this sort involving the Russian adjective-forming suffix /-ast-/ is shown in 94; note further that the feminine inflectional suffix /-a/ is normally stress-attracting (cf. *molo-dá* ‘young:NOM.SG.F’). Compare also Melvold 1989: 70–4 on the Russian derivational suffix /-ač-/ , which behaves similarly.

(94) Accentual Dominance in Russian: /-ast-/ (cf. Revithiadou 1999: 221)

- a. borod-ást-a ‘heavily-bearded.NOM.SG.F’
(cf. *borod-á* ‘beard.F.NOM.SG’, NOM.PL *bórod-y*)
- b. gorl-ást-a ‘loud-mouthed.NOM.SG.F’
(cf. *górl-o* ‘throat.N.NOM/ACC/SG’, DAT.SG *górl-u*)
- c. jazyk-ást-a ‘sharp-tongued.NOM.SG.F’
(cf. *jazyk-ú* ‘tongue-DAT.SG’, NOM.PL *jazyk-í*)

What is remarkable about the behavior of *-ást-* is the fact that it evidently attracts stress regardless of the type of base to which it attaches. In this respect, *-ást-* is decidedly different from the inflectional suffixes of the Russian noun (see overviews on their stress patterns, with analysis, in Melvold 1989: 12–27 or Alderete 2001a: 82–90), which all exhibit both stressed and unstressed allomorphs; whether a given nominal inflectional suffix receives primary stress or not depends on both its own prosodic properties and those of the stem. In 94, *-ást-* attaches to nominal stems that can be analyzed as unaccented (/borod-/ ‘beard’ with mobile stress in its paradigm), accented (/gorl- ‘throat’, with stress always on the stem), and unaccentable or post-accenting (/jazyk(-) ‘tongue’, with stress always on the inflectional suffix), but primary stress always emerges on the suffix *-ást-*, never its base, nor on any inflectional suffixes following it. The behavior of *-ást-* evidently cannot be reduced to “leftmost wins”, as in Cupeño (otherwise, one might find ^x*górlasta*), nor “rightmost wins”, as in Italian (otherwise, one might find ^x*borodastá*).

This type of behavior is often referred to “accentual dominance”, and is well documented for Russian, Lithuanian, Modern Greek, and Vedic Sanskrit (cf. 4.2.4 in Chapter 4). An affix like Russian *-ást-* ignores the prosodic preferences of any other material and imposes its own preference (i.e., for primary stress to fall on *-ást-*) on the entire phonological string. Among one major approach to this phenomenon, morphemes of the *ást-* type are treated as *cyclic*: certain morphemes (“dominant” morphemes) may “induce a new prosodic (*lexical accent*) cycle and a grid mark in that cycle” (Bogomolets 2020: 17), thus resulting in primary stress assigned to the outermost derivational layer within the domain of cyclicity. Determining whether or not a morpheme induces a cycle and the application of stress rules is the approach adopted by many, e.g., Chung 1983, Halle and Vergnaud 1987, Melvold 1989, Czaykowska-Higgins 1993, Schwayder 2015, Kiparsky 2018a.

Under one interpretation (cf. Halle and Vergnaud 1987: 86–7), the stresses or grid marks lexically carried by morphemes inside the cycle-inducing morpheme, or assigned on earlier cycles, are not carried over to the new cycle; hence, the position of stress in the new cycle created by the dominant morpheme is either determined by the preference of that morpheme, or by the default stress rule. An alternative interpretation is that “dominant” morphemes induce *erasure* of all other pre-existing stress marks; Kiparsky (2014) and Kiparsky (2018a) also seems to adopt this “accent-erasure rule” interpretation of dominance. Melvold (1989: 50–60), for instance, explicitly divorces dominance and cyclicity, arguing that Russian possesses derivational suffixes such as the abstract-noun forming suffix *-ost-* (e.g., *mólodost-* ‘youth’) are cyclic, but do not trigger the failure to carry over lexical accents from an inner cycle.⁷⁸ Russian *-ást-* would then count as a dominant morpheme that attracts stress; the Ancient Greek examples seen in (86) ($\beta\alpha\sigma\acute{\iota}\lambda\epsilon\iota\alpha$ [basíleia] built with /-ia-/ to $\beta\alpha\sigma\lambda\epsilon\acute{\upsilon}\varsigma$ [basileús]) above could be explained as containing a dominant unaccented derivational suffix that induces default word prominence. Among analyses of dominance effects as due to cyclicity, whether a given morpheme induces a cycle or not is furthermore usually taken to be a lexical property, i.e., a diacritic or feature like [+/-dominant] (Bogomolets 2020: 266–7 is explicit on this point). In a Russian-like language, a four-way typology then emerges between dominant and non-dominant (or “recessive”) and marked (i.e., accented, pre-accenting, etc.) and unmarked morphemes, as shown for Russian based on analyses in Melvold 1989 in 95.

(95) Typology of Stress Dominance and Lexical Markedness (exemplified by Russian)

	DOMINANT	RECESSIVE
MARKED	ADJ <i>-áč-</i>	NOM.SG.F <i>-á</i>
UNMARKED	(NML.M /-En’-/) ^a	NML.F <i>-ost-</i>

^aMelvold’s notation /E/ indicates a *jer*-vowel. Melvold (1989: 75) indicates that this derivational suffix is the only example of an unaccented but stress-erasing affix in the morphology of Russian that she could find, and that it is also unproductive, with few types to be found in the language. Since all of the nouns seemingly containing this suffix exhibit initial stress, they might all simply constitute cases of default stress assigned to “demorphologized” forms, in the sense of Probert 2006b (i.e., loss of morphological structure results in failure of special lexical prosody determined by affixes to apply). Given these observations, one might wonder whether actively stress-erasing derivational suffixes exist in Russian specifically, or indeed any language whatsoever. The Ancient Greek $\beta\alpha\sigma\acute{\iota}\lambda\epsilon\iota\alpha$ -type, for instance, will be analyzed in Chapter 6 as indeed constituting an application of default word-level prosody, but in such a way that requires no erasure of the underlying lexical prosody of its bases.

Alternatives to diacritic dominance have been developed in Alderete 2001a and Revithiadou 1999. A core idea pursued in Alderete 2001a was that lexical accent systems in which prominence appears to be drawn away from the root in complex stems are subject to the

⁷⁸Melvold’s argument runs as follows: since Russian possesses a cyclic rule that assigns stress to the leftmost syllable if no lexical accents are present, but which cannot apply on the innermost cycle (i.e., the root), the fact that unaccented roots like /molod-/ ‘young’ have build nouns in *-ost-* with non-mobile stress on the leftmost syllable shows that /-ost-/ triggers stress assignment to the initial syllable on the constituent [[mólod] -ost-]. As a result, when accented inflectional suffixes are attached, e.g., DAT.PL /mólodost-ám/, the leftmost accent wins, giving *mólodostam*. If *-ost-* were dominant in Halle & Vergnaud’s sense, then lexical accents of the base, as in *gorbátost-* ‘condition of being humpback’ should disappear, yielding a stem ^X*górbatost-*.

effects of ANTIFAITHFULNESS constraints. Under Alderete's analysis of stress in Cupeño, for instance stress in that language tends to occur towards the left edge due to special faithfulness to the lexical accents borne by roots (contrast 76 above). In contrast, where effects of accentual "dominance", preaccentuation, or postaccentuation are observable (e.g. in Russian or Japanese, treated in Alderete 2001a: 193–264), the shift of stress on the surface away from the position preferred by a lexical accent is attributed to high-ranking antifaithfulness constraints. However, given that cases of supposed "root-controlled" accent in which both roots and other affixes may be lexically marked are probably best reanalyzed as simple lexical accent systems (cf. Bogomolets 2020: 22), a major motivation for the use of antifaithfulness in the analysis of lexical accent systems is obviated. Furthermore, because antifaithfulness constraints have been observed to pose considerable theoretical difficulties in themselves (cf. Trommer 2006), to avoid their usage is desirable.

Revithiadou (1999) adopts just the opposite tactic, attributing dominance effects in Russian and Modern Greek to special faithfulness constraints that hold over the HEADS of words containing morphologically complex stems. In practice, identifying the head of a morphologically complex stem is, in non-polysynthetic languages with extensive inflectional morphology – typical in Indo-European languages – fairly straightforward: the morphological head of the word is the morpheme that is responsible for determining the basic morphosyntactic properties and behavior of the entire word, that is, its part of speech and subcategorization frame, what inflectional endings it takes and in what sort of agreement phenomena it participates. Revithiadou (1999: 20) similarly observes: "Derivational suffixes are almost always heads because they define the lexical category, class or gender of the derived form, e.g. *angel-os* (noun) 'angel' > *angel-ik-os* (adjective) 'of angels.'" See further discussion on this point in Sandell 2015a: 176–81 and Yates 2017: 25–9, which arrives to the same major point: the outermost derivational morpheme of a morphologically complex stem is usually to be identified as the head of the word; inflectional morphemes, in contrast, never constitute heads.

In accord with her usage of an autosegmental representation of lexical accents, Revithiadou (1999: 28) formulates the relevant HEADFAITH (LexicalAccent) constraint as in 96, though a reformulation in terms of ANCHOR constraints to refer to foot edges contained in the lexical entries of morphological heads would easily be possible. In Revithiadou's conception, the prosodic systems of Russian and Modern Greek are generally characterized by the abstract constraint ranking shown in 97, which holds that faithfulness to lexical accents sponsored by morphological heads is privileged over other lexical accents, which in turn generally takes precedence over the prosodic markedness constraints that generate the default stress. Yates (2017: Ch. 4) has mustered compelling data and arguments in favor of counting Hittite among such *head-dependent lexical accent systems*, and Patseva (2017) has done the same for Bulgarian.

(96) HEADFAITH (LA)

A lexical accent sponsored by a morphological head in the input has a correspondent in the output (MAX-PROM(Head)).

A lexical accent hosted by a morphological head in the output has a correspondent in the input (DEP-PROM(Head)).

(97) *Head-dependent system with lexical accents*: HEADFAITH (LA) $\gg$ FAITH (LA) $\gg$ DEFAULT STRUCTURE $\gg$ HEADSTRESS (LA) = Modern Greek, Russian, Hittite

The basic behavior of dominance effects implemented through HEADFAITH is illustrated for Russian *gorlásta* ‘loud-mouthed.NOM.SG.F’ (see 94b. above) in the tableau below. Under the assumption that prosodic markedness constraints prefer stress on the initial syllable (so Kiparsky and Halle 1977: 219–27, Melvold 1989: 16–8), stress on the the head, *-ást-*, is motivated purely by its status as the morphological head of the word, since the prosodic markedness constraints would favor candidate b., with initial stress, and all candidates tie on the simple faithfulness constraint, MAX-PROM.

(98) /górl-ást-á/ $\rightarrow$ [gor.(‘las.ta)] ‘loud-mouthed.NOM.SG.F’ (cf. Revithiadou 1999: 221)

/górl-ást-á/		MAX-PROM(Head)	MAX-PROM	ALL-Ft-L	FtBIN
a.	☞ gor.(‘las.ta)		**	*	
b.	(‘gor.las).ta	*!	**		
c.	gor.las.(‘ta)	*!	**	**	*

In systems with head-dependent stress, the invocation of HEADFAITH, assuming a basic ranking like the one in 98 (MAX-PROM(Head) $\gg$ MAX-PROM $\gg$ PROSODIC MARKEDNESS) thus predicts that morphological heads should be compatible the following effects:⁷⁹

- imposition of stress in the position preferred by some lexical accent sponsored by the head.
- stress in a position that minimizes violations of faithfulness to lexical accents sponsored by non-heads, and, in case of multiple lexical accents, of prosodic markedness constraints (= simple lexical accent)
- stress in a position that minimizes violations of prosodic markedness constraints (= default stress).

⁷⁹In practical terms, the ranking CULMINATIVITY $\gg$ MAX-PROM $\gg$ MAX-PROM(Head) $\gg$ PROSODIC MARKEDNESS would always produce identical outcomes, since under either ranking, an input with multiple lexical accents would tie on MAX-PROM, whose conflict would be resolved by MAX-PROM(Head). Since MAX-PROM(Head) would then always be winner-preferring, standard constraint ranking algorithms (see, for instance, 3.2.1.1) will discover the MAX-PROM(Head) dominates MAX-PROM.

Effect a. should occur when the phonological realization of the head contains any sort of lexical accent; effect b. obtains when the head has no lexical accent, but other lexical accents are present in the input string; effect c. obtains when neither head nor any other morphemes sponsor lexical accents. In this regard, HEADFAITH makes a prediction that differs crucially from the use a diacritic feature [+/-dominance]: morphological heads should not normally condition any genuine “erasure” of lexical accents on their bases, imposing default word prosody. Given that the head faithfulness hypothesis allows for a clear link between independently justifiable morphological structure and prosodic behavior, it may be theoretically preferred to a diacritic feature [+/-dominance]. The crucial empirical question then is: do unmarked (i.e., unaccented) [+dominant] derivational suffixes really exist at all? Although I have not conducted a comprehensive survey of systems exhibiting dominance effects, at least Melvold’s Russian data, as well as the data from Vedic Sanskrit (Chapter 4) and Ancient Greek (Chapter 6), appears open to analyses that do not require unaccented [+dominant] derivational suffixes, which would fit well with seeing dominance effects as resulting from special faithfulness.

The relevance of morphological headedness has also been argued by Revithiadou to go beyond effects of prosodic faithfulness, and instead instantiate itself via the markedness constraint stated in 99 (cf. (Revithiadou 1999: 28)). This constraint penalizes the absence of (primary) stress on a syllable peak coextensive with segmental material corresponding to the head of a morphologically complex word.

- (99) HEADSTRESS
Morphological Heads are Stressed

While evidence for the activity of HEADSTRESS seems relatively secure, at least for languages with polysynthetic morphology (mainly Salishan and Yupik languages, e.g., Norton Sound Yupik and Thompson Salish), cases of interaction between HEADSTRESS and lexical accents are not so readily forthcoming. Revithiadou (1999: 22–3) mentions exactly two possible cases: Tahltan (Athabaskan, British Columbia) and Hua (East New Guinea, Gorokan). For neither of these cases is the data really sufficient to determine with certainty what interaction of lexical properties and HEADSTRESS might yield the reported prosodic patterns.

Consider the data from Tahltan in 100. According to Revithiadou’s analysis, each root either receives a high tone on its initial syllable or is marked for high tone. Whether the difference in position of high tone between, e.g., [lédih] ‘sweet’ and [keyéh] ‘town’ is entirely lexical (i.e., both H-toned syllables are marked lexically as such) or whether one is a default is unclear, however. Besides the further rhythmically determined high tones that occur on alternating syllables to the right of a root’s high tone (in 100e.-g.) – which supposedly show that H is not culminative in the language – 100g. shows a high tone on a deitic prefix, which in Revithiadou’s view constitutes a head. Whether this additional high tone constitutes a lexical tone/accent or an accent that is assigned due to HEADSTRESS is, however, not clear. Furthermore, Nater (1989: 30–1) disputes the view of Cook (1972) (taken over by Revithiadou), that high tone is not culminative; his representations of the data suggest that the leftmost lexical H is the only H in the word, and, in accord with Cook 1972: 233, default H falls on the leftmost syllable of a polysyllabic word.

- (100) Tahltan High Tone (after Revithiadou 1999:23; URs mine; forms converted to IPA based on information in Nater 1989: 25–8; supposed morphological head in g. underlined)
- a. /ʔedih/ → [ʔédih] ‘sweet’
 - b. /keyéh/ → [keyéh] ‘town’
 - c. /na:tθ’et/ → [ná:tθ’et] ‘it has fallen off’
 - d. /ta:q’áʔ/ → [ta:q’áʔ] ‘needle’
 - e. /hode-se:h/ → [hódesé:h] ‘I talk’
 - f. /hodéθi:-de:h/ → [hodéθi:dé:h] ‘we talk’
 - g. /kiθ-hede:s-de:l/ → [kíθhéde:sdé:l] ‘they three or more run’

Finally, the most recent descriptions in Alderete and Bob 2005 and Alderete 2014 claim that every stem (root plus non-inflectional suffixes) has an obligatory stress, which will surface as primary stress. Prefixes behave like proclitics: they are obligatorily footed and stressed, and have a secondary stress; the prosody of prefixed words is like the prosody of compounds in the language. I conclude that Tahltan likely does not head-stress effects. Since the interpretation of word-level prosody in Hua seems similarly unclear and disputed (cf. van Zanten and Dol 2011: 131), I tentatively assume, *pace* Revithiadou, that head-stress and lexical accent may not, in fact, be permitted to interact in a single prosodic system.

2.5 Computational Perspectives on Stress Assignment

Just as stress assignment occupies an enormous role in the phonological literature generally, a number of investigations into the computational nature of stress assignment processes have been conducted. The questions with which this literature has been concerned are somewhat different from, and in part orthogonal to, the issues in the phonological analysis of stress assignment that have been treated up to this point. Much the research on the computational nature of stress assignment processes has focused on determining their complexity within the Chomsky hierarchy (in particular, establishing that they are *regular*, as has been argued to be the case for phonological processes generally; see further Kaplan and Kay 1994, Graf 2010) or what types of functions particular stress patterns instantiate (e.g., sequential vs. non-sequential).⁸⁰ Much of this work is also somewhat orthogonal to broader investigations into the *learnability* of stress systems, though Heinz 2009 constitutes an important exception; work on learnability in parametric and especially constraint-based

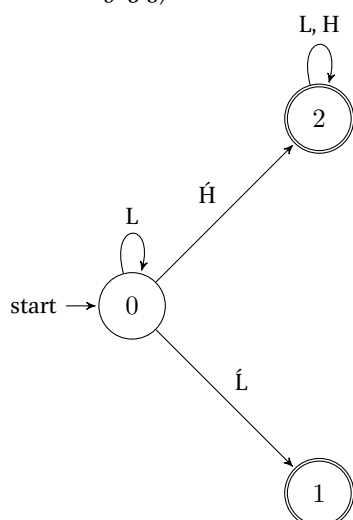
⁸⁰In addition, potentially important insights into the classes of constraints that are necessary or desirable in constraint-based grammars have emerged from this work. In particular, several have touched upon the issue that admitting gradient constraints, or perhaps even allowing categorical constraints to be violated an unbounded number of times, results in a model with greater formal power than that needed to express phonological processes (being of (sub)-regular complexity). Constraints of the ALIGN family (cf. 2.3.3 above) have, in particular, been singled out as problematic (cf. Heinz 2009: 324). See Eisner 1997, Karttunen 1998, McCarthy 2003, or Isardi 2009 for discussion along these lines. I acknowledge these issues, but will continue throughout to operate with ALIGN constraints where necessary on account of their familiarity and the absence of clear alternatives in some cases.

frameworks will be the focus of the following chapter. As a transition into a larger treatment of the learning and learnability of stress, I briefly situate some major claims concerning the computational nature of stress assignment processes in the larger context of the present work.

Worth mentioning is that, to my knowledge, very little work on the computational complexity of stress assignment, the categorization of stress rules in terms of e.g., the Chomsky Hierarchy, or the learnability of stress systems has concerned itself with lexical stress patterns. To some extent, this research gap is self-explanatory: certain types of lexical stress patterns (e.g., the Cupeño type; see Yates 2017: Ch. 2, and, in agreement with Yates, Bogomolets 2020: 140–5) are not fundamentally different from unbounded weight-sensitive systems. Cupeño, then, can be characterized as a default-to-same system, where lexical marking (“accented morpheme”) plays the role that a heavy syllable does in a stress system like that of Selkup (cf. 73 above). Where more different types of lexical marking (e.g., accented vs. preaccenting or post-accenting) and considerations of morphological structure (head faithfulness or head stress) also play a role, little in the way of computational investigation has been undertaken. Intuitively, more different types of entities to which stress is sensitive increases the computational complexity of a grammar, perhaps to the extent that certain hypotheses about the computational nature of stress rules might require revision, at minimum. See, for instance, the problems created by the Hindi stress system of Kelkar 1968, with its supposed unusual ternary weight contrast ($V < \{VV, VC\} < VCC$), for the characterization of all stress systems as 1-1 neighborhood distinct in Heinz 2009.⁸¹

⁸¹The characterization of a rule system as j - k neighborhood distinct refers to the neighborhoods (i.e., incoming and outgoing paths, whether the state is a start or end state) in a finite-state acceptor (FSA). A finite-state acceptor is then 1-1 neighborhood distinct if no state in the acceptor has identical incoming and outgoing environments. For example, even unbounded stress patterns, like the one represented in the FSA shown below, satisfy the property of being 1-1 neighborhood distinct.

- (1) A FSA Diagram of an Unbounded Stress Pattern (“primary stress on leftmost heavy, else rightmost”; cf. Heinz 2009: 315)



Bjorkman and Dunbar 2016 constitutes an exception to this observation, in which two lexical accent systems, one from Chamorro (Chung 1983) and one from Nez Perce (Crook 1999), are considered in light of the computational generalization that all stress patterns of any nature, like phonological processes more generally, should be finite-state. The core claim of Bjorkman and Dunbar (2016: 351) is that “while phonology is sensitive to some aspects of morphological structure, true constituency is not used, as would be predicted by the phonological cycle: the structure seen by phonology is *nonisomorphic* to the full morphological structure in the sense of Scheer 2010.” This claim accords with the argument advanced much earlier by Melvold (1989) concerning stress in Russian, namely, that cyclicity does not adequately capture dominance effects (see above).

In the Chamorro case, default stress is penultimate, but a lexically accented prefix that is attached to the root alone or a stem with root and suffix(es), such that the prefix morphologically outscopes all suffixes, will receive stress; if, however, a prefix is outscoped by any suffix, anywhere in the morphological structure, stress reverts to the default penultimate stress. In a subpattern of verbs in Nez Perce, by contrast, the attachment of any lexically accented suffix will result in attraction of primary stress to this suffix, regardless of morphosyntactic scope relations, even though, when verbs of the relevant subpattern have only lexically accented prefixes, the *leftmost* lexical accent is realized as primary stress. In effect, the attachment of a suffix seems to trigger reversion to the dominant overall pattern of verbal stress, which is to assign primary stress to the *rightmost* lexical accent.⁸² Examples (101) and (102) contrast the scope-sensitive behavior of Chamorro with the scope-insensitive behavior of Nez Perce.

- (101) Chamorro: Stress attracted to lexically accented prefix only if it outscopes all suffixes (see Bjorkman and Dunbar 2016: 353)
- a. [mantika] → *mantika* ‘fat’
 - b. [[mí-mantika]-ña] → *mimantikáña* ‘more abounding in fat’
 - c. [á-[kwentus-i]] → *ákwentusi* ‘to speak to one another’
- (102) Nez Perce: Primary stress falls on rightmost lexical accent if any lexically accented prefixes are present (verbal subpattern; see Bjorkman and Dunbar 2016: 354–5)
- a. [nées-[sepée-[siléew-[[cúukwe]-cee]]]] → *néessepèsilèwcùkwece*
PL.OBJ-CAUS-by.seeing-know-INCOMPLETE
‘I make you know by seeing’
 - b. [hii-[nées-[sepée-[[páay]-núu]-cee]]]] → *hìnàsàpapàynóoca*
3-PL.OBJ-CAUS-arrive-toward-INCOMPLETE

Given the types of interactions between lexical accent and morphological structure found in Chamorro and Nez Perce, Bjorkman and Dunbar infer that stress patterns like the invented one shown in (103) do not exist. This pattern, which they term “Winner by Height, If Special” (WHIS) characterized by allowing the relative scope of two lexical accents to determine

⁸²The Nez Perce stress system as a whole is thus more intricate than the Chamorro system, since it allows two different patterns among verbs, but shows attraction of stress to nominal roots, in case a nominal form otherwise has no lexical accents.

which of the two surfaces as primary stress. In a., the suffix /tá/ outscopes /tú/ and therefore receives the stress; in b., the prefix /rú/ outscopes /tá/ and therefore receives stress. The supposed non-existence of WHIS patterns leads Bjorkman & Dunbar to further conclude that lexical stress patterns, or stress patterns sensitive to morphological structure, also fall within the domain of finite-state computations, and probably even merely a subset thereof.

(103) Winner by Height, If Special (Bjorkman and Dunbar 2016: 355)

- a. [ti [[[tú be] tá] to]] → *titubetáto*
- b. [ti [[rú [be tá]] to]] → *tirúbetato*

The lexical accent system of Vedic Sanskrit, as analyzed in 4.3, might though constitute a counterexample to Bjorkman and Dunbar's claim that no stress systems simultaneously sensitive to lexical accent and morphological scope exist. In Vedic, the lexical accent of a derivational suffix that outscopes all other derivational suffixes seems to categorically determine the position of stress.⁸³ However, determining whether the Vedic case constitutes a genuine counterexample is difficult to establish, since prefixes are uncommon in Vedic (arguably, most possible prefixes can be treated as first members of a compound), and derivational suffixes devoid of some kind of stress-attracting property are rare to non-existent. Clearly demonstrating that the Vedic system shows features of Dunbar and Bjorkman's WHIS pattern may not be possible for empirical reasons.

What is most important for the larger claim is that “finite-state automata [which are sufficient for representing aspects of language that are of regular complexity] cannot identify arbitrary constituents” (Bjorkman and Dunbar 2016: 357). The crucial question is then whether identifying the morphological head (typically the outermost derivational suffix) indeed requires identifying an arbitrary constituent. In a Vedic form like [[[á [[ris] tá]] *tāti*] bhís] *ariṣṭátātibhis* ‘security:INST.PL’ (cf. *áriṣṭa*- ‘unharméd’), the position of primary stress is determined by the preaccenting property of the derivational suffix *-tāti-*: *-tāti-* is special, and outscopes the root /ris-/, derivational suffix /-tá-/, and prefix /á-/, but is outscoped by the inflectional suffix /-bhís/.⁸⁴ Stress assignment thus appears to depend upon being able to identify the specific constituent containing *-tāti-*, more specifically, the last constituent at the stem-inflection boundary or the constituent that establishes the syntactic category of the word.

In light of the good evidence discussed above and further in Chapters 4 and 6 for effects of head dominance in Vedic Sanskrit and Ancient Greek, the ultimate question is whether the property of being a head, and its capacity to control the stress of the entire word depending upon its lexical properties, poses a challenge for the characterization of stress as

⁸³Lexically accented inflectional suffixes that occur in such cases would morphosyntactically outscope the derivational suffixes, but their lexical accents are either “weaker” than the lexical accents of (most) derivational suffixes, or simply ignored.

⁸⁴That the stress of *ariṣṭátātibhis* does not depend upon the lexical accent of /-tá-/ is shown by the fact that preaccenting /'-tāti-/ conditions stress on the preceding syllable also when that syllable has no lexical accent, e.g., [[[[jya-] -istha-] -tāti-] -m] *jyeṣṭhátātīm* ‘supremacy:ACC.SG’. See 4.2.3 and 4.3.4 on preaccenting affixes in Vedic.

either strictly regular, 1-1 neighborhood distinct, or able to be represented using subsequential functions alone (see Koser and Jardine 2020 on the latter point).⁸⁵ What import the effects of head dominance ultimately hold for the adequate computational characterization of stress systems thus remains to be explored in detail.

2.6 Some Outstanding Questions and Problems in the Study of Stress Systems

In the immediately foregoing 80 pages, an introduction to the phonological analysis of stress assignment from a multitude of perspectives has been carried out. In doing so, an attempt at comprehensiveness was made, describing how word-level prominence relations (“stress”) may arise not only through the construction of prosodic constituents according to number of syllables and syllable weight, but also by reference to other sorts of phonological elements (in particular tone; 2.4.1.2) and lexically specified or morphologically determined prosodic structure (2.4.2). At (3), I laid out a tentative typology of stress systems that could result from the combinatorial possibilities of factors that have been found to drive the assignment of word-level prominence relations. Whether some of the apparent gaps in this typology – especially as concerns combinations of morphologically sensitive with morphologically insensitive factors – constitute unlearnable systems, accidental gaps, or are the product of faulty theoretical constructs remains to be investigated more thoroughly.

Much of the chapter (2.2 and 2.3) aimed to present reader the uninitiated reader with sufficient context and tools to understand the well-trodden research grounds of “phonologically predictable” stress systems characterized by the absence of prespecified information about prominence relations in their lexical entries and general indifference to morphological structure. Precisely because such systems are typologically common and the same principles typically govern a “default” stress in systems that refer to additional factors like tone, lexical diacritics, or morphological structure, grasping their workings is utterly essential for much of the discussion in Chapter 3 and understanding the analyses in Part II. I focused on one influential parametric treatment of stress, Hayes 1995, and some typical Optimality-Theoretic constraints; both parametric and constraint-based approaches to stress continue to be lively topics of research, both in the domain empirical analysis and the learnability of phonological systems, as well as in the description of prosodic change (cf. Lahiri 2015). In Part II, the constraints introduced at 2.3 will shape the substance of many analyses, though reference to parameters and parametric descriptions of stress patterns will occasionally continue to be employed as well. Despite the wish to be comprehensive, not every theoretical alternative to the analysis of stress systems (e.g., the Simplified Bracketed Grid of Halle and Isardi 1995, the “primary first” model of van der Hulst (1996, 2009), or the foot-free constraints

⁸⁵A (*p*-)subsequential function can be represented by a (*p*-)subsequential finite state acceptor, which can generate an additional output string from a final state in the FSA (Mohri 1997: 4). Koser and Jardine (2020) themselves show that certain stress patterns, in particular default-to-same-side quantity-sensitive patterns, cannot be captured with subsequential functions. Given the descriptive similarity of simple lexical accent systems, as in Cupeño and Modern Italian, that have a same-side default stress, these patterns, too, must then be computationally non-subsequential.

for quantity-insensitive stress of Gordon 2002) could be given due justice here.

As regards factors influencing stress placement beyond syllable count and syllable weight, many questions remain open. While more and more evidence for the influence of lexical tone on stress continues to emerge, other factors that have been considered, such as vowel sonority, are disputed or may be reduced to a subtype of syllable weight. How to properly analyze lexical and morphologically sensitive stress systems constitutes another area of uncertainty and continuing investigation. In the representation of “lexical accent” (lexically specified information on prominence relations), the choice between an autosegmental representation or a representation in terms of foot structure is difficult to empirically determine, since most cases appear amenable to either representation. On the question of effects of “accentual dominance”, I will hold to the “Headmost Wins” model proposed in Revithiadou 1999, in part because it seems to fit well the facts of the Indo-European languages with which I am most familiar. One major alternative, the “root-controlled accent” model developed in Alderete 2001a, was explicitly rejected.⁸⁶

Going forward, the analyses developed in Part II will, to some extent, serve to show how well, or not, these tools for the analysis of stress systems can accommodate the facts of some older Indo-European languages, and how readily they can be employed in the modeling of prosodic change. In the upcoming chapter, I will elucidate how the constraint-based approach to the analysis of stress systems supports a credible approach to their learnability, and discuss how such models of phonological learnability may potentially be leveraged to gain insights into diachronic changes in aspects of word prosody.

⁸⁶I have not devoted space to an exposition of further alternatives, namely, the use of different diacritic degrees of strength (as in Vaxman 2016), especially through the employ of Gradient Symbolic Representations (GSR; see Smolensky and Goldrick 2016 for background, Medvedeva et al. 2022 and Zimmerman et al. 2023 on their application to lexical accent). As will be argued in Chapter 4, reference to lexical left and right foot edges, plus morphological headedness, is sufficient to capture all types of non-compound stress in Vedic Sanskrit. In particular, the ranking of constraints referring to different foot edges and foot edges with respect to morphological headedness can produce different degrees of dominance effects. Precisely because GSRs allow for continuous numeric scales, they are powerful and flexible, seemingly able to accommodate any degree of gradience in phonological phenomena. Insofar as the more restrictive predictions of a foot or autosegmental representation plus headedness have not been carefully explored in a non-superficial way (the claims made in Medvedeva et al. 2022 about Vedic Sanskrit are, in my view, false, and presumably arise from second-hand engagement with the data), I consider it premature to assume that a less restrictive view of the representation of lexical accent is necessary.

CHAPTER 3

Learnability, Acquisition, and Change in Constraint-Based Grammars: Implications for and Applications to Metrical and Lexical Stress

3.1 Introduction and General Considerations

Since the ultimate goal of the present work is to obtain a better understanding of changes in prosodic systems generally and stress systems specifically, we require tools for representing the transmission and learning of stress grammars. Under the assumption that first-language acquisition, particularly in its early stages, plays a central role in phonological change (in support of some version of this perspective, see, e.g., Hale 2008) an understanding of what must be acquired (i.e., what must learners know about the prosody of their target language?) and how it might plausibly be learned (i.e., how does a phonological grammar and lexicon emerge from the raw data?) is necessary.¹ Furthermore, the grammar and lexicon constructed during the acquisition of an L1-phonology will show effects in conditions of language contact, from the incorporation of loanwords to prolonged conditions of bilingualism, which may also prove relevant in diachronic perspectives. To this end, the present chapter aims to outline two distinct areas of research: on the one hand, work on the learnability of phonological grammars and lexicons (3.2), and on the other, the acquisition of L1- and L2-prosody, as well as psycholinguistic experimentation on differences in perception between speakers possessing different language backgrounds (3.3).

The chapter is then rounded out with a look at current approaches to the computational modeling of language change, in particular in the domain of (prosodic) phonology (3.3.6).

¹The extent to which language change of all varieties emerges among children or older speakers, under what types of learning conditions, remains a difficult and debated question. On the one hand, learners not yet equipped with any “complete” grammar or adult-like language competence (i.e., young children) intuitively strike many as a natural locus for change, since the space for reanalysis of ambient data seems ample. On the other hand, the view expressed by Bloomfield (1931 [1984]: 476), “every speaker is constantly adapting his speech habits to those of his interlocutors”, would allow for speakers to exhibit changes throughout their adult lives (see, e.g., Harrington et al. 2019 on subtle individual-level phonetic drift). And numerous sociolinguistic studies (e.g., Sankoff and Blondeau 2007) bear out change in the usage of phonological variants among adult speakers. I incline towards an overall conception whereby young learners are responsible for the introduction of *novel* variants via reanalysis, but adult speakers that are not “native speakers” of a grammar containing some novel variants (i.e., who did not themselves inductively arrive to that variant) play a role in diffusing and further transforming certain innovative patterns, in part because their capacity to master the novel pattern is not as strong or their exposure to the novel pattern is less extensive (cf. Labov 2007 and Stanford and Kenny 2013 on the diffusion of the Northern Cities Vowel Shift).

Doing so will satisfy an overarching goal, namely, to show how work from learnability theory in the domain of phonology specifically can be combined with research on agent-based models of language evolution and change. In this way, I aim to develop the general architecture of a model that can simulate diachronic changes in phonological grammars consisting of violable constraints and lexical items. Some work in this direction is already present in Staubs 2014 and Ito and Feldman 2022, which will be reviewed below. Work in Chapters 5 and 7 may be regarded as a continuation and cross-pollination of elements of that existing work: the use of constraint grammars (absent in Ito and Feldman 2022) to capture changes involving concrete linguistic data (absent from Staubs 2014).

Before proceeding to broadly characterize the issues to be addressed in under the heads of phonological learnability and acquisition, respectively, I would be remiss in not acknowledging that concrete *phonological* learning must rest, to a considerable degree, on *phonetic* learning. Matters such as how a learner converges on a set of phonetic categories out of the extensive variability observable in the phonetic realization of phonological entities (see, on aspects of this issue, Fowler 1981, Smith 1992 Pierrehumbert 2001) will not be discussed here. I largely take for granted that learners engaged in phonological acquisition in the terms laid out below have been able to induce their language-specific phonological categories, are able to mostly filter out irrelevant phonetic variation (though this sort of variation can indeed constitute an important locus for the seeds of phonological change), and generally learn veridically (i.e., that the learning data are largely processed and perceived as intended by the speaker; but see 3.3.5.3 below).

The foremost purpose of this chapter is to introduce approaches to and algorithms for the learning of constraint-based grammars. On the basis of the overview provided in Section 3.2, the application of these learning algorithms to the investigation of diachronic phonological change in conjunction with agent-based models (introduced in Section 3.3.6) will be explored. Before delving into any detailed descriptions, the object of learning itself must be clarified: what must be learned in order to construct an adequate portrait of the phonology of a language?

In terms of phonological knowledge, Albright and Hayes (2011: 661–2) describe three main types that adult speakers possess, which child learners must acquire, and which a complete model of phonological learning ought to be able to acquire on the basis of realistic, positive data alone: phonotactics (what phonological sequences are licit), alternations (how underlying sounds are regularly transformed in pattern-governed fashion), and free variation (what options exist for the realization of a sound, when more than one output given the same input is permitted). The acquisition of the last of these types of phonological knowledge, that of free variation, will not be dealt with in much detail, beyond acknowledging that types of sociolinguistically (Labov 2001) or lexically conditioned (Zuraw 2000, Zuraw 2009) phonological variation are very much real (cf. Coetzee and Pater 2011), the architecture of phonological grammars must be able to accommodate such variation.

Knowledge of phonotactics and alternations, apart from any free variation, and how to reproduce the observable patterns, constitutes the major categorical issues. The standard view of a simple generative phonology is that a production-oriented speaker is able to gen-

erate a surface phonological representation that can be phonetically implemented from the combination of a LEXICON and a GRAMMAR. The phonological lexicon consists of the underlying phonological material that is associated with (i.e., “spells out”) semantic, syntactic, and morphosyntactic information or features; the phonological grammar – usually treated as a set of ordered rewrite rules (as Chomsky and Halle 1968) or as a set of ranked or weighted constraints (Legrendre et al. 1990, Prince and Smolensky 1993 [2004]) – manipulates the underlying phonological representations (URs) to produce surface representations (SRs), which may not be fully faithful (i.e., identical to) the underlying representations. As an example, consider the toy language $\mathcal{A}1$ and phonological derivations given in (1).

(1) Phonological Lexicon and Grammar of $\mathcal{A}1$

‘1.SG.capybara’ /ka-peti/	‘capybara’ /peti/	Lexicon UR
ka'peti ka'pe:ti ka'pe:t	pe'ti pe'ti: –	Stress Assignment Stress-to-Weight Post-Tonic Syncope
[ka'pe:t] ‘my capybara’	[pe'ti:] ‘capybara’	SR

In the above example, the phonological lexicon consists of two elements (/ka-/ ‘1.SG’ and /peti/ ‘capybara’), while the grammar consists of three processes (stress assignment to the peninitial syllable, vowel lengthening under stress, and deletion of post-tonic vowels). Problematic for learners, however, is that neither the lexicon nor the grammar are directly accessible: both must be inferred and induced from the surface representations (and in real life, from the specific phonetic implementations of the surface forms). One may immediately note that, in language $\mathcal{A}1$, the posited UR /peti/ has two different realizations: [‘pe:t] and [pe’ti:].

As any student schooled in the art of morphophonological analysis learns, the development of a complete set of hypotheses concerning the morphology and phonology in a language able to adequately account for attested patterns of allomorphy and alternation requires several distinct but dependent pieces: 1) an analysis of the internal morphological structure of each word-form; 2) a set of underlying representations corresponding to the phonological spell-out of each morpheme; 3) a system of interacting rules or constraints that generates the attested data (whether categorical or variable) out of the hypothesized underlying representations. As Cotterell et al. (2015: 433) rightly observe, the entire procedure is “difficult even when we are told which morphemes are expressed by each word, because the unknown underlying forms of the morphemes must cooperate properly with one another and with the unknown phonological rules to produce the observed results. Because of these interactions, we must reconstruct everything jointly.”

In the above example, phonological and morphophonological learning can be divided into three steps: PHONOTACTIC learning, MORPHOLOGICAL learning, and MORPHOPHONOLOGICAL learning. These steps in the larger portrait of phonological learning will continue to be relevant throughout this chapter. The first step, phonotactic learning, consists essentially of the learning of constraints that serve to rule out illicit surface structures; phonotactic learning is reasonably straightforward when it involves categorical restrictions that apply

across the board in a language, regardless of any morphological or phonological boundaries. In the example of Language $\mathcal{A}1$, under the assumption that the learner can already segment prosodic words and distinguish syllables bearing main stress from unstressed syllables, the STRESS-TO-WEIGHT rule (“stressed vowels must be long”) could be learned at this phase, if long vowels in the language occur exclusively under main stress (i.e., there are no word like [pe:’ti:], in which an unstressed long vowel occurs).² In the event that long vowels are freely permitted in a language, the acquisition of STRESS-TO-WEIGHT would need to wait until the phase of morphophonological learning, since a learner would need to distinguish between underlying long vowels (i.e., long vowels that always surface as such) and underlying short vowels that undergo lengthening. Learning the process of POST-TONIC SYNCOPE in $\mathcal{A}1$, meanwhile depends on knowledge about paradigmatic relations: the learner cannot decide that there is such a process until it recognizes that [ka’pe:t] and [pe’ti:] (among other forms) contain the same morpheme for ‘copybara’, whose phonological UR should be posited as /peti/. Thus, morphological segmentation must take place before the morphophonological work of constructing (ideally unitary) URs and processes to derive surface allomorphs can be carried out.

Learning the processes of STRESS ASSIGNMENT in $\mathcal{A}1$ can potentially take place as part of phonotactic learning as well. Let us take $\mathcal{A}1$ to have a quantity-insensitive stress system, which assigns primary stress to the penultimate syllable. In that case, a set of phonotactic markedness constraints can be induced to penalize a main stress anywhere other than the second syllable, and to penalize a second syllable without main stress (cf. Hayes and Wilson 2008: 407–11). Similarly, the set of grid-based constraints employed in Gordon 2002 could be ranked using input-output mappings from URs identical to SRs and obtain penultimate stress (cf. Hayes 2004, Prince and Tesar 2004, and the end of 3.2.1.1 below on such a phase of phonotactic learning). Even with the assumption of foot structure, a learner could apply Robust Interpretive Parsing (RIP; see 3.2.2 below) to discover the correct ranking of stress-related constraints. However, if main stress is not perfectly predictable based on its surface distribution, e.g., the language contains words [’pe:ti] and [pe’ti:] (again assuming that long vowels only occur under stress), the acquisition of distinct underlying features (in this case, lexical stress or underlying differences in vowel length) would be necessary to learn a grammar and lexicon able to generate stress correctly.

As this toy example should illustrate, how the learner should go about testing combinations of constraint-ranking or rule system plus lexicon in order to discover the specific combination that is guaranteed to correctly reproduce the learning data is far from straightforward. As discussion in 3.3 below will further emphasize, a certain conflict exists between the goals of research into phonological learnability and the goals of understanding and explaining phonological change. The former is concerned with being able to guarantee that a given combination of grammar and lexicon can be perfectly acquired, i.e., transmitted in a

²In principle, given the one-to-one coincidence of primary stress and long vowels on the surface, the learner could equally explain the data by assuming that all vowels are underlyingly long and all unstressed vowels undergo shortening. At least in a constraint-based grammatical architecture, this alternative could be ruled out by imposing the selection of a UR and grammar that maximizes harmony, i.e., would have the fewest total violations between UR and SR.

lossless fashion; that latter, however, presupposes that some “imperfection” in transmission necessarily occurs. The interesting question is then: where and how can potential “imperfection” in transmission can lay the seeds of phonological change?³ Thus, a fuller grasp of some mechanisms of phonological learnability should help to determine where potential “leaks” that might allow for change could creep in.

3.2 Learnability in Phonological Theory

Learnability as treated in this section presumes that phonological grammars are characterized in terms of conflicting constraints on the well-formedness of outputs and faithfulness to the underlying representations assumed by the learner, as in Optimality Theory. Some of the same fundamental questions of phonological learnability have been explored using alternative characterizations of phonological grammars, e.g., as input-output rewrite rules or as the setting of parameters; see, for instance, Gildea and Jurafsky 1996 or Albright and Hayes 1999 and Albright and Hayes 2003 on the induction of (morpho)phonological rules, or Dresher and Kaye 1990, Frank and Kapur 1996, Dresher 1999, or Nazarov and Jarosz 2017 on parameter setting. While reference to the setting of parameters may equally allow for a neat characterization of changes in grammars (cf. Roberts 2007 with respect to syntax, Lahiri 2015 with respect to stress), research on parameter setting in phonology is perhaps less developed than work on constraint ranking, presumably for the reason that many phonological phenomena do not admit of a perspicuous characterization in terms of parameters. Notably, to my knowledge, only stress assignment (including lexical accent; see Dresher 2016 and 3.2.2.3 below) has received significant attention as a phonological phenomenon learnable via parameter setting.⁴

As per the discussion of the toy example in (1) above, I treat below both issues in the learnability of phonological grammars and the learnability of their phonological representations. The former problem consists essentially in determining the relationships between constraints, given the constraints themselves and input-output mappings. Falling under the latter problem are so-called “hidden” structures: underlying lexical representations and the “invisible” prosodic structures (syllable boundaries, foot structure) present in a fully specified phonological output. The treatment of hidden structure problems then culminates in discussion of the learnability of lexical accent specifically, which involves the simultaneous

³As will be discussed more fully in Chapter under 3.3 below, one might reasonably ask whether all change in *phonological* systems involves *phonetic* changes (e.g., though coarticulation; cf. Ohala 1993b, Solé and Ohala 2010) – effectively, changes in the structure of the primary linguistic data – that lead to the apprehension and acquisition of innovative grammars. For agent-based modeling of how phonetic variants may translate into novel phonological categories, at least at the segmental level, see Harrington et al. 2018 and 3.3.7 below. A contention made in at 3.3 is that certain phonological changes – in particular involving stress grammars – are not (always) readily attributable to phonetic variation and should instead be motivated through reanalysis of the overt data at the phonological level, more akin to explanations applied to syntactic change (Kroch 1989, Zimmermann 2017).

⁴Given the extensive range of theoretically possible segmental alternations, one might rightly wonder what, if any parameters are supposed to exist that could govern the presence or absence of a specific type of allophonic or morphophonemic alternation in a language.

acquisition of both underlying prosodic information and surface prosodic structure.

3.2.1 Learning Constraint Grammars

In this section, I introduce some influential approaches to the learning of constraint-based phonological grammars. Again, these approaches do not pretend exactly reproduce the totality of phonological acquisition, nor to induce the constraints themselves; the constraints are either presumed to be innate, or their induction is left as a problem separate from the ranking or weighting of the constraints.⁵ I begin with an introduction to the major algorithms for determining a strict ranking of constraints in Optimality Theory (Prince and Smolensky 1993 [2004]), before turning to other types of constraint grammars that calculate winners other than through the strict domination of constraints and that can generate patterns of free variation; the focus there will lie with so-called Maximum Entropy Grammars.

3.2.1.1 Constraint Demotion

In this section, I largely follow the explanations presented in Tesar and Smolensky 2000: Ch. 3 & 8. The earliest expositions of constraint demotion algorithms may be found in Tesar and Smolensky 1993 and Tesar 1995. Further presentations of versions of constraint demotion algorithms are found in Hayes 2004: 169, Smolensky et al. 2006: 489–95 and Becker 2009: 164. Pedagogical expositions of constraint demotion are available in Kager 1999: Ch. 7 and McCarthy 2002: 202–20. Further possible variations on constraint demotion, employing different biases in the initial state or the application of demotion, are treated in Hayes 2004, Prince and Tesar 2004, and Yates and Zukoff 2018: 241–50.

(Mutl-)Recursive Constraint Demotion The first necessary ingredient for the ranking of constraints is data in the form of a tableau and means of comparing the violation profiles of winners and losers. The notation of tableaux in a comparative format, marking constraints as “winner-preferring” (**w**), “loser-preferring” (**l**) or “equivalent” (**e**) is due to Prince 2002. This type of notation in the discussion of ranking arguments and learnability has gained currency in preference to the mark-data pairs format originally employed by Tesar and Smolensky. Consider the sample tableau in 2a.: under a representation of winner-loser relations using mark-data pairs, the table in 2b. results; using the Comparative Tableau format of Prince, 2c. is produced (cf. Prince 2002: 271–2 for this example).⁶ The cancellation of marks in 2b. indicates where candidates tie or partly tie on constraints, e.g., winning candidate A is left with

⁵For one fairly successful implementation of the induction of markedness constraints from surface data, see Hayes and Wilson 2008, in which phonotactic markedness constraints are induced and their weights determined based on the frequency of occurrence of phonological sequences in the learning data.

⁶In the Princian notation, the usage of the tilde ($\sim$) may be understood as functionally the opposite of the mathematical precedence operator ($\prec$). The latter means that the candidate to the left of the operator is a suboptimal parse of a given input $\mathcal{S}$, whereas the candidate to the right is the optimal parse of $\mathcal{S}$. Conversely, the candidate to the left of the tilde constitutes the optimal parse. One could also employ the mathematical succession symbol, $\succ$, and this is indeed sometimes done (e.g., in Becker 2009).

one more violation of C_3 than losing candidate B, but A and C tie on this constraint, since they have the same number of violations. In the case of a comparative tableau, violations shown as asterisks can effectively be converted to positive numbers, and the $1 \times n$ matrix corresponding to a losing candidate can be subtracted from the $1 \times n$ of the winning candidate; any 0 values become **e**, positive numbers become **L** (the optimal candidate violates the constraint more than the losing candidate), while negative violations become **w** (the losing candidate violates the constraint more than the optimal candidate).

(2) Representation of an OT Tableau Using Mark-data Pairs and Comparative Tableaux

a. Original Data Tableau (with ranking)

		C_1	C_2	C_3
a.	A	*		***
b.	B	*	*!	**
c.	C	**!	*	***

b. Mark-data Pairs after Cancellation

$loser \prec winner$	$marks(loser)$	$marks(winner)$
$B \prec A$	$*C_1 *C_2 *C_3 *C_3$	$*C_1 *C_3 *C_3 *C_3$
$C \prec A$	$*C_1 *C_1 C_2 *C_3 *C_3 *C_3$	$*C_1 *C_3 *C_3 *C_3$

c. Comparative Tableau

	C_1	C_2	C_3
$A \sim B$	e	w	L
$A \sim C$	w	w	e

Examining 2b. and 2c., one can deduce the necessary constraint ranking: C_3 has uncancelled violations of the winner in 2b., and is marked as **L** (“loser-prefering”) in 2c., and therefore must be ranked below both C_1 and C_2 ; otherwise, candidate B could win, if C_3 were ranked above C_2 . Conversely, since the B and C have uncancelled violation marks on C_2 and C_3 in 2b., and are marked as **w** in 2c., these must dominate C_3 . When many tableaux, with many winner $\sim$ loser pairs to consider are involved, however, algorithmic implementation becomes helpful in determining the right rankings. How do the mark-data pairs or a set of comparative tableaux then feed into an algorithm for determining a constraint ranking?

First, note that the basic Recursive Constraint Demotion (RCD) is a batch algorithm: it has access to the totality of the learning data (also referred to as the *Support*) at all times; it does not operate “myopically”, adjusting a grammar in response to a single datum or some subset of the data.⁷ The algorithm makes progress by progressively identifying constraints

⁷I will consider RCD to essentially be a batch algorithm on the basis of this property, even though learning

that are not violated by winners (or cancelled out), then installing the constraints that prefer winners strata above constraints that (sometimes) prefer losers. Again, 2 reflects a very simple case: RCD sees that C_1 and C_2 are not present in $marks(winner)$ (A does not violate C_2 , and its violation of C_1 is cancelled out because both losers also violate C_1 at least once), or equivalently, that C_1 and C_2 are both winner-prefering (only **w** occurs in the columns of C_1 and C_2). Therefore, C_1 and C_2 are installed in the highest stratum of constraints. Then, because either C_1 and/or C_2 occurs in the $marks(loser)$ of the comparisons between $B \prec A$ and $C \prec A$, or equivalently, because either C_1 and/or C_2 contains a **w** in the comparisons $A \sim B$ and $A \sim C$, the *rows of these comparisons are removed from the Support*. At this point, the Support is empty, and the ranking $\{C_1, C_2\} \gg C_3$ is able to account for the data. This “standard” version of RCD will recursively place constraints containing only **w**’s or **e**’s in their columns into a stratum above remaining unranked constraints, but below constraints that have already been ranked.

The recursive creation of constraint hierarchies can be illustrated more clearly by adding another constraint and a further candidate to the example in 2. From this point forward, I will exclusively employ the Comparative Tableau format.

(3) Updated Extension of (2)

a. Original Data Tableau (with ranking)

		C_1	C_2	C_3	C_4
a.	A	*		***	*
b.	B	*	*!	**	
c.	C	**!	*!	***	
d.	D	*		****!	

b. Comparative Tableau (full Support)

	C_1	C_2	C_3	C_4
$A \sim B$	e	w	L	L
$A \sim C$	w	w	e	L
$A \sim D$	e	e	w	L

c. Comparative Tableau (after one pass of RCD)

occurs recursively, and the ranking is not derived from a single optimization procedure in one fell swoop. RCD differs from the algorithm that Tesar and Smolensky (1993: 15) call “Batch CD”, in that the latter applies the “Core CD” algorithm (demoting any loser-prefering constraints to the stratum immediately below winner-prefering and neutral constraints), while RCD does not. Batch CD, though, must also be run iteratively to find a complete ranking. Essentially for the reason that Tesar and Smolensky (1993: 19) conclude that “the real difference between Batch CD and RCD is slight”, and because both provably converge, Batch CD has been mostly ignored in subsequent literature, and is indeed not included in Tesar and Smolensky 2000.

	C_1	C_2	C_3	C_4
$A \sim D$	e	e	w	L

Ranking the constraints C_{1-4} given winner A and losers B, C, and D now requires two steps. First, as before, C_1 and C_2 are placed in the highest stratum because their columns contain only **w**'s or **e**'s; at this point, the comparisons $A \sim B$ and $A \sim C$ are removed from the support, leaving only $A \sim D$. Because the **L** in the column of C_3 is no longer present in the Support, C_3 may be placed in the next-highest stratum, since it now contains only a **w**. The Support is then empty and C_4 is left in the lowest stratum, yielding the constraint hierarchy $\{C_1, C_2\} \gg C_3 \gg C_4$.

These descriptions of the procedure practically applied in RCD in the above descriptions can be put in terms of the informal algorithm in 4 after Prince 2002: 291, converted into somewhat more formal pseudo-code in 5, after Becker 2009: 163.⁸

(4) RECURSIVE CONSTRAINT DEMOTION (Informal Algorithm; cf. Prince 2002: 291)

Given a comparative tableau containing unranked constraints,

1. Locate all constraint columns lacking **L**;, i.e., containing only free **w** and/or **e**.
2. Place all of these constraints in a stratum ranked just below all previously ranked strata (or below an “empty” stratum if no constraints have yet been ranked).
3. Remove from consideration all constraints just ranked, as well as the ranking arguments (i.e., rows in the comparative tableau) that these constraint satisfy (i.e., which have a **w** in one of the constraints just ranked).
4. Go back to 1., applying it to remaining columns until there are no more ranking arguments (i.e., all rows have been removed from the comparative tableau).

(5) RECURSIVE CONSTRAINT DEMOTION (Pseudo-Code; cf. Becker 2009: 163)

Given a Support S ,

Given a set of constraints in S , *not-yet-ranked constraints*,

H := a new constraint hierarchy.

While S is not empty:

- a. *current-stratum* := all the constraints in *not-yet-ranked constraints* that contain no **L** in their column in S .⁹

⁸The algorithm described by Becker, however, behaves slightly differently from Prince's adaptation of Tesar & Smolensky's algorithm: Becker's version of RCD installs only constraints that contain at least one **w** in their column, never constraints that contain only **e**'s in their column. The result will be that columns consisting exclusively of **e**'s at some stage are not put in higher strata, but instead tend to sink to the lowest stratum, along with loser-preferring constraints.

⁹Becker's pseudo-code here reads:
current-stratum := all the constraints in *not-yet-ranked constraints* that have at least one **w** and no **L**'s in their column in S .

- b. If *current-stratum* $\neq \emptyset$,
 - i. remove winner-loser pairs that are assigned a **w** by any constraint in *current-stratum*.
 - ii. place *current-stratum* in the next stratum in H , below all existing strata.
 - iii. remove the constraints in *current-stratum* from *not-yet-ranked-constraints*.

Put *not-yet-ranked-constraints*, in the next stratum in H , below all existing strata.
Return H .

Before considering other varieties of constraint demotion that have been developed in the literature, two issues warrant further consideration. First, in what regards does Becker (2009)'s version of RCD lead to different results? Second, how does RCD potentially interface with diachronic changes in constraint rankings, or, under what conditions can RCD applied by a learner $\mathcal{B}$ obtain a different grammar than a learner $\mathcal{A}$?

As to the first question, it may be partly answered by slightly altering the example in (3), removing one violation on C_1 from candidate C . The comparative tableau will then look as in 6a. The difference between the “standard” RCD algorithm and Becker’s version hinges upon the treatment of **e**’s in the comparative tableau. Here, C_1 contains now exclusively **e**. Under “standard” RCD as laid out in 4 and 5, C_1 and C_2 will first be ranked, because their columns contain no **L**’s; subsequently, C_3 is placed in the next stratum (because its only **L** in the pair $A \sim B$ has been removed), leaving C_4 in the lowest stratum. Under Becker (2009)’s version, only C_2 can be first ranked, because only its column contains at least one **w** and no **L**’s; subsequently C_3 is placed in a stratum below C_2 (it contains a **w** and no **L**’s); finally, C_1 and C_4 are left over, because the Support is now empty, and so are assigned to the lowest stratum.

(6) Comparison of RCD: Treatment of **e**.

a. Comparative Tableau (full Support)

	C_1	C_2	C_3	C_4
$A \sim B$	e	w	L	L
$A \sim C$	e	w	e	L
$A \sim D$	e	e	w	L

- b. Ranking under Standard RCD: $\{C_1, C_2\} \gg C_3 \gg C_4$
- c. Ranking under Becker (2009)’s RCD: $C_2 \gg C_3 \gg \{C_1, C_4\}$

Another variation shows what occurs if C_1 contains all three relationships (imagine that C_1 is violated twice by winning candidate A , once by B , three times by C , and twice by D). In this case, C_2 is still exclusively winner-preferring, and so under both versions of RCD is ranked first. C_2 explains the pairs $A \sim B$ and $A \sim C$. Under standard RCD, since C_1 now contains an **e**, it will be placed in the next stratum with winner-preferring C_3 , leaving C_4 in

the lowest stratum. Conversely, Becker's RCD will not rank C_1 on the basis of its **e**, with the result that C_1 will again ultimately be placed in the lowest stratum at the end of the ranking procedure. See 7 below.

(7) Comparison of RCD: Treatment of **e** 2.

a. Comparative Tableau (full Support)

	C_1	C_2	C_3	C_4
$A \sim B$	L	w	L	L
$A \sim C$	w	w	e	L
$A \sim D$	e	e	w	L

b. Ranking under Standard RCD: $C_2 \gg \{C_1, C_3\} \gg C_4$

c. Ranking under Becker (2009)'s RCD: $C_2 \gg C_3 \gg \{C_1, C_4\}$

Note that in all of the examples described here, C_1 was not truly necessary for determining the winner: C_2 and C_3 were sufficient. The fundamental difference between “standard” and Beckerian RCD thus consists in the fact that the former ranks constraints that are “useless”, but not actively harmful to a winner, as high as possible (i.e., at whatever pass of the algorithm where the constraint has no **L** in its column), whereas the latter will rank them as low as possible, i.e., in the lowest stratum. Becker's version of RCD is thus, in a sense, more sensitive to the utility of constraints. The utility or relative importance of particular constraints plays a strong and more explicit role in MIRCD (see on Most-Informative Constraint Demotion below) and in forms of constraint grammars involving weighted constraints.¹⁰

Error-Driven Constraint Demotion An online and error-driven implementation of constraint demotion, in contrast to the batch-processing RCD algorithm discussed above, is also introduced in Tesar and Smolensky 2000: 50–3; 90–100. The fundamental distinction between RCD and EDCD consists in the fact that EDCD operates with just a single mark-data pair / comparative tableau at a time. While RCD, because it has a view on the totality of the system at once, is guaranteed to always find a ranking and indeed always discover the same ranking, assuming that any consistent ranking able generate all winners exists. EDCD, on the other hand, is not able to discover whether a consistent ranking exists, or not, because it is restricted to the evaluation of just a single winner-loser pair at a time. Furthermore, certain constraint violation profiles may result in the repeated cyclic demotion of constraints (Dresher 1999), or grammar learned may be a less-restrictive superset grammar compatible with the data, rather than the more restrictive subset grammar that actually generated the data (Tesar and Smolensky 2000: 100; see also Albright and Hayes 2011: §3.3 on the subset problem). Moreover, the general thrust of the criticism made by Dresher (1999: 59–63) with

¹⁰Note also that, in an example like the ones portrayed in 6 or 7, a Maximum Entropy Grammar (cf. (3.2.1.2) below) will find a weight of 0 for C_1 , which further confirms that intuition that C_1 does no real work in determining winners.

respect to Robust Interpretive Parsing / Constraint Demotion (see further Section 3.2.2 below) is potentially applicable to EDCD as well.

Because learning under EDCD is fundamentally *error-driven* (cf. generally Wexler and Culicover 1980 on error-driven learning), learning – that is, updates or changes to a constraint ranking – occurs only when the output of the learner’s current grammar differs from the winner (i.e., a learning datum) for some input I . In order to conduct learning, the learner must first observe a learning datum p and identify p as the possible output parse of some given input I . Then, the learner generates its own optimal parse p' using its current constraint hierarchy $\mathcal{H}$. If $p \neq p'$, then the learner will apply Constraint Demotion to the comparative tableau consisting of $p \sim p'$; if $p = p'$, then no change to the current constraint hierarchy $\mathcal{H}$ is applied.

Let us illustrate the operation of EDCD in practice with Tableau 8a. below, with the final desired constraint ranking. Before learning commences, the constraints may be assigned to an arbitrary initial hierarchy (assuming that no learning has yet occurred), or perhaps left unranked in a pool.¹¹ Assume the arbitrary initial hierarchy $C_4 \gg C_3 \gg C_1 \gg C_2$, represented in 8b.

Under this constraint ranking, the learner’s optimal parse will be B; when the learner is exposed to a surface instance of A as a positive learning datum, it will associate A with the input able to generate A, B, C, D, and generate B as its optimal parse, whereupon it will determine that $A \neq B$, and so generate a comparative tableau to which constraint demotion may be applied. Following the basic Constraint Demotion algorithm (Tesar and Smolensky 2000: 95), C_4 and C_3 must now be demoted immediately below C_2 : the former two constraints are both loser-preferring, while C_2 is winner-preferring. Since the stratum below C_2 was already occupied by C_1 , C_1 , C_3 and C_4 are now unranked with respect to one another, and both A and D are potential optimal parses of I (the stratum below C_2 contains five total violation marks). When the learner again receives the output A, its parse may be either A (in which case CD is not applied, and the constraint rankings remain unchanged) or its parse may be D, in which case an error occurs and CD is applied. The result will be the demotion of C_4 below C_3 . The resulting hierarchy will then have A as its optimal parse, and therefore, no changes to the constraint ranking can occur as a result of encountering the output A as a parse of I : the learner’s optimal parse p' and the data will always agree.

¹¹The latter choice engenders potentially more difficulty for the learner in generating its own optimal parse p' , since candidates may readily tie.

(8) Illustration of EDCD

a. Reproduction of Tableau 3a.

		C ₁	C ₂	C ₃	C ₄
a.	 A	*		***	*
b.	B	*	*!	**	
c.	C	**!	*!	***	
d.	D	*		****!	

b. Arbitrary Initial Hierarchy: $C_4 \gg C_3 \gg C_2 \gg C_1$

I		C ₄	C ₃	C ₂	C ₁
a.	A	*!	***		*
b.	 B		**	*	*
c.	C		***!	*	**
d.	D		****!*		*

c. Comparative Tableau $A \sim B$

	C ₁	C ₂	C ₃	C ₄
$A \sim B$	e	w	L	L

d. New Hierarchy: $C_2 \gg C_1, C_3, C_4$

I		C ₂	C ₃	C ₄	C ₁
a.	 A		***	*!	*
b.	B	*!	**		*
c.	C	*!	***		**
d.	 D		****!		*

e. Comparative Tableau: $A \sim D$

	C ₁	C ₂	C ₃	C ₄
$A \sim D$	e	e	w	L

f. New Hierarchy: $C_2 \gg C_3, C_1 \gg C_4$

I		C_2	C_3	C_1	C_4
a.	 A		***	*	*
b.	B	*!	**	*	
c.	C	*!	***	**	
d.	D		****!	*	

EDCD in this case has successfully converged upon the crucial rankings $C_2 \gg C_3$ and $C_3 \gg C_4$. C_1 , simply because it was ranked below C_2 in the initial hierarchy, finds itself in the same stratum where it began, immediately below C_2 . However, had the ranking $C_1 \gg C_2$ held in the initial state, EDCE would never have had reason to demote C_1 (C_1 is never loser-preferring), thus would have found itself in the top stratum, above C_2 after learning terminated. EDCE should, in general, converge on a hierarchy compatible with all winners present in the positive data, but some non-convergence issues have been pointed out (Tesar 1998, Drescher 1999: 56–63): a learner may become “stuck”, cycling between the demotion of constraints without finding a ranking that produces the winners. Without considering the issue too deeply here, many such purported convergence problems of EDCE may result from examples considering too little data or employing too few constraints; the use of “mark-pooling ties”, as seen for the undifferentiated stratum below C_2 in 8d. above can lead to genuine non-convergence (see Boersma 2009).

A few variants on EDCE are conceivable. One choice is be made, discussed in Tesar and Smolensky 2000: 131, note 3, is whether an observed parse is to be repeatedly used as a learning datum until no further change to the constraint ranking occurs. Another choice is how to decide when to terminate learning, since the learner cannot in principle know whether it has converged on the target grammar; Tesar and Smolensky (2000: 100) assert that convergence to a grammar consistent with all positive evidence should occur after exposure to maximally $N(N - 1)$ informative examples, where N is the number of constraints.¹² At this point, we can write an informal procedure of EDCE, based on Tesar and Smolensky 2000: 99–100, with inspiration from Prince and Becker, and a crucial improvement contributed by Boersma 2009.¹³

¹²An “informative example” is a winner-loser pair that results in the demotion of at least one constraint.

¹³The crucial improvement added by Boersma consists in the evaluation of constraint hierarchies that are not fully ranked using *permuting ties*, so that constraints with an indeterminate relative ranking are made into a random totally ranked hierarchy at evaluation time. For instance, a hypothetical constraint hierarchy $C_1 \gg C_2, C_3 \gg C_4$ could have the total hierarchy $C_1 \gg C_2 \gg C_3 \gg C_4$ or $C_1 \gg C_3 \gg C_2 \gg C_4$ at evaluation time.

(9) **ERROR-DRIVEN RECURSIVE CONSTRAINT DEMOTION** (Informal Algorithm; cf. Boersma 2009: 673; 680)

Given a set of positive surface forms associated with an input and an initial constraint hierarchy,

1. The learner receives an input-output pair (I, O) .
2. The learner finds its own optimal output(s) over the input I , by doing the following:
 - a. The learner takes all strata in its current constraint hierarchy that are not fully ranked and imposes a random total ranking of those constraints, while maintaining all other existing ranking.
 - b. The learner establishes which output(s) are optimal under the selected total ranking.
 - c. In case more than one output is optimal, a single output from this set is randomly chosen as a winner.
3. If the learner's output from I , O' differs from the received learning datum, O , do the following:
 - a. Construct a comparative tableau with O as winner and O' as loser.
 - b. Locate the highest-ranked constraint column that contains a **W**, C_w
 - c. Locate all constraint columns that contain a **L** and which are not already ranked below C_w .
 - d. Place all the constraint columns identified in the immediately preceding step in a stratum below C_w .
4. Go back to 1., repeating until O' and O have been registered as identical for each input-output pair in the learning data or $N(N - 1)$ informative examples have been seen.

Other Variants of Constraint Demotion Further variants on batch-processing MRCD or the online EDCD consist essentially in heuristics, which can be implemented formally, to find constraint hierarchies that more successfully escape the problem of subset grammars, or which can adequately acquire the phonotactics of a language without recourse to data involving alternations. The two issues are in fact related, and the issue of phonotactic learning is especially relevant for the present enterprise, since all types of stress grammars that do not include lexical accent should be acquisible through phonotactic learning alone. I first briefly describe the subset problem and the notion of phonotactic learning as part of phonological acquisition before outlining some of the specific adjustments proposed to constraint demotion algorithms.

The so-called subset problem in learnability concerns that fact that exposure to licit forms of a language does not directly indicate what is ill-formed, with the result that a grammar constructed on the basis of positive data alone may be insufficiently restrictive. In other

words, the positive data support both a less-restrictive as well as more-restrictive grammars. A specific desired more-restrictive grammar falls within a subset of the more-restrictive grammars compatible with the less-restrictive grammar. See already Baker 1979 and Angluin 1980, and the detailed treatment by Hale and Reiss (2003). In the context of standard Optimality Theory, grammars with more higher-ranked markedness constraints dominating the corresponding faithfulness constraints are more restrictive: fewer types of surface patterns will be licit. In employing RCD or EDCD as described above to rank a constraint set on the basis of positive data alone, faithfulness constraints frequently remain highly ranked, since no positive data compels their demotion below a markedness constraint that would ensure the restrictiveness of the grammar.

This issue in its OT instantiation may be illustrated through the toy example in 10. Consider a language that allows [+spread glottis] (aspirated) plosives, but not in combination with the feature [+voice]. On the surface, stops in sequences like [pi], [bi], [p^hi] are grammatical, while *[b^hi] is not. Ideally, a specific markedness constraint $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ should ban a stop like [b^h] on the surface, but because the learner receives no positive evidence that an underlying /b^hi/ might be realized instead as [bi] or [p^hi], in violation of IDENT-[voice] or IDENT-[spread glottis], it cannot rank $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ over IDENT-[voice] or IDENT-[spread glottis]. Indeed, the licit plosives in [bi] and [p^hi] suggest that IDENT-[voice] and IDENT-[spread glottis] are generally respected.

Assuming a procedure of PHONOTACTIC LEARNING, whereby a learner attempts to discover as much as possible about the rankings of its phonology without reference alternations (inspired by the realities of phonological learning in babies and young children, who cannot yet successfully segment morphemes; cf. Hayes 2004: 159–64, Albright and Hayes 2011: 667–70), each surface output is taken to be identical with a UR. In 10, each licit plosive may be compared with an alternative candidate having a different value for [voice] or [spread glottis]. As the comparative tableau shows, since each type of presumed underlying plosive always surfaces faithfully, the two IDENT constraints are always winning-preferring. Since no voiced aspirates occur on the surface, $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ is always winner-preferring, too; however, RCD will rank all three exclusively winner-preferring constraints in a constraint above the two other markedness constraints. The result is that, if an input like /b^hi/ is then considered, both the fully faithful candidate [b^hi] and alternatives are considered equally viable, as shown in 10c. Although it would be desirable to exclude voiced aspirates altogether, a learner equipped with either version of RCD considered above will have acquired a subset grammar that could (sometimes) permit voiced aspirates.

(10) Phonotactic Grammar without Voiced Aspirates

a. Comparative Tableau

		*[+voice]	*[+spread glottis]	*[+spread glottis, +voice]	IDENT-[voice]	IDENT-[spread glottis]
a.	/pi/ → pi ~ bi	w	e	e	w	e
b.	/pi/ → pi ~ p ^h i	e	w	e	e	w
c.	/bi/ → bi ~ pi	L	e	e	w	e
d.	/bi/ → bi ~ b ^h i	e	w	w	e	w
e.	/p ^h i/ → p ^h i ~ pi	e	L	e	e	w
f.	/p ^h i/ → p ^h i ~ b ^h i	w	e	w	w	e

b. Ranking under RCD:

{ IDENT-[voice], IDENT-[spread glottis], * $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ } $\gg$ { *[+voice], *[+spread glottis] }

c. Input: /b^hi/

/b ^h i/	* $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$	IDENT-[voice]	IDENT-[spread glottis]	*[+voice]	*[+spread glottis]
a. b ^h i	*!			*	*
b. bi			*!	*	
c. p ^h		*!			*

d. Desired Ranking Hierarchy:

* $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ $\gg$ { IDENT-[voice], IDENT-[spread glottis], } $\gg$ { *[+voice], *[+spread glottis] }

The fundamental issue in the case at hand is how to try to ensure that * $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$ comes to dominate the respective faithfulness constraints. For this simple example, one relatively easy heuristic, frequently proposed and adopted in the literature (e.g., Smolensky 1996, Gnanadesikan 2004), will suffice: in an initial ranking hierarchy, all markedness constraints should dominate all faithfulness constraints (MARKEDNESS $\gg$ FAITHFULNESS bias). Given an initial ranking hierarchy

{ * $\left[\begin{smallmatrix} +\text{spread glottis} \\ +\text{voice} \end{smallmatrix} \right]$, *[+voice], *[+spread glottis] } $\gg$ { IDENT-[voice], IDENT-[spread glottis] },

on the basis of the support in 10a., the two simple markedness constraints can be demoted

below the two faithfulness constraints, producing the desired hierarchy stated in 10a., which will categorically exclude voiced aspirates.¹⁴

More detailed are the proposals in Hayes 2004 (Low Faithfulness Constraint Demotion; LFCD) and Prince and Tesar 2004 (Biased Constraint Demotion; BCD) that apply more stringent criteria for installing a constraint in a new stratum. The specific criteria proposed by Hayes and Prince and Tesar may be summarized as follows (cf. Hayes 2004: 178–81, Prince and Tesar 2004: 258–68), phrased following Hayes:

1. **Avoid Losers:** install constraints that are not loser-preferring first.
2. **Favor Markedness:** install markedness constraints first.
3. **Favor Activeness:** install constraints containing at least one **w** in the support before constraints containing only **e**.¹⁵
4. **Favor Specificity:** install constraints that have a subset of the violations of another constraints violations first.
5. **Favor Autonomy / Find Minimal Faithfulness:** install faithfulness constraints that are able to exclude losing candidates alone first / install faithfulness constraints first that will make the most markedness constraints available for ranking in subsequent creation of hierarchies.

In both Low Faithfulness and Biased Constraint Demotion, the first three criteria are present and applied in that order when selecting constraints eligible for building a stratum. **Favor Specificity** has no analogue in Biased Constraint, but Prince and Tesar 2004: 271–8 observe that RCD will rank more specific markedness constraints above more general ones without any further provisions. The final criterion, which deals with unraked faithfulness constraints, reflects different approaches take by Hayes versus Prince and Tesar. These criteria, implemented algorithmically (see Prince and Tesar 2004: 269–70 for pseudo-code of Biased Constraint Demotion), demonstrably help to escape the subset problem in many concrete cases and allow learners to make much more progress towards discovering necessary rankings in phonotactic learning.

One further possible adjustment to RCD, involving a specifically quantitative principle, has been proposed in Zukoff 2017b: 112–22 and Yates and Zukoff 2018: 246–59. The algorithm proposed by Yates and Zukoff, Maximially Informative Constraint Demotion (MIRCD), also contains the principle of **Favor Activeness**, but decides which constraints to install in a new stratum first on the basis of how many **w** are present in a constraint's column in the support: constraints containing more total **w** (and no **L**) are installed first. In the specific cases considered by Yates and Zukoff, concerning the history of patterns of reduplication in ancient Anatolian languages, the most interesting example (Pre-Hittite II/III) involves installing an

¹⁴Technically, the RCD algorithm would need to be adjusted so as to first place all markedness constraints with no **L** in their column in a stratum, and then all faithfulness constraints without any **L** in another pass of stratum-building.

¹⁵This principle is also effectively present in the version of RCD from Becker 2009 discussed above.

base-reduplicant faithfulness constraint (CONTIGUITY-Base-Reduplicant) before a markedness constraint (*Poorly Cued Repetition) because the former contains more winner-marks in the support. Insofar as CONTIGUITY-Base-Reduplicant is considered a faithfulness constraint, both LFCD and BCD would install the winner-preferring markedness constraint first. MIRCD overall can be considered a further extension and strengthening of the principle **Favor Activeness**. Whether actually necessary in some cases, and whether “activeness” of this sort can be used to outweigh **Favor Markedness**, or whether other biases might be at work (installing base-reduplicant or other output-output faithfulness constraints before markedness constraints) remains unclear (cf. discussion in Yates and Zukoff 2018: 256–7).

3.2.1.2 Phonological Variation and Constraint Ganging: On the Way to Maximum Entropy Grammars

All of the grammars and grammar-learning algorithms considered thus far presume two properties: 1) the winning outputs are categorical, i.e., more than one winning output should not be permitted; 2) winners are determined by reference to constraint domination alone, where violation of higher-ranked constraints eliminates losing candidates. In this section, I outline some alternative architectures for constraint-based grammars that differ from Standard OT in terms of at least one of those properties. Particular focus will be given to so-called Maximum Entropy Grammars (MaxEnt); the section concludes with discussion of some reasons to believe that MaxEnt grammars perhaps represent that best currently available means to formalize constraint-based grammars generally, in particular in connection to language change.

Constraint Ganging A theoretical antecedent of Optimality Theory, Harmonic Grammar (Legrendre et al. 1990; see also applications in Smolensky and Legrendre 2006), differs from OT as laid out in Prince and Smolensky 1993 [2004] in one crucial way: in contrast to Optimality Theory, constraints are not strictly ranked, such that the violation of any higher-ranked constraint is necessarily fatal in the evaluation of a candidate, but rather *weighted*. This weighting of constraints allows for the possibility of so-called “constraint ganging”, whereby a candidate A that violates a higher-weighted constraint may nevertheless prove more harmonic than some competitor B, if that competitor violates more lower-weighted constraints, resulting in an overall higher disharmony score $\mathcal{H}$. As shown in the schematic example in 11, candidate A wins over candidates B and C, despite violating the highest-weighted constraint α (weight = 3), because the competitors violate more of the lower-weighted constraints β (weight = 2) and γ (weight = 1), with the consequence that the harmony, $\mathcal{H}$, of candidate A is overall lowest.¹⁶ The harmony scores are calculated by multiplying the weight of each constraint by the number of violations of that constraint (number of viola-

¹⁶ Alternatively, the weights can be assigned negative values, so that the Harmony ($\mathcal{H}$) values are then negative numbers, and the winning candidate then has the highest harmony (as opposed to lowest disharmony). In 11, the winner would have a $\mathcal{H}$ -value of -3, while all losing candidates would have a smaller harmony, -4. I will consistently operate with positive values for constraint weights and consider the lowest positive number value of $\mathcal{H}$ to mark the winning candidate.

tions $\times$ Weight_{constraint}), then summing the values across all constraints for that candidate; this is represented mathematically in 12. I operate with positive numbers throughout; thus the candidate with the lowest $\mathcal{H}$ (i.e., closest to 0, which would mean the violation of no constraint) is selected as the winner. If a constraint is not violated by a given candidate, the relevant cell in the tableau is left blank (= 0). This procedure is most common, but see Hayes 2017 on further possibilities, particularly when “noise” for producing variation in outputs is desired (see also below).

(11) Schematic Tableau in Harmonic Grammar

			α	β	γ
a.		3 A	1×3		
b.		4 B		2×2	
c.		4 C		1×2	2×1
d.		4 D	1×3		1×1

(12) $\mathcal{H}(x) = \sum_{i=1}^{i=n(\text{constraints})} w_i f_i(x)$

The (dis)harmony $\mathcal{H}$ of a candidate x , is the sum of the products of the weight of a constraint, w_i by the number of violations of that constraint by the candidate x , $f_i(x)$.

Constraint ganging appears to be a desirable property in cases where either no plausible constraint is available that could be invoked to exclude losers alone is available, or crafting a more complex, specific constraint constraint, which has a subset of the violations of some simpler constraints, is undesirable. In general, many OT analyses that invoke more specific markedness constraints could be reformulated in terms of Harmonic Grammar, employing only simpler markedness constraints. Japanese provides one classic case of the first sort, where an emergent interaction of markedness constraints generates a complex pattern of phonotactic well-formedness for voiced and geminate stops. Specifically, what must be explained is the distribution of the following: obstruent voicing at compound boundaries (*rendaku*), the restriction on multiple voiced obstruents internal to morphemes (“Lyman’s Law”), and the occurrence of voiced geminate stops. See Ito and Mester 2003a, Nishimura 2003, Kawahara et al. 2006, and Kawahara and Sano 2014 for treatments of the interaction of these phenomena; data below is drawn from those sources.

First, *rendaku* (“sequential voicing”) frequently causes an underlying voiceless obstruent to surface as the corresponding voiced obstruent when it appears as the first segment in the second member of a compound, as in 13.¹⁷ In order to generate *rendaku*, a markedness

¹⁷The conditions governing the application of *rendaku* constitute a complex phonological problem in their own right, subject to myriad phonotactic preferences as well as the lexical preferences for certain lexemes to trigger or undergo *rendaku*. See Tanaka 2017 for one recent approach. For simplicity, I assume that *rendaku* could regularly apply in all of the examples given below.

constraint penalizing voiceless obstruents at compound boundaries (which will always fall between two underlying [+sonorant] segments, due to other properties of Japanese phonotactics) must outrank IDENT-[voice], as illustrated in the corresponding tableau.

- (13) *Rendaku* in Japanese Noun–Noun Compounds
- /matu-ke/ → *matuge* ‘eyebrow-hair, eyebrow’
 - /ume-su/ → *umezu* ‘plum vinegar’
 - /kuma-te/ → *kumade* ‘bear-hand, rake’

- (14) Rendaku ≫ IDENT-[voice]

kuma-te		Rendaku	IDENT-[voice]
a.	kumate	*!	
b.	 kumade		*

One major condition under which the application of *rendaku* is categorically blocked is when the second member of a compound, whose initial segment could theoretically undergo *rendaku*, already contains a voiced obstruent. This blocking effect is attributable to a more general phonotactic restriction against multiple voiced obstruents internal to a morpheme (*[D...D]_{morpheme}, classically termed “Lyman’s Law” (Lyman 1894; see also Vance 2022: Ch. 5 on Lyman’s original work), which holds at leads in the native (*yamato*) lexicon of Japanese. As the examples in 15 show, Lyman’s Law appears to block *rendaku*, which implies the ranking of the markedness constraint instantiating Lyman’s Law over the markedness constraint driving *rendaku*, as illustrated in the tableau 16 deriving *tekagami* ‘hand-mirror’. Similarly, voiced geminate stops do not appear in native Japanese lexemes, which may be seen as a further instantiation of Lyman’s Law, or an independent markedness constraint *VOICEDGEMINATE.¹⁸

- (15) Non-application of *Rendaku*

- /te-kagami/ → *tekagami* ‘hand-mirror’ (^Xtegagami)
- /ito-sugi/ → *itosugi* ‘thread-cedar, cypress’ (^Xitozugi)
- /tya-tubo/ → *tyatubo* ‘tea jar’ (^Xtyatubo)

- (16) Lyman’s Law ≫ Rendaku ≫ IDENT-[voice]

te-kagami		Lyman’s Law	Rendaku	IDENT-[voice]
a.	 tekagami		*	
b.	tegagami	*!		*

¹⁸If both the absence of voiced geminate stops and more than one voiced obstruent within a morpheme reflect the same phonotactic restriction, it must be encoded as a ban on the association of more than one consonant segment on a timing tier (see 1.2.5 in Chapter 1) to a [+voice, –sonorant] segment within a given morpheme.

However, violations of Lyman's Law appear in loanwords (17). Taken at face value, this fact is incompatible with the ranking Lyman's Law $\gg$ Rendaku $\gg$ Ident-[voice]. In order to ensure the correct winner in 17, a special faithfulness constraint, ranked above Lyman's Law, would need to be invoked. Such a faithfulness constraint would need to refer, however, to lexically specific properties.

(17) Lyman's Law Violations in Loanwords

- a. /gibu/ $\rightarrow$ *gibu* 'give' (X gipu)
- b. /bagu/ $\rightarrow$ *bagu* 'bug' (X baku)
- c. /dagu/ $\rightarrow$ *dagu* 'Doug' (X daku)

(18) Lyman's Law $\gg$ Rendaku $\gg$ IDENT-[voice] fails!

bagu		Lyman's Law	Rendaku	IDENT-[voice]
a.	☺ bagu	*!		
b.	☹ baku			*

The admissibility of Lyman's Law violations in loanwords further carries over to voiced geminates, which are generally permitted. However, when an additional voiced stop occurs within the lexeme, the geminate is then optionally devoiced, as seen in 20. This behavior speaks to a gradient well-formedness effect: two voiced stops (either as part of a geminate or not) are permissible, but three crosses a boundary, such that repair becomes desirable for some speakers (see the experimental results of Kawahara and Sano 2014, which point to gradient judgments on the ill-formedness of Lyman's Law-violating sequences).

(19) Voiced Geminates in Loanwords

- a. /sunobbu/ $\rightarrow$ *sunobbu* 'snob' (X sunoppu)
- b. /reddo/ $\rightarrow$ *reddo* 'red' (X retto)
- c. /eggu/ $\rightarrow$ *eggu* 'egg' (X ekku)

(20) (Optional) Devoicing of Voiced Geminates

- a. /doggu/ $\rightarrow$ *dokku* $\sim$ *doggu* 'dog'
- b. /gebberusu/ $\rightarrow$ *gepperusu* $\sim$ *gebburusu* 'Göbbels'
- c. /baddo/ $\rightarrow$ *batto* $\sim$ *baddo* 'bad'

In order to simultaneously generate the contradictory patterns presented by the failure of *rendaku* in Lyman's Law environments but the violation of Lyman's Law in loanwords, constraint ganging is a possible recourse. Since the losing candidate in a compound violates both IDENT-[voice] and Lyman's Law, the ganging of those two constraints may be invoked to overcome the otherwise powerful markedness constraint, RENDAKU, as shown in 21. Where an input contains two voiced obstruents, as in 22, the greater weight of IDENT-[voice] rules out the application of Lyman's Law. Finally, the optional devoicing of voiced obstruents can be attributed to the effect of a further, lower-weighted constraint penalizing surface voiced geminates, so as to produce a tie between faithful and unfaithful candidates.

$$(21) \text{ IDENT-[voice] (3) + Lyman's Law (2) > Rendaku (4)}$$

te-kagami		Rendaku	IDENT-Voice	Lyman's Law	*VoicedGeminate
a.	5 tegagami		3	2	
b.	4 tekagami	4			

$$(22) \text{ IDENT-[voice] (3) > Lyman's Law (2)}$$

bagu		Rendaku	IDENT-Voice	Lyman's Law	*VoicedGeminate
a.	2 bagu			2	
b.	3 baku		3		

$$(23) \text{ Lyman's Law (2) + *VoicedGeminate (1) = IDENT-[voice] (3)}$$

doggu		Rendaku	IDENT-Voice	Lyman's Law	*VoicedGeminate
a.	3 dokku		3		
b.	3 doggu			2	1

Thus, the employ of weighted constraints in this case allows for an analysis that does not require recourse to lexically specific constraints, lexical strata (e.g., a lexically encoded distinction between loanwords and non-loanwords), or other morphological information. More generally, analyses allowing constraint ganging can also substitute the effect of so-called constraint conjunction, at least in many instances; in Standard OT, a conjoined constrained is a higher-ranking constraint whose violation is triggered when the conditions for violation of two lower-ranking constraints hold (Ito and Mester 2003a, Smolensky 2006). In many cases, the additive effect allowed by constraint ganging renders constraint conjunction superfluous (Potts et al. 2010, Pater 2016); see now, however, Shih 2017 on cases that call for “super-additivity”, i.e., where the effect of violating two constraints is greater than their simple additivity.

Constraint weights in a Harmonic Grammar with categorical outputs (i.e., no variation or pseudo-variation, like for voiced geminate stops in Japanese above) may be determined straightforwardly by solving a system of linear equations, computationally, through linear programming. Such an approach is described in Potts et al. 2010 and implemented in software such as Praat (Boersma and Weenink 1992–2023). Such “Linear OT” solves linear inequalities into which a tableau with violation profiles can be converted, and is guaranteed to find a set of viable weights, if one exists, assuming no token variation in the data (i.e., every distinct input has only one possible output).

Various algorithms exist for calculating the weights of constraints for a given set of data. For the analysis presented below, I calculated weights using both Linear OT (Potts et al. 2010) in Praat (Boersma and Weenink 1992–2023) and Maximum Entropy (MaxEnt) Grammar with the software of Hayes et al. 2009a. I assumed no genuine token variation in the data (i.e., every distinct input has only one possible output), under which Linear OT is guaranteed to find a set of viable weights, if one exists. Considering the winner $\sim$ loser pairs in 17 and 21 above, the inequalities below must hold true.¹⁹ When the system of equations is solvable, the weights W of the constraints must be such that the disharmony $\mathcal{H}$ of the winning candidate is less than the corresponding disharmony of each losing candidate.

- *bagu* $\sim$ *baku* (17): $(1 \cdot W(\text{Lyman's Law}) < (1 \cdot W(\text{IDENT}[\text{voice}]))$
- *tekagami* $\sim$ *tegagami*: $(1 \cdot W(\text{Rendaku}) < (1 \cdot W(\text{Lyman's Law}) + (1 \cdot W_{\text{IDENT}[\text{voice}]})$

Finding the weights is then an optimization problem and an algorithmic problem: how do we define what the best weights are, and what procedure will lead us to the optimal solution? Stated in 24 is the objective function proposed by Potts et al. (2010), which provides the orientation for the optimization problem. The function given below will find the smallest value for the sum of the weights, while holding to some minimum non-zero weight and enforcing a minimum margin of separation between winning and losing candidates.²⁰ For an algorithmic implementation, the authors of Linear OT employ the simplex algorithm; see the references in Potts et al. 2010: fn. 5. Packages for solving systems of linear equations in a variety of high-level programming languages, (e.g., R, Python, Matlab) can also be applied to a data structure representing the relevant linear inequalities, given the objective function in 24.

(24) Objective Function for Linear OT (Harmonic Grammar Weights)

- a. Minimize $\sum_{i=1}^n w_i$
- b. Subject to: $w_i > 0$
 $\mathcal{H}(\text{Loser}_a) - \mathcal{H}(\text{Winner}_a) \geq 1; \mathcal{H}(\text{Loser}_b) - \mathcal{H}(\text{Winner}_b) \geq 1; \dots$

Worth noting is that the type of problems posed by subset grammars and phonotactic learning have not, to my knowledge, been directly dealt with in the context of Harmonic Grammar as opposed to Standard OT. When applied to the simple phonotactic learning problem discussed at the end of the preceding section, where non-occurring breathy voiced stops should be excluded, Linear OT will behave precisely like RCD, weighting both faithfulness constraints ($W = 2$) above all three markedness constraints ($W = 1$), at least if the minimum weight allowed and minimum separation are 1. In that case, an input with a breathy voiced stop, /b^hi/, cannot exclude faithful [b^hi] as a potential output, as shown

¹⁹The reader should mentally fill in a 0 multiplied by the weight of each constraint not violated by a candidate.

²⁰The latter two parameters may be freely chosen, though in practice a value of 1 for both will always yield a solution, if it exists.

in 25. Thanks to constraint ganging, the markedness constraint $\begin{bmatrix} +\text{voice} \\ +\text{spread glottis} \end{bmatrix}$ need not be weighted above the faithfulness constraints: with larger minimum weights (e.g., 2) but a small minimum separation (1), the larger number of markedness constraints can rule out the faithful candidate $^x[\text{b}^{\text{hi}}]$ as ungrammatical, as shown in 26. The question of what algorithmic procedure, perhaps inspired by LFCD or BCD, could be appended to Linear OT in order to exclude overly permissive grammars, perhaps taking advantage of cumulative constraint interaction, is a question for further research.

(25) $/\text{b}^{\text{hi}}/ \rightarrow ???$

$/\text{b}^{\text{hi}}/$	IDENT-[voice]	IDENT-[spread glottis]	*[+voice,+sg.]	*[+voice]	*[+sg.]
a. 3 $\rightarrow$ $[\text{b}^{\text{hi}}]$			1×1	1×1	1×1
b. 3 $\rightarrow$ $[\text{bi}]$		2×1		1×1	
c. 3 $\rightarrow$ $[\text{p}^{\text{hi}}]$	2×1				1×1

(26) $/\text{b}^{\text{hi}}/ \rightarrow ^x[\text{b}^{\text{hi}}]$

$/\text{b}^{\text{hi}}/$	IDENT-[voice]	IDENT-[s. g.]	*[+voice,+sg.]	*[+voice]	*[+sg.]
a. 6 $[\text{b}^{\text{hi}}]$			2×1	2×1	2×1
b. 5 $\rightarrow$ $[\text{bi}]$		3×1		2×1	
c. 5 $\rightarrow$ $[\text{p}^{\text{hi}}]$	3×1				2×1

Phonological Variation and the Gradual Learning Algorithm While the question of cumulation constraint interaction as opposed to strict weighting is primarily a theoretic question, since empirically many phenomena that seem to call for constraint ganging might be dealt with through some other extension of grammars with strict ranking (such as constraint conjunction), the issue of *variation* in phonological outputs is an empirical problem that must be stared down directly. Variable outputs of phonological grammars are ubiquitous (see generally Coetzee 2011), from different rates of coronal stop deletion in English (Coetzee 2004: Ch. 5) to nasal substitution in Tagalog (Zuraw 2000, Zuraw 2009); psycholinguistic tasks involving the production of *wug*-items or ratings of their well-formedness have repeatedly pointed towards the probabilistic selection of multiple (relativ more or less) well-formed outputs (e.g., Hayes et al. 2009b on vowel harmony in Hungarian). One early attempt to combine variable outputs with constraint-based grammars was developed in Anttila 1997, which allows for variation in the choice of rankings at evaluation time. Below, I introduce the Gradual Learning Algorithm for Stochastic OT and Maximum Entropy Grammar as means of modeling variation that remain in use, and whose behaviors continue to be explored in

the literature (see, e.g., Jarosz 2016 and Boersma and Pater 2016 for recent developments involving the GLA).

The Gradual Learning Algorithm (GLA), introduced in Boersma 1997 and Boersma and Hayes 2001 (see also a brief introduction in Apoussidou 2007: 40–3)), is an online, error-driven learning algorithm (cf. generally Rosenblatt 1958 on the Perceptron algorithm and Wexler and Culicover 1980): fundamentally, this means that the grammar of a learning agent is updated (changed in some fashion) in response to each further single token of data that it encounters (as in EDCD discussed above). In the implementation of online error-driven learning for constraint-based grammars, the basic idea is that the tokens from a “teacher” (some source of forms that correspond to the “correct” forms in the target grammar) are compared with the predicted output of the “learner” for a given input. If the learner’s output differs from the token of data provided by the “teacher”, then the ranking-value of constraints is correspondingly adjusted (cf. Boersma and Hayes 2001: 53):

- The ranking value, r , of constraints violated by the teacher’s learning datum are **decreased** (i.e., violations of those constraints are made less painful, thereby making the candidate associated with the learning teacher’s learning datum more harmonic).
- The ranking value, r , of constraints violated by the learner’s preferred output candidate are **increased** (i.e., violations of those constraints are made more painful, thereby making the learner’s preferred winner less harmonic).

In short, the GLA will increase the ranking value of winner-preferring candidates, while decreasing the ranking value of loser-preferring candidates. How much, precisely, the ranking-values (or weights, in a Harmonic Grammar context; see Boersma and Pater 2016 on learning harmonic grammars with the GLA) are adjusted is specified as a learning rate (in the context of Boersma’s work, this learning rate is called PLASTICITY).²¹ As adapted from Jarosz 2016: 204 the update rules for the GLA in its respective OT and HG implementations are as follows, where L_i stands for violation profile the most harmonic output of the learner’s grammar, which is a candidate that differs from the learning datum, and its violation profile (W_i):

(27) OT-GLA Update Rule

$$r_{new} = r_{old} + \Delta r_i, \text{ where } \Delta r_i = \epsilon \times \text{sign}(L_i - W_i)$$

(28) HG-GLA Update Rule

$$w_{new} = w_{old} + \Delta w_i, \text{ where } \Delta w_i = \epsilon \times (L_i - W_i)$$

Here, r or w is a numeric vector with indices $1 \dots n$, where n is the number of constraints in the grammar, corresponding to the ranking values or weights of all constraints in the grammar. The subscripts *old* and *new* indicate the ranking values or weights prior to and after

²¹ Another parameter that must be set in a run of the GLA is the PLASTICITY DECREMENT, which lowers the learning rate (i.e., makes learning slower) after exposure to some prespecified number of tokens.

the addition of a numeric vector δr_i or δw_i of length n which contains the corresponding amounts by which each constraint's ranking value or weight must be increased or decreased. The subscript i corresponds to a given input form in the grammar; thus, the changes in ranking values or weights are based on a particular tableau. To calculate the changes in ranking values or weights, the numeric vector L_i (the violation profile of the learner's preferred candidate in tableau i) is subtracted from the numeric vector W_i (the violation profile of the datum provided by the teacher in tableau i), and multiplied by a learning rate ϵ . Whenever a constraint is not violated by the learner and is violated by a teacher (or violated more times by the teacher than by the learner), a negative number results, which will lead to a decrease in ranking value / weight of that constraint; the inverse occurs when a constraint is violated by the learner but not by the teacher, and 0 results just in case teacher and learner agree on the violation for a given constraint, leading to no change. The major difference between the OT and HG implementations of the GLA is that the former transforms all values greater than 1 or less than -1 to 1 and -1 , thereby regularizing the rate of constraint updates.

In order to generate variation in outputs, Stochastic OT, or Noisy Harmonic Grammar, adds some different amount of random “noise” to the ranking value of each constraint; this noise is sampled from a normal distribution with a mean of 0 (cf. Boersma and Hayes 2001: 48–9).²² The result of this “noise”, when applied to constraints with relatively close ranking values, is that their relative rankings may readily differ each time a winner is to be selected for a given output, thereby giving rise to variable outputs. The farther apart the ranking values of two constraints, the less likely that the noise added at evaluation time will create a different ranking or materially affect the relative total disharmony of candidates.

In case the learning data contains multiple surface forms O_1 and O_2 in free variation associated with some input I , exposure to each of the different “winners” (O_1 and O_2) with error-driven updates of constraint ranking values or weights will tend to cause the ranking values of the constraints that distinguish O_1 and O_2 to become sufficiently close that noise added at evaluation time can result in either output. Depending upon the relative proportions of O_1 and O_2 in the learning data, the ranking values of the distinguishing constraints may be relatively close (e.g., if both outputs are equally likely) or more distant (e.g., if O_2 is relatively infrequent, occurring one-tenth as often as O_1).

Although the GLA is able to successfully model many attested patterns of free variation in phonology (see, e.g., in Boersma and Hayes 2001), gradient patterns in productivity (Albright and Hayes 2006), and has been claimed to successfully reproduce stages of phonological acquisition (Boersma and Levelt 2000, Curtin and Zuraw 2001, Tessier 2007, Jesney and Tessier 2011), some patterns of variation of been discovered for which the GLA cannot find a solution, i.e., will not converge on a stable set of ranking values / weights (Pater 2008, Magri 2012). In large part for this reason, namely, that proof of convergence for the GLA involving data exhibiting free variation has not been provided, alternatives with a more sound and

²²In its original conception the “noise” is in fact a value selected from the normal distribution with mean equal to the ranking value of the constraint; the standard deviation used in Boersma and Hayes 2001 and by default in Boersma and Weenink 1992–2023 is 2.0. See Hayes 2017 for a comparison of the performance of several different implementations of Noisy Harmonic Grammar.

better understood mathematical grounding have been sought.²³

Maximum Entropy Grammars Such an alternative exists, and has been extensively developed in the context of phonological research, in the form of log-linear models. The specific type employed is maximum entropy modeling (MaxEnt), where the model maintains as much uncertainty as possible (Manning and Schütze 1999: 589–97); that is, their entropy remains maximally high given the inclusion of as much information as possible from the available data. Or, in the helpful formulation of Hayes and Wilson (2008: 386), “maximizing entropy is in fact equivalent to maximizing the probability of the observed forms given the constraints.” MaxEnt models are equivalent to multinomial logistic regression – there is no mathematical difference, merely a different label (Jurafsky and Martin 2009: 193–207). Similarities between the choice of winning candidates in OT/HG and classification problems were first pointed out by Goldwater and Johnson (2003) and Jäger (2007), all of whom appealed to the capacity of MaxEnt models to 1) capture cumulative constraint interaction, 2) model free variation, and 3) benefit from learning algorithms for determining weights that have known convergence properties (cf. Goldwater and Johnson 2003: 117–8, Hayes and Wilson 2008: 385–9). The goal of this section is to describe how MaxEnt models determine candidate probabilities once constraint weights have been established, an objective function for finding optimal constraint weights, and illustrate a MaxEnt grammar at work in a case involving free variation. I do not go into the algorithms (in particular, the Conjugate Gradient favored by Goldwater and Johnson 2003: 117–8, Hayes and Wilson 2008: 385–9) used for finding weights from complete data (i.e., batch learning), but do illustrate practically how weights for a MaxEnt model using the `optim()` function in R, and how an on-line version of MaxEnt employing the perceptron update rule (or Stochastic Gradient Ascent) may be implemented. I refer the reader to White 2013: 95–101, Staubs 2014: Ch. 1, Nazarov 2016: 57–62, Jarosz 2019: 69–75 for other helpful overviews of MaxEnt models from the perspective of phonological learning.

As an example to illustrate the behavior of probabilistic outputs under a MaxEnt Grammar, consider a language, partly inspired by Gothic (see Miller 2019: 26–8, Goering and Sandell Accepted (2021)), in which underlying voiced obstruents /b, d, g, z/ may be realized as corresponding fricatives [β, ð, γ, z] when following a [–consonantal] segment, and further as voiceless fricatives [ɸ, θ, x, s] at the right edge of a prosodic word. Assume further that, in the environment [–cons.]_____ω, underlying /b, d, g, z/ is realized (here exemplified with the labials) as the following sounds at the following rates: [b] (0.1), [β] (0.2), [ɸ] (0.7), [p] (0.0).²⁴ The tableau in 29 shows the number of violations of the candidates (or *classes*) for a set of four relevant constraints (or *features*), given an input /ab]_ω/. The constraints (features) consist mainly of two typical markedness constraints (*[–cons.][+voi., –cont.] and *[+voi., +cont.]]_ω=), which penalize voiced stops following [–consonantal] segments and which penalize voiced fricatives word finally, and three markedness constraints of the *MAP

²³Boersma and Pater (2016) claim to provide proof of convergence for HG-GLA in cases where each input maps to a single, categorical output, i.e., where no free variation is involved.

²⁴These probabilities are freely invented for the purposes of illustration, but Gothic certainly did exhibit allophonic alternations of the sort outlined assumed here.

family (Steriade 2008, Zuraw 2007), which penalize the mapping of sounds of one natural class on to another. Here, the constraints $*\text{MAP}(\text{b}, \beta)$, $*\text{MAP}(\text{b}, \phi)$, and $*\text{MAP}(\text{b}, \text{p})$ stand for penalties on the mapping of underlying voiced stops onto voiced fricatives, voiceless fricatives, and voiced stops, respectively.²⁵ The standard IDENT constraints are included in 29 for comparison.

$$(29) \quad /ab]_{\omega}/ \rightarrow [a\phi], [a\beta], [ab]$$

$/ab]_{\omega}/$			$*[-\text{cons.}][+\text{voi.}, -\text{cont.}]_{\omega}$	$*[+\text{voi.}, +\text{cont.}]_{\omega}$	$*\text{MAP}(\text{b}, \beta)$	$*\text{MAP}(\text{b}, \phi)$	$*\text{MAP}(\text{b}, \text{p})$	IDENT-[voice]	IDENT-[cont.]
a.	0.1	$ab]_{\omega}$	1						
b.	0.2	$a\beta]_{\omega}$		1	1				1
c.	0.7	$a\phi]_{\omega}$				1		1	1
d.	0	$ap]_{\omega}$					1	1	

The question now is how the particular pattern of constraint violations seen in 29 – the predictor features – can be exploited to yield the candidate (class) probabilities given here. Once weights for each of the constraints are known, the probability of a given output candidate y given the input x , n constraints, and a normalization term Z (to ensure that probabilities sum to 1) is defined in 30; the calculation of the normalization term Z is given in 31. The portion of 30 without the normalization term is the exponentiated negative harmony, $\mathcal{H}$ (cf. 12 above), of the candidate y ; $\exp(-\mathcal{H})$ may also be called the MaxEnt value of a candidate (Hayes and Wilson 2008: 384).²⁶ The normalization term Z is then the sum of the MaxEnt values of the output candidates for the input x . In other words, to find the probability of a candidate y , the (dis)harmony of each candidate is calculated by multiplying its violation profile $f_i(y, x)$ by the constraint weights w_i , then its MaxEnt value is obtained by taking the inverse exponential of $\mathcal{H}$, and summing the MaxEnt values to obtain Z , whereupon the specific MaxEnt value of y is divided by Z to yield a probability between 0 and 1.

²⁵The use of $*\text{MAP}$ here is necessary in order to exclude the realization of underlying voiced stops as corresponding voiceless stops. Were standard faithfulness constraints of the IDENT family employed, a candidate with $[\text{p}]$ would harmonically bound the candidate in 29 with greatest probability, $[a\phi]$ (0.7). Both in Standard OT or any stochastic grammar model, it would therefore be impossible to exclude $[ap]$ as a winner; in a MaxEnt model, $[ap]$ and $[a\phi]$ would be assigned equal probability. See White 2013: 31–47, 101–6 on the analysis of “saltatory” phonological patterns, where a candidate that is featurally closer to the input than (the) winner(s) is not permitted.

²⁶The reason that the inverse (dis)harmony must be taken is to ensure that candidates with the lowest positive (dis)harmony will receive the highest amount of probability mass. Alternatively, if all constraint weights themselves are made negative numbers, then multiplying the (dis)harmony value by -1 is unnecessary.

- (30) Maximum Entropy Probability

$$P(y|x) = \frac{1}{Z} \exp(-\sum_{i=1}^n w_i f_i(y, x))$$
- (31) Maximum Entropy Normalization Term Z

$$Z = \sum_{y \in Y(x)} \exp(-\sum_{i=1}^n w_i f_i(y, x))$$

In the example at hand, the probability of candidate *c.* above, $[a\phi]_\omega$, may then be expressed as $0.7 = \frac{\exp(-\mathcal{H}(a))}{Z}$. What must then be determined is the (dis)harmony of each candidate. Where all constraint weights are zero, the candidate probabilities are all equal: because the constraints (features) do not help to predict the likelihood of selecting any outcome, every candidate has the same $\mathcal{H}$ and thus the same MaxEnt value and probability. The table in 32 shows the weights (rounded to three decimal places) for all constraints shown in 29 using two different methods: on the left are weights found using the MaxEnt Grammar Tool of Hayes et al. 2009a, on the right weights found using the `optim()` function in R using the optimization method L-BFGS-B (Byrd et al. 1995) with lower bound of 0 and upper bound of 50.²⁷ To reflect the candidate probabilities, integers of 10, 20, 70, and 0 were provided as the frequencies of the outputs.

- (32) Example Weights for MaxEnt Grammar based on 29²⁸

Constraint	MaxEnt Grammar Tool	<code>optim()</code>
*[−cons.][+voi., −cont.]	1.945	8.011
*[+voi., +cont.]] _ω	0.626	3.339
IDENT-[voice]	0	4.232
IDENT-[cont]	0	0.64
*MAP(b, β)	0.626	3.339
*MAP(b, φ)	0	1.1931
*MAP(b, p)	12.539	22.526

²⁷The objective function used for finding optimal weights is the maximization of the summed log probability of each candidate (where $P(c) \leq 1$) given the empirical likelihood of that candidate and some hypothetical set of weights, w . This objective function is defined in 1, and can be thought of simply as minimizing the log probability of the weights given the data. No Gaussian regularizer was employed here. Compare further Goldwater and Johnson 2003: 113 or Wilson 2006: 956–7. If the candidate probabilities are changed to negative numbers (by multiplying by -1 in the objective function below), then the objective is to be minimized rather than maximized.

- (1) Objective Function for Optimizing Weights of Maximum Entropy Grammars
 Maximize $PLOG$, where

$$PLOG = \sum_{n=1}^n \log(w(y_{(n)}|x_{(n)}))$$

²⁸The difference in weights reported here is due to the total absence of a Gaussian regularizer in the objective function used with the `optim()` function, whereas the MaxEnt Grammar tool always employs a Gaussian regularizer, even if with small $\lambda = 0$ and large $\lambda^2 = 10000$. When the objective function maximizing the logarithmic probability of the data provided to `optim()` includes a Gaussian regularization term, and weights are confined to the range 0–50, the MaxEnt Grammar tool and `optim()` using L-BFGS-B discover near-identical weights.

In both sets of weights, the two constraints assigned the highest weights are $*[-\text{cons.}][+\text{voi.}, -\text{cont.}]$ and $*\text{MAP}(\text{b}, \text{p})$, which are uniquely violated by the two candidates with the lowest probabilities. Likewise, the two constraints uniquely violated by candidate $\text{b.}, [\text{a}\beta]_\omega$, $*[+\text{voi.}, +\text{cont.}]_\omega$ and $*\text{MAP}(\text{b}, \beta)$ are assigned identical weights, reflecting the fact that the work done to diminish that probability of that candidate is divided equally over those two constraints. To round out this example, I give the (dis)harmony $\mathcal{H}$ and MaxEnt value for each candidate in 29 using the weights found with the MaxEnt Grammar Tool; the probability of each candidate is then its MaxEnt value divided by the sum of all MaxEnt values for the input (Z). As can be seen, a larger (dis)harmony translates into a smaller MaxEnt value and lower share of probability. In this case, the predicted probabilities match the empirical probabilities within hundreds of thousands of a decimal place.

(33) (Dis)harmony and MaxEnt Value of Candidates in 29

Candidate	$\mathcal{H}$	MaxEnt Value	Empirical Probability
$\text{ab}]_\omega$	1.945902	0.1428583	0.1
$\text{a}\beta]_\omega$	1.252758	0.2857157	0.2
$\text{a}\phi]_\omega$	0	1	0.7
$\text{ap}]_\omega$	12.539394	0.000003582699	0

Thus, once a set of optimal weights has been determined, the calculation of candidate probabilities for some given input (underlying representation) is straightforward. When all data of all input to output mappings is available to the learner, allowing that data to be processed in a batch fashion, existing optimization algorithms are guaranteed to discover the optimal set of weights. Suitable algorithms, discussed in Goldwater and Johnson 2003, Wilson 2006, or Hayes and Wilson 2008 include the Conjugate Gradient algorithm of Press et al. 1992 or the L-BFGS-B algorithm of Byrd et al. 1995. In general, the weights to be found will maximize the probability of the observed data ($P(D)$), which can be calculated by multiplying the probabilities currently predicted by the model for each output given its input. $P(D)$ may then conveniently be converted to the objective function discussed under note 27 above. The search space of this objective (or better, the log probability of the observed data given some set of weights) is convex, and therefore an optimum can be found as long as the algorithm always proceeds “upward” in its search.

(34) Function for Maximizing the Probability of Observed Data

$$P(D) = \prod_{n=1}^n P(y_n|x_n)$$

However, as Staubs (2014: 54) observes, maximization of probability of the data given the same proportional relationship between the data (e.g., whether the probabilities for the example treated in this section are based on token frequencies of $\{1, 2, 7\}$ or $\{10, 20, 70\}$, or $\{100, 200, 700\}$) can become more categorical – when more total observations are present, maximizing of the objective function will yield different weights. Thus, while the optimization algorithms mentioned above will indeed always provably converge on the same set of weights for specific given frequencies in the learning data, the weights found depend on the scaling of the data, not merely the relative frequencies.

As an alternative to a batch learning MaxEnt model (cf. also batch learning with RCD or Linear OT discussed above), MaxEnt grammars can also be learned in an online fashion (cf. online learning with EDCD or the GLA above). Such an alternative can employ Stochastic Gradient Ascent (SGA), as first proposed by Jäger (2007), where the rule used for the updating of constraint weights is then effectively identical with the GLA for Harmonic Grammar (see 28 above). As in other online learning models, learning, that is, updates to constraint weights, takes place only when the candidate chosen by the learner’s grammar differs from the datum provided by a teacher’s grammar. Recall from above that new weights are calculated by subtracting the violation profile of the learner’s optimal candidate from the violation profile of the learning datum, multiplying this difference by a specified learning rate, then adding this value to the current constraint weights. In the context of a MaxEnt grammar to be learned, the crucial difference in comparison to EDCD or the GLA is in the selection of an output by the learner: the learner determines the probabilities of each candidate under its current grammar according to the (dis)harmonies and MaxEnt values that its current constraint weights yield. As with the GLA, constraints violated by the “loser” (the learner’s output) but not by the “winner” (the empirical learning datum) will increase in weight; when the reverse holds the constraint’s weight will decrease, with a lower bound set at 0.

The learning procedure for MaxEnt Stochastic Gradient Ascent may be described with the procedure in 35. According to Jäger (2007: 7), convergence obtains when the difference between the empirical expectation value of each constraint and the learned expectation value of each constraint is 0. Furthermore, Staubs (2014: 38) observes that, because MaxEnt SGA is practically equivalent to perceptron learning, the convergence proof of Novikoff (1962) holds: when the candidates all constitute linearly-separable classes, convergence in the long run is guaranteed. In the case of learning where the inputs are fully specified, and any coherent noisy harmonic grammar would be learnable, the candidates should constitute linearly-separable classes (cf. the convergence proof for Noisy Harmonic Grammar in Boersma and Pater 2016). Conversely, when the inputs are not fully known – i.e., when hidden structure is involved – convergence to the global maximum is not guaranteed; the learner may become stuck in a local maximum, though the weights learned will normally provide a better fit to the learning data with respect to the starting position.

(35) Pseudo-Code of Constraint Weight Learning under MaxEnt SGA

Given a set of input-output pairs,

Given an initial vector of constraint weights, w

Given an empirical frequency distribution of outputs, O

1. Sample an output o from O
2. Find the input i associated with o
3. Calculate the predicted output distribution of outputs under current constraint weights for the input i using 30, i.e., find the MaxEnt value and probability for each o in from i given current weights.
4. Sample an output o_{learner} from the learner’s distribution of outputs.

5. If $o_{\text{learner}} \neq o_{\text{teacher}}$:
 $w_{\text{new}} = w_{\text{old}} + \Delta w_i$, where $\Delta w_i = \epsilon \times (L_i - W_i)$
6. If $P(o)$ in $O = P(o_{\text{learner}})$ in O_{learner} for all O :
 Stop

To illustrate some performance differences between batch learning of MaxEnt Grammars and MaxEnt SGA, I report here the results for learning based on the data represented in 29, with the empirical probabilities given there (concretely, 10, 20, 70 and 0 tokens each). The table below shows the weights learned for each constraint and the predicted probability of each candidate after sampling 50000 tokens from the empirical distribution, given a learning rate of 0.01 and an initial weight of 0 for all constraints.²⁹ The overall performance may be assessed by calculated the sum squared error (SSE), i.e., summing the squared difference between empirical and predicted probabilities; that the SSE improves with exposure to more data is empirically confirmed in this case: between tests sampling 20000, 50000, and 80000 tokens, respectively, the SSE decreases from 0.0045 to 0.0015 to 0.0003.

(36) MaxEnt SGA for the Pseudo-Gothic Grammar of 29: 50000 Sampled Tokens, $\eta = 0.01$

a. Constraint Weights Learned

Constraint	Weight
*[−cons.][+voi., −cont.]	2.19
*[+voi., +cont.]] _ω	0.84
IDENT-[voice]	0.16
IDENT-[cont]	0.19
*MAP(b, β)	0.84
*MAP(b, φ)	0.09
*MAP(b, p)	6.02

b. Predicted Candidate Probabilities

Candidate	Predicted Probability
ab] _ω	0.122695922
aβ] _ω	0.168967960
aφ] _ω	0.706066279
ap] _ω	0.002269839

3.2.1.3 Is there a Correct Approach to Phonological Learnability?

The preceding discussion has examined varieties of constraint grammars and algorithms that can be implemented in order to learn them, and shown that the major available approaches differ along one of three dimensions:

²⁹In this specific case performance might be improved by implementing a Markedness $\gg$ Faithfulness bias, by setting initial weights for markedness constraints to some integer greater than 0. Further worth noting is that, since the learning rate is set to 0.01, learning weights more exactly than to two decimal places is not possible. A smaller learning rate would allow for more fine-grained weights, but learning would proceed more slowly.

- whether the model can generate variable outputs given the same input (e.g., Standard OT vs. Stochastic OT);
- whether constraints are strictly ranked or weighted, allowing for cumulative effects or not (e.g., Standard OT vs. Harmonic Grammar);
- whether learning proceeds in a batch or online fashion (e.g., RCD or Linear OT vs. EDCD or SGA).

I adopt the perspective advocated by, e.g., Hayes (2017: 1), that variation is pervasive in linguistic data, and learning patterns of structured variation is a necessary part of tackling phonological learnability. Taking a further cue from variationist sociolinguistics and variationist approaches to language change (cf. Kroch 1989, Zimmermann 2017: 2–6, Stadler 2017: 28–33), I assume that the presence of variation in learning data typically precedes change – either from one categorical pattern to another, or one variable pattern to another variable pattern. That is, I doubt in practice that two different diachronic states exhibiting two distinct categorical outputs are not usually mediated by a phase of variation. These principles alone means that some version of Stochastic OT, Noisy Harmonic Grammar, or MaxEnt grammar – which can of course model any data exhibiting purely categorical behaviors – are adequate.

On the question of whether cumulative constraint interaction constitutes a desirable feature, the experimental comparison of (Noisy) Harmonic Grammar and Stochastic OT conducted by Boersma and Pater (2016) delivers useful testimony. The authors compared the success of different on-line learning models (EDCD, OT-GLA and HG-GLA) in learning 124 metrical stress grammars involving hidden structure (see immediately below). They found that Harmonic Grammars always successfully converged on the correct grammar more often than EDCD or OT-GLA, while Noisy Harmonic Grammars performed best overall always converging for 108 languages (clearly learnable languages), sometimes converging for 5 languages (difficult to learn languages), and never converging for 11 languages (perhaps unlearnable languages). By this criterion of performance, grammar models that permit cumulative constraint interaction appear better equipped.

The final question, that of batch versus online learning, is perhaps the most difficult to straightforwardly resolve. At this juncture, it is clear that the batch algorithms (e.g., Conjugate Gradient, L-BFGS-B) available for determining constraint weights under MaxEnt have an advantage from a mathematical perspective: convergence on a grammar, even where the data involve variation, is guaranteed. The perceptron learning algorithm applied in HG-GLA or MaxEnt SGA may fail to converge where the data involve variation or hidden structure (cf. Boersma and Pater 2016 with respect to HG-GLA). At the same time, some have explicitly argued that on-line learning models are linguistically more realistic; Jäger (2007: 10), for instance, declares that “it [viz. GLA, perceptron update, SGA] is arguably a better model of human language acquisition than the off-line training algorithms for ME models that are standardly used”. Indeed, on questions of acquisition order of more marked phonological structures where the empirical patterns of acquisition order exhibited by children should, ideally, be reproducible by a learning model, only the GLA/SGA (Boersma and Levelt 2000,

Jäger 2007: 7–9) and Error-Selective Learning (Tessier 2007) have been demonstrated to do so (though the latter cannot produce variable patterns or match frequencies). However, the possibility of equipping a batch MaxEnt learner with some biases through a Gaussian prior (cf. Wilson 2006) means that the phenomena of acquisition order might partly be reproduced, if it can be shown that data including fewer overall tokens (but the same general frequencies) exhibit similar patterns. To my knowledge, such a test has not previously been conducted, so I fill this gap here, again on the basis of the data on Dutch syllable structure from Boersma and Levelt 2000.

In the analysis developed here, four markedness constraints were employed to penalize certain syllable structures, and an all-purpose faithfulness constraint: *CODA (“no syllable codas”), ONSET (“no syllables without onsets”), *COMPLEXCODA (“no codas with more than one consonant”), *COMPLEXONSET (“no syllables with more than one onset consonant”), and FAITH (“no changes to inputs”). In a complete adult grammar, FAITH must outweigh all markedness constraints, so that all syllable inputs are realized faithfully. For this experiment, candidates the fully faithful candidate for each syllable type and competitors with stepwise less-marked types; e.g., for /CCVCC/, the alternative candidates consist of [CCVC], [CCVC], [CCV], and [CV].

Following Jäger 2007, the initial weights or preferred weights (μ) for the constraints were set to 10 for the four markedness constraints and 0 for the one faithfulness constraint; following White 2013: 108, the σ^2 was set to 0.6. Then, weights were calculated using the MaxEnt Grammar Tool (Hayes et al. 2009a) for four different total numbers of tokens (50, 100, 500, and 2000), with the number of tokens for each syllable type based on their relative frequencies in data of child-directed speech from Boersma and Levelt 2000, given in the table in 37. Thus, when 50 total tokens were used, there were 22.405 tokens of CV, 16.025 of CVC, etc. The overall results of the tests are reported in 38, where the proportion of fully faithful outputs for each syllable type is given for each total number of tokens, and 39 shows the weight of FAITH in each test.

(37) Relative Proportion of Syllable Types in Child-Directed Dutch Speech

Syllable Structure	Proportional Frequency
CV	44.81%
CVC	32.05%
VC	11.99%
V	3.85%
CVCC	3.25%
CCVC	1.98%
CCV	1.38%
VCC	0.42%
CCVCC	0.26%

(38) % Correct of Each Dutch Syllable Type

Syllable Structure	Correct (50)	Correct (100)	Correct (500)	Correct (2000)
CV	0.999	0.999	0.999	0.999
CVC	0.526	0.745	0.938	0.983
VC	0.125	0.25	0.883	0.968
V	0.213	0.39	0.889	0.968
CVCC	0.031	0.167	0.718	0.916
CCVC	0.028	0.141	0.656	0.892
CCV	0.054	0.189	0.699	0.908
VCC	0.012	0.141	.716	0.916
CCVCC	0.003	0.087	.7122	0.837

(39) Weight of FAITH

50 Tokens	100 Tokens	500 Tokens	2000 Tokens
6.628	7.694	9.328	10.435

The results above show that marked structures are typically realized unfaithfully in the presence of little data, but progressively more faithfully as more data becomes available, and the weight of FAITH gradually becomes higher. As intuition concerning MaxEnt grammars and the use of priors would suggest, something approximating the gradual acquisition of more complex syllable types in Dutch can be modeled through a sort of gradual batch learning. Since a high weight assigned to FAITH and relatively lower weights assigned to the markedness constraints with respect to the mean in the prior, 10, will translate into successful acquisition of more marked structures, the presence of more total data allows for the priors to be gradually overridden. Note further that the probabilities of faithful realization consistently roughly correspond to the reported acquisition order for syllable types by Dutch children (Levelt and van de Vijver 1998, Levelt and van de Vijver 2004), which is partly sensitive to the number of markedness violation incurred and partly sensitive to frequency of syllable type.

In concrete terms, one might imagine a child periodically updating her grammar in batch fashion after having received a new batch of further data; the grammar might even be updated after every new datum, with the priors and relative frequencies determining which structures can be realized most faithfully. As long as no data is maliciously withheld, the learner will converge on a grammar that faithfully reproduces the adult grammar and the adult outputs (which is also true for the GLA/SGA in this particular case). A prediction made by the MaxEnt grammar is that the child will, at some intermediate stage of acquisition, sometimes correctly faithfully generate marked structures, but sometimes not; Stochastic OT makes similar predictions. The conclusion at this juncture is that batch processing MaxEnt grammars, equipped with suitable learning biases, can reproduce order-of-acquisition phenomena just as well as on-line learning algorithms.

One further empirical point in favor of finding winning candidates by calculating MaxEnt values is the fact that MaxEnt reproduces sigmoid curves; such sigmoid curves are typical of logarithmic functions. As Hayes (2021) argues, Maximum Entropy Grammars are most well-suited to the reproduction of sigmoid curves in general, and the sigmoid curves repro-

duced by MaxEnt grammars match the sigmoid curves observable in the language data more closely than alternatives (e.g., Noisy Harmonic Grammar). Specifically, in cases where candidates of different inputs violate a constraint that can be violated in a scalar fashion to different degrees, the probability of those candidates takes on a sigmoid shape. See, for instance, the analysis of vowel harmony in Tommo So in McPherson and Hayes 2016. There is thus a good case to be made that Maximum Entropy Grammars come closest to approximating the cognitive architecture (language-specific or not) that is involved in the learning and production of language.

To summarize the above discussion, the MaxEnt architecture presents, at present, perhaps the most compelling option for the modeling of phonological grammars.³⁰ Whether a MaxEnt model is most realistically to be implemented through a form of batch or on-line processing remains, however, an unresolved question, even if, as discussed above, batch processing presents some advantages concerning convergence properties.

One aspect that has not been considered very explicitly, however, is the testimony of language change. Famously, Kroch (1989) observed sigmoid curves in instances of syntactic change from English, Portuguese, and other languages. More recent studies (Blythe and Croft 2003: 279–80, Stadler et al. 2016) point to other instances of or show similar sigmoid curves in other domains of language as well. Compare also discussion on appearance of sigmoid curves in cases of language change in Zuraw 2003, Zimmermann 2017, and Stadler 2017. As Hayes (2021) argues, Maximum Entropy Grammars are most well-suited to the reproduction of sigmoid curves in general (cf. also the comparison to types of Noisy Harmonic Grammar, which can generate sigmoid-like curves as well, but not as closely as MaxEnt). The fact that diachronic changes of a typical sort, where an innovative variant B arises in competition with conservative variant A, increases in frequency first slowly, then rather quickly, allowing variant A to persist for a relatively long time before completely ousting it, fits well with logistic functions in general.

Nonetheless, an important question is whether batch MaxEnt learners are able to learn too well, happily matching frequencies, perhaps allowing variation to always persist indefinitely. This point – the ability of particular models of phonological learnability to adequately capture phonological, in particular prosodic, change – will be revisited at 3.3.3 below. First, an introduction to the learnability of “hidden” phonological structures is necessary, as well as to problems involved in the acquisition and change of grammars with lexically contrastive stress.

³⁰One further innovation that might be considered would be the use of mixed-effects logistic regression, rather than standard logistic regression (= MaxEnt). Mixed-effects models might, for instance, be used to capture lexically specific effects simultaneously alongside the weights of general phonological constraints. See Zymet 2019 for an exploration of the performance of standard MaxEnt and a mixed-effects model to the application of velar palatalization in Slovenian.

3.2.2 “Hidden” Structures in Phonology: Learning Prosodic Structures and Underlying Representations

All of the approaches to phonological learnability treated in the preceding section share the same relatively modest objective: the discovery of a ranking of constraints, or weight for constraints, that can reproduce specified phonological outputs (i.e., surface representations, including any and all prosodic structures) given fully specified phonological inputs (i.e., underlying representations) and the constraints themselves. That many significant aspects of phonological learning are not covered by the methods discussed above is evident. Which further, crucial questions must be answered by a learned may be partly summarized in the following five questions:

1. How do learners determine what sounds are contrastive ($\sim$ phoneme learning)?
2. How do learners determine which sequences of sounds are permitted ($\sim$ learning of phonotactics)?
3. How do learners acquire the constraints governing the relationship between inputs and outputs (or alternatively, between surface forms)?
4. How do learners acquire the ranking of constraints governing prosodic structures without overt correlates?
5. How do learners acquire underlying representations (or alternatively, the structure of relationships between surface forms)?

Progress has been made on all of these questions, with work on distributional learning substantially informing the first (phoneme learning; see, e.g., Maye et al. 2002 or Hitczenko and Feldman 2022 on distributional learning), and Hayes and Wilson 2008 constituting a considerable advance on questions two and three; the procedure of phonotactic learning discussed above also offers indications of how general restrictions on licit phonological structures might be learned. Questions four and five, meanwhile, represent particularly tenacious problems, because their acquisition requires making hypotheses about a grammar, but the correct grammar cannot be inferred until the prosodic structures and URs – neither of which are directly observable – are known (cf. Albright and Hayes 2011: 675). In other words, the structure of underlying representations and the structure of prosodic representations lacking in direct phonetic correlates (weight- or tone-bearing units, syllables, feet, prosodic word edges) is *hidden* from the learner.³¹

The fundamental difficulty in such hidden structure problems consists in the fact that they are difficult for the learner to crack. Without prior knowledge of either the correct grammar (i.e., correct constraint ranking/weighting) or the hidden structure, the learner does not

³¹The distribution of prosodic elements with direct (if not unambiguous) phonetic correlates, e.g., stress or tone, may in part be captured through the learning of phonotactic structures. For instance, in quantity-insensitive stress systems in which the position of primary stress can be described by reference to some number of vocalic peaks with respect to a word edge (assuming that the prosodic boundary of the word is already identified), then phonotactic constraints on (dis)allowed stress positions might suffice.

necessarily know whether the unexpected character of a learning datum is due to an incorrect grammar or incorrect representations; Dresher (1999: 54–8) refers to this conundrum as the “credit problem” (cf. Jarosz 2019: 75–6). Consider, for example, that a learner encounters two (paradigmatically related) surface forms, [ˈpan:a] and [paˈna:ju]; without any knowledge of the grammar, the position of the primary stress could be attributed to a number of different conceivable combinations of grammar and/or lexicon: is stress always penultimate, always on heavy syllables, or perhaps on peninitial syllables while avoiding final syllables?

Since diachronic change in stress systems, in particular stress systems involving a substantial lexical or morphological component is a central focus of the present work, how the relevant prosodic structures and underlying representations might be acquired is likewise of great interest. In the following, I first describe the procedure or Robust Interpretive Parsing as an approach to the learning of hidden prosodic structure in conjunction with constraint grammars, then turn to some approaches to the induction of underlying representations in the face of phonological alternations, then finally consider the conjoint problem posed by the acquisition of stress assignment in lexical accent systems.

3.2.2.1 Metrical Structure, Foot Structure, and Robust Interpretive Parsing

This section aims to introduce methods developed for circumventing the hidden structure problem with respect to prosodic phonology, in particular metrical structure (foot structure). The primary technique to be discussed is ROBUST INTERPRETIVE PARSING (RIP, first described in Tesar and Smolensky 1998), which can be applied to the learning of both hidden metrical structure and underlying representations alongside the learning of a constraint grammar.³² For brief, relatively non-technical introductions to models developed and results obtained concerning the learning of hidden structure, see Albright and Hayes 2011 or Jarosz 2019: 75–81.

Under RIP for the learning of hidden prosodic structures, winners are not specified in advance, but rather overt forms provided from training data are used to evaluate pairs of overt forms and surface forms deriving from a single UR. Alternatively, for the learning of underlying representations, RIP evaluates pairs overt forms and phonological URs deriving from a single string of underlying morphemes. In this introduction to RIP and alternative algorithms for learning from overt forms proposed in Jarosz 2013 and Jarosz 2016 (cf. also Biró 2013) I will devote attention to and exemplify the algorithms using foot structure.

The standard examples of the use of RIP in the literature concern the learning of hidden phonological structures, such as feet or syllables, that do not have direct phonetic exponents. In this literature, square brackets are used to mark *overt* forms [OVERT], i.e., the learning data, backslashes for *surface* forms containing hidden structures /SR/, and underlying phonological representations are shown between pipes |UR|. Tableaux thus show an input UR between pipes, and candidates consist of a mapping [OVERT] ⇒ /SR/, where the

³²Exactly how RIP can be used to determine URs from overt forms will be treated in detail below, where finding a hidden *lexical* representation is treated as a problem similar to finding hidden prosodic structure in a surface representation.

latter contains hidden structure. To consider the general learning problem, consider an overt form $[\sigma'\sigma\sigma]$ with a primary stress on the second syllable of a trisyllabic word. Such an overt form is compatible with an analysis as either a left-aligned iamb $/(\sigma\sigma)\sigma/$ or a right-aligned trochee $/\sigma(\sigma\sigma)/$; feet are not present in the overt form $[\sigma'\sigma\sigma]$, and either foot structure can be built from the UR $|\sigma\sigma\sigma|$ without any underlying prosodic structure.³³ The goal of RIP in treating such prosodic hidden structure is to determine which SR with which constraint ranking best matches the set of overt forms in the data. All usual implementations of RIP depend upon an error-driven, online model of grammar learning (e.g., EDCD, OT/HG-GLA, SGA; see above).

Learning hidden structure using RIP consists essentially of three steps, once an overt form is provided to the learner: first, matching a learning data (overt form) to the most harmonic parse (surface form) for that form under the learner's current grammar; second, generating the winning candidate using standard production-directed parsing from the UR that matches the overt form; third, comparing the outputs from the first and second steps and updating the grammar with some constraint ranking or weighting algorithm if the SRs from the first two steps differ. In the first formulation of RIP in Tesar and Smolensky 1998 and Tesar and Smolensky 2000, online Error-Driven Constraint Demotion (EDCD) is applied to update a constraint ranking (see under 3.2.1.1 above). RIP runs iteratively until the most harmonic parse of the overt form and the production-generated parse are identical.

As an example, consider a learner $\mathcal{L}$ that currently possesses a grammar with the ranking given in (40)a., which then encounters the overt form $[\sigma'\sigma\sigma]$ (compare similar examples in Jarosz 2013: 35–6 and Jarosz 2016: 207–9). This overt form can be directly assigned to the UR $|\sigma\sigma\sigma|$, which has the following possible production-directed parses in terms of primary stress and foot structure given in (40)b. (assuming exclusively binary feet). Of the four possible candidates, only b. and c. constitute robust parses, because only they can generate the expected overt structure, with stress on the second/penultimate syllable; these are given in (40)c.; finally, all this information is summarized in the tableau in (40)d.

- (40) Grammar and Parses of $\mathcal{L}$ Encountering $[\sigma'\sigma\sigma]$
- a. Current Grammar: ALL-FT-L $\gg$ TROCHAIC $\gg$ IAMBIC $\gg$ ALL-FT-R
 - b. Possible Production-Directed Parses: $/(\sigma'\sigma)\sigma/$, $/(\sigma'\sigma)\sigma/$, $/\sigma(\sigma'\sigma)/$, $/\sigma(\sigma'\sigma)/$
 - c. Possible Robust Parses: $/(\sigma'\sigma)\sigma/$, $/\sigma(\sigma'\sigma)/$

³³Technically, syllable boundaries would need to be determined as part of interpretive parsing as well. Issues of learnability posed by fundamental ambiguities in foot structure were first addressed in the parametric cue-based learner of Dresher and Kaye 1990, further built upon in Dresher 1999, and applied to lexical stress systems in Dresher 2016; further recent research into parameter-based learning of metrical structure has been conducted by Pearl 2011 and Nazarov and Jarosz 2017.

d. Tableau:

	$ \sigma\sigma\sigma $	ALL-Ft-L	TROCHAIC	IAMBIC	ALL-Ft-R
a. 	$[\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$			*	*
b. RIP	$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$		*!		*
c.	$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$	*!		*	
d.	$[\sigma\sigma'\sigma] \Rightarrow /(\sigma\sigma')\sigma/$	*!	*		

Among the two possible robust parses, candidates b. and c. the most harmonic one is candidate b., because candidate c. violates higher-ranked ALL-Ft-L. The robust parse, candidate b., is, however, different from the most harmonic production-directed parse (POP) under the current grammar, candidate a., marked with the pointing hand. These two candidates must then be compared, treating candidate b. as the winner and candidate a. as the loser. When using Constraint Demotion, a winner-loser mark-data pair is created, as shown in (41), with the ultimate result that all the loser-preferring constraints will be demoted below the winner-preferring constraints.³⁴

(41) Mark-Data Pair of Robust Parse versus Production-Oriented Parse

$ \sigma\sigma\sigma $	ALL-Ft-L	TROCHAIC	IAMBIC	ALL-Ft-R
$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/ \sim [\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$	e	L	W	e

Since TROCHAIC is loser-preferring while IAMBIC is winner-preferring, the update to the grammar will then be to demote TROCHAIC to the stratum below IAMBIC. After this update, the most harmonic production-oriented parse will be candidate b., identical with the robust interpretive parse. Without other updates to the grammar that would result in a different most harmonic production-oriented parse, encountering the overt form $[\sigma'\sigma\sigma]$ will not result in any further changes to the grammar, because the most harmonic robust and production oriented parses will be identical (candidate b.). The new constraint hierarchy and the correct derivation of candidate b. as winner by both the RIP and POP is illustrated in 42. Compare Tesar and Smolensky 2000: 60–2 for a similar example involving multiple steps of constraint demotion to obtain identity of the RIP and the POP.

³⁴Recall from 3.2.1.1 above that a “loser-preferring” constraint is a constraint that assigns more violations to the winner than to the loser, while a “winner-preferring” constraint assigns more violations to the loser than to the winner. In the case at hand, IAMBIC is winner-preferring because it assigns more violations to the candidate $[\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$.

(42) RIP = POP after Constraint Demotion

	$ \sigma\sigma\sigma $	ALL-Ft-L	IAMBIC	ALL-Ft-R	TROCHAIC
a.	$[\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$		*!	*	
b.  , RIP	$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/$			*	*
c.	$[\sigma'\sigma\sigma] \Rightarrow /(\sigma'(\sigma\sigma))/$	*!	*		
d.	$[\sigma\sigma'\sigma] \Rightarrow /(\sigma(\sigma'\sigma))/$	*!			*

In principle, the generation of RIPs, POPs, and subsequent constraint demotion could continue as long as the most harmonic RIP differs from the most harmonic POP on a given overt form (cf. the example Tesar and Smolensky 2000: 59–61). In the present case, a single iteration of parsing and constraint demotion sufficed in order to obtain a ranking hierarchy under which the RIP and POP of the overt form $[\sigma'\sigma\sigma]$ are identical.³⁵

Beginning with the original tests of RIP/EDCD on hidden metrical structure conducted in Tesar and Smolensky 1998 and Tesar and Smolensky 2000: Ch. 5, that RIP/EDCD frequently fails to converge on the correct grammar, and that some grammars may not be learnable at all, has been apparent. Namely, the learner can encounter the data in such a way that some constraints are endlessly demoted below one another with each instance of learning (see also the discussion in Drescher 1999). Apoussidou and Boersma (2004) and Apoussidou

³⁵Although the RIP algorithm is defined so as to operate with the Constraint Demotion algorithm, considering what might occur with Recursive Constraint Demotion poses an interesting problem. One approach would be to throw out the existing ranking entirely, then generate mark-data pairs comparing the winning robust parse to both the winning production-oriented parse and any alternative robust parses. Given the mark-data pairs

$ \sigma\sigma\sigma $	ALL-Ft-L	TROCHAIC	IAMBIC	ALL-Ft-R
$[\sigma\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/ \sim [\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$	e	L	W	e
$[\sigma\sigma\sigma] \Rightarrow /(\sigma'\sigma)\sigma/ \sim [\sigma\sigma\sigma] \Rightarrow /(\sigma\sigma)\sigma/$	W	L	W	L

RCD would produce the ranking ALL-Ft-L, IAMBIC $\gg$ ALL-Ft-R, TROCHAIC; this ranking obtains from the fact that ALL-Ft-L and IAMBIC are exclusively winner-preferring, while ALL-Ft-R, TROCHAIC are exclusively loser-preferring. This ranking would likewise result in a match of RIP and POP on the overt form $[\sigma'\sigma\sigma]$. T

In turn, the Maximally Informative Recursive Constraint Demotion (MIRCD) algorithm of Yates and Zukoff 2018, given the mark-data pairs immediately above, would produce the ranking IAMBIC $\gg$ ALL-Ft-L $\gg$ ALL-Ft-R, TROCHAIC; this ranking obtains from the fact that IAMBIC contains more winners than ALL-Ft-L.

Employing (a variety of) the RCD algorithm with RIP, however, will result in very different behavior than using online EDCD. Whereas online EDCD will, when faced with new data, update rankings in a compatible direction, RCD will undertake a total re-ranking, with the result that, in order for RCD to work with RIP, a given overt form must be available, that, when paired with some POP and alternative parses, will result in a ranking that will allow for agreement between RIPs and POPs for all distinct overt forms in the data. Precisely because RCD is designed to converge on a ranking given a batch of data, intuitively, there it seems reasonable to think that RCD combined with online RIP might often fail to converge; an alternative would be to consider parses on all distinct overt forms in the data before applying RCD to rank constraints, then repeating if RIPs and POPs do not agree for all URs. That is, a batch version of RIP could perhaps be undertaken by populating the Support with further RIPs each time that a new overt form becomes available.

(2007: Ch. 3–4) then further demonstrated that RIP could be combined with the GLA, but the tests of convergence carried out by Boersma and Pater (2016) ultimately revealed that RIP, whether used with Tesar and Smolensky's (2000) original version of EDCD, the version of Boersma 2009, or OT-GLA, performs suboptimally on problems of hidden metrical structure; of these implementations, RIP/EDCD applying pooling ties as per Tesar and Smolensky 2000 seemed to perform best, obtaining convergence for ~60% of grammars tested. Instead, only with HG-GLA, with added noise in constraint evaluation, does RIP arguably perform reasonably well, successfully learning more than 88% of grammars (cf. also Jarosz 2013: 39–40).³⁶ Although this result might be interpreted as offering further support for a grammatical architecture employing weighted constraints, Jarosz (2013, 2016) subsequently showed, using the same set of test languages, that the comparatively poor performance of RIP/OT-GLA could be improved upon by adjustments to the algorithm for the determination of the robust interpretive parse and the production-oriented parse. The novel versions proposed by Jarosz, Resampling Robust Interpretive Parsing (RRIP) and Expectation Interpretive Parsing (EIP), whether used with OT-GLA or HG-GLA, perform comparably to or better than RIP/HG-GLA with noise in terms of correctness and convergence speed.³⁷

Since I concluded at the end of 3.2.1.3 above that MaxEnt grammars likely represent – on the basis of mathematical, empirical, psychological, and historical evidence – the best available tool for the modeling of phonological grammars, how exactly variants of RIP un-

³⁶More recently, Prickett and Pater (2019) have shown that some problems with the learnability of particular stress patterns in the set of 124 languages originally used in Tesar and Smolensky 2000 likely has to do with the constraint set itself. For comparability, all tests involving these metrical stress grammars have employed the same constraint set, but new tests conducted by Prickett and Pater with additional or different constraints suggest that overall performance may be improved. See also Pater and Prickett 2022 for further applications.

³⁷This note summarizes how Jarosz's RRIP and EIP differ from RIP. Under original the original RIP model, as exemplified at 41 above, for each overt datum d received, the learner finds the relevant UR, obtains a robust interpretative parse (RIP), obtains a production-oriented parse (POP), and updates the grammar if the RIP $\neq$ POP. While the RIP and POP that are compared are ultimately fully structured, they will only differ if their overt forms differ in some respect. Jarosz (2013, 2016) points out that the learner in fact could decide to update by first comparing its overt optimal output to the overt form of the learning datum, then generating a robust parse only based on the datum only if the two overt forms are not identical. Using OT-GLA, different temporary grammars (see discussion of the GLA above) for evaluation could then be used to generate the learner's optimal form, on the one hand, and the RIP of the learning datum, on the other: this is RRIP. In other words, grammar sampling occurs twice, not only once.

However, when using both original RIP or RRIP with OT-GLA, employing a two-step parsing process means that the probability of generating a given parse is not the same as the probability of producing such an output under the learner's current stochastic grammar (Jarosz 2013: 49). For instance, given the four constraints used in 41 above, if the two constraints on edge-orientation of feet have the same ranking value, both above the constraints on foot shape, and TROCHEE above IAMB then the stochastic grammar as applied to a word / $\sigma\sigma\sigma$ / would generate initial stress or peninitial stress, with a trochaic foot, 50% of the time. In generating a robust interpretive parse for the datum [$\sigma'\sigma\sigma$], the restriction to overt forms with stress on the second syllable means that the parse [$(\sigma'\sigma)\sigma$], with an iambic foot aligned with the left edge, becomes an option, even though the learner in production would never generate such an output under its current grammar. In other words, RIP does not effectively use the knowledge of the already established ranking TROCHEE $\gg$ IAMB, and might even undo a correct ranking that has already been learned. Instead, if the robust interpretive parse is found by selecting a parse from the probability of a parse given the current grammar and the overt learning datum, no mismatch in probabilities between parsing and production should occur. This procedure constitutes Expectation Interpretative Parsing (EIP).

der Stochastic OT or Noisy Harmonic Grammar perform is, for the present project, not of greatest relevance. Jarosz’s results and improved versions of RIP may, though, hold equally for SGA MaxEnt. Indeed, when a MaxEnt grammar architecture is employed, Expectation Maximization (in the sense of Dempster et al. 1977) is straightforward to implement: both the robust interpretive parse and the production-oriented parse can be sampled from the same probability distribution of possible winners under the current grammar that are compatible with the received learning datum. Compare the discussion on this point in Staubs 2014: 9 and Jarosz 2013: 52–3.³⁸ Thus, as long as the learner’s current candidate probabilities are used for selecting the robust parse, and only the probability mass of candidates matching the overt form of the learning datum are used, Expectation Maximization will ensue: the selection of the robust interpretive parse in an online learner always makes use of the probabilities under the learner’s current grammar.

As Jarosz (2015: 4) notes nearly all work on constraint-based approaches to the learning of structural ambiguity operates with an online, error-driven model (Tesar and Smolensky 2000: Ch. 4, Jarosz 2013, Staubs 2014, Boersma and Pater 2016). Given, however, that the question of whether MaxEnt grammars perform better or possess more psychologically plausibility when learned in online or batch fashion could not be resolved above, worth considering here is how and whether a version of RIP could be implemented in a batch setting. Two attempts have been made thus far: Expectation Drive Pairwise Ranking in Jarosz 2019 (building upon the idea of partially ordered grammars of Anttila 1997, Anttila 2002) and a batch MaxEnt model in Prickett and Pater 2019 and Pater and Prickett 2022 (building on Pater et al. 2012). Both approaches share the feature of using Expectation Maximization (cf. above) to discover the constraint rankings or weights and fully specified representations that maximize the likelihood of the overt forms. In the Maximum Entropy context, this means that all fully specified representations that are consistent with the overt representations are considered simultaneously (e.g., for the overt form [L 'L L], the foot parses [(L 'L) L] and [L ('L L)] would be considered, while [(L L) L] and [L (L 'L)] would not): given some initial weights, a grammar is selected deterministically that corresponds to the probabilities of the compatible overt forms (Pater and Prickett 2022: 4).

Consider, for example, a language without quantity sensitivity, stressed monosyllables, primary stress on the initial syllable in disyllabic words, but otherwise stress on the peninitial syllable. A typical analysis might posit left-aligned iambs, with monosyllabic feet in monosyllables and disyllables (to avoid stress on final syllables). Given only words between 1 and 4 syllables, a total of eight distinct robust interpretive parses (considering only primary stress) will be compatible with the overt forms of different lengths, consisting of ['L] (1), ['L L] (2), [L 'L L] (3), [L 'L L L] (3), and thus a total of 18 different combinations of robust interpretive parses are compatible with this data, assuming that there is no variation. Given this data alone, the phonologist might posit one of several possible combinations of foot parses able to generate this data, using the prosodic markedness constraints introduced in Chapter 2; four options are listed here, though more exist.³⁹

³⁸Indeed, Jarosz directly points out that Maximum Entropy Grammars are “not subject to the parsing–production mismatch problem” that occurs under RIP or RRIP, and which motivated her development of EIP.

³⁹Specifically, using all constraints listed in Table 2.1 in Chapter 2, except for those irrelevant to quantity-

- a. Iambic feet unless final, $\text{PARSE-}\sigma \gg \text{NONFINALITY-}\mathcal{F}$:
 $[('L)], [('L L)], [(L 'L) L], [(L 'L) L L]$
- b. Iambic feet unless final, $\text{NONFINALITY-}\mathcal{F} \gg \text{PARSE-}\sigma$:
 $[('L)], [('L) L], [(L 'L) L], [(L 'L) L L]$
- c. Trochaic feet, $\text{NONINITIALITY-}\mathcal{F} \gg \text{NONFINALITY-}\mathcal{F}$:
 $[('L)], [('L) L], [L ('L L)], [L ('L L) L]$
- d. Monomoraic feet:
 $[('L)], [('L) L], [L ('L) L], [L ('L) L L]$

Many phonologists would share the intuition that options a. and b. represent the most plausible choices, with c. and d. representing typologically less likely options, given the rarity of systems with active NONINITIALITY and systems that do not regularly strive for binary feet. A batch implementation of RIP under MaxEnt can further demonstrate that options a. and b. are the most plausible: given all 18 possible combinations are foot parses that are compatible with the overt forms, the parses shown in a. and b. allow for convergent grammars that have the highest (and very similar) negative log-likelihood (-0.06671 for a. vs. -0.05844 for b.).⁴⁰

insentitive systems without secondary stress (i.e., WSP, RH-CONTOUR, LEFTMOST/RIGHTMOST, PARSE-C, and *CLASH, 12 different combinations are possible that can generate foot structures compatible with the overt forms with near 100% probability. Of those 12 options, six strict rankings are available.

⁴⁰The relevant constraint rankings found through MRCD that produce grammars a. and b. are:

- a. $\text{GRWD} = \text{PRWD ALL-Ft-L}, \text{ALIGN-L}(\omega, \mathcal{F}), \text{PARSE-}\sigma \gg \text{NONFIN-}\sigma, \text{FTBIN}, \text{ALL-Ft-R}, \text{NONFIN-Ft}, \text{ALIGN-R}(\omega, \mathcal{F}), \text{NONINITIALITY-}\mathcal{F} \gg \text{IAMB} \gg \text{TROCHEE}$
- b. $\text{GRWD} = \text{PRWD ALL-Ft-L}, \text{ALIGN-L}(\omega, \mathcal{F}), \text{IAMB} \gg \text{NONFIN-}\sigma, \text{NONFIN-Ft}, \text{NONINITIALITY-}\mathcal{F} \gg \text{ALIGN-R}(\omega, \mathcal{F}), \text{FTBIN}, \text{ALL-Ft-R}, \text{PARSE-}\sigma \gg \text{TROCHEE}$

The optimized weights for these constraints, found using `OPTIM()` in R with method L-BFGS-B, $\mu = 0$ and $\sigma^2 = 10000$, are reported in the table below:

Constraint	Weight under Parse a.	Weight under Parse b.
NONFIN- σ	10.97	4.97
FTBIN	0.99	0
GRWD = PRWD	2.66	5.23
ALLFT-L	2.74	1.21
ALLFT-R	0.91	0
IAMB	5.49	6.85
TROCHEE	0	0
NONFIN- $\mathcal{F}$	0	6.52
ALIGN-L($\omega, \mathcal{F}$)	5.31	6.44
ALIGN-R($\omega, \mathcal{F}$)	4.27	0
NONINIT- $\mathcal{F}$	0	0
PARSE- σ	6.31	5.99

The major factors accounting for the difference in log-likelihood between parse a. and parse b. are likely the need for higher average constraint weights under parse a. and the higher number of constraints given a non-zero weight (nine vs. seven). In a sense, model represented by the grammar of parse b. is less complex.

The architecture of such a batch learner is relatively straightforward: given the overt forms, the candidates corresponding to each UR (e.g., /L L L/) compatible with that overt form are identified as possible winners, then all logically possible combinations of winners are enumerated,⁴¹ and each possible combination of licit winners is tested, where constraints are optimized using the L-BFGS-B algorithm, saving information on the weights, negative log-likelihood, and convergence status for each combination. As reported above, the negative log-likelihood was greatest for the combination of overt forms in a. and b.; highest negative log-likelihood could thus be used as a criterion for identifying the robust parses and grammar that agree with the phonologist's intuitions. Strictly following this criterion would result in choosing parses and grammar b.: left-aligned iambs with final consonant extrametricality. In order to find optimal plausible parses, it is necessary first to analytically exclude all candidates that directly conflict (i.e., where the data is categorical, the winners with 0 probability), then consider all possible combinations of winners, which in the categorical case is the product of the number of possible winners in each tableau.⁴² Although enumerating and testing the likelihood of all possible combinations of fully parsed winners compatible with the overt forms may be rather computationally intensive, it is in principle possible.

Utilizing an online MaxEnt learner along the lines of the one described in Staubs 2014: Ch. 1 (compare brief discussion above) obtains similar results for the case currently under discussion. Based on the four overt forms (['L], ['L L], [L 'L L], and [L 'L L L]), any grammar trained on a sufficient number of tokens with a learning rate between 0.01 and 0.5 converges on a grammar that predicts the parses [('L)], [(L 'L) L], and [(L 'L) L L] with greater than 99% probability; only for the disyllabic input, /L L/, is there ambiguity, with some simulations predicting the parse [('L) L], others [('L L)]. 1000 simulations with a learning rate of 0.01 converge towards [('L) L] in 391 cases, but [('L L)] in 609 cases.⁴³ In short, both batch

⁴¹For enumerating the possible combinations of licit winners, I employed a Monte Carlo method, sampling one winner from the set of licit winners for each input 1000 times, then reducing the set of samples to all unique cases. In order to ensure that all unique cases were sampled at least once, a *post hoc* control was employed to check that the number of unique samples was equal to the number of logically possible combinations, in this case, 18.

⁴²In a worst-case situation with categorical outputs, considering longer words of up to seven syllables, secondary stress, and quantity-sensitivity, an overt form might be compatible with up to 16 possible parses. In the original Robust Interpretive Parsing Data of Tesar and Smolensky, there are 54 URs, meaning that there are might be 16⁵⁴ (!!!) possible combinations of compatible parses; in practice, the situation is not nearly so grave, since shorter words have many few possible compatible parses (e.g., at most two for disyllables, at most three for trisyllables, at most five for four four-syllable words). Considering four-syllable words alone, the situation is becomes worrisome (450,868,486,864,896 – over 450 trillion possible combinations), but perhaps not wholly unmanageable (reported computing times are variable, but finding all possible combinations and optimizing over them would probably take less than one year).

⁴³In this small simulations, tokens of any length occurred with equal likelihood in the learning data. This result is due to differences in order of exposure to the data: since three- and four-syllable forms support giving more weight to FTBIN over PARSE-σ, early exposure to more three- and four-syllable tokens places the learner on a track to acquire a grammar that will generate the disyllabic parse [('L L)]. More generally, the fact that an error-driven learner can distinct grammars that generate the overt data equally well replicates the point made by Apoussidou (2007: Ch. 5) based on the stress pattern of Pintupi: order of exposure to learning data in error-driven learners affects the exact grammar acquired.

and online MaxEnt learners appear able to cope largely successfully with hidden prosodic structure (compare, again, simulations in Staubs 2014: Ch. 2 and Pater and Prickett 2022).

However, one must bear in mind that the learning of hidden prosodic structure is concerned with discovering the correct ranking or weighting of markedness constraints alone. Hence, the learner must merely find which overt structures never occur, for which it receives direct evidence, and determine a ranking or weighting of constraints that can assign prosodic structure able to correctly generate the overt forms. The lexical properties of the inputs, beyond their segmental composition, are irrelevant. In the case of the next type of hidden structure to be discussed, underlying representations, the problems are more severe.

3.2.2.2 Underlying Representations

Standard generative (going back to earlier structuralist; cf. Nida 1949: Ch. 2) approaches to morphophonological analysis take the principle of “one meaning, one form” very seriously: for a given (bundle of) semantic or morphosyntactic feature(s), a single underlying phonological form should be assumed to spell out that (bundle of) feature(s). The lexicon and/or morphological component of the grammar accesses connections between either pairs of underlying phonological form and meaning, or rewrite rules that insert phonological material in the context of given semantic/morphosyntactic features.

Traditional manual phonological analysis requires the phonological underlying forms to be “puzzled out” by hand. In case a morpheme has but one major exponent in surface forms (i.e., exhibits no allomorphy), the segmental and prosodic string that exceptionlessly seems to express given semantic/morphosyntactic features can be adopted as the phonological UR.⁴⁴ In case a morpheme exhibits more than one major exponent in surface forms (i.e., exhibits allomorphy), standard analytical procedure looks to the phonological environment of each allomorph, and treats as the UR the allomorph that occurs in a variety of contexts that cannot be captured by a natural class, or simply the allomorph occurring in the greatest diversity of contexts (in case the context for each allomorph can be described using a natural class, or multiple allomorphs occur in contexts that do not constitute natural classes).⁴⁵ This work, of course, assumes that segmentation of morphemes has already been successfully carried out.

Although RIP can successfully learn prosodic structures, how it should generalize to the learning of underlying representations is less straightforward, precisely because faithfulness constraints come into play. Consider a language with word-final devoicing of obstruents, such as Polish: since underlying /d/ and /t/ always map to [t], a robust interpretive parse will always map overt [t] to underlying /t/, since that mapping would imply no violations of either markedness or faithfulness; mapping overt [t] to underlying /d/ would satisfy the relevant markedness constraints, but necessarily incur some faithfulness violations. As Jarosz (2015:

⁴⁴By “major exponent”, I mean variants that contain phonetic distinctions that are predictable and rule-governed, and which typically can be perceived as distinct, if not necessarily contrastive, by native speakers.

⁴⁵This basic procedure does not immediately transfer to cases in which composite URs need to be “cobbled together” out of various properties that never occur simultaneously on a single surface allomorph.

12) puts it: “This is another instance of the Parsing Production Mismatch (Jarosz 2013), albeit a much more systematic and devastating one.”

Given the principle of the Rich Base, particular URs cannot be excluded *a priori*, and given no constraints on URs, a UR that will incur no fatal markedness violation and that is identical to the overt form will always harmonically bound a UR that differs from the overt form. Without equipping a learner with additional mechanisms to puzzle out URs, the paradigmatic alternation in the voice feature of the stem-final plosive as well as vowel height seen in, e.g., Polish [lut] ‘ice:NOM.SG’ versus [lodi] ‘ice:NOM.PL’ cannot be learned as derived from an underlying stem /lod-/, since the mapping [lut] $\Rightarrow$ /lut/ would be more harmonic than the mapping [lut] $\Rightarrow$ /lod/. Assuming that phonotactic learning has successfully converged on the ranking

$$* \begin{bmatrix} -\text{sonorant} \\ -\text{voice} \end{bmatrix}]_{\omega}, * [o] / \text{---} \begin{bmatrix} +\text{voice} \\ -\text{nasal} \end{bmatrix}]_{\omega} \gg \text{IDENT}[-\text{voice}], \text{IDENT}[-\text{high}],$$

then parsing [lut] as corresponding to underlying /lut/ is the most harmonic choice, as illustrated in 43. This particular case, which involves opacity, is particularly challenging, since the condition that motivates raising of /o/ to [u] is destroyed on the surface (counterbleeding on environment). The UR that would be posited under a traditional analysis, /lod-/ is two steps removed from the overt form, and does not admit of a direct solution via RIP

(43) UR = SR is maximally harmonic

ice:NOM.SG		*[-son., +voi]] _ω *[o] / --- [+voi., -nas.] _ω IDENT[-voice] IDENT[-high]	
a.	[lut] $\Rightarrow$ /lut/		
b.	[lut] $\Rightarrow$ /lud/	*!	
c.	[lut] $\Rightarrow$ /lot/		*!
d.	[lut] $\Rightarrow$ /lod/	*!	*

Proposals to account for paradigmatic alternations (allomorphy) may be divided to two broad approaches, following Albright and Hayes 2011: approaches employing underlying representations, and approaches employing surface mappings. Under approaches of the latter type accept only surface representations and define the learning task as the acquisition of mappings between paradigmatically related word-forms; perhaps the most well-developed model of this sort is the Minimal Generalization Learner (Albright and Hayes 1999, Albright 2002b, Albright and Hayes 2003). Approaches that endorse underlying to surface mappings (the former type), on the other hand, may in turn be subdivided into three groups: 1) detection of inconsistency in the UR posited for a lexical item or morphosyntactic feature bundle (Tesar 2014 and related work); 2) expectation maximization applied to all possible combinations of grammars and underlying representations given the overt forms (Jarosz 2006); 3) markedness constraints over underlying representations (Apoussidou 2007). The second of these UR-learning approaches is not, at present, computationally tractable, since it exhaustively evaluate all combinations of possible constraint rankings and combinations of

underlying representations (cf. Tesar 2014: 251);⁴⁶ I therefore will not discuss it further here. Since I will further operate throughout the present work with classical underlying representations, I will not present a detailed exposition of any model of surface mappings, but refer the reader to the references above on the Minimal Generalization Learner, as well as discussion in Sandell 2015a (esp. and Bowers 2015 (esp. 82–91) on its behaviors and the predictions of the associated “Single-Surface Base Hypothesis”.⁴⁷ Properties and the workings of each of the other remaining two models (constraints over underlying representations and inconsistency detection) will be summarized below.

Constraints over Underlying Representations If the example in 43 is simplified to a case involving only alternation in obstruent voicing (e.g. Polish [klup] ‘club:NOM.SG’ vs. [klubi] ‘club:NOM.PL’), learning an underlying representation for the singular form, instantiating obstruent devoicing and thus incurring a faithfulness violation, could be approached as a matter of acquiring a constraint ranking under which a seemingly fully faithful candidate *does*, in fact, violate some higher-ranking markedness constraint. Apoussidou (2007: Ch. 6), building upon suggestions in Boersma 2001, proposes that relationships between the generative semantic and morphosyntactic components of the grammar with the interpretive phonological component are subject to evaluation by markedness constraints that penalize specific form-meaning pairings.⁴⁸ In general terms, this type of markedness constraint could be defined (following Apoussidou 2007: 170) as in 44; a specific example based on the Polish obstruent-voicing alternation is given in 45.

(44) Lexical Markedness Constraints on Underlying Representations:

Assign one violation * if the semantics ‘ $\mathcal{S}$ ’ or the morphosyntactic feature bundle ‘ $\mathcal{M}$ ’ is realized as the phonological string [X], or in parsing, if the phonological string [X] is interpreted bearing the semantics ‘ $\mathcal{S}$ ’ or the morphosyntactic feature bundle ‘ $\mathcal{M}$ ’.

⁴⁶A similar, but perhaps slightly more tractable approach could be the one outlined above for batch MaxEnt learner of hidden structure. Since permutations of constraint rankings need not be exhaustively evaluated in that case, only the combinations of URs must be enumerated. Nevertheless, some criteria would be needed to restrict the set of possible URs to be considered.

⁴⁷Bowers (2015) goes indeed argues that paradigms that would traditionally be analyzed as involving composite underlying representations, where no single surface allomorph contains all properties to be attributed to the UR, pose a challenge for the Single-Surface Base hypothesis. Namely, in such cases, paradigmatic leveling should normally be predicted. In the specific case of alternations in obstruent voicing and vowel reduction found in Russian, which have supposedly coexisted stably for ca. 700 years, the Single Surface Base hypothesis would seem to predict leveling and loss of at least one of the relevant alternations (Bowers 2015: 60–2).

⁴⁸Similar in spirit to the constraints over URs proposed by Boersma and Apoussidou is the constraint USELISTED introduced by Zuraw (2000: 49–53). The major distinction between the constraint in 45 and USELISTED is that the latter explicitly penalizes parses or productions that are morphologically complex: “The input portion of a candidate must be a single lexical entry (1 violation if not true).” However, USELISTED can easily be repurposed to refer to smaller form–meaning bundles (e.g., in Sandell 2017, and thus can be made effectively equivalent to the constraint type used by Apoussidou.

(45) Lexical Markedness Constraints for the Semantics ‘club’ in Polish

- a. *|klup-| ‘club’: Assign one violation * if the meaning ‘club’ is connected to the phonological string |klup-|.
- b. *|klub-| ‘club’: Assign one violation * if the meaning ‘club’ is connected to the phonological string |klub-|.

In production, the selection of the UR |klub-| to yield the input /klub/ for ‘club:NOM.SG’ will be assured if *|klup-| outranks *|klub-|. As illustrated in 46, the two fully faithful outputs, candidates b. and c., lose as a result of violation of high-ranking markedness constraints, *[-son., +voi] _ω or *|klup-|, respectively; the winner, candidate a., is unfaithful in the feature voice with respect to its input, /klub/, but satisfies both the phonotactic markedness constraint and the higher-ranked UR-markedness constraint.⁴⁹

(46) ‘club:NOM.SG’ ⇒ /klub/ → [klup]

‘club:NOM.SG’		*[-son., +voi] _ω	* klup-	IDENT-[voi]	* klub-
a.	 /klub/ → [klup]			*	*
b.	/klub/ → [klup]		*!		
c.	/klub/ → [klub]	*!			*

The crucial ranking *|klup-| ≫ *|klub-| can be acquired via error-driven robust interpretive parsing, regardless of the order in which a learner is exposed to the data. First, since obstruent voicing is contrastive outside of word-final position (cf. [kot]~[koti] ‘cat’), IDENT-[voice] must dominate a phonotactic markedness constraint like *[+voice][-voice][+voice]. Then, given the plural form [klubi], only the UR |klub-| can be used to generate the overt form, which requires the ranking *|klup-| ≫ *|klub-|. Finally, the singular form [klup] necessitates the ranking *[-son., +voi] _ω ≫ IDENT-[voice].

Simulations conducted by Apoussidou on the acquisition of contrastive stress for two stems and two inflection endings in Modern Greek constitute a successful proof of concept: robust interpretive parsing can learn hidden structure in the form of underlying representations, provided that constraints able to differentiate between the URs considered in the parses are available. Subsequent work employing URs over underlyingly representations has been successfully extended to MaxEnt grammars in Pater et al. 2012, Smith 2015, Nelson 2019, and Tan 2022.⁵⁰ This further work has also demonstrated that UR constraints can success-

⁴⁹Not further considered here, or in other work on the use of constraints over underlying representations as far as I am aware, is the option of the null parse, in which ‘club’ would be realized as |∅| → [∅], thereby satisfying all constraints. In order to preclude this possibility, invoking a constraint such as REALIZE MORPHEME (Kurusu 2001; see Zukoff and Sandell 2015 for an application to Gothic verbal morphology) would be necessary to penalize providing no phonological substance to some given semantics or morphosyntactic feature bundle.

⁵⁰Much of this other work structures the UR constraints slightly differently, namely, by penalizing the non-usage rather than used of some phonological input. For example, a constraint “{club}= /klup- /” would be violated by all candidates employing a UR other than /klup- / (recall, under Apoussidou’s implementation, it is preceisely the candidates that employ the UR /klup- / that incur a violation of *|klup- /).

fully handle a variety of further problems, including the variable selection of allomorphs (especially in cases of phonological suppletion), underspecification, morpheme segmentation, and grammars that support rich bases (i.e., which can generate outputs with appropriate probabilities given the known data).

A fundamental theoretical objection to which this entire approach is open is expressed by Tesar (2014: 251–2): using constraints over URs “will explode the number of constraints for systems with even relatively modest assumptions about the number of possible underlying forms”, with the consequence that “the number of possible rankings is now greater than the number of possible lexica”. Tesar’s second concern, the size of the search space for the correct ranking, should in principle away in a MaxEnt implementation, where a global or at least local optimum is guaranteed to be found relatively quickly; even if the grammar contains hundreds of thousands of constraints, optimization is not prohibitively difficult. On the other hand, the sheer number of constraints necessary is worrisome, if the search space for even a single UR is subject to the principle of the rich base and thus theoretically infinite. Since the cases discussed in the literature cited above employ UR constraints crafted by linguists – and no work of which I am aware has considered lexica with more than eight total UR constraints – the problem of constraint induction and selection remains to be solved.

Existing work on the induction of phonotactic markedness constraints (see especially Hayes and Wilson 2008) shows that the induction of large numbers of constraints is a tractable issue. Ensuring that the search space for UR constraints is finite would require adopting some global restrictions. Virtually all cases of allomorphy reduce to one of three types: 1) the UR is equivalent to one of the surface forms; 2) the UR is a composite of different properties found across multiple surface forms; 3) the UR is suppletive (I take cases of apparent underspecification to involve suppletive UR selection). Licit UR constraints could thus be restricted to actually occurring surface allomorphs and combinations of their sites of alternation. Given the three major allomorphs typically acknowledged for the productive noun plural suffix of English, {[*-z*], [*-s*], [*-iz*]}, the URs to be considered should be limited to /*-z*/, /*-s*/, /*-iz*/ (i.e., the three surface allomorphs) plus /*-is*/ (a composite UR comprised of the alternations $\emptyset \sim [i]$ and $[z] \sim [s]$ seen in the surface allomorphs). In practice, the total number of UR constraints would depend upon the number of surface allomorphs and their phonological similarity.⁵¹ Nelson (2019: 64–6) has even made progress on induction of URs from unsegmented data, applying only the restriction that the UR is a non-empty string (and implicitly that the UR is equivalent to some surface allomorph). In short, although the implementation of constraints over underlying representations quickly exceeds what the analyst by hand can manage, successful computational implementation is likely within the realm of possibility, though further investigation involving larger datasets is necessary.

Inconsistency Detection, Contrast Pairs and the Output-Driven Learner An alternative line of investigation into UR selection, pursued mainly by Bruce Tesar and students (Tesar

⁵¹For instance, if the English plural allomorphs were {[*-z*], [*-s*], [*-tim*]}, assuming string alignment where the only segment of [*-s*] or [*-z*] aligns with the first or last segment of [*-tim*], the URs to be considered would be: /*-z*/, /*-s*/, /*-tim*/, /*-zim*/, /*-sim*/, /*-tiz*/, /*-tis*/.

2003, Tesar 2004, Merchant 2008, Akers 2012, Tesar 2014, as well as Bowers 2015), perhaps more closely resembles the traditional procedure for UR selection. Briefly summarized, an OT grammar and corresponding lexicon of underlying representations can be built through the following procedure (see Tesar 2014: 274–6; 335–6; 376–7):

1. PHONOTACTIC LEARNING with MRCD is applied to Biased Constraint Demotion, produces an initial constraint ranking.
2. After morpheme segmentation, each set of surface forms for a given morpheme is examined. An INITIAL LEXICON is constructed, with feature values set for all non-alternating components of a morpheme's phonological realization.
3. Further constraint rankings and need for the use of CONTRAST PAIRS are discovered by conducted single-form learning on all individual word forms.
 - In single-form learning, for a given surface form, the UR that is most different from the SR in terms of unknown features is used to derive the surface form. For example, if ['pa-ka] has found non-alternating vowel length for the morpheme on the left, but stress and vowel length alternations are found at all other sites, the most different conceivable UR for ['pa-ka] is /pa-'ka:/.
4. One by one, CONTRAST PAIRS are built and used to set unknown underlying features of morphemes and find further constraint rankings. The algorithm begins with contrast pairs that have the fewest number of unset underlying features.
 - A contrast pair is a pair of two surface forms that are identical in all but one morpheme.
 - Where the contrast pair contains allomorphy, INCONSISTENCY DETECTION is used to identify UR+grammar combinations (given known rankings and known URs) that would be inconsistent, i.e., cause MRCD to fail to discover a valid ranking.
 - Setting unknown feature values is achieved by testing their consistency with known rankings.
5. The algorithm terminates when no further underlying features can be set.

In other words, the entire procedure relies on initial phonotactic learning to discover constraint rankings that must hold true of all identity maps, then employs inconsistency detection to identify different and mutually incompatible paradigmatic patterns that indicate different underlying structures; temporary underlying values may then be tested to discover possible underlying feature values consistent with the totality of the rankings and URs that have already been established.

Since extensive, step-by-step exemplification of this learning procedure is available in Tesar 2014: Ch. 7–8, I refer the interested reader there (for a helpful summary of the central ideas developed in Tesar’s book, see also Magri 2014).

Of present relevance is principally whether the Output-Driven Learner sketched above presents an alternative compatible with the theoretical commitment to Maximum Entropy grammars adopted above. In theory, the algorithms described by Tesar could be paired with a grammar using weighted constraints, but where the grammar generates a probability distribution over candidates, how to implement inconsistency detection is not obvious. Indeed, as soon as probability distributions are generated, attempting expectation maximizing becomes tempting, and if the total number of URs to be considered is adequately limited by heuristics considered above, a global evaluation of the enumerated options may not be prohibitively difficult, at least if URs can be selected independently of one another, or further heuristics (e.g., generalization of one UR to other morphemes implicated in the same pattern of alternative) may be employed as well. While Tesar’s Output-Driven Learner merits further research, particularly under a model of phonological grammar more straightforwardly compatible with capturing variable phenomena.

3.2.2.3 On the Learnability of Lexical Accent

Considerable progress has been made in understanding and successfully treating hidden structure problems, especially concerning prosodic structure, where several successful approaches are available. Underlying representations remain a more difficult nut to crack, but both the use of constraints on underlying representations and the use of contrast pairs represent plausible paths forward. In closing this section, I briefly consider the particular interest that lexical accent systems of the sort briefly introduced at the opening of Chapter 1 and described in greater detail at 2.4.2 hold for the learning of hidden structure.

Modeling the learnability of lexical accent has been central to research on the parallel learning of phonological grammars and underlying representations, since, as per Drescher (2016: 231), “lexical accent presents in a particularly sharp form some basic problems that learning models must overcome.” What is particularly challenging, from the perspective of learnability, is that acquiring such systems potentially requires dealing with both sorts of hidden structure discussed above: determining the correct metrical representations *and* discovering the correct underlying representations. For instance, given a (quantity-insensitive) percept $[\sigma' \sigma \sigma]$, primary stress on the penultimate syllable might not only be attributed to one of multiple different foot parses, but also due to faithfulness to lexical stress, overriding the preferences set by a ranking of prosodic markedness constraints. For the larger question of prosodic change, understanding how a learner reacts, given the fact that a she cannot know at the outset whether the language at hand requires lexical stress or can generate its stress patterns through the interaction of markedness constraints alone is of import for the larger project.

Although the PAKA-World of employed by Tesar on the learning of underlying representations (see immediately above) is indeed concerned with the learning of lexical stress

and the ranking of prosodic markedness constraints, it does not, however, deal with stress arising from hidden foot structure, and thus only really concerns hidden structure arising from underlying representations. Apoussidou (2007: 187–93) constitutes perhaps the only detailed attempt in the literature to model the simultaneous learning of both URs and foot structure via simulation, and the results obtained have interesting implications. Apoussidou’s model consists of fairly standard prosodic markedness constraints, plus four faithfulness constraints on lexical accents (MAX/DEP(Root) ‘don’t delete/insert an accent on a root’ and MAX/DEP(Affix) ‘don’t delete/insert an accent on a suffix’), and 12 UR constraints (two per morpheme, one with a lexical accent and one without, e.g., *|‑a| and *|‑á|). The overt training data contain just six forms of Modern Greek, shown in (47).

(47) Overt Forms in the Double Hidden Structure Simulation of Apoussidou 2007: 187–93

[ˈɣondola]	‘gondola:F.NOM.SG’	[ˈθalasa]	‘sea:F.NOM.SG’	[asˈtrayalos]	‘ankle.M.NOM.SG’
[ˈɣondolon]	‘gondola:F.GEN.PL’	[θalaˈson]	‘sea:F.GEN.PL’	[asˈtrayaˈlon]	‘ankle.M.GEN.PL’

The basic phonological analysis of the patterns instantiated in these overt forms is uncomplicated. Where no lexical accents are present, stress is assigned to the rightmost syllabic trochee, given final syllable extrametricality; [(ˈθa.la.)⟨sa⟩] and [as.(ˈtra.ɣa).⟨los⟩] instantiate this pattern. Where only one lexical accent is present, it receives the lexical stress: [(θa.la).(ˈson)], [as.(tra.ɣa).(ˈlon)] and [(ˈɣon.do).⟨la⟩] instantiate this behavior. Finally, where two lexical accents are present in the input, the one that best accords with the prosodic markedness constraints wins; [(ˈɣon.do).⟨lon⟩] reflects this behavior, with the lexical accent of the stem, /ɣónɗol‑/, winning over the lexical accent of the inflectional ending GEN.PL /‑ón/.

Learning a ranking of markedness, faithfulness, and lexical constraints through RIP under Stochastic OT is entirely successful: nine learners trained on the six overt forms always produce the expected primary stress. However, for five of the six overt forms, (all except [ˈɣondola], where the preferences of the prosodic markedness constraints and faithfulness constraints perfectly coincide), more than one candidate could win in production with some probability, after training. In the two cases of final stress, [θalaˈson] and [astrayaˈlon], an iambic foot ([θa.(la.ˈson)]), contrary to the preferred trochaic foot) and a monosyllabic foot ([θa.la.(ˈson)]), contrary to preferred binary feet) compete. In the three other cases of antepenultimate stress, [ˈɣondolon], [ˈθalasa], and [asˈtrayalos], different URs prove to be plausible, with the less likely alternative occurring in 25–40% of evaluations in most simulations.⁵² This general result indicates that different learners could come to generate identical overt

⁵²The competition between multiple URs where none is expected may be a consequence of Apoussidou’s constraint set. Since the constraint set contains both MAX/DEP(Root) and (MAX/DEP(Affix)), the overt form [ˈɣondolon] can be explained by ranking MAX(Root) ≫ MAX(Affix). Unnecessarily distinguishing between root-faithfulness and affix faithfulness is the source of a problem described by Apoussidou (pp. 191–2) in the performance of her simulations, namely, that the overt form [ˈɣondolon] is sometimes given the parse /(ɣón.do).l-on/, treating the GEN.PL suffix as unaccented. With only a single MAX-PROM in the grammar, *|‑on| could more freely outrank MAX-Prom, and leave prosodic markedness constraints to ensure that the correct output results.

forms via different grammars. Furthermore, those different grammars, acquired on the basis of one set of learning data, could lead to different outputs in novel contexts; for instance, a learner more likely to select the stressless UR /-on/ of the GEN.PL might be more likely to generate non-final stress when producing novel genitive plural forms. To what extent the variability in grammar acquired depends upon specific details of Apoussidou's experimental design (i.e., choice of constraints, use of the GLA with Stochastic OT) is not clear; replication with different design features would be desirable.⁵³

Overall, the acquisition of a lexical accent system *à la* Apoussidou amounts to learning three crucial pieces of information about the grammar and lexicon:

- Lexical representations carry prosodic marking of some kind, and faithfulness constraints referring to those prosodic marks are relevant in the grammar.
- Lexical representations can contrast in prosodic marking, and ranking of the corresponding UR constraints amounts to learning the correct lexical property.
- Parsing of foot structure in cases where no underlying prosodic marks can be present (because violation of the corresponding faithfulness constraints would create an incorrect surface form) must lead to a ranking of prosodic markedness constraints that generates a default pattern.

Keep in mind, moreover, that learning of all this information reduces to the learning of constraint ranks given some set of overt forms (where morphological segmentation and the induction of UR constraints is taken for granted, i.e., provided by the analyst).

Diametrically opposed in orientation on several crucial theoretical choices is the approach outlined in Dresher 2016 (compare also the brief treatment of lexical stress in Halle and Isardi 1995: 415–8). First, Dresher's analysis of metrical phonology is parametric, rather than based on constraint interaction; second, the construction of prosodic constituents consists in process application, rather than being determined by an interplay of markedness and faithfulness; third and finally, underlying prosodic properties – lexical accent – are to be induced from contrast to already determined (“set”) lexical features. In this latter respect, the notion of how lexical stress alternations should be handled (i.e., where URs “sit” in the grammar) and acquired bears fundamental similarities to Tesar's (2004, 2014) work.

⁵³The simulations with GLA + Stochastic OT in Apoussidou 2007: Ch. 4–5 on the purely metrical stress systems of Classical Latin and Pintupi, where different learning agents likewise acquired different prosodic grammars, is to my mind a strong indication that the selection of constraints and the architecture of Stochastic OT itself probably permits more variation than desired. Such purely metrical stress systems ought to contain fewer ambiguities than any lexical stress system, and the stress pattern of Pintupi in particular (syllabic trochees, primary stress on the initial syllable, secondary stress on every footed odd syllable, no degenerate feet) ought, one might hope, be regularly acquired with a consistent grammar (i.e., the right analysis should be fairly obvious, given the right data and constraints). Indeed, precisely a Pintupian-type stress system appears, in the simulations conducted by Staubs (2014: 88–98), to be extremely stable: its probability of transitioning to a different stress grammar, given the same data, is virtually 0. A Pintupi-like system is also similar to the *only* stress grammar (primary initial stress, no secondary stress) consistently correctly acquired by the Naive Parameter Learner of Yang 2002 in Pearl 2011 (cf. Nazarov and Jarosz 2017: 9).

As mentioned under 2.4.2, the view taken by Dresher on lexical accent in the context of the Simplified Bracketed Grid model is that lexical accent functions essentially like a heavy syllable in a quantity-sensitive system, with the difference that, in quantity-sensitive systems, a heavy syllable projects a left or right bracket for building feet onto line 0 of the grid *by rule* that refers to heavy syllables, whereas syllables with lexical accents carry a left or right bracket *as part of their lexical representation*. In this respect, Dresher's view on the lexical representation of lexical accent resembles the view argued by Özçelik (2014) to hold for Turkish or Yates (2020) to hold for Vedic Sanskrit (see also Chapter 4): lexical accent is underlying foot structure.⁵⁴ A typology of unaccented, accented, preaccenting, and postaccenting syllables after Dresher (2016: 234) is shown in (48).

(48) Representation of Lexical Accent under the Simplified Bracketed Grid

a. Accented Syllable (left-headed foot)

x	Line 1
x (x x	Line 0
σ (σ σ	Syllables

b. Accented Syllable (right-headed foot)

x	Line 1
x x) x	Line 0
σ σ) σ	Syllables

c. Preaccenting Syllable

x	Line 1
x)x x	Line 0
σ)σ σ	Syllables

d. Postaccenting Syllable

x	Line 1
x x(x	Line 0
σ σ(σ	Syllables

In much the same way that learning the grammar and lexicon of a lexical accent system under a constraint-based model with UR constraints consists in learning the ranking of those constraints, learning the grammar and lexicon of a system using SBG amounts to setting several prosodic parameters (similar, though not identical, to metrical stress parameters discussed in 2.2) and the lexical constituent edges borne by particular potential syllable heads. The relevant parameters in SBG are listed in (49); see generally Halle and Isardi 1995. Algorithms for learning constraint rankings or weightings have been explored in detail above; learning parameter setting is a rather different enterprise, dependent upon attempting to set parameters based on *cues*, which may require comparing the data from multiple word forms (e.g., the cue for quantity sensitivity should consist in identifying words of the same number of syllables with different stress patterns); see Dresher and Kaye 1990 for a proposal as to

⁵⁴ An alternative explicitly excluded by Dresher (2016: 233) is that a syllable might have a lexical prosodic mark such as “A” (accented) or “U” (unaccented) similar in nature to L (light syllable) and H (heavy syllable), where a rule could then refer to the mark in the way that quantity-sensitive systems supposedly refer to the H to project brackets onto line 0 of the grid.

the order in which parameters for metrical stress should be set.⁵⁵ While prosodic structures are hidden, just as assumed by constraint-ranking approaches, a parameter-setting model takes them to become gradually less hidden as progressively more parameters are successfully set.⁵⁶

- (49) Principal Metrical Stress Parameters in Simplified Bracketed Grid Theory
- a. Quantity Sensitivity: Yes/No
If yes: Project a {left/right} bracket to the {left/right} of a heavy syllable on line o.
 - b. Line o Edge Marking: Project a {left/right} bracket to the {left/right} of the {left/right}-most element on line o.
 - c. Iterative Constituent Construction: Yes/No
If yes: Project a {left/right} bracket for each pair of elements on line o, starting at the {left/right} edge.
 - d. Line o Head Marking: Constituents on line o project their {left/right}-most element onto line 1.
 - e. Line 1 Head Marking: Project a {left/right} bracket to the {left/right} of the {left/right}-most element on line o.

Setting parameters is based on responses to specific *cues*, where the learner is assumed to be able to compare all types of prosodic patterns available in the learning data and to be able to reason about the significance of some contrast, or lack thereof, in a fashion similar to the analyst. That is, given the toy forms consisting of stems and inflectional suffixes in (50) (modeled on Russian stress patterns), Dresher allows for the learner to reason in the following fashion to arrive to the necessary lexical representations.

- (50) Toy Lexical Stress Data in Dresher 2016: 243–4 (underlining marks a lexical constituent bracket, acute marks primary stress)
- a. Accented + Accented: goróv-a, dólóv-a
 - b. Unaccented + Accented: kolov-á
 - c. Postaccenting + Accented: molov-á
 - d. Accented + Unaccented: goróv-u, dólóv-u
 - e. Unaccented + Unaccented: kólov-u
 - f. Postaccenting + Unaccented: molov-ú

⁵⁵Dresher (2016: 239–41) further allows that the setting of parameters may be dependent upon frequency of patterns in the data, but does not elaborate on their exact role in cue-based learners. For an explicit attempt to incorporate probabilities into parameter setting, see Nazarov and Jarosz 2017.

⁵⁶While Robust Interpretive Parsing never allows for the overt data to be enriched with hidden prosodic structure in the course of learning, the learning of particular rankings or weightings serves to make particular parses of the data more likely or even certain, and is thus practically equivalent to the setting of a parameter in a cue-based learner.

First, since all the data contains forms with identical syllabic structures (CV.CV.CV), but different stress patterns, quantity-sensitivity cannot be used to explain differences in stress, and therefore the language is taken to be quantity insensitive. The learner is then allowed to sort stems themselves – presumably able to look across all paradigmatic forms of stems – based on whether they always bear stress on the stem (e.g., *goróv-*), never bear stress on the stem (e.g., *molov-*), or exhibit two stress allomorphs (e.g., *kolov-~kólov-*). From this contrastive comparison, the learner can conclude that stems like *goróv-* must be marked with a lexical accent (either a *ˈ* or a *ˌ*, depending upon what parameters for constituent construction are ultimately set), and that stems like *molov-* must be postaccenting (i.e., marked as */molov(-/*, with a constituent edge to the left of the stem). Next, in comparing the behavior of affixes in comparison each other, the learner discovers that some affixes appear with stress only after the postaccenting stems (e.g., *-u~-ú*) and these must therefore be unaccented; the remaining affixes are accented. At this point, the learner has discovered via contrast pairs that the language possesses a three-way lexical stress contrast in stems, and a two-way contrast in inflectional suffixes; only the exact marking of the accented stems and endings is uncertain. In short, learning the core of lexical properties is made to look easy, if the learner is assumed to be capable of reasoning about the lexical significance of contrasts in a global fashion, rather than myopically and naively.

The learner then proceeds to set the other parameters relevant for stress. Since stress can fall on initial and final syllables, extrametricality parameters are turned off. Since there is no secondary stress, iterative constituent construction is also turned off. Given knowledge of the tentative lexical representations, the learner finds only initial stress in forms consisting exclusively of unaccented morphemes (e.g., *kólov-u*), the learner can conclude that line 0 head marking is on the left, and from forms with accented morphemes only (e.g., *goróv-a*) can thereby conclude that accentedness is equivalent to a left constituent edge (in order to accord with line 0 head marking on the left). Similarly, main stress is found to fall in constituent-sized windows at the left edge, meaning the line 1 constituents are left-headed (i.e., set as LLL); finally, constituents on line 0 are filled out by placing right edge brackets (i.e., RRR) for (pairs of) elements.

Since a version of Dresher’s parameter learning for lexical stress has not been implemented computationally, explicitly comparing it to the performance of RIP with lexically specific constraints is difficult. More generally, however, the work of Nazarov and Jarosz (2017) indicates that, while parameter-setting can constitute a successful approach to the learning of metrical stress grammars, the learning model must crucially be stochastic, and must be able to compare the outputs of its own grammars to the overt learning data. These fundamental design features (comparison of learner’s parse and actual output, probabilistic outputs) are thus shared between a successful parameter learner and RIP.

At present, then, the use of constraints over underlying representations, combined with Robust Interpretive Parsing and a stochastic grammar model, as applied by Apoussidou (2007) (cf. also Tan 2022: 59–62 for simulations using PAKA-world systems), constitutes the clearest, and most easily implementable approach to the simultaneous construction of metrical stress grammars and learning of lexical stress. Furthermore, adopting constraints over underlying representations allows for a clear distinction between phonotactic learning and the

learning of alternations. How exactly the phase of phonotactic learning plays out in lexical stress systems, particularly in light of the possibility for change in stress systems, will now guide the discussion in the second part of this chapter.

3.3 Prosodic Change: Considerations of Learnability, Acquisition, and Psycholinguistics

The discussion of phonological learnability up to this point, including the learnability of lexical stress, has proceeded from one basic premise: a viable model of learnability should be capable of correctly acquiring and reproducing any attested synchronic phonological system (or at least the partial grammars explicitly examined). The predominant interest in the correct learning of a great diversity of phonological patterns that occurs throughout the literature on learnability mostly sets aside the question at the forefront of the historical phonologist's mind: when and where does change in phonological grammars come about? I formulate in (51) what I term the core conundrum of phonological learnability from a diachronic perspective: can new or different grammars be acquired without any overt changes in the structure and patterns present in the learning data? Put in more concrete terms with respect to lexical stress: if a language possesses lexically contrastive stress, why should a stress grammar without lexically contrastive stress emerge out of fundamentally similar data?

(51) The Core Conundrum of Diachronic Phonological Learnability

- a. Any synchronically attested phonological grammar (plus lexicon) must be acquirable and diachronically faithfully transmissible: what is learnable is transmissible.⁵⁷
- b. But phonological grammars (and lexica) are subject to change diachronically (i.e., are presumably not always transmitted faithfully): transmission is not always faithful.

The “core conundrum” as formulated here bears evident similarity to the classic actuation problem of language change as formulated by Weinreich et al. (1968) (see most recently the overview in Yu 2023): why should a particular change occur in a particular language at a particular time, but not in some other language seemingly having the underlying conditions, or in that particular language at some other point in time? Ideally, a well-designed model of language change in general, or of phonological change in particular (focusing on prosody and stress) must be able to account for instances of both change and stability.

In this section, I first aim to make explicit why changes relating to the phonological grammars and lexica responsible for determining the position of prosodic prominence constitute

⁵⁷For a convincing case study of phonological change driven by opacity, see the treatment of Odawa in Bowers 2015: Ch. 5. Bowers demonstrates that patterns of rhythmic syncope can create patterns of alternation in surface percepts that render it impossible for the original combination of phonological grammar + lexicon responsible for generating the data to be faithfully acquired. In such cases, restructuring of some kind seems guaranteed.

an especially challenging problem. Having clarified aspects of this issue, I show concretely under 3.3.3 how, at least in some grossly simplified cases, certain models of phonological learnability are potentially adequate for simulating change in metrical stress grammars or the loss of lexical stress via reanalysis. Under 3.3.4, I then further review research in the acquisition and perception of stress and “pitch accent”, with the aim of uncovering particular factors, especially learning biases, that might impact the (mis)acquisition of a prosodic grammar and lexicon. I close the chapter at 3.3.6 with a brief introduction to iterated agent-based learning, first introduced at 3.3.3, as a potential means of profitably modeling prosodic change at the level of phonological grammars. These considerations will ultimately serve to inform the design of simulations undertaken in Chapters 5 and 7.

3.3.1 What is a “Stress Shift”, Anyway?

The terms “stress shift” or “accent shift” are not infrequently encountered in historical treatments of prosody, where they usually designate a change in the position of word-level prosodic prominence, both at a systemic level, or the level of individual lexemes, hopefully under clearly definable phonological or morphological conditions.⁵⁸ In this usage, the persistent left-edge stress reconstructed for Proto-Germanic (see 7) may constitute a “stress shift” vis-à-vis Proto-Indo-European, but so could “Wheeler’s Law” in Proto-Greek, seemingly dependent on a pattern of syllable weight (see below), or the accentuation of Ancient Greek nouns built with the suffix *-σι-* [*-si-*] (< PIE **[-ti-]*). To make a first attempt at more clearly characterizing the notion of “stress shift”, I leave cases that strongly interface with morphological structure out of consideration. Cases in which a “stress shift” is either systemic or limited to some phonologically defined environment are then the topic of this section.

To better grasp why understanding changes in stress (~ prominence) assignment poses a particular challenge, briefly reviewing the *communis opinio* on segmental sound change (~ emergence and loss of segmental processes) may be helpful. First, coarticulation and variation in implementation at the phonetic level is ubiquitous, and results in a large pool of subphonemic variants of segments (of which speakers may possess some awareness; see German et al. 2013, Reinisch et al. 2013, Reinisch and Mitterer 2016). This large pool of variants may be phonologized in various ways (phonemic split or merger, allophone-creating or neutralizing processes) through hyper- or hypocorrection (Ohala 1981, Ohala 1989, Ohala 1993b, Ohala 1993a, Ohala 2003, Labov 1994). Novel phonological processes, which themselves tend to generate variation at the phonological level exhibit many different trajectories: they may become established as obligatory (rather than variable) allophonic or morphophonemic processes; their phonological domain of application may become broader, and result in a change “going to completion” without any conditioning environment; they may become established as transmissible sociolinguistic variable; or they may simply die out. See Kiparsky 2003b, Kiparsky 2015, and Kiparsky 2016 on these potential trajectories. The

⁵⁸I employ “stress/accent shift” in this section in an exclusively diachronic sense. I thus exclude cases of synchronic processes, such as the well-known (apparent) movement of word-level stress in phrases in English for the purposes of avoiding stress clash (e.g., *thirteén* in isolation, vs. *thirteen mén*; see Shattuck-Hufnagel et al. 1994 for a review of accounts of this particular phenomenon).

important point is that segmental sound change fundamentally involves the acquisition of a phonological process out of a pool of phonetic variation, whereby the originally mechanical (coarticulation, minute phonetic implementation) ultimately becomes psychological.

While an essentially similar account might apply to some varieties of prosodic change, e.g., tonogenesis via reattribution of acoustic effects of glottal state contrasts (see Kingston 2005 on Athabaskan, for instance), that it holds in the case of changes in grammars of stress assignment is less obviously the case. At least in languages with culminative and obligatory stress, a phonological algorithm that can generate the actually occurring pattern(s) of primary (and secondary) stress is presumably acquired. If a change occurs in the stress patterns found between an earlier and a later stage of a language, this change cannot straightforwardly be attributed to anything like phonologization of phonetic variation: the language already possessed stress, and a coherent stress assignment algorithm could have been induced from the data.

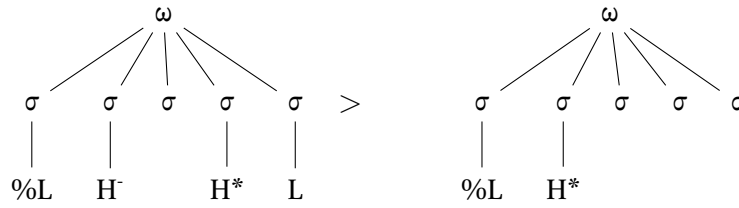
One type of change in which other aspects of prosody may impact the acquisition of a grammar of prominence assignment may be found in some dialects of Basque.⁵⁹ In particular, the word-initial boundary tone %LH⁻, supposedly reconstructable for Common Basque, may have engendered changes in the assignment of prominence. For instance, in some Western Basque dialects (e.g., Elorrio; see further Hualde 2003b) the %LH⁻ has been reinterpreted as consistent prominence on the peninitial syllable. In other words, the fact the marking of the word-level prominence (H*L) and a boundary tone (%LH⁻) shared a pitch target allowed for reinterpretation of the latter as representing prominence.⁶⁰ Thus, while the position of a word-level pitch accent was initially unbounded in Common Basque, in the affected Western Basque dialects, the position of prominence has become entirely predictable. The variety of development illustrated in 52 shows, for a hypothetical five-syllable prosodic word, that the H⁻ component of the boundary tone may be reinterpreted as a H* associated with the peninitial syllable, resulting in total disappearance of the original H*L pitch accent.⁶¹

⁵⁹The essential word-prosodic properties of the last common ancestor of all Basque dialects may be reconstructed as essentially identical with North Bizkaian dialects, in particular the Gernika-Gexto subtype (Hualde 2022: 34–5). The word prosody of these dialects is characterized by four major features: 1) a distinction between lexically accented and unaccented lexemes; 2) accented lexemes receive a default word-level pitch accent on the leftmost accented syllable, which is always non-final; 3) unaccented lexemes only surface with a nuclear pitch accent in phrase-final position, on their final syllable; and 4) all words present a %LH⁻ initial boundary tone.

⁶⁰See Hualde et al. 2002: 574–5 on these facts, where the authors further propose that contact with Spanish, which instead shows a %L initial boundary tone, was a crucial motivating factor; accommodation to the Spanish boundary tone thus has an impact on the nature of the pitch accent or the syllable to which it is assigned.

⁶¹Incidentally, Egurtzegi and Elordieta (2022: 13) propose that the L component of the H*L pitch accent found in Northern Bizkaian may have arisen from an original word-level L% boundary tone.

(52) Reinterpretation of Boundary Tone with Initial Rise as Peninitial Pitch Accent (Western Basque)



If this interpretation of the prosodic history of Western Basque dialects is correct, then, at least in languages with consistent word-level pitch accents (or “prominence tones”), such as most varieties of Swedish and Norwegian or Egyptian Arabic, reinterpretation of word-level boundary tones may result in a systemic “prominence shift”. Note, however, the role of language contact with Spanish proposed by Hualde et al. (2002) as a crucial actuating factor behind this change. Indeed, if a bias towards interpreting higher F_0 as marking prominence exists, then any language having regular word-level boundary tones with a High component, could, in principle, undergo a reinterpretation of an intonational boundary High as marking the element with word-level prominence.

In light of cases such as Western Basque, one perhaps not unreasonable hypothesis might be that true reanalysis with respect to stress systems internal to the phonology itself does not, in fact, ever occur. That is, unless relevant, potentially disambiguating data are truly “maliciously withheld” (in the words of Tesar and Smolensky 1998: 246) from a learner, or the structure of the data has phonetic properties that are amenable to some reinterpretation, a grammar for the assignment of metrical prominence should *always* be accurately acquirable and diachronically transmissible. To be sure, speculation that relative frequency or infrequency of certain stress patterns, combined with the overlap in predictions made by distinct grammars, might allow for a reanalysis of the grammar (see, e.g., Probert 2012 on the right edge stress window of Ancient Greek, or Lahiri 2015 on different stress grammars in the history of Latin), is common enough. However, demonstrating that such cases really constitute a “mislearning” of the same surface data, as opposed to being driven by some change in the structure of the learning data itself, is empirically challenging. It might turn out to be the case that changes in phonological grammars are always motivated by the phonetic substance of the learning data, much as segmental sound changes via hyper- or hypocorrection are preceded by sub-phonological phonetic developments.

The purpose of the following two brief case studies is to present credible arguments against this strong view, and instead suggest that changes in phonetic substance or the potential novel reinterpretation of preexisting surface features need not underlie every prosodic change concerning the position of word-level prominence. The first, on Latin, merely offers confirmation and substantiation of a widely held assumption: when two stress grammars make strongly overlapping predictions, one is open to reinterpretation as the other. In such cases, precisely the frequency of potentially disambiguating forms is crucial: the data need only be accidentally withheld, or be sufficiently infrequent as not to be able to contradict a learner’s preexisting hypothesis. The second, on a limited “accent shift” in Proto-Greek (“Wheeler’s Law”), is argued to be comprehensible in terms of an innovative constraint rank-

ing of a standard markedness constraint (PARSE- σ) with respect to faithfulness to the position of prominence in underlying representations (MAX-PROMINENCE).

3.3.1.1 From Old Latin to Classical Latin

In contrast to the case of Wheeler’s Law just mentioned, the differences in stress assignment between the well-known Classical Latin system and the Old Latin (Plautine) system examined at 2.2.7.1 has not usually been framed in terms of a mechanical, surface movement of prosodic prominence, but instead considered as a matter of a change in parameter setting or constraint rankings (see Jacobs 2003a and Jacobs 2003b on the latter). In terms of surface outputs, the two stress grammars produce nearly identical outputs, probably differing only in four-syllable words of the shape $/\sim\sim\sim\sigma/$ and $/-\sim\sim\sim\sigma/$, where the Old Latin system generates preantepenultimate stress, but the Classical system antepenultimate stress (cf. (28) under 2.2.7.1).⁶² Framed concretely, for two- to five-syllable words, when the weight of the final syllable is ignored, 30 different patterns can occur, and in all but the two just named shapes, primary stress will be assigned to the antepenult ($28/30 = 93.\bar{3}\%$).

To obtain, as a first approximation, a sense of the relative frequencies of words of different prosodic shapes Latin, I processed a small corpus of 72580 words from the *De bello gallico* of Caesar (up to the end of Book 7, chapter 71), and outfitted it with macrons to mark vowel length using the Python script available from <https://github.com/Alatius/latin-macronizer>. Scripts written in R were then used to convert each word to abstract sequences of heavy (H) and light (L) syllables, marking all final syllables as indifferent (σ). Table 3.1 gives the token frequencies of all occurring prosodic shapes up to five syllables in length (shapes of frequency rank 27 and 32–38 constitute six-syllable words and are suppressed here). Monosyllables (frequency ranks 2 and 82) have 13182 tokens (18.2%), while words of more than five syllables have 1020 tokens (1.4%).⁶³ The two prosodic shapes for which the Old and Classical rules make different predictions are given in bold, and together constitute just 1.3% of all tokens (977).

To determine the likelihood that a quantity-sensitive stress grammar permitting only a three-syllable window, as in Classical Latin, might be induced from data containing tokens of preantepenultimate stress, as in Old Latin, simulations using Robust Interpretive Parsing were carried out in Praat (Boersma and Weenink 1992–2023). A distribution of token frequencies for two- to five-syllable forms based on the data from Caesar described above was provided, with the weight difference in final syllables was maintained (i.e., every final syllable is either “H” or “L”) and primary stress was coded manually, according to the presumed Old Latin stresses; these constitute a total of 59 distinct prosodic shapes (one possible five-syllable pattern with a light final syllable did not occur). Using the function `Create metrics grammar...`, tableaux with possible parses of overt forms containing only pri-

⁶²Recall that the Old Latin grammar appears to forbid degenerate feet, but to allow stress clash. The two shapes with preantepenultimate stress are thus parsed as $[(\sim\sim)\sim\langle\sigma\rangle]$ and $[(\sim\sim)(\sim\sim)\sim\langle\sigma\rangle]$. In all other cases, the rightmost foot will begin on the antepenultimate syllable.

⁶³Files containing these frequencies, used as inputs to the simulations in Praat discussed below, are available at <https://github.com/rpsandell/Sandell-LMU-Habilitation/tree/main/Chapter%203/Latin%20Stress>.

Frequency Rank	Prosodic Shape	Token Frequency
1	H σ	13708
2	H	13181
3	L σ	8487
4	HH σ	6545
5	HL σ	6092
6	LH σ	3774
7	LL σ	3063
8	HLH σ	2524
9	HHH σ	2016
10	HLL σ	1846
11	HHL σ	1736
12	LLH σ	1290
13	LHH σ	1114
14	LHL σ	1103
15	LLL σ	932
16	HLLH σ	538
17	HHLH σ	531
18	HLHL σ	460
19	HLHH σ	404
20	LHLH σ	332
21	LLLH σ	329
22	HHHL σ	245
23	LLHL σ	240
24	HHHH σ	236
25	LLHH σ	221
26	LHHL σ	168
28	HHLL σ	109
29	LLLL σ	109
30	LHHH σ	97
31	LHLL σ	85
38	HLLL σ	45

Table 3.1: Token Frequencies of Prosodic Shapes (1–5 Syllables) in the First 72580 Words of *De bello gallico*

# Training Tokens	% Correct	[ˈ ~ ~ ~ σ]?	[– ˈ ~ ~ ~ σ]?
100000	~ 99%	~99%	0%
90000	~ 95%	~99%	0%
80000	~91%	~99%	0%
70000	~93	~97%	0%
50000	~88%	~92%	0%
40000	~85%	~38%	0%
30000	~ 85%	~33.5%	0%
20000	~92%	~23%	0%
10000	~89%	~12%	0%
8000	~92%	~7%	0%
5000	~84%	~1.5%	~0.1%
1000	~83%	0%	~1%

Table 3.2: Fraction of Correct Parses and Performance on Preantepenultimate Stress for Stochastic OT Grammars trained on the Preclassical Latin stress pattern

mary stress, marked by “1” such as [H H₁ L H] → /(H₂) (H₁) L H/ were automatically generated, along with their violation profiles for 14 constraints.⁶⁴

The grammar was trained using the surface distribution of Old Latin primary stresses, using Stochastic Optimality Theory (Boersma and Hayes 2001) with an evaluation noise of 2, and standard Robust Interpretive Parsing (Tesar and Smolensky 2000: Ch. 4).⁶⁵ All constraints were assigned a positive ranking value of 100 in the initial state. Table 3.2 reports the fraction of correct parses (i.e., the proportion of possible inputs for which the grammar learned generates an overt form identical to the overt forms in the training data), given different amounts of training data, and reports the proportion tokens for words of the shapes / ~ ~ ~ σ/ and /– ~ ~ ~ σ/ predicted to have primary stress on the antepenultimate syllable. The proportions there show the best performance on 10 runs, using an evaluation noise of 2 in both training and testing.⁶⁶

The relationship between the amount of training data and proportion of correct parses is not always positive; results vary by simulation, such that performance may be worse given more tokens. In particular, because the correlation between primary stress and heavy syllables is strong (in the training data, primary stress occurs on a heavy syllable in 74.9% of tokens), WSP maintains a sufficiently high ranking value with respect to NONFINALITY-σ that some proportion of tokens are generated with stress on heavy final syllables.⁶⁷ The specific

⁶⁴The constraints included were: *CLASH, *LAPSE, ALL-Ft-R/L, MAIN-L/R, ALIGN-ω-L/R, TROCHAIC, IAMBIC, NONFINALITY-σ, PARSE-σ, and WSP. See 2.3 on these constraints and their definitions.

⁶⁵This combination of grammar model and parsing algorithm are used here precisely because they are known to generally exhibit worse performance than alternatives (Jarosz 2013, Boersma and Pater 2016). If the Old Latin stress pattern is regularly correctly learned with such a model, it should not be regarded as posing particular learnability challenges, at least given the constraint set employed.

⁶⁶Occasionally, performance was much worse than the best result reported here, sometimes even below 50% at relatively large numbers of tokens.

⁶⁷Anecdotally, I report here that the performance of the alternative Resampling Robust Interpretive Parsing

performance on words of the shape $/\sim\sim\sim\sigma/$, does appear to consistently improve with larger amounts of training data. Because more data supports stress closer to the right edge of the word, and words shorter than four syllables are mostly ambiguous as to the status of ALIGN- ω -L, a parse generating antepenultimate stress, $[\sim(' \sim) \sigma]$, as in Classical Latin, is preferred until relatively larger amounts of data have been encountered. The range between 40000 and 50000 tokens seems to constitute the bend in the S-shaped curve, where learning transitions from mostly generating $[\sim(' \sim) \sigma]$ to instead mostly generating $[(\sim \sim) \sim \sigma]$.

Preantepenultimate stress in five-syllable forms of the shape $/-\sim\sim\sim\sigma/$, on the other hand, is consistently not correctly learned. Given that this pattern is, in fact, the rarest among all two- to five-syllable words, this behavior is perhaps unsurprising. This behavior is though more likely a consequence of the constraint inventory made available: since feet of the shape ('H L) are not penalized (no equivalent of RH-CONTOUR is present), the ranking of PARSE- σ above ALL-FT-L means that the parse $[(\sim \sim) (' \sim) \langle \sigma \rangle]$ is preferred. However, when the constraint FOOTBIMORAIC (“no foot should contain more than two morae”) is included as well, although overall performance improves, a parse $[(\sim \sim) (' \sim) \sim \sigma]$ still fails to be generated. Perhaps training data with overt secondary stress, indicating the licitness of *CLASH violations is necessary. Alternatively, the behavior found here may constitute evidence that acquiring a grammar that generates the output $[(\sim \sim) (' \sim) \sim \langle \sigma \rangle]$ – primary stress on a light preantepenult in a five-syllable word – is difficult, if not impossible. The statistical evidence for the parse $[(\sim \sim) (' \sim) \sim \sigma]$ from Sturtevant (1919: 242–3) merits reconsideration.⁶⁸

The crucial overall conclusion from the simulations undertaken here is that the frequency distribution of prosodic shapes in Latin is such that evidence for the Old Latin stress rule is indeed comparatively weak. Consequently, the learning trajectory predicts mostly antepenultimate stress in words of the shape $/\sim\sim\sim\sigma/$ until a substantial amount of data has been seen, while also largely correctly generating stress words of other shapes (when not partly derailed by overzealous WSP). In other words, while preantepenultimate stress is certainly learnable via RIP, it does not emerge quickly and easily. These results are thus compatible with the interpretation that a learner might maintain an early grammar hypothesis, if accidentally deprived of a sufficient number of tokens with preantepenultimate stress. I conclude that the learning trajectory for the acquisition of metrical stress through frequency relations in words shapes constitutes a sufficient basis for understanding why the Old Latin stress rule was likely to develop into the Classical rule.

3.3.1.2 Wheeler’s Law in Proto-Greek

Consider now the case of so-called “Wheeler’s Law” in the history of Ancient Greek (Wheeler 1885: 60–104; cf. also Collinge 1985: 221–3, Probert 2006b: 88–96 and Gunkel 2013). This “accent shift” is usually taken to have occurred in Common Greek. The empirical basis for the

and Expected Interpretive Parsing, at least in their implementation in Praat, is not always better on this data than standard RIP. Compare the discussion of the similar Classical Sanskrit stress pattern at 5.2.5.

⁶⁸Note further that the similar Classical Sanskrit stress pattern treated in Chapter 5 has antepenultimate stress in this same configuration, and instead has preantepenultimate stress in five-syllable words only in the configuration $/\sigma-\sim\sim\sim\sigma/$, parsed as $[\sigma(' \sim) \sim \langle \sigma \rangle]$.

phenomenon consists of lexemes and disyllabic suffixes that have cognates in Vedic Sanskrit, where the prominence in Vedic stands on a final syllable, but in Ancient Greek, the high tone instead stands on the penultimate syllable. Under the standard interpretation, Vedic preserves the original position of the prominence, while the Greek position of the high tone constitutes an innovation. *Paradebeispiele* are cited in (53), showing the reconstructed PIE form and the attested form in 1st Millennium Attic-Ionic Greek. The relevant phonological generalization appears to be that, when a sequence of two light syllables stood at the right edge of the word (given final consonant extrametricality) and were preceded by a heavy syllable, and prominence originally stood on the final syllable, it moved one syllable to the left onto the penult, thus: *[H L ́] > [H ́ L L].⁶⁹ Note that any prominence on a final syllable must be ascribed to a lexical prosodic feature, since PIE and Greek at any stage would have preferentially assigned prominence to a syllable farther to the left.

- (53) Wheeler's Law in Ancient Greek: PIE *[H L ́] > Greek [H ́ L L]
- a. *[poikílos] > ποικίλος [poikílos] 'multi-colored:NOM.SG' (cf. Vedic *peśará-*)
 - b. *[dedeg-məh,nós] > δεδεγμένος [dedegménos] 'awaiting, expecting:NOM.SG' (cf. Vedic PART.PERF.MID suffix *-āná-*)
 - c. *[teh,likós] > τηλίκος [te:líkos] 'so great'
 - d. **Contrast:** χαμαλός [k^ht^hamalós] 'on the ground' (not ^xχαμάλος)

Mechanically, the description [H L ́] > [H ́ L L] adequately characterizes the facts, but says nothing about whether some other prosodic feature might have been reanalyzed as high pitch on a penult, or what might have changed in a grammar of prominence assignment to allow for a diachronically unfaithful development.⁷⁰ Since many – perhaps most – examples of Wheeler's Law involve forms of four or more syllables, explaining the innovative High pitch on the penult as an initial boundary tone, besides the fact the no independent evidence for such a boundary tone is available, is not credible. Conversely, retraction to avoid crowding with a final word-level boundary tone is intuitively plausible, but why such retraction would not apply to words with a light antepenult and High on the final as well (i.e., [L L ́]) is not self-evident.

A different approach might assume that Wheeler's Law somehow improves metrical structure. Assume that High pitch, at the relevant phase of Proto-Greek, also signals the metrically prominent syllable.⁷¹ Avoidance of prominence on a final syllable (high-ranking NONFINALITY-σ) would be intuitively plausible, but also not compatible with the data, since

⁶⁹Wheeler's original formulation, in his chapter heading (p. 60 in Wheeler 1885) is: "Daktylisch ausgehende Oxytona werden zu Paroxytona".

⁷⁰See further Probert 2006b: 88–96 on apparent exceptions to Wheeler's Law: both cases of apparent over-application and underapplication occur in 1st Millennium Greek. Cases of underapplication are largely explicable as younger formations no longer subject to the grammar that induced the "shift". Cases of overapplication are explicable through lexical restructuring, where variants of the same suffix originally created by Wheeler's Law were leveled out.

⁷¹This assumption may not be true for (all) Greek dialects of the 1st Millennium BCE. See generally 6, in particular the "Sauzet-Golston" model of the Ancient Greek recessive accent explicated at 6.2.2.1.

High tone was allowed on final syllables outside the Wheeler's Law configuration. If, however, syllables are assumed to be parsed into moraic trochees (or at least that at least two feet close to the right edge of the word were admitted), and that no more than one violation of ALL-Ft-R was permitted, a difference between words of the shape $/\sigma_o H L L/$ and words of the shape $/\sigma_o L L L/$ emerges.⁷² In the former case, assigning stress to the final light syllable in $[\sigma_o H L 'L]$ would imply a $[\sigma_o H L ('L)]$. In the latter case, primary stress on the final light syllable still results in a degenerate foot, but the two light syllables immediately to its left make a well-formed foot, within the window allowed by ALL-Ft-R $\times 2$, thus $[\sigma_o (L L) ('L)]$.⁷³

Wheeler's Law can therefore be understood as an improvement in terms of number of syllables parsed into feet: a parse $[\sigma_o H ('L L)]$ can improve over $[\sigma_o H L ('L)]$ on PARSE- σ , but a parse $[\sigma_o L ('L L)]$ does not improve over $[\sigma_o (L L) ('L)]$. (54)a. shows that, in a word consisting of the shape $/H L L \acute{L}/$, ὀρφανικός 'orphaned', maintaining the lexical prominence on the final syllable best satisfies PARSE- σ , without constructing any feet too far from the right edge of the word or forming illicit feet of the shape (H L), which would violate RH-CONTOUR. In turn, (54)b. illustrates that a word of the shape $/L H L \acute{L}/$, δεδεγμένος 'expecting', can improve the number of syllables parsed if a binary, rather than degenerate, foot is constructed.⁷⁴

(54) Wheeler's Law driven by PARSE- $\sigma \gg$ MAX-PROMINENCE

a. $/orp^han-ikó-s/ \rightarrow [or.(p^ha.ni).('kó)\langle s \rangle]$

$/orp^han-ikó-s/$		ALL-Ft-R $\times 2$	RH-CONTOUR	TROCHEE	WSP	PARSE- σ	MAX-PROM	FtBIN	NONFINALITY- σ	ALL-Ft-R
a.	$or.(p^ha.ni).('kó)\langle s \rangle$				*	*		*	*	*
b.	$or.p^ha.('ní.ko)\langle s \rangle$				*	**!	*			
c.	$(,or.p^ha).('ní.ko)\langle s \rangle$	*!	*!				*			**
d.	$or.('p^há.ni).ko\langle s \rangle$				*	**!	*			*
e.	$or.p^ha.ni.('kó)\langle s \rangle$				*	**!*		*	*	
f.	$(,or).(p^ha.ni).('kó)\langle s \rangle$	*!	*!					*	*	*****

⁷²See 6.2.3 for motivation of the restriction on feet in Ancient Greek – here projected back to Proto-Greek – to a small window at the right edge of the word, such that more than one violation of ALL-Ft-R is fatal.

⁷³Note that the repair of $/L L L/$ sequences in prehistoric Greek (the so-called “Tribrach Law” of de Saussure 1884) applies only when at least one further syllable follows the $/L L L/$ sequence in a word. See most recently Steriade 2018 on this phenomenon.

⁷⁴See the tableaux with more candidates on abstract prosodic forms at in Wheeler's_Law_2.txt in the GitHub repository at <https://github.com/rpsandell/Sandell-LMU-Habilitation> (Chapter 3).

b. /de-deg-menó-s/ → [de.deg.(‘mé.no)⟨s⟩]

/de-deg-menó-s/		ALL-Ft-R × 2	RH-CONTOUR	TROCHEE	WSP	PARSE-σ	MAX-PROM	FtBIN	NONFINALITY-σ	ALL-Ft-R
a.	de.deg.(‘mé.no)⟨s⟩				*	**	*			
b.	de.(deg.me).(‘nó)⟨s⟩		*!			*		*	*	*
c.	de.deg.(me.‘nó)⟨s⟩			*!	*	**				
d.	de.(deg).(‘mé.no)⟨s⟩	*!			*	**	*			**
e.	(de.deg).(‘mé.no)⟨s⟩	*!			*		*			**
f.	de.(‘dég.me).no⟨s⟩		*!			**	*			*

This account adds an explanatory layer to this particular stress shift, making it comprehensible in terms of an improvement to prosodic structure, but still fails to explain why the necessary ranking $\text{PARSE-}\sigma \gg \text{MAX-PROM}$ should have come about in Proto-Greek. Given a reconstruction like *[poĩkílos], the speakers of Proto-Greek who generated the innovative *[poĩkílos] must have been exposed to the older form, but opted to act unfaithfully, generating with their grammar a form with prominence in a position contrary to the data that they received. Moreover, as argued above, there is no solid reason to believe that any factor specifically would attract stress to a light penult or repel stress from a final V or VC syllable: innovative word prominence generated via the phonological grammar, contradicting the learning data, seems to have occurred. This situation describes the puzzle of lexical stress loss discussed immediately below.

The two cases just considered from the prosodic (pre)histories of Latin and Ancient Greek strongly suggest that some “shifts” in the position of word-level prominence, at least where just a small subset of word shapes is concerned, can be attributed to the phonological grammar proper. That is, no change in or possible reinterpretation of the surface percepts can be readily held responsible for the emergence of an innovative metrical (viz. lexical) stress grammar. Instead, the innovative grammars may be attributed to the way in which the preexisting data is, or can be, interpreted. While the fundamental actuation problem of language change is not thereby remotely resolved, that the possibility change, as opposed to unerring stability, may be seen as inherent in the structure of the learning data, constitutes a small victory.

3.3.2 The Puzzle of Lexical Stress Loss

As discussed under 3.2.2.3 above, the crucial difference between the learning of a straightforward metrical stress pattern and the acquisition of a lexical accent system is that the latter also requires a theory of acquisition of underlying representations, and, in the case of constraint-based learning, faithfulness constraints come into play. For instance, one can readily see that stress assignment in PIE, and one of its descendants, Early Proto-Germanic, must be dependent upon lexically prespecified prosodic information, for the straightforward

reason that there existed forms that were identical in terms of other prosodically relevant properties (syllable count and syllable weight), as example (55) shows.

(55) Stress Minimal Pair in PIE and Early Proto-Germanic⁷⁵

- a. 'foot:GEN.SG' */ped-és/ → *[pe.dés] > Early Proto-Gmc. *[fe.'tes] (>> later *['fo:.tiz])
- b. 'foot:NOM.PL' */pod-es/ → *[pó.des] > Early Proto-Gmc. *['fo.tez] (>> later *['fo:.tiz])

No purely markedness-driven stress-assignment algorithm can simultaneously generate both *[pe.dés] with final stress and *[pó.des] with initial stress; both inputs would necessarily have the same output.⁷⁶ At some point in the course of learning, the learner must be able to conclude that the lexical entries of the morphemes involved in these words have particular prespecified properties in the input to the phonology, such that the surface contrast can be generated. The relatively “early” acquisition of a grammar that permits the production of minimal stress pairs stands out in the context of much work on learnability in Optimality Theory, which assumes an initial stage of so-called “phonotactic learning” (see 3.2.1.1 above).⁷⁷ Already during the phase of phonotactic learning, assuming a model of Error-Driven Constraint Demotion plus Robust Interpretive Parsing for learning constraint rankings, a learner ought to quickly conclude that a constraint like MAX-PROMINENCE must dominate any prosodic markedness constraints that could be used to explain why stress falls on the initial syllable in a EPGmc. *['fo.tez], but on the final in *[fe.'tes].

The example employed in Tesar 2014: 336–8, on phonotactic learning of a language with lexically contrastive stress, where the both the surface word forms [pá.ka] and [pa.ká] occur, results in the acquisition of the (sub)-ranking IDENT-[stress] >> MAINLEFT, MAINRIGHT as soon as the learner has encountered a token of [pá.ka] and a token of [pa.ká].⁷⁸ If the

⁷⁵The difference in vowel quality of the root */pod-/~/ped-/ is to be ignored here. The homophonous later Proto-Germanic form, *['fo:.tiz], contains a third stem allomorph, [fo:t-], originally found only in the nominative singular, leveled through the paradigm. The GEN.SG suffix also had an allomorph [-ez] (> [-iz]), due to Verner's Law, which ultimately displaced the allomorph [-es] (> [-is]).

⁷⁶In principle, one could invoke, for the example at hand, preferential attraction of stress to a particular vowel (e.g., stress to [o]); I take for granted that this analytical alternative would not prove very profitable.

⁷⁷Recall that “phonotactic learning” takes place prior to morphological segmentation: the learner attempts to learn as much about constraint rankings from licit outputs via an identity map, where the input is phonologically identical to the output; cf. generally Prince and Tesar 2004 and Hayes 2004. Compare the brief explanation of Tesar (2003: 169): “The learner, lacking initial knowledge of the underlying forms, is presumed to adopt, for each observed word, a hypothesized phonological input that is identical to the observed surface form of the word. The learner's goal is to arrive at a constraint ranking that maps each observed word (as input) to itself (as surface form), but that generates as few other surface forms as possible.”

⁷⁸In the process of phonotactic learning, given an initial state where MARKEDNESS >> FAITHFULNESS, if the learner first encounters [pá.ka], MAINRIGHT will be demoted below MAINLEFT, to a stratum above IDENT-[stress]. After the learner encounters [pa.ká], however, the learner's stored learning data (the so-called “support”) will see that both MAINLEFT and MAINRIGHT are violable by winners, while IDENT-Stress is never violated by a winning output, and therefore must outrank the two markedness constraints. Note that also a purely error-driven learning algorithm, like Error-Driven Constraint Demotion (as opposed to a batch-learner like Biased Constraint Demotion or Multi-Recursive Constraint Demotion, assumed in this example), will also eventually arrive one of the rankings IDENT-[stress] >> MAINLEFT >> MAINRIGHT or IDENT-[stress] >> MAINRIGHT >> MAINLEFT, depending upon the order in which the data were encountered.

learning data contains such examples, and they are processed veridically by the learner, the learner will discover that faithfulness to lexical stress must be highly ranked, and the ultimate result down the line in the course of learning will be the construction of a phonological lexicon with prosodic representations of lexical accent.

The tableaux in 56 illustrates the learning trajectory in miniature that constraint ranking might take, given an initial MARKEDNESS $\gg$ FAITHFULNESS bias, if the learner is exposed to the types *[fe.'tes] and *[fo.tez] in that order. This procedure, assumes, of course, that the percepts *[fo.tez] and *[fe.'tes] are always interpreted veridically. In 56a., only candidates c. and d. are licit robust parses that match the overt form, and c. is more harmonic. The learner's production-oriented parse is, however, candidate a., which violates only IAMB in the top stratum. Since the RIP violates TROCHEE and NONFINALITY- σ , these must be demoted below all constraints that prefer the winner (i.e., would cause the loser [(fo.'tes)] to lose, namely, IAMB and MAX-PROM. When the learner then encounters the percept [fo.tez], in 56b., its production-oriented parse is now [(fo.'tez)], with stress identical to the RIP of the preceding tableau; the RIP in this case is, however, [(fo.tez)], which results in the demotion of loser-preferring IAMB below winner-preferring MAX-PROM, TROCHEE, and NONFINALITY- σ . Because MAX-PROM now dominates TROCHEE, NONFINALITY, and IAMB, when the learner again encounters either [fo.'tes] or [fo.tez], the remaining markedness constraints in the top stratum (ALL-Ft-L, ALL-Ft-R, FTBIN) will exclude monosyllabic parses, and MAX-PROMINENCE will be decisive, such that the RIP and POP for these overt forms will be identical and nothing further will be learned.

(56) Phonotactic Learning of Lexical Stress in Proto-Germanic:

Initial State: TROCHEE, IAMB, ALL-Ft-L, ALL-Ft-R, FTBIN, NONFINALITY- σ $\gg$ MAX-PROMINENCE

- a. Percept [fo.'tes] $\Rightarrow$ UR /fo'tes/ $\rightarrow$...
 Robust Interpretive Parse: [(fo.'tes)]
 Production-Oriented Parse: [(fo.'tes)]

/fo'tes/		TROCHEE		IAMB		ALL-Ft-L		ALL-Ft-R		FTBIN		NONFINALITY- σ		MAX-PROMINENCE	
a.	 ('fo.tes)			*										*	
b.	(fo).tes						*		*					*	
c.	RIP (fo.'tes)	*										*			
d.	fo.(tes)					*			*		*				

New Constraint Hierarchy:

IAMB, ALL-Ft-L, ALL-Ft-R, FTBIN $\gg$ MAX-PROMINENCE $\gg$ TROCHEE, NONFINALITY- σ

- b. Percept ['fo.tez] $\Rightarrow$ UR /'fotez/ $\rightarrow$...
 Robust Interpretive Parse: [('fo.tez)]
 Production-Oriented Parse: [('fo.tez)]

		IAMB	ALL-Ft-L	ALL-Ft-R	FtBIN	MAX-PROMINENCE	TROCHEE	NONFINALITY-σ
/'fotez/								
a.	RIP ('fo.tez)	*						
b.	('fo).tez			*	*			
c.	☞ (fo.'tez)					*	*	*
d.	fo.('tez)		*		*	*		*

New Constraint Hierarchy:

ALL-FT-L, ALL-FT-R, FTBIN $\gg$ MAX-PROMINENCE $\gg$ TROCHEE, NONFINALITY- σ
 $\gg$ IAMB

The behavior of EDCD/RIP during phonotactic learning in the face of data containing contrastive surface prominence makes a strong prediction: as long as the data are perceived veridically, lexical stress contrasts ought *always* to be maintained, synchronically, and more importantly, also diachronically. That is, because constraint demotion is swift and categorical faithfulness constraints that preserve contrasts in prosodic prominence will inevitably come to be ranked above many prosodic markedness constraints. In many concrete instances, such behavior is indeed desirable: the fact that languages with stress systems like Vedic Sanskrit are acquisible at all constitutes direct evidence that surface stress contrasts are not wantonly ignored by learners. Furthermore, certain historical cases, such as Latin to Italian or Ancient to Modern Greek (on the latter, see also 6.4), suggest that learners may also readily acquire new lexically specified prosody in order to be able to faithfully reproduce the position of stress in the learning data, when stress becomes algorithmically unpredictable. Such developments, where surface outputs in terms of stress seem to have been faithfully maintained, constitute an instance of what Lahiri (2015) terms “pertinacity of prosody”, with the “same outputs, [but] different grammar [and lexicon]”. The cases of *loss* of lexical stress contrasts diachronically, as seen for later Sanskrit or later Proto-Germanic, or any other Indo-European language that has not maintained the original lexically contrastive prominence system of Proto-Indo-European (cf. Halle 1997: 275; 310), are on the other hand puzzling.

The situation is perhaps only marginally better under the assumption of an error-based learner that gradually updates constraint weights by a small amount in response in each processed learning datum (e.g., the Gradual Learning Algorithm [GLA]; cf. Boersma and Levelt 2000, Boersma and Hayes 2001). With such an approach to the learning of constraint weights, assuming an initial state where all markedness constraints outrank all faithfulness constraints (cf. Prince and Tesar 2004), the markedness constraints that could produce a consistent fixed stress pattern will be demoted (and depending upon the exact update rule

and rate of learning, MAX-PROMINENCE perhaps promoted) with virtually every token processed. Simulation with the GLA in Boersma and Weenink 1992–2023 indicates that perhaps a few hundred tokens are sufficient to arrive to the categorical ranking MAX-PROM $\gg$ MAIN-LEFT, MAINRIGHT, TROCHEE, IAMB, using Tesar’s simple PAKA example. Until the learner, in the course of phonotactic learning, achieves the appropriate high ranking of MAX-PROM, its outputs on the particular lexical items will exhibit variation in the position of primary stress; nevertheless, the correct ranking of MAX-PROM, for a language with contrastive lexical stress, seems to be obtained comparatively early in the course of learning.

That phonotactic learning under EDCD/RIP or similar models designed to learn a correct ranking or weighting of constraints evidently cannot adequately account for the loss of contrastive lexical stress is not necessarily a weakness; EDCD/RIP is intended to be able to always discover a grammar that can reproduce the learning data. The incompatibility of EDCD/RIP with this type of phonological change is, however, representative of the behavior of models of learnability surveyed in 3.2.1.1 and more broadly. At least in the limit, Stochastic OT under the GLA, or MaxEnt grammars, if not equipped with any sort of biases, and given a sufficient amount of data, will normally be able assign weights to prosodic faithfulness constraints that ensure the faithful transmission of lexically contrastive stress. Given credible evidence for the possibility of changes in phonological grammars without changes in the phonetic substance of the learning data, two crucial questions present themselves:

- Does this specific type of prosodic change, i.e., loss of lexically contrastive stress, require a preceding change in the overt phonetic substance of the learning data?
- If not, in what fashion can some model of phonological learnability be crafted so as to admit of this type of change?

The case of Wheeler’s Law above suggests that, prosodic systems lexically contrastive stress may undergo at least small-scale alterations, without any obvious changes to the substance of the learning data. Therefore, a good-faith attempt ought to be made to discover what models, assumptions, and parameters are potentially compatible with either the gradual or sudden disappearance of special prosodic marking in the lexicon.

3.3.3 The Relation of Learnability and Change: Further Possibilities for Reanalysis

If the phonological percepts themselves have not changed, and the phonological data is not subject to any sort of systematic misperception, then the frequency distribution of phonological patterns might constitute the most likely source for the emergence of novel stress patterns diachronically. The further relevant question is then how “subtle differences in the relative frequencies of possible output forms generated by the grammar” (White 2013: 187) could or should give rise to reanalysis and change.

For independent reasons discussed at 3.2.1.3 above, Maximum Entropy grammars constitute the computationally most attractive model of phonological learnability considered here. One question left open there was whether constraint weights should be determined through an on-line learning algorithm (like Stochastic Gradient Ascent) or a batch algorithm.

The fact that the convergence to single grammar under on-line Stochastic Gradient Ascent is not guaranteed in the face of data exhibiting free variation – that is, it may become stuck in a local optimum – may perhaps be seen as a valuable feature, rather than a bug, from the perspective of language change. Indeed, Staub (2014) exploits this feature: a learning agent may fall into a local optimum and thereby posit a grammar different from the grammar that generated the data on which it was trained. The likelihood of encountering particular local optima in the face of particular data, when chained over generations of learning, yields Staub’s predictions on the likelihood of a learner ending up with a given grammar; see further below on Staub’s work. In short, MaxEnt SGA presents a computationally, and perhaps psychologically, plausible model for the emergence of a novel grammar out of the data generated by a different grammar.

An important question is therefore whether the behavior of batch learning algorithms can be managed in such a way as to allow for language change as well. To test the possibility of conditions under which batch MaxEnt grammars might yield a phonological change, I develop here a simple case example involving lexically contrastive stress. I show here that a simulation of intergenerational transmission (simple agent-based iterated learning with single agents; see 3.3.7 below) can indeed allow for the gradual disappearance of contrastive stress, given the right assumptions about biases in the constraint inventory, biases governing constraint weights, and the frequency distribution of stress patterns.

Consider the following case: a language possesses only disyllabic prosodic words, only open syllables, with no distinctions in syllable weight, but does have lexical contrasts in stress. Thus, the lexical entry of word prosody may be either $/\sigma \sigma/$ or $/\sigma' \sigma/$; if the language is acquired perfectly, all types with underlying initial stress, $/\sigma \sigma/$, should always be generated as a token with initial stress. In successful transmission of a grammar, learners will acquire a ranking with faithfulness to lexical stress (here, MAX-PROMINENCE) above markedness constraints that would penalize initial stress (e.g., IAMB, potentially NONINITIALITY- $\mathcal{F}$) or final stress (e.g., NONFINALITY- σ , TROCHEE, potentially NONFINALITY- $\mathcal{F}$). Assume further that lexemes with initial stress are at least twice as frequent as lexemes with final stress, e.g., 200 types versus 100 types, and the token frequency distributions across types of different stress are equivalent.

Given a robust set of prosodic markedness concerning metrical stress taken from those discussed in Section 2.3 of Chapter 2, if all markedness constraints have some non-zero weight, candidates exhibiting final stress will always be somewhat less harmonic, since violations of NONFINALITY- σ have no parallel violations in candidates with initial stress. Admitting NONFINALITY- σ to the constraint inventory thus already biases a learner against acquiring grammars that generate and/or faithfully preserve final stress. Since the foot structure of winning candidates is invisible, both candidates $[(\sigma \sigma)]$ and $[(\sigma') \sigma]$ may be considered winners for inputs with initial stress. A tableau with four production-oriented parses for the input $/\sigma \sigma/$ is shown in 57. The constraint inventory itself is biased in favor of the binary parse, $[(\sigma \sigma)]$, but $[(\sigma') \sigma]$ is compatible with the overt form $[\sigma \sigma]$ and hence marked as a (potential) winner (which would obtain if the ranking NONFIN- $\mathcal{F} \gg$ FTBIN, PARSE- σ were discovered).

(57) Output Parses for the Input /'σ σ/

/'σ σ/		MAX-PROM	LAMB	TROCHEE	FTBIN	PARSE-σ	ALL-Ft-L	ALL-Ft-R	NONFIN-σ	NONFIN-f	NONINIT-f
a.	 ('σ σ)		*!							*!	*!
b.	 ('σ) σ				*!	*!		*!			*!
c.	(σ 'σ)	*!		*!					*!	*!	*!
d.	σ ('σ)	*!			*!	*!	*!		*!	*!	

Given a Gaussian prior like the one used in the discussion of Dutch syllable structure at 3.2.1.3 above ($\mu = 10$ for markedness constraints, 0 for the single faithfulness constraint employed, MAX-PROM; $\sigma^2 = 0.6$), and only 300 total tokens (200 initial stress, 100 final stress, equally distributed among the two possible parses), the weights learned will predict initial stress in lexemes with initial stress fully 99.6% of the time, but incorrectly generate initial stress in words with lexical final stress in 18.5% of cases. When the total number of tokens increases to about 2800 (2000 initial stress, 800 final stress), the error-rate in words with final stress decreases to about 2.71%. This behavior is predicated upon the use of the Gaussian prior and a difference in initial weights: because MAX-PROM initially has a weight of 0, more data with final stress is needed to ensure that MAX-Prom receives a sufficiently high weight that the probability of generating an output $[('σ σ)]$ or $[('σ) σ]$ given the input $/σ 'σ/$ becomes virtually 0. That is, because of the bias encoded through initially high weights for markedness and low weights for faithfulness, exposure to a relatively large amount of data is needed for MAX-PROM to obtain a sufficiently high weight, and constraints penalizing final stress a sufficiently low weight, so that the input $/σ 'σ/$ is realized just as faithfully as the input $/ 'σ σ/$.

Thus, biases in terms of the constraint inventory, relative frequencies in the lexicon, and MARKEDNESS $\gg$ FAITHFULNESS in the initial states, make lexically encoded final stress more difficult to acquire. Based on these results, one might conclude that such a language could exhibit a gradual attrition of lexemes with final stress. Under the assumption that learning in a generation terminates after 2800 tokens, then 2.71% of tokens stress on the final syllable would shift to production as initial stress in a subsequent generation; this slightly larger number of tokens with initial stress, when provided to a learner with the same biases, will acquire a grammar that produces yet slightly higher rates of error on lexemes with final stress, 2.86%. Iterated over 50 generations, the decrease in number of tokens with final stress across each generation is approximately linear, but the proportion of error, i.e., the rate of change, is logarithmic: between generations 1 and 29, error on tokens with final stress increases from $\sim 2.7\%$ to $\sim 10.8\%$, but rapid change appears between about generations 35 and 42: the error rate increases from $\sim 17.9\%$ to over 99%. Figures 3.1 and 3.2 below illustrate the change in tokens predicted to have initial or final stress, and the error rate on tokens originally belonging to final stress (or, in other words, the proportion of tokens originally having final stress instead assigned initial stress).⁷⁹

⁷⁹Tableaux and R code for this simulation are available in the accompanying GitHub repository under Chap-

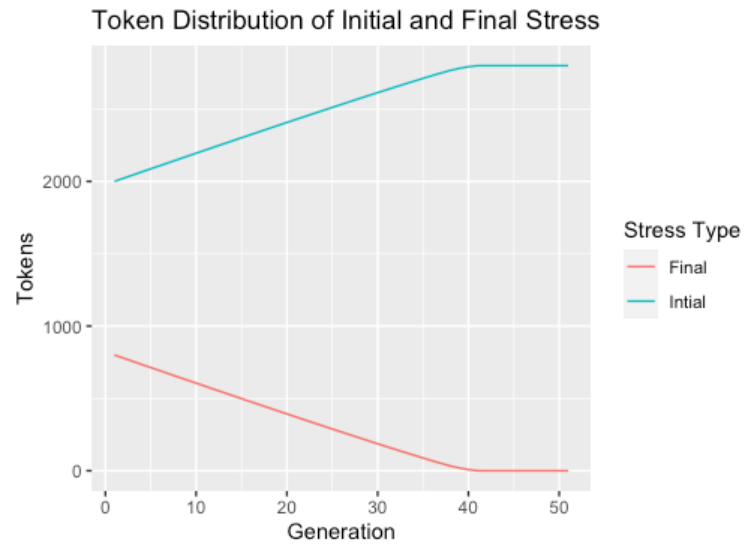


Figure 3.1: Number of Tokens with Initial and Final Stress in Each Generation

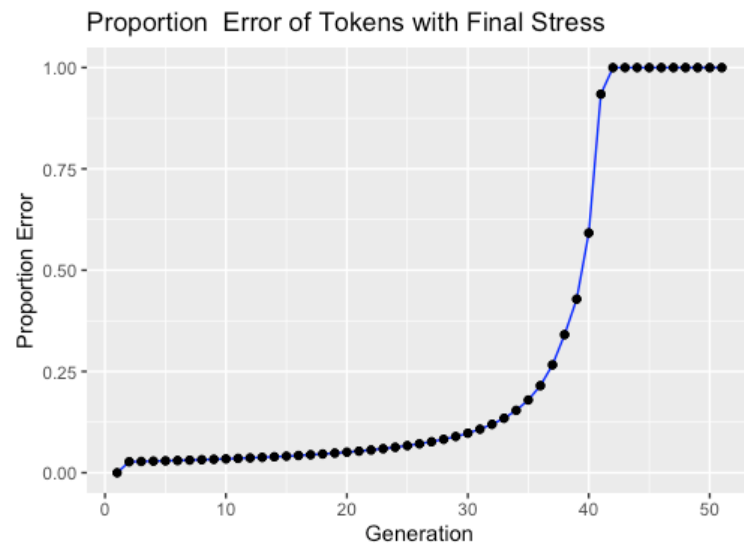


Figure 3.2: Proportion of Error on Tokens with Original Final Stress ($n = 800$ at $G = 0$)

An inverse trend occurs naturally in the weight of MAX-PROM. In the first generation, this constraint receives a relatively high, active weight (over 8.3, from an initial weight of 0). With each successive generation, since fewer tokens contradict the statistical default initial stress pattern, the weight of MAX-Prom is lower, and a collapse in its effect corresponds precisely to the virtual disappearance of final stress: between generations 35 and 42, its weight falls from over 7.2 to under 1. This behavior matches the intuition that, the less token variation in terms of lexical stress is present in the grammar, the less need there is for a faithfulness constraints like MAX-PROM to play a role: markedness alone takes over.

Although this small simulation constitutes a massive simplification of the overall learning problem, it does show, in the abstract at least, that the presence of biases – whether in the frequency distributions, the constraint set, or learning of constraint weights – might suffice as an explanation for a diachronic shift from a grammar and lexicon with lexically contrastive stress to a grammar with a fixed-stress pattern. In particular, one might intuitively expect for preexisting, perhaps entirely accidental patterns in terms of frequency relations in the learning data to be reinforced over generations. For instance, if stress on final syllables is significantly underrepresented, as in the toy case developed here, learners should – as this simulation has shown – be able to posit grammars that make possible the production of tokens with some alternative position of stress for word forms that *ought* to consistently exhibit final stress.

The sort of general account suggested here presumes, however, that learners with adult(-like) phonological competence will generate patterns of free variation in word stress, at least among lexical items belonging to one of the original classes. A crucial empirical question is then whether speakers and learners tolerate and generate free variation in prosodic properties at all. Hypothetically, seeming token variation might exist only at the population level, due to type variation at the level of individual speakers, which would imply that speakers arrange their grammars and lexica so as to make token variation impossible. To make this distinction concrete, consider the stress patterns generated for four lexical items by two speakers of the toy language treated here: both speakers have categorical initial stress on lexical items A and B, both have categorical final stress on lexical item C, but speaker 1 has categorical final stress on lexical item D where speaker 2 has categorical initial stress; at the psychological level of grammars, there is then no variation, but at the population level, lexical item D will exhibit token variation. Brief examination of two attested cases of change involving word prosody, one from English and another from South Kyungsang Korean, may prove instructive regarding this question.

3.3.3.1 Stress in English Disyllabic Noun–Verb Pairs

English possesses numerous pairs of segmentally (underlyingly) identical nouns and verbs with two underlying syllable peaks. In terms of their relative stress patterns, such pairs may be sorted into three classes, shown in 58: both members of the pair exhibit initial stress; both members of the pair exhibit final stress; or the noun exhibits initial stress, while the

ter 3/Simple Lexical Stress.

verb shows final stress.⁸⁰

(58) Patterns of Stress in English Disyllabic Noun–Verb Pairs

- a. **Initial Stress** {1, 1}: N *á*nchor vs. V *á*nchor
- b. **Final Stress** {2, 2}: N *revi*ew vs. V *revi*ew
- c. **Alternating Stress** {1, 2}: N *í*mport vs. V *imp*ort

At least at a population level, variation in the stress “class” of noun–verb pairs can and demonstrably does change over time, and the particular class membership of certain lexical pairs may vary by dialect.⁸¹ Building on earlier work by Sherman (1975) and Phillips (1984), Sondregger and Niyogi (2013) (see also Sondregger 2009) document stability and change in disyllabic noun–verbs pairs based on stress reported in dictionaries between c. 1600 and 2000 CE. In a list of 110 noun–verb pairs with a token frequency of at least 1 per million in the British National Corpus, stability is the rule: only 13 (11.8%) of the pairs in that list attest any change. Based on a larger list of 149 pairs from Sherman 1975, consultation of dictionaries revealed four possible trajectories of change: initial > alternating; alternating > initial; alternating > final; and final > alternating. Among these four trajectories, final > alternating was by far most common. Note that a conceivable pattern with final stress on the noun, but initial stress on the verb, never comes about. Occasionally, dictionaries report on variation in stress pattern or provide no certain stress, which may be interpreted as awareness of population-level or speaker-level variation.

Further crucial present-day spoken-language data was gathered by Sondregger and Niyogi from a radio corpus based on 34 stories on the American National Public Radio (originally reported in full in Sondregger 2009: 76–77). Tokens of the nouns *perfume*, *research*, *address* and *increase* from 31 speakers (18 male; 25 North American), if produced by a speaker at least five times, were collected. Importantly, this data exhibits both inter-speaker and intra-speaker variation. For instance, some North American speakers (M1) produced *résearch* exclusively with initial stress; others produced both initial or final stress (e.g., M7 had five tokens of initial stress, and four of final stress); yet other produced only final stress (M10). While not large in quantity, this data constitutes credible evidence that adult speakers, at least in a language with contrastive lexical stress, tolerate inter-speaker variation and generate intra-speaker variation in word prosody.

At least as far as the most common type of diachronic development in noun–verb pairs is concerned, final > alternating, evidence also exists of a relationship between token frequency and change in stress pattern. Specifically, as the token frequency of the noun–verb pair declines, the stronger the relationship between variable or initial stress reported in dictionaries for the noun. This behavior suggests, in line with stress patterns in disyllabic nouns

⁸⁰Productive Ø-derivation allows for the ready creation of novel denominal verbs in English, which as a rule, remain identical to the nominal base in terms of segmental and word-prosodic properties, and thus fall into the {1, 1} class.

⁸¹According to my own intuitions, some pairs are acceptable with more than one pattern, e.g., *research* seems acceptable with both initial stress or alternating stress. This intuition is borne out by the data Sondregger and Niyogi’s spoken English corpus described below.

in English more generally, that final stress is exceptional, and, when not supported by sufficiently high token frequency, the expected or default pattern (initial stress) may apply.⁸² Since most words exhibiting that alternating pattern may be analyzed as containing a bound prefix preceding a (usually bound) root, final stress (= stress on the root) may likewise instantiate the historically old requirement in Germanic that roots in verbal forms bear stress (see discussion and formalization in Chapter 7); alternatively, the relevant prefixes may be treated as post-accenting or unstressable.

Roughly speaking, the three attested patterns and their diachronic dynamics are ultimately amenable to an explanation in terms of direction of derivation, regularization of stress, and change in morphological structure:

- The {1, 1} pattern (initial stress) mostly consists of original nouns alongside denominal verbs; rare cases of {1, 1} arising from older {1, 2} (*V subjéct* > *súbject*) may be attributed to failure to perceive the verb as prefixed.
- The {2, 2} pattern (final stress) mostly consists of original verbs alongside deverbal nouns, where the noun is faithful to the stress of the verb.
- The {1, 2} pattern (alternating stress) mostly consists of original verbs alongside deverbal nouns, where the stress of the noun has been regularized.
- Rare cases of {1, 1} > {1, 2} (e.g., *V cómbat* > *combát*) constitute cases of an original noun alongside denominal verb, where the verb has been reanalyzed as containing a prefix, which induces root stress on the verb.⁸³

Regardless of the correct account and motivation of diachronic shifts in stress among English noun–verb pairs, the crucial conclusions from Sondregger & Niyogi’s work are essentially the two reported above:

- Intra-speaker variation in the production of contrastive word prosody may coexist alongside inter-speaker variation.
- Token frequency at the level of individual lexemes may influence impact the maintenance of exceptional word-prosodic patterns against exceptional patterns.

3.3.3.2 South Kyungsang Korean

Recent detailed phonetic and phonological investigations into the word prosody of South Kyungsang Korean point to a change in progress, which involves the merger and loss of lexically contrastive pitch patterns. These developments have been described in detail by Lee and Zhang (2014), Lee and Jongman 2015 and Lee et al. 2016 (cf. also overview of South

⁸²See Probert 2006b: Ch. 5–10 for similar claims concerning changes in the accentuation of nouns in Ancient Greek.

⁸³Alternatively, final stress in the verb might, in some cases, reflect younger borrowing of the verb by some speakers from a Latin or Romance source, creating competition with the segmentally identical denominal verb.

Kyungsang Korean's word prosody in and its history in Kenstowicz et al. 2008), from both phonetic and phonological perspectives. In most earlier reports in the literature, South Kyungsang (as well as the closely related North Kyungsang) possesses lexically contrastive prominence signaled by differences in Fo contours across the word, perhaps properly characterized as a restricted tone system. The recent acoustic investigations suggest the attrition and loss of these contrast contours among younger speakers, who often exhibit word-level intonational contours more similar to those found in the Seoul dialect; overall, for all lexical accent classes, younger Kyungsang speakers exhibit reduced acoustic contrast between the classes and more delayed Fo peaks (Lee and Jongman 2015: 29).

Descriptively, three classes exist for monosyllabic nominal stems and four for disyllabic nominal stems, distinguished in the analysis of Lee and Zhang (2014) by type of tone (non-spreading /H/ vs. spreading /H⁺/) and its underlying association or alignment with respect to syllable peaks. The surface patterns, whose distinctions become fully evident only when suffixes are added to stems, are shown in 59 and 60 (examples are adapted from Lee and Zhang 2014: 73–4 and Lee and Jongman 2015: 17). For instance, /mal^{H+}/ 'speech' has a spreading /H⁺/ tone aligned with its right edge (~ peninitial syllable peak), which results in a surface rising tone in the monosyllable, a L-H sequence when a single further syllable peak is added, and a L-HH sequence with a two surfurther syllable peaks are added.⁸⁴ The absence of the pattern seen with *kéwul* 'mirror' among monosyllables is attributed to the impossibility of underlying association or alignment with a penultimate peak.

(59) Pitch Patterns of Monosyllabic Nominal Stems in South Kyungsang Korean

Initial H ⁺ (H ₁)			Ø (H ₂)			Peninitial H ⁺ (R)		
Form	Gloss	Tone	Form	Gloss	Tone	Form	Gloss	Tone
múl	'water'	H	mál	'horse'	H	mǎl	'speech'	R
múl-í	'water-NOM'	H-H	mál-i	'horse-NOM'	H-L	mal-í	'speech-NOM'	L-H
múl-póta	'water-than'	H-HL	mal-póta	'horse-than'	L-HL	mal-pótá	'speech-than'	L-HH
múl-ímyen	'water-if'	H-HL	mál-imyen	'horse-if'	H-LL	mal-ímyén	'speech-if'	L-HH

(60) Pitch Patterns of Disyllabic Nominal Stems in South Kyungsang Korean

Initial H ⁺ (HH)			Ø (LH ₂)			Peninitial H ⁺ (LH ₁)			Penult H (HL)		
Form	Gloss	Tone	Form	Gloss	Tone	Form	Gloss	Tone	Form	Gloss	Tone
kúlím	'painting'	H	palám	'wind'	LH	salám	'person'	LH	kéwul	'mirror'	HL
kúlím-i	'-NOM'	HH-L	palám-i	'-NOM'	LH-L	salám-í	'-NOM'	LH-H	kéwul-i	'-NOM'	HL-L
kúlím-pota	'-than'	HH-LL	palám-póta	'-than'	LH-HL	salám-póta	'-than'	LH-HL	kéwul-pota	'-than'	HL-LL
kúlím-imyen	'-if'	HH-LL	palám-imyen	'-if'	LH-LL	salám-imyen	'-if'	LH-HL	kéwul-imyen	'-if'	HL-LL

This inventory of underlying contrasts, while recoverable from distinctions across suffixed forms, is neutralized in some instance. In particular, toneless stems (*mál* 'horse', *palám* 'wind'), when unsuffixed, neutralize with initial H^{H+} among monosyllables (*múl* 'water'), but with peninitial H⁺ among disyllables (*salám* 'person'). The distinction between toneless (LH₂) and peninitial H⁺ (LH₁) stems also disappears on the surface when disyllabic suffixes with underlying H (/ -póta/) are added: both are realized as LH-HL.

⁸⁴The analysis of Lee and Zhang 2014 in fact treats H and H⁺ tones as underlyingly linked to moras, which may be floating, in the case of monosyllabic stems with peninitial H⁺.

In acoustic data on the realization of pitch patterns gathered from four male speakers in by Lee and Zhang (2014), three of the speakers exhibited some intra-speaker variation, and across all four speakers, inter-speaker variation could be found. For instance, Lee & Zhang's speaker 1 realized $\emptyset$ (H2/LH2) and peninitial H⁺ (R/LH1) forms with disyllabic suffixes attached variably, both with the expected pitch patterns and with patterns that are suggestive of some confusion or incipient merger of the toneless and peninitial H⁺ on stems, or perhaps between toneless and H-bearing disyllabic suffixes. Specifically peninitial H⁺ (R) monosyllables with any disyllabic suffix were realized only 20% of the time as expected L-HH, but 80% of the time as L-HL, which is instead expected on $\emptyset$ (H2) monosyllables with H-bearing disyllabic suffixes (*mal-póta* 'horse-than'). Toneless (LH2) disyllables with an toneless disyllabic suffix, meanwhile, were realized by this speaker in 60% of cases as LH-HL, which is expected for peninitial H⁺ (LH1) disyllables (*salám-ímyen* 'person-if'), and 40% of the time with expected LH-LL. Speaker 3 exhibited a different pattern of variation on the same word types as Speaker 1, while Speaker 4 showed a strong tendency for neutralization of the lexical distinction between toneless and H-bearing suffixes (i.e., different disyllabic suffixes were largely realized with the same pattern). The crucial observation for present purposes is that either the grammar or lexicon of these speakers must be structured to permit at least type variation; in Lee & Zhang's test, only one token of each word form from each speaker was gathered, for which reason it is impossible to say whether this speaker would produce genuine token variation as well.

In short, Lee and Zhang (2014) have shown that there is a tendency to neutralize the distinctions between the R/HL1 (peninitial H⁺) and H2/LH2 ($\emptyset$) pitch patterns. That is, at the population level, at least, some speakers have signs of exhibit a class mergers, while others do not. At the level of individuals, Lee and Zhang (2014: 97) report that the merger of these classes among speakers that show it is not yet complete: the contrast is still present between some lexemes, but has been lost for others. What remains unclear is the extent to which some (adult) speakers of South Kyungsang exhibit token variation on lexemes, rather than type variation between lexemes, further crystalizing into population-level differences between speakers with and without this particular lexical contrast.

Whether the changes reported are due primarily to internal or external factors is unclear. Lee et al. (2016: 326) point out that the developments may well be due in large part to increasing contact with and a perceived prestige associated with the Seoul dialect; indeed, the acoustic properties found in the word prosody of younger South Kyungsang very much appear to show movement towards the properties of the Seoul system (compare the schematic pitch contours in Lee and Jongman 2015: 30). Nonetheless, the partial indistinguishability of certain lexical pitch properties in certain contexts might well promote lexical restructuring; indeed, North Kyungsang has undergone a complete merger of the toneless (H2/LH2) and peninitial H⁺ (R/LH1) classes (Lee and Zhang 2014: 107). External and internal influence may ultimate both be at work. Lee et al. (2016: 349) ultimately conclude that the major generational differences observed between older and younger South Kyungsang speakers – reduced Fo scaling and peak delay – are due to the adoption of a (phrase-level) initial low boundary tone from the Seoul dialect, but that differences in lexical pitch properties are still intact. The intra- and inter-speaker variation found among the speakers in Lee

and Zhang 2014, meanwhile, suggests that speakers have or are in the process of restructuring lexical representations, likely due to inherent language-internal surface neutralizations (though these might be promoted by the acoustic changes brought about by the addition of the initial low boundary tone).

One final observation concerns the tonal behavior of disyllabic suffixes in the data of Lee et al. 2016. There, innovative, younger Kyungsang speakers appear to have systematically neutralized the difference between *-pota* and *-imyen* on tonless stems, while older, conservative Kyungsang speakers exhibit token variation under the same conditions (Lee et al. 2016: 340–3). This testimony offers convincing evidence of variation in realization of a prosodic pattern, token-by-token within speakers, preceding the establishment of restructured forms or a novel grammar.

3.3.3.3 Variation (Probably) Precedes Change in Lexical Prosody

The toy simulation conducted above to illustrate a possible trajectory for change in the lexicon and grammar of a language with contrastive lexical stress crucially assumed that variation in the position of stress could be produced on a token-by-token basis. That is, speakers (learning agents) were taken to be able to produce either [pa'ka] or ['paka], given a lexical entry /pa'ka/, with prespecified stress; the lexical restructuring of /pa'ka/ as /paka/, without prespecified stress, was necessarily mediated by token-level variation.

The next fundamental question to be addressed concerns the possible *source* of variation. Two possible sources have already been mentioned in discussion above. On the one hand, the trajectory of prosodic acquisition, given the specific structure of the data, might result in the production of novel variants; the simulation on Old Latin stress is compatible with this option. On the other hand, external factors – in effect, language contact – might introduce new data into an environment; at least some aspects of the ongoing change in South Kyungsang Korean are plausibly interpreted as accommodation of aspects of the international phonology of Seoul Korean. To help better grasp how variation might emerge, even when data are assumed to be essentially free of variation, an examination of the acquisition and perception of word-level prosody is in order. The ultimate objective is to more firmly establish what sort of assumptions should be encoded in a computational model of the acquisition of stress grammars, as well as how learning agents of different linguistic backgrounds (i.e., L1 vs L2 speakers) might be appropriately handled in such a model.

3.3.4 The Acquisition and Psycholinguistics of Prominence Assignment

The acquisition of many aspects of prosody – especially in contrast to morphology – is proverbially precocious. While adult-like competence in, for instance, the production of exceptional inflectional forms (e.g., past tenses of strong verbs in Germanic languages) is not obtained until later elementary-school age, most core aspects of prosody, in both perception and prosodic, seem to be well established by 5–6 years of age, at latest. Some aspects, in particular sensitivity to metrical structures and prominence marking (stress or *F₀* contrasts in languages with culminative tone), may emerge extremely early. For instance, infants as

young as three days old have been shown in head-turning paradigms to exhibit sensitivity to the predominant rhythmic pattern (i.e., trochaic or iambic) of the principal ambient language (Christophe et al. 1994). In terms of fundamental sensitivity and preferences, Bhatara et al. (2018) find that 9-month-old infants possess the ability to discriminate between lexically contrastive prosodic features of virtually any kind (stress, pitch contours, tone) present in their L1, and already seem to prefer the stress pattern of their L1 by 6 months. At the onset of word use at around one year of age, children in different linguistic environments already control suprasegmental features in distinct ways: DePaolis et al. (2008) found that American-English-, Finnish-, French- and Welsh-acquiring children exhibit different production patterns with regard to duration, intensity and Fo by the time that first words are produced. A meta-review on milestones in the acquisition of prosody by Rusiewicz and Esteve-Gibert (2018) concurs that clear evidence for a distinction in production and perception between stressed and unstressed syllables emerges between 8 and 14 months, and that the full repertoire of pitch accents and boundary tones provided by the intonational phonology is in place as early as 18 months, at latest by 24 months.

In this section, I review the evidence concerning the acquisition of prosody concerning the assignment of prominence at the word level. I first consider fixed metrical stress patterns (3.3.4.1), then contrastive lexical stress (3.3.4.2) in first language acquisition, before considering psycholinguistic evidence on differences between native speakers of languages pertaining to different word-prosodic types (3.3.4.3). Particularly the latter evidence may have crucial implications for assumptions concerning the phonological representation of prosodic properties, and in turn bears on the interpretation of results concerning L2 prosodic acquisition. The overarching question throughout is: how early, and how accurately, are children able to reproduce the word-level stress patterns of their L1, and, in the case of L2 acquisition, how effectively can L2 learners acquire different prosodic grammars and lexica?

3.3.4.1 L1 Metrical Stress

Surprisingly little direct research seems to have been conducted into the emergent stress patterns in children acquiring a language with entirely or mostly predictable metrical stress. This gap has two comprehensible bases: 1) in the production of early words, truncation that limits word size to one or two syllables may inhibit the analysis of stress patterns whatsoever at this phase; 2) by the time that longer words of three or more syllables emerge, the predominant stress rule seems to be strongly established. As to the second point, consider the investigation of Łukaszewicz and Rozborski 2008: *all* three-year-old children acquiring Polish as L1 with neurotypical development *exceptionlessly* place primary stress on the penultimate syllable of a word.⁸⁵ Such a result accords with the comment of Tessier (2009: fn. 21): “It seems to be anecdotally acknowledged that the position of word stress is one of the first things that children acquire in their phonology: that children make few errors in phonolog-

⁸⁵ Łukaszewicz and Rozborski’s study, did, however, entirely exclude words with exceptional antepenultimate stress (pg. 269), so it is unclear whether L1 Polish-acquiring children at that age also command exceptional antepenultimate stress, or whether the relevant lexemes (e.g., *uniwersytet* ‘university’, *historycy* ‘historians’) exhibit any tendency to be regularized to the predominant penultimate pattern.

ical stress, and overcome them much sooner than, e.g., segmental or featural errors.”⁸⁶

The outputs of Finnish-acquiring children also provide some indications not only of early accurate primary stress acquisition, but accurate acquisition of secondary stress and trochaic foot structure as well. Specifically, the parsing of three- and four-syllable words seems to have an impact word truncation: trisyllables are often subject to truncation of the final syllable at a stage at which quadrisyllables are produced without any truncation. Under the assumption that trisyllables are parsed as $[('σσ)σ]$, but quadrisyllables as $[('σσ)(σσ)]$, with the consequence that the unstressed and unparsed final-syllable in trisyllables continues to be subject to truncation for longer in the course of acquisition than the third syllable of a four-syllable word (Torvelainen 2007, described in Saaristo-Helin et al. 2011: 355; Kulju and Savinainen-Makkonen 2008). While the leftmost syllable, with primary stress, of most words is usually not subject to truncation, to attribute its preservation to knowledge of the position of primary stress, as opposed to knowledge of word boundaries and a preference to preserve initial syllables, is a slight leap. Under the hypothesis of prosodic bootstrapping (Gleitman and Wanner 1982, Thiessen and Saffran 2003, survey in Höhle 2009), whereby prosodic information, such as the acoustic correlates of stress and intonation, is employed to identify syntactic constituents, awareness of stress, at least in the Finnish case, could be assumed to precede knowledge of prosodic word boundaries.

Even among bilingual children, the respective word-stress systems appear to be well differentiated by about the age of 3, though effects of prosodic transfer may appear in children under the age of 3. Among younger children, apparent cases of transfer in stress assignment during production appeared among German-Spanish bilingual children at ages 1;0 and 2;6 (though both of these languages possess extensive lexical stress). While monolingual German-acquiring children exhibit truncation of words $[σ'σ]$ to $[σ]$, Spanish-acquiring monolinguals do not; in the study of Arias and Lleó 2009, the bilinguals were found to often apply to the German truncation strategy to Spanish forms with final stress. As concerns older children the study of Zembrzuski et al. (2020) carried out a nonce-word repetition task with children ages 4–7 being raised in the UK with Polish-speaking parents; for these children, the authors considered Polish to be the L1 and English as the L2. At this age, children acquiring both a language with a predominant metrical stress pattern (Polish) and extensive lexical stress (English) are fully able to neatly and accurately apply the expected stress patterns. Neither was transfer of penultimate stress (applying the Polish algorithm) in the English production task, nor quantity-sensitivity (applying English constraints) the Polish production task, in evidence. Remarkably, this absence of prosodic transfer in terms of word stress occurs alongside effects of segment-level transfer effects.

In their survey of research into the acquisition of rhythm, Post and Payne (2018) present evidence suggesting that quantity-sensitivity – at least in production – emerges more slowly in the course of acquisition. In other words, early outputs from children acquiring languages with quantity-sensitive rhythmic patterns, like the West Germanic languages, tend to exhibit fewer effects of this quantity sensitivity (e.g., vowel reduction). This observation might be

⁸⁶Tessier here is referring primarily to languages with contrastive lexical stress, like English, which one might intuitively assume to require more data, and thus more time, to acquire accurately.

interpreted as aligned with the view that quantity-insensitive stress systems are less computationally complex, more easily acquired, and may well represent a default hypothesis at the outset of acquisition.

Again, while the evidence on the acquisition of phonologically predictable stress rules is not extensive, available studies suggest that such rules are in place and employed to achieve adult-like performance in the position of prominence, by the age of three, at the latest. The Finnish evidence concerning syllable truncation suggests that knowledge of the position of primary and secondary stress – which is entirely predictable – is in place even earlier, since the application of truncation depends upon stress assignment, and syllable truncation in Finnish-acquiring children tends to disappear almost entirely by about the age of three (Torvelainen 2007, Saaristo-Helin 2009). That a stress pattern of low complexity might be learned relatively early further accords with the psycholinguistic evidence on stress perception discussed under 3.3.4.3.

3.3.4.2 L1 Lexical Stress and Lexical Pitch Accent

In accord with the behaviors predicted by the phase of phonotactic learning in Optimality-Theoretic grammar acquisition (cf. the end of 3.2.1.1 above), the literature on prosodic development in children acquiring a first language with surface contrasts of the sort ['paka] vs. [pa'ka] indicates that successful acquisition of the stress grammar in languages possessing lexically prespecified stress is also probably achieved by about the third year of life, and almost certainly by the fourth. This general result has been observed for Spanish (Graham 1977, Hochberg 1988), Dutch (Fikkert 1994, Zonneveld and Nouveau 2004), and English (Klein 1984, Demuth 1996, Kehoe and Stoel-Gammon 1996).⁸⁷ Representative is the summary of Hochberg (1988: 704): “Moreover, the fact that 3-year-olds appeared to have as complete a command of the Spanish stress as the 5-year-olds indicates the precocity of this aspect of phonological acquisition.”

Potentially indicative of the early lexical representation of stress, at least among children acquiring languages with contrastive word stress, or that stress rules apply accurately to a UR before or in parallel with segmental realization, is the fact that, in early word production between ages 1 and 3, unstressed syllables are frequently not realized. This effect is well-established for West Germanic languages (English, Dutch, German: Allen and Hawkins 1978; Fikkert 1994; Gerken 1994; Wijnen et al. 1994), but also for Hebrew (Galit 2002: 72–8). In all of these languages, the preference for preserving a syllable with can be distinguished from potential faithfulness to initial or final syllable; e.g., in Hebrew. words with final stress preserve only the final syllable, while words with non-final stress preserve the stressed syllable and the final syllable. One of the most detailed longitudinal descriptions of phases of stress-pattern acquisition remains Fikkert 1994: Ch. 6, on Dutch-acquiring children between the ages of 1;4 and 2;9. With respect to disyllabic words, words with initial stress are always produced with initial stress. Words with final stress, on the other hand, exhibit four typical

⁸⁷A useful summary of some major accounts of prosodic acquisition, with emphasis on stress is presented in Kehoe and Stoel-Gammon 1997. Kehoe (2018) offers an updated examination of this literature, with more discussion of languages outside the Germanic family.

phases (pp. 206–7):

1. Truncation of the initial unstressed syllable: the disyllable is reduced to a monosyllable, with only the stress final syllable being realized (e.g., [ˈraf] for adult [ʃiːˈraf] ‘giraffe’).
2. Trochaicization: both syllables of the disyllable are pronounced, but primary stress is shifted to the initial syllable (> [ˈʃiːraf]).
3. Level stress: both syllables are realized with acoustically similar degrees of stress (transcribed by Fikkert as primary stress on both syllables, > [ˈʃiːˈraf]).
4. Adult-like: both syllables are realized, and primary stress falls on the final syllable (> [ʃiːˈraf]).

In many children, this development is fairly rapid, occurring in a span of just a few months. One of Fikkert’s subjects, Robin, had exclusively productions in the first stage up to age 2;0, then the second at 2;1, the third from 2;2, and consistently adult-like outputs from 2;4 (Fikkert 1994: 202–3).

Nonetheless, the preservation of syllables with primary stress might alternatively be interpreted merely as an effect of perceptual salience, where syllables under stress as well as at prosodic boundaries would be more salient; see Kehoe and Stoel-Gammon 1997: 139 for this view. At minimum then, syllables that have primary stress in adult productions are then most readily encoded in early lexical representations.⁸⁸ And in contrast to the evidence from West Germanic, Spanish, Italian, and Hebrew, in which a syllable with word-level prominence seems exceptionlessly to be preserved when truncation occurs, data from Japanese learners (ages 1;5 – 2;1) shows that, while syllables associated with the lexical H* pitch accent are usually preserved, deletion of those syllables seems to occur more readily than in languages possessing canonical stress (Ota 2006: 284–5). Unclear, however, is how exactly all possible factors (e.g., preservation of edge syllables, preservation of long vowels) interacts with syllables bearing H* to generate the attested patterns of truncation in Japanese-acquiring children.

While virtually no evidence of errors in the position of stress among children acquiring Finnish- or Polish-like stress systems is reported, for languages with lexical stress, there exists a small amount of evidence pointing towards “overregularizations” in the position of stress, or in the regularization of the overall stress pattern by applying segmental alterations of the word, especially among somewhat younger children. Hochberg (1988: 690) cites a dictionary study of four Spanish-acquiring children in the age range 2;3–3;4 (Giraldo 1971: 340), which

⁸⁸For instance, the experiment of Curtin et al. (2005) with seven-month-old English-acquiring infants suggests that infants indeed represent stressed syllables different than unstressed syllables. In this experiment, the infants were familiarized with an artificial language, then presented with English sentences containing the “words” (i.e., sequences that might have been segmented as words) from the artificial language, as well as other nonce syllable sequences that were not present in the artificial language. Overall, the infants listened significantly longer utterances containing newly learned “words” with initial stress (see 3.3.5.3 below on the “trochaic bias”).

shows some occasional shifts of antepenultimate to penultimate stress in words whose last three syllables are of the form [CV(C)CVCV], e.g., child [to.'ma.go] for adult [es.'to.ma.yo] 'stomach'. Zonneveld and Nouveau (2004: 378–9), in a study involving 3- and 4-year-old speakers of Dutch, found no raw “stress shifts”, and among a total of 678 output tokens from the children, and only *four* instances of a segmental change resulting in a change of stress pattern from a “less regular” to a “more regular” pattern (though *no* instances whatsoever of changes in the opposite direction). In the same paper (Zonneveld and Nouveau 2004: 380–8), in a study designed around the repetition of novel “wug” items, error rates were much higher, and tended overwhelmingly to shift the stress or adjust the shape of the word from a more marked to a less marked pattern; error rates were lowest among wugs already exhibiting a maximally unmarked stress pattern. Hochberg (1988: 690–8) reports an essentially identical result from a similar experiment with Spanish-acquiring children in the age range 3;7–5;11: wugs with “irregular” stress were more likely to be altered in some fashion, and were usually altered so as to make their stress pattern more “regular”.

Some data on the acquisition of prosody at the lexical level in relation to larger phrasal prosody is available from Japanese. Ota (2003a), analyzing data from two children in the corpus of de Boysson-Bardies and Vihman 1991 (age 1;6) and three from Ota 2003b (ages 1;8–2;1), found principally that the pitch contour associated with lexical accent at the word level is acquired earlier than phrasal intonation.⁸⁹ That is, in Tōkyō Japanese, there exists a contrast between lexically accented and lexically unaccented words, with the former having a global fall from H* to L, the latter having only a rise from L to H between the first and second syllable. In general, children reproduced the global fall on lexically accented words with greater accuracy, often failing to produce the global rise found on unaccented lexemes. Among the five children, Ota (2003a: 375) distinguishes three phases of acquisition: 1) “no tonal structure” in one of the youngest; 2) “lexical pitch accent only” among two children; and 3) “emergence of phrasal pitch” among the two oldest children. Of particular interest for present purposes is that fact that, already among children in the “lexical pitch accent only”-phase, distinctions in the position of the H* and fall to L are reliably reproduced; in other words, Japanese-acquiring children appear to possess knowledge of lexical prosodic contrasts and be able to accurately generate them – at least in the comparatively frequent lexemes in their active vocabulary – before the age of two.

Taken together, the results of these experiments and studies might suggest that, in a real-world context, very low-frequency lexical items might undergo a stress shift or sporadic segmental alteration in order to improve the prosodic output of the word (cf. Sondregger and Niyogi 2013 on the diachrony of English noun–verb pairs above). In effect, if the learner’s “registration” of a lexeme’s phonological properties in the lexicon is incomplete, then apparent “deviations” might not incur a (sufficiently strong) faithfulness violation, thus allowing less marked structures to emerge.⁹⁰ The situation is somewhat tenuous, however: the word

⁸⁹Research the acquisition of lexical pitch accents is generally wanting, only Japanese having been much studied at all (cf. Bhatara et al. 2018: 47).

⁹⁰The usual interpretation of in the literature combining phonological learnability with data from language acquisition (e.g., Levelt and van de Vijver 2004, Tessier 2009) is that the relevant faithfulness constraints are still too low-ranked in the grammar to prevent less marked structures from being produced.

must be frequent enough that it is learned and transmitted intergenerationally (rather than forgotten and lost from the lexicon), but not so frequent that the learner can construct a clear representation of all segmental and prosodic properties of the lexeme, and, in doubt, produces a “similar-sounding” word that conforms to the less marked prosodic pattern of the language. Such change might accord with the observation of Probert (2006b), that low-token-frequency lexemes of Ancient Greek were more likely to exhibit a shift from oxytone (high tone on final syllable) to recessive accentuation.⁹¹ This hypothesis requires that lexical representations of segments tend to be more accurate than lexical representations of word prosody, despite the fact that children seem to exhibit more accuracy in the production of prosodic features, broadly speaking, at any earlier age (compare also the study of Zembrzuski et al. 2020, where Polish-English child bilinguals exhibited segmental transfer effects, but almost no transfer effects in word stress).

Overall, one observes a tension between the fact that children simultaneously acquire, with relatively high accuracy, both the “default” stress preferences of their language and instance of deviance from the default pattern. The fact that errors – whether in the form of stress shifts or segmental adjustments – *may*, even to a limited extent, occur, indicates that accurate acquisition of stress in all lexemes is far from instantaneous. Insofar as errors are more likely to occur among unknown or infrequent lexemes, such behavior speaks in favor of a gradual acquisition of a secure lexical entry. Such behavior can be modeled in an OT grammar via some variety of constraints over underlying representations (Apoussidou 2007: Ch. 6, Pater et al. 2012), where the correct UR must be ranked / weighted with respect to the other constraints in the grammar. Thus, while some room for misacquisition of stress among individual lexical items in the larger context of a lexical accent system exists, the vast majority of lexical items are likely to be accurately acquired with the “correct” prosodic specifications. Furthermore, in the face of the early age at which both metrical stress rules and lexical stress contrasts are established, these are evidently aspects of the phonology, or of the grammar and lexicon more broadly, that do not require an enormous amount of data. Given, though, that information about word-level prosody is contained in virtually every utterance – as opposed to particular intonational tunes, morphological material, or syntactic structures – that children have received a sufficient quantity of data to perform similarly to adults might not be so surprising.⁹² Insofar, though, as simply “more data” is assumed to permit the accurate acquisition of any learnable stress grammar, change in stress grammars via reanalysis comes to seem perhaps unintuitive – at least if learners are presumed to have sufficient access to primary linguistic data to permit accurate acquisition at all. To be sure, the frequency of phonological patterns in the ambient data influences their rate of acquisition

⁹¹In the Ancient Greek context, that no prosodically regularizing segmental changes to oxytones have been observed is perhaps not surprising, since high tone on the final syllable, regardless of its weight or segmental content, is always irregular. A conceivable regularization could be to transform a paroxytone [...*ῶ*], with high tone on a light penult followed by a light final, into an regular recessive form by making the final syllable heavy, i.e., [...*ῶ*] > [...*ῶ*]. Whether credible examples of this sort of change in Greek are attested remains to be investigated.

⁹²At the same time, the study of Arciuli and Ballard 2017 found that measurable differences in the acoustic composition of stressed syllables between older children (8-11 years old) acquiring Australian English and adults are still present.

(Roark and Demuth 2000), but it should then only be in cases in which the quantity of data is insufficient, or the data itself suffused with variation that cannot be properly reproduced by any grammar, that the emergence of a novel grammar should be expected to occur.

3.3.4.3 “Stress Deafness”

Of potential import for the question of prosodic change is how L2 speakers of a language deal with the prosody of the L2, as well as how the prosody of loanwords is treated by speakers of the receiving language. Both in cases in which loanwords enter a language, or when L2 speakers acquire and speak their L2, the question of stress perception becomes relevant. To what extent can speakers of the receiving language accurately preserve the stress pattern of a loanword? Or, to what extent can L2 speakers accurately acquire reproduce stress patterns of the target language? The behavior of L2 speakers becomes especially relevant in the situation that tokens that they produce might potentially serve as learning data for further generations acquiring the language as an L1. Intuitively, accurate reproduction of stress on loanwords or in L2 acquisition might be assumed to depend upon the stress grammar of a speaker's L1:

- If a language has lexically unpredictable stress and the position of stress is unbounded (e.g., Vedic Sanskrit), the position of stress in any loanword can / should be faithfully maintained. In L2 acquisition, L1-speakers of such languages should be able to accurately reproduce any stress patterns (whether rule-governed or lexical).
- If a language has lexically unpredictable stress within a window (e.g., Attic-Ionic Greek), the position of stress in any loanword can be faithfully maintained within the window, but not outside the window. Similarly, in L2 acquisition, L1-speakers of such languages should at least be able to produce stress contrasts within the window defined by the grammar of their L1.
- If a language has phonologically predictable fixed stress (e.g., Classical Latin, Finnish), stress in loanwords will be ignored; all loanwords will be subject to stress regularization. By the same token, one might expect for L1-speakers of such languages to apply, by default, the stress algorithm of their L1, and perhaps struggle to produce other stress patterns.

That these intuitive predictions are not completely borne out in practice is shown, with respect to loanwords, by the fact that some languages with predictable metrical stress do possess lexemes with exceptional stress (e.g., Polish).

How speakers with different types of L1 stress systems has, in fact, been the subject of a fair amount of psycholinguistic and some neurolinguistic investigation, with particular reference to the phenomenon of so-called “stress-deafness” (Dupoux et al. 2001, Peperkamp and Dupoux 2002, Peperkamp 2004 Dupoux et al. 2008, Skoruppa et al. 2009, Peperkamp et al. 2010, Domahs 2012, Skoruppa et al. 2013). Speakers who are “stress-deaf” experience considerably greater difficulty identifying stressed syllables in an experimental setting. Interestingly, among languages with algorithmic word-level stress patterns, there are some marked

differences. The experiments in Peperkamp and Dupoux 2002 indicate that languages may be classed into at least three groups:

- Languages whose L1 speakers are “stress-deaf”, e.g., French, Finnish, Hungarian. These languages have fixed-stress, left- or right-edge systems or no word-level stress at all (e.g., a phrase-level boundary prominence in French; Jun and Fougeron 2002).
- Languages whose L1 speakers are not really “stress deaf”, e.g., Polish, Macedonian. These languages have algorithmic stress, but of a somewhat more complex sort (e.g., penultimate or antepenultimate primary stress), and often a small category of loaned lexemes with lexical stress deviating from the algorithmic pattern. L1 speakers of these languages are not as good at stress identification as L1 speakers of languages with lexically contrastive stress, but closer in performance to lexically contrastive speakers than to “stress-deaf” speakers.
- Languages whose L1 speakers are definitely not “stress deaf”, e.g., Spanish, Russian. L1 speakers of these languages possess lexically contrastive stress.

Thus, in broad terms, speakers of languages such as Spanish can generally correctly identify syllables as “stressed”; conversely, languages such as French are quite bad at this task. In Dupoux et al. 2001, across all experiments and experimental designs, error rates among L1 French speakers in the identification of stress were always statistically significantly higher than error rates among L2 Spanish speakers.⁹³ Remarkably, indications of “stress deafness” are available for children as young as nine months of age: Skoruppa et al. (2009) found, in one experiment, that infants in a Spanish-speaking environment could distinguish between [‘CVCV] and [CV‘CV] sequences, but infants in a French-speaking environment did not. Yet more interesting with respect to phonological representations are the results of a second experiment in the same study: on exposure to a segmentally identical sequence with different stress, [‘pima] versus [pi‘ma], the French-acquiring infants did exhibit a statistically significant novelty effect; Skoruppa et al. draw the conclusion that the problem, for French acquiring infants, thus lies with the ability to encode differences in stress phonologically, not the ability to perceive the acoustic correlates of stress.

Along these lines, Peperkamp (2004) argues that the distinction in performance between L1 speakers of languages of the French or Finnish-type, which result in native speakers with stress deafness, and L1 speakers of languages of the Polish/ Macedonian-type depends upon whether the speakers are able to readily encode prosodic information in their mental lexicon. That L1 speakers of some languages should be able to encode prosodic information in lexical representations while L1 speakers of other languages cannot is in turn a function of

⁹³For both L1 French and L1 Spanish speakers, error rates increase as a function of word length in syllables. For L1 Spanish speakers, correct identification of stress on 2- and 3-syllable words was, depending upon task conditions, often at or near ceiling, and remained well above chance in words of all lengths. The performance of L1 French speakers generally remained at or slightly above chance for words with four or fewer syllables, but at or below chance in words with five or six syllables.

prosodic acquisition, namely, whether the redundant encoding of stress in the mental lexicon was necessary in early phases of acquisition, prior to the complete acquisition of an algorithmic stress rule. Peperkamp argues that this difference in the process of prosodic acquisition is a matter of whether stress in a given language is SURFACE-OBSERVABLE or not; in Finnish, stress is surface-observable, whereas in Polish, it is not.

- (61) SURFACE-OBSERVABLE: A phonological regularity is *surface-observable* if it can be inferred from bare utterances, that is, utterances in which word boundaries are not marked.⁹⁴

The distinction then supposedly lies in how easily the stress assignment algorithm of language can be determined; the stress-assignment rule in French or Finnish can be inferred without knowledge of the boundaries of content words (cf. Staubs 2014: Ch. 2 for computational simulations indicating that edgemost stress algorithms are learnable more quickly and accurately than other patterns). In Polish, on the other hand, the stress-assignment rule is not surface observable: learners must be able to parse content words first. In the case of French, because the edge of an utterance always corresponds to the edge of a prosodic word as well, learners may infer from the systematic occurrence of utterance-final stress that stress is word-final, too. In Polish, on the other hand, because the language possesses monosyllabic content words, the final three syllables of an utterance could appear as [$\sigma'\sigma$], [$\sigma\sigma\#\sigma$], or [$\sigma\#\sigma\sigma$]; thus, stress may be either final or penultimate in an utterance. Peperkamp's proposal is then that, if the stress algorithm of a language is surface-observable, then it can be acquired pre-lexically, i.e., before infants are capable of successfully segmenting sequences of content words. The further implication is that learners of Polish-like languages must have early redundant encoding of stress in the mental lexicon, before they are able to acquire their language's predictable stress rule. As a result, lexical exceptions also permissible, because the capacity to encode stress lexically generally persists. In other words, French-learning babies "deactivate" the ability to phonologically encode stress differences very early, because they can predict stress by rule early in acquisition, whereas Polish-learning babies still need to attend to stress differences, because they need more time in order to deduce the stress rules of the language.

The larger claim is then that learners of Polish-like languages have early redundant encoding of stress in the mental lexicon, before they are able to acquire their language's predictable stress rule; lexical exceptions are thus permissible, because stress can be encoded lexically. In effect, French-learning babies "deactivate" the ability to hear stress differences very early, because they can predict stress by rule early in acquisition, whereas Polish-learning babies still need to attend to stress differences, because they need more time to deduce their stress rules. Seen from the perspective of phonotactic learning alone, applying phonotactic learning to utterance-level Finnish data, for instance, provides no evidence that MAX-PROMINENCE need dominate any markedness constraints; applied to utterance-level Polish

⁹⁴In fact, the prosody of French is probably best analyzed as applying right-edge stress only at the level of the phonological phrase, and not the level of the prosodic word, but the surface-observability criterion is still met, since the right edge of an utterance will similarly correspond to the right edge of the last phonological phrase in the utterance.

data, it might appear that stress is not wholly predictable, and that faithfulness constraints therefore have a role to play. Differences are evident in the treatment of loanwords then, as well: languages like French and Finnish do not permit any lexical exceptions to their stress rules, whereas languages like Polish or Macedonian do: loanwords can be borrowed with faithful preservation of the position of stress.

The distinction between Polish-like and French-like systems then becomes evident in loanwords: in French or Finnish, the stress pattern of all loanwords is adapted to the universal default stress pattern, while Polish or Macedonian speakers may faithfully maintain the stress of the loaning language. When it comes to the production of lexemes in an L2 by an L1 Finnish speaker, then, the expectation should be that this speaker would find it difficult *not* to apply the default prosodic pattern of their L1.⁹⁵ The potential implication for language change via language contact is that an L1 Finnish speaker acquiring an L2 with lexically-contrastive, unbounded stress (e.g., Sanskrit), would introduce more tokens of word-initial stress for all lexemes. Logically, the greater the number of L1 Finnish speakers shifting to this L2, the greater the potential for a learner to encounter only initially stressed tokens of some lexeme.

The existence of “stress-deafness” among L1-speakers of particular languages holds potential implications for prosodic change. If speakers of such languages experience difficulties in 1) perceiving the position of stress in an L2 with lexically contrastive stress and 2) struggle to avoid producing stress patterns other than the pattern found in their L1, then L2-speakers could potentially introduce more variability into the prosody of a speech community. More concretely, an effect of the presence of L2-speakers of a Sanskrit-like language having an L1 exhibiting fixed left-edge stress could be to introduce more tokens of left-edge stress into the learning data of subsequent generations. Logically conceivable are scenarios in which the prosodic characteristics of the primary linguistic data are substantially altered through either the addition of loanwords (cf. the case of English as described in Fikkert et al. 2006, Lahiri 2015: 231–43 or German as described by Speyer 2009) or tokens produced by L2 speakers with different prosodic grammars.

3.3.4.4 The Acquisition of L2 Prosody and Testimony from Artificial Grammar Learning

To what extent can L2 speakers accurately produce and perceive stress patterns differing from their L1? Theoretically, the L2 speakers might be able to engage in some production strategies that render their stress production native-like, even if they cannot fully acquire the prosodic grammar and lexicon of the L2. Kijak (2009) carried out experiments on the perception of stress in a L2 with a wide array of different L1 speakers acquiring a diversity of L2. While Kijak’s results for the most part accord with the predictions made by the model in Peperkamp and Dupoux 2002 and Peperkamp 2004, L1 Czech speakers acquiring L2 Polish constituted an interesting exception, in that they could accurately perceive Polish stress, despite having acquired an L1 with surface-observable stress (primary stress is

⁹⁵Interestingly, there may be a precisely reverse tendency among speakers of languages with lexically contrastive stress: Revithiadou and Lengeris (2016) found that native speakers of Modern Greek perceived stress on ~ 75% of tokens when listening to real Greek words whose acoustic cues to stress were suppressed.

Czech is assigned to the first syllable of the prosodic word). Kijak's conclusion is then that surface-observability does not necessarily lead to stress deafness; in the specific Czech case, the ability to perceive stress differences well may be due to the fact that "in Czech stress fulfills an important demarcative function which has been shown to be crucial in speech segmentation" (Kijak 2009: 320).

At the same time, Kijak's (2009) examination of stress discrimination capabilities among L2 speakers at different skill levels seems to support view that the effects of L1 prosody will persevere in an L2.

The models [viz., Peperkamp & Dupoux's model of stress deafness and the stress typology model of Altmann and Vogel 2002] predict that once the perceptual ability for stress is shaped in L1 it remains more or less unaffected throughout life: this includes L2 perception, the success of which is a direct projection of general (or L1) perception ability. Unfortunately, due to (the availability of) unequal numbers of participants per proficiency level within and across L1s in our test we could not provide definitive evidence for or against this claim. However, the trends observed in our data showed that subjects indeed progressed little, if at all, within each language group, and the success rates at the beginners' level predicted roughly the overall success rate of the whole language group, in accordance with the models' claims.

On the other hand, experimental results reported in Altmann 2006 lead to the conclusion that capabilities in L2 perception and L2 production of (English) stress do not necessarily go hand-in-hand: "speakers of languages that performed poorly in the perception task were still able to display fairly target-like production, while speakers of languages that performed target-like in perception displayed poor production of English nonce words regarding stress placement. Hearing stress and articulating stress are independent from each other" (Altmann 2006: 159). Hence, an L1 French speaker might display stress-deafness in a perception task, but nevertheless have developed a production strategy that allows her to generate the lexical stress of some L2 with a certain degree of accuracy. Indeed, Altmann found that native French speakers, who employ stress, but have no lexical prosodic contrasts, could produce the position of stress in English more accurately than native Japanese speakers, who do have lexical prosodic contrasts; in perception, the performance was reversed (Altmann 2006: 140–6). Nevertheless, Altmann's French speakers did exhibit an a greater tendency for producing word-final stress, which indicates that their L1 leads to a greater likelihood of producing a certain stress pattern.

However, because Altmann's experiment involved nonce words without any discernible morphological structure, the Spanish speakers exhibited a perhaps surprising result: they did not apply a pattern like the pattern of the native English speakers, and were less "English-like" than French or Arabic speakers. However, this experiment perhaps reveals a fundamental issue in testing performance of L2 stress production: with nonce words, speakers of L1s that have lexical stress or make use of morphological structure cannot use those strategies to truly learn the stress pattern. Consequently, the overall results and conclusions drawn by

Altmann about L2 stress production may not be wholly reliable. One interpretation is that such an experiment merely reveals the predominant default pattern, and that the default stress rules of English and French or Arabic are more similar than the default stress rules of English and Spanish or Japanese.

Indeed, Kijak (2009: 321) found that, on production tasks using actual lexemes of Polish, L2 learners having French or Czech (predicted to be stress deaf by Peperkamp & Dupoux) as their L1 performed substantially worse than native speakers of German, Spanish, Italian, Russian, or English. The implication in this case is that, because French and Czech speakers either cannot lexically store prosodic information, or have a much more restricted repertoire of permissible stress patterns in their L1, to accurately produce a different stress pattern presents a challenge for them. Note though, that although French and Czech speakers performed worse on a production task than L1 speakers of languages with lexical stress, the total amount of accurate tokens produced was still relatively high: ~75% for L1 French speakers and ~65% for L1 Czech speakers (Kijak 2009: 291–305). Nonetheless, because of the lower overall accuracy, such L2 speakers risk introducing a considerable amount of noise into learning data, generating not only correct patterns and patterns that conform to their L1, but sporadically producing yet other patterns as well.

Yet another study conducted with adult L1 speakers of Polish having extensive experience with English (Porzuczek and Rojczyk 2017) found that these L2 English-learners exhibited a tendency to generate stress on the initial syllable of the word. The typical penultimate stress pattern of Polish is not, however, surface-observable, and L1 speakers of Polish have been found to not to be genuinely “stress deaf”. The strategy adopted by most participants in Porzuczek & Rojczyk (2017)’s study suggests that a new default stress rule for the L2 may be acquired, which may be overapplied in cases in which the learner has not (yet) successfully constructed a lexical representation with exceptional stress.

Similarly, the study of Archibald (1993) on the L2 acquisition of English stress by L1 speakers of Polish and Hungarian offers broad evidence for transfer effects from the metrical phonology of the L1. Just as in Altmann’s (2006) study, Archibald found that the ability to perceive correct stress patterns generally outstripped the capacity of the L1 Polish and Hungarian speakers to generate the correct stress patterns in production. Among L1 Hungarian speakers, Archibald attributes most errors in production to either a tendency to select the leftmost foot in the word, rather than the rightmost, as the site for initial stress, or to ignore coda consonants as a source for syllable weight. Among L1 Polish speakers, many errors in production could be explained by a failure to construct quantity-sensitive feet, and a failure to observe extrametricality of final syllables. The perhaps unsurprising result, then, is that L1 speakers of languages with (largely) metrically predictable stress exhibit some tendency, at least, to carry the metrical properties of their L1 over into their L2, at least in production. From the diachronic point of view, the behavior in production is particularly interesting and relevant, since the outputs from L2 speakers could equally enter into a synchronic pool of variation, or serve as primary linguistic data for (some) learners.

Although disentangling the relationship between the capacities of stress perception and stress production is evidently difficult, as the works of Altmann 2006 and Kijak 2009 show,

the general indications seem to be that L1 speakers of languages having surface-observable stress patterns indeed produce more variable outputs, and depart from the stress of the target L2 more often. These linking strands of research on L2 stress perception and production ultimately give good reason to believe that L1 speakers of Finnish-like language with “surface-true” stress would, when speaking an L2 with unbounded lexical stress, not infrequently depart from the correct lexical stresses. If the production experiments of Altmann and Kijak constitute an approximative guide, the L2 speakers with L1 Finnish would have some probability – perhaps at least chance – of producing the correct lexical stress. In the event that their production strategy or lexical information fails to arrive to the correct lexical stress, they would be most likely to produce word-initial stress, or more rarely, some alternative pattern.

3.3.5 Learning Biases and the Computational Modeling of Prosodic Acquisition and Change

The overview of investigations into the first- and second-language acquisition word-level prosody (more specifically stress) revealed two broad tendencies: first, in L1 acquisition, stress is acquired early (by c. three years), accurately, and also overcomes transfer effects among bilinguals at an early age; second, transfer effects from the L1 appear in L2 prosodic acquisition, and L1 speakers of languages whose speakers tend to be “stress deaf” may be yet more likely to exhibit transfer effects, or produce errors in terms of lexical stress. The results of L1 acquisition studies suggest that great stability – diachronic persistence – of word-level prosodic structures is perhaps to be expected, as long as the structure of the learning data remains substantially unchanged (e.g., no segmental sound changes affecting syllable count or syllable weight have occurred). With respect to L2 acquisition, on the other hand, the apparent difficulties that at least “stress-deaf” speakers encounter in the accurate production of novel stress patterns has the potential to constitute a source of variation in the data that a learner receives concerning the word-prosody of her target L1.

The final matters to be addressed in this chapter concern the question of possible *biases* relevant to the acquisition of stress systems and how, exactly, the models of learnability and assumptions about prosodic acquisition can be implemented in a computational model. Although substantial empirical data speaks in favor of the capacity of language-learners to engage in frequency matching behavior, thereby reproducing the variation encountered in their data (cf. Hayes et al. 2009b: 826), especially if one of the variants comes to be associated with some social factor (which in turn may allow for diachronic transmission of a sociolinguistic variable; cf. generally Labov 2001), the phonological acquisition process may be constrained by a number of *BIASES*, which might weaken the possibility of the acquisition of variation, on the one hand, and perhaps favor the acquisition of a specific variant along, on the other. In general terms, the phonological learning process may be constrained by types of *INDUCTIVE BIASES* (Briscoe 2000, Griffiths and Kalish 2007, Kalish et al. 2007), which may be subdivided into the following varieties:⁹⁶ For more on the issue of learning biases in language generally, including their mathematical implementation, see Culbertson et al.

⁹⁶Since the present investigations do not directly engage with of coarticulation or other sorts of phonetic-level variation, the role of *CHANNEL BIAS* is not directly considered here.

2013.

- **SUBSTANTIVE BIAS:** the physical properties of speech sounds (i.e., their substance) render certain phonological patterns more or less difficult to justify for the learner (Wilson 2006, Pater and Moreton 2012, Moreton and Pater 2012). For instance, a preference for palatalization of velars preceding [+high] and/or [+front] vowels, and a corresponding dispreference for palatalization preceding [+low] and/or [+back] vowels.
- **ANALYTIC BIAS:** some inherent cognitive predispositions with respect to phonological grammars that “facilitate the learning of some phonological patterns and inhibit that of others” (Moreton 2008: 84, Bennett 2014).
- **CATEGORICITY BIAS:** learners prefer patterns that exceptionless generalizations to patterns that admit of irregularities and variation (Pierrehumbert 2001, Wedel 2006, Kirby and Sondregger 2015).⁹⁷

Taken together, these types of inductive biases would serve to favor the acquisition of certain phonological structures over others; the substantive biases would prefer the acquisition of phonetically natural over phonetically arbitrary patterns, while the categoricity biases would support the acquisition of limited contrasts, disfavoring variation and larger numbers of phonological contrasts. Experimental work using iterated learning (see further below) has demonstrated that even quantitatively small biases may have qualitatively substantial effects, in particular in terms of a drive towards “regularization” (Real and Griffiths 2009). Furthermore, other studies involving younger children suggest that children exhibit an even stronger tendency towards the regularization of phonological variants than adults (Kam and Newport 2005, Kam and Newport 2009).

What exactly would these sorts of biases look like at the level of word prosody, and stress specifically? The very inclusion of constraints like **WEIGHT-TO-STRESS PRINCIPLE** or ***CLASH**, for instance, as a presumably universally available markedness constraints may be seen as the implementation of substantive biases: the association of stress with syllables that already possess greater duration exploits that duration as a cue to prominence, while the avoidance of stress on adjacent syllables heightens their contrast and helps to maintain rhythmic alternations between prosodically more and less prominent elements. However, were no plausible substantive (phonetically grounded) motivation for a constraint like WSP available, but its effects still were evident across the typology of stress systems, then its presence in the constraint inventory would constitute an analytic bias. An further analytic bias with respect to stress patterns might conceivably be a preference to interpret feet as trochaic rather than iambic, where both parses might be possible, e.g., to parse a [L 'L L] as trochaic [L ('L L)] rather than iambic [(L 'L) L], or consider the parse [L L ('H)] as preferable to [L (L 'H)]. At the same time, the Iambic/Trochaic Law (Hayes 1995: 79–85; but cf. Kager 1993) might be

⁹⁷ Both the notion “island of reliability” for morphophonological rules in discussed in Albright 2002b, Albright 2002a, and Albright and Hayes 2003, as well as the conception of the “tolerance principle” in Yang 2016 and Yang 2017 might also be viewed as reflecting a categoricity bias: rules or patterns with fewer exceptions are more likely to be productively employed.

understood as a sort of substantive bias as well, concerning how contrasts in intensity or duration are preferentially interpreted with respect to stress.⁹⁸ Finally, an analytic bias in favor of certain particular stress patterns may exist, insofar as the constraints assumed to be generally available can be permuted into more different rankings that will produce a particular stress grammar.

3.3.5.1 The Typology of Quantity-Insensitive Stress Patterns

At the typological level, the emergence and frequency of a particular stress pattern may correspond to the innate propensity of learners to acquire a particular pattern. If certain varieties of stress patterns can be acquired more quickly and easily, then one might *a priori* expect for such patterns to recur cross-linguistically more often than expected. In this context, “more quickly and easily” should be understood as meaning that overall less data is required in order to converge upon that grammar, in particular when that grammar is indeed the target. According to the database of Heinz 2007, the distribution quantity-insensitive stress patterns looks as follows:

- (62) Fixed-stress languages with counts from the Stress Pattern Database of Heinz 2007 (after Staubs 2014: 20)

	From left	From right
Distance 0	'σσσσσσσσ initial 69 languages	σσσσσσσσ' final 74 languages
Distance 1	σ'σσσσσσσσ peninitial 12 languages	σσσσσσσ' penultimate 60 languages
Distance 2	σσ'σσσσσσ postpeninitial 0 languages	σσσσσ' antepenultimate 8 languages

From the typology, one may observe the following trends: first, as has long been recognized, stress at the edge (initial- or final syllable) is predominant. However, the distribution of edge-orientation is very uneven ($\chi^2 = 25.39, p < 0.001$), which suggests that distance from edge and the edge itself are not independent of one another – note the infrequency of peninitial systems vis-à-vis penultimate systems. Finally, the infrequency of peninitial systems and absence of postpeninitial systems is suggestive of a right-edge bias: 81 left-edge systems vs. 142 right edge systems ($p < 0.001$ under Exact Binomial Test).

In the context of the acquisition of prosody, and stress systems specifically, the computational simulations carried out in Staubs 2014: Ch. 1 (cf. also Staubs 2013 for a briefer summary of similar results) show that the typology of stress patterns may follow from biases that

⁹⁸In practice, however, in simulations on the phonological learnability of stress systems, this seeming substantive bias is most easily implemented analytically, by not allowing GEN to propose candidates in quantity-insensitive languages with anti-iambic feet of the shape (H 'L).

learners exhibit in the acquisition of stress grammars. Specifically, grammars that result in typologically more common stress patterns are apparently acquired more quickly and more accurately than typologically less common patterns.⁹⁹ The fact that, among fixed-stress systems, left-edge and right-edge systems are typologically more common than patterns with stress falling farther from word edges, is then arguably a consequence of the inductive biases that favor the easy acquisition of such edge-aligned systems.¹⁰⁰ That more data is needed to unambiguously conclude that a grammar should generate, e.g., antepenultimate stress, is a natural consequence of the fact that patterns allowing stress farther from a word edge are more confusable: a sequence [*ʔσσσ*] is compatible, for instance, with both an initial-stress or antepenultimate-stress grammar. One issue (see Staubs 2014: 35–6), however, is that peninitial stress might appear just as learnable as penultimate stress, contrary to the typology seen in (62); reference to some further substantive or analytic bias is necessary to account for this imbalance (see further on NONFINALITY) below.

Overall, Staub's results appear to correspond nicely to the hypotheses concerning the basis for stress deafness in Peperkamp and Dupoux 2002 and Peperkamp 2004: the most common stress systems are acquisible more quickly, which would support the notion that a phonological rule for the correct production of stress can be activated before lexical representations of word prosody need to be stored. One might also further hypothesize that systems with fixed-edge stress might be most likely to emerge out of the surface variation in the position of stress in the event that a learner fails to acquire lexical stress contrasts altogether.

3.3.5.2 Natural and Unnatural Stress Rules: Evidence for Analytic Biases?

While the typology of quantity-insensitive stress patterns itself strongly indicates that certain analytic biases interact to produce inductive biases in favor of the acquisition of certain stress patterns, typology does not directly show what the limits of learning with respect to stress grammars are. Namely, are certain conceivable stress patterns not merely disfavored (like antepenultimate stress), but indeed impossible to acquire altogether? And if certain patterns cannot be acquired, how should such unlearnable patterns be excluded from the domain of grammatical hypotheses? These questions go to the heart of stress typologies and the design of constraint inventories intended to capture them: which patterns should be excluded, and do certain constraint inventories risk overgeneration (predict non-existent, impossible patterns) or undergenerate (fail to predict attested or learnable patterns).

These questions have been approached experimentally most directly by Carpenter in a

⁹⁹Here, “more quickly and more accurately” means that computational learning agents general acquired exposure to fewer tokens of data before converging upon the correct grammar, and were less likely to mistakenly induce a grammar that would result in any outputs different from the outputs present in the learning data. For example, a learner presented with data in favor of fixed right-edge stress usually arrives to a constraint ranking able to generate right-edge stress exclusively sooner than, e.g., a learner presented with data in favor of antepenultimate stress would be able to arrive to a constraint ranking able to generate exclusively antepenultimate stress.

¹⁰⁰Note that Staubs' work considered exclusively fixed-stress systems without any quantity sensitivity.

series of experiments (Carpenter 2006, Carpenter 2010, Carpenter 2016b, Carpenter 2016a). As a whole, the results from Carpenter’s experiments on the learning of stress patterns in artificial languages parallels the result obtained in Hayes et al. (2009b) concerning the status and productivity of Hungarian vowel harmony patterns: “unnatural” rule or constraints may be induced, but may require more data to give them support, in contrast to natural (markedness) constraints, which are presumably made available through domain-specific knowledge about language, and need not be induced from the ambient data. I summarize here the major findings of each of these specific experiments:

- Carpenter 2006: Ch. 2–3, Carpenter 2010: Adult speakers of both (North American) English and (Quebec) French performed significantly better at learning a (purportedly) natural stress pattern, in which primary stress is assigned to the leftmost syllable with a low vowel, else the leftmost syllable, where the only vowels used were [a], [æ], [i], and [u].¹⁰¹ All vowels were equalized for duration and intensity, so that subjects could not rely on direct acoustic cues. Subjects assigned to a group trained on an unnatural pattern, in which primary stress was assigned to the leftmost syllable with a high vowel, else the leftmost syllable, performed less well. In the case of the natural pattern, position of stress (initial, final, or medial) did not affect performance; in the case of the unnatural pattern, forms for which initial stress was expected were more often produced correctly, suggesting that the subjects trained on the unnatural pattern tended to generalize a natural pattern of initial stress.
- Carpenter 2006: Ch. 4, Carpenter 2016b: Adult speakers of (North American) English and (European) French performed significantly better at learning a stress rule that is attested (“stress the leftmost heavy syllable, else the leftmost syllable”) than an unattested (and presumably unnatural) stress rule (“stress the leftmost light syllable, else the leftmost”). English speakers perform better than French speakers at this task, on both the natural and unnatural language conditions, though the performance of French speakers remains above chance.
- Carpenter 2016a: 9- and 10-year-old children acquiring North American English were trained on artificial languages like those used in Carpenter 2010. In contrast to adult speakers of English and Quebec French, who performed statistically significantly better than chance for both the natural and unnatural artificial language, children did not. However, among the subjects assigned to the unnatural group, errors were significantly influenced by the options: if a word contained both a high and low vowel, subjects were significantly more likely to select stress on the low vowel, despite having been trained on a pattern that preferentially assigned stress to high vowels.

¹⁰¹The assumption that stress being attracted to lower (= more sonorous) vowels constitutes a natural pattern, as Carpenter assumes, has though more recently been called into question (Shih 2018b, Bowers 2019). If it is indeed that case that “sonority-driven stress” does not genuinely exist, then attempts at replicating Carpenter’s results become crucial. Alternatively, if apparent cases of sonority-driven stress are open to reinterpretation as weight-driven stress, as proposed in Shih and de Lacy 2019 and McCollum 2020, then Carpenter’s results in her artificial languages with sonority-driven stress can be considered fundamentally parallel to her results for weight-driven stress.

One perhaps unintuitive and surprising result for the entire enterprise of research into linguistic learning biases came to light in a study by Gerken and Bollt (2008) with 9- and 7.5-month-old monolingual English-acquiring infants. In two experiments, the older infants exhibited the capacity to generalize a pattern assigning stress to heavy syllables once presented with at least three exemplars, but could not generalize an unnatural stress pattern (“assign stress to syllables starting with /t/”) from such a small number of tokens. Conversely, the younger, 7.5-month-old infants, were successful at generalizing the unnatural pattern. However, the pattern assumed by Gerken and Bollt (2008) to be “unnatural” may not in fact be so: in Pirahã, for instance, syllables having voiceless obstruents in their onset preferentially attract stress (see Topintzi 2010: 77–83 for an analysis of stress assignment in Pirahã that treats voiceless obstruents as weight-bearing). The crucial difference between the 9- and 7.5-month-old infants might consist in what they have learned about the effects of segments on syllable weight in their ambient language: younger infants are still open to the hypothesis that voiceless obstruents create heavy syllables, while 9-month-old infants have conclusively rejected this hypothesis, and have become insensitive to it, in much the same way that French-acquiring infants become insensitive to contrasts in the position of stress.

3.3.5.3 Some Specific Biases on the Learning of Stress Patterns

While Carpenter’s artificial grammar learning experiments offer strong grounds to believe that stress grammars that would require a constraint like ANTI-WSP are likely unlearnable, and should therefore be excluded by an innate bias, this line of investigation does not offer any particular indications as to what sort of patterns, among the learnable ones, might be privileged in acquisition in some way. The fundamental question here is whether either analytic or substantive biases facilitate or render more difficult the acquisition of certain actually occurring stress patterns. I here consider evidence for three specific biases, privileging trochaic feet, non-final stress, and stress on heavy syllables, then close this section with a short treatment of how these biases could be implemented in a learning model.

The “Trochaic Bias” Recall from the brief summary of stress acquisition in Dutch from Fikkert 1994 above that Dutch-acquiring children pass through a phase in which disyllables with final stress like [ʃi:.'raf] ‘giraffe’ are instead realized with initial stress. This phase of prosodic development among children acquiring West Germanic languages, combined with an apparent preference among infants for listening to trochaic (strong-weak) rhythmic patterns has led to claims that an innate “trochaic bias” might exist: humans might prefer trochaic structures to iambic structures. Typologically, among languages with clear secondary stress, trochaic systems are much better represented than iambic systems (153 vs. 31 languages in Goedemans and van der Hulst 2013b, though there are some evident areal and genetic patterns).

First, among adult speakers of English, preferences for trochaic feet have emerged in stress perception tasks, where the likelihood of misperception is higher for disyllables with final (iambic) stress, rather than unambiguously trochaic initial stress (Kelly and Bock 1988,

Kelly 1989).¹⁰² Since, however, unambiguous trochaic structures are predominant in English, the performance of adults may well simply reflect the acquired preferences of their prosodic grammar. The notion of such a trochaic bias in the acquisition of stress goes back to at least to work of Allen and Hawkins (1978, 1979). (cf. also the discussion Fikkert 1994: 188–92). Among nine-month-old English-acquiring infants, Jusczyk et al. (1993) found a listening preference for trochaic stress patterns, as did Bijeljac-Babic et al. (2016) for six-month-old French-German bilingual infants. Some studies of stress acquisition of French (Höhle et al. 2009), also seem to point trochaic preference emerging in a language that exhibit predominantly iambic stress (as opposed to, e.g., Dutch, English, or Spanish, where the statistically predominant patterns are trochaic).

Similarly, Adam and Bat-El (2009) presents a study of a single Hebrew-acquiring child that points towards a supposed trochaic bias in stress patterns. Namely, in the age period 1;2–1;7, after the emergence of the first lexical items, lexemes with clear iambic stress – mainly disyllables with final stress – were often produced with initial stress, no stress, or reduced to a monosyllable. In contrast, words with a possible trochaic parse were more faithfully reproduced. Moreover, the child attempted to produce more lexemes with trochaic stress, where trochaic stress is the correct target, in the first months of active production, even though iambic stress is considerably more frequent in the Hebrew lexicon.

Finally in the survey of Kehoe (2018) on the relationship between word truncation and possible trochaic preferences, the following behaviors have been observed in the literature:

- All West Germanic languages exhibit robust evidence for an early “trochaic bias”.
- Catalan (Prieto 2006) and Hebrew (Ben-David 2012) learners seem to pass through a phase in which forms with final stress are realized with initial stress. Particularly for Hebrew, this result is unexpected, since iambic stress is statistically predominant in the language.
- Portuguese (Santos 2006, Correia 2009) and Greek-acquiring children (Tsakosta 2004), on the other hand, seem to simultaneously allow for both trochaic and iambic feet, with no clear trochaic bias.
- In Japanese, disyllables are rarely truncated at all, and monomoraic forms may even be augmented to become bimoraic (Ota 2001).

On the basis of these results, I tentatively assume that there may well exist a universal preference for trochaic foot structures, which can be practically implemented via the ranking *TROCHEE* $\gg$ *IAMB* in the initial state of phonological acquisition. Attempts to satisfy *TROCHEE* (or to satisfy both simultaneously) may lead to the production of forms with level stress or truncation.

¹⁰²This specific result, however, cannot be straightforwardly disentangled from a possible dispreference for final stress.

NONFINALITY Staubs (2014: 113) poses an important question with respect to stress learning in particular and the acquisition of NONFINALITY effects in particular: could learning be impacted by some propensity (= bias) of the learner to misperceive, misinterpret, or simply ignore certain types of stress patterns? Given that, cross-linguistically, the construction of feet so as to avoid alignment with the right edge of the prosodic word occurs not infrequently, one might reasonably wonder whether some cross-linguistically common features or innate biases predispose learners towards the acquisition of active NONFINALITY effects, whether in the form of avoidance of stress on the final syllable (NONFINALITY- σ) or alignment of the (head) foot with the right edge of the word (NONFINALITY-F).¹⁰³ Gordon (2008), Gordon (2011a), Gordon and Roettger (2017) and Roettger and Gordon (2017) point towards some phonetic factors (a substantive bias) that might predispose learners to avoidance of stress on final syllables, in particular the avoidance of high pitch accents due to tonal crowding, since final syllables will tend to exhibit pitch lowering, especially in phrase- or utterance-final contexts. Further supporting a NONFINALITY bias is the typological rarity of peninitial stress and virtual absence of post-peninitial stress – such systems would regularly assign stress to final syllables in di- and trisyllabic words. Moreover, in at least one typological survey of stress systems (Goedemans and van der Hulst 2013a), systems with stress on penultimate syllables were found to be substantially more common than final stress (in other typological surveys, final-stress systems are somewhat more common than penultimate-stress systems).

In proposing to account for NONFINALITY as a cross-linguistically common phenomenon, Staubs (2014: 113–32) suggests “abstracting [NONFINALITY effects] to a simple notion that there is some bias for stress to be missed—or ignored—by the learner when it would create final stress forms. This bias has a natural interpretation within the learning framework discussed in this work: input data, normally sampled from the teacher and provided to the learner, is not always given to that learner unaltered. Instead, the “channel” affects the learner’s simulated perception of the form, giving it non-veridical forms as targets for learning.” Under a tonal crowding account of final-stress avoidance, a high pitch accent /H*/ is most likely to be retracted to the penult, resulting in perception of stress on the penult.¹⁰⁴

¹⁰³A completely parallel NONINITIALITY effect probably does not exist; indeed, the evidence for the necessity of NONINITIALITY in the analysis of metrical stress systems whatsoever is disputed, and plausible examples are rare. However, there are reasons to think that NONFINALITY pressures may result from both the avoidance of stress on final syllables *and* the alignment of a foot with the right edge of a word, whereas NONINITIALITY only comes in the form of avoidance of foot edge alignment (cf. Kennedy 1994); see discussion at 2.3.4. Staubs (2014: 120–1)’s simulations support the absence of a constraint NONINITIALITY- σ from the universal inventory.

¹⁰⁴It should not escape the readers’ notice that precisely such tonal crowding might play a role in the assignment of pitch and its interface with stress in Ancient Greek (compare specific proposals in Gunkel 2014 and Ito and Mester 2017). Furthermore, stress is consistently banned from final syllables in both Classical Latin, as well as in words of three or more syllables in Classical Sanskrit (see 5.2.3). This combined evidence is suggestive of some factor in the inherited prosody of Indo-European languages that makes stress – perhaps usually associated with a /H*/ or /H*L/ pitch accent – on final syllables unattractive. Note also that, in the emergence of fixed-stress patterns in many Slavic languages, as treated by Bethin (1998) and Baerman (1999), a crucial common development – sometimes observable across closely related dialects with more or less restrictive stress patterns – is a ban on stress in final syllables. Indeed, among all Slavic varieties with fixed-stress patterns, I am not aware of any that have developed persistent final stress out of the inherited Proto-Slavic lexical accent system.

A reasonable alternative, from the perspective of error-driven learning and Robust Interpretive Parsing in particular, would be for the learner to perceive the stress on the token as their grammar would have generated it, thereby strengthening their existing grammar. A neutral alternative would be to randomly assign stress to another, non-final syllable. Staubs concludes that all three approaches would result in penultimate stress in many cases, given how short most word forms are: in disyllables, all three approaches would result in stress surfacing on the penult; in trisyllables, a randomized stress assignment would place stress on the penult in half of the cases; if stress is assigned according to the learner's grammar, stress could instead consistently shift from the final to the initial syllable (though the learner's grammar might change in the course of acquisition and then generate the other alternative stress position).

The simulations undertaken by Staubs demonstrate that the relative typological frequencies of different fixed-stress patterns can emerge diachronically (i.e., by transmission of data from “teachers” to “learners”) over generations of learning. Fit to different typological frequencies of fixed-stress patterns, as found in different surveys, may be obtained by modulating the constraint set (i.e., with or without NONINITIALITY- σ) or the probability that final stress be misperceived (good results were obtained with a probability around 10%); in general, adopting either reassignment of misperceived stress to penultimate syllables or random reassignment worked well. What Staubs' simulations indicate, then, is that 1) assuming the activity of NONFINALITY- σ (and NONINITIALITY-Foot in stress grammars is reasonable, and/or 2) assuming that word-final stress is liable to be sometimes misperceived (i.e., not always learned veridically) fits the typological distribution of stress patterns.

For a lexical accent system, the possibility of non-veridical perception of some specific type of learning tokens can quickly have some substantial impact: among rarer lexemes or rarer case endings that would have borne stress on the final syllable, the probability of encountering the lexeme with stress elsewhere than one the “correct” final syllable is well above zero. For instance, among the 10002 distinct lemmata (lexical stems) in the *R̥gveda*, 53% occur only once; oxytone words among the rarer part of the lexicon might well (with a small but non-insignificant probability) be learned as having a different stress pattern. Diachronically, the prediction is then made that infrequent lexemes with final stress should exhibit a tendency to develop non-final stress, in the context of a grammar with lexical stress contrasts. Perhaps yet more important for the course of learning is the relationship between MAX-PROMINENCE and NONFINALITY: if a ranking NONFINALITY- $\sigma \gg$ MAX-PROMINENCE, for instance survives beyond the phase of phonotactic learning, into childhood, then words with final stress might continue to be generated with non-final stress.

Equipping the constraint inventory with both NONFINALITY- σ and NONFINALITY-f will, given an initial MARKEDNESS $\gg$ FAITHFULNESS bias, result in a relatively strong bias against forms with word-final stress. In an initial state with all markedness constraints equally ranked or weighted, in a trisyllabic form $/\sigma \sigma \sigma/$, for instance, a right-aligned iambic parse $[\sigma (\sigma ' \sigma)]$ will be harmonically bounded by any parses with initial or medial stress, since only $[\sigma (\sigma ' \sigma)]$ violates NONFINALITY- σ . The very imbalance in the constraint inventory would then constitute an analytic bias that makes the acquisition of final stress less likely.

A further question is whether NONFINALITY- $\mathcal{F}$, in particular, requires a strong bias in order to facilitate the learning (or diachronic emergence) of patterns that frequently produce antepenultimate primary stress, such as quantity-insensitive antepenultimate stress or the well-known Classical Latin pattern. Since constraints like ALL-FT-R and ALIGN- ω -R counterbalance NONFINALITY- $\mathcal{F}$, initial states without a bias in favor of NONFINALITY- $\mathcal{F}$ will effectively be biased in favor of feet aligned with the right edge of the word, at least under models that permit constraint ganging. This question is discussed further and tested through simulation at 5.3.2.2 and 5.3.2.3.

WEIGHT-TO-STRESS or Quantity-Insensitivity? Some approaches to the learnability of metrical stress, particularly parameter-based approaches, treat quantity-insensitive stress systems as a sort of default or baseline. For instance, in the parametric learning model proposed by Dresher and Kaye (1990) (cf. Dresher 1999, Dresher 2016), the learner requires cues for the potential relevance of syllable weight in stress patterns before considering hypotheses other than quantity-insensitive ones. On the other hand, the tendency for prominence and duration (created either via sonorous codas or non-sonorous onsets; see Topintzi 2010 and Ryan 2014 on the contribution of onsets to syllable weight) to often go hand-in-hand is potentially suggestive of a bias of some kind. The question is whether a substantive bias leads learners to interpret syllables with greater duration as also being the site of metrical prominence, leading to grammars with WSP effects, or whether an analytic bias predisposes learners to assign metrical prominence to syllables identified as phonologically heavier in the context of the language being acquired. Both possibilities might be true: a substantive bias might favor the emergence of quantity-sensitive stress diachronically, at least where the language's phonology provides evidence for contrastive syllable weight in some form, while an analytic bias would favor the acquisition of quantity-sensitive stress, and perhaps even provide some more support for unbounded stress systems.

One argument for the potential need for an analytic bias in favor of an active WSP comes from the existence of unbounded quantity-sensitive stress systems generally, and Default-to-Same Side (DTS) stress systems in particular (see 2.4.1). Heinz (2009: 309–10), for instance, notes the challenge that unbounded stress systems pose for proposals of stress learnability, due to their non-local nature. Furthermore, as discussed at 2.5, Koser and Jardine (2020) have shown that DTS patterns are especially complex among stress systems as a consequence of being computationally non-subsequential.¹⁰⁵ Typologically, quantity-insensitive stress patterns are indeed more common: Goedemans and van der Hulst (2013c) count 281 quantity-insensitive versus 219 weight-sensitive stress patterns – though the latter also include cases

¹⁰⁵Interestingly, Carpenter's (2016) study precisely employed DTS patterns ("stress the leftmost heavy/light syllable, else the leftmost syllable"). While Carpenter's subjects performed significantly above chance on this pattern, it would be of interest to know whether DTO or bounded quantity-sensitive patterns are learned to a higher degree of accuracy. Also worth noting is that the computational simulations at 5.3 more easily learn DTS than DTO patterns. This behavior fits the intuitions of most phonologists, and the factorial typologies discussed in Baković 1998, that quantity-sensitive default-to-opposite patterns require a more articulated constraint hierarchy. There thus remains a disconnect between Koser and Jardine's formal result and intuitions of phonologists, as well as empirical results obtained through simulation here.

of diacritic weight (such as Cupeño’s DTS pattern; see 2.4.2.1). Although weight-sensitive patterns are thus somewhat less frequent, the crucial question is whether they are nonetheless overattested: does the simplicity bias in favor of quantity-insensitive systems need support to be overcome? Does the inclusion of WSP (with some non-zero initial weight suffice), or must the learner assume a more active role for this constraint at the outset of learning?

Perhaps speaking in favor of an persistently active role for the WSP are the results of experiments with speakers of Brazilian Portuguese undertaken in Garcia 2017a (see also Garcia 2017b and Garcia 2019). Based on lexical statistics alone, one might expect for Portuguese speakers to acquire a (presumably unnatural) “anti-WSP” constraint, or perhaps at least to demote WSP sufficiently to neutralize its effects. In the Portuguese lexicon of words with three or more syllables (see (63), based on Table 2 in Garcia 2019), words with a final or penultimate syllable tend strongly (overwhelmingly in the latter case) to place primary stress on the final syllable, whereas words with a heavy antepenult repel primary stress: the rate of antepenultimate stress in HLL-words is lower than the rate of antepenultimate stress in LLL words. In an auditory forced choice task of contrasting target pairs with either antepenultimate and penultimate stress, or penultimate and final stress, the preferences of Portuguese speakers match the preference for stressed heavy syllables in the penult and final, but do not reproduce the seeming stress-repelling effect of heavy antepenultimate syllables. Instead, stressed heavy antepenultimate syllables were preferred to stressed light penultimate syllables in words of the shape HLL; the preferences of the speakers are thus contrary to the lexical statistics of the language on the point. Thus, in addition to Carpenter’s (2016) evidence that supports the easier learnability of weight-sensitive stress in experimental settings for English speakers, Garcia’s investigations indicate that a preference for stressed heavy syllables (at least among speakers of languages with quantity-sensitivity) remains even in the face of data to the contrary.

(63) Stress Patterns Found in the Four Most Common Weight Profiles of Words with 3 or More Syllables in the Portuguese Lexicon ($N = 130,555$; see Garcia 2014 derived from Houaiss et al. 2001)

Weight Profile	Stress	<i>n</i>	%
LLL	Antepenultimate	16562	24
	Penultimate	49641	71
	Final	3374	5
HLL	Antepenultimate	3281	21
	Penultimate	11813	76
	Final	511	3
LHL	Antepenultimate	10	< 1
	Penultimate	26579	98.9
	Final	390	1.1
LLH	Antepenultimate	119	1
	Penultimate	4852	26
	Final	13430	73

From the internal perspective of constraint inventories, the WSP might also require some

support in the form of a bias, since its effects may be counterbalanced by PARSE- σ or RH-CONTOUR. For instance, where /L H L/ sequences arise, if trochaic feet are already known to be favored, the parse [L ('H L)] violates a constraint against feet of the form ('H L) (RH-CONTOUR), but the parse [L ('H) L] incurs an additional violation of PARSE- σ . Given the experimental evidence discussed here that speak to the naturalness of WSP effects, and the fact that quantity-sensitive systems are not rare, despite their greater computational complexity, speaks clearly in favor of a bias towards stressed heavy syllables in some form. That this bias is at least partly substantive seems likely, but in the present context, must be implemented analytically through the usage of the WSP.

Summary on Specific Biases I thus consider three particular stress biases plausible: a preference for stress on heavy syllables (however defined, language-specifically); a preference for trochaic feet; and avoidance of stress on final syllables. In the constraint inventory with which I will generally operate in Part II, at least the third of these biases corresponds to a markedness constraint that does not have a parallel opposing constraint. Hence, in an initial state where MARKEDNESS dominates FAITHFULNESS, and all markedness constraints are ranked equally and candidates with stress on final syllables will incur greater penalties than other alternatives. A question that can perhaps be tested through computational experimentation is whether better performance obtains if the constraints NONFINALITY- σ and WSP are additionally weighted above other markedness constraints, and given a bias that prefers to maintain a higher weight. Precisely this sort of implementation would seem to be necessary to produce the trochaic bias, since TROCHEE is counterbalanced by IAMB in the constraint inventory. How biases implemented with respect to TROCHEE, NONFINALITY, and WSP impact the learning of stress grammars in a concrete case will be addressed at 5.3.2.3.

3.3.6 Towards Modeling Phonological Change: The Outlook for Computational Simulation

In Chapters 5 and 7, iterated agent-based learning simulations (see 3.3.7) will be employed to model the transmission of stress grammars between learning agents equipped with Max-Ent grammars (cf. 3.2.1.2). The objectives of those simulations are to attempt to emulate possible pathways of prosodic development (specifically, in the stress system) between two historical points in the history of a language, and will be discussed in more detail at the relevant points in the respective chapters (5.3 and 7.4). Some of most direct antecedents to the type of simulations to be used are found in Staubs 2014 (compare also Hughto 2020) and Ito and Feldman 2022, though the structure and ends of the particular simulations that I design differ from either. In contrast to Staubs 2014, the concern in this work lies with historical change, rather than typology; in contrast to Ito and Feldman 2022, the focus lies with stress grammars encoded with constraints of the sort encountered in Chapter 2. A small body of other research has also sought to apply iterated learning to questions of language change (e.g., Harrington et al. 2018, Harrington et al. 2019, Beguš 2020), but differs more substantially in either simulation design or level of linguistic analysis, and so will be discussed only more briefly below.

3.3.7 Agent-Based Learning: Basic Concepts

Descriptions of the basics of agent-based simulations and iterated learning as applied to linguistic data can be found in, e.g., Kirby and Hurford 2002 or Wedel 2015.¹⁰⁶ In their fundamental conception, agent-based iterated learning models are simple: a learning agent equipped with some sort of pattern induction device receives data in some form, from which it learns and generalizes, before generating its own outputs. In the simplest sort of simulation design, a single agent in an initial generation (A_1) receives some seed data, and then generates data for another single agent in a successive generation (A_2), which in turn generates data for learning for a further single agent in the next generation (A_3), and so on. Transmission and learning proceeds across generations (often thousands) until a maximum number of generations is reached or some prespecified condition is reached by an agent. Figure 3.3 shows this simple structure of transmission; in effect, the learner for generation G becomes the teacher for the learning agent of generation $G + 1$. Figure 3.4 visualizes a more complex network of learning agents, in which multiple agents initialized at the outset of a generation learn from each other as well as from the “teacher” agents of the preceding generation. Other types of non-iterated agent-based learning, in which one or more agents simply act as teachers to each other after training on some seed data can be used as well (see Wedel 2012 for one example of an agent-based model of this sort).

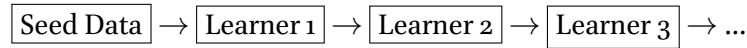


Figure 3.3: Basic Design of Single-Agent Iterated Learning (cf. Staubs 2014: 14 or Wedel 2015: 153)

Agent-based iterated learning models have been employed extensively in the investigation of the *emergence* of structure in language out of unstructured data. In the context of simulations on the emergence of language structure, so-called “bottlenecks” are observed as a result of the need to draw inferences from unseen data (Brighton 2002, Kirby 2002, Brighton et al. 2005, Smith et al. 2003).¹⁰⁷ Similar “bottlenecks” may apply in the context

¹⁰⁶The term “iterated learning” for the study of dynamic systems, especially in the areas of language evolution, typology, and change, is often attributed to Kirby (1999).

¹⁰⁷The results of these simulations have further been provided with mathematical grounding by, e.g., Griffiths and Kalish 2007, Real and Griffiths 2009, and Rafferty et al. 2014, who have observed that, when learners are assumed to apply Bayesian inference, their posterior distributions over languages should come to approximate their prior distributions over languages, where the posterior probabilities are calculated using Bayes’ rule (here, $P(l)$ is the prior bias for a language and $P(d)$ is the probability of the observed data), shown in (1). Given that iterated learning constitutes a Markov chain described by the transition probabilities between any possible state and any other state that the learner may find itself in, the Markov chain’s stationary distribution will be the learners’ prior distribution over the hypotheses (cf. Rafferty et al. 2014: 1409–11).

(1) Posterior probability of $P(l|d)$ given by Bayes’ rule

$$P(l|d) = \frac{P(d|l)P(l)}{\sum_{l'} P(d|l')P(l')}$$

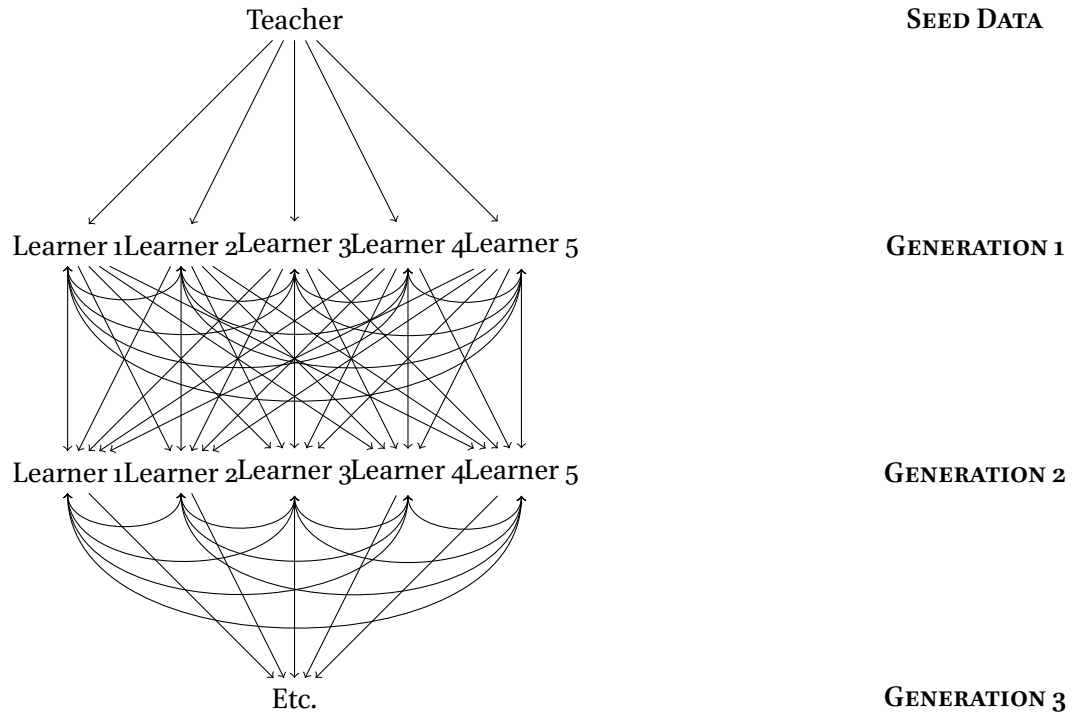


Figure 3.4: Design of Multi-Agent Iterated Learning with Agent Interaction (cf. Sandell and Zukoff 2017)

of phonology as well, if a teacher agent finds the need to generate a form with an underlying phonological structure that has not been encountered in its learning data. For instance, productive word-formation, inflection, or adoption of loanwords might result in the creation of a segmental sequence or prosodic shape to which the agent must then assign structure, i.e., determine how to realize a sequence, assign primary stress, etc. The choice made by the agent on the basis of the phonological grammar induced from its learning data may not correspond to the choice that its teacher(s) would have made. In short, since each successive agent is not guaranteed to encounter the same learning data in the same order with the exact same frequencies (in particular, low-probability data may not be transmitted at all), a learner may construct different linguistic structures from the data. Insofar as the learning agents behave probabilistically (i.e., generate outputs according to some probability distribution), relatively small changes in the likelihood of a given output may accumulate or “snowball” over the course of successive generations, with the result that the outputs after tens or hundreds of generations may look substantially different from the outputs generated by the initial teacher or the distributions found in the seed data.

The models of grammar assumed for a learning agent to possess are open to the analyst. Possible choices include, *inter alia*, connectionist networks (e.g., Hare and Elman 1995), exemplar storage plus a classification algorithm, such as k -nearest-neighbors (e.g., Pierrehumbert 2001, with a handcrafted classification rule), or memory-free updates to the parameters

of a classification model (e.g., MaxEnt with stochastic gradient ascent, as in Staubs 2014). In large part, the choice of grammar model depends upon the level of linguistic analysis: exemplar models are frequently encountered in the phonetic literature on sound change, while MaxEnt grammars are represented in the phonological literature. Depending upon exactly what type of grammar model is adopted, choices concerning further parameters must be made in simulation design, though virtually all applications of iterated learning require that some number of learning tokens to which a learning agent should be exposed per generation be specified. Furthermore, again depending upon the model, a given learning datum may or may not impact its learning of a phonetic categorical, morphophonological rule, phonological grammar, etc.; for instance, under Robust Interpretive Parsing for the learning of hidden metrical structure (cf. 3.2.2.1), a given learning datum may or may not induce any change in a learner's current phonological grammar.

3.3.8 Uses of Iterated Learning in the Study of Linguistic Typology and Language Change

One of the earliest usages of iterated agent-based learning (in fact *avant la lettre*) is found in the study of Hare and Elman (1995) on the development of the past tense system between Old and Modern English. Using a connectionist network based on the backpropagation algorithm of Rumelhart et al. (1986), Hare and Elman first model the attraction subclasses of weak class I (Ib and Ic) into weak class II during the Old English period, then further the tendency for strong (mostly “irregular”) verbs to adopt weak (“regular”) past tense inflection. This model incorporates factors of phonological similarity, type frequency of patterns, and token frequency of individual lexemes, and even captures the “irregularization” of original weak verbs due to their strong phonological similarity with a robust strong verb pattern (cf. the similar notion of “Islands of Reliability” for English past tense formations in Albright and Hayes 2003). In the main, that Hare and Elman's model has the capacity to generate errors with some probability that may be taken as learning data for subsequent generations of agents allows for errors to accumulate over generations and yield trajectories that broadly resemble the attested changes.

In the domain of phonetics and phonology, exemplar models with learning agents (whether iterated or not), usually employing data in the form of reduced elements of the phonetic signal (e.g., absolute values of formant values for vowels) have been used to examine the dynamics of phonetic category construction as well as a range of sound changes, both complete and in progress. Examples of this work include: the treatment of chain shifts in vowel inventories, with particular reference to the Northern Cities Shift (Labov et al. 1972) in Stanford and Kenny 2013; /u/-Fronting in Standard British English (Harrington and Schiel 2017, Harrington et al. 2018); and sibilant inventories, especially the development of Polish sibilants between ca. 1200 and 1500 in Boersma and Hamann 2008. Most recently, Beguš (2020) has employed generative adversarial networks (GANs) trained on raw acoustic data to model a hypothetical merger of aspirated and unaspirated allophones of voiceless stops in English (in effect, a form of rule loss).

Of most direct relevance for the present enterprise is the employ of iterated learning to study (hypothetical) phonological change above and beyond cases of segmental sound

change. Here, two studies, Staubs 2014 and Ito and Feldman 2022 constitute effective examples of how iterated agent-based learning can enrich our understanding of phonological grammars. In terms of overall design and conception, the simulations employed by Staubs, since the learning agents essentially consist of MaxEnt grammars equipped with markedness constraints on prosodic structure, will have direct implications for the design of simulations at 5.3 and 7.4. Ito and Feldman's paper, meanwhile, addresses a concrete case of prosodic change, much as my treatments of Indic and Germanic will seek to do. Since some of Staubs' work on the typology of quantity-insensitive stress systems and exploration of biases (in particular with respect to NONFINALITY ; see 3.3.5.3) has been mentioned at several points above, and will frequently serve as a point of reference for in discussion in later chapters, I focus here on Ito and Feldman's paper as an example of the study of prosodic change via agent-based iterated learning.

Ito and Feldman (2022) examine historical changes in the assignment of Sino-Korean words in the Korean lexicon to one of several distinct pitch accent classes. The change in assignment of pitch accent class is handled by the authors as a species of analogical change sensitive to phonological environments, somewhat similar in conception to the view on analogical change taken by Albright (Albright 2002b, Albright 2008b). Specifically, the paper aims to demonstrate that changes in pitch-accent class attested between the Middle Korean of the 15–16th centuries and two present-day dialects (South Kyungsang and Yanbian; see 3.3.3.2 above on the former) can be simulated through iterated learning models that learn the specific phonotactics governing each pitch-accent class. Three agent-based iterated learning simulations are employed to answer the following main question: can iterated learning with phonotactic constraints capture the general patterns of change attested in the historical record?

Before discussing Ito and Feldman's simulations and their results, some background on the (historical) prosody of Korean is needed. Much like the lexicon of Japanese (cf. Ito and Mester 2003a), the Korean lexicon may be divided into a native stratum, a stratum of borrowings from Middle Chinese around the 9–10th centuries, and a stratum of recent (primary English) loanwords. The Sino-Korean lexicon of Middle Chinese possessed three contrastive tonal patterns in disyllabic words, predictable on the basis of the tone in the Middle Chinese source and the coda segment in the Sino-Korean form itself: L(ow) (< Middle Chinese mid level tone) R(ising) (< Middle Chinese mid rising or Middle Chinese high falling), or H(igh) (< Middle Chinese mid rising, Middle Chinese high falling, or Middle Chinese mid level if the syllable had a coda voiceless stop).¹⁰⁸

The analysis undertaken by the authors used 8564 two-syllable Sino-Korean nouns as a basis. For all four tonal patterns on disyllables (HX, LH, LL, and RX), South Kyungsang and Yanbian have regular, expected correspondences; for instance, Yanbian should normally

¹⁰⁸Concerning the reconstruction of the values of the four traditional Middle Chinese tones (*level*, *rising*, *departing*, and *entering*), which can be reduced to an underlying contrast between just three tones (where *entering* is an allophone of *level*), Pan and Zhang (2015: 84) state: "It is commonly accepted that the *pingsheng* is with a level contour, the *shangsheng* a high rising tone, the *qusheng* a falling tone, and the *rusheng* a checked tone (a syllable that ends with a stop consonant). Thus, their tonal values may be reconstructed as 133, 135, 151, and 13?, respectively."

continue rising (LH) and level low (LL) as such, and has generally merged HX and RX as falling HL, giving three present-day classes. However, a considerable proportion of Sino-Korean lexemes (30% in South Kyungsang, 37% in Yanbian) attest a different tone in the present-day language than what the known Middle Korean tone would predict. In attempting to identify factors that would either promote or discourage change in tone on a given lexeme, Ito and Feldman point out the following factors:

- Certain underlyingly distinct tones are neutralized in their citation forms (compare the tables for South Kyungsang under 3.3.3.2), and the larger of the two classes subject to neutralization (by type frequency) tends to attract more members.
- Individual morphemes that have a higher type frequency (i.e., recur in more distinct lexemes) tend to remain diachronically stable.

Learners in each generation were equipped with a model able to make predictions about the tonal class of a lexeme or morpheme based on phonotactic generalizations. In order to allow for the possibility of change, in each generation, the lexical knowledge of tonal classes would first be passed on faithfully, but then the learning agent would “forget” the tone of 5-25% of morphemes in the data (treating each occurrence of a morpheme in the lexicon as independent), and forced to predict the tone for the forgotten morphemes based on the morphemes whose tone was remembered. The new lexicon with complete tones would then be passed on to the learning agent of the next generation. Three different types of models were tested for allowing the learning agents to predict the tonal class of a morpheme: likelihood a syllable of a particular (C)(G)V(G)(C) to belong to a given tonal class; phonotactic constraints induced using a version of the learner of Hayes and Wilson 2008 to predict the likelihood of a syllable to have a particular tone under a MaxEnt grammar with the constraints; a combination of the two models. These three models were compared to a baseline model, in which a forgotten morpheme would be assigned a tone based on the type frequency of tonal classes in the training data.

The best-performing models after 20 generations of iteration were found, for both dialects, to generate similar rates of (mis)correspondence to the regular phonological development of tonal patterns. Overall, all three models with phonotactic knowledge performed better than the baseline model. Overall, the combination model, which equipped the learner with the ability to learn both full syllable shapes and their likelihoods of having particular tones and general phonotactic constraints was able to achieve the best fit to the present-day distributions of tonal classes in the two dialects with the comparatively smallest number of model parameters (as assessed by Bayes’ Information Criterion). Considerable variability was found, however, among individual lexemes across multiple runs; the question here is whether the simulations should be able to capture the specific diachronic behavior of morphemes, and not just the broad outlines of change.

The crucial empirical innovation in Ito and Feldman’s paper, distinct from much work on iterated learning and computational work on language change in general, is that the authors employ real (or realistically reconstructed) data: the inputs to the first generation of agents consist in actual lexemes of Middle Korean with their actual known phonological

characteristics, and the performance of the model can be measured against the actual outcomes in later stages of the language. Studies in Chapters 5 and 7 share this sort of empirical grounding with Ito and Feldman's work, with the difference that the changes there will involve stress grammars, rather than the lexical properties of morphemes.

3.4 Summary on Learning, Learnability, and Prosodic Change

This chapter has presented a wide-ranging overview of the scholarship on the learnability of phonological grammars, with a specific focus on constraint-based models and the learnability of metrical and lexical stress grammars. The introduction to basic constraint-ranking algorithms in 3.2.1.1 furnishes the reader with the necessary background to understand how constraint rankings presenting many of the analyses in Part II are determined. Given the prevalence of variation at the phonological level, however, models of phonological grammars able to accommodate variable phenomena, and Maximum Entropy models in particular, are to be preferred (3.2.1.2). Beyond the learning of rankings or weightings of phonological constraints when both surface outputs and an analysis of the underlying phonological structures are given, how to approach problems of so-called “hidden structure” – that is, the learning of ambiguous or invisible prosodic structures, as well as underlying representations – was treated under 3.2.2. Issues of hidden structure learning are of particular importance to the present work, since it is precisely concerning with phonological change relating to such invisible or ambiguously transmitted prosodic information, namely, foot structure and the underlying representation of lexical accent. Since the induction of underlying representations still constitutes a not fully tractable problem given unstructured learning data, attempts to tackle changes in prosodic structure in Part II will primarily concentrate on metrical structure assigned to partly prosodified strings.

In this chapter's second half, questions of prosodic change, specifically change in the grammars for the assignment of stress, in relation to phonological learnability and prosodic acquisition, took center stage. Driving the discussion there was the pair of observations that I term the “Core Conundrum of Diachronic Phonological Learnability”, stated at (51): why are demonstrably learnable phonological systems sometimes mislearned? Of particular relevance for the study of prosodic change in older Indo-European languages is the issue of lexical stress: if lexically contrastive stress can be acquired without particularly difficulty, why does it sometimes make way for fixed metrical stress systems?

Although grammars for the assignment of stress may well be affected by other aspects of prosody, e.g., the reinterpretation of boundary tones as pitch accents in some varieties of Basque, in many historically attested “stress shifts” among the Indo-European languages, such factors are either unknown or unlikely. Instead, some changes in the position of stress can be adequately captured as cases of reanalysis at the phonological level: the learning trajectory of a stress grammar results in different outputs with respect to the target grammar. I argued that the emergence of the Classical Latin Stress grammar and the phenomenon of Wheeler's Law in Proto-Greek are attributable to rankings of constraints that represent an intermediate stage on the way to target grammars: ALL-FT-R $\gg$ ALIGN- ω -L in the Latin case,

and $\text{PARSE-}\sigma \gg \text{MAX-PROMINENCE}$ in the Greek case. Simulation of the Latin case substantiated this argument.

To lay the groundwork for research into cases of transformation of lexical stress systems into fixed metrical stress systems in Chapters 5 and 7, closer consideration of how lexical stress might be mislearned and fail to propagate itself was necessary. A small simulation at 3.3.3 showed that, at least in principle, lexically contrastive stress supported by prosodic faithfulness constraints can undergo attrition and loss, in much the same way that competition among variants in other domains of the grammar may be resolved in favor of a single variant. An overview of two known cases of prosodic change, in English and Kyungsang Korean, established that prosodic changes, at the lexical level at least, are indeed most likely preceded by a phase of variation. Finally, an examination of the current state of research on L1 and L2 acquisition of prosody, including the phenomenon of “stress deafness”, was undertaken in order to have a frame of reference in developing assumptions about the initial state in prosodic acquisition, and how speakers of different linguistic backgrounds might interact.

This concludes Part I, which was concerned with providing the necessary background on the phonological study of prosodic structure (Chapter 1), stress assignment (Chapter 2), and learnability and change in phonological grammars (Chapter 3). In Part II, these methods will be applied to specific problems in the synchrony and diachrony of the Indic, Greek, and Germanic branches of the Indo-European language family:

- Chapter 4 treats the synchronic grammar of stress assignment in Vedic Sanskrit, and attempts to shed light on some aspects of its intonational phonology as well.
- Chapter 5 attempts to reconstruct a synchronic metrical stress grammar that emerged in the later histories of (some) Indic languages, and attempts to account for the emergence of such a system out of the older Vedic system, in part with support from agent-based iterated learning simulations.
- Chapter 6 surveys problems in the word prosody of 1st Millennium Attic-Ionic Greek, seeks to develop a new analysis thereof, develops an account of the slightly divergent word prosody of Doric Greek, and describes the relationship of word prosody in 1st Millennium Greek to its Proto-Indo-European ancestor and Modern Greek descendant.
- Chapter 7 works out a more precise and differentiated reconstruction of word-level prosody in Proto-Germanic, before seeking to explain its emergence out of Proto-Indo-European word prosody, again by reference to results from agent-based iterated learning simulations.
- Chapter 8 summarizes the major results of Part II and lays out future directions on corpus-based and computationally supported research into prosodic change.

Part II

**Case Studies in the Analysis of Prosodic
Systems of Older Indo-European
Languages**

CHAPTER 4

Stress and Tone in Vedic Sanskrit: A Synchronic Analysis of Intonation, Lexical Stress, and Metrical Structure

4.1 Introduction and Preliminary Observations

This chapter attempts a comprehensive analysis of stress assignment and its interactions with intonation in Vedic Sanskrit, principally on the basis of the oldest Vedic text, the *R̥gveda* (RV); in cases of variation internal to the RV, I give precedence to evidence from the portions of the RV standardly regarded as relatively more archaic, i.e., the so-called Family Books (RV Books II–VII). In the larger context of this book, the analysis developed here serves two functions: first, it provides a direct point of departure for discussion of minor prosodic changes observable within the Vedic period, as well as the development of the stress system of Classical Sanskrit, to be treated in detail in Chapter 5; second, because the prosodic system of Vedic is very similar in crucial respects to the prosodic system of Proto-Indo-European, a clearer portrait of Vedic prosody helps to lay the groundwork for an understanding of the emergence of the Proto-Germanic stress system out of Proto-Indo-European (the primary subject of Chapter 7). For the purposes of documenting, understanding, and explaining the networks of prosodic changes in the earliest-attested Indo-European languages, the grammar of Vedic prosody constitutes an indispensable touchstone. From a theoretical perspective, Vedic prosody is similarly of great interest insofar as its testimony bears on debates concerning the representation of lexical accent and the assignment of stress in morphological interface systems, furnishing evidence from a language with particularly rich and extensive systems of inflectional and derivational morphology, as well as various types of compounds and kinds of prosodic morphology at best much more weakly represented in other Indo-European languages with lexical accent (i.e., reduplication and infixation).

Contemporary analysis of stress assignment in Vedic essentially goes back to Kiparsky 1973, with modifications and extensions developed over the subsequent decades in Kiparsky and Halle 1977, Kiparsky 1984, Kiparsky 2010a, Kiparsky 2014, Kiparsky 2016, and Kiparsky 2018a; Kiparsky's approach to Vedic prosody, following the presentation in the more recent publications, will be introduced in Section 4.2 below. In terms of empirical coverage, Kiparsky's body of work is, at this writing, the most comprehensive. Less comprehensive treatments in large part inspired by Kiparsky's work, but recast in terms of Optimality Theory (or the theoretically adjacent Harmonic Grammar) have also been developed over the past decade, represented by Frazier 2007, Marston 2009, Sandell 2015a: Ch. 7, Yates 2017: 179–88, and Yates 2020; the present chapter will, to a large extent, represent an extension and improvement upon the analyses proposed in Sandell 2015a: Ch. 7 and Yates 2020. Calabrese

(2019: Ch. 4), on the other hand, has reworked Kiparsky's proposals employing the metrical grid model of Halle and Isardi 1995. Various issues in Vedic stress and intonation have been recently discussed in Gunkel 2017 (intonation of vocatives), Kümmel 2021 (cruxes/cruces of stress assignment in the Kiparskian model), and Keydana 2021 (partial analysis of stress assignment in terms of Network Morphology; cf. Corbett and Fraser 1993).

A considerable gap in all of the analyses cited is the failure to establish – insofar as is possible – the metrical stress grammar of Vedic. In large part, this gap naturally arises from the fact that stress assignment in Vedic is dependent upon lexical and morphological factors to such a degree that the purely metrical component can seemingly be reduced to one general preference, namely, primary stress should fall as close to the left edge of the prosodic word as possible, which is compatible with a number of distinct stress grammars. In addition to constructing the largely morphological interface grammar of stress that derives the facts of Vedic primary stress, this chapter will strive to discover the constraint ranking that governs “default” stress assignment in the absence of lexical or morphological considerations, or at least, what sort of stress grammar phonotactic learning without lexical or morphological information would be most likely to converge upon.

In the remaining portions of this introductory section, I first provide a brief introduction to our sources on Vedic prosody (4.1.1), then discuss what is known about the phonetics of Vedic stress (4.1.2), before addressing the question of whether the type of word-level prominence seen in Vedic is in fact best approach as from the perspective of stress, “pitch accent”, or tone, including there some discussion on discernable aspects of Vedic intonation (4.1.3). By way of transition, I demonstrate the insufficiency of a purely metrical analysis of Vedic stress assignment and illustrating some stress basic patterns (4.1.4). Section 4.2 then works through the analysis of Vedic stress assignment developed across Kiparsky's publications, explaining its machinery and assumptions while illustrating its efficacy across most stress patterns in inflection, derivation, and compounding. Section 4.3 then takes up some of the insights from Sandell 2015a: Ch. 7 and Yates 2020 to reanalyze the facts introduced in Sections 4.1 and 4.2 in a comprehensive Optimality Theoretic grammar; this grammar deliberately attempts to keep the number of active constraints to a minimum. In the final major portion, Section 4.4, however, an attempt is made to more precisely determine the rankings of constraints on metrical structures in Vedic responsible for the default (non-morphological) stress rule of the language.

I do not pretend to total empirical comprehensiveness (i.e., the capacity to predict why every token in the Vedic corpus, and in the RV is accented as it is), but do believe that the analysis presented in Section 4.3 can offer empirical comprehensiveness when also applied, after closer investigation, to further other morphological categories largely left out of consideration here (e.g., *vrddhi*-derivation). I do, however, intend to argue that the account offered in this chapter is comparatively more complete and preferable to existing analyses (though one's mileage may vary, depending upon one's theoretical predisposition). Most importantly for the purposes of this work, I argue that the portrait of Vedic prosody worked out in this chapter enables a clearer and more successful understanding of its subsequent historical developments, to be treated in Chapter 5.

4.1.1 Brief History of the Transmission of Vedic Prosody

The data to be examined and analyzed in this chapter stem directly from editions of Vedic texts printed with accentuation beginning in the second half of the 19th Century and continuing to the present day, their data then entering into the grammatical and lexicographical works on which I have primarily relied (e.g., Grassmann 1872 [1976], Macdonell 1910, Debrunner 1954); see Jamison and Witzel 1992 [2003]: 5–27 on the multitude of Vedic texts and related philological work. The printed editions of Vedic texts rest upon a manuscript tradition in India that itself is likely not more than about 1000 years old, and in fact the oldest known manuscripts lack any system of accentual marking (cf. Witzel 1997: 259). This manuscript tradition in turn rests upon a much older oral tradition, dating back to the period of the composition of the Vedas (~ 1500–1000 BCE;). This oral tradition was evidently still robust in the 19th century, but has been in serious decline throughout the 20th century; unfortunately, the available Vedic textual editions did not and have not taken account of the oral tradition, even though the manuscript tradition is “commonly known” to be “remarkably inferior to that of the oral tradition” (so Jamison and Witzel (1992 [2003]: 25)). For the purposes of the study of Vedic prosody, the available material is more than sufficiently coherent to construct a solid phonological analysis, though one wonders whether occasional apparent contradictions are an artifact of the manuscript transmission.

The number of accented Vedic texts is much smaller than the total number of available texts, with accented manuscripts having been produced essentially only for the most important parts of the Vedic canon, namely the four Vedas (*Ṛgveda*, *Atharvaveda*, *Sāmaveda*, and *Yajurveda*) in both *saṃhitāpāṭha* and *padapāṭha*, the *Taittirīya Brāhmaṇa* (and its *Āraṇyaka*), and *Śatapatha Brāhmaṇa* (along with the *Bṛhadāraṇyaka Upaniṣad*), and quotations from the *saṃhitās* in otherwise unaccented *sūtra*-texts sometimes provide these quotations with accents.¹ This total amount of prosodically relevant material is indeed substantial, and already the *Ṛgveda* alone would be sufficient for arriving to mostly reliable general conclusions. Nevertheless, the accented texts are biased towards the oldest period of the Vedic language, for which reason possible changes in prosody during the Vedic period, whether affecting individual lexemes, classes of words, or the system as a whole, are harder to detect.

In the indigenous grammatical tradition, aspects of prosody are treated in the *prātiśākhya*s, relatively short treatises concerning the pronunciation of given Vedic texts, which are available for three of the four Vedas (missing only for the *Sāmaveda*), though one each is available for two different *saṃhita* texts of the *Yajurveda* (the TS and VS, respectively); see Whitney 1869 for a detailed overview. Extensive discussion and rule systems for Vedic accentuation are found in the grammarians Pāṇini (cf. von Böhtlingk 1845), Kāṭjājana, and Patañjali, which essentially agree with the statements made in the *prātiśākhya*s; cf. Haug 1874: 4–5.² There, three general types of accent, usually understood to correspond to differences

¹See Jamison and Witzel 1992 [2003]: 16–8 for a synopsis of Vedic texts.

²That Pāṇini could systematically state rules for the placement of stress in Vedic that essentially agree with the accents in the manuscript tradition could be interpreted in multiple ways: the prosodic grammar of Vedic exhibited relatively few changes prior to a (relatively rapid?) transformation into the Classical Sanskrit system, or; Pāṇini's rules privileged the accentuation of the oldest parts of the Vedic canon, as described in the

in pitch (see further below) are distinguished: *anudātta* ‘not elevated’ (further distinguished into regular *anudātta*, *anudāttatara* ‘more non-elevated’, and *pracaya* ‘(slightly) increasing’), *udātta* ‘elevated’, and *svarita* ‘sounded’ (further distinguished into dependent and independent types). The *udātta* is universally now accepted as representing an obligatory and culminative prominence (in most cases; see further 4.1.3), and is represented in transliteration with an acute, e.g., *bhāvati*, *matīm*, etc.

Although the available *prātiśākhya*s agree on the different types of accent, how these different types are orthographically represented varies from text to text. The most detailed treatment of the different systems of accent marking found in Vedic texts is found in Haug 1874: 21–107, whose division into four schools is followed by Macdonell (1910: 78–80; 1916: 448–53); cf. also Beguš 2016: 2 and Witzel 2020. Of the four accent-marking types, the system found in the ŚB is the most divergent, and perhaps gives indications of developments towards the novel stress system that emerged towards the end of the Vedic and beginning of the Classical period (see Macdonell 1910: 79); discussion of the testimony of the ŚB will be left for the following chapter (5.1.2), since it potentially attests to emergent prosodic changes during the Vedic period. The other types of accent marking are fundamentally similar, insofar as they, in the normal case, somehow indicate exactly one *udātta* per unit word (presumably the ω_{Max}). In fact, the most common notational system, used in manuscripts of the RV and AV, for instance, does not mark the syllable with *udātta* at all, but rather marks the syllable with *anudātta* (with a horizontal stroke below) and *svarita* (with vertical stroke above). The systems found in the two saṃhitās of the Yajurveda (Matraiyāṇisaṃhita and Kāṭhakaśaṃhitā) and Sāmaveda are more explicit: the former mark *udātta* with a vertical stroke above and (different types of) *svarita* with other diacritics; the latter writes numerals above the syllables, using 3 for *anudātta*, 1 exclusively for *udātta*, and 2 usually for *svarita* (but also for *udātta* in words without *svarita*). Since no *prātiśākhya*s corresponding to the texts having these latter systems of accent marking are available, these systems are understood only from the manuscripts themselves, though they do in certain respects better match the statements of Pāṇini (cf. below and Wackernagel 1896a: 285).

In the early reception of and teaching of Sanskrit grammar and texts in the West, even basic features of Vedic prosody were at best incompletely grasped: the grammar of Colebrooke (1805: 9–10) knows the terms *udātta*, *anudātta* and *svarita*, but fails to clearly explain them, while the earliest published excerpts of the RV published in Europe (Rosen 1830) omit accent marks (though the underlying manuscripts contain them). Beginning with von Böhtlingk 1845, in which Pāṇini’s rules were systematically applied to generate accented nominal and verbal paradigms, knowledge of Vedic prosody gradually improved, owing to study of the *prātiśākhya*s (Whitney 1869), fieldwork in India (Haug 1874), and comparison the prosody of other Indo-European languages (Bopp 1854). In general, at most very minor improvements or adjustments can be made to the descriptions available in Wackernagel 1896a: 281–94 and Macdonell 1910: 76–82.

*prātiśākhya*s, or; the rules of accentuation in Pāṇini’s grammar were (in part) relied upon for producing accented manuscripts. Most likely is that Pāṇini knew the *prātiśākhya*s and crafted his rules of Vedic accentuation partly in accord with them.

4.1.2 Phonetic and Phonological Properties of Vedic Stress

As mentioned above, the Vedic *prāṭisākhya*s and Pāṇini distinguish essentially three types of syllable prosody: *anudātta*, *udātta*, and *svārīta*. The standard interpretation applied to these terms, based on their etymologies and brief definitions supplied in the *prāṭisākhya*s and Pāṇini, is that they designate differences in pitch (fundamental frequency). Pāṇini (1.2.29) defines the *udātta* as “a vowel spoken with raised voice”, the *an-udātta* is the opposite of the *udātta*, while the *svārīta* is called *samāhāraḥ* ‘combination [of the others just defined]’ (cf. Whitney 1869: 23 on identical or nearly identical statements in the *prāṭisākhya*s of the TS and VS). In terms of a phonological tonal representation, the three basic Vedic accents, following Pāṇini, could then be designated Low (*anudātta*), High (*udātta*), and Falling (*svārīta*). The *udātta*, corresponding to a pitch peak, might further be identified as the locus of prosodic prominence, based on how these differences in pitch are typically distributed within prosodic words.

Assuming three types of tones, Low (L), High (H), and Falling (F) contained in the inventory of Vedic tones, Vedic surface forms could, in principle, exhibit 3^2 (= 9) different tone bigrams. However, the set of two-tone sequences that are licit in Vedic is limited to just four of those nine possible combinations. Each of these tone bigrams and a statement on their well-formedness is presented in Table 4.1. I include here reference to one further restriction as well: falling tones are not permitted at the left edge of the prosodic word in isolation.³ Conversely, any tone may occur at the right edge of the prosodic word.

Tone Bigram	Licit?
LL	Yes
LH	Yes
LF	No
HL	No
HH	No
HF	Yes
FL	Yes
FH	No
FF	No
ω [F	No

Table 4.1: Bigram Combinations of Vedic Surface Tones

These tonotactic restrictions effectively mean that the the high tone may only be preceded by a low tone and immediately followed by a falling tone (if any tone follows it). The full set of licit surface tonal sequences can then be captured with the following structural description: $[L_0H(F L_0)]_{\omega}$. This structural description states that a prosodic word must contain

³Within a *pada*, which may be taken as a non-minimal phonological phrase ($\bar{\varphi}$ or $\varphi_{[+max]}$), when a word bears H on its final syllable, the first syllable of following word will be F, if it is not itself H. Strictly speaking then, F is banned only at the left edge of an intonational phrase.

exactly one high tone, which may be preceded by an unbounded sequence of low tones to its left, and optionally by a falling tone and an unbounded sequence of low tones to its right.

To make this description explicit, the tableau in example (1) presents all theoretically possible combinations of a three tone sequence given the Vedic inventory; the combinations that are licensed are given at the top in candidates a.–c. and marked with ✱; losing options are excluded by tonotactic markedness constraints; candidates excluded by violations of OBLIGATORINESS or CULMINATIVITY (i.e., absence of or occurrence of more than one H) are omitted. Hence, a tonotactic constraint *HH (“no sequences of two high tones”) is unnecessary in the constraint inventory, since any candidates containing any such sequences would be independently excluded by CULMINATIVITY (but see the following section on instances that at least superficially violate OBLIGATORINESS or CULMINATIVITY).

(1) Tableau of Licit Prosodic Words Containing Three Tones

/INPUT/	*LF	*HL	*FH	*FF	* _ω [F
a. ✱ H F L					
b. ✱ L H F					
c. ✱ L L H					
d. L F H	*!				
e. H L L		*!			
f. L H L		*!			
g. F H F			*!		*!
h. H F F				*!	
i. F L H					*!
j. F H L		*!	*!		*!
k. F F H			*!	*!	*!

The constraint *HL ensures that a high tone is always followed by a falling tone, if the high tone is not the final tone in the word. *LF ensures that a falling tone never occurs except where a high tone occurs immediately to its left. *FF ensures and *FH together ensure that only a low tone may occur to the right of a falling tone; while *_ω[F bans falling tones from occurring as the first tone in the word. Together, these five tonotactic markedness constraints correctly generate three facts about the distribution of pitch in Vedic Sanskrit:

- the position of the high tone within the prosodic word is totally unrestricted.
- low tones are unrestricted **except** from the position immediately to the right of the high tone.
- the falling tone is restricted precisely to the position immediately to the right of the high tone.

Statements b. and c. together show that the one context in which falling tones are permitted and from which low tones are banned constitute a complementary distribution. On this basis, one may conclude that the falling tone may be treated merely as an allophone of the unmarked low tone, in the position immediately to the right of the high tone.⁴ The major seeming exception to such a treatment comes from words that appear with a *svārīta* directly following an *anudātta*, i.e., with a so-called “independent” *svārīta*; the vast majority of instances of such independent *svārīta* are attributable to a post-lexical process of glide formation (identifiable as post-lexical since it freely applies across prosodic word boundaries) where the *udātta* was associated with a high vowel.⁵ A word with independent *svārīta* contains the same pitch targets as a word with dependent *svārīta*, merely temporally compressed into a shorter duration, i.e., largely onto just one syllable, with the *udātta* rise interpolated between the syllables, perhaps onto the glide, which very frequently precedes an independent *svārīta* (Beguš 2016: 10).⁶

Phonologically, the interpretation of Pāṇinian word prosody is straightforward and attractive: the *udātta* corresponds to culminative and obligatory prominence, and derivation of all pitch targets in the word is predicatable, following automatically from the position of the *udātta*. However, the prosodic notational system employed in the RV and AVŚ (among other Vedic texts) and corresponding descriptions in their *prātisākhya*s create some seeming difficulties for this interpretation: the *svārīta* contains the highest pitch in the word prior to a pitch fall, though the *udātta* is nevertheless higher than the *anudātta*. This description corresponds to the impressionistic description of pitch in RVic recitation (in Maharashtra and Gujarat) reported by Haug (1874), who questions the practice of regarding the *udātta* as the accent proper (i.e., prominence) of the word. If it were the case that the syllable with *svārīta* were to be regarded as the locus of word prominence in Vedic, both the synchronic analysis of Vedic accentuation and direct diachronic comparison with other Indo-European languages would encounter a considerable complication.

Fortunately, the fact that some schools of Vedic recitation, and perhaps originally some Vedic dialects, aligned the word-level pitch peak with the *svārīta* is both readily explicable from the point of view of intonation and almost certainly constitutes a historical innovation (as concluded already by Wackernagel 1896a: 286). First, insofar as peak *F₀* occurred with the *svārīta* in some Vedic schools or dialects, this feature likely reflects the common intonational phenomenon of so-called peak delay: the *F₀* stemming from a pitch accent peaks somewhat after the TBU with which the peak accent is associated (cf. Silverman and Pierre-

⁴Further analysis might suggest the need to treat at least some falling tones as underlying — certain morphemes appear to have the property of placing the high tone one syllable to their left, which could be understood on the grounds that such morphemes are specified as falling, and thus determine where the high tone must occur. Since I will ultimately pursue a stress-oriented rather than tone-oriented analysis of Vedic word prosody, this suggestion will not be further developed.

⁵Furthermore, in the texts of the RV although this glide formation is graphically represented, the meter indicates that the high vowel must still count as a syllable nucleus in most instance. In the oldest Vedic, at least, the phenomenon of independent *svārīta* is thus marginal at best.

⁶As Beguš (2016: 11) points out, the pitch rise in cases of independent *svārīta* also likely begins somewhat earlier, which concurs with the different marking given in the SV to the *anudāttara* syllable preceding an independent *svārīta* (3ra vs. usual 3).

humbert 1990, Arvaniti et al. 1998, D’Imperio 2001, Arvaniti and Fletcher 2020: 88); compare similarly Beguš 2016: 3. Thus, for the Vedic schools and their manuscripts that obtained an Fo peak on the *svarita* syllable, the *udātta* may still be treated as the word-level prominence; peak delay may be regarded as an aspect of the post-lexical intonational phonology or phonetic implementation, and is irrelevant for the analysis of prominence calculation. Such a distinction between underlying and surface pitch in evaluating Vedic prosody is similarly endorsed by Scharf (2012: 411), who argues that “rules in various *prātiśākhya*s and other linguistic treatises... transform underlying accent to surface accent that is accurately represented by graphic symbols in received manuscripts of the *saṃhitās* that these *prātiśākhya*s describe”. Similarly, Witzel (2020: 265) applies instrumental analysis to the recitation of the both the *saṃhitapāṭha* and *padapāṭha* of the RV in Kerala recorded by Staal 1961, and observes that there the Fo peak occurs on the *udātta* in recitation of the *padapāṭha* (which “undoes” post-lexical phonological processes generally), but on the *svarita* in recitation of the *saṃhitapāṭha*.⁷

As to the matter of historical innovation of peak delay, the zone of innovation may be approximately localized to a north central (Witzel suggests western Bihar in specific origin) area of the Indian subcontinent. First, the systems described by Pāṇini and the *Vājasaneyiprātiśākhya*, which are in near perfect agreement with one another, represent geographic extremes: Pāṇini came from the far northwest, while Vājaseneyi school of the Yajurveda was in the far east of the subcontinent (Videha, North Bihar); cf. Witzel 2020: 269. Meanwhile, an increasing number of manuscripts of various Vedic texts (including parts of the RV and AV) from other geographic extremes (Kashmir, Nepal, Kerala, Tamil Naḍu, i.e., in the north and south) exhibit accent-marking practices closer to those of the MS and KS, i.e., marking the *udātta* explicitly (see Witzel 2020: *passim*). Geographically separate regions thus preserved the original Vedic intonational system without peak delay.

For the purposes of phonological analysis of word-level prominence, these facts mean that the acute given in transcription as representation of the *udātta* employed in grammatical and lexicographical works, as well as textual editions, may be taken over without serious worry, and form the input to phonological analysis. Given that this pitch peak is, as stated above, typically obligatory and culminative in a prosodic word, how should such a prosodic system be characterized, and in what ways might it interact with intonation?

4.1.3 Vedic Word-Level Prominence: Pure Intonation, Pitch-Accent, or Tone?

First, it is relatively clear that the Vedic language does not exhibit even a minimally privative contrast in lexical tone, as typically posited for e.g., Stockholm Swedish or Tōkyō Japanese. The *position* of the prosodic prominence is contrastive (see example (6) below), like stress in Russian, English, or Italian, but there is no need to invoke a distinction between lexemes bearing lexically specified pitch (such as H* in Swedish, under the analysis of Riad 1998, Riad 2014) and lexemes that exhibit only a default prominence intonation associated with the

⁷Though the pitch peak on *udātta* in the *padapāṭha* recitation might also be interpreted as a historical archaism in the face of the spread of new recitational practice, which is the interpretation that Witzel (2020: 280) himself applies.

metrically prominent syllable (such as HL* in Swedish). Every “accentable” word receives the same type of intonation, perhaps to be described as H* or H*L: a high tone associated with the metrically prominent syllable followed by a low tone. All other syllables are realized by default with lower pitch (*ekāśruti* ‘monotone’), which does not imply the association of any phonologically specified tone. There does, however, exist a distinction between “accentable” (with H) and “unaccentable” (without H) words, where many (though not all) of the latter class represent function words (pronouns, particles), that might rightly be treated as clitics.

From this point of view, one might well class Vedic among the intonation-only languages and attribute the emphasis placed on word-level pitch in the indigenous grammatical tradition as an effect of the tendency to regard pitch as a major correlate of stress in stress *perception* (cf. 1.3.3.1 in Chapter 1). On the other hand, Vedic texts exhibit little evidence for the role of different intonational tunes: the same lexical item consistently exhibits the *udātta* in the same position; it is not the case that a lexeme occasionally appears without a *udātta* because its stressed syllable is aligned with an L as part of a particular tune. A partisan for Vedic prosody as representing a language with stress, not “pitch-accent” or tone, might argue that the *udātta* simply does not really reflect pitch, or perhaps corresponds to a basic intonation applied to words produced in isolation. Against these hypotheses speak, in the first case, the indigenous testimony concerning word prosody centered around the pitch excursions from *anudātta* (low) to *udātta* (high) to *svarita* (fall back to low), and in the second case, that Vedic texts so assiduously reflect so much segmental sandhi, that one might have expected for (clear) differences of intonation to have been transmitted as well.⁸

In languages of the Vedic type, in which prosodically prominent syllables seem to be distinguished from non-prominent syllables principally through Fo excursions, and the conditions of CULMINATIVITY, OBLIGATORINESS, and PRIVATIVITY seem to be (almost) exceptionlessly satisfied, one confronts the question of how pitch should be phonologically represented, if at all. Under a primarily “accentual” view of prosodic prominence as represented by, e.g., van der Hulst (1999, 2011, 2012), some sort of Fo excursion associated with “accent” in a language like Vedic or even Tōkyō Japanese, constitutes a mere phonetic implementation, in the same fashion that “accent” in English or Dutch may be implemented by increases in duration and amplitude. The opposite representational extreme is to treat many lexical entries as bearing tone, /H/ in the Vedic case; H tone is, however, culminative and obligatory, potentially resulting in deletion of some underlying /H/ or insertion of non-underlying /H/, after selecting one H-bearing syllable as the prosodic head of the word, and banning H from all non-heads.⁹

Treating Vedic word prosody in this latter fashion would make it akin to the “tone-driven stress” reported for Golin (Bunn and Bunn 1970) or Tibetan (Meredith 1990); cf. de Lacy 2002:

⁸Logically, however, the representation of differences in the realization of segments does not necessarily entail that intonation would be similarly carefully transmitted.

⁹Such an interpretation of Vedic is suggested in passing by Yip (2001: 531–2) and de Lacy (2002: 2) on the basis of Kiparsky and Halle 1977. Kiparsky & Halle are not, in fact, quite so explicit, but the formulation of their rules of METATONY and DEACCENTUATION (Kiparsky and Halle 1977: 209–10) imputed to Vedic Sanskrit lends itself to the interpretation that Vedic lexical entries should be understood as having underlying tone, and Kiparsky (2014) has suggested as much more directly (cf. further below).

2.¹⁰ Tone-driven stress systems assign stress to a particular syllable based (partly) on conditions arising from lexical tone (see the overview under 2.4.1.2 in Chapter 2). The third major representational option for Vedic has been suggested in the preceding paragraph: prominence (i.e., stress) is calculated independently of any tone or intonation, but a default /H*/ (or perhaps /H*L/) prominence tone (to be understood in much the same sense as the prominence tone proposed for varieties of Swedish and Norwegian by Riad (1998), Riad 2014), inserted at the level of each prosodic word and almost always associated to the stressed syllable.¹¹

4.1.3.1 An Intonational Account of “Unaccented” Words

Evidence for the latter interpretation may exist in the form of at least three well-defined and well-attested instances in Vedic where the effects of intonation potentially shine through (see Macdonell 1916 [1993]: 465–9 for extensive exemplification of these phenomena). As will be argued below, these cases can largely be understood as instance in which another intonational tone may override the default prominence tone /H*/ and prevent the latter from being realized. These phenomena may be described as follows:

- absence of any high tone on finite verbs in main clauses; but...
- regular appearance of the high tone on the expected syllable in phrase-initial (*pāda*-initial) position;¹²

¹⁰The distribution of tone in both Golin and Tibetan, is, however, very different from the kind of distribution seen in Vedic. In Golin, for instance, contrastive /H/ and /L/ tones are evidently lexical, and neither tone is either culminative or obligatory; stress is, however, attracted to the rightmost H in a word, else the final L (Bunn and Bunn 1970: 4–5). Likewise, Meredith (1990: Ch. 2) analyzes Tibetan as possessing an inventory of four contrastive lexical tones: low rising, low falling, high level, and high falling (Meredith 1990: 54–7). The interactions between tone and metrical structure further posited by Meredith are fairly intricate, amounting, in a first round of structure construction, to insertion of /H/ on foot heads, deletion of rising or falling contours on foot tails, and total deletion of tone on unparsed syllables (Meredith 1990: 65). Stress is then (in nouns) preferentially assigned to the leftmost heavy, high-toned syllable, else the leftmost heavy syllable, else the leftmost high-toned syllable, else the leftmost syllable (Meredith 1990: 89–95). The important point for present purposes is that the “tone-driven stress” of Golin or Tibetan bears little direct resemblance to Vedic prosody.

¹¹The primary distinction between word prosody in Vedic and most Scandinavian languages would then consist in the *absence* of any lexical tone in Vedic that might result in surface contrasts. Riad (1998: 93–4) reports (referring to Gårding 1977 and Bruce and Gårding 1978) that the Swedish dialect of Helsinki lacks lexical tone (“accent 2”) and instead only shows a /H*/ prominence tone in prosodic words. If such a phonological characterization of the prosody of Helsinki Swedish is correct, then it and Vedic may well reflect phonologically similar word-prosodic systems, characterized by lexical stress and a consistent word-level pitch accent associated to stress. Hellmuth (2007) also describes the word prosody of Egyptian Arabic in a very similar fashion: the domain of intonational pitch accent distribution in Egyptian Arabic is the ω , meaning that every ω in the language receives a pitch accent (analyzed by Hellmuth as LH*) associated with its (metrically determined) stressed syllable (cf. Hayes 1995: 67–71 on the calculation of stress in Egyptian Arabic). Distinguishing both most varieties of Swedish and Norwegian and Egyptian Arabic from most West Germanic languages is then the apparent obligatoriness of pitch accents at the level of the prosodic word, as opposed to the phonological phrase. Vedic then patterns with Swedish/Norwegian/Egyptian Arabic in terms of its distribution of pitch accents.

¹²I tentatively hold that *pāda*-initial position constitutes the left boundary of a recursive or maximal

- absence of any high tone on vocatives generally;¹³ but...
- regular appearance of high tone on the first syllable of vocatives at the beginning of a *pāda*.¹⁴

The lattermost case admits of a relatively clear intonational interpretation: phrases are marked with an initial boundary tone, %H, which regularly associates to the first syllable of a word in phrase-initial position (so Gunkel 2017; cf. similarly Klein 1992: 2; 86–7), and may constitute the *udātta* of that word if no H* is present. The fact that finite verbs in phrase-initial position can show an *udātta* on the same syllable on which an *udātta* occurs when the verb is in a subordinate clause (e.g., *kṛṇóti* ‘makes’ in phrase-initial position and in a subordinate clause) in turn suggests that the *udātta* or absence thereof, on verbs at least, is of an intonational nature more generally: phrase-initial finite verbs might be assumed to associate the %H tone to their metrically prominent syllable, finite verbs in subordinate clauses (can) bear default prominence tone (probably H*), while finite verbs in main clauses otherwise receive a different, syntactically motivated intonation (probably L*, perhaps overphonologized from phrase-final L%; cf. generally Padgett 2015 on phenomena of overphonologization). However, the fact that phrase-initial vocatives precisely do not surface with an *udātta* in the position expected from other case forms, but rather always on the initial syllable (e.g., *pítar* ‘father:VOC.SG, not ^x*pítár*; cf. ACC.SG *pítáram*) must mean that either metrically or intonationally distinct from all other morphological words. At least four hypotheses are conceivable:

- vocatives build “stressless” (maximal) prosodic words, in violation of OBLIGATORINESS (~ HEADEDNESS).

phonological phrase, $\tilde{\varphi}$ or $\varphi_{[+max]}$, given evidence for smaller phonological phrases larger than one word but smaller than the *pāda* (cf. Jamison 2010). In this case, the hemistich could be considered equivalent to the minimal intonational phrase, $\iota_{[+min]}$. That true *visarga*-formation (creation of [h], *visarjanīya*, as opposed to assimilation in place of [s] or [r] to a following voiceless obstruent) does not occur at the end of every *pāda*, but rather at the end of a hemistich or stanza indicates that this boundary corresponds to a prosodic domain larger than the *pāda*. The proposed tentative correspondences between morphological or metrical units with prosodic structures in Vedic would then be as follows:

Morphological / Metrical Unit	Prosodic Structure
Morphological Word	$\omega_{[+min]}$
Morphological Word + Clitic	$\tilde{\omega}$ / $\omega_{[+max]}$
Compound	$\tilde{\omega}$ / $\omega_{[+max]}$
2 Prosodic Words	$\varphi_{[+min]}$
<i>pāda</i>	$\tilde{\varphi}$ / $\varphi_{[+max]}$
Hemistich	$\iota_{[+min]}$
Stanza	$\tilde{\iota}$ / $\iota_{[+max]}$

¹³This suppression of high tone holds throughout entire vocative phrases, e.g., $\acute{a}$ [*rājānā mahāṛtasya gopā | śindhupatiḥ kṣatriyā*]_{VOC} *yātam arvāk* ‘O you two kings, herdsmen of great truth, lords of the rivers and rulers – travel here to this place’ (RV 7.64.2ab).

¹⁴Note in the example immediately above that *śindhupati*, although occurring within a sequence of vocatives, exhibits the phrase-initial (*pāda*-initial) high tone.

- vocatives are effectively clitics, building prosodic words with a neighboring content words.
- vocatives undergo “barytonesis”: stress is assigned to the initial syllable (cf. Fortson 2010: 116, 128, and *passim*, attributed by Gunkel (2017) to Benfey (1845: 907)).
- an L* vocative intonation is generally compatible with simultaneous realization of a boundary %H.

Under the second hypothesis, non-initial vocatives might not provide any further information on intonation, since low tone throughout the remainder of the prosodic word after an *udātta* and *SVARITA* would be expected. This hypothesis is, however, unlikely, insofar as absence of high tone applies throughout vocative phrases (cf. fn. 13). Under the third hypothesis, which Gunkel (2017) explicitly argues against, vocatives would undergo both a change in metrical structure (stress assigned to the initial syllable) and perhaps receive a special intonation (L*) when not *pāda*-initial, otherwise the initial-syllable stress occurs; but any change in metrical structure is superfluous for explaining a difference in pitch if a %H boundary tone is admitted. Under the first hypothesis, vocatives need not receive any “special” intonation, but merely fail to realize the default word-level intonation, because they lack a stressed syllable to which a starred tone may associate; however, vocatives would then (perhaps) constitute virtually the only stressless prosodic words in Vedic.¹⁵

The fourth hypothesis constitutes a purely intonational account as well: vocatives are to be realized with a L* tune, which gives the appearance of *udātta* suppression, but phrase-initial vocatives are realized with both %H and vocative L*.¹⁶ Thus, a phrase-initial vocative with non-initial stress would have the intonational contour %H L*, while a vocative with lexical stress on the first syllable would preferentially realize %H (or conceivably crowd both %H and L* onto the first syllable). Were it indeed the case, though, that vocative L* could be obligatorily associated to a metrically prominent syllable also hosting %H (though one might have expected evidence for a *pāda*-initial vocative *svarita*). This latter interpretation would be consistent with the general principle that metrically prominent syllables attract intonation, but pitch accents (starred tones) generally take precedence over boundary tones (cf. Arvaniti and Fletcher 2020: 84–6).

Let us adopt the tentative assumption that the phonologically stress-like aspects of Vedic prosody indicate specification of a metrically prominent syllable to which a default prominence intonation, H*, is attracted in the normal case. Assuming further that the phonological realization of VOCATIVE in Vedic is L*, the intonational behavior of vocatives and $\tilde{\varphi}$ -initial

¹⁵Gunkel (2017), following Banti 1984: 68, suggests that some instances of the pronoun/demonstrative *asya* ‘of this one here’ may also be unaccented. Compare further Macdonell (1910: 81): “all oblique cases formed from the demonstrative pronoun *a-*, when used merely to replace a preceding substantive” are without *udātta*.

¹⁶Insofar as Vedic vocatives are assumed to be completely “flat” in intonation, i.e., realized with L monotone (*ékaśruti*) throughout, as their transmission suggests, such a description appears somewhat unusual in comparison to vocative intonations in many present-day European languages, which typically involve some more complex intonational patterns, often involving some type of rise or fall in pitch in the course of production. See Pešková 2019 for a comparison of vocative intonation in Czech, Spanish, and Italian, with reference to vocative intonation in other European languages.

boundary %H may be modeled in an OT grammar fragment employing constraints on the association of intonational elements proposed in Gussenhoven 2004: Ch. 8 as follows:

- Multiple H within the domain of a prosodic word are banned (OCP(H, ω)). Consequence: realization of both %H and prominence H* is not permitted.
- A stressed syllable should be associated with some tone ($\sigma \Leftarrow T$).
- Realization of vocative L* is preferred to the realization of prominence H* (MAX-L* $\gg$ MAX-H*)
- Realization of boundary %H at the beginning of a non-minimal phonological phrase is preferred to the realization of either vocative L* or prominence H* (MAX-H- $\varphi_{[-min]}$ $\gg$ MAX-L* $\gg$ MAX-H*)
- %H- $\varphi_{[-min]}$ should be realized as close to the left edge of a non-minimal phonological phrase as possible (ALIGN-Left-H- $\varphi_{[-min]}$)
- Multiple tones may not be associated with a single tone-bearing unit (NOCROWD), or alternatively, no contour tones are permitted (NOCONTOUR).
- NOASSOCIATION, which penalizes the association of phonological tones with TBUs, outranks MAX-H*, thus resulting in the realization of the H* prominence tone only when it is the only tone in the input.

With respect to vocatives, there are then three relevant cases: 1) vocatives outside of $\varphi_{[-min]}$ -initial position; 2) vocatives with non-initial stress in $\varphi_{[-min]}$ -initial position; 3) vocatives with initial stress in $\varphi_{[-min]}$ -initial position. In the first case, vocative L* is simply associated with the stressed syllable in preference to default H*, while H* is deleted; in the second case, the boundary tone %H is associated with the first syllable of the vocative (= first syllable of the $\varphi_{[-min]}$), and L* is associated to the stressed syllable; in the third case, the boundary tone is associated to the leftmost syllable (= the stressed syllable), in preference to L*.¹⁷ The only peculiarity of this grammar is that it would appear to privilege the realization of a boundary tone, %H- $\varphi_{[-min]}$, above the realization of starred tones under some conditions.

The tableaux in (2)–(4) illustrate the derivation of these patterns. I contrast there a stem with non-initial lexical stress, *devá*- ‘god’, with a stem having initial lexical stress, *bhrátar*- ‘brother’. I further assume that OBLIGATORINESS is inviolable and therefore do not include candidates without stress; the underlying stress of the stem is marked with an IPA primary stress diacritic [ˈ].

¹⁷ Alternatively, in the third case, the leftmost (= stressed) syllable might host both the boundary H- $\varphi_{[-min]}$ and H*, if NOCONTOUR rather than NOCROWD is decisive, but the pitch is the same either way.

- (2) Phonological Derivation of Vocative Intonation in Vedic 1: Non-initial stress, not in $\varphi_{[+max]}$ -initial position

/de'va, H*, L*/		$\sigma \Leftarrow T$	MAX-H _i	MAX-L*-VOC	NOASSOC	ALIGN-L-H- $\varphi_{[+max]}$	MAX-H*
a.				*			*
b.				*!	*		
c.		*!		*			
d.					**!		

The outcome in (2) is determined by the preference for realizing vocative L* above prominence H*. Wherever L* is not associated, MAX-L*-VOC excludes the candidate (candidates b. and c.). Candidate d., meanwhile, is ruled out by the ranking of NOASSOCIATION above MAX-H*: as long as the association of the stressed syllable to a tone can be satisfied, the realization of further tones is superfluous. Candidate a., realizing vocation L* on the stressed syllable thus emerges as the winner.

When the vocative stands at the beginning of a $\varphi_{[+max]}$, the %H boundary tone is also present in the derivation. As long as both MAX-H- $\varphi_{[+max]}$ and MAX-L*-VOC outrank NOASSOCIATION and MAX-H*, both H- $\varphi_{[+max]}$ and L* will be realized, and in preference to prominence H*. These considerations alone exclude candidates b.-d. in (3), where candidate d. further fails to associate a tone to the stressed syllable. The choice between candidates a. and e. is then regulated by ALIGN-L-H- $\varphi_{[+max]}$, which prefers for the boundary %H tone to stand as close to the left edge of the maximal phonological phrase as possible.

- (3) Phonological Derivation of Vocative Intonation in Vedic 2: Non-initial stress, $\varphi_{[+max]}$ -initial position

/de'va, H*, L*, H- $\varphi_{[+max]}$ /		$\sigma \Leftarrow T$	MAX-H _i	MAX-L*-VOC	NOASSOC	ALIGN-L-H- $\varphi_{[+max]}$	MAX-H*
a.					**		*
b.		*!		**			
c.			*!	*	*	*	*
d.		*!	*	*			*
g.				**	*!		*

The derivation of vocative forms with initial stress at the beginning of an $\varphi_{[+max]}$ is then essentially the same as in (3), the only difference being that L* in this case associates to an unstressed syllable instead, since ALIGN-Left-H- $\varphi_{[+max]}$ ensures that the leftmost syllable, which incidentally bears stress, is associated with the %H boundary tone. Assuming that crowding of tones is excluded, ALIGN-Left-H- $\varphi_{[+max]}$ prefers candidate a. to candidate b. in (4), while the violation of MAX-L*-VOC in candidate c. rules it out.

In vocatives containing more than two syllables, the association of $H-\varphi_{[+max]}$ will always be fixed to the leftmost syllable. Where stress is non-initial, L^* is predicated to associate to the stressed syllable, wherever it happens to fall. Where, exactly, L^* associates when the initial stressed syllable happens to be occupied by the $\%H-\varphi_{[+max]}$ boundary tone is less clear; depending upon edge-alignment preferences of L^* , the obvious candidates would be the second syllable (left alignment) or final syllable (right alignment).

(4) Phonological Derivation of Vocative Intonation in Vedic 3: initial stress, $\varphi_{[+max]}$ -initial position

ition

/'bhrātar, H*, L*, H-φ _[+max] /		σ ⇐ T	MAX-H _i	MAX-L*-VOC	NOASSOC	ALIGN-L-H-φ _[+max]	MAX-H*
a.	H-φ_[+max] 'σ bh r ā L* σ t a r				**		*
b.	L* 'σ bh r ā H-φ_[+max] σ t a r				**	*!	*
c.	H-φ_[+max] 'σ bh r ā σ t a r			*!	*		*

Can this analysis be extended to the intonational behavior of finite verbs? The crucial difference in comparison to vocatives is that *udātta* consistently falls on their stressed syllable, rather than the leftmost syllable, when occurring in *pāda*-initial ($\varphi_{[+max]}$ -initial) position. Since morphosyntactically identical forms in different syntactic configurations (i.e., in subordinate clauses) show *udātta*, to presume the realization of a tone competing with the prominence tone is not as straightforward as in the case the vocative, where a special intonation may be assumed (cf. Ladd 1978 or Gussenhoven 2004: 313–5 on the English vocative intonation and Kubozono and Mizoguchi 2019 on vocative intonation in Tōkyō Japanese).¹⁸

¹⁸The results of Kubozono's acoustic analysis show that Tōkyō Japanese has three distinct, pragmatically conditioned, vocative intonations, though in all cases, the distinction in F_0 between “accented” and “unaccented” words is maintained.

If a L* intonation were assumed to be present in the case of finite verbs in main clauses, the difference in position of a H in $\varphi_{[+max]}$ -initial position can be motivated only by assuming that a distinct MAX constraint holds over this particular L* (MAX-L*-FINITE).

Thus, if L*-VOCATIVE is distinguished from L*-FINITE, with different MAX constraints holding over the two, and the realization of L*-VOCATIVE is assumed to be obligatory except where crowding would result (i.e., on monosyllables), then $\varphi_{[+max]}$ -initial vocatives may be associated with two tones, as long as both MAX-H_i and MAX-L*-VOCATIVE outrank NOASSOCIATION. ALIGN-L- $\varphi_{[+max]}$ will then prefer for the boundary to occur closer to the left edge. Granting that L*-FINITE and prominence H* are both dominated by NOASSOCIATION, their realization would be licensed only in order to provide a stressed syllable with a tone. The tableau under (5), with the verb *kṛnóti* ‘make:NPST.3SG’, illustrates this analysis: the boundary tone H- $\varphi_{[+max]}$ is associated to the stressed syllable to satisfy ' $\sigma \Leftarrow T$ ', since realization of H- $\varphi_{[+max]}$ is required regardless, while the realization of the other tones in the input is not obligatory.

(5) Intonation of Finite Verbs: non-initial stress, $\varphi_{[+max]}$ -initial position

/kr'noti, H*, L*-FIN, H- $\varphi_{[+max]}$ /		' $\sigma \Leftarrow T$	MAX-H _i	MAX-L*-VOC	NOASSOC	ALIGN-L-H- $\varphi_{[+max]}$	MAX-L*-FIN	MAX-H*
a.								
				*	*	*	*	*
				**!		*		
				**!			*	
d.		*!						
				*		*	*	*

The alternative to the assumption of an articulated ranking of MAX constraints holding over

specific tones would be to treat finite verbs as *deaccented*, with the consequence that prominence tone H* is not provided in the input. Compare Riad 2014: 262–4 on deaccented material, including most finite verbs, in Swedish. The fundamental difference between the assumption of a “special finite verb intonation” outlined above and deaccentuation in finite verbs is that, in the latter case, DEP-Tone must be assumed to outrank $\sigma \Leftarrow T$: no non-underlying tones are inserted in order to provide a tone for the stressed syllable.¹⁹

4.1.3.2 Violations of CULMINATIVITY?

Regardless of exactly how the absence of *udātta* among vocatives and non-finite verbs is best explained, the important point for present purposes is that analyses are available that do not necessarily require the *destressing* of such words: they may be assumed to always possess a metrically prominent syllable, if not a pitch accent associated with that syllable. In contrast to the cases of prosodic words that appear to lack *udātta* are sequences of linearly adjacent and syntactically closely associated elements with two *udātta*, traditionally described as having a double accent, i.e., two high tones, seemingly in violation of CULMINATIVITY. Such double-accented forms are of three types (cf. Macdonell 1910: 81):

- a. infinitives built with the suffix *-tavái* (e.g., *ápabhartavái* ‘to take away’);
- b. coordinative compounds in the oldest Vedic, with two members inflected in the dual (e.g., *mitrá-váruṇā* ‘Mitra and Varuṇa’);
- c. so-called syntactical compounds, in which the first member is inflected for case and modifies the second member (e.g., *brhás-páti-* ‘lord of prayer’)

The two compound types (b. and c.) may unproblematically be regarded as containing two separate prosodic words, each of which naturally bears stress and is liable to receive prominence H*; see further Section 4.2.5. An similarly obvious explanation for the infinitives in *-tavái* is wanting, but to reckon with a recently grammaticalized affix that still behaves as an independent prosodic word is not implausible.²⁰

What gives Vedic the character, then, of a “pitch-accent” language is the existence of a default word-level prominence tone, H*, and the relatively sparsity of (discernible) alternative intonational tunes that might override, displace, or lower this H* such that a prosodic word would be transmitted without an indication of raised pitch associated with some particular syllable. As discussed above under footnote 11, the relatively dense occurrence of intonational pitch accents, consistently appearing at the ω -level, is nicely paralleled by Egyptian Arabic, which however, exhibits fully predictably metrical stress; the position of stress in

¹⁹As Riad (2014: 263) notes, West Germanic languages typically exhibit deaccentuation of any material already given in the discourse. Swedish and prosodically similar North Germanic varieties maintain more intonational contrasts even when a constituent is already given in the discourse.

²⁰Perhaps also worth noting is that such infinitives are very often followed by the enclitic emphatic particle *u* (18/23 instances in the RV). Whether *u* might have the effect of (usually) conditioning an accent on a preceding syllable would require further closer investigation.

Vedic (and association site for H*), conversely, is not predictable on the basis of surface phonological properties alone. This latter feature will be the focus of investigation in the remainder of this chapter.

4.1.4 Vedic Stress is Metrically Unpredictable

Before delving deeper into a treatment of stress assignment in Vedic, I begin with a short demonstration of the evident insufficiency of any analysis of word-level prosodic prominence in this language exclusively in terms of metrical stress. Furthermore, metrical constraints seemingly play but a marginal role in determining the position of stress in any given word form. As observed already in Chapter 1 at example (1), Vedic exhibits both some genuine minimal stress pairs (shown in (6); see Poucha 1943: 131–5 for a collection of 88 genuine minimal pairs distinguished solely by accent position found in the RV and AV) and numerous word forms of identical prosodic shape in terms of syllable count and pattern of syllable weight (shown in (7)); note that all closed syllables in Vedic count as heavy, and that consonant clusters are generally syllabified to avoid complex syllable margins and obey the sonority-sequencing principle as much as possible (see Cooper 2015: Ch. 2–3).

(6) Minimal Stress Pairs in Vedic

- a. *bráhmanasya* [brá:ɦmənəsɟə] ‘pious:M.GEN.SG’ vs.
bráɦmanasya [bra:ɦmənəsɟə] ‘Brahman:M.GEN.SG’
- b. *ápasā* [əpəsa:] ‘work:N.INST.SG’ vs. *apásā* [əpəsa:] ‘active, industrious:INST.SG’
- c. *jaraṇá* [jərəná:] ‘old age:F.NOM/ACC.SG’ vs. *jaránā* [jərəná:] ‘dry wood:F.NOM/ACC.SG’

(7) Prosodic Contrast Pairs in Vedic

- a. Two-syllable: *úpa* [úpə] ‘to(wards), by’ vs. *utá* [utá] ‘and’
- b. Four-syllable: *puróhitam* [puró:ɦitəm] ‘domestic priest:M.ACC.SG’
vs. *purútāmam* [puru:táməm] ‘much:CMPV.M.ACC.SG’
- c. Six-syllable: *abhimātiṣāhás* [əb^hima:tiṣa:həs] ‘conquering enemies:NOM.PL’ vs.
ánipadyamānam [ənipədyəma:nəm] ‘not falling down:ACC.SG’

The minimal stress pairs in (6) make evident that lexical contrasts in prosodic specification must exist at some level in Vedic; no prosodic or segmental differences can be invoked in such cases, and, unlike the well-known minimal stress pairs in English, or stress assignment in languages such as Lenakel (Austronesian; Lynch 1974, Lynch 1978), does not depend on syntactic category. The further prosodic contrast pairs show that the extent of lexical specification and morphological control of stress must be wide-ranging in Vedic: stress is not limited to a window at either edge of the word, no consistent rules of foot parsing can generate the primary stress, and syllable weight plays no decisive role in determining the position of the stress. The position of stress is “unbounded” in much the same fashion as it can be in weight- or tone-sensitive systems, but the property to which stress is attracted should be defined either as an abstract prominence (e.g., a grid mark) or a prosodic constituent (a foot edge).

The questions in terms of phonological processes and representations to be answered with respect to Vedic are then the following:

- What patterns of intra- and interparadigmatic stress regularly occur in Vedic?
- What type of prosodic representations are compatible with or necessary for determining stress assignment (and possibly for driving other phonological processes)?
- What default rules of stress assignment apply in the absence of prespecified prominence? Can any stress rules override prespecified prominence?

4.1.5 Morphological Categories and Morphological Structure in Vedic

Given that the morphology of a word is frequently decisive for determining the position of the stress in Vedic, a general picture of morphological structures will facilitate comprehension of the analyses presented in Sections 4.2 and 4.3 below.

The first central division is between inflectable and uninflectable words: nouns (including pronouns), adjectives (including numerals), and verbs are all inflectable, while remaining word categories (adverbs, adpositions, conjunctions, and other particles) are uninflectable. Adverbs may be morphologically complex; words belonging to the other indeclinable categories are typically morphologically simplex and generally not longer than two syllables.

On the noun (and adjectives, which are noun-like), the inflectional categories of case and number, when phonologically realized, appear as a portmanteau suffix; nouns and adjectives thus never contain more than a single inflectional affix, which is always a suffix. The stem of a noun may be morphologically simplex (root nouns), or contain one or more derivational suffixes. Derivational suffixes used for building nouns and adjectives are traditionally divided into “primary” affixes, which may only attach directly to a root, and “secondary” affixes, which may attach to either roots or to morphologically complex stems. The inflectional suffixes found in the noun and adjective are also traditionally divided into “strong” and “weak” endings: affixes realizing NOM.SG, ACC.SG, NOM/ACC.DU, and NOM.PL are “strong”, while suffixes realizing all other case and number combinations are “weak”. This distinction is relevant to the form that the stem in consonant-final (athematic) stems assume.

Verbs similarly realize the inflectional categories of person, number, and voice as a portmanteau suffix; separate suffixes may occur for the realization of modal features; a prefix *á-* marks past tense, and, more rarely a suffix *-syá-* occurs to mark future tense.²¹ The formation of different aspectual stems of the verb, traditionally referred to as “present”, “aorist”, and “perfect” belong to the domain of derivation (see Dahl 2010 or Dahl 2011, and most recently, Hollenbaugh 2021, on the semantics of tense and aspect in Vedic). The basic template for nominal forms in Vedic is thus simply STEM-INFLECTION, while the verb exhibits a potentially

²¹ The future in *-syá-* arguably has not, in Vedic Sanskrit, yet grammaticalized into an inflectional affix, but instead still is to be classed among derivational suffixes.

more complex structure, though the majority of verbal forms in early Vedic likely exhibit but a single inflectional suffix: (PAST)-STEM-(FUTURE)-(MOOD)-PERSON/NUMBER/VOICE.

Stem-formation in Vedic is both fairly complex and diverse. The central and only obligatory component of an inflectable stem is the root, and Vedic possesses both nouns and verbs with stems equivalent to roots (root formations); adjectives are largely derived from nominal or verbal roots or stems. Vedic possesses a large array of noun- and adjective-forming suffixes, which may be divided into “primary” suffixes, which may only derive a stem directly from a root, and “secondary” suffixes, which derive stems from existing stems (including morphologically simplex stems). The range of stem-formation processes for verbs is somewhat more restricted and may be briefly sketched here; the tense–aspect stems are primary, built directly to roots, while other verbal stems are generally secondary.

1. Perfect Stem: Reduplication; reduplicant of the shape (C)a-

2. Aorist Stem:

- Stem = Root
- Suffixes: -a-, -s-, -iṣ-, -ṣiṣ-, and -sa-

3. Present Stem:

- Stem = Root (II)
- Suffixes: -a- (I), -á- (VI), -ya- (IV), -no-/-nu- (V)
- Prefixing reduplication (III) with reduplicant of the shape (C)V-
- Infix: -n(á)- (VII, IX)

4. Valency Changing:

- Causative Present: suffix -áya-
- Causative Aorist: prefixing reduplication with reduplicant of the shape (C)V(:)-
- Passive: suffix -yá-

5. Other Secondary:

- Denominative: suffix -yá-
- Intensive: reduplication with reduplicant equivalent to a bimoraic foot (cf. (46) in Chapter 1)
- Desiderative: prefixing reduplicant of the shape (C)i-, plus suffix -s-

Compounding in Vedic is very productive, and a wide variety of types of compounds are attested (see 4.2.5 below). Besides proper compounds consisting of two stems, Vedic also exhibits synthetic compounds, which are stems derived via attachment of a derivational suffix from a stem built by compounding. That is, Vedic knows both proper stem compounds of the sort [[X]-[Y]] as well as synthetic compounds of the sort [[[X]-[Y]]-Z], where Z is a derivational suffix taking morphological scope over the compound built by X and Y. This distinction proves crucial in accounting for the stress found in many compounds.

4.2 Major Phenomena and the Kiparskian Analyses

Now equipped with basic notions of prosodic and morphological structure in Vedic, I proceed to first illustrate the patterns and mechanisms of stress assignment in Vedic on the basis of the set of prosodic rules and representations developed in Kiparsky 2010a, Kiparsky 2014, and Kiparsky 2018a. The first and most basic distinction necessary is the difference between underlying ACCENT and surface ICTUS, as termed by Kiparsky. A phonological string may contain multiple underlying accents, stemming either from phonological properties of lexical entries and or generated by rule, and these accents may be manipulated by rules; a prosodic word exhibits only one surface ictus, which corresponds to exactly one of the accents or the only accent in a string, or is assigned by rule at the end of a derivation in which no accents are present or have otherwise been created. Since, as discussed in 4.1.3, the Vedic word-level high tone (= Kiparsky's ICTUS) essentially satisfies a phonological definition of stress (cf. (1.3.3.2) in Chapter 1), I will employ the term STRESS instead of ICTUS. Kiparsky is not always explicit about the representation of ACCENT, though Kiparsky (2014: 4) proposes that, for Vedic at least, "accent is lexically represented as High tone ($\acute{e}$ = $/^H e/$)"; for practical purposes, as far as I can determine, whether accent in Vedic is represented as a diacritic mark (à la Revithiadou 1999 or Alderete 2001a) or lexical tone does not materially impact the workings of Kiparsky's accentual rules (at least insofar as both would constitute autosegments).

4.2.1 Accentedness, the BASIC ACCENTUATION PRINCIPLE, and ZERO-GRADE Vowel Deletion

A principal lexical distinction assumed is that between "accented" and "unaccented" morphemes; the former are effectively stress-preferring, while the latter are stress-neutral. When combined with (a) rule(s) that determine how strings containing multiple accented morphemes or strings without any accented morphemes are to be treated, the position of stress can be predicted. Just such a rule is the BASIC ACCENTUATION PRINCIPLE (BAP), stated in the version found in Kiparsky 2014: 4 in (8), where "domain" should be taken to mean the maximal prosodic word, ω -Max. For the sake of explicitness, I translate Kiparsky's prose statement of the BAP into two rules in (9), named ACCENT SELECTION and DEFAULT ACCENT; there, $\acute{\sigma}$ represents an accented or high-toned syllable, σ any unaccented or toneless syllable. The ordering of ACCENT SELECTION and DEFAULT ACCENT is not crucial, given the formulation

of the rules, but I conventionally assume the former to apply before the latter.²²

(8) BASIC ACCENTUATION PRINCIPLE (BAP)

- The leftmost High toned syllable of a domain retains High tone, the others get low tone.
- If there is no High toned syllable, put High tone on the leftmost syllable.

(9) Formalization of the BAP

a. ACCENT SELECTION

$$\dot{\sigma} \rightarrow \sigma / \dot{\sigma} \sigma_0 \text{ --- }]_{\omega\text{-Max}} \text{ (iterative)}$$

An accented/high-toned syllable loses its accent/high tone if any syllable to its left within the maximal prosodic word bears an accent/high tone. This rule applies iteratively until no targets remain.

b. DEFAULT ACCENT

$$\sigma \rightarrow \sigma' / \omega\text{-Max}[\text{---}\sigma_0]$$

The syllable at the left edge of a maximal prosodic word in a string of unaccented/toneless syllables receives a high tone.

4.2.1.1 Accentual Mobility

The rules in (8) and (9), plus the lexical distinction between accented and unaccented phonological entries, can generate the distinction between MOBILE and IMMOBILE inflectional paradigms among monosyllabic stems. Indeed, monosyllabic inflectional stems (more precisely, stems that are monomorphemic and monosyllabic in their weak cases; see further discussion below) are the only type of stem in Vedic that exhibits genuine intraparadigmatic mobility of the stress between the stem and an inflectional ending (as opposed to paradigms with “pseudo-mobility” or “secondary mobility” due to vowel $\sim \emptyset$ alternations; cf. further below). Thus, in the mobile root nouns, stress appears on the first syllable (root syllable) in strong case forms, under the assumption that the relevant inflectional suffixes are unaccented, but on the second syllable (the inflectional suffix), assuming that those inflectional suffixes are accented. The lexical entries for inflectional suffixes of athematic nouns are given in (10), and for non-past person/number suffixes of the verb in (11). Derivations in (12) and (13) illustrate the application of the rules of the BAP; alternations in vowels of the root may be ignored at present. As illustrated there, DEFAULT ACCENT applies only in the leftmost column where ACCENT SELECTION does not apply, because no underlying accents enter the derivation. Three of the four configurations result in stress on the initial syllable, whereas only one (unaccented stem plus accented suffix) results in stress on the final syllable.²³

²²The formalization of DEFAULT ACCENT cannot be simplified to simply target the first syllable of the word without also examining the properties of all other syllables. If such a rule applied after ACCENT SELECTION, all initial syllables would receive an accent, and violations of CULMINATIVITY would be rampant; if such a rule applied before ACCENT SELECTION, it would feed ACCENT SELECTION and result in regular stress on the leftmost syllable.

²³As will become clearer throughout this section, the role for DEFAULT ACCENT is in fact relatively minor, actively applying virtually only in the relevant cases illustrated in 12 and 13. In fact, the analysis adopted by

(10) Lexical Entries of Vedic Athematic Nominal Inflectional Suffixes²⁴

UNACCENTED		ACCENTED	
NOM.SG	/-s/	INS.SG	/-á/
ACC.SG	/-am/	DAT.SG	/-é/
NOM/ACC.DU	/-āu/	ABL/GEN.SG	/-ás/
NOM.PL	/-as/	LOC.SG	/-í/
		INS/DAT/ABL.DU	/-bhyám/
		GEN/LOC.DU	/-ós/
		ACC.PL	/-ás/
		INST.PL	/-bhís/
		DAT/ABL.PL	/bhyás/
		GEN.PL	/-ám/
		LOC.PL	/-sú/

(11) Lexical Entries of Verbal Non-Past Person/Number Suffixes²⁵

UNACCENTED		ACCENTED	
1SG	/-mi/	1DU	/-vás/
2SG	/-si/	2DU	/-thás/
3SG	/-ti/	3DU	/-tás/
		1PL	/-más(i)/
		2PL	/-thá(na)/
		3PL	/-ánti/

(12) Mobile and Immobile Stress in the Noun

MOBILE		IMMOBILE		
'yoke-ACC.SG	'yoke-INS.SG'	'cow-ACC.SG'	'cow-INS.SG'	
/yuj-am/	/yuj-á/	/gáv-am/	/gáv-á/	UR
—	—	—	gávā	ACCENT SELECTION
yújam	—	—	—	DEFAULT ACCENT
[yújam]	[yujá]	[gávam]	[gávā]	SR

(13) Mobile and Immobile Stress in the Verb

MOBILE		IMMOBILE		
'eat-3SG	'eat-3PL'	'fashion-3SG'	'fashion-3PL'	
/ad-ti/	/ad-ánti/	/táks-ti/	/táks-ánti/	UR
—	—	—	táksanti	ACCENT SELECTION
ádti	—	—	—	DEFAULT ACCENT
[átti]	[adánti]	[táṣṭi]*	[táksati]	SR

Kiparsky (2014, 2018) of "preaccenting" inflectional suffixes for those suffixes tentatively labeled "unaccented" under (10) and (11) ultimately renders DEFAULT ACCENT essentially superfluous.

²⁴Kiparsky (2018a) in fact regards the inflectional endings labeled here as unaccented as *preaccenting*; see 4.2.3 below on pre-accentedness and more discussion of the prosodic properties of these inflectional suffixes.

²⁵See the immediately preceding note.

This pattern of stress mobility, though readily analyzed, is in fact relatively uncommon in Vedic: intraparadigmatic stress movement between some syllable of the stem and an inflectional suffix is essentially limited to root formations of the type seen above, plus active participles built with a suffix *-ánt-* ~ *-át-* ~ *-at-*.²⁶ Instead, the majority of all inflectable Vedic lexemes exhibit an IMMOBILE stress pattern: stress remains on the same syllable throughout the entire inflectional paradigm, never being attracted to the “accented” inflectional endings. How this behavior is to be treated will be taken up in the following section. First, a closer examination at productive patterns of vowel alternation in Vedic is necessary.

4.2.1.2 VOWEL DELETION (Zero-Grade Ablaut) and VOWEL STRENGTHENING

Under a traditional conception of Sanskrit morphology, any morpheme (in most cases, corresponding prosodically to a single syllable) may theoretically exhibit one of three possible vowel gradation: *loka* (zero grade), full grade (*guṇa*), or lengthened grade (*vr̥ddhi*). A morpheme in its zero-grade state typically has no syllable nucleus, a nucleus corresponding to a high vowel, a nucleus corresponding to a syllabic liquid *r* (rarely *l*), or shows a short *a* /ə/; a morpheme in its full-grade state always contains a short *a* /ə/, which may coalesce with an underlying glide to produce a long mid vowel, *e* or *o*, or an abstract segment /ⁱ/ on so-called *seṭ*-roots to produce a long vowel *ā* /a:/, or show a nasal which disappears in the corresponding zero grade; a morpheme in its lengthened-grade state contain a long *ā* /a:/, which may produce a diphthong *ai* or *au*. The typical correspondences seen in Vedic Sanskrit are given in 14; note that for all of these patterns, mirror-image versions with a sonorant consonant preceding the *a* in the full grade exist, which are identical in their zero-grade states.

(14) Sankrit Vowel Gradation (Ablaut)

Zero Grade (<i>loka</i>)	Full Grade (<i>guṇa</i>)	Lengthened Grade (<i>vr̥ddhi</i>)
<i>i</i>	<i>e/ay</i>	<i>ai/āy</i>
<i>u</i>	<i>o/av</i>	<i>au/āv</i>
<i>r̥</i>	<i>ar</i>	<i>ār</i>
<i>a</i>	<i>aN</i>	<i>āN</i>
<i>i</i>	<i>ā/a</i>	<i>ā</i>

The reader will notice that some of the partial paradigms derived in (12) and (13) exhibit vowel gradation, e.g., ACC.SG *gávam* vs. DAT.SG *gáve*, giving the appearance of alternation between lengthened- and full-grade states. Other root nouns descriptively exhibit different patterns, e.g., ‘dog:M’ appears to show an alternation between lengthened grade (*svān-ā* ‘dog-NOM/ACC,DU’, *sānu* ‘back:NOM/ACC.SG’) and zero grade (*śún-as* ‘dog-GEN.SG’, *snú-ṣu* ‘back-LOC.PL’). Among root present formations in verbs, /ad-/ ‘eat’ and /tāk-/ ‘fashion’ shown above are in fact atypical; most common is to find an alternation between full and zero grade, e.g., *é-ti*, *y-ánti* to /ay-/ ‘go’. In root formations, among both nouns and verbs, if the stem exhibits vowel gradation whatsoever, the following generalization holds: the surface allomorph found preceding accented inflectional suffixes will always be “weaker” (full

²⁶Genuine stress mobility is arguably to be found among certain present stems showing a nasal infix, as well as among present stems built with reduplication; these categories will be examined more closely under 4.2.3.

grade or zero grade) than the surface allomorph found preceding the inflectional suffixes listed as unaccented above (lengthened grade or zero grade). Based on this observation, and the observation that similar patterns of vowel alternation occur in derivational suffixes immediately preceding inflection (e.g., *-tar-* ~ *-tr-*, *-vant-* ~ *-vat-*), Kiparsky formulates a process of vowel deletion in terms of the usual Indo-European antecedents of Sanskrit *a*, i.e., PIE **e* and **o*.

- (15) ZERO-GRADE Ablaut (Kiparsky 2018a: 136; cf. Kiparsky 2010a: 145)
 Apophonic *e, o* → ∅ before an accented morpheme, unless a non-deletable vowel intervenes.²⁷

That ZERO GRADE can apply to underlying short *a* in Vedic is clear, but whether it can also apply to underlying long *ā* is less clear. Kiparsky provides no derivations that clearly indicate deletion of /a:/, and suggests two possible sources for instances of surface *ā*: 1) inherently accented roots lengthen the root vowel in their strong stem, (Kiparsky 2018b: 137); 2) unaccented suffixes cause morphemes to be strengthened to a bimoraic size in the course of cyclic foot construction (Kiparsky 2018b: 140). Given the relative paucity of accented roots generally, the suggestion in 1) is difficult to substantiate, and synchronically in Vedic, exceptions in both directions exist: at least one root present seems to undergo strengthening, even though it exhibits mobile stress and hence cannot be accented (3.SG *mārṣti* vs. 3.PL *mṛjānti* to *mṛj-* ‘wipe’),²⁸ while at least one noun must be accented but shows no lengthening to *ā* (*nár-as* ‘man-NOM.PL vs. *nár-ā* ‘man-INST.SG).

Systematic examination of the patterns of vowel alternation in root nouns indicates that they are generally compatible with the following analysis: 1) underlying /ā/ is not subject to ablaut; 2) underlying /a/ is lengthened to *ā* in when followed by an unaccented morpheme, but deleted when followed by an unaccented morpheme, provided that a well-formed onset would result. First, root nouns that appear in the zero grade in the strong cases (like *yuj-* above) simply maintain zero grade throughout. The evidence for non-deletion of /ā/ comes from *vāc-* ‘voice:F’, which remains as *vāc-* throughout its paradigm (e.g., *vāc-í* ‘voice-LOC.SG’), where the failure to undergo zero grade cannot be phonotactically motivated. Given that underlying /ā/ is unaffected by ZERO GRADE, underlying /a/ may undergo deletion in weak cases but strengthening to [ā] in strong cases; such as approach fits the alternation *dyāv-as* ‘sky-NOM.PL ~ *dīv-í* ‘sky-LOC.SG’, if the UR is assumed to be /dyav-/ , the underlying /a/ is subject to either lengthening or deletion; that this stem indeed contains an /a/ is supported by

²⁷Although Kiparsky does not exactly define what constitutes a “non-deletable” vowel, a PIE *e* or *o* vowel that cannot undergo deletion must be intended, since Kiparsky offers derivations showing the application of ZERO GRADE across intervening high vowels.

²⁸Under some views, the zero-grade *r* seen in *mṛjānti* is a younger formation, replacing an expected **mār-jati*, parallel to *tákṣati*. Note, however, that *mṛjānti* is already attested in the RV. Insofar as the alternation *mārṣti*~*mṛjānti* seems to have existed as a synchronic paradigm in Vedic, either some paradigm members (presumably the strong forms) would have to constitute lexicalized exceptions, or the presence of lengthened-grade *ā* in the strong stem must be accounted for other than by the accentedness of the root (i.e., no ^X/márj-/).

forms of the adjective *abhi-dyav-* ‘going towards heaven’.²⁹ In the case of *nar-* ‘man’, an underlying zero-grade stem /nṛ-/ may be assumed, with a phonotactically motivated full grade preceding all vowel-initial case forms; stress regularly occurs on the root, except in the GEN.PL *narām* / *nṛnām* (see further below on stress in the GEN.PL).³⁰

Regardless of how patterns of vowel alternation in Vedic are best to be analyzed — and to what extent their burden falls on (opaque) phonological processes versus the morphology — it is sufficient for explicating the other components of the Kiparskian analysis of Vedic prosody to know the following: 1) apparent instances of vowel deletion are generally assumed by Kiparsky to be due to the rule in 15, and 2) application of 15 may constitute evidence of a lexically accented morpheme, if unambiguous evidence from the position of stress itself in inflection and derivation is wanting.

4.2.2 Secondary Mobility and the OXYTONE RULE

Before exploring motivations for the predominance of immobility in paradigmatic stress in the Vedic, it is necessary to distinguish between the genuine stress mobility seen in 12 and 13 from cases of pseudo-mobility, which Kiparsky generally refers to as “secondary mobility”. Secondary mobility mostly occurs when the strong forms of a paradigm exhibit stress on the rightmost syllable of the stem (e.g., *pitár-am* ‘father-ACC.SG), but in the weak forms, stress appears to move from the stem onto the inflectional suffix (e.g., *pitṛ-é* ‘father-DAT.SG). The crucial difference between genuine and pseudo-mobility is, however, that, in the latter case, the stem is also shortened by one syllable by a process of vowel deletion; in cases where vowel deletion applies but a syllable nucleus remains, stress remains on the final syllable of the stem (e.g., *pitṛ-bhis* ‘father-INST.SG). In many paradigms, such as the forms of *pitár-* ‘father’ just mentioned, whether such secondary mobility means that the underlying accent of an accented morpheme surfaces as stress, or that an accent reassociates to the right, is unclear; both treatments would yield the same result.

However, two other types of rarer stress shift, exemplified in the first instance by two *u*-stems, *sānu-* ‘back’ and *dāru-* ‘wood’, and in the second instance by an *i*-stem *arí-* ‘friend’, indicate that secondary mobility involves rightward shift of an accent; cf. Kiparsky 2018a: 140.³¹ In the former type, deletion of the *á* in the stem results in stress on the following

²⁹Already in the RV, gives indications of occasionally being treated as an *u*-stem /di-u-/ (or similar), given the (relatively uncommon, 6 × RV) GEN.SG *dyós* (vs. GEN.SG *divás*, 426 × RV) and regular INST.PL *dyúbhis* (19 × RV, never **dyubhís*).

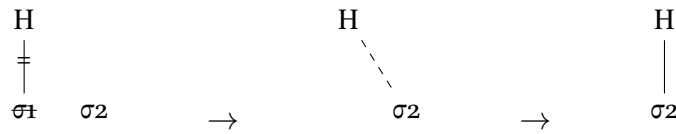
³⁰Against an accented stem would speak the adjectival derivative *nṛvánt-*, since underlyingly accented bases regularly maintain stress when a derivative in *-vánt-* is built. Many of the possible examples in the RV, however, may rather be interpreted as adverbs in *-vát-*, which would attract the stress. See generally Barth 2018 on the relationship between adverbials in *-vát-* and adjectives in *-mánt-/vánt-*. Barth does, however, explicitly argue that an adjective *nṛvánt-* served as a basis for creation of the adverbial suffix *-vát-*, which, if correct, would imply that a genuine adjective *nṛvánt-* with suffixial stress must have existed. One possible account of the overall situation is that the root was historically unaccented (whence *nrvánt-*), but in Vedic was ultimately reanalyzed as accented (/nṛ-/), thereby productively generating weak case forms with stress on the root syllable. See further 4.3 for an account of the rather unusual accent behavior of *nṛ-*.

³¹The most explicit available treatment of stress in these Vedic lexemes is given in Yates 2020: 262–3, and

u, not on the inflectional suffix (e.g., *snú-ṣu* ‘back-LOC.SG’, not ^x*snu-ṣú*), while in the latter, glide formation before vowel-initial inflectional suffixes results in stress on the final syllable (*ary-ás* ‘friend-NOM.PL’, not ^x*áry-as*).³² Kiparsky is not explicit in the formalization of such an accent shift, but given the representation of “accent” as high tone (cf. above) and the definition in 16, a formalization as in 17 would be possible. The result is that, at any step in the derivation, if the syllable that hosts a high tone is lost, the high tone will reassociate to the nearest syllable to the right.³³ The derivations in 18 illustrate secondary mobility induced by vowel deletion and glide formation, assuming for the moment underlying accentedness of the stems in each case.³⁴

- (16) SECONDARY MOBILITY (cf. Kiparsky 2014: 9): When a syllable is eliminated, its accent shifts to the right.

- (17) SECONDARY MOBILITY



- (18) Derivations involving SECONDARY MOBILITY

‘back-NOM/ACC.SG’ /sánu-Ø/	‘back-LOC.SG’ /sánu-sú/	‘friend-ACC.SG’ /arí-m/	‘friend-NOM.PL’ /arí-as/	Gloss UR
—	snu-su	—	—	ZERO GRADE
—	—	—	aryas	GLIDE FORMATION
—	snúsú	—	aryás	SECONDARY MOBILITY
—	snúsu	—	—	ACCENT SELECTION
—	—	—	—	DEFAULT ACCENT
[sánu]	[snúṣu]	[arím]	[aryás]	SR

With the distinction between paradigms with genuine and pseudo-mobility now clearly established, I turn to the question of why paradigmatic stress mobility should be relatively rare in Vedic, and mostly absent among polysyllabic stems. A facile answer would be assume that polysyllabic stems, and the derivational affixes used to build them in most cases, accidentally happen to be accented, in which case the BAP alone would be sufficient to account for the absence of mobility. That such an assumption is insufficient is shown by the contrast

will be discussed further in 4.3.

³²Exact parallels to *arí-* that might constitute a counterexample are not forthcoming, mainly because such *i*-stems typically exhibit vowel *insertion* preceding vowel-initial inflectional suffixes, rather than glide formation, in which case stress remains on the rightmost stem syllable, e.g., *matí-m* ‘mind-ACC.SG’ vs. *matáy-as* ‘mind-NOM.PL’.

³³Again, in the most common type of secondary mobility, the high tone from a deleted syllable might simply fail to be reassociated, since the following syllable in instances like *pitré* will already be associated with a high tone.

³⁴For the moment, I leave VOWEL DELETION as an unspecified process that deletes an underlying /ā/ or /a/; this process is discussed and labeled (ZERO GRADE) under 15.

in stress behavior between the *i*-stem *arcí-* ‘flame’ and the *a*-stem *devá-* ‘god’: both nouns exhibit persistent stress on the rightmost syllable of the stem in their respective paradigms (e.g., INST.PL *arcíbhis* and *devébhis*), but when a further adjective with the suffix *-mant-*/*-vant-* is built, stress shifts to the adjective-forming suffix when attached to the *i*-stem, but remains on the suffix *-á-* of the *a*-stem, thus *arcimánt-* ‘having flames’ but *devávant-* ‘surrounded by gods’.³⁵ This fact indicates that *arcí-* and *devá-* must, underlyingly, be prosodically distinct; whatever exact properties the suffix *-mant-* / *-vant-* is assumed to have, the BAP alone cannot account for why *arcí-* and *devá-* should exhibit apparently identical stress in their basic forms, but different stress patterns in a derived form. Assuming, for the moment, that *-mant-* / *-vant-* bears a lexical accent, the derivations in 19 show that the BAP would predict either persistent stress on the base or a consistent shift to the derivational suffix.

(19) Invalid Derivations of *arcimánt-* and *devámant-*

Unaccented Stem		Accented Stem		
/arc-i-mánt-am/	/dev-a-vánt-am/	/arc-í-mánt-am/	/dev-á-vánt-am/	UR
—	—	arcímantam	devávantam	ACCENT SELECTION
[arcimántam]	^X [devavántam]	^X [arcímantam]	[devavántam]	SR

The missing ingredient would appear to be either a process that deletes accents under certain conditions, and so enables a stress shift, or a process that inserts accents on unaccented stems, and thus potentially neutralizes the underlying distinction between stems with an accent and stems without an accent. Kiparsky’s analysis adopts the latter approach, employing the rule given in 20, ordered in a derivation before the BAP. In this way, an unaccented stem /arc-i-/ is always provided with an accent, thus resulting in stress immobility in its paradigm, while stress shift to the right in *arcimánt-* is a consequence of the unaccentedness of /i-/.

(20) OXYTONE RULE (OR)

$$\sigma \rightarrow \acute{\sigma} / [\dots\sigma ___]_{\text{Stem}} \text{ Infl}$$

“...a final accent is assigned to *all* inflected polysyllabic stems.”

The OXYTONE RULE may then be regarded as the essential factor resulting in the regularity of intraparadigmatic stress immobility in Vedic: stems that bear a lexical accent in virtue of the lexicon naturally never show mobility, while all unaccented polysyllabic stems should receive an accent at the right edge of their stems. The typology of stress patterns in inflection predicted by the OR and BAP together is thus threefold:

1. IMMOBILE: stems with an underlying accent, which always realize the (leftmost) stem accent in preference to accents of inflectional suffixes.
2. OXYTONE: polysyllabic stems without and underlying accent, which always exhibit stress on the rightmost syllable of the stem (modulo SECONDARY MOBILITY).

³⁵See Sandell 2016 on the distribution of the suffix allomorphs *-mant-* and *-vant-*. I argue there that the prosodic properties of the input to suffixation in fact partly influence the selection of the suffix allomorph.

3. MOBILE: unaccented, monosyllabic stems, which exhibit stress on either the first syllable (by DEFAULT ACCENT) or realize the accent of an inflectional suffix.

A further attractive feature of this analysis is that it correctly predicts the behavior of certain compounds, specifically, exocentric possessive compounds with a noun as second member (*bahuvrīhi*-compounds), which always possess polysyllabic stems, and show regular stress according to the leftmost accent of the compound members, or else belong to the oxytone pattern, if all compound members are unaccented (cf. Kiparsky 2010a: 172–3, Lundquist and Yates 2018: 2127–8). This general accentual behavior holds for such possessive compounds, regardless of the part of speech of the first member (noun, adjective, or preverb). In 21, some examples involving the noun *-jihvá-* ‘tongue’ and their corresponding URs are provided, where it should be assumed that */jihvá-/* is accented (an *a*-stem like *devá-*);³⁶ 22 then presents some cases in which both compound members must be treated as unaccented.

- (21) Accentuation of *bahuvrīhi*-compounds with second member *-jihvá-* tongue.
 - a. */[híraṇya- + jihvá-] m/ → híraṇyajihvam* ‘having golden tongues:ACC.SG’
The simplex *híraṇya-* ‘golden’ must be accented */híraṇya-/*; all *bahuvrīhi*-compounds in the RV with *híraṇya-* as a first member show stress on the first syllable (see Grassmann 1872 [1976]: 1666–9).
 - b. */[su + jihvá-] as/ → sujihvās* ‘having lovely tongues, speaking well:NOM.PL’
The adverbial *su* ‘well, good’ must be unaccented */su/*; compounds of all types with *su* as first member are nearly always stressed on the second member.
 - c. */[agni + jihvá-] as/ → agnijihvās* ‘having Agni as a tongue’
For */agni-/*, cf. *agnivánt-* ‘located at the fire’
- (22) Oxytone *bahuvrīhi*-compounds
 - a. */[su + pad-] ī/ → supád-ī* ‘having lovely feet-F.NOM.SG’
For */su/*, cf. above; for */pad-/*, compare *padvánt-* ‘footed’.
 - b. */[śiti + pad-] as/ → śitipád-as* ‘having white feet-NOM.PL’
For unaccented */śiti-/* (compound form of */śvit-i-/*), compare *śviticí-* ‘whitish’
 - c. */[tri + vrt-] am/ → trivṛt-am* ‘threefold-ACC.SG’
For */tri-/* ‘three’, cf. INST.PL *tribhís*; for */vrt-/* ‘turn’, cf. INST.SG *vṛtá*

4.2.3 Issues for the OXYTONE RULE, Excess Mobility, and Preaccenting Morphemes

Although Kiparsky (2018a: 143) sees the OR as being due to “prosodic, rather than morphological causes”, Vedic does know specific two exceptions in the noun (of which Kiparsky is aware; cf. Kiparsky 2010a: 155), as well as several broader classes of derived stems in the verb, whose stress patterns cannot be immediately brought into accord with the OR. The two apparent exceptions in the noun, *púmāṃs-* ‘man’ and *pánthā-* ‘path’, have been discussed in detail in Sandell 2015a: 181–4; see there for full paradigms. The fundamental problem with these

³⁶Note that *-jihvá-* is restricted to compounds in Vedic, substituting the independent substantive *jihvá-* ‘tongue:F’.

two stems is that their disyllabic strong stems exhibit stress on the initial syllable (ACC.SG: *púmāṃsam*, *pánthām*), while their weak case forms present genuine stress mobility to the accented weak case suffixes, with the stem undergoing vowel deletion to become a single syllable in size (GEN.SG: *pumsás*, *pathás*). Under the assumption that the OR ought to apply to all inputs in which the stem contains at least two potential syllable nuclei, the strong stems with stress on the leftmost syllable would seem to require a lexical accent associated with the leftmost vowel, e.g., /púmāṃs-/; precisely such an accent is, however, incompatible with the weak stems, which would be predicted to maintain stress on the first syllable. Kiparsky thus opts to diacritically exempt these stems from the application of the OR; Sandell (2015a: 183) suggests that the OR should be sensitive morphological structure, perhaps instead of or in addition to prosodic structure, with only *polymorphemic* stems potentially triggering the OR.³⁷

Similarly problematic, in that their stems fulfill the structural description of the OR but instead exhibit stress mobility, are active participles built with a suffix *-ánt-*, as well as Class V (nasal suffix) presents, Class IX (nasal infix to *seṭ*-root) presents, the reduplicated perfect, the optative to athematic verbs, and the numeral *aṣṭá* ‘eight.’³⁸ All of the just named categories share a common accentual pattern: the strong stem bears stress on the rightmost syllable of the stem, while the weak stem, although it still contains a vowel on that could bear stress on the same syllable when the inflectional suffix begins with a consonant, consistently cedes stress to an accented inflectional suffix.³⁹ Example 23 illustrates this pattern in the categories just mentioned, which I will term “excessive” mobility, since it ought to be excluded by the OR; cf. Kümmel 2021.⁴⁰

³⁷As observed in Sandell (2015a: *loc. cit.*), *púmāṃs-* ‘man’ and *pánthā-* ‘path’ are synchronically morphologically isolated in Vedic, and the latter shows signs of being converted into an *i*-stem /path-i-/. A further possible argument in favor of the morphological simplicity of *pánthā-* ‘path’, at least, is that the weak stem presents a double zero-grade, with deletion not only of the non-high vowel nearest to an accented inflectional suffix, but also nasal deletion. Under the analysis of zero-grade ablaut in Kiparsky 2018a, the application of vowel deletion is blocked when a non-high vowel/nasal has been parsed into a bimoraic foot on an earlier cycle in the derivation. That the first syllable of *panthā-* remains transparent to vowel deletion indicates that no cycle has occurred before inflection. If the OR is assumed to apply only to stems that have undergone a cycle because of a word-formation process, *púmāṃsam* and *pánthām* would naturally escape its application. In terms of Lexical Phonology, one might see the OR as a stem-level process, assigning an accent to the rightmost syllable at the stem level; morphologically simplex stems would, however, not undergo any stem-level phonology.

³⁸Likewise problematic, but in slightly different ways, are reduplicated (Class III) present stems. Particularly difficult for their analysis is that the endings 3PL /-ánti/, /-ántay/, /-ánta/ behave as though they are unaccented, surfacing as *-ati*, *-ate*, *-ata* in reduplicated presents. Furthermore, while many reduplicated presents usually exhibit stress on the leftmost syllable (reduplicant), e.g., *bíbharti* ‘bear:NPST.3SG’, some instead show stress on the root, e.g., *juhóti* ‘sacrifice:NPST.3SG’, without any other evidence for difference in accentedness of the roots.

³⁹Class VII nasal infix presents of the type 3SG *yunák-ti*, 1PL *yuñj-ánti*, on the other hand, are consistent with application of the OR, since vowel deletion in such cases eliminates a syllable in the stem.

⁴⁰The distribution of stem allomorphs for the participle in *-ant-* is as in the noun elsewhere; in the present stems and the perfect, the strong stem occurs before ending of the active singular, the weak stem elsewhere; in the optative, the 1/2PL active also employ the strong stem.

(23) Excessive Stem–Suffix Mobility in the Finite Verb

Root	STRONG ACT.3SG	WEAK (consonant) MID.3SG	WEAK (vowel) ACT.3PL	Category
kṛ- ‘make’	<i>caḱāra</i>	ACT.1PL <i>caḱṛmā</i>	<i>caḱrūr</i>	Perfect
yuj- ‘join’	<i>yunákti</i>	<i>yuníkté</i>	<i>yuñjánti</i>	Class VII Present
pu ⁱ - ‘purify’	<i>punáti</i>	<i>puníté</i>	<i>punánti</i>	Class IX Present
su- ‘press’	<i>sunóti</i>	<i>sunuté</i>	<i>sunvánti</i>	Class V Present
kṛ- ‘make’	<i>ḱṛnuyát*</i>	<i>ḱṛnvítá*</i>	<i>ḱṛṇuyúr*</i>	Optative (Class V Present)

(24) Excessive Stem–Suffix Mobility: Active Participle of *duh*- ‘milk’

- a. NOM.PL *duhántas*
- b. DAT.SG *duhaté*
- c. INST.PL *duhádbhis**
- d. N.NOM/ACC.SG *duhát**

(25) Excessive Stem–Suffix Mobility: ‘eight’

- a. NOM/ACC.SG *aṣṭá(v)*
- b. INST.PL *aṣṭābhís*

Just as in the case of *púmams*- and *pánthā*- above, the stress seen in the strong stem of (23)–(25) can neither be attributed to the OR, nor to a lexical accent of the stem-forming morphology; in either case, a paradigm with immobile stress would be predicted.⁴¹ However, the sort of tense-aspect morphology seen in 23, as well as the participles, may be regarded as non-cyclic (so Kiparsky 2018a: 140–1), thus exempting these verbal stems from application of the OR, if this rule is assumed to be sensitive to cyclicity, i.e., to apply after the derivation of the stem, before inflection. From the perspective of the morphosyntax, the fact that morphologically complex nominal stems appear to undergo cyclic derivation, while many verbal stems do not (see 4.2.4 on verbal morphology that does apparently undergo cyclic derivation), could be attributed to the fact that the morphology involved in constructing nominal or adjectival stems is (potentially) category-changing, i.e., a category head N or ADJ is introduced. The derivational suffixes of 23 and 24, on the other hand, are precisely *not* category-changing, being compatible exclusively with verbal roots.

To exempt both the range of verbal morphology seen here, as well as *púmams*- and *pánthā*- from application of the OR is then no great obstacle: the latter nouns are root-like, while the verbal morphology does not trigger the stem-level phonology where the OR applies. Prosodically, the problem then becomes not why evidently accented inflectional suffixes attract stress, but why the rightmost syllable of the stem should be stressed in the strong stem. The evidence from secondary mobility (cf. 16 above) seems to exclude treating, e.g.,

⁴¹ In the perfect in particular, a lexical accent on the root can generally be excluded, since the vast majority of verbal roots in Vedic are demonstrably unaccented. To attribute a post-accenting property to the reduplicative prefix, which would induce stress on the following syllable, would be able to correctly generate the strong stem (*caḱāra*) and pre-vocalic weak stem (*caḱrūr*), but still not the pre-consonantal weak stem (*caḱṛmā*, not ^X*caḱṛma*).

the nasal suffix *-no/-nu-* as lexically accented; stress on the antepenultimate syllable in pre-consonantal weak stems (thus ^X*sunúte*) would be predicted.⁴²

Here it becomes necessary to introduce a third type of prosodic property, namely, preaccentedness. In the domain of nominal derivation, certain suffixes consistently condition the appearance of stress on an immediately preceding syllable, for instance, *a*-stems with stress on the root (e.g., *bhára-* ‘carrying’), and neuter *s*-stems built with the suffix *-as-* (e.g., *máнас-* ‘mind’). That the stress in such cases is distinct from left-edge stress created by the BAP is evident from the appearance of stress in the same position in adjectives derived with *-mant-/vant-* or when such nouns occur as the second member of a *karmadhāraya*-compound (e.g., *máнас-vant-* ‘full of mental energy’, *pra-máнас-* ‘careful’ [AVŚ]). The GEN.SG of *máнас-*, *má-nasas* is, for instance, derived from /man-’as-ás/ by ACCENT SELECTION. Kiparsky is nowhere explicit as concerns the exact lexical representation of preaccents. Continuing with the proposed representation of accent as /H*/ , a preaccent must similarly introduce an /H*/; however, a floating accent might instead preferentially associate with the leftmost syllable (at least under a constraint-based perspective, where left edge stress represents a preference in the grammar, rather than application of a rule), barring other rules imposing locality of accent association. Furthermore, in light of 16, association of a /H*/ to a potential syllable nucleus to the right should be excluded. For this reason, I suggest viewing preaccents in the Kiparskian system as bitonal /H* L/ where the /L/ component is preassociated to a syllable nucleus on the right, thereby compelling the /H*/ component to associate to the nearest syllable nucleus to the left.

If derivational affixes may exhibit preaccentedness, inflectional affixes could as well, at least in principle. Kiparsky therefore proposes that the singular non-past inflectional suffixes of the verb, those of both present and perfect stems, are preaccenting (cf. Kiparsky 2018a: 136). The inputs to, e.g., *sunóti* and *sunuté*, would thus be /su-naw-’ti/ and /su-naw-tái/, respectively, without the application of any further accentual rules. For most of the categories listed in 23–25, attributing preaccentedness to the inflectional suffixes found in the strong stem is unproblematic. However, the representation of preaccents as /H*L/ does not fit as nicely with the non-syllabic suffixes */-m/*, */-s/*, */-t/* found in the strong stem of optatives.⁴³ In addition, where the optative suffix occurs in its full-grade form alongside an accented inflectional suffix, stress remains on the optative suffix (e.g., *śṛṇuyáma* ‘crush:NPST.OPT.1PL’),

⁴²Nevertheless, it is not entirely clear that that secondary mobility functions precisely as Kiparsky envisions. As discussed above, the stress on accented inflectional suffixes in most cases of secondary mobility (*pitré* type) is non-diagnostic between reassociation of an accent and loss of an accent via vowel deletion. Though the *aryás*-type certainly requires reassociation of an accent, the reassociated accent could be an accent introduced by the OR, rather than a lexical accent. In *snúsu*, however, if Kiparsky’s (2018) conception of vowel deletion is correct, the cycle on which an oxytone accent is introduced in a stem /sān-u-/ should simultaneously render the first vowel /ā/ immune to vowel deletion, which suggests that this stem, too, is not subject to the OR. Nonetheless, a stem with two lexical accents /sān-ú-/ cannot be excluded either (cf. Yates 2020: fn. 10). The point is that to adjudicate between rightward accent reassociation and outright accent deletion in cases of secondary mobility remains difficult.

⁴³Under a different representation of lexical accent, specifically, as prespecified foot structure, this issue seems less troubling – a purely consonantal affix could equally introduce a foot boundary. However, intuitively more satisfying would be to assume that only morphological material containing potential syllable nuclei would ever need to assert a preference as to whether it serves as a foot head or foot tail.

which is in principle compatible only with a lexically accented suffix /-yáⁱ-/.⁴⁴ In the active participle, to assume preaccenting case endings in the noun for the strong cases would correctly generate the stress (e.g., /duh-ant'-am/ → *duhántam*), but encounters the same potential issue regarding the representation of preaccentedness for the non-syllabic NOM.SG /-s/, and more worrisomely, would incorrectly predict non-initial stress in the strong cases of *púmaṃs-* and *pánthā-* (/pumaṃs'-am/ → ^x*pumáṃsam*).⁴⁵ As far as the numeral 'eight' is concerned, a stem /astā-/ combined with a preaccenting NOM/ACC.SG /-'au/ or accented /-bhís/ would yield attested *astā(v)* and *aṣṭābhís*. However, *bahuvrīhi*-compounds with 'eight' as a first member suggest that the stem was interpreted as having a lexical accent, e.g., *aṣṭā-vandhura-* 'having eight chariot seats'. Given these issues, although the assumption of preaccenting inflectional suffixes can resolve most cases of excessive mobility, some a residue of irresolvable or contradictory prosodic behaviors nevertheless remains.⁴⁶

One final remark on the role of preaccenting inflectional suffixes in the prosodic lexicon and grammar of Vedic under the Kiparskian analysis is warranted. If all inflectional suffixes that were in 2.1 above initially taken to be unaccented are instead assumed to be preaccenting, the consequence is that *all* inflected word forms will contain at least one underlying accent. Consequently, the DEFAULT ACCENT component of the BAP, which assigns an accent to the leftmost syllable and thereby generates initial-syllable stress, becomes irrelevant: all forms whose stress would be attributed to DEFAULT ACCENT (essentially, the strong stems of root formations) would then be expected to produce stress on the first syllable due to the preaccenting inflectional suffix (cf. derivations in 12–13 above). The role of the BAP is then effectively reduced to ACCENT SELECTION alone. Likewise, the activity of the OXYTONE RULE in the grammar is somewhat reduced, since the stress at the right-edge of the stem in strong cases in polymorphemic nouns could again be attributed to preaccenting inflectional suffixes; in paradigms of unaccented stems like /ma-ti-/ 'mind', the OR is nevertheless needed in order to ensure that stress does not shift to the weak cases of such stems. The fundamental principles underlying Kiparsky's overall approach to Vedic accentuation as described here thus seem to require a certain amount of redundancy.

⁴⁴Regardless of whether the non-application of zero-grade ablaut in such cases is an innovation of Indo-Iranian or not (see Hoffmann 1968), the position of the stress ought to be productively derivable. An unaccented optative suffix can only be preserved if the stress in the 1PL and 2PL is assumed to somehow depend on the position of the stress seen in the persons singular, or that the optative somehow exceptionally selects for unaccented allomorphs of the 1PL and 2PL, which would only be required for precisely such athematic optatives.

⁴⁵A last escape for these two nouns would be to assert the availability of accentually distinct strong and weak stems, e.g., strong /púmaṃs-/ vs. weak /pumaṃs-/. The same manouver could be employed for the active participle as well, i.e., strong /-ánt-/ vs. weak /-at-/.

⁴⁶One general alternative for all the cases of excessive mobility would be to adopt accented derivational suffixes (e.g., /-náu-/ , /-ánt-/), and to assume that lexical accents /H*/ do not undergo reassociation when their host in the input is deleted, and instead are lost altogether. Such an analysis necessitates, however, a different treatment of secondary mobility (cf. discussion above), at least for the rarer types.

4.2.4 Accentual Dominance and Affix Typology

The exegesis of Kiparsky's analysis of Vedic stress up to this point has operated with a distinction between unaccented, accented, and preaccenting morphemes, where the conflict between multiple underlying accents is resolved by the ACCENT SELECTION component of the BAP, resulting in stress that corresponds to the leftmost lexical accent. In cases in which the string contains no underlying accents – which, under Kiparsky's analyses, ought to be a very rare occurrence – a default stress on the initial syllable should be assigned, or stress should appear at the right edge of the stem in many morphologically complex stems. That ACCENT SELECTION as defined in 9 is insufficient is, however, demonstrated by many cases in which not the leftmost accent surfaces as stress, but rather stress appears on the outermost (typically rightmost) derivational morpheme. The following discussion examines affixes that appear to impose a specific stress on their bases and considers possible underlying distinctions in morphological structure or cyclic process interaction that might be responsible for such behavior.

4.2.4.1 The Morphologically Dominant Affixes

The affixes implicated in the behavior of stress imposition almost exceptionlessly bear stress themselves, or almost exceptionlessly cause stress to appear on the preceding syllable, when they are present and clearly constitute the outermost element of the stem. Such affixes include the past passive participle in *-tá-*, the passive in *-yá-*, as well as abstract nouns in *-tā-* that condition stress on the preceding syllable. Examples of these three suffixes in derivation are given in 26; in all cases, the accent introduced by the rightmost suffix determines the stress in the output.

(26) Stress-Determining Suffixes

- a. /táks-tá-m/ → *taṣṭám* 'fashioned:ACC.SG' (cf. *tákṣati*)
- b. /rú-rudh-sa-yá-māna-/ → (*apa*)-*rurutsyámāna-* 'being desired for expelling' (cf. *rúrutsamāna-*; Heenen 2006: 214, Kulikov 2012: 699 on this rare passive to a desiderative, attested at KS 10.9)^{47, 48}
- c. /bándh(-)u-tā/ → *bandhútā* 'relationship:NOM/ACC.SG' (cf. *bándhu-*)⁴⁹

⁴⁷Note that the reduplicated desiderative typically exhibits stress on its initial syllable, thus indicating a lexically accented reduplicant; cf. Macdonell 1910: 387–90.

⁴⁸I follow in this example the accentuation printed by Kulikov 2012: 699. Heenen (2006: 214), on the other hand, prints the form without an accent, and the text of the KS available at <http://titus.uni-frankfurt.de/texte/etcs/ind/aind/ved/yvs/ks/ks.htm> based on the edition of von Schroeder 1900–1910 also prints no accent. I am inclined to believe that Kulikov's printed accent is indeed the expected accent, but the form with accent is probably a ghost.

⁴⁹Whether the root /bandh-/ 'bind' is accented or not is difficult to say, though the Class IX present *badhnāti* might speak against such an assumption. The stress of the noun *bándhu-* 'connection, relation' is probably best explained as a case of initial stress in "demorphologized" stems (cf. Lundquist 2015a on initial stress in *tí*-stems in general and Sandell 2016 on *śákti-* 'power, ability' in particular). Substantives derived with a suffix /-u-/ in Vedic are uncommon (cf. Debrunner 1954: 473–4, though most still exhibit oxytone stress. For the purposes of the present example, it is only relevant that the stem /bándhu-/ be accented, whether as a property of the root

Such affixes, which can impose a different accent than the leftmost one in the input, must evidently possess some further property, distinct from their lexical accent. For Kiparsky, this property is DOMINANCE, which he conceptualizes as a diacritic that triggers the application of an accent-deletion process that removes any accents to the left of the dominant morpheme (“dominant suffixes erase the inherent accent of the stem to which they are added”, so Kiparsky (2010a: 171)). Affixes which lack this property, including virtually all inflectional affixes, are instead labeled as RECESSIVE, and do not trigger any accent deletion by the rule stated in 27. Since Kiparsky suggests that Vedic may possess unaccented derivational suffixes, and indeed phonologically null dominant affixes (Kiparsky 2010a: 169; 173) the relevant rule of accent deletion must be conditioned by the diacritic feature itself, rather than a lexical accent to the right of an accent potentially subject to deletion.⁵⁰

(27) ACCENTUAL DOMINANCE

$H \rightarrow \emptyset / __ X_0 [+dominant]$ (iterative)

A high tone (lexical accent) is deleted if an element with the feature [+dominant] occurs anywhere to its right. The rule applies until no targets remain.

The derivations in 28 illustrate the application of the rule in 27 to lexical accents introduced both by roots or other derivational affixes, and contrasts the effects of the BAP and OXYTONE RULE. Dominant morphemes are marked there with underlining, e.g., /-tá-/. The form in the rightmost column, *tákṣati*, repeated from 13, shows the typical effect of “leftmost accent wins” when no dominant suffix is present; in the corresponding past participle derived from the same root, the dominant suffix /-ta-/ results in deletion of the underlying accent of the root. The noun *gárbha*- ‘womb’, meanwhile, is built with the preaccenting suffix /-’a-/, whose accent wins over accents introduced by the OR or present in inflectional

or the entire stem.

⁵⁰Kiparsky (2014: 15) indicates that the agent noun suffix that regularly occurs with stress, *-tár-* (e.g., *codayítár-* ‘impeller’), might be dominant and unaccented, and Kiparsky (2016: fn. 18) makes this view explicit. Since the application of 4.2.4 potentially allows for an accent assigned by the OXYTONE RULE to emerge on the dominant suffix, if accented suffixes must trigger zero-grade ablaut, then the fact that *-tar-* does not (cf., e.g., *kartár-* ‘doer, agent’) might speak in favor of its unaccentedness. On the basis of the position of the stress alone, whether a suffix is accented or not is difficult to determine. Diagnostic, as usual, would be an adjective in *-mant-/vant-* or the occurrence of such a noun in *-tár-* as the first member of a possessive compound, but examples from Vedic of either are not forthcoming. Note further that, because Kiparsky treats the causative suffix *-áya-* as properly built with an unaccented suffix /-ay-/ plus a preaccenting tense-aspect suffix /-a-/, *-tár-* forms built to causative stems like *codayítár-* are not indicative of any application of ZERO GRADE. However, if such tense-aspect morphology is non-cyclic, causatives ought not to impose stress on an accented base, but at least one causative to an intensive, *varīvarj-áy-ant-i* (AVŚ 12.15.22a) ‘make to be twisting about:F.NOM.SG’ suggests here that the lexical accent of a causative suffix wins out over the lexical accent of the intensive prefix, consistent with the scope of the form.

One might also further observe that a connection between root ablaut and the prosodic properties of a root-attaching derivational suffix in Vedic is tenuous at best. In the case of /-ti-/, for which the evidence of its unaccentedness is clearer than in the case of *-tar-*, the root often exhibits zero grade, where phonotactically licit, e.g., *matí-* ‘mind’, *kṣití-*, *stutí-*, *śrutí-*. Admittedly, certain accented derivational suffixes do regularly condition zero grade of the root (especially past-participle forming /-tá-/), but I do not regard root ablaut behavior as a secure indicator of a morpheme’s prosodic properties.

suffixes; when the dominant accented suffix /-tvá-/ is added, the accent of /-’a-/ is deleted and stress surfaces on -tvá-.

(28) Application of ACCENTUAL DOMINANCE

‘womb-LOC.SG’ /garb-’a-í/	‘pregnancy-ACC.SG’ /garbh-’a-tv́a-m/	‘fashion-PPP-ACC.SG’ /táks-tá-m/	‘fashion-3PL’ /táks-ánti/	Gloss UR
—	garbhatvám	takstám	—	ACCENTUAL DOMINANCE
gárbháí	—	—	—	OXYTONE RULE
gárbhai	—	—	táksanti	ACCENT SELECTION
—	—	—	—	DEFAULT ACCENT
[gárbhe]	[garbhatvám]	[táʃti]*	[táksati]	SR

In general, the common property of dominant morphemes in Vedic is arguably connected to the OXYTONE RULE, or, at least, the two phenomena might share a common basis: cases in which the OR applies and cases in which accentual dominance applies nearly always involve the determination of a lexical category (e.g., noun for abstracts in -tv́a-, adjective for root-attaching superlatives in -iṣṭha-, verb for denominatives in -yá-) or a change in argument structure on verbs (passive -yá-, causative -áya-, reduplication in desideratives and intensives). In terms of morphosyntactic structure, these types of heads adjoin “closer” to the root. In contrast, the derivational affixes that both fail to trigger a cycle and allow for application of the OXYTONE RULE, as well as any inflection, encode morphosyntactic information that stands “farther” from the root. Again, when framed in terms of Lexical Phonology, ACCENTUAL DOMINANCE and the OXYTONE RULE may be seen as stem-level processes that (can) apply when a stem-level cycle is triggered.

On this point, Kiparsky admits of recessive derivational suffixes, in particular adjectival -mant-/vant-, which could be compared with the Level 2 derivational suffixes of English that do not affect the stress of their bases (e.g., adverbial -ly; cf. Kiparsky 2014: 16). However, -mant-/vant- is perhaps the only piece of derivational morphology in Vedic that arguably fulfills the criteria for triggering a stem-level cycle, yet evidently does not impose an accent on its base (cf. gómant- from lexically accented /gáv-/). Two possibilities suggest themselves, depending upon where, exactly, ACCENTUAL DOMINANCE and the OXYTONE RULE should be located in the lexical phonology of Vedic, and whether or not diacritic specifications of dominance and recessivity are desirable:

1. Level-Distinction: ACCENTUAL DOMINANCE is an automatic process that applies only at Level 1, erasing all but the rightmost accent on the stem at this level. The OXYTONE RULE applies at both Level 1 and Level 2, or it may be assumed that stems with Level 1 affixes are also submitted to the Level 2 phonology. Affixes that are evidently not subject to the OR (e.g., participial -ant-) only enter into the word-level phonology, which is the domain of the BAP.
2. Dominant-Recessive Distinction: Category- and argument-structure-changing derivation belongs to Level 1, where most affixes are marked as [+dominant] and trigger ACCENTUAL DOMINANCE, whereas few others (in particular, -mant-/vant-) are [–dominant];

the OXYTONE RULE also applies at this level. Other derivation (e.g., participial *-ant-*) belongs to Level 2, where no particular processes pertaining to accentuation apply. Inflectional affixes only enter into the word-level phonology, where the BAP applies.

These two conceptions of the system of word-formation and its interaction with accentuation in Vedic are a consequence of attempting to strictly associate certain aspects of morphosyntactic structure with accent-changing processes. Such an attempt requires distinguishing at least *-mant-/vant-* from non-cyclic derivation on the one hand (because *-mant-/vant-* appears to satisfy the relevant criteria for cyclic derivation), but also from most other affixes involved in cyclic derivation, on the other (because it does not induce ACCENTUAL DOMINANCE). For the purposes of analysis of the accent, the choice made by Kiparsky is simpler: there is not a direct correspondence between morphosyntactic structure and accentuation, but simply a diacritic distinction between [+dominant] and [–dominant]. [–dominant] affixes trigger neither ACCENTUAL DOMINANCE nor the OXYTONE RULE (they belong to Level 2), whereas [+dominant] affixes undergo both (they belong to Level 1). Under this account, [+dominant] suffixes that introduce neither an accent nor a preaccent will nevertheless regularly surface as stressed, because they delete other accents on their bases and undergo the OR. Strictly root-attaching */-ti-/* and the agent-noun suffix */-tar-/* (cf. fn. 50) may constitute precisely such derivational suffixes.

4.2.4.2 Genitive Plural *-ām* and Suffix Typology

A potential point in favor of a purely diacritic motivation for accentual dominance lies with the prosodic behavior of the genitive plural suffix, */-(n)-ām/*. That this suffix is lexically accented is evident from the expected accentual mobility in unaccented root nouns, e.g., *padām* ‘foot:GEN.PL’. Peculiar to this suffix, however, is that fact that it not infrequently attracts stress on OXYTONE stems (e.g., *kṣītī-nām* ‘habitation-GEN.PL’), as well as on some stems that otherwise appear to be lexically accented (e.g., *nar-ām* ‘man-GEN.PL’). Since a stem like */ksi-ti-/* should receive a stem-edge accent by the OR, it would appear that the genitive plural suffix in such cases causes erasure of the accent inserted by the OR. If */-(n)-ām/* is indeed dominant, as suggested by Kiparsky (2010a: fn. 5), then accentual dominance would be a potential property of inflectional affixes as well, and render any structurally motivated account of the distinction between dominant and recessive affixes virtually hopeless.

However, essentially all cases of such excessive mobility in the genitive plural involve sonorant-stem nouns that could, without the intrusive nasal seen between the stem and *-ām*, exhibit the most common type of secondary mobility. That is, a virtual ^x*kṣityām* could have been generated. Perry and Vaux (2018) therefore argue that the stress seen in forms like *kṣītīnām* ‘abode:GEN.PL’ reflects the expected stress generated via glide formation and secondary mobility; a readjustment rule then results in the insertion of */-n-/* at the stem-inflection boundary, after stress assignment, thereby generating the appearance of a form in which an accent on the stem has been deleted.⁵¹ As Perry and Vaux (2018: 281–2) correctly

⁵¹ An argument against the account proposed by Perry and Vaux (2018) could be that, among the oxytone *i-* and *u-*stems, one commonly finds full-grade *-áy-* or *-áv-* preceding vowel-initial case suffixes. Since no sec-

observe in the case of *nadīnām* ‘river:GEN.PL’, where no secondary mobility would be predicted, the genitive plural suffix occurs with stress (without nasal insertion, one might have expected ^x*nadīyām*, since the stem contains an underlying long vowel, /nadī-/).

4.2.5 Compound Accentuation and Deaccentuation

Discussion of *bahuvrīhi*-compounds (adjectival exocentric possessives) above showed that the combination of BAP and OR together correctly predict the stress pattern seen there: stress corresponds to the leftmost accent of the stem (BAP), otherwise falling on the rightmost syllable of the stem (OR). A complete account of Vedic prosody should be able to supply an account of stress assignment for all of the rich and diverse array of compound types found in Vedic. See Macdonell 1910: 155–77 or Macdonell 1916 [1993]: 265–82 for a survey of all varieties of compounds found in Vedic and Macdonell 1910: 91–7 for an overview of accent in compounds; compare further also Whitney 1889 [1960]: 480–515. In this section, I first review the analysis of Vedic compound accentuation presented by Kiparsky 2010a: 172–6, but, because of its laconic character, then examine the stress patterns all compound varieties and subtypes, following the traditional functional classification given in Macdonell 1916 [1993]: *loc. cit.*⁵²

The approach to the assignment of stress in composition suggested by Kiparsky allows for a division into two classes from the point of view of morphological structure and prosodic

ondary mobility occurs in such cases, *a*-insertion must occur prior to SECONDARY MOBILITY, at least; the problem is then to motivate why *a*-insertion never applies preceding genitive plural /-ām/.

⁵²For convenience, I provide here an outline of the Vedic compound types after Macdonell, though the presentation below will not strictly adhere to this order:

- I. Verbal Compounds (in which the compound is a V)
- II. Nominal Compounds (in which the compound is a N or A)
 1. Coordinative (*dvandva*) Compounds
 2. Determinative (*tatpuruṣa* in general) Compounds
 - a. Dependent (proper *tatpuruṣa*, including *upapada* [synthetic]) Determinatives
 - b. Descriptive (*karmadhāraya-tatpuruṣa*, including subtypes with negatives and numeratives) Determinatives
 3. Possessive (*bahuvrīhi*) Compounds
 - a. Descriptive *karmadhāraya* Possessives
 - b. Dependent *tatpuruṣa* Possessives
 4. Governing Compounds
 5. Syntactical Compounds
 6. Iterative (*āmredita*) Compounds

behavior:

1. “Regular” Compounds of the morphological structure [[X] [Y]] (stem compounds): prosodically, these compounds are subject to the OR and BAP.
Generally subject to this behavior are exocentric possessives, descriptive determinatives, iteratives, governing compounds (probably), and certain proper dependent determinatives.
2. Compounds with a compounding affix (probably always suffix) of the structure [[[X] [Y]] Z] (synthetic compounds): prosodically, the stress of the compound is determined by the lexical accent of the compounding affix (i.e., ACCENTUAL DOMINANCE erases all other accents).
Generally subject to this behavior are coordinatives and synthetic dependent determinatives, but also some descriptive determinatives with nominal heads and some exocentric possessives.

This analysis of prosody in compounds is attractive insofar as it does not introduce additional rules of stress assignment peculiar to composition: stress is predictable based on application of the same rules developed on the basis of stress assignment in the inflection and derivation of nouns and verbs. In the vast majority of instances, given the right assumptions about the constituent morphological structure of the compound, the expected stress will follow. By and large, the assumption of compounding suffixes that scope over the entire compound in the case of synthetic compounds is unproblematic – the result, in most cases, is a second member that is otherwise unattested as a simplex, and the suffix imposes precisely the same stress otherwise found in such nominal derivatives. There are, however, numerous cases in which *no* overt compounding suffix appears, yet the stress does not follow from the lexical accents of its members, but rather typically assigns stress to the rightmost syllable of the stem. Fully consistent in this behavior are the coordinatives (see further 4.2.5.2), which show stem-final stress but no compounding suffix. Kiparsky does not address the coordinatives specifically, but, in the case of descriptive determinatives and possessives that show similar stem-edge stress, proposes that such compounds contain segmentally null compounding affixes that trigger ACCENTUAL DOMINANCE, with the consequence that oxytone stress surfaces. This particular proposal is consistent with the rest of the prosodic grammar, but a clearer understanding of the segmentally null compounding suffix(es), under which conditions that should be employed and why, calls for deeper elucidation.

4.2.5.1 Iterative Compounds

Relatively straightforward, but also interesting for purposes of determining larger prosodic structures, is stress assignment in iterative compounds (*āmreḍitās*) of the type *divé-dive* ‘every day’. Both members of the compound are inflected, and stress invariably occurs on the first member, wherever stress would fall in the word.⁵³ Interestingly, iteratives with clearly

⁵³Pāṇini in fact does not group *āmreḍita* compounds with other types of compounds, instead regarding them as repeated instances of the same word, where the second instance receives low pitch (*Āṣṭadhyāyī* 8.1.1–

fully unaccented inputs (e.g., the strong cases of *i*- or *u*-stems with oxytone stress) are essentially absent in the RV (see the collection in Klein 2003); the one possible example is *agním-agnim* ‘Agni always’. If the stem /agni-/ is indeed unaccented, the iterative compound shows that the OR must apply to each stem separately, and not merely to a stem built out of inflected grammatical words. In virtually all other instances of iteratives in the RV, the stress derives directly from a lexical accent on the first member (whether as part of the stem or an inflectional suffix). I therefore tentatively assume that each member of an iterative compound builds a minimal prosodic word in which the OR and probably BAP applies, and that a recursive prosodic word ω_{Max} is built of the two minimal prosodic words, and the BAP applies (again); cf. Kiparsky 2010a: 176 (“reduplicative (*āmreḍita*) “compounds” go directly to the BAP”).

(29) Stress in Iterative Compounds

- a. / [agni-m] + [agni-m] / → / agnīm + agnīm / → *agnīm-agnim*
For the (likely) unaccentedness of /agni-/ , cf. *agnijihvá-* ‘having a tongue like Agni’
- b. / [áhar] + [áhar] / → / [áhar + áhar] / → *áhar-ahar* ‘day by day’
For the accentedness of /áhar-/ , cf. *áharjāta-* ‘born in the day’ (AVŚ 3.14.1, 5.28.12)
- c. / [div-é] + [div-é] / → / [divé + divé] / → *divé-dive* ‘day by day’
div-~ dyu- is a mobile root noun.

4.2.5.2 Coordinative Compounds

The oldest type of endocentric coordinative (*dvandva*) compound, constituting about three-fourths of all tokens in the RV, in which two nouns inflected in the dual both bearing an *udatta* (e.g., *mātára-pitārā* ‘mother and father’) stand adjacent to one another, is wholly unproblematic, one may safely assume that both elements, at this phase, constitute wholly independent grammatical and (maximal) prosodic words. This is simply coordination of nominals without a phonologically overt conjunction (like *utá* or *ca*). Later, when genuine stem compounding to build endocentric copulatives (beginning tendentially from younger parts of the RV) emerges, the nearly exceptionless stress pattern is for stress to occur on the final syllable of the stem. Descriptively, it appears as through any lexical accents of the stem are erased and the OR assigns an accent to the right edge of the stem. Kiparsky (2010a: 173) proposes precisely such an analysis for certain *bahuvrīhi* compounds that behave similarly in terms of stress, namely, that a “dominant zero compositional suffix” induces application of ACCENTUAL DOMINANCE, then allowing an accent subsequently assigned by the OR to appear as stress. The noun *keśa-śmaśrú-* (AVŚ 8.2.17), in which both members are lexically accented, instantiates such apparent erasure of stem accents without the addition of any overt compounding suffix. The question arises here whether it is licit to assume that the spell-out of a morpheme may consist solely in the application of a process; alternatively, one might propose that the relevant compositional affix is realized as a floating high tone /H*/

8.1.3, von Böhtlingk 1887 [1964]: 428; cf. Klein 2003: 773, fn. 2). In the *padapāṭha* text, however, iteratives are separated with *avagraha*, exactly like the members of other compound types (cf. Macdonell 1910: 91).

that associates to the right edge of the stem, perhaps applying similar rightward association of floating H assumed by Kiparsky for SECONDARY MOBILITY. Or, perhaps the application of ACCENTUAL DOMINANCE and OXYTONE RULE is simply an expected part of the relevant stem-level phonology that applies upon the insertion of such a segmentally empty compositional affix.

(30) Derivation of Endocentric Copulative Compounds	
/[[kéśa-śmaśru]-Ø ^[+Dominant]]/	'hair and beard'
kesá-śmaśru-	ACCENTUAL DOMINANCE
keśa-śmaśrú-	OXYTONE RULE
—	BAP
<i>keśaśmaśrú-</i>	SR

4.2.5.3 Determinative Compounds

In determinative (endocentric) compounds of both the dependent (*tatpuruṣa*) and descriptive (*karmadhāraya*) types, the position of the stress exhibits some degree of variation: in the RV and AV, the stress may occur on either the first or the second member (as in possessive compounds), though there is already a tendency for stress to fall on the second member, especially the rightmost syllable of the stem (just as in copulative compounds). Generally speaking, this broad class of compounds provides evidence both for the direct application of the BAP to the compound, selecting the leftmost lexical accent of either the first or second member, as well as for the apparent deletion of underlying accents and oxytonesis. Further complicating the analysis of stress in determinatives is the fact that many constitute synthetic (*upapada*) compounds built with a further overt derivational suffix, while certain classes of first or second members appear to trigger a pattern of accentuation contrary to an expected general pattern.

Proper *tatpuruṣa* and *upapada* Determinatives Determinative (endocentric) compounds in which the first member is syntactically dependent upon (governs) the second member (the head) most commonly exhibit stress on the second member of the compound, on the rightmost, penultimate, or even antepenultimate syllable of the stem. That stress does not always occur on the final syllable of the stem excludes an the type of analysis just applied to *dvandva* compounds above. In terms of syntactic relation, one typically finds a first member as the accusative complement of a verbal head (e.g., *haviṣ-ád-* 'oblation-eat = eating the oblation', *agnim-indh-á-* 'fire-kindling') or a genitive specifier of a nominal head (e.g., *udameghá-* 'water-shower = shower of water', *deva-māná-* 'god-abode = abode of the gods'), but all types of syntactic relations are possible. Morphologically, the second members of such compounds are very diverse, listed in 31.

- (31) Second Compound Members in Dependent Determinatives: Stress on Second Member
- as a pure verbal root, extended with suffix *-t-* when otherwise vowel-final (e.g., *varuṇa-dhrú-t-* 'deceiving Varuṇa');

- b. as the simplex noun itself (e.g., *jīva-loká-* ‘living being-world = world of the living’);
- c. as a verbal noun built with suffix *-á-* (e.g., *hasta-grbh-á-* ‘hand-grasping’);⁵⁴
- d. as a verbal noun built with suffix *-ana-* (e.g., *amitra-dámbh-ana-* ‘enemy-injuring’);
- e. as a verbal noun built with suffix *-van-* (e.g., *soma-pā-van* ‘soma-drinking’);
- f. as a verbal noun built with suffix *-man-* (e.g., *āśu-hé-man-* ‘swiftly speeding’);
- g. as a verbal noun built with suffix *-thá-* (e.g., *putra-kṛ-thá-* ‘son-making = procreation of sons’);
- h. as an adjective built with suffix *-ín-* (e.g., *uktha-saṃs-ín-* ‘praise-uttering’);
- i. as an adjective built with suffix *-i-* (e.g., *pathi-rákṣ-i-* ‘protecting the path’);
- j. as an action noun built with suffix *-ya-* (N) / *-yá-* (F) (e.g., *deva-hú-ya-* ‘god-invocation = invocation of the gods’);
- k. as an action noun built with suffix *-ā-* (e.g., *māmṣ-bhikṣ-ā-* ‘request for flesh’);
- l. as an agent noun built with suffix *-aka-* (e.g., *abhi-króś-aka-* ‘one who cries out’);
- m. as an agent noun built with suffix *-snú-* (e.g., *abhi-śocayī-ṣṇú-* ‘causing heat’);
- n. as an agent noun built with suffix *-tár-* (e.g., *ṇṛ-pā-tṛ-* ‘protecting men’);
- o. as an agent noun built with suffix *-ú-* (e.g., *rāṣṭra-dīps-ú-* ‘wishing to injure the country’);
- p. as a gerundive in *-íya-*, as well as *--anīya-* or *-énya-* (e.g., *nīvi-bhāry-íya-* ‘to be worn in a skirt’)

In all but the first two of these cases, one can identify a segmentally overt compositional suffix, which imposes the same type of stress as simplexes built with such affixes: these are evidently synthetic compounds of the structure [[X] [Y] Z], where Z is an accented or preaccenting derivational suffix (so Kiparsky 2010a: 175–6). The stress on the final stem syllable of the first two cases, with a pure verbal root or simplex noun, may be assumed to contain a segmentally null compositional suffix that results in the same sort of regular oxytone stress as among the *dvandva* compounds. In the form *jīvalokám* (RV 10.18.08a) for instance, the input is /jīvá-loká-/, which requires that either the underlying accent of the morphological head (/loká-/ ‘being-world’) is selected, or that an oxytone stress is imposed on the entire compound.

However, the stress pattern of another group of dependent determinatives instead usually shows stress on the first member (where it would bear stress as a simplex), but on the second member when the first member certainly has no lexical accent. Second members conforming to this pattern include the following:

(32) Determinative Compounds with 1st or Second Member Stress

⁵⁴Subsumed under this group may be a large number in which the second member corresponds to the a present stem built with a root, with truncation of the present stem’s theme vowel if thematic (Classes I, IV, VI, X); cf. Macdonell 1910: 160–1.

- a. verbal nouns with suffix *-ti-* (e.g., *devá-hūti-* ‘invocation of the gods’ vs. *ṛju-nīti-* ‘correct leading’)
- b. past participles with suffix */-tá-/* or */-ná-/* (e.g., *devá-hita-* ‘ordained by the gods’ vs. *uda-plutá-* ‘swimming in the water’)
- c. primary comparatives in *-īyaṃs-* (e.g., *prāti-cyavīyāṃs-* ‘pressing closer against’) and primary superlatives in *-iṣṭha-* (e.g., *ā-gamiṣṭha-* ‘coming quickly’)

The stress pattern seen in this group is identical to the stress pattern typical of most *bahuvrīhi* compounds (cf. 4.2.2): the leftmost lexical accent is realized as stress, or stress appears on the rightmost syllable when no lexical accents are present in the stem.⁵⁵ This behavior is evident from a rare example of a compound with two participles in *-ta-* /*kṛtá* + *dvistá-* → *kṛtá-dviṣṭa-* ‘hating what has been done’ (AVŚ), on the one hand, and a form with a */-ti-/* as a second member preceded by an unaccented first member, *ṛju-nīti-* (RV 1.90.01a) ‘correct leading’ ← */ṛju + nīti-* / on the other.

In other cases in which the second member is an independently existing adjective, and no external compositional suffix is added, stress in fact largely occurs on the first member of the compound instead, apparently selecting the leftmost lexical accent available, which usually happens to fall on the first member. This behavior fits with the behavior of the participles in */-tá-/* or */-ná-/*. Examples may be found for adjectives formed with all manner of derivational suffixes (see further examples in Macdonell 1910: 167).

(33) Dependent Determinatives with Adjectival 2nd Member

- a. */-u-/*: *máde-raghu-* ‘quick in exhilaration’ ← */máda-i-raghu-/*; but also *an-āśú-* ‘not swift’ ← */an-āśū-/*, where *āśú-* must be accented (cf. *āśú-heṣas-* ‘having neighing swift one (=horses); a name of the Ásvins’ (RV 8.10.12))
- b. */-á-/*: *suté-kara-* ‘active in (offering) pressed stuff (i.e., Soma)’ ← */sutá-i-kará-/*
- c. */-rá-/*: *tanú-śubhra-* ‘shining in body’ ← */tanú-subhrá-/*

A handful of instances in which the stress occurs on a first member that should otherwise be identified as unaccented, however, suggest that application of the BAP (specifically, DEFAULT ACCENT) *without* application of the OXYTONE RULE may have been permitted in determinatives without an adjectival second member. If so, then an ideal analysis should simultaneously be able to explain oxytonesis in *bahuvrīhi*-compounds of the type in 22 as well as coordinatives under 4.2.5.2, while also excluding default oxytonesis in determinatives with adjectival second members. Some relevant cases are given in 34.

(34) Determinatives with (unexpected) 1st Syllable Stress

- a. *tríaruṇa-* ‘Name of a man (red in three places)’ (RV 5.27.1d) and *tríaruṣī-* ‘red in three (parts of the body)’ (RV 8.46.22d) present stress on the first member */tri-/*, which must however be unaccented (INST.PL *tribhís*).

⁵⁵Since the primary comparatives and superlatives occur exclusively in composition with propositions and *śám-*, all of which are probably accented, there are no examples in which the lexical accent of the second member surfaces in this type.

- b. *sú-priya*- ‘very dear’ (AVŚ 7.38.2c); note that *su*- is consistently without stress in *bahuvrīhi*-compounds in which it is a first member (e.g., *su-mānas*-).
- c. *dúr-hita*- ‘faring ill, doing badly’; note that *dus*- is consistently without stress in *bahuvrīhi*-compounds (e.g., *dur-matī*- ‘weak-minded’), and may also be unstressed in descriptive determinatives (*dur-matī*- ‘bad disposition of mind, envy’), even in cases in which the second member is unaccented (like /ma-ti-/ ‘mind’)

The latter example, *dúr-hita*-, may be interpreted as an instance of the generalization that determinatives with a participle in /-tá-/ as the second member nearly always show stress on the first member. However, such a generalization would best be interpreted as a sort of *overgeneralization* arising from the fact that such compounds usually have stress on the first member due to the BAP (ACCENT SELECTION). But more to the point, the stress of all compounds built with the monosyllabic adverbs *a*- ‘not’ (see further below), *su*- ‘well’, and *dus*- is complex, and, at first glance at least, contradictory. Indeed, within Vedic the same compound is sometimes attested with different stresses (e.g., RV *a-bhrātár*- vs. AVŚ *a-bhrātár*- ‘having no brothers’), which likely indicates that different Vedic speakers at different points in time applied different morphology in the construction of these compounds. Poucha (1943: 137) lists ten more cases from the RV (four internal to the RV) and AVŚ of adjectives built with *a(n)*- with stems of at least three syllables that exhibit stress on either the first or final syllable of the stem.⁵⁶

Most difficult to explain are examples that present stress in a position that neither appears to follow from the lexical accents of the compound members nor can result from the OXYTONE RULE or DEFAULT ACCENT. One such instance is *vidmanāpas*- ‘working with wisdom’. Given that the second member is the *s*-stem adjective *apas*- and the first member is the noun *vidmánā*-, the types of compound accentuation seen thus far could produce ^X*vidmánāpas*- (leftmost lexical accent), ^X*vidmanāpás*- (oxytone stress), or perhaps ^X*vidmanāpas*- (leftmost syllable stress).⁵⁷ Such cases are sufficiently rare that they may be left aside (though their incorporation into a principled account would be desirable, if possible). Some similar-looking cases involving possessive compounds, in which the lexical accent of a second member would be expected to produce a compound with oxytone stress, but instead exhibit a sort of retraction of stress onto the penultimate syllable of the stem include *su-vīra*- ‘rich in heroes’ (cf. *vīrá*- ‘man’) and *su-gāndhi*- ‘sweet-smelling’ (cf. *gandhá*- ‘smell’; but also *su-gandhī*-).

***karmadhāraya-tatpuruṣa* Determinatives** Unlike the dependent determinatives, descriptive determinatives are comparatively rarely constructed with a compositional suffix of some kind; in the usual case, the second member of the compound is always a noun or adjective that may independently occur as such. Nevertheless, some descriptive determinatives are

⁵⁶The four forms in the RV are: *án-ugra*- ~ *an-ugrá*- ‘not strong, weak’; *á-prāśasta*- ~ *a-prāśastá*- ‘not praise-worthy’; *á-śrama*- ~ *a-śramá*- ‘tireless’; *á-saścat*- ~ *a-saścát*- ‘not following’.

⁵⁷In this particular case, the stress of the corresponding *s*-stem noun, *apas*- ‘work’ might conceivably play a role. In principle, the only way in which the stress of *vidmanāpas*- could be generated would be through a preaccenting compositional suffix /-as/, but just such a suffix builds nouns, not adjectives.

built with a suffix for a verbal noun, which acts as a compositional suffix and induces stress on the second member of the compound:

(35) Synthetic Descriptive Determinatives

- a. verbal noun built with *-van-* (e.g., *raghu-pát-van-* ‘flying swiftly’)
- b. verbal noun built with *-man-* (e.g., *vīlu-pát-man-* ‘flying mightily’)
- c. verbal noun build with suffix *-i-* (e.g., *tuvi-śván-i-* ‘roaring mightily’)
- d. pure verbal root (e.g., *īśana-kṛt-* ‘acting as a ruler’)

Stress on the final or penultimate syllable of the stem thus represents the general pattern when the second member is a verbal noun.

Conversely, in many RVic examples of descriptive determinatives, ACCENT SELECTION appears responsible for generating the stress, yielding stress on the first member or a non-oxytone stress on the second member, if the second member is an adjective or substantive. Forms such as *úlūka-yātu-* ‘owl-demon’ (cf. *úlūka-*) and *pūrṇā-mas-* ‘full moon’, *stóma-taṣṭa-* ‘fashioned as a hymn of praise’ (cf. *stóma-* ‘praise-hymn’) fit this pattern. The consistency with which descriptive compounds built with privative *a(n)-* exhibit stress on the initial syllable suggests that /á(n)-/ is accented and receives stress according to ACCENT SELECTION (cf. Macdonell 1910: 94; e.g., *á-kṛta-* ‘not done’, *á-citti-* ‘thoughtlessness’). By the same token, descriptive compounds with *duṣ-* ‘bad’ or *su-* ‘good’ as a first member are but rarely accented in the RV; some instances with clearly unaccented second members exhibit oxytone stress (e.g., *du-ṣtutí-* ‘bad song of praise’, *su-matí-* ‘good mind; kindness’).⁵⁸

Somewhat puzzling, however, are some forms with stress on the first member that does not conform to the stress of the simplex, e.g., *madhyám-dina-* ‘midday’ (cf. *mádhyā-* ‘middle’), as well as all descriptive determinatives with *vísva-* ‘all’ as a first member, e.g., *vísva-mānuṣa-* ‘every man’. Note, however, that the same peculiarity holds for *bahuvrīhis* will *vísva-* as a first member. In effect, the compositional form of *vísva-* has a different lexical accent from the simplex.⁵⁹

⁵⁸With *duṣ-*, the only two examples where the initial syllable is stressed are: *dúṣ-vidvas-* ‘of evil intentions’ (RV 7.94.12b) and *dúr-hita-* ‘badly placed, in a bad situation’ (RV 8.19.26d). With *su-*, many more forms in the RV attest initial stress: *sú-gmīya-* ‘bringing luck’, *sú-grathita-* ‘well-connected’, *sú-taṣṭa-* ‘well-fashioned’, *sú-dhita-* ‘well-placed’, *sú-nīti-* (alongside *su-nīti-* and *su-prānīti-* ‘leading well’), *sú-pūta-* ‘well-explained’, *sú-pūrṇa-* ‘well-filled’, *sú-prīta-* ‘very pleased’, *sú-bhṛta-* ‘well-carried’, *sú-mahat-* ‘very noble’, *sú-mita-* ‘well-measured’, *sú-miti-* ‘good establishment’, *sú-yata-* ‘well-directed’, *sú-yukta-* ‘well-yoked’, *sú-vidvas-* ‘knowing well’, *sú-vipra-* ‘praying seriously’, *sú-śiṣṭi-* ‘good instruction’, *sú-śrāta-* ‘well-cooked’, *sú-śruta-* ‘very famous’, *sú-ṣuta-* ‘well-pressed’, *sú-ṣuti-* ‘easy birth’ (for **sú-sūti-*), *sú-ṣūta-* ‘well-generated’, *su-ṣtutá-* ‘well-praised’, *sú-saṃśita-* ‘well-sharpened’, *sú-saṃskṛta-* ‘nicely equipped’, *sú-samidha-* ‘well-ignited’, *sú-samubdha-* ‘well-bound’, *sú-sampiṣṭa-* ‘completely smashed’, *sú-sammṛṣṭa-* ‘well-groomed’, *sú-saha-* ‘nice connection’, *sú-snuṣa-* ‘having beautiful daughters-in-law’, *sú-huta-* ‘well-poured’, *sú-hotṛ-* ‘sacrificing well’. At least among descriptive compounds having a past participle as a second member, stress on *sú-* is the normal pattern in the RV, if not without exception totally; stress on *sú-* when the second member is a *ti*-stem is possible, if not predominant; finally, stress on *sú-* with other types of second member is exceptional.

⁵⁹I would cautiously suggest that the lexical accent of the compositional form is original, the simplex perhaps having undergone retraction due to demorphologization and initial stress in “demorphologized forms”.

To some extent already in the RV, descriptive determinatives exhibit an increasing tendency to place stress on the rightmost syllable of the stem, as in coordinatives, regardless of the lexical accents of the compound members. Compare the descriptive *hiraṇya-piṇḍá-* ‘lump of gold’ with the *bahuvrīhi hiraṇya-keśa-* ‘having golden hair’. Macdonell (1910: 93) characterizes this oxytone stress as the regularly expected pattern in descriptive determinatives, which then constitutes the rule post-RV, with the exception of 1) some synthetic descriptives with a preaccenting suffix, 2) privatives in *a-* with initial stress, and 3) forms with a past participle in *-tá-* as second member. The accentual behavior of descriptive determinatives overall best fits an original behavior similar to possessives, with stress determined primarily by ACCENT SELECTION, later preferring a null compositional suffix, but preserving remnants of the older rule in exceptional morphological stipulations.

4.2.5.4 Possessive Compounds

The stress of exocentric possessive compounds (*bahuvrīhis*) has already been addressed at the end of Section 4.2.2: stress falls on the leftmost lexically accented syllable, otherwise at the right edge of the stem. See examples under 21. Like descriptive compounds, possessives do occasionally appear to be built with an overt compositional affix, thereby conditioning stress on the final or penultimate syllable of the stem, rather than on the leftmost lexical accent (compare examples given by Kiparsky 2010a: 173).

The only appreciable exceptions to this behavior are found among possessives with either the privative *a(n)-* or a numeral as the first member. In the case of privative *a(n)-*, in contrast to descriptive compounds with such first members, stress most commonly appears on the final syllable of the stem, regardless of the lexical accent of the second member; see examples in 36 and contrasting examples with stress on the first member in 37). Similarly, compounds with a numeral as a first member (e.g., *dví-* ‘two’, *trí-* ‘three’, *pāñca-* ‘five’) frequently show oxytone stress as well, though in some instances, this oxytone stress may be due to a compositional suffix *-á-* (e.g., *tri-vandhurá-* ‘three-seated’, *pañca-yojaná-* ‘distance of five Yojanas’; cf. Macdonell 1910: 174). In yet other cases, the second member receives stress by ACCENT SELECTION in accord with the unaccentedness of the first member (e.g., *dvi-dhára-* ‘forming two streams’, *tri-nábhi-* ‘having three naves’). In principle, all instances of oxytone stress in these cases can be generated via deaccentuation as in coordinatives; the fundamental problem is to establish when and why deaccentuation of the stem and oxytonesis ought to apply and when not. I regard this issue as going somewhat beyond an adequate description of Vedic prosody; important is that the phonological grammar be capable of generating the patterns, even if the relevant morphological conditions are not fully understood or idiosyncratic.

(36) Possessive Compounds with Privative *a(n)-* and Oxytone Stress

- a. *a-phalá-* ‘unfruitful’ (cf. *phála-* ‘fruit’)
- b. RV *a-bhrātár-* ‘brotherless’ (cf. *bhrātár-* ‘brother’; but also *a-bhrātár-* in the AVŚ)
- c. *a-balá-* ‘not possessing strength’ (cf. *BÁLA-* ‘strength’)
- d. *a-bandhú-* ‘having no relatives’ (cf. *bándhu-* ‘relative’)

(37) Possessive Compounds with Privative *a(n)*- and Non-Oxytone Stress

- a. *a-vīra*- ‘childless’ (cf. *vīrá*- ‘man, hero’)
- b. *á-kṣiti*- ‘imperishable’ (cf. *kṣití*- ‘ruin, destruction’)
- c. *a-śéśas*- ‘without offspring’ (cf. *śéśas*- ‘offspring’)
- d. *á-mṛtyu*- ‘deathless’ (cf. *mṛtyú*- ‘death’)

4.2.5.5 Governing Compounds

Comparatively infrequent in the earliest Vedic are compounds that syntactically have the structure of a prepositional phrase or verbal phrase with direct object; the head of the compound (i.e., preposition or verb) always occurs on the left. The latter sort (VPs) presents three subtypes, depending upon the form of the first member: an active participle in its weak stem (e.g., *tarád-dveśas*- ‘overcoming enemies’), a thematic present stem (e.g., *trasá-dasyu*- ‘terrifying the foes’), or a verbal noun in *-ti*- (e.g., *púṣṭi-gu*- ‘rearing cattle’).

Among the prepositional governing compounds, stress occurs either on the preposition, on the syllable that bears stress in the case of disyllabic prepositions (e.g., *ánu-path-a*- ‘along the path’) or on an ostensible compounding suffix, virtually always *-a-* or *-ya-* (e.g., *ati-rātr-á*- ‘lasting overnight’, *prati-kām-íya*- ‘being in accordance with desire’). Both of these stress patterns are readily compatible with types of compound stress already discussed above: the former represents selection of the leftmost lexical accent, the latter represents the accentual dominance of the outermost derivational suffix. That the former stress pattern is found when no compounding suffixes is added is to be expected, e.g., *áti-avi*- ‘going over the wool’ (*á*- ‘sheep’), *á-pathi*- ‘on the path’ (*páthā/i*- ‘path’), *upári-martya*- ‘being above men’ (*mártya*- ‘mortal’). Conversely, one might have expected for the compounding suffix to always determine the stress when present, though such is evidently not always the case: a form such as *tiró-ahn-íya*- ‘being beyond a day’ exhibits stress on the prepositional head rather than the apparent compositional suffix *-ya-*. To bring such cases into accord with the general analysis of compound stress, one must assume either additional unaccented allomorphs of the suffixes *-a-* and *-ya-*, or accept that the second member does not represent a proper synthetic compound, but rather an alternative derivative in *-a-* or *-ya-* unattested as an independent noun in early Vedic.⁶⁰

The verbal governing compounds, meanwhile, almost never contain a compounding suffix (*śikṣa-nar-á*- ‘helping men’ being one rare exception), and always show stress on the first member of the compound, regardless of which type of first member is found. Given that the stem-forming suffix of Class VI presents must be lexically accented */-á-/*, that examples all present stress on that suffix vowel is unsurprising.⁶¹ Somewhat more remarkable is the

⁶⁰The second option may well represent the correct choice, since the second member sometimes is ultimately attested as an independent form in some cases. Possibly some speakers made the inference from synthetic governing compounds in *-a-* or *-ya-* that the second member should be a noun built with such a suffix, and so productively derived such nouns to serve in a compound of the structure [[X] [Y]].

⁶¹Whether the suffix */-á-/* is indeed present in these cases is not entirely clear. The form *trasá-dasyu*- mentioned above is the only example of this type in which the alternative accentuation on the first syllable of the first member would be possible (cf. Macdonell 1910: 176, and the root $\sqrt{tras}$ ‘tremble’ indeed builds a Class I

consistent stress on the participial suffix *-át-*, which, in its weak form, is usually unstressed, ceding stress to lexically accented case endings; compounds such as *bharád-vāja-* ‘carrying the prize’ might thus support treating (an allomorph of) the active participle *-ant-/at-* as originally lexically accented.⁶² Most remarkable of all are the governing compounds with *ti-* stem first members: as discussed at length above, other derivatives to such nouns, as well as *bahuvrīhi*-compounds, clearly demonstrates that such stems generally possess unaccented stems. Nevertheless, four of the six types found in the RV shows stress on the suffix *-ti-*.⁶³ Given that the entire compound is inflected, not both members (contrast the formation of iterative compounds under 4.2.5.1 above), assuming a prosodic structure [[ω] [ω]]_{ω-Max} for governing compounds generally seems incongruous. To all lights, the stress assigned via the OXYTONE RULE in the *ti*-stem appears to be inherited by the compound, but how the available stress rules could license such behavior is unclear; at 4.3.6 below, I propose that stress assigned to the morphological head of a word can be inherited by the head of a compound as well.

4.2.5.6 Summary on the Accentuation of Compounds

From the point of view of stress, Vedic compounds may be sorted into the following groups:

1. Compounds that apply the lexical accent of a segmentally overt compositional suffix (e.g., most descriptive compounds with a verbal noun as second member, a handful of *bahuvrīhis*, prepositional governing compounds).
2. Compounds that select for the leftmost lexical accent in the input (many *bahuvrīhis*, all iteratives, originally determinatives with a nominal or adjectival head, possibly verbal governing compounds).
3. Compounds that apply oxytone stress in the absence of any lexical accents (some *bahuvrīhis*, some descriptive compounds).
4. Compounds that ignore the lexical accents of the stems and instead regularly assign oxytone stress (*dvandvas*, some *bahuvrīhis*, an innovative pattern in determinatives with nominal head).

present with stress on the root, thus *trása-*. If indeed the type in *-át-* is historically secondary to the type in *-á-* (see the following note), the fact that the root in the latter type exhibits zero grade, where possible (e.g., *rdhád-ray-* ‘increasing wealth’, *vidád-vasu-* ‘winning goods’) would support reconstructing the Indo-European present suffix **[é]/[o-]* for the formation of verbal governing compounds.

⁶²Given that such compounds with a participle as a first member are unknown in older Indo-European languages outside Indic and Iranian, whereas the type with a present stem as first member is paralleled (e.g., in Greek, the type *φέρέ-οικος* [p^heré-oikos] ‘carrying its house = snail’) some scholars have proposed that the participial governing compounds are historically younger, emerging from reanalysis of cases in which a *-t-* was inserted after the final vowel of the first member as containing the active participle suffix; see Renou 1940: 217–20, Lowe 2015: 270.

⁶³The first members *rīti-* ‘streaming’, *vīti-* ‘enjoying’, and *vṛṣṭi-* occur as independent nouns, with stress on the suffix as expected; a first member *dāti-* is otherwise unattested, and the first member *púṣṭi-* shows a stress contrary to the independent noun *púṣṭi-* ‘raising’, but occurs there in a personal name *púṣṭi-gu-*, which might be subject to stress retraction in names.

Patterns 2 and 3 may be unified as stress patterns that apply the rules of stress calculation for nominal accent more generally; these compounds do not obviously diverge from general rules of stress assignment. Patterns 1 and 4, meanwhile, are explicable by reference to accentual dominance as a rule that erases all lexical accents to the left of a morphological element with the feature [+dominant]: the synthetic compounds with overt derivational suffixes thus realize their lexical accents, while types with regular oxytone stress follow from application of the oxytone rule to a stem whose lexical accents have been erased (or are ignored). All of these processes, perhaps with the exception of segmentally null dominant affixes, are paralleled in stress assignment in the inflectional and derivational morphology of the language.⁶⁴

To a substantial extent, then issues regarding the accentuation of compounds are perhaps best regarded as matters of word formation and syntax than as matters of phonology. In large part, inasmuch as a particular morphological structure or syntactic relations between the compound members can be justified, the particular stress seen in the compound can be justified as well.

4.2.6 Summary

The Kiparskian analysis of Vedic accentuation may be qualitatively assessed as largely successful, insofar as it is capable correctly reproducing the attested data in many cases and making non-trivial predictions about novel data. That is, it is not necessarily to lexically specify the stress of every distinct lexeme — an approach that would perfectly fit the learning data, but be incapable of generalizing to novel data — nor to classify lexemes into different arbitrary categories associated with particular stress patterns. Rather, the stress patterns are emergent from the components assumed to be present in a given word form. In this summary, I distinguish between aspects of the Kiparskian analysis that appear to be very successful in correctly explaining some part of the data from aspects that seem less certain.

1. Prosodic Specification in Inflection. Minimally, a contrast between stress-attracting (“accented”) and stress-neutral (“unaccented”) morphemes is necessary to produce genuine MOBILE patterns in inflectional paradigms. Whether stress-rejecting (“preaccenting”) inflectional morphemes are also necessary is less clear, because the BAP suffices to generate initial stress on its own; the suffixes of the perfect active may constitute the best candidates for preaccenting inflectional morphemes. At minimum, however, the stem *nṛ-* ‘man’ does not fit into this binary distinction since it behaves partly like a stress-attracting stem (GEN.SG *náras*, like *gávas*), but not in the GEN.PL (*narám*; contrast *gávām*), and the overall rarity of stress-attracting stems means that it is difficult to say with certainty whether more stems with the same behavior as *nṛ-* existed or not.
2. Prosodic Specification in Suffixing Derivation. Minimally, a contrast between stress-attracting (“accented”) and stress-rejecting (“preaccenting”) derivational suffixes is

⁶⁴Compare, however, Kiparsky (2010a: 166–70) on so-called “internal derivation”, whereby two segmentally identical nominal stems with different accentuation are treated as derived from one another.

necessary to generate word classes with regular stress on the suffix, on the one hand, and word classes with regular stress on the preceding syllable to the left, on the other. Stress-neutral derivational suffixes almost certainly exist as well, given the propensity of certain derivational suffixes to cede stress to further derivational suffixes and/or to stress-attracting roots or prefixes. However, the confounding effects of the OXYTONE RULE and ACCENTUAL DOMINANCE make the choice of analysis in specific cases unclear.

3. THE BASIC ACCENTUATION PRINCIPLE (cf. 8 and 9). A word-level preference for stress towards the left edge of the word securely explains at least three major stress patterns in Vedic: 1) genuine stress mobility in the inflectional paradigms of root formations, where the root is unaccented, but immobility where the root is accented; 2) stress that corresponds to the leftmost lexical accent in exocentric possessive compounds (cf. Section 4.2.5.4 and example 21); 3) regular stress on the first compound member in iterative compounds (cf. 4.2.5.1).
4. Secondary Mobility (cf. 17). The rightward re-association of de-linking lexical accents in patterns like *pitáras* : *pitré* or *arím* : *aryás* makes empirically correct predictions. The rule itself is, however, stipulative, and appears at odds with the general preference, encoded in the BAP, for stress to tendentially occur towards the left edge of the prosodic word.
5. THE OXYTONE RULE. In the inflection of polysyllabic nominal and adjectival stems, including compounds, stress always remains on the stem: there are no inflectional paradigms with persistent stress on inflectional suffixes, and with the exception of the two nominal stems *púmaṃs-* ‘man’ and *pánthā-* ‘path’, no polysyllabic stems exhibit mobility of stress between the stem and inflectional suffixes. The fact that Kiparsky explicitly ties the application of this rule simply to polysyllabicity, however, results in further overgeneration among polysyllabic verbal stems.

At a broader systematic level, the rules and representations proposed by Kiparsky are very successful in describing stress in both the inflection and derivation of nouns and adjectives, including most varieties of compounds. Among nouns and adjectives, the OXYTONE RULE makes almost entirely correct predictions, and a distinction between accented and unaccented inflectional suffixes would suffice. ACCENTUAL DOMINANCE amounts to the imposition of the stress preferred by an accented or pre-accenting suffix; all derivational suffixes, if accented or preaccenting, could be treated as dominant, while unaccented outermost derivational suffixes would only potentially surface with stress via the OXYTONE RULE. Only potentially problematic would be certain compounds (mainly *bahuvrīhis* with unaccented monosyllabic first members, but also some determinative compounds with nominal heads) that appear with stress on the last syllable of the stem, where it appears that the OXYTONE RULE has overridden a lexical accent closer to the left edge of the word (cf. Kiparsky 2010a: 173–4).⁶⁵ In short, stress in nouns and adjectives is largely predictable, and can func-

⁶⁵ Insofar as the accent of the outermost derivational suffix is assumed to always win, such compounds might

tion well even with less lexical specification (no preaccenting inflectional suffixes, no dominant/recessive distinction).

Among verbs (including the active participle in *-ant-*), the system of rules and representations functions less well, mainly for the reasons discussed in detail under 4.2.3: many polysyllabic verbal stems must effectly be exempted from the OR, but accounting for their pattern of stem-edge-to-ending mobility (e.g., 3SG.NPST *punāti* : 2PL.NSPT *punītā* ‘purify’) results in the introduction of preaccenting inflectional suffixes, among both verbal and nominal inflectional affixes. As discussed above, the use of preaccenting suffixes with non-syllabic endings (e.g., 3SG *-t*) is questionable, and depends on lack of specificity surrounding the representation of preaccentedness. At all events, allowing for the stress in a form like *sunāt* ‘press:3SG to be generated from /su-nā-’t/ is not compatible with any of the possibilities for the representation of lexical accent discussed at 2.4.2.2.

4.3 Vedic Stress in Optimality Theory: Faithfulness to Underlying Foot Edges

The analysis of Vedic stress assignment developed by Kiparsky presented above obtains large amount of empirical coverage; inasmuch as one can accept its machinery and assumptions, it can successfully generate most types of stress pattern found in Vedic. Nevertheless, the aim to provide a great degree of empirical coverage has left aside a number of theoretical questions, and the overall model arguably includes a number of redundancies and complications, that, with some changes in terms of representations and analytical model, might be treated more straightforwardly. The two major differences of approach here will be to operate with lexically specified foot boundaries (following Inkelas 1994, Özçelik (2014), Yawney 2018, and Yates 2020, *inter alia*), rather than abstract lexical accents, and with constraints that refer to structure inherited from the morphology (i.e., differences between heads, stems, and inflection), rather than explicitly interleaving the application of phonological processes and the construction of morphological structure. In contrast to the analysis proposed in Sandell 2015a: Ch. 7, the employ of harmonic grammar and constraint ganging proves unnecessary, given the of prosodic faithfulness argued to be independently necessary.

The crucial innovations of the analysis to be presented here, in comparison to similar analyses proposed in Sandell 2015a: Ch. 7 and Yates 2020, are the following:

- a wider array of markedness constraints concerning metrical structure is employed; that is, the well-formedness of foot structure, rather than merely the position of the most prominent syllable, is considered in greater detail.

be taken to have a segmentally null compounding suffix with a prosodic specification that attracts stress to the right edge of the stem (e.g., floating /H*/ , foot head, or similar). Given the independent existence of compounding suffixes that attract stress in Vedic, such an assumption is entirely possible and furthermore does not differ much from Kiparsky’s own assumption of “dominant zero compositional suffixes”. The difference would be that such compositional suffixes would not be *prosodically* null: their H* would be preferentially realized, just like the H* sponsored by an accented derivational suffix. Preventing this floating H* from associating elsewhere, other than the right edge of the stem, would still require explanation.

- stem-edge stress in nominal formations – including certain compounds – is argued to result in from the need to best satisfy conflicting preferences concerning the position of stress with respect to the leftmost or rightmost syllable, for stress to fall on a syllable peak projecting material belonging to the morphological head of the word, for the right edge of the word to be aligned with the right edge of a foot, and in some instances, to satisfy the anchoring of foot edges.
- a lexical distinction between a total of four types of prosodic constituent edges.

This analysis claims to account for the same set of facts handled well by the Kiparskian analysis, but succeeds in treating them in a single parallel constraint-based grammar, and does without any processes of accentual “erasure”.⁶⁶ Furthermore, I contend that the absence of persistent stem stress in many primary verbal formations is due fundamentally to differences in morphological construction: virtually all polysyllabic stems, whether nominal, adjectival, or verbal should be assumed to contain a category-specifying head. This morphosyntactic property, which applies to virtually all word-formation suffixes, justifies the employ of constraints of the type HEADFAITH and HEADSTRESS, as originally developed by Revithiadou (1999). The usage of such constraints, when combined with appropriate lexical representations, not only suffices to capture basic patterns of stress mobility in inflection, dominance effects in derivation, and “secondary mobility” (as variously achieved by Kiparsky, Sandell, and Yates), but also provides a coherent specification of preaccenting morphemes.

Finally, outfitting the present analysis with a fuller array of prosodic markedness constraints is of crucial importance for further investigation into the later history of stress assignment in Sanskrit, to be undertaken in the following chapter. Namely, since stress in Classical Sanskrit is fully predictable and metrically determined, understanding which metrical constraints interact to generate the “default” stress of Vedic – or more precisely, what stress pattern would be predicted under which prosodic conditions, in the absence of lexical accents – can serve to more clearly bridge the gap between Vedic stress prosody and Classical Sanskrit stress prosody.

Before beginning with the details of the optimality theoretic analysis, a brief review of the distinct types of stress patterns found in Vedic inflectional and derivational morphology, so that the success of the overall model can, in part, be assessed in terms of how many patterns in can explain without reference to special lexical properties (i.e., lexical constraints or special listed allomorphs). Each pattern is given a name, each subtype within the more

⁶⁶The one point on which the expected stress cannot be readily reproduced concerns the stress of unaccented *bahuvrihi*-compounds, and similar compounds without any overt compounding suffix that appear with stress on the rightmost syllable of the stem. As it turns out, the expected behavior can be generated if the stem of the compound is three syllables or longer, and it is assumed that a segmentally null compounding suffix corresponds to a left foot boundary inserted between the stem and inflectional ending. A form like *śītipādas* ‘having white feet:NOM.PL (cf. 22.b.) would thus have the underlying structure [[[[śīti] [pad]] (∅)] as]. Alternatively, if the a segmentally null compounding affix can somehow mark the final stem syllable as a foot head (i.e., effectively create a structure [[[[śīti] [(pad)] ∅] as]), then such stem-edge stress in compounds poses no issues whatsoever.

general patterns is exemplified with a single stem, and the type frequency of the subpatterns is anecdotally observed.

1. Inflectional Accentuation

- (a) Genuine Mobility: stress may be attracted from a stem to one or more inflectional inflectional suffixes
 - i. Monosyllabic Full Mobility: stress falls on all stress-attracting inflectional suffixes; type *pádam* ~ *padé* or *átti* ~ *adánti* — reasonably well-attested.
 - ii. Monosyllabic Limited Mobility: stress attracted only to GEN.PL -*ám*; type *náram* ~ *narám* — limited to the stem *nṛ-* ‘man’.
 - iii. Polysyllabic Full Mobility: stress falls on all stress-attracting inflectional suffixes, else the first syllable of a disyllabic stem; type *púmāṃsam* ~ *pumśi* — limited to the two stems *púmāṃs-* ‘male’ and *páthā-* ‘path’.
- (b) Secondary Mobility: stress shifts one syllable to the right when a syllable peak hosting stress is lost
 - i. Type 1: glide formation provokes mobility; type *arím* ~ NOM.PL *aryás* ‘friend’ — limited to this one stem.
 - ii. Type 2: deletion of the first vowel in a disyllabic stem results in mobility; type *sánu* ~ *snúṣu* ‘back’ — limited to *sánu-* ‘back’ and *dāru-* ‘wood’.
 - iii. Type 3: deletion of a vowel preceding a stem-final sonorant results in mobility to a vowel-initial inflectional suffix; type *pitāram* ~ *pitré* ~ *pitṛbhis* ‘father’ — reasonably well-attested.
- (c) Oxytone Limited Mobility: stress remains on the rightmost syllable of the stem, except in the GEN.PL; type *matím* ~ *matíbhis* ~ *matīnām* — reasonably well attested.
- (d) Exceptional Mobility (“true hysterokinetic”): stress shifts between the penultimate syllable and final syllable in polysyllabic stems, even where the rightmost syllable of the stem contains a vocalic peak able to host stress.
 - i. Active Participle: stress falls on allomorph -*ánt-*, or on allomorph -*át-* before consonant-initial inflectional endings, but on inflectional suffixes if vowel-initial (e.g., -*at-é*) — found just with the active participle suffix -*ant-*.
 - ii. Polymorpehmic Finite Verbs (Nasal Presents, Optative): stress falls on final syllable of the stem, but shifts to any and all inflectional suffixes that appear with stress in the type *átti* ~ *adánti* — found with Class V (-*nó* ~ -*nu-*) and Class IX nasal presents (-*ná-* ~ -*nī-*), as well as the optative in -*yá-* ~ -*ī-*.

- (e) Inflectional Immobility: stress remains on the same syllable / morpheme in all inflected forms.
 - i. Monosyllabic immobility: stress always falls on the root; type *gávam* ~ *gavé* or *táṣṭi* ~ *tákṣati* — attested for perhaps less than a dozen nominal or verbal roots.
 - ii. Polysyllabic immobility: stress always falls somewhere on the stem, usually as determined according to dominant derivational accentuation.

2. Derivational Accentuation

- (a) Dominant Accentuation: outermost derivational suffix determines the stress of the stem and excludes genuine paradigmatic mobility; in further derivation, these suffixes cede stress to other dominant suffixes, but not to recessive derivational suffixes.
 - i. Accented Dominance: stress occurs on the syllable corresponding to the derivational suffix; type *devá-* ‘god’ or *devatvá-* ‘godliness’ — generally very common.
 - ii. Preaccenting Dominance: stress occurs on the syllable immediately preceding the syllable corresponding to the derivational suffix; type *máнас-* ‘mind’ or *bandhútā-* ‘relation’ — reasonably well-attested.
- (b) Recessive (Oxytone) Accentuation: stress may occur on such a derivational suffix only if it is the outermost suffix and not preceded by an accented or preaccenting prefix, root, or other derivational suffix to its left; genuine paradigmatic mobility does not occur except to the GEN.PL suffix *-ām*.
 - i. Root derivation: type *matí-* ‘mind’ — reasonably well attested.
 - ii. Stem derivation: type *arcimánt-* — reasonably well-attested.

3. Compound Accentuation

- (a) Synthetic Compounds: stress of the compound is determined by an overt derivational suffix scoping over the compound; to be understood as a type of dominant derivational accentuation.
 - Typical of descriptive compounds and prepositional governing compounds.
- (b) Leftmost Wins: Stress occurs on the leftmost of any suffixes that trigger dominant accentuation.
 - Typical of *bahuvrīhis*, iteratives, less commonly on determinatives with nominal or adjectival stress.

- (c) Oxytone Compounds: stress occurs on the rightmost syllable of the compound, without any overt compounding affix.
- i. Monosyllabic Second Member: a comparatively rare type of *bahuvrīhi* or descriptive compound.
 - ii. *dvandvas*
 - iii. *bahuvrīhis* with first-member *a-*
 - iv. Determinatives with nominal head

In the following, a single constraint ranking will, given appropriate representations, be able to account for most of the types enumerated above. Outstanding problems will remain, however, with instances of limited paradigmatic mobility (i.e., where stress shifts only to the GEN.PL ending *-ām*), excessive mobility, and compounds with oxytone stress. In the first two cases, I will argue that the historical background of the relevant derivational suffixes provides some insight, and motivates the inclusion of a special lexical constraint in the case of the GEN.PL and the use of multiple listed allomorphs with different prosodic properties in the case of excessive mobility. As regards compounds with oxytone stress, I follow the reasoning that, since some such compounds (specifically, all except those of the type *śiti-pád-*) necessitate the employ of a segmentally null compounding affix that behaves as a morphological head, the few *bahuvrīhis* whose stress Kiparsky attributes to the operation of the OXYTONE RULE (i.e., like *siti-pád-*) may be handled in the same fashion. Through these means, complete descriptive coverage of Vedic stress assignment can be achieved.

4.3.1 Constraints, Lexical Representations, and Genuine Mobility

The analysis of Vedic stress assignment to be explicated in the following sections consists a handful of standard prosodic markedness constraints governing foot structure, supplemented with specific prosodic faithfulness constraints, and one further markedness constraint militating in favor of the marking of stress on morphological heads. Since lexical accent will be represented as underlying foot structure, following the proposal of Yates 2020, these additional prosodic faithfulness and markedness constraints will belong to the ANCHOR and ALIGN families, which demand faithfulness in the position of underlying prosodic edges relative to vocalic peaks, or call for the alignment of given morphological constituent edges with prosodic constituent edges. The constraints that will be encountered are listed in 4.2, divided in four classes: prosodic markedness, prosodic faithfulness, morphology-prosody interface constraints, and other constraints on segments; the constraints relating to the morphology-prosody interface are also classed markedness and faithfulness constraints. Ultimately, the powerful role of morphology-prosody interface constraints in the grammar means that few prosodic markedness constraints are necessary overall, and play but a marginal role in the grammar.

Two brief remarks on these constraints and the candidates to be considered are necessary. First, the constraints of the ANCHOR family are to be understood as two constraints each, that is, one holding over left foot boundaries and one holding over right edges. Second,

Prosodic Markedness	Prosodic Faithfulness	Morphology-Prosody Interface	Other
CULMINATIVITY TROCHEE ALL-Ft-L ALL-Ft-R FTBIN NONFINALITY- σ PARSE- σ ALIGN($\mathcal{F}$, L, σ_{Hd} , L)	ANCHOR-L/ R_f ⁶⁷ ANCHOR-L/ R_f -HEAD	ANCHOR-L/ R_f -HEAD ALIGN($\mathcal{F}$, L, σ_{Hd} , L)	*VV MAX-V

Table 4.2: Constraints Employed in the Optimality-Theoretic Analysis of Vedic Lexical Stress

feet in Vedic appear to be “truly quantity insensitive” (in the sense of Kager (1992)), meaning that FTBIN and PARSE- σ strive to create consistently disyllabic feet, with the constraints WSP and RH-CONTOUR (generally excluded here) evidently being so low ranked that candidates containing feet of the shapes (‘HH), (‘LH), or (‘HL) will be preferred to candidates containing monosyllabic but bimoraic feet of the shape (‘H). Finally, the fact that a ranking ALL-Ft-L/R $\gg$ PARSE- σ obtains in the grammar means that only a single foot per prosodic word will be constructed. To date, no unambiguous evidence for foot structure driving segmental processes in Vedic has been uncovered, and thus only a single foot per ω can and should be assumed.

As in the exposition of Vedic stress patterns under Section 2 above, I begin with a treatment of the patterns of genuine stress mobility (Type 1.1) found in nouns and verbs with morphologically simplex stems, in the vast majority of cases constituting a single syllable. Rather than operating with lexical accents in the form of abstract prominence or prespecified lexical high tone, will regard certain morphemes containing at least one potential syllable peak (normally a vowel) as being prespecified for foot structure, with a left or right foot boundary. Morphemes marked with a left foot boundary are, in most instances, functionally equivalent to accented morphemes, while morphemes marked with a right foot boundary are typically equivalent to preaccenting morphemes, though the later sort will first be introduced in detail under 4.3.4. Thus, given the list of unaccented and accented inflectional endings in the noun and verb from (10) and (11), the lexical representation of the accented morphemes may be rewritten as, e.g., DAT.PL /-(e/ or NPST.3PL /-(anti/.

Just as mobility to the final syllable in paradigms with genuine stress mobility was attributed to the lexical accent of certain inflectional endings above, here final stress is fundamentally due to faithfulness to the position of the foot boundary: the position of a foot boundary should not be shifted away from its position in the input. Following Yates (2020: 259–60), the relevant constraint belongs to ANCHOR family, extended here to apply to both left and right foot boundaries, defined in 38. The very similar ANCHOR constraint holding over the head of a morphologically complex word is also given here, in 39, but its role and necessity in the grammar will first be motivated in more detail below.

(38) ANCHOR-L/ R_f : The left/right edge of every foot i in the input corresponds to the

left/right edge of a foot i in the output. Assign one violation (*) if any syllable peak intervenes between f_i in the input and f_i in the output.⁶⁸

- (39) ANCHOR-L/R-HEAD _{f} : The left/right edge of a foot i in the input sponsored by a morpheme that is a category-determining head of a morphological word corresponds to the left/right edge of a foot i in the output. Assign one violation (*) for each syllable peak that intervenes between f_i in the input and f_i in the output.

Nominal paradigms that exhibit full stress mobility (Type 1.1.1), of the type ACC.SG *pádam* : LOC.SG *padí*, mainly demonstrate that the prosodic faithfulness constraint ANCHOR-L _{f} must dominate nearly all prosodic markedness constraints with the exception of TROCHEE and CULMINATIVITY. Given a root or stem like /pad-/ , unmarked for any prosodic constituency, either a disyllabic trochaic foot is constructed, as in 40, or when an accented inflectional ending like LOC.SG /-(i/ is present, ANCHOR-L _{f} decides the shape of the foot and the position of primary stress. Indeed, the winner [pa.(‘di)] in 41 is very telling, since it entails violations of several lower-ranked markedness constraints (FTBIN, NONFINALITY- σ , ALL-Ft-L, and PARSE- σ), whereas the competing candidate [(‘pa.di)] violates only ANCHOR-L _{f} . Justification for the articulated ranking of the prosodic markedness constraints will become clear in the course of this section.

- (40) /pad-am/ ‘foot-ACC.SG’ → [(‘pa.dam)]

/pad-am/	TROCHEE	ANCHOR-L _{f}	FTBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  (‘pa.dam)							
b. (pa.‘dam)	*!						
c. (‘pa).dam			*!			*	*
d. pa.(‘dam)				*!	*		*

- (41) /pad-_i(i/ ‘foot-LOC.SG’ → [(pa._i(‘di)]

⁶⁸ A non-categorical version of this constraint, allowing for multiple violations in case multiple syllable peaks intervened between f_i in the input and f_i in the output was also tested, but proved to produce no substantively different outcomes. Thus, this constraint is violable only up to the number of underlying foot edges in the input.

/pad-(i/	TROCHEE	ANCHOR-L _f	FTBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.  pa _i ('di)			*	*	*		*
b. _i ('pa.di)		*!					
c. _i (pa.'di)	*!	*					
d. _i ('pa).di)		*!	*			*	*

Identical considerations apply to polysyllabic but monomorphemic stems, such as *pumāṇs-* ‘male’. When the input contains no underlying prosodic structure, the prosodic markedness constraints are decisive, producing a binary trochaic foot with primary stress on the leftmost syllable. When, however, an inflectional suffix is marked with a left foot edge, stress shifts onto the final syllable in order to satisfy the higher-ranking prosodic faithfulness constraint. Ablaut in such cases, i.e., deletion of a stem vowel, may be seen as a means of improving prosodic structure: candidate d. in 43 incurs additional violations of ALL-Ft-L and PARSE-σ, which are avoided by applying deletion of the medial underlying vowel, /a:/.

(42) /puma:ms-am/ → [(‘pu.mã:).sam]

/puma:ms-am/	TROCHEE	ANCHOR-L _f	FtBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.  (‘pu.mã:).sam						*	*
b. pu.(‘mã:).sam					*!		**
c. pu.mã:.(‘sam)				*!	**		**
d. pu.(‘mã:sam)					*!		*

(43) /puma:ms-_i(as/ → [pũ:._i(‘sas)]

/puma:ms-(as/	TROCHEE	ANCHOR-L _f	FtBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ	MAX-V
a.  pũ:. _i (‘sas)				*	*		*	*
b. _i (‘pũ:).sas		*!				*	*	*
c. _i (‘pũ:sas)		*!						*
d. pu.mã: _i (‘sas)				*	**!		**	

The grammar fragment shown in the immediately preceding tableaux would also be sufficient to account for intraparadigmatic stress immobility of the type ACC.SG *gávam* : DAT.SG *gáve* (Type 1.5.1). Given representations /(gav-am/ and /(gav-(e/, respectively, the stress on the initial syllable of the latter form will follow simply from NONFINALITY-σ, if ANCHOR-L_f alone does not decide a winner. Indeed, the winner [(‘ga.ve:)] outperforms losing^x[ga.(‘ve:)] on most of the relevant prosodic markedness constraints. However, for reasons to be expounded upon below, the representation ultimately to be assigned to the root *gav-* ‘cow’ will in fact be /(gav)-/, with a prespecified left and right foot edge. In order to understand why a morpheme like /(gav)-/ must be marked as bearing a “strong” accent — a fully specified foot centered on a single vocalic peak — a more detailed exploration of the properties of PREACCENTING morphemes as well as dominance effects in derivation will be necessary. At this juncture, however, a typology of prosodic lexical specification may provisionally be presented, and its effects previewed.

The table in 4.3 shows a four-way distinction in terms of prespecified foot structure: morphemes marked with both a left and right foot edge (“strong accented”), morphemes marked only with a left edge (“weak accented”), morphemes marked only with a right edge (“preaccenting”), and prosodically unmarked or neutral morphemes (“unaccented”). That Vedic can and should, I will argue, be treated as having a representational distinction between partly and fully specified feet distinguishes it from the treatment of lexically specified stress in Turkish for instance, where the facts seem to be analyzable employing either fully speci-

fied feet (Özçelik 2014) or only partly specified feet (Yawney 2018). In the course of the following sections, I will demonstrate that these distinctions, when combined with the activity of constraints on head dominance modeled on the work of Revithiadou (1999), are sufficient to capture essentially most patterns of Vedic stress assignment.

	Left Edge	No Left Edge
Right Edge	STRONG ACCENTED /(gav-)/ ‘cow’, PPP /-(ta)-/, GEN.PL /-(ām)/	PREACCENTING PERF.3SG /-a)/, NML /-as)/
No Right Edge	WEAK ACCENTED LOC.SG /-(i)/, ADJ /-(mant-)/, /(nr-/ ‘man’	UNACCENTED NOM.PL /-as/, /pad-/ ‘foot’

Table 4.3: Four-Way Typology of Underlying Prosodic Specification

The distinctions given in Table 4.3 partly map on to the accentual typology developed in Kiparsky 2014: 15–6. First, the strong accented morphemes here correspond both to accented roots as well as many [+dominant] accented derivational suffixes in Kiparsky’s work. The preaccenting morphemes mostly correspond straightforwardly to morphemes analyzed as preaccenting by Kiparsky, but no distinction between dominant and recessive preaccents will prove necessary.⁶⁹ The weak accented morphemes, in turn, largely represent affixes identified by Kiparsky as recessively accented morphemes or as unaccented suffixes that can receive stress via the OXYTONE RULE; likewise, the morphemes classified here as unaccented are also considered unaccented by Kiparsky.. The work achieved by Kiparsky’s distinction between dominant and recessive affixes, is, in the main achieved through constraints on head faithfulness, i.e., ANCHOR-L/R-HEAD_F.

The distinction posited here between “strong” and “weak” accented morphemes, where preaccenting morphemes prove effectively to be “strong” as well, is that the latter receive stress only when they constitute the morphological head of the stem, are the leftmost foot edge in a string with only “weak” accents, or the only foot edge in a string. “Strong” accents or preaccents, on the other hand, always impose their preferences over the preference of a “weak” accent; what both of these types have in common is that they both contain right foot edges, and thus a higher ranking of ANCHOR-R_F in the grammar proves sufficient for capturing this relatively greater “strength”. Finally, even unaccented morphemes have the chance to attract stress in virtue of HEADSTRESS (i.e., ALIGN-(_F, L, σ_{Hd} , L), provided that no lexically specified foot edges are present in the underlying string.

To close this section, consider the tableaux in 44, 45, and 46, which introduce the further prosodic faithfulness constraints ANCHOR-R_F and the lexically specific constraint ANCHOR-L_F-GEN.PL. In 44, the stress on the initial syllable may be derived essentially through ANCHOR-R_F alone, which not only ensures in this case that stress must occur on the initial syllable, but results in a prosodically deficient foot. The succeeding sections will gradually build up to a justification for this analytical decision. In 46, meanwhile, we encounter a rare, perhaps

⁶⁹The only suffixes claimed by Kiparsky to be “recessive preaccenting” are inflectional person-number suffixes, which can equally be treated as unaccented (and indeed must be, in the cases of non-syllabic endings), plus the subjunctive formant /-a)/, which appears to consistently behave as expected in verbal inflection.

unique, instance of a root specified with a “weak” accent, /(*nr*-/, which cedes stress to the GEN.PL suffix /-(a:m/, which is subject to a special lexical faithfulness constraint that projects the left foot edge of this inflection ending in preference to other left foot edges. The ranking $\text{ANCHOR-R}_f \gg \text{ANCHOR-L}_f\text{-GEN.PL}$, however, ensures that roots with “strong” accents like /(*gav*)-/ never exhibit any mobility to any inflectional suffix, including the GEN.PL.⁷⁰ Thus, the inflection of the root /(*nr*-/ ‘man’ provides crucial evidence for the difference between strongly and weakly accented roots: whereas /(*gav*-)/ ‘cow’ maintains stress on the root syllable even in the GEN.PL (*gávām*), /(*nr*-/ ‘man’ instead cedes stress to strong accented /-(ām)/ (thus *narām*), but not to any of the weakly accented inflectional suffixes (thus, e.g., INST.PL *nṛbhis*). The distinction repeats itself in the respective derivatives in *-mant*-/*-vant*-. this derivational suffix is but weakly accented, i.e., /-(vant-/ ~ /-(mant-/, with the consequence that we find *gómant*- but *nṛvánt*-; see further 4.3.3 on these suffixes.

(44) /_i(*gav*)_{j-k}(*e*/ ‘cow-DAT.SG’ → *gavé* [_i(‘*ga*)_j.*ve*:])

/i(<i>gav</i>) _{j-k} (<i>e</i> /	TROCHEE	ANCHOR-R _f	ANCHOR-L _f	FtBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ	MAX-V
a.  i(‘ <i>ga</i>) _j . <i>ve</i> :			*	*			*	*	
b. i(‘ <i>ga.ve</i> :) _j		*!	*						
c. i(<i>ga.ve</i> :) _j	*!	*	*		*				
d. <i>ga</i> - _k (‘ <i>ve</i> :) _j		*!	*		*	*		*	

(45) /_i(*nr*-_j(*bhis*/ ‘man-INST.PL’ → *nṛbhis* [(‘*nṛ*.*bhis*)])

/i(<i>nr</i> - _j (<i>bhis</i> /	TROCHEE	ANCHOR-R _f	ANCHOR-L _f	FtBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.  i(‘ <i>nṛ</i> . <i>bhis</i>)			*					
b. <i>nṛ</i> - _j (‘ <i>bhis</i>)			*		*!	*!		*
c. i(<i>nṛ</i>). <i>bhis</i>			*	*!			*	*

⁷⁰If one were to operate with a more abstract level of prosodic “strength”, one could entertain the notion that /*gav*-/ ‘cow’ and /-(a:m/ ‘GEN.PL’ are strongly accented, /*nr*-/ ‘man’ and other potentially stress-attracting inflectional suffixes are weakly accented, and that stress appears on the leftmost strong accent, else the leftmost weak accent, else the leftmost syllable. Encoding strength more concretely in terms of foot structure unfortunately cannot predict expected *narām* simply by specifying the GEN.PL as /-(a:m)/: from a UR /(*nr*-(a:m)/, the winning candidate ^X[(‘*na.ra*:m)] would be predicted, since both it and the expected alternative [na.(‘*ra*:m)] satisfy ANCHOR-R_f , while equally violating ANCHOR-L_f , leaving markedness constraints like ALL-Ft-L and NONFINALITY-σ to adjudicate in favor of ^X[(‘*na.ra*:m)].

(46) $/_i(nr_j(a:m/ \text{ 'man-GEN.PL}' \rightarrow nar\acute{a}m [na.(^{'ra:m})]$ ⁷¹

$/_i(nr_j(a:m/$	TROCHEE	ANCHOR- R_{τ}	ANCHOR- L_{τ} -GEN.PL	ANCHOR- L_{τ}	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  $na_j(^{'ra:m})$				*		*	*		*
b. $_i(^{'na.ra:m})$			*!	*					
c. $_i(^{'na}).ra:m$			*!	*	*			*	*

⁷¹ An alternative form with the GEN.PL allomorph $-n\acute{a}m$, $n\eta\eta\acute{a}m$, is in fact more frequent, but the n -less allomorph still attested for this lexeme is surely older.

$$(47) \quad /_i(\text{gav})_j\text{-}_k(\text{a:m}/ \text{'cow-GEN.PL'} \rightarrow \text{gávām} [_i(\text{'ga})_j.\text{va:m}]$$

$/_i(\text{gav})_j\text{-}_k(\text{a:m}/$		TROCHEE	ANCHOR- R_τ	ANCHOR- L_τ -GEN.PL	ANCHOR- L_τ	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.	$_i(\text{ga})_j.\text{va:m}$			*	*	*			*	*
b.	$_i(\text{'ga}.\text{va:m})$		*!	*	*					
c.	$\text{ga}_k.\text{(va:m)}$		*!		*		*	*		*

Under this general approach, *right* foot boundaries prove to be generally decisive, and impose the position of a foot constituent more strongly than left foot boundaries. Before developing a more precise picture of stress assignment in derivation, the remaining major patterns of stress assignment in inflection will be filled in with a treatment of secondary mobility in the following section.

4.3.2 Secondary Mobility

Recall from 4.2.2 that certain lexemes and stem classes in Vedic exhibit a pseudo-mobile pattern of stress conditioned by ablaut: when a vowel that otherwise would bear the stress is deleted, stress instead appears on the nearest remaining vowel to the right. Under the Kiparskian analysis, this rightward movement of stress is due to the rule of SECONDARY MOBILITY (see 16 above), which reassigns a lexical accent from a deleted syllable peak to the next syllable peak to its right. Following Yates (2020), patterns of secondary mobility should be broken down more precisely into three types (see also the discussion under 4.2.2):

- Type 1: due to glide formation, shifting stress onto an inflectional suffix, whether accented or not; limited essentially to the lexeme *arí-* ‘friend’ (ACC.SG *arí-m*, NOM.PL *ary-ás*).⁷²

⁷²The crucial form NOM.PL *aryás* is well-attested (16 × RV, though not all in the sense ‘friend’). Yates (2020: fn. 9) takes note of a further hapax (RV 10.106.3b) *paśvā* ‘herd animal:NOM.DU, which would appear to instantiate the same pattern of secondary mobility. The stress of this form may be problematic, however, insofar as the inflectional suffix of the NOM.DU seen here /-ā/ is unaccented, and the stem *paśú-* would appear to be underlyingly unaccented as well (compare *paśumánt-* ‘having (many) herd animals’ and *paśu-ráksi-* ‘protector of herd animals’. *paśvā* might, however, be explicable under a suitable interpretation of HEADSTRESS (see below); alternatively, the lexeme’s UR might have been restructured from /paś-(u-/ to /paś(u)-/ by some Vedic speakers, though the only other possible indication thereof in Vedic literature is the compound *paśú-vrata-* ‘acting like herd animals’ (MS 1.8.7).

Macdonell (1910: 81) mentions at least one case that would contradict the apparent rightward shift of stress due to glide formation seen in *aryás*, namely, *mítryaḥ* ‘friendly:NOM.PL.M (RV 2.6.7c) (vs. ACC.SG *mítríyam* at RV 5.85.7a and *a-mítríyam* ‘unfriendly’ at 9.61.20a). The *udatta* on the initial syllable in *mítryaḥ* might be attributed to the stress of the base *mítra-* ‘friend’, though the derivational suffix *-íya-* should normally be expected to attract stress.

- Type 2: due to vowel deletion of the first vowel in a disyllabic stem, shifting stress onto the second syllable of the stem; limited to essentially two lexemes, *sánu-* ‘back’ (weak stem in LOC.SG *snú-ṣu*) and *dāru-* ‘wood’ (weak stem in INST.SG *drú-ṇā*).⁷³
- Type 3: due to vowel deletion in the final syllable of a polysyllabic stem, shifting stress onto an accented inflectional suffix; well-represented in Vedic among nominal stems terminating in *-r* (e.g., *pitár-* ‘father’) and *-n* (e.g., *ukṣán-* ‘ox’).

Here, I summarize and present the analysis of Yates (2020), who proposes that patterns of vowel deletion in Types 2 and 3 and the particular pattern of rightward stress shift found in Vedic can be given a coherent and unified account when “accentedness” is treated as the left edge of a trochaic foot. In general, the specific lexemes and stem classes that exhibit secondary mobility in inflection are compatible with an analysis involving a “weak” accent, i.e., a left foot edge alone.⁷⁴ The essence of the analysis consists in the following behaviors: the Type 1 pattern follows directly from the constraint ranking seen above, given a lexical representation /ar(i-/⁷⁵; similarly, the disyllabic stems of Type 2 keep the leftmost of two left foot edges, but improve harmony on ALL-Ft-R via vowel deletion; finally, vowel deletion in Type 3 enables two underlying left foot edges to correspond to a single foot edge on the surface, thereby enabling better satisfaction of ANCHOR-L_F.

The NOM.PL *aryás*, with stress on the second syllable, emerges as a winner from the input /ar(i-as/ given that vowel hiatus is dispreferred (*VV) and can be unproblematically resolved by glide formation (in violation of IDENT-[syllabic]). An alternative ^X[(‘aryas)], moving the left edge of a foot to the left edge of the word, while more harmonic with respect to ALL-Ft-L, violates higher-ranked ANCHOR-L_F and so fails. This outcome is illustrated in 48.

⁷³The noun *jānu-* ‘knee’ plausibly belongs to this type as well, but is only attested in its NOM/ACC.SG form *jānu*; the expected zero-grade stem *jñu-* appears in compounds (e.g., *abhi-jñú-* ‘up to the knees’), but none are fully probative as to stress behaviors.

⁷⁴Since adjectives in *-vant-* to ablauting *n*-stems of the type *ukṣán-* are unknown, it is difficult to decisively determine whether this suffix is “weak” /-(an-/- or “strong” /-(an-)/-. If a “strong” suffix were to be assumed, I believe that the loss of syllabicity seen in, e.g., *ukṣnás* ‘ox:GEN.SG’ would result in an evaluation of ANCHOR-R_F with respect to the vowel to its right. That is, from an underlying /uks-(an)-(as/, an output [(‘uk).ṣan] would violate ANCHOR-R_F, but [uk.(‘ṣan)] would not. As the data stands, to assume a “weak” suffix /-(an-/- is equally possible and produces no potential issues. Likewise, identifying the suffix *-tár-* as “weak” or “strong” is difficult, since the only adjectives in *-mant-* or *-vant-* found in Vedic to such stems are *pitṛmant-* (AV, VS, ŚB), *pitṛvant-* (RV), *mātṛmant-* (AV, ŚB), *mātṛvant-* (Cāṇakya), and *gopitṛmant-* (Kauṣītaki Upaniṣad II.1). The latter two forms maintain stress on the suffix *-tṛ-*, suggesting a strong accent (though note that they do not occur in saṃhitā texts), while the former three are found with both stress on *-tṛ-* and *-mánt-* ~ *-vánt-*. I infer from this situation that Vedic speakers themselves were unsure whether /-tar-/ was “weak” /-(tar-/- or “strong” /-(tar-)/-, but the fact that secondary mobility obtains in the inflection of such stems regardless supports my assumption concerning the evaluation of ANCHOR-R_F.

⁷⁵The position of the stress in the adjectival karmadhāraya *arí-dhāyas-* ‘nourishing the faithful’ (RV 1.126.05b) might rather support a representation /ar(i)-/, with a “strong” accent. Since an originally segmentable *i*-stem almost surely etymologically underlies *arí-*, and many *i*-stems in Vedic in fact have “weak” accents, the “strong” accentedness of this lexeme would likely reflect a restructuring of its UR.

(48) /ar_i(i-as) → [ar_i(^hyas)]

/ar _i (i-as)	*VV	ANCHOR-R _f	ANCHOR-L _f	FTBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ	IDENT-[syllabic]
a.  ar _i (^h yas)					*	*		*	
b. a _i (^h ri.as)	*!					*		*	
c. i(^h ar.yas)			*!					*	

The disyllabic stems belonging to Type 2 may be assumed to possess lexical representations with a left foot edge preceding the first vocalic peak, thus /(<sa:nu-/ and /(<da:ru-/. The behavior of stress in the inflection of these forms contrasts nicely with the behavior of the similarly disyllabic (and similarly probably monomorphemic) form stems *púmāms-* ‘man’ and *pánthā-* ‘path’: the latter allow for stress to move onto the final syllable when ablaut occurs, whereas the two *u*-stems always maintain stress on the first syllable. When combined with an accented inflectional suffix, e.g., LOC.PL /(<sa:nu-(su/, at least one violation of ANCHOR-L_f (as in *gāve* above) is inevitable in order to satisfy CUMULATIVITY. Given the deletion of the leftmost vowel, the choice between stress on the first and second syllable is essentially decided by FTBIN (in this particular case) or ALL-Ft-L (in, e.g., INST.SG *snúnā*): stress on the final syllable will always perform worse on these constraints. The choice between a candidate with and without vowel deletion (e.g., [(<sa:nu).su]) is regulated here by ALL-Ft-R: fewer syllables allows for the word’s only foot to align with the right edge of the word.⁷⁶ Candidate a. in 49 thus satisfies the constraints on foot edge alignment.

(49) /i(<sa:nu-j(su/ → [i(<snu.ʃu)]

/i(<sa:nu-j(su/	CULMINITIVITY	ANCHOR-R _f	ANCHOR-L _f	FTBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ	MAX-V
a.  i(<snu.ʃu)			*					*	*
b. snu-j(<ʃu)	*!		*	*	*	*		*	*
c. i(<sa.nu).ʃu			*				*!	*	
d. i(<snu)-j(<su)	*!			**	*				

⁷⁶One effect of ALL-Ft-L and ALL-Ft-R in Vedic may well be to reduce the total number of syllables in a word, wherever possible, in order to reduce the number of violations to these constraints. Such alignment-optimizing deletion cannot, however, constitute a complete and correct account of Vedic vowel ~ ∅ alternations, since vowel deletion occurs even when the total number of syllables is not ultimately reduced (e.g., in pre-consonantal weak-case forms of sonorant stems), and I suspect that instances where the syllable count could be licitly reduced, but yet vowel deletion does not occur, could be enumerated as well.

Although Types 1 and 2 of secondary mobility are comparatively rare in Vedic, it is their evidence that is in fact decisive in pointing towards a rightward movement of stress arising from lexical prosody, since the vowels that appear with stress in these types are unlikely to be specified as accented. Indeed, an accented inflectional suffix NOM.PL /-(as/ is to be categorically excluded, since this suffix, outside of *aryás*, never surfaces with stress; similarly, the second vowel of *sānu-* and *dāru-* should probably be regarded as unaccented, since a representation like /(*sān*(u-/ would otherwise be unique in the language.⁷⁷ Type 3 is rather more ambiguous, because stress in this type always surfaces on the vowels of endings that must be assumed to be accented, since they can attract stress in genuinely mobile nominal paradigms.

In the view of Yates (2020: 10–11), the vowel deletion seen in Type 3 forms such as *pitré* ‘father:DAT.SG’ or *ukṣṇás* ‘ox:GEN.SG’ is motivated by better satisfaction of ANCHOR-L_F, at the cost of violating MAX-V; in other words, at least some instances of ablaut are potentially attributable to the optimization of faithfulness relations in foot structure. Yates assumes that forms belonging to Type 3 generally present the underlying prosodic structure /Root-(Derivation-(Inflection)/, where both derivational and inflectional suffix are accented (i.e., specified with a left foot edge). Given such a representation, the presence of an accented inflectional suffix can indeed motivate vowel deletion in order to simultaneously satisfy ANCHOR-L_F and at least one other constraint that outweighs MAX-V. In example 50, the output with stress on the final syllable allows for the one left foot edge on the surface to stand in correspondence with two underlying foot edges, thereby avoiding very costly violations of ANCHOR-L_F. Since all other candidates violate ANCHOR-L_F at least once, whether they exhibit vowel deletion or not, they are less harmonic than the winner.

(50) /pit_i(ar_j(e:/ → *pitré* [pit.(‘re:)]

/pit _i (ar _j (e:/		CULMINITIVITY	ANCHOR-R _F	ANCHOR-L _F	FtBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ	MAX-V
a.	pit _{i,j} (‘re:)					*	*		*	*
b.	pi _i (‘ta.re:)			*!			*		*	
c.	i(‘pit).re:			*!*				*	*	*
d.	pi.ta _j (‘re:)			*!		*	**			

In paradigms with secondary mobility of Type 3, stress is expected to remain on the penultimate (= stem-final) syllable in cases where a syllable remains, e.g., INST.PL *pitṛbhis* [pi.(‘tṛ.b^his)], through the expected interaction of ANCHOR-L/R_F and ALL-Ft-L, parallel to the behavior seen in 45.

⁷⁷ Compare, however, the proposal of stems with multiple lexical accents of the *furnaris*-type in Modern Greek at (91) in Chapter 2.

Yates then goes somewhat farther, arguing that lexical representations with foot edges are not only compatible with the stress behavior of secondary mobility and that they can motivate some patterns of vowel deletion, but that an autosegmental representation of lexical accent as per Revithiadou (1999) and Alderete (2001b) will in fact fail to capture these three patterns. According to Yates' claim, the specific representations and constraints employed by Revithiadou would predict leftward reassociation of stress in Types 1 and 3, at least. Precisely because Type 3 constitutes the only well-attested and undeniably productive sort of secondary mobility, to account for at least its behavior in a stress grammar is indispensable; whether Types 1 and 2 might be due to further lexical peculiarities cannot be entirely excluded.

To understand to the force of Yates' argument, a brief review of the constraints employed by Revithiadou and Alderete to govern faithfulness relations between lexical accents is necessary (see also 2.4.2.2 in Chapter 2 for exact definitions). Their constraints, MAX-PROM, DEP-PROM, and *FLOP-PROM, penalize the deletion, insertion, and reassociation, respectively, of lexical accents underlyingly associated with some underlying vocalic peak. Revithiadou and Alderete differ, however, in how violations of *FLOP-PROM are assessed. Under the definition of Revithiadou (1999: 52–3), *FLOP-PROM is violated only if the vocalic peak associated with a prominence in the input is also present in the output; in case the vocalic peak is absent in the output, the prominence originally autosegmentally linked to it in the input may freely reassociate without penalty. Under the definition of Alderete 2001b: 461, *FLOP-PROM is violated whenever a given prominence *i* is associated with some vocalic peak in the output other than the one to which *i* was originally associated, regardless of whether the original peak is present in the output or not.⁷⁸

As long as constraint like ALL-FT-L or ALIGN-(Prom, L, ω , L) is assumed to be decisive in Vedic for adjudicating conflict between lexical accents of equivalent morphological status and for assigning default prominence (an assumption that has been taken over in every constraint-based treatment of Vedic stress, to directly reproduce Kiparsky's BAP), secondary mobility of Type 1 (NOM.PL *aryás*) becomes impossible under either interpretation of *FLOP, and one of either Type 2 (by Alderete's definition) or Type 3 (by Revithiadou's definition) becomes impossible to correctly generate, depending upon which interpretation of *FLOP is employed. Such impossibility of rightward reassociation of autosegments holds under both strict Optimality Theory or Harmonic Grammar: no ranking or weighting of the relevant constraints can generate the desired winners. I therefore present the tableaux below with strict rankings determined using Multi-Recursive Constraint Demotion (cf. 3.2.1.1 in Chapter 3) based on forms of the types seen in 4.3.1 above, plus one *bahuvrīhi*-compound (*híranya-jihvam* ← /híranya-jihvá-m/) and forms of Types 2 and 3 (*snuṣú* and *ukṣnás*); two rankings are given in 51.⁷⁹ These two rankings are in fact identical; only the interpretation of *FLOP

⁷⁸By Alderete's definition, violation of *FLOP-PROM (his NO-FLOP-PROM) results from the severing of an underlying linkage between prominence and a sponsor; in the case of stress prominence, where spreading should be categorically excluded, the severing of a linkage can then result from either reassociation or from deletion of the sponsor.

⁷⁹A form longer than the essentially two-syllable forms relevant to genuine stress mobility and secondary mobility in Vedic is necessary to establish that no preference for feet aligned with the right edge of the word is

affects whether Type 2 or Type 3 is correctly generable, while the given ranking will make false predictions for the other type.

(51) Constraint Rankings for Autosegmental Lexical Accents in Vedic

- a. *FLOP after Revithiadou (1999) (can generate Type 2): *FLOP, *VV, CULMINATIVITY, OBLIGATORINESS $\gg$ MAX-PROM, DEP-PROM $\gg$ ALL-Ft-L $\gg$ ALL-Ft-R
- b. *FLOP after Alderete (2001b) (can generate Type 3):
*FLOP, *VV, CULMINATIVITY, OBLIGATORINESS $\gg$ MAX-PROM, DEP-PROM $\gg$ ALL-Ft-L $\gg$ ALL-Ft-R

The tableaux in 52 show that the ranking of constraints under an autosegmental interpretation of lexical accent cannot generate the form *aryás* whatsoever. Since the lexical accent presumably associated with the second vowel, /arí-/ , in the input is forcibly reassocciated due to glide formation, the result is equal (non)-violations of *FLOP for all candidates that do not also violate *VV. Consequently, the winner ALL-Ft-L is determined by ALL-Ft-L (when *FLOP is not violated) or *FLOP itself (when violated), thereby causing expected [ar.(‘yas)] to lose to ^X[(‘ar).yas].

(52) Output of NOM.PL /arí-as/ (Autosegmental Lexical Accent)

- a. No Violations of *FLOP (underlying vocalic peak no longer vocalic)

/arí-as/	*VV	*FLOP	MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
a. ☹ ar.(‘yas)					*!	
b. a.(‘ri.yas)	*!					
c. • (‘ar).yas						*
d. (‘ar).yas			*!	*!		

- b. Violations of *FLOP through Reassociation

/arí-as/	*VV	*FLOP	MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
a. ☹ ar.(‘yas)		*!			*	
b. a.(‘ri.yas)	*!					
c. (‘ar).yas		*!				*
d. • (‘ar).yas			*	*		*

at work, which could exclude candidates with initial stress in Type 1 (i.e., ^X[(‘ar).yas]).

The predictions then diverge concerning Types 2 and 3. In Type 2, since an input like LOC.PL /sá:nu-sú/ contains two lexical accents, at least one violation of MAX-PROM must occur in order to satisfy CULMINATIVITY. If the strong lexical accent underlyingly associated with the first vowel of /sá:nu-/ may reassociate without violation of *FLOP, then ALL-Ft-L correctly selects for ['snu.su] as the winner. If, on the other hand, any reassociation is expected to result in violations of *FLOP, unexpected ^x[snu.('su)] will be predicted to surface, since the lexical accent of /-sú/ can then realize stress without incurring a violation of higher-ranked *FLOP.

(53) Output of LOC.PL /sá:nu-su/ (Autosegmental Lexical Accent)

a. No Violations of *FLOP (underlying vocalic peak deleted)

/sá:nu-sú/		CULMINATIVITY		*FLOP		MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
		*VV							
a.	☞ ('snu.ʃu)					*			
b.	snu.(ʃu)					*		*!	
c.	('snu.ʃu)					**!	*!		
d.	('snu).(ʃu)	*!						*	*

b. Violations of *FLOP through Reassociation

/sánu-sú/		CULMINATIVITY		*FLOP		MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
		*VV							
a.	☹ ('snu.ʃu)			*!		*			
b.	☛ snu.(ʃu)					*		*	
c.	('snu.ʃu)					**!	*!		
d.	('snu).(ʃu)	*!						*	*

Precisely inverse predictions hold with respect to Type 3, which ought to be expected: stress appears on the leftmost syllable in Type 2, but rightmost syllable in Type 3. Hence, when violations of *FLOP through any reassociation are applied, the candidate that realizes the lexical accent of the inflectional suffix in a form like GEN.SG /uksán-ás/ (→ [ukʃ.(ʃnas)]) will be selected as the winner, in virtue of avoiding violations of *FLOP.⁸⁰ Conversely, when reassociation of the strong lexical accent on the stem would be permitted without cost, ALL-Ft-L would be expected to instead militate in favor of undesired ^x[(ʃukʃ).nas], with initial stress. These outcomes are shown in 54.

⁸⁰For the purposes of exposition of the present argument, I present the stem /uksán-/ as morphological simplex, with lexical accent on its second vowel. This particular lexeme, and many other forms exhibiting Type 3 secondary mobility, might instead be analyzed as morphologically complex stems without any lexical accents, which receive stress by the OXYTONE RULE or HEADSTRESS as described below. If all cases of Type 3 secondary mobility could be attributed to effects of HEADSTRESS, then forms like [ukʃ.(ʃnas)] from underlying /uks-an-ás/ would be expected, satisfying both HEADSTRESS and MAX-PROM (~ ANCHOR-L/R_F). As long as at least one stem showing Type 3 secondary mobility demonstrably must be treated as underlying /C₀VC₀VR-/ , the fundamental problem with Type 3 for an autosegmental analysis of lexical accent remains.

(54) Output of GEN.SG /uksán-ás/ (Autosegmental Lexical Accent)

a. No Violations of *FLOP (underlying vocalic peak deleted)

/uksán-ás/		CULMINATIVITY		*VU	*FLOP	MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
a.	☹ ukṣ.(nas)					*		*!	
b.	ukṣ.(nas)					*		*!	
c.	☹ ('ukṣ).nas					*			*
d.	('ukṣ).nas					**!	*!		*

b. Violations of *FLOP through Reassociation

/uksán-ás/		CULMINATIVITY		*VU	*FLOP	MAX-PROM	DEP-PROM	ALL-Ft-L	ALL-Ft-R
a.	☹ ukṣ.(nas)					*		*	
b.	ukṣ.(nas)				*!	*		*	
c.	('ukṣ).nas				*!	*			*
d.	('ukṣ).nas					**!	*!		*

The proper analysis of secondary mobility thus evidently, as shown by Yates (2020), poses a near-insuperable problem for the autosegmental analysis of lexical accent in Vedic, at least without some fundamental revision of assumptions concerning the edge orientation of Vedic stress, which do not appear promising.⁸¹ At best, one might adopt Alderete's definition of *FLOP in order to account for Type 3, and then attempt to wave away the rarer Types 1 and 2 as constituting special cases of lexicalized stress allomorphy. However, insofar as phonological representations involving prespecified foot structure can also successfully accommodate the stress facts in derivation as well, then the usage of such representations may be regarded as affording better empirical coverage of the Vedic data.

⁸¹One might be tempted to suggest that stress in the absence of a lexical accent in Vedic is in fact results in a right-aligned trochee (thus producing largely penultimate stress, as in, e.g., *pádam*), but on the leftmost lexical accent. Vedic would then be analyzed as exhibiting a species default-to-opposite stress (outside of morphologically more complex forms): the leftmost special element receives stress, else the rightmost trochaic foot. The fundamental problem with such a treatment is twofold: first, insofar as Vedic builds only ever constructs a single foot per prosodic word, multiple special elements are not expected to result in multiple feet in the output, with the consequence that ALL-Ft-L/R and HEAD-Ft-L/R (cf. the analysis of default-to-opposite systems in Baković 1998) will have identical violation profiles and cannot be used to generate different behaviors in inputs with and without lexical accents; second, insofar as the preferred stress is always to the leftmost lexical accent, leftward reassociation of an autosegment might be predicted in Types 1 and 3 of secondary mobility, in any case. A reanalysis of the Vedic facts as a default-to-opposite system thus seems likely to fail.

One further alternative not considered by Yates (2020), and not employed whatsoever by Revithiadou (1999) or Alderete (2001a), is that autosegmental lexical accents might also be governed by UNIFORMITY, which militates against the coalescence of underlying material on the surface. That is, underlying lexical accents might be permitted to merge into a single prominence on the surface, such that the one surface prominence would then stand in a correspondence relationship with two underlying lexical accents. If UNIFORMITY-PROM is ranked below MAX/DEP-PROM, then, under Revithiadou's interpretation of *FLOP, Type 3 can produce the expected winner by allowing the prominence projected from the deleted vowel to coalesce with the prominence on the final syllable. It ultimately turns out, however, that a line of analysis using UNIFORMITY to generate both Type 2 and Type 3 can only succeed if ALL-FT-R outranks ALL-FT-L, but such a ranking is evidently incompatible with the initial stress of unaccented disyllabic roots (e.g., *púmāṃsam*), as well as the seeming preference for stress closer to the left edge in array of other forms. I therefore do not pursue an investigation of possible analysis involving UNIFORMITY-PROM any further.

4.3.3 Morphological Headedness and “Strong” versus “Weak” Accents: Dominance Effects as HEADFAITH

I now turn to the treatment of forms containing morphologically complex stems, beginning with nominal and adjectival forms whose outermost (i.e., rightmost) derivational suffixes may be analyzed as either “strong” or “weak” accented (i.e., marked with a left foot edge). The stress of all stems built with either a “strong” or “weak” accented derivational suffix is largely immobile in inflectional paradigms, but differs in the genitive plural, giving rise to the difference between pattern 1.3 and pattern 1.5.2. The principal behaviors to be captured are thus as follows:

- In inflection, stems whose stem stress is determined by a “weak” lexical accent exhibit paradigmatic mobility to GEN.PL -*ám*.
- In derivation, a stem built with a suffix having a “weak” lexical accent receives stress only when the string contains only weakly accented or unaccented morphemes.
- In derivation, a stem built with a “strong” derivational suffix always exhibits stress on that suffix.

Most clearly representing the category of derivational suffixes with a “weak” lexical accent is the adjective forming suffix *-mant-* / *-vant-*, first discussed under 4.2.2 above. There, following Kiparsky's presentation, I showed that treating this suffix as lexically accented /-mánt- / -vánt-/ can, in combination with the BASIC ACCENTUATION PRINCIPLE and OXY-TONE RULE, correctly generate the prosodically variable behavior of this suffix, as well as the prosodically variable behavior of certain bases: it surfaces as unstressed on lexically accented bases (e.g., *gómant-* ‘having cows’, *devávant-* ‘surrounded by gods’), but as stressed on bases without a lexical accent (e.g., *padvánt-* ‘having feet’, *arcimánt-* ‘having flames’). The stressed or unstressedness of *-mant-* / *-vant-* itself can fall out from the BAP, while bases like

arci- ‘flame’ can cede stress to *-mant-* / *-vant-*, granting that their stress is due to the OXYTONE RULE. The translation of these behaviors and Kiparsky’s analysis of such forms into a constraint-based analysis is not, however, entirely straightforward.

First, *-mant-* / *-vant-* is virtually unique in Vedic in being a derivational suffix that alters the syntactic category of its base ($N \Rightarrow A$) without imposing a particular stress on the stem (i.e., on the suffix itself or the preceding syllable). In Kiparsky’s terms, *-mant-* / *-vant-* is recessive (as opposed to dominant), because it does not erase the lexical accents of its base. However, as discussed under 4.2.3, the correlation between accentual dominance and the capacity to establish the syntactic category of the base (or change the argument structure of a verb) is near perfect. For instance, inflectional endings (including feminine *-ī*), verbal stem-forming suffixes, (secondary) comparatives and superlatives (on which see now Onishi 2021), the true participles in *-ant-* and *-(m)āna-* (verbs with nominal agreement), and diminutives (*-ka-*) do not impose stress on their bases, may only attract it in case their bases are unaccented, and do not impose a syntactic category on their bases. Other category-changing noun- or adjective-forming suffixes are dominant, in Kiparsky’s sense; one might, therefore, have expected *-mant-* / *-vant-* to belong to this latter group. To account for this discrepancy, accentual dominance is treated by Kiparsky as a lexical property: an affix is marked as [+/-dominant], with [+dominant] affixes triggering the application of accent erasure, as formalized in 4.2.4.

Second, directly rewriting the OXYTONE RULE (repeated in 55 below for convenience) as a constraint that reproduces its behavior poses some difficulties if stress is taken to depend on foot structure, rather than abstract prominence. If stress is interpreted simply as pure prominence, an alignment constraint as in 56 might suffice. In a prosodic system that categorically prefers trochaic feet, however, consistently aligning the strong portion of a foot with the right edge of the inflectional stem requires a rather more convoluted formulation, as in 57. Specifically, part of the alignment constraint would need to refer to a prosodic category (a syllable) standing in a particular relation to morphological structure (the right edge of the stem), which seems to run counter to the spirit of generalized alignment constraints.

(55) OXYTONE RULE (OR)

$\sigma \rightarrow \acute{\sigma} / [\dots\sigma ___]_{\text{Stem Infl}}$

“...a final accent is assigned to *all* inflected polysyllabic stems.”

(56) ALIGN-($\acute{\sigma}$, R, Complex-Stem, R)

The right edge of a syllable with stress prominence must align with the right edge of a (morphologically complex) stem.⁸² Assign one violation (*) if the syllable with stress prominence is not aligned with the right edge of a stem.

(57) ALIGN-(f , L, σ]_{Complex-Stem, L})

The left edge of foot must align with the left edge of the syllable standing at the right edge of the (morphologically complex) stem. Assign one violation (*) if the left edge

⁸²Recall from the discussion of the disyllabic but morphologically simplex stems *pánthā-* ‘path’ and *púmāṇs-* ‘male’, that a version of stem oxytonesis that does not overgenerate in nominal accentuation must refer to the morphological complexity of the stem.

of the syllable standing at the right edge of the stem is not aligned with the left edge of a foot.

To circumvent the problems posed by the exceptional behavior of *-mant* / *-vant-*, Sandell (2015a: 176–89) proposed that Vedic exhibits a species of HEADSTRESS: *-mant* / *-vant-* was argued to surface with stress in purely virtue of being the head of a morphologically complex word, where the lexical representation of the suffix lacks a lexical accent, i.e., is /-mant-/ ~ /-vant-/. As developed by Revithiadou (1999: 26–31), HEADSTRESS constraints militate in favor of the attraction of prosodic prominence to the head of a morphologically complex structure, in preference to a default pattern generated by markedness constraints, and perhaps in preference to faithfulness to lexical accents.⁸³ I state Revithiadou’s original version of HEADSTRESS in 58a. and a more specific version designed to refer to foot structure in 58b.

However, as I will illustrate in more detail in the following section, to account for Vedic stress in terms of underlying foot structure, while seeking to obtain certain stress patterns through a constraint like HEADSTRESS_{F-L} (shown below) encounters one insuperable problem: HEADSTRESS does not ever result in the imposition of stress on a preaccenting head. Namely the forms *mánas* and *matím* are both disyllabic and of the shape /LH/, but the former contains a preaccenting suffix /-as/; a non-existent ^X*manás* [ma.(‘nas)], however, could satisfy both HEADSTRESS while remaining faithful to an underlying right foot edge. If, however, HEADSTRESS is violable in order to better satisfy prosodic markedness constraints (*mánas* [(‘ma.nas)] satisfies ALL-FT-L and NONFINALITY-σ), then a form like *matím* cannot be generated: under an analysis involving HEADSTRESS, the derivational suffix /-ti-/ is treated as unaccented to allow it to cede stress to the suffix *-mant* / *-vant-*.

(58) HEADSTRESS Constraints

- a. HEADSTRESS (Revithiadou 1999: 28; 190)
Morphological heads are stressed.
- b. HEADSTRESS_{F-L} (= ALIGN-($\mathcal{F}$, L, σ_{Head} , L)): The left edge of a foot must be aligned with the left edge of a syllable containing a syllable nucleus built from segmental material expressed by the head of the morphological word (σ_{Head}). Assign one violation (*) if a syllable peak intervenes between σ_{Head} and the left edge of a foot.⁸⁴

⁸³As indicated in discussion above, I assume that the relevant variety of headedness applicable in Vedic, holding across both HEADSTRESS and HEADFAITH (to be discussed further below), is the capacity to alter the syntactic category in a complex stem. Headedness constraints do not, in Vedic, hold over roots (i.e., morphologically simplex stems) or affixes that do not affect the syntactic category of their bases.

⁸⁴I am indebted to Tony Yates for formulating a first version of this constraint in his seminar on Indo-European Word Prosody in the Spring of 2020 at the University of California, Los Angeles. The NOM/ACC.DU form *paśvā́*, mentioned in footnote 66 above, might speak in favor of a different interpretation of HEADSTRESS, which would allow for HEADSTRESS_{F-L} to hold over consonants projected from the word’s morphological head as well. In the case of *paśvā́*, since the head /-u-/ surfaces as a glide, the status of σ_{Hd} is applied to the syllable containing that glide. Other instances of secondary mobility do not, however, seem not to require the activity of HEADSTRESS_{F-L}, since they virtually always contain a lexically accented derivational suffix, and can thus be analyzed essentially as under 4.3.2 above.

However, as I will illustrate in more detail in the following section, to account for Vedic stress in terms of underlying foot structure, while seeking to obtain certain stress patterns through a constraint like HEADSTRESS_{F-L} (shown below) encounters one insuperable problem: HEADSTRESS does not ever result in the imposition of stress on a preaccenting head. Namely the forms *mánas* and *matím* are both disyllabic and of the shape /LH/, but the former contains a preaccenting suffix /-as/; a non-existent $^X\text{manás}$ [ma.(‘nas)], however, could satisfy both HEADSTRESS while remaining faithful to an underlying right foot edge. If, however, HEADSTRESS is violable in order to better satisfy prosodic markedness constraints (*mánas* [(‘ma.nas)] satisfies ALL-Ft-L and $\text{NONFINALITY-}\sigma$), then a form like *matím* cannot be generated: under an analysis involving HEADSTRESS , the derivational suffix /-ant-/ is treated as unaccented to allow it to cede stress to the suffix *-mant* / *-vant*-. For this reason, an analysis centered around HEADSTRESS must be abandoned.⁸⁵ Instead, the distinction between consistently stress-attracting derivational suffixes like past participle *-tá-* /-(ta)-/ and stress-variable derivational suffixes like *-tí-* /-(ti)-/ can be captured through a representational distinction between morphemes with fully specified feet (“strong” accents), i.e., /(V)/, and morphemes with only a left foot edge (“weak” accents), i.e., /(V)/.

Given a representation of *-mant-* $\sim$ *-vant-* as /-(mant-/ $\sim$ /-(vant-/ , stress on such adjectives built to unaccented roots, like /pad-/ ‘foot’, or “strong” accented roots, like /(gav)-/ ‘cow’, can fall out directly from the ranking $\text{ANCHOR-R}_f \gg \text{ANCHOR-L}_f \gg \text{ALL-Ft-L}$, where ALL-Ft-L would prefer the leftmost underlying foot edge, from among the root, derivational suffix, or inflectional ending. Illustrative derivations to this effect are shown in 59 and 60, where simply the leftmost foot edge can be held responsible for the position of stress (though ANCHOR-R_f determines the precise prosodic constituency in 60, deciding between candidates a. and d.). This relatively straightforward analysis is complicated by the one root with a weak accent, /(nr -/ ‘man’: we find *nránt-*, not $^X\text{nránt-}$. Given a UR /(nr -(vant-/ , some other factor must be invoked to explain why, in this case, the left foot edge associated with the derivational suffix /-(vant-/ is preserved in the output.

(59) /pad-_i(vant-am/ ‘having feet-ACC.SG’ $\rightarrow$ *padvántam* [pad-_i(‘van.tam)])

/pad- _i (vant-am/		ANCHOR-R_f		ANCHOR-L_f		FTBIN		$\text{NONFINALITY-}\sigma$		ALL-Ft-L		ALL-Ft-R		$\text{PARSE-}\sigma$	
a.	 pad.(‘van.tam)								*					*	
b.	(‘pad.van).tam		*!								*		*	*	

(60) /_i(gav)_{j-k}(mant-₁(as/ ‘having cows:GEN.SG’ $\rightarrow$ *gómatas* [_i(‘go:)_j.ma.tas])

⁸⁵A further minor weakness of an analysis centered around the effects of HEADSTRESS is its incapacity to explain the stress of the adjective *nránt-* ‘having men’. When only a binary distinction between accented and unaccented morphemes is assumed, *nr-* ‘man’ ought to be classed with the accented morphemes, given the fact that stress occurs on the root in its inflectional paradigm (except in the GEN.PL), which leads to the prediction that an adjective derived with /-vant-/ should surface as $^X\text{nránt-}$.

$i(\text{gav})_j\text{-}_k(\text{mant-})_l(\text{as})$		ANCHOR- R_f	ANCHOR- L_f	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.	 $i(\text{'go:})_j.\text{ma.tas}$		**				**	**
b.	$\text{go:}_k(\text{'ma.tas})$	*!	**			*		*
c.	$\text{go:}.\text{ma}_l(\text{'tas})$	*!	**		*	**		**
d.	$i(\text{'go:}.\text{ma}).\text{tas}$	*!	**				*	*

In effect, a form like *nṛvánt-* ‘rich in men’ — alongside innumerable other similar cases in Vedic — suggests that the prosodic specifications of the morphological head of the word are privileged over those of non-heads. Such behavior fits well with the phenomenon of **head faithfulness**, as attributed by Revithiadou (1999: Ch. 4) to Modern Greek and Russian or by Yates (2017: 152–68) to Hittite (see also discussion in 2.4.2.3 in Chapter 2). Constraints of the type HEADFAITH were developed by Revithiadou to enforce faithfulness to the prosodic properties of morphological heads in the output. Faithfulness to a foot edge associated with a morphological head may similarly be invoked in cases such as *arcimánt-* (NOM/ACC.DU *arcimántā*; compare INST.PL *ṛṣṭimádbhis* ‘having spears’ for the weak stem), where the base, *arcí-*, must, like /-(nr-/, be prespecified with a left foot edge, thus /arc-(i-/. To this end, a constraint intended to mimic the effects of Revithiadou (1999)’s HEADFAITH-type constraints might be conceived of as a species of ANCHOR constraint, parallel to ANCHOR-L/ R_f , but enforcing special faithfulness when the foot edge is associated with the word’s head; such a constraint is repeated in 61.

- (61) ANCHOR-L/ R_f -HEAD: The left/right edge of a foot i in the input sponsored by a morpheme that is a category-determining head of a morphological word corresponds to the left/right edge of a foot i in the output. Assign one violation (*) for each syllable peak that intervenes between f_i in the input and f_i in the output.

Where only left foot edges in the input are concerned, to rank ANCHOR- L_f -HEAD above ALL-Ft-L in the grammar is sufficient to generate the correct outputs. As shown in 62 and 63, when two candidates tie on basic ANCHOR- L_f , and one of the left foot edges is sponsored by a category-changing derivational suffix, ANCHOR- L_f -HEAD, rather than ALL-Ft-L, resolves the conflict. Those two tableaux illustrate the determination of stem stress where the base contains a “weak” lexical accent to which a derivational suffix with a “weak” accent is attached. Indeed, the ranking ANCHOR- L_f $\gg$ ANCHOR- L_f -HEAD helps to improve the harmony of winning candidates in forms like *gómatas* ‘having cows:GEN.SG’ [$(\text{'go:}).\text{ma.tas}$], shown in 64, where the left foot edge sponsored by the head, /-(mant-/, is not faithfully anchored, due to the ranking ANCHOR- R_f $\gg$ ANCHOR- L_f -HEAD. This tableau thus illustrates an effect that will become crucial in the treatment of preaccenting morphemes below: the faithful anchoring of right foot edges, whether sponsored by the head of a word or not, outranks the faithful anchoring of left foot edges throughout.

(62) $/i(nr_j(vant_{Hd}-am/ \text{ 'rich in men-ACC.SG'} \rightarrow nrvántam [nr_1('van.tam)]$

$/i(nr_j(vant-am/$	ANCHOR- R_f	ANCHOR- L_f	ANCHOR- L_f -HEAD	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  $nr_j('van.tam)$		*				*		*
b. $i('nr.van).tam$		*	*!				*	*
c. $nr_1('van).tam$		*				*	*!	**

(63) $/arc_i(i_j(mant_{Hd}-a:/ \text{ 'having flames-NOM/ACC.DU'} \rightarrow arcimántā [ar.ci_1('man.ta:)]$

$/arc_i(i_j(mant_{Hd}-a:/$	ANCHOR- R_f	ANCHOR- L_f	ANCHOR- L_f -HEAD	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  $ar.ci_j('man.ta:)$		*				**		**
b. $('ar.ci).man.ta:$		**!	*				**	**
c. $ar_i('ci.man).ta:$		*	*!			*	*	**

(64) /_i(gav)_{j-k}(mant_{Hd-l}(as/ 'having cows:GEN.SG' → *gómatas* [_i('go:)_j.ma.tas]

		CULMINATIVITY	ANCHOR-R _F	ANCHOR-L _F	ANCHOR-L _F -HEAD	FTBIN	NONFINALITY-σ	ALL-Ft-L	ALL-Ft-R	PARSE-σ
	_i (gav) _{j-k} (mant _{-l} (as									
a.	 _i ('go:) _j .ma.tas			**	*				**	**
b.	go:. _k ('ma.tas)		*!	**				*		*
c.	go:. _i ma. _i ('tas)		*!	**	*		*	**		**
d.	_i ('go:. _i ma).tas		*!	**	*				*	*
e.	_i ('go:) _{j-k} ('ma.tas)	*!		*				*	**	

In many primary nominal and adjectival stems built with suffixes such as *-i-* (seen in *arcí-* 'flame', the base of *arcimánt-*) above or *-ti-* (e.g., *matí-* 'mind', as well as many in *-u-* (e.g., *bahú-* 'much') or *-tu-* (e.g., *ketú-* 'light'), paradigmatic stress typically belongs to pattern 1.3: stress occurs on the syllable of the derivational suffix in all inflectional forms with exception of the GEN.PL. Stress can thus be calculated for such stems exactly as for the paradigm of /(*nr-* 'man', shown in the tableaux in 45 and 46: the weak lexical accent of the stem in, e.g., INST.PL /ma-(ti-(bhis/ wins out over the weak lexical accent of the inflectional suffix (both in virtue of ANCHOR-L_F-HEAD and ALL-Ft-L), thus *matíbhis*, but cedes stress to the lexically exceptional GEN.PL, thus *matínām*. In derivation, these suffixes also cede stress to *-vánt-* or *-mánt-*, since ANCHOR-L_F-HEAD prefers the left foot edge projected from the head in, e.g., /ke:-(tu-(mant-/ → *ketumánt-* 'bright'). Compare the tableau in 63 for an illustration of this effect. To repeat, in the paradigms of words headed by a derivational suffix marked with only a left foot edge, stress remains on the stem in nearly all inflectional forms, since ANCHOR-L_F-HEAD prefers faithfulness to the lexical foot boundary of the head morpheme, and ALL-Ft-L prefers a foot closer to the left edge.⁸⁶ The behavior of stress attraction to the derivational suffix *-mant-* ~ *-vant-* and stress attraction to the GEN.PL is consistently parallel: if a stem cedes stress to *-mant-* ~ *-vant-*, it also cedes stress to the GEN.PL, and vice-versa; if a stem does not cede stress to *-mant-* ~ *-vant-* (e.g., *krátumant-* from *krátu-* 'power'), it does not cede stress to the GEN.PL (thus *krátūnām*).⁸⁷ This analysis then further predicts that adjectives in

⁸⁶A handful of feminine forms in *-ī-* to adjectives in *-mant-* / *-vant-* might, at first glance, appear to contradict the generalization that non-category changing affixes cannot attract stress away from an unaccented head: *gomatīm* (RV 8.24.30d, 10.75.06c), *śaśvatīṣu* (RV 8.60.17c), *āśvāvatīm* (RV 10.97.7a), *somāvatīm* (RV 10.97.7a, immediately following *āśvāvatīm*). The first case is readily explicable as indeed containing a nominalizing head: it is not an adjective, but rather a name for a tributary of the Indus; similarly, *śaśvatī-* (as against 17 × *śaśvatī-*) should also be understood as involving a nominalization (cf. the translation of Jamison and Brereton (2014), 'each and every (clan)'); the last two cases also arguably constitute nominalizations, being the names of specific types of plants the 'horse-possessor' and 'soma-possessor' (cf. again the translation of Jamison and Brereton (2014) of the relevant passage). The suffix *-ī-* in these cases should therefore be regarded as derivational, not inflectional: an effect of HEADFAITH may and should be invoked.

⁸⁷In this regard, the identification by Kiparsky (2010a) of the GEN.PL as being [+dominant], and therefore stress-attracting, cannot be correct. First, as already observed by Perry and Vaux (2018: 281–3), being [+dominant] should entail that /-(ām/ always attracts stress from all stems, at least within Kiparsky's system of accen-

-mant- ~ *-vant-* with stress on this suffix should cede stress to the GEN.PL, but relevant data is sparse.⁸⁸

Meanwhile, in the case of lexically accented derivationally suffixes that virtually always attract the stress (e.g., abstract-noun forming /-(tva)-/, possessive-adjective forming /-(in)-/, past participle-forming /-(ta)-/), faithfulness to a right foot edge, and further faithfulness to the right and left foot edges projected from the morphological head, results in consistent stress attraction. Thus, no morphemes need be specified as [+dominant]: it is sufficient to know whether or not a given affix constitutes the head of the word or not.⁸⁹ More specifically, when a derivational suffix like /-(in)-/ contains the only underlying right foot edge in the input, stress would be licit only on *-ín-* itself or the immediately preceding syllable, since alternative candidates would violate ANCHOR-R_F; in case other morphemes to the left of the head also contain right foot edges, undominated ANCHOR-R_F-HEAD will ensure that it is precisely the right foot edge associated with the head that remains in the output. The parallel ANCHOR constraints holding over left foot edges then in turn serve to guarantee that the left foot edge sponsored by the head is maintained in the output in preference to other underlying left foot edges, as already demonstrated in cases like *arcimántā* in 63 above.

Example 65 illustrates a simple case in which “strong” lexically accented /-(ta)-/ overrides the “strong” lexical accent of a root, /-(taks)-/, to its left (contrast the output *gómatas* in 64 above, where an weakly accented head does not surface with the stress). The virtual unboundedness of this effect is evinced through *bahuvrīhi*-compounds having an overt compositional suffix (cf. 4.2.5 above and 4.3.6 below), which functions as a head. In *sūryadevatīyā* ‘having the sun as a deity:F.NOM.SG’ (MS 3.7.10.x, 4.5.5.bz), the underlying morphological and

tual rules. Conversely, under the present approach, if /-(a:m/ were assessed for violations of ANCHOR-L_F-HEAD, like a derivational suffix, it could attract stress from a root like /-(nr)-/, but not from a derivational suffix like /-(ti)-/: competing candidates ^X*matínām* and *matínám* would in that case have equal violations on ANCHOR-L_F and ANCHOR-L_F-HEAD, leading ALL-FT-L and NONFINALITY-σ to resolve the conflict in favor of unexpected ^X*matínām*. One way or another, the stress behavior of the GEN.PL ending is explicable only through special lexical properties.

⁸⁸A systematic search of the RV and AVŚ revealed the following potentially relevant forms: *ātmanvátām* (AVŚ 19.27.8c), *padvátīnām* (RV 1.152.3a), *yātumátīnām* (RV 1.133.3b), and *revátīnām* (RV 10.180.1d). All other GEN.PL forms to the relevant class of adjectives show stress on the derivational base. The three RVic forms share the property that they contain the feminine agreement suffix *-ī-*, which would cause stress on the ending *-nām* to fall yet farther from the left edge. *ātmanvátām*, however, appears to plainly contradict the prediction made here. Nonetheless, the fact that mobility to the GEN.PL suffix is precisely limited to mono- and disyllabic stems, as far as I am able to determine, may indicate that the additional violations of ALL-FT-L incurred by stress on the final syllable in stems with three or more syllables outweighs faithfulness to the left foot edge of the GEN.PL. Such an observation fits with the more general observation that inflectional stress mobility is generally disallowed in polysyllabic stems. A functioning harmonic grammar analysis to this effect was tested, which indicates that *ātmanvátām* can indeed be accommodated in a phonological analysis that allows for constraint ganging effects.

⁸⁹The constraint-based equivalent to a rule like in 4.2.4 would be to employ lexically indexed constraints (following Pater 2010). The relevant lexically marked derivational suffixes (dominant suffixes) would introduce an effectively undominated version of ANCHOR-R/L_F indexed to those morphemes, thus inducing stronger faithfulness to precisely their lexically specified feet. In practical terms, the analysis would function almost identically to the present analysis centered around HEADFAITH, but has the decided disadvantage of making stress attraction dependent upon an arbitrary lexical property, rather than a predictable morphological configuration.

prosodic structure of the compound is [[[sūrya)-] [dev(a)-tā)-]] (iya_{Hd})-], and both compound members contain at least one prespecified foot edge; nevertheless, the prespecified left and right foot boundaries borne by the head, /-(iya)-/, determines the position of the stress, as illustrated in 66. In both 65 and 66, candidates in which a foot is constructed to place stress anywhere other than on the syllable corresponding to the head of the word will not have fewer violations of basic ANCHOR-R_f and ANCHOR-L_f, while additionally violating at least one of the corresponding HEADFAITH anchoring constraints. Consequently, the candidate preserving the lexically specified foot edges of the head will always constitute the most harmonic candidate.

(65) /_i(taks)_{j-k}(ta_{Hd})_{l-m}/ → *taṣṭám* ‘fashioned:ACC.SG’ [taṣ.(‘tam)]

		ANCHOR-R _f -HEAD ANCHOR-R _f ANCHOR-L _f ANCHOR-L _f -HEAD FTBIN NONFINALITY-σ ALL-Ft-L ALL-Ft-R PARSE-σ								
/ _i (taks) _{j-k} (ta _{Hd}) _{l-m} /										
a.	taṣ. _k (‘tam) _l		*	*			*	*		*
b.	_i (‘taṣ.taṣ) _l		*	*	*!					
c.	_i (‘taṣ) _j .taṣ	*!	*	*					*	*

(66) /su:r-ya)_i-de:v-j(a)_k-ta:)_{l-m}(ya_{Hd})_n-a:/ → [sūr.ya.de:va.(‘ti.yā)]⁹⁰

		ANCHOR-R _f -HEAD ANCHOR-R _f ANCHOR-L _f ANCHOR-L _f -HEAD FTBIN NONFINALITY-σ ALL-Ft-L ALL-Ft-R PARSE-σ								
/su:r-ya) _i -de:v-j(a) _k -ta:) _{l-m} (ya _{Hd}) _n -a:/										
a.	(‘su:r.ya) _i .de:va.ti.yam	*!	***	**	*					****
b.	su:r.ya.de:j(‘va) _k .ti.yam	*!	**	*	*	*		****	**	*****
c.	su:r.ya.de:va. _m (‘ti.yam) _n		***	*				*****		*****
d.	su:r.ya.de:j(‘va.ti) _k .ti.yam	*!	***	*	*			***	*	*****

A small typology of the behavior of stress with respect to “strong” and “weak” accented morphemes may now be enumerated, in particular as pertains to formations containing multiple suffixes. Combinations of “strong” and “weak” morphemes, with or without the rightmost of those morphemes playing the role of morphological head, are shown in Table

⁹⁰The head suffix /-(ya)-/ surfaces with the allomorph -*īya*- here, arguably due to a preference for a disyllabic rather than monosyllabic foot (usually, the allomorph -*īya*- is found after a sequence VCC or $\bar{V}C$). The /a:/ of the suffix /-ta:)/ is subject to morphological truncation. This truncation complicates how, exactly, faithfulness to ANCHOR-R_f with respect to the right foot boundary *l* should be evaluated. I assume that making the vowel /i/ in the suffix -*īya*- into the weak part of a foot, as in candidate d., constitutes movement of the foot edge across a syllable; conversely, candidate b. is assumed to satisfy ANCHOR-L/R_f with respect to *l*, since the foot edge’s position has not shifted with respect to potential syllable nuclei.

4.4, assuming that /X/ and /Y/ each represent morphemes containing exactly one underlying syllabic nucleus. As the table indicates, the only circumstances in which stress will surface on the second morpheme, Y, are when X is weak and Y constitutes the head, or when Y is both strong and a head. In cases where the head is weak and at least one strong morpheme occurs to its left, stress will be predicted to occur as far to the left as possible while minimizing violations of ANCHOR-R_F and ANCHOR-L_F.

First Morpheme	Second Morpheme Non-Head		Second Morpheme Head	
	Weak	Strong	Weak	Strong
Weak	/X-(Y-/	/X-(Y)-/	/X-(Y _{Hd}	/X-(Y _{Hd})-/
	[('XY)]	[('XY)]	[X('Y)]	[X('Y)]
Strong	/X)-(Y-/	/X)-(Y)-/	/X)-(Y _{Hd} -/	/X)-(Y _{Hd})/
	[('X)Y]	[('X)Y]	[('X)Y]	[X('Y)]

Table 4.4: Typology of Combination of “Strong” and “Weak” Accented Morphemes

Let us now summarize the principal distinctions between an analysis employing a HEAD-FAITH constraint and its benefits vis-à-vis the Kiparskian model outlined in Section 2:

- Reference to HEADFAITH (here, ANCHOR-R/L_F-HEAD) enables the definition of a morphosyntactic natural class to which stress-attraction effects apply more strongly in Vedic: category-changing heads of a morphological word. Arbitrarily excluding certain derivational suffixes (in particular, *-mant-* ~ *-vant-*) from the class of items with which they appear to share morphosyntactic properties is unnecessary, if the suffix is assumed to have a “weak” lexical accent, like many inflectional suffixes.⁹¹
- Identifying the types of formations where HEADFAITH is not active, namely, morphologically simplex nouns and adjectives, as well as most verbs, allows for successful reproduction of the effects of the OXYTONE RULE without encountering the potential same issues of overgeneration that the OR may produce.
- An appeal to morphological headedness at least partly obviates the need to distinguish between lexically specified “dominant” and “recessive” derivational suffixes: “dominant accented” derivational suffixes are simply suffixes with a “strong” lexical accent

⁹¹Comparative evidence is good for regarding *-mant-/vant-* as unaccented, or weak in its capacity to attract stress: adjectives in Greek built with the cognate suffix *-εις*, GEN *-εντος* exceptionlessly exhibit recessive accentuation (cf. Chapter 6), and in Hittite, the evidence from adjectives in *-want-* best lends itself to the interpretation that the suffix is unaccented in that language (see Maier 2013 for most of the data and Bondi 2020 for an assessment and analysis). How exactly the lexical prosody of PIE **-/ǵent-/* and the stress grammar of PIE should be reconstructed so as to generate the facts of the three languages is not entirely clear. I would cautiously suggest the reconstruction of an unaccented **-/ǵent-/*, which could attract stress through HEADSTRESS (cf. the constraints in (58)). Such a reconstruction assumes that HEADSTRESS is no longer active in any of the daughter languages, but its effects were reanalyzed as a “weak” lexical accent in Indic.

that may act as morphological heads (see the following section on “dominant preaccenting” suffixes); “recessive” suffixes are either non-heads (the majority of cases), or constitute heads with a “weak” accent or no accent whatsoever (though whether this lattermost cases actually exists is unclear).

Finally, note that, even in a constraint-based model allowing for some derivational architecture (e.g., Stratal OT; Kiparsky 2010b), constraints referring to morphological headedness probably cannot be eliminated from an analysis. In the Stratal OT conception involving a division of the phonology into Stem, Word, and Post-Lexical levels, the Stem Level would encounter seeming contradictions: there are certain stems (with dominant derivational suffixes) that effectively prefer the rightmost lexical accent (i.e., that of the head), but others (mainly certain compounds) that evidently prefer the leftmost lexical accent. Excluding the relevant compound types from the stem-level phonology would appear unprincipled, for which reason headedness would still need to be invoked at the stem level in order to account for dominance effects. Overall, I see no easy escape from the need to reference morphological headedness in Vedic prosody.⁹²

4.3.4 Underlying Foot Tails and Dominant Preaccentedness

The discussion up to this point has demonstrated that the following effects in Vedic stress assignment can be derived with the assumption of a distinction between stress-neutral (mostly roots), weakly stress-attracting (i.e., specified with a left foot edge) and strongly stress-attracting morphemes (i.e., prespecified with a left and a right foot edge), as well as special faithfulness to the prosodic properties of category-changing morphological heads (HEADFAITH:

- edge-to-edge stress mobility between stem and inflectional ending in morphologically simplex nominal and verbal stems (Type 1.1)
- secondary mobility of stress due to vowel deletion, attributable to harmony-maximizing satisfaction of ANCHOR-L_F (Type 1.2)
- attraction of stress to weakly stress-attracting category-changing heads when appended to bases with no lexically prespecified feet only only left foot edges (Type 2.2, producing in inflection Type 1.3).
- accentual dominance as an effect of the constraints ANCHOR-L/R_F-HEAD above the effects of basic ANCHOR-L/R_F (Type 2.1.1, producing in inflection Type 1.5.2).

Largely tacitly ignored, however, have been purely *preaccenting* morphemes of the sort evidently seen in neuter *s*-stems such as *mán-as*- ‘mind’, first introduced under 4.2.3. The

⁹²I reject the notion that dominance effects might involve anti-faithfulness constraints (as proposed by Alderete (2001a)) and explicitly argued for Vedic data by Frazier (2007)). The reason is straightforwardly that I hold that anti-faithfulness constraints produce generally undesirable effects in constraint grammars generally; see arguments to this effect in Apoussidou 2003 (concerning lexical accent in Modern Greek) and Trommer 2006.

first crucial issue to resolve is the precise prosodic representation of Vedic preaccentedness, especially with regards to underlying foot structure, for which two possibilities immediately present themselves: 1) Vedic preaccenting morphemes resist foot parsing, i.e., they have structures like /)-as-/, placing a right foot edge to the left of the morpheme; 2) Vedic morphemes are prespecified as the weak member of a trochaic foot, i.e., they have structures like /-as)-/, placing a right foot edge to the right of the morpheme. Intuitively, the latter is more promising, insofar as it fits with the position of right foot edges assumed above for “strong” accented morphemes, and indeed, further phenomena discussed here will prove to speak in favor of the second option as well.

4.3.4.1 Determining the Representation of Vedic Preaccents

In a prosodic system assumed to consistently construct trochaic, disyllabic feet (of the shapes LL, LH, or HL, or even HH), if a morpheme marked as preaccenting has no potential syllable nucleus, either representation will produce the same result: stress should fall on the second vowel to that morpheme’s left. No such behavior of non-syllabic morphemes is found in Vedic, which is best interpreted to mean that non-syllabic morphemes are simply stress-neutral (as one might intuitively expect).⁹³ If, however, a potential syllable nucleus is present in the morpheme’s UR, the types of representations make distinct predictions: if the morpheme’s syllable peak resists foot parsing, the expected behavior would be like that described for non-syllabic morphemes – stress should appear two syllables to the left of the stress-rejecting morpheme, when possible; if, on the other hand, the morpheme’s syllable peak is marked as a preferred foot tail, then stress will be expected to fall on the immediately syllable to its left.

Easy-to-account-for cases of preaccentedness are evident in the inflection of the verbal category of the so-called perfect. All perfect stems are minimally disyllabic, and are always followed by syllabic inflectional suffixes.⁹⁴ In non-past active singular forms, stress appears on the penultimate (root) syllable; in middle, dual, and plural forms, stress falls on the inflectional ending. One thus finds to $\sqrt{stu}$ ‘praise’, e.g., NPST.2SG *tu-ṣṭó-tha*, NPST.3SG *tu-ṣṭáv-a*, NPST.3PL *tu-ṣṭuv-úr*, NPST.MID.3SG *tu-ṣṭuv-é*. Given an unaccented stem, the inflectional endings of the active singular may be treated as preaccenting, with a right foot edge to the right of the syllable nucleus, thus 2SG /-tha)/, 3SG /-a)/, while other endings are accented, e.g., 1PL /-(ma/.⁹⁵ In all of these cases, the surface stress matches the only underlying lexical accent:

⁹³Recall that Kiparsky (2015) proposes that certain non-syllabic inflectional affixes, e.g., NOM.SG /-s/, PST.3SG /-t/, are preaccenting, which he interprets to mean that they can induce stress on the nearest syllable peak to their left. Given the representation of lexical accentuation of as prespecified foot structure, for non-syllabic affixes to produce such behavior seems impossible. Assuming an autosegment representation of lexical accent likewise does not immediately explain why the floating lexical accent introduced by /-s/ or /-t/ should dock to the nearest syllable nucleus, rather than some syllable nucleus farther to the left, as ALL-FT-L (vel sim.) would prefer.

⁹⁴An exception to the property of stem syllabicity is found in the perfect weak stem of many roots of the shape /C₁aC₂-/; see Sandell 2015a: Ch. 8, Sandell 2017, and Zukoff 2017a: Ch. 5 for details on and analysis of stem formation in this subtype.

⁹⁵In the corresponding past tense (plurperfect), the strong accented inflectional prefix /-(a)-/ always receives

2SG /RED-kar-tha/ → *cakártha*; 1PL /RED-kr-(ma/ → *cakrmá*.⁹⁶ Satisfaction of high-ranking ANCHOR-R_f or ANCHOR-L_f will unproblematically generate these outcomes.

The empirical data on stress-resisting affixes proves to offer some further indications as to which representation of preaccentuation should be adopted. First, there are certain preaccenting suffixes, e.g. nominalizer *-tā-*, primary comparative *-īyas-*, or primary superlative *-iṣṭha-*, which always build a heavy syllable, but never bear the stress themselves. Examples with the primary comparative and superlative suffix in 67 show instances in which the syllable to the left of the suffix is heavy and stressed. Thus, it cannot be the case that a representation like /-īyas-/ can ever result in stress on this suffix itself. Such data is compatible with three interpretations: 1) FTBIN in Vedic is satisfied only by strictly disyllabic feet (i.e., Vedic foot structure is, like Estonian, “truly quantity insensitive” in the sense of Kager 1992); 2) penalties to dispreferred foot structures of the shapes ('HH) or ('LH) always weaker than violations of PARSE-σ, thus similarly leading to disyllabic feet when stress occurs anywhere except on the final syllable; 3) or preaccentedness entails stress-rejecting representations like /-īyas-/.

(67) Stress in Primary Comparatives and Superlatives to Heavy Roots

- a. *nédīyas-* ‘nearer’ and *nédīṣṭha-* ‘nearest’ (^X*nédīyas-*, ^X*nédīṣṭha-*)
- b. *mámhīyas-* ‘more generous’ and *mámhīṣṭha-* ‘most generous’
(^X*mámhīyas-*, ^X*mámhīṣṭha-*)

Since several of the most prototypical preaccenting suffixes are strictly root-attracting (neuter *-as-*, agentive *-tar-*, the comparative and superlative seen above), and roots are typically monosyllabic, probabative examples that would disambiguate between the possible

stress, as expected. More interesting are forms of the perfect injunctive, i.e., perfect stems with past tense endings identical to those found in the imperfect or injunctive of present stems. The number of accented forms in the RV that might constitute perfect injunctives is few: 3SG *cākán* (7x), MID.3PL *pīpáyanta* (3x), 3SG *rārán*, 2DU (*un*)*ninītháh*, 3PL *cákramanta*, 1PL *mamṣjmá*, MID.3PL *cakṣpánta*, MID.3PL *vavṛdhánta* (2x). Stressed as expected here are the two non-third-person forms, 2DU (*un*)*ninītháh* and 1PL *mamṣjmá*, where an accented inflectional ending receives the stress. The MID.3PL forms are decidedly non-uniform, showing stress on the first syllable in one case (*cákramanta*), the root in one case (*pīpáyanta*), and thematic vowel preceding the ending *-nta* in two cases (*cakṣpánta*, *vavṛdhánta*). Finally, two 3SG forms, *cākán* and *rārán* show root stress. The ending of the MID.3PL forms is itself remarkable, in that a thematic vowel is consistently inserted; in some cases, this *-a-* might be the marker of the subjunctive, which itself is to be analyzed as preaccenting /-a)/. Given that thematic vowels in the verbal system are consistently accented or preaccenting, *cákramanta* is all the more notable; I wonder whether an original **cákramata*, without thematic vowel, might underlie this form, whose stress would then simply reflect default left-edge stress. Finally, all tokens of the two 3SG forms, with one exception (RV 2.11.3a), are found in relatively younger portions of the RV (Books 1 and 10), and might reflect a surface analogical generalization applied in a rare inflectional category, namely, that perfects should have root stress when not stressed on an ending. Note that the stress of such forms also superficially matches the position of stress in many other PST.3SG forms without prefix /-(a)-/, in which cases the stress on the final syllable is to be attributed to an accented stem-forming suffix (see 3.7 below).

⁹⁶One peculiarity in the stress pattern of the perfect to the root $\sqrt{taks}$ ‘fashion’ is that stress shifts to the accented inflectional suffixes, even though the root must be treated as strong accented /-(taks)-/ (cf. NPST.3PL *tákṣati*, PTCP.ACT *tákat-*. Expected would be PERF.3PL ^X*tatákṣur*, not actually occurring *tatakṣúr*).

underlying positions of a right foot boundary are not abundant. However, a few *bahuvrīhi*-compounds with first member *su-* ‘good, well’ and a neuter *s*-stem as second member, indicate that /-as-/- must be the correct representation: stress always remains on the first syllable to the left of *-as-*, and does not shift onto *su-*, as a representation /)as-/ would predict. Illustrative data is given in 68; for the unaccentedness of *su-*, compare further the tripartite *bahuvrīhi*-compound *su-prá-cetas-* ‘very wise’, where stress occurs in faithfulness to the leftmost right and left foot edges (i.e., the UR is /su-[(pra)-[cet-as)]]-/.⁹⁷

- (68) *bahuvrīhi*-compounds containing first member *su-* with neuter *s*-stem as second member
- a. NOM/ACC.DU *su-vācas-ā* ‘very eloquent (having beautiful speech)’
 - b. NOM.PL *su-vārcās* ‘having a beautiful shine’
 - c. ACC.SG *su-bhōjas-am* ‘very nourishing’

The combined evidence of stress behavior in the context of preaccenting morphemes thus leads to two important conclusions, which support the assumptions adopted throughout: first, feet in Vedic are to be regarded as strictly quantity-insensitive: disyllabic feet are built to better satisfy PARSE- σ across the board, even where feet of the shape (‘HH) or (‘LH) would result; second, preaccenting morphemes must be represented as foot tails that include a potential syllable nucleus of a morpheme’s phonological representation. This representational assumption matches perfectly the assumption made for the representation of “strong” accenting morphemes, and thus enables a pleasing symmetry in the prespecified prosodic structures of Vedic (see Table 4.3 above).

4.3.4.2 Dominance Effects with Preaccenting Heads

In the preceding section, “strong” accented morphemes proved able to consistently attract stress away from “weak” accented in virtue of the ranking $\text{ANCHOR-R}_f \gg \text{ANCHOR-L}_f \gg \text{ANCHOR-L}_f\text{-HEAD}$, which leads to the prediction that preaccenting morphemes are effectively “strong” as well. This prediction is confirmed by the position of stress in stems built with the “weak” suffix *-mant-* $\sim$ *-vant-* to bases containing a preaccenting specification. Some examples are shown in 69, and a derivation of *yáśasvati* ‘famous:M.LOC.SG’ is given in 70. There, the winning candidate a. simultaneously satisfies ANCHOR-R_f and best satisfies ALL-FT-L and $\text{PARSE-}\sigma$, despite being unfaithful to the left foot edge projected from the head, /-(vant-/-.

- (69) Preaccenting Base + Weak Suffix $\Rightarrow$ Faithfulness to Base Stress
- a. /man-as)-(vant-/ $\rightarrow$ *mánasvant-* [(‘ma.nas).vant-] ‘wise’ (cf. *mánas-*)
 - b. /yaç-as)-(vant-/ $\rightarrow$ *yáśasvant-* [(‘ya.ças).vant-] ‘famous’ (cf. *yáśas-*)

⁹⁷One might suggest that *su-* is stress-rejecting, that is, post-accenting. Speaking against *su-* as strictly stress-rejecting, however, are *karmadhāraya*-compounds of the type *sú-kṛta-* ‘well-made’, in which *su-* does bear stress.

- (70) c. /bandhu)-(mant-/ → *bándhumant-* [(‘ban.dhu).mant-] ‘rich in relatives’
(cf. *bándhu-*)
d. /ras-a)-(vant-/ → *rásavant-* ‘strong; tasty’ (cf. *rása-*)
yaç-as)_{i-i}(vant)_{Hd-k}(i/ ‘famous-M.LOC.SG’ → *yásasvati* [(‘ya.ças)_i.va.ti]

What occurs when a preaccenting head is attached to a base whose stress is also determined by a right foot edge? Potentially relevant data come from derivational suffixes that may attach to other derived stems or compounds, and which appear to regularly condition stress on the immediately preceding syllable. Outside of compounds (especially determinative compounds) having an overt compounding suffix for producing a verbal noun (e.g., *soma-pá-van*- ‘soma-drinking’, with preaccenting *-van-*), a set of related abstract-forming suffixes are found in potentially relevant configurations: *-tā-*, *-tāt-*, and *-tāti-*. Of particular relevance are examples like those given in 71, in which the base form exhibits stress on the initial syllable, but shifts to the peninitial syllable when one of these suffixes is attached; the examples cited there are exhaustive for the RV.

Precisely cases of the sort seen in the example above are telling, since the fixed stress on the initial syllable of the base word throughout its paradigm indicates that the lexeme must be marked with either an “strong” accented first vowel (e.g., /*(ba)*ndhu-/) or preaccenting

second vowel (e.g., /bandhu-/). As the examples here indicate above, an input /(ba)ndhu-tā)/ or /bandhu)-ta:)/ can other generate the desired output [ban.(‘dhu.tā)] only if faithfulness to the right foot edge projected from the head /-tā-/ is able to outweigh the benefits to either placing stress closer to the left edge (^X*bándutā*-) or placing stress on the morphological head syllable itself, σ_{Head} (^X*bandhutá*-). Under the present analysis, the latter alternative, ^X*bandhutá* ‘relation:F.NOM/ACC.SG’ will be excluded due to more violations of ALL-Ft-L and NONFINALITY- σ ; in longer forms such as *jyeṣṭhātātīm*, the winner is selected over an alternative with stress on the derivational suffix again in virtue of ALL-Ft-L. See the tableaux in 72 and 73.

(72) /bandhu)_i-ta:)_{Hd})/ ‘relation:NOM/ACC.SG’ → *bandhūtā* [ban.(‘dhu.tā)]

/bandhu) _i -ta:) _{Hd})/	ANCHOR-R _f -HEAD	ANCHOR-R _f	ANCHOR-L _f	ANCHOR-L _f -HEAD	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  ban.(‘dhu.ta:) _j		*					*		*
b. (‘ban.dhu) _i .ta:	*!	*						*	*
c. ban.dhu.(‘ta:)		*				*!	**!		**

(73) /jya:(i)_istha-ta:)_jti)_{Hd}-m/ ‘supreme lordship-ACC.SG’ → *jyeṣṭhātātīm* [jye:ṣ.(‘ṭha.tā).tim]

/jya:(i) _i stha-ta:) _j ti) _{Hd} -m/	ANCHOR-R _f -HEAD	ANCHOR-R _f	ANCHOR-L _f	ANCHOR-L _f -HEAD	FtBIN	NONFINALITY- σ	ALL-Ft-L	ALL-Ft-R	PARSE- σ
a.  jyeṣ.(‘ṭha.tā).tim		*					*	*	**
b. (jyeṣ).ṭha.tā.tim	*!	*						***	***
c. jyeṣ.ṭha.(‘tā.tim)	*!	**					**		**
d. jyeṣ.ṭha.(‘tā).tim		*					**!	*	***

To fill out the typology of stress assignment in derivation, consider further the case of *kavyātā* ‘wisdom’. The position of the stress again indicates the the preaccenting head, /-tā-/ is decisive in determining the stress of the entire stem, achieving its preference despite the fact that it is preceded by *two* strong accented suffixes. That is, the UR should be taken as /kav-(i)-(a)-tā-/), since this noun is derived from the adjective *kavyá*- ‘wise’, which must be built with a suffix /-(a)-/, in turn derived from the noun *kaví*- ‘sage’, which contains a strong accented suffix /-(i)-/, as indicated by the possessive compound *kaví-kratu*- ‘having the strength of a sage’. The winning candidate a. in 74, *kavyātā* emerges as the winner because the only alternative that also satisfies ANCHOR-R_f-HEAD performs worse on ANCHOR-L_f and ALL-Ft-L. In evaluation of the candidates, I assume that glide formation effectively induces a rightward shift of the foot edges (cf. the treatment of secondary mobility above)

originally associated with the suffix /-(i)-/, such that the underlying edges *i* and *k*, and *j* and *l* can be anchored to the same edge in the output. For this reason, the winning candidate a. incurs no violations of ANCHOR- $L_{\mathcal{F}}$.

$$(74) \quad /kav_{-i}(i)_{j-k}(a)_{l-ta:_{Hd}})_m/ \text{ ‘wisdom:NOM/ACC.SG’} \rightarrow kav\acute{y}at\bar{a} \ [kav.(^{\prime}ya.ta:)]$$

$/kav_{-i}(i)_{j-k}(a)_{l-ta:_{Hd}})_m/$		ANCHOR- $R_{\mathcal{F}}$ -HEAD	ANCHOR- $R_{\mathcal{F}}$	ANCHOR- $L_{\mathcal{F}}$	ANCHOR- $L_{\mathcal{F}}$ -HEAD	FTBIN	NONFINALITY- σ	ALL-FT-L	ALL-FT-R	PARSE- σ
a.	$kav_{i,k}(^{\prime}ya.ta:)_m$		**					*		*
b.	$kav_{i,k}(^{\prime}ya)_{j,l}.t\bar{a}$	*!	*	*				*	*	**
c.	$(^{\prime}kav.ya)_{j,l}.t\bar{a}$	*!	*	**					*	*
d.	$kav.ya.(^{\prime}t\bar{a})_m$		**	*!*				**		**

This survey of stems constructed with preaccenting derivational suffixes again demonstrates that the position of stress in a derived stem is controlled by the prosodic preferences of the morphological head of the word. When the head morpheme is lexically specified as /-(X)-/, stress falls on the head syllable throughout the paradigm (Type 2.1.1); when the head morpheme is lexically specified as /-X)-/, stress falls on the immediately preceding syllable. In most cases, the ranking of the four anchor constraints is sufficient to correctly predict stress assignment; in some cases involving preaccenting heads, the lower-ranking prosodic markedness constraints ALL-FT-L and NONFINALITY- σ may play a minor role in choosing between a monosyllabic and disyllabic foot, always militating in favor of disyllabic feet. Only when the head of a derived stem lacks marking with a right foot edge does the head fail to impose its prosodic preference, though this situation is effectively restricted to the (abundant) possessive adjectives in *-mant-* ~ *-vant-*. In the following excursus, I provide further motivation for the need to operate with the four-way underlying prosodic distinction between “strong” accented, “weak” accented, preaccenting, and unaccented morphemes, with dominance effects achieved through HEADFAITH, rather than other tempting alternatives.

4.3.4.3 Excursus: Why HEADFAITH and not HEADSTRESS?

Sandell (2015a: 176–89) developed a partial analysis of stress assignment in Vedic that differed from the present analysis in at least three crucial respects: first, the prosodic properties of lexical entries were assumed to be autosegments, following Revithiadou (1999: Ch. 2); second, Sandell argued that stress attraction to morphological heads treated here as having “weak” lexical accents could be generated through use of HEADSTRESS, thus assigning stress to a lexically unaccented morphological head; third, accentual dominance could be generated as an effect of constraint ganging in a harmonic grammar model, that is, simultaneous faithfulness to underlying lexical accents and satisfaction of HEADSTRESS could predict the dominance effects shown by “strong” accented morphemes like past participle *-tá-*. While the treatment of Vedic accentuation in Sandell 2015a did not investigate preaccenting affixes

deeply, it could be extended without difficulty to dominance effects shown by preaccenting derivational suffixes by calling upon HEADFAITH as well. As long as preaccents are conceptualized as an autosegment marked with a weak “.” foot tail, faithfulness to this lexical accent, when sponsored by a morphological head, would generate dominant preaccenting effects. A three-way prosodic distinction, between stress-attracting morphemes marked with “*”, preaccenting suffixes marked with “.”, and unaccented suffixes could then adequately capture most of the data. Unaccented derivational suffixes could receive stress, when appended to unaccented stems, in virtue of HEADSTRESS.

That the analysis of Sandell 2015a was not simply revised and extended in the present treatment has three main justifications. First, a three-way lexical prosodic distinction, plus effects of HEADSTRESS and HEADFAITH, is insufficient for distinguishing between the types of paradigmatic stress seen in monosyllabic stems, i.e., full mobility in stems like *pad-* ‘foot’, mobility only to the GEN.PL in stems like *nṛ-* ‘man’, and immobility in stems like *gav-* ‘cow’. In particular, since *nṛ-* keeps stress where *pad-* cedes it (contrast *nṛbhis* vs. *padbhís*), it cannot be unaccented or preaccenting; however, since the GEN.PL, *narám* contrasts with GEN.PL *gávām*, *nṛ-* and *gav-* require different representations as well.⁹⁹ Second, and more pressingly, the assumption that lexical accents constitute autosegments is unable to neatly explain the facts of secondary mobility, as discussed above. The consequence of this issue, namely, the assumption that lexical accents should be represented as foot edges, brings a further, most severe complication: no grammar, whether employing strictly ranked constraints or weighted constraints, in which stress on the morphological head of stems like *ma-tí-* or *pad-vánt-* is taken to depend on stress attraction to an unaccented morpheme, can prove compatible with the behavior of preaccenting derivational suffixes. In other words, as long as lexical accent is modeled representationally in terms of foot structure, and one adopts lexical representations like /ma-ti-m/ ($\rightarrow$ *matím*) and /man-as/ ($\rightarrow$ *mánas*), irreconcilable conflicts in the ranking or weighting of constraints result.

Comparative tableaux given in (75) demonstrate why the particular set of assumptions that would be needed to translate the analysis of Sandell 2015a into an analysis with lexically specified feet proves to be impossible. I refer the reader back to (58) for a definition of the constraint HEADSTRESS_{F-L}. A specific, more restrictive version of FTBIN, FT = ($\sigma\sigma$), is employed in the comparative tableaux to demonstrate further that a specific preference for disyllabic feet in Vedic cannot rescue an analysis that crucially depends on HEADSTRESS. In order to render the tableaux more legible, the four ANCHOR constraints are compressed into two. First, item c., indicates that basic ANCHOR-L_F must outrank the relevant prosodic markedness constraints. Comparison of violations in the pairs of items a. and b., however, reveals them to be diametrically opposed: the loser in a., ^X[('ma.tim)], incurs exactly the same violations in exactly the same quantities as the winner in b., [('ma.nas)], and vice-versa. Hence, no ranking or weighting of the four constraints ALL-FT-L, PARSE- σ , and FTBIN ($\sim$ FT = ($\sigma\sigma$)) exists. Furthermore, since the prosodic structures of [ma.tim] and [ma.nas] are oth-

⁹⁹While oxytone stems of the *matí-* type show paradigmatic mobility parallel to *nṛ-* ‘man’, they can be accounted for by ranking special faithfulness to the lexical accent of the genitive plural above HEADSTRESS, under this approach. Since HEADSTRESS does not apply to monomorphemic stems, root stress throughout the paradigm of *nṛ-*, if it is assumed to be unaccented, cannot be explained.

erwise identical (a light syllable followed by a heavy syllable), no other conceivable prosodic markedness constraint could possibly distinguish between the two winner-loser pairs. Fundamentally, the issue boils down to the fact that, although /-as)-/ is presumably marked with a right foot edge, faithfulness to this foot edge is maintained, regardless of whether a disyllabic (winning *mánas*) or monosyllabic foot (losing ^x*manás*) is constructed. Therefore, prosodic markedness constraints like ALL-FT-L must prefer the winner, *mánas*. However, since item a., with winner *matím*, depends upon HEADSTRESS_{F-L} to generate the correct outcome, this constraint, violated by the winner of b., would presumably outrank the prosodic markedness constraints.

(75) Comparative Tableau: *matím* ‘mind:ACC.SG’, *mánas* ‘mind:NOM/ACC.SG’, *padí* ‘foot:LOC.SG’

		ANCHOR-L/R _F	ANCHOR-R/L _F -HEAD	ALL-FT-L	HEADSTRESS _{F-L}	PARSE-σ	FT = (σσ)
	/ma-ti _{Hd} -m/						
a.	ma.(‘tim) ~ (‘ma.tim)			1 vs. 0 (L)	0 vs. 1 (W)	1 vs. 0 (L)	1 vs. 0 (L)
	/man-as _{Hd}) _i /						
b.	(‘ma.nas) _i ~ ma.(‘nas) _i			0 vs. 1 (W)	1 vs. 0 (L)	0 vs. 1 (W)	0 vs. 1 (W)
	/pad-(i/						
c.	pa.(‘di) ~ (‘pa.di)	0 vs. 1 (W)		1 vs. 0 (L)		1 vs. 0 (L)	1 vs. 0 (L)

Hence, an irresolvable quandry results, unless some further lexical properties can be attributed to the input of either *matím* or *mánas*, so that the prosodic faithfulness constraints can distinguish the winners. The easiest analytical move, pursued to its logical conclusion in the sections above, is to associate a left foot edge with the lexical entry of the derivational suffix *-ti-*, thus /-(ti-/. The revised comparative tableaux shown in (76) incorporate this analysis, with the result that the ANCHOR constraints are now exclusively winner-preferring, allowing them to be ranked above the prosodic markedness constraints and HEADSTRESS_{F-L}. The latter, in turn, can be ranked below the markedness constraints, which are winner-preferring for item b.¹⁰⁰

(76) Revised Comparative Tableau: *matím* ‘mind:ACC.SG’, *mánas* ‘mind:NOM/ACC.SG’, *padí* ‘foot:LOC.SG’

¹⁰⁰Incidentally, in a full set of tableaux including HEADSTRESS_{F-L}, MRCD ranks HEADSTRESS_{F-L} above PARSE-σ.

		ANCHOR-L/ R_f	ANCHOR-R/ L_f -HEAD	ALL-Ft-L	HEADSTRESS $_f$ -L	PARSE- σ	Ft = ($\sigma\sigma$)
	/ma-(ti _{Hd} -m/						
a.	ma.(‘tim) ~ (‘ma.tim)	0 vs. 1 (W)	0 vs. 1 (W)	1 vs. 0 (L)	0 vs. 1 (W)	1 vs. 0 (L)	1 vs. 0 (L)
	/man-as _{Hd}) _i /						
b.	(‘ma.nas) _i ~ ma.(‘nas) _i			0 vs. 1 (W)	1 vs. 0 (L)	0 vs. 1 (W)	0 vs. 1 (W)
	/pad-(i/						
c.	pa.(‘di) ~ (‘pa.di)	0 vs. 1 (W)		1 vs. 0 (L)		1 vs. 0 (L)	1 vs. 0 (L)

Nonetheless, a different representational choice could salvage the role of HEADSTRESS in Vedic prosody. Specifically, one could assume that preaccenting morphemes are not merely equipped with a right foot edge, but are explicitly marked on the stress grid as the weak part of a foot, as per the autosegmental representation of lexical accent used in Revithiadou 1999 (see discussion in 2.4.2.2). This approach could be applied as follows: if /-as)-/ is specified as a “.” on the stress grid, i.e., an unstressed syllable, and this property must be faithfully preserved in the output (through a species of HEADFAITH), then a candidate like ^X[(‘ma.nas)] is easy to exclude. However, this tactic would constitute a departure from the theoretical assumptions employed up to this point, following Özçelik (2014) and Yates (2020), that lexical marking of prosodic constituency should be sufficient to predict exceptional stress, without invoking autosegmental relations between segments and the stress grid. Given that precisely those kinds of autosegmental relations have been shown to make false predictions concerning secondary mobility in Vedic — at least when employed without specification of prosodic constituency as well — the treatment of lexical stress in Vedic would find itself needing to encode both stress directly in grid representations *and* the constituent structure of those stresses. At minimum, the choice to simultaneously represent both stress and feet results in more complex representations, and must allow for constraints that refer both to the stress relations on the grid as well as metrical constraints on foot structure. Again, from the point of view of theoretical parsimony, being able to dispense with at least one set of representations and constraints — be it the “top down” representation of stress or the “bottom-up” representation of prosodic constituency — would be desirable.

4.3.5 Revisiting Cruces: The Genitive Plural Ending and Exceptional Mobility

Two particularly recalcitrant aspects of Vedic accentuation were introduced under 4.2.3 and 4.2.4.2 above: the pattern labeled “exceptional mobility” (Type 1.4), where mobility stress mobility between the stem edge and inflectional suffixes in morphologically complex polysyllabic stems obtains, and the attraction of stress to the genitive singular plural in -(n)*ām* among certain stem types (namely, Types 1.1.2 and 1.3), where otherwise no stress mobility to inflectional suffixes occurs. In the course of the present analysis, the particular tendency of the genitive plural to attract stress has been handled through a lexically specific constraint. The objective of this section is to argue that both the exceptional stress-attracting property of the genitive plural suffix and the stems involved in exceptional mobility have fundamentally historical explanations: namely, these behaviors were explicable under the analysis

developed in this section earlier in the history of Vedic (most likely, in Proto-Indo-Iranian). However, subsequent phonological changes rendered the motivation for accentual mobility in both cases opaque, with the consequence that, in the case of the genitive plural, a lexically specific constraint came into being, and in cases of exceptional mobility, multiple allomorphs with different prosodic properties were acquired.

4.3.5.1 The Ending of the Genitive Plural

A historical explanation of the exceptional “strength” of the genitive plural inflectional ending $/(n)\text{-}\bar{a}m/$, able to attract stress away from other “weak” accented morphemes, is relatively straightforward. First, metrical evidence from Vedic and the closely related Avestan indicates that the ending $\text{-}\bar{a}m$ was indeed originally disyllabic, and thus to be reconstructed for Proto-Indo-Iranian as $*[-a.H\check{a}m]$. The accentuation of the historically related ending $\text{-}\acute{\omega}n$ in Greek likewise points to an originally disyllabic suffix, with higher pitch on the non-final vowel, thus $*[-\acute{o}.H\check{o}m]$; the circumflex (falling) accent of Greek is, outside of monosyllables and a synchronic accentual rule ($\sigma\omega\tau\eta\rho\alpha$ -Rule), almost entirely to be attributed to historical vowel contraction (see Chapter 6 for further details on Greek). Identical considerations apply to the Lithuanian genitive plural suffix $\text{-}\acute{u}$: the circumflex (rising) accentuation in that language is likewise to be derived historically from a contracted sequence $*[-\acute{o}Hom]$. A reconstructed ending $*[-\acute{o}H\check{o}m]$ is further compatible with the genitive plural suffixes found in both the athematic and thematic declensions of virtually all older Indo-European languages (Fortson 2010: 118; 129.¹⁰¹ A standard internal reconstruction of the ending posited for PIE, $*[-\acute{o}H\check{o}m]$, analyzes it as containing (the) derivational suffix(es ending in) $*[-(C)\acute{o}\text{-}]$ plus an original inflectional suffix $*[-(H)\acute{o}m]$. Thus, the ending $*[-\acute{o}Hom]$ would have originally belonged to thematic nouns with stress on the stem-final syllable; and ending $/-\acute{o}\text{-}H\acute{o}m/$ was ultimately reanalyzed as $*[-\acute{o}H\acute{o}m]$, either independently in many daughter languages or already in PIE itself.

The crucial assumption required in order to be able to smoothly generate the Vedic facts a representation of the ending $*[-oHom]$ with two lexical accents (= left foot edges), that is Pre-Vedic $*[-(a)H(am)]$, reflecting the etymological origin of the suffix as two lexically accented suffixes.¹⁰² The special “strength” granted to the genitive plural suffix through the

¹⁰¹In many Indo-European branches (e.g., Italic, Anatolian), the ending found is completely ambiguous between $*[-om]$ with short $*[o]$, $*[-\acute{o}m]$ with long $*[\acute{o}]$, and disyllabic $*[-oHom]$. The ending found in Celtic (Proto-Celtic $*[-om]$ > Old Irish nasalizing mutation) and Slavic (OCS $-\bar{o}$), however, would seem to reflect a PIE $*[-om]$, but such an ending might be attributed to either an original athematic ending $*[-om]$ applied also to the thematic declension in those branches, or potentially shortening of long $*[\acute{o}]$ before nasals in final syllables.

¹⁰²The exact course and chronology that restructuring of the phonological representations must have taken to produce such an outcome is somewhat tricky. In the internally reconstructed original state, there existed one inflectional ending, GEN.PL $/(H)\acute{o}m/$, which surfaced with prominence, thus $[-\acute{o}m]$, on unaccented bases allowing mobility; when attached to a derivational suffix in $*[-(C)\acute{o}\text{-}]$, the UR $/(C)\acute{o}\text{-}(H)\acute{o}m/$ must have surfaced as $*[-\acute{o}(H)\acute{o}m]$. In PIE, PIIr., and Pre-Vedic, as long as instances of the original athematic GEN.PL. $*[-\acute{o}m]$ were present on the surface at the time when vowel contraction began to apply to the complex ending $*[-\acute{o}Hom]$ > $*[-\acute{o}m]$, speakers deriving this surface $*[-\acute{o}m]$ would presumably still have taken it from an underlying $*[-\acute{o}(-)H\acute{o}m]$, marked with two lexical accents. For a subsequent generation of learners, for whom the contracted

constraint ANCHOR- L_f -GEN.PL is in fact equivalent to an additional violation of ANCHOR- L_f . I invite the reader to compare the outcomes of the tableaux in 46 and 47, removing ANCHOR- L_f -GEN.PL and instead adding another violation of ANCHOR- L_f to candidates in which stress does not fall on the inflectional ending; the outcomes will remain the same, and correct. In effect, a lexical representation with *two* left foot boundaries, /-(a(am/ or /-((ām/ would prove adequate to explain the stress behavior of this inflectional suffix. Furthermore, precisely such a representation accords with the intuition of Perry and Vaux (2018) that mobility to the genitive plural constitutes a species of secondary mobility, if one accepts the insight of Yates (2020) that Type 3 secondary mobility results from the correspondence of two underlying left foot boundaries with a single left foot boundary in the output. In short, the historical etymological complexity of the inflectional ending -(n)ām can be leveraged to justify a lexical representation with multiple left foot edges, or equivalently, a lexically specific constraint that functions like an additional violation of ANCHOR- L_f .

4.3.5.2 Excess Mobility

Recall from Section 4.2.3 that Vedic presents a number of stem classes – essentially restricted to the verbal system – in which stress alternates between the rightmost stem syllable and the final syllable (i.e., an inflectional ending), despite the fact that the stress could always remain on the penultimate syllable throughout the paradigm. In other words, the behavior of forms such as Class V presents (e.g., 3.PL *sunó-ti*, 2.PL *sunu-té*) contrasts with nominal and adjectival stems generally, where stress always remains on the stem (unless the final stem syllable is lost, inducing Type 3 secondary mobility). The crux of the issues is that such excessive mobility is not expected if a stem-forming suffix such as /-naw-/ is lexically accented, /-(naw-/ (whether it counts as a category-changing head or not): from a UR 2.PL /su-(nu-(te/, ^X*sunúte* would be expected, with either ANCHOR- L_f or ALL-FT-L militating in favor of the left foot edge associated with /-(naw-/ ~ /-(nu-/. At the same time, stress other than on the initial syllable is unexpected if the stem is prosodically unmarked (e.g., 3.SG ^X*súnoti*) would be expected. Identical considerations and predictions result, regardless of whether the lexical accents are conceptualized as foot edges or autosegments. I repeat here for convenience the table in 23 above, in which all but the second row presents a paradigm with clear excessive mobility.

(77) Excessive Stem–Suffix Mobility in the Finite Verb

ending *[-óm] was no longer transparent, the paradigmatic stress patterns that speakers possessed of an etymologically transparent UR /-ó(-)Hóm/ would have generated, would have induced the new generation to find an explanation for those patterns, either in the form of a UR like */-((ām/ or a lexically specific constraint.

Root	STRONG ACT.3SG	WEAK (consonant) MID.3SG	WEAK (vowel) ACT.3PL	Category
kṛ- ‘make’	<i>caḱāra</i>	ACT.1PL <i>caḱṛmā</i>	<i>caḱrūr</i>	Perfect
yuj- ‘join’	<i>yunákti</i>	<i>yunkté</i>	<i>yuñjánti</i>	Class VII Present
pu ¹ - ‘purify’	<i>punáti</i>	<i>punīté</i>	<i>punánti</i>	Class IX Present
su- ‘press’	<i>sunóti</i>	<i>sunuté</i>	<i>sunvánti</i>	Class V Present
kṛ- ‘make’	<i>kṛnuyát*</i>	<i>kṛnvūtá*</i>	<i>kṛnuyúr*</i>	Optative (Class V Present)

The verbal categories given in the table here all show the same pattern: stem-edge stress in strong inflectional forms, but suffixal stress in weak inflectional forms. The pattern found in the active participle built with *-ant-~at-* presents fundamentally the same problem, but with the additional complication that stress remains on the suffix *-át-* if the inflectional suffix is consonant-initial or word-final; compare the examples in 78, repeated from 24 above.

- (78) Excessive Stem–Suffix Mobility: Active Participle of *duh-* ‘milk’^{103, 104} The active participle thus presents three surface allomorphs, stressed *-ánt-* and *-át-* alongside unstressed *-at-*.

- a. NOM.PL *duhántas*
- b. DAT.SG *duhaté*
- c. INST.PL *duhádbhis**
- d. N.NOM/ACC.SG *duhát**

Before continuing the discussion, two of the categories listed in 77 can be removed, since their patterns are easily explicable: the Perfect in the first row and the Class VII Present in the second row. As demonstrated under 4.3.4 above, the stress mobility found in the perfect is unproblematically generable given preaccenting endings in the strong inflectional forms (e.g., 3SG /-a/) and “weak” accented endings in the weak inflectional forms (e.g., 3PL /-(ur/). The Class VII present, in turn may be analyzed as containing an accented infix, /-(na-/ but since its weak allomorph is non-syllabic, Type 3 secondary mobility to an inflectional ending occurs. Thus, only the Class V and IX presents, the optative, and the active participle truly instantiate exceptional mobility.

¹⁰³The distribution of stress in the participle is thus in fact similar to the most common sort of secondary mobility, as also observed in Perry and Vaux (2018: 283–5). Under the analysis proposed by Perry and Vaux, secondary mobility involving an accented suffix /-án-/ is indeed at work; the dental stop found throughout the paradigm is due to a later readjustment rule that inserts a /-t-/ after a non-perfect participle preceding any phonological material.

¹⁰⁴Perry and Vaux (2018: 284, fn. 20) further suggest that the same set of rules and representations (accented suffix, secondary mobility, *t*-insertion as readjustment) accounts for the stress of adjectives in *-mant-/vant-* as well, since the distribution of stress is similar. The crucial difference, though, is that stress in these adjectives never surfaces on the inflectional suffix, instead always remaining on the suffix *-mant-/vant-* (cf. *padváte*, *dyumáte*, etc.). Hence, no secondary mobility can apply in these forms; whether rule of *t*-insertion might be responsible for the dental segment is in part orthogonal, since the assignment of stress to a morphological head might apply after such a readjustment rule. An OT equivalent to such a readjustment rule may be found in a lexically-specific constraint ranking introduced by the active participle that allows for epenthesis of a dental stop in the environment /V__ŷ/.

The solution adopted for stress in the perfect opens the question of whether treating the inflectional endings found in the strong forms as preaccenting could be viable for all further cases as well. Indeed, Kiparsky (2018a: 136) proposes to generate the stem-edge stress in strong cases through preaccenting inflectional suffixes (see again the discussion under 4.2.3). A right foot edge would, in fact, obtain the desired effect when the inflectional ending is syllable (/su-naw-ti/ → *sunóti*), but not, however, if the ending is non-syllabic (/kr-nu-ya:t/ → ^x*kṛnúyāt*, not expected *kṛnuyát**). At least under the present approach, then, generating exceptional mobility via preaccentuation is a non-starter.

Could the introduction of autosegmental floating accents instead be viable? Presumably, a non-syllabic affix that introduces a floating strong lexical accent would result in leftmost stress in Vedic. That is, a representation like OPT.3SG /kr-nu-ya:t/ (where boldface here indicates a floating strong lexical accent sponsored by the suffix /-t/) might be expected to associate to the leftmost syllable on the surface. Perhaps one means of obtaining the desired result is to operate with a DOMAIN constraint of the sort discussed by Myers and Carleton (1996) and Revithiadou (1999: 53–5): in the cases treated there, a markedness constraint is proposed, *DOMAIN, which penalizes association of a lexical accent within the domain that it sponsors. A converse faithfulness constraint, FAITH-DOMAIN, would penalize cases in which a lexical accent is not realized within the domain that sponsors it. In effect, this would mean that the strong lexical accent sponsored by e.g., /-t/ in Vedic ought to be associated with a syllable containing this /-t/; if the floating lexical accent is associated to a syllable not containing the morpheme that sponsored it, then FAITH-DOMAIN is violated. However, precisely the same constraint would then predict suffix on the inflectional ending if syllable: 3SG /su-naw-ti/ would be predicted to generate ^x*sunotí*, rather than expected *sunóti*.

Further complicating this matter is the fact that, in the noun, the NOM.SG suffix /-s/ frequently (in any consonant-final stem) has no clear segmental exponent: this suffix is subject to deletion whenever a consonant precedes it. In such cases, whether FAITH-DOMAIN still holds is less evident, since the segmental material that establishes the domain is absent. One could alternatively see the domain as a sort of abstract bracket with which a syllable boundary should be aligned, but in this case, the solution begins to look similar to the use of foot edges. All things considered, treating the “strong” inflectional suffixes as preaccenting in order to account for excess mobility seems unlikely to succeed, once preaccentedness is fitted out with an explicit formal representation.

A final maneuver, which could indeed successfully account for much of the data, would be to treat the syllabic suffixes that appear with stem-edge stress, e.g., 3SG.NPST *-ti* or ACC.SG *-am* as preaccenting (thus /-ti/), /-am/), but non-syllabic endings like 3SG *-t* or NOM.SG *-s* as “weak accented” (thus /-(t/) or /-(s/), such that then an input like /kr-nu-ya:-(t/ or /-ant-(s/ would assess violations of ANCHOR-L_F with respect to the nearest underlying vowel. That is, an output like *kṛnuyát* would satisfy ANCHOR-L_F, while alternatives with stress occurring further to the left would violate it.¹⁰⁵ Even a segmentally null inflectional ending /-(/ for the

¹⁰⁵In the event that also the expected winner, *kṛnuyát*, should also be understood as violating ANCHOR-L_F, one might invoke a non-categorical definition of the constraint, such that one violation would be incurred for each vocalic peak intervening between the foot edge’s correspondent in the output and its position in the input. This alternative definition of ANCHOR will not lead to any false predictions among the other patterns

NEUT.NOM/ACC.SG could be invoked to generate forms like *dhṛṣát* ‘emboldening’ (e.g., RV 8.62.05a) out of underlying /dhṛs-at-/. Only the pre-consonantly weak cases forms of the participle, where stress remains on the suffix *-át-* would then remain unexplained. Nevertheless, arbitrarily positing underlying left foot edges for non-syllabic endings but underlying right foot edges for syllabic endings seems intuitively unsatisfying.

A less arbitrary approach, which would again falter only on the consonant-initial weak case forms of the active participle (e.g., INST.PL *uśádbhis* ‘desiring’), would be to see a role for HEADSTRESS in Vedic stress assignment. The further assumption required would merely be that the stem-forming affixes implicated in exceptional mobility, /-ant-/, /-naw-/, /-ya:-/, and /-na-/, be granted the status of morphological heads; in terms of filling in the typology of lexical prosody, doing so is attractive, since these affixes would then constitute the only unaccented derivational suffixes in the language, which would otherwise be absent altogether. Given the constraint HEADSTRESS_{F-L} from 58, plus a ranking ANCHOR-L_F ≫ HEADSTRESS_{F-L} ≫ ALL-Ft-L, the strong forms like *sunót* or *duhántas* emerge because they do not violate HEADSTRESS_{F-L}, while weak forms like *sunutá* or *duhanté* win because they satisfy ANCHOR-L_F. The tableaux in 79 and 80 present derivations of these outcomes.¹⁰⁶ False predictions would come about only in the consonant-initial weak case forms of the participle, where an input /duh-ant-(bhis/ would be expected to yield ^x*duhadbhís*, not actual *duhádbhis*.*.

(79) /duh-ant_{Hd}-as/ ‘milking-NOM.PL’ → *duhántas* [du.(‘han.tas)]

/duh-ant _{Hd} -as/		ANCHOR-R _F -HEAD							
		ANCHOR-R _F				ANCHOR-L _F -HEAD			
		ANCHOR-L _F				HEADSTRESS _{F-L}			
		ALL-Ft-L				ALL-Ft-R			
		PARSE-σ							
a.	du.(‘han.tas)						*		*
b.	(‘du.han).tas					*!		*	*

treated throughout.

¹⁰⁶Note that this analytical scenario requires a specification of FtBIN as being fulfilled only by disyllabic feet. The reason is that in forms like *jyeṣṭhátātim*, the ranking ALL-Ft-L ≫ HEADSTRESS_{F-L} would be necessary to generate the correct position of the stress (given UR /jyā-i)stha-tā)ti_{Hd}-m/, unless disyllabic feet are explicitly enforced; that is, FtBIN will rule out the candidate with stress on the head, [jyeṣ.ṭha.(‘ta:).tim]. In that case, the ranking ANCHOR-L_F ≫ FtBIN ≫ HEADSTRESS_{F-L} ≫ ALL-Ft-L will fit the facts of exceptional mobility and encounter no contradictions in other forms.

(80) /duh-ant_{Hd}-(e:/ ‘milking-DAT.SG’ → *duhaté* [du.ha.(‘te:)]

/duh-ant _{Hd} -(e:/		ANCHOR-R _f -HEAD	ANCHOR-R _f	ANCHOR-L _f	ANCHOR-L _f -HEAD	HEADSTRESS _{f-L}	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.	☞ du.han.(‘te:)					*	**		**
b.	du.(‘han.te:)				*!		*		*
c.	(‘du.han).te:				*!	*		*	*

Maximally arbitrary but phonologically easiest and descriptively complete would be to admit of accented and unaccented allomorphs for each of the relevant stem-forming suffixes, and in the active participle, three distinct allomorphs, /-(ant-/ , /-(at-/ , and /-at-/ . The selection of allomorphs is not readily phonologically predictable and must then necessarily be driven by morphosyntactic features. That is, the respective allomorphs do not occur in fully predictable prosodic contexts (see 81 below): while the context of unstressed *-at-* is indeed predictable, the accented allomorphs *-ánt-* and *-át-* overlap. Given the correct allomorph inserted in the correct morphosyntactic context, generating the correct stress is trivial. Why, however, should distinct competing allomorphs be available, rather than a (prosodically) unitary UR?

(81) Distribution of Allomorphs of Active Participle *-ant-*

	Syllabic Context	Lexical Prosody of Following Morpheme
<i>-ánt-</i>	Open or closed syllable	Unaccented
<i>-át-</i>	Closed syllable	Accented or unaccented/absent
<i>-at-</i>	Open syllable	Accented

The fundamental assumption made is that history creates these conflicts that can lead to the acquisition of multiple listed allomorphs. Specifically, the availability of such allomorphs is *diachronically* may be attributed to the dispreference for certain types of syllable peaks to bear stress in Proto-Indo-European and Proto-Indo-Iranian, namely, [+syllabic, +consonantal] segments and the epenthetic vowel (PIE) *[ə] ~ (PIIr.) *[i]. The table in 82 presents Proto-Indo-European *transponat* reconstructions based on the Vedic forms of nasal presents to laryngeal-final roots, the optative to a nasal present, and the active participle; what should stand out in such paradigms is that the stress falls on the final syllable when the penultimate syllable contains either a syllabic nasal or a schwa as its syllable nucleus, and the final syllable is a lexically accented inflectional ending, usually with a full vowel.¹⁰⁷

¹⁰⁷The 3PL of athematic optative forms is, for PIE, to be properly reconstructed with an ending *[-ih₁ent]; am uncertain as to where the position of the prosodic prominence fell, since other languages are non-diagnostic between [-ih₁ent] or *[-ih₁ént]. For a pre-consonantal form of the active participle, I give the preform of the endless N.NOM/ACC.SG.

(82) Reconstruction of Stems with Excess Mobility

	STRONG	WEAK (consonant)	WEAK (vowel)	
* /peuH-/ ‘purify’	*[pu.néH.ti]	*[pu.nəH.tóy]	*[pun.Hén.ti]	Nasal Present
* /sker-/ ‘cut’	*[skr.nu.yéh ₁ t]	*[skr.nu.yəh ₁ .tó]	*[skr.nu.ĩf]	Optative
* /d ^h eṽg ^h -/ ‘meet’	*[d ^h u.g ^h én.tes]	*[d ^h u.g ^h ñt]	*[d ^h u.g ^h ñ.téy]	Act.Part.

Under the assumption that the relevant stem-forming suffixes were originally lexically accented, stress on this suffix, e.g., *[-ént-], *[-né-], etc., is expected in combination with unaccented roots. When such suffixes instead occurred in their zero grade forms preceding accented inflectional endings, the fact that ANCHOR-L_F would be violated at least once by any candidate allowed for markedness constraints militating against stress on syllable peaks like *[ŋ] or [ə] to prefer stress on the lexically accented inflectional ending.¹⁰⁸ The weight or ranking of such a constraint need merely outweigh the effect of ALL-Ft-L, since stress occurs one syllable farther to the right. The tableau in 83 illustrates this effect for a reconstructed DAT.SG of an active participle, *[d^hu.g^hñ.téy], under the assumption that morphological headedness constraints play no active role in these formations.

		ANCHOR-L _F *[ŋ]/*[ə]		
		ALL-Ft-L		
/d ^h eṽg ^h - _i (ent- _j (e _i /				
(83)	a.  d ^h u.g ^h ñ _j (téy)	*		**
	b. d ^h u _i (g ^h ñ.téy)	*	*!	*
	c. (d ^h u.g ^h ñ).téy	**!		

The stress of the neuter NOM/ACC.SG on a zero-grade suffix [-ñt] will follow from the original accentedness of the suffix (i.e., /-(ent-/), but the stress preceding consonant-initial endings will not; again, an INST.PL ^X[-ñtb^hís] would be expected. Two possible accounts can justify the actual state of affairs. Perhaps the allomorph [-ñt-] > [-át-] found in the N.NOM/ACC.SG comes to be selected in preconsonantal contexts as well, since they share a common prosodic environment (i.e., will build a closed syllable on the surface). Or, given that all of the consonantal-initial case endings found in Vedic are generally assumed to reflect relatively late grammaticalizations of cliticized particles (see Jasanoff 2009 on the instrumental plural, specifically), these suffixes probably originally stood outside the domain of stress assignment, and the stressed allomorph -át- may be a last remnant of this situation.¹⁰⁹

¹⁰⁸ Given that the same effect seems to have held for *[u], to judge from the Class V presents of the *sunóti* type as well, one might be inclined to invoke sonority-based stress, or a subtle WSP effect: given lower sonority and/or shorter duration of [+syllabic] segments [ŋ], [ə], or [u], stress will preferentially be assigned to vowels like [e] or [o] with higher sonority and/or longer duration. Compare Kobayashi 2004: 22–4 and Byrd 2015: 75–9 for discussion of sonority effects in the selection of syllabic nuclei in PIE.

¹⁰⁹ Why, exactly, a trace of an original unstressedness these endings should have remained in the active participle in particular is unclear.

Whatever the ideal treatment of excess mobility may be, this discussion should impress upon the reader that its behavior is simply not straightforwardly with any thoroughly developed grammar of Vedic stress. One way or another, to a greater or lesser extent, recourse to lexical exceptionality, morphological readjustment rules (as per Perry and Vaux 2018), or similar devices seems unavoidable.

4.3.6 Stress in Compounds

With an account of dominance effects in category-changing derivation now clearly established, we can return to the behavior of stress in compounds, following the descriptions provided in 4.2.5 above. Recall from the description there that a distinction between “regular” compounds of the structure [[X] [Y]] and synthetic compounds with a compounding suffix [[[X] [Y]] Z] was necessary to adopt. The accentuation of compounds of the latter type, ubiquitous among the descriptive compounds, also found among the possessives, and characteristic of prepositional governing compounds, is straightforward to accommodate in terms of head faithfulness: since the overt compounding suffix may be regarded as a category-determining head, its prosodic properties are imposed upon the entire stem; recall also that such suffixes were treated by Kiparsky as dominant, erasing other lexical accents to their left. Similarly, the stress behavior observable for many possessive compounds, all iterative compounds, many determinative compounds, and probably most verbal governing compounds, whereby stress corresponds to the leftmost “strong” lexical accent (described under 4.2.5 simply as “leftmost lexical accent”), falls out unproblematically from the interaction of ANCHOR-R_f and ALL-FT-L that results in selection of the leftmost underlying right foot edge, under the assumption that no considerations of morphological headedness play an active role in the calculation of stress for such compounds. Importantly, if the first member contains only a “weak” lexical accent, i.e., a left foot edge, (e.g., / (nr-caks-as) - / → *nṛcákṣas*- ‘seeing men’), then stress falls on the second member, in accord with an underlying right foot edge found there.

Less straightforwardly explicable under the present analysis, on the other hand, are compounds that present stem-edge stress (i.e., on the rightmost syllable preceding inflectional suffixes, e.g., *śiti-pád*- ‘white-footed’), either in the absence of any underlying lexical accents (as with some possessives and descriptive compounds) or as a consistent rule (as is the case for all copulative compounds, possessive compounds with privative *a*- or a numeral as first member, and as an historically innovative pattern among determinative compounds with nominal heads). Under Kiparsky’s account of stress in compounds, these cases are attributable either to direct application of the OXYTONE RULE on unaccented strings, or to the addition of a segmentally null but [+dominant] compounding suffix, which conditions erasure of other lexical accents and allows for subsequent determination of stress by the OXYTONE RULE. Since the OR has no direct analogue under the analysis developed in this section, its effects being instead reproduced through HEADFAITH, a different sort of explanation may be necessary.

Relevant sorts of possessive and descriptive compounds with unaccented stems indicate that the rightmost stem syllable in such cases is treated as the strong member of a foot. For

example, NOM.PL /*śiti-pad-as*/ ‘white-footed’ must be parsed as [śi.ti.(‘pād.as)]. One might attempt to argue that, in the case of an unaccented descriptive determinative such as *ṛju-nīti-* ‘correct leading’, an inheritance of headedness is at work: *nīti-* leading, constitutes the semantic and morphological head of the compound, and the head of that word is, in turn, the suffix /-(ti-/), thereby leading to the parse [ṛ.ju.nī.(tí-] in order to satisfy ANCHOR-L_f-HEAD (given a UR /rj-(u-nī-(ti-/). In other words, the property of headedness percolates upwards from the head of the head. Similar considerations could apply to verbal governing compounds whose first member is a noun in *-ti-*: in such forms (e.g., *vr̥ṣṭí-dyāv-* ‘raining the sky (i.e., causing the sky to rain)’), the compounded (i.e., the phrasal head) falls on the left, and “weak” accented /-(ti-/ constitutes the head of that word, thus attracting stress. The tableaux in 84 and 85 contrast the application of “head percolation” in a determinative compound with the absence of a need for a headedness effect in a *bahuvrīhi* of the presumed structure [[X] [Y]].

(84) /rj_i(u-ni:~j(t_{Hd}-m/ ‘good-leading-ACC.SG’ → *ṛjunītim* [ṛ.ju.(‘ni:tim)]

/rj _i (u-ni:~j(t _{Hd} -m/		ANCHOR-R _f -HEAD	ANCHOR-R _f	ANCHOR-L _f	ANCHOR-L _f -HEAD	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.	ṛ.ju.(‘ni:tim)			*		**		**
b.	ṛ.(‘ju.ni:).tim			*	*!	*	*	**
c.	(‘ṛ.ju).ni:tim			**!	*		**	**

(85) /_i(nr-caks-as)_j-as/ ‘seeing men-M.NOM.PL’ → *nṛcákṣas* [nṛ.(cák.ṣa)_j.sas]

_i (nr-caks-as) _j -as		ANCHOR-R _f -HEAD	ANCHOR-R _f	ANCHOR-L _f	ANCHOR-L _f -HEAD	ALL-Ft-L	ALL-Ft-R	PARSE-σ
a.	nṛ.(cák.ṣa) _j .sas			*		*	*	**
b.	(‘nṛ.cak).ṣa.sas		*!	*			**	**
c.	nṛ.cak.(‘ṣa).sas			*		**!	*	***

The possessive compounds of the *śiti-pád-* type, however, resist explanation in terms of reference to either an underlying foot edge or head attraction of any sort. If, following the traditional understanding of *bahuvrīhis* as as null adjectivizations of descriptive determinatives, one were to regard the second member /pad-/ as a head, one might invoke attraction to it in virtue of HEADSTRESS, necessarily outranking ALL-Ft-L. However, any such head attraction is obviously contradicted by the numerous cases in which ALL-Ft-L, rather than HEADSTRESS, chooses between two underlying “strong” lexical accents.

Here, one must first admit that the *siti-pád-* type (i.e., *bahuvrīhis* whose stress under the Kiparskian account can be explained directly via the OXYTONE RULE) are not abundant. Sec-

ond, with the exception of *śiti-pád-* itself, all oxytone *bahuvrīhis* in the RV where the second member must lack a lexical accent are found with just five particular first members: *dvi-* ‘two’, *tri-* ‘three’, *su-* ‘good’, *dus-* ‘bad’ and privative *a(n)-*, where *a(n)-* and *tri-* in particular tend to trigger an otherwise unexpected oxytone stress. Some examples are given in 86.

(86) Possessive Compounds with Oxytone Stress

- a. *śiti-pád-* ‘white-footed’ (RV)
- b. *dvi-pád-* ‘two-footed’ (RV)
- c. *dus-pád-* ‘having bad feet’ (RV)
- d. *su-pád-* ‘having beautiful feet’ (RV)
- e. *tri-stúbh-* ‘threefold praise (name of a meter)’
- f. *a-bandhú-* ‘having no relatives’ (cf. *bándhu-*)
- g. *tri-vandhurá-* ‘three-seated’ (cf. *vandhúra-*)

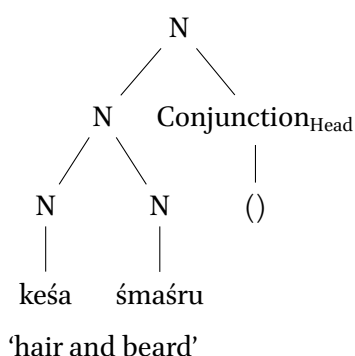
With these cases may be contrasted at least a handful of *bahuvrīhis* with first member in *a(n)-* exhibiting stress on the initial syllable to second members in /-(ti-/: *á-kṣiti-* ‘having no ruin = imperishable’ (RV 1.40.4b, 8.103.b, 9.66.7c; cf. *kṣiti-* 5 × AVŚ), *án-ānubhūti-* (RV 6.47.17c) ‘having no loyalty’, *á-diti-* ‘lack of possessions’ (RV 4.2.11d, 1.152.6d, 10.87.18b).¹¹⁰ Another interesting example is *híri-śmaśru-* ‘having a golden-colored beard’; if *śmaśru-* is treated rather as unaccented /śmaśru-/, then the barytone stress of the compound might be comprehensible. These cases at least constitute possible counterexamples to the presumption that stem-edge stress in *bahuvrīhis* reflects a default phonological process. For this reason, I would suggest that the examples of the sort seen in 86 contain a segmentally null compounding affix consisting of a “strong” lexical accent attached to the right edge of the stem, such that the rightmost vowel of the stem then constitutes the relevant point of reference for ANCHOR-L/R_F-HEAD. Whether any total consistency in the application of such a null compounding affix in *bahuvrīhis* is to be discovered exceeds the scope of the present investigation.

Further support for a segmentally null compounding affix is also furnished by coordinative compounds of the type of the type *keśa-śmaśrú-* ‘hair and beard’, for which stem-edge stress is systematically attested. At least in the coordinative type, an appeal to headedness of the members is a non-starter, which means that such compounds must somehow genuinely compel the assignment of stress to the rightmost stem syllable, while simultaneously ignoring any underlying stress preferences contained in the compound members. In the context of the analysis developed in this section, the only viable manoeuvre is assume the insertion of a foot edges subject to head faithfulness, i.e., ANCHOR-L/R_{F-Head}, which would be functionally equivalent to the null dominant compounding suffix proposed by Kiparsky in such cases. A segmentally null compounding suffix thus appears difficult to escape.

¹¹⁰There may be an argument to be made on the basis of comparative data that these forms with barytone stress are older, or at least contain older, inherited second members. *á-kṣiti-* famously has a near-match in Greek ἄφθιτος ‘imperishable’, and the parallel *-ti-stem φθίσις also exists; the same applies to Vedic *bhūti-* (= Gk. φύσις) and *díti-* (= Gk. δόσις).

Note, however, that the desired stress will result only if the rightmost vowel of the stem is marked with a left and right foot head, i.e., it is treated as having a “strong” lexical accent. Arguably, given the morphological structure of the stem of coordinative compounds in 87, the compounding left foot edge necessarily “adopts” the nearest vowel to its left: a left foot edge cannot mark a vowel to its right that is has not yet been inserted in the morphological derivation (i.e., via inflectional rules). In other words, the morphological stem in 87 is phonologically linearized as /keśa)śmaś(Headru)-/

(87) Hierarchical Structure of Vedic Coordinative Compounds



To summarize, treating Vedic compound stress without reliance on the phonologically sensitive OXYTONE RULE introduces a need for considerable further reliance on morphological and syntactic relations between the compound members. Given the larger costs brought with the OXYTONE RULE – discussed above and reviewed again in the following section – the further assumptions suggested here as a means to accomodate compound stress must first be shown to be either empirically false or theoretically unmotivated.

4.4 The Default Stress Grammar of Vedic

Before concluding this chapter with a summary of the major results obtained in the preceding Section 4.3 and implications for research to come in the following chapter, worth considering now is how the “default” metrical stress grammar of Vedic should be characterized. That is, if the morphology-prosody interface constraints simply were to “vanish” from the phonological grammar of Vedic, or, in cases in which they are irrelevant, what type of prosodic outputs would be expected. In part, this question has been addressed under 4.3.1 above, where inputs without any lexically specified prosody, such as /puma:ms-am/ ‘male-ACC.SG’, were discussed; this discussion may now be enriched in light of a final total ranking of the constraints first listed in Table 4.2, given in (88)s.¹¹¹

¹¹¹This ranking was obtained using the MRCD algorithm in the software of Becker et al. 2007 and confirmed with an implementation of the same algorithm, in the version described by Becker 2009, written by the author in R (see under Chapter 3 at <https://github.com/rpsandell/Sandell-LMU-Habilitation>). A file containing the all candidates considered, in OT-Soft format, `Vedic_Str_Wk_Stress.txt` is available under

(88) Final Constraint Ranking for the Analysis of Section 4.3:

TROCHEE, CULMINATIVITY, *VV, ANCHOR-R_f-HEAD $\gg$ ANCHOR-R_f $\gg$ (ANCHOR-L_f-GEN.PL), ANCHOR-L_f $\gg$ ANCHOR-L_f-HEAD, FTBIN $\gg$ ALL-Ft-L, NONFINALITY- σ $\gg$ WSP, ALL-Ft-R, HEADSTRESS_{f-L} $\gg$ PARSE- σ , MAX-V

When the morphology-prosody interface constraints and other purely segmental constraints are removed, the ranking in 89 of prosodic markedness constraints alone remains. Any other garden-variety prosodic markedness constraints may sit in the lowest stratum or a stratum below PARSE- σ . The range of predicted prosodic parses and stress outcomes made by this grammar is entirely straightforward, and indeed, uninteresting: a disyllabic (in virtue of FTBIN $\gg$ WSP $\gg$ PARSE-SYLL $\gg$ RH-CONTOUR [latter not given]) trochaic foot will be built at the left edge of the word, leaving all other syllables outside this foot unparsed (in virtue of ALL-Ft-L $\gg$ ALL-Ft-R $\gg$ PARSE- σ); stress will then consistently emerge on the initial syllable of the prosodic word.

(89) Final Constraint Ranking of Prosodic Markedness Constraints Alone:

TROCHEE, CULMINATIVITY $\gg$ FTBIN $\gg$ ALL-Ft-L, NONFINALITY- σ $\gg$ WSP, ALL-Ft-R $\gg$ PARSE- σ

In accord with all of the preceding discussion, the role of prosodic markedness in Vedic is indeed demonstrably minimal; rather, lexically specified prosodic preferences and morphological structure are decisive in most instances. The few active prosodic markedness serve essentially just to limit the creation of foot structure and to ensure that the single foot per ω takes on the appropriate shape.¹¹²

This overall result, however, from the perspective of investigations into variation in the accentuation of specific lexical items of Vedic, is satisfying and offers further confirmation for those results. First, Poucha (1943) observed a general tendency, internal to the RV and between the RV and AVŚ, for later variants of the same word to be more likely to exhibit stress on a syllable closer to the left edge. More recently, treatments of stress in *ti*-stems by Lundquist (2015a) and Sandell (2016) advanced the argument that etymological *ti*-stems that come to be treated as morphological simplexes would tend to present persistent stress on the initial syllable of their paradigms; Sandell pointed in particular to the case of *śakti*- $\sim$ *śakti*- ‘power, ability’ (with derivative *śātīvant*- ‘able, skilled’) internal to the RV, while Lundquist dealt in particular with RV *matī*- ‘mind’ versus ŚB *māti*-. See also 5.1.1.

That lexemes like *śakti*- develop persistent stress on the initial syllable of their paradigms helps to shed further light on a phenomenon of paradigmatic stress in Vedic that, in the analysis above, was taken merely to be an epiphenomenon of the fact that most stems contain

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¹¹²Other means of ranking the same constraints make yet more evident that, even among the constraints considered, only TROCHEE, ALL-Ft-L, and ALL-Ft-R are of non-negligible relevance in choosing winning candidates. Specifically, a ranking found using the MIRCD algorithm of Yates and Zukoff (2018) consigns all prosodic markedness constraints except TROCHEE, ALL-Ft-L, and ALL-Ft-R to the lowest stratum, while a weighting of constraints obtained using the MaxEnt Grammar Tool of Hayes et al. 2009a (with $\mu = 0$, $\sigma^2 = 10000$) assigns a weight of zero to all prosodic markedness constraints except TROCHEE, ALL-Ft-L, ALL-Ft-R and FTBIN (the lattermost however close to 0).

at least one lexical accent: stress only shifts to inflectional endings (again, except for the GEN.PL) if the stem is always monosyllabic (e.g., *pad-*) or has a monosyllabic allomorph (e.g., *púmāms-*, with allomorph *pums-*). Under an analysis in terms of Harmonic Grammar, persistent stem stress on the leftmost syllable is predicted, even if the stem lacks any lexical accents, as a consequence of additional violations of ALL-FT-L incurred by candidates that satisfy ANCHOR-L_F by attracting stress to the final syllable. In this regard, the general failure of stress to be attracted to stress-preferring inflectional suffixes in polysyllabic stems finds explanation in the general prosodic preferences of the language (cf. also Sandell (2015a: 207–9), where lack of intraparadigmatic mobility in polysyllabic stems was treated as an effect of Output-Output faithfulness). The tableaux deriving INST.PL *sáktibhis* below shows that, given a polysyllabic base, multiple violations of ALL-FT-L can outweigh a single violation of ANCHOR-L-FT.¹¹³

(90) /çakti-(bhis/ ‘ability-INST.PL’ → *sáktibhis* [(‘çak.ti).bhis]

/çakti-(bhis/		ANCHOR-L _F	ALL-FT-L	PARSE-σ
a.	8 (‘çak.ti).bhis	7 × 1		1 × 1
b.	9 çak.ti.(‘bhis)		4 × 2	1 × 1

This single result indicates that a harmonic grammar formalism allowing for constraint gang-ing effects is likely most appropriate as a means of capturing a fuller range of stress phenomena in Vedic, even if the vast majority of cases are explicable using strictly ranked constraints, as was done above.

Finally, important to recognize is the fact that substantially different choices concerning the representation of lexical accent and the analysis of key stress assignment phenomena in Vedic would not seriously impact the default metrical stress grammar, at least as far as foot structure is assumed to be constructed and relevant at all. Under an analysis in which Vedic lexical accent were instead treated as an autosegmental association of vocalic peaks to strong or weak lexical accents (“*” and “.”), the same preferences in terms of markedness would govern the prosodic outputs would come to the fore: stress would preferentially surface on a disyllabic trochee maximally close to the left edge of the word.¹¹⁴ Only in the event that Vedic prosodic structure were to be analyzed without foot structure altogether (i.e., as tone or pure prominence), all constraints relevant to foot structure would simply fall away; in that case, some version of CULMINATIVITY would necessarily still hold, and ALL-FT-L would

¹¹³Constraint weights were calculated using Linear OT in Boersma and Weenink 1992–2023. Weights were solved for all the constraints 88, using the candidates on which the analysis in Section 3 was based, plus the candidates given in 90 below.

¹¹⁴Compare the analysis of Modern Greek Stress in Revithiadou 1999: 84–127 for the translation of autosegmental lexical accent into foot structure in phonological outputs, governed by prosodic markedness constraints.

be replaced by a similar alignment constraint, preferring alignment of the most prominent syllable or high tone with the left edge of the word. The core insights (“no more than one prominence per ω ; prominence ideally leftward”) thus remain constant under any analysis.

4.5 Conclusions and Prospects

This chapter has undertaken a comprehensive survey of the synchronic prosody of Vedic Sanskrit, with a predominant focus placed on the calculation of the position of word-level prosodic prominence, which is characterized by a heavy dependence on morphological structure and lexical specification. Beginning from an overview of what is known and can be established concerning the phonetics of Vedic prosody and evidence for some effects of intonational phonology in Section 4.1, I first carried out an extensive exposition of the analysis of Vedic accentuation that has been developed by Kiparsky (particularly its most recent forms) in 4.2, before proposing a (partly) novel analysis in Section 4.3, and finally seeking to draw conclusions on the default stress grammar of the language in Section 4.4. In the course of the chapter, I have advanced the following major claims:

- The position of word-level prosodic prominence in Vedic can and probably should be characterized in terms of stress, which attracts a prominence tone, H^* or perhaps H^*L : the distributional restrictions found on this prominence essentially fit the criteria of canonical stress languages, and apparent deviations (“accentless” words) largely admit of explanation by reference to intonation (see Section 4.1.3).
- The Kiparskian analyses of Vedic accentuation obtain admirable empirical coverage, but suffer from several worrisome defects: 1) an inexplicitness concerning the representation of lexical accent (especially regarding preaccentuation); 2) the tendency of some core phonological rules (the OXYTONE RULE) to overgenerate, thereby necessitating recourse to lexical exceptionality and/or the (non)-application of processes as an effect of cyclicity; 3) an abstract distinction between DOMINANT and RECESSIVE affixes (see especially the discussion in 4.2.3 and 4.2.4).
- The major stress patterns initiated in intraparadigmatic inflection as well as interparadigmatic derivation in Vedic can be generated in a Optimality Theoretic Grammar that characterizes lexical accent in terms of prespecified foot structure and special faithfulness to the underlying prosodic properties of (category-changing) morphological heads. A small residue of lexical exceptionality remains, which can be attributed to historical factors.
- By default, prosodic words in Vedic contain a single, quantity-insensitive, disyllabic, trochaic foot that strives to stand as close to the left edge of the word as possible. Higher-ranking prosodic faithfulness constraints, however, result in deviance from this unmarked prosodic form in many, indeed statistically the majority, of outputs in Vedic.

While much discussion has been devoted throughout this chapter to questions of the proper and necessary representation of lexical accent in Vedic, especially the choice between prespecified foot structure vs. an autosegmental prosodic role, the choice is somewhat “six of one, half a dozen of another”. While operating with prespecified feet indeed facilitates the analysis of secondary mobility (see 4.2.2 and 4.3.2) and may be leveraged in some instances to account for patterns of vowel deletion, it complicates the analysis of preaccentuation effects, thereby necessitating more lexical prosodic distinctions (i.e., the distinction between “strong” accented, “weak” accented, preaccenting, and unaccented morphemes proposed in 4.3.1). Since no set of representational choices and grammar considered here is able to neatly handle the patterns of excess mobility (see 4.3.5.2), it remains to be seen whether the analysis proposed in Section 4.3 can be modified so as its empirical coverage with fewer subsidiary assumptions.

Looking ahead to the case studies to come in Chapters 5, 6, and 7, the results obtained and theoretical devices introduced hold implications for each of the further language (branches) to be investigated (Classical Sanskrit, Greek, and Germanic). The concrete constraint grammar developed here to capture most of the patterns in Vedic stress assignment will be explicated contrasted in 5 with the stress grammar that generates the stress patterns of Classical Sanskrit, and raise the important (and vexing) question of how and why the properties of lexical accent and head dominance failed to be acquired later in the history of Indic. Near-identical questions recur with respect to the prosodic development of the Germanic languages in 7, only there with Proto-Indo-European as the point of reference. In contrast, features of lexical accent and head dominance will prove to still play a role in the prosodic system(s) of Attic-Ionic (and Doric) Greek, though there, the interactions between stress, tone, and intonation at the level of the prosodic word will prove to be even more intricately intertwined than the Vedic phenomena treated under 4.1.3. Now, a look into the development and change of the Vedic prosodic system, on the way to the metrical stress system of Classical Sanskrit, can commence.

CHAPTER 5

From Lexical Stress to Metrical Stress: Prosodic Change in Sanskrit

Building upon the results of the preceding Chapter 4, this chapter concerns itself with changes in the prosodic system of Sanskrit during the course of the Vedic and especially into the Classical period. As was explained in 4.1.1, the fact that the direct marking of prosody in the manuscripts of Vedic texts is largely limited to the very earliest compositions, testimony concerning any transitional prosodic system of later Vedic is relatively meager. Possible indications may be gathered from apparent changes in the stress of individual lexemes, as well as the comparatively divergent system of prosodic marking used in the *Śatapathabrāhmaṇa*. These developments and their possible significance are briefly surveyed and assessed in 5.1. The core of the chapter is built in Sections 5.2 and 5.3, in which first the system of stress assignment in Classical Sanskrit is reconstructed and analyzed, and then an attempt to computationally model the emergence of such a system out of the older Vedic system is attempted.

5.1 Potential Prosodic Developments in the Vedic Period

Among the Vedic texts outfitted with systems of word-prosodic marking, testimony concerning the position of stress (as signaled by the *udātta*) in individual words and some larger word classes is not entirely uniform: variation in the position of stress (*udātta*) is found in numerous individual lexemes, and in some instances, yet wider changes in the stress pattern of a morphological category may be discerned. In addition, the system of prosodic marking employed in the *Śatapathabrāhmaṇa* diverges from the systems found in the other Vedic texts in so many substantial ways that one rightly wonders whether the prosody of the ŚB must not reflect some innovative prosodic grammar with respect to the systems seen in the RV and AV, for instance. While the significance of this material as a whole is difficult to assess, and does not seem to always point to a single coherent change in lexical properties or a prosodic grammar, it suffices to demonstrate that the prosodic grammar described in the preceding chapter was not entirely static and stable.¹ Most tantalizing would be to find possible forerunners of the stress rule established for later Sanskrit under 5.2. Unfortunately, the evidence from lexical variation in the RV and AV is best interpreted otherwise, while the

¹Indeed, given the established dialectal diversity underlying the various Vedic schools (see generally Witzel 1989), complete uniformity in the prosody of every lexeme would be surprising. However, because variation in position of the *udātta* is attested within one and the same text, the prosodic variation cannot be attributed straightforwardly to any dialectal variation.

evidence from the ŚB does not allow for any confident conclusions.

5.1.1 Changes in Lexical Stress and “Demorphologization”

Already Poucha (1943) carried out a systematic survey of segmentally identical forms occurring in the RV and AVŚ that are attested with the *udātta* on different syllables.² Not of concern for present purposes are true minimal pairs, in which the differences in stress also corresponds to a difference in lexical semantics; in some instances, the two forms are accidentally segmentally identical (e.g., *ákra*- ‘inactive’ vs. *akrá*- ‘banner’), while in other instances, a morphological relationship between the pair exists (e.g., *ápas*- ‘work, action’ vs. *apás*- ‘active’). More interesting are cases in which clearly the same lexical item occurs with the *udātta* in more than one position, e.g., *ákṣi*- ~ *akṣí*- ‘eye’ or *trípad*- ~ *tripád*- ‘three-footed’. Such stress-variants are found internal to one or the other text (*ákṣi*- ~ *akṣí*- both in the RV) or across the two texts (only *tripád*- occurs in the RV, *trípad*- and *tripád*- occur in the AVŚ). These cases constitute a total of 92 lexemes, but I exclude one case of an original dual *dvandva* with two *udāttas* in the RV (*agnīsomáu*) as against a single stem-edge *udātta* in the AVŚ (*agnīsomáu*; cf. 4.2.5.2 above), leaving 91 total forms.

Although Poucha is interested in establishing for each individual case which variant is chronologically younger, his assumptions on the internal chronology of the RV cannot be taken over with confidence. While differences in position of stress between the RV and AVŚ might be attributed to chronology, since the latter is certainly younger than the former, the decisive effect of morphological structure on stress in Vedic means that different stresses may be attributed to different morphological analyses applied by different speakers. Only in case the variant data were truly uniform in their behavior from the point of view of metrical stress – which they are not – might chronology and prosodic change at the level of the grammar make for a convincing explanation. Ultimately, many of the pairs gathered by Poucha are probably best explained by reference to the notion of “demorphologization” (Probert 2006b), or the application of innovative compound stress rules. Beyond cases explicable in these ways, one can perhaps make out a tendency to avoid stress on final syllables.

Table 5.1 summarizes the patterns based on position of the *udātta* and number of (stem) syllables in the data gathered by Poucha (1943), remaining agnostic as to the direction of any “stress shift” in general or in specific cases. For the purposes of labeling the data, two-syllable words have only an initial and final syllable, three-syllable words have an initial, penultimate, and final syllable, while four-syllable words also have a peninitial syllable. In order to properly assess the potential emergence of any broader tendencies in prosodic markedness, data that could be otherwise explained must be excluded.

Among the 34 disyllables, at least four forms, *varṣmán*- ‘height’, *śaktí*- ‘power’, *nṛtí*- ‘dance’, and *puṣṭí*- ‘nourishment’ probably reflect a shift $\sigma\acute{\sigma} > \acute{\sigma}\sigma$ due to loss of morphological structure: the suffixes *-ti*- and *-man*- (MASCULINE) typically attract stress, but when interpreted as morphological simplexes, prosodic faithfulness constraints no longer hold, and prosodic

²Poucha’s data was gathered on the basis of lexical resources available to him, principally Grassmann 1872 [1976] and Whitney 1881.

Stress Positions	Two Syllables	Three Syllables	Four Syllables
Final : Initial	34	23	2
Final : Penultimate	–	6	4
Final : Peninitial	–	–	5
Penultimate : Peninitial	–	–	1
Penultimate : Initial	–	15	
Peninitial : Initial	–	–	1
Sum	34	44	13

Table 5.1: Frequencies of Variant Stress in the RV and AVŚ (data from Poucha 1943)

markedness constraints generate initial stress. See already Lundquist 2015b on this development among Vedic *-ti*-stems generally, Sandell 2016 on *śakti*- specifically, and Sandell 2015a: 210 on masculine *man*-stems.

Meanwhile, the most common morphological configuration among trisyllables with either initial or final stress are adjectival determinative compounds built with privative *a(n)*- as the first member. Recall from discussion under 4.2.5.3 and 4.2.5.4 that negative *a(n)*- usually does not receive stress, with some exceptions (mainly privative compounds with a past passive participle as second member). In many of these cases, interpretation of the compound as morphologically simplex could lead to the assignment of initial stress, and in a few instances, the semantics suggests loss of morphological transparency as well (e.g., *a-ghniyá*- ‘not to be slain > cow, bull’; *arasás*- ‘juiceless > powerless’). Except perhaps in one instance, demorphologization may hold here, as well. Where the AVŚ has *a-stṛtá*- ‘invincible’ as against *á-stṛta*- in the RV, the form with final stress might reflect the expected lexical accent of a past passive participle, without application of the special subrule holding over privatives to past participles. The same explanation may apply to two four-syllable forms with *a*- as well (*ásapatna*- ‘without a rival’, and *ápraśasta*- ‘not praiseworthy’, though note that the former is a *bahuvrīhi*).

Given the frequency with which “demorphologization” can be invoked as a plausible explanation for variants with initial stress, at least, the safest way to exploit Poucha’s data to intuit any genuine traces of innovations in terms of prosodic markedness is to consider solely only forms that are likely morphologically simplex in any case, and for which a chronology can be established. These criteria leave at most perhaps 5 forms: *arbuda*- ‘a mythical monster’, *udumbala*- ‘kind of tree; light brown’, *turīya*- ‘fourth’, *gandhvara*- ‘name of a group of mythical creatures’, and *paraśu*- ‘axe’. These five cases do all share a leftward retraction between the older and younger variant, but not in a fully coherent way. *arbudá*-, *udumbalá*-, *gandhvará*-, and *paraśú*- all likely originally have final stress, and in three of these cases, the younger form presents initial stress (*árbuda*-, *gándhvara*- and *páraśu*-), but younger *udumbála*- instead has stress on a light penult. RVic *turíya*- ‘fourth’ stands alongside *túrīya*-, where the position of stress moves from a heavy penult to a light initial syllable. Most generously then, *árbuda*-, *gándhvara*- and *páraśu*- and *udumbála*- might be interpreted as indicative of a tendency to avoid final stress (NONFINALITY-σ), but hardly seems to enforce itself throughout the lexicon.

In sum, variant stress found in early Vedic almost certainly provides evidence for different stress patterns that might emerge depending upon how Vedic speakers interpreted the morphological structure of the inputs. Clear evidence for broader markedness-governed stress preferences is, however, lacking. The plausible cases of left-edge stress plausibly attributed to “demorphologization” nevertheless have the effect of decreasing the amount of data exhibiting stress at the right edge of the word; the total number of cases is probably too small to seriously impact the overall structure and distributions of stress in the Vedic period. See further the data on the frequencies of stress patterns in the RV under 5.2.4.

5.1.2 The Prosodic System of the *Śatapathabrāhmaṇa*

The *Śatapathabrāhmaṇa* is unique among Vedic texts that employ some system of word-prosodic marking. Recall from the introduction under 4.1.1 that the system most commonly found (e.g., for the RV and AV) leaves the syllable with *udātta* (= stressed syllable, site of association of starred tone) unmarked, but places a vertical stroke above the following syllable with *svarita* (if that syllable is not itself *udātta*), and a horizontal stroke below the preceding syllable with *anudātta* (if it is neither *udātta* nor *svarita*). The diacritic representation of word-prosody in the ŚB, however, is distinct from the representation used in any other accented Vedic text: it employs exclusively the horizontal stroke below an *akṣara*, and in many cases, the position of the horizontal stroke appears to correspond to a syllable that would be *udātta* in other Vedic texts. Example (1) (taken from Cardona 1993: 25) reproduces (in romanization) VS 17.72 and the parallel passage in ŚB 9.2.3.34; acute above a vowel in the VS passage marks *udātta* as usual, while a horizontal line below a vowel in the ŚB passage indicates a syllable orthographically marked as such. Superficial comparison of this passage shows that the ŚB’s horizontal stroke usually matches the *udātta* of the VS, but not always: of the 11 *udātta* syllables in the VS passage, 10 are marked with the horizontal stroke in the ŚB passage; in one sequence of two syllables (*bhāsāntārikṣam* vs. *bhāsāntarīkṣam*) the ŚB’s horizontal stroke falls one syllable earlier than the VS’s *udātta*. Remarkable in this specific instance is that a “mismatch” between *udātta* and horizontal stroke occurs at the one place where two syllables with *udātta* occur adjacent to one another.

- (1) *Vājasaneyisaṃhitā* 17.72 vs. *Śatapathabrāhmaṇa* 9.2.3.34
- | | |
|--------------------------|-----------------------------|
| pr̥ṣṭhé pr̥thivyā́ḥ sīda | pr̥ṣṭhe pr̥thivyāḥ sīda |
| bhāsāntārikṣam ā́ pr̥ṇa | bhāsāntarīkṣam ā pr̥ṇa |
| jyótiṣā dívam út tabhāna | jyotiṣā divam út tabhāna |
| téjasā díśa úd dṛm̐ha | tejasā dīśa ud dṛm̐hetyevaṃ |
| | hyeṣa etat sarvaṃ karōti |

‘sit on the back of the earth, cross over to the sky with light, shine out towards heaven with brightness, firm up the quarters [of the sky] with brilliance...’

The statistical preponderance with which the usual Vedic *udātta* corresponds to the the ŚB’s horizontal stroke has lead most western grammarians to conclude that this horizontal stroke indeed, in most cases, signals a syllable with *udātta*, that is, word-level prominence

and higher pitch, just as in the RV; cf. Weber 1867, Macdonell 1910: 80 or Wackernagel 1896a: 252, and further Cardona 1993: 35–9 for a convenient collection of statements in the literature on the prosodic system of the ŚB. Under this conception, the ŚB differs from the RV or AV primarily just in the way in which the syllable with word-level prominence is signaled, much as the *saṃhitās* of the YV uncontroversially use the vertical stroke above an akṣara to mark the *udātta*. The relationship between the prosodic-marking system of the RV and AV and that of the ŚB is, though, treated explicitly in one grammatical text, the *Bhāṣikasūtra*; sūtras 1.5–18 explain how the prosody (or prosodic marks) of the RV/AV system can be transformed into the prosody (or prosodic marks) of the ŚB (see Kielhorn 1867). The fundamental uncertainty in the interpretation of the relevant sections of the *Bhāṣikasūtra* concerns its usage of the terms *udātta* and *anudātta*: the question is whether these terms are meant to describe *pitch*, i.e., high and low pitch, or whether they merely describe graphic devices, i.e., lack of any prosodic diacritic versus horizontal stroke below. Weber (1867: 434) formulates this difference explicitly with respect to the view of Kielhorn (1867): “Nach meiner Ansicht handelt es sich...um eine graphische Bezeichnung der Accente, nach Kielhorn dagegen um deren Aussprache...” The view that the *Bhāṣikasūtra* refers to the realization of pitch has most recently been defended by Cardona (1993: 1; 24).

If Cardona’s interpretation (which seems to mostly align with interpretations found among other Indian grammarians) is correct, then, assuming that the position of obligatory and culminative prominence in the ŚB is, for most word forms, no different than the prominence found in the RV or AV, prominence is either associated with low pitch (perhaps HL*) or perhaps not associated with a word-level pitch accent at all. Ultimately of greatest interest for present purposes is then whether the deviations in position of the ŚB’s *anudātta* versus the RV’s *udātta* allow for some coherent interpretation that signals a stronger role for markedness constraints regulating metrical structure. These differences may be summarized as follows (cf. Macdonell 1910: 80):

1. Where the RV would have a sequence of two or more syllables with *udātta*, only the rightmost is marked with the horizontal stroke in the ŚB. In longer sequences of *udātta*, if the ŚB produced an alternating pattern, one might interpret such a behavior as clash avoidance. As it stands, this behavior more closely resembles a strict OCP effect, with adjacent identical tones of some kind (depending upon whether horizontal stroke is assumed to signal L or H) on the tonal tier being banned.
2. When vowel contraction across word boundaries would produce two adjacent syllables with *udātta* in the RV, the ŚB places the horizontal stroke on the preceding syllable within the same word. Thus, where the RV has *evágnír* (= *evá agníḥ*), the ŚB would write *evagnīṛ*.
3. Two adjacent horizontal strokes are permitted, though, across the morphological word boundary within a compound, though not systematically. In these cases, the RV would have only a single *udātta* in the compound. Likewise, sequences of preverb + verb that have only a single *udātta* in the RV appear with two horizontal strokes in the ŚB.

4. What would appear as an independent *svarita* in the RV always appears with no mark in the ŚB, and in such cases, the ŚB places a horizontal stroke below the preceding syllable, regardless of whether this preceding syllable would be *udātta* or *anudātta*. If the preceding syllable would have an *udātta* in the RV anyway, the correspondence in the ŚB's system is normal. Where the preceding syllable would be *anudātta* in the RV's system, what must be explained is then the placement of stress in the ŚB on the syllable that is marked as *anudātta* in the RV. To be clear, in the RVic system, when a *svarita* immediately follows an *anudātta*, where exactly the stress occurs is unclear.
5. In the ŚB, before a pause (likely the end of an intonational phrase), a syllable that the ŚB would normally mark with a horizontal stroke receives an entirely different orthographic marking, namely, three dots under the syllable. This practice is not, however, wholly consistent.
6. In words of four or more syllables in which the RV would have an *udātta* on the initial syllable, the ŚB marks both the initial syllable and either the final or penultimate syllable with a horizontal stroke. Some examples suggest that the position of the second, unexpected horizontal stroke towards at the right edge of the word may be regulated by syllable weight: on the final syllable if it contains a long vowel or coda consonant, on the penult if the final contains only a short vowel.

In particular, features 3. and 6., which both involve the marking of two syllables in one word, are evocative of aspects of the Classical Sanskrit stress rules to be discussed in the following section. Specifically, with respect to compounds (3.), the stressed syllable of the rightmost member of the compound in Classical Sanskrit seems to regularly receive the primary compound stress. More generally, words in Classical Sanskrit regularly align primary stress towards the right edge of the prosodic word; however, stress on final syllables is typically avoided, contrary to what the placement of a second horizontal stroke in the ŚB would suggest. The occurrence of multiple horizontal strokes within longer prosodic words lends itself to the interpretation that both a primary and secondary stress is present, whereas the prosody of the RV and AV presents no evidence for secondary stress. Since indications of secondary stress are available for Classical Sanskrit, seeing a development away from lexical stress towards metrically regulated stress in these behaviors is tempting.

In order to fully and properly assess the value and significance of the ŚB's testimony for the history of Indic prosody, a fuller evaluation of the material is needed than is possible given the current state of the secondary literature. Nonetheless, at least Leumann (1892: 50) was of the conviction that the combination of phenomena in prosodic marking found in the ŚB points towards the development of a stress system regulated by syllable weight, for which evidence will be presented in the following sections.

5.2 Stress Assignment in Classical Sanskrit: Testimony, Reconstruction, and Analysis

The central objectives of this section are, on the one hand, to gather data and reconstruct the position of stress in Classical Sanskrit as far as possible, and on the other, to provide a formalized analysis of a rule for the assignment of primary stress. I open with a brief discussion of what conclusions should and should not be drawn from the virtual absence of direct testimony concerning word prosody in Classical Sanskrit. Instead, the likely position of primary stress in Classical Sanskrit must be largely reconstructed on the basis of indirect testimony concerning segmental developments in the Middle Indic (Prākṛit) and Modern Indic languages; here, where certain segmental changes (e.g., syncope) do or do not apply can be interpreted as a potential indicator of which syllable bore primary stress in the respective Sanskrit form. As already claimed by Jacobi (1893), much of the relevant data on segmental developments, where available, can be interpreted so as to align with the few available traces of direct testimony. After sorting the data and drawing generalizations from it in 5.2.2.1, possible connections to stress rules of Modern Indic varieties are considered under 5.2.2.2. With core generalizations in hand, an Optimality-Theoretic analysis is proposed in 5.2.3; the properties of the relevant phonological grammar are then directly compared with the phonological grammar developed for Vedic Sanskrit in the preceding Chapter 4.

Before delving into the direct and indirect indicators of stress in Classical Sanskrit, the relationship between the Vedic and Classical language must be problematized. A genuine complication in discussing any sort of language change supposedly occurring between Vedic and Classical Sanskrit is the fact that the latter is, to some extent, an artificial construct. At the time of the composition of Pāṇini's grammar, comparatively few individuals were natively acquiring a language like that described by Pāṇini in a naturalistic setting; most speakers of Indic languages, by the middle of the 1st Millennium BCE (600–500 BCE), regularly spoke an idiom probably much more similar to the Middle Indic varieties first directly attested after 300 BCE (i.e., the Prākṛits represented in the Aśokan "Rock Edicts"; see Bloch 1950, Schneider 1978 and Anderson 1990), and later in their own literary and grammatical traditions. Oberlies (2007: 163) goes so far as to state that "the MIA [Middle Indo-Aryan] languages are not younger than ('classical') Sanskrit. And a number of their morphophonological and lexical features betray the fact that they are not direct descendants of Rigvedic Sanskrit, the main basis of 'Classical' Sanskrit; rather they descend from dialects which, despite many similarities, were different from Rigvedic and in some regards even more archaic". The evidence of some phonologically predictable stress rule to be discussed here is therefore probably rather to be understood as some stress rule that developed in some varieties of "vulgar Indic" during the 1st Millennium BCE, perhaps already predominating in at least some Old Indic dialects on the path to their Middle Indic descendants. This stress rule likely came to be applied to the pronunciation of Sanskrit some time after the Vedic period. For convenience, I will however continue to speak of the "Classical Sanskrit stress rule", though this prosodic development may well precedes the historical boundary between Vedic and the grammatically fixed Classical Sanskrit by some centuries; compare further the discussion in Burrow 1973: 115.

5.2.1 (Absence of) Direct Testimony

Direct sources for the word-prosodic system of Indic languages following upon the Vedic language are meager at best. First, already most non-*saṃhitā* texts of Vedic, as well as manuscripts of the epics *Mahābhārata* and *Rāmāyana*, to say nothing of Classical Sanskrit texts, are completely devoid of any diacritic indications as to stress, tone, or intonation. Furthermore, as Wackernagel (1896a: 283; 296–7) observes, Pāṇini and his interpreters in the Indian grammatical tradition (e.g., Patañjali, Śāntanava) exclusively discuss the accentuation of Vedic, whereas later grammarians of Sanskrit (e.g., Kātantra) say nothing concerning accentuation whatsoever. Only in the *Phīṣūtra* of Śāntanava (edition by Kielhorn 1866) do we find a (perhaps accidental) exception: although Śāntanava purports to offer rules for the accentuation of Vedic, some of his rules do not match the facts of Vedic, but instead appear compatible with a predictable metrical stress system. Of particular note in this regard is Rule II.19, which in the translation of Kielhorn (1866: 36) states: “Die schwere Sylbe eines überzweisybligen Wortes [ist udātta], auf die eine oder zwei leichte Endsylben folgen”.³ See also Kazama 1975: 53–6 for discussion of the relation between the *Phīṣūtra* and a possible Classical Sanskrit stress rule.

This virtual silence concerning word prosody in both the textual and indigenous grammatical sources for Classical Sanskrit has occasionally led some scholars to assume that Classical Sanskrit and the Prākṛits simply lacked word stress or word-level pitch accents altogether (e.g., Bloch 1915: §32). Such an inference, however, mistakenly takes the absence of evidence as evidence of absence and would furthermore inhibit any uniform explanation for some segmental facts assembled below. Furthermore, the first-hand knowledge of Sanskrit won by Martin Haug and Georg Bühler through study with *paṇḍits* in India gives direct indications that it was the practice of their teachers and informants to regularly produce stress on each word, apparently according to the number of syllables and their weight. The earliest such description to appear in print is in Haug 1874: 99 (quoted just below), though Bühler (1883: Schrifttafel) offers many more informative examples. Bühler’s description of stress patterns in the pronunciation of Classical Sanskrit were, apparently, taken over by further western pedagogical works of Sanskrit in (e.g., Perry 1885: 229–30, Whitney 1889 [1960]: 96, Thumb and Hauschild 1958: 43–4, Gonda 1963: 10, Mayrhofer 1978: 25–6), though these often provide less detail or reinterpret his rules. Whitney (1889 [1960]), in fact, does not really present a version of Bühler’s generalizations at all, but instead merely states that “it is a general habit with European scholars to pronounce Sanskrit words according to the rules of the Latin accent.”

Wenn nun behauptet worden ist, dass das Sanskrit, wenn gesprochen, keinen Accent mehr zeige, sondern ganz monoton klinge, so ist diess nicht ganz richtig. Gewisse Sylben werden auch jetzt noch beim Sprchen mit mehr Nachdruck als andere hervorgehoben; es sind vorzugsweise die langen Sylben, denen sich

³“The heavy syllable of a word with more than two syllable is *udātta*, upon which one or two light final syllables follow.” Compare further Jacobi 1898: 567–8 and his list of further specific cases in which Śāntanava prescribes an *udātta* that is at odds with directly attested Vedic forms.

der Ton zuwendet; auch wird er mehr nach vorne geworfen, bei mehrsyllbigen Worten gern auf die drittletzte Sylbe. So wird z. B. *madhyándina*, *vājasanéyi*, *taittiríya*, *déva*, *bráhmana*, *kārayati*, *pratīṣṭha*, *ágni*, *gáta*, *dátta* u. s. w. gesprochen.⁴

5.2.1.1 The Testimony of Bühler and Haug

Since it remains the most complete accounting of a possible rule system for the word prosody of Classical Sanskrit, a detailed exposition of the principles expounded in Bühler 1883 is warranted. First, Bühler makes explicit that “alle indischen Brahmanen gebrauchen...bei der Aussprache des Sanskrit einen Ictus-Accent, der von dem...musikalischen Accente (*svara*) [des Veda] zu unterscheiden ist.”⁵ Besides the handful of forms cited in the quotation from Haug just above, Bühler provides about 30 further examples, all of which are marked with primary stress (acute) and some with a secondary stress (grave, only pretonic). Three fairly superficial observations about Bühler and Haug’s data permit some initial hypotheses about the nature of the Classical Sanskrit stress rule to be formulated:

- No minimal stress pairs occur in the data (i.e., no words of identical prosodic shape but different stress pattern) $\Rightarrow$ stress is not lexically contrastive or sensitive to morphological structure.
- Secondary stress, when given, only occurs to the left of the primary stress $\Rightarrow$ primary stress falls on the rightmost foot of the prosodic word (undominated RIGHTMOST).
- Stress rarely falls on a final syllable, and then only in disyllables with a light initial syllable and long-voweled final (i.e., [L $\bar{V}$ (C)]) $\Rightarrow$ extrametricality of some form likely holds.

I reproduce here the four major rules given by Bühler, translated and expanded upon, with glosses given to the Sanskrit examples, along with the prosodic shape of the form.⁶ Below, I summarize the larger generalizations that can be drawn from this data.

1. In simple verbs and their derivatives, the syllable of the root is preferentially accented, but the accent never goes back farther than the fourth syllable, and rarely farther back than the third. The accent can only stand on the fourth-to-last syllable when the antepenult and penult are light, and on the antepenult when the penult is light.

⁴“If it has been claimed that Sanskrit, when spoken, no longer exhibits any accent, but rather sounds entirely monotone, this is not entirely correct. Certain syllables are still at present emphasized in speech with greater force; it is preferentially heavy syllables towards which the tone turns; it is also happily thrown further forward, onto the antepenult in the case of polysyllabic words. Thus, for instance, are [the following words] spoken: ...”

⁵“all Indian Brahmins use...a stress-accent in their pronounciation, which is...to be distinguished from the musical accent (*svara*) (of Vedic).”

⁶Forms below marked with a superscript preceding ^V mean that the position of stress reported by Bühler is identical with the position of the *udātta* in Vedic. Words not so marked disagree with the position of the Vedic *udātta*, or are unattested in accented Vedic texts.

- (2) Stress on the Preantepenult
- a. *dúhitaram* ‘daughter:ACC.SG’ = [‘L L L VC]
- (3) Stress on the Antepenult
- a. ^V*kāraṇāt* ‘doing:ABL.SG’ = [‘L L V:C]
- b. *kṣípasi* ‘hit:PRES.2SG’ = [‘L L V]
- c. ^V*bódhati* ‘observe:PRES.3SG’ = [‘H L L]
- d. ^V*náśyatha* ‘perish:PRES.2PL’ = [‘H L L] (or [‘L L L]? see 2. below)
- e. *dúhitā* ‘daughter:NOM.SG’ = [‘L L H]
- (4) Stress on the Penult
- a. *karaṇéna* ‘doing:INST.SG’ = [L L ‘H V]
- b. *bodhāvaḥ* ‘observe:PRES.1DU’ = [H ‘H VC]
- c. ^V*kṣipāmaḥ* ‘hit:PRES.1PL’ = [L ‘H VC]
- d. *duhitṛṇām* ‘daughter:GEN.PL’ = [L L ‘H V:C]
2. Nominal derivatives mostly maintain the accent of the base word (“des ursprünglichen Wortes”) while observing the restrictions above. Clusters with *y* /j/ and *v* /v/ usually do not make a closed (and thus heavy) syllable.
- (5) Sets of Nominal Derivatives
- a. *rāṅku-* ‘a species of deer’ vs. *rāṅkava-* ‘made from the hair of a *rāṅku-*’
- b. *gārga-* ‘name of a sage’ vs. *gārgyaḥ* ‘patronymic from *garga-*:NOM.SG’: *gārgyāyanī* ‘patronymic from *gārgya-*:F’
- (6) Stress on antepenult where penult has a C + glide cluster
- a. *prābalyam* ‘superiority of power:ACC.SG’ = [‘H L VC]
- b. *úktatvāt* ‘being spoken:ABL.SG’ = [‘H L V̄C]
3. In verbs and verbal derivatives that are attached to prepositions, as well as in augmented (i.e., with prefix *a-*) and reduplicated forms, and throughout inflection, the accent moves back (i.e., towards the left edge) if the root or stem syllable is light. Polysyllabic prepositions preserve their accent as a secondary accent.
- (7) Stress to the left of a (light) root syllable
- a. ^V*á-gamat* ‘come:IMPF.3SG’ = [‘H L VC]
- b. ^V*á-natam* ‘bend down:IMPF.1SG’ = [‘H L VC]
- c. *anú-ṣṭhitam* ‘performed:ACC.SG’ = [L ‘H L VC]
- d. ^V*á-gamat* ‘go:IMPF.3SG’ = [‘L L VC]
- e. ^V*á-kṣipat* ‘hit:IMPF.3SG’ = [‘L L VC]
- f. *bíbhrta* ‘bear:IMPF.2PL’ = [‘L L V]
- g. *túṣṭuma* ‘praise:PERF.1PL’ = [‘H L V]
- h. *tástima* ‘stand:PERF.1PL’ = [‘H L V]

- i. *tástḥau* ‘stand:PERF.1SG’ = [‘H V:’]⁷
- (8) Stress on a heavy root syllable
- a. *ut-kṛṣṭam* ‘attracted:ACC.SG’ = [H ‘H VC]
- b. *nir-úktam* ‘uttered:ACC.SG’ = [L ‘H VC]
- c. *a-jívat* ‘live:IMPF.3SG’ = [L ‘H VC]
- d. *a-yát* ‘go:IMPF.3SG’ = [L ‘V:C]
- e. ^V*bibhárti* ‘bear:PRES.3SG’ = [L ‘H V]
- f. ^V*tuṣṭáva* ‘praise:PERF.3SG’ = [L ‘H V]
- g. ^V*jagáu* ‘go:PERF.1SG’ = [L ‘H]
- (9) Secondary stress on a prepositional prefix
- a. ^V*ùpa-gácchati* = ‘approach:PRES.3SG’ [L L ‘H L V]
- b. ^V*ùpa-gámatam* = ‘approach:aor.2du’ = [L L ‘L L VC]
4. In compounds, each member keeps its own accent, apart from when the first member is a monosyllable, with the accent of the of the head (“Haupttheil”) being the stronger.
- (10) First member is polysyllabic
- a. *rāja-púruṣam* ‘royal minister:ACC.SG’
- b. *pàrvata-śikharā-kāram* ‘maker of jagged mountains:ACC.SG’
- (11) First member is monosyllabic⁸
- a. *ún-mukham* ‘raising one’s face:ACC.SG’
- b. *díg-gajam* ‘a mythical elephant:ACC.SG’
- c. *pra-śíṣyam* ‘the pupil of a pupil:ACC.SG’

Setting aside stress assignment in compounds, the prosodic patterns present in Bühler’s and Haug’s data are thus as in Table 5.2, organized by the structure of the final syllable and number of syllables (2–5). The two cases of stress on a final syllable are marked in bold.

Bühler’s data, if correct and taken at face value, potentially offers valuable and fairly detailed information on the position of stress in words up to five syllables in length, though the number of non-compound forms with four or more syllables is limited. Note further that the smaller number of forms given by Haug agree completely with Bühler’s provisions. These data can all be neatly summarized for words of various length with five statements:

- If a word contains exactly one heavy non-final syllable, primary stress falls on that syllable.

⁷In this case, the root syllable is in fact heavy, *-sthau*. This example thus contradicts Bühler’s generalization that stress should only occur to the left of a root syllable if the root syllable is light.

⁸What Bühler does not explicitly say, but his examples of compounds with monosyllabic first members imply, is that stress is assigned as though the word were not a compound: primary stress occurs on the first syllable of a trisyllabic word with light penult in *únmukham* and *díggajam*. Given that *y* supposedly forms part of a complex onset, one might expect *prá.śi.ṣyam*, with stress on the first syllable, since the penult would in that case be light.

Final Syllable	Two-syllable	Three-syllable	Four-syllable	Five-syllable
Ů	([L Ů]) ([H Ů])	[L L Ů] [H L Ů] [L 'H Ů]	[L L 'H Ů] ([H L L Ů]) ([H L 'H Ů])	[L L 'H L Ů] ([H L L 'H Ů])
ŮC		[L L ŮC] [L 'H ŮC] [H L ŮC] [H 'H ŮC]	[L L L ŮC] [L 'H L ŮC]	[L L 'L L ŮC]
Ů̃	[L 'Ů̃] [H Ů̃]	[L L Ů̃]	[H 'H L Ů̃]	
Ů̃C	[L 'Ů̃C]	[L L Ů̃C] [H L Ů̃C]	[L L 'H Ů̃C]	

Table 5.2: Prosodic Patterns and Stress in the Data of Bühler 1883: Schrifttafel (cf. Perry 1885: 229–30) and Haug 1874 (given in parentheses)

(12) Penult: *ka.ra.'ne.na* 'doing:INST.SG' = [L L 'H Ů]

(13) Antepenult: *a.'nuṣ.ṭhi.tam* 'performed:ACC.SG' = [L 'H L ŮC]

- If a word contains two (presumably also more) heavy non-final syllables, primary stress falls on the rightmost of those.

(14) Penult: *bo.'dhā.vaḥ* 'observe:PRES.1DU' = [H 'H VC]

(15) Antepenult: *gār.'gyā.ya.nī* 'patronymic from *gārgya*:-F' = [H 'H L Ů̃]

- In four-syllable words, primary stress falls on the initial syllable if all non-final syllables are light.

(16) Preantepenultimate stress: *'du.hi.ta.ram* 'daughter:ACC.SG' = [L L L ŮC]

- In a five-syllable word, primary stress falls on the antepenult if all non-final syllables are light.

(17) Stress on Antepenult in 5σ: *u.pa.'ga.ma.tam* 'approach:AOR.2DU' = [L L 'L L ŮC]

- In a disyllable with light penult and long-voweled final, primary stress falls on the final syllable.

(18) Stress on a final syllable with long vowel: *ja.'gau* 'go:PERF.1SG' = [L 'Ů̃]

While the stress patterns reflected in Bühler's data appear internally coherent and would allow for a relatively straightforward analysis in terms of predictable metrical stress, several potential issues and objections create uncertainties. First, a number of prosodic shapes (e.g., [H H]) are not exemplified whatsoever, and the absence of further inflected forms having a heavy penult among prefixed verbs makes it difficult to unambiguously distinguish between thoroughgoing metrical stress and a morphologically conditioned stress on the root. Second, the position of the stress reported by Bühler agrees in a not insubstantial number of cases with the position of the Vedic *udātta* (excluding the five compounds, and three lexemes unattested in Vedic, 13/29); hence, while the number of stark discrepancies between the Vedic *udātta* and reported Classical Sanskrit stress make it unlikely that any serious amount of direct inheritance of the Vedic word prominence is at work here, more data in order to exclude this possibility would be desirable. Finally, and most worrisomely, the pattern of primary stress assignment that can be inferred here is virtually identical with the stress pattern found in some Modern Indic languages, in particular Hindi (see 2.2.6). Although nothing whatsoever is known about the linguistic background of the speakers of Sanskrit on whom Bühler relied, it is at least *a priori* thinkable that the stress grammar of the native Modern Indic language(s) known by Sanskrit speakers in India in the 19th century may have directly influenced the prosody of their Classical Sanskrit, especially given that, as discussed above, the grammatical tradition makes no provisions for non-Vedic prosody.⁹

5.2.1.2 Existing Analyses and a Provisional Parametric Stress Grammar

The question of how to analyze patterns (similar to) those found in Bühler and Haug's data has recently been approached by Keydana (2016) and Vaux (2021).¹⁰ The analysis presented by Keydana is ultimately wanting in several respects; Vaux's treatment, on the other hand, correctly generates the data that it wants to explain, but makes assumptions that diverge from what is reported by Bühler, instead using data from the stresses printed in editions of the Clay Sanskrit Library, whose editors evidently have a slightly different understanding of stress rules for Classical Sanskrit than the ones implied by Bühler's data (which, I repeat, appears to be the primary source for all pedagogical statements on Classical Sanskrit stress).

Keydana's treatment is a simple OT analysis given by the ranking in 19. This analysis predicts that feet are obligatorily binary and trochaic, and that all heavy syllables are footed; when syllable weight plays no role, feet will stand as close to the left edge of the word as possible.¹¹ One first issue is that no provision for avoiding stress on final syllables is encoded: hence, any final syllable containing a long vowel, if there are no other heavy syllables in the word, would be predicted to bear stress. Second, in case a word contains more than one heavy syllable, since all heavy syllables should be stressed, no provision is made for deter-

⁹Conversely, that the stress grammars of modern Indic languages have been affected by borrowings from Sanskrit is possible as well: many, perhaps most, non-compound formations of more than three syllables in Hindi are Sanskrit borrowings. See the discussion surrounding issues of Hindi stress under 2.2.6.

¹⁰Thanks to Bert Vaux for sharing his unpublished work and for discussion of the data.

¹¹As it stands, the four constraints and their ranking predict that only a single foot per word will be built. Since Keydana says nothing about forms that might have secondary stress or more than one foot, I leave this issue aside.

mining which should receive primary stress. In a sequence containing up to four light syllables, stress is correctly predicted to fall on leftmost syllable, and likewise, in words with a light final syllable and only one heavy syllable, stress will be assigned to the only heavy syllable. In words of five or more syllables, however, primary stress is in no way held within a four-syllable window: a word like *u.pa.ga.ma.tam* would instead be predicted to have stress on its initial syllable (^X*u.pa.ga.ma.tam*).

- (19) Constraint ranking for the analysis of Classical Sanskrit Stress in Keydana 2016:
FTBIN, WSP, TROCHEE ≫ ALL-Ft-L

The analysis in Vaux 2021, meanwhile, is a successful implementation in the framework of Halle and Isardi 1995, but makes two assumptions contrary to what is reported by Bühler:

- Words of five or more syllables with no heavy non-final syllables receive stress on the penult (thus *a.rá.pa.ca.na* ‘collective of the five Buddhas’, contrary to Bühler’s ^V*ù.pa.gá.ma.tam*).¹²
- Stress never falls on the final syllable, regardless of its weight (contrary to Bühler’s *a-yát* and *jagáu*).

Except in these cases, Vaux’s analysis is adequate for handling Bühler’s (and Haug’s) data. Briefly summarized: left constituent brackets are projected to the left of non-final heavy syllables on line 0 of the grid; a right constituent bracket is projected to the left of the rightmost syllable; iterative constituent constraint proceeds from right to left, allowing crucially for *ternary* (trisyllabic) constituents; the leftmost syllable of a constituent is then projected onto line 1; and finally, on line 1, a right constituent bracket is placed to the right of the rightmost grid mark, and the rightmost grid mark in this constituent is projected as the head onto line 2. An illustrative derivation of *aṃsáphalaka*- ‘shoulder-blade’ is given in (20), where Vaux assumes a preantepenultimate stress; I would instead assume the correct stress, compatible with Bühler’s data, to be *aṃsaphálaka*- instead. Crucial to the entire analysis is the construction of ternary constituents and the avoidance of including final syllables in constituents. Indeed, because of how extrametricality is implemented in the Simplified Bracketed Grid model, as an “all or nothing” parameter, cases like *jagáu*, with final stress, must either be suppressed or attributed to non-phonological factors.

- (20) Derivation of *aṃsaphalaka*- ‘shoulder-blade’ in the SBG analysis of Vaux 2021

a. Bracket Projection: H → (H
 (× × × × ×
 H L L L L
 aṃ sá pha la ka

¹²It may be the case that the Clay Sanskrit Library editors have mistakenly interpreted Bühler as saying that verbs may have stress on the root, but nouns of four or more syllables will regularly have preantepenultimate stress, if no heavy syllables are present.

- b. Edge Marking: right bracket to the left of the rightmost grid mark
- | | | | | | | |
|----|----|-----|----|----|---|---|
| (| × | × | × | × |) | × |
| H | L | L | L | L | | |
| aṃ | sa | pha | la | ka | | |
| si | | | | | | |
- c. Iterative Constituent Construction: $R \rightarrow L$; avoid $(\times \times)$
- | | | | | | | | |
|----|----|-----|----|----|---|---|---|
| (| × | (| × | × | × |) | × |
| H | L | L | L | L | | | |
| aṃ | sa | pha | la | ka | | | |
- d. Line 1 Projection and Edge Marking: leftmost grid marks of constituents projected, then right bracket to the right of the rightmost grid mark
- | | | | | | | | |
|----|----|-----|----|----|---|---|---|
| × | × | | | | | | |
| (| × | (| × | × | × |) | × |
| H | L | L | L | L | | | |
| aṃ | sa | pha | la | ka | | | |
- e. Line 2 Projection: rightmost grid mark of constituent projected
- | | | | | | | | |
|----|----|-----|----|----|---|---|---|
| | × | | | | | | |
| × | × | | | | | | |
| (| × | (| × | × | × |) | × |
| H | L | L | L | L | | | |
| aṃ | sa | pha | la | ka | | | |

In light of the fact that Vaux's analysis both requires ternary constituents and does not faithfully correspond to the most secure data available – the direct reports of Bühler – an alternative analysis is in order.¹³ Provisionally, Bühler's examples appear compatible with the assumption that primary stress is assigned to the rightmost moraic trochee, treating final syllables as extrametrical in most configurations. Only in the configuration $[L \bar{V}(C)]$ is final stress permitted, presumably because a parse $[('L) H]$ would violate FTBIN, while a parse $[('L H)]$ would violate both NONFINALITY- $\mathcal{F}$ and WSP. In longer sequences of light syllables, such as *ūpagācchati*, a left-to-right parsing into two moraic trochees, $[(,u.pə).('gəc).c^hə.(ti)]$, producing a secondary stress and a primary stress on the rightmost foot, fits the available data. A parametric stress grammar, employing parameters from Hayes 1995 (see 2.2) can be captured with the grammar in 21.

(21) Parametric Stress Grammar for Classical Sanskrit

- a. **Foot Shape:** Moraic Trochee ($\mathcal{F} = -$ or $\sim$)
- b. **Extrametricality:** Final Syllable (but not if only a single light syllable would remain – look-ahead problem)
- c. **Direction of Parsing:** Left-to-right

¹³Ternary constituents are generally admissible in SBG, and thus do not pose a theory-internal problem. One larger point of Vaux's presentation is that the pattern reported for Classical Sanskrit with preantepenultimate stress is certainly rare, and the empirical basis of it and similar systems (e.g., Pre-Classical Latin discussed under 2.2.7.1) requires better grounding.

d. End Rule: Right

Stated in prose, this grammar constructs feet and assigns stress as follows: “Form moraic trochees from left to right, ignoring the final syllable (unless only a single light syllable would remain); assign primary stress to the rightmost foot”. Illustrative derivations are shown in 22. In all cases except *jagáu*, final syllable extrametricality removes the last syllable from foot construction; all heavy syllables or adjacent pairs of light syllables then build feet from left to right, which results in penultimate light syllables remaining unparsed in cases like [(‘du.hi).ta.<ram>] or [(,u.pa).(<gac>).cha.<ti>]. The rightmost foot then receives primary stress, thereby producing configurations like [(,u.pa).(<gac>).cha.<ti>].

(22) Illustrative Derivations of Bühler’s Classical Sanskrit Stress Data

Syllabified Input	ja.gau	bi.bhṛ.ta	ut.kṛṣ.tam	du.hi.ta.ram	u.pa.gac.cha.ti
Final σ ExM	—*	bi.bhṛ.<ta>	ut.kṛṣ.<tam>	du.hi.ta.<ram>	u.pa.gac.cha.<ti>
Moraic Trochees L-to-R	ja.<gau>	(bi.bhṛ).<ta>	(ut).(kṛṣ).<tam>	(du.hi).ta.<ram>	(u.pa).(<gac>).cha.<ti>
End Rule Right	ja.(<gau>)	(‘bi.bhṛ).<ta>	(ut).(<kṛṣ>).<tam>	(‘du.hi).ta.<ram>	(,u.pa).(<gac>).cha.<ti>

Although adequate as a description of Bühler’s non-compound data, this analysis suffers from three general issues. First, empirically, two potentially interesting prosodic shapes are missing: [L ṼC] (disyllable with a light initial syllable followed by a closed final syllable with a short vowel) and [L L L L L σ] (word of six or more syllables with only light non-final syllables). Depending upon whether the former is stress as [‘L ṼC] or [L ‘ṼC], it is possible to determine whether final consonant extrametricality applies or not. In the latter, the parametric analysis here would predict the footing and stress pattern [(,L L) (‘L L) L σ], with preantepenultimate stress, but because an OT analysis can invoke ALIGN-L(ω , F) to generate a word-initial foot in forms like [(‘du.hi).ta.<ram>], the relative ranking of ALL-FT-L and ALL-FT-R could predict [(,L L) L (‘L L) σ] antepenultimate stress in this configuration instead. Next, there is a theoretical issue: the way in which extrametricality is encoded here results in a look-ahead problem, because inputs like /jagau/ must know that applying final syllable extrametricality would ultimately result in building a degenerate foot, ^x[(‘ja).gau]. This issue can also be overcome in an optimality-theoretic implementation. Finally, the most serious problem is of an epistemological nature: to what extent can we trust Bühler and Haug’s data, especially if purported to reflect the stress rules for a language as spoken 2000 years ago or more, for whose prosody we lack any direct reports. I turn first to this latter problem, and thereby also deal with the first, empirical question. In Section 5.2.3, a constraint-based analysis of Bühler’s (and additional) data resolves the outstanding theoretical issue.

5.2.2 Reconstructing the Position of Classical Sanskrit Stress

The concerns about the total reliability of the (relatively meager) concrete data furnished by Haug and Bühler means that independent confirmation of just such a stress pattern should be sought elsewhere. Thankfully, some traces of just such a clearer evidential basis have been delivered by Jacobi (1893) (cf. also Jacobi 1898), whose data and claims will be discussed in

detail below (5.2.2.1). Jacobi argued that patterns of syncope, vowel lengthening/shortening, and consonant gemination found in Pāli and the Prākṛits (and in some cases, in Sanskrit itself) vis-à-vis their Sanskrit preforms can be best understood if primary stress was assumed to stand in the position predicted by Bühler's generalizations. Jacobi's claims were, however, disputed by Pischel (Pischel 1897, Pischel 1898), who instead argued that much of the data mustered by Jacobi can or should instead be attributed to stress in the position of the Vedic *udātta* (Jacobi 1898 is an attempt to refute Pischel's arguments). Pischel (1900: 48) ultimately classifies the Prākṛits into two groups on the basis of their word-prosodic systems: one group in which the position of the Vedic *udātta* is preserved (in particular, Mahārāṣṭrī, Ardhamāgadhī, and Jainmahārāṣṭrī), and another in which an innovative stress rule predominated (in particular, Śaurasēnī, Māgadhī, Dhakkī).¹⁴ Jacobi's position is largely followed by Grierson (1895a) and partially by Turner (1916), who derive further developments into Modern Indic languages (especially Hindi and Gujarātī) from maintenance of a primary stress on the syllable of a Sanskrit preform generated according to Bühler's principles, though Turner also argues that Māhārāṣṭrī, at least, possessed a distinct word-prosodic; the evidence of the Modern Indic languages is reviewed under 5.2.2.2.¹⁵

5.2.2.1 Evidence from Segmental Change: From Sanskrit to Middle to Modern Indic

In this section, I organize and evaluate the data treated in Jacobi 1893.¹⁶ The data are organized according to prosodic shape and the phonological developments that the examples (potential) exhibit in comparison to their Sanskrit preforms. Most extensive and illuminating are the cases of syncope, which are given first in the tables, followed by cases of other vowel changes, and finally consonant gemination. Note, however, not every form mentioned by Jacobi is reproduced here.

Ultimately, much of the data mustered by Jacobi is unfortunately open to multiple interpretations. First, since the position of the Vedic *udātta* sometimes accidentally agrees with the position of stress predicted by the Classical Sanskrit stress rule, any segmental changes

¹⁴Pischel writes: "Der Accent von Mahārāṣṭrī, Ardhamāgadhī, Jainmahārāṣṭrī, dem poetischen Apabhraṃśa, vorausichtlich auch Jainaśaurasēnī, entspricht wesentlich dem vedischen. Da auf ihm die Schwächung und der Ausfall, sowie die Steigerung der Vocal beruht, ferner die Verdoppelung der Consonanten in bestimmten Fällen, kann er nicht rein musikalisch, sondern muss vorwiegend expiratorisch gewesen sein. In Śaurasēnī, Māgadhī, Dhakkī is der Accent des classischen Sanskrit nachweisbar, der mit dem des Latein meist übereinstimmt." Pischel thus seems not to doubt that the patterns reported for Classical Sanskrit by Bühler are correct, but merely that the innovation of a predictable metrical stress rule affected all of post-Vedic Indic, as argued by Jacobi.

¹⁵See also Ollett 2012: 274 for a brief discussion of possible relationships between particular Prākṛits and Modern Indic languages in the context of their prosodic systems.

¹⁶A practical philological problem in working with Jacobi's material is that Middle Indic forms are often labeled with as an undifferentiated "Prākṛit". By and large, Jacobi is probably employing the term "Prākṛit" in the sense meant by most indigenous grammarians, "the best Prākṛit", i.e., Māhārāṣṭrī (cf. Pischel 1900: 1). To some extent, but not in every case, reference to Pischel 1900 allows for more precise designation. Insofar as one would ideally like to limit the data to examples from Prākṛits where total agreement exists that the Classical Sanskrit Stress rule applied in the ancestor language, one must allow for some uncertainty or admit that the data presented here as evidence may require later revision.

in such forms could equally be attributed to the *udātta*. Second, even where the *udātta* and the predicted stress of the C. Skt. rule do not align, some cases of syncope could be interpreted as resulting from stress in either positions (e.g., a medial syncope [L L L] > [L L], assuming either initial or final primary stress). Finally, because Jacobi's data primarily stem from Māhārāṣṭrī, where many consonants are often lost intervocalically, distinguishing between a genuine syncope and vowel contraction is, in many cases, impossible. Nonetheless, despite these issues, a handful of apparently valid datapoints remain, which indeed appear to substantiate Bühler's reports.

Syncope of Short Vowels All of the cases discussed here involve the apparent syncope of a short vowel in a Sanskrit form containing three or more syllables. The historically younger forms thus all contain at least one syllable fewer, though sometimes show yet fewer syllables due to loss of intervocalic consonants and vowel contraction. In each table, I provide the Vedic form with its *udātta*, if attested, a "Classical Sanskrit" form with stress assigned according to Bühler's principles, a reconstructed form showing just the application of syncope, finally the younger (later Sanskrit, Pāli, or Prākṛit) form, followed by a gloss. To clearly distinguish the Vedic and the Classical accent here, I employ the acute for prominence in Vedic forms, but an IPA stress mark in Classical forms. All of Jacobi's nominal and adjectival examples are given in a citation (stem) form, without inflectional endings; since the weight of the final syllable in words of more than two syllables seems to have been irrelevant, this practice makes no difference in most cases. Many of Jacobi's examples are taken from the 12th century Prākṛit grammar of Hemacandra (edition of Pischel 1877), where most material is to be understood as Māhārāṣṭrī. When not otherwise marked here, the form is either Māhārāṣṭrī or of uncertain appurtenance. Forms with a textual attestation in parentheses are found in Sanskrit, not Prākṛit.

(23) Sanskrit Forms of the Shape [L L σ]

Vedic	Classical	Syncopated	Attested Younger	Gloss
—	'ba.da.ra-	badra-	<i>bora</i>	'jujube tree'
—	'ka.da.la	kadla-	<i>kela</i>	'banana tree'
<i>surabhí-</i>	'su.ra.bhi-	surbhi-	<i>subbhi</i>	'fragrant'
<i>durabhí-</i>	'du.ra.bhi-	durbhi-	<i>dubbhi</i>	'bad-smelling'
<i>navatí-</i>	'na.va.ti-	navti-	<i>nauī</i>	'ninth'
<i>gavayá-</i>	'ga.va.ya-	gavya-	<i>gaua</i>	'species of ox'
<i>lavaná-</i>	'la.va.na-	lauṇa-	<i>loṇa</i>	'saline'
—	la.ya.ṇa	laiṇa-	<i>leṇa-</i>	?
<i>bhávati</i>	'bhavati	bhauti	<i>bhoti</i>	'become:PRES.3SG'
<i>ádhr̥ti-</i>	'a.dhr̥ti-	adhti-	<i>addhii</i>	'laxity'
<i>bhaginī-*</i>	'bhaginī-	bhagnī-	<i>bhagnī-</i> (Dvirūp.)	'sister; woman'
—	'ku.na.dī	—	Pāli <i>kunnadī-</i>	'name of a small river'
—	'pra.ka.ṭa-	—	M., Mg. <i>pāḍa</i>	'evident'
<i>mūsala-</i>	'mu.sala-	—	<i>mūsala-</i>	'pestle'
<i>subhága-</i>	'su.bha.ga-	—	<i>sūhava-</i>	'lucky'
<i>āpagā-</i>	'a.pa.gā-	—	<i>āpagā-</i> (Dvirūp.)	'turning away'
—	'cu.lu.ka-	—	<i>cūluka-</i> (Dvirūp.)	'vessel or hand for holding water'
<i>nāraka-</i>	'na.ra.ka-	—	<i>nāraka-</i> (Dvirūp.)	'hell'
<i>pūruṣa-</i>	'pu.ru.ṣa-	—	<i>pūruṣa-</i> (Dvirūp.)	'man, male'
<i>vīvidha-</i>	'vi.vi.dha-	—	<i>vīvidha-</i> (Dvirūp.)	'piercing'
<i>sukṛtá-/súkṛta-</i>	'su.kṛ.ta-	—	<i>sūkṛta-</i> (Dvirūp.)	'good deed/well done'
—	'spha.ti.ka-	—	<i>spātika-</i> (Dvirūp.)	'quartz'
<i>hārīta-</i>	'ha.ri.ta-	—	<i>hārīta-</i> (Dvirūp.)	'yellowish'

Here, all of the forms adduced by Jacobi, except for Pāli *kunnadi* and the Sanskrit variant *bhagnī* (Dvirūp.) seem to be Māhārāṣṭrī. In many all the cases where a Vedic form is known, its prominence fell on the final syllable of the stem, e.g., *surabhí-* 'fragrant'. In two instances, the Vedic form had initial prominence, *bhávati* (> *bhoti*), and *ádhr̥ti-* (> *addhii-*). For this set of data, syncope of a medial open syllable in Māhārāṣṭrī could be conditioned by initial stress exclusively (with Jacobi) or by either initial or final stress (per Pischel).

The properly Sanskrit form *bhagnī* is particularly telling, since its Vedic accent (if such a form existed in Vedic) would certainly have been *bhaginī** (compare *aśvínī-* or *vajínī-* [RV]), with prominence on the syllable ultimately subject to syncope. This form alone constitutes a good indication that stress was shifted to a syllable other than the original medial syllable with *udātta*. Similarly, the Māhārāṣṭrī form *sūhava-* 'lucky', with a lengthened vowel in the first syllable, suggests stress on the first syllable, in contrast to medial prominence in Vedic *subhága-*.¹⁷ These two forms, at least, offers an indication that stress in Classical Sanskrit at the time of composition of the *Dvirūpakoṣas* (12th c. CE) was not on the medial syllable of

¹⁷ Pischel (1900: 58) proposes instead, however, that the long *ū* of *sūhava-* is due to "analogy" with the semantic opposite *dūhava-* 'unlucky', where the *ū* results from compensatory lengthening out of the Skt. *dur-bhaga-*.

bhaginī. This form one form, at least, offers some independent support for a stress pattern [‘L L σ] in Classical Sanskrit. Among the numerous forms with a lengthened initial syllable taken from the *Dvirūpkoṣas*, the Vedic prominence, where known, fell on the initial syllable; those data are thus non-probative.

The following table presents four- and five-syllable forms mentioned by Jacobi, where, penultimate, antepenultimate, and preantepenultimate syllables are all light.

(24) Sanskrit Forms of the Shapes [L L L σ] and [L L L L σ]

Vedic	Classical	Syncopated	Attested Younger	Gloss
<i>katháyati</i>	'ka.tha.ya.ti	kath.ya.ti	M. <i>kahei</i> , Ś. <i>kadhedu</i>	‘tell:CAUS.PRES.3SG’
—	'a.nu.bha.va-	—	Pāli <i>ānubhava-</i>	‘experience’
—	'pra.va.ca.na-	—	AMg. <i>pāvayaṇa-</i>	‘recitation’
<i>ásam̐yata-</i>	'a.sa(m̐).ja.ta-	—	Pkt. <i>assam̐jaya</i>	‘not kept together’
—	'sa(m̐).na.kha.'pa.da-	sanakhpada-	AMg. <i>saṇapphaya-</i>	‘having feet with nails in contact’
—	a.pa.'sa.ra.ti	avsarati	M. <i>osarī</i>	‘retreat:PRES:3SG’
—	u.pa.'ha.si.ta-	uvhasita	M. <i>ūhasia</i>	‘laughed at’
—	m̐.ta.'pi.t̐.ka	—	Pāli <i>matapittika-</i>	‘having a dead father’

In the cases where vowel lengthening or consonant gemination occur, prominence in Vedic on the initial syllable is either attested or likely. The development of causatives built with the Vedic suffix *-áya-*, which appears as *-e-* in all Prākṛits, is disputed: Jacobi (1893: 578) claims that syncope led to *-ya-*, whereupon the glide merged with the vowel to *e*; Pischel (1900: 114–7) instead claims that a sequence *-ya-* [-jə-] became [i], and thus when preceded by *a* as in *-aya-*, the new diphthong *-ai-* ultimately became *e*. If Jacobi’s view is correct, given that the palatal glide usually induced palatalization of preceding stops in the Prākṛits, syncope in a virtual causative ['ka.tha.ya.ti] must have postdated the palatalization of stops (cf. Jamison 2009: 38). In the case of the compound *sa(m̐)-nakha-pada-*, some discrepancy arises: Jacobi (1893: 575) supposes stress on the fourth-to-last syllable, such a stress would be at odds with the evidence from other [L L L L σ] words. Jacobi seems to disregard that the form is a nominal compound, where, according to Bühler’s principles, the primary stress should have been on the final member, *pada-*; the expected stress pattern would be *sānakhapáda-*, with secondary stress on the first syllable and primary on the penult (cf. Pischel 1900: 112).

Two five-syllable Māhārāṣṭrī verbal forms exhibit syncope of the preantepenult: *osarāi* and *ūhasia*. In the case of the finite verb *osarāi*, in any use of such a form in Vedic, no prominence would have fallen on the syllable that shows syncope: the prepositional prefix would have, in a main clause, borne prominence (*ápasarati*), but the verbal stem itself would have in a subordinate clause (*apasáratī*; cf. Macdonell 1910: 106–7). A Vedic preform for *ūhasia* would likewise probably have borne prominence on the initial syllable (cf. *úpa-k̐pta-* ‘ready, equipped’). Pāli *matapittika-*, in turn, is a nominal compound, so does not necessarily provide evidence for the stress pattern of five-syllable forms, though for its second member, *-pittika-*, it suggests stress at variance with the likely prominence of a Vedic form, which ought

to have been *pitṛka*-. The evidence here is thus overall compatible with Bühler's rules and examples, but largely equivocal in terms of attesting any change vis-à-vis Vedic.

Next, I consider Jacobi's forms containing a heavy penult, of any length and overall prosodic pattern. A general problem with many of these examples is that they involve a consonant that must have been lost intervocalically, whereupon vowel contraction, rather than syncope, could equally well explain the form. For instance, loss of intervocalic *k* in *kumbhakāra* 'potter' would yield **kumbhaāra*, with M. *kumbhāra*- resulting from syncope (so Jacobi 1893) or vowel contraction (so Pischel 1900: 72). Both explanations are fundamentally uncertain, since contraction of *a* and *ā* occurs unpredictably.

(25) Sanskrit Forms of the Shapes [σ , H σ]

Vedic	Classical	Syncopated	Attested Younger	Gloss
<i>kumbhākāra</i> - — — <i>praticīnā</i> <i>pariveśā-</i> <i>udicīna</i> —	kum.bha.'kā.ra- su.var.ṇa.kā.ra- pā.da.'pī.ṭha- pra.tī.'cī.na pa.ri.'ve.śa- u.dī.'cī.na bhāṇ.ḍā.'gā.ra-	kumbh(k)āra- suvarṇ(k)āra- pā(d)pīṭha- prat(c)īna parveśa- ud(c)īna bhāṇḍ(g)āra-	M. <i>kumbhāra</i> - M. <i>sonāra</i> - M. <i>pāvīḍha</i> - paḍīṇa Hindi <i>parosa</i> M. <i>uḍīna</i> M., Skt. <i>bhāṇḍāra</i> -	'potter' 'a weight of gold' 'stool' 'turned towards' '(offering of) food' 'turned towards north' 'treasury'
— — — <i>ārāti</i> - <i>prādeśā-</i>	ā.'gā.ra- ā.'ro.ga- cā.'tu.da.śa- ā.'rā.ti- prā.'de.śa-	— — — — —	Pāli, Skt. <i>agāra</i> - Pāli <i>aroga</i> - Pāli <i>catuddasa</i> - Skt. <i>arāti</i> - Skt. <i>pradeśa</i> -	'apartment' 'one of seven suns' 'forty' 'enemy' 'span of thumb and forefinger'

In this set, some examples could point to syncope of a pretonic light antepenult (e.g., *kum.bha.'kā.ra* > *kumbhāra*), while others would imply the syncope of a heavy antepenult (at least of long *ī*, e.g., *pra.tī.'cī.na*). With the exception of Hindi *parosa* 'food', all of the data supposedly showing syncope is equivocal; *pariveśa*-, in turn, although originally nominal compound, may well have been regarded as morphologically simplex, and thus valid for showing penultimate stress in a [L L 'H σ] word. On the other hand, the considerable number of Pāli and Classical Sanskrit forms of the original shape [H H σ] that exhibit shortening of the initial first syllable indeed speak in favor of stress on the heavy penult in such cases. Particularly telling is the pair of Vedic *ārāti* 'enemy' (MS) versus *arāti* (Raghuvamśa 12.89), where the syllable with *udātta* in Vedic shows shortening.

At the same time, some further Sanskrit-internal pairs cited by Jacobi (1893: 581), consisting almost entirely of trisyllabic words, exhibit shortening of a long vowel in the penult, where the antepenult is also heavy. For example, a variant *kirmīra*- exists alongside *kirmīra*-. These examples should be sorted into two groups: cases where the antepenult is closed, and thus always heavy (as long as no segments are deleted), and cases with an open, long-voweled antepenult. As in the instances of [H H σ] > [L H σ], the motivation for shortening may be

attributed to the avoidance of *CLASH. For the majority of these cases, since the antepenult cannot be readily made light, the penult is shortened instead. Other cases of penultimate shortening attested in Sanskrit where the antepenult is light may be seen as creating more harmonic forms that better satisfy PARSE- σ and ALIGN-L(ω , f). See the following section 5.2.3 for the role of these constraints in the Classical Sanskrit Stress grammar.

(26) Sanskrit forms with variants having a light penult

Closed Penult	Gloss	Open Penult	Gloss
<i>kirmīrá-</i>	‘variegated’	<i>pārāvata-</i>	‘remote, distant’
<i>gāṇḍīva-</i>	‘bow’	<i>śyāmāka-</i>	‘millet’
<i>dūstūra-</i>	‘thorn-apple’		
<i>bhallūka-</i>	‘dog; shell’		
<i>hiṇḍīra-</i>	‘pomegranate’		

In the next table, I examine the forms with a heavy antepenultimate syllable and light penultimate syllable, of any length and overall prosodic pattern.

(27) Sanskrit Forms of the Shape [σ_0 H L σ]

Vedic	Classical	Syncopated	Attested Younger	Gloss
<i>návamālikā-</i>	na.va.'mā.li.kā-	naumāliā-	M., Ś. <i>nomāliā</i>	‘PN’
—	a.va.'lam.bi.ta-	aulambita	M. <i>olambia</i>	‘hanging down’
—	pra.ti.'ve.śa.ka-	pratveśaka	Hindi <i>paḍosa</i>	‘neighborhood’
—	'pū.ta.ra-	pūtra-	M. <i>pōra-</i>	‘an aquatic animal’
<i>bhājana-</i>	'bhā.ja.na-	bhājna-	M. <i>bhāṇa-</i>	‘share, division’
—	kā.'lā.ya.sa-	kālā(y)sa-	M. <i>kālāsa-</i>	‘iron’
<i>pacāmāhe</i>	pa.'cā.ma.he	pacāmhe	Ś. <i>pacamhe</i>	‘cook: PRES.MID.1PL’
<i>jīvita-</i>	'jī.vi.ta-	jīvta-	M. <i>jīya-</i>	‘living; life’
<i>palāyita-</i>	pa.'lā.yi.ta-	palā(y)ta-	JM. <i>palāya-</i>	‘flown, fled’
<i>vaiśvānarā-</i>	vais.'vā.na.ra-	vaiśvānra-	<i>vaiśvāndara-</i> (Sīyāḍoṇī Inscr.)	‘universal’
<i>sūnāra-</i>	'sū.na.ra-	sūnra-	Skt. <i>sundara-</i>	‘glad; beautiful’
—	*śi.'liṃ.mu.ca-	śiliṃca-	Skt. <i>silīnca-</i> (Dvirūp.)	?
<i>śalāvāhana-</i>	sā.la.'vā.ha.na-	sālvāhana-	M. <i>sālāhaṇa-</i>	‘PN’

Of particular note in this set is the verbal form with the MID.1PL ending Ved. *-māhe* ~ *-mahe*. Jacobi (1893: 576) points out that Pāli and Śaurasenī verbs containing this inherited ending have a syncopated form, *-mhe* or *-mha*, respectively. Given that also Pischel accepts application of the Classical Sanskrit stress rule for Śaurasenī, and the Vedic ending shows prominence on the vowel evidently subject to syncope, verbs containing this ending in Śaurasenī can be assumed to have always borne stress on a syllable to the left of the penult, as Bühler’s rules would predict.

The last set of data consists of four and five syllable forms in which the fourth-to-last syllable is heavy, but penult and antepenult are light.

(28) Sanskrit Forms of the Shape $[(\sigma) H L L \sigma]$

Vedic	Classical	Syncopated	Attested Younger	Gloss
<i>pūgaphalá-</i>	'pū.ga.pha.la-	pūgphala-	M. <i>poppphala</i> , Ś. <i>poppphalī</i>	'betel-nut'
—	'rā.ja.ku.la-	rā(jk)ula-	JM., Ḍh. <i>rāula</i>	'royal family'
<i>devákula-</i>	'de.va.ku.la-	deu(k)ula-	M., Ś, Mg. <i>deula</i>	'temple'
—	'vyā.ka.ra.ṇa-	vyā(k)raṇa-	JM. <i>vāraṇa</i>	'seperation'
—	'prā.vā.ra.ṇa-	prāvraṇa	<i>pāuraṇa</i> (Hem. I.175)	'cloak'
<i>hiraṇyavatī-</i>	hi.'ran.ya.va.tī-	hiraṇ(y)vatī-	Skt. <i>hiraṇvatī-</i> (MBh.)	'possessing gold'
<i>hiraṇyamāya-</i>	hi.'raṇ.ya.ma.ya-	hiraṇ(y)maya-	Skt. <i>hiraṇmaya-</i> (TS)	'golden'

Here, in particular the two Sanskrit forms *hiraṇvatī-* and *hiraṇmaya-* are quite valuable: these seem to clearly show syncope of a light antepenult, as does the form *poppphala/poppphalī* (so also Pischel 1900: 112).¹⁸ These three cases may be classed together as syncope of a post-tonic light syllable. The remaining forms are again equivocal, where, because of the loss of intervocalic consonants, the reduction in syllables may be attributed either to syncope or vowel contraction.

Summary If interpreted in the most generous fashion possible, the data assembled by Jacobi purporting to represent cases of syncope in Middle Indic varieties (and Māhārāṣṭrī in particular) could potentially admit of the following general developments:

- An immediately posttonic light syllable is subject to syncope in trisyllabic forms with stress on the initial syllable: $['H/L_a L_b \sigma] > [H/L_a \sigma]$
- An immediately pretonic light syllable is subject to syncope in words of four or more syllables, if stress fell a heavy penult: $[\sigma_1 L 'H \sigma] > [\sigma_1 'H \sigma]$
- An immediately pretonic syllable is subject to syncope in words of five or more syllables, stress falls on a heavy antepenult: $[\sigma_1 L_a 'H/L_b L_c \sigma] > [\sigma_1 'H/L_b L_c \sigma]$
- An immediately posttonic light syllable is subject to syncope in words of four or more syllables, if stress fell on the preantepenult: $[\sigma_o 'H/L L_a L_b \sigma] > [\sigma_o 'H/L L_b \sigma]$.

However, the general validity of syncope for all Prākritis (in particular Māhārāṣṭrī) conditioned by the surface position of primary stress as assigned by Bühler's rules for Classical Sanskrit is rather doubtful. Among words of the shape $[L L \sigma]$, genuine examples of syncope occur, but since the Vedic *udātta*, where known, in virtually all cases fell on the unsynco-pated initial or final syllable, syncope in such cases need not be attributed to the Classical Sanskrit rule. In virtually all other cases, alternative possible purely segmental developments, in particular consonant lenition and vowel contraction, could be responsible for the attested Prākritic forms.

¹⁸Zachary Rothstein-Dowd points out to me that *hiraṇmaya-*, at least, might be open to an alternative account, namely, the (morphological) deletion of a (perceived) suffix *-ya-* in a first compound member (so-called "Schulz's Law").

What remains of Jacobi's collection are rather the few Śauraseni cases that attest syncope, some Sanskrit forms attesting syncope, a number of Sanskrit and Pāli forms that exhibit vowel lengthening or consonant gemination, and the Sanskrit cases of vowel shortening. Unfortunately, the cases of variant Sanskrit forms are often known purely from grammatical texts, such as the Dvirūpakoṣas, for which reason placing the variants in chronological relation to each other is difficult. Nonetheless, even the more limited number of such forms can somewhat restrict the the search space of a possible stress grammar. In the main, these examples point to stress on a heavy syllable or the leftmost syllable: they are compatible with Bühler's rules, but also, e.g., with an unbounded weight-prominence grammar: stress the (?-most) heavy, else the leftmost syllable. If the cases of vowel shortening in initial syllables such as Ved. *árāti*- alongside Skt. *arāti*- are admitted, then also a default-to-opposite grammar (rightmost non-final heavy syllable, else leftmost) would be possible.

(29) Credible Instances of Syncope in Sanskrit and Śauraseni

Sanskrit Prosodic Shape	(Vedic) Preform	Attested Form	Gloss
[L H L L σ]	<i>híraṇyavatī-</i>	Skt. <i>hiraṇvatī-</i> (MBh. 6×)	'possessing gold'
[L H L L σ]	<i>híraṇyamáya-</i>	Skt. <i>hiraṇmaya-</i> (TS) ^a	'golden'
[H L L σ]	<i>pūgaphala-</i>	Ś. <i>popphalī</i>	'betel-nut'
[L H L σ]	<i>pacāmáhe</i>	Ś. <i>pacamha</i>	'cook:PRES.MID.1PL'
[H L σ]	<i>sūnára-</i> (RV)	Skt. <i>sundara-</i>	'glad; beautiful'
[L L σ]	<i>bhagínī-</i> *	bhagnī (Dvirūp.)	'sister; woman'

^aThis latter variant, with apparent syncope, is very well attested in Vedic texts; the TS citation is merely its earliest occurrence.

5.2.2.2 Inheritance of Stress Rules in Modern Indic Languages

Above, the survey of data from Jacobi 1893 revealed most of the easily available evidence from the Prākṛits to be equivocal: for the Māhārāṣṭrī forms that made up the bulk of the data claimed to exhibit syncope, syncope could either be attributed to the position of Vedic prominence, or a set of segmental changes independent of stress could be held responsible for the disappearance of a syllable. Only in a few cases, largely consisting of forms found in Sanskrit texts, or Pāli, or Śauraseni Prākṛit, could relatively unambiguous evidence for stress on a syllable at odds with an attested or likely Vedic stress be established.

Instead, relatively more secure evidence for the validity of Bühler's Classical Sanskrit stress rules may be found in Modern Indic languages. Grierson (1895b) offers a number of correspondences between Sanskrit preforms and inherited Hindi forms, where the syllable predicted to bear stress by the Classical Sanskrit rule both escapes any syncope and bears stress in Hindi. Examples are shown in the table under 30. Stress in the Sanskrit preforms is marked with an IPA stress mark; Hindi forms are represented as by Grierson, with an acute marking primary stress; numbers in parentheses indicate the page in Grierson 1895b.

(30) Stress in Sanskrit > Stress in Hindi (Grierson 1895b)

Sanskrit Prosodic Shape	Sanskrit Preform	Hindi	Gloss
[L L σ]	'pra.tha.maḥ	<i>páhlā</i>	'first' (141)
[H ṼC]	'vid.yut	<i>bījī</i>	'lightning' (141)
[H H σ]	kā.'sī.saṃ	<i>kasís</i>	'green iron sulphate' (142)
[H L σ]	'kū.pa.kaḥ	<i>kūā</i>	'a well' (144)
[H H L σ]	abh.'yañ.ji.taḥ	<i>bhījā</i>	'anointed' (145)
[L Ṽ]	'ba.la-	<i>bál</i>	'strength' (147)

Overall, Grierson's data likewise indicate that primary stress fell on the rightmost non-final heavy syllable, else on initial syllable.

The survey of Turner (1916) shows for Gujarātī in more detail that the application of Bühler's rule to a Sanskrit preform will correctly predict which non-final syllable will ultimately be preserved. According to Turner (1916: 213), the position of primary stress in Gujarātī itself typically remains identical with the stress that was assigned by the Classical Sanskrit rule; under this conception, stress in Gujarātī would regularly occur on the penult, and exceptionally on the final syllable (cf. also Turner 1921: 342–4, who admits of shifts of stress to the initial syllable, particularly in verbal paradigms).¹⁹ The table in 31 reproduces the overview given in Turner 1916: 214.

(31) Sanskrit Stress > Gujarātī Stress

Sanskrit Prosodic Shape	Sanskrit Preform	Gujarātī	(Skt.) Gloss
[H σ]	'grā.maḥ	'gām	'village'
[L σ]	'jal.am	'jaḷ	'water'
[H L σ]	'mat.ku.naḥ	'mā.kaṇ	'bug'
[L L σ]	'ka.ma.laḥ	'ka.maḷ	'lotus'
[H L L σ]	'gar.bhi.ṇi.kā*	'gābh.ṇi	'pregnancy'
[L L L σ]	'ku.ma.la.kaḥ	'kum.ḷo	'pot'
[L H σ]	a.'raṇ.yam	'rān	'wilderness'
[H H σ]	sau.bhāg.yam	so.'hāg	'success'
[L H L σ]	ka.'thā.ṇi.kā	ka.'hā.ṇi	'story'
[H L H σ]	cak.ra.'vā.kaḥ	cak.'vā	'Cakra bird'
[L L H σ]	pri.ya.'kā.raḥ	pi.'ār	'congenial'
[L L H L σ]	pri.ya.'kā.ra.kaḥ	pi.ā.ro	'agreeable'

The data above match the generalization that primary stress fell on the rightmost non-final heavy syllable, and that stress on the pre-antepenult could occur. One important detail

¹⁹Setting aside the claims of sonority-driven stress made for Gujarātī (e.g., in de Lacy 2002 and de Lacy 2006, now refuted by Shih 2018b and Bowers 2019), some other scholarship has claimed that primary stress in Gujarātī consistently falls on initial syllables (Pandit 1958, Patel and Mody 1960). The phonetic study of Bowers (2019) supports the claim of regular initial stress. Below, I report stress and vowel length for Gujarātī as given by Turner (1916), though these prosodic features may not hold for (all) speakers of the present-day language (or even have held at the time of Turner's work a century ago). The patterns of syncope, taken as a whole, do though seem to suggest that primary stress fell on the syllable indicated as stressed by Turner at some point in the history of the language.

that that allows for more precise determination of Classical Sanskrit prosody, is the form *jalam* ‘water’: here, stress occurs on the initial syllable of a word with a closed final syllable. If the final syllable counted as heavy, stress on the initial syllable would contradict cases like *ayāt* given by Bühler. However, since the two instances of the stress pattern [L H] in Bühler contain a long vowel in the final syllable, it is possible that the final segment was extrametrical, thus rendering the final syllable in forms like *jalam* effectively light. I therefore assume that the parse [(‘ja.la)⟨m⟩] held in such cases.

In addition, Turner (1916: 217) mentions some four-syllable forms of the shape [H L L σ], and two five-syllable forms of the shape [L H L L σ], all supposedly continuing a form with stress on the heavy preantepenult. These examples are compatible with stress on a heavy pre-antepenult, given that intervocalic [m] underwent lenition to a nasalized fricative [ṽ] when in the onset of a stressed syllable (so Turner). In turn, nasalization ultimately appears on the first syllable in Gujarātī (32a-b.); in contrast, when stress fell elsewhere, [m] remains as such into Gujarātī (32c-d.).²⁰

- (32) Syncope of light antepenult (+ later contraction from loss of [ṽ] < [m]) in Gujarātī
- a. *samárpayati* (caus. to *saṃ-√ṛ* ‘join together’) > *saṽáryai* > G. *sōpe*
 - b. **samárgḥaka-* ‘cheap’ (dim. to attested *samargha-*) > *saṽákha-* > G. *sōgho*
 - c. *pāñcamakaḥ* ‘fifth’ > G. *pāc̣mo* (cf. Hindi *pāc̣wā*)
 - d. *sáptamakaḥ* ‘seventh’ > *sātmo* (cf. Hindi *sātwā*)

The examples in (33), showing words of five or more syllables, are also compatible with the supposed Classical Sanskrit pattern, but the cases of [L L L L σ] are technically ambiguous between primary stress on the initial syllable and the antepenult. The forms given here are given by Turner (1916: 223–4). G. *parjāle* presents the only instance of a non-compound form containing more than five syllables, but unfortunately has a heavy preantepenult, and thus cannot help to determine where stress might have fallen in six-syllable words with only light non-final syllables.

(33) Development of Words of Five or More Syllables into Gujarātī

Sanskrit Prosodic Shape	Sanskrit Preform	Gujarātī	(Skt.) Gloss
[L H/L? H L L σ]	pa.ri.'jvā.la.ya.ti ^a	par.'jā.ḷe	‘burn:CAUS.3SG’
[L L L L σ]	,pa.ri.'ṇa.ya.ti	'par.ne ^b	‘lead around; marry:CAUS.3SG’
[H H L L σ]	prak.'ṣā.la.ya.ti	pa.'khā.ḷe	‘cleanse:CAUS.3SG’
[H H L L σ]	pras.'thā.pa.ya.ti	pa.'ṭhā.ve	‘send:CAUS.3SG’
[L H L L σ]	ca.túr.gu.ṇa.kaḥ*	'cōg.ṇo	‘having four qualities’
[H H L H σ]	yaj.ño.pa.'vī.tam	janōī	‘investiture with sacred thread’
[H L H L σ]	śī.ta.'kā.la.kaḥ	śī.yā.ḷo	‘cold season’

^aSyllabification unclear.

^bTurner argues for a later retraction of stress to the initial syllable.

²⁰A somewhat more natural interpretation of these developments is that, in forms like *sáptamakaḥ*, the post-tonic syllable underwent syncope, creating a C + [m] cluster, which was preserved; intervocalic [m], on the other hand, underwent lenition. The lenition of [m] is thus still diagnostic of the position of stress.

The ultimate conclusion to be drawn at this juncture is the following: at least some subset of post-Vedic Indic languages, including either the Old Indic variety that formed the basis for Classical Sanskrit (or whose word prosody was ultimately adopted for Sanskrit), developed a grammar of stress assignment that essentially accords with the rules described by Bühler (1883). In the view of Turner (1916: 229–30) the largely parallel developments in terms of of syncope and apocope seen in Hindi, Sindhi, Punjabi, and Bengali suggest that the ancestors of those languages, too, knew the Classical Sanskrit stress rule.

The only particular stress pattern given by Bühler that lacks some independent confirmation is the pattern [L 'H], that is, primary stress on a final heavy syllable in a disyllabic form with a light penult, as in *ayāt* or *jagáu*. Since just such a stress pattern poses no real obstacles to a coherent metrical stress grammar, I will assume that Bühler is correct on that point as well. Finally, one relatively minor point not obviously substantiated by available data concerns the treatment of glides, *y* and *v*, in clusters. Bühler asserts that complex onsets are formed in such cases; examples like Ved. *hiraṇyavatī* > Skt. *hiraṇvatī*, Skt. *vidyut* > Hindi *bījī*, or Skt. *aranyam* > Gujarātī *rāṇ* would suggest instead that the C + glide sequence creates a closed syllable. I tentatively suggest that the syllabification of C + glide clusters in Classical Sanskrit was, perhaps unsurprisingly, dependent upon the exact consonant; at least a nasal may not have made a licit complex onset with a following *y*.

5.2.2.3 Evidence from Meter?

Before developing a constraint-based analysis of the Bühlerian stress system, worth mentioning is some further evidence from the domain of metrics that might marginally support such a stress grammar. First, Kazama (1975: 62) undertook a pilot study on coincidence of stress predicted by Bühler's rules and verse ictus in the cadence of *śloka*s of Epic Sanskrit.²¹ In the 1323 verses surveyed, stress and verse ictus coincided in 1010 cases (76.3%), but did not in 313 cases (23.6%); examples of coincidence or its lack are shown in 34. While the rate of stress–ictus is obviously well above chance ($p < 0.001$, Exact Binomial Test), a more careful study should control for the likelihood of placing a word of the shape $[\sigma_0 \text{ H L } \sigma]$ in a cadence, as opposed to a $[\text{H } \# \text{ L } \sigma]$ sequence.

- (34) Stress–Ictus Coincidence in Epic Sanskrit *śloka* (ictic syllable underlined; stress predicted by the rules of Bühler 1883 and Sandell 2023 marked with acute)

- a. Match: ...*tásya* *mānuṣaḥ*
- b. Mismatch: ...*naréṣu* *ca* *nálo* *vāraḥ*

The study of Fairbanks (1992) serves as a further possible indication of the relationship between Classical Sanskrit stress and stress in some Modern Indic languages. Fairbanks demonstrated that versification in Hindi depends upon assigning word stress again to rules essentially identical to Bühler's. Given the correctness of Fairbanks' study, stress in Hindi and the Classical Sanskrit stress rule would seem to match nicely. Importantly, most longer

²¹Thanks to Bert Vaux for making me aware of Kazama's study and for furnishing me with a copy of the relevant article.

word forms in Fairbanks' corpus – many three-syllable and virtually all four-syllable words – are themselves borrowings from Sanskrit.

5.2.3 The Classical Sanskrit Stress Rule: Analysis

The objective of this section is to sketch a constraint-based matrical stress grammar capable of reproducing the stress patterns reported for Classical Sanskrit by Bühler and partly substantiated by independent data surveyed in the immediately preceding sections. First, I again summarize the major properties of this stress grammar, and then before developing its constraint-based implementation, remark on some outstanding lacunae in the data or uncertainties in their interpretation are in order.

5.2.3.1 Summarizing the Patterns

The stress pattern that can be deduced for Classical Sanskrit based on Bühler's reports, essentially substantiated by the data from Jacobi 1893 (if not the full extent of Jacobi's claims) and Turner 1916 looks as follows:

- Disyllables bear primary stress on the first syllable, unless the first syllable is light and the final syllable contains a long vowel (i.e., [L 'V:(C)])
- Words of three or more syllables bear primary stress on the rightmost heavy non-final syllable *up to the forth-to-last* (preantepenult); in other words, a four-syllable right edge stress window exists.
- Words of three or four syllables in which penult, antepenult, and preantepenult are all light bear primary stress on the initial syllable.
- Words of five syllables in which penult, antepenult, and preantepenult are all light bear primary stress on the antepenult.

Unfortunately completely absent from the available data are any forms of six or more syllables that are not nominal compounds, and which contain no non-final heavy syllables. Here, there are two possibilities, depending upon the rankings of ALL-Ft-R and ALL-Ft-L: antepenultimate or preantepenultimate stress. If the ranking ALL-Ft-R $\gg$ ALL-Ft-L holds, then it is predicted that stress will always fall on the antepenult in a word of five or more syllables, if all syllables in the window are light. If, on the other hand, the ranking ALL-Ft-L $\gg$ ALL-Ft-R holds, then primary stress will fall on the antepenult in words with an odd number of syllables, but on the preantepenult in words with an even number of syllables.

5.2.3.2 Analysis of the Classical Sanskrit Stress Rule

Throughout the presentation of the analysis given here I, employ, where possible, the same forms given by Bühler (1883) (listed under 5.2.1 above), with reference to parallel supporting

datapoints. The use of the foot-based non-finality constraints is in fact crucial to deriving the correct outcomes, in particular, explaining why primary stress occurs on final syllables in the configuration [L 'V:(C)], but is otherwise avoided. Specifically, in order to generate final syllable extrametricality in all words of three or more syllables, but to allow for either initial or final stress in disyllables, NONFINALITY- $\mathcal{F}$ must dominate PARSE-C, which in turn dominates PARSE- σ . This ranking means that final syllable extrametricality is the preferred method for avoiding feet at the right edge of the word, but, where necessary, final consonant extrametricality can be invoked in order to provide a binary foot, as in [('ja.la)⟨m⟩] ‘water’.

Disyllabic Forms In disyllables, primary stress always falls on the initial (penultimate) syllable, unless the final syllable contains a long vowel and the initial syllable is light. Taking account of the presence or absence of a final consonant, the two general patterns are thus: ['σ V(C)] versus [L 'V:(C)]. Final consonants must be treated as extrametrical, given the testimony of Gujarātī *jaḷ* < Skt. ['ja.lam]: if the final syllable behaved as heavy, the stress pattern [L 'VC] would be expected. Unfortunately, Bühler’s data show no words of the shape [L 'VC], and examples of Sanskrit [L 'V:(C)] inherited into Prākṛits or Modern Indic were not readily forthcoming.

The [L 'V:(C)] pattern, seen in *a-yāt* ‘go:IMPF.3SG’ and *^Vjagáu* ‘go:PERF.1SG’, alone is very informative, since it shows that attraction of stress to a heavy syllable (WSP) can outweigh avoidance of primary stress on final syllables, where the right edge of the foot is aligned with the right edge of the word (NONFINALITY- σ). Candidates that could produce stress on the initial light syllable, like b. and c. in 35, must either violate a constraint like FOOTBINARITY or point to the ranking of WSP over constraints that would pull a foot to the left edge of the prosodic word. Since a candidate with iambic stress, as in 35d., would satisfy the left-oriented constraints, but obviously fails in cases like [('ja.la)⟨m⟩] (in 36), which must have trochaic stress, TROCHEE can be safely ranked as undominated. The derivation of [('ja.la)⟨m⟩] in (36) further shows that final consonant extrametricality helps to ensure a foot that satisfies foot binarity (see candidate d.). For reasons clarified below, PARSE-Segment, violated by winning candidate 36a., must outrank WSP, which in turn entails the use of NONFINALITY- $\mathcal{F}$, outranking PARSE-C, in order to exclude candidate 36b. In effect, the use of final consonant extrametricality is permitted only to help to satisfy undominated FTBIN.

(35) Skt. /jagau/ → [ja.('gau)] go:PERF.1SG’

/jagau/		TROCHEE	FTBIN	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-FT-L	PARSE- σ
a.	 ja.('gau)				*	*	*	
b.	('ja).gau		*!	*				
c.	('ja.gau)			*!				
d.	(ja. 'gau)	*!						

(36) Skt. /jalam/ → [(‘ja.la)⟨m⟩] ‘water:ACC.SG’

/jalam/	TROCHEE	FTBIN	NONFIN- $\bar{f}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \bar{f}$)	ALL-FT-L	PARSE- σ
a.  (‘ja.la)⟨m⟩				*					
b. ja.(lam)			*!			*	*	*	*
c. (ja.’la)⟨m⟩	*!			*					
d. (‘ja).lam		*!			*				*

Cases in which the initial syllable of a disyllabic word is heavy and the final syllable contains a short vowel fall out straightforwardly: WSP is sufficient to ensure that primary stress is assigned to the initial syllable, as in *rāṅkum* ‘species of deer:ACC.SG’ or *grāmah* ‘village’ (> G. *gām*, as shown in (37). In the event that the initial syllable is heavy and the final contains a long vowel, but no final consonant, avoidance of alignment of the foot with primary stress with the right edge of the word will prefer initial stress, as is the case for *tāsthau* ‘stand:PERF.3SG’, shown in (38).²² Parsing both heavy syllables is again ruled out by the violation of NONFINALITY- $\bar{f}$; further below, violation of *CLASH in sequences of heavy syllables will prove to be admissible.

(37) /raṅkum/ → [(‘raṅ).ku(m)] ‘deer:ACC.SG’

/raṅkum/	TROCHEE	FTBIN	NONFIN- $\bar{f}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \bar{f}$)	ALL-FT-L	PARSE- σ
a.  (‘raṅ).kum					*				*
b. ran.(‘kum)			*!			*	*	*	*
c. (‘ran.ku)⟨m⟩	*!			*					

(38) /tasthau/ → [(‘tas).thau] ‘stand:PERF.1SG’

/tasthau/	TROCHEE	FTBIN	NONFIN- $\bar{f}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \bar{f}$)	ALL-FT-L	PARSE- σ	*CLASH
a.  (‘tas).thau					*				*	
b. tas.(‘thau)			*!		*	*	*	*	*	
c. (,tas).(‘thau)			*!		*	*	*	*	*	*

²²That the Sanskrit long diphthongs ⟨au⟩ [a:u] and ⟨ai⟩ [a:i] do not act on the surface as vowel plus glide sequences is demonstrated by the fact that a consonant may occur word finally after these diphthongs (e.g., *dyauh* ‘day:NOM.SG’), though Sanskrit otherwise allows no consonant clusters word finally.

No disyllabic forms containing a final syllable with a long vowel and a following consonant are present in the survey of data above, but other polysyllables having such a configuration indicate that the final syllable did not receive stress (e.g., *duhitṛnām* [(du.hi).('tṛ).na:m] 'daughter:GEN.PL'). The absence of primary stress on the final syllable is due to the high ranking of both NONFINALITY- $\mathcal{F}$ and PARSE-C; again, only satisfaction of FTBIN can compel a violation of one of these constraints.

Polysyllables with Only Light Non-Final Syllables In both three- and four-syllable forms containing only light non-final syllables, primary stress occurs on the initial syllable in part due to ALIGN-L($\omega, \mathcal{F}$): the left edge of a word must be aligned with the left edge of a foot. In case the word contains exactly three syllables, only the parse [('L L) σ] is viable, since alternatives satisfying FTBIN would violate either NONFINALITY- $\mathcal{F}$ or PARSE-C. Hence, in the tableaux deriving *kṣīpasi*, *ágamat*, and *káranāt* immediately below, all with primary stress on the initial syllable, the losing candidates are all excluded due to violations of either NONFINALITY- $\mathcal{F}$ or PARSE-C.

(39) /ksipasi/ → [('kṣi.pa).si] 'hit:PRES.2SG

/ksipasi/	TROCHEE	FTBIN	NONFIN- $\mathcal{F}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-FT-L	PARSE- σ	*CLASH
a.  ('kṣi.pa).si					*				*	
b. kṣi.('pa.si)			*!				*	*	*	

(40) /agamat/ → [('a.ga).mat]

/agamat/	RIGHTMOST	TROCHEE	FTBIN	NONFIN- $\mathcal{F}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-FT-L	PARSE- σ	*CLASH
a.  ('a.ga).mat						*				*	
b. a.('ga.ma)(t)					*!			*	*	*	
c. a.('ga.mat)				*!		*		*	*	*	
d. (a.ga).('mat)				*!			*		**		

(41) /karanāt/ → [('ka.ra).nāt]

/karanāt/		RIGHTMOST	TROCHEE	FTBIN	NONFIN- $\mathcal{F}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-Ft-L	PARSE- σ	*CLASH
a.	𑂔𑂗𑂢𑂰 ('ka.ra).ṇāt						*				*	
b.	ka.('ra.nā)⟨t⟩					*!	*					
c.	ka.('ra.nāt)				*!		*		*	*	*	
d.	(,ka.ra).('nāt)				*!			*		**		

The crucial role of ALIGN-L($\omega, \mathcal{F}$) truly emerges in quadrisyllabic forms like *dūhitaram*, the derivation of which is shown in (42). Primary stress on the antepenult (candidate b.) would indeed be viable, but the violation of ALIGN-L($\omega, \mathcal{F}$) renders it a losing option; the possibility of parsing the string into two feet, or parsing a final heavy syllable as a foot (candidates c. and d.) again encounter violations of high-ranking PARSE-C or NONFINALITY- $\mathcal{F}$. Vowel lengthening in the initial syllable of Pāli *ānubhava*- ‘experience’ substantiates initial stress in words of this shape, as does initial stress in Gujarātī *kumlo* ‘lotus’ (< Skt. *kāmalakah*).

(42) /duhitaram/ → [(‘du.hi).ta.ram] ‘daughter:ACC.SG

/duhitaram/		RIGHTMOST	TROCHEE	FTBIN	NONFIN- $\mathcal{F}$	PARSE-C	WSP	NONFIN- σ	ALIGN-L($\omega, \mathcal{F}$)	ALL-Ft-L	PARSE- σ	*CLASH
a.	𑂔𑂗𑂢𑂰𑂢𑂰𑂢𑂰 ('du.hi).ta.ram						*				**	
b.	du.('hi.ta).ram						*		*!	*!	**	
c.	(,du.hi).('ta.ra)⟨m⟩					*!			*	**		

When the number of syllables is extended to five, the contiguous sequence of four light syllables beginning from the left edge allows room for two feet without risk of a violation of either PARSE-C or NONFINALITY- $\mathcal{F}$. ALIGN-L($\omega, \mathcal{F}$) is then readily satisfied, and primary stress is predicted to fall on the antepenult by adding RIGHTMOST to the highest stratum. Since ALIGN-L($\omega, \mathcal{F}$) can be satisfied in the presence of two feet, RIGHTMOST serves to select the rightmost foot in the word as the locus of primary stress. A prefixed verbal form such as *ūpagāmatam* ‘approach:AOR.2DU’, for which Bühler even further indicates a secondary stress, illustrates this pattern. In turn, ranking ALIGN-L($\omega, \mathcal{F}$) and PARSE- σ above ALL-Ft-L and ALL-Ft-Rt ensures that secondary stress is generated in such cases. Syncope of a pre-tonic short vowel, [i], in Hindi *paḍosa* ‘neighborhood’ (< Skt. *prativésaka*-) likewise points to antepenultimate stress.

(43) /upagamatam/ → [(,u.pa).('ga.ma).tam] 'approach:AOR.2DU'

/upagamatam/		RIGHTMOST	TROCHEE	FTBIN	NONFIN-F	PARSE-C	WSP	NONFIN-σ	ALIGN-L(ω, f)	PARSE-σ	ALL-Ft-L	ALL-Ft-R
a.	☞ (,u.pa).('ga.ma).tam						*			*	**	***
b.	('u.pa).(,ga.ma).tam	*!					*			*	**	***
c.	u.(,pa.ga).ma.tam						*		*!	***!	*	**
d.	u.pa.(,ga.ma).tam						*		*!	***!	**	*
e.	('u.pa).ga.ma.tam						*			***!		***

Polysyllables with Exactly One Heavy Non-Final Syllable Among most polysyllabic forms containing exactly one heavy non-final syllable, the usual high-ranking constraints PARSE-C, NONFINALITY-F, and WSP will conspire to ensure that the single heavy syllable is footed. When the single heavy syllable falls within a window up to the preantepenult, avoidance of violations of FTBIN or *CLASH will further ensure that the foot built around that heavy syllable is the rightmost foot in the word, and thereby eligible for primary stress. In most three- and four-syllable forms with a single heavy syllable, e.g., *bódhati* 'observe:PRES.3SG', *kṣipāmaḥ* 'hit:PRES.1PL' or *kathānikā* 'chat:NOM.SG' (> G. *kahāṇi*), no space for a secondary stress is available. The single light syllable between the left edge of the word and the heavy syllable in some cases leads to violations of ALIGN-L(ω, f) in order to satisfy higher-ranked WSP. In four-syllable words of the shape [L L H σ], e.g., *karaṇēna* 'doing:INST.SG' or *duhitṛṇām* 'daughter:GEN.PL', on the other hand, secondary stress is predicted on the initial syllable (though no direct indications of such secondary stress are available here). Further cases of stress on a heavy preantepenult are substantiated by Gujarātī *gābhṇi* 'fetus' (< *garbhini**) and *kēsri* (< *keśarikaḥ**). The tableaux in 44 and 45 illustrate cases of primary stress assigned to the single heavy syllable in three- and four-syllable words.

(44) /ksipāmas/ → [kṣi.(,pā).maḥ] 'hit:NPST.1PL'

/ksipāmas/		RIGHTMOST	TROCHEE	FTBIN	NONFIN-F	PARSE-C	WSP	NONFIN-σ	ALIGN-L(ω, f)	PARSE-σ	ALL-Ft-L	ALL-Ft-R
a.	☞ kṣi.(,pā).maḥ						*		*	**	*	*
b.	('kṣi.pā).maḥ						***!			*		*
c.	kṣi.pā.(,maḥ)				*!		*	*	*	**	**	

Candidate f. in (45) is of particular interest, since it shows that primary stress cannot be attracted to long vowels in final syllables where a violation of NONFINALITY-F could be

avoided by failing to parse the rightmost consonant. Again, the ranking $\text{FtBIN} \gg \text{NONFIN-L} \gg \text{PARSE-C} \gg \text{WSP}$ keeps stress off of final syllables except in cases like [ja.(‘gau)], and keeps final consonants parsed into syllables, except in cases like [(‘ja.la)<m>].

(45) /duhitṛnām/ → [(‘du.hi).(‘tṛ).nām] ‘daughter:GEN.PL’

/duhitṛnām/	RIGHTMOST	FtBIN	NONFIN-F	PARSE-C	WSP	NONFIN-σ	ALIGN-L(ω, F)	*CLASH	PARSE-σ	ALL-Ft-L	ALL-Ft-R
a. (‘du.hi).(‘tṛ).nām					*				*	**	***
b. (‘du.hi).(‘tṛ).nām	*!				*				*	**	***
c. (‘du.hi).(‘tṛ).(‘nām)			*!			*		*		*****	***
d. (‘du.hi).tṛ.nām					**!				**		**
e. du.hi.(‘tṛ).nām					*		*!		**!*	**	*
f. (‘du.hi).(‘tṛ).(‘nā)<m>				*!		*		*		*****	***

The five-syllable forms with exactly one heavy syllable illustrate further why the metrical stress grammar of Classical Sanskrit has a four-syllable window, rather than unbounded prominence-driven stress: if a heavy syllable occurs up to the fourth-to-last syllable, it blocks the construction of further feet to its right. The tableau deriving stress on the heavy preantepenult of *hirāṇyavatī* ‘possessing gold:F’ (contrast attested Vedic *hīraṇyavatī*) is given in (46). The avoidance of stress clash (which, if permitted, would further lead to primary stress on the antepenult) in this case shows that *CLASH must outrank PARSE-σ.

(46) /hiranyavatī/ → [hi.(‘raṇ).ya.va.tī]²³ ‘golden:F.NOM.SG’

/hiranyavatī/	RIGHTMOST	FtBIN	NONFIN-F	PARSE-C	WSP	NONFIN-σ	ALIGN-L(ω, F)	*CLASH	PARSE-σ	ALL-Ft-R	ALL-Ft-L
a. hi.(‘raṇ).ya.va.tī					*		*		*****	***	*
b. hi.(‘raṇ).ya.va.(‘tī)			*!			*	*		***	***	*****
c. hi.(‘raṇ).(‘ya.va).tī					*		*	*!	**	*****	***
d. (‘hi.raṇ).(‘ya.va).tī					**!				*	*****	**
e. (‘hi).(‘raṇ).ya.va.tī		*!			*		*		***	*****	*

²³Empirically, whether the immediately following light syllable was parsed into a foot with the heavy syllable cannot be determined. A candidate [hi.(‘raṇ.ya).va.tī] could, if necessary, be excluded through high-ranking RH-CONTOUR.

A heavy syllable on the fifth-to-last syllable on the other hand, would allow for a disyllabic foot to be constructed around the antepenult and penult, without incurring any violations of *CLASH. Consequently, primary stress should fall on the antepenult in such cases. Unfortunately, no clear examples of a form of the shape [H L L L σ] occur in the data examined in the preceding section; nevertheless, that primary stress would occur on the antepenult, thus yielding [(,H) L ('L L) σ] in such cases, seems likely.

Polysyllables with Multiple Heavy Non-Final Syllables Polysyllabic words containing more than one heavy non-final syllable are largely straightforward to derive. The overall necessary ranking RIGHTMOST $\gg$ WSP $\gg$ ALIGN-L(ω , $\mathcal{F}$), *CLASH will result in stress on all heavy syllables, whether in clash or not, with the rightmost foot then selected for primary stress. The necessity of admitting stress clash to avoid violations of WSP follows from the ranking $\gg$ WSP $\gg$ ALIGN-L(ω , $\mathcal{F}$) established for disyllabic words of the shape [L 'V:(C)] established above: given that stress falls on the obligatorily heavy syllable in such forms (see 35 above), stress on a heavy syllable is more important than stress on an initial syllable (cf. also 46). In forms like *bodhā́vaḥ* below, where the two rightmost syllables are both heavy, WSP violations are avoided only when both syllables receive stress, creating a violation of *CLASH; the alternative candidates in (47)b. and c. both violate WSP, although c. avoids violations of both ALIGN-L(ω , $\mathcal{F}$) and *CLASH.

(47) /bodhāvas/ → [(,bo).(‘dhā).vaḥ]

/bodhāvas/	RIGHTMOST	NONFIN- σ	PARSE-C	WSP	NONFIN- σ	ALIGN-L(ω, σ)	*CLASH	PARSE- σ	ALL-Ft-R	ALL-Ft-L
a.  (,bo).(‘dhā).vaḥ				*			*	*	***	*
b. bo.(‘dhā).vaḥ				**!		*		**	*	*
c. (‘bo).dhā.vaḥ				**!				**	**	
d. (‘bo).(,dhā).vaḥ	*!			*			*	*	***	*

In contrast when the initial syllable is light and immediately followed by a heavy syllable, it can remain unfooted, as in (46) above. When no stress clash results, due to a light syllable intervening between two heavy syllables, primary stress again surfaces on the rightmost heavy, as seen for *śītakālakah* (> G. *śīyālo*) in (48); light syllables trapped between two heavy syllables remain unstressed, since providing them with stress would result in violation of either FTBIN or WSP (in addition to a further *CLASH violation). Much same applies to a form containing three heavy syllables, *yajñopavītam* (> G. *janōi*), given in (49), where stress is permitted in clash, and the primary stress necessarily falls on the rightmost heavy syllable; the stressed non-final heavy syllables (long vowels) are preserved diachronically.

(48) /śītakālas/ → [(,śī).ta.(‘kā).laḥ] ‘cold season:NOM.SG’

/śītakālas/	RIGHTMOST	NONFIN- σ	PARSE-C	WSP	NONFIN- σ	ALIGN-L(ω, σ)	*CLASH	PARSE- σ	ALL-Ft-R	ALL-Ft-L
a.  (,śī).ta.(‘kā).laḥ				*				**	*****	**
b. śī.ta.(‘kā).laḥ				**!		*		***	*	**
c. (,śī).(‘ta.kā).laḥ				**!			*	*	*****	*
d. (‘śī).ta.(,kā).laḥ	*!			*				**	*****	**

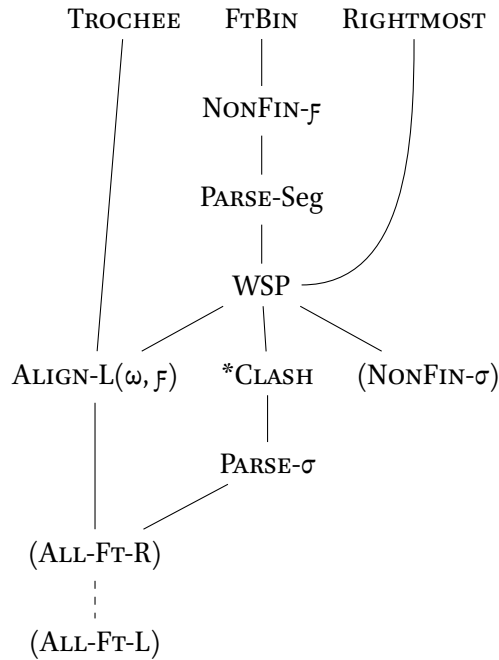
(49) /yajñopavītam/ → [(,yaj).(ñō).pa.(‘vī).tam] ‘investiture with sacred thread:ACC.SG’

/yajñopavītam/	RIGHTMOST	NONFIN- σ	PARSE-SEG	WSP	NONFIN- σ	ALIGN-L(ω, σ)	*CLASH	PARSE- σ	ALL-Ft-R	ALL-Ft-L
a.  (,yaj).(,ñō).pa.(‘vī).tam				*			*	**	7	4
b. (,yaj).ñō.pa.(‘vī).tam				**!				***	5	3
c. yaj.(,ñō).pa.(‘vī).tam				**!		*		***	4	4
d. (,yaj).(,ñō).(‘pa.vī).tam				**!			**	*	7	3

Final Total Ranking and Commentary Based on the illustrative tableaux and derivations of the forms treated in above, the total ranking given in (50) obtains; constraints given in parentheses could be excluded without changing the winning candidates or the rest of the ranking. The Hasse diagram below it reflects the strictly necessary ranking relationships. Note, however, that the lack of six-syllable forms consisting solely of light syllables leaves the relative ranking of ALL-Ft-L and ALL-Ft-R undetermined. The ranking ALL-Ft-R $\gg$ ALL-Ft-L is shown, but the opposite would be compatible with the known data. Likewise unknown is whether feet of the shape ('H L) are permitted or not, i.e., what ranking holds between PARSE- σ and RH-CONTOUR; above, I chose to leave light syllables unparsed and exclude RH-CONTOUR from consideration.

- (50) Final Total Constraint Ranking for the Stress Grammar of Classical Sanskrit
TROCHEE, FOOTBINARITY, RIGHTMOST $\gg$ NONFINALITY- $\mathcal{F}$ $\gg$ PARSE-Segment $\gg$ WEIGHT-TO-STRESS $\gg$ (NONFINALITY- σ), ALIGN-L(ω , $\mathcal{F}$), *CLASH $\gg$ PARSE- σ $\gg$ (ALL-FEET-R) $\gg$ (ALL-Ft-L)

- (51) Hasse Diagram of Final Total Ranking for Classical Sanskrit Stress



The top stratum, consisting of TROCHEE, FOOTBINARITY, and RIGHTMOST ensures that every foot, if disyllabic, is trochaic, that all feet are minimally disyllabic or bimoraic, and that the rightmost foot in words containing multiple feet bears the primary stress. The next two strata, consisting of NONFINALITY- $\mathcal{F}$ and PARSE-Segment, respectively, conspire to keep the final syllable of the word unfooted: violation of the former is permitted only in cases like [ja.('gau)] (cf. 35), in order to satisfy FTBIN; the latter is violated mainly in cases like [(('ja.la)<m)] (though also [a.('yā)<t)], where foot binarity must be satisfied and a violation of NONFINALITY- $\mathcal{F}$ can be avoided. The fact that these two constraints are relatively high-ranked means that all words of three or more syllables do not need to violate either, since

FTBIN can in such cases always be satisfied. Then, in words containing any heavy syllables to the left of the final syllable, WSP will compel their footing, and RIGHTMOST will select the rightmost non-final heavy syllable for primary stress.

Constraints in the strata below WSP are decisive only in forms of four or more syllables without any heavy non-final syllables, as illustrated in (42) and (43). In particular, ALIGN-L(ω , f) is responsible for the occurrence of primary stress on the initial syllable in words of the shape /L L L σ / (cf. (42)), while *CLASH blocks the footing of an /L L/ sequence immediately to the right of a heavy syllable on the preantepenult, thereby helping to establish the four-syllable stress window (cf. (46)). Finally, PARSE- σ favors the construction of more feet in longer words. NONFINALITY- σ , ALL-FT-L and ALL-FT-R play no crucial role in deriving the forms treated above.

5.2.4 Comparison to Vedic Stress

At the level of the phonological grammar governing the position of primary stress, the prosody of Vedic Sanskrit treated in the preceding chapter and resulting grammar (see the constraint ranking in (88), repeated for convenience here) appears to share very little with the stress grammar corresponding to the Classical Sanskrit stress rule. Beyond the features of canonical stress systems (CULMINATIVITY, OBLIGATORINESS, etc.), both grammars share a strong preference for trochaic (TROCHEE) and binary feet (FTBIN) – though even the latter is outweighed by constraints on the morphology-prosody interface in Vedic. Moreover, both of these preferences are so common typologically that reference to any historical relationship is hardly necessary. Furthermore, as observed in 4.4, the ranking of prosodic markedness constraints for Vedic would simply predict left-edge stress throughout the grammar. In terms of surface outputs, consistent left-edge stress and the Classical Sanskrit grammar are sufficiently far removed from one another in their predictions that attempting to discern any necessary historical relationship between them looks like a non-starter.²⁴

(52) Final Constraint Ranking for the Analysis of Section 4.3:

TROCHEE, CULMINATIVITY, *VV, ANCHOR- R_f -HEAD $\gg$ ANCHOR- R_f $\gg$ (ANCHOR- L_f -GEN.PL), ANCHOR- L_f $\gg$ ANCHOR- L_f -HEAD, FTBIN $\gg$ ALL-FT-L, NONFINALITY- σ $\gg$ WSP, ALL-FT-R, HEADSTRESS- f -L $\gg$ PARSE- σ , MAX-V

If the prosodic grammars of Vedic and Classical Sanskrit have little to do with one another, the relationship between their prosodic systems must be entirely mediated by surface

²⁴Lahiri (2015: 225–6) suggests that the predictions of an initial-stress grammar with final consonant extrametricality that build only perfect moraic trochees is sufficiently similar to the Old or Classical Latin stress rule that the former might be reanalyzed as the latter. That both types of stress grammar make identical overt predictions in disyllables and trisyllables, with the exception of words of the shape /H H σ /, is apparent. In terms of token frequencies, though, /H H σ / words do not constitute a tiny group of exceptions in Latin, and words of four or more syllables, for which the two types of grammars make largely different predictions, still constitute about 20% of all tokens in a Latin corpus (compare Table 3.1). A development from a fixed-initial stress grammar to a pattern like the Old Latin (> Classical Latin) or Classical Sanskrit pattern is thus far from trivial, and given the ease with which fixed-initial stress is learned and maintained diachronically, perhaps surprising.

outputs (as one would expect anyway): simply “switching off” the prosodic faithfulness constraints of Vedic would lead to fixed-initial stress, which obviously did not occur. However, the activity of head dominance in the stress grammar of Vedic does allow one to observe one substantial commonality in outputs: given that Vedic is mainly suffixing in its morphology, head dominance there tends to pull stress towards the right edge of the word, often the penultimate or antepenultimate syllable, while the combined effects of RIGHTMOST (and WSP) do much the same in the Classical Sanskrit grammar. This observation leads one to wonder whether the particular instantiation of metrical stress seen in Classical Sanskrit plausibly reflects a reinterpretation of the effects of head dominance in Vedic.

To this end, a statistical comparison of the relative frequencies of word-prominence in Vedic with the predicted stress patterns of the Classical Sanskrit stress rule is worthwhile. While categorical predictions for the Classical Sanskrit stress rule may be made on the basis of the grammar in (50) above, the frequencies of word prominence in Vedic for words of any given prosodic shape must be empirically discovered. Therefore, I first gathered the token frequencies of all words bearing an *udātta* in the *R̥gveda* (RV). The ultimate goal is to compare the accuracy of some conceivable metrical stress grammars with the data from the RV.

Data collection on the frequencies of the *udātta* in words of different prosodic shapes in the RV were collected using a tokenized *padapāṭha* text furnished by Dieter Gunkel and Kevin Ryan. Using a script written in R, regular expressions were employed to extract tokens with particular combinations of number of syllables, syllable weight (heavy vs. light syllable, and, for final syllables, short or long vowel and presence or absence of a final consonant),²⁵ and position of stress by syllable. Overall statistics on the token frequency of stress by word length and syllable position are shown in Table 5.3; underlying those frequencies are a total of 545 distinct prosodic patterns. Tables of token frequencies for two-, three-, four- and five-syllable words are provided in the appendix at the end of this chapter. The total number of tokens present in the corpus used was 164766; of those, 129459 tokens have exactly one *udātta* (the data shown in 5.3), 34959 tokens have no *udātta*, while a mere 348 tokens contain multiple *udāttas*. I further excluded 1335 tokens containing an independent *svarita* that cannot be restored to a form with an *udātta*.²⁶

Two general patterns in the Vedic distribution of the *udātta* merit comment. First, stress on final syllables is persistently *underrepresented*, and this underrepresentation becomes stronger as word length increases. Stress is thus statistically significantly more likely to fall elsewhere, rather than on the initial syllable, under the assumption that stress would be equally distributed among all syllable positions; see Table 5.4. Already in three- and four-syllable words, that fact that stress tends to remain on an inflectional stem throughout the paradigm, and that most inflectional suffixes add a syllable, means that final stress will not

²⁵Note that, because of regular phonological processes, consonant clusters in word-final position in Sanskrit almost never occur. The small handful remaining in the data used were transformed to [V(:)C].

²⁶Since the total amount of data containing such independent *svaritas* is small, it does not affect the overall picture substantially. Depending upon how the evidence of the ŚB is interpreted (see 5.1.2 above), these forms could potentially be understood as having prominence either on the syllable with the independent *svarita*, or on the preceding syllable.

# σ / Stress Position	1	2	3	4	5	6	7	No <i>udātta</i>	Total
1 σ	20629							9706	30335
2 σ	27168	25103						10961	63232
3 σ	16566	14440	8133					10276	49415
4 σ	4732	3736	3863	1804				2338	16473
5 σ	905	837	617	462	121			322	3264
6 σ	70	72	74	48	32	9		13	318
7 σ	12	8	1	8	1	—	—	2	32
8 σ	3	—	—	—	—	—	1	1	5
Total	70085 (49456) ^a	44196	12688	2322	154	9	1	33624	163079 ^b

Table 5.3: Token Frequencies of Position of Stress (*udātta*) by Syllable in the *Ṛgveda*

^aThe value in parentheses is the total without monosyllabic forms.

^bThis total figure deviates from the total count of tokens (164766) in the RV by 1687 tokens. Most of these tokens contain either an independent *svarita* (1335 tokens) or contain more than one *udātta*. The remaining 4 missing tokens must contain some peculiar sequences accidentally excluded in searches, or represent mere arithmetical errors. In any event, these tokens altogether constitute only about 1% of the total tokens in the RV.

	2-syllable	3-syllable	4-syllable	5-syllable
Expected % (proportional distribution)	0.5	0.3	0.25	0.2
Empirical % Final Stress	48%	20.7%	12.8%	4.1%
<i>p</i> -value (1-sample proportions test)	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
χ^2 -value	81.5	2775	1128.3	463.11

Table 5.4: Statistical Underrepresentation of Final Stress in the *Ṛgveda* (by Token)

occur often; among five-syllable syllables, since most of these are compounds, stress is determined either by a lexical stress of a compound member or particular compound stress rules (see 33).

Second, stress exhibits a slight tendency towards overrepresentation on heavy syllables, especially when the final syllable bears stress. This tendency is true for stress in all positions in two- and three-syllable words, and holds in five-syllable words except when stress falls on the initial syllable; in four-syllable words, stress co-occurs with heavy syllables in initial and final syllables, though not when stress is on the penult or peninitial syllable. See the tables and statistical tests in Appendix B. From these two distributional tendencies inherent in the stress data from the *Ṛgveda*, one can perhaps see two aspects of the Classical Sanskrit stress grammar: avoidance of stress on final syllables and preferential assignment of primary stress to heavy syllables. In Vedic, a categorical restrictions are hardly present, but the overlap may not be identical.

Based on the simplistic simulations under 3.3.2 in Chapter 3, one might hypothesize that a metrical stress grammar already capable of correctly generating most (i.e., above 50%) of tokens correctly might have a natural advantage in the long run. The data in Table 5.3 shows clearly that no simple quantity-insensitive stress grammar would have such an advantage

in performance. Excluding wholly uninformative one-syllable tokens, an initial-stress grammar performs best, and is statistically significantly better than its closest competitor, antepenultimate stress.²⁷

- (53) Proportion correctly predicted *udātta* under quantity-insensitive stress systems (Token Frequencies)

	Initial	Peninitial	Antepenult	Penult	Final
% Tokens Correct	45.4%	40.4%	44.2%	42.3%	32.3%

Likewise, at least at the level of token frequency, some conceivable grammars sensitive to syllable weight and/or incorporating extrametricality do not successfully generate a true majority of Vedic stresses either. Here, the best results are obtained by either a default-to-same side left-edge grammar with final syllable extrametricality, or the Classical Sanskrit grammar. These two grammars make largely similar predictions, diverging only when multiple non-final heavy syllables are present, as well as in many five-syllable words. The proportion of correct predictions of some such grammars is indicated in (54).

- (54) Proportion correctly predicted *udātta* under quantity-sensitive stress systems (Token Frequencies)

	DTS Left	DTS Right	Left iamb	Right trochee	C. Skt Rule
% Tokens Correct	44.4%	37.3%	42.1%	39.6%	42.6%

The data reported up to this point is based on the *token* frequencies of prosodic patterns found in the RV, and many of the simulations in 5.3 below will take this token-frequency data as a starting point. Perhaps more relevant for the acquisition of a stress grammar could be the relative *type* frequencies of patterns, where the data from relatively longer forms (> 3 syllables) is less likely to be drowned out by the sea of less informative data from shorter forms. Whether token frequency, type frequency, or some other derivative statistic is better employed as a basis for the acquisition of a metrical stress pattern is unknown; I know of no research that concretely bears on this point, and the relationship between token and type frequency in the determination of productivity of linguistic patterns and their exceptions remains a difficult question (see an overview in Sandell 2015a: Ch. 1–2, and specific proposals for the quantitative assessment of productivity in Baayen 1989, Albright 2002a, and Yang 2016). To at least enable a comparison of potentially differing productions, Table 5.5 reports the type frequencies in the syllabic position of the *udātta*, where a type is counted as any distinct, fully inflected word form in the *padapāṭha* text of the RV. This data consists of a total of 31593 distinct types, including 25508 types with a single *udātta*, 76 types with multiple *udāttas*, 5559 types with no *udātta*, and 450 types with an independent *svarita*.

²⁷The contingency table below compares an initial stress grammar and an antepenultimate stress grammar: $\chi^2 = 32.318, p < 0.001$.

	Correct	Incorrect
Initial	49456	59370
Antepenultimate	48136	60690

# σ / Stress Position	1	2	3	4	5	6	7	No <i>udātta</i>	Total
1 σ	205							95	300
2 σ	2600	2770						1219	6589
3 σ	4467	3643	2520					3088	13718
4 σ	2315	1808	1978	979				1025	8105
5 σ	545	517	434	356	96			121	2069
6 σ	58	57	61	40	25	9		8	258
7 σ	10	4	1	5	1	–	–	2	23
8 σ	3	–	–	–	–	–	1	1	5
Total	10203 (9998) ^a	8798	4995	1379	123	9	1	5559	31067 ^b

Table 5.5: Type Frequencies of Position of Stress (*udātta*) by Syllable in the *R̥gveda*

^aThe value in parentheses is the total without monosyllabic forms.

^bTo obtain the total of 31593 distinct types, add 450 (independent *svarita* count) and 76 (multiple *udātta* count).

	2-syllable	3-syllable	4-syllable	5-syllable
Expected % (proportional distribution)	0.5	0.3	0.25	0.2
Empirical % Final Stress	51.6%	23.7%	13.8%	5.2%
<i>p</i> -value (1-sample proportions test)	<i>p</i> = 0.01055	<i>p</i> < 0.001	<i>p</i> < 0.001	<i>p</i> < 0.001
χ^2 -value	5.32	442.88	470.73	417.5

Table 5.6: Statistical Underrepresentation of Final Stress in the *R̥gveda* (by Type)

In this type-frequency data, one feature immediately stands out: the underrepresentation of final stress found for the token-frequency data is much less severe, and in fact reversed for disyllables, where types with final stress somewhat outnumber types with initial stress. Table 5.6 reports the relative proportion of final stress by type in two-to-five-syllable words and the extent of any deviation from an equal distribution. Examining the relationship between position of *udātta* and syllable weight by type in the RV yields the frequencies shown in (55). χ^2 -tests indicate that final stress is statistically significantly overrepresented when the initial syllable is light (and conversely, overrepresented on the initial syllable when heavy) as well as when the final syllable contains a long vowel (see the contingency tables in (56). From these distributions overall, it is tempting to see a forerunner of the particular pattern of extraprosodicity at work in the Classical Sanskrit pattern: no stress on final syllables in words with more than three syllables, but final stress is permitted in disyllables if the initial is light and the final contains a long vowel.

(55) Distribution of Stress Position and Syllable Weight in Disyllabic Words in the *R̥gveda* (Type Frequencies)

Shape / Stress Position	1	2	SUM
L.ṽ	68	58	7114
L.ṽ	128	234	3939
L.ṽC	368	338	9823
L.ṽC	113	200	1617
H.ṽ	243	79	4107
H.ṽ	454	536	5853
H.ṽC	794	813	15375
H.ṽC	432	512	4443
SUM	27168	25103	52271

(56) Distribution of Stress Position in Disyllables by Weight of Initial Syllable and Length of Final Vowel (Type Frequencies)

a. Stress Position by Weight of Initial Syllable

	Initial H	Initial L	Total
Initial Stress	1923	677	2600
Final Stress	1940	830	2770
Total	3863	1507	5370

$$\chi^2 = 10.043, p = 0.0015$$

b. Stress Position by Vowel Length in Final Syllable

	Final V:(C)	Final V(C)	Total
Initial Stress	1127	1473	2600
Final Stress	1482	1288	2770
Total	2602	2761	5370

$$\chi^2 = 54.968 p < 0.001$$

Considering type counts rather than token counts also produces different relative performances of quantity-sensitive and quantity-insensitive metrical stress grammars. Overall, any given fixed-stress pattern correctly accounts for a smaller number of types, since the relatively good performance of virtually any grammar on disyllables, combined with the high token frequency of disyllabic forms, does not drive correct rates of prediction up. The general tendencies are otherwise similar to the predictions made on the basis of token frequency data, and the relative rankings of each stress grammar would appear to be the same: fixed-initial stress performs best, followed by a default-to-opposite left-oriented system, then antepenultimate stress, then the Classical Sanskrit pattern. This data further supports the notion of initial stress as the default pattern in Vedic, though only patterns that frequently predict final stress appear to perform substantially worse.

(57) Proportion correctly predicted *udātta* under quantity-insensitive stress systems (Type

Frequencies)

	Initial	Peninitial	Antepenult	Penult	Final
% Tokens Correct	39.1%	34.5%	36.7%	33.7%	24.9%

- (58) Proportion correctly predicted *udātta* under quantity-sensitive stress systems (Type Frequencies)

	DTS Left	DTS Right	Left iamb	Right trochee	C. Skt Rule
% Tokens Correct	37.1%	32.1%	35.1%	31.9%	35.8%

5.2.5 Is the Classical Sanskrit Stress Grammar Learnable?

A first crucial point to establish is the learnability of the stress grammar posited above for Classical Sanskrit.²⁸ Evidently, a ranking of the constraints assumed in Section 5.2.3 can indeed generate the expected patterns. However, because the simulation designs here include slightly different and involve somewhat different constraints, and not every logically possible parse has been included in the candidate set, testing the learnability of the Classical Sanskrit pattern from the perspective of categorical outputs is necessary. These tests ensure that the agent-based simulations carried out below theoretically could converge on a constraint weighting capable of generating the Classical Sanskrit primary stress pattern.

First, the same tableaux created for the agent-based learning simulations described immediately below were simply marked with one parse corresponding to the Classical Sanskrit data as the winning candidate, and submitted to the MRCD ranking algorithm. Some parses for five-syllable words were necessarily inferred, since unambiguous data was not always available.²⁹ The ranking found, reported in (59) substantially agrees with the ranking developed by hand above, and shows that a categorical ranking of the Classical Sanskrit pattern does exist for all 2–5 syllable words.

- (59) MRCD ranking of constraints used in agent-based simulations capable of generating the Classical Sanskrit pattern
 FTBIN, TROCHEE, RH-CONTOUR $\gg$ NONFINALITY- $\mathcal{F}$ $\gg$ PARSE-C $\gg$ WSP $\gg$ NONFINALITY- σ , ALIGN- ω -L, *CLASH $\gg$ PARSE- σ , ALIGN- ω -R, RIGHTMOST $\gg$ NONINITIALITY- $\mathcal{F}$, IAMB, ALL-FT-L, ALL-FT-R, LEFTMOST

Furthermore, the Classical Sanskrit pattern is itself of some theoretical interest, for two reasons: first, its data seem to constitute a genuine case of a four-syllable window, since stress can fall on the final syllable, at least in a limited number of cases (contrast the Old Latin pattern discussed in 2.2.7.1, which also allows stress on the preantepenult, but never on the final syllable); second, the existence of four-syllable windows is disputed, and their

²⁸Files for simulation in both Praat and R discussed in this section are found at <https://github.com/rpsandell/Sandell-LMU-Habilitation> under Chapter 5.

²⁹I assume, in accord with the analysis above, that five-syllable words maintained the four-syllable stress window, placing stress on the rightmost non-final heavy syllable within that window, or else on a light antepenult. Thus, the rightmost non-final heavy syllable determines the position of primary stress, except in words of the shape $/\sigma L L L \sigma/$, where the parses $[(,H) L ('L L) \sigma]$ and $[(,L L) ('L L) \sigma]$ win.

claimed non-existence sometimes used to make claims about the validity of certain metrical constraint families or their interactions (cf. Pater 2009: 1017–21, Kager 2012); compare also the skepticism expressed in Vaux 2021.³⁰ Therefore, establishing whether the Classical Sanskrit pattern can be successfully learned from data without variation via Robust Interpretive Parsing after exposure to a finite amount of data is a non-trivial question.

To this end, the tableaux formatted for Robust Interpretive Parsing were fed overt forms consisting solely of the same winning candidates submitted to MRCD above, with all overt forms being equally probable (so that no crucial data from longer words was accidentally or maliciously withheld). This simulation was conducted in Praat (Boersma and Weenink 1992–2023) using, comparing standard Robust Interpretive Parsing, Resampling Robust Interpretive Parsing, and Expected Interpretive Parsing (Jarosz 2013), using Stochastic OT and Noisy Harmonic Grammar, with default settings. Prior to any learning, with all ranking values set to 100 (i.e., markedness initially high), only In final evaluation, only c. 27% of production-oriented parses generate an overt form that matches the data. After learning on a total of 400000 tokens, the proportion of correct parses increased to between ~90% (in the worst case) and ~96%, depending upon simulation, grammar model, and sampling procedure. Some representative results are shown in 60.³¹ Exact ranking values learned frequently differed from the ranking found by MRCD, with winners selected by hand, resulting in somewhat different foot structures, but usually an identical position of primary stress.

(60) Proportion of correct parses after learning from categorical overt forms

	RIP	RRIP	EIP
Stochastic OT	93.2%	91.47%	91.64%
Noisy HG	95.7%	95.8%	95.8%

In the best performing simulations under Noisy Harmonic Grammar, 114/118 input types generate the expected Classical Sanskrit primary stress with greatest probability. The one major false prediction is found in four-syllable words of the shape /L L L σ/, for which antepenultimate stress, [L ('L L) σ], rather than preantepenultimate stress, [('L L) L σ], is predicted. Under the same ranking values learned, if at least one non-final heavy syllable is present, stress is allowed to move to the preantepenult, thus [L ('H) L L σ], but not the initial syllable in five-syllable words (*[('H) L L L σ]. The four-syllable stress window appears to be learnable, or at least learnable under optimal conditions.

A further test of the learnability of the Classical Sanskrit stress pattern in an online setting was conducted using a MaxEnt learner under Stochastic Gradient Ascent with Robust

³⁰Romanian might also have a genuine four-syllable window (Roca 1999: 676), and some further languages may behave similarly in permitting preantepenultimate stress, like Old Latin (see 2.2.7.1) or even between the final and preantepenultimate syllables, as in Palestinian Arabic (Hayes 1995: 125–30).

³¹The proportion of correct parses reported here is without any evaluation noise. Interestingly, the use of evaluation noise tended to diminish the proportion of correct parses. Interestingly, at least under Stochastic OT, the RRIP and EIP algorithms appear to perform slightly worse than standard RIP, and not appreciably better than standard RIP under Noisy HG. This behavior, unexpected given the results in Jarosz 2013, may point to some errors in the tableaux used.

Interpretive Parsing coded by the author (see immediately below for details of the simulation design). The same overt 2–5 syllable forms containing only primary stresses matching the grammar as analyzed above were used as training data, with each type of prosodic shape having an equal probability of being sampled as a learning token. 40 simulations were conducted in which initial constraint weights were set to 5 and the learning rate was set to 0.1, with different amounts of training data between 2000 and 80000 tokens being used. The fit to the predictions at the end of the simulation was assessed with reference to the Sum Squared Error (SSE) between the probabilities of overt forms that would be generated by the categorical grammar in (59) and the probabilities predicted by the learner at the end of the simulation. The SSE for 15 other categorical stress grammars (e.g., fixed initial stress, default-to-opposite left, right-edge trochees) was used as a comparison. In the worst case, when trained on only 2000 tokens, the Classical Sanskrit pattern had an SSE of 17.54, and only one other categorical pattern considered, the Classical Latin pattern, had a better SSE (14.9). In all simulations trained on 4000 or more tokens, the Classical Sanskrit grammar obtained an SSE of less than 10, and was the best-performing among all categorical stress grammars considered; a quantity sensitive right-edge default-to-opposite pattern with final syllable extrametrical was always the second-best performing pattern. When trained on 20000 or more tokens, the Classical Sanskrit patterns always obtained virtual categoricity, with an SSE less than 1. The weights of all constraints at the end of the best-performing simulation (72000 tokens) are shown in (61). While not identical with the strictly-ranked constraint grammar above, the broad outlines are similar: WSP, FTBIN, TROCHEE, NONFINALITY- $\mathcal{F}$, and PARSE-C receive high weights, as does ALIGN- ω -L; RIGHTMOST, *CLASH, and NONFINALITY- σ have lower, but still substantial weights, while ALL-FT-L/R are low-ranked, and other constraints play almost no role. The only major difference concerns the relatively low weight learned for RH-CONTOUR, which is undominated in the strictly-ranked grammar.

- (61) Constraint weights learned after training on 72000 tokens of categorical Classical Sanskrit data (MaxEnt under SGA, $\eta = 0.1$)

Constraint	WEIGHT
PARSE-C	14.30
PARSE- σ	0.00
FTBIN	11.90
NONFINALITY- σ	4.70
NONFINALITY-f	13.70
NONINITIALITY-f	0.00
WSP	15.20
TROCHEE	10.30
IAMB	0.10
ALL-Ft-L	2.80
ALL-Ft-R	0.20
LEFTMOST	2.30
RIGHTMOST	7.70
ALIGN- ω -L	11.70
ALIGN- ω -R	0.10
RH-CONTOUR	1.30
*CLASH	6.20

Overall, these results suggest that the Classical Sanskrit stress grammar constitutes a boundary case, perhaps on the edge of properly learnable stress systems: the constraint interactions capable of producing the pattern are sufficiently intricate, and the surface distribution of primary stress sufficiently ambiguous, that good, even near-perfect performance in learning via RIP can be obtained, but certainly not as quickly or straightforwardly as would be the case for relatively simpler stress grammars.³² One mitigating condition may be the fact that the overt forms in the training data, and all candidates, lack any indication of possible secondary stress, which would help to disambiguate between parses, and thereby more clearly signal the target grammar. The further important result here is that the test of the MaxEnt SGA RIP learner has been validated as able to learn the Classical Sanskrit pattern, at least when provided with unambiguous data.

5.3 Modeling the Emergence of a Metrical Stress Grammar

Considering the prosodic grammar of Vedic Sanskrit on the one hand, and the prosodic grammar of an ancestor of Gujarātī, on the other, which probably possessed the “Classical Sanskrit stress rule”, at least, one may speak of a “total replacement change” *at the level of phonological grammar and lexicon*: the lexicon is free of prespecified lexical stress, and requires a ranking of prosodic markedness constraints completely different than the Vedic

³²In particular, the behavior of stress in disyllables, dependent on vowel length and consonant extrametricality, arrives to near-categoricity only comparatively late, and for /L V:(C)/ forms, even in the best performing simulation, the alternative of initial stress still receives a probability above 1%.

ranking.³³ Presumably, all Vedic speakers who produced the Ṛgvedic corpus possessed a grammar and lexicon similar to the one worked out in Chapter 5; conversely all speakers of the Gujarātī ancestor possessed a stress grammar like the one given in 50. This total replacement of one phonological grammar with another, and of lexical entries regularly marked for properties relevant to stress assignment with lexical entries devoid of such marking, is however only partly visible on the surface. Some – indeed many – lexemes and their inflectional forms will accidentally bear prominence in the same position (cf. Lahiri’s (2015) discussion of ambiguity in stress grammars).

The overall problem treated here is substantially simplified by excluding the acquisition of a complete lexicon and the learning of specific lexical properties that would accompany the acquisition of a lexical accent system (see 3.2.2.3). This choice is grounded in reasons discussed in Chapters 3: the acquisition of a predictable metrical stress grammar without lexical exceptions implies that, at the stage of phonotactic learning, learners have already largely concluded that special faithfulness to underlying prosodic properties is superfluous. Stated more explicitly, the learners do not acquire a ranking under which some faithfulness constraint such as IDENT-[stress], MAX-PROM, or ANCHOR-R/L_F dominates many or all prosodic markedness constraints. Most interesting to observe, then, is the acquisition of phonotactic grammars using Robust Interpretive Parsing where learning agents do not simply explain the data by positing, e.g., a left foot edge in every lexical representation and assigning ANCHOR-R/L_F an untouchably high weight. Instead, the interesting question is whether learners can escape from the trap of purely lexicalized stress, and if so, what shape(s) some innovative metrical stress grammar(s) might assume.

The second part of this question can be addressed by simply denying learning agents access to prosodic faithfulness constraints whatsoever. Without access to a constraint like MAX-PROM, but faced with stress data similar to the Vedic patterns, a learner can acquire a grammar capable that generates variation, when equipped with a suitable grammar model (e.g., Stochastic OT or MaxEnt): and input /L H V/ could be realized as [‘L H V], [L ‘H V] or [L H ‘V], all with some non-zero probability. However, raw Vedic data on the position of prominence is shot through with so much variation, that even a grammar that considers a large number of possible parameters (i.e., prosodic markedness constraints) may not be able to perfectly match the frequencies of the data. Given the usual behavior of iterated learning models (see 3.3.7), I expect that learning over multiple training generations will ultimately produce either near categorical outcomes, or some pattern of stable variation.

The first part of the question is fundamentally more difficult. As discussed under 3.3.2, as long as learners have the ability to make use of lexical contrasts in stress, enforced by the activity of some faithfulness constraint(s), the default prediction is the successful maintenance of lexical features. That is, if lexically contrastive stress is pervasive in the overt learning data, one might predict for a constraint like MAX-PROM to obtain a high weight already at the phase of phonotactic learning. Consequently, if learners posit prosodic information

³³The sense of “total replacement change” here is perhaps in a somewhat different than the definition proposed by Zimmermann 2017: 2. Zimmerman defines three features for “common replacement changes”: binarity (two variants compete with one another), completeness (one variant eventually completely ousts the other), and measurability (a record quantitatively attesting to the historical course of development).

in the UR of every lexeme, such that, all lexemes of the shape /H L L H V₁/, even if words of this shape are relatively rare and stress on final light syllables is relatively rare, will surface as [H L L H V₁], faithfully maintaining the lexical stress of the learning data, due to the activity of MAX-PROM. Once MAX-PROM or its practical equivalent is found during phonotactic learning to constitute an important component of the language's phonology, the capacity to generate lexically contrastive stress is baked in.

The trap that the easy acquisition of lexically contrastive stress poses is not easy to overcome. However, standard assumptions in the learnability literature concerning the relative ranking of markedness and faithfulness constraints offers some grounds for hope: a low ranking of faithfulness, but high ranking of markedness, in the initial state, may mean that MAX-PROM does not always obtain a weight sufficiently high to overcome the effects of markedness. Given the raw Vedic data, one might imagine, for instance, that stress on final syllables might become rarer over time: because stress is relatively underrepresented on final syllables, the weights of NONFINALITY will not often be demoted, and might be able to (partly) overcome the effect of MAX-PROM at the conclusion of learning.

The simulations conducted below are intended to test how a stress system might develop over time, and any set of assumptions can enable an escape from the trap of lexical faithfulness. The larger objective is to test the extent to which a stress grammar may develop purely as a result of phonological reanalysis. The only factor permitted to change will be the probability of stress on a given syllable in a word of a given prosodic shape. Fundamentally, the goal is to provide computational data on the question of whether phonological change can occur independently of change in the structure of the learning data itself.

These simulations also serve to test the interactions between some frequently invoked prosodic markedness constraints, in the intergenerational transmission of systems assumed to be quantity sensitive.³⁴ The constraints employed (see below) partially overlap with the set used by Staubs (2014: Ch. 2), but also differ in some crucial respects, specifically, by including constraints such as WSP and RH-CONTOUR that assess violations that refer to syllable weight. Thus, potential learning biases in which weight distinctions play a role (e.g., a preference for weight and stress to align; cf. 3.3.5) can be concretely assessed as well.

5.3.1 Learning Architecture and Simulation Design

The general learning architecture to be adopted for the simulations described below is Maximum Entropy Grammar with weights learned by Stochastic Gradient Ascent, in an iterated learning model.³⁵ See the relevant sections of Chapter 3 and Staubs 2014: Ch. 1 on this model. Since the goal is to acquire a stress grammar from overt forms, where prosodic constituency is hidden from the learner, Robust Interpretive Parsing is adopted (see 3.2.2.1). Whenever constraint weights were to be updated, the Perceptron update rule (see step 5. in 35) was

³⁴The simulations here then, simplify the learning problem in one crucial respect: the learner need not discover that distinctions in syllable quantity play a role in the language's prosody.

³⁵Code and data are available in the GitHub repository at <https://github.com/rpsandell/Sandell-LMU-Habilitation> under Chapter 5.

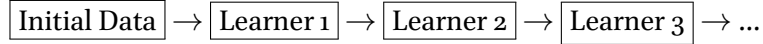


Figure 5.1: Basic Design of Single-Agent Iterated Learning

applied, sometimes with differences in learning rate for markedness and faithfulness constraints. Simple iterated agent models of single teacher-learner pairs are used, as schematically illustrated in Figure 5.1: beyond the first generation, the learning agent of the previous generation serves as the teacher for the subsequent generation.

Thus, in the first generation of learning, overt forms are sampled from the empirical distribution of overt forms in the Ṛgvedic data (e.g., [H₁ L V:C]). In each subsequent generation, the sampling procedure is slightly more complex: first, a prosodic shape without stress is sampled from the empirical distribution of the RV (e.g., [H L V:C]), so these probabilities remain constant across generations (i.e., no sort of “lexical change” that would alter this aspect of the data is permitted); then, an overt form with stress is generated based on the probabilities of candidates for that prosodic shape in the preceding generation (e.g., [H₁ L V:C]). In this way, the likelihood of being exposed to words of particular prosodic shapes, ignoring stress, is assumed to remain constant across generations, but the likelihood of encountering particular patterns of primary stress on those words may vary from generation to generation, depending upon the constraint weights acquired in the preceding generation.

Tableaux for all actually occurring two- to five-syllable prosodic shapes in the RV composed of the primitives L (non-final light syllable), H (non-final heavy syllable), V (final short vowel), V: (final long vowel), C (final consonant), and 1 (marking primary stress after any of L, H, V or V:) were crafted partly by hand and partly by scripting in R. Since not all logically possible combinations (especially for five-syllable words) are actually attested in the Ṛgveda, the total number of inputs (118) considered here is smaller than the number combinations of the primitives that could be generated. Each tableaux contains a UR, e.g., /L L H L VC/, and candidates consisting of an overt form identical to the UR but with primary stress, e.g., [L L H₁ L VC] mapping to a parse with foot structure, e.g., [(L L) (H₁) L VC]. In order to ensure matching of Vedic tokens to candidates, the parses contain only an indication of primary stress, even where multiple feet are present. In the case of parses like [(L L) (H₁) L VC], two sets of violations are possible: one with a violation of TROCHEE and CLASH (i.e., [(L L₂) (H₁) L VC]), the other with a violation of IAMB (i.e., [(L₂ L) (H₁) L VC]).

For two- and three-syllable words, all logically possible foot parses, including final consonant extrametricality when stress was assigned to the final or penult, were included. For four- and five-syllable words, many possible parses involving degenerate feet consisting of a single light syllable, or parses with final stress, were not generated. Parses with a single foot consisting of two heavy non-final syllables, i.e., (H H), were excluded altogether. (62) shows the 21 parses considered for the input /H H H H VC/.

(62) Parses of input /H H H H VC/ included in simulations

Overt Form	Parse
[H ₁ H H H VC]	(H ₁) H H H VC
[H H ₁ H H VC]	H (H ₁) H H VC
[H H H ₁ H VC]	H H (H ₁) H VC
[H H H H ₁ VC]	H H H (H ₁) VC
[H H H H V ₁ C]	H H H H (V ₁ C)
[H H H H V ₁ C]	H H H H (V ₁)C
[H H H H ₁ VC]	H H H (H ₁ V)C
[H H H H V ₁ C]	H H H (H V ₁)C
[H H H H ₁ VC]	H H (H) (H ₁) VC
[H ₁ H H H VC]	(H ₁) (H) H H VC
[H H ₁ H H VC]	(H) (H ₁) H H VC
[H H ₁ H H VC]	H (H ₁) (H) H VC
[H H H ₁ H VC]	H (H) (H ₁) H VC
[H H H ₁ H VC]	H H (H ₁) (H) VC
[H H ₁ H H VC]	(H) (H ₁) (H) H VC
[H H H ₁ H VC]	H (H) (H ₁) (H) VC
[H H H H ₁ VC]	(H) H H (H ₁) VC
[H H H ₁ H VC]	(H) (H) (H ₁) (H) VC
[H H H H ₁ VC]	(H) (H) (H) (H ₁) VC
[H H ₁ H H VC]	(H) (H ₁) (H) (H) VC
[H ₁ H H H VC]	(H ₁) (H) (H) (H) VC

Constraints employed are enumerated in (63); see Section 2.3 for definitions. Violation profiles were, again, mainly input by hand for two- and three-syllable forms, and mainly generated by scripting for four- and five-syllable forms. With the exception of MAX-PROMINENCE (not included in all simulations; see further below), which evaluates faithfulness to lexical stress in an input, all constraints used are markedness constraints. One difference in the constraint set employed for these simulations with respect to the constraint set discussed in 2.3 and used in the analysis of the Classical Sanskrit grammar above concerns LEFT-/RIGHTMOST. While I have usually treated this constraint as a categorical constraint that evaluates the position of primary stress over multiple feet, the implementation here counts violations in the position of primary stress with respect to the beginning or end of the word. Using this different interpretation was necessary because many of the parses used in the simulations contain only a single foot, and hence the effect of attraction towards an edge would otherwise be left exclusively to ALL-FT-L/R.

(63) Constraints employed in agent-based learning simulations

- a. PARSE-C and PARSE- σ
- b. FOOTBINARITY
- c. NONFINALITY- σ , NONFINALITY- $\mathcal{F}$, and NONINITIALITY- $\mathcal{F}$
- d. WEIGHT-TO-STRESS PRINCIPLE

- e. RH-CONTOUR
- f. TROCHEE and IAMB
- g. ALL-FT-L/R
- h. LEFT-/RIGHMOST
- i. ALIGN- ω -L/R
- j. *CLASH
- k. MAX-PROMINENCE (only in simulations including lexical stress)

The first set of simulations described under 5.3.2 ignore lexical stress altogether. That is, the model is not equipped with any prosodic faithfulness constraints like MAX-PROM that would enable faithful mappings like $/L\ L_1\ L/ \rightarrow [L\ (L_1\ L)]$ to satisfy the faithfulness constraints. The constraint set with which the learning agents are equipped consists exclusively of prosodic markedness constraints. Three variants of this first simulation are tested: 1) markedness low; 2) markedness high; 3) specific markedness biases. Under the first variant, all constraints have an initial weight of 0; consequently, candidates that have more total violations than other candidates with fewer total violations are nevertheless equally probable in production in the initial state. Under the second variant, all constraints have an initial weight of 10, with the result that candidates with more total violations will be less harmonic and thus much less probable in production in the initial state. Under the third variant, all constraints are given a non-zero initial weight of at least 5, but some are initially weighted higher than others, with a weight of 10; this last condition reflects possible specific biases in prosodic acquisition, discussed under 3.3.5.

The second set of simulations does include the faithfulness constraint MAX-PROM, and a much larger number of inputs as well: one tableau is included for each possible position of stress in each prosodic shape. Thus, the single tableau for the input $/L\ H\ L\ L\ VC/$ in the simulations without lexical stress multiplies into five tableaux, for the inputs $/L_1\ H\ L\ L\ VC/$, $/L\ H_1\ L\ L\ VC/$, $/L\ H\ L_1\ L\ VC/$, $/L\ H\ L\ L_1\ VC/$, and $/L\ H\ L\ L\ V_1C/$. The total number of tableaux then increases from 118 to 502, where all candidates are the same as in the first set of simulations, with the difference that pairs of overt form and parse that do not match the posited underlying lexical stress will be assigned a violation of MAX-PROM, in addition any relevant markedness constraints. That is, given an overt form $[L\ H_1\ L\ L\ VC]$, the learner will find the input $/L\ H_1\ L\ L\ VC/$; production-oriented parses are permitted to diverge in the position of lexical stress, e.g., to generate $[L\ H\ (L_1\ L)\ VC]$ from the input $/L\ H_1\ L\ L\ VC/$, but will incur a violation of MAX-PROM.

As intuitively would be expected, when exposed to overt forms that do not potentially conform to a single coherent metrical stress pattern, the weight of MAX-PROM will inexorably rise, so that, given sufficient data, MAX-PROM alone will be able to largely determine the output for any given input with prespecified lexical stress. This behavior constitutes the trap of lexical stress, related to the “core conundrum” of phonological change discussed under Section 3.3: if a grammar with high faithfulness must be posited during the phase of phonotactic learning in order to adequately explain the data, how can change (via reanalysis of the data alone) come about whatsoever? The role of faithfulness must therefore be managed such a

fashion that both change and stable transmission are possible.

First, clarifying the reason behind the inexorable rise in the weight of MAX-PROM is necessary. Recall that, in using Stochastic Gradient Ascent, weights are updated by increasing the weight of constraints violated by the learner, but not by the teacher, and decreasing the weight of constraints violated by the teacher, but not by the learner. In simulations involving lexical stress, the teacher's candidate will never violate MAX-PROM (since only parses in which the position of stress is identical with the overt form constitute possible winners), but the learner's candidate always will (otherwise, teacher and learner would both have parses corresponding to overt forms with stress in the same position, and no learning would need to occur). Thus, MAX-PROM is guaranteed to increase by some amount on each instance of update in weights (i.e., each time the overt form generated by the learner's parse would differ from the overt form generated by the teacher's parse), alongside any changes in weight to markedness constraints. Intuition suggests that, if the learning rate for MAX-PROM is too high, lexical stress will always be stably transmitted; if it is too low, the effect will be equivalent to the constraint's absence; if it is moderate, the intergenerational transmission of lexical stress may be similar, but not identical, so that lexical stress might gradually disappear in the long run. The lattermost choice seems tempting at first glance, but might make the unfortunate prediction that lexically contrastive stress should always gradually die out, seemingly contrary to the testimony of languages such as Greek or Russian, which have maintained grammars with lexically contrastive prominence – if in altered guises – for thousands of years.

A further option could be to tie the effect of MAX-PROM to the token frequency of lexical items, somewhat along the lines conceived for the constraint USE-LISTED in Zuraw 2000. Conceptually, the cost for violation of MAX-PROM in case of greater lexical support for a given lexical stress should be greater, and provide correspondingly less support for a higher base weight of MAX-PROM. Imagine that a given lexical item, lex_i has a known token frequency known to the learner of 100 and the prosodic pattern /L H1 V/. In calculating the probability of candidates, candidates that violate MAX-PROM, such as [(L1 H) V] could be assessed on MAX-PROM as $\text{Violations}(\text{MAX-PROM}) = 1 \times \log(\text{freq}(lex_i))$, so that $\text{Violations}(\text{MAX-PROM}) = 4.61$ in this case. In updating the weight of MAX-PROM, the violation of MAX-PROM (which will always be equal to 1) could be divided by $\log(\text{freq}(lex_i))$, so that, if the base learning rate for MAX-PROM is 0.01, a violation of MAX-PROM by a learner will result in an increase of only ~ 0.0022 , rather than 0.01.³⁶ In practice, when applying a logarithmic transformation of violations to MAX-PROM based on token frequency, lexical items observed only once or twice will weaken the effect of MAX-PROM ($\log(1) = 0$, $\log(2) = 0.69314$) in determining the learner's winner, but strengthen it when the token frequency of a lexeme exceeds 2.³⁷ Such a practice

³⁶The motivation for applying a log transformation to token frequency lies with the general psychological evidence that humans and other animals are sensitive to frequency by magnitudes of difference roughly similar to the logarithmic transformation of the raw frequency: the difference between 1 and 10 tokens is perceived as much greater than the difference between 91 and 100 tokens. Humans certainly have an intuitive sense of a frequency difference in the first case, but not in the latter.

³⁷To avoid division by 0, when the learning rate is divided by the log of a lexeme's token frequency, and the lexeme's token frequency is 1 or 2, the maximum learning rate will be used, i.e., $\eta/\log(1)$ and $\eta/\log(2)$ are transformed to 0.1.

may mean that the weight of MAX-PROM will tend to remain lower overall, and the weights of markedness constraints will be shaped more strongly by lower-frequency lexical items: because unfaithful parses will tend to have lower probabilities, as the token frequencies of the lexemes in the lexicon grow, learners will more often choose a parse with stress matching the stress of the overt form from the teacher.

In the final two sets of simulations, I test the behavior of both of these possibilities. In lexical stress simulation one (5.3.3.1), the learner is given only abstract prosodic shapes, and associates them with abstract lexical entries conforming to that shape and the position of primary stress. The crucial decision is the learning rate employed for MAX-PROM relative to the prosodic markedness constraints. In lexical stress simulation two (5.3.3.1), the learner must maintain a lexicon with the token frequencies of specific (fully inflected) lexical items, in order to calculate the temporary violation profiles of MAX-PROM its learning rate. Again, in order to maintain a stable lexicon – rather than a shrinking lexicon, where less frequent items would be less likely to be transmitted – the original list of word forms is maintained as a background lexicon of sorts. In the first generation of learning, agents are exposed directly to overt forms sampled from this lexicon, in order to construct their own lexica and grammars. In each subsequent generation, for the purposes of creating overt forms, the agents of $G - 1$ (where G is the current generation), whose constraint weights are fixed, are exposed to forms sampled from the background lexicon and asked to generate a production-oriented parse for it. Consider a genuinely occurring Vedic form, *śvaśruvām* ‘brother-in-law:GEN.PL’, which occurs in the RV only once; if sampled from the background lexicon during learning for $G = 2$, the frequency list for the agent will not necessarily receive the overt parse [H L V:1C], but could instead receive the overt parse [H1 L V:C], if the teaching agent would generate such an output for the UR //H L V:1C//. Since teaching agents might generate different overt forms for the same lexeme, the learning agents keep track of different token frequencies for different possible overt forms produced for the same lexeme.

5.3.2 Simulations without Lexical Stress

5.3.2.1 Simulation Set 1: No Biases, Markedness Low

In this first set of simulations, the weights of all prosodic markedness constraints mentioned above are initially set to 0 at the beginning of each generation of learning. Consequently, every candidate for each UR is initially equiprobable (e.g., all four parses considered for the input /L1 L/ will have a probability of 0.25 at the beginning of a learning generation). In most individual simulations, a single-agent generation was trained on 10000 tokens; learning rates (η) of 0.01, and 0.1 were tested, and differences in overall behavior are reported below.³⁸ In general, higher learning rates increase the likelihood that near-categorical grammars emerge in individual generations, but also increase the likelihood that the grammar will move away from a potentially stable, categorical state. I hypothesize here that the particular instability of grammars both intra- and intergenerationally is due to the fact that candidates that would

³⁸The choice of number of tokens for training per generation was motivated substantially by processing time. Ideally, more efficient code would allow for testing of larger amounts of learning data per generation.

be harmonically bounded given non-zero weights have some significant probability of being selected as teachers' or learners' parses at the beginning of each generation.

To obtain an initial sense of how the overall simulation design (i.e., choice of constraints, candidates, quantity of data, learning rate) would interact with the overt data from the RV, a first set of five single-generation simulations was carried out. The weights of all 17 constraints for each single-generation simulation are reported in 64, along with an the mean weight across all five simulations. By and large, patterns of weights induced tend to exhibit a sort of balance between constraints that favor stress closer to the left edge and constraints that favor stress closer to the right edge (compare, for instance, the fact that LEFTMOST often has a higher weight than RIGHTMOST, but ALL-FT-R and ALIGN- ω -R tend to outrank their left edge counterparts). Since the overt data from the RV contain a slight bias towards stress closer to the left edge, many single-generation simulations learn grammars that slightly favor higher weights for constraints that would place stress close to the left edge, but not always. As hypothesized above, the simulation design and grammar model does not permit the learner to fully match the empirical frequencies found in the RVic data; 65 shows the predicted frequencies of initial and final stress in disyllabic words of different prosodic shapes in single-generation simulations 1 and 4 (see the constraint weights in 64).

- (64) Constraint weights learned in five single-generation simulations (initial weight = 0, learning rate = 0.01, learning tokens = 20000)

Constraint	Weight-1	Weight-2	Weight-3	Weight-4	Weight-5	Mean
PARSE-C	0.17	0.54	0.90	0.13	0.37	0.422
PARSE- σ	0.08	0.36	0.00	0.39	0.00	0.166
FTBIN	0.27	0.56	0.77	0.69	0.11	0.48
NONFIN- σ	0.14	0.07	0.00	0.07	0.02	0.06
NONFIN-f	0.25	0.07	0.27	0.13	0.32	0.208
NONINIT-f	0.9	0.4	0.6	0.4	0.2	0.5
WSP	0.07	0.08	0.18	0.37	0.31	0.202
TROCHEE	1.77	1.30	1.27	1.14	1.23	1.342
IAMB	0.78	0.41	0.93	0.50	1.08	0.74
ALL-FT-L	0.44	0.18	0.10	0.40	0.29	0.282
ALL-FT-R	0.25	0.21	0.18	0.59	0.11	0.268
LEFTMOST	0.60	0.55	0.48	0.54	0.32	0.498
RIGHTMOST	0.04	0.06	0.12	0.14	0.07	0.086
ALIGN- ω -L	0.17	0.40	0.36	0.64	0.06	0.326
ALIGN- ω -R	0.29	0.74	0.65	0.58	0.49	0.55
RH-CONTOUR	0.71	0.66	0.35	0.04	0.30	0.412
CLASH	0.18	0.03	0.06	0.01	0.04	0.064

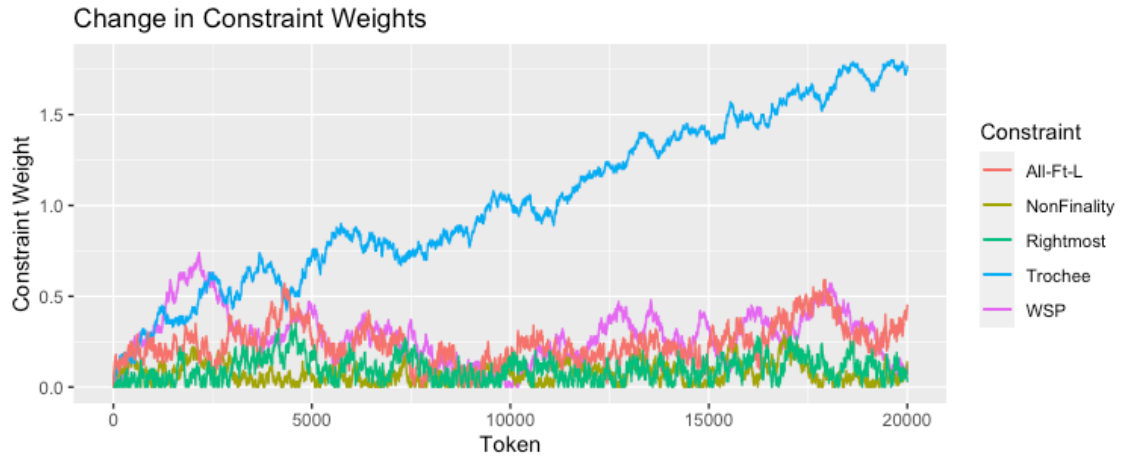


Figure 5.2: Change in Constraint Weights in Generation 1 of Simulation 1.1 (20000 tokens, $\eta = 0.01$)

- (65) Token frequencies of disyllabic words in the Ṛgveda and predicted tokens frequencies in two single-generation simulations

	Ṛgveda		Weight-1		Weight-4	
	Initial	Final	Initial	Final	Initial	Final
/L V/	4609	2505	5494	1720	4416	2698
/L V:/	1788	2151	911	3028	1464	2475
/L VC/	4654	5169	2398	7425	3951	5872
/L V:C/	445	1172	350	1267	516	1101
/H V/	2746	1361	3037	1070	3186	921
/H V:/	2926	2927	2349	3504	2927	2926
/H VC/	8137	7238	4799	10576	6508	8867
/H V:C/	1863	2580	1547	2896	1127	3216

Within single simulations, the fact that the data do not point to any coherent metrical pattern (or pattern of variation) is evident that the evolution of weight in the course of the single generation is unstable. For most constraints, weights tend to move up and down in a zig-zag fashion, rather than linearly increasing or remaining stable after some number of instances of learning. Figure 5.2 shows, as an example, the change in weights of WSP, NONFINALITY- σ , TROCHEE, ALL-Ft-L, and RIGHTMOST during a single generation; with the exception of TROCHEE, the weights jump about wildly in response to the disparate Vedic stress data.

As a means of testing the fit of a grammar learned to a concrete possible stress grammar, I calculate the predicted probabilities of each possible overt form given each input, then take the calculate the sum squared error (SSE) between the current predictions and the prediction of a given possible stress grammar. I consider 14 distinct possible stress grammars, which hardly exhaust the realm of possibility, but represent a reasonable cross-section

of options. The stress grammars compared are five quantity-insensitive systems (initial, peninitial, final, penultimate, and antepenultimate), six unbounded weight-sensitive systems (default-to-same-side and default-to-opposite left and right, plus two additional right-oriented systems with final syllable extrametricality) and three other quantity-sensitive patterns (the Classical Sanskrit pattern, the well-known Classical Latin pattern, and left-edge-oriented iambs). Intuitively, the lower the probability assigned to parses that would produce surface forms that would not occur under a given grammar, the lower the SSE, which is indicative overall fit. For instance, if a set of weights produces for the UR //H L V:// the possible overt forms [H₁ L V:], [H L₁ V:], and [H L V₁], through whatever parses, with probabilities of 0.8, 0.05, and 0.1, respectively, the SSE for a quantity-insensitive initial-stress grammar would be 0.0525, since the probabilities of the overt forms under that grammar would be 1, 0, and 0, respectively.

The three crucial questions going forward, which apply equally to the second and third sets of simulations in this section are the following:

1. Do single-agent learners converge on a metrical stress grammar with (near) categorical outputs in a finite number of learning generations?
2. If so, how long does convergence on a categorical grammar take, and what types of grammars tend to be learned (e.g., initial stress, final stress, default-to-same side weight-sensitive, Classical Sanskrit)?
3. To what extent do any categorical grammars relate to or depend upon the initial Vedic frequency distributions?

The third question may be addressed in part already by considering the probabilities of particular outputs given the initial weights to which the constraints are set at the beginning of each learning generation. When all constraints have an initial weight of 0, the likelihood of an output with stress in a particular position depends, in part, upon the number of possible parses considered with stress in that position. Among two- or three-syllable words, outputs with final stress are generally just as probable as candidates with initial stress, or are more probable, thanks to the inclusion of more different candidates incorporating violations of final consonant extrametricality; in words of three or more syllables, outputs with medial stress are more probable, since more different parses are available; among five syllable words, final stress is least probable. Thus, if the initial probability distribution at the beginning of each generation shapes the frequency of outputs in the long run, patterns with final or medial stress might be expected to predominate. The potential effect of the available parses can be assessed by the fit of the initial probabilities to the 14 possible stress grammars mentioned above.

In 10 simulations of 50 generations, the SSE for each of the 14 stress grammars was calculated on the predictions at the end of each generation. Simulations 1.a-e. employed a learning rate of 0.01, while 1.f-j. used a faster learning rate of 0.1. Table 5.8 reports the grammar with the best fit at the end of the 50th generation and the SSE for that grammar, as well as the second-best grammar and the fit of the Classical Sanskrit grammar. In many cases,

Grammar	Initial State SSE
QI Initial	101.74
QI Peninitial	78.72
QI Final	99.24
QI Penultimate	85.20
QI Antepenultimate	83.07
DTS Left	90.47
DTS Right	92.89
DTO Left	90.76
DTO Right	92.59
DTS Right + Extrametricality	83.66
DTO Right + Extrametricality	86.04
Classical Latin	83.39
Classical Sanskrit	84.64
Left-Edge Iambs	90.40

Table 5.7: Sum Squared Error of 14 Categorical Stress Grammars before Learning (all weights = 0)

the best-fit grammar in the 50th generation is not the lowest SSE achieved for any grammar in any generation, and may indeed be worse than the best SSE in the first generation. For example, in simulation 1.b., a QI initial grammar achieved an SSE of 49.71 in generation 1, and 40.41 in generation 22, while the lowest SSE achieved by any grammar in any generation was in that simulation was 39.67 (left iambs, generation 35). The best-performing grammar during the course of the entire simulation is reported in Table 5.9.

As Table 5.9 shows, near-categorical outputs were occasionally obtained at some point during the simulation. For instance, in simulation 1.f, a QI Initial grammar obtained an SSE of 0.011; indeed, in this simulation, the QI initial grammar maintained a SSE below 10 between generations 6 and 19. With the exception of simulations 1.d and 1.j, the grammar with the best SSE obtained was always identical to the best or second-best performing grammar in generation 50. The behavior of simulation 1.j. in particular calls for further comment, since only in this simulation did the overt forms expected under the Classical Sanskrit stress rule at any point during the simulation. (66) shows the constraint weights learned at the end of generation 5 in simulation 1.j, sorted by weight. While hardly identical to the strict ranking discovered for the Classical Sanskrit pattern (cf. 50), the weights given in (66) result in active effects of WSP, NONFINALITY- $\bar{f}$, and FtBIN, while preferring stress closer to the right edge than the left edge (RIGHTMOST $\gg$ LEFTMOST). During the course of simulation 1.j., the Classical Sanskrit grammar and similar quantity-sensitive, right-edge-oriented grammars that tendentially avoid final stress (DTS/O-R-FSM, Classical Latin rule) performed best between generations 2 and 30. This behavior suggests that the combination of quantity-sensitivity and a right-edge orientation, while less likely to emerge, may do so.

Simulation #	Best-Fit Grammar in Gen. 50	SSE	2nd-Best Grammar in Gen. 50	SSE	C. Skt. Rank in Gen. 50	SSE
1.a	QI Penult	45.88	DTS-R-FSM	67.61	5/14	69.25
1.b	QI Initial	64.92	Left Iambs	73.21	6/14	91.38
1.c	QI Penult	28.09	DTS-R-FSM	67.39	5/14	81.40
1.d	DTS-L	61.49	DTO-L	64.30	6/14	68.31
1.e	QI Initial	14.03	Left Iambs	60.72	7/14	144.61
1.f	QI Initial	45.16	DTS-L	55.65	6/14	164.97
1.g	DTS-L	46.69	Left Iambs	53.99	6/14	156.67
1.h	QI Initial	61.63	DTS-L	53.24	6/14	162.71
1.i	QI Penult	33.26	DTS-R-FSM	116.12	5/14	142.18
1.j	DTO-L	62.14	DTS-L	64.10	7/14	108.52

Table 5.8: Best-Fitting Stress Grammars in Simulation 1 at Generation 50 ($\eta = 0.01$ for a-e., 0.1 for f-j., 50 Generations)

Simulation #	Best-Fit Grammar	SSE	Generation #
1.a	QI Penult	41.10	43
1.b	Left Iambs	39.67	35
1.c	QI Penult	28.09	50
1.d	DTS-R	34.46	31
1.e	QI Initial	14.03	50
1.f	QI Initial	0.011	11
1.g	DTS-L	20.37	47
1.h	QI Initial	1.95	19
1.i	QI Penult	21.15	11
1.j	C. Skt.	29.91	5

Table 5.9: Best-Fitting Stress Grammars in Simulation 1 at any Generation

(66) Constraint Weights in Simulation 1.j, Generation 5

Constraint	Weight
NONFINALITY- $\mathcal{F}$	4.1
WSP	3.9
PARSE-C	3.2
FTBIN	2.4
NONINITIALITY- $\mathcal{F}$	2.1
RIGHTMOST	1.9
PARSE- σ	1.8
ALL-Ft-L	1.7
IAMB	1.7
NONFINALITY- σ	1.6
RH-CONTOUR	1.4
ALIGN- ω -L	1.3
TROCHEE	1.0
LEFTMOST	0.7
ALL-Ft-R	0.5
ALIGN- ω -R	0.3
*CLASH	0.1

Although the total sample size considered here, 10 simulations, is small, some tentative conclusions may be drawn about the effect of the frequency distributions in the seed data (the prominence frequencies in the RV) and the effect of learning rate. First, although it is possible for the learning agents to converge on near-categorical stress grammars in any given generation, the only type of grammar for which categoricity is found to emerge is the simple QI Initial type. Furthermore, because of the equiprobability of all candidates at the outset of learning in each generation, and the variation inherent at the outset, neither stable inter-generational transmission of a pattern, nor monotonic convergence towards a categorical pattern over the course of the simulation, guaranteed. At the lower learning rate, $\eta = 0.01$, the semblance of an approximately monotonic movement towards a particular stress grammar can be seen; Figure 5.3 shows the SSE for the QI Penult and DTS-R-FSM patterns over 50 generations. If the simulation continued over a larger number of generations, convergence on a categorical pattern might eventually have resulted.

Second, some clear tendencies emerge across all or many of the simulations, which are most easily attributable to the initial frequency distribution. Particularly striking is the fact that patterns that would consistently or frequently generate stress on the final syllable (QI Final, DTS-R, DTO-R) only rarely appear competitive; only in simulation 1.d. did the DTS-R pattern achieve a comparatively good SSE value in the course of the simulation. The failure to acquire patterns that would generate final stress is straightforwardly explained by the comparatively low frequency of final stress in the RVic data (cf. 5.3). Precisely because there is no penalty against final stress through NONFINALITY at the beginning of a generation, this behavior can only be explained by reference to learning from the data. Furthermore, when a clearly right-edge-oriented pattern is dominant in any given generation, it is nearly always of a type that avoids final stress: QI Penult or DTS-R-FSM. Conversely, a marked tendency for

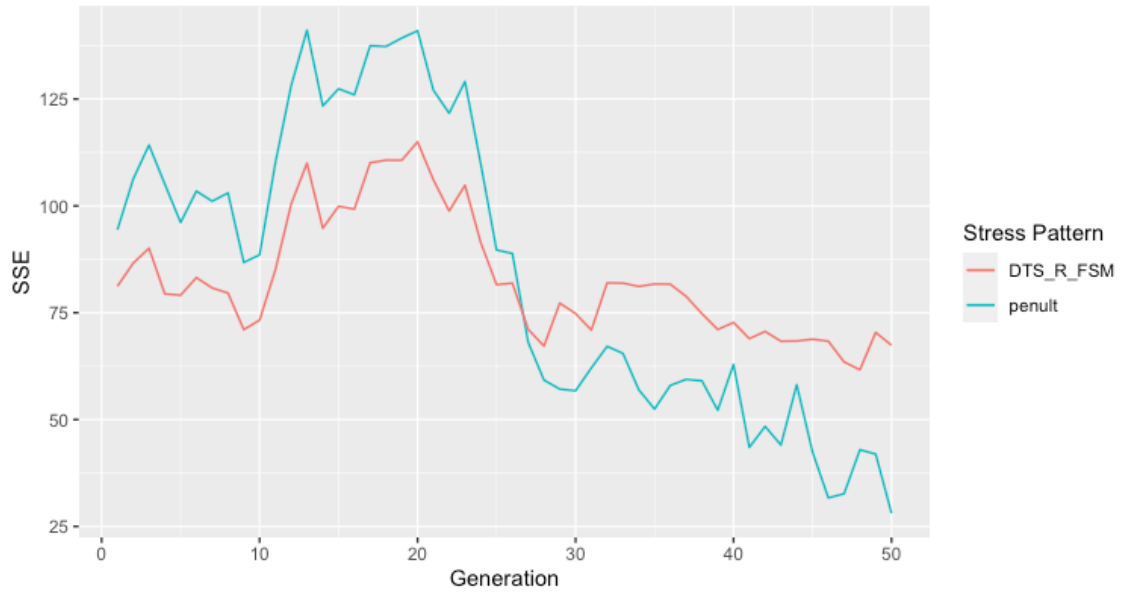


Figure 5.3: SSE in Simulation 1.c for QI Penult and DTS-R-FSM Patterns

grammars that place stress at or near the left edge (QI Initial, DTS/O-L, Left lambs) to emerge is in evidence: in 7/10 simulations, one of these four patterns is the best-performing in generation 50 and/or best-performing at any point during the simulation. Finally, the fact that the Vedic data do not exhibit a robust correlation between prominence and syllable weight means that a quantity-sensitive pattern could sometimes outperform quantity-insensitive patterns, but was statistically less likely to do so.

To summarize with respect to the three questions posed above, the simulations conducted here suggest the following answers:

1. Single-agent learners can under the conditions imposed here can converge on near-categorical outputs given initially highly variable data. However, evidence of this behavior is available only for one of the most easily learnable patterns (QI Initial stress), and only when the learning rate was relatively higher ($\eta = 0.1$).
2. On the rare occasions that near-categorical grammars emerged, they could come about after ca. 10 generations, and were invariable of the initial-stress type.
3. A clear avoidance of final stress and preference for left-edge oriented patterns suggest that the Vedic frequency distributions indeed shape the course of learning.

5.3.2.2 Simulation Set 2: Markedness High

The following set of simulations differs from the previous set only respect to the settings of the prosodic markedness constraints at the beginning of each learning generation: these are

set to a non-zero weight, 5. Consequently, parses with a higher total number of markedness violations will receive a lower probability prior to any learning. For instance, due to the combined effects of PARSE- σ and FTBIN, on the one hand, and NONFINALITY- σ and NONFINALITY- $\mathcal{F}$ on the other, only the parses $[(L_1 L) L]$ and $[L (L_1 L)]$ receive any substantial probability (each 49.8%) for the input $//L L L//$ in the initial state.

To again provide some first indications of the effects of the unaltered RVic frequency distributions on constraint weights, (67) reports the weights learned in the first generation of five simulations.³⁹ In terms of the relative rank of the constraints, the results are similar to (67) in some respects, but noticeably different in others. TROCHEE, for instance, tends to rise in weight from a 0 starting point, and remain relatively high with respect to a non-zero starting point; *CLASH is generally unaffected, whether initial weights are high or low. On the other hand, while ALIGN- ω -R showed a moderate propensity to rise from an initial weight of 0, it tends to not move from an initial non-zero weight.

(67) Constraint weights learned in five single-generation simulations (initial weight all constraints = 5, learning rate = 0.01, learning tokens = 10000)

Constraint	Weight-1	Weight-2	Weight-3	Weight-4	Weight-5	Mean
PARSE-C	5.50	4.70	4.60	4.80	4.90	4.90
PARSE- σ	0.10	0.30	0.40	0.60	0.00	0.28
FTBIN	5.20	4.90	2.10	5.30	4.90	4.48
NONFINALITY- σ	0.30	0.50	0.10	0.50	0.40	0.36
NONFINALITY- $\mathcal{F}$	3.90	4.90	5.40	5.50	4.40	4.82
NONINITIALITY- $\mathcal{F}$	1.60	0.70	0.60	0.80	1.20	0.98
WSP	0.20	0.20	0.30	0.10	0.70	0.30
TROCHEE	3.80	3.70	4.00	3.50	4.20	3.84
IAMB	2.10	2.20	1.70	1.50	2.40	1.98
ALL-Ft-L	4.00	3.10	2.90	3.70	3.10	3.36
ALL-Ft-R	1.40	1.00	1.50	1.90	1.30	1.42
LEFTMOST	1.20	1.60	1.70	1.80	1.40	1.54
RIGHTMOST	3.40	2.70	2.70	2.30	3.10	2.84
ALIGN- ω -L	1.20	1.10	0.50	0.80	1.00	0.92
ALIGN- ω -R	5.00	4.00	5.60	5.30	5.00	4.98
RH-CONTOUR	1.30	0.80	0.80	0.10	2.00	1.00
*CLASH	5.00	5.00	4.80	5.00	5.00	4.96

The fact that many candidates are more heavily penalized, or indeed harmonically bounded, in the initial state, also means the the initial fit to certain grammars is substantially better or worse as compared to models with all markedness constraints initially set to 0. In examining the initial state SSEs for each of the 14 stress patterns taken into consideration, grammars that are more likely to place stress away from word edges, especially the right edge, have lower SSEs in the initial state. Furthermore, thanks to the effect of WSP, quantity-sensitive

³⁹Since the results of the previous set of simulations indicated that a relatively higher learning rate is needed to converge on near-categorical outputs within a single generation of learning, $\eta = 0.1$ is employed throughout.

patterns fit the initial probabilities better. Two additional categorical stress stress pattern are included here and from this point forward for balance: right-edge trochees, which predict stress on final heavy syllables, else on the penultimate syllable; and a DTS Left pattern with extrametricality of the final syllable, though not in disyllables (this latter pattern is very similar similar to the C. Skt. pattern, but with left rather than right orientation).

Grammar	Initial State SSE
QI Initial	171.70
QI Peninitial	130.10
QI Final	196.45
QI Penultimate	81.92
QI Antepenultimate	128.72
DTS Left	101.96
DTS Right	166.55
DTO Left	106.02
DTO Right	162.48
DTS Right + Extrametricality	82.44
DTO Right + Extrametricality	88.77
Classical Latin	96.74
Classical Sanskrit	90.99
Left-Edge Iambs	132.22
Right-Edge Trochee	171.6194
DTS Left + 3σ Extrametricality	99.37

Table 5.10: Sum Squared Error of 16 Categorical Stress Grammars before Learning (all weights = 5)

As in the first set of simulations above, 10 simulations in total were tested, using faster and slower learning rates. The major difference with respect to Simulation 1 is that clearer effects of convergence towards a categorical grammar over the generations, and stable maintenance of a categorical pattern are often to be found. Both of these tendencies are evident in simulation 2.g., for instance, in which the SSE for a QI initial grammar is already lowest in generation 1, falls below 1 by generation 5, and remains the dominant, essentially categorical output. As in the preceding section, Table 5.12 shows the two best-performing grammars at the termination point, and Table 5.13 shows the grammar with the best claim to categoricity at any point during the simulation. That grammars that produce comparatively less token variation are being learned is apparent from the fact that alternative grammars are often substantially worse; in 2.b and 2.c, for instance, the second-best categorical grammar has a much higher SSE than the best.

Two tendencies stand out in these results, which clearly correlate with the learning rate. With a higher learning rate, $\eta = 0.1$, there is a tendency for categorical initial-stress grammars to emerge; simulations 2.g, 2.h., and more weakly, 2.i., reflect this pattern. Simulation 1.f., on the other hand, reflects a case in which a right-edge orientation and quantity-sensitivity in-

Grammar	SSE-1	SSE-2	SSE-3	SSE-4	SSE-5
QI Initial	133.73	94.26	165.19	37.55	58.92
QI Peninitial	94.03	87.71	144.00	107.80	84.28
QI Final	97.18	137.65	67.46	168.29	141.06
QI Penultimate	73.79	81.41	88.47	128.35	108.79
QI Antepenultimate	102.45	99.71	150.61	121.75	105.89
DTS Left	104.23	109.55	137.08	91.45	79.90
DTS Right	78.95	127.25	40.22	156.72	129.99
DTO Left	105.14	113.22	137.48	96.65	84.38
DTO Right	78.04	123.58	39.83	151.52	125.51
DTS Right + Extrametricality	89.90	108.61	111.90	122.57	108.66
DTO Right + Extrametricality	97.71	112.15	121.26	116.38	107.32
Classical Latin	93.95	111.53	120.65	133.55	116.92
Classical Sanskrit	93.54	114.30	118.16	126.59	111.86
Left-Edge Iambs	110.29	101.50	151.56	81.65	70.75
Right-Edge Trochee	76.06	120.67	33.30	155.75	128.91

Table 5.11: Sum Squared Error of 16 Categorical Stress Grammars after 1 Generation of Learning (all weights = 5)

stead emerged. Meanwhile, 2.j constitutes a case in which a high weight of RIGHTMOST tends to prefer primary stress on the right edge, but in words above three syllables in length, the likelihoods are far from categorical, with the result that no grammar provides a particularly good fit. This overall pattern of behavior matches the results obtained from the first set of simulations: the seed data in generation 1 most readily facilitate the learning of patterns with non-final stress, oriented towards the left edge of the word.

With a lower learning rate, 0.01, a clear tendency for convergence towards right-edge patterns with avoidance of final stress appears. Figures 5.3.2.2 and 5.3.2.2 show the evolution of SSE for right-edge-oriented patterns with NONFINALITY effects (QI Penult, DTS-R-FSM, DTO-R-FSM, C. Lat. rule, C. Skt. rule) in simulations 2.b and 2.e, where the latter assigns more weight to WSP. This behavior is perhaps most readily explained as emergence of the least marked grammars in the initial state: the five right-edge patterns with extrametricality are also the five with the lowest SSEs in the initial state of each generation (cf. Table 5.10). This behavior indicates that the learning rate $\eta = 0.01$, combined with only 10000 tokens per generation, offers the learning agents too little opportunity to learn from the data. Although this behavior is not entirely guaranteed, as simulation 2.a. demonstrates (see Figure 5.3.2.2), comparing the SSEs of the QI Initial, QI Penult, DTS-L-3 σ FSM, and DTS-R-FSM patterns in that simulation), the pattern of behavior indicates that either a higher learning rate or more learning data per generation is needed to ensure that the initial state does not unduly influence the outcomes.

To provide more representative statistics on the probability of certain best-fit grammars, again with the caveat that the best-fit grammar at the end of a simulation may not necessarily

Simulation #	Best-Fit Grammar in Gen. 50	SSE	2nd-Best Grammar in Gen. 50	SSE	C. Skt. Rank in Gen. 50	SSE
2.a	DTS-L-3sFSM	36.04	QI Initial	44.27	7/16	136.82
2.b	QI Penult	13.27	DTS-R-FSM	81.17	4/16	100.69
2.c	QI Penult	39.224	DTS-R-FSM	94.82	3/16	107.64
2.d	QI Penult	33.68	DTS-R-FSM	71.76	3/16	87.01
2.e	DTS-R-FSM	48.29311	QI Penult	49.45	3/16	57.96
2.f	DTS-R	18.87	Right Trochees	25.07	6/16	197.23
2.g	QI Initial	< 0.001	DTS-L-3 σ FSM	91.77	8/16	191.69
2.h	QI Initial	< 0.001	DTS-L-3 σ FSM	91.75	8/16	191.47
2.i	QI Initial	38.35	DTS-L-3 σ FSM	39.2992	7/16	164.56
2.j	QI Penult	79.3	DTS-R-FSM	89.5	4/16	94.53

Table 5.12: Best-Fitting Stress Grammars in Simulation 2 at Generation 50 ($\eta = 0.01$ for a-e., 0.1 for f-j., 50 Generations)

Simulation #	Best-Fit Grammar	SSE	Generation #
2.a	QI Initial	30.13	36
2.b	QI Penult	11.71	49
2.c	QI Penult	21.34	30
2.d	QI Penult	25.57	43
2.e	QI Penult	33.66	40
2.f	Right Trochees	4.80	28
2.g	QI Initial	< 0.001	24
2.h	QI Initial	< 0.001	45
2.i	QI Initial	2.29	4
2.j	DTS-L-3 σ FSM	31.25	3

Table 5.13: Best-Fitting Stress Grammars in Simulation 2 at any Generation

correspond to a categorical stress grammar, I ran 100 further simulations of 50 generations, again with 10000 tokens per generation and the higher learning rate of $\eta = 0.1$. The number of simulations in which a given grammar had the lowest SSE at generation 50 or lowest SSE at any point overall during the simulation is reported in Table 5.14. While the QI Initial and Penult patterns, especially the former, tend to emerge most robustly, other quantity sensitive patterns, in particular patterns that have substantial overlap with the Classical Sanskrit rule, such as Right Trochees and DTS-R-FSM, are not poorly represented.

To what extent does convergence on a categorical grammar really occur within 50 generations? In exactly 80% of these 100 simulations, the lowest SSE in generation 50 was lower than the lowest SSE of any of the 16 grammars prior to any learning (81.92), and a better SSE with respect to the starting point was achieved at some point in the course of every simulation (in the worst case, the lowest SSE was 67.41). In exactly half of all simulations, the best

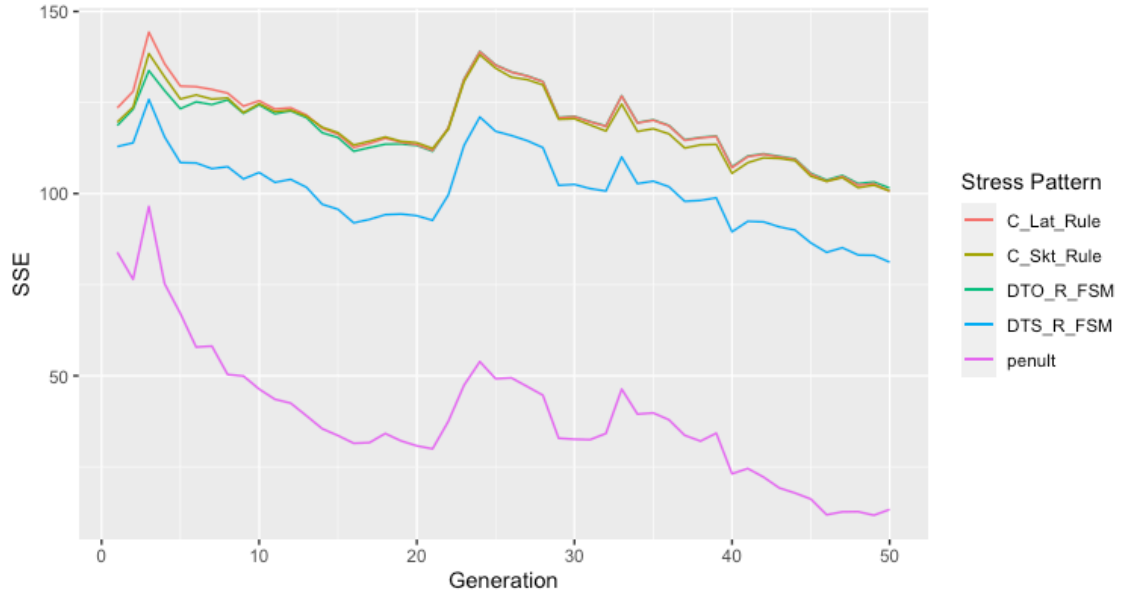


Figure 5.4: SSEs of Right-Edge-Oriented Grammars in Simulation 2.b

SSE at generation 50 was lower than 50, and in 29 simulations, it was lower than 20. Thus, in somewhat less than one-third of all simulations, a grammar with strong similarities to one of the categorical stress grammars considered had been learned at the termination of the simulation.

One of these cases included the Classical Latin stress pattern (in simulation 2.k.71, SSE at generation 50 = 8.79), which shares many similarities with the Classical Sanskrit pattern (often the second-best grammar throughout 2.k.71); the weights learned in generation 50 that yield this relatively good fit to the Classical Latin pattern are shown in (68). The relatively low weights for $\text{PARSE-}\sigma$ and $\text{ALIGN-}\omega\text{-R}$, combined with high weights on $\text{NONFINALITY-}\sigma/\text{f}$ allow for final syllable extrametricality, while a non-zero weight for WSP, and the relative ranking of TROCHEE , ALL-FT-R and RIGHTMOST above their counterparts results in substantial probability being assigned to Classical-Latin-like parses. Divergences from the Classical Latin pattern here are due to the effect of $\text{ALIGN-}\omega\text{-L}$, which, in combination with WSP, results in too much probability mass being assigned to primary stress on a heavy syllable to the left of the antepenult.

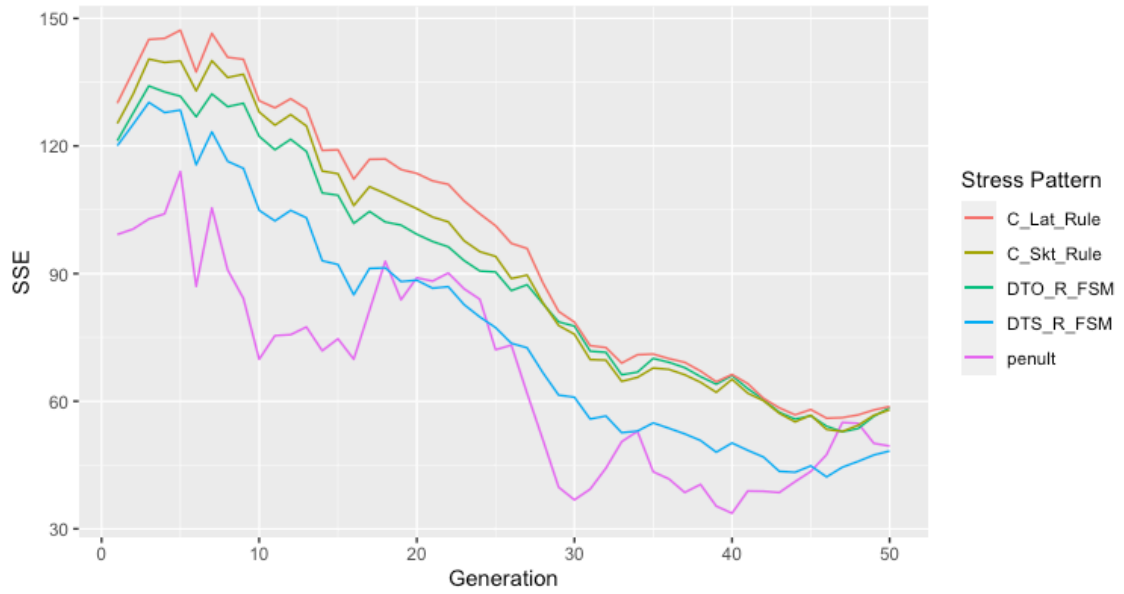


Figure 5.5: SSEs of Right-Edge-Oriented Grammars in Simulation 2.e

(68) Weights Learned in Generation 50 of Simulation 2.k.71 (best-fit grammar = Classical Latin Pattern)

Constraint	Weight
PARSE-C	8.90
PARSE- σ	1.20
FTBIN	5.60
NONFINALITY- σ	7.60
NONFINALITY-f	8.20
NONINITIALITY-f	4.80
WSP	3.50
TROCHEE	7.40
IAMB	4.20
ALL-Ft-L	2.90
ALL-Ft-R	3.90
LEFTMOST	4.10
RIGHTMOST	5.90
ALIGN- ω -L	5.20
ALIGN- ω -R	0.60
RH-CONTOUR	5.10
*CLASH	5.20

That the data received in the first generation substantially shapes the direction of learning subsequently, as might be intuitively expected, follows from the fact that, in slightly more

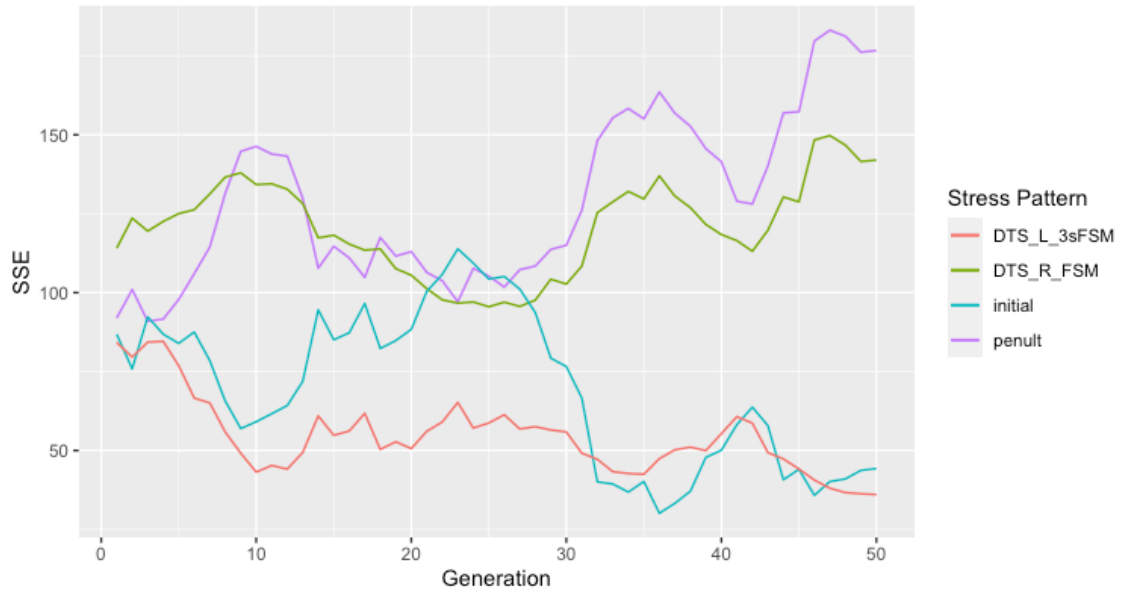


Figure 5.6: SSEs of Initial and Penultimate Grammars in Simulation 2.a

than half of all runs in simulation 2.k (52/100), the best-performing grammar in generation 1 is the same as the best-performing grammar at any point during the run. This number, 52%, almost certainly underestimates this relationship, since grammars that predict similar overt forms often trade places in the course of a simulation. Further worth noting is that the types of best-performing grammars at the end of generation 1 do bear a clear resemblance to the distribution of stress in the *R̥gveda*: 37% are closest to the QI Initial pattern, 46% are similar to a frequently non-initial but frequently non-final pattern (QI Peninitial, QI Penult, Initial Iambs, Right Trochees), 15% unbounded quantity-sensitive (DTS-L, DTS-L-3σFSM, DTO-R), and 2% QI Final.

What general conclusions concerning the diachronic transmission of stress patterns or the present simulation design can be drawn from the results above? First, the fact that the QI initial pattern is robustly supported, and indeed appears to be the only pattern that is stably acquired and transmitted intergenerationally, both confirms results in earlier scholarship and indicates a “founder’s effect” at the outset of the simulations. Conversely, the QI Final pattern, which is in principle just as readily learnable, very rarely performs well, which may be attributed to both the effects of NONFINALITY-σ in the constraint inventory and the low frequency of final stress in the seed data from the RV. That other patterns, and especially quantity-sensitive patterns, emerge as top-performing grammars less frequently may be attributed simultaneously to their greater complexity (i.e., more constraint interactions are needed to generate the pattern correctly) and their greater confusability (i.e., their overt forms are more easily misparsed in ways compatible with different grammars). This hypothesis would match the results obtained in Staubs 2014: 94–8: stress in windows larger

Grammar	n Best SSE in Generation 50	n Best SSE Overall
QI Initial	22	42
QI Penult	22	21
Right Trochees	16	12
DTS-L-3 σ FSM	15	9
DTS-R-FSM	7	1
QI Peninitial	6	1
Left Iambs	5	9
DTS-L	2	3
QI Antepenult	1	0
C. Lat Rule	1	1
C. Skt Rule	1	0
DTO-L	1	0
Final	1	0

Table 5.14: Simulation 2.k: Best-Fit Grammars over 100 Simulations (50 generations, 10000 tokens per generation, $\eta = 0.1$)

than three, and even stress in a three-syllable windows, tends to be mislearned intergenerationally, whereas one- or two-syllable windows are much more stable in transmission. Staubs' result refers, however, to quantity-insensitive patterns alone.

In terms of simulation design, a major factor perhaps to be revised is the amount of learning data provided in each generation. Precisely because patterns like DTS-L-3 σ FSM or the Classical Latin rule are comparatively complex, they likely require more data in absolute terms to be learned accurately, even when the data contain no variation to begin (compare the simulations on Old Latin stress at 3.3.1.1). At the same time, providing more data gives the learner a better chance of finding constraint weights that enable perfect frequency matching. For instance, if training on 50000 were sufficient to allow a learner to converge on constraint weights that very closely match the frequency distributions in the Vedic data, then that same frequency matching grammar stands a good chance – perhaps too good a chance – of being stably transmitted diachronically (cf. Niyogi 2009 on such non-bifurcating dynamics in the face of variation). Further simulation may be able to determine how much data would result in undesirable “overlearning”, i.e., the tendency to maintain *a priori* improbable variable stress patterns over an indefinite number of generations.

In practice, it may be necessary or desirable to implement a *categoricity bias* in the learning model that penalizes high rates of token variation in outputs. Kirby and Sondregger (2015), for instance, have shown that including a categoricity bias in the form of a simple Gaussian prior, when two categorical variants are available, facilitates stable maintenance of one of the two variants in iterated learning, and tends to push learners towards categorical usage of one or the other variant, if substantial variation should arise. In the present context of primary stress outputs, each input has a number of different possible overt forms equal to the number of syllables. Thus, a given input (e.g., /H L VC/) with n syllables has a categorical output when the probabilities of the three possible overt forms have yielded the maximum

possible value for σ^2 given n (e.g., 0.5773503 when $n = 3$). A learner then might conceivably decide to carry out an update to its constraint weights if doing so would improve the level of categoricity in the grammar as a whole, i.e., increase $\sum_x \sigma^2(x)$ where x is an input. Another very crude categoricity bias could take the form of a raw boost given to the probability of the candidate with greatest likelihood of production after learning has terminated. I leave the implementation of this or some other form of a categoricity bias in iterated-learning MaxEnt models to future investigations.

5.3.2.3 Simulation Set 3: Specific Prosodic Biases

This last set of simulations tests the effects of biases in favor of a higher weight for certain markedness constraint on the overall behavior of learning. As discussed at 3.3.5, reasons to assume the existence of substantive and/or analytical biases supporting higher weights for NONFINALITY, WSP, and TROCHEE can be mustered. In a batch learning MaxEnt model, a bias in favor of higher weights for these constraints would apply a Gaussian prior with a mean higher than the mean for other markedness constraints, and a standard deviation lower than the other markedness constraints (cf. 3.2.1.2). In an online MaxEnt model, such as the one employed here, a bias can be implemented as a higher initial weight for the biased constraints and a slower learning rate, so that these constraints are demoted in weight less quickly. Concretely, a bias in favor of NONFINALITY- σ , NONFINALITY- $\mathfrak{f}$, WSP, and TROCHEE is implemented here by setting the initial weight of these constraints to 10, while other constraints are set to 5, and these constraints are equipped with a slower learning rate, one-half the η used for all other constraints (e.g., 0.05 instead of 0.1).

As with the previous sets of simulations, I first report in the tables under Table 5.15 and (69) the SSEs for 16 categorical grammars in the initial state before learning and the weights found for the constraints after one generation. The new initial state SSEs very much reflect the biases: quantity-insensitive grammars fit much less well overall, and especially quantity-sensitive grammars having a right-edge orientation but final-syllable extrametricality in most cases (DTS Right + Extrametricality, DTO Right + Extrametricality, Classical Latin, and Classical Sanskrit) present the best fit. Given that relatively intricate stress patterns like the Classical Sanskrit rule seem virtually not to arise in the unbiased simulations, biases supporting some of the constraints that play an active role in such systems may be necessary to explain their existence whatsoever.

Grammar	Initial State SSE
QI Initial	133.01
QI Peninitial	133.12
QI Final	201.34
QI Penultimate	103.33
QI Antepenultimate	118.98
DTS Left	81.58
DTS Right	170.67
DTO Left	87.07
DTO Right	165.19
DTS Right + Extrametricality	43.76
DTO Right + Extrametricality	39.30
Classical Latin	67.00
Classical Sanskrit	49.10
Left-Edge Iambs	108.17
Right Trochees	183.78
DTS Left + 3σ Extrametricality	73.56

Table 5.15: Sum Squared Error of 16 Categorical Stress Grammars before Learning (biased constraints weight = 10, all others = 5)

- (69) Constraint weights learned in five single-generation simulations (initial weight all unbiased constraints = 5, initial weight all unbiased constraints = 10, learning rate unbiased constraints = 0.1, learning rate biased constraints = 0.05, learning tokens = 10000)

Constraint	Weight-1	Weight-2	Weight-3	Weight-4	Weight-5	Mean
PARSE-C	8.80	9.80	9.50	10.50	9.90	9.70
PARSE- σ	0.30	0.30	0.00	0.60	0.40	0.32
FTBIN	1.20	0.30	1.00	2.00	0.40	0.98
NONFINALITY- σ	0.30	0.35	0.00	0.15	0.20	0.20
NONFINALITY-f	8.20	8.30	8.00	7.85	7.95	8.06
NONINITIALITY-f	0.60	0.90	0.70	0.60	0.80	0.72
WSP	0.35	0.15	0.50	0.35	0.35	0.34
TROCHEE	8.40	10.10	9.10	8.90	8.60	9.02
IAMB	1.00	3.60	2.10	1.80	2.50	2.20
ALL-Ft-L	4.10	1.80	3.70	4.50	2.10	3.24
ALL-Ft-R	0.90	1.20	0.80	1.00	0.80	0.94
LEFTMOST	0.80	2.50	1.30	0.90	1.50	1.40
RIGHTMOST	3.70	2.60	3.20	3.50	2.30	3.06
ALIGN-WD-L	0.20	0.50	0.60	0.30	0.30	0.38
ALIGN-WD-R	9.10	8.30	8.40	7.90	8.00	8.34
RH-CONTOUR	6.00	6.40	6.80	6.20	7.10	6.50
CLASH	4.70	4.90	5.30	4.50	4.70	4.82

A look at the SSEs of the 16 categorical grammars for the first generation suggests that, given the learning rates adopted for the constraints and the amount of learning data made available, the learner is able to largely overcome the initial bias in its attempt to match the frequencies in the Vedic data: none of the individual simulations here carries over a clear preference for a right-oriented quantity-sensitive pattern, and in many cases the SSE of the best categorical grammar is worse than any of the grammars favored by the biased weighting in the initial state. This result further indicates that the emergence of the Classical Sanskrit stress pattern cannot be not an immediate, direct response to the distribution of word-level prominence frequencies as the appeared in the RV, if lexical stress were simply to have vanished. I hypothesize, however, that the biased constraints will enable the Classical Sanskrit or similar patterns to establish themselves more frequently when learning is iterated over many generations.

Table 5.17 reports, as usual, the best-performing grammar after 50 generations of transmission. Perhaps surprisingly, the biased constraints seem to have little impact on the outcomes: again, the QI Initial and Penult patterns tend to perform best, and only the former is truly learned categorically. At least with the parameters adopted for this set of simulations ($\eta = 0.1$, 10000 tokens per generation), the Vedic seed data seems to sufficiently impact the course of learning over generations so as to render the emergence of stress patterns other than QI Initial, QI Penult, or at the outside, perhaps Right Trochees, rather unlikely.

To confirm that a lower learning rate would indeed result in learners too too in thrall to the initial state at the outset of each generation, and thus in the long run tend to converge towards the grammars with the lowest SSEs in the initial state (see Table 5.15), a further set of 10 simulations with $\eta = 0.01$ was run. These results are reported in Table 5.18. As expected, the categorical grammar with lowest SSE in generation 50 is, in most cases, one of

Grammar	SSE-1	SSE-2	SSE-3	SSE-4	SSE-5
QI Initial	103.97	75.47	40.96	49.28	80.79
QI Peninitial	85.82	83.96	106.07	91.06	78.26
QI Final	106.36	128.72	149.67	151.49	121.91
QI Penult	77.64	88.30	130.88	117.31	90.70
QI Antepenult	106.23	93.59	114.85	104.95	93.40
DTS Left	104.51	78.65	55.40	70.31	76.47
DTS Right	99.05	119.94	138.49	139.83	112.11
DTO Left	106.81	81.36	59.95	75.22	78.78
DTO Right	96.75	117.23	133.94	134.93	109.80
DTS R + Extrametricality	99.39	90.17	108.59	104.87	86.93
DTO R + Extrametricality	105.74	90.93	101.08	99.22	87.63
Classical Latin	106.29	96.49	120.09	112.00	93.44
Classical Sanskrit	105.95	93.03	107.78	104.78	88.86
Left Iambs	96.97	74.20	53.74	61.69	72.20
Right Trochee	92.73	121.38	142.54	141.84	114.08
DTS Left + 3 σ Extrametricality	104.83	75.54	49.18	63.42	74.53

Table 5.16: Sum Squared Error of 16 Categorical Stress Grammars after 1 Generation of Learning (initial weights and learning rates same as for 69)

the right-oriented quantity sensitive patterns, especially DTS-R-FSM: in 7/10 of these simulations, DTS-R-FSM performs best, and, in 9/10 cases, the C. Skt. pattern stands at rank 2 or 3. Simulation 3.q. presents an interesting case in which the DTS-R-FSM reaches a minimum SSE of 22.49 in generation 27, after which a “flip” in directionality begins to occur, with the result that left-oriented patterns, in particular DTS-L-3 σ FSM, improve in performance. This sort of behavior might be explained if the DTS-R-FSM pattern has reached a floor in performance, but because the pattern has not become genuinely categorical – indeed, substantial token variation continues to be predicted for inputs with no heavy syllables – a small shift in the data sampled could result in a change in relative constraint weights controlling edge orientation. This “crossover” of right-oriented and left-oriented quantity-sensitive patterns in 3.q is shown in Figure 5.7.

Since simulations 3.k–t strongly indicate that the biased constraints very much limit the range of grammars that come into contention when the learning rate is low, I again ran 100 further simulations of 50 generations, again with 10000 tokens per generation and the higher learning rate of $\eta = 0.1$, to gain a more complete picture of how the biases in favor of WSP, TROCHEE, and NONFINALITY shape outcomes. Again, the number of simulations in which a given grammar had the lowest SSE at generation 50 or lowest SSE at any point overall during the simulation is reported in (5.19). In comparison to the set of 100 simulations without specific biases, some similarities, but also differences, in comparison to the results shown in (5.14) above. First, the QI Initial, QI Peninitial, DTS-L-3 σ FSM, and Right Trochee patterns are again among the most-frequent best-fitting grammars, but perhaps surprisingly, the QI

Simulation #	Best-Fit Grammar in Gen. 50	SSE	2nd-Best Grammar in Gen. 50	SSE	C. Skt. Rank in Gen. 50	SSE
3.a	QI Initial	< 0.001	DTS-L-3 σ FSM	91.63	8/16	191.72
3.b	QI Initial	< 0.001	DTS-L-3 σ FSM	86.32	8/16	187.87
3.c	QI Penult	50.83	Right Trochee	80.94	7/16	145.33
3.d	QI Initial	< 0.001	DTS-L-3 σ FSM	91.78	8/16	191.84
3.e	DTS-L-3 σ FSM	31.83	Left Iamb	57.73	7/16	165.06
3.f	Right Trochee	34.88	DTO-R	58.43981	7/16	187.61
3.g	Right Trochee	49.39	DTS-R	75.10	7/16	192.31
3.h	QI Penult	16.75	DTS-R-FSM	91.13	3/16	108.54
3.i	QI Penult	99.38	Right Trochee	152.56	12/16	211.18
3.j	QI Penult	87.13	Peninitial	110.46	4/16	136.88

Table 5.17: Best-Fitting Stress Grammars in Simulation 3 at Generation 50 ($\eta = 0.1$, 50 Generations, 10000 tokens per generation)

Final pattern more often performs comparatively well. This latter result must be due to the WSP bias yielding some cases in which the frequent availability of heavy final syllables results in the reanalysis of a Right Trochee pattern as a QI final pattern. Indeed, in three runs, 3.u.70, 30.u.71, and 3.u.91 the QI Final pattern is indeed effectively categorical; Categoricality, assessed here as an SSE of less than 5, obtains in 33 of the 100 runs ($2 \times$ for DTS-L-3 σ FSM, $2 \times$ for QI Penult, $3 \times$ for QI Final, and $26 \times$ for QI Initial). The frequency with which categoricity obtains in this set of simulations is higher compared to the set in (5.14), though not statistically significantly so (33 versus 27). Also notable is that, in the biased set of simulations, the range of top-performing grammars is more restricted (10 versus 13). Overall, the greater facility with which quantity-insensitive patterns are learned shines through: since they are less readily confused with other patterns, their intergenerational transmission is less error-prone and thus more stable.

One factor in the simulations that has not thus far been explicitly considered is the quantity of learning data provided to the agents. If the seed data represented a categorical metrical stress pattern, particularly a more complex one, less data would certainly increase the likelihood that the pattern be misacquired as a consequence of missing or insufficient data. Where the initial data contains significant variation, on the other hand, less data might facilitate the acquisition of categorical grammars, since the learning agents will have less data on which to calibrate a frequency-matching grammar. However, the learners will, at the same time, have less chance to overcome the bias of the initial state. 100 more runs using the biased initial state were carried out with $\eta = 0.1$, but only 2500 tokens per generation. The best-performing grammar in generation 50 and overall for each of these runs is reported in Table 5.20.

Forcing the learning agents to generalize from total data in each generation does indeed significantly affect the likelihood that a given grammar will perform best in generation 50 or at any point during the simulation run: Fisher's exact test comparing the values in the

Simulation #	Best-Fit Grammar in Gen. 50	SSE	2nd-Best Grammar in Gen. 50	SSE	C. Skt. Rank in Gen. 50	SSE
3.k	DTS-R-FSM	43.51	C. Skt.	54.37	2/16	54.37
3.l	DTS-R-FSM	39.62	C. Skt.	49.13	2/16	49.13
3.m	DTS-R-FSM	25.04	C. Skt.	31.13	2/16	31.13
3.n	DTS-R-FSM	44.06	QI Penult	50.65	3/16	54.20
3.o	QI Penult	49.82	DTS-R-FSM	53.14	3/16	64.03
3.p	DTS-R-FSM	46.61	DTO-R-FSM	47.34	3/16	53.33
3.q	DTS-L-3 σ FSM	41.74	DTS-L	49.26	6/16	74.62
3.r	DTS-R-FSM	42.22	DTO-R-FSM	50.58	3/16	52.29
3.s	QI Penult	42.98	DTS-R-FSM	43.38	3/16	55.33
3.t	DTS-R-FSM	30.49	DTO-R-FSM	33.10	3/16	36.37

Table 5.18: Best-Fitting Stress Grammars in Simulation 3 at Generation 50 ($\eta = 0.01$, 50 Generations, 10000 tokens per generation)

columns “ n Best SSE in Generation 50” and “ n Best SSE Overall” in (5.19) and (5.20) returns $p < 0.001$ in both cases. In particular, the right-edge-oriented patterns exhibit better performance when less training data is provided. This result is unsurprising, since the biased constraints have the effect of making the right-oriented patterns better-performing in the initial state; the trend towards left-oriented patterns like QI Initial and DTS-L-3 σ FSM is a response to the learning data in generation 1.

What remains the case is that grammars that exhibit small stress windows, and especially quantity-insensitive patterns, tend more often to emerge and have better chances of becoming stable. One further grounds for this behavior could be the fact that patterns such as QI Initial, QI Penult, and QS Right Trochees is that they can be distinguished from other patterns *even in disyllabic forms*. Other patterns can be distinguished from these patterns only in forms of three or more syllables, and may require yet longer forms to obtain information on certain behaviors. Hence, patterns in which stress should regularly occur on the antepenult or postpeninintial syllable are difficult to learn when there is not enough data from forms of three, and especially four or more syllables, relative to the frequency of disyllables. The last set of simulations without regard to lexical stress therefore serves to test to what extent employing a different frequency distribution for the sampling of training data might affect the course of learning.

5.3.2.4 Simulation Set 4: Specific Prosodic Biases with Type Frequencies

This set of simulations reports on results of learning using the type frequencies of lexemes in the RV with particular prosodic shapes, rather than token frequencies; compare again the data in Tables 5.3 and 5.5 for these frequencies. Note in particular that, whereas disyllables comprise nearly half (48.1%) of all 2–5-syllable tokens in the RVic data, they make up less than one quarter of all 2–5-syllable types (21.5%). Hence, if prosodic acquisition is assumed

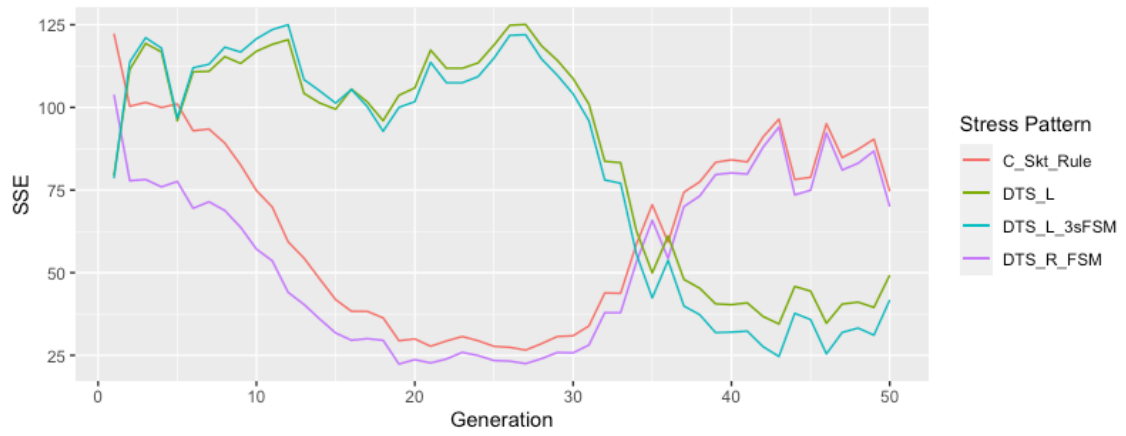


Figure 5.7: Crossover of Edge Orientation in Simulation 3.q

Grammar	n Best SSE in Generation 50	n Best SSE Overall
QI Initial	29	39
QI Penult	18	17
DTS-L- 3σ FSM	16	11
Right Trochees	15	12
QI Final	8	10
QI Peninitial	8	8
DTS-L	3	2
DTS-R	1	0
Left Iambs	1	0
C. Lat Rule	0	1
DTS-R-FSM	0	0
DTO-R-FSM	0	0
C. Skt. Rule	0	1

Table 5.19: Simulation 3.u: Best-Fit Grammars over 100 Simulations (50 generations, 10000 tokens per generation, $\eta = 0.1$ for unbiased constraints)

Grammar	<i>n</i> Best SSE in Generation 50	<i>n</i> Best SSE Overall
QI Initial	19	24
QI Penult	26	32
DTS-L-3 σ FSM	16	11
Right Trochees	23	25
QI Final	0	0
QI Peninitial	1	3
DTS-L	2	2
DTS-R	1	0
Left Iambs	0	0
C. Lat Rule	1	1
DTS-R-FSM	9	1
DTO-R-FSM	1	0
C. Skt. Rule	0	1

Table 5.20: Simulation 3.v: Best-Fit Grammars over 100 Simulations (50 generations, 2500 tokens per generation, $\eta = 0.1$ for unbiased constraints)

to depend upon the type, rather than token frequencies of patterns in the data, learners, at least in the Vedic case, should be able to gather relatively more from three- and four-syllable forms, since there are more distinct types among words of that length. The crucial difference in the set of simulations performed here is simply that each token of potential learning data is drawn from the frequency distribution of word types as reflected in Table 5.5, rather than word tokens.

To determine whether the use of the type frequency distribution for the seed data would impact the likelihood with which particular varieties of stress grammars tend to emerge, once again 100 simulations were run. Under the assumption that the use of the specific constraint biases for WSP, TROCHEE, and NONFINALITY- σ/F would increase that likelihood that patterns favoring high weights for these constraints would tend emerge and potentially stabilize, these biases were instituted, as in the preceding set of simulations. As above, Table 5.21 shows the number of runs in which a particular stress grammar had the lowest SSE in generation 50, and the number of runs in which that grammar had the lowest SSE during the simulation run.

As in the previous sets of simulations, the quantity-insensitive patterns are best represented, though the QI peninitial pattern seems perform more robustly under these conditions. If anything, quantity-sensitive patterns appear to be at a greater disadvantage given the type frequencies as seed data: the five QI patterns represent 3/4 of the best overall SSE, and nearly 2/3 of the best SSE in generation 50. This result is may be due to the fact that all fixed-stress patterns have a worse fit the raw RVic type frequency data than to the raw token frequency data, thereby leaving the more easily-acquired QI patterns more opportunity to naturally emerge.

However, in a handful of cases, a clear, relatively stable preference for a right-edge quantity-sensitive pattern has come about. Simulation 4.a.55, for instance, achieves an SSE of 3.83 for

Grammar	n Best SSE in Generation 50	n Best SSE Overall
QI Initial	24	35
QI Penult	22	18
DTS-L-3 σ FSM	6	1
Right Trochees	18	18
QI Final	2	3
QI Peninitial	17	19
DTS-L	0	0
DTS-R	1	0
Left Iambs	1	0
C. Lat Rule	0	2
DTS-R-FSM	6	4
DTO-R-FSM	1	0
C. Skt. Rule	1	0
QI Antepenult	1	0
DTS-R	1	0

Table 5.21: Simulation 4.a: Best-Fit Grammars over 100 Simulations (50 generations, 10000 tokens per generation, $\eta = 0.1$ for unbiased constraints)

the Classical Latin pattern (generation 33). Simulation 4.a.73, in which the Classical Sanskrit pattern has the best SSE in generation 50 (30.31), in fact seems to have settled into the stable transmission of pattern not otherwise considered, similar to DTS-R-FSM and the Classical Sanskrit rule: stress on the rightmost non-final heavy syllable if a non-final heavy syllable is available; stress on the final heavy syllable if no other heavy syllable is available; variation between penultimate (more likely) and antepenultimate (less likely) in forms with no heavy syllables.

These results suffice to show that the type frequencies of prosodic patterns in the RV also do not particularly strongly favor the emergence of patterns similar to that of the Classical Sanskrit stress rule. At the margins, right-oriented QS patterns may be slightly more likely to perform well; compare the four instances of DTS-R-FSM and two instances of the Classical Latin rule as the best SSE overall here, versus one instance of the Classical Latin rule and one instance of the Classical Sanskrit rule as the best SSE overall in simulation 3.u. above.

From the point of view of language history, the sum results of the four sets of iterated-learning simulations conducted here suggest that the distribution of prominence frequencies in the RV alone does not constitute an adequate basis for explaining the ultimate emergence of the Classical Sanskrit pattern. Deterministically simulating emergence of the Classical Sanskrit pattern, or simulating a reasonably good likelihood of its emergence, will evidently require recourse to other assumptions. From the point of view of computational modeling of stress grammars using constraints, the present results confirm both the general tendency for MaxEnt grammars to prefer simpler patterns (Moreton and Pater 2012) and the specific claim that quantity-insensitive fixed-stress patterns with smaller stress windows are more easily learned (Bane and Riggle 2008, Staubs 2014). Novel results have, though, been

obtained with respect to three points:

- First, simulations with MaxEnt learners employing Stochastic Gradient Descent have convincingly shown that quantity-sensitive patterns indeed pose greater learnability challenges and are more difficult to transmit stably between generations.
- Second, simpler stress patterns will tend to emerge even given initial data containing considerable variation.
- Third, the effects of the stress distributions at the outset tend to persist throughout the simulation, rendering even very simple patterns, such as QS Final, that have little support in the original data, unlikely to emerge.

5.3.3 Simulations with Lexical Stress

This latter set of two simulations adds the possibility for the acquisition of lexically listed stress during phonotactic learning. As described at the end of 5.3.1 above, 502 tableaux for RIP are included, each corresponding to a UR with primary stress marked on a particular syllable. In both of these simulations, the learning of lexical stress in the course of phonotactic learning is implemented through the inclusion of the faithfulness constraint MAX-PROMINENCE. To ensure that lexical stress be *learned*, rather than being encoded from the start, a typical MARKEDNESS $\gg$ FAITHFULNESS bias is implemented for the initial state of each generation. Since the effects of implementing specific prosodic biases in 5.3.2.3 did not clearly increase that likelihood of producing the historically attested situation, I simplify the initial state for these simulations, setting all markedness constraints to a weight of 5. Finally, besides the 16 fixed-stress patterns whose fit to the grammars learned was assessed throughout the above simulations, I also include a pure lexical stress grammar, in which the winning candidate in each tableau has primary stress on the same vocalic peak as in the input. Note that, because the total number of tableaux is greater in these simulations, the SSEs reported, even where outputs are near-categorical, will be higher overall.

Two major questions will guide the assessment of simulation results:

- Does the amount of lexical stress (i.e., probability assigned to overt forms matching the lexical stress in the URs) decrease or remain stable intergenerationally?
- Does the inclusion of lexical stress affect which fixed-stress grammars perform best relative to the grammars learned?

5.3.3.1 Simulation Set 5: η Low for MAX-PROMINENCE

In this set of simulations, the learning rate η for all markedness constraints will be set to 0.1, while the rate for MAX-PROM will be correspondingly lower, 0.01. While the “perfect lexical stress” grammar stands the best chance of being acquired with great accuracy in the first generation of learning, for it to do so, its weight must rise from 0 to a comparatively high

weight, relative to the markedness constraints. Table 5.22 shows the SSE of the categorical stress grammars and the “perfect lexical stress” grammar in the initial state, with MAX-PROM weighted at 0; in relative terms, the fit of the fixed-stress grammars should be identical to the fit in the initial state of Simulation 2, with the difference that SSEs are necessarily higher overall here. Thus, as in simulation 2, many right-edge oriented patterns with final extrametricality have the best fit in the absence of data, and the “perfect lexical stress grammar” fits the initial-state predictions comparatively poorly.

Grammar	Initial State SSE
QI Initial	765.27
QI Peninitial	576.15
QI Final	851.92
QI Penultimate	370.32
QI Antepenultimate	564.98
DTS Left	451.39
DTS Right	725.71
DTO Left	463.04
DTO Right	714.05
DTS Right + Extrametricality	369.74
DTO Right + Extrametricality	396.44
Classical Latin	436.29
Classical Sanskrit	439.13
Left-Edge Iambs	590.50
Right Trochees	756.92
DTS Left + 3σ Extrametricality	439.08
Perfect Lexical Stress	635.73

Table 5.22: Sum Squared Error of 16 Categorical Stress Grammars before Learning (biased constraints weight = 10, all others = 5)

In one single-generation simulation, the weights obtained after exposure to 10000 tokens are then as in (70). These weights might be interpreted as the weights for the stress grammar of Vedic that could be acquired through phonotactic learning, with the markedness constraints providing the default stress. There, the weight of MAX-PROM exceeds the weight of all markedness constraints with the exception of PARSE-C. As expected, the SSE of the Perfect Lexical Stress pattern has become very low, sinking to 16.06, while the SSEs of nearly all fixed-stress patterns of increased, all exceeding 700 at the end of the generation. Figure 5.8 shows, also as predicted, that the weight of MAX-PROM consistently rises throughout the simulation, since no data exists that would contradict it; for comparison, the weights of WSP, TROCHEE, ALL-FT-L and RIGHTMOST are included as well. Notice in particular that the weight of TROCHEE rises initially, as a constraint that is evidently helpful in accounting for much of the data (compare Figure 5.2 above), but later falls as the increased weight of MAX-PROM allows for more of the data to be explained by recourse that constraint alone.

- (70) Constraint weights learned in one single-generation simulation (initial weight markedness constraints = 5, initial weight MAX-PROM = 0, learning rate markedness constraints = 0.1, learning rate MAX-PROM = 0.01, learning tokens = 10000)

Constraint	Weights
PARSE-C	6.90
PARSE- σ	0.50
FTBIN	4.40
NONFINALITY- σ	0.10
NONFINALITY-f	3.40
NONINITIALITY-f	3.50
WSP	0.10
TROCHEE	4.20
IAMB	3.20
ALL-Ft-L	4.70
ALL-Ft-R	2.60
LEFTMOST	3.20
RIGHTMOST	4.50
ALIGN- ω -L	3.80
ALIGN- ω -R	4.80
RH-CONTOUR	1.60
*CLASH	5.00
MAX-PROM	5.77

Although the weight obtained by MAX-PROM after 10000 tokens is sufficient to yield outputs in which each lexically stressed UR will map to an overt form with primary stress in the same position, a residue of discrepancy and variation remains. For instance, given the input //H V₁//, the grammar with weights in (70) will nevertheless produce an overt form with stress on the initial syllable, [H₁ V], in 2.5% of cases. This residual unfaithfulness could lay the seeds for the gradual demise of lexical stress as controlled by MAX-PROM in the long run.

Iterated simulations over 50 generations of learning will now test the behavior of MAX-PROM in particular, and its role in potentially shaping the potential emergence of other fixed-stress grammars. A first simulation indicates that while some movement away from the original lexical stress likelihoods indeed occurs, this change is very slow: in generation 1 of simulation 1.a., the input //H V₁// assigned a probability of 2.5% to the unfaithful output [H₁ V]; in simulation 5.b, in generation 50, the input //H V₁// assigns the unfaithful output [H₁ V] a probability of 11.7%.⁴⁰ While some fixed-stress patterns do exhibit some marked improvement in their SSEs over the course of the simulation, such improvements are largely marginal and fleeting: Figure 5.9 shows the SSEs for the Perfect Lexical Stress, QI Penult, QI Initial, and Classical Sanskrit patterns, where all always remain far from obtaining cate-

⁴⁰The crucial difference between this simulation and the simple simulation at 3.3.2 is that the simulation here, besides operating in an online rather than batch setting, does not directly reassign unfaithful outputs as evidence for some alternative.

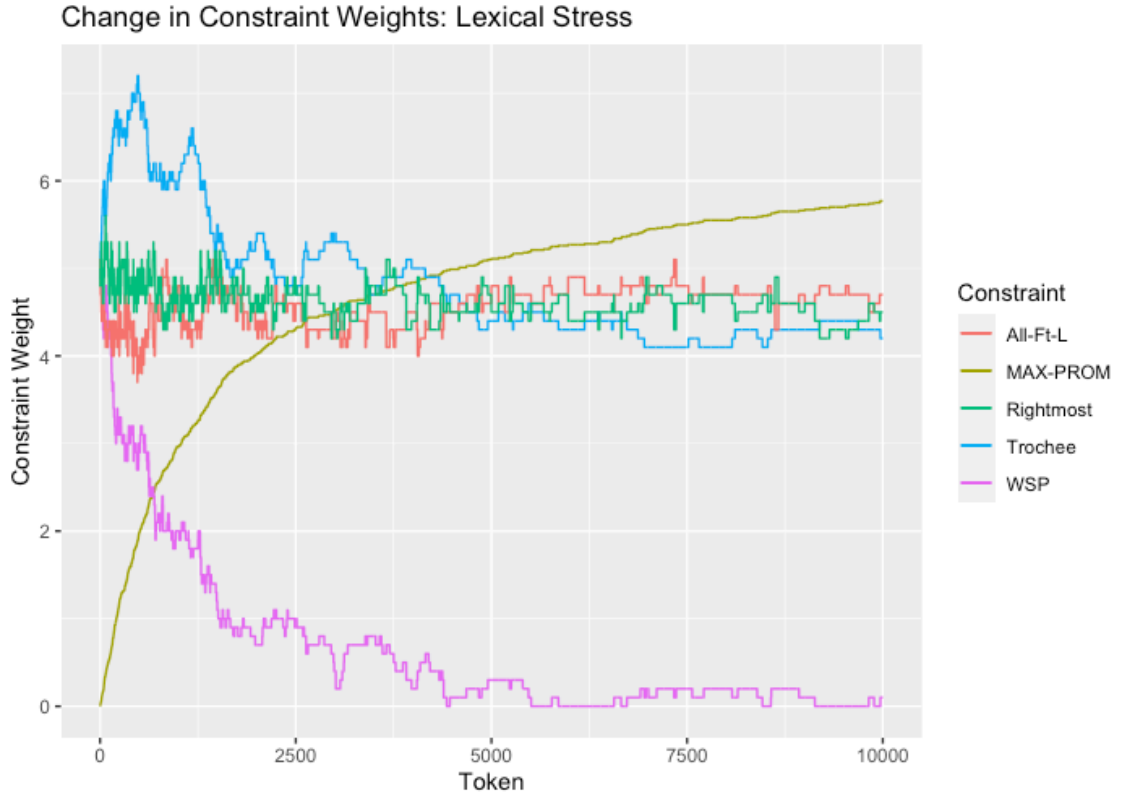


Figure 5.8: Change in Constraint Weights in Generation 1 of Simulation 5.a (10000 tokens, $\eta = 0.1$)

goricity. Evidently, the combination of learning data and learning rate allows the agents to hew too faithfully to unpredictable stress patterns for any true break to occur: the weight of MAX-PROM is indeed highest in generation 1 (5.82) but falls to a low of only 4.4 (in generation 32), ending in generation 50 at a weight of 5.12.

By contrast, when the quantity of learning data is limited to merely 5000 tokens per generation, the weight learned for MAX-PROM declines precipitously within 50 generations, and begins to approach 0 within 100 generations. Figures 5.10 and 5.11 show the increase (= worse performance) of the SSE for the Perfect Lexical stress pattern over the course of two simulations, one in which right-edge-oriented patterns tend to show improvement, the other in which left-edge-oriented patterns tend to become better performing than the pure lexical pattern. When iterated learning is allowed to continue for 200 generations, MAX-PROM indeed becomes ineffectual: in simulation 5.e., its weight first falls below 0.2 in generation 54, and never exceeds 0.38 in any subsequent generation. In the same simulation, convergence on a near-categorical QI peninitial pattern also obtains, where, for all inputs, with the exception of disyllables ending in / $\check{V}$ /, stress on the second syllable from the left is predicted above 95%. Compare the SSEs for the QI initial, QI peninitial, left iambic, right trochaic, and Classical Sanskrit patterns over the course of this simulation in 5.12; note in particular the relative stability that sets in around generation 120, after which the SSEs of these patterns

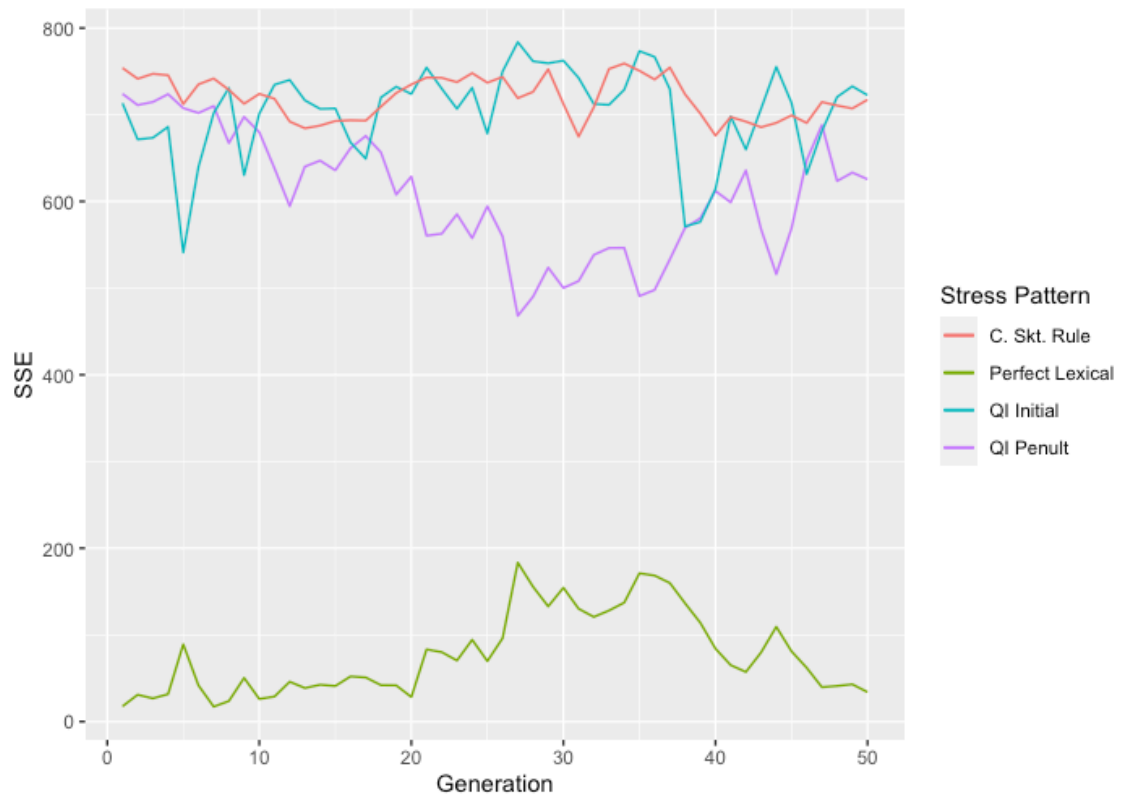


Figure 5.9: Change in SSEs in Simulation 5.b (10000 tokens per generation, 50 generations, $\eta = 0.1$)

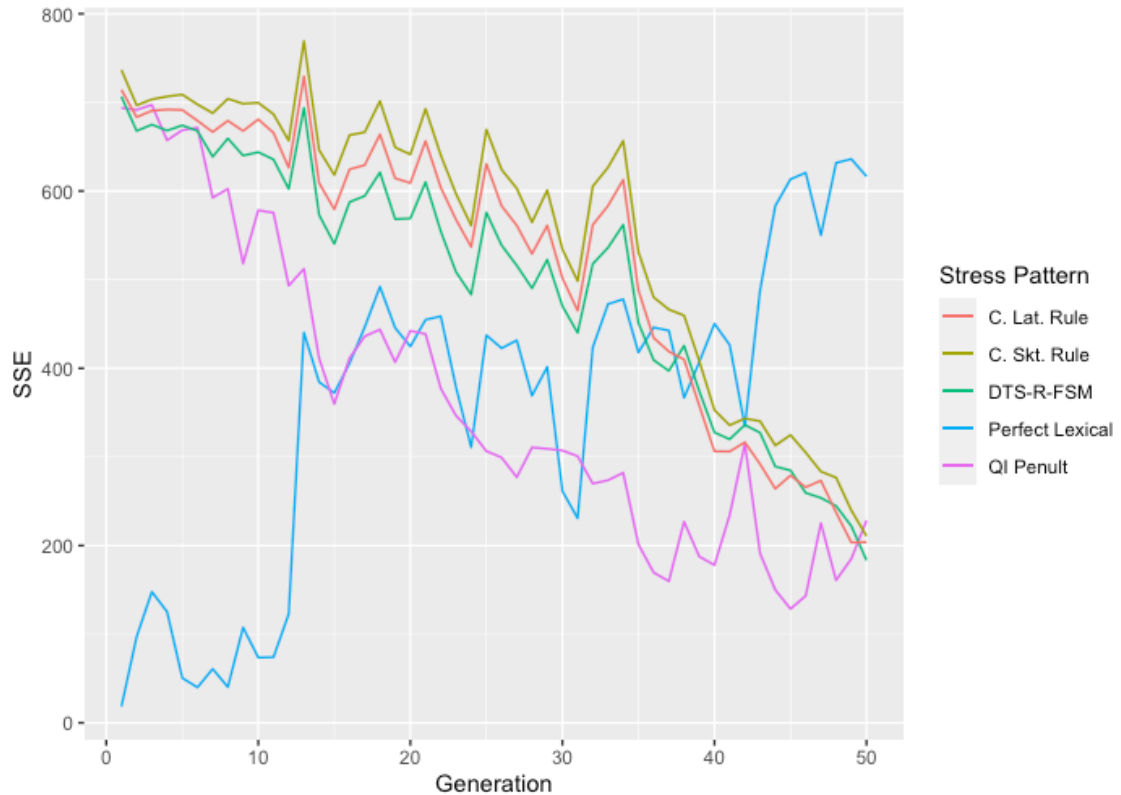


Figure 5.10: Change in SSEs in Simulation 5.c (5000 tokens per generation, 50 generations, $\eta = 0.1$)

exhibit little subsequent change from generation to generation.

These results demonstrate that, at least in principle, if relevant prosodic faithfulness constraints fail to obtain sufficiently high weights during the phase of phonotactic learning, and if what is not learned during early phonological acquisition cannot be overcome through further updates to weights and acquisition of the right underlying representations for individual morphemes later, variation will enter the data. That variation will then propagate itself over generations, and support the gradual erosion of lexically contrastive stress in favor of fixed-stress. To put it simply: lexically contrastive stress, under the present model assumptions about learning, requires more sheer data, just as relatively more complex fixed-stress patterns do.

Translated into real-world terms, the loss of lexically contrastive stress where less learning data is provided might suggest that prosodic acquisition in a linguistically impoverished environment might result in the decline of lexical stress. Exactly what social context could yield such an impoverished environment is, though, unclear, and given the facility with which lexically contrastive stress seems to be acquired (cf. 3.3.4.1), whether learning data could become sufficiently impoverished in all but the most extreme conditions is unknown.



Figure 5.11: Change in SSEs in Simulation 5.d (5000 tokens per generation, 100 generations, $\eta = 0.1$)

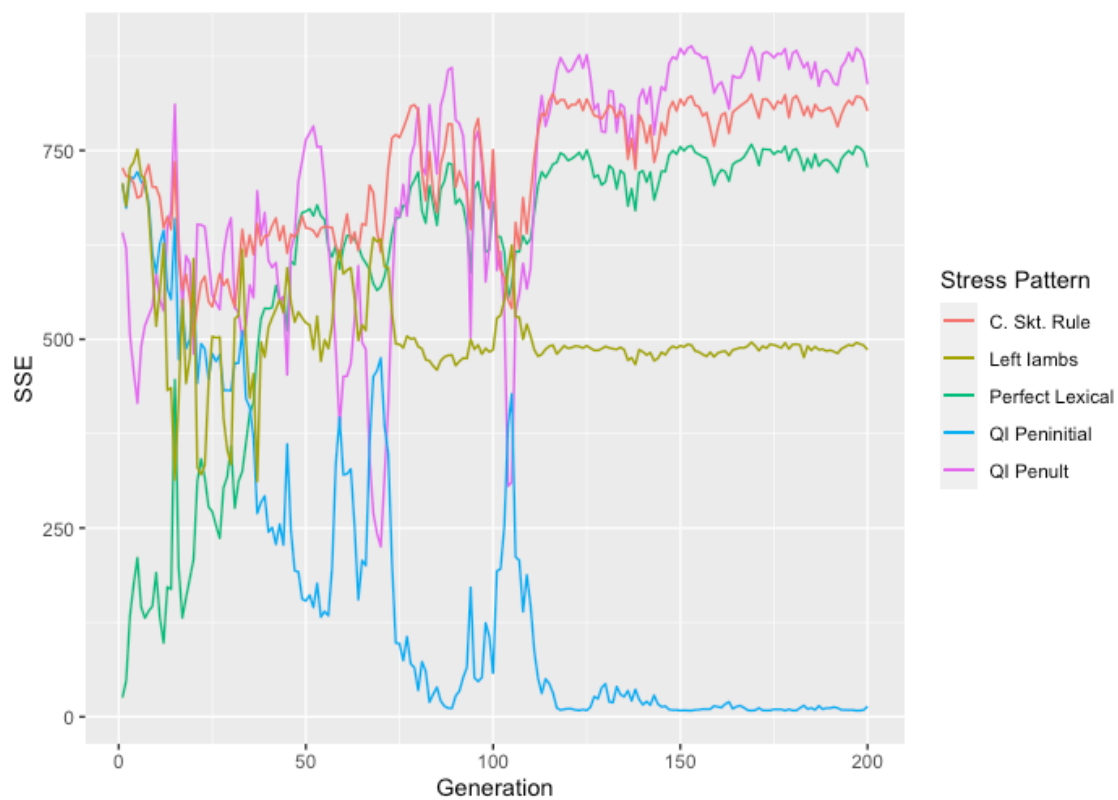


Figure 5.12: Change in SSEs in Simulation 5.e (5000 tokens per generation, 200 generations, $\eta = 0.1$)

5.3.3.2 Simulation Set 6: η for MAX-PROMINENCE Dependent on Token Frequency

Intuitively, the model employed here fundamentally should serve as a stronger break on weight increases for MAX-PROM, while simultaneously providing enough learning data per generation to allow for the potentially stable transmission of even relatively complex stress patterns. First, because of the way in which the update rule is adjusted and constraint violations are calculated based on the natural logarithm of token frequencies, a lexical item must be encountered at least twice to potentially result in an increase in the weight of MAX-PROM. Second, because, when a lexeme has a higher token frequency, violations of MAX-PROM become more costly, the likelihood that a learner's parse will deviate from a teacher's parse diminishes, thereby leading to fewer updates in constraint weights overall. Faithfulness to lexical stress is maintained in this case less by a relatively high weight learned for MAX-PROM and more by frequency of lexemes. However, when a teacher agent must produce a lexeme that it itself has encountered rarely (or in some cases, perhaps not at all), the likelihood of an error in production is correspondingly higher.

To first compare the behavior of MAX-PROM in the first generation of learning with its behavior in the preceding set a simulations using a fixed learning rate of 0.01, Figure 5.13 shows the change in weights for MAX-PROM relative to the markedness constraints WSP, TROCHEE, ALL-Ft-L and RIGHTMOST. First, note that the weight achieved by MAX-PROM after 10000 tokens is lower than in simulation 4.a: only 4.38 versus 5.77. Furthermore, in contrast to the relatively steady growth of MAX-PROM in generation 1 simulation 4.a, which quickly comes to exceed the weights of most markedness constraints (in 4.a., after around 4000 tokens), the growth of MAX-PROM in 6.a is negligible at the beginning (for the first c. 200 tokens), and occasionally punctuated by longer plateaus of little to no growth. These plateaus represent periods in which either previously unseen lexemes are first encountered, or lexemes with an already relatively high token frequency are sampled again. On hypothesis that is not borne out, however, is that the adjustment to the violation profile of candidates based on token frequency would result in fewer updates overall; instead, likely because the weight of MAX-PROM is held lower for longer, the learner's overt forms more often diverge from the learning data, resulting in more total constraint updates during the course of the simulation (typically 500–1000 updates in the simulations in 4; in 6.a, 3657 weight updates occurred). Holding the weight of MAX-PROM lower for longer thus does give the learner more opportunities to discover potential regularities in metrical stress patterns.

As in simulation set 5, it can be assumed here that permitting the weight of MAX-PROM during phonotactic learning to substantially exceed the weight of all markedness constraints will result in faithful intergenerational transmission of pure lexical stress. Since, however, the reference to token frequency results in slower growth in the weight of MAX-PROM during a simulation, relatively more data can be provided to learning agents without guaranteeing the maintenance of lexical stress.

Simulation 6.b allowed for iterated learning to continue for 50 generations. The change in SSEs for the perfect lexical stress pattern and the three generally best-fitting categorical patterns (QI Initial, QI Peninitial, DTS-L-3 σ FSM) are shown in (5.14), and (5.15) compares the development of the constraints MAX-PROM, ALL-Ft-L and WSP over the 50 generations.

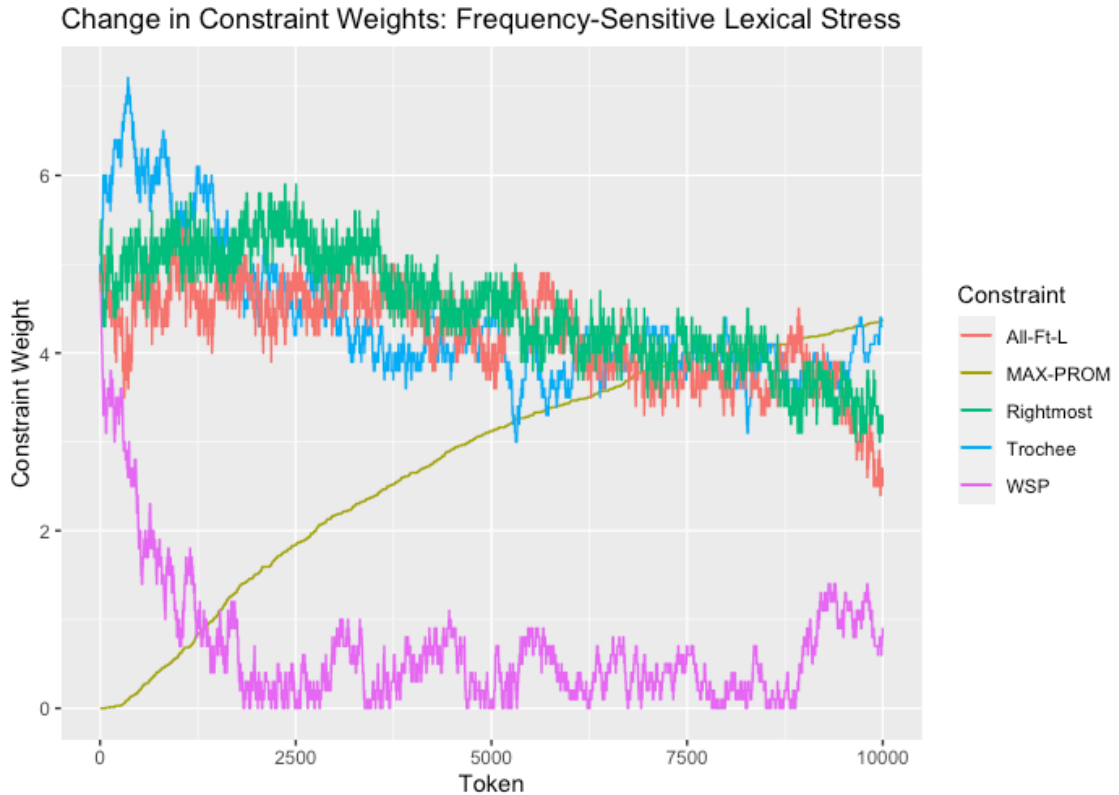


Figure 5.13: Change in Constraint Weights in Generation 1 of Simulation 6.a (10000 tokens, η for markedness constraints = 0.1, base η for MAX-PROM = 0.01)

Two interesting and notable behaviors may be observed here concerning the weight of MAX-PROM, which in turn directly impacts the performance of the perfect lexical stress pattern: 1) as in the previous set of simulations, the weight of MAX-PROM does decline over the course of many generations but 2) in contrast to simulation 5.e., which saw a relatively stable decline in the weight of MAX-PROM until reaching near 0, or simulation 5.d., in which MAX-PROM saw periods of stability punctuated by periods of decline, resulting in an overall downward trajectory, MAX-PROM in this case shows a precipitous fall from a relatively stable weight of ~ 3 to a weight below 1 in a few generations. That is, MAX-PROM's behavior in simulation 6.b roughly resembles an S-curve, which is perhaps to be expected, since its updates to its weight during a given generation depend upon the natural logarithm of a lexeme's token frequency.

While 50 generations in this case was not sufficient to allow for the emergence of a stable fixed-stress pattern, unlike in 5.e., the trend towards a left-edge-oriented pattern is discernible, and eventual convergence on a pattern like QI Initial or QI Peninitial would be unsurprising, given the results of earlier sets of simulations. The major difference in behavior that can be achieved by tying the weight of a lexical faithfulness constraint to lexical frequencies would appear to be the relatively swift and sudden decline of lexical stress intergenerationally; such behavior would accord better with what is known about the usual trajectories of language change (see relevant discussion under 3.2.1.3 and 3.3).

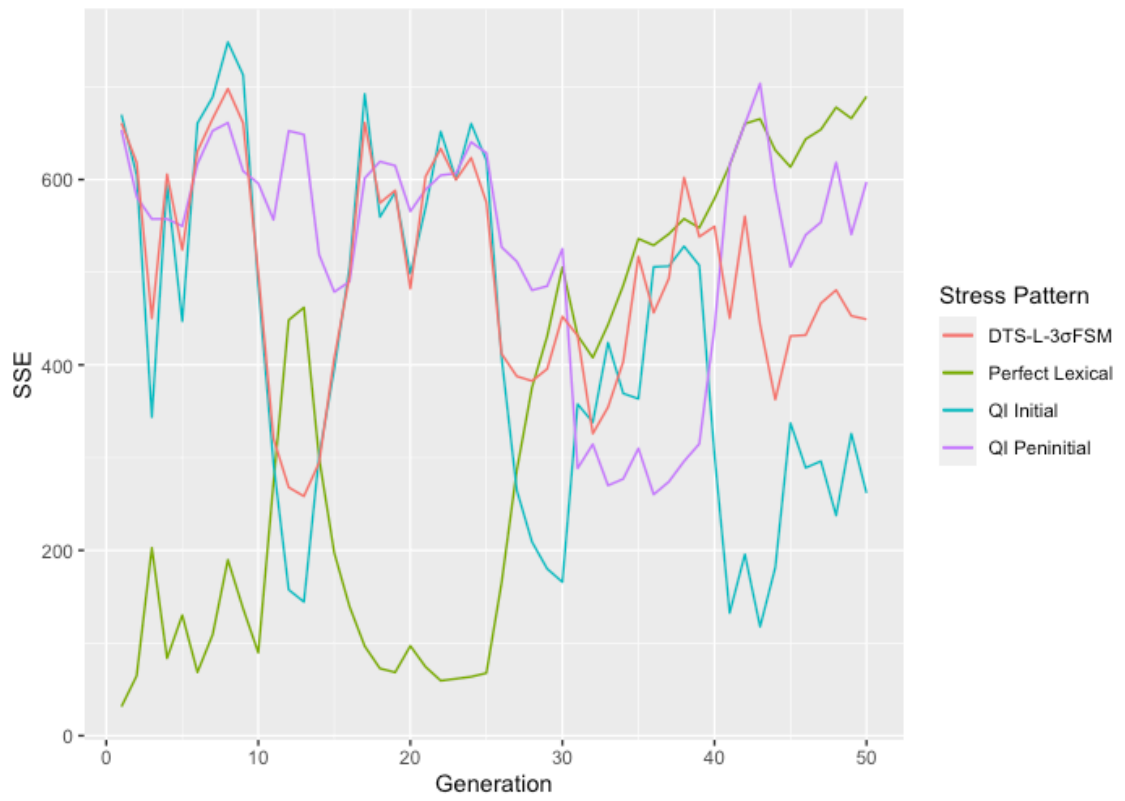


Figure 5.14: Change in SSEs during Simulation 6.b
(10000 tokens per generation, η for markedness constraints = 0.1, base η for MAX-PROM = 0.01)

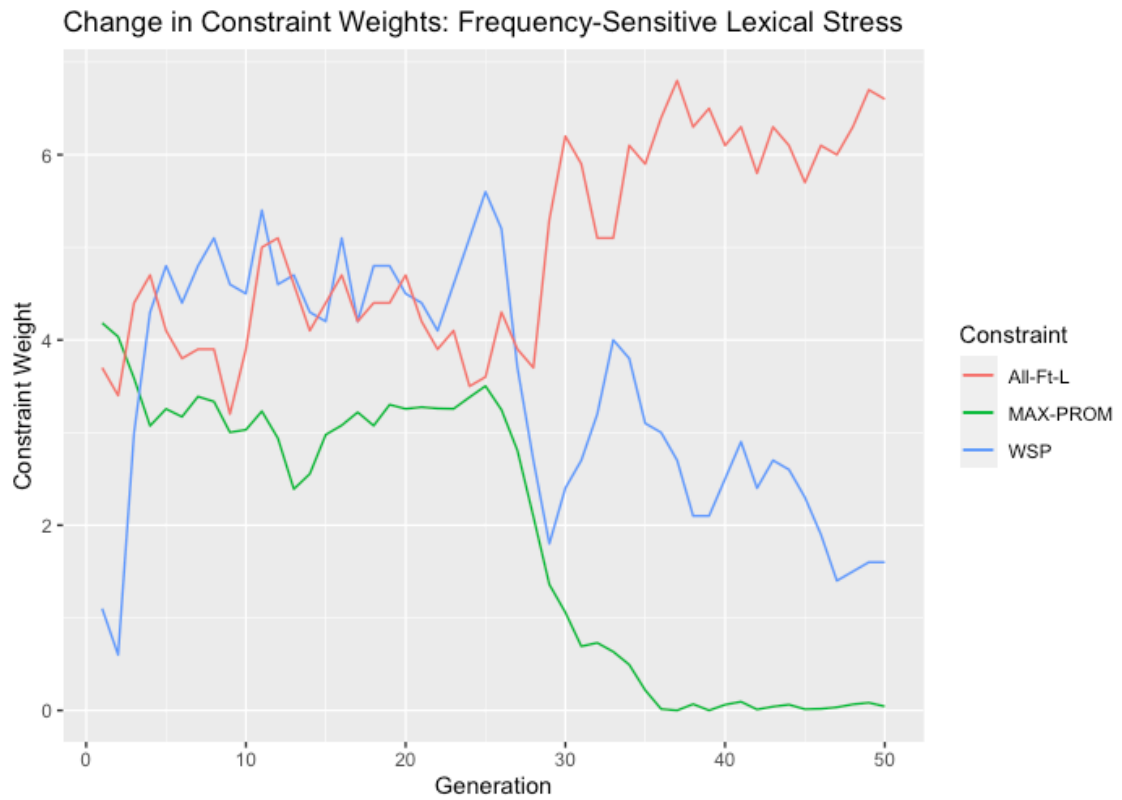


Figure 5.15: Change in Constraint Weights during Simulation 6.b
(10000 tokens per generation, η for markedness constraints = 0.1, base η for MAX-PROM = 0.01)

5.3.4 Global Interpretation of Simulation Results

From the perspective of the objectives stated at the outset of this section, namely, to attempt to trace the trajectory of prosodic change between Vedic and Classical Sanskrit through simulation, the results obtained are somewhat sobering: at least with the models of learning and simulation adopted (single-agent learning of MaxEnt grammars), no very clear relationship between the data from the RV (Section 5.2.4) and the Classical Sanskrit stress pattern emerged. More specifically, the empirical distributions of word-level prominence in the RV did not prove to shape the acquisition of phonotactic stress grammars so strongly that the Classical Sanskrit pattern would prove likely to emerge, under any of the various conditions explored above. To some extent, this outcome is unsurprising: the prominence distributions, at the token and type level, do not correspond well to any fixed-stress pattern whatsoever, let alone the fairly intricate Classical Sanskrit pattern specifically.

That the initial Vedic data were entirely irrelevant for learning in the long run proved not to be the case: the comparative rarity of final stress and relatively robust representation of stress at the left edge of the word seemed to promote the emergence of stress patterns without final stress or with stress regularly assigned closer to the left edge of the word. Across the board, the results obtained here also reproduced, using constraints and candidates that make the acquisition of quantity-sensitive stress patterns possible, some results of earlier studies involving iterated learning with MaxEnt grammars, in particular the simplicity bias (compare similar findings in Staubs 2014 and Hughto 2020). Specifically, patterns that require the interaction of fewer constraints are learned more quickly (i.e., through exposure to less data), and therefore more likely to emerge and be transmitted accurately. With respect to stress patterns, the quantity-insensitive patterns and quantity-sensitive patterns with a small window (right trochees, left iambs) proved to have best chances of performing well. Only in very rarely did unbounded patterns or patterns with a relatively larger stress window, such as the Classical Latin or Classical Sanskrit patterns, appear to achieve near-categorical performance in a given generation or learning, and even less rarely succeeded in propagating themselves accurately.

Worth mentioning in this context is also the rarity with which unbounded default-to-opposite patterns tended to perform well, given the finding of Koser and Jardine (2020) that such patterns are computationally simpler than unbounded default-to-same patterns. That DTO-L/R patterns usually perform poorly must be a consequence of how such patterns are captured with the constraint set used here, making them relatively more complex. An open question is thus whether DTO stress patterns can be modeled in constraint-based grammars in a fashion that would indeed render them simpler to learn.

In the simulations here, specific biases on some prosodic markedness constraints that were considered (TROCHEE, WSP, NONFINALITY- $\sigma/\bar{\sigma}$), at least in the implementation here, did not seem to be sufficient to have any substantial effect on the outcome of learning under reasonable parameters. Even though these specific constraints would appear to favor the learning of quantity-sensitive right-edge systems, the Vedic data, with its (slight) predominance of left-edge stress, could overcome these biases, unless either learning rates were low or data exposure was limited.

Simulations including the possibility of learning lexically specific stress (at least over abstract prosodic shapes) during phonotactic learning likewise confirm that establishing lexical stress via a simple faithfulness constraint, in case no fixed-stress pattern is available to be discovered, is unproblematic for a MaxEnt SGA learner. Indeed, given a sufficiently high learning rate and sufficient amount of learning data, as long as learning is assumed to always be veridical, matching the frequencies of stress patterns, or learning the lexical stress associated with some lexical, poses few obstacles. This result only further reinforces the conundrum of learnability with respect to lexically specific stress (see (51) in Chapter 3): if constructing lexical entries with prespecified stress (or other prosodic properties) is largely unproblematic, and phonotactic learning will support high weights for relevant faithfulness constraints where a contrast exists, what motivates a failure to learn?

Many options for future exploration in similar directions that have not been explored here do, however, remain open. Two major features that can and should be implemented in future tests of learnability for quantity-sensitive and lexically specific stress are the following:

- **Multi-Agent Iterated Learning:** Each generation includes multiple learning agents that act as teachers to each other as well as teachers to a subsequent generation. Based on results in Kirby and Sondregger 2015, a multi-agent model (which has been tested to a limited extent with constraint grammars in Sandell and Zukoff 2017 and Sandell 2021) should be able to better simulate effects of stability, transition, and change.
- **Learning of Underlying Representations:** Simulations here have been restricted to phonotactic learning and have largely operated with abstract prosodic shapes, rather than segmentally concrete lexemes. A more comprehensive and realistic model for the acquisition of lexically specific stress should include a means for learning underlying representations parallel to the weighting of general markedness and faithfulness constraints (see discussion at 3.2.2.2).

Finally important to emphasize is that the inability of the model employed here to simulate the effects of a specific diachronic change may be due to additional relevant factors not considered by the model whatsoever. For instance, changes in the productivity of certain morphological categories in Indic during the 1st Millennium BCE would also have a decided impact on the frequency of certain stress patterns in the lexicon. In particular, that the athematic nominal and verbal categories of Vedic Sanskrit, which were prosodically characterized by stress mobility in their paradigms, became practically extinct in Middle Indic, having lost ground to thematic categories that frequently presented stress on the penultimate or antepenultimate syllable (depending on inflectional form), may have had a considerable impact on the prosodic development of Indic languages as well. Integrating a parallel model of morphological productivity, able to combine the transmission, birth, and death of lexical items, into a model of phonological learning, constitutes another possible line of research.

5.4 Conclusions: Successes and Challenges

This chapter has explored the continuing prosodic history of Indic beyond the earliest attested situation represented by the Vedas, treated in the preceding chapter. I began with a brief search for traces of possible innovations or changes in word prosody during the Vedic period, but came up largely empty-handed. Although a handful of lexemes and perhaps at least one derivational suffix (-*ti*-) show changes in the position of prominence individual, no large-scale, coherent patterns are clearly identifiable. Similarly, while the deviant aspects of prosodic marking in the *Śatapāthabrāhmaṇa* are potentially suggestive of the presence of secondary stress and clash avoidance, their precise interpretation remains too uncertain without undertaking a further detailed philological investigation.

Much more profitable was the attempt at 5.2 to reconstruct a constraint grammar governing the assignment of primary stress in some Indic languages after the Vedic period (the “Classical Sanskrit stress rule”). Some meager indications from variant Sanskrit forms, Middle Indic languages, and most of all segmental developments into Modern Gujarātī, were found to support the stress patterns reported for the pronunciation of Classical Sanskrit in 19th century sources. The stress pattern that can be discerned from the available data is indeed complex and typologically rare, but can be captured with the standard inventory of prosodic markedness constraints used throughout the present work, and seems to constitute a learnable pattern, as confirmed by computational simulation. To summarize this pattern:

- In disyllabic words, place primary stress on the first syllable, unless the first syllable is light *and* the final contains a long vowel/diphthong. [Initial syllable stress certain for disyllables with a heavy initial syllable or light open final syllable; final stress likely when the final syllable contains a long vowel or diphthong.]
- In words of three or more syllables, place primary stress on the rightmost non-final heavy syllable *up to the fourth-to-last* syllable. [Certainly correct if the penult or antepenult is heavy, likely correct for a heavy preantepenult.]
- In three- or four-syllable words, place primary stress on the initial syllable if penult and antepenult are light. [Certainly correct for three-syllable words, likely correct for four syllable words.]
- In words of five or more syllables, place primary stress on the antepenult if penult, antepenult, and preantepenult are all light. [Likely correct for five syllable words; behavior in words of five or more syllables is presently a conjecture.]
- Secondary stress likely occurs only in words of five or more syllables, assigned to the initial syllable.

The remainder of the chapter was concerned with investigating the learnability of the Classical Sanskrit stress pattern, and then determining whether a direct line of development

could be drawn between it and the distribution of word-level prominence in Vedic (specifically based on data from the *R̥gveda*). Initial simple simulations employing Robust Interpretive Parsing established that pattern can indeed be learned from overt forms without variation. In attempting to consider how this pattern might have arisen, agent-based iterated learning with MaxEnt grammars was employed. In these simulations, agents were seeded with data based on the frequency distributions of prosodic prominence in the *R̥gveda*, and the fit of the stress grammar learned to a number of different fixed-stress patterns was assessed, testing various model parameters (weight of constraints in the initial state, learning rate, amount of learning data). Results indicate that the Classical Sanskrit pattern would be statistically unlikely to emerge from the Vedic prominence distributions, but the simulations have successfully replicated results concerning the learnability of similar stress systems obtained using a similar model design but different data. More generally, the overall model behaviors suggest that single-agent iterated learning is likely not a suitable tool for capturing the class of diachronic change considered in this chapter, namely, the emergence of a (relatively complex) fixed-stress pattern out of unpredictable lexical stress.

In providing further grounds to believe that some Indic languages and Classical Sanskrit itself developed a right-edge-oriented quantity-sensitive stress pattern, another cornerstone for comparative considerations of how the related Indo-European languages developed in their word prosody has been laid. Furthermore, in confirming the learnability of this pattern, stress systems with four-syllable windows ought to be considered to lie within the range of acquisible stress patterns. A task for the future remains, however, to better understand the historical conditions that facilitate the emergence of such a comparatively difficult-to-learn system.

Appendix A: Distributions of the Position of the *udātta* in the Ṛgveda

- (71) Distribution of Stress Position and Syllable Weight in Disyllabic Words in the *Ṛgveda*
(Token Frequencies)

Shape / Stress Position	1
ṽ	7012
ṽ	4764
ṽC	7432
ṽC	1425
SUM	20633

- (72) Distribution of Stress Position and Syllable Weight in Disyllabic Words in the *Ṛgveda*
(Token Frequencies)

Shape / Stress Position	1	2	SUM
L.ṽ	4609	2505	7114
L.ṽ	1788	2151	3939
L.ṽC	4654	5169	9823
L.ṽC	445	1172	1617
H.ṽ	2746	1361	4107
H.ṽ	2926	2927	5853
H.ṽC	8137	7238	15375
H.ṽC	1857	2580	4443
SUM	27168	25103	52271

(73) Distribution of Stress Position and Syllable Weight in Trisyllabic Words in the *R̥gveda*
(Token Frequencies)

Shape / Stress Position	1	2	3	SUM
L.L.ṽ	229	148	75	452
L.L.ṽ	1353	885	1063	3301
L.L.ṽC	2084	1653	1178	4915
L.L.ṽC	311	310	456	1077
L.H.ṽ	1472	1547	89	3108
L.H.ṽ	492	542	660	1694
L.H.ṽC	1698	1170	1051	3919
L.H.ṽC	993	598	913	2504
H.L.ṽ	338	182	28	548
H.L.ṽ	1539	1501	663	3703
H.L.ṽC	1750	2322	989	5061
H.L.ṽC	321	158	207	686
H.H.ṽ	2183	1575	13	3771
H.H.ṽ	245	120	159	524
H.H.ṽC	1092	1298	304	2694
H.H.ṽC	315	341	171	827
SUM	16415	14350	8019	38784

(74) Distribution of Stress Position and Syllable Weight in Four-syllable Words in the *R̥gveda*
(Token Frequencies)

Shape / Stress Position	1	2	3	4	SUM
L.L.L.ṽ	8	7	2	0	17
L.L.L.ṽ	69	33	52	11	165
L.L.L.ṽC	114	84	116	22	336
L.L.L.ṽC	28	12	4	23	67
L.L.H.ṽ	605	340	322	2	1269
L.L.H.ṽ	56	41	66	96	259
L.L.H.ṽC	391	353	382	283	1409
L.L.H.ṽC	74	65	37	80	256
L.H.L.ṽ	195	50	81	4	330
L.H.L.ṽ	331	239	273	143	986
L.H.L.ṽC	615	518	891	258	2282
L.H.L.ṽC	90	47	41	42	220
H.L.L.ṽ	4	21	1	0	26
H.L.L.ṽ	35	27	36	18	116
H.L.L.ṽC	93	71	124	21	309
H.L.L.ṽC	22	29	6	18	75
L.H.H.ṽ	98	73	87	1	259
L.H.H.ṽ	34	25	27	33	119
L.H.H.ṽC	136	129	170	49	484
L.H.H.ṽC	27	20	13	14	74
H.L.H.ṽ	296	118	183	3	600
H.L.H.ṽ	153	238	125	82	598
H.L.H.ṽC	459	523	322	209	1513
H.L.H.ṽC	258	248	85	164	755
H.H.L.ṽ	35	3	19	0	57
H.H.L.ṽ	46	54	41	28	169
H.H.L.ṽC	112	74	109	78	373
H.H.L.ṽC	27	22	4	24	77
H.H.H.ṽ	37	30	63	0	163
H.H.H.ṽ	55	9	14	12	90
H.H.H.ṽC	82	69	116	37	304
H.H.H.ṽC	23	15	33	12	83
SUM	4608	3587	4845	1767	13807

(75) Distribution of Stress Position and Syllable Weight in Five-syllable Words in the *R̥gveda*
(Token Frequencies)

Shape / Stress Position	1	2	3	4	5	SUM
L.L.L.L.Ṽ	0	0	1	0	0	1
L.L.L.L.Ṽ	1	0	0	0	0	1
L.L.L.L.VC	1	3	1	0	0	5
L.L.L.L.VC	1	1	0	0	0	2
L.L.L.H.Ṽ	14	16	12	14	1	57
L.L.L.H.Ṽ	2	0	0	0	0	2
L.L.L.H.VC	14	4	4	4	0	26
L.L.L.H.VC	0	2	1	0	1	4
L.L.H.L.Ṽ	4	4	20	0	0	28
L.L.H.L.Ṽ	21	22	29	15	11	98
L.L.H.L.VC	73	16	61	29	11	190
L.L.H.L.VC	11	1	1	2	3	18
L.H.L.L.Ṽ	5	1	2	0	0	8
L.H.L.L.Ṽ	2	2	4	1	1	10
L.H.L.L.VC	12	10	11	2	1	36
L.H.L.L.VC	2	1	1	0	0	4
H.L.L.L.Ṽ	0	0	0	0	0	0
H.L.L.L.Ṽ	1	3	1	1	0	6
H.L.L.L.VC	6	11	0	3	2	22
H.L.L.L.VC	1	1	0	0	0	2
L.L.H.H.Ṽ	95	19	52	82	1	249
L.L.H.H.Ṽ	0	0	5	1	1	7
L.L.H.H.VC	32	11	22	28	2	95
L.L.H.H.VC	9	5	0	0	0	14
L.H.L.H.Ṽ	37	24	15	26	1	103
L.H.L.H.Ṽ	15	18	5	3	4	45
L.H.L.H.VC	83	72	35	27	15	232
L.H.L.H.VC	27	15	16	3	5	66
H.L.L.H.Ṽ	5	20	2	25	0	52
H.L.L.H.Ṽ	3	3	0	1	1	8
H.L.L.H.VC	8	11	2	4	1	26
H.L.L.H.VC	1	1	5	1	0	8
L.H.H.L.Ṽ	0	0	0	0	0	0
L.H.H.L.Ṽ	3	3	2	0	4	12
L.H.H.L.VC	4	3	3	2	3	15
L.H.H.L.VC	1	1	0	0	0	2
H.L.H.L.Ṽ	9	19	10	4	0	42
H.L.H.L.Ṽ	129	131	26	16	3	305
H.L.H.L.VC	90	202	137	46	18	493
H.L.H.L.VC	4	6	11	0	7	28
H.H.L.L.Ṽ	0	3	1	0	0	4
H.H.L.L.Ṽ	15	3	0	0	0	18
H.H.L.L.VC	6	9	2	1	1	19
H.H.L.L.VC	1	1	0	0	0	2
L.H.H.H.Ṽ	17	22	2	20	1	62
L.H.H.H.Ṽ	0	2	1	0	0	3
L.H.H.H.VC	12	7	9	11	0	39
L.H.H.H.VC	2	0	3	0	0	5
H.L.H.H.Ṽ	33	44	43	23	0	143
H.L.H.H.Ṽ	4	1	0	0	0	5
H.L.H.H.VC	17	18	3	15	1	54
H.L.H.H.VC	0	0	1	1	4	6
H.H.L.H.Ṽ	17	12	5	15	0	49
H.H.L.H.Ṽ	4	5	2	2	2	15
H.H.L.H.VC	22	16	20	14	2	74
H.H.L.H.VC	5	7	6	2	2	22
H.H.H.L.Ṽ	0	1	0	0	0	1
H.H.H.L.Ṽ	4	1	3	2	1	11
H.H.H.L.VC	0	2	8	0	0	10
H.H.H.L.VC	0	0	0	0	1	1
H.H.H.H.Ṽ	1	10	5	5	0	21
H.H.H.H.Ṽ	2	0	2	1	3	8
H.H.H.H.VC	5	5	4	2	1	17
H.H.H.H.VC	0	0	0	1	0	1
SUM	893	831	617	455	116	2912

Appendix B: Relation between *udātta* and Syllable Weight in the *R̥gveda*

Stress in the RV exhibits a slight tendency towards overrepresentation on heavy syllables, at least as assessed by token frequency.

- This tendency holds consistently among 2- and 3-syllable words: heavy syllables in words of this length are significantly more likely to have the *udātta*.
- It also holds consistently among final syllables across the board: heavy final syllables are more likely to bear an *udātta*.
- Among 5-syllable words, it holds only for the penult and antepenult.
- Among 4-syllable words, stress on a *light* penult or antepenult is overrepresented.

(76) Disyllables

a. Initial Syllable

	1st Heavy	1st Light
Stressed	15514	11449
Unstressed	13852	11087

 $\chi^2 = 20.9, p < 0.001$

b. Final Syllable

	Final Heavy	Final Light
Stressed	21135	3804
Unstressed	19721	7332

 $\chi^2 = 1081.8, p < 0.001$

(77) Trisyllables

a. Initial Syllable

	1st Heavy	1st Light
Stressed	8776	8632
Unstressed	10031	12338

 $\chi^2 = 121.64, p < 0.001$

b. Medial Syllable

	Medial Heavy	Medial Light
Stressed	6850	6849
Unstressed	11850	12584

 $\chi^2 = 7.9025, p = 0.005$

c. Final Syllable

	Final Heavy	Final Light
Stressed	7814	205
Unstressed	23091	7674

 $\chi^2 = 1968, p < 0.001$

(78) Quadrisyllables

a. Initial Syllable

	1st Heavy	1st Light
Stressed	1737	2871
Unstressed	3571	5661

 $\chi^2 = 1.2209, p = 0.2692$

b. 2nd Syllable

	2nd Heavy	2nd Light
Stressed	1377	2210
Unstressed	4693	5560

$\chi^2 = 58.533, p < 0.001$ — stressed light syllable overattested!

c. 3rd Syllable

	3rd Heavy	3rd Light
Stressed	2045	1800
Unstressed	6196	3805

$\chi^2 = 88.249, p < 0.001$ — stressed light syllable overattested!

d. Final Syllable

	Final Heavy	Final Light
Stressed	1757	10
Unstressed	9329	2711

$\chi^2 = 467.81, p < 0.001$

(79) Five-syllable words

a. Initial Syllable

	1st Heavy	1st Light
Stressed	393	500
Unstressed	1057	924

$\chi^2 = 21.144, p < 0.001$ – stressed light syllable overattested!

b. 2nd Syllable

	2nd Heavy	2nd Light
Stressed	256	575
Unstressed	644	1405

$\chi^2 = 0.0799, p = 0.7773$

c. 3rd Syllable

	3rd Heavy	3rd Light
Stressed	463	154
Unstressed	1496	718

$\chi^2 = 12.286, p < 0.001$

d. 4th Syllable

	4th Heavy	4th Light
Stressed	331	125
Unstressed	1154	1141

$\chi^2 = 75.289, p < 0.001$

e. Final Syllable

	Final Heavy	Final Light
Stressed	109	4
Unstressed	1983	816

$$\chi^2 = 33.969, p < 0.001$$

CHAPTER 6

Ancient Greek: The Interplay of Stress, Tone, and Intonation

6.1 Introduction

This chapter devotes itself to the word-level prosody of the Greek language in the 1st Millennium BCE across dialects and diachrony. The primary descriptive objectives are thus the following:

- to further develop and fill in missing parts of a systemic analysis of prominence calculation for the Attic-Ionic dialect of Ancient Greek.
- to properly describe the prosodic system of the Doric dialect of Ancient Greek, insofar as available evidence permits secure conclusions.
- to precisely characterize the prosodic differences across Greek dialects of the 1st Millennium BCE in terms of phonological grammars;
- and finally, briefly outline the major innovations and changes in the prosodic grammar and lexicon of Greek between Ancient Attic-Ionic via the Koiné and Modern Greek.

These synchronic descriptions will lay the groundwork for further investigations into of prosodic change. In this chapter, two diachronic lines of development will be considered: 1) the prosodic differences between Attic-Ionic and Doric emerging from their nearest common ancestor; 2) the development of the Modern Greek system of prominence calculation out of the Attic-Ionic system by way of the Koiné. Discussion at the end of this chapter will survey the major diachronic differences that emerge between Ancient and Modern Greek; we will see that the diachronic trajectory of Greek is in broad respects similar to the trajectory adopted by many Romance languages: the position of some overt indicator of prominence remains diachronically largely unchanged at the lexical level, but the rankings of constraints and lexical marking on particular morphemes change in response to one another, as well as in response to independent changes in segmental prosody (specifically, the loss of contrastive vowel length). In this respect, the long-term history of Greek prosody attests to diachronic pertinacity of the type described by Lahiri (2015) as “same outputs, different grammar [and different lexicon]”. That is, throughout the history of the language, learners have systematically preserved largely similar surface outputs, undertaking whatever alterations in terms of the lexicon or grammar might be necessary in order to account for the position of the prosodic prominence in the learning data.

The core of the chapter follows in the next section, 6.2, devoted to the prosody of the well-known Attic-Ionic dialects; the objective there is nothing less than a comprehensive phonological treatment of the totality of its word-level prosody, both rule-governed, lexical, and involving enclitics. Since most existing literature is either essentially descriptive in nature, or, in theoretically oriented analyses, usually only a subset of the relevant phenomena and patterns is addressed, a new, more extensive analysis is necessary. Section 6.3 then considers the prosody of other 1st Millennium Greek dialects whose prosody was known to diverge from that of Attic-Ionic in some way, especially Doric; that section concludes with considerations of how to reconstruct the prosody of Proto-Greek. The chapter then draws to a close with a brief look at the relationship between the grammars and lexicons of Attic-Ionic prosody, on the one hand, and Modern Greek, on the other.

6.2 Ancient Attic-Ionic

A fully adequate and complete description of the system for the assignment of prosodic prominence for Ancient Attic-Ionic, despite centuries of philological and linguistic research, remains a desideratum. At present, even seemingly fundamental questions are still debated, such as how the word-level prosodic prominence was phonetically realized, or whether the language should be analyzed as possessing a metrical stress system or some sort of tonal system, which is unclear at first glance. In addition to these basic questions arise further problems of the extent to which the language possessed lexically specified prosodic structure and/or lexical tone, as well as how these matters of prominence realization interact with the language's intonational phonology. All of these issues will be addressed in this section in detail, departing from empirical groundwork that is, thankfully, relatively uncontroversial. The focus lies on the position of a word-level high pitch, H, directly indicated in texts produced beginning in the second century BCE.¹ I begin in 6.2.1 with a survey of the patterns of H distribution and its realization on different syllable types in both free-standing orthotonic words and prosodic words with clitics. Following this overview of the evidence, I summarize some existing attempts at analysis of those patterns at 6.2.2, which will then inform the novel analysis worked out under 6.2.3.

6.2.1 Attic-Ionic Word Prosody: Empirical Grounding

In order to clearly grasp the particular pitfalls posed by Attic-Ionic prosody in attempting any phonological analysis, a thorough survey of the relevant data is needed. This survey is structured as follows:

¹Orthographic devices for indicating the position of different pitches in writing of Greek were likely invented at Alexandria in the early second century BCE, and indeed first appear in papyri datable to that period. The systematic usage of these devices in the production of Greek manuscripts is not documented before the ninth century CE (!). Knowledge of how words of particular prosodic shapes are to be accented in a text thus rests, in large part, on grammatical works concerned with prosody going back to the latter part of the 1st Millennium BCE. See Probert 2006b: Ch. 1 for a thorough survey of the nature of the evidence for Ancient Greek accentuation, which, in its totality, may be regarded as fairly secure.

- the distribution of high tone.
- “recessively accented” words.
- monosyllabic stems and paradigmatic high tone mobility.
- intraparadigmatic high tone mobility.
- the accentuation of clitics.

Based on this survey, Section 6.2.1.7 attempts a typological characterization of Attic-Ionic in terms of stress, pitch accent, and tone, based on the discussion in 1.3.3.2 in Chapter 1.

6.2.1.1 The Distribution of High Tone

With reference to the position and distribution of a HIGH TONE (H) within a prosodic word – whatever the source of that H may be, and ignoring clitic phenomena for the moment – certain fundamental facts concerning the domain within which H may surface may be outlined as in 1.²

- (1) Restrictions on the position of H in Attic-Ionic
 - A. A minimal prosodic word ($\omega_{[+min]}$) – equivalent to an orthotonic word without any proclitics or enclitics – contains at least one and at most one high tone (H).
 - B. Only $\left[\begin{array}{c} +\text{syllabic} \\ -\text{consonantal} \end{array} \right]$ segments may host this H.³
 - C. In syllables with a long vowel, [V:], or diphthong, [VV], the H may be realized as part of a falling tone (HL) sequence (“circumflex”) or as H alone (“acute”).⁴
 - D. The position of the H is restricted by the LAW OF LIMITATION (LoL; also *Dreimorengesetz*)
 1. In prosodic words of three or more syllables (containing three or more vocalic peaks), the H is limited to occurring within a window of the three rightmost syllables.
 - a. Grammatical: [ánthrō:pos] ‘man:NOM.SG’, [ant^hró:po:] ‘man.GEN.SG’, [podô:n] ‘foot.GEN.PL’

²The basic data on the position and distribution of word-level high tone are presented systematically in Probert 2003; the same facts, with more extensive discussion of the history of scholarship, are reviewed in Probert 2006b: Ch. 2–3.

³See discussion under 6.2.1.5 for possible evidence that syllables containing [+sonorant] coda segments may also have been capable of supporting more than one tone. Since I will adopt the position that Attic-Ionic possessed no rising tone (LH) on any syllables of any shape, a word’s H tone will always be temporally aligned with (part of) a vowel.

⁴This observation touches already on a debated point, namely, whether the “acute” accent on bimoraic syllable nuclei was a high tone alone or a rising tone LH. I assume for the moment that “acute” is H alone, not rising LH, but the question will be addressed further throughout the succeeding sections.

- b. Ungrammatical: ^X[póluant^hraks] ‘having much coal:NOM.SG’, ^X[poíkilot^hriks] ‘having dappled hair/feathers:NOM.SG’
2. If the final syllable contains a long vowel, [V:], certain diphthongs, [VV], or a vowel followed by two⁵ consonants, [VCC], then the H must fall on the penultimate or final syllable.⁶
- a. Grammatical: [ant^hró:po:], [poluánt^hraks]
- b. Ungrammatical: ^X[ánt^hró:po:], [polúant^hraks]
3. If the final syllable contains merely a short vowel, [V̆], or short vowel followed by a single consonant, [V̆C] (including most diphthongs in [-oi] and [-ai]), then the H may fall on any of the antepenultimate, penultimate, or ultimate syllable.
- E. If the penult contains a long vowel, [V:], or diphthong [VV], and the final syllable contains a short vowel, [V̆], followed by any number of consonants, the penult may *not* contain an acute (σωτήρα RULE / LEX σωτήρα).
- a. σωτήρα ← /sɔ:-té:r-a/ ‘savior.ACC.SG’
- b. πολυπῖδαξ ← /polu-pi:dak-s/ ‘having many fountains.NOM.SG’
- c. κήρυξ ← /kɛ:ruk-s/ ‘herald.NOM.SG’⁷

⁵Theoretically also more than two, but in practice the only consonant cluster sequences allowed at the right edge of a word in Attic-Ionic are [ks] and [ps].

⁶This restriction is violated in a few instances, in particular the GEN.SG of *i*-stems such as πόλις [pólis] ‘city’, e.g., GEN.SG πόλεως [póleɔ:s]. Historically, such forms contained a penultimate syllable with a long vowel and a final syllable with a short vowel, thus *[póle:os], in which the position of the high tone obeys the conditions here.

⁷While frequently invoked as an example for a circumflex in a long-voweled penult followed by a final syllable that is heavy due to the consonant cluster, the vowel length in the final syllable of the nominative, and in turn the accentuation of this form, κήρυξ, is not altogether certain. West (1990: XLVIII), for instance, argues for the accentuation *κήρυξ instead (and similarly *φοῖνιξ instead of usually accepted φοῖνιξ), on the basis of the fact that we know the vowel υ (~ ι) to be long in other case forms of the relevant words (thus, e.g., GEN.SG κήρυκος, φοῖνικος). If the last vowel of the stem in these two lexemes were indeed long, then an acute on the penult would be expected, contrary to the claim of Herodian (in the unreliable reconstruction of Lentz 1867: 352) that the vowel is short in the nominative.

The first important premise is that words exhibiting final cluster [-ks] or [-ps] mostly have concordant metrical and accentual evidence concerning the length of the vowel in the final syllable: words like πῖδαξ ‘well’, DAT.SG πῖδαξι have a short ᾱ (circumflex in the nominative and metrical scansion in other case forms agree), while words like θώραξ ‘armor’, DAT.SG θώραξι have a long ᾱ (acute in the nominative and metrical scansion in other case forms agree). Similar contrastive pairs can be found words having a mid vowel in the final syllable (αἶθοψ ‘fiery, sparkling’ vs. κώνωψ ‘gnat, mosquito’; βοῦκληψ ‘cattle-thief’ vs. σκώληξ ‘(earth)worm, grub’). However, for words that occur with an orthographic ⟨υ⟩ or ⟨ι⟩ preceding final [-ks] or [-ps], there do not seem to be any examples of polysyllables in which the stem is known to have a long vowel throughout the entire paradigm: there are no words of 1st Millennium Greek with a long vowel or diphthong in the penult, which can distinguish between acute and circumflex, and a securely long [i:] or [u:] in the nominative singular as confirmed by an undisputed acute. Following Probert (2003: 84), I therefore take Herodian’s claim to be genuine: κήρυξ and φοῖνιξ indeed had short vowels in their final syllables and a circumflex on the VV penult, and paradigmatically showed an alternation between [i]/[u] and [i:]/[u:]. Given the apparent absence of a contrasting word like **θήρυξ [purely invented], with persistent long high vowel throughout the paradigm, I tentatively assume that long high vowels in Ancient Attic-Ionic were shortened at least in word final position before two consonants (if not perhaps in closed syllables more broadly). The monosyllabic ῖξ ‘grub’ (GEN ἱξός), if it indeed has persistent

For the sake of clarity, the position of H as permitted by the LAW OF LIMITATION is summarized again in 2.

(2) LAW OF LIMITATION

a. Acute Final:

- i. λιγυρός [ligurós] ‘shrill:M.NOM.SG’
- ii. νεκροβαστάξ [nekrobastáks] ‘corpse-bearing:NOM.SG’
- iii. λιγυρούς [ligurós:s] ‘shrill:M.ACC.PL’

b. Circumflex Final: ψυχικῶν [psu:k^hikô:n] ‘spiritual:GEN.PL’

c. Acute Penult: ἀνθρώπους [anthró:po:s] ‘man:ACC.PL’ (^X[anthrô:po:s], ^X[ánt^ho:po:s])

d. Circumflex Penult:

- i. σωτήρα [sɔ:tê:ra] ‘savior:ACC.SG’ (^X[sɔ:tê:ra]),
- ii. πολυπίδαξ [polupî:daks] ‘having many springs:NOM.SG’ (^X[polupídaks], ^X[polúpi:daks])

e. Acute Antepenult:

- i. ἄνθρωπος [ánt^hro:pos] ‘man:NOM.SG’
- ii. λιθόβουλος [lit^hóbo:los] ‘struck with rocks:M.NOM.SG’ (^X[lít^hobo:los])

As observed already in Misteli 1868, the prosodic weight of the penultimate syllable appears to be irrelevant, while the weight of the final syllable is decisive. The Law of Limitation and the σωτήρα Rule together reveal an interesting tension, since the former is clearly sensitive to syllable weight contributed by consonants in the final syllable ([V:] = [VCC]), whereas the latter is not ([V̌] = [VC_o]). Thus, for word forms with the recessive accent (see further below), the LEX σωτήρα is relevant just in case the final syllable contains the configuration [V̌CC] – in all other cases, the H will either fall on the antepult or the final will contain two vocalic morae and thus not condition the application of the LEX σωτήρα.

The examples shown under 1 and 2 above reveal that the position of the H is essentially “free” within the confines specified by the LoL, which in and of itself demonstrates that prosodic prominence (tentatively assuming that H reflects prosodic prominence in some fashion) must be at least partly *lexically* regulated, either through lexically specified stress (if stress always attracts or manifests itself as H) or tone (if some morphemes are assumed to bear lexical tone). Minimal pairs distinguished only by the position of the H, in a polysyllabic word, like those shown in 3, make evident that the assignment of word-level H cannot reflect the workings of a metrical stress algorithm alone; some sort of prosodic structure must be lexically prespecified (as already assumed for the example under 1.E.a.). Surface minimal pairs distinguished only by a difference in acute versus circumflex accentuation on a long vocalic syllable nucleus as in 4, while comparatively rare, are also attested; note that this

long [i:], including in the nominative, might then suggest that high-vowel shortening would be restricted to polysyllabic stems.

For practical purposes, this means that I take the accentuation in NOM.SG κῆρυξ and φοῖνιξ, with circumflex and short vowel in the final syllable, to be correct. The proper underlying representation of the form may, however, be /kɛ:ru:k-s/ (and similarly, /p^hoini:k-s/), with application of high vowel shortening creating the necessary conditions for the circumflex in the penult.

contrast is limited to final syllables (including monosyllables), and many instances of falling tone (though not all) are attributable to synchronic or historically recent processes of vowel contraction.

- (3) Attic-Ionic Minimal Pairs in Position of H
 - a. τόμος [tómos] ‘slice, cut’ vs. τομός [tomós] ‘cutting, sharp’
 - b. σκάφη [skáp^hɛ:] ‘trough; light boat’ vs. σκαφή [skap^hɛ:] ‘digging’
- (4) Attic-Ionic Minimal Pairs in Type of Tone
 - a. χῆρ [kê:r] ‘heart:NOM.ACC.SG’ vs. κήρ [ké:r] ‘fate.NOM.SG’
 - b. ἀγοράς ‘meeting place:ACC.PL’ vs. ἀγοράς ‘id:GEN.SG’⁸

One major task in the analysis of Attic-Ionic word-level prosody is thus to account for the distribution of lexical and purely phonological prominence marking. Across the lexicon, the major options for the syllable position of a high tone in open word classes (nouns, adjectives, verbs) are essentially as follows:

- in finite verbal forms, the high tone is almost always recessive (see below).⁹
- in nouns, adjectives, and non-finite verbal forms with a monosyllabic stem (which are always morphologically simplex), the high tone may (and regularly does) shift between the stem and the inflectional suffix (“genuine mobility”).
- in nouns, adjectives, and non-finite verbal forms with a polysyllabic stem (which are often, but not always, morphologically complex), the high tone may be “recessive” or fixed on the last syllable of the stem.
- in nouns, adjectives, and non-finite verbal forms with polysyllabic stems, where the number of syllables increases in inflectional forms, the high tone may necessarily shift one syllable to the right in order to adhere to the Law of Limitation (“pseudo-mobility”), or change from acute to circumflex in accord with the LEX σωτήρᾱ (cf. 1)

6.2.1.2 The “Recessive” Accent

One point of general agreement in scholarship on Attic-Ionic prosody is that the so-called “recessive” accent constitutes a phonological “default”, which determined where and how

⁸Note, however, that this latter minimal pair, like virtually all instances of a circumflex in a final syllable to a polysyllabic stem, is to be derived via post-lexical vowel contraction after assignment of a High tone to the penult (see Noyer 1997 and Probert 2010 on such derivations). The uncontracted forms may sometimes be found in Homer (GEN.PL ἀγοράων *Il.* 2.275, *Od.* 4.818).

⁹The only systematic exception is the IMPV.2SG.AOR.MID in the class of aorists built with the derivational suffix /-ε/-, e.g., λαβοῦ ‘take (for yourself)!’. A handful of further IMPV.ACT.2SG forms (e.g., εἰπέ ‘speak’) are likewise non-recessive, while the PRES.IND. forms of ‘be’ (εἰμί) and ‘say’ (φημί) generally behave as disyllabic enclitics. See a full list of exceptions (with some minor differences between Attic and Ionic, and alternatives attested in grammarians, in Probert 2003: 42.

prosodic prominence was realized, in the absence of any lexical (or perhaps paradigmatic or morphosyntactic) information that might call for prominence to be realized differently (for a demonstration of this point on the basis of changes in prosody at the lexical level, see Probert 2006a and Probert 2006b). In short, a word form is said to be “recessively” accented when its high tone stands as close to the left edge of the word as permitted by the LoL. In words containing three or more [+syllabic] segments, the high tone thus occurs on different syllables according to the weight of the final syllable, and leads to “pseudo-mobility” of the high tone between the penult and antepenult within paradigms, as illustrated in the paradigm for ‘man’ /ant^hrɔ:p-o-/ below (extrametrical word-final segments that do not contribute to syllable weight are marked with ⟨⟩). There, the high tone falls on the antepenult when the final syllable contains a short vowel, but the penult when the final contains a long vowel.¹⁰

(5) ‘man’ /ant^hrɔ:p-o-/

	SG	PL
NOM	[ánt ^h rɔ:po(s)]	[ánt ^h rɔ:po(i)]
GEN	[ant ^h rɔ:po:]	[ant ^h rɔ:po:n]
ACC	[ánt ^h rɔ:po(n)]	[ant ^h rɔ:po:s]
DAT	[ant ^h rɔ:po:i]	[ant ^h rɔ:poi(s)]
VOC	[ánt ^h rɔ:pe]	[ánt ^h rɔ:po(i)]

If the term “recessively accented” is applied to any word whose H is temporally aligned with the leftmost vocalic mora of a word and followed by at least one low tone (L), the following types count as “recessive”:

- Monosyllables with circumflex, e.g., πῦρ [pû:r] ‘fire:NOM/ACC.SG’.
- Disyllables or longer with acute penult (when final has [V:]) or circumflex penult (when the final contains a short vowel); cf. (2)c. and d. above
- Trisyllables with a light final having acute on the antepenult; cf. (2)e.

Hence, all words in which the final syllable bears high tone (whether acute or circumflex) and all words in which the penult bears high tone when the final syllable is light ([-V̇(C)]) are to be characterized as “non-recessive”. Since broad consensus exists that either metrical structure or general constraints on the distribution of prominence are responsible for the recessive pattern, further consideration of it is left to 6.2.2 below.

Two broad categories of exceptions to recessive accentuation in finite verbal forms exist; in both types, the high tone surfaces closer to the right edge of the word than would otherwise be expected. In the first instance, vowel contraction that seemingly applies after assignment of high tone produces a circumflex on a final or penultimate syllable where an acute on

¹⁰In principle, the lexeme could be lexically marked as /ánt^hrɔ:p-o-/, with shift of the underlying high tone to the penult enforced by the LoL. However, because any lexical marking on an antepenultimate or preantepenultimate [+syllabic] segment will be either occasionally or systematically neutralized by the LoL, positing such a lexical diacritic would be superfluous from the point of view of learnability, given that any antepenult high tone can be accounted for by the principles of recessive high tone assignment.

the antepenult or penult would normally occur. This rather common category of exception is readily analyzed under any version of Lexical Phonology, if word-level prosody is taken to belong to the word-level phonology, but vowel contraction to the post-lexical phonology (see further Noyer 1997 or Probert 2010 on other cases in which vowel contraction appears to apply after assignment of high tone). The second, statistically rarer class of exceptions consists of verbs whose rightmost syllable contains a short vowel and is coextensive with root, preceded by prefixes adding at least two syllables; in such cases, high tone always falls on the penult, rather than antepenult (cf. Vendryes 1904 [1945]: 129, Probert 2003: 47). A principled explanation for this behavior is less obvious, though Steriade (2014: 3) rightly points out that, because verbal forms usually have at least one suffix following the root, configurations like those in 6b. come about only rarely, while cases of recessive high tone on the first syllable to the left of the root syllable are commonplace. This restriction could then be viewed as a morphologically sensitive phonotactic constraint: no high tone farther than one syllable to the left of the root.¹¹ I consider both of these phenomena to stand outside the core aspects of Attic-Ionic word prosody, which can be dealt with through subsidiary assumptions, and thus will not be analyzed in any further detail.

(6) Exceptions to “Recessive” Accentuation in Finite Verbs

- a. ἐφιλοῦμεν [epʰilô:men] ‘love:IMPF.1PL’ ← /e-pʰile-o-men/ (not ^x[epʰilo:men])
- b. ἀπόδος [apódos] ‘give up:AOR.IMPV.2SG’ (not ^x[ápodos]),
παρέσχον [paréskon] ‘hand over:AOR.1SG’

6.2.1.3 Monosyllabic Stems and Paradigmatic High Tone Mobility

Three facts about lexemes with a monosyllabic stem merit observation: 1) when the prosodic word is monosyllabic, it must contain a long vowel (e.g., σκῶρ [skô:r] ‘dung’), diphthong, or two coda consonants (e.g., θρίξ [tʰríks] ‘hair’); 2) when the prosodic word is monosyllable, its accentuation may be either acute (e.g., κήρ [ké:r] ‘fate’) or circumflex (e.g., κῆρ [kê:r] ‘heart’); 3) certain inflectional suffixes (expressing the case features GEN or DAT) always attract high tone away from the stem, as shown for the paradigms of ‘foot’, ‘mouse’, ‘nose’, and ‘goat’ below. These behaviors and surface contrasts call for immediate comment, since their significance for the larger prosodic grammar has perhaps not been sufficiently appreciated.

(7) High Tone Mobility and Acute-Circumflex Contrast in Monosyllabic Stems

¹¹ Particularly if prefixes are assumed to stand outside the ω_{+min} (compare the structures proposed for prefixed verbs in Proto-Germanic in Chapter 7), and, with Golston (1990), recessive high tone regularly appears one syllable to the left of the primary stress (via a LH* default word intonation), then a prosodic structure like [apó [(‘do)s] _{ω_{-min}}] _{ω_{-max}} might suffice to capture this phenomenon.

	‘foot’	‘mouse’	‘nose’	‘goat’
NOM.SG	[pó:s]	[mû:s]	[rí:s]	[aík:s]
ACC.SG	[pód-a]	[mû:n]	[rí:n-a]	[aíg-a]
GEN.SG	[pod-ós]	[mu-ós]	[rí:n-ós]	[aig-ós]
DAT.SG	[pod-í]	[mu-í]	[rí:n-í]	[aig-í]
NOM.PL	[pód-es]	[mú-es]	[rí:n-es]	[aíg-es]
ACC.PL	[pód-as]	[mú-as]	[rí:n-as]	[aíg-as]
GEN.PL	[pod-ô:n]	[mu-ô:n]	[rí:n-ô:n]	[aig-ô:n]
DAT.PL	[po-sí]	[mu-sí]	[rí:-sí]	[aik-sí]

First in need of comment is the alternation in vowel length seen in the paradigms of ‘foot’ and ‘mouse’, which show long vowels in monosyllabic forms (NOM.SG ποús [pó:s], μûς [mû:s]; ACC.SG μûν [mû:n]). The stems of these two lexemes must be analyzed as containing a short vowel, thus /pod-/ and /mu-/, since the vowel length distinction is contrastive in disyllabic forms (NOM.PL πόδες [pódes] vs. ῥῖνες [rí:nes]). Lengthening of the underlying short vowel in monosyllables having but a single word-final consonant thus follows the assumption of a minimal word requirement on content words and the assumption of final consonant extrametricality; see further Golston 1991: 73–100.¹² Exceptions to a bimoraic minimal word requirement on content words are limited to a small handful of 2.SG imperatives of the shape CVC (e.g., δός [dós] ‘give’ and θές [thés] ‘put’). Further remarkable is that monosyllabic finite verbal forms – which in principle should be “recessively” accented – almost exceptionally exhibit the circumflex (e.g., βῆ [go:AOR.3SG], θοῦ [put:AOR.IMPV.MID.2SG]); compare Vendryes 1904 [1945]: 111 and see Probert 2003: 42 for some exceptions to recessivity in the Ancient Greek verb.¹³

Second, in contrast to the behavior of such monomorphemic, monosyllabic stems in Vedic Sanskrit, *all* monosyllabic stems in Attic-Ionic exhibit mobility of the high tone between the stem and certain inflectional suffixes; compare the descriptions and analyses at 4.2.1.1 and 4.3.2. Such intraparadigmatic high tone mobility extends even to stems that have become monosyllabic within the attested history of Greek through vowel contraction (cf. Kim 2002a: 82–3); contrast Homeric disyllabic ῥῖ-ος ‘sheep:GEN.SG’ with Attic monosyllabic οἰ-ός. While Vedic possesses a contrast between accented and unaccented monosyllabic stems (contrast ending-accented *pad-é* ‘foot:DAT.SG’ with stem-accented *gáv-e* ‘cow-DAT.SG’), in Attic-Ionic one instead finds a distinction is between monosyllables which may have a

¹²The minimal word requirement here is understood as the requirement that the ω dominate a branching structure above the level of the segment. Practically, this means that a word must contain at least two syllables or, if there is only one syllable, that the single syllable be bimoraic. I do not take the MWR to be a requirement for a well-formed bimoraic foot, ($\bar{\sigma}$), though this effect obtains in monosyllables.

¹³The exceptions to recessive position of high tone belong to just two categories:

- MID.IMPV.2SG of thematic aorists: βαλοῦ [balô:] ‘throw!’, λιποῦ [lipô:] ‘leave!’, etc.
- Five particular ACT.IMPV.2SG thematic aorists: εἰπέ [e:pé] ‘speak!’, ἐλθέ [eltʰé] ‘go!’, εὗρέ [euré] ‘find!’, ἰδέ [idé] ‘behold!’, λαβέ [labé] ‘take!’ (latter two Attic only).

circumflex on the stem syllable (e.g., σκῶρ, σκατός (later also σκωρός) ‘dung’; μῦς, μύς, μύες ‘mouse’; κῆρ, κήρος ‘heart’;¹⁴ αἶξ, αἶγός, αἶγες ‘goat’) and those that only show an acute (e.g., πούς, ποδός, πόδες ‘foot’; ῥίς, ῥινός, ῥίνες ‘nose’; κήρ, κηρός, κήρες ‘fate’). The paradigms in 7 above represent the four possible patterns:

- ‘foot’-Type: long vowel with acute in the monosyllabic nominative singular (πούς), short vowel with acute in other strong cases (ACC.SG πόδα), acute on the inflectional suffix of oblique cases (GEN.SG ποδός).
- ‘nose’-Type: long vowel or diphthong with acute in monosyllabic strong cases (NOM.SG ῥίς), but circumflex in polysyllabic strong cases (NOM.PL ῥίνες due to the σωτήρα Rule, ACC.SG ῥίνα), acute on the inflectional suffix of oblique cases (GEN.SG ῥινός).
- ‘mouse’-Type: long vowel with circumflex in the monosyllabic nominative singular (μῦς), short vowel with acute in other strong cases (NOM.PL μύες), acute on the inflectional suffix of oblique cases (GEN.SG μύός).
- ‘goat’-Type: long vowel or diphthong with circumflex in all strong cases (NOM.SG αἶξ, NOM.PL αἶγες), acute on the inflectional suffix of oblique cases (GEN.SG αἶγός).

Regardless of whether the monosyllabic nominative (nominative/accusative in the case of neuters) presents an acute or circumflex, the oblique case suffixes /-os/, /-i/, /-ɔ:n/, /-si/, and /-oin/ always bear an acute (-ός, -ί, -σί) or circumflex (-ῶν, -οῖν) accent. Just as the etymologically related case suffixes of Vedic (e.g., GEN.SG /-ás/, LOC.SG /-í/) similarly attract the stress and high pitch when paired with unaccented stems, it is attractive to posit that these consistently accented inflectional suffixes likewise have a lexical prosodic property that attracts the accent. This property could be either lexical stress or foot structure, if high pitch is assumed to be attracted to stress; alternatively, it could be lexical high pitch, which attracts stress, if stress is assumed to be present whatsoever. I will tentatively designate these suffixes as lexically specified for a stress-attracting starred high tone, H*, which presents as circumflex in the GEN.PL -ῶν [-ỗn] and GEN./DAT.DU -οῖν [-oî̃n] (see 6.2.1.6 and 6.2.3.6 for discussion of how possibly to account for the falling tone in these endings). The case suffixes /-n/~/-/a/, /-es/, /-as/, and /-e/, which never bear high tone, can be assumed to bear no lexical specification. This pair of analytical choices follows from the basic assumptions concerning the “recessive” accent above: since the high tone by default could fall on the penult in a disyllable, if it falls on the final syllable, a lexical property must be at work.

¹⁴This noun is worth noting precisely because it is unique in being a monosyllable arising from contraction that does not exhibit mobility between the stem and inflectional suffixes. The lack of mobility transmitted in the textual tradition must be a consequence of the belief that the form did arise from contraction, as the artificial disyllabic form κέαρ shows. κέαρ does not occur in Homer, first appearing in Pindar, Bacchylides, and the Attic tragedians. Probably the circumflex of the NOM./ACC.SG, whatever its reason for being, led to the belief that the vowel was a product of contraction, and hence should not show mobility in its paradigm. See Chantraine 1968–1980 [2009]: 478–9, who considers the circumflex κέαρ to be a false archaism, and the DAT.SG κῆρι to follow from it. In contrast, Hoenigswald (1997) considers the accentuation of κῆρι to be an archaism, reflecting an original endingless locative *[kê:r], or an originally unaccented dative singular inflectional ending.

The surface contrast in monosyllables between acute and circumflex extends beyond nouns to monosyllabic adverbs and particles (although most particles are proclitic) as well. Although most non-proclitic forms show a circumflex (e.g., *νῦν* ‘now’, *πῶς* ‘how?’), some have an acute (*πρῶ* ‘early (in the day)’, *μή* ‘not’, *χρή* ‘it is necessary’); see Probert 2003: 123.¹⁵ At least at first glance, a lexical distinction, represented with an underlying prosodic feature, like tone, would be attractive; indeed, the minimal pair *κῆρ* ‘heart’ versus *κῆρ* ‘fate’ seems to virtually require a lexical distinction.¹⁶ A derived distinction attributable to underlying differences in e.g., vowel length or segmental composition must be excluded, however. Olander (2007) has collected and sorted the data on monosyllabic nominal stems, and his data appear to admit of two major generalizations:¹⁷

1. Non-neuter stems with at least one consonant at the edge of the stem usually show an acute in their nominative singular. Given that non-neuter nouns have an underlying inflectional ending /-s/, such forms would have at least two underlying consonants at the edge of the word.

- (8) /VC₁C/ receives acute
- a. /klɔ:p-s/ → [kló:p-s] ‘thief’
 - b. /pod-s/ → [pó:s] ‘foot’

2. Neuter stems and non-neuter stems with no stem-edge consonant usually show a circumflex in their nominative singular. Given that neuter nouns have no segmental inflection suffix in their NOM./ACC.SG, in these cases, just a single consonant separates the syllable nucleus from the edge of the word.

- (9) /V(C)/ receives circumflex
- a. /skɔ:r/ → [skô:r] ‘dung’
 - b. /mu-s/ → [mû:s] ‘mouse’

These two generalizations would then be compatible with the accent assignment rules given in 10. Although Olander’s observations fit most monosyllabic stems with a plausible Indo-European etymology, rules like in 10 synchronically have too many exceptions to constitute a viable approach to the acute versus circumflex contrast.¹⁸ Besides cases of vowel

¹⁵On *μή* with acute as an emphatic, non-proclitic form, cf. Probert 2003: 136.

¹⁶In this particular case, a complicating factor is that *κῆρ* ‘heart’ continues a PIE stem */kerd-/ with an additional consonant, which may segmentally distinguish it from *κῆρ* ‘fate’.

¹⁷Olander himself formulates these generalizations in purely historical terms (“monosyllabic words ending in two consonants in Proto-Greek have acute tone in Attic, while those ending in fewer than two consonants have circumflex tone”), but can be expressed synchronically without difficulty.

¹⁸Olander himself formulates these patterns in a fashion that could be interpreted as faithfulness to a paradigmatic base: “If the accusative is disyllabic, the nominative has acute tone...if the accusative is monosyllabic, the nominative has circumflex tone.” The accusative of non-neuter consonant-final stems will always be disyllabic, and thus be expected to have an acute when it contains a short vowel; when the accusative contains a long vowel, however, it will be circumflex due to the Lex *σωνῆρα* as in *ῥῖνα* [rí:na] ‘nose:ACC.SG’, though the corresponding nominative is acute, *ῥίς* [rí:s]. The accentuation of the nominative cannot, therefore, directly

contraction that have produced a pitch contour opposite that predicted by 10 (e.g., *πρῶν* ‘foreland’, *οἷς* ‘sheep’; cf. Olander 2007: 3–4), the quantifiers *εἷς* ‘one’ and *πᾶς* ‘all’, historically continuing *[séms] and *[pánts], have an unexpected circumflex, as do the nominatives *αἰξ* ‘goat’, *γλαῦξ* ‘owl’ and *πῶξ* ‘hare’ (at least in Attic), while *θῶς* ‘jackal’ (GEN. *θῶος*) and *Τρῶς* ‘Trojan’ (GEN *Τρῶος*) have an unexpected acute.¹⁹ Note further that some particles, such as *χρή* ‘it is necessary’, also do not obey this potential synchronic generalization concerning the distribution of acute and circumflex.

(10) Accent Assignment Rules for Monosyllables

a. Acute Accent Assignment in Monosyllables

$$V \rightarrow \dot{V} / {}_{(\omega)}[C_0 \text{---} C_2]_{(\omega)}$$

The only vowel in a prosodic word is assigned an acute accent if it is followed by two or more consonants.

b. Circumflex Accent Assignment in Monosyllables

$$\{V_i, VV\} \rightarrow \hat{V}_i / {}_\omega[C_0(C)]_\omega$$

The only long vowel or diphthong in a prosodic word is assigned a circumflex accent if it is followed by one or no consonants.

In short, positing some underlying prosodic distinction will prove necessary to generate the distinction between acute and circumflex monosyllables. On the strength of the evidence of monosyllabic verbs, which almost exceptionlessly show a circumflex (cf. above), the acute would appear to constitute the marked case. While assuming that all nominal roots of Ancient Greek are unmarked for stress, but that some inflectional endings are (e.g., DAT.SG /-í/) is sufficient to account for intraparadigmatic *mobility* of high tone (compare an analysis to this effect in Kim 2002b: 92–5, Kim 2002a: 81–3), it does not suffice to handle the contrast between the acute and circumflex in monosyllables. For precisely this reason, specifying one type of surface tonal pattern as lexical marked seems necessary. Since a stress-attracting, starred high tone, H*, was proposed above for the high-tone attracting inflectional endings, attributing the same property to acute stems like /pód-/ ‘foot’ and /rí:n-/ ‘nose’ will serve to limit the inventory of underlying lexical prosodic contrasts, but suffice to generate the surface contrasts.

One other type of limited intraparadigmatic mobility is found in 1st declension ($\bar{a}/\eta$ -stem) nouns. Such nouns, when recessive in all other inflectional forms, nevertheless cede

constitute an instance of faithfulness to the surface accent found in the corresponding accusative. Besides the invocation of faithfulness to a paradigmatic base, it is difficult to see how a difference in number of syllables in the accusative could directly influence the accentuation of the nominative. Moreover, this pattern, as so described by Olander, may well be a merely epiphenomenal, since the forms with monosyllabic accusatives are either a) neuters, which in their nominatives have only a single underlying consonant at the right edge, or b) vowel-final stems, which of course have no consonants at their right edge. Hence, reference to the underlying differences in segmental structure between the acute and circumflex monosyllables appears more promising.

¹⁹In the case of the quantifiers, their high token frequency might have promoted lexical restructuring, i.e., lexical storage as simply /e:s/ and /pa:s/ rather than /en-s/ and /pant-s/, despite paradigmatic alternations that support URs with the underlying consonants. The words with a diphthong followed by at least one consonant consonant, like αἴξ ‘goat’, are all attested with acute outside Attic or in the κοινή (Schwyzer 1938 [1953]: 377).

the high tone to the ending of the genitive plural; see the paradigm of *χώρα* ‘village’ in 11. This case, like other similar cases of surface non-recessivity in verbal forms, can be attributed to vowel contraction after assignment of the recessive high tone. Thus, the GEN.PL [kɔ:mɔ̃:n] is derived from underlying /kɔ:m-ε:-ɔ:n/, with H assigned to the penult as the recessive accent, yielding /kɔ:mé:ɔ:n/, whereupon vowel contraction gives a circumflex final syllable in [kɔ:mɔ̃:n].²⁰ See also the end of the preceding section on such cases of postlexical vowel contraction.

(11)	‘village’ /kom-ε:-/	
	SG	PL
NOM	[kɔ́:mɛ:]	[kɔ̃:ma(i)]
GEN	[kɔ́:mɛ:(s)]	[kɔ:mɔ̃:n]
ACC	[kɔ́:mɛ:(n)]	[kɔ́:ma:(s)]
DAT	[kɔ́:mɛ:i]	[kɔ́:mai(s)]
VOC	[kɔ́:mɛ]	[kɔ̃:ma(i)]

6.2.1.4 High Tone in Derivation

In morphologically complex stems, or stems containing at least two syllables, two basic patterns of high tone distribution are found:

- High tone is recessive, exhibiting “pseudo-mobility” according to the LoL (compare the paradigm of *ἄνθρωπος* ‘man’ in 5)
- High tone is persistent on the same derivational suffix, which can be equivalent to the final syllable (acute) or the penult (as acute before a long-voweled final, but circumflex before a short-voweled final).

As recognized clearly in Steriade 1988: 279–81, the rightmost or outermost derivational suffix in a morphologically complex nominal or adjectival stem either allows for recessive

²⁰As Dieter Gunkel reminds me, the feminine forms of thematic adjectives (1st class adjectives), e.g. do not produce a circumflex as a contraction product in the GEN.PL when the corresponding masculine form is properly recessive (see Probert 2003: 58–9). Contrast, for example, forms of *δίκαιος* ‘just, right’ with forms of *καλός* ‘beautiful’ given below. The crucial difference between the two types of adjectival accentuation depends upon the stem being oxytone or not; when the base masculine stem is oxytone (like /kal-ó-/), an oxytone feminine formant must be used (thus /kal-é:-/). Further issues in the correct application of accent assignment and vowel contraction in adjectives I leave as a problem of interface with the morphology.

- | | |
|-----|--|
| (1) | <i>δίκαιος</i> [díkaios] ‘just, right’ |
| a. | Masculine: NOM.SG <i>δίκαιος</i> [díkaios], GEN.PL <i>δικαίων</i> [dikaíɔ:n] |
| b. | Feminine: NOM.SG <i>δικαία</i> [dikaía:], GEN.PL <i>δικαίων</i> [dikaíɔ:n] |
| (2) | <i>καλός</i> [kalós] ‘beautiful’ |
| a. | Masculine: NOM.SG <i>καλός</i> [kalós], GEN.PL <i>καλῶν</i> [kalô:n] |
| b. | Feminine: NOM.SG <i>καλή</i> [kalé:], GEN.PL <i>καλῶν</i> [kalô:n] |

accent throughout the paradigm, attracts high tone to itself, if attached to a base that would exhibit a non-recessive high tone, conditions recessive accentuation. In 12a., the derivational suffix /-eú-/ attracts high tone to the final syllable; in 12b., the new rightmost derivational suffix /-ia-/ does not attract high tone, but does not allow the high-tone attracting property of /-eú-/ to emerge, either; instead, high tone is recessive. Further examples in 13 illustrate additional cases of derivational suffixes attract high tone where the recessive accent could occur father to the left; finally, the derivation of *πενθεριδεύς* [pent^herideús] ‘brother-in-law’ in 14 demonstrates the usual dominance of the rightmost derivational suffix. See further Kuryłowicz 1958: 130-7 for a list of derivational suffixes organized by their tendency to attract high tone or condition recessive accentuation (for the latter, see especially pg. 136).²¹) from original *ἡδυλός, with H-attracting -λό-, and thus do not constitute genuine counterexamples to a binary distinction in accentual properties of derivational suffixes. See further the discussion in Probert 2006b: 212–20.

- (12) An outermost derivational suffix either a) attracts the high tone or b) induces recessive accentuation
- a. /basil-eú-s/ βασιλεύς [basileús] ‘king:NOM.SG’ – recessive ^xβασίλευς [basíleus] does not occur.
 - b. /basil-eú-ia/ βασίλεια [basíleia] ‘queen:NOM.SG’ – no persistence of high tone from the derivational base to produce ^xβασιλεῖα [basileîa].
Recessive accent instead applies.
- (13) Derivational suffixes with lexical /H*/
- a. DIM.M /-ísko-/: ἀστερ-ίσκο-ς [asterískos] ‘little star’
 - b. ADJ /-ró-/: σιγῆ-ρό-ς [sigē:rós] ‘silent’
 - c. AGT.M /-eú-/: χαλκ-εú-ς [k^halkéus] ‘coppersmith’
 - d. AGT.M /-té:r-/: σω-τήρ [sɔ:té:r] ‘savior’
 - e. NML /-íd-/: ἀσπίδ-ος [aspídos] ‘small.shield-GEN.SG’
- (14) /pent^her-ó-íd-eú-s/ → πενθεριδεύς [pent^herideús] ‘brother-in-law’ (father.in.law-NML-NML-M.NOM.SG)
Compare πενθερός [pent^herós] ‘father-in-law’

Such attraction of prosodic prominence – here equating surface H with the position of stress – to the outermost derivational suffix looks precisely like a case of “accentual dominance” in the sense of Kiparsky and Halle 1977 or Kiparsky 2010a. As seen in 4, a nearly identical phenomenon is at work in Vedic Sanskrit; there, special faithfulness to the prosodic properties of morphological heads was invoked: HEADFAITH constraints, modeled on Revithiadou 1999 (see 2.4.2.3 in Chapter 2). A competition between some lexical prosodic property and the recessive accent – however the latter is itself to be analyzed – can fundamentally

²¹Worth mentioning in this connection are examples in which a suffix of the shape /-CṼ-/ apparently preserves the position of the High tone, rather than conditioning recessive accentuation. Kuryłowicz (1958: 136) cites *δριμύλος* ‘sharp’ (< *δριμύς* ‘sharp’) and *ἡδυλός* ‘sweet’ (< *ἡδύς* ‘sweet’). Such cases are best explained by the application of Wheeler’s Law (retraction of acute in words of the shape – ∪ ∪; cf. 3.3.1.2

account for the alternation in position of high tone seen between βασιλεύς [basileús] and βασίλεια [basíleia] in 12: if the outermost derivational suffix is not high-tone attracting, the default recessive accent applies (cf. Sandell 2015a: 190–2).

Further analytically appealing would be to assume that high-tone attracting derivational suffixes possess the same underlying prosodic property as the inflectional suffixes that receive high tone in monosyllabic stems (see above), since both categories of affix regularly surface with acute (modulo the σωτήρα Rule, and with the exception of the circumflex-bearing endings GEN.PL -ῶν [-ô:n] and (the rare) GEN/DAT.DU -οῖν). Reference merely to the rightmost of any prominence attracting morphemes, that is, treating Attic-Ionic as a simple lexical accent system like Cupeño or Italian (cf. 2.4.2.1), will fail on multiple fronts, as the analysis at will make clear 6.2.3. In particular, the absence of mobility to high-tone attracting endings, and the emergence of default recessive accentuation in βασίλεια-type forms would be unexpected – simple application of a “rightmost wins” principle will not suffice.

6.2.1.5 Ancient Greek Enclitics: Factual Outline

Since a comprehensive treatment of the Attic-Ionic word prosody will necessarily bring the effects of prosodically deficient lexemes – enclitics – into play, this section presents a brief overview of the accentual effects that result from encliticization. Crucially, maximally long sequences of enclitics with their successive accentual effects are surveyed (cf. now Probert 2016) here as well, since these facts must also be brought to account, although existing formal treatments of Attic-Ionic accentuation have largely considered only the behavior of orthotonic words with a single enclitic.

Table 6.1 lays out the possible combinations of orthotonic words with a single enclitic by their accentuation and weight of the final syllable with enclitics of different sizes, and a description of the outcome. For a description of these facts, see Probert 2003: 147–8).

The tabular overview in 6.1 makes evident the following facts: 1) the weight of the final syllable is relevant only when the host has a circumflex on the penult; 2) monosyllabic and disyllabic enclitics condition different behaviors only when the penult bears an acute, or the penult has a circumflex and the final syllable of the host is heavy. This latter configuration – a penult with circumflex but heavy final syllable due to the consonant cluster – is also precisely the configuration in which the position of the high tone is restricted to the penult by the LAW OF LIMITATION, but must occur as a circumflex if the penult contains a long vowel or diphthong (traditionally ascribed to the LEX σωτήρα).

Nonetheless, a somewhat fraught question, for which no extensive investigation has yet been undertaken, concerns the effect of enclitics on orthotonic words when the orthotonic word is of the shape and accentuation of κῆρυξ [kê:ruks] ‘herald’, i.e., a circumflex on the penult followed by a heavy final with a short vowel. According to the standard doctrine, represented in Table 6.1, such words receive no secondary acute on their final syllable, in contrast to forms like οἶκος [oîkos] ‘house’, where the final syllable is light. This standard doctrine is represented in the epitomes of major ancient works on Ancient Greek accentuation. In the epitome of Pseudo-Arcadius to Herodian’s *De prosodia catholica* (160.6–17), it

Host	Enclitic	Host+Enclitic	Host Accent	Host Final σ Weight ^a	# Clitic σ ^b	Results
πατήρ	τις	πατήρ τις	Final Acute	L/H	1/2	Host unchanged; no accent on clitic
καλῶς	ἐστι	καλῶς ἐστι	Final Circumflex	L/H	1/2	Host unchanged; no accent on clitic
μεγάλος	τις	μεγάλος τις	Penult Acute	L/H	1	Host unchanged; no accent on clitic
φίλος	ἐστι	φίλος ἐστί	Penult Acute	L/H	2	Host unchanged; acute on 2 σ of clitic
οἶκος	τις	οἶκός τις	Penult Circumflex	L	1/2	Acute on host final σ
κῆρυξ	τις	κῆρυξ τις	Penult Circumflex	H	1	Host unchanged; no accent on clitic
κῆρυξ	ἐστι	κῆρυξ ἐστί	Penult Circumflex	H	2	Host unchanged; acute on 2 σ of clitic
ἄνθρωπος	τις	ἄνθρωπός τις	Antepen. Acute	L/H	1/2	Acute on host final σ

Table 6.1: Accentual behavior of host + single enclitic combinations in Attic-Ionic

^a Here, H and L stand for “heavy” and “light”, respectively.

^b Enclitic lexemes in Ancient Greek never exceed two syllables in length.

is explicitly stated that words like φοῖνιξ and κῆρυξ receive no secondary acute on their final syllable when an enclitic follows (see the edition of Roussou 2018).²² However, words of this shape sometimes appear in manuscripts with a secondary acute (e.g., κῆρύξ τις in the scholia vetera to Il. 24.149, κῆρύξ τε in Thucydides 1.29.4, Φοῖνιξ φησι in the scholia vetera to Il. 1.261), just like words of the shape οἶκος. The doctrine as prescribed by Herodian is, on its face, less intuitive: since enclitic accentuation mostly only attends to the position of the high tone (no secondary acute unless the penult is circumflex or the antepenult is circumflex), one might expect for κῆρύξ-type and οἶκος-type words to behave identically. Cases like κῆρύξ τις

²² The relevant section of text (with Roussou’s translation) reads:

ήνικά εὐρεθῇ ἢ προηγουμένη λέξις ἢ προπαροξύτονος ἢ παροξύτονος τροχαϊκῇ ἀπὸ τόνου, ὡς ἐπὶ τοῦ ‘τυφθέντα’ (ἀπὸ γὰρ τοῦ τόνου ἐπὶ τὴν λήγουσαν ἀναποδίζων τροχαῖον εὐρίσκει) ἢ προπερισπωμένη μὴ ἔχουσα τὴν τελευταίαν θέσει μακράν, τότε ἀναπέμπουσι τῇ προηγουμένη λέξει, οἶον· ‘Ἀρκεσίλαός τε’, ‘Ὀμηρός ἐστι’, ‘γυναῖκές εἰσι’, ‘τυφθέντά τε’. πρόσκειται ‘μὴ ἔχουσα τὴν τελευταίαν θέσει μακράν’ διὰ τὸ ‘φοῖνιξ’ καὶ ‘κῆρυξ’. τούτων γὰρ προηγουμένων οὐ γίνεται ἔγκλισις· ‘φοῖνιξ ἐστί’, ‘κῆρυξ ἐστίν’ If the preceding word is proparoxytone, or is paroxytone and trochaic from the accent to the end, as in τυφθέντα (for one who goes back from the accent to the final syllable finds a trochee), or is perispomenon and doesn’t have its final syllable long by position, then it sends the accent back onto the preceding word, as in Ἀρκεσίλαός τε, Ὀμηρός ἐστι, γυναῖκές εἰσι, and τυφθέντά τε. The words ‘and doesn’t have its final syllable long by position’ are included because of the words φοῖνιξ and κῆρυξ. For when these precede there is no throwing back of the accent: φοῖνιξ ἐστί, κῆρυξ ἐστίν.

might be explained away as overgeneralization by later speakers who did not command the grammar natively and did not know every detail of the teachings on enclitic accentuation. On the other hand, without a systematic philological assessment of the problem, we cannot be wholly certain whether the distinction drawn by Herodian was faulty; in absence to the contrary, though, I assume that it is correct and must be accounted for.²³

To the cases in 6.1, worth further discussion is the claim in Herodian (i. 563 2–4) that words having an acute on the penult, but whose penult was heavy, indeed received a further acute on their final syllable (cf. Probert 2003: 148–9), e.g., ἄλλός τις, ἐνθά ποτε, rather than the alternatives of no change or acute on the second syllable of the enclitic. Probert (2003: 149) offers the plausible interpretation (with comparison to the behavior of tone in Lithuanian) that this subrule applied only when the penultimate syllable contained a coda segment that was [+sonorant], and thus capable of hosting tone.²⁴ That is, ἄλλός τις would be effectively equivalent to οἰκός τις. If Probert’s interpretation is correct, the consequences for our understanding of Attic-Ionic prosody as a whole would be important, since [+sonorant] codas (~ syllables containing them) should then generally be interpreted as contour-tone-licensing units, and that the orthographic acute on a penultimate syllable with a [+sonorant] coda could be understood as a circumflex, e.g., [ál̥.los].²⁵ Furthermore, for words with an acute antepenult and [+sonorant] coda like ἄνθρωπος [ántʰrɔːpos], the implication is that the H tone belongs to both morae of the syllable; if the H were linked only to the second mora of the syllable [an], then the [a] would not bear H, and to orthographically mark it as such might be surprising. The conclusion is then that an acute on a bimoraic syllable should not be understood as a rising tone LH, but rather as a level H (in agreement with Steriade 2014: 7); see 6.2.1.6 for further reasons to believe that an acute on a long vocalic nucleus was a level, not rising tone (*pace* Jakobson 1937 [1971] and many in his wake, e.g., Jasanoff 2017: 11).

The exact prosodic pattern found in sequences of multiple enclitics, in contrast, is debated. The standard view presented in many handbooks and grammars (cf. Probert 2003: 152–3), is that each enclitic in the sequence received an acute on its final syllable, with the exception of the last enclitic in the sequence, which received no high tone. The clitic sequence εἰ ποὺ τίς τινα (Thucydides 4.47.3), with proclitic εἰ followed by the enclitics ποὺ τίς and τινα, could exemplify this rule. Probert (2016), however, finds that the the following iterative rules (what she terms the “Revised Götting-Barrett System”) best accord with re-

²³As an aid to further investigation of the problem, I list here forms of κῆρυξ-type words that are printed in any editions used in the *Thesaurus Linguae Graecae* (Pantelia 2021) with secondary acute: κῆρύξ, διωρύξ, μεσηλίξ, ἀφῆλίξ, εὐῆλίξ, Ἰππώνάξ, κλιμάξ, μιλᾶξ, μεῖράξ, λαϊλάψ, Φοῖνίξ, χοῖνίξ. The occurrence of secondary acute in such cases is decidedly in the statistical minority, but at present it is impossible to say how much the production of medieval manuscripts and modern editions on this point has been influenced by Herodian’s doctrine.

²⁴One of the examples for this subrule given by Herodian, ἔστι τις (with non-enclitic ἔστι) shows a penultimate syllable with a [–sonorant] coda, but Probert suggests that “this rule applied in reality only to words in which the penultimate syllable had a liquid or nasal consonant (or possibly σ) after the short vowel.” Probert further mentions that West’s edition of the *Iliad* (West 1998) prints an acute on the final syllable in cases where the penult ends in a liquid, nasal, or σ, but does not when the penult ends in a stop (e.g., ὄφρα κε at *Il.* 2.440).

²⁵Under the interpretation that the syllable constitutes the relevant TBU in Ancient Greek, then syllables with [+sonorant] codas would be licit bearers of contour tones, not only syllables with long nuclei (i.e., long vowels and diphthongs).

marks on the accentuation of enclitics in the scholia to Homer, as well as the accentuation of enclitics in the Venetus A manuscript of the *Iliad*.

(15) Revised Göttling-Barrett System (per Probert 2016, with her examples)

Each time a further enclitic word is added to a an existing prosodic word:

- I. If the sequence preceding the new enclitic would have an acute or circumflex on its last syllable in isolation, this sequence retains its accent before an enclitic (with no change of an acute to a grave). No accent appears on the new enclitic.

a. /timé:n per/ → [timé:n per] ‘honor:ACC.SG though’

b. /kaí te/ → [kaí te] ‘and’

- II. If the sequence preceding the new enclitic has a circumflex on its second-to-last syllable, or an acute on its third-to-last syllable, it retains this accent and acquires an additional acute on its last syllable before an enclitic. No accent appears on the new enclitic.

· /ê: pe: / → [ê: pé:] ‘or where’ ⇒ /ê: pe: me/ → [ê: pé: me] ‘or where 1SG.ACC

- III. If the sequence preceding the new enclitic has an acute on its penultimate syllable, then...

- i. ...if this sequence ends in a trochaic (– ∪) pattern, it retains its accent and acquires an additional acute on its last syllable before an enclitic. No accent appears on the new enclitic.

a. /timé:n per/ → [timé:n per] ⇒ /timé:n per moi/ → [timé:n pér moi] ‘honor:ACC.SG though 1SG.DAT’

b. /kaí te/ → [kaí te] ⇒ /kaí te me/ → [kaí té me] ‘and 1SG.ACC’

- ii. ...otherwise, this sequence simply retains its accent before an enclitic. If the enclitic is monosyllabic it appears without an accent; if the enclitic is disyllabic it appears with an acute on its second syllable (or grave, if a non-enclitic word follows without intervening punctuation), or a circumflex in the case of τινόιν (DAT.DU) or τινῶν (GEN.PL).

· /kaí te/ → [kaí te] ‘and’ ⇒ /kaí te me/ → [kaí té me] ‘and 1SG.ACC’
⇒ /kaí té me p^hε:si/ → [kaí té me p^hε:sí] ‘and 1SG.ACC say:3SG.PRES’

Throughout the subrules of this system, the final enclitic in a sequence avoids bearing an high tone, except when it is disyllabic and the penultimate syllable of the preceding sequence bears an acute (III.ii). This fact is entirely in accord with the accentuation that results from simple orthotonic plus single clitic sequences shown in 6.1. As the analyses of enclitic accentuation below will show, the final or only enclitic in a sequence is subject to the avoidance of either high tone on adjacent syllables or stress clash. The difference in sequences of multiple enclitics is that high tone (acute) on adjacent syllables is apparently

permitted is permitted to arise just in case an existing sequence is of a trochaic shape (III.i): [kaí té me].

Why trochaic sequences preceding another enclitic should condition different accentual behavior is not immediately clear. On either a metrical or tonal account of enclitic accentuation, the shape of the newly added enclitic itself – whether heavy or light, mono- or disyllabic – should be more relevant than the sequence of heavy and light syllables immediately preceding the new enclitic. Again, as suggested by Probert (2003: 148–9), the underlying basis for the behavior of a “trochaic sequence” may be due to the possibility that syllables closed by liquids, nasals, and perhaps /s/ as well, in fact had circumflex intonation (i.e., could bear falling tone); one might even extend this assumption farther to cases like /timé:n per moi/, assuming that the final nasal of /timé:n/ can host a low tone, and thus the resulting [timé:n pér moi] would exhibit no clash of high tone. The treatment of /kaí te me/ → [kaí té me] appears more directly contradictory, since it would be theoretically possible to write a circumflex “χαῖ”; without any prescribed rule of changing an acute on a final syllable to a circumflex when followed by a light enclitic, the writing accentuation of the orthotonic form might have been retained by default. Note furthermore that, if the Lex σωτήρα were presumed to apply within enclitic sequences, not just within orthotonic words, then an output [kaî te] would be expected.

Especially if one presumes that the Lex σωτήρα operated in cases like /kaí te/ and /timé:n per/ to yield [kaî te] and [timé:n per], then the addition of a further enclitic to give sequences [timé:n pér moi] and [kaî té me] would simply be parallel to cases like [oîkós tis] – an further high may occur on the syllable following a syllable with falling tone. However, under the intonational account of the Lex σωτήρα offered by Gunkel (2013) (see further under 6.2.3 below), if the tonal assignments in a sequence of enclitics were calculated in parallel (i.e., from /kaí te me/ to [kaî té me], with no intermediate step [kaî te] alone), there would be no opportunity for a low boundary tone, L_{ω} % to condition retraction of the high tone, because the second-to-last enclitic would never come into consideration for hosting a low boundary tone. Given that recessively accented orthotonic words must be assigned the recessive accent prior to the addition of any enclitics, it is credible to hold that the calculation of prosody in enclitic sequences proceeds derivationally, with a new optimal output being calculated with the addition of each further enclitic, taking over the output from the preceding derivational step. Similarly, the occurrence of multiple high tones within the domain of single lexical words perhaps shows that CULMINATIVITY is violable in the grammar applicable above $\omega_{[+min]}$.

6.2.1.6 Word-Level Tone and its Representation: Typological and Empirical Arguments concerning the Nature of “Acute” and “Circumflex”

That every orthotonic word of Attic-Ionic without any clitics exhibits at least one and at most one H tone is virtually undisputed; in 6.2.2 and 6.2.3, I systematically discuss how this fea-

ture might be treated through constraints on the distribution of stress and tone together.²⁶ However, besides a word-level H, further tones have been posited as integral or preferred components of Attic-Ionic word prosody. Furthermore, given the assumption of tonal autosegments, a tone-bearing unit (i.e., mora or syllable) must be specified. This section considers typological arguments and empirical evidence concerning the existence, nature, and representation of word-level level tones, with particular reference to the tone-to-tone mappings found in settings of Ancient Greek music.²⁷

All of the preceding discussion has adopted the assumption that syllables bearing an acute or circumflex accent had higher pitch than surrounding syllables within the same prosodic word. This assumption follows from descriptions of accent provided by grammarians in antiquity (see, in brief, Probert 2003: 3–15, and in deeper detail, Probert 2006b: 15–45), the difference between syllables bearing an acute or circumflex and unaccented syllables is characterized primarily as one of pitch: accented syllables in general exhibit higher pitch. The terms used by the ancient grammarians, ὀξύς ‘acute’ (lit. ‘sharp’) and ὀξύβαρεια ‘circumflex’ (lit. ‘sharp-deep’), point towards a pitch peak relative to the unaccented syllables, which are labelled βαρύς ‘grave’ (lit. ‘heavy, deep’), as are acutes in final syllables that become graves.²⁸ That pitch on syllables marked with an acute or circumflex was relatively higher is further confirmed by tone-to-tone mappings in settings of Ancient Greek music: in the Delphic hymns (documents 20 and 21 of Pöhlmann and West 2001), the single accented syllable of a word is never set to a note lower than any other syllable in that word (see West 1992: 197–200, Devine and Stephens 1994: Ch. 4, Probert 2003: 19–21, Probert 2006b: 47–8).

For most types of syllable structures, an appropriate tonal representation in terms of high (H) and low (L) tones for accented syllables is fairly clear. For the circumflex, which is restricted to syllables containing two vocalic morae (VV; i.e., syllables with a long vowel or diphthong as nucleus), an autosegmental–metrical representation as a high–low (HL) sequence straightforwardly captures the falling pitch universally assumed to characterize it. This HL tonal sequence may either have one tone each linked to one mora, as most descriptions, going back to Jakobson 1937 [1971] (applying to Ancient Greek notions developed in Jakobson 1931 [1962]), seem to assume, or two tones linked to a single syllable; compare the representation in Jasanoff 2017: 11, shown in 16a., or its alternative with the syllable as a TBU in 16(b).²⁹

²⁶The one real exception to seeming obligatoriness and culminativity of H depends on the interpretation of words whose acute would fall on the final syllable and which have instead a grave accent when immediately followed by another orthotonic word in a sentence. If their H is deleted or never associated, then culminativity of H tone would seem to be violated under some circumstances.

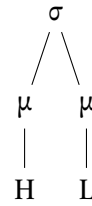
²⁷I am deeply indebted to Dieter Gunkel for sharing his corpus of Ancient Greek music (based on West 1992 Pöhlmann and West 2001), his unpublished work, and extensive discussion on tone-to-tone mappings in the Delphic hymns. Discussion on the nature of the circumflex below is based on joint work with Dieter Gunkel in progress.

²⁸Some readers may be more familiar with the term περισπόμενος ‘bent, round’ with reference to the circumflex, but this term technically refers, in origin, to the shape of the graphical mark employed for the circumflex, not a description of the pitch contour of the circumflex accent; see Allen 1968 [1987]: 125, fn. 25.

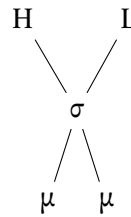
²⁹Jasanoff’s autosegmental–metrical representations in fact omit any L tones, apparently tacitly assuming that the L tones are inserted by default rules in the intonation of prosodic words. Such a representation might well be appropriate for cases of genuinely underlying H tones, if underlying differences in the association of H

(16) Autosegmental–Metrical Representation of HL Circumflex

a. TBU = μ



b. TBU = σ



In syllables containing but a single vocalic mora, meanwhile, only an acute is permitted, which may be represented as a lone H tone linked to the single mora or to the syllable node. Uncertainty persists, however, concerning the representation of acute accents on VV-syllables. Although most current descriptions (e.g., Allen 1968 [1987]: 125, Szemerényi 1996: 73, Probert 2003: 15–16, Probert 2006b: 58, Jasanoff 2011: 11) imply or explicitly provide a representation for the acute as in 17, in which the first mora is linked to a L tone and the second to a H, an alternative, shown in 18, in which a single H spreads across two morae, or is linked to a syllable dominating two vocalic morae.³⁰ Against a bitonal representation of the acute, Steriade (2014: 7) takes the acute to be a level high tone (“HH = acute (ὦ)”) for reasons motivated in part by the accentuation enclitics,³¹ while Devine and Stephens (1994: 444–5) seem to view both circumflex and acute as involving a high-to-low pitch movement (“it is the High-Low movement of the acute or circumflex that triggers catathesis [downstep]”).

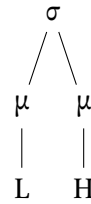
tones to TBUs are assumed to be responsible for (some) instances of contrastive acute and circumflex on the surface. Here, I give what I take to be a surface representation for the circumflex.

³⁰Jasanoff’s explanation of his representations implies that he believes that, in cases where a H tone is underlyingly linked to the second of two vocalic morae, the surface output will be a rising tone (“On a long vowel or diphthong, there are two theoretical possibilities, depending on whether the tonal peak occurs relatively early or relatively late in the timing of the syllable. In the latter case, the resulting *rising contour* [emphasis mine] is phonologically and notationally identified with the acute accent on short vowels”).

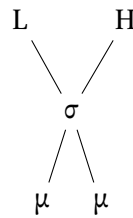
³¹Specifically, Steriade reasons that, in words containing an acute penult with long-voweled final syllable, e.g., δαίμων [daímɔ:n], the addition of an acute on the final syllable in an enclitic sequence ought to be permitted if the acute were a LH sequence and avoidance of adjacent high tones on the tonal tier partly governs enclitic accent. That is, a rising contour [LH] ought to allow ^Xδαίμων ἐστι [daímɔ:nesti], contrary to fact (δαίμων ἐστι).

(17) Autosegmental–Metrical Representation of LH Acute on Bimoraic VV-Syllable

a. TBU = μ

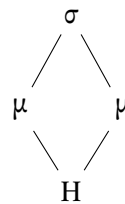


b. TBU = σ

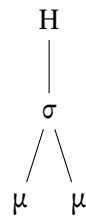


(18) Autosegmental–Metrical Representation of H Acute on VV-Syllable

a. TBU = μ



b. TBU = σ



In the following, I first survey the typology of tone and similar “pitch-accent” systems, which casts doubt on a LH representation for the acute, then muster empirical evidence from musical settings in two Delpic Hymns (i.e., the Paeans of Athenaios and Limenios; Pöhlmann and West 2001 [=DAGM], documents 20 and 21) that speaks decisively against such a representation.

The Nature of the Acute If the surface autosegmental–metrical representations of circumflex and acute are accepted, then Ancient Greek would know a two-way contrast between

falling tone (circumflex) and rising tone (acute), at least on VV-syllables. As such, Ancient Greek would then possess weight-sensitive contour tones, comparable to Hausa, though the latter does not permit rising tones (see Gordon 2001: 436).³² However, the presence of contour tones in a language usually presupposes the presence of level tones, and languages with just two (surface) tones typically contrast the level tones, [H] and [L] on all syllables of all structures, not only a rising tone and a falling tone (Yip 2002: 24–5). Hence, the absence of a level H tone on VV-syllables in Ancient Greek might be somewhat surprising.

Conversely, if the acute on a VV-syllable is taken to represent a level H tone instead, the Ancient Greek situation would fit within the broader typology. The survey of 198 (tone) languages in Zhang (2002) found 39 without any rising tones, but only 3 lacking in a surface falling tone. Furthermore, the positional limitation of the Ancient Greek circumflex to final syllables (modulo the $\sigma\omega\tau\eta\rho\alpha$ Rule) follows from the sensitivity of contour tones to syllable duration, given final syllables are often subject to phonetic lengthening effects; in some languages, this additional duration provides the necessary conditions for licensing of contour tone (e.g., Kikuyu; cf. Clark 1983 and Zhang 2004: 928–31). Kubozono (2017), in a survey of Japanese dialects, finds that syllable-level rising tones are permitted only in Nagasaki Japanese; in other dialects, where pitch accents, default tones, and boundary tones might create a rising tone, it is repaired. For instance, in the Tōkyō dialect, the boundary tone $\%LH_\omega$ regularly deletes its L component in words whose first syllable is bimoraic, with a V: or VR nucleus that could host multiple tones; see examples in 19. Similarly, on the view that loanwords regularly assign H* to the antepenultimate mora, if the antepenultimate mora is the second mora of a syllable, H* is retracted by one mora to produce a falling tone rather than rising tone (cf. also Kubozono 1989 on this point).

- (19) Boundary Tone $\%/L\ H/ \rightarrow [\%H]$ in Tōkyō Japanese
 - a. [tó:.kʲo:] ‘Tokyo’, not ^X[tǒ:.kʲo:]
 - b. [kaí.dan] ‘staircase’, not ^X[kaǐ.dan]
- (20) Retraction of H* from Antepenultimate to Preantepenultimate Mora
 - a. [rón.don] ‘London’, not ^X[roñ.don]
 - b. [saí.da:] ‘cider’, not ^X[saǐ.da:]

Finally, worthy of comment are features of word-level tone in Serbo-Croatian and Lithuanian, which, like Ancient Greek, have been traditionally taken to exhibit a two-way distinction between falling and rising tone in stressed syllables. Precisely these descriptions seem to have informed Jakobson’s (1937) analysis of the acute on VV-syllables as a rising tone. Among the Serbo-Croatian dialects, clearer phonological description of what entities stand behind the labels “rising” and “falling”, as well as better instrumental measurements of syllable prosodies, indicate that syllable level rising-tone probably does not exist. In Standard

³²The definition of contour tone employed here is the occurrence of multiple tonal targets within a syllable, regardless of whether those tonal targets may be phonologically decomposed or not, and regardless of whether the syllable or mora is the relevant TBU for the language. Note that Yip (2002: 28) considers the “true contour tone languages” to be just those languages in which the contours constitute “single phonological objects” or languages in which multiple tones are permitted to attach to a single TBU.

Neo-Štokavian Serbo-Croatian, for instance, the terms “falling” and “rising” refer to a pitch contour spanning two syllables, where “falling” contour has the peak on the earlier syllable, “rising” contour has peak on the later syllable (cf. the recent analysis in Zec and Zsiga 2022). Meanwhile, the detailed acoustic data collected by Langston (2006: 30–70) (see especially p. 40–2) from seven different Čakavian dialects points unambiguously to a distinction between a level high (H = “Čakavian acute”) and falling (HL) on stressed syllables containing long vowels, with no rising tone being present.³³ Finally, as far as Lithuanian is concerned, exactly what acoustic correlates reliably distinguish “acute” (traditionally, falling tone) and “circumflex” (traditionally, rising tone) remains unclear. Instrumental measurements reported in Pakerys 1982, Dogil 1999, and Kelly 2013 point to considerable interspeaker variation, though agree on one thing: Fo contours do not distinguish the two accent types; rather, syllable duration or the timing of an intensity peak sometimes emerge as significantly different.

In sum, the considerations above indicate that a contrast between H (level high) and HL (falling) on VV-syllables for Ancient Greek would fit well into the typological distributions of contour tone; a contrast between LH (rising) and HL (falling), on the other hand, if not impossible, would be unusual. More directly telling is the testimony from Ancient Greek music, under the assumption that pitch movement in the music strove to match pitch targets of the phonological tones (cf. Devine and Stephens 1994: 162–70 on such a tendency cross-linguistically, or McPherson and Ryan 2018 for a detailed study of the relationship between tone and tune in the music of Tommo So). To this end, I examined all instances of melisms (i.e., instances where a single syllable in the text is set to multiple pitches in the tune) on VV-syllables in two of the longest and most complete examples of Ancient Greek non-strophic vocal music: the Paeans of Athenaios and Limenios (DAGM 20 and 21). This data admits of two major conclusions:

- first, syllables with circumflex accent are significantly more likely to be set to a melism at all than acute, grave, or unaccented syllables, which are set to melisms at approximately the same rate; from this fact, I infer that the circumflex syllables contain multiple phonological tonal targets, while acute, grave, or unaccented syllables contain at most one.
- second, while circumflex melisms are consistently (perhaps exceptionlessly) set to falling melisms, acute and unaccented VV-syllables may be set to either rising or falling melisms; from this fact, I infer that the fully specified HL contour of the circumflex is compatible only with a falling melism, while acute, grave, or unaccented syllables are compatible with either a rising or falling because they are specified for but a single tone.

DAGM 20 and 21 contain 57 reasonable secure examples of a melism set to a VV-syllable,

³³Langston himself is somewhat cautious in interpreting his results with respect to traditional descriptions, such as that given in Belić 1909; he allows that an earlier rising tone may have become a level high tone.

	Circumflex	Acute	Unaccented	Grave	SUM
Melism	28 (~64%)	6 (18.75%)	21 (~16%)	2 (~15%)	57 (~26%)
Non-melism	16 (~26%)	26 (81.25%)	109 (~84%)	13 (~85%)	164 (~74%)
SUM	44	32	130	15	221

Table 6.2: Proportion of Melismatic and Non-Melismatic VV-Syllables in DAGM 20 and 21 by Accent

	Circumflex vs. Acute	Circumflex vs. Grave	Circumflex vs. Unaccented	Acute vs. Grave	Acute vs. Unaccented
χ^2-Value	13.337	9.4024	34.326	0.0019615	0.0077885
p-Value	< 0.001	0.002167	< 0.001	0.9647	0.9297

Table 6.3: Results of Pairwise χ^2 -Tests Applied to Data from Table 6.2

alongside 164 instances of non-melismatic VV-syllables.³⁴ As the values in Table 6.2 indicate, only about one-fourth (26%) of all VV-syllables are set to melisms, with an apparently higher rate of melismatic setting being applied to circumflexes (64%), but an approximately similar lower rate (less than 20%) in VV-syllables of any other accent types (cf. already Devine and Stephens 1994: 173 [“the circumflex is most frequently set to a melism”], but without any precise descriptive statistics).

The relative rates of melismatic versus non-melismatic setting are statistically significantly different when the circumflex is compared with any other accent type, as pairs of Pearson’s χ^2 -test applied to data from Table 6.2 demonstrate ($p < 0.0022$ in all cases). Conversely, the rates of melismatic setting are not statistically significantly different between acute or grave and unaccented syllables ($p > 0.9$ in all cases). These results are reported in Table 6.3.

In 44 of the 57 VV-melisms considered above, the exact pitch for both notes of the melism can be read. From these 44 datapoints, trends in terms of the directionality and interval of the melism emerge: first, the average interval of the melism on circumflex syllables (c. 3 semitones) is larger than on acute, unaccented, or grave VV-syllables (1.5-2 semitones); second, while circumflex melisms are nearly or perhaps always set to a falling melism, acute/grave and unaccented syllables exhibit some variation, with both varieties showing a rising melism in about two-thirds of cases (4/6 and 11/17, respectively), or falling in one third. No difference in proportion of rising vs. falling melism exists between unaccented on the one hand or acute/grave on the other ($p = 1$, Fisher’s Exact Test). These data are summarized in Table 6.4. Although the small total number of acute and grave melisms limits the confidence with which conclusions can be drawn, the data at least do not unambiguously speak in favor of a bitonal pitch target on acute and grave VV-syllables: the average interval of the melism in semitones of such syllables is closer to the average interval of the melism on unaccented syl-

³⁴Of these 57 melisms, in 12 cases it is evident that the syllable was set to a melism, though one of the pitches cannot be discerned; in one case, both pitches cannot be read, but the inscription nevertheless points to the presence of two notes.

	Circumflex	Acute	Unaccented	Grave	Acute + Grave
# Rising	1 ^a	2	11	2	4
# Falling	20	2	6	0	2
Mean Interval	37.53 Hz; 3.245 semitones	-14.535 Hz; 2.25 semitones	-4.946 Hz; 2.176 semitones	-21.295 Hz; 1.5 semitones	-16.788 Hz; 2 semitones
Median Interval	41.63 Hz; 3 semitones	-0.39 Hz; 1.5 semitones	-14.69 Hz; 2 semitones	-21.295 Hz; 1.5 semitones	-13.105 Hz; 1.5 semitones

^aThis single instance of a circumflex syllable set to a rising melism occurs at DAGM 20.18 and may well be an error. See the apparatus to line 20 in DAGM: 66, as well as West 1992: 291, who considers the conjecture of Reinach 1909: 155, that the Db is rather an A to be “probable”. This token would thus involve a fall of four semitones, rather than a rise of four semitones. Pöhlmann (1970: 140) adopts a general rule that melisms on circumflex syllables should always be descending.

Table 6.4: Directionality and Interval of Melism in VV-Syllables by Accent Type

lables, and the rate of variation in directionality is higher than among circumflex syllables (33% falling melisms on acute/grave versus only 4.8% rising melisms on circumflex [assuming that the single instance of a rising circumflex melism is not an error]). Furthermore, even though the likelihood that an acute or grave VV-syllable be set to a rising melism is statistically significantly greater than for a circumflex melism ($p < 0.01$, Fisher’s Exact test), the same applies to differences between circumflex and unaccented VV-syllables ($\chi^2 = 12.973$, $p < 0.001$, Pearson’s χ^2 -test).

In conclusion, when taken together, both typological considerations and evidence from tone-tune settings on VV-melisms in DAGM 20 and 21 most readily support the assumption that the Ancient Greek acute was consistently realized as a level high tone, H, on all syllable types.

The Low Boundary Tone and the Grave Besides the seemingly obligatory high tone in content words of all shapes, sizes, and structures, the tendency of this H to occur, by default, away from the right edge of the prosodic word on the penultimate or antepenultimate syllable has inspired speculation as to the presence of other tones that might intervene between the H and word’s right edge. Beginning with Allen (1973: 231–4; 246), followed then by Devine and Stephens (1994: 154; 180) and subsequently adopted in some form in many phonological analyses of the recessive accent (see below), a “contonation” consisting of a pitch peak, H, plus a fall to a low pitch, L, has been posited. Devine and Stephens (1994: 154; 180; 189; 191) go further, and posit that, in addition to the HL “contonation”, a low boundary tone, L%, that would preferentially associate with the rightmost TBU (in their view, mora), of the prosodic word. Based on patterns of pitch movement in musical settings, Devine and Stephens (1994: 189) propose a general, default accentual excursion with rise to H, sharp fall from H to L in the contonation, then a further slight fall from L to L%; see Figure 6.1.

Typologically, the regular occurrence of a boundary tone at the edge of a maximal prosodic word (encompassing, in Ancient Greek, one orthotonic word and any enclitics) would not be wholly surprising. Turkish may even, by default mark both the initial and final syllable of a

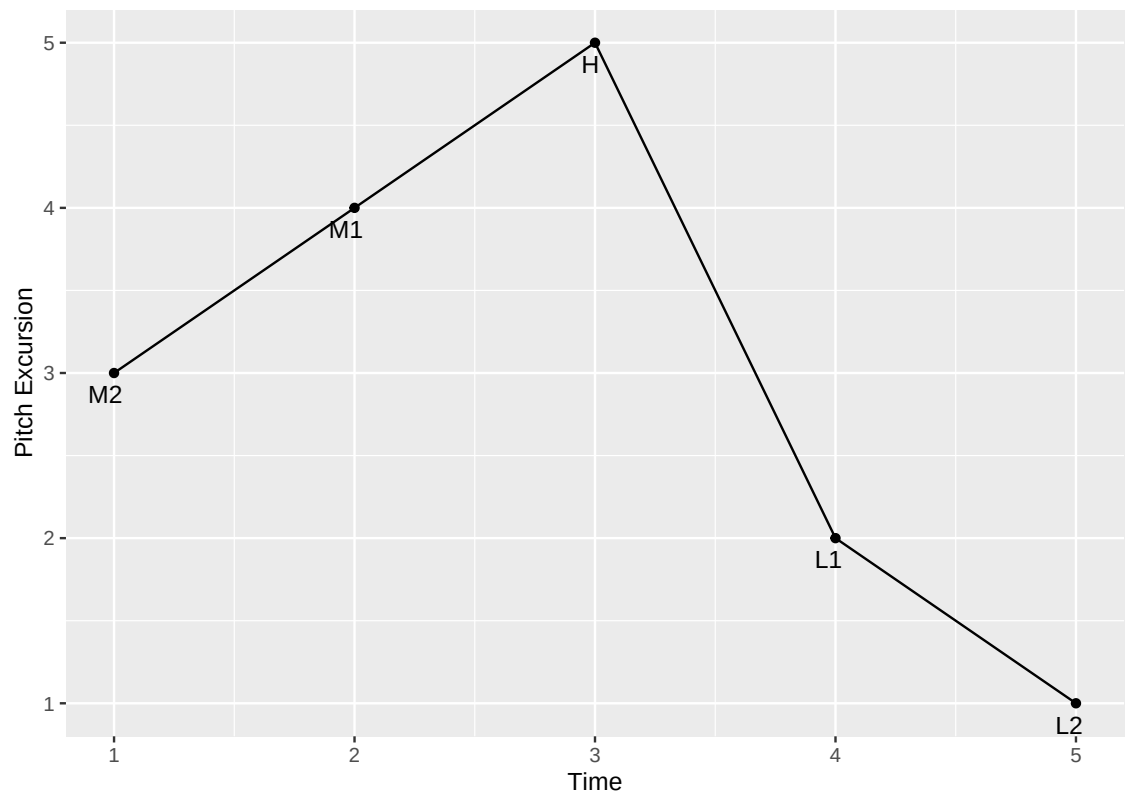


Figure 6.1: Default Ancient Greek “Accentual Excursion” (after Devine and Stephens 1994: 189)

prosodic word with boundary tones: an essentially obligatory %L at the beginning (Ipek and Jun 2013) and, if the analysis of Özçelik (2014: 234–8) is correct, a H% at the end (in prosodic words lacking lexically specified stress). Compare further the obligatory initial boundary tone, %LH standardly posited for Tōkyō Japanese mentioned above.

In the context of Ancient Greek, the assumption of a low boundary tone potentially buys a unified explanation for several facts. For one, as argued by Gunkel (2015), the $\sigma\omega\tau\eta\rho\alpha$ Rule (only falling tone may occur on a VV-penult before a final with a short vowel) may be readily understood as avoidance of adjacent H and $L_\omega\%$ on the tonal tier, in order to allow for adequate time to fall from the Fo peak and reach the trough set by the $L_\omega\%$; Gunkel refers to parallel instances of “temporal shifting” of tonal targets conditioned by boundary tones in Chichewa (Myers 1999) and Chickasaw (Gordon 2008).³⁵ In addition, the conversion of an acute final syllable in an orthotonic word alone or on the last syllable of an enclitic sequence to a grave (so-called “lulling”; cf. Probert 2003: 16–18; 36–8 or Goldstein 2015: 57–9) might constitute the lowering or deletion of a H (acute) when occurring or competing to occur on the same syllable as $L_\omega\%$.³⁶

Once again, the tone-to-tune associations in Ancient Greek music may potentially offer further support for the assumption of the $L_\omega\%$. Gunkel (2022) examines the position of a *trough*, that is, the lowest pitch, between two H tones on adjacent orthotonic words or clitic sequences without any intervening large prosodic boundaries (e.g., phonological or intonational phrase, plausibly signaled by punctuation) in the two Delphic hymns discussed above (DAGM 20 and 21). Gunkel finds that about 72% of troughs (43/60) occur on the final mora or syllable, a figure which drops to c. 61% (22/36) when all cases are excluded in which the position of the trough on the final mora/syllable might be attributed to some other factor.³⁷ Furthermore, in cases in which the trough does not fall on the final syllable/mora, it is then nearly always on the penultimate syllable/mora.

As to the question of how a potential $L_\omega\%$ interacts with final acute syllables, Devine and Stephens (1994: 180–4) argued, again on the basis of pitch contours in Ancient Greek music, that syllables with a grave have a lower pitch target than syllables with an acute, but a higher pitch target than unaccented syllables (i.e., syllables without acute, circumflex, or grave). On this basis, they concluded that the grave represents a lowered high pitch, distinct from either acute or unaccented syllables. Given that an acute (H) plus $L_\omega\%$ sequence aiming to associate with a final syllable evidently does not produce a contour tone similar to a circumflex (HL), one might suspect that Devine and Stephens’ “lowered High tone” constitutes a Mid tone (M) resulting from blending the H and $L_\omega\%$ targets. However, Gunkel and Ryan (2022), reassessing a trisyllabic sequences σ + Grave + σ in DAGM 20 and 21, conclude

³⁵Differently, Steriade (2014) considers the $\sigma\omega\tau\eta\rho\alpha$ Rule to be an epiphenomenon, arising from a default realization of stress on a penult with a VV nucleus as falling tone; see further under 6.2.2.3 below.

³⁶The lowering hypothesis has generally been more widely accepted in the scholarship on Greek prosody: compare Ehrlich 1906: 573, Ehrlich 1912: 250–62, Allen 1973: 245–8, Devine and Stephens 1994: 180–4, and Probert 2006b: 59. The alternative view, that the grave represents deletion of the acute (H tone), is represented in Meillet 1905–6 and Grammont 1948: 388–90.

³⁷The particular cases excluded include instances in which another type of $L(*)$ tone might fall on the final mora/syllable (e.g., a circumflex on the final syllable), as well as cases in which an unaccented final syllable is “sandwiched” between two syllables with a high tone.

that the behavior of the grave is not statistically distinguishable from the behavior of unaccented syllables, which instead speaks in favor of the grave representing deletion of a High tone. Deletion of H would also seem to accord with the perspective of the ancient grammarians, who designated both the grave as substitute for acute and unaccented syllables βαρύς [barús] ‘low’. Note, however, that early papyri found with prosodic indications usually employ the following practice: marking all unaccented syllables and the grave symbol, but the “lulled” final acute is not marked at all, perhaps suggesting that it was perceived as different from either the high acute and the low unaccented syllables (see Probert 2015: 936; 938).³⁸ On balance, deciding between the competing hypotheses for the nature of the grave on an empirical basis is not possible at present.

On the whole, although the existence of a word-level boundary tone, $L_{\omega}\%$, in Ancient Greek (Attic-Ionic specifically) cannot be regarded as wholly certain, convergent evidence from accentual phenomena and tendencies in tone-tune mappings make its presence at least plausible. Perhaps most importantly, assuming its presence may facilitate an overall analysis of Attic-Ionic word prosody, as argued in 6.2.2 and 6.2.3 below.

The Tone-Bearing Unit and the Derivation of the Circumflex A basic assumption in virtually all work on Greek prosody, including work that argues for the presence of word-level stress alongside tone, is that the mora (μ) constitutes the tone-bearing unit, i.e., the level of prosodic structure to which a phonological tone must “dock”. The major alternative would be to treat the syllable (σ) as the host for phonological tones. This analytical choice is motivated primarily by analysis of the circumflex: given that the circumflex constitutes the apparent only syllable-level contour tone, HL, it can be analyzed as consisting of two level tones, H and L, if the TBU is the mora; furthermore, under the assumption that L tone is inserted on all moras not associated with an H, at least some instances of circumflex can be treated as arising from specification of H on the leftmost mora of a syllable with two vocalic morae (cf. 16a.) above. This choice follows typical reasoning surrounding the choice of TBU in phonological analysis: if monomoraic syllables are permitted to host maximally one tone, but bimoraic syllables (or certain types of bimoraic syllables) may host two tones, then the TBU must be the mora, not the syllable (cf. Hyman 1985, Yip 2002: 73). Since at least vowels containing a long vowel or diphthong evidently contrast in their capacity to host multiple tones on the surface, but – supposedly – syllables containing only a short vowel do not (i.e., can only exhibit the acute), reference to the mora would seem sensible.

Three arguments, one theoretical, one empirical, and one typological may be mustered that throw the place of the mora as the TBU of Ancient Greek into question. First, one line of research into the distribution and analysis of contour tone generally (Zhang 2002, Zhang 2004, Kaplan (2020)) questions the appropriateness and adequacy of reference to the mora as TBU. Specifically, Zhang argues that contour tone is sensitive to the duration of syllable rime phonetically, rather than the mora count of a syllable phonologically; the relatively

³⁸Specifically, Probert (2015: 936) states: “In accented papyri earlier than the third century AD, the most normal practice is to avoid writing any accent on the final syllable of an oxytone word in the sentence—as if the pitch peak is not striking enough to merit an acute sign but not weak enough to merit a grave.”

fine-grained distinctions that languages may observe in terms of a syllable's capacity to host contour tone (see the survey in Gordon 2001) bears this point out. Kaplan, in turn, analyzes the tone systems of Adhola (Nilotic) and Ikalanga (Bantu) as cases where the distribution of contour tone would satisfy the usual conditions motivating the selection of the mora as TBU, but in which tone spreading refers to the syllable instead. Since the contour tone distribution can be motivated by reference to syllable duration instead, reference to the mora for treating tone in Adhola or Ikalanga is superfluous. On the empirical side of Ancient Greek, meanwhile, that the circumflex – which is indeed restricted to bimoraic syllables – is unique in having multiple phonological tones associated within a single syllable is uncertain. If, for instance, the grave accent is correctly understood as H followed by L% on a word-final syllable, the fact that the grave can occur even on short vowels with no coda consonants (e.g., ποδι [podì]) would indicate that syllable-level contour tone is not even restricted to heavy syllables (even if strongly dispreferred).³⁹ Finally, if the prosody of Attic-Ionic indeed includes stress as a component of its word prosody, as is assumed by many analysts (Allen 1973, Steriade 1988, Golston 1990; differently Ito and Mester 2017, Revithiadou 2018), then, given that stress is a property usually attributed exclusively to syllables (compare 1.3.3 and 2.1), one might expect for stress and tone to interact at the level of the syllable.

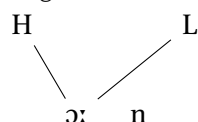
In light of the above considerations, whether the mora or the syllable should be regarded as the TBU of Ancient Greek must be considered uncertain. However, in the event that the syllable is taken as the TBU, deriving at least a subset of surface circumflexes presents complications. Unproblematic would be circumflexes induced by the σωτήρα Rule, since such cases might be handled as transformations of an underlying acute into a circumflex and/or assignment of Allen's (1973) entire HL "contonation" to a single syllable with VV-nucleus. Similarly unproblematic would be circumflexes transparently due to post-lexical vowel contraction (cf. 6 and 11 above). Less straightforwardly explained without expanding the inventory of possible lexical prosody from just the H* considered above for roots like /pód-/ 'foot' and inflectional suffixes like GEN.SG /-ós/, however, is the genitive plural suffix as seen in monosyllabic stems, e.g., ποδῶν [podô:n], as well as the GEN.DAT.DU suffix -οῖν [-oîn]. Namely, if the GEN.PL suffix has the segmental UR /-ɔ:n/, three motivations of its surface circumflex are then conceivable: a lexical L tone, transformed into a HL tone; a lexical HL tone; or some lexical prosodic property that causes the HL "contonation" to associate to this syllable. Under any of these three interpretations, an additional prosodic property would need to be allowed into lexical entries. One perhaps overly facile escape from the assumption of additional lexical prosody would be to interpret both endings as containing multiple underlying vowels, with the first associated to a lexical H, thus yielding circumflex after vowel contraction. Under such an interpretation, the lexical entry for the GEN.SG ending could look as in

³⁹If indeed both circumflex and grave syllables contain at least two tonal targets, one must assume that the HL of the circumflex on a VV-syllable (e.g., ποδῶν 'foot:GEN.SG') and the H+L% on a VV-syllable (e.g., σωτήρ 'savior:NOM.SG') must result in (phonetically) different pitch profiles; otherwise one might have expected for both types to have been seen as prosodically identical. Alternatively, the grave on final VV-syllables might reflect the phonological knowledge that such a syllable would present as acute when followed by an enclitic (i.e., non-final in the ω_{max}, e.g., τμήν περ) or at the end of an (maximal) intonational phrase (possibly due to a H% boundary tone).

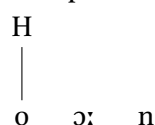
21b. rather than a.⁴⁰ Given segment input like /pod-oɔ:n/, specifying the inflectional suffix as bearing a lexical tone would, in fact, not even be necessary, since the presence of the three underlying potential syllable peaks with a long final would allow for application of the recessive accent in a way that would place H tone on the second-to-last vowel, thus virtual (word-level) [podóɔ:n] → (post-lexical) [podô:n].

(21) Conceivable Lexical Entries for the GEN.PL Ending -ôv

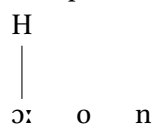
a. Single Vowel



b. Multiple Vowels: [ǞV:]



c. Multiple Vowels: [V:Ǟ]



While posting a bivocalic sequence for the genitive plural suffix specifically (and genitive/dative dual as well) is abstract, it takes a free ride on processes that are independently justifiable and do not require recourse to abstract representations, while avoiding the assumption of additional underlying contrasts. Furthermore, while no justification for a plausible synchronic representation, both the genitive plural and genitive/dative dual suffixes of Greek continue inherited disyllabic endings, GEN.PL */-oHǝm/ and GEN/DAT.DU */-oǝsin/ (see also discussion of the Vedic genitive plural ending at 4.3.5.1), and for the latter, a disyllabic form -oĩv [-oĩ.in] is directly attested in Homer (e.g., ποδοĩv, *Od.* 16.6; see Rix 1976 [1992]: 156–7; 160). The fact of being originally disyllabic at least provides a historical explanation as to why these two inflectional suffixes exceptionally present a circumflex; the synchronic analytical choice is merely whether to directly encode the tonal property or to recapitulate the historical vowel contraction.

The particular peculiar pattern of high tone mobility in several monosyllable nouns known to have had a historically disyllabic stem might offer further support for treating the GEN.PL and GEN/DAT.DU as synchronically underlyingly bivocalic. Consider the paradigm of παĩς [paĩs] ‘child’ in 22: as with other monosyllabic stems (cf. 7 above), genitive and dative endings attract the H tone, but not the GEN.PL or GEN/DAT.DU, where instead an acute appears

⁴⁰One of the two vowels in option b. would need to be /ɔ:/, in order to yield the contraction product [ɔ:] on the surface; an underlying sequence /-oon/ would contract to [-o:n].

on the stem. Its stem must be reconstructed as *[pawid-], making it originally disyllabic, of the prosodic shape $\sigma \sigma$ when followed by a vowel.⁴¹

(22) Attic-Ionic Paradigm of παῖς [paĩs] ‘child’: Partial H Mobility

	SG	DU	PL
NOM	παῖς	παῖδε	παῖδες
ACC	παῖδα	παῖδε	παῖδας
GEN	παιδός	παιδοιν	παίδων
DAT	παιδί	παιδoin	παισί

If the entire paradigm of nouns like παῖς exhibited recessive H tone on the stem, like all other polysyllabic nominal stems, an account would be easy: the stems are underlyingly polysyllabic (e.g., /pa.id-/), and thus do not permit mobility of H to inflectional endings (compare /murme:k-/ ‘ant’ or /onuk^h-/ ‘nail’, with DAT.SG μύρμηκι [múme:ki] and ὄνυχι [ónuk^hi], respectively). Similarly, if we found GEN.PL ^Xπαίδων, the behavior of H tone would be identical to that of normal monosyllabic stems. Such “partial mobility” indicates that the GEN.PL and DAT.PL endings have some property that causes a recessive H tone to be preferred to high tone on the ending. Perhaps the confluence of an underlyingly polysyllabic stem that contracts to a diphthong and an underlyingly bivocalic inflectional suffix enables the pattern of “partial mobility”. While this phenomenon is not easily brought into line with an account of regularly monosyllabic H tone mobility and recessive accentuation, it may, at least, constitute further evidence of the peculiar prosodic status of the GEN.PL and GEN/DAT.DU endings.

6.2.1.7 “Pitch Accents” and Metrical Stress: Towards the Typological Position of Attic-Ionic Prosody

A standard assumption concerning the prosodic typology of Ancient Attic-Ionic Greek is that it possessed a “pitch-accent” system for building prominence relations at the level of the prosodic word. Probert (2003: 3), at the outset of her general guide to Ancient Greek accentuation simply declares: “The accent of ancient Greek was a **pitch accent**.” Similarly, Gunkel (2014: 7) states: “In this system [viz. the accentual system of Ancient Greek texts], phonological, morphological, and lexical factors conspire to associate a **pitch accent** with one syllable of each lexical word.” Neither Probert (2003) nor Gunkel (2014) specify what properties of Attic-Ionic allow the language’s prosody to be so classified; they may have in mind some traditional assumption that “pitch” as opposed to “stress” accent is a matter of phonetic substance (the former realizing abstract “accent” as a pitch excursion alone, the latter realizing abstract “accent” through a combination of pitch, duration, and amplitude); compare the discussion in 1.3.3.2. Probert (2006b: 55–7) does, however, review some definitions of “pitch accent” (e.g., Jakobson 1931 [1962]: 166–71, Ladd 1996: 156), and correctly

⁴¹Most of the other stems exhibiting this behavior can likewise unproblematically be traced back to a disyllabic stem; see Probert 2003: 74 for a list and Kim 2002b: 93 for discussion. Kim (*loc. cit.*, fn. 26) takes the recessive accent in the GEN.PL to be the preservation of an archaism, i.e., the position of H when the stem was still disyllabic. Kim’s analysis thus amounts to the selection of different stem allomorphs: a monosyllabic stem (/paid-/) in most cases, a disyllabic stem (/pa.id-/) in the GEN.PL and GEN/DAT.DU.

recognizes that the prosodic systems of Japanese, on the one hand, and Swedish, on the other, which have both been labeled as “pitch accent systems” must be distinguished from one another; in Probert (2006b: 57)’s view, whether Attic-Ionic used pitch as the sole realization of “accent” (as, supposedly, in Japanese, following Beckman 1986), or possessed “pitch-differentiated stress” (as, supposedly, in Swedish), is difficult to decide. Elsewhere, explicit parallels are often drawn with other languages that ostensibly possess “pitch-accent” systems, in particular the systems of varieties of Japanese and Basque (cf. Devine and Stephens 1994: 211–15, Steriade 2014: 5–9, Ito and Mester 2017: 2–6). Indeed, Devine and Stephens (1994: 214) conclude that the Attic-Ionic “pitch accent” should be regarded as fundamentally akin to the “pitch accent” of Japanese.⁴²

Most problematically for the application of the term “pitch accent” to Attic-Ionic is the critique developed by Hyman (2009), who argues that that no coherent category of “pitch-accent” languages, all having some unified set of prosodic properties, actually exists (see also 1.3.3.3). Rather, the canonical “pitch-accent” languages, such as Japanese, contrast with canonical “stress-accent” languages, such as English or Russian, insofar as they typically lack at least one of the four properties listed in (2), which tend to characterize “stress-accent” languages (cf. generally Beckman 1986 on the acoustic correlates in English). In addition to those properties listed in 23, one may add the further but observation of Gordon (2014), that intonational pitch accents (i.e. “starred” tones, e.g., H*; cf. Beckman and Pierrehumbert 1986) tend to be placed on the same prosodic constituent marked for stress.

(23) Properties of “stress-accent” languages

- a. OBLIGATORINESS: every prosodic word has at least one most prominent prosodic constituent (typically a syllable).
- b. CULMINATIVITY: every prosodic word has at most one most prominent prosodic constituent (typically a syllable).
- c. PRIVATIVITY: a given constituent (usually a syllable) either has prominence or lacks it, such that the prosodic constituent may be labeled either “stressed” or “unstressed”.
- d. METRICALITY: prosodic prominence is positionally restricted (typically by reference to foot structure).⁴³

Hyman (2009: 220) observes that the literature on pitch accent seems to conceive of

⁴²“The prosodic system of Japanese illustrates how a system of durational rhythm can coexist with an independently patterned pure pitch accent. Such a system provides a closer typological parallel to Greek than the other systems discussed in this chapter” (Devine and Stephens 1994: 214). Perhaps the greatest oversight here, however, is how to explain the sort of stress window constituted by the LAW OF LIMITATION – the position of high tone in Ancient Attic-Ionic is highly restricted (as in many metrical stress systems), but is largely not positionally limited in Japanese. Ito and Mester (2017)’s account is thus an improvement, but, as argued below, ultimately suffers from the need to assume a bond between the intonational prominence and a boundary tone, effectively dressing up the edge restriction in a tonal guise.

⁴³More flexibly, in the light of the analysis for Vedic Sanskrit presented in Chapter 4, one could posit determination of stress by reference to foot structure, even where the foot structure is not strongly restricted to the domain near some word edge.

“pitch-accent languages” as languages having highly defective or restricted tone, which exhibit at least one or more of the properties listed in 23. The fundamental difficulty in picking out the class of pitch-accent languages comes from the fact that, as Hyman demonstrates, languages whose prosody calls for an explicitly tonal analysis may also exhibit one or more of these properties. Thus, while canonical “stress languages” will exhibit all of the properties in 23, and canonical “tone languages” may well lack all of those properties, many others may exhibit some combination of those properties, while either exhibiting several lexical tonal contrasts (making them tonal languages) or having only very restricted tone.

Attic-Ionic uncontroversially possesses at least the properties of OBLIGATORINESS and CULMINATIVITY with respect to the distribution of word-level H (cf. 1 above).⁴⁴ In this respect, one may contrast Attic-Ionic with Tōkyō Japanese and Haya (Bantu). While Attic-Ionic Greek requires that every (minimal) prosodic word contain at most and least one H (OBLIG(H) and CULM(H) appear to be undominated), Tōkyō Japanese does **not** require a falling intonation (H*L) that realizes “accent” on every prosodic word; in Haya, meanwhile, more than one underlying /H/ may realized on the same prosodic word.

(24) OBLIGATORINESS and CULMINATIVITY in Attic-Ionic Greek

- a. Grammatical (exactly oneH present): ποδός ‘foot.GEN.SG’, πολυπίδαξ [polupî:-daks] ‘having many fountains.NOM.SG’, λύετε [lúete] ‘untie.PRES.2PL’
- b. Ungrammatical (OBLIGATORINESS violated through absence of H): ^Xποδος, ^Xπολυπιδαξ, ^Xλυετε
- c. Ungrammatical (CULMINATIVITY violated through presence of more than one H): ^Xπόδος, ^Xπολυπίδαξ, ^Xλύετέ

(25) Absence of OBLIGATORINESS in Tōkyō Japanese

- a. Grammatical (H*L present): [tamágo-to] ‘egg-and’
- b. Grammatical (H*L absent): [sakana-to] ‘fish-and’

(26) Absence of CULMINATIVITY in Haya (cf. Hyman and Byarushengo 1984: 56, Hyman 2009: 223): /ó-mu-tí/ ‘tree’

- a. Grammatical (only one H surfaces in isolation): [omúti] ‘tree’
- b. Grammatical (both underlying H surface when embedded in a phrase): [akagul ómutí gwange] ‘he bought my tree’

The distribution of word-level H, on the surface, similarly satisfies PRIVATIVITY: no more than one H is found in any given prosodic word (without enclitics). Underlyingly, consid-

⁴⁴For the moment, the effects of enclitics, which may condition the appearance of multiple H on vocalic segments belonging to the same morphological word, e.g. ἄνθρωπός τις ‘some man’, are ignored. The precise domain for the incorporation of such enclitics is disputed, but it must be some unit either larger than the prosodic word (e.g., the clitic group, under a non-recursive view of prosodic cateory domains) or larger than the minimal prosodic word ($\omega_{[+min]}$), under the MIN-MAX recursive model of prosody of Ito and Mester 2009 and Ito and Mester 2012; see discussion in 1.2.2. Goldstein (2015: 49-68) explicitly argues that the prosodically deficient lexemes of Greek, the clitics (as opposed to the postpositives), are adjoined to a minimal prosodic word and thus build a recursive maximal prosodic word. One may then understand the properties of OBLIGATORINESS and CULMINATIVITY as properties of the $\omega_{[+min]}$ built by non-clitic lexemes.

ering the discussion on monosyllabic stems above, a privative lexical distinction in tone, between /H/ and /Ø/, is likely sufficient. Finally, the Law of Limitation (see 2) constitutes a strong indication of an effect of METRICITY: the fact that H tone is restricted to a window at the right edge of the window may be taken as a *prima facie* indication that prosodic constituency or constraints on the distribution of stress with respect to the right edge of the word play an important role. In light of these considerations, categorizing Ancient Attic-Ionic among languages possessing stress is appealing; the open question is whether doing so is analytically necessary or desirable.

6.2.2 Existing Analyses of Attic-Ionic Prosody

Here, I will first summarize and review the existing phonological analyses of Attic-Ionic word-level prosody, organized into three groups: the related metrical analyses of Steriade 1988, Sauzet 1989, and Golston 1990; the tonal analysis of Ito and Mester 2017, with extensions and improvements in Revithiadou 2018 and Revithiadou 2019; and the updated metrical analysis of Steriade 2014 (including a grammar for tonal realization of stress).⁴⁵ Many of these analysis focus on the so-called “recessive” accent, which is to be considered the phonological “default” word-prosodic pattern of Attic-Ionic realized in the absence of any lexically specified prosody that might call for prominence to be realized differently. Precisely because the recessive accent may illustrate the workings of the phonological grammar relatively independently of information from the lexicon or morphology, a satisfactory analysis of the recessive accent builds the cornerstone of a broader analysis of Attic-Ionic word prosody. While these analyses do usually attempt extensions to enclitic accentuation, the lexically determined prosody of monosyllabic nominal stems and in derivation is, with the exception of Steriade’s work, largely ignored. Different in perspective is Kim 2002b: Ch. 3, which is concerned more with the Indo-European inheritance in Greek prosody, and thus more with the lexically determined aspects of prosody than default rules.

6.2.2.1 The Metrical Stress Analyses

The series of papers from Steriade (1988), Sauzet (1989), and Golston (1990) build successively on one another and share a common set of assumptions. Perhaps most important among these assumptions is that the position of the Attic-Ionic H may be explained by reference to metrical structure. Specifically, all three analyses of recessive accent assume the following (with OT interpretation given in parentheses):

- (27) Shared analytical assumptions of Steriade 1988, Sauzet 1989, and Golston 1990
 - a. Word-final consonants are extrametrical (NONFINALITY- $\mathcal{F}$ $\gg$ PARSE-C)
 - b. Feet are trochaic (TROCHEE)
 - c. Feet are constructed from right to left (ALL-FEET-RIGHT)

⁴⁵For references to older treatments of the recessive accent of Attic-Ionic without any formal apparatus, cf. Probert 2006b: 103–12; some of the analyses treated under here are also briefly reviewed by Probert (2006b: 112–23).

- d. The rightmost foot receives the primary prosodic prominence (RIGHTMOST)

These analyses diverge, however, with respect to the quantity-sensitivity of the feet, syllable extrametricality, and the pitch (or intonational) realization of the primary prosodic prominence:

- Steriade and Sauzet take the feet to be quantity insensitive, thus syllabic trochees, whereas Golston operates with moraic trochees.
- Sauzet allows for the construction of degenerate monosyllabic feet.
- Steriade stipulates that final light syllables – [ǃ] or [ǃC] – are further subject to final syllable extrametricality.
- Steriade holds that the foot receiving primary prosodic prominence receives an H, whereas Sauzet and Golston presume a tonal sequence HL*: the L* tone docks to the stressed syllable, leaving the H tone to be aligned with the immediately preceding syllable, where possible.

Consider that this HL* pitch accent supposedly occurs in a language where the default pitch both before and after the pitch accent is L. This situation looks like the mirror image of Tōkyō Japanese and Northern Bizkaian Basque, where the pitch accent is best treated as H*L, with long stretches of high pitch occurring before the pitch accent due to a pitch rise at the beginning of a prosodic word (i.e., a %LH boundary tone). In languages like Vedic Sanskrit and Attic-Ionic Greek, where syllables by default have low pitch, either H* or HL* pitch accents make sense for marking the turning point of prominence; in languages like Tōkyō Japanese and Northern Bizkaian Basque, where high pitch is the norm across most of a word, only H*L is appropriate to marking a turning point.

An intuitively appealing feature of Steriade’s analysis is that the metrically most prominent syllable is marked by H; as Ito and Mester (2017) point out (referring to de Lacy 2002; see also 2.4.1.2), metrical prominence marked tonally typically takes the form of H or a contour in which H aligns with the metrically prominent syllable. This satisfying alignment of H with stress (or realization of stress as H) comes, however, at the cost of applying both final-consonant and final-syllable extrametricality, just in case the final syllable would be light. The derivation of NOM.SG ἄνθρωπος, through the application of extrametricality rules and syllabic trochee construction, is illustrated in (11). Unfortunately, the feeding of FINAL LIGHT SYLLABLE EXTRAMETRICALITY by FINAL CONSONANT EXTRAMETRICALITY is not possible under a NONFINALITY approach to extrametricality (cf. generally Hyde 2011), because the segments constituting the final light syllable are not final in the domain of the prosodic word if followed by another consonant. If NONFINALITY were to truly apply to last syllable in the domain of the ω, then to distinguish between [ǃC] final syllables, on the one hand, and [V:(C)] or [VCC] finals, on the other, would be impossible – all should be designated as extrametrical, and forms like, e.g., GEN.SG ἄνθρωπου [ántʰrɔ:po:] should result.⁴⁶

⁴⁶That this result holds for Optimality Theory and Harmonic Grammar, with extrametricality ef-

(28) Metrical Derivation of ἄνθρωπος under Steriade 1988

/ant ^h rɔ:pos/	UR
an.t ^h rɔ:pos	SYLLABIFICATION
an.t ^h rɔ:po⟨s⟩	FINAL CONSONANT EXTRAMETRICALITY
an.t ^h rɔ:⟨po⟨s⟩⟩	FINAL LIGHT SYLLABLE EXTRAMETRICALITY
(an.t ^h rɔ:).⟨po⟨s⟩⟩	SYLLABIC TROCHEES RIGHT-TO-LEFT
(ˈan.t ^h rɔ:).⟨po⟨s⟩⟩	END RULE RIGHT
[án.t ^h rɔ:pos]	SR

The proposal originating in Sauzet 1989, that the default word intonation is HL*, eliminates the need to posit the metrical invisibility of final light syllables. Instead, in a form like [án^hrɔ:pos], a syllabic trochee can be built at the right edge of the word, this rightmost trochee is designated as the head foot of the word, and the L* part of the HL* contonation docks to the stressed syllable, thereby resulting in the appearance of the high tone on the nearest TBU to the left. The result is the appearance of prominence realized as H on the antepenult, although stress, in fact, is posited to fall on the penult.

(29) Metrical Derivation of ἄνθρωπος under Sauzet 1989

/ant ^h rɔ:pos/	UR
an.t ^h rɔ:pos	SYLLABIFICATION
an.t ^h rɔ:po⟨s⟩	FINAL CONSONANT EXTRAMETRICALITY
an.(t ^h rɔ:po)⟨s⟩⟩	SYLLABIC TROCHEES RIGHT-TO-LEFT
an.(ˈt ^h rɔ:po)⟨s⟩⟩	END RULE RIGHT
[án.t ^h rɔ:pos]	SR

However, the assumption that the preferred foot structure of Attic-Ionic was a syllabic trochee – if a syllabic trochee is assumed to have the structural constituency as defined in Hayes 1995: 63; 71 (see also 4 in Chapter 2) – will make incorrect predictions about the position of the high tone, unless degenerate monosyllabic feet are permitted in the event of a heavy syllable at the right edge of the word. Namely, a syllabic trochee in the Hayesian foot inventory will, in Attic-Ionic words of three syllables or longer, always assign stress to the penult, and thereby generate appearance of the high tone on the antepenult. Sauzet instead allows for a syllabic trochee to be built around either one heavy syllable or a heavy–light sequence, meaning that the rightmost foot will be the final syllable if the final syllable is heavy.

(30) Hayesian SYLLABIC TROCHEE

$$\begin{array}{cc} (\times & .) \\ \sigma & \sigma \end{array}$$

facts implemented by relative rankings of NONFIN- $\check{f}$, NONFIN- σ , PARSE-C, and PARSE- σ , is shown by the tableaux available in the file GK_FSEXM.TXT in the accompanying GitHub repository at <https://github.com/rpsandell/Sandell-LMU-Habilitation> under Chapter 6. Fundamentally, simultaneous antepenultimate stress in the configuration / $\sigma \sigma \check{V}C$ / but penultimate stress in the configuration / $\sigma \sigma \check{V}CC$ / is not generable because, in the former, ALL-FT-Lis loser-preferring while PARSE-C is winner preferring, while in the latter, the reverse is true.

(31) Alternative weight-sensitive SYLLABIC TROCHEE (Sauzet 1989)

Form $\begin{array}{c} (\times) \\ \text{—} \end{array}$ $\begin{array}{c} (\cdot) \\ \smile \end{array}$ otherwise $\begin{array}{c} (\times) \\ \text{—} \end{array}$

As Golston (1990: 73) observes, both the type of syllabic trochee employed by Sauzet or a moraic trochee will correctly predict the position of the Attic-Ionic word-level high tone under the assumption of a default intonation HL*. If one wishes to hold to the more constrained typological inventory of feet proposed in Hayes 1995, then to adopt the moraic trochee as the preferred foot shape of Attic-Ionic is necessary (insofar as feet are to be employed at all; cf. further discussion in the immediately following subsections).

Several further phonological and morphophonological phenomena in Attic-Ionic, as well as apparent rhythmic preferences in poetic meter, might speak in favor of the moraic trochee as a unit of prosodic organization. First, as argued in Golston 1991: 73–84, content words of Attic-Ionic are almost exceptionlessly bimoraic, which could be interpreted as an effect of a prosodic word headed by a minimally bimoraic foot (see also discussion under 6.2.1.3). This minimal word requirement is most evidently at work in vowel length alternations that it conditions on monosyllabic stems of the underlying shape /CṼ(C)/, illustrated in 32; incidently, the same process is only explicable and justifiable if word-final consonants are treated as extrametrical. The GEN.SG forms in 32 stand for all other inflectional forms of such nouns, where a short vowel always occurs, while the monosyllabic NOM.SG requires lengthening of the underlying short vowel.

(32) Minimal word requirement and vowel lengthening in monosyllables

NOM.SG		GEN.SG		Gloss
πῦρ	[pû:r]	πυρός	[purós]	‘fire’
μῦς	[mû:s]	μυός	[muós]	‘mouse’
πούς	[pó:s]	ποδός	[podós]	‘foot’

In addition, Gunkel (2011) has shown that nouns built with the derivational suffix -μα(τ)- [-ma(t)-] exhibit a diachronic tendency to select root allomorphs containing a short vowel, replacing older forms employing a root allomorph with a long vowel or diphthong (see further data under 45 at 1.2.3 in Chapter 1). One conceivable advantage of NOM/ACC.SG χύμα [(kʰú.ma)] ‘liquid, stream’ over χεῦμα [kʰeûma] ‘id.’, prosodically speaking, is that the former can build precisely one well-formed bimoraic foot and parse all syllables into a foot; the latter, χεῦμα [(kʰeû).ma] or [(kʰeû.ma)], will leave the final light syllable unparse or create a trimoraic foot that violates RH-CONTOUR; see generally Zuraw 2018 for a comparative survey of such “trochaic shortening” exemplified across Austronesian languages.⁴⁷ Finally, the systematic equivalence of one heavy and two light syllables throughout all Ancient Greek poetic meters (cf. generally West 1982), as well as apparent tendencies in the organization and distribution of sequences of light and heavy syllables (see Allen 1973: 333, Golston and

⁴⁷ Another less intuitive, but nevertheless possible, interpretation of the replacement of χεῦμα by χύμα in the history of Greek is that the parse [(kʰeû).ma] is motivated by a ranking WSP ≫ NONINITIALITY-*f*; but if NONINITIALITY-*f* in turn generally outranks FTBIN and NONFINALITY-*f*, a parse [kú.(ma)] would be more harmonic.

Riad 2000 Golston and Riad 2005, or the summary in Golston 2014) across various metrical forms, together support the organization of syllables into quantity-sensitive trochaic units.

As applied in practice, foot construction and intonational assignment at the word level would look as exemplified in (33); a * marks the stressed syllable to which the L* portion of the HL* contonation will link, while the acute marks the position of the H tone.⁴⁸ I apply here exhaustive parsing into moraic trochees, but a single foot near the right edge of the word would suffice.

(33) Foot construction and intonational assignment per Golston (1990)

A. Heavy final syllables build a bimoraic foot, receive stress, and produce a high tone on the penult.

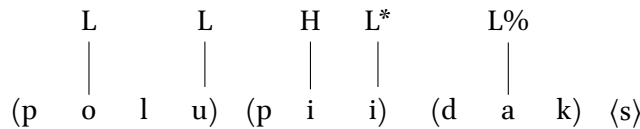
1. (άν).(θρώ).(*που) [(an).(t^hró:).(*po:)] ‘man:GEN.SG’
2. (πολυ).(άν).(*θρακ)⟨ς⟩ [(po.lu).(án).(t^hrak)⟨s⟩] ‘having much coal:NOM.SG’

B. Light final syllables remain unparsed when preceded by a heavy penult, or build the weak part of a foot when preceded by a light penult. The result, high tone on the antepenult, remains the same.

1. (άν).(*θρω).πο⟨ς⟩ [(án).(t^hró:).po⟨s⟩] ‘man:NOM.SG’
2. έ(δυ.νά).(*με.θα) [e.(du.na).(*me.t^ha)] ‘be able:IMPF.MID.1PL’

In recessively accented forms like πολυπίδαξ ‘having many wells.NOM.SG’, the workings of the LEX σωτήρα must result in a partial misalignment between the preferred stress-to-HL* -intonation mapping. According to Gunkel (2013), H on the TBU immediately preceding the TBU linked to low boundary tone L% attached to the last TBU of the ω requires more time for the transition from H to L%, so that both tones are durationally adequate. A structure like [(po.lu)(píi).(*dak)⟨s⟩] – with acute penult – would also potentially result in crowding of L* and L% onto the last vocalic mora. The L* is thus retracted onto the penultimate TBU resulting in [(po.lu)(pîi).(*dàk)⟨s⟩], as represented autosegmentally in (34).

(34) Retraction of HL* to avoid crowding of L% on final tone-bearing unit



⁴⁸ An unexplained aspect of this overall analysis is why no prosodic words with only L* occur (i.e., in which L* associates to an initial stressed syllable and H does not associate at all). Since, under this general line of analysis, CULMINATIVITY ought to be satisfied by stress, further invoking CULMINATIVITY(H), to require each prosodic word to contain at least one H tone seems stipulative. If the HL* contonation were really a bitonal pitch accent, i.e., H+L*, one might well expect for the H portion to fail to associate when there is no syllable to the left of the stressed syllable, given that the default contonation might not associate at all in non-recessively accented words. Another interpretation is that the HL* contonation is a unitary contour tone, where the * applies to the entire complex, but which is regularly split over two syllables in order to avoid the creation of syllable-level contour tones on the surface (perhaps NOCONTOUR ≫ NOSPREAD).

Precisely what occurs in forms like ἀστυάναξ [astuánaks] ‘lord of the city:NOM.SG’ is not entirely clear, and might hinge on the question of whether [+sonorant] codas were licit tonal hosts (see discussion under 6.2.1.5 above). Regardless, other forms, such as ποικιλόθριξ [poik-ilót^hriks] ‘spotted’ (Euripides, *Alcestis* 584; not ^Xποικίλοθριξ) show that, in the event of a heavy final syllable, the high tone categorically cannot move to the antepenult. The combined Golston–Gunkel account here would thus need to assume either that the L% tone is unrealized that a sequence HL% is tolerated, or that L* and L% are crowded onto the final vocalic mora.

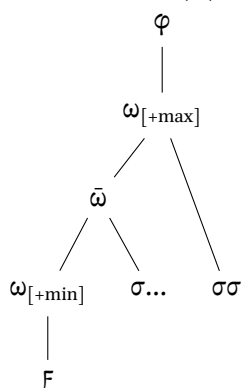
Before turning to a treatment of the prosodic behavior of enclitics given the analysis of Golston 1990, an objection leveled in Ito and Mester 2017: 15 should be mentioned at this juncture. Specifically in response to Sauzet and Golston’s treatment of Attic-Ionic stress as attracting a HL* tone, Ito and Mester assert that “the cross-linguistically solid link between H tones and footheads (de Lacy 2002) is severed, *contra naturam*. There are certainly intonational pitch accents whose designated tone is L, but we are here not dealing with special intonational markers with specific meanings, but with the basic word prominence marker of a language.” In other words, if pitch is the primary or only real phonetic correlate of Attic-Ionic stress, we might expect for it to be raised Fo, not lowered Fo. Such is the situation in, e.g., Nubi (Gussenhoven 2006): high pitch alone – without increases in intensity or duration – arises from stress. Nubi thus has a H* pitch accent linking to stress. To Ito and Mester’s objection, one might further add that the lexical accents of Attic-Ionic seem to require an underlying H, which would mean that underlyingly lexically unaccented strings realize stress differently than lexically accented strings (cf. 6.2.1.3 above on a possible lexical H).

For the moment, I will simply mention that this objection, while superficially attractive, is not entirely compelling. In particular, if one follows the analysis of Central Swedish word prosody advanced in Riad 2014, many features seen there would seem to parallel the assumptions needed under Golston’s analysis of Attic-Ionic (compare also Kristoffersen 2000: Ch. 9 on the similar system of Eastern Norwegian). Fundamentally, the question is the same question with which Devine and Stephens (1994: 211–5) and Probert (2006b: 55–7) wrestle: should the treatment of tone in Attic-Ionic be seen as more “Japanese-like” or more “Swedish-like”? The entire question will be considered again in greater detail at 6.2.4.

Enclitics under the Moraic Trochee Analysis The essence of Golston’s moraic trochee analysis is also compatible with the basic patterns of orthotonic word plus enclitic, as shown in Table 6.1 under 6.2.1.5. Concerning the larger prosodic structure of orthotonic + enclitic sequences, I follow Goldstein (2015: 50–1) in taking Attic-Ionic enclitics to be prosodically deficient syllables that cannot project their own ω , but which adjoin to a host $\omega_{[+min]}$ and project a recursive prosodic word, which may recur iteratively until the final enclitic is reached, at which point the maximal projection ω_{max} is created (cf. also Revithiadou (2018: 15), who independently posits the same structure). See discussion under 1.2.2 in Chapter 1, with reference to Selkirk 1995, Ito and Mester 2009, Kabak and Revithiadou 2009b, and Ito and Mester 2012 on prosodic word recursion. Schematically, the prosodic structure between the foot in an orthotonic word and level of the phonological phrase, φ , looks as depicted in 35, where a theoretically unlimited number of intermediate projections $\bar{\omega}$ adjoining to enclitics of the

shape σ or $\sigma\sigma$ could occur. Introducing another (primary) stress on successive projections of ω would parallel the situation in Neapolitan Italian described in (37) under 1.2.2.

(35) Prosodic word (ω) recursion with enclitics



Here, it can be taken for granted that orthotonic words construct a $\omega_{[+min]}$, and that the occurrence of at least one H is obligatory in that domain. When it occurs alone, an orthotonic word constitutes only a $\omega_{[+min]}$, since it dominates no other ω . The L% thus could be either a boundary tone at either the $\omega_{[+min]}$ specifically or the highest projection of a ω obtained. The latter option – L% assigned to the final TBU of the extended projection – potentially accounts for the general avoidance of H on the enclitic, or the final enclitic in a sequence of enclitics. The H that may occur on the final syllabic of a disyllabic enclitic, such as *ἐστί*, could potentially reflect a lexical H, or a H% occurring in a complex prosodic word (perhaps competing for realization with a L%).⁴⁹

Let us now first summarize again some of the facts discernable from Table 6.1:

- Monosyllabic enclitics, whether heavy or light, do not bear H immediately after an orthotonic word.
- Monosyllabic enclitics, whether heavy or light, may condition an acute on the final syllable of a host.
- Disyllabic enclitics, whether light–light or light–heavy, may condition acute on the final syllable of a host, or bear an acute / circumflex on their second syllable.

Under Golston's (1990) treatment of enclitics, he posits that all enclitics generally sponsor a floating H* tone, which reassociates to the final syllable of the orthotonic word or the

⁴⁹Revithiadou 2018: 16–17 proposes that enclitics are often incorporated into the ω of the orthotonic word, thus forming a $\omega_{[+min]}$ together with it, but disyllabic enclitics may condition recursion and receive a H% tone. Whether allowing for enclitics of different size to build entirely different prosodic structures with respect to the $\omega_{[+min]}$ is paralleled elsewhere is unclear. Insofar as the prosodic structures above the level of the ω are dependent upon or map onto morphological or syntactic structures, it would be surprising to find an instance in which the properties of a sub- ω constituent influence the construction of constituents above the $\omega_{[+min]}$.

final syllable of a disyllabic enclitic. The essential question is what type of prominence tone the recursive prosodic word would insert: HL* (like the $\omega_{[+min]}$) or a H*? Here, I suggest that the assumption of a distinct intonational H* is unnecessary for monosyllabic enclitics, if they are permitted to construct degenerate monosyllabic feet. However, given the H tone that can surface on the final syllable of disyllabic enclitics, assuming a lexical H* associated with their second syllable becomes necessary, which is suppressed under some conditions. The distribution of H in forms with simple enclitics can then be governed essentially by three undominated types of OCP constraint: *HH, *LLL, and *CLASH. The first, which militates against adjacent H sequences, is assumed by most analyses of Attic-Ionic enclitics; the second militates against long sequences of L tones preceding the boundary of a prosodic word, and the last penalizes adjacent stressed syllables.

Looking farther afield to sequences of multiple enclitics, *HH (= OCP(H)) does, however, seem to be violated, specifically, when the last two syllables preceding the new enclitic form a heavy–light (trochaic) sequence (cf. *καί τέ με*). An easy escape from this issue might be to assert that *HH applies only within the bounds of the $\omega_{[+min]}$. While a trochaic sequence at the edge of a ω in the input to a further ω might affect H on new enclitics, the same principle cannot be at work in orthotonic words, since words of the shape X–light–light with acute on the antepenult (e.g., *πέδιλον* ‘sandal’) will accept a further acute on the final syllable (*πέδιλόν ἐστι* ‘it’s a sandal’).

The attested sequence *καί τέ με φησὶ* might be explained as follows: without H on either *τέ* or *μέ*, a sequence LLL will result, but in order for H to occur on *μέ*, the syllable *φη* would need to be footed. *καί τέ με φησὶ* (*Iliad* 1.521, Venetus A),⁵⁰ in fact, best satisfies *CLASH and PARSE-σ – there are no adjacent stressed syllables –, while also satisfying *LLL, at the cost of violating *HH. However, *HH is really only necessary to exclude HH sequences in words with acute on the penult and a short final syllable, such as *φίλος τις* [*p^hílos tis*] or *μεγάλα τινα* [*mégala tina*] – other cases can be handled by *CLASH alone. Rather than accounting for *φίλος τις* through *HH, we might treat it as instantiating optimization of foot structure and tone: it becomes [*p^hi(*los.ti)<s>*]; or, quite simply, hosting a L(*) on the final syllable makes the final syllable unavailable for association to another H, which would yield a contour tone over a short vowel.⁵¹

Furthermore, the distinction in enclitic behavior between *οἶκός τινος* and *κῆρυξ τινός* can be handled by reference to stress clash, but not by *HH. Since both forms contain a circumflex on the first syllable, an L would intervene between two H. As shown in 36, in disyllabic words of the shape heavy–light, the final unstressed light syllable allows for the enclitic to build a stress foot and condition H on the final syllable of the orthotonic word; in word of the shape heavy–heavy, conversely, *CLASH would be violated if both the final syllable of the host and the first syllable of the enclitic were to bear stress.

⁵⁰ Most editions of the *Iliad* (Allen 1931, West 1998) in fact print *καί τέ μέ φησι*, with acute on *μέ* and without an accent on the second syllable of *φησι*, according to the traditional handbook system of accentuation in multiple enclitic sequences. See above, however, on the “Revised Götting-Barrett System” of Probert 2016, which I take to be correct.

⁵¹ Since words like *μεγάλος*, at least, are non-recessive, their tonal profile under Golston’s analysis is less clear. If it merely has lexical H and no other tone, then a blocking analysis is excluded.

(36) *Clash in Disyllabic Words with Enclitics

- a. (*οἶ).κός(ς) (*τι.νο)(ς) ‘of some house’
- b. ^X(κῆ).(*ρύκ)(ς) (*τι.νο)ς ‘someone’s herald’ – violates *CLASH, since two stressed syllables stand adjacent to one another (Correct: κῆρυξ τινός)

In turn, *HH may be invoked for cases in which an acute occurs on the penult of the orthotonic word, as shown in 37. In the case of disyllabic enclitics, whether they exhibit H tone on their final syllable in cases where a secondary acute on the orthotonic word would not be permitted depends on whether the orthotonic word’s H tone is on the final syllable or not. In such cases, a competition exists between the avoidance of either a stress lapse or tonal lapse (*EXTENDED LAPSE-RIGHT or *LLL(%)]_ω) at the right edge of the (recursive) prosodic word and association of a L% at the edge of the (recursive) prosodic word.⁵² In 38a., H surfaces on the final syllable of the disyllabic clitic, since a stress clash would otherwise result (cf. 36b.), but in 38b., stress clash is no concern, and secondary acute is preferred to acute on the final syllable; finally, 38c., which has an acute on the final syllable of the orthotonic word, is not at risk of violating *EXTENDED LAPSE-RIGHT or *LLL(%)]_ω

(37) *HH blocks Secondary Acute

- a. ^X(*φί.λό)(ς) (*τι)(ς) – violates *HH (Correct: φίλος τις).
- b. ^Xμ.ε.γά.λά (*τι.να)(ς) (Correct: μεγάληα τινά)

(38) *EXTENDED LAPSE-Right or *LLL(%)]_ω prevents lack of H within the rightmost three syllables

- a. ^X(κῆ)(*ρυκ)(ς) τινος (Correct: κῆρυξ τινός)
- b. ^X(*οἶ)κο(ι) τι(*νῶν) counting the L of the circumflex and two subsequent syllables (Correct: οἶκοί τινων)
- c. ^Xπα(*τρί) τι(*νί) – violation of (ALIGN-R(L%, ω_{max}) unmotivated by *HH, *CLASH, or *LLL(%)]_ω).

Thus, cases in which secondary acute does occur on the final syllable of the host, the association of another HL* tone can be regulated as in orthotonic words, assuming that even a monosyllabic enclitic may build a monomoraic foot in order that a $\bar{\omega}$ or $\omega_{\max}$ satisfies HEAD-EDNESS by directly dominating at least one foot.⁵³ In case the clitic is disyllabic, a binary foot may be constructed without issue. When, however, H surfaces on the second syllable of a disyllabic enclitic, implying that stress falls on that syllable, this is due to either *CLASH or *HH blocking the secondary acute, but *LLL(%)]_ω (~ *EXTLAPSE-R) requiring a stress or high tone within the final three syllables. Looking ahead, the effects resulting from *HH and *LLL(%) could likewise be generated under a purely tonal analysis of Attic-Ionic word

⁵²These two constraints also can be said to hold true of orthotonic words in isolation, since an H is always one of the last three tones in a word, and, under Golston’s moraic trochee analysis, stress always occurs on the final or penultimate syllable, meaning that no sequences of more than two stress syllables occur).

⁵³Keep in mind that the very existence of disyllabic clitics in Attic-Ionic shows that clitic-hood and the capacity to build a minimally bimoraic foot must be unrelated. Clitics are simply lexemes that cannot project their own $\omega_{[\text{+min}]}$.

prosody. Nevertheless, the pattern found with hosts of the shape $\kappa\eta\rho\upsilon\xi$ [$k\hat{e}:ruks$] (circumflex on a disyllable preceding a heavy final syllable) is difficult to explain without reference to avoidance of stress clash.

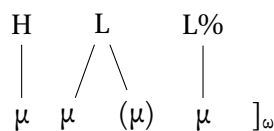
- (39) A foot within the clitic is preferred
- a. $(*\circ\hat{\iota})\kappa\acute{o}\langle\varsigma\rangle (*\tau\iota)\langle\varsigma\rangle$ – FOOTBINARITY violated, perhaps to satisfy preference for $\bar{\omega}$ to dominate a ϕ (i.e., STRICT LICENSING).
 - b. $(\acute{\alpha}\nu)(*\theta\rho\omega)\pi\acute{o}\langle\varsigma\rangle (*\tau\iota\nu\omicron)\langle\varsigma\rangle$
- (40) Acute on the rightmost syllable of a recursive prosodic word is a last resort to satisfy $*LLL(\%)]_{\omega}$
- a. $(\kappa\eta)(*\rho\upsilon)\xi\tau\iota(*\nu\acute{o})\varsigma$ – *CLASH avoided, $*LLL(\%)]_{\omega}$ sequence avoided.
 - b. $(*\phi\acute{\iota}\lambda\omicron)\varsigma\tau\iota(*\nu\omicron)\varsigma$ – *HH avoided, $*LLL(\%)]_{\omega}$ sequence avoided.

6.2.2.2 A Tonal Analysis: Ito and Mester 2017

The metrical approaches to the recessive accent surveyed and illustrated above would imply that Ancient Attic-Ionic indeed fulfilled the criteria for stress outlined in Hyman 2009, and therefore should not be classed as a pitch-accent language. In stark contrast to the foregoing analyses stands the recent treatment of Ito and Mester (2017), which would strip the need for metricality from the accentual calculus of Attic-Ionic, and thereby place it in the class of languages such as Japanese, which exhibit a tonal realization of accent.⁵⁴

Considering the recessive accent alone, Ito and Mester depart from the presumption that Attic-Ionic preferentially constructed an intonational pattern (or “word melody”) as in (41). This intonational pattern combines two elements observable in Attic-Ionic pitch patterns, following both Allen (1973: 231–4; 246) and Devine and Stephens (1994: 180): a HL “contonation” (i.e., the need for low pitch to follow high pitch) and a low boundary tone, L% (on the occurrence of L% tones crosslinguistically, cf. Féry 2017: *passim*, Gordon 2000, Gordon 2014; but note that cases of right edge marking at the level of the prosodic word, minimal or maximal, seem comparatively less common).

(41) Intonational Pattern of the Attic-Ionic Recessive Accent



A low boundary tone, L%, is assigned to the rightmost mora of a prosodic word, and should be preceded by a low tone, L, which may spread over two morae within the same syllable, and which is then preceded by a high tone, H.

⁵⁴Note, however, that the literature on the prosody of Japanese is itself divided on whether the pitch patterns of the language are best analyzed tonally or accentually, though the accentual view is predominant (Japanese literature). While OBLIGATORINESS is evidently not a feature Japanese prosody (cf. 25 above), a role for METRICITY of any sort is uncertain, just as in Attic-Ionic. See now Poppe 2015 for a rich cross-dialectal survey of potential interactions of metrical and tonal constraints in Japanese.

In Ito and Mester's view, this intonational pattern can be obtained directly through the workings of constraints applying to tones: no metrical structure is necessary to predict the position of the H. In other words, the pattern is “irreducibly tonal in nature”. Ito and Mester thus imply that HL+L% is the basic (or default) tonal pattern that should try to assert itself in any given prosodic word, though their analysis does not try to take account of prosodic words consisting of only two morae (such as $\mu\acute{\omega}\varsigma$ [mû:s] or $\pi\acute{o}\upsilon\varsigma$ [pó:s]), in which the complete intonational pattern cannot be realized (at least not without tonal crowding), or the interaction of the constraints giving rise to the HL+L% sequence with apparently lexical high tones (e.g., $\pi\alpha\tau\acute{\epsilon}\rho$ [paté:r]) that place the high tone much closer to the right edge of the prosodic word.⁵⁵

Under this analysis, a tonal anti-lapse constraint, NOLAPSE-L%/μ, combined with an alignment constraint ALIGNRIGHT-L%/ω, then serves to reproduce the effects of the “Law of Limitation” by fixing the L% to a mora at the right edge of the word. NOLAPSE-L%/μ forces the L% tone to associate with no more than one mora, while ALIGNRIGHT-L%/ω penalizes the association of this tone to any but the final mora of the ω.⁵⁶ The interaction of these two constraints accounts for the placement of H on the antepenult of a word such as $\acute{\epsilon}\delta\upsilon\nu\acute{\alpha}\mu\epsilon\theta\alpha$ ‘we were able’, as opposed to the preantepenult ($^X\acute{\epsilon}\delta\upsilon\nu\acute{\alpha}\mu\epsilon\theta\alpha$).⁵⁷

(42) 1PL.IMP.F.ACT.IND $\acute{\epsilon}\delta\upsilon\nu\acute{\alpha}\mu\epsilon\theta\alpha$ ‘we were able’, not $^X\acute{\epsilon}\delta\upsilon\nu\acute{\alpha}\mu\epsilon\theta\alpha$

a. Grammatical: L% on exactly one mora at the right edge of the prosodic word:

	L		L		H		L		L%
YES:	e	d	u	n	a	m	e	t ^h	a

b. Ungrammatical: L% linked to two morae in violation of NOLAPSE-L%/μ:

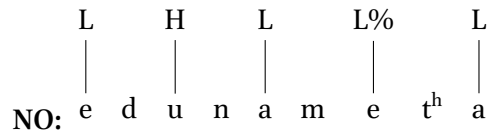
	L		H		L		L%
NO:	e	d	u	n	a	m	e t ^h a

⁵⁵Ito and Mester do, however, treat the constraint ALIGNLEFT-H/ω as lexically indexed to apply only to words belonging to the classes of recessively accented words. As Revithiadou (2018: fn. 9) correctly points out, this lexical indexation of the constraint that compels maximally leftward association of the high tone is superfluous if non-recessively accented words are taken to bear a lexical high tone (e.g., /an^hrɔ:p-o-s/ has no lexical marking, whereas /paté:r/ bears an underlying high tone). One might further add that considering application of the constraints generating the recessive accent in some way to be lexically restricted completely vitiates the general demonstration of Probert (2006b), that the recessive pattern is the “default” prosodic pattern of the language, and thus surfaces precisely in the *absence* of any lexically prespecified prosodic information.

⁵⁶Technically, such an ALIGN constraint could be interpreted as to incur more violations the farther it stands from the preferred edge. In this case, because all candidates considered by Ito & Mester that occur any violation of this constraint lose, a categorical versus gradient interpretation of violations is irrelevant.

⁵⁷I presume here that Ito and Mester would assign a L to all morae preceding the mora to which the H associates, though they nowhere indicate this explicitly. Any tones assumed to be present, or not, preceding the H, are irrelevant.

- c. Ungrammatical: L% not aligned with the right edge of the prosodic word, in violation of ALIGNRIGHT-L%/ω



At this juncture, one might reasonably question whether the constraint *NO*LAPSE-L%/μ is really an appropriate tool, or at least appropriate term, in the domain of autosegmental phonology. Constraints of the family LAPSE are constraints on tonal tiers, which may be viewed as a version of the OCP, in that they serve to penalize stretches of phonologically identical material. In the analysis of prosody, LAPSE and EXTENDEDLAPSE have typically been employed with respect to stress, to militate against sequences of stressless syllables with respect to some word edge (cf. Gordon 2002, building on Selkirk 1984 and an OT implementation in Elenbaas and Kager 1999). As illustrated schematically in (43), a sequence of more than one syllable without stress violates simple *LAPSE, while a sequence more than two stressless syllables violates the further *EXTENDEDLAPSE (while two adjacent stressed syllables would violate *CLASH). The usage of constraints like *LAPSE and *EXTENDEDLAPSE in the domain of metrical stress is well-established, but within the tonal domain, they would appear to do much of the same work as the OCP.

- (43) Violation of *LAPSE and *EXTENDEDLAPSE with respect to stress at a right word edge

- a. Violates *LAPSE-Right: [σ'σσσ]
- b. Violates *EXTENDEDLAPSE-Right: ['σσσσ]

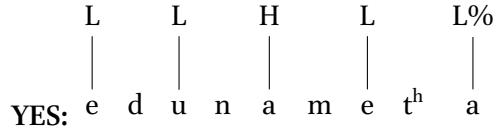
Nevertheless, one may point to a handful of other analyses in the literature in which *LAPSE has been applied in order to regulate tonal configurations: Zoll 2003: 243–7 (on Kukuya and Hausa), Ito and Mester 2013 (on Japanese), and Downing and Mtenje 2017: 218 (on Chichewa). However, *LAPSE in these analyses appears to penalize rather different configurations. On the one hand, in Zoll 2003: 243–7, it penalizes the association of a L tone adjoined to more than one TBU; such configurations could also potentially be penalized by CRISPEGE if a tone crosses syllable or word boundaries. On the other hand, as in Ito and Mester 2013, *LAPSE militates against sequences of more than one L not interrupted by some other tone; this effect could be more appropriately captured by a constraint such as OCP-L. Perhaps the extension of LAPSE constraints to the tonal domain is perhaps meant to parallel the behavior of an unmarked tone to lack of stress; however, in the specific Greek case at hand, it is applied to a specific marked tone, L%. A constraint such as *NOSPREAD-L% (“do not associate L% to multiple TBUs”) ultimately seems more apt for describing the necessary restriction that L% associate only with a single mora.

Yet other possible tonal configurations could allow for the H to migrate to a preantepenultimate syllable, if tones may associate across syllable boundaries or if the HL contonation were permitted to skip over a mora, leaving it without a tone. The two constraints

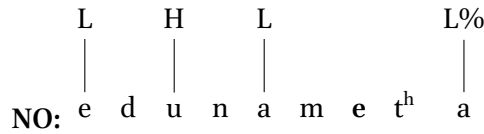
CONTIGUITY-T and CRISPEDGE- σ /T force every TBU within a string of tones to be associated with a tone and prevent association of tones across syllable boundaries, respectively.

(44) 1.PL.IMPF.ACT.IND ἐδύναμεθα ‘we were able’, not ^Xἐδύναμεθα

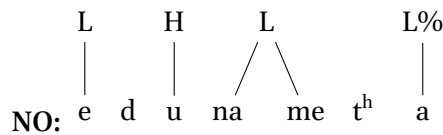
- a. Grammatical: No tones are associated across syllable boundaries and no morae are left without association to a tone.



- b. Ungrammatical: Penultimate syllable not linked to a tone in violation of CONTIGUITY-T



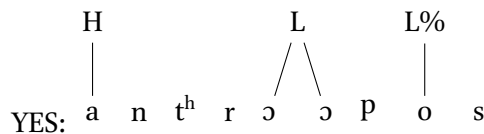
- c. Ungrammatical: L linked to penult and antepenult in violation of CRISPEDGE- σ /T



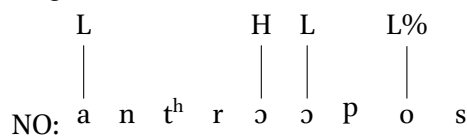
Finally, ALIGNLEFT-H/ ω forces the H of the HL contour tone to be placed as close to the left edge of the word as possible, while satisfying all other constraints. This constraint only plays a role when the final syllable contains but a single TBU, which will host the L%, but the penult contains two TBUs, which could host the HL contonation within the same syllable, giving rise to a circumflex accent.

(45) ‘man:NOM.SG’ ἄνθρωπος, not ^Xανθρωπος

- a. Grammatical: H is associated with the leftmost possible mora of the word, while NO LAPSE-L%/ μ , ALIGNRIGHT-L%/ ω , CONTIGUITY-T, and CRISPEDGE- σ /T are satisfied.



- b. Ungrammatical: H stands one mora farther from the left edge of the word.



The four constraints exemplified under (42) and (44) must always be satisfied; violations of ALIGNLEFT-H/ ω are to be limited, but its violation is not fatal. The total ranking is as

shown in (46).⁵⁸

- (46) Ranking for Ito & Mester's Analysis of Recessive Accent:
 NO LAPSE-L%/μ, ALIGNRIGHT-L%/ω, CONTIGUITY-T, CRISPEDGE-σ/T ≫ ALIGNLEFT-H/ω

Nevertheless, there exist further conceivable tonal configurations – i.e., other candidates – that are not taken into consideration by Ito and Mester. Specifically, their analysis assumes that the HL contonation and the low boundary tone L% must be directly adjacent to one another, without the intervention of other L tones between HL and L%. However, no constraint in the analysis directly penalizes a sequence [HLLL%] at the right edge of the word. Configurations like the one shown in (47) are thus not directly ruled out by any of the constraints in (46) above.

- (47) ^xάνθρῳπου ^x[ántʰrɔ:po:] 'man:GEN.SG'
- | | | | | | |
|---|--------|-----|---|---|-----|
| H | | L | | L | L% |
| | | / \ | | | |
| a | n tʰ r | ɔ | ɔ | P | o o |

In (47), an additional L stands between the L% and the HL, allowing the H to appear closer to the left edge of the word, and thereby incurring fewer violations of ALIGNLEFT-H/ω than the expected winner, άνθρῳπου [ántʰrɔ:po:]. If all syllables/morae are assumed to be associated with a tone (TBU ← T ≫ NOASSOCIATION) limiting the insertion of further L tones through, e.g., DEP-L will be insufficient, since the winner άνθρῳπου [ántʰrɔ:po:] would bear an L on the initial vocalic mora; the losing candidate represented in (45) and the desired winner would contain the same total number of tones. Alternatively, one might appeal to an OCP constraint, like OCP-L, in order to penalize sequences of L tones uninterrupted by an H. Granting the ranking OCP-L ≫ ALIGNLEFT-H/ω, winning winning άνθρῳπου [ántʰrɔ:po:] will indeed be preferred to losing ^xάνθρῳπου [ántʰrɔ:po:], since the winner avoids any LL sequence. In words of a shape like ἐδυνάμεθα, on the other hand, an LL sequence would be unavoidable, whether the H is associated with the vocalic mora projected by the penultimate or antepenultimate syllable, as shown in (48).⁵⁹

- (48) Preantepenult ^xἐδύναμεθα, or antepenult ἐδυνάμεθα

a. LL sequence at left edge = antepenult H

L	L	H	L	L%
e	d	u	n	a
		m	e	tʰ
				a

⁵⁸Both Ito and Mester 2017 and Revithiadou 2018 do not explicitly show the division of the constraints into strata, but their tableaux (and the Recursive Constraint Demotion algorithm) make evident that all constraints but ALIGNLEFT-H/ω are undominated, ranked above it.

⁵⁹Perhaps the best recourse, as Ito and Mester's own diagrams imply, is to assume the TBUs preceding the H simply have no tonal associations.

b. LL sequence between H and L% = preantepenult H

L		H		L		L		L%
e	d	u	n	a	m	e	t ^h	a

If neither DEP-L nor OCP-L can adequately enforce adjacency of the low boundary tone and the high–low contonation, the final recourse would be to employ a tonal phonotactic constraint *LLL%, which would specifically penalize the type of configuration seen in (48)b. While observationally adequate, the use of such a phonotactic constraint effectively stipulates that the HL contonation and L% must be directly adjacent to one another. In this regard, the analysis would then become somewhat parochial.⁶⁰ One could alternatively directly stipulate that the tonal sequence HLL% as a whole constitutes the recessive accent, and is not to be subdivided; under this assumption, the inclusion of a boundary tone in the supposed realization of word accent would be unusual, if not unprecedented. The overall impression is that the fixed complete sequence HLL% must be regulated by some markedness constraint that explicitly encodes the relation of (part of) the HL to the L%, but the position of the HL with respect to the L% is perhaps difficult to attribute to the emergent interaction of independently justifiable constraints, depending upon exactly what assumptions are made concerning any tonal associations beyond the HL+L% sequence.

Granting for the moment that the obligatory adjacency of HL + L% can be explained without difficulty, all cases in which the final syllable contains either a short vowel alone or is followed by a single consonant ([V̥(C)], like ἀνθρώπος [ántʰrɔːpos]) or a long vowel / diphthong ([VV]), like ἀνθρώπου [ántʰrɔːpoː], are correctly predicted by Ito and Mester’s tonal constraints. In turning to forms containing a short vowel followed by two consonants in the final syllable, such as πολυπῖδαξ [polupî:daks] ‘having many wells:NOM.SG’ or ἀστυάναξ [astuánaks] ‘lord of the city:NOM.SG’, however, potential problems arise. Namely, under the assumption that tones link only to [+syllabic, –consonantal] morae (or at least [+sonorant] morae), the analysis would incorrectly predict ^Xπολύπῖδαξ [polúpi:daks] and ^Xἀστυάναξ [as-túanaks], with H on the antepenult, as illustrated in (49): all four higher-ranking constraints are satisfied, and the H stands closer to the left edge of the word.⁶¹

(49) Antepenult H predicted if only three TBUs are present in the last three syllables

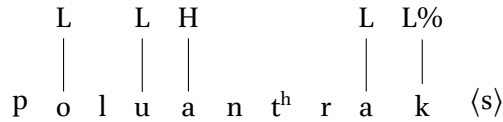
L				H		L		L%
a	s	t	u	a	n	a	k	⟨s⟩

⁶⁰Keep in mind, though that exactly this constraint, *LLL%, was invoked at (38) above as a possible motivation behind aspects of enclitic accentuation.

⁶¹The correct accentuation of πολυπῖδαξ, with circumflex penult, may be interpreted as a consequence of the LEX σωτήρα, for which Ito and Mester have no account, but which will be addressed further below. Alternatively – and more likely, given that such compounds are mostly regularly recessively accented – is that the HL sequence in such cases associates entirely to the penult to produce a falling tone.

The solution adopted by Ito and Mester to this issue is to presume that *all* morae, regardless of their sonority, make for licit tonal hosts, at least of the L% tone. In this fashion, the penult acute of a form like πολυάνθραξ [poluántʰraks] ‘having much coal’ can be correctly generated.

(50) L% links to a final consonantal, [–sonorant, –continuant] mora



To account for the (syllabic position) of the high tone in words like πολυπίδαξ [polupîdaks] and πολυάνθραξ [poluántʰraks] as under (50) has two potentially serious drawbacks. First, the assumption that voiceless stops may bear intonation is in itself phonetically highly dubious: voicelessness “virtually precludes the possibility of realizing intonational information” (Gordon 2014: 100; see also Barnes 2002, Myers 2011). Nevertheless, at the purely phonological level, nothing precludes linking a tone to a voiceless obstruent; phonetically, however, this tone will have no direct cue. The situation in Attic-Ionic would therefore be rather different than, the behavior of HL tones in Luganda, for instance. In Luganda, lexemes such as [ˈkkùla] ‘treasure’ align a HL tone with the leftmost vowel of the word, leaving the H component of the contour tone essentially phonetically unrealized. In this case, there is, however, still a phonetic trace of the effectively unrealized low tone resulting from slightly lower realization of the high tone (Ladefoged and Johnson 2011: 259).⁶² In the Attic-Ionic case, since no [+syllabic] or even [+sonorant] segment would follow the L% on an obstruent at the right edge, the reasonable presumption is that such an analysis predicts that L% would remain entirely phonetically uncued in such cases.

Second, and perhaps more seriously, if a L% is indeed assigned to a mora projected from an obstruent in the final syllable in forms like πολυπίδαξ, then the intonational account of the Lex σωτήρα (Gunkel 2013, Gunkel 2015) would be completely vitiated.⁶³ Under Gunkel’s

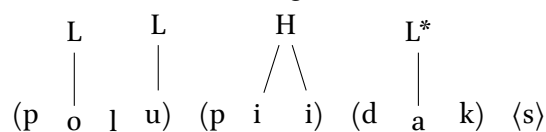
⁶²Revithiadou (2019) also points to some further instances in which syllables containing short vowel plus obstruent rhymes may licitly host HL tones, the L component being associated to the obstruent (as in Thai; Lee 2008: 179–82) or salvaged through lengthening of the vowel, thereby allowing both parts of the HL tone to associate to a vowel (as in Hausa; cf. Gordon 2001). Revithiadou tentatively suggests that the vowel in final syllables of forms like πολυάνθραξ [poluántʰraks] or ποικιλόθριξ [poikilótʰriks] may have been subject to lengthening. To my knowledge, there exists no independent evidence for short vowel lengthening in [VCC] finals in Attic-Ionic, but even if there were such lengthening, then one might expect for the prosodic contrast between [V:(C)] and [V̌CC] finals to be neutralized, which is not the case. Speaking against the assumption of such vowel lengthening is the behavior of [V̌CC] in meter (specifically, the dactylic hexameter of Homer): in the investigation of Ryan (2019: 149–52), V̌CC sequences behave effectively as lighter than VVC sequences. Furthermore, I report anecdotally here than the [V̌ps] or [V̌ks] sequences that can end a word seem to usually be avoided in the second metron of a foot in the hexameter, where the second metron prefers relatively “heavier” sequences; a more systematic investigation is needed.

⁶³See discussion under 6.2.3 for reasons to doubt that some or all cases of the Lex σωτήρα have anything to do with a L%. Insofar as Ito and Mester assume the presence of a L% anyway, attributing the workings of the Lex σωτήρα to the boundary tone seems reasonable

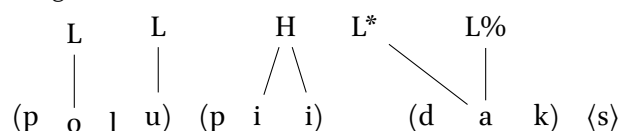
treatment, the motivation for the Lex $\sigma\omega\tau\eta\rho\alpha$ is precisely that the only vocalic mora available in the final syllable must host the low boundary tone, L%. For the initial assignment of H, Gunkel follows the analysis of Golston 1990 (see above). Just in case the H part of the HL* contonation is forced to occur on the penultimate syllable, it prefers to appear as a circumflex so as to allow the L* portion to surface fully before the L% tone and thereby avoid the crowding of pitch transitions. Gunkel regards the result as the retraction of H by one TBU (with μ as TBU); such H retraction such to avoid of crowding with a L% is also in evidence in Chickasaw (Gordon 2003, Gordon 2008) and Chichewa (Myers 1999: 225–33).⁶⁴ The derivation of $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$ would thus precede in (51).

- (51) Derivation of $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$ [polupî:daks] under Gunkel's tonal crowding avoidance hypothesis

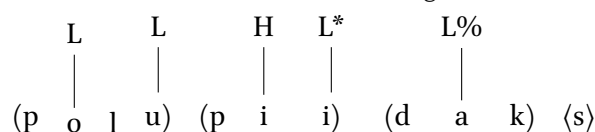
- a. Foot formation and assignment of HL* contonation



- b. Assignment of L%



- c. Retraction of HL* to avoid crowding of L% on final mora

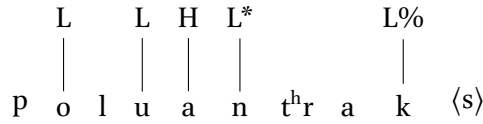


To account for the LEX $\sigma\omega\tau\eta\rho\alpha$ per Gunkel, the only further component needed would be a constraint that penalizes the association of multiple tones to the same TBU, such as *MULTIPLE(T) in Peng (2013: 371)'s analysis of Yoruba tone or *CONTOUR (Yip 1999); compare also NOCROWD and NOCONTOUR in Gussenhoven 2004: Ch. 8. One might reasonably ask why retraction of H fails occur in forms like $\pi\omicron\lambda\upsilon\acute{\alpha}\nu\theta\rho\alpha\zeta$ or $\acute{\alpha}\sigma\tau\upsilon\acute{\alpha}\nu\alpha\zeta$ to avoid tonal crowding. At first glance, forms like $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$ do not exhibit preferred position of the H with respect to vocalic morae, but yet do not require retraction by an additional mora into the antepenultimate syllable. Perhaps the most straightforward answer is that, in a form like $^X\pi\omicron\lambda\upsilon\acute{\alpha}\nu\theta\rho\alpha\zeta$ or $^X\acute{\alpha}\sigma\tau\upsilon\acute{\alpha}\nu\alpha\zeta$, the high tone would stand outside the domain permitted by the LAW OF LIMITATION (LoL). If the LoL is interpreted so as to enforce association of H to the first syllable to the left of the rightmost bimoraic foot, the forms with antepenult H like $^X\pi\omicron\lambda\upsilon\acute{\alpha}\nu\theta\rho\alpha\zeta$ or $^X\acute{\alpha}\sigma\tau\upsilon\acute{\alpha}\nu\alpha\zeta$ are impermissible, because, if parsed as [(po.lu).(an).(t^hrak)<s>], the H would then

⁶⁴A phonological constraint driving the LEX $\sigma\omega\tau\eta\rho\alpha$ that does not refer to L% at all can be invented (e.g., * $\mu\acute{\mu}\mu$ employed in Kiparsky 2003a), but insofar as a phonetic grounding for the phonological process is desirable, Gunkel's intonational account might preferable (see also Devine and Stephens 1994: 155).

be associated to the left of the second-to-last foot. Alternatively, if the analysis of Steriade 2014 (see 6.2.2.3 below) is adopted, then ranking *LAPSE RIGHT- $\bar{\sigma}$ (“no sequences of stressless syllables [=syllables without H] if the final syllable is heavy”) above a constraint like MULTIPLE(T) will exclude tonal retraction in such cases.⁶⁵

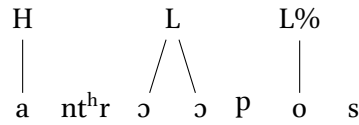
- (52) L% links to a final consonantal mora



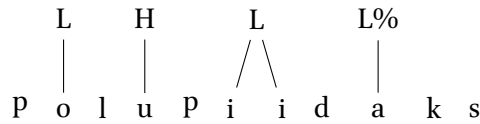
Such an assumption does not help, however, in cases like $\acute{\alpha}\sigma\tau\upsilon\acute{\alpha}\nu\alpha\zeta$, since the [+sonorant] nasal would belong to the onset of the final syllable: the penult hosting the H consists of a short vowel alone.

In the end, an intonational account of the LEX $\sigma\omega\tau\eta\rho\alpha$ is then difficult, perhaps impossible, to bring into alignment with Ito and Mester’s tonal analysis of the recessive accent. In particular, if the two analyses are combined, then forms like $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$ are incorrectly predicted to show high tone on the antepenult, thus $^X\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$ (contrast $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$), as illustrated in (54).

- (53) HL+L%: $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$



- (54) HL+L%: $^X\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$

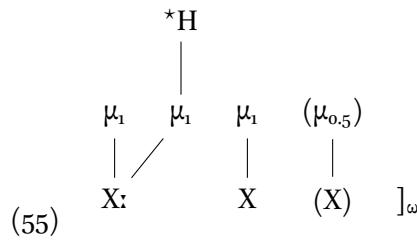


Salvaging the Lex $\sigma\omega\tau\eta\rho\alpha$ with Gradient Symbolic Representations Revithiadou (2018) equally recognizes that Ito and Mester’s analysis lacks a ready account of the facts usually attributed to the LEX $\sigma\omega\tau\eta\rho\alpha$, but attempts to rescue the claim that L% may associate to a weight-bearing obstruent by recourse to lexically gradient weight of such morae (following the model of Smolensky and Goldrick 2016), allowing them to host tone or not under different conditions. The essence of the account offered by Revithiadou runs as follows:

- Moras projected by consonants underlyingly have an *activation level* (AL) of 0.5, which is sufficient to contribute to syllable weight, but insufficient to host tone.

⁶⁵If the interpretation of syllables having [+sonorant] codas offered in Probert 2003: 150 is correct, $\pi\omicron\lambda\upsilon\acute{\alpha}\nu\theta\rho\alpha\zeta$ might even reflect the LEX $\sigma\omega\tau\eta\rho\alpha$, too, assuming that the ultimate tonal pattern in such a word form is [poluáñt^hraks], with L associated to the [n], or HL associated to the heavy syllable dominating [an].

- Consonantal moras may be “strengthened” through (some sort of) DEP violation to increase their activation level, when it is preferable for a consonant to host L%, due to the ranking ALIGNRIGHT-L%/ω ≫ DEP.
- The LEX σωτηρᾱ is driven by a markedness constraint that penalizes association of a H* to a non-head mora when followed by the last mora in the ω having a full activation level of 1 (shown as μ₁); the relevant constraint is shown in (55).⁶⁶



- *CRISPEGE-σ/T is violated to allow the L of the contonation to associate across a syllable boundary, with both the vocalic mora of the final syllable and the second mora of the penult in a form like κῆρυξ ‘herald’.

One immediate issue is that the markedness constraint in (55) is not evaluated by Revithiadou as a proper markedness constraint: the losing candidate with acute ^xκῆρυξ is still interpreted by Revithiadou as assigning L% to the vocalic mora projected from the [k], which is given full activation weight in order to make it able to host a tone. If the mora projected from the [k] in the output has full activation weight, then the H linked to the second mora of the penult would no longer stand next to the last mora of the word with full activation weight. (55) seems then to be a markedness constraint evaluated over the UR (where the final mora indeed does not have a full activation weight of 1), which is in principle illicit. Further debatable is the assumption that an acute represents a tonal LH; as discussed under 6.2.1.6, it seems more likely that the acute is to be understood as level H, in which case the motivation for the constraint in (55) would require reformulation, if it can be salvaged at all.

Revithiadou (2019) changes tactics somewhat. In a modified analysis with the same theoretical machinery, Revithiadou assumes that an activation weight of 0.8 is sufficient to host tone, and thus the total moraic activation in the final syllable of a word like κῆρυξ [kê:ruks] becomes 1.8 (1 from the short vowel, 0.8 from the [k]); the penult, on the other hand, has a moraic activation weight of 2 (1 from each mora of the long vowel), making it “heavier” than the final syllable. Revithiadou then takes the LEX σωτηρᾱ to be an instantiation of the WEIGHT-TO-STRESS principle: because the penult is heavier than the final, the stress (= the

⁶⁶(Revithiadou assumes here that long vowels with acute are rising, LH, not level H associated to both morae, contrary to the musical evidence for the latter representation discussed under 6.2.1.6 above. Note further the similarity of this constraint to Kiparsky (2003a)’s *μῦ.μ, which likewise more or less just directly penalizes the specific prosodic configuration that appears to be repaired by application of the Lex σωτηρᾱ, without any deeper motivation.

entire HL contonation) prefers to associate with the penult, giving rise to a circumflex.⁶⁷ This manouver has intuitive appeal – prominence should fall on heavier syllables – and can in fact also be implemented using the weight-to-stress mapping constraints of Ryan 2019: Ch. 2, without recourse to gradient representations (see 6.2.3.4 below).

To view the LEX σωτήρα as merely an instantiation of WEIGHT-TO-STRESS does not, on the other hand, transparently account for its effects in words that perhaps bear a lexical high tone, like the *Paradebeispiel* σωτήρα [sɔ:té:ra] itself. In forms like /sɔ:té:r/, which presumably carry an underlying lexically specified high tone that surfaces as acute in a final syllable or penult followed by a final with a long vowel (NOM.SG σωτήρ [sɔ:té:r], GEN.PL σωτήρων [sɔ:té:rɔ:n]), but circumflex in a penult followed by a short vowel (ACC.SG σωτήρα), a lexically prespecified high tone must be reassociated, so as to link only to the leftmost mora of the penult. Precisely because words with lexically specified high tone need not have a HL contonation (i.e., they have only H*, not H*L or HL*), the retraction of the lexically specified high tone would seem to instantiate the workings of some further markedness constraint, perhaps *HL% (no directly adjacent H and L% tones on the tonal tier), if a L% is assumed to be present.

As far as enclitic accentuation is concerned, Revithiadou (2018, 2019) deviates from the treatments in Steriade 1988 or Golston 1990 by positing that monosyllabic and disyllabic enclitics have different prosodic structures: the former are regularly incorporated into a single prosodic word with the host, whereas the latter may either be incorporated (in the instances where a secondary acute occurs) or build a recursive prosodic word (in the instances in which a H may be associated to the final syllable of the disyllabic enclitic). Revithiadou (2018: 15) suggests that the H occurring on the final of a disyllabic enclitic, as in, e.g., δαίμων τινός [daímon tinós] ‘someone’s *daimōn*’ is a H% boundary tone inserted only in recursive prosodic words. Arguably one problem with viewing this type of acute as a H% boundary tone is that it is also subject to becoming a grave when not followed by another enclitic or non-final in a sentence; if grave accentuation is due to a L% competing with or coalescing with a H tone, then both boundary tones would need to be assumed to occur at the same edge in such cases. Furthermore, in cases in which a monosyllabic enclitic occurs after a disyllabic one (e.g., δαίμων τινός τε [daímon tinós te] ‘and someone’s *daimōn*’), if the prosodic structure is then [[δαίμων]_ω τινός τε]_ω, then either the presumed H% is associated with the penult, rather than final syllable of the recursive prosodic word, or must be displaced by a further HL introduced by the addition of the incorporating monosyllabic enclitic, τε.

All things considered, four general objections may be raised against the purely tonal account of Attic-Ionic word prosody:

- The strict adjacency of the HL contonation and L% boundary tone must be stipulated:

⁶⁷Curiously, Revithiadou’s tableau contrasting winning [kê:ruks] with losing [ké:ruks] shows, in the winner, the L% associated to the obstruent [k], and no tone associated to vocalic [u]. The winner would thus seem to violate CONTIGUITY-T; if the L% in winning [ké:ruks] were instead associated to the mora of vocalic [u], then ALIGNRIGHT-L%/ω would be violated. The problem then arises that violations of CONTIGUITY-T and ALIGNRIGHT-L%/ω are otherwise fatal in Ito and Mester’s account. To allow the mora projected from [k] to remain below the critical activation threshold does not help either, because would predict that recessive words with [Ǟ(C)] and [ǞCC] in the final syllable would behave identically, which they obviously do not.

how to best prevent HL from associating as far left as possible, but not away from the L%, is unclear. Under any account invoking stress, without or without foot structure, the position of H(L) is easier to regulate.

- Generating $\sigma\omega\tau\eta\rho\alpha$ -type accentuation in words ending in [-ks] or [-ps] requires reference to subtle distinctions in the activation level of morae, and the analytically complex machinery of Gradient Symbolic Representations.
- Generating $\sigma\omega\tau\eta\rho\alpha$ -type accentuation in non-recessive stems like $\sigma\omega\tau\eta\rho$ itself remains unexplained.
- The treatment of enclitic accentuation assumes different prosodic structures without evidence independent of accentuation, and does not immediately obviously extend correctly to sequences of multiple enclitics.

6.2.2.3 Steriade 2014

In an unpublished presentation from 2014, Steriade presents the most comprehensive analysis of Attic-Ionic word-level prosody since Steriade 1988. Steriade's analysis argues that the position and realization of the high tone are determined through independent calculi. Under this approach, Attic-Ionic can be considered to be effectively a stress system comparable to Modern Greek, in which prominence is obligatory, cumulative, privative, and determined by metrical principles. Furthermore, stress and the syllable containing a High tone are always coextensive; thus, primary stress is considered to fall on the antepenult in forms like $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$. Once the position of the stress is established through a set of constraints used for stress systems (based on a quantity-sensitive version of the inventory employed in Gordon 2002), a separate set of constraints controls the tonal realization of the prominence as HH (acute) or HL (circumflex). In contrast to Non-recessive accentuation, meanwhile, is controlled by prosodic faithfulness to underlying accents, with preference given to the rightmost of any such accents.

The position of stress is primarily regulated by the constraints and ranking given in (56). In underlyingly unaccented strings, the IDENT constraints play no role; NONFINALITY EXTENDEDNONFINALITY keep stress away from the final syllable or penultimate syllable, while EXTENDEDLAPSE-Right and *LAPSE-Right- $\bar{\sigma}$ prevent stress from being placed beyond the antepenult or penult. In other words, stress is repelled from the right edge while being kept within a three-syllable window. The table in 6.5 explains the effects of the constraints in regulating the position of stress for recessively accented words.

- (56) Basic Stress Ranking of Steriade 2014 EXTENDEDLAPSE-Right, *LAPSE-Right- $\bar{\sigma}$ $\gg$ IDENT-Accent, IDENT-Accent-Right $\gg$ NONFINALITY $\gg$ EXTENDEDNONFINALITY⁶⁸

⁶⁸Steriade (2014:5) shows a Hasse diagram with NONFIN $\gg$ EXTNONFIN, but no data or tableau that she discusses appears to require this ranking. A better ranking would perhaps be EXTNONFIN $\gg$ IDENT-Accent, IDENT-Accent-Right $\gg$ NONFINALITY.

Weight of Final σ	Weight of Penult σ	Stress Position	Why?
Heavy	Heavy or Light	Penult	Only EXTENDEDNONFINALITY violated; Antepenult would violate *LAPSERIGHT $\bar{\sigma}$.
Light	Heavy	Antepenult	All constraints satisfied; Pre-antepenult would violate EXTLAPSERIGHT.
Light	Light	Antepenult	All constraints satisfied Pre-antepenult would violate EXTLAPSERIGHT.

Table 6.5: Constraint interactions generating recessive accentuation in Steriade 2014

Thus, the Law of Limitation is controlled by the two LAPSE constraints: *EXTENDEDLAPSERIGHT and *LAPSERIGHT $\bar{\sigma}$; the former penalizes a sequence of more than two unstressed syllables at the right edge of the word, while the latter penalizes a sequence of more than one unstressed syllable just in case the final syllable is heavy.⁶⁹ The latter constraint is an innovation, but it or a similar family of constraints may be necessary for capturing quantity-sensitive stress without reference to feet. A form like GEN.PL $\acute{\alpha}\nu\theta\rho\acute{\omega}\pi\omicron\upsilon$ rather than $^X\acute{\alpha}\nu\theta\rho\omega\pi\omicron\upsilon$ (with high tone on the antepenult, as in the NOM.SG) is thus excluded because it violates *LAPSERIGHT $\bar{\sigma}$;⁷⁰ any possible forms with high tone farther to the left than the antepenult would violate *EXTENDED LAPSE RIGHT.

When, meanwhile, a lexical accent is present, the ranking of IDENT-Accent, IDENT-Accent-Right above the NONFINALITY constraints means that lexical stress will surface faithfully within the three-syllable window, with the rightmost such lexical accent winning out, as illustrated in 57. This treatment of lexically specified stress is, however, perhaps inadequate in two respects. First, it must assume that the inflectional suffixes that attract high tone in monosyllabic stems (i.e., that are seemingly accented, e.g., GEN.SG /-ós/) cannot attract high tone in polysyllabic stems; otherwise, an input GEN.SG /astér-ós/ would appear as $^X\acute{\alpha}\sigma\tau\epsilon\rho\acute{\omicron}\varsigma$ [asterós]. Second, how cases in which addition of an unaccented derivational suffix triggers application of recessive accentuation ($\beta\alpha\sigma\iota\lambda\epsilon\acute{\upsilon}\varsigma \rightarrow \beta\alpha\sigma\iota\lambda\epsilon\iota\alpha$, not $^X\beta\alpha\sigma\iota\lambda\epsilon\iota\alpha$) are to be resolved is unclear.

(57) Rightmost lexical accent wins

- a. $\acute{\alpha}\sigma\tau\acute{\epsilon}\rho\text{-}\omicron\varsigma$ [astér-os] $\leftarrow$ /astér-os/ ‘star-GEN.SG’
- b. $\acute{\alpha}\sigma\tau\epsilon\rho\text{-}\iota\sigma\chi\text{-}\omicron\varsigma$ [aster-ísk-os] $\leftarrow$ /astér-ísk-os/ ‘star-DIM-NOM.SG’

Although the constraints EXTENDEDLAPSE-Right, *LAPSE-Right, and NONFINALITY have

⁶⁹Without explicit recourse to final segment extrametricality as a device, the set of configurations that count as heavy in final syllables must be stipulated as {V:C₀ VC₂}.

⁷⁰Here, forms like GEN.SG $\pi\acute{\omicron}\lambda\epsilon\omega\varsigma$ ‘city’, arising through quantitative metathesis, violate LAPSERIGHT $\bar{\sigma}$ on the surface. Since underlying lexical accents of stems normally shift rightwards to accommodate *LAPSERIGHT $\bar{\sigma}$, forms like $\pi\acute{\omicron}\lambda\epsilon\omega\varsigma$ cannot be readily attributed to the ranking of faithfulness to lexical accents over *LAPSERIGHT $\bar{\sigma}$. Accounting for such exceptions appears to require recourse to either a derivational model, in which assignment of the high tone precedes QM from a UR /pole:-os/, or Output-Output correspondence can maintain the high tone on the first stem syllable, where it would be assigned in, e.g., NOM.SG $\pi\acute{\omicron}\lambda\iota\varsigma$. An ostensible problem with the latter approach might be that it would incorrectly predict the maintenance of the high tone on the same syllable of a stem throughout, resulting in e.g., $^X\acute{\alpha}\nu\theta\rho\omega\pi\omicron\upsilon$.

been demonstrated by Gordon (2002) to help produce correct results for quantity-insensitive stress systems, the other two constraints, EXTENDEDNONFINALITY (“avoid stress on the last two syllables”) and LAPSE-Right- $\bar{\sigma}$ (“no more than one stressless syllable at the right edge, if the final is heavy”) are innovative. The innovation of EXTENDEDNONFINALITY is itself is probably not necessary, and the typological predictions made by LAPSE-Right- $\bar{\sigma}$, meanwhile, are unclear.

In Gordon 2002: 504, penultimate stress is derived through the ranking *EXTENDEDLAPSE-Right $\gg$ ALIGN(X_1 , L) $\gg$ ALIGN(X_1 , R): a lapse of more than two syllables at the right edge is excluded, but stress should, at the same time, stand as far to the left as possible.⁷¹ In order to allow for mobility of high tone in monosyllabic stems (πόδα $\sim$ ποδός), NONFINALITY must generally outrank IDENT-Accent; something further must dominate IDENT-Accent in order to prevent faithfulness to accented endings of third declension nouns with polysyllabic stems (e.g., GEN.SG ῥοιθος, not ^Xῥοιθος). Instead of EXTENDEDNONFINALITY, ALIGN(X_1 , L) can be employed, if Harmonic Grammar-style constraint ganking is permitted. . Under such an analysis, a single violation of IDENT-Accent-IO is less harmonic than a single violation of ALIGN(X_1 , L), but two or more violations of ALIGN(X_1 , L) are less harmonic than a single violation of IDENT-Accent-IO. Thus, GEN.SG [podós] is preferred because it satisfied IDENT-Accent-IO, but GEN.SG [órni^hos] emerges as a winner because it avoids violation of ALIGN(X_1 , L) altogether. Arithmetically, this result extends to any word with more than three syllables and a lexical accent on the rightmost vowel: two additional violations of ALIGN(X_1 , L) will be less harmonic than a single violation of IDENT-Accent-IO. In terms of analytical coverage, a Harmonic Grammar analysis using ALIGN(X_1 , L) can probably account for the Attic-Ionic data just as well as NONFINALITY and EXTENDED NONFINALITY together; the benefit of the former analysis may be that it does not need to invoke an explicit dispreference for the prominence to fall on the penultimate syllable.

*LAPSE-Right- $\bar{\sigma}$ plays an essential role in generating the effects of the LAW OF LIMITATION, since it simulates the restriction of stress to the final or penult when the final is heavy in the same way that construction of a moraic trochee does in Golston’s (1990) analysis of the recessive accent. While not necessarily a fatal objection, the explicit reference to syllable weight in *LAPSE constraints has not been tested in terms of the typology of its predictions. For instance, if *LAPSE-R/L- $\bar{\sigma}$ is admitted, and a parallel *EXTENDEDLAPSE-R/L- $\bar{\sigma}$ as well, some potentially odd-seeming stress systems might result. Consider the constraint ranking given in 58, where *EXTENDEDLAPSE-R/L- $\bar{\sigma}$ penalizes stress farther to the left than the antepenult just in case the final is heavy. The result would be a system that places primary stress on the penult in two syllable words, the antepenult whenever the final is heavy in longer words, else the initial syllable in words with no heavy syllables. More generally, stress patterns in which the position of stress is restricted by a neighboring heavy syllable, rather than stress being directly attracted to the heavier syllable, may well exist (Steriade views Romanian and Spanish as possessing similar window restrictions, where a heavy final limits stress to the penult), though whether they must be analyzed in such a fashion is unclear.⁷² Thus, the questions

⁷¹Gordon (2002: 498) defines ALIGN(X_1 , L/R) as follows: “Every grid mark of level 1 is aligned with the left/right edge of level 0 grid marks in a prosodic word.”

⁷²With regard to the situation in Romanian and Spanish, both languages do seem to permit antepenultimate

to be investigated regarding the usage of quantity-sensitive *LAPSE constraints are whether they achieve the expression of stress generalizations more effectively than, e.g., quantity sensitive feet, and second, whether they lead to worrisome overgeneration in the typology of stress.

$$(58) \text{NONFINALITY} \gg * \text{EXTENDED LAPSE-R-}\bar{\sigma} \gg \text{ALIGN}(X_i, L) \gg * \text{LAPSE-R-}\bar{\sigma}$$

Putting aside, for the moment, the question of whether the position recessive accent should be most appropriately driven by (EXTENDED) NONFINALITY or simply ALIGN, we can consider the other portion of Steriade's analysis, namely, how the accent is realized, once its position is calculated. Here, the crucial claim is that, while the realization of the accent itself is a high tone, the tonal constraints of the phonology prefer for this H to be signaled by a preceding and following L. Thus, within a syllable with two TBUs (two vocalic morae), a contour HL should ideally be realized. To derive a HL default, three constraints are at work:

- **H//L on 'σ**: penalizes any accents syllable without a HL or LH contour (thus, presumably any accented short syllables with just H or any long accented syllable with HH).
- $\mu_1 > \mu_2$: penalizes any case in which the second mora is more prominent than the first, i.e., HL or HH is preferable to LH.
- **ALIGN $\acute{\mu}$ -H** penalizes misalignment between tonal and moraic prominence: if a given mora in a syllable is marked as the prominent one, it ought to bear the H (thus: LH on [$\acute{\mu}\mu$] or HL on [$\mu\acute{\mu}$] violate this constraint).

The result is that, in final syllables and in stressed long penults followed by only one vocalic mora in the ultima, the default HL (=circumflex) will emerge; this makes the $\sigma\omega\tau\eta\rho\alpha$ Rule look epiphenomenal, at least in recessive words like $\chi\eta\rho\upsilon\zeta$ or $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\zeta$. Where a stressed penult is followed by an ultima with two vocalic morae, *LAPSE RIGHT $\bar{\Sigma}$ has likewise forced the stress onto the penult rather than the antepenult, or when the final is light, stress can be retracted to the preantepenult. In both of these cases, the stressed syllable surfaces with level H ($\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$, $\alpha\nu\theta\rho\acute{\omega}\pi\omicron\upsilon$), rather than the falling, default HL. Consequently, Steriade must create a sort of mirror-image of the $\sigma\omega\tau\eta\rho\alpha$ Rule: the falling contour is banned if mora than one mora follows within the prosodic word. In effect, rather than just banning H from the configuration [σ_0 .VV.V], HL is banned from the configuration [σ_0 .VV].⁷³ To summarize: in forms

stress in a small set of final [-VC] words (e.g., Rom. *máximum*, Sp. *régimen*), which however tend to belong to a technical register of the language (see Roca 1999: 674–6 for examples and discussion).

⁷³Steriade (2014: 8) does rightly observe, in arguing for the default status of the circumflex on VV-nuclei and the epiphenomenality of the $\sigma\omega\tau\eta\rho\alpha$ Rule, that the $\sigma\omega\tau\eta\rho\alpha$ Rule does not apply as a consequence of adding enclitics: an oxytone like $\sigma\omega\tau\eta\rho$ 'savior' does not become perispomenon, e.g., $^X\sigma\omega\tau\eta\rho\ \tau\epsilon$ 'and savior'. While this fact is fatal is attributing grave formation – which applies at the end of orthotonic words and clitic sequences equally – and the $\sigma\omega\tau\eta\rho\alpha$ Rule to the same causal factor (e.g., L%), nothing necessarily speaks against limiting the domain of the $\sigma\omega\tau\eta\rho\alpha$ Rule to the minimal prosodic word. That is, the prosodic boundary in a structure [$\sigma\omega\tau\eta\rho$] _{ω -min} $\tau\epsilon$] _{ω -max} could block the alternation to circumflex. Incidentally, if one holds that (some) enclitics are incorporated into the minimal prosodic word, rather than always building a recursive prosodic word, as in Revithiadou 2018, non-application of the $\sigma\omega\tau\eta\rho\alpha$ Rule in a case like $\sigma\omega\tau\eta\rho\ \tau\epsilon$ would be impossible to explain altogether.

like πολυπίδαξ [polupí:daks], HL is permitted on the penult because only a single vocalic mora follows; conversely, in forms like ἀλώπηξ [aló:pe:ks], more than one vocalic mora follows, and the penult therefore receives HH (LH would violate ALIGN $\acute{\mu}$ -H, so instead H//L on 'σ is violated).

A further consequence of treating HL as a default is that the acute in final syllables in the nominative and accusative in many animate nouns is seen as problematic. Words with accent on the final syllable must have a lexical accent, so that Steriade considers the realization of an acute in words such as φυγή 'flight' or ποιμήν 'shepherd' to be an issue indicates that she takes the lexical accent to be just that: a diacritic marking or foot edge that indicates in which syllable the accent should be realized. If one were to operate with lexically specified tone linked to a specific TBU (or perhaps even a lexical diacritic already linked to a given mora), the acute in such cases could be attributed to tonal faithfulness. Interesting to note among the third declension nouns with acute in the final syllable is that these are, for the most part if not entirely, stems with a length alternation between their (mono- or disyllabic) nominative singular and (di- or trisyllabic) forms in other cases: cf. ποιμήν : ποιμένος; πούς : ποδός. In such cases, the underlying stem vowel is arguably short, in which case it could only bear an acute, and is lengthened either to satisfy a minimal word requirement or through compensatory lengthening. Stems like /ri:n-/ 'nose', however, which maintain a long vowel throughout the paradigm, have an acute in their monosyllabic nominative (ῥίς), expected mobility to accented inflectional endings (ῥινός), and circumflex in disyllabic strong cases (ῥίνα, ῥίνες), as per the Lex σῶτηρα. In disyllabic stems, length alternation is less the rule, and non-alternating stems certainly may have acute on their final syllable in monosyllabic forms, but in monosyllabic stems, many forms are affected by such an alternation.

Finally, the treatment of enclitic accentuation works mostly well, in that the same constraints used for generating H in orthotonic words can be invoked to generate H on clitics, granting that CULMINATIVITY violations are allowed within the extended prosodic word, and that immediately adjacent H tones are not permitted (OCP(H)). However, two not insubstantial objections may be raised. First, no account of the difference in behavior between words like οἶκος and words like κήρυξ with respect to enclitics is available: both are predicted to show a secondary acute on the final syllable of the orthotonic word, which would be contrary to fact for ^xκήρύξ τις 'some herald'. Second, that no H falls on a disyllabic enclitic in instances where an H or HL falls on the final syllable of the orthotonic word (πατήρ ἐστι) must be encoded by brute force: a markedness constraint against H on an enclitic.

6.2.2.4 Summary of Existing Accounts

As the discussion above has aimed to impress upon the reader, a coherent account of Attic-Word prosody constitutes a non-trivial problem. In part, the difficulties are empirical: we do not know, for instance, what properties the grave accent had, and which boundary tones were really present in which prosodic constituents. Although the testimony of tone-tune settings (see 6.2.1.6 above) inspires confidence about the nature of the acute and circumflex, no other sorts of tones belonging to the ω or larger prosodic constituents have been securely confirmed or disproven. Much of the analytical work undertaken thus inevitably rests on a

certain amount of guesswork, attempting to cobble together pieces that result in descriptive adequacy while obtaining typological plausibility and analytical elegance.

The existing approaches to the analysis of prosody in Attic-Ionic offered by Sauzet/Golston, Steriade, Ito & Mester, and Revithiadou all carry certain empirical, theoretical, and typological advantages, but simultaneously exhibit some difficulties on each account. In the case of the work from Sauzet/Golston and Ito & Mester, one issue is that their analyses of the “recessive” accent are not situated in the larger context of cases in which the accent is not recessive; ideally, a systematic account of Attic-Ionic prosody ought to invoke some of the same principles in generating both the default recessive accent and other cases. Furthermore, not all potential combinations of relevant prosodic shapes are always discussed. Steriade achieves much more empirical coverage, addressing accentual mobility in monosyllabic stems and the persistence of the high tone on the final or penultimate syllable in paradigms of particular lexemes or classes of words built with the same derivational suffixes. The constraints invoked to do so (e.g., POLYSYLLABIC, ACCENT OBLIQUE), however, are somewhat unsatisfying; they give the impression of stipulatively generating the correct outcomes by brute force, rather than through generally applicable principles. In this sense, while Steriade’s analysis takes good account of the lexical accent subsystem of Greek, its workings look almost independent of the grammar responsible for the recessive accent.

To close, I summarize below the structural differences in recessive accentuation posited by the three main accounts summarized above, indicating which structures may be problematic or unaddressed. Perhaps surprisingly, explicit accounts of the association structures in disyllabic words are sometimes lacking in the tonal account of Ito & Mester and Revithiadou. Note as well that the Gradient Symbolic Representations employed by Revithiadou allow for coda obstruents to act as tone-bearing in case the final vowel is short. Golston’s moraic trochees with HL* pitch accent, for their part, do not directly explain recessive words with σωτήρα-Rule structures, like πολυπῖδαξ; perhaps a transformation of πολπῖδαξ → πολυπῖδαξ after linking of the pitch accent is assumed.

As far as H mobility in monosyllables is concerned, only Steriade has seriously addressed this phenomenon. Likewise, the position of H in derived words with multiple derivational affixes has only been seriously considered by Steriade. Again, the lack of interface between lexically specified aspects of prosody and default word prosody must be remedied.

6.2.3 Towards a Revised Analysis of Attic-Ionic Word Prosody

In the following, I develop a novel analysis of Attic-Ionic word prosody that attempts to draw from some of the insights of the earlier analyses surveyed above while also taking account of the empirical indications of prosodic structure offered by evidence from Ancient Greek music and meter. Some of these earlier insights include:

- the role of the moraic trochee as an organizing unit of suprasyllabic prosodic structure, as further empirically supported by phenomena in meter and allomorph selection (Golston 1990, Golston and Riad 2000, Golston and Riad 2005, Gunkel 2010: Ch. 1-2);

Prosodic Shape	Example	Golston 1990	Ito & Mester / Revithiadou	Steriade 2014
[V̌.V̌(C)]	φίλος	(^h p ^h i.lo)⟨s⟩, L* delinked to prevent crowding	Unaddressed	'p ^h i.lo default H on V̌
[V̌.V̌CC]	κόραξ	kó.([*] 'rak)⟨s⟩	Unaddressed; L% associated to [k]?	'kó.raks, default H on V̌
[V̌.V̌VC ₀]	πίθηξ	pí.(^h t ^h ε:k)⟨s⟩	pi ^H .t ^h ε: ^{LL%} ks	'pí.t ^h ε:ks, default H on V̌
[VV.V̌C ₀]	κῆρυξ	(kε:).(^h ruk)⟨s⟩	kε: ^{HL} .ru ^{L%} ks tone-bearing [k] avoided	'kε:.ruks default HL on VV
[VV.V̌VC ₀]	θώραξ	(t ^h ɔ:).(^h 'rak)⟨s⟩	t ^h ɔ ^H .ra: ^{LL%} ks	't ^h ɔ:.raks no HL when followed by more than one vocalic mora
[σ ₁ .V̌.V̌(C)]	ἄλκιμος 'mighty'	(al).(^h 'ki.mo)⟨s⟩	a ^H l.ki ^L .mo ^{L%}	'al.ki.mos, default H on V̌
[σ ₁ .V̌.V̌VC ₁]	ἀλκίμους	(al).ki.(^h 'mo:)⟨s⟩	al.ki ^H .mo: ^{LL%} s	al.'ki.mo:s no HL when followed by more than one vocalic mora
[σ ₁ .VV.V̌(C)]	ἄνθρωπος	(an).(^h t ^h rɔ:).po⟨s⟩	a ^H n.t ^h rɔ: ^L .po ^{L%} s	'an.t ^h rɔ:.pos default H on V̌
[σ ₁ .VV.V̌VC ₁]	ἀνθρώπου	(an).(t ^h rɔ:).(^h 'po:)⟨s⟩	an.t ^h rɔ: ^H .po: ^{LL%} s	'an.t ^h rɔ:.po: no HL when followed by more than one vocalic mora
[σ ₁ .V̌.V̌CC]	παιδότης	(pai).do.([*] 'trip)⟨s⟩	pai.do ^H .tri ^L p ^{L%} s	pai.'dó.trips default H on V̌
[σ ₁ .VV.V̌CC]	πολυπιδάξ	(po.lu).(pi:).([*] 'dak)⟨s⟩	po.lu.pi: ^{HL} .da ^{L%} ks	po.lu.'pí:.daks default HL on VV

Table 6.6: Prosodic structures predicted by three analyses of recessive accentuation

- variability in the treatment of words ending in [-VCC] as targets for primary stress or tonal linking (Revithiadou 2018);
- HL as a default tonal pattern on syllables with primary stress with VV nuclei (Steriade 2014).

6.2.3.1 Preliminaries: TBUs, Foot Structure, and a Tonal Inventory

Before delving into the details of the analysis, some underlying assumptions concerning prosodic structures and the relationship between tone and those structures require some explanation. The relevant questions to be discussed are the following:

- how does tone relate to stress, if stress is present, and what is the tone-bearing unit?
- what foot structures, if any, are present, and how do they relate to the calculation of primary stress and tonal association?
- what lexical tones are present, and which intonational prominence tones (= starred tones, pitch accents) are inserted?
- what sort of boundary tones occur with which prosodic constituents?

Relation of Tone and Stress A major point of contrast with many of the analyses considered above will concern the tone-bearing unit (TBU) and basic presentation of Greek prosody.⁷⁴ Namely, I will assume that the TBU of Attic-Ionic was the syllable (σ), not the mora (μ); this assumption brings into line the status of stress and tone in the language: both are assumed to be present, and both are properties associated to or realized on syllables. In this fashion, word-level pitch accents (starred tones) may be understood to prefer to stand in a one-to-one correspondence with stress, given that some or all of the mapping constraints between stress and tone given in 59 prove to be highly ranked in the grammar.

- (59) Stress-to-Tone and Tone-to-Stress mapping constraints over pitch accents, tone, and stress
- a. $T^* \rightarrow '\sigma$ (starred tone to primary stress): Assign one violation * for every starred tone in the output that is not associated with a syllable bearing primary stress.
 - b. $'\sigma \leftarrow T^*$ (primary stress to starred tone): Assign one violation for * each syllable bearing primary stress in the output that is not associated with a starred tone.
 - c. $'\sigma \leftarrow T$ (primary stress to tone): Assign one violation for * each syllable bearing primary stress in the output that is not associated with any tone.

⁷⁴Sauzet (1989) and Golston (1990) are not wholly explicit on what sort of TBU they presume. Rules for the association of the HL* pitch accent (see Golston 1990: 72–4) indicate that it is to associate to a metrically prominent syllable, not a metrically prominent mora or mora within a metrically prominent syllable, but Golston's prose often speaks of association of tones to morae.

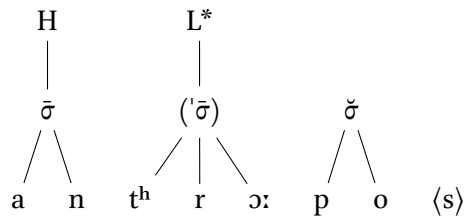
- d. $\sigma \leftarrow T$ (stress to tone): Assign one violation * syllable bearing any degree of stress in the output that is not associated with any tone.

Given, then, that intonational prominence tones (pitch accents) are inserted by the formation of particular prosodic structures (namely, the ω), or that certainly morphemes are lexically preassociated to starred tones, the placement of tone and stress can amount to either linking of starred tones to stress, or the attraction of stress to the position of a starred tone. The same inventory of constraints, combined with OBLIGATORINESS and CULMINATIVITY of stress, will ensure that maximally one pitch accent per word occurs.

What, then, is the source of stress? On account of the arguments from meter and allomorph selection (Golston 1990, Golston and Riad 2000, Golston and Riad 2005, Gunkel 2010: Ch. 1-2), moraic trochees occurring towards the right edge of the word seem most attractive. Foot structure is then the main device through which the effects of the Law of Limitation are achieved. As far as alternatives are concerned, syllabic trochees are not viable for the reasons argued in Golston 1990, imposing a window through quantity-sensitive *LAPSE (Steriade 2014) constraints is too untested, and assuming an absence of stress altogether (Ito and Mester 2017, Revithiadou 2018) is implausible in the face of the metrical evidence and evident OBLIGATORINESS and CULMINATIVITY.

As assumed by Sauzet and Golston, operating with moraic trochees requires the assumption of an HL^* prominence tone (pitch accent) in recessively accented words for cases in which H occurs as far to the left as allowed by the Law of Limitation. In a word like $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$ [án.tʰrɔː.pos], given extrametricality of the final consonant, any model of relationships between syllable weight and stress (see Ryan 2019: Ch. 2 for a survey) that also grants that stress tends to prefer to be closer to the right will select the medial syllable [tʰrɔː] as the locus of primary stress. As example 60 shows, a mutual association between L^* and primary stress is satisfied in such cases.

(60) $T^* \Leftrightarrow \sigma$ in recessively accented words



Ultimately, it will turn out that the constraints governing mappings between stress and tone are likely undominated in the grammar. Taken together, these constraints mean that all syllables bearing stress must be associated with some tone, and any starred tone (pitch accent) in the output must be associated with the primary stress, and every primary stress must be associated with a starred tone; the undominated ranking of $T^* \rightarrow \sigma$ thus ensures that no prosodic word contains more than one pitch accent as a consequence of likewise undominated CULMINATIVITY.

Foot Structure Considering the distribution of H in recessively accented words can then serve as a diagnostic of the position of primary stress. Whenever the final syllable of a word with recessive accentuation contains a long vowel or heavy diphthong (i.e., a VV-nucleus), level H tone (acute) inevitably occurs on the preceding syllable, regardless of the weight of the penult; therefore, primary stress may be inferred to fall on the final syllable in such cases, associating with the the L* portion of the prominence tone (e.g., ἀνθρώπου [an.(^hrɔ́:).(‘po:)]). Likewise, if the penult contains a VV-nucleus and the final is of the structure $\check{V}(C)$, as in 60, H tone falls on the antepenult. These basic cases indicate that, if multiple feet are built at all, the rightmost receives primary stress. Similarly, when both final and penult contain a short vowel and the final is not heavy, H tone falls on the antepenult (e.g., ἄλκιμος [ál.(‘ki.mo)<(s)>] ‘mighty’, δαίδαλος [daí.(‘da.lo)<(s)>]), regardless of the weight and composition of the antepenult; this distribution indicates that no foot’s right edge may stand more than two syllables from the end of the word.

The situation becomes more interesting and complex when the final syllable is of the shape $\check{V}CC$. Here, based on the distribution of H and L*, stress may be interpreted as distributing itself differently between penult and antepenult. When the penult contains only a short vowel, whether heavy ($\check{V}C_1$) or light ($\check{V}$), the penult is acute (e.g., κόραξ [kó.(‘rak)<(s)>] ‘crow’, ἄντυξ [án.tuk<(s)>] ‘rim’), regardless of the weight of the antepenult (e.g., ὠμόθριξ [ɔ:.mó.(‘^hrik)<(s)>] ‘having a fierce crest’).⁷⁵ The ready interpretation of these cases is then that the primary stress falls on the final syllable, heavy due to the consonant cluster. However, if the penult contains a VV-nucleus the penult instead exhibits a circumflex; one interpretation, which I will adopt, is that the penult in such cases attracts the primary stress in violation of RIGHTMOST (assuming that the $\check{V}CC$ final is footed) because such nuclei are durationally longer and thus relatively heavier. Words such as κήρυξ [(‘kê:).(‘ruk)<(s)>] and πολυπίδαξ [po.lu.(‘pî:).(‘dak)<(s)>] thus place the entirety of the prominence tone HL* on the VV-nucleus of the penult.⁷⁶ To this end, the MAIN → VV, defined in (61), may be invoked (Ryan 2019: 36–7).⁷⁷

- (61) MAIN → VV: Primary stress should fall on a long vowel or diphthong. Assign one violation * if a syllable containing a long vowel or diphthong is footed and does not bear the primary stress in a word.

Thus, the two strongest preferences in the grammar of Attic-Ionic foot construction are the avoidance of feet whose right edge stand more than two syllables from the right edge of

⁷⁵ ὠμόθριξ is a *hapax legomenon*, found only in Lycophon, *Alexandra* 340.

⁷⁶ That the H does not associate to the antepenult, yielding ^Xπολύπιδαξ, in such cases, must be explained and motivated. A proposal is developed under the full analysis of recessive accentuation below.

⁷⁷ I crucially deviate from Ryan’s definition by referencing foot structure. While in recessively accented words primary stress on the antepenult or other syllables farther to the left when neither the penult nor final syllable contains a VV-syllable can be avoided by the ranking ALL-Ft-R × 2 ≫ MAIN → VV, issues would then potentially arise in lexically accented forms with H* on a light final syllable, but a VV penult, e.g., σιγήρός ‘silent’. In such cases, I assume that the heavy penult remains unfooted (in violation of WSP) in order to avoid a violation of higher-ranking MAIN → VV. See further discussion under 6.2.3.4 below.

the word (ALL-FT-R $\times$ 2), on the one hand, and avoidance of foot structures other than (–) or (–◌), on the other.⁷⁸ Violations of RH-CONTOUR (penalizing (‘–◌)) are strictly excluded, and violations of WSP are avoided within the two syllable stress window. The constraint *CLASH is included to illustrate that stress clash is permitted when a sequence of two heavy syllables ends a ω ; assuming that all heavy final syllables build a foot proves a necessary ingredient in explaining the Law of Limitation, and may prove necessary to account for some aspects of enclitic accentuation (see 6.2.3.7 below). Of next greatest importance is the preference to assign primary stress to heavy syllables with VV nuclei via the constraint MAIN $\rightarrow$ VV.MAIN $\rightarrow$ VV is satisfied when primary stress falls on a long vowel or diphthong. Ranking this constraint above the usual WSP would seem to place Ancient Greek among languages that exhibit the weight scale $V < VC < VV$ for purposes of stress, including Chickasaw, Finnish, Hupa, Klamath, Mam, Tamil, among others (see Ryan 2019: 22–9 on such weight scales and their exemplification).

For all combinations of penultimate and final syllable containing at least one heavy ([VV] or [VCC]) syllable, or a sequence of two [V̆] syllables at the right edge, the constraint ranking given in Table 6.7 below will always produce primary stress on one of the last two syllables. These combinations are all licit; their violation profiles constitute the least marked prosodic structures to be constructed in the absence of preassociated lexical tone.

Exactly what foot structures result in disyllabic words without any VV-nuclei a VCC final syllable is less immediately clear. Given the preferential association between L* and primary stress in recessively accented words, words of the shape φίλος [pʰi.lo(s)] either break the association T* $\Leftrightarrow$ ‘σ (if a binary foot [(‘pʰi.lo)(s)] is built) or violate FOOTBINARITY (if a foot [pʰi.(‘lo)(s)] is built). Due to phonotactic constraints on tonal distribution and faithfulness constraints over underlying tone, both metrical structures prove to be compatible with the larger treatment of word prosody.

Tonal Inventory and Tonal Phonotactics The immediately preceding discussion has argued that the assumption of moraic trochees built at the right edge of a minimal prosodic words necessitates the assumption of prominence tone consisting of a high-low sequence, where the low portion preferentially associates to stressed syllables. I take this intonational prominence tone (“pitch accent”) to be inserted with the construction of each and every $\omega_{[+min]}$ constituent, which means, in cases of non-recessive accentuation, HL* competes with (at least one) lexical tone; furthermore, the presumption of unitary underlying representations throughout the lexicon means that some recessively accented words may realize HL* in preference to (one or more) underlying lexical tones in certain prosodic configurations.

Besides the HL* prominence tone, lexical entries must be able to be marked for level high tone, H*, since as observed under 6.2.1.3, acute and circumflex are minimally contrastive in monosyllables, and this surface contrast is not reducible (synchronically, at least) to any other segmental or prosodic property. Hence, the underlying representation of, e.g., /rí:n-/ ‘nose’ is marked as H*, whereas /aig-/ ‘goat’ is not, as shown in 62. The surface circumflex in

⁷⁸ ALL-FT-R $\times$ 2 is included as a separate constraint to simulate the self-ganging of this constraint in a Harmonic Grammar implementation, which is necessary to obtain the desired results.

		ALL-Ft-Right × 2	RH-CONTOUR	MAIN → VV	WSP	ALL-Ft-Right	RIGHTMOST	*Clash
a.	Ť.('ŤC)⟨C⟩							
b.	Ť.('VV)⟨C⟩							
c.	('VV).Ť⟨C⟩					*		
d.	('VV).('ŤC)⟨C⟩					*	*	*
e.	(,VV).('VV)⟨C⟩					*		
f.	ŤC.('Ť.Ť)⟨C⟩				*			
g.	VV.('Ť.Ť).⟨C⟩				*			
h.	ŤC.Ť.('VV)⟨C⟩				*			
i.	ŤC.('VV).Ť⟨C⟩				*	*		
j.	ŤC.(,VV).('VV)				*	*		
k.	VV.Ť.('VC)⟨C⟩				*			
l.	Ť.Ť.('VV).('VC)⟨C⟩					*	*	*
m.	VV.('VC).Ť⟨C⟩				*	*		

Table 6.7: Distribution of Stress and Foot Structure in Recessively Accented Words

unmarked monosyllables is thus nothing more than the HL* prominence tone. The availability of H* in lexical entries means that it can be posited as well for all word forms that exhibit non-recessive accentuations; the lexeme /sɔ:tér-/ , built with the agent noun suffix /-tér-/ , presumably has an H* , whereas recessively accented monosyllables presumably exhibit the default prominence tone, HL* .

(62) Underlying H* vs. ∅ Contrast in Monosyllables

a. H* Present

H*

|

r i: n- 'nose'

b. No H*

∅

|

ai g- 'goat'

(63) Underlying H* vs. ∅ Contrast in Polysyllables

a. H* Present

H*

|

s ɔ: t ε: r- 'savior'

b. No H*

p o l u p i: d a k 'having many wells'

A crucial unanswered question concerning the HL* prominence tone – never addressed in other presentations of Attic-Ionic prosody that presume it – is whether it constitutes a binary pitch accent consisting of two level tones (thus properly H+L*) or a unary falling pitch accent (thus HL*). This question is not trivial, since it decides what violations hold when H and L* are distributed over multiple syllables (i.e., whether NOSPREAD is violated or not), and what faithfulness violations might be incurred if the prominence tone were not fully associated to TBUs. In short, if this pitch accent is unary, the fact that H and L* always appear together may be attributed to the operation of a single faithfulness constraint MAX-HL*: either the entire tone is present in the output, or not. The assumption of a unary tone would require, for cases like 60, that the appearance of H and L* on separate TBUs be due to the spreading of the tone. A binary pitch accent, H+L*, would incur no violations of NOSPREAD, but in principle allow for outputs in which only one of L* or H are realized; the fact that some H exceptionlessly occurs on the surface could then be attributed to relatively high ranking MAX-H.⁷⁹ Given that H and L* more often and not surface on adjacent syllables (constrained by CONTIGUITY-T), a binary pitch accent H+L* will be more harmonic for most outputs.

⁷⁹I am reluctant to invoke a constraint such as OBLIGATORINESS(H), which would simply require every prosodic word to have a high tone. In principle, OBLIGATORINESS should be reserved for stress.

Cases in which lexical H* exhibits alternations between level high tone and falling tone, conditioned by the $\sigma\omega\tau\eta\rho\alpha$ Rule, must in turn involve either an alteration or replacement of underlying H*. A replacement of the underlying, preassociated H* with the default H+L* would require that the vowel associated with H* also be marked for lexical stress; otherwise, given a ban on level H* in a penult preceding a $\check{V}$ final syllable, default stress and prominence tone would yield $^X\sigma\acute{\omega}\tau\eta\rho\alpha$. In order to avoid positing an unnecessary excess of lexical specification of prosody, H* alone can be invoked to attract stress, with the consequence that $\sigma\omega\tau\eta\rho\alpha$ -alternations cannot be attributed to the H+L* default being attracted to the same syllable. Hence, the falling tones in penultimate syllables that alternate with level high tones must be attributed to the insertion of another L tone (violating DEP-L) or transforming the level H* into a unitary falling tone, H*L.

The more fundamental problem of $\sigma\omega\tau\eta\rho\alpha$ -alternations is how to motivate them without recourse to essentially *ad hoc* constraints (e.g., * $\mu\mu\mu$ in Kiparsky 2003a), if they are not to be explained by an alternation in tone assignment between (marked) H* and (unmarked) H+L*. Appeals to a L% boundary tone, as in Gunkel 2013 and Revithiadou 2018 (following suggestions in Devine and Stephens 1994: 180–4), help provide a substantive motivation, but suffer from lack of a secure empirical basis (see discussion of Gunkel and Ryan 2022 under 6.2.1.6 above), and even in analytical terms, assuming a consistent L% at the level of the $\omega_{[+min]}$ (see 6.2.3.2 below) creates problems for understanding the distribution of the grave accent (an issue which leads to issues for the analyses in Revithiadou 2018 and 2019, where the grave is ignored). Perhaps the best that can be achieved, at present, is to formulate a phonotactic constraint on the distribution of tone in the language that simultaneously has general validity, can motivate $\sigma\omega\tau\eta\rho\alpha$ -alternations, and perhaps other patterns of pitch movement as well, such as H-mobility in monosyllabic stems.

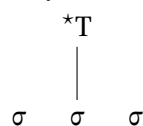
When the syllable is taken to be the tone-bearing unit of Attic-Ionic, the following restriction on tone may be assumed: every minimal prosodic word in which the pitch accent is non-final must obligatorily exhibit either a rise between two specified tonal targets or a fall between two specified tonal targets. The two tonotactic configurations shown in (64) are thus illicit because they exhibit only a single, non-final level tone in a word in which another tone could be hosted (either on another syllable, or perhaps as a contour tone within a syllable). These two configurations can then be unified, and their ungrammaticality expressed with the tonotactic markedness constraint given in (65) (*NOFALL $_{\omega}$)

(64) Two ungrammatical tonotactic configurations of Attic-Ionic

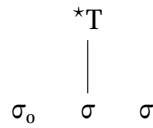
a. Disyllable with level tone on initial syllable and no tone on final



b. Trisyllable or longer with level tone on penult and no tone earlier or later



- (65) Tonotactic markedness constraint banning absence of HL contour in the $\omega_{[+min]}$ contour: $^*\text{NOFALL}_{\omega}$



Assign one violation * for each minimal prosodic word (ω_{ω}) in which only a single level tone associated to a non-final syllable is found.

Banning these two configurations amounts, effectively, to a brute-force specification of what tonal patterns are permitted to occur, without any substantive motivation. Moreover, given the inventory of lexical and intonational tones adopted here, only rarely does the markedness constraint in 65 play a potentially significant role, namely, in $\sigma\omega\tau\eta\rho\alpha$ -type words with a lexical H^* (thus /sɔ:tɛ:r-/), and in monosyllabic stems with lexical H^* to which another syllable is added (e.g., DAT.SG /pód-í/); what both configurations share is that they risk realizing a level $H(^*)$ tone on a non-final syllable without any following fall to L. In recessively accented words, the HL^* prominence tone, plus constraints governing the distribution of stress and tone, suffice to avoid the banned configuration without explicitly declaring it ungrammatical. Repairing an instance of 65 can be achieved in one of three ways: 1) inserting another tone (thereby violating DEP-T); 2) transforming an underlying level tone into a surface contour tone; or 3) ensuring that a level tone appears on a final, rather than non-final, syllable. Which of these options is appropriate under which conditions will be worked out below.

Boundary Tones In the analysis developed below, I do not assume the presence of any boundary tones at the level of the $\omega_{[+min]}$ or any larger prosodic constituent. This choice is made on the basis of the considerable uncertainties involved, and that, in the event that any boundary tones later obtain a more secure empirical basis, effects attributed here to phonotactic constraints referring to prosodic constituents may be reformulated with respect to boundary tones. Before considering the effects that the assumption of boundary tones might have on the derivation of pitch patterns on monosyllabic words, let us briefly review the claims made concerning boundary tones in the literature and problematize them:

1. Devine and Stephens (1994: 180–4) believed to have found evidence for a further low pitch target on the final syllable of words with H on the antepenult in Ancient Greek music. However, Gunkel and Ryan (2022) have recently reevaluated this claim with a larger corpus of musical materials, and, having analyzed it with greater statistical sophistication, find no statistically significant differences between syllables potentially associated with L% and unaccented syllables.
2. Ito and Mester (2017) and Revithiadou (2018) include a L% boundary tone at the right edge of the $\omega_{[+min]}$ as an essential component in generating the behavior of recessive and enclitic accentuation. Revithiadou further posits a H% boundary tone at the right edge of recursive prosodic words. A L% tone is, however difficult to integrate in a

coherent phonological analysis of pitch patterns on monosyllables (see below), and a H% at the edge of a $\omega_{\max}$ is seemingly contradicted by the fact that H on the last syllable of disyllabic enclitics is also subject to transformation into a grave (e.g., $\kappa\eta\rho\nu\xi\acute{\epsilon}\sigma\tau\iota$).

3. Gunkel (2013) has attributed the $\sigma\omega\tau\eta\rho\alpha$ Rule to avoidance of adjacent H and L%, but a persistent L% at the right edge of all minimal prosodic words, regardless of whether or not enclitics follow, is difficult to square with the failure of final syllable acutes to become grave when enclitics follow.

In short, given its uncertain empirical basis, a L% tone at the right edge of the minimal prosodic word may create more problems than it solves. If the $\sigma\omega\tau\eta\rho\alpha$ Rule may be motivated via the tonotactic constraint in 65, the formation of graves may similarly be motivated by a phonotactic restriction referring to the edges of prosodic constituents: H may not occur immediately before the right edge of a $\omega_{\min/\max}$ if immediately followed by the left edge of another ω (at any level of recursion) as shown in 66.⁸⁰ The grave indicates presence of stress but absence of H tone. It may then indicate either a H* that has been changed to a L*, or that a syllable bears stress but no tone.

- (66) Tonotactic markedness constraint banning H at the right edge of ω preceding ω
- $$\begin{array}{c} (*H]_{\omega} \omega [] \\ *H \\ | \\ \sigma \quad \omega] \quad \omega [\end{array}$$

Largely unproblematic, on the other hand, would be to assume a H% boundary tone at the right edge of intonational phrases, on the basis of the fact that grave formation does not apply in this context (cf. Goldstein 2014): the presence of H% would precisely license the surface acute allowed at the right edge of intonational phrases. Nonetheless, the evidence of Hagel 1994/95: 93–100 that oxytone words (words with final acute) are avoided in the last two feet of the dactylic hexameter of Homer might suggest overall intonational declination and perhaps a Low target at the right edge of the ι (see further the discussion in Bozzone 2021: 74–8). What boundary tones, exactly, might be present at the levels of the ϕ or ι is uncertain, but also not crucial for present purposes.

6.2.3.2 Lexical H* and Default H+L* in Monosyllables

This section is concerned with the prosody of monosyllables, represented here with the NOM.SG forms of various nouns.⁸¹ While generating the desired patterns is almost trivial

⁸⁰Such a markedness constraint might be motivated if the ω were assumed to have an *initial* boundary tone of some sort: %L might trigger tonal assimilation, while %H might trigger tonal dissimilation; no relevant investigations on the status of potential initial boundary tones in Ancient Attic-Ionic have been conducted.

⁸¹Throughout, I borrow from Gussenhoven 2004: Ch. 8 constraints on tonal realization, tonal association, and that enforce the association between stressed syllables and starred tones in particular.

given the representations and markedness constraints assumed here, one further objective is to show that a L_{ω} boundary tone creates considerable complications.

As discussed under 6.2.1.3 above, Attic-Ionic must possess a surface lexical contrast between stems with level high tone, H^* , and falling tone, $H+L^*$; compare again the minimal pair $\kappa\acute{\eta}\rho$ [$k\acute{e}:r$] ‘fate’ versus $\kappa\hat{\eta}\rho$ [$k\hat{e}:r$] ‘heart’. Given reasons to believe that the falling tone is the automatic or default tone on monosyllables with a VV-nucleus, lexemes like / $k\acute{e}:r$ / and / $p\acute{o}d$ -/ may be assumed to bear a lexical high tone, H^* , associated with the potential syllable nucleus. Lexemes such as /aig-/ ‘goat’ or /mu-/ ‘mouse’, on the other hand, would have no lexical specification and thus receive falling tone by default. This $H+L^*$ tone, which surfaces as a circumflex in unmarked monosyllabic forms, is the “contonation” posited by Allen (1973), assumed to be inserted in every minimal prosodic word; when both the intonational HL^* and the lexical H^* are present in an input, they compete for realization.⁸² Assuming that unassociated tones undergo stray deletion, the preservation of lexical H^* in monosyllables like $\kappa\acute{\eta}\rho$ [$k\acute{e}:r$] may be attributed to several factors: preservation of underlying associations (IDENT-Association), preservation of a H^* (MAX- H^*), or avoidance of contour tones (NOCONTOUR). If any one of these constraints dominates the realization of $H+L^*$ (MAX-H, which also applies to H^* , and MAX- L^*), the lexical H^* will surface faithfully, as shown in 70; when no H^* is present, $H+L^*$ is allowed to associate instead. Since, in the development of this analysis as a whole, H^* will be the only underlyingly associated tone that is posited, IDENT-Association and MAX- H^* will always have the same violation profiles; I employ MAX- H^* throughout.⁸³

- (67) NOCONTOUR: TBUs should not be associated with multiple distinct tonal targets. Assign one violation * for each instance of HL or LH associated to a TBU.
- (68) MAX-H: Every level high tone, H, in the input corresponds to a tone in the output. Assign one violation * for each H in the input that does not stand in a correspondence relationship to a tone in the output.
- (69) MAX- L^* : Every starred level low tone, L^* , in the input corresponds to a tone in the output. Assign one violation * for each L^* in the input that does not stand in a correspondence relationship to a tone in the output.
- (70) Faithful Realization of Lexical H^* in Monosyllables: / $k\acute{e}:r$, $H+L^*$ / $\rightarrow$ [($k\acute{e}$):(r)]

⁸²Recall from above that I treat the $H+L^*$ contonation as a binary tone, which may be realized over multiple TBUs (syllables), but cannot be only partly realized; empirically, the $H+L^*$ contour, when it occurs, is always complete.

⁸³For the sake of simplifying autosegmental representations in tableaux, only the segmental, syllabic, and tonal tiers are shown. Feet, syllable weight, and stress are marked by parentheses and diacritics on syllables. ($\acute{\sigma}$), for instance, stands for a footed heavy syllable with primary stress. In underlying representations, vowels preassociated with a H^* tone are marked with an acute; other initially unassociated tones are separated from the segmental UR by commas. Thus / $k\acute{e}:r$, $H+L^*$ / indicates H^* preassociated to the vowel / $\acute{e}$ /, and a tone $H+L^*$ is additionally present in the input.

/ké:r, HL/		MAX-H*	MAX-H	NOCONTOUR	MAX-L*
a.					
			*		*
b.					
		*!	*	*!	

Granting that $H+L^*$ is always made available with the construction of each $\omega_{[+min]}$, and perhaps each successive recursion of the prosodic word, it will associate to the only syllable of a monosyllable, yielding a contour tone, when no preassociated H^* is present. That $H+L^*$ is associated at all, rather than producing a toneless form, or inserting a different level tone (H or L) indicates that NOCONTOUR and NOASSOCIATION (“do not associate TBUs any with tones”) are dominated by constraints penalizing stressed syllables without tonal association ($\sigma \leftarrow T$, and here more specifically $\sigma' \leftarrow T^*$; cf. 59 above) and insertion of new tones (DEP-T). Furthermore, it must be the case that MAX-H dominates NOCONTOUR; otherwise, the binary pitch accent would be realized as simple L^* . Similarly, one of $\sigma \leftarrow T^*$ or MAX- L^* must dominate NOCONTOUR; otherwise, the binary pitch accent would be realized as simple H.

- (71) NOASSOCIATION: TBUs should not be associated with tones. Assign one violation * for each TBU on the surface that is associated with at least one tone.
- (72) Faithful Realization of Intonational $H+L^*$ in Monosyllables: /mu-s, $H+L^*$ / $\rightarrow$ [(‘mû:)<s>]

/mu-s, HL/		$\sigma \leftarrow T^*$	MAX-H	DEP-T	NoCONTOUR	MAX-L*	NoASSOC
a.					*		*
b.		*!	*!			*	
c.				*!		*	*
d.		*!				*	*

To account for the realization of acute as grave at the right edge of a ω when followed by another ω , the constraint $^*H]_{\omega \omega}$ introduced at 66 must be invoked. If this markedness constraint dominates $MAX-H^*$, $MAX-H$, and any and all constraints requiring (stressed) syllables to be associated with (starred) tones, a toneless output might be predicted. On the other hand, if $^*H]_{\omega \omega}$ merely dominates IDENT-T, all other just mentioned constraints may be satisfied; in that case, underlying H^* would be transformed into L^* . However, because $^*H]_{\omega \omega}$ could equally be satisfied by realizing the default $H+L^*$ contour instead – which is evidently licit, since circumflex monosyllables are permitted at the juncture between two ω (see 73a.) – a toneless monosyllable must be ruled out, as the comparative tableaux in 73b. shows. Rather, grave formation must be taken to involve the adjustment of H^* to L^* through violation of IDENT-T.⁸⁴ Tableaux 73a. and c. are compatible with one another.

(73) Grave formation via $^*H]_{\omega \omega}$

a. Circumflex $H+L^*$ permitted in the context $]_{\omega \omega}[: /mu-s, H+L^*]_{\omega \omega} [...] \rightarrow [(^{\circ}mu:^{H+L^*})(s)]$

⁸⁴This conclusion is not wholly unproblematic, since it suggests that syllables with grave would still contain a Low target, rather than being interpolated between two H targets, as the evidence in Gunkel and Ryan 2022 suggests. Without clearer data on the extent to which syllables bearing L^* as part of $H+L^*$ might be subject to pitch interpolation in musical settings, the presence of L^* here cannot be rejected out of hand.

/mu-s, H+L*] _{ω ω} [.../		*H] _{ω ω} [	MAX-H*	MAX-H	'σ ← T*	NoCONTOUR	MAX-L*	IDENT-T	NoASSOC
a.									*
				*!					*
				*!	*!	*			

- b. Comparative Tableaux: toneless monosyllable (deletion of H*) is incompatible with circumflex monosyllable⁸⁵

		MAX-H	MAX-L*	NoASSOC	NoCONTOUR	'σ ← T*
a.	/kɛ:r, H+L*] _{ω ω} [.../					
	('kɛ:)(r) ~ ('kɛ: ^{H+L*})(r)	L	L	W	W	L
a.	/mu-s, H+L*] _{ω ω} [.../					
	('mu: ^{H+L*})(s) ~ ('mu:)(s)	W	W	L	L	W

- c. Underlying H* transformed into L*: /kɛ:r, H+L*]_{ω ω}[.../ → [('kɛ:^{L*})(r)]

⁸⁵For typographic simplicity and considerations of space, tones associated with a syllable nucleus are represented here with a superscript adjacent to the licensing segment.

/kɛ:r, H+L*] _ω [...]/		*H] _ω	MAX-H*	MAX-H	'σ ← T*	NoCONTOUR	MAX-L*	IDENT-T	NoASSOC
a.	$ \begin{array}{c} H^* \\ \\ ('σ) \\ / \quad \backslash \\ k \quad \epsilon: \quad r \end{array} $	*!		*			*		*
b.	$ \begin{array}{c} H \quad L^* \\ \quad \\ ('σ) \\ / \quad \backslash \\ k \quad \epsilon: \quad r \end{array} $		*!	*		*			*
c.	$ \begin{array}{c} L^* \\ \\ ('σ) \\ / \quad \backslash \\ k \quad \epsilon: \quad r \end{array} $			*			*	*	*
d.	$ \begin{array}{c} H^*L \\ \\ ('σ) \\ / \quad \backslash \\ k \quad \epsilon: \quad r \end{array} $			*		*!	*	*	*
e.	$ \begin{array}{c} \\ \\ ('σ) \\ / \quad \backslash \\ k \quad \epsilon: \quad r \end{array} $		*!	**!	*!		*		

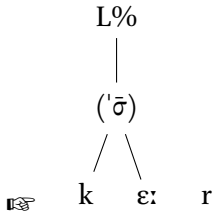
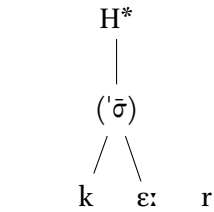
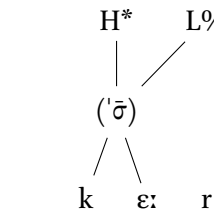
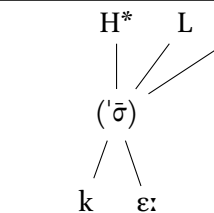
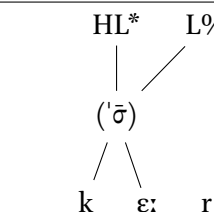
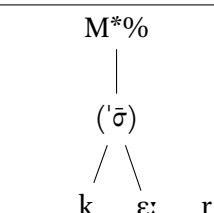
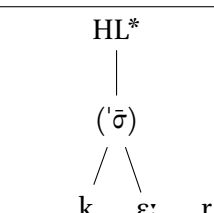
How would this analysis differ if the grave accent were instead attributed to the effects of a low boundary tone aligned with the right edge of the $\omega_{\max}$, $L_{\omega-\max}$ %. Given that this presumed boundary tone does not result in a contour tone on monosyllables with acute, the potential association of H^* and $L_{\omega-\max}$ % to the same syllable evidently has a different effect than associating $H+L^*$ to a syllable: $H^* + L_{\omega-\max}$ % yields a grave (on either $\check{V}$ - or VV -syllables), whereas $H+L^*$ yields a circumflex (allowed on VV -syllables [or possibly $V[+sonorant]$ -syllables] only). If one follows Gunkel (2013) in penalizing the adjacency of H and $L\%$ on the tonal tier, violation of this sequence might be repaired by epenthesis of L . However, tonal epenthesis is evidently not a licit solution given the input /kɛ:r, $L\%$, HL /, either; again a falling tone would be predicted. One factor motivating the avoidance of a tritonal sequence $H^*LL_{\omega-\max}$ on a (heavy) syllable could be tonal crowding: the number of tones exceeds the available duration of licit segmental tonal hosts. NOCROWD, defined in 74, will penalize the two tonal targets associated to a V -syllable or three associated to a VV -syllable.

- (74) NOCROWD: The number of tonal targets associated to a TBU must not exceed the duration of licit tonal hosts in the TBU. Assign one violation * for each tonal target above 1 associated with a syllable containing a short vowel, or above 2 for each syllable containing a long vowel or diphthong.⁸⁶

Instead, two options are then conceivable for generating the grave: either the H^* and $L_{\omega\text{-max}}\%$ undergo coalescence to yield a Mid tone (M), with a pitch target lower than H but higher than L, or the H^* is deleted to allow for association of $L\%$ with the final syllable rather than H^* . As discussed under 6.2.1.6 above, deletion seems to better accord with the most recent assessment of the musical evidence (Gunkel and Ryan 2022), but even a boundary target remains uncertain. In terms of constraints, the matter hinges on the relative rankings of MAX- H^* , MAX- $L\%$, and UNIFORMITY-T: lower-ranking MAX- H^* allows for deletion, whereas lower-ranking UNIFORMITY-T allows for coalescence. Considering, at a first pass, only cases of inputs like /ké:r, HL*, L%/ (alongside /ké:r, H+L*/ and /mu-s, H+L*/ shown above), a ranking that generates deletion of the lexical H^* and realization of the $L_{\omega\text{-max}}\%$ exists, shown in 75.

⁸⁶This definition of NOCROWD deviates from the definition given in Gussenhoven 2004: 149, which penalizes the association of more than one tone to one TBU. If one wishes to hold strictly to that definition, then NOCROWD must be low-ranked in the present grammar, and a special version of it, NOCROWD($\check{V}$) must be invoked to penalize the association of multiple tones to a TBU, σ , whose nucleus contains just a short vowel. Compare, for instance, the constraint *CONTOUR/SHORT in Li 2003: 178.

(75) Grave as H* Deletion: /kɛ:r, HL*, L%/ → [(^{L%}kɛ:r)(r)]

/ké:r, HL*, L%/		MAX-L%	*H*-L%	DEP-L	NOCROWD	UNIFORMITY(T)	MAX-H*	NOCONTOUR	MAX-L*
a.							*		*
b.		*!							*
c.			*!					*	*
d.				*!	*!			*	*
e.					*!		*	*	
f.						*!			*
g.		*!					*	*	

From the perspective of the larger phonological analysis, however, the hypothesis of H* deletion to make way for L% presents serious difficulties: if underlyingly high-toned monosyllables delete their H* to accomodate L%, why do underlying unmarked monosyllables like /mu-s/ always exhibit a contour tone, regardless? Consider the comparative tableaux for the inputs /ké:r, HL, L%/ and /mu-s, HL, L%/ in 76: given the available constraints and candidates, no ranking exists that can generate both [('ké:^{L%})(r)] and [('mu:^{HL*})(s)]. In other words, that both grave and circumflex occur in the prosodic context where L_{ω-max}% is present is seemingly inexplicable, if the grave represents high tone deletion.

(76) Comparative Tableaux: Grave as Deletion of H* Incompatible with Circumflex Monosyllables

	MAX-H*	MAX-HL	MAX-L%	NOCONTOUR
/ké:r, HL*/ ~ ('ké: ^{HL*})(r)				
a. ('ké: ^{H*})(r) ~ ('ké: ^{HL})(r)	W	L	e	W
/ké:r, HL*, L%/				
a. ('ké: ^{L%})(r) ~ ('ké: ^{H*})(r)	L	e	W	e
b. ('ké: ^{L%})(r) ~ ('ké: ^{HL*})(r)	e	L	W	W
/mu-s, HL*, L%/				
a. ('mu: ^{HL*})(s) ~ ('mu: ^{L%})(s)	e	W	L	L

This comparative tableau assumes, however, that the contour tone seen on the surface of forms like μûς [mû:s] represents the H+L* contonation alone, without L_{ω-max}%. The winning output cannot, however, violate NOCROWD to satisfy MAX-L%; otherwise, a tritonal output like [('ké:^{H*LL%})(r)] would be permissible, too. Quite simply, because all of the outputs available for the input /mu-s, HL*, L%/ are also available to the input /ké:r, HL, L%/, the only way in which the latter can generate an output different from the former is via faithfulness to its lexical H*. This line of phonological analysis leads to the conclusion that the grave does not represent deletion of an acute H*. The remaining options are then either preservation of H* and L_{ω-max}L% via violation of UNIFORMITY(T), yielding a lowered High tone (as proposed by Devine and Stephens (1994: 181–4)), or to abandon the notion of a word-level right-edge boundary tone altogether, and seek a different account of the grave (and the Lex σωτῆρα) altogether.

The tableaux in 77 and 78 illustrate that tonal coalescence to M*% on monosyllables with underlying H* and circumflex on unmarked monosyllables are compatible with one another. In 78, realization of the falling H+L* tone may be given precedence in order to satisfy association of syllables with primary stress with a starred tone (σ' ← T*), rather than with a boundary tone alone; coalescence, in this case, is excluded, since the result would be a H-L*% sequence.⁸⁷

⁸⁷ One further candidate not considered in 78 is the complete coalescence of the H+L* tone and the L_{ω-max}%. If such a coalescence is taken to involve two violations of UNIFORMITY-T, then a Harmonic Grammar analysis

(77) H* Coalesces with L_{ω-max} %: /kɛ:r, HL*, L_{ω-max} %/ → [(^{M*}kɛ:r)⟨r⟩]

can generate the categorical winners, by allowing two violation of UNIFORMITY-T to outweigh one violation of MAX-L%.

/kɛ:r, H+L*, L%/		MAX-H*	*H* L%	DEP-L	NO CROWD	'σ ← T*	MAX-L%	NO CONTOUR	MAX-L*	UNIFORMITY-T
a.	$ \begin{array}{c} \text{L\%} \\ \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $	*!				*!			*	
b.	$ \begin{array}{c} \text{H*} \\ \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $						*!		*	
c.	$ \begin{array}{c} \text{H*} \quad \text{L\%} \\ \quad / \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $		*!					*	*	
d.	$ \begin{array}{c} \text{H*} \quad \text{L} \quad \text{L\%} \\ \quad / \quad / \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $			*!	*!			*	*	
e.	$ \begin{array}{c} \text{H} \quad \text{L*} \quad \text{L\%} \\ \backslash \quad \quad / \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $	*!			*!			*		
f.	$ \begin{array}{c} \text{M*}\% \\ \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $								*	*
g.	$ \begin{array}{c} \text{H} \quad \text{L*} \\ \backslash \quad \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{k} \quad \text{ɛ:} \quad \text{r} \end{array} $	*!					*	*		*

(78) H+L* defeats L_{ω-max} %: /mu-s, H+L*, L_{ω-max} %/ → [('mu:^{HL*})<s>]

/ké:r, HL*, L%/		MAX-H*	*H*-L%	DEP-L	NO CROWD	σ' ← T*	UNIFORMITY-T × 2	MAX-L%	NO CONTOUR	MAX-L*	UNIFORMITY(T)
a.	$ \begin{array}{c} \text{L\%} \\ \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{m} \quad \text{u:} \quad \text{s} \end{array} $					*!			*		
b.	$ \begin{array}{c} \text{H} \quad \text{L}^* \quad \text{L\%} \\ \quad / \quad \backslash \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{m} \quad \text{u:} \quad \text{s} \end{array} $				*!			*			
c.	$ \begin{array}{c} \text{M}^*\% \\ \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{m} \quad \text{u:} \quad \text{s} \end{array} $						*!				**
d.	$ \begin{array}{c} \text{H} \quad \text{L}^*\% \\ \quad / \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{m} \quad \text{u:} \quad \text{s} \end{array} $		*!					*			*
e.	$ \begin{array}{c} \text{H} \quad \text{L}^* \\ \quad / \\ (' \bar{\sigma}) \\ / \quad \backslash \\ \text{m} \quad \text{u:} \quad \text{s} \end{array} $				*!		*	*			

All things considered, empirical traces of lowered High tone as the realization of the grave are even weaker than evidence for Low tone in contexts where the grave occurs. From the point of view of the constraint grammar involved, neither a ban on H-L% sequences nor a ban on H at the juncture between prosodic words is simpler. I therefore regard the derivation of L* out of underlying H* as preferable to the derivation of M*% out of underlying H* + L%, given our present state of knowledge.

6.2.3.3 Tonal Association in Recessive Accentuation

Having dispensed with a L% boundary tone, the next issue to be considered is the association of the H+L* prominence tone in recessively accented words, given the constraints on metrical structure sketched in Table 6.7 above. Indeed, deriving the desired tonal pattern in most recessively accented forms is then largely trivial: the L* portion associates to the rightmost heavy syllable (or penult, if penult and final are light) and the H portion associates to the immediately preceding syllable. In this fashion, both CONCATENATE (cf. Gussenhoven 2004: 150–1; see further below) holding over the bitonal H+L* pitch accent and NOCONTOUR are satisfied. In a verbal form such as *μανθάνω* [man.t^há.(ˈnɔː)] ‘, in which presumably stress falls on and the only foot is built around the final syllable, none of the prosodic markedness constraints mentioned under Table 6.7 are violated. Likewise, all constraints on the association between stress and tone given under 59 are perfectly satisfied, since, all stressed syllables are associated with tone, primary stress is associated with a starred tone, and the starred tone is associated with the syllable bearing primary stress. That H tone does not occur farther to the left (e.g., ^x*μανθάνω* [mán..t^ha.(ˈnɔː)]) nor on the final syllable, yielding a contour tone (thus ^x*μανθανῶ* [man..t^ha.(ˈnôː)]) is due to the effects of CONCATENATE (cf. Gussenhoven 2004: 150–1) and NOCONTOUR, which must merely outrank NOASSOCIATION, since more than one syllable is associated with tone. The combination of constraints in (59) and Table 6.7, plus CONCATENATE, NOCONTOUR, and NOASSOCIATION, suffice to reproduce the major results of Golston 1990.

Just two major cases call for further comment, since their tonal associations do not fall out straightforwardly from the same basic pattern: cases in which a penult with a long vocalic nucleus is followed by a heavy final syllable (e.g., *κήρυξ* [kêːruks] ‘herald’ or *πολυπῖδαξ* [polupîːdaks] ‘having many wells’), and disyllabic forms in which both syllables are light (e.g., *φίλος* [p^hílos] ‘friend’). In fact, disyllables like *κήρυξ* (pattern d. in Table 6.7) are unproblematic: given the ranking of T* → ‘σ (starred tone to primary stress) ≫ ‘σ → T (any tone to any stress), plus domination of NOCONTOUR by both MAX-H, MAX-L*, and IDENT-T, as well as an undominated LINEARITY-T (forbidding metathesis of the linear tonal sequence in the input), outputs like *κήρυξ* as the realization of default H+L* emerge. See the tableau in 79.

(79) NOM.SG /kɛːruk-s, H+L*/ → [kêːruks]

/kɛ:rʊk-s, H+L*/		LINEARITY-T	MAX-H	T* → 'σ	MAX-L*	IDENT-T	'σ ← T	NOCONTOUR	NOASSOCIATION
a.							*	*	*
b.				*!					*
c.						*!	*		
d.									
e.									

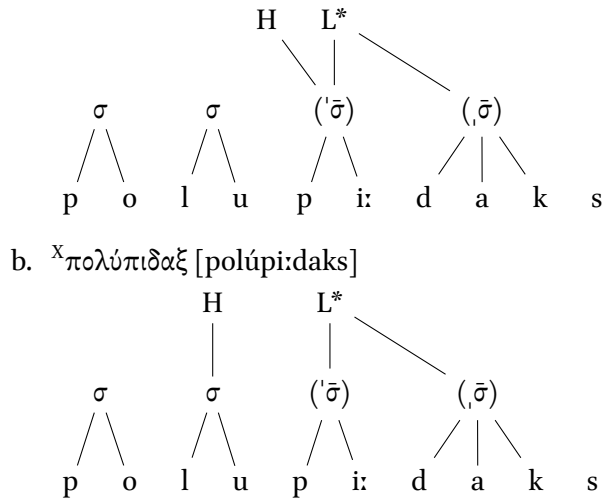
A gratifying aspect of recessive accentuation by reference to foot structure and stress is that the effects of the Law of Limitation are nicely reproduced in virtually all prosodic shapes, without recourse to subsidiary constraints. However, the constraints on tonal association introduced thus far prove insufficiently restrictive for words like *πολύπιδας* [polupî:daks]: given the foot and stress pattern [po.lu.('pi:).(,dak)<s>], L* ought to associate to the penult with primary stress; however, then an association of H to the antepenult, as in forms like *ἄνθρωπος*, would be predicted, since expected *πολύπιδας* violates NOCONTOUR, whereas a non-existent ^X*πολύπιδας* violates no constraints other than 'σ → T – as in winning candidate a. in 79 – NOASSOCIATION. Since both expected *πολύπιδας* and losing ^X*πολύπιδας* would seem to violate 'σ → T, but the latter incurs no violations of NOCONTOUR, the ranking NOCONTOUR ≫ NOASSOCIATION will predict that unexpected ^X*πολύπιδας* wins.

Notable, however, is that the word contains two feet, as opposed to just the single foot found in cases like [an.(thrɔː).po(s)], and another stress to the right of the primary stress, as opposed to cases like an.(thrɔː).(poː)]. Two solutions are conceivable for the problem posed by πολυπιῖδαξ, then: reference to a restriction of the position of H with respect to foot structure (i.e., a direct encoding of the Law of Limitation) or a restriction on the positioning of the bitonal pitch accent in the configuration (‘σ’).(σ). If one opts for the former option, the Law of Limitation might be stated as follows: the tonal window is maximally two feet, such that no tone may be realized outside a window of two feet; if a word contains a single foot, tones may be associated to its left; if it contains two feet, no tones may be associated farther to the outside of the second (leftmost) foot. In case of the latter option, one might assume that L* spreads over the final and penult in order to satisfy both T* → ‘σ and σ → T (and that such spreading would occur in forms like κῆρυξ as well), and consequently, H cannot associate with the antepenultimate syllable.

The latter option is in principle to be preferred, since it does not require any more or less explicit encoding of the Law of Limitation as a markedness constraint, but allows it to be emergent from the interaction of constraint on metrical and tonal structure.⁸⁸ In order to better motivate and understand why the tonal configuration in (80)a. is permitted, but (80)b. is not, a more precise definition of the constraint CONCATENATE is needed. I take violations of CONCATENATE to result whenever one or more TBUs with which Tone A of a bitonal sequence is associated are not directly adjacent to one or more TBUs with which Tone B is associated; see the definition in 81. Concretely, this means that a bitonal sequence, such as H+L*, may only be realized in one of three ways (assuming both parts are associated to a TBU): as a contour tone on one TBU (= both tones associated to the same TBU); as two level tones on immediately adjacent TBUs (both tones associated to separate TBUs); or as a level tone on one TBU and contour tone on an immediately adjacent TBU (= spreading of one tone over two adjacent TBUs); in the event that the bitonal sequence distributes itself over three or more TBUs, as in (80)b., the result is ungrammatical.

- (80) Tonal associations in grammatical πολυπιῖδαξ vs. ungrammatical πολύπιδαξ, given spreading of L*
- a. πολυπιῖδαξ [polupî:daks]

⁸⁸Evidence for the activity of such an explicit encoding of the Law of Limitation as a ban on H outside the second foot from the right (i.e., *H[F][F]) might be found if one could cite any forms with lexical H* outside the range of permitted by the Law of Limitation, where instead recessive accent occurs, and the recessivity of the form cannot arguably be attributed to any other factor. Numerous compounds with a first member having H* and a sufficiently long second member (e.g., βραδύ-θάνατος ‘dying slowly’ from βραδύς ‘slow’ and θάνατος ‘death’) would meet the basic structural conditions, but compound accentuation in Attic-Ionic is insufficiently well understood (see now Probert 2021 for a helpful synoptic overview) to draw secure conclusions; recessive accentuation in such cases might be attributable to other particular metrical or tonal rules. Possible configurations involving derivational suffixes without lexical H* are nigh impossible to find, since most such “unaccented” suffixes (e.g., abstract noun forming -σι-, nominalizing -οσι-) attach almost exclusively to roots. The result is that such chains of multiple unaccented derivational suffixes following a base with H* seem to be virtually non-existent.



- (81) **CONCATENATE:** All TBUs to which tones in a bitonal sequence are associated must be directly adjacent to one another. Assign one violation * if the tones any of the tones in a bitonal sequences are associated to non-contiguous TBUs.⁸⁹

Now equipped with the undominated constraint **CONCATENATE**, $T^* \rightarrow$ and ${}_{\sigma} \leftarrow T$ may be newly ranked into the same stratum, crucially dominating **NO SPREAD**, **NO CONTOUR**, and **NO ASSOCIATION**. Winning *πολύπιδαξ* then emerges in the tableau in (82). Candidate a. wins because it satisfies all stress-to-tone association constraints, incurring violations only of the lower-ranked set of **NO SPREAD**, **NO CONTOUR**, and **NO ASSOCIATION**. Candidate b., which associates the $H+L^*$ pitch across three syllables, fatally violates **CONCATENATE**, avoiding only a violation of **NO CONTOUR** in return. Finally, candidate c. (cf. candidate a. in 79) loses because it leaves the stressed final syllable without association to a tone, despite avoidance of violations of both **NO CONTOUR** and **NO SPREAD**.

- (82) /polupi:dak-s, $H+L^*$ / $\rightarrow$ [po.lu.(^ˈpî:).(dàk).(s)]

⁸⁹This definition departs substantially from both the letter and spirit of the constraint with the same name introduced in Gussenhoven 2004: 150; perhaps a more appropriate, distinct name could be devised. Concretely, the important effect of the constraint is to assign a violation when a bitonal pitch accent distributes itself over three or more TBUs (in the present context, syllables).

/polupi:dak-s, H+L*/		CONCATENATE	MAX-H	T* → 'σ	'σ ← T	MAX-L*	IDENT-T	NO SPREAD	NO CONTOUR	NO ASSOCIATION
a.								*	*	**
b.		*!								***
c.					*!				*	

Thus, all patterns containing at least one heavy syllable between the penultimate and final syllable with recessive accentuation are accounted for, as well as words containing at least three syllables with a sequence of two light syllables at the right edge of the word (e.g., ἐδυνάμεθα [e.du.ná.(mè.tʰa)] ‘is.able:1.PL.PRS.MID’). What, however, should occur in the event that a disyllabic word consists of just two light syllables is less obvious. Whereas forms like πολυπίδαξ seemed relatively clear in terms of prosodic constituency, but less clear with regard to tonal association, recessively accented forms like φίλος [pʰí.los] ‘friend’ present questions and problems in terms of both tonal association and metrical structure. Since disyllabic forms are further exceptional in presenting paradigms with genuine mobility of High tone (cf. 6.2.1.3 above) regardless of syllable count and weight (e.g., NOM.PL πόδες [pódes] versus GEN.SG ποδός [podós]), how to analyze disyllabic recessive forms and High tone mobility should be treated together.

6.2.3.4 HEADFAITH to H* in Derivation and the Lex σωτήρα

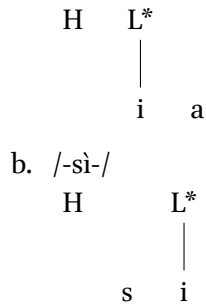
Before, however, the complex of issues involved in the tonal and metrical structures of disyllables can be adequately addressed, a detour into the subsystem of High tone shift in derivation is necessary. This detour will help to clarify the conditions under which lexical H* sponsored by certain morphemes is preferentially realized over the default H+L* prominence tone.

As correctly observed already by Steriade (1988: 279–81), morphologically complex nominal and adjectival formations with one or more (parsable) derivational suffixes mostly exhibit High tone on the rightmost or outermost suffix, or alternatively condition application of recessive accentuation on a base that was not recessively accented (cf. also Probert 2006b: 144–8 and 6.2.1.4 above for examples). Treatments of Attic-Ionic prosody in derived stems

thus almost universally assume that suffixes like /-ísko-/ ‘DIM’ (e.g., ἀστερίσκος [asterískos] ‘little star’) are prosodically marked and attract high tone (here, presumably linked underlyingly to H*), whereas suffixes like /-io-/ ‘DIM’ (e.g., πόδιον [pódion]) ‘little foot’) are prosodically unmarked and seem to condition recessive accentuation. Probert (2006b: 148) nicely points out the following accentual pair built to πούς [pós:s] ‘foot:NOM.SG’ (GEN.SG ποδός), which has a marked stem that results in an acute (UR /pód-/): when the unmarked suffix /-io-/ is attached to the base /pód-/ , thus /pód-io-/ , the result is recessively accented πόδιον [pódion] ‘little foot’; when, on the other hand, the marked derivational suffix /-ikó-/ is attached, thus giving /pód-ikó-/ , the result is ποδικός [podikós] ‘pertaining to a metrical foot’, with H on the marked vowel. In terms of lexical accent systems, affixes like /-ísko-/ might be classified as “dominant accented”, whereas suffixes like /-io-/ are either “dominant unaccented” or simply “unaccented”.

A first important claim to be advanced here is that derivational suffixes that induce recessive accent, such as /-ia-/ in (83) (repeated from (12) above), simply allow the default constraints on foot construction, stress, and association of the H+L* pitch accent to apply. They neither erase other underlying H* pitch accents, nor do are they underlyingly associated with pitch accents of their own. The view that no accentual erasure occurs follows from related discussion in 2.4.2 in Chapter 2 and 4.3.3 in Chapter 5, to the effect that “dominant unaccented” morphemes have at best weak empirical support and questionable theoretical undergirding. Indeed, as I will argue here, recessive accentuation in forms like βασίλεια can be derived without assuming that the underlying H* of the suffix /-eú-/ is erased; rather, a virtual ^Xbasileía simply does not constitute the optimal candidate. Second, assuming that derivational suffixes like /-io-/ or /-ia-/ any sort of direct pitch accent association with (one of) the vowels in an unmarked derivational suffix is fundamentally to be dispreferred on the grounds that it introduces more underlying constraints, and moreover would lead to empirically false results in at least some cases: for instance, the representations for the suffixes /-ia-/ and /-si-/ would be adequate in cases in which H associates to the immediately preceding syllable (like βασίλεια [basíleia]) but not in which the H appears farther to the left (e.g., ἄλωσις [haló:sis] ‘capture’). That such suffixes introduce a floating H+L*, whose realization is preferred to the realization of the H* sponsored by inner morphemes, would be viable (see Sandell 2015a: 190–2 for a sketch of an analysis to this effect), but is perhaps superfluous, given that the default word prosody provides a H+L* pitch accent in any case.

- (83) An outermost derivational suffix either a) attracts the high tone or b) induces recessive accentuation
- a. /basil-eú-s/ βασιλεῦς [basileús] ‘king:NOM.SG’ – recessive ^Xbasileus [basíleus] does not occur.
 - b. /basil-eú-ia/ βασίλεια [basíleia] ‘queen:NOM.SG’ – no persistence of high tone from the derivational base to produce ^Xbasileía [basileía].
Recessive accent instead applies.
- (84) H+L* representation for unmarked derivational suffixes
- a. /-ia-/



Furthermore, the existence of non-High-attracting derivational suffixes combined with the fact that inflectional suffixes outside of monosyllabic stems as a rule do not surface with High tone, excludes the possibility that the patterns present in Attic-Ionic can be treated as a simple lexical accent system (cf. 2.4.2.1 on Cupeño and Modern Italian). For instance, if a constraint like ALIGN-(H*, R, ω, R) (“place any H* as close to the right edge of the word as possible;”) were the governing principle at work, one would expect for a string like /asp-íd-ós/ ‘small.shield-GEN.SG containing the derivational suffix /-íd-/ and GEN.SG inflectional suffix /-ós/ to surface as ^Xἀσπιδός [aspídós], not actual ἀσπίδος [aspídos], in which the H* of the derivational suffix determines where H occurs in the word.

Hence, as suggested already in Sandell 2015a: 190–2, the regular “accentual dominance” of most derivational suffixes in Ancient Attic-Ionic may be attributed to the workings of HEADFAITH, i.e., the faithfulness to prosodic properties sponsored by the morphological head of the word (see Revithiadou 1999: 26–31). The effect of HEADFAITH-H* employed here is functionally identical to the HEADFAITH constraints applied to foot edges in Chapter 4 (see example (61) there), differing only in that it refers to the lexical H* that may be sponsored by a morphological head instead. As in Vedic, I assume that this constraint holds mainly over category-changing or category-determining affixes, and perhaps not of simple roots; this assumption is backed up by the extensive argumentation and demonstration in Probert 2006b to the effect that lexical items that come to be treated as morphologically simplex tend to adopt recessive accentuation.⁹⁰ Assuming that simple roots marked with H*, such as /pód-/ ‘foot’ or /rí:n-/ ‘nose’, do not fall within the purview of HEADFAITH-H* facilitates the treatment of high tone mobility in disyllables, though it is technically possible to assume that HEADFAITH-H* holds over *all* H* sponsored by a stem; see further discussion under 6.2.3.5.

- (85) HEADFAITH-H*: A H* tone in the input sponsored by a morpheme that is a category-determining head of a morphological word corresponds to a tone in the output. Assign one violation * for each such H* in the input that is not associated with the same tone linked to the same TBU in the category-determining head in the output.

In cases in which a string contains multiple derivational affixes marked with H* within the stress domain of the Law of Limitation (i.e., the final two syllables), HEADFAITH-H* en-

⁹⁰Crucially, this assumption on the applicability of HEADFAITH-H* assumes a generally high level of morphological parsability throughout the language. This parsability then necessarily extends to some lexemes whose (synchronic) morphological complexity might reasonably be doubted, e.g., kinship terms like πατήρ (ACC.SG πατέρα).

sures that the outermost wins, even where other prosodic markedness constraints (WSP, MAIN $\Rightarrow$ VV, ALL-FT-R) would be better satisfied. In such cases, the H* sponsored by the head is evidently preferred to realization of the default H+L*, even if the constraints on primary stress placement would make the same choice.⁹¹ Thus, In cases like σωτήρ ‘savior:NOM.SG’ or ἀσπίδος ‘small.shield:GEN.SG’, the constraints on foot structure primary stress placement are ideally satisfied, and lead to realization of H* on the syllable with primary stress, thereby also satisfying all constraints to stress-to-tone mapping. 86a. and b. thus merely speak to the ranking HEADFAITH-H* $\gg$ MAX-H, MAX-L*. More interesting are cases like 86c., σιγηρός ‘silent’, in which the H* of the suffix /-ró-/ is evidently faithfully realized, but a several foot structures and stress placements are conceivable, depending what divergences between T* to stress mapping might be allowed.

(86) Foot structure and stress in forms with effects of HEADFAITH

- a. /sɔː-téːr/ $\rightarrow$ [(sɔː).(‘téː)(r)]
- b. /asp-íd-ós/ $\rightarrow$ [as.(‘pí.do)(s)]
- c. /sig-éː-ró-s/ $\rightarrow$ [si.geː.(‘ró)(s)] (or [si.(‘geː).(ró)(s)] or [si.(geː).(‘ró)(s)]?)

But in fact, given what has already been established about the rankings of constraints on foot structure for generating recessive accentuation (see the preceding section and Table 6.7), in particular the rankings MAIN $\rightarrow$ VV $\gg$ WSP $\gg$ ALL-FT-R, and $\sigma \leftarrow T \gg$ NOSPREAD, the only permissible candidate that preserves H* in satisfaction of HEADFAITH and holds to the established rankings is [si.geː.(‘ró)(s)], which continues to indicate that all stressed syllables must be associated with tone ($\sigma \leftarrow T$), and any starred tones realized on the surface must be associated with primary stress ($T^* \rightarrow \sigma$).⁹² Winners like [si.(‘geː).(ró)(s)] also helpfully indicate that preservation of H* sponsored by heads may also allow for violations of FOOTBINARITY. The tableau in (87) summarizes these ranking arguments: what is crucial is that the VV-syllable is not footed, thereby violating WSP, but leaving it unfooted avoids a violation of higher-ranked MAIN $\rightarrow$ VV. Thus, candidates c.-e. all incur fatal violations of high-ranked stress-to-tone and tone-to-stress constraints, or do not place primary stress on a footed VV-syllable; between candidates a. and b., only a. satisfies HEADFAITH-H*, despite its violation of WSP and FTBIN.

(87) /sig-éː-ró-s/ $\rightarrow$ [si.(‘geː).(ró)(s)] ‘silent:NOM.SG’

⁹¹For this reason, it is empirically difficult to determine in cases in which a outermost derivational suffix is marked with H* and followed by an inflectional suffix containing a long vowel (e.g., GEN.PL forms like σωτήρων or ἀσπίδων) faithfully realize the H*, or whether stress is placed on the long vowel in the final syllable, and H+L* is realized instead.

⁹²I exclude the further option [si.(géː).(‘ró)(s)], with spreading of H* to a heavy penult having secondary stress on empirical grounds, although it, too, could be a viable winner. There is no evidence, to my knowledge, that High pitch on the final syllable would spread leftwards to a heavy penult. On the other hand, that the L* portion of the H+L* prominence tone associates to the penult with primary stress in [si.(‘geː).(ró)(s)] cannot be readily excluded, and indeed would predicted to win, if taken into consideration. If the lexical H is indeed a starred tone, a type of CULMINATIVITY might hold that prevents that realization of both the H* sponsored by the head and the default L*.

/sig-é:-ró-s, H+L*/		$\sigma \leftarrow T$	$T^* \rightarrow \sigma$	MAIN $\rightarrow$ VW	HEADFAITH-H*	WSP	FtBIN	MAX-H*	ALL-Ft-R	$\sigma \leftarrow T^*$	NOSPREAD	MAX-L*
a.						*	*	*				*
b.				#!				*	*			*
c.		#!	#!				*	*	*	*		*
d.		#!	#!				*	*	*			*
e.			#!				*	*	*		*	*

Given this ranking, forms like βασιλεια fall out readily, once the behavior of the Lex σωτηρα is understood more precisely. In short, in cases like βασιλεια from underlying /basil-eú-ia/, recessive H+L* competes with the lexical H* of a non-head. However, faithful realization of the non-head's H* would further require transformation of underlying H* into a contour tone in order to satisfy the Lex σωτηρα, understood here as satisfaction of a tonotactic markedness constraint that penalizes H when non-final and neither preceded nor followed by a Low tone (see 65 above). To preserve the H* of the non-head would thus require a violation of either IDENT-T (assumed here, since violations of this constraint were invoked

above to generate “lulling”, i.e., transformation of underlying H* to L*) or DEP-T.⁹³ In such cases, the ranking IDENT-T (“no change in pitch type”) ≫ IDENT-Association (“no severing of underlying autosegmental associations”) is crucial: preservation of the underlying H* of /-eú-/ would require violation of IDENT-T in order to satisfy *NOFALL_ω. A potential violation of *NOFALL_ω is thus repaired in the case at hand by replacing a lexical tone with the default H+L* prominence tone. In other inflectional forms of this word type that have a long vowel in the final syllable, e.g., DAT.PL βασιλείαις [ba.si.(leí).(‘ai)s] (see example (89)), the acute that appears on the penult must likewise reflect the H portion of the H+L* prominence tone.

(88) NOM/ACC.SG /basil-eú-ia/ → [ba.sí.(‘leí).a]

/basil-eú-ia, H+L*/		*NOFALL _ω	σ ← T	T* → ‘σ	MAIN → VV	HEADFAITH-H*	WSP	FtBIN	MAX-H	ALL-Ft-R	IDENT-T	IDENT-Assoc	MAX-L*
a.									*	*		*	
b.									*	*	*!		*
c.		*!							*	*			*

(89) DAT.PL /basil-eú-ia-is/ → [ba.si.leí.(ai)(s)]

⁹³To the best of my knowledge, there are not transparent cases of synchronic derivation in which a lexically accented derivational suffix containing a short vowel, e.g., /-íd-/ is followed by an unaccented derivational suffix containing a short vowel that conditions recessive accentuation. That is, there are no clearly synchronically derivable forms like ^Xᾗσπιδον ‘pertaining to a small shield’ built with unaccented /-on/ to a base with accented /-íd-. All of the clear instances in which recessive accentuation results in a secondary derivative where one can distinguish between preservation of lexical H* and recessive accentuation would require application of the σωτῆρα Rule if the H* were faithfully preserved.

/basil-eú-ia-is, H+L*/		*NOFALL _ω	σ ← T	T* → 'σ	MAIN → VV	HEADFAITH-H*	WSP	FTBIN	MAX-H	ALL-Ft-R	IDENT-T	IDENT-Assoc	MAX-L*
a.									*	*		*	
b.		*!	*	*					*	*			*
c.			*!	*!					*	*	*		*
d.			*!						*	*	*		*

A larger prediction is then that repairs other than replacement of H* with H+L* to satisfy *NOFALL_ω should occur only when the H* is further protected by HEADFAITH-H*. When the H* of the head is sponsored by a VV sequence, the preferred repair to *NOFALL_ω is the creation of a contour tone through a violation of IDENT-T, thereby creating a unitary contour tone H*L, as shown in (90). Candidates a.-c. all preserve the lexical H*; candidate d. is excluded by its violation of HEADFAITH-H*. The fully faithful candidate, a., violates *NOFALL_ω, however. The relative ranking of DEP-T and IDENT-T then decide between candidates b. and c.

(90) Lex σωτήρα 1: /H*/ → [H*L] (/sɔ:-tɛ:r-a/ → [sɔ:.(tɛ:).ra])

/sɔː-té:r-a, H+L*/		*NOFALL _ω	σ ← T	T* → 'σ	MAIN → VV	HEADFAITH-H*	MAX-H	ALL-Ft-R	DEP-T	IDENT-T	IDENT-Assoc	NOCONTOUR	MAX-L*
a.		*!					*	*					*
b.							*	*		*		*	*
c.							*	*	*!				*
d.						*!	*	*		*			*

When, on the other hand, the syllable corresponding to the morphological head contains only a short vowel (e.g., ACC.SG πατέρα to πατήρ), a contour tone on the short vowel is excluded by *CONTOUR/SHORT ($\sim$ NOCROWD). In such cases, *NOFALL_ω must instead be satisfied by epenthesis of another Low tone.⁹⁴ Thus, in the tableau in (91), the fully faithful candidate, a., is again ruled out by the requirement for a tonal fall in the prosodic word, and candidate d. cannot win due to its violation of HEADFAITH-H*. Candidate c., [pa.(tê*.ra)], violates lower-ranked IDENT-T in creating a contour tone H*L, but thereby incurs a violation of the higher-ranked markedness constraint NOCROWD ($\sim$ *CONTOUR/SHORT). Hence, candidate b., showing L epenthesis, wins.

(91) Lex σωτήρ 2: /H*/ → [H* + L] (/patér-a/ → [pa.(té.ra)])

⁹⁴Two other alternatives are conceivable, and viable constraint rankings could be found. Under one alternative, the L* is preserved in all instances in which a non-final H* is preserved, and associated either to the VV-penult along with the H*, yielding a contour tone (σωτήρ), or to the unstressed final syllable (πατέρ). This alternative analysis would entail a relatively higher ranking of MAX-L* and lower ranking of T* → 'σ, but has the disadvantage that then starred tones can freely occur on unstressed syllables, and might allow for the default L* to always associated somewhere in word. A second alternative for words of the πατέρα type specifically is that they, too, instantiate creation of a contour tone H*L, which is however spread over two syllables and thereby avoids a violation of *CONTOUR/SHORT.

/pa-tér-a, H+L*/		*NOFALL _ω	NO CROWD	'σ ← T	T* → 'σ	HEADFAITH-H*	MAX-H	ALL-FT-R	DEP-T	IDENT-T	IDENT-ASSOC	NO CONTOUR	MAX-L*
a.	H* σ ('σ σ) p a t e r a	*!					*	*					*
b. 	H* L σ ('σ σ) p a t e r a						*	*	*				*
c.	H*L σ ('σ σ) p a t e r a						*	*		*		*	*
d.	H L* σ ('σ σ) p a t e r a					*!	*	*			*		

Likewise, when the final syllable contains a long vowel and is thus perform footed and stressed (e.g. GEN.PL σωτήρων to σωτήρ), a Low tone is epenthesized that both prevents violation of *NOFALL_ω and provides the stressed final syllable with a tone.⁹⁵ In this case, the fully faithful candidate in (92)a. has two fatal problems: a minimal prosodic word without a falling contour, and a stressed syllable without tone; the winner, candidate b., resolves both problems through its violation of DEP-T. Not assigning stress to the final long vowel is not permitted, since WSP outranks ALL-FT-R, for which reason repairing the problems with candidate a. by violation of IDENT-T alone, as in candidate c., is not permitted. Finally, employing recessive accentuation instead, as in candidate d., is ruled out by HEADFAITH-H*.

(92) Lex σωτήρα 3: /H*/ → [H* + L] (/sɔ:-té:r-ɔ:n/ → [sɔ:.('té:r).('ɔ:)n])⁹⁶

⁹⁵ Again, if spreading of unitary H*L may be assumed to yield the surface acute on the VV-penult, violation of IDENT-T and NOSPREAD could be invoked instead. However, given that forms like πολυπίδαξ must spread a L* tone and still show a contour tone on the VV-penult, I find it unattractive to think that the unitary H*L would behave differently in its effects of tonal spreading.

⁹⁶ I ignore for the moment issues concerning the lexical representation of inflectional suffixes containing a long vocalic nucleus; see discussion under 6.2.3.6.

/sɔ:-tér-ɔ:n, H+L*/		*NOFALL _ω	σ ← T	T* → σ	HEADFAITH-H*	MAX-H	WSP	ALL-Ft-R	DEP-T	IDENT-T	IDENT-Assoc	NOCONTOU	MA
a.		*!	*!			*	*	*					*
b.						*	*	*	*				*
c.						*	**!			*		*	*
d.				*!		*	*	*			*		

Under this analysis, the set of forms traditionally classified as motivated by the Lex σωτήρα and the set of forms calling for a repair via violations of IDENT-T or DEP-T diverge from one another. According to the analysis above, disyllables like κήρυξ or πίδαξ simply faithfully associate the default H+L* prominence tone to the syllable bearing primary stress; they satisfy *NOFALL_ω, but need not incur any violations of either faithfulness constraints to do so. Similarly, forms like βασιλεια, in which realization of an underlying H* would necessitate a faithfulness violation to satisfy NOFALL_ω, H+L* is realized in preference to lexical H*. However, in case a lexical H* must be preserved due to HEADFAITH-H*, violations of IDENT-T (σωτήρα) or DEP-T (πατέρα, σωτήρων) step in to ensure that a High-to-Low pitch contour occurs somewhere in the minimal prosodic word.

6.2.3.5 High Tone (Im)Mobility, Recessive Disyllables without Heavy Syllables, and Trochaic Shortening

The finding of the analysis above, namely, that lexical H* is preserved mainly when protected by HEADFAITH-H* then allows for a fairly straightforward explanation of why polysyllabic stems without any lexical H* in the stem regularly exhibit recessive accentuation. In the paradigms of third-declension stems like μύρμηξ ‘ant’ (DAT.SG μύρμηκι) or πομπόλυξ

(DAT.SG *πομπόλυκι*) ‘bubble’, there is neither an advantage in terms of prosodic markedness nor in terms of faithfulness to lexical prosody to generating a pattern other than the recessive pattern. This observation holds trivially if the inflectional suffixes for GEN and DAT are assumed to be unmarked, and follows for all case endings with a short vowel (*/-ós/*, */-í/*, */-sí/*), given that FOOTBINARITY is ranked in the same stratum or above MAX-H and above IDENT-ASSOCIATION.⁹⁷ That is, an output [pom.po.lu.(^lki)], in which the lexical H* of the DAT.SG ending is preserved, is not superior to assigning the default foot structure and default H+L* intonation, that is, [pom.pó.(^lù.ki)]. Candidates preserving more than one starred tone, such as 93c., are ruled out by T* → ‘σ, given that CULMINATIVITY ensures that only a single primary stress remains. Similarly, preserving the lexical H* on unstressed syllable is harmonically even worse, since the result is primary stress without tone, as in d. Finally, spreading of the lexical H* is excluded by the ranking NOSPREAD ≫ IDENT-ASSOC, shown in candidate e.

(93) ‘bubble:DAT.SG’ /pompoluk-í/ → [pom.pó.(^lù*.ki)]

⁹⁷ Deriving the correct tonal pattern in genitive plural forms and the prosodically similar genitive/dative dual is not so straightforward. See the following section.

/pompoluk-í, H+L*/		$\sigma \leftarrow T$	$T^* \rightarrow ' \sigma$	MAX-H	FTBIN	ALL-FT-R	$' \sigma \leftarrow T^*$	NOSPREAD	IDENT-ASSOC	MAX-L*
a.				*					*	
b.				*	*!					*
c.			*!		*	*				
d.		*!	*	*			*			*
e.				*				*!		*

Of course, if all inflectional endings in polysyllabic stems are simply treated as unmarked (i.e., without H*), then recessive accentuation will be automatic on unmarked stems, and FTBIN can then be removed from the grammar altogether. Candidates like [pom.po.lu.([']kí)] would be eliminated by low-ranking MAX-L*, in any case. Such allomorphy could be trivially generated in the morphology by assuming vocabulary insertion rules that provide marked allomorphs with H* only in the event that the stem contains a single potential syllable nucleus, as shown in 94 for DAT.SG.⁹⁸

⁹⁸ Alternatively, a readjustment rule that adds an H* to any inflectional ending bearing GENITIVE or DATIVE features if the stem meets the phonological structural description shown in 94a.

(94) Allomorphy of DATIVE SINGULAR in 3rd Declension Nouns

- a. [+DATIVE, -PLURAL] $\leftrightarrow$ /-í/ // [C_o[+syllabic]([-consonantal])C_o]_{Stem}____
b. [+DATIVE, -PLURAL] $\leftrightarrow$ /-i/ // *elsewhere*

Given that the distinction between mobile and immobile paradigms is, however, fundamentally phonological, however, being able to derive persistent immobility of high tone in most case forms of polysyllabic stems through the interaction of phonological constraints is attractive, since recourse to allomorphy among inflectional suffixes can then be kept to a minimum. If the application of recessive accentuation, creating paradigms with no H tone mobility (or only pseudo-mobility) is then the prosodic default, what lexical or properties distinguish monosyllabic stems such that their inflectional paradigms can exhibit genuine H tone mobility?

Recall from the discussion under 6.2.1.3 above that third declension monosyllabic nominal stems may be constructed from both stems (e.g., /pód-/ ‘foot’ vs. /mu-/ ‘mouse’) and inflectional endings (e.g., NOM.PL /-es/ vs. GEN.SG /-ós/) that contrast in marking with H*. As reference to the paradigms in (7) shows, repeated immediately below for convenience, these two properties are independent and may freely combine. Important to note, however, is that the inflectional endings marked with H* seem to always determine the position of the single H, regardless of the marking of the stem. In this regard, root nouns of Greek differ from root nouns of Vedic, where a variety of different stress patterns occur, insofar as Greek presents only a single overall prosodic pattern: H on the stem when the ending is not marked with H*, H on the ending when it is marked. The underlying H* vs. $\emptyset$ contrast on stems can be diagnosed only from the monosyllabic forms of the paradigm.⁹⁹

(95) High Tone Mobility and Acute-Circumflex Contrast in Monosyllabic Stems

	‘foot’	‘mouse’	‘nose’	‘goat’
NOM.SG	[póːs]	[mûːs]	[ríːs]	[aík-s]
ACC.SG	[pód-a]	[mûːn]	[ríːn-a]	[aíg-a]
GEN.SG	[pod-ós]	[mu-ós]	[riːn-ós]	[aig-ós]
DAT.SG	[pod-í]	[mu-í]	[riːn-í]	[aig-í]
NOM.PL	[pód-es]	[mú-es]	[ríːn-es]	[aíg-es]
ACC.PL	[pód-as]	[mú-as]	[ríːn-as]	[aíg-as]
GEN.PL	[pod-ôːn]	[mu-ôːn]	[riːn-ôːn]	[aig-ôːn]
DAT.PL	[po-sí]	[mu-sí]	[riː-sí]	[aik-sí]

Operating under the presumption that faithfully realized H* is also diagnostic as to the position of primary stress as a consequence of undominated T* $\rightarrow$ ‘σ, all of the inflectional endings with acute have but one viable interpretation in terms of foot structure: a degenerate foot, e.g., /pód-í/ $\rightarrow$ [po.(‘dí)] or /mu-sí/ $\rightarrow$ [mu.(‘sí)]. When the stem is unmarked, as

⁹⁹Disyllabic forms of the ‘nose’-type with circumflex on the stem, but marked underlyingly with H*, show the effect of the Lex $\sigma\omega\tau\eta\rho\alpha$; compare the tableau under (90) above.

for /mu-/ ‘mouse’ or /aig-/ ‘goat’, shift of stress and preservation of the lexical H* of the inflectional ending is directly predicted as long as at least one other markedness constraint outranks FOOTBINARITY. Since all of the forms considered thus far are compatible with ranking $\sigma \leftarrow T^*$ high in the grammar, this constraint will suffice to rule out disyllabic trochaic parses like [(‘mu.si)] that associate H+L* with the initial and final syllables, respectively; I take for granted that TROCHEE is undominated in the grammar, so that an iambic parse, [(mu.‘si)] is also not permitted.¹⁰⁰ The mismatch in association between starred tone thus fundamentally prevents the formation of a well-formed trochee, and associating only the starred tone L* is inhibited by the ranking MAX-H $\gg$ MAX-L*. Finally, ranking IDENT-ASSOCIATION above MAX-L* captures why the lexical H* of the inflectional ending wins out over the recessive accent. The basic generalization is then: in cases where a well-formed trochee would result in an unacceptable mismatch in terms of tone-to-stress and stress-to-tone associations, the lexical tone of a morpheme parsed into a degenerate foot is preferred. These candidates and ranking considerations are expressed in 96.

(96) /mu-sí, H+L*/ $\rightarrow$ [mu.(‘sí*)] ‘mouse:DAT.PL’

¹⁰⁰Even admitting iambic feet would change nothing substantially in the analysis, as far as what can be overtly determined is concerned: one could substitute iambic feet in many cases where here degenerate feet are assumed.

/mu-sí, H+L*/		$\sigma \leftarrow T$	$\sigma \leftarrow T^*$	$T^* \rightarrow \sigma$	MAX-H	FtBIN	IDENT-ASSOC	MAX-L*
a.	$ \begin{array}{c} H^* \\ \\ \begin{array}{cc} (' \sigma & \sigma) \\ / \quad \backslash & / \quad \backslash \\ m \quad u & s \quad i \end{array} \end{array} $	*!	*!	*!	*			*
b.	$ \begin{array}{cc} H & L^* \\ & \\ \begin{array}{cc} (' \sigma & \sigma) \\ / \quad \backslash & / \quad \backslash \\ m \quad u & s \quad i \end{array} \end{array} $		*!	*!	*		*	
c.	$ \begin{array}{c} L^* \\ \\ \begin{array}{cc} (' \sigma & \sigma) \\ / \quad \backslash & / \quad \backslash \\ m \quad u & s \quad i \end{array} \end{array} $				***!		*	
d. 	$ \begin{array}{c} H^* \\ \\ \begin{array}{cc} \sigma & (' \sigma) \\ / \quad \backslash & / \quad \backslash \\ m \quad u & s \quad i \end{array} \end{array} $				*	*		*
e.	$ \begin{array}{cc} H & L^* \\ & \\ \begin{array}{cc} \sigma & (' \sigma) \\ / \quad \backslash & / \quad \backslash \\ m \quad u & s \quad i \end{array} \end{array} $				*	*	*!	

Provided that monomorphemic stems are not subject to the effects of HEADFAITH-H*, mobility in short-voweled stems marked with H* is predicted as well, as a consequence of *NOFALL_ω, NOCROWD, and DEP-T: realizing H* on the initial syllable alone to give an output like ^X[(pódi)] will violate NOFALL_ω, transforming the H* of the stem into a contour tone H*L will violate NOCROWD, and inserting another L tone on the final syllable will violate DEP-T. Given that these three constraints outrank FtBIN, mobility to the H*-marked ending is again predicted, as illustrated by the tableau in 97

(97) DAT.SG /pód-í, H+L*/ → [po.(dǐ*)] 'foot:DAT.SG'

/pód-í, H+L*/		NOFALL _ω	NOCROWD	MAX-H	DEP-T	IDENT-T	FTBIN	IDENT-ASSOC	MAX-L
a.	$ \begin{array}{c} H^* \\ \\ (\sigma \quad \sigma) \\ / \quad \backslash \quad / \quad \backslash \\ p \quad o \quad d \quad i \end{array} $	*!		**				*	*
b.	$ \begin{array}{c} HL \\ \\ (\sigma \quad \sigma) \\ / \quad \backslash \quad / \quad \backslash \\ p \quad o \quad d \quad i \end{array} $	*!		**		*		*	*
c.	$ \begin{array}{c} H^* \quad L \\ \quad \\ (\sigma \quad \sigma) \\ / \quad \backslash \quad / \quad \backslash \\ p \quad o \quad d \quad i \end{array} $			**	*!			*	*
d. 	$ \begin{array}{c} H^* \\ \\ \sigma \quad (\sigma) \\ / \quad \backslash \quad / \quad \backslash \\ p \quad o \quad d \quad i \end{array} $			**			*	*	*
e.	$ \begin{array}{c} H \quad L^* \\ \quad \\ \sigma \quad (\sigma) \\ / \quad \backslash \quad / \quad \backslash \\ p \quad o \quad d \quad i \end{array} $			**			*	**!	

Rather more complicated are the H*-marked stems containing long vowels, such as /rí:n- / ‘nose’ or /t^hé:r- / ‘beast’. Recall from the tableau under (87) that violation of WSP when a long vowel occurs in a penult could only be motivated by faithfulness to the H* of a morphological head. Since an inflectional ending like DAT.SG /-í/ is not subject to HEADFAITH-H*, and WSP must outrank IDENT-T, another constraint would be needed that militates against primary stress and/or High tone on initial syllables. A constraint like NONINITIALITY- $\mathcal{F}$ would fit, but confronts the problem that disyllables like $\kappa\eta\rho\nu\xi$ or $\pi\hat{\iota}\delta\alpha\xi$, seemingly parsed as [(‘kê:).(rùk)<s)] and [(‘pî:).(dak)<s)], must allow for primary stress on an initial long vowel, showing that both MAIN $\rightarrow$ VV and WSP would necessary outrank NONINITIALITY- $\mathcal{F}$, were the latter to be included in the grammar. Explaining why expected [ri:.(ní)] wins

over $^X[(\text{'r}\acute{\text{i}}\text{:}).\text{ni}]$ thus poses a serious challenge.

The admittedly *ad hoc* solution that can be mustered to account for this issue is to invoke a constraint that was explicitly argued against above, namely, the preference for the alignment of H^* pitch accents with the right edge of the prosodic word. Above, in the discussion of HEADFAITH- H^* , I argued that a constraint like ALIGN- (H^*, R, ω, R) , defined in 98, is insufficient to account for the effects of High tone shift in derivation. Those arguments remain valid, but data like $\acute{\text{p}}\text{iv}\acute{\text{i}}$, if indeed correctly interpreted as $[\text{ri}:(\text{'n}\acute{\text{i}})]$, suggest that ALIGN- (H^*, R, ω, R) ultimately does have a role to play in the grammar. If ranked in the same stratum with WSP below HEADFAITH- H^* , ALL-Ft-R, being ranked above FTBIN, will ensure that $^X[(\text{'r}\acute{\text{i}}\text{:}).\text{ni}]$, with initial stress and contour tone, loses to $[\text{ri}:(\text{'n}\acute{\text{i}})]$. ALIGN- (H^*, R, ω, R) could also be invoked in (97) above to derive the winner there as well, though was not necessary to do so.

- (98) ALIGN- (H^*, R, ω, R) : Any H^* pitch accent on the surface should be aligned with the rightmost available syllable in the word. Assign one violation * for each syllable that intervenes between the right edge of the (minimal) prosodic word and any H^* pitch accent in the output.¹⁰¹

- (99) $/\text{r}\acute{\text{i}}:\text{n}-\acute{\text{i}}, H+L^*/ \rightarrow [\text{ri}:(\text{'n}\acute{\text{i}}^*)]$ 'nose:DAT.SG¹⁰²

¹⁰¹Crucially, this constraint must refer exclusively to H^* tones, not H tones generally. An identical alignment constraint evaluated over the position of all H tones would necessarily be too low-ranked in the grammar to produce the necessary effect in forms like $\acute{\text{p}}\text{iv}\acute{\text{i}}$.

¹⁰²A further *ad hoc* assumption necessary to derive the expected winner and to rule out recessive accentuation in forms like the present one is to allow for self-ganging of IDENT-ASSOC, such that two violations thereof result in a greater penalty than any single violation of WSP.

/rí:n-í, H+L*/		NOFALL _ω	T* → 'σ	IDENT-ASSOC × 2	ALIGN-R(H*, ω)	WSP	MAX-H	ALL-Ft-R	IDENT-T	FTBIN	IDENT-ASSOC
a.	$ \begin{array}{c} H^* \\ \\ (' \bar{\sigma}) \quad \sigma \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad i \end{array} $	*!			*		**	*			*
b.	$ \begin{array}{c} H^*L \\ \\ (' \bar{\sigma}) \quad \sigma \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad i \end{array} $				*		**	*!	*		*
c.	$ \begin{array}{c} H^* \\ \\ (' \bar{\sigma}) \quad \sigma \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad i \end{array} $		*!				**	*			*
d.	$ \begin{array}{c} H^* \\ \\ \bar{\sigma} \quad (' \sigma) \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad i \end{array} $					*	**			*	*
e.	$ \begin{array}{c} H \quad L^* \\ \quad / \\ (' \bar{\sigma}) \quad \sigma \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad i \end{array} $		*!				**	*		*	**

Given the acceptability of degenerate feet in order to ensure exact correspondence between starred tones and primary stress, the only possible winner in cases in which an unmarked stem containing a short vowel combines with an unmarked inflectional ending, e.g., NOM.PL $\mu\acute{o}\epsilon\varsigma$ [múes], is the candidate with a degenerate foot, [mú.('è)(s)]. The complete H+L* prominence tone is realized, and only building feet such that primary stress falls on the final syllable will permit the strict preservation of stress-to-tone and tone-to-stress relations. The tableau in 100 shows that any candidate other than [mú.('è)(s)] violates either an undominated stress-to-tone or tone-to-stress constraint, or MAX-H. Identical considerations apply to any other inputs of the same prosodic shape belonging to different inflectional classes, e.g., φίλος 'friend'.

(100) /mu-es, H+L*/ → [mú.('è)(s)] 'mouse:NOM.PL'

/mu-es, H+L*/		T* → 'σ	'σ ← T*	MAX-H	DEP-T	FTBIN	MAX-L*
a.							
b.		*!	*!				
c.				*!			
d.		*!			*		*

The admission of degenerate feet into the prosody of Ancient Attic-Ionic then has further serious implications for the understanding of Gunkel's (2010: Ch. 1, 2011) evidence for apparent trochaic shortening in Attic-Ionic. For convenience, I provide in (101) data that was mentioned in Chapter 1 under example (45). These data indicate that nouns built with the suffix -μα(τ)- [-ma(t)-] preferentially selected a short-voweled root allomorph when it was available, frequently replacing older forms that contained an allomorph with a long vowel or diphthong. Thus, given a UR like /{x^heu-, k^hu-}, -ma(t)-/, in which the phonological grammar can choose between one of two available allomorphs, prosodic considerations prefer the short-voweled allomorph, thus producing [k^húma]. Under Gunkel's interpretation, the choice of the short vowel allomorph constitutes a garden-variety case of trochaic shortening, where a parse [(^hk^hú.ma)] would be preferred over [(^hx^heû).ma] on the constraints ALL-FT-R and PARSE-σ, given that RH-CONTOUR excludes a parse [(^hx^heû.ma)].

(101) Trochaic Shortening in Ancient Attic-Ionic Greek

Other Nominal Form	Gloss	Older $\mu\alpha$ -stem	Younger $\mu\alpha$ -stem	Gloss
k ^h ú-sis	‘(act of) pouring’	x ^h eû-ma	k ^h ú-ma	‘that which is poured’
heure-sis	‘discovery’	heure:-ma	heure-ma	‘invention’
klí-sis	‘inclination’	klí:-ma	klí-ma	‘direction, region’
rú-sis	‘flow’	reû-ma	rú-ma	‘stream, current’
ap ^h -aire-sis	‘taking away’	par-aíre:-ma	ap ^h -aire-ma	‘tribute’
pó-sis	‘drink(ing)’	pô:-ma	pó-ma	‘drink’
súm-bá-sis	‘going together’	bε:-ma	para-súm-ba-ma	‘secondary circumstance’
p ^h óre-sis	‘wearing (of clothes)’	p ^h óre:-ma	p ^h óre-ma	‘that which is carried/worn’
aná-t ^h e-sis	‘dedication’	aná-t ^h ε:-ma	aná-t ^h e-ma	‘that which is dedicated’
dó-sis	‘(act of) giving’	*dô:-ma	dó-ma	‘gift’
én-e-sis	‘injection’	hê:-ma	én-e-ma	‘injection’
dé-sis	‘binding’	hupó-dε:-ma	dé-ma	‘band, rope’
krí-sis	‘decision’	krí:-ma	krí-ma	‘(matter for) decision’
*épse-sis	‘(act of) boiling’	épse:-ma	épse-ma	‘that which is boiled’
sk ^h é-sis	‘condition’	sk ^h ê:-ma	sk ^h é-ma	‘form, figure’
el-lip-és	‘omitting’	leím-ma	leím-ma	‘remains’
haimo-r:ag-ía	‘hemorrhage’	rε:g-ma	rε:g-ma	‘break, tear’
lab-éz	‘grip, hold’	lê:m-ma	lê:m-ma	‘that which is received’
dák-os	‘bite, sting’	dê:g-ma	dê:g-ma	‘bite, sting’
sú-zdug-os	‘paired’	zdeûg-ma	zdeûg-ma	‘something used for joining’
pág-ε:	‘thing that fastens’	pê:g-ma	pê:g-ma	‘thing that is fastened’

Assuming that selection of the short-voweled allomorph instead yields a parse with a degenerate foot, thus [k^hú.(‘mà*)] is perhaps surprising only in the sense that it is not absolutely harmony-improving with respect to the alternative [(‘x^heû).ma]: the expected winner still avoids violations of ALL-Ft-R, but incurs a violation of FtBIN instead. Again, the need to perfectly align starred tones and stress is the driving factor behind the deviant foot structure. The tableau in (102) illustrates that, by choosing a short-voweled allomorph, the winner [k^hú.(‘mà*)] can be relatively harmony-improving, given the ranking ALL-Ft-R ≫ FtBIN, since no violation of WSP will be incurred.

(102) /{x^heu-, k^hu-}, -ma(t)-, H+L*/ → [k^hú.(‘mà*)] ‘what is poured:NOM/ACC.SG’

/{x ^h eu-, k ^h u-}, -ma(t)-, H+L*/	RH-CONTOUR			WSP	ALL-Ft-R	FtBIN
	T*	‘σ	‘σ ← T*			
a.  k ^h ú.(‘mà*)						*
b. (‘k ^h eû.*).ma					*!	
c. (‘k ^h ú.mà*)		*!	*!			
d. (‘k ^h eû.*).ma	*!					
e. k ^h eú.(‘mà*)				*!		

6.2.3.6 Summary and the Representation of the Genitive Plural Suffix Revisited

The analysis developed in the preceding sections for the prosody of minimal prosodic words of Ancient Attic-Ionic is indeed intricate, adheres to a few governing principles that are responsible for most of the major phenomena seen in the grammar. These governing principles take the form of undominated constraints, whose major effects are summarized in (103).

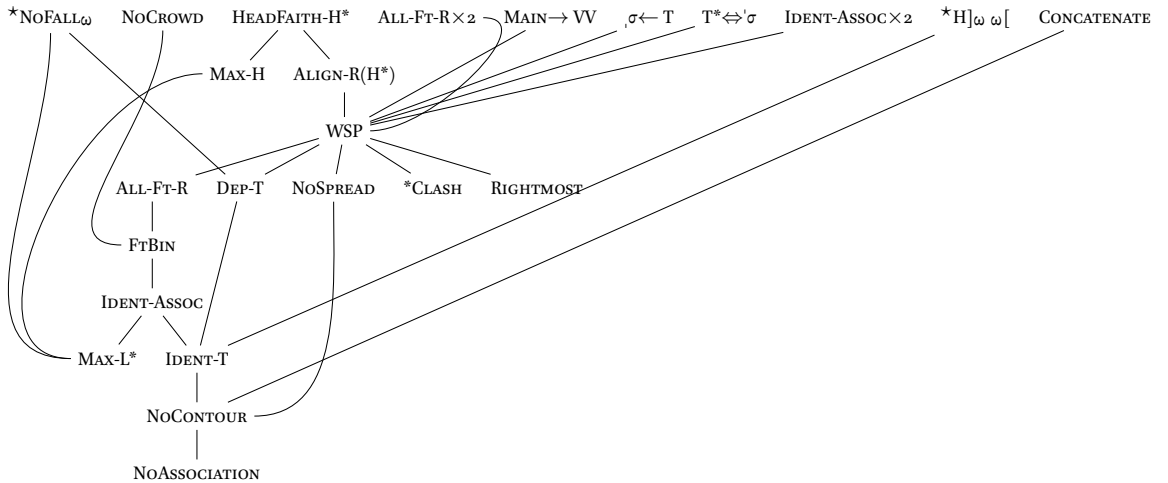
(103) Undominated constraints in Ancient Attic-Ionic word prosody

- a. $\sigma \rightarrow T$: Every stressed syllable in the input must be associated with some tone. The number of stressed syllables cannot normally exceed the number of tones in the output. Satisfaction of this constraint may trigger spreading of tone in some instances.
- b. $T^* \Leftrightarrow \sigma$: starred tones and primary stress are in a one-to-one relationship. No primary stress may lack a starred tone, and no starred tone may occur in the output if not associated with a syllable bearing primary stress.
- c. $*H]_{\omega} \omega$: A High tone may not be final in a (maximal) prosodic word if immediately followed by another prosodic word within the same intonational phrase. This constraint triggers grave formation, understood here as lowering of H^* to L^* (violating IDENT-T)
- d. $*NOFALL_{\omega}$: A word containing at least two syllables in which a H or L tone is non-final must contain a fall from H to L within the minimal prosodic word. This constraint is responsible for triggering violations of IDENT-T and DEP-T, in configurations in which a H^* tone must be preserved on a non-final syllable.
- e. NOCROWD ($\sim$ NOCONTOUR / SHORT): A syllable TBU dominating a single V may not host a contour tone $\sim$ be associated with multiple tonal targets.
- f. RH-CONTOUR: Trochaic feet of the shape ($\bar{\sigma} \sigma$) are banned. This ban motivates the limited trochaic shortening via selection of short-voweled root allomorphs.
- g. MAIN $\rightarrow$ VV: Main stress should fall on a syllable with a long vowel or diphthong, provided that it is footed. This constraint is responsible for the placement of primary stress on a non-rightmost foot in forms with a VV penult preceding a $\check{V}CC$ final, such as $\kappa\hat{\eta}\rho\nu\check{\xi}$.
- h. CONCATENATE: Tones in binary pitch accents must be associated exclusively on immediately adjacent TBUs. This constraint prevents the spreading of the binary $H+L^*$ sequence across more than two total syllables, providing a restriction on tonal distribution for the Law of Limitation in one case (type $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\check{\xi}$) not already covered by constraints on foot structure.
- i. ALL-FEET-RIGHT $\times 2$: No foot may be constructed more than two syllables from the right edge of the word. This constraint ensures that primary stress is always final or penultimate.
- j. IDENT-ASSOCIATION $\times 2$: If at least two lexical H^* tones are present in the input, at least one must be preserved over the H tone of the default $H+L^*$ intonation.

Beyond these undominated constraints, the further probably undominated HEADFAITH-H* plays an essential role in ensuring that the lexical H* tones projected from morphological heads remain in the output, much as in Vedic Sanskrit (but with foot edges instead). Similar in effect, and ultimately necessary to derive one subcase of genuine high tone mobility in monosyllabic stems (see (99) above), is a preference for H* tones to occur as close to the right edge of a word as possible (ALIGN-R(H*, ω); in practice, these only surface on initial and final syllables, in accord with the restriction of primary stress to initial and final syllables. Ranking MAX-H high in the grammar as well ensures that every ω will surface with a H tone, either from a morpheme's lexical entry or from the H+L* default.

The crucial ranking relationships that can be established from a comprehensive consideration of candidates similar to the ones reflected in the tableaux in this section are shown in the Hasse diagram in (104).¹⁰³ The core rankings, which produce the foot structures, stress, and tonal associations responsible for recessive accentuation, follow mainly from the constraints that must dominate WSP, and which in are dominated by WSP; the somewhat AD HOC character of CONCATENATE, invoked in order to prevent H tone from associating too far to the left in forms like $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\xi$, is reflected in the fact that this constraint need only dominate relatively low-ranked NOCONTOUR, and otherwise plays no further role in the grammar. The lexical H* subsystem emerges from the domination relationships proceeding down from undominated HEADFAITH-H*, and the $\sigma\omega\tau\eta\rho\alpha$ Rule, treated above mainly as satisfaction of the tonotactic markedness constraint *NOFALL $_{\omega}$, follows from its domination of DEP-T, in turn dominating IDENT-T.

(104) Hasse diagram of crucial constraint rankings for Attic-Ionic word prosody



One particular pattern that will not, however, emerge from the present ranking is the correct accentuation of genitive plural forms in polysyllabic stems.¹⁰⁴ In monosyllabic stems,

¹⁰³These relationships were determined manually based on elementary ranking conditions generated using Becker et al. 2007. A file formatted for OT-Help and OT-Soft with the candidates and their violations is available at github.com/Sandell-LMU-Habilitation, under Chapter 6.

¹⁰⁴I will speak here exclusively of the GEN.PL ending with the two accentual allomorphs $-\omega\nu$ and $-\hat{\omega}\nu$, but all considerations apply equally to the much less frequently attested GEN/DAT.DU ending $-\sigma\iota\nu \sim -\hat{\sigma}\iota\nu$.

the ending $-\acute{\omega}\nu$ exhibits genuine mobility, just like the endings GEN.SG $/-\acute{o}s/$, DAT.SG $/-\acute{i}/$, and DAT.PL $/-\acute{s}\acute{i}/$. This mobility in forms like $\pi\omicron\delta\acute{\omega}\nu$, $\mu\upsilon\acute{\omega}\nu$, $\alpha\acute{\iota}\gamma\acute{\omega}\nu$, or $\rho\acute{\iota}\nu\acute{\omega}\nu$ is correctly predicted, given that the ending is marked with a lexical H^* , and furthermore, $\text{ALIGN-R}(H^*)$ need not be invoked, as it must be for cases like DAT.SG $\rho\acute{\iota}\nu\acute{i}$ seen above, since the final syllable in $\rho\acute{\iota}\nu\acute{\omega}\nu$ is itself heavy. This combination of being heavy and lexically marked becomes a problem in unmarked polysyllabic stems: the appearance of recessive accentuation in forms like $\mu\acute{\upsilon}\rho\mu\eta\chi\acute{\iota}$ ‘ant:DAT.SG’ could be successfully derived above via the ranking $\text{FTBIN} \gg \text{IDENT-ASSOC} \gg \text{NOCONTOUR}$: because realizing the presumed H^* of the ending $/-\acute{i}/$ would require building a degenerate foot, recessive accentuation applies instead. However, since the ending $/-\acute{\hat{\sigma}}:n/$ does not risk incurring any violation of FTBIN, the ranking in 104 predicts that this ending should *always* surface with High tone, contrary to fact; see the tableau for the input $/\text{mur}\epsilon:k-\acute{\hat{\sigma}}:n, H+L^*/$ in 105, where a winner $^X\mu\upsilon\rho\mu\eta\chi\acute{\omega}\nu$ is incorrectly predicted.

(105) $/\text{mur}\epsilon:k-\acute{\hat{\sigma}}:n, H+L^*/ \rightarrow ^X[\text{mur}.,(\text{m}\epsilon:).('k\acute{\hat{\sigma}}:)(n)]$ (expected: $[\text{mur}.,(\text{m}\acute{\epsilon}:).('k\grave{\hat{\sigma}}:)(n)]$)

$/\text{mur}\epsilon:k-\acute{\hat{\sigma}}:n, H+L^*/$		$T^* \rightarrow ' \acute{\sigma}$	MAX-H	WSP	RIGHTMOST	NO SPREAD	FTBIN	IDENT-ASSOC	NO CONTOUR
a.			*						*
b.			*					*!	
c.			*	*!				*	
d.			*		*!	*!		*	*

Between the predicted winner, a., and unexpected loser, b., agreement exists in terms of stress placement and footing, i.e., in this configuration, default constraints and lexical H* agree in placement of primary stress. The inactivity of FTBIN predicts that underlying associations should be preserved. One viable solution within the context of the present analysis is indeed to reckon with multiple accentual allomorphs of H-tone-attracting inflectional suffixes, as represented under (94 above: monosyllabic stems receive the allomorph with H*, whereas polysyllabic stems (including historically contracted stems, like /pa.id-/ ‘child’; see the end of 6.2.1.6 above) receive an allomorph without H*. Compare the allomorphy rules in 106. While not ideal, the emergence of such allomorphy seems diachronically plausible via a reanalysis: given that many, probably most, polysyllabic stems have a derivational suffix that determines the position of H tone on the surface, any H*-marked inflectional suffixes would occur without H-tone in such forms. Stem polysyllabicity and marking of a morphological head with H* could thus be conflated, leading to the assumption on the part of learners that H*-marked inflectional suffixes also possessed unmarked allomorphs. By consistently employing the unmarked allomorphs in polysyllabic stems, the overall phonological harmony of the expected winners is also increased (i.e., no faithfulness violations for failure to realize inflectional H* are incurred).¹⁰⁵

(106) Allomorphy of GENITIVE PLURAL in Ancient Greek 3rd Declension Nouns

- a. [+GENITIVE, +PLURAL] $\leftrightarrow$ /-ó:n/ // [C_o[+syllabic]([−consonantal])C_o]_{Stem}_____
- b. [+GENITIVE, +PLURAL] $\leftrightarrow$ /-ɔ:n/ // *elsewhere*

A further outstanding issue is how and why the genitive plural ending surfaces with a circumflex, rather than acute, when it bears H tone at all. In the context of the present constraint grammar, the emergence of a circumflex falls out from the ranking MAX-H $\gg$ MAX-L* $\gg$ NOCONTOUR: the L* portion of the H+L* intonation is associated to the long-voweled inflectional ending marked with H* (candidate e. in 107). Associating two starred tones with one stressed syllable seems intuitively odd, but is not penalized by the inventory of tone-to-stress and stress-to-tone constraints. As illustrated for ῥῑνῶν ‘nose:GEN.PL’ in (107), the choice between candidates d. (with acute) and e. (with circumflex) is determined by the preference for realizing L* and thereby creating a contour tone.¹⁰⁶ Thus, the inventory of lexically marked tone can be maintained as a privative contrast between H* and $\emptyset$.

(107) Underlying /-ó:n/ surfaces as [-ô:n]

¹⁰⁵ Another possibility altogether, not without its own insubstantial cost, is to posit that the genitive plural suffix is not marked with H*, but rather L*. When inserted into the present grammar, however, such a representation predicts that genitive plural forms of monosyllabic stems will exhibit consistent H tone on the initial syllable, rather than circumflex on the final.

¹⁰⁶ I assume that the H portion of H+L* does not associate instead, despite the higher ranking of MAX-H, due to the OCP. That a falling H*+L* tone is created, rather than a rising L*+H* is likely due to a ban on rising tones in the language. Compare 6.2.1.6 above for the evidence that Attic-Ionic possessed no rising tones.

/rɪːn-óːn, H+L*/		*NOFALL _ω	T* → 'σ	MAX-H	ALIGN-R-H*	WSP	RIGHTMOST	IDENT-ASSOC	IDENT-T	MAX-L*	NOCONTOUR
a.	$ \begin{array}{c} H^* \quad H^* \\ \quad \\ (\bar{\sigma}) \quad (' \bar{\sigma}) \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad o: \quad \langle n \rangle \end{array} $	*!		*	*					*	
b.	$ \begin{array}{c} H^* \\ \\ (' \bar{\sigma}) \quad \bar{\sigma} \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad o: \quad \langle n \rangle \end{array} $	*!		**	*	*		*		*	
c.	$ \begin{array}{c} H^*L \\ \\ (' \bar{\sigma}) \quad \bar{\sigma} \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad o: \quad \langle n \rangle \end{array} $			**	*!	*		*	*	*	*
d.	$ \begin{array}{c} H^* \\ \\ \bar{\sigma} \quad (' \bar{\sigma}) \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad o: \quad \langle n \rangle \end{array} $			**		*		*		*!	
e.	$ \begin{array}{c} H^* \quad L^* \\ \quad / \\ \bar{\sigma} \quad (' \bar{\sigma}) \\ / \quad \backslash \quad / \quad \backslash \\ r \quad i: \quad n \quad o: \quad \langle n \rangle \end{array} $			**		*		*			*

To summarize more broadly, a look at the intricate network of constraint relationships posited here, and contrasted with other existing analyses of Attic-Ionic word prosody under 6.2.2, leads to the perhaps dispiriting conclusion that no very secure interpretation of the facts is possible. The problem is not, however, that the available data is necessarily faulty in some way – though uncertainties at the margins exist – but rather that it is open to too many different possible interpretations. Without a secure knowledge of exactly where primary stress fell and precisely what tonal targets were present in which prosodic words of which shapes, analysts are forced to make educated guesses, thus encouraging a flowering of diverse analyses. The best way forward is to continue to exploit the best available substitute that

we have for pitch tracks: restrictions on tonal distribution within meter (cf. Hagel 1994/95) and in the corpus of Ancient Greek music. Only with advances in the knowledge of which intonational contours and boundary tones must be present under what circumstances can real further progress be made.

6.2.3.7 The Accentuation of Single and Multiple Enclitic Sequences

To conclude the larger treatment of Attic-Ionic word prosody, I propose a cursory analysis of the placement of additional accents in enclitic sequences. I refer the reader to the description of the (likely) facts under 6.2.1.5 above. First, let us summarize what occurs when just a single enclitic follows an orthotonic word. When the final syllable itself bears any H tone, any tone on the immediately following enclitic, whether mono- or disyllabic, is forbidden; in these cases, the final syllable has both H tone and stress. When the final syllable is presumably footed and L-tone bearing, no secondary acute on the orthotonic word is permitted (because the final syllable is already occupied by tone), but H on the second syllable of disyllabic enclitics may occur. Finally, if the final syllable is neither footed nor tone bearing, both secondary acute in case of monosyllabic enclitics and H on disyllabic enclitics may occur. To summarize:

- Final syllable has primary stress and H(L) tone (e.g., πατήρ [pa.('té:)r], καλῶς [ka.('lô:)s])
⇒ no secondary acute or extra H on disyllabic enclitics (πατήρ τις, πατήρ ἐστί).
- Final syllable is part of a foot, whether it bears stress or not, and associated with L(*) tone (e.g., μέγας [me.('gá.lò)s], φίλος [pʰí.('lò)s], κήρυξ ('kê:).(rùk)s])
⇒ no secondary acute, but extra H on disyllabic enclitics permitted (κήρυξ τις, κήρυξ ἐστί).
- Final syllable is neither footed nor tone bearing (e.g., οἶκος [('oî).kos], ἄνθρωπος [án.('thrɔ:).pos])
⇒ secondary acute on the orthotonic word, no extra H on disyllabic enclitics (οἶκος τις, οἶκος ἐστί).

In this fashion, why words like κήρυξ should build a natural class with words like φίλος or μέγας is clear: the final syllable is footed and low-tone bearing. Conditions on enclitic accentuation may then perhaps best be formulated exclusively with respect to tone: when the final TBU bears an H, no tone is permitted on the first enclitic; when the final TBU is associated with an L, then no H is permitted on the first syllable of the enclitic; when the final TBU is associated with no tone, an H appears on the final syllable of the orthotonic word. That the restrictions on secondary H are not primarily governed by stress and foot structure would seem to follow from two facts: words like μέγας [me.('gá.lò)s] and οἶκος [('oî).kos] are identical in terms of stress placement, but behave differently with respect to secondary acute; the shape of the enclitic beyond the number of syllables (one or two) is irrelevant (i.e., light and heavy monosyllabic enclitics behave identically, and disyllabic enclitics, regardless of the pattern of light and heavy syllables, behave identically).

Thus, two constraints govern the distribution of H in single enclitics, and one further constraint will be necessary to account for the behavior of multiple enclitics. The two essential

constraints refer to tone: first, a ban on a sequence of three or more TBUs bearing only L or no tone (*LLL) preceding the right edge of a (recursive) prosodic word, and second, a ban on a rising sequence (i.e., L or $\emptyset$ + H) over a (recursive) right prosodic word boundary (*L]_ω H).¹⁰⁷ These two constraints mean that, in the first enclitic after an orthotonic word, a further H is superfluous when H already appears on the last syllable of the orthotonic word, while the second means that, when a further H is necessary, it may only associate before the]_{ω-min} boundary or on the second syllable of a disyllabic enclitic. This latter constraint is *ad hoc*, but captures a descriptively true fact: any additional H in enclitic sequences never occurs on the first syllable after the orthotonic word. Furthermore, in multiple enclitic sequences, further new H is introduced in ways that largely conform to this constraint as well (e.g., [[χαί] τεε] ⇒ [[[χαί] τέ] με], not ^Xχαί τε μέ, which would violate *L]_ω H). These two constraints are defined in (108) and (109).

- (108) *LLL: A contiguous sequence of three or more TBUs bearing only L tone or no tone is not permitted. Assign one violation * for each such sequence in the output.
- (109) *L]_ω H: A sequence of L tone and H tone on adjacent TBUs between one (recursive) prosodic word and the next recursive prosodic word, is not permitted. Assign one violation * for each such sequence in the output.

In the cases in which the secondary acute appears on the final syllable of an orthotonic word, I assume that a new secondary stress is created, rather than a primary stress, and so CULMINATIVITY_ω is not violated. The only stress within a new recursive prosodic word layer is, on the other hand, perforce primary (as in Neapolitan Italian; see the discussion at 16 under 1.2.2), and thus induces a violation of CULMINATIVITY_ω, defined in (110).

- (110) CULMINATIVITY_ω: A non-minimal prosodic word (ω_[-min]) must contain at one one primary stress. Assign one violation * for each primary stress in a non-minimal prosodic word that exceeds 1.

Assuming that a H tone is made available by the construction of a recursive prosodic word,¹⁰⁸ single enclitics are derived as in Table 6.8 (using the enclitics τις ‘some’ and ἐστὶ ‘is’ for exemplification). Where no violation of *LLL violation would result, no secondary enclitic appears; when such a violation would result, but the final syllable of the host is already occupied by a Low tone, CULMINATIVITY_ω is violated in a way that also respects *L]_ω H. Finally, when the final syllable of the orthotonic word is available, a new secondary stress is created in the preexisting ω_[+min], whether or not a violation of WSP results. Thus, all cases in which a secondary acute appears, either on the orthotonic word or a disyllabic enclitic, are

¹⁰⁷ A functional explanation for such a constraint might make reference to the grave accent: since grave formation creates a L* on the final syllable, and most other orthotonic words also bear L(*) on the final syllable one way or another, a rise L–H is usually indicative of the boundary between two orthotonic words, rather than between orthotonic word and enclitic. This constraint thus effectively reserves a “turn” from a pitch trough to a new peak to the boundary between separate orthotonic words.

¹⁰⁸ Equivalent would be the assumption that an H tone is epenthesized in violation of DEP-T in order to provide the necessary H, and a stressed syllable is created to provide it with a suitable host.

Input	Output	Constraint Considerations
/pa.(^h té:r), tis/	[[pa.(^h té:r)] tis]	No *LLL violation
/pa.(^h té:r), esti/	[[pa.(^h té:r) es.ti]	No *LLL violation
/ka.(^h lò)s, esti/	[[ka.(^h lò:s) es.ti]	No *LLL violation
/p ^h í.(^h lò)s, tis/	[[p ^h í.(^h lò)s] tis]	No *LLL violation
/p ^h í.(^h lò)s, esti/	[[p ^h í.(^h lò)s] es(^h tí)]	Satisfies *LLL and *L] _ω H; violates CULMINATIVITY _ω
/('oî).kos, tis/	[[('oî).(kó)s] tis]	Satisfies *LAPSE
/án.(^h thrò:).pos, tis/	[[án.(^h thrò:).(pó)s] tis]	Satisfies *LLL and *L] _ω H; satisfies CULMINATIVITY _ω
/án.(^h thrò:).pos, esti/	[[án.(^h thrò:).(pó)s] es.ti]	Satisfies *LLL and *L] _ω H; satisfies CULMINATIVITY _ω , but violates WSP

Table 6.8: Derivation of Single Enclitic Sequences

driven by potential *LLL violations, with one exception: when words of the shape οἶκος are followed by a single enclitic, since *LLL would not be violated if no secondary acute appears. To account for sequences like οἶκος τις ‘some house’, *LAPSE (defined in 111), ranked crucially below CULMINATIVITY_ω, may be invoked: thus, stress lapses at the right edge of (recursive) prosodic words may be repaired only if they can be achieved without a violation of CULMINATIVITY_ω (i.e., without placing stress on the enclitic). The general constraint ranking *LLL, *L]_ω H ≫ CULMINATIVITY_ω ≫ *LAPSE ≫ WSP is then sufficient.

- (111) *LAPSE: A sequence of more than one unstressed syllable at the right edge of a prosodic word of any kind is not permitted. Assign one violation * if a sequence of two or more stressless syllables occur at the right edge of a prosodic word at any level of recursion.

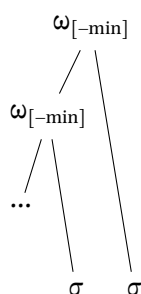
Perhaps surprisingly, in contrast to most analysis of enclitic accentuation, OCP-High (as invoked in, e.g., Golston 1990 or Steriade 2014) plays no role here. In short, OCP-High is both superfluous and insufficient for handling the accentuation of single enclitics. Under the present analysis of orthotonic words, a L tone occurs after the H in *all* orthotonic words (including those with acute penult, like πατέρα), except those with an acute final syllable (like πατήρ). OCP-H would thus be relevant in forms like πατήρ alone, yet forms with a final circumflex like ποδῶν ‘foot:GEN.PL’ behave identically in terms of enclitic accentuation. Furthermore, OCP-High is evidently violated in sequences of multiple enclitics, both the traditional handbook interpretation and the “Revised Göttling-Barrett” (RGB) model discussed under 6.2.1.5 above (e.g., sequences like καί τέ με φησί, where two adjacent H tones appear on the surface).

When further enclitics are added, the major difference in comparison to single enclitics is that, the next enclitic in the sequence may or may not exhibit a further secondary acute, depending upon what rules for multiple enclitic accentuation are considered correct. Under the traditional handbook rules, every further enclitic except the last in a sequence receives a secondary acute. Under the RGB model, the addition of a further acute depends upon the

distance between the rightmost H and the right edge in the input preceding the new enclitic, as well as the weight of the last two syllables in the input preceding the new enclitic.¹⁰⁹ Other models largely refer to lexical exceptions, or refer to sequences of syllable weight in different ways. I present here an analysis that largely reproduces the traditional handbook rules, but better handles the behavior of disyllabic enclitics, for which the manuscript evidence strongly suggests that they received a secondary acute when preceded by a monosyllabic enclitic (e.g., *καί τέ με φησί*), but not when preceded by a disyllabic enclitic (e.g., *οὐ ἐθέν ἐστι*), and that a monosyllabic enclitic followed by a disyllabic enclitic typically did not receive a secondary acute (e.g., *ἄ μοι ἐστι*).

The sequence *καί τέ με φησί* is thus overall instructive: ^X*καί τε με* is not allowed, although *LLL would not be violated, and with the addition of *φησι*, a sequence of two syllables without H arises, but across a]_ω boundary. Assuming that H is indicative of further stress on enclitics, the correct generalization appears to be that a sequence of more than one unstressed enclitic is not permitted.¹¹⁰ In a sequence of two unstressed enclitics, two layers of non-minimal prosodic words directly dominate only a (sequence of) syllables. That is the higher non-minimal prosodic word would directly dominate only (a) syllable(s) on its right branch, and dominate another non-minimal prosodic word on its left branch, which in turn also dominates only (a) syllable(s) on its right branch. This structurally apparently illicit configuration is shown in (112). This ill-formed structure necessarily also violates *LAPSE, but because CULMINATIVITY_ω must dominate *LAPSE (otherwise, disyllabic enclitics would always receive stress after an orthotonic word), the configuration in (112), referring to non-minimal prosodic words, must additionally be marked as ungrammatical.

(112) Ill-Formed Sequence of Non-Minimal Prosodic Words through Cliticization



¹⁰⁹Specifically, if H tone is relatively far from the right edge (circumflex penult or acute antepenult), a secondary acute appears on the last syllable of the input; or, if the input ends in a trochaic (– ◡) sequence, and the final syllable of the input is not itself H-tone bearing, then a secondary acute appears on the last syllable of the input; finally, newly added disyllabic enclitics always receive a secondary acute on their final syllable if they do not condition a secondary acute on the preceding sequence according to the already named conditions. All of the conditions for the RGB system correctly generate orthotonic + single enclitic sequences, and then treat the newly created sequence like an orthotonic word when a new enclitic is added.

¹¹⁰I tacitly grant here that a non-underlying tone H* is inserted, and the insertion that H* is preferred to L* on account of a constraint like de Lacy (2002)'s *HD/L. Alternatively, one could assume that the construction of a ω[-min] results in the insertion of a H*, which is practically equivalent to assumptions regarding H* tones on enclitics in Golston 1990.

Since sequences of more than four enclitics are, to my knowledge, unattested, the degeneracy of (112) might be understood as due to the effects of a version HEADEDNESS, which penalizes “level skipping” in the prosodic hierarchy (see 18 in Chapter 1). In (112), the lower $\omega_{[-min]}$ likely dominates another $\omega_{[-min]}$ or the orthotonic word ($\omega_{[+max]}$). The illicit sequence would then contain two $\omega_{[-min]}$ that do not directly dominate any instance of the immediately lower prosodic category, the $\mathcal{F}$: they dominate instances of the same category (ω) and a lower category (σ). There are thus two “headless” $\omega_{[-min]}$. The constraint in (113) gives a precise definition of the conditions for violation.

- (113) HEADEDNESS $\times 2$: More than prosodic word without a head (i.e., that does not directly dominate a foot) is banned. Assign one violation * if two or more ω in the output do not directly dominate at least one $\mathcal{F}$.

Satisfying HEADEDNESS $\times 2$ is done in such a way as to respect $*L]_{\omega} H$: the new stress and H are placed in positions that do not result in a rise in pitch over a single right word boundary. Hence, in constructing the larger sequence $\kappa\alpha\iota\ \tau\acute{\epsilon}\ \mu\epsilon\ \varphi\eta\sigma\acute{\iota}$, we do not find, e.g., $^X\kappa\alpha\iota\ \tau\epsilon\ \mu\acute{\epsilon}\ \varphi\eta\sigma\acute{\iota}$ or $^X\kappa\alpha\iota\ \tau\acute{\epsilon}\ \mu\epsilon\ \varphi\eta\sigma\acute{\iota}$ instead. To summarize, the accentuation of all single enclitic sequences is correctly predicted by the constraint ranking given in 114, and many multiple enclitic sequences fall out correctly as well. The subranking WSP $\gg$ DEP-T $\gg$ FtBIN is taken over from the treatment of orthotonic words and does not crucially determine any aspects of enclitic accentuation. The tableaux in 115a. and b. illustrate how the accentual sequences $\kappa\alpha\iota\ \tau\acute{\epsilon}\ \mu\epsilon$ and in turn $\kappa\alpha\iota\ \tau\acute{\epsilon}\ \mu\epsilon\ \varphi\eta\sigma\acute{\iota}$ are generated; violations of WSP, FtBIN and DEP-T are assessed only with respect to the outermost enclitic, whereas CULMINATIVITY $_{\omega}$ and HEADEDNESS is assessed over the entire sequence.

- (114) Constraint ranking governing enclitic accentuation: $*LLL$, HEADEDNESS $\times 2$, $*L]_{\omega} H$ $\gg$ CULMINATIVITY $_{\omega}$ $\gg$ $*LAPSE$, OBLIGATORINESS $\gg$ WSP $\gg$ DEP-T $\gg$ FtBIN

- (115) Accentuation of multiple enclitic sequences

a. $/[(['\kappa\alpha\iota'])\ te] + \mu\epsilon/ \rightarrow [[(['\kappa\alpha\iota'])\ ('t\acute{\epsilon})]\ me]$

$/[(['\kappa\alpha\iota'])\ te] + \mu\epsilon/$		$*LLL$	HEADEDNESS $\times 2$	$*L]_{\omega} H$	CULMINATIVITY $_{\omega}$	$*LAPSE$	OBLIGATORINESS	WSP	DEP-T	FtBIN
a.	$[[(['\kappa\alpha\iota'])\ ('t\acute{\epsilon})]\ me]$				*		*		*	*
b.	$[[(['\kappa\alpha\iota'])\ te]\ me]$		*!			*	**			
c.	$[[(['\kappa\alpha\iota'])\ te]\ ('m\acute{\epsilon})]$			*!					*	*

b. /[[[(['kaí)] ('té)] me] + p^hε:si/ → [[[[(['kaí)] ('té)] me] p^hε:(sí)]

/[[[(['kaí)] ('té)] me] + p ^h ε:si/	*LLL	HEADEDNESS×2	*L _ω H	CULMINATIVITY _ω	*LAPSE	OBLIGATORINESS	WSP	DEP-T	FtBIN
a.  [[[[(['kaí)] ('té)] me] p ^h ε:(sí)]				**		*	*	*	*
b. [[[[(['kaí)] ('té)] me] p ^h ε:si]	*!	*!		*		**	*		
c. [[[[(['kaí)] ('té)] me] ('p ^h é:).si]			*!	**				*	
d. [[[[(['kaí)] ('té)] ('mé)] p ^h ε:si]				**	*!	*	*	*	*

Although much of this analysis has an *ad hoc* flavor, it is largely empirically adequate: it is evidently capable of handling sequences of orthotonic + single enclitic correctly, and it performs relatively well against the 70 examples of multiple enclitic sequences in Books 1–3 of the *Iliad* in the Venetus A manuscript collected by Probert (2016). Overall, the predictions made by the grammar in (114) are not as good as the “Revised Göttling-Barrett” (RGB) developed in Probert 2016 (see discussion under 6.2.1.5 and above), though better than any other model considered by Probert: 48/70 multiple enclitic sequences are correctly predicted (vs. 53/70 by the RGB model, and at most 47 by any other model).¹¹¹ The major difference in the predictions of the grammar in (114) and the RGB model concerns sensitivity to syllable weight: the RGB model allows for a further secondary acute on a monosyllabic light enclitic (e.g., τι or τε) only when the preceding syllable is heavy (e.g., ὅς τέ νυ), when that the immediately preceding heavy syllable is acute or circumflex. The grammar in (114), on the other hand, predicts that, in any sequence of two or more enclitics, one of the last two must have stress (and High tone), for instance predicts ²οὐδέ τί πω where the Venetus A has οὐδέ τι πω (which would violate HEADEDNESS×2).

6.2.4 The Swedish Synthesis

In the discussion of existing accounts of the recessive accent under 6.2.2 and 6.2.3.3, I argued that data are more readily compatible with a metrical analysis, rather than an analysis purely in terms of tone. As mentioned briefly at the end of Section 6.2.2.2, proponents of a purely tonal analysis of Greek prosody (Devine and Stephens 1994, Ito and Mester 2017, Revithiadou

¹¹¹Worth pointing out, however, is that the performance between different models of multiple enclitic accentuation is not, in fact, statistically significantly different based on token counts. The better performance of the RGB model relative to other models in the 70 token sample is mainly due to the number of cases where the RGB model predicts no secondary acute on a further enclitic. For example the RGB model is not significantly better in performance than the grammar in 114, or any of the models labeled as B (Bi, Bii, Biii) by Probert (two-sample tests for equality of proportions return $p > 0.05$ in these comparisons). Given that some prosodically identical sequences in the 70 token sample are accented differently (e.g., οὐδέ τι μιν vs. χειρί τέ μιν), a larger sample needs to be assessed in order to obtain statistically reliable results. I am, ultimately, somewhat pessimistic about the prospects of successfully determining the correct original patterns for multiple enclitic accentuation: the data may be too distorted by two millennia of (mis)interpretations of ancient grammatical doctrine on the question.

2018) reject the notion that the prosodic prominence should be realized as a (H)+L* tone.¹¹² Here, however, one must be attentive to distinguish between the *realization* of stress per se and the *intonation* that stress attracts. In contrast to the view expressed in Steriade 2014, which sees the *realization* of stress itself as tonal, the Sauzet–Golston analysis, which was built upon here, does not see stress as realized through pitch at all, but merely that stress *attracts* the default word-level intonational pattern H+L*, made accessible by the construction of the minimal prosodic word. Nevertheless, the issue remains that the lexically specified prosody, seems to require H* tones, which then stands in contrast to the L* intonation attracted to stress when no lexical H* is present. The question is whether the co-occurrence of lexical H*, realized under primary stress, and default H+L*, attracted to primary stress in the absence of H*, should be permitted within one prosodic grammar. Note further that the situation proposed for Ancient Attic-Ionic is also distinct from the situation in Vedic or Egyptian Arabic, where word-level stress and a default word-level intonation co-occur in the vast majority of cases.

The short answer is probably “yes”, and ultimately, the closest typological parallels for the Attic-Ionic prosodic system are perhaps represented by some present-day varieties of Scandinavian languages (Central Swedish, Eastern Norwegian). Some particulars of Central Swedish, in particular, are strikingly similar to the analysis of Attic-Ionic prosody that results from a marriage of the Sauzet–Golston analysis of the recessive accent with the analysis of lexical accent presented under 6.2.3. Five features characterize the Central Swedish system, for instance:

- Swedish feet are typically bimoraic and trochaic, though a prosodic word is limited to having exactly one foot (Riad 2014: 134).
- In the absence of lexical stress (foot edges under Riad’s account), a foot is constructed around and stress assigned to the rightmost bimoraic sequence in a prosodic word (Riad 2014: 200).
- Morphemes may be specified for bearing foot edges with respect to their segmental content, thereby influencing the placement of stress (Riad 2014: 201–10).
- Morphemes may also be specified for a lexical tone, H* (“Accent 2”), which is realized with the word-level pitch accent H*+L (Riad 2014: 184; 233–5) when under the stress (whether lexical or default).
- Prosodic words without a lexical tone receive a pitch accent H+L* (“Accent 1”) linked to their stressed syllable (Riad 2014: 184–88).

Thus, in Riad’s account of C. Swedish prosody, four factors interact: 1) algorithmically calculated stress; 2) lexically-specified stress (underlying foot structure); 3) lexically specified

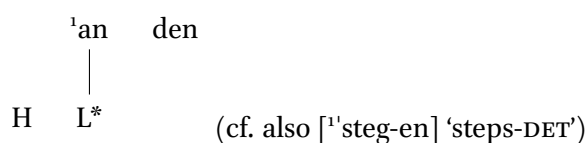
¹¹²So Ito and Mester (2017: 15): “The cross-linguistically solid link between H tones and footheads (de Lacy 2002 [see also 2.4.1.2, RS]) is severed, *contra naturam*. There are certainly intonational pitch accents whose designated tone is L, but we are here not dealing with special intonational markers with specific meanings, but with the basic word prominence marker of a language.”

tone; 4) default intonation (prominence tone). With respect to inventories of lexical tone and default word-level intonation, Central Swedish and Ancient Attic-Ionic would be practically identical; the C. Swedish system additionally includes lexically-specified stress, which is likely unnecessary under an account of Attic-Ionic that adopts a relatively strict association between starred tone and stress. I conclude that Ito & Mester's strong expectation that stress should regularly associate with high tone ultimately appears unwarranted in the face of the Swedish facts.

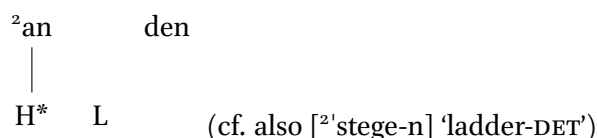
The distinction between surface "accent 1" and "accent 2" ("acute" and "grave" in older literature in C. Swedish) is, on Riad's account, due to lexical marking of tone for accent 2, while accent 1 simply reflects default intonation associated to stress (whether determined by default or by lexical stress). Minimal pairs in the strict sense are difficult to find; the classic segmental minimal pairs, shown in (116) (cf. also Féry 2017: 188–94), in fact have distinct morphological structures. Accent 2 must be realized on the syllable with primary stress, and must be followed by at least one further syllable; the consequence of these two restrictions is that accent 2 may reassociate from a host morpheme to a syllable in another morpheme that receives the primary stress, and that accent 2 is never found on monosyllables or on forms with ultimate stress.

(116) Central Swedish Accent 1 vs. Accent 2: Privative ¹¹³

a. Central Swedish Accent 1 (*and-en* 'duck-DET')



b. Central Swedish Accent 2 (*ande-n* 'spirit-DET')



However, besides the lexical sources of accent 2 – marked on either roots or suffixes – all words with more than one stress (typically compounds) are obligatorily assigned accent 2 on the syllable with primary stress. The basic tonal patterns or melodies are H*+L for accent 2, but H+L* for accent 1. Under focus, the fundamental difference – just a lexically specified H* – becomes clear, since Accent 1 shows L*+H, but Accent 2 H*LH; under focus, both accentual types contain the sequence LH (which is then also followed by L% at the end of a word), but in accent 2 forms, a H* precedes the LHL% sequence.

Central Swedish also manifests some restrictions on H* that are reminiscent of the tonotactic constraint *NOFALL_ω posited at (65) and grave formation. Specifically, C. Swedish H*

¹¹³Historically, the tonal word accent of "Accent 1" found in some North Germanic (Central Swedish, Eastern Norwegian) varieties is related so-called *stød* (a prosodically distributed creaky voice or glottal stop) in other varieties (Standard Danish).

must obligatorily be followed by a L tone on a following syllable; consequently, monosyllabic words of C. Swedish all show L*, never H*. This fact accords with the interpretation of the $\sigma\omega\tau\eta\rho\alpha$ Rule at 6.2.3.4 as L-tone insertion ($\sim$ contour tone creation), and the transformation of any word-final lexical H* into surface L* (“lulling” = grave formation) when not final in an intonational phrase.

Overall, in comparison to Riad’s treatment of Swedish, the Attic-Ionic system is perhaps even simpler since it probably does not require any lexically-specified stress. Rather, the morphemes bearing H* may attract stress in order to ensure its realization, but otherwise, stress is algorithmically calculated. In that sense, it is most appropriate to say conclude that Ancient Attic-Ionic fulfills the criteria for a stress language (per Hyman 2009), because the placement of pitch in the language regularly depends upon stress, although the language possesses lexical tone (like Central Swedish). Thus, a more canonical “pitch-accent” language such as Tōkyō Japanese, insofar as it lacks OBLIGATORINESS and does not regularly employ metrical principles for determining prosodic prominence, is not an ideal parallel for Attic-Ionic.

Two appreciable differences between Attic-Ionic and C. Swedish emerge in the behavior of tones, however. First, in contrast to Attic-Ionic’s rigorous preservation of at least one H per $\omega_{[+min]}$, C. Swedish allows H to float ($\sim$ be deleted); see the representation of *and-en* ‘duck-DET’ just above. Second, whereas Attic-Ionic allows for multiple H tones (arguably multiple tonal accents) in recursive prosodic words, C. Swedish permits maximally one tonal accent per maximal prosodic word (Riad 2014: 120).

Finally, some last typological objections against the tonal analysis of Attic-Ionic word prosody are available, too. In case the analysis of Ito and Mester 2017 can be salvaged, typological parallels for a severely restricted tone system (obligatory, culminative, and privative H) seem to be rarer than stress-dependent tone systems of the North Germanic or Balto-Slavic types, if not altogether non-existent. Nubi (Gussenhoven 2006) may represent such a system, but only if its H assignment is *not* to be attributed to metrical structure (which it might well be). If Nubi H does regularly depend on metrical structure, then it would simply instantiate yet another case of stress-dependent tone, perhaps comparable to Vedic Sanskrit.

6.3 Prosody in Other Ancient Greek Dialects

Although information on the prosody of the many varieties of Greek spoken in the 1st Millennium BCE is scanty in comparison to the data on Attic-Ionic, enough data – either in the form of statements or examples given by later grammars or papyri transmitted with accentuation – is available for at least a couple of other dialects so as to make explicit comparison with the Attic-Ionic system possible. Nevertheless, one must bear in mind that the statements of grammarians are hardly complete and sometimes dubious, while the testimony from papyri is fairly limited. Occasionally, scholia to manuscripts of an author who was known to be of non-Attic origin discuss the accentuation of a word in the author’s presumably native dialect.

Traditional classification of the Ancient Greek dialects recognized three major groups,

Attic-Ionic, Aeolic, and Doric, plus Arcado-Cypriot. A newer evolving consensus recognizes a major division between East Greek (including Attic-Ionic, the Aeolic dialects, and Arcado-Cypriot) versus West Greek (divided into Doric and Northwest Greek). Both the terms “Aeolic” and “Doric” are somewhat problematic. “Doric” has a traditional usage corresponding to “West Greek”, although statements about “Doric” in Greek grammarians perhaps largely mean the West Greek dialects of the Peloponnese (such as Spartan and Corinthian). The existence of an “Aeolic” subfamily at all has been repeatedly questioned (cf. Parker 2008, Miller 2014: 219), and statements from grammarians about “Aeolic” probably largely intend Lesbian, for which the greatest amount of literature has been preserved. Here, I confine discussion of Aeolic prosody to the dialect of Lesbian, and base the treatment of “Doric” prosody on relatively coherent indications found in grammarians and a handful of poets of the 1st Millennium BCE (see details below).

6.3.1 Lesbian (Aeolic)

The facts of Lesbian word prosody may be summarized very briefly; for present purposes, the examples in Probert 2003: 159–60 (see there for references to the philological background, as well as Probert 2006b: 72–3) are sufficient. In short, all prosodic words without enclitics in Lesbian exceptionlessly exhibited recessive accentuation, placing the word-level H precisely where the relevant constraints of Attic-Ionic would predict for it to occur. When enclitics are present, the same principles of enclitic accentuation as in Attic-Ionic appear to hold.

(117) Contrast: Attic-Ionic and Lesbian Accentuation

Attic	Lesbian	Gloss
φαίνεται μοι	φαίνεται μοι	‘seem:PRS.3SG.MID 1SG.DAT’
θεοῖσιν	θέοισιν	‘gods:DAT.PL’
άνήρ	ώνηρ	‘man:NOM.SG’
ήδύ	ᾶδύ	‘sweet’

The first example, φαίνεται μοι [p^haí.ne.taí moi], shows a finite verb with recessive accentuation just as in Attic-Ionic, followed by an enclitic pronoun that results in a further acute on the final syllable of the orthotonic word, just as in Attic-Ionic. Important to note here also is that the word-final diphthong, [-ai] in [p^haí.ne.tai] continues to count as light, just as in Attic. In the next example, where Attic-Ionic exhibits a lexical accent that results in a H closer to the right edge of the word than the recessive accent would generate, Lesbian shows precisely the accent that would be expected under recessive accentuation: H on the penult given a short final (θέοισιν), acute on a long penult given a long final (ώνηρ), but circumflex on a long penult given a short final (ᾶδύ) in a disyllabic form.¹¹⁴

¹¹⁴Recall from above that the circumflex accent on initial syllables in disyllabic forms of the shape like κήρυξ or ηδύ / ᾶδύ here is attributable to association of the H+L* pitch accent with the stressed VV-syllable. Since any forms in which genuine transformation or replacement of underlying H* in Attic (true Lex σωτήρα) have three or more syllables, such forms will have H on the antepenult in Lesbian. There is thus no Lex σωτήρα as such in Lesbian.

These data readily permit the conclusion that Attic-Ionic and Lesbian possessed the same basic grammar for the construction of metrical structure and intonation: stress is assigned to the rightmost moraic trochee (modulo the effect of MAIN $\rightarrow$ VV in $\pi\omicron\lambda\upsilon\pi\acute{\iota}\delta\alpha\xi$ -type words), and the default word-level intonational pattern is H+L*. The fundamental difference between the two dialects in terms of word-level prosody is that Attic-Ionic is replete with lexical H* pitch accents, whereas Lesbian evidently seems to lack them. Assuming that the last common ancestor of Attic-Ionic and Lesbian possessed a word-prosodic grammar as attested in Attic-Ionic, the question is then how to account for the seeming wholesale loss of lexical accents (or their subordination to the default phonological algorithm, which amounts to the same result); compare also Probert 2006b: 143. This problem is of precisely the same nature as the problem discussed in Chapters 5 and 7 concerning the development of Classical Sanskrit out of Vedic and Proto-Germanic out of PIE word-level prosody: what factors bring about the seeming wholesale excision of lexical accent from a language's lexicon?

While the particular details in the history of Lesbian must be distinct from the Germanic or Indic cases, the basic explanation is probably the same: prosodic patterns other than the default recessive pattern became sufficiently rare so as to render the acquisition of a high ranking for constraints militating in favor of the faithful realization of lexical accents (under the present analysis, lexical H* tone) impossible. As discussed in the Germanic case, increased quantities of primary linguistic data in favor of a default stress pattern may have resulted, on the one hand, from the “demorphologization” of morphologically complex lexemes whose derivational morphemes bore lexical accents, and on the other, decreased productivity of derivational processes that would have resulted in non-recessive accent.

6.3.2 Doric

In contrast to the situation of Lesbian, for which the available data can readily be reconciled with a single overarching principle, the Doric data are considerably more heterogeneous.¹¹⁵ A superficial comparison of Attic-Ionic forms with their attested Doric counterparts seems to frequently point towards the high tone occurring one syllable closer to the right edge of the word in Doric, or to show an acute where Attic has a circumflex in a penultimate syllable.¹¹⁶ I leave differences in the accentuation of monosyllabic forms (e.g., Attic-Ionic $\sigma\kappa\acute{\omega}\rho$

¹¹⁵The statements of the Alexandrian grammarians, which form the core of the Doric data, are collected in Ahrens 1843: 26–35 (reproduced in Strockis 2011, where a largely unhelpful phonological interpretation is offered). Some papyri of works of the lyric poets Alcman (Sparta; 7th c. BCE), Stesichoros (Sicily; 7–6 c. BCE) and Ibycus (Samos; 6 c. BCE) contain a number of “Doric” accents, and a smaller number of words with presumably “Doric” accents are attested in papyri of Pindar. Finally, the manuscript transmission of the *Bucolica* of Theocritus contains a handful of seemingly “Doric” accents as well. The data from the primary sources are helpfully collected in Molinos Tejada 1990 (Theocritus), Teijero 1993 (Stesichoros, Ibycus, Alcman), and Hinge 2006 (thorough treatment of the material from Alcman).

¹¹⁶The reader should also bear in mind that the accentual material from the “Doric” authors mentioned in the preceding note should not be expected to present a unified portrait of Doric prosody, because both the lyric poets and Theocritus employed a “mixed” dialect, combining Doric features with Attic-Ionic features; see generally Cassio 1993, as well as Cassio 2007 (on Alcman), Tribulato 2016 (on the lyric poets), and Vessella 2016 (on Theocritus). Given that Molinos Tejada (1990: 1–19) has shown that the Theocritus-papyri transmit distinct

‘dung’ versus Doric σχώρ) out of consideration, on the assumption that these differences are due to lexical properties (i.e., whether the stem is marked with H*) that lack any systematic synchronic motivation.

(118) Contrast: Attic-Ionic versus Doric Accentuation

Attic	Doric	Gloss
ἄγγελοι	ἀγγέλλοι	‘messenger:NOM.PL’
ἔλεγον	ἐλέγον	‘collect:IMPF.3PL’
παῖδες	παίδες	‘children:NOM.PL’
ἄνθρωποι	ἀνθρώποι (n.b., not *ἄνθρώποι)	‘man:NOM.PL’

If one considers just the verb Att.-Ion. ἔλεγον in contrast with Dor. ἐλέγον, for which recessive accentuation may be assumed, at least two different distinct analyses suggest themselves:

- The default word intonation of Doric was H* assigned to the stressed syllable, but feet were otherwise constructed just as in Attic-Ionic, thus [e.(‘lé.go)⟨n⟩].
- Word-final consonants (including the “light” diphthongs, like NOM.PL /-oi/) are not extrametrical, but the default word intonation was H+L*, as in Attic-Ionic, thus parsed as [e.lé.(‘gòn)] (in contrast to Attic-Ionic [é.(‘lè.go)⟨n⟩])

Both hypotheses are compatible with much of the Doric data; both could correctly generate the Doric accent in (118). Particularly well-attested are words of the morphological categories listed in (119), which often exhibit “Doric” accentuation the papyri and manuscripts, in agreement with the statements from grammarians (cf. Molinos Tejada 1990: 1–19, Teijero 1993: 156–62, and Probert 2003: 160–2).¹¹⁷

(119) Regular differences in accentuation in Doric, by morphological category

- 3.PL aorist and imperfect forms ending in -ον, -εν, or -αν have an acute penult in Doric where Attic has an acute antepenult, or an acute penult where Attic has a circumflex penult.

Doric	Attic	Gloss
ηλύθον	ἦλυθον	‘obtain’ 3.PL.IMPF
ἦνθον	ῆλθον	‘obtain’ 3.PL.AOR
εδώκαν	ἔδωκαν	‘give’ 3.PL.AOR

accents for words that are otherwise morphologically and phonologically identical, this indicates that some the papyri were not consistently accented according to Doric rules – some words transmit Doric prosody, but others transmit Attic-Ionic prosody instead. Therefore, only attested word forms that attest accentuation differing from their Attic-Ionic counterparts should be brought to account; words accented just like their Attic-Ionic counterparts, if identical in syllable count, distribution of syllable weight, and morphology to forms attested with distinct “Doric” accentuation can be assumed to have been transmitted with an Attic-Ionic accent.

¹¹⁷Doric forms are printed here as the papyri and manuscripts.

- b. All morphological categories that end in an inflectional suffix -οι or -αι (e.g., inflectional endings of non-past middle forms, NOM.PL forms of the 1st and 2nd declensions)

Doric	Attic	Gloss
δραμέται	δραμεῖται ^a	‘run:FUT.MID.3SG’
α]σέυμαι	άσεύμαι	‘sing:FUT.MID.1SG’
γεραϊτάτοι	γεραίτατοι	‘oldest:NOM.PL.MASC.’
αρίστοι	ἄριστοι	‘best:NOM.PL.M’
δυστάνοι	δύστανοι	‘unlucky:NOM.PL.M’

^aNote that the surface circumflex in the penult of such Attic future forms is due to a vowel contraction, i.e., δραμέεται > δραμεῖται.

- c. The NOM.PL. of the 3rd declension in -ες.

Doric	Attic	Gloss
παῖδες	παῖδες	‘children’
γυναί[κες	γυναικες	‘women’
πονέυντε[ς	πόνευντες	‘hard-working:PTCP.PRS.NOM.PL’

- d. The ACC.PL. of the 3rd and 1st declensions in -ᾶς (in the 1st declension never -ᾱς, in contrast to Attic).

Doric	Attic	Gloss
χείρας	χείρας	‘hands’
παῖδας	παῖδας	‘children’
Μοίρας	Μοῖρᾶς	‘Fates’

- e. Infinitives in -αι, -εν, -αι, -μεν, and -σθαι (cf. Hinge 2006: 122-7 on the Doric, Gunkel 2014: 11 on the “preaccenting” behavior of these suffixes)

Doric	Pseudo-Attic ^a	Gloss
αμύναι	ἀμύναι	‘defend’ AOR.INF.ACT
ακούσαι	ακούσαι	‘hear’ AOR.INF.ACT
ἦμεν	ἦμεν	‘be’ PRS.INF.ACT.
φάιμεν	φαίμεν	‘appear’ PRS.INF.ACT
μῶμέσθαι	μωμέσθαι	‘criticize’ PRS.INF.MED

^aThe Attic forms given here, when not actually attested, are constructed for the purposes of accentual comparison.

One interpretation offered to account for much of these data (e.g., Probert 2006b: 162) is that the Lex σωτήρα was not active in Doric, which would account for cases in which Attic shows a circumflex on a penultimate syllable where Doric shows an acute. However, to simply declare the Lex σωτήρα invalid for Doric does not explain precisely why it was invalid. Did Doric simply admit of H(*) without a succeeding fall to L? Furthermore, non-application of the LEX σωτήρα does not immediately explain the movement of the high tone one syllable to the right in forms like πονέυντε[ς, αρίστοι, or εδῶκαν. In the latter two cases, the shift

of position has been attributed to consistent “heaviness” of diphthongs, on the one hand, and an underlying suffix /-V-nt/ in 3.PL forms, where word-final /t/ is deleted after accent assignment (cf. Probert 2003: 161). πονεύντε[ς ‘hard-working’ would still require some further justification.

Various other forms attested in Alcman that have a word-final [-s] or [-n], beyond the morphological categories listed above, and which exhibit the same distribution of H: acute instead of circumflex on a penult, and acute on the penult instead of the antepenult. Also remarkable here is that forms that could and should have H on the antepenult in Attic-Ionic form instead have H on the penult when the word ends in [-Ṽs/n]. The verb συρίσδες is instructive in this regard: a form accented by Attic-Ionic rules would appear as ^xσύρισδες (note that the corresponding συρίζει has acute on the penult due to the heavy final syllable).

(120) Further Doric Forms in [-s] or [-n] from Alcman

Doric	Attic	Gloss
λιμνάτις	λιμνήτις	‘swamp-dwelling’
μακαρίτις	μακαρίτης	‘recently dead (adj.)’
Φιλίνον	Φιλίνον	Personal Name
ώμες	ῶμεν	‘be:PRES.SUBJ.1SG’
στάθος	στήθος	‘breast’
θήλυς	θήλυς	‘woman’
άινος	αἶνος	‘story’
συρίσδες	συρίζει	PRES.3SG ‘whistle’

Interestingly while Doric forms ending in [-ṼC] regularly show non-application of the Lex σωτήρα, the rule seems generally to hold when a word ends in simple [-Ṽ]. The only attested exceptions to this generalization are listed in 121.

(121) Exceptions to application of the Lex σωτήρα in Ṽ-final words in Doric

- a. άιγα [sic] ‘goat:ACC.SG (1×, Alcman-papyrus)
- b. παιδα ‘child:ACC.SG (2×, Alcman; 1× Ibycus; 2× Theocritus)¹¹⁸

Although in principle άιγα [sic] and παιδα are the Doric *lectiones difficiliores* that should be brought into agreement with a general explanation, the rarity of Lex σωτήρα violations in words ending in [-Ṽ], despite the frequency of words with the relevant prosodic shape, might be taken to mean that the Lex σωτήρα did operate as in Attic in words ending in simple [-Ṽ], and that άιγα and παιδα constitute false hyperdoricisms. That is, when providing papyri or manuscripts with accents, some scribes or copyists familiar with certain Doric patterns of accentuation falsely generalized that that a circumflex penult in Attic should regularly show an acute penult in Doric.

Finally, generalizations from the few [-ṼCC] forms attested in Doric authors are difficult to make. Clear failures to observe the Lex σωτήρα in such words occur only in Pindar.

¹¹⁸Theocritus has παιδα (as in Attic) much more often, while Ibycus also has 1× παιδα.

(122) [-VCC] in Doric authors observes the LEX σωτήρα (?)

- a. Theocritus: ὦναξ ‘master’ (1×), Ἰππῶναξ ‘Hipponax’ (2×) (never ^xῶναξ, ^xἸππῶναξ)
- b. Pindar: κᾶρυξ ‘herald’ (1×),¹¹⁹ ὦναξ (2×) vs. κᾶρυξ (3×), Φοῖνιξ (1×)¹²⁰
- c. Bacchylides: κᾶρυξ (1×)

Unfortunately, no relevant forms ending in [-VCC] seem to be preserved in the Doric lyric poets. Although ὦναξ, Ἰππῶναξ, and κᾶρυξ can equally be Attic forms (with trivial “doricization” of η to α̃ in the latter), it should be mentioned here that the κᾶρυξ and Φοῖνιξ seem to reflect editorial choices, whereas κᾶρυξ is attested in unedited fragments. κᾶρυξ and Φοῖνιξ might reflect an editorial attempt to further “doricize” the forms, under the assumption that the Lex σωτήρα generally does not hold in Doric.¹²¹ Alternatively, their acute might be due to analogical reintroduction of long [u:] and [i:] into the nominative singular from the other case forms (cf. Attic GEN.SG κήρυκος, φοῖνικος), with failure to apply high-vowel shortening.

Since the evidence from [-VCC] forms is equivocal, we are left with general non-application of the Lex σωτήρα in words ending in [-V{s, n}], but its largely regular application in words simply terminating in [-V̌]. Assuming that, for Doric words ending in [-V̌], the justification for the Lex σωτήρα is the same as in Attic-Ionic developed above (in words like παῖδα, association of H+L* to the stressed syllable; in words like πατήρα, the creation of a falling tone to satisfy the constraint *NOFALL_ω defined in 65), the non-observance of the Lex σωτήρα in words ending in [-V̌{s, n}] potentially be explained through the following further assumptions:

- word-final consonants are never extrametrical: all [-VC_i] syllables, including all final diphthongs that count as light in Attic, instead make a heavy syllable
- MAIN → VV plays no role: Doric applies the weight scale V < {VC_i, VV} (in contrast to the V < VC < VV weight scale of Attic-Ionic)

Lack of final extrametricality can thus simultaneously account for lack of the LEX σωτήρα in [-V̌{s, n}] forms and movement of the high tone one syllable to the right in other forms. Forms such as παῖδες or ἐδώκαν would assign primary stress to the final syllable on account of RIGHTMOST, associate L* to the final syllable, and H to the penult; the latter would have the foot structure [e.(dó:).('kàn)], presuming L* linked to the final foot, just as in Attic-Ionic. Indeed, the forms κᾶρυξ and Φοῖνιξ discussed just above are correctly predicted under this analysis as well: the former is parsed as [ká:.('rùks)]. To summarize: words whose final syllable simply contains a short vowel should behave accentually in Doric just as in Attic; words with a final consonant should always have primary stress on the final syllable, and thus regularly

¹¹⁹This form occurs only in a fragment of an otherwise unknown poetic work of Pindar. See Maehler and Snell 1975 (part 2, fragment 140a, line 68).

¹²⁰Note, however, that in the lexemes κήρυξ and φοῖνιξ, a certain amount of confusion seems to have existed concerned the vowel length of the final syllable, leading to NOM.SG forms with an acute on the first syllable.

¹²¹Which is not to say that the editorial choices in the Pindaric odes – based on Byzantine manuscripts – are false, but merely that the transmission of κᾶρυξ is less direct than the transmission of κᾶρυξ.

show H on the penult (when recessively accented). The interpretation of words with three or more syllables that have a VV-penult with circumflex in Attic (e.g., λιμνήτις or μακαρίτης just above) is less clear. Since the Attic forms must be marked with lexical H*, if the Doric forms are so marked too (e.g. as /makar-íti-s/), then the foot structure and tonal assignment must be interpreted as, e.g., [ma.ka.(‘rí*).(tís)], where stress on the final syllable requires insertion of L in order to satisfy $\sigma \leftarrow T$ and *NOFALL_ω. Alternatively, every Doric form considered thus far can be treated as properly recessive, with no lexical H* whatsoever. While Doric must have still possessed forms marked with H* – sometimes differently than Attic (e.g., φρατήρ ‘clansman’ vs. Attic φράτηρ) – lexical restructuring might have reduced the role for H* in the language.

Further evidence from the domain of word minimality also speaks in favor of word-final consonants as weight-bearing in Doric. First, two separate Doric sources point to a with short [o] in the NOM.SG of ‘foot’: fragment 59 from the lyric poets (Page 1962) contains the form πός, while Hesychius (3008) transmits the Laconian NOM.SG as πόρ.^{122, 123} Recall that Attic πούς [po:s] is due to lengthening of underlying /o/ to satisfy a bimoraic minimal word requirement; assuming that Doric possesses such a requirement, too, the absence of a long vowel in Doric πός / πόρ suggests that the final consonant was weight-bearing. Similarly, at Pindar *Olympia* 2.85, the form πάν ‘all’ with metrically secure short [a] (πάν precedes a vowel-initial word) occurs, suggesting that the vowel was not lengthened, in contrast to Attic πᾶν.¹²⁴

To interpret the accentual testimony of Doric as evidence for lack of final-consonant extrametricality in this dialect thus goes a long way towards explaining the accentual divergences between Attic-Ionic and Doric. In particular, it gives a unified account of why word forms ending in [-Ńn], [-Ńs], or a diphthong force the high tone one syllable closer to the right edge of the word (from antepenult to penult), and seem fail to apply the Lex σωτήρα. This account does not, however, explain the entirety of attested accentual differences, in particular in the monosyllabic forms Dor. γλαύξ ‘owl’, παῖς ‘child’, and σκῶρ ‘dung’, which show an acute where Attic attests a circumflex.¹²⁵

Of further potential difficulty for the interpretation offered here is the claim of the grammarian Apollonius Dyscolus, that the 1SG of an imperfect or aorist in Doric was accented as in Attic (Molinos Tejada 1990: 7). That is, the corresponding IMPF.1SG to the Dor. 3PL ἐλέγον above would have been ἔλεγον, as in Attic. If indeed true, then lack of final consonant extrametricality from the prosodic grammar of Doric is alone not sufficient – or would even seem

¹²²Hesychius line 4515 reads πῶς· πός. ὑπὸ Δωριέων, but this makes no sense, since the Attic nominative is of course πούς; this line is therefore often amended (cf. Liddell et al. 1925–1940: s.v. πούς) to read πός· πούς. ὑπὸ Δωριέων ‘πός· ‘foot’ among the Dorians’, which would be sensible and accord with the rest of the evidence.

¹²³The NOM.SG βῶς ‘cow’, with long [ɔ:], is attested in both Alcman and Theocritus. In contrast to Attic-Ionic βοῦς, which shows lengthening of /bo-s/ to [bo:s] in order to satisfy a bimoraic minimal word requirement, the Doric βῶς may well be an archaism, directly continuing PIE *[gʷōs].

¹²⁴The line *Olympia* 2.85 reads: ἐς δὲ τὸ πᾶν ἐρμάνεων. On the text and scansion, see Maehler and Snell 1975: 6–11.

¹²⁵If the account of Olander (2007) is on the right track regarding the historical origins of the distinction between acute and circumflex in monosyllables, and that the creation of such a distinction post-dates a loss of final consonant extrametricality in Doric, then all monosyllabic forms with at least one final consonant would be predicted to show an acute.

to make incorrect predictions for 1SG imperfects and aorists, at least. Given the comparative infrequency of 1SG forms as compared to 3PL forms, and the issue of persistent Atticisms in texts in “Doric” (cf. the footnote at the beginning of this section) there is no particular reason to take this doctrine of Apollonius Dyscolus too seriously (though it may have then impacted the textual transmission). In light of forms like *συρίσδες* (under 120) and *πονεύντε[ς]*, the position of the high tone on the penult is difficult to explain unless the word-final [s] is weight-bearing. Despite uncertainties, the single major parametric difference that most easily captures Attic-Ionic vs. Doric differences must final consonant extrametricality, or, constraint-based terms, the relative rankings of NONFINALITY- $\mathcal{F}$ and PARSE-C.

Concretely, the number of different surface patterns of H placement in Doric is then fewer in comparison to Attic-Ionic. In words with no lexical H*, if the final syllable is heavy (i.e., contains a VV- or VC-rhyme of any sort), primary stress will fall on the final syllable and H will occur by default on the penult. If the final syllable is light (i.e., contains a V-rhyme), primary stress will fall on the penult, and H will occur by default on the antepenult. Lexical H* may still result in H and primary stress on a final syllable, and presumably could on a penult as well. However, because a final VC-rhyme will always count as heavy, words potentially having lexical H* on a penult are predicted to behave like words with a final VV-rhyme for purposes of the Lex *σωτήρα*.¹²⁶

6.3.3 The Prosodic Diversification of Greek

On their faces, both the Lesbian and Doric prosodic systems might be characterized as simplifications vis-à-vis the Attic-Ionic system. In Lesbian, the distinction between words with lexical H* and lexically unmarked forms has been abandoned entirely; hence, prosodic faithfulness constraints like HEADFAITH-H* play practically no role in the grammar. In Doric, conversely, the grammar of prosodic markedness is relatively less complex: all final [VC₁] syllables count as heavy, and the scale for syllable weight applied is likewise less articulated: $V < \{VC, VV\}$, as opposed to the finer-grained $V < VC < VV$ of Attic-Ionic. Since both the Lesbian and Doric word-prosodic systems result in neutralizations in the surface position of high tone where Attic-Ionic allows for distinctions, it follows that a prosodic grammar similar to that of Attic-Ionic should be reconstructed for the ancestor of these 1st Millennium BCE dialects.

Furthermore, the combined evidence of Attic-Ionic, Lesbian, and Doric, despite their differences, point towards some common reconstructable prosodic features that constitute innovations vis-à-vis the word-prosodic grammar of Proto-Indo-European. In particular, the change of edge orientation from the left edge (as in Vedic Sanskrit and Hittite) to the

¹²⁶One need not accept the particular conception of the Lex *σωτήρα* as a repair to the constraint *NOFALL_ω advanced here in order to come to the same conclusion. As long one expects for an acute on a VV-penult to surface as a circumflex when the final syllable is heavy, the Doric facts in forms like *μακάρτις* ‘recently dead’ will be correctly predicted. One does, however, to allow that the circumflex on the penult in Attic-Ionic forms like *κῆρυξ* or *καταίτις* ‘leather helmet’ (*Il.* 10.258) is *not* due to the Lex *σωτήρα* proper that causes alternations between lexical acute and a circumflex, but rather due to the association of all of the H+L* contour to the syllable with primary stress.

right edge, and the restriction of stress to the rightmost moraic trochee (module effects of MAIN $\rightarrow$ VV in Attic), as opposed to the unbounded stress of Vedic and Hittite, constitute striking developments. Although Probert (2012) has pointed out that the majority of word types in the inherited lexicon of Proto-Greek probably accidentally adhered to the Law of Limitation; in the paradigms of thematic verbs with a disyllabic stem (e.g., $\tau\rho\acute{\epsilon}\phi\omicron$ - ‘turn’), assumed to have persistent root stress (due to the preaccenting suffix /-’o-/), as in Vedic, the number of paradigmatic members that would not accidentally conform to the Law of Limitation is marginal (effectively limited to forms with a VV or disyllabic inflectional ending, e.g., 3DU in $\tau\eta\nu$, MID.3PL in $\mu\epsilon\theta\alpha$, IMPF.MID.ISG $\mu\eta\nu$; cf. Probert 2012: 168–71). A further test undertaken by Probert (2012: 172–3), based on reconstructing a plausible PIE accentuation for a forms in *Iliad* 13.1–9, finds that 42/46 (91.3%) orthotonic words there accidentally conform to the Law of Limitation.¹²⁷ However, why the evidence for the possibility of stress outside the domain of final two syllables ($\sim$ high tone outside of the rightmost three syllables) should or could have been simply ignored is less clear, especially given that only words of three or more syllables are potentially disambiguating whatsoever.

Considering the data on the token frequency of positions of the *udātta* in the *R̥gveda* given in Chapter 5, fully 89.3% of 1–5 syllable accented tokens ($\frac{115208}{128964}$) in the *R̥gveda* show high tone in a position that conforms to the Attic-Ionic Law of Limitation (i.e., up to the antepenult, if the final does not contain a VV-nucleus). This descriptive statistic is remarkably similar to the proportion of LoL-conforming tokens reconstructed by Probert in her small excerpt from the *Iliad*. However, the majority of 5-syllable tokens have stress on the initial or peninitial syllable ($\frac{1724}{2912} = 59.2\%$), and a healthy number of 4-syllable tokens ($\frac{5732}{13807}$) have high tone on the initial syllable or an antepenult with VV-final. Thus, among the longer forms that can provide more information about the grammar, evidence for high tone outside a restricted right edge window is not inconsiderable. Here, a closer consideration of exactly what types of word forms in Vedic exhibit high tone in a position that disobeys the Greek Law of Limitation requires investigation: do these mostly reflect morphological categories or formations that are absent or unproductive in Greek?

At the level of the grammar, the basic difference in the grammar between PIE as represented by Vedic Sanskrit in Chapter 4 and Proto-Greek can be characterized by a change in the relative rankings of ALL-FT-L and ALL-FT-R, and especially the great restriction resulting from the self-ganging of ALL-FT-R in Greek. In PIE, ALL-FT-L keeps stress close to the left edge of the word; in Proto-Greek ALL-FT-R $\times$ 2 ($\sim$ *LAPSE) categorically bans stress outside of a two-syllable window. Conversely, Vedic and Greek share the faithfulness to prosody of morphological heads as a central component of their grammars, one which overrides virtually all other preferences in terms of prominence placement. These core aspects of rankings are contrasted in (123).

(123) PIE to Proto-Greek Prominence Grammars

¹²⁷ Of the four exceptions to the LoL reconstructed by Probert (* $\text{A}\beta\iota\omega\nu$ ‘without a living, nomad:GEN.PL’, * $\delta\iota\kappa\alpha\iota\sigma\tau\alpha\tau\omega\nu$ ‘just:UPER.GEN.PL’, * $\alpha\nu\theta\rho\omega\pi\omega\nu$ ‘man:GEN.PL’, * $\alpha\theta\alpha\nu\alpha\tau\omega\nu$ ‘immortal:GEN.PL’), two (* $\text{A}\beta\iota\omega\nu$ and * $\alpha\nu\theta\rho\omega\pi\omega\nu$) are non-confirming only due to the inflectional suffix with a VV-nucleus. Thus, only two stems in Probert’s sample are completely non-LoL-conforming altogether.

- a. PIE: HEADFAITH $\gg$ MAX-Prominence $\gg$ ALL-Ft-L $\gg$ ALL-Ft-R, WSP
Lexical accents are generally preserved anywhere in the word; prominence towards the left edge is preferred.
- b. Proto-Greek: ALL-Ft-R $\times 2$ $\gg$ HEADFAITH $\gg$ MAX-Prom $\gg$ WSP $\gg$ ALL-Ft-R $\gg$ ALL-Ft-L
No stress outside the rightmost two syllables; lexical accents (H*) are allowed only within the two-syllable window.

One other crucial aspect of prosodic reconstruction that can be inferred from the testimony of Greek is the status of final consonant extrametricality. First, obtaining high levels of overlap between the position of H inherited from PIE and the Greek Law of Limitation requires that final consonant extrametricality be a feature of Proto-Greek prosody. Furthermore, the usual interpretation of “Wheeler’s Law” (see discussion under 3.3.1.2) requires that final [-VC] syllables be treated as light in Proto-Greek as well. In addition, independent of arguments in favor of final consonant extrametricality from indications of a bimoraic minimal word requirement in PIE itself (see Byrd 2015: 113–7), since the status of final consonant extrametricality cannot be readily determined for Vedic, no concrete evidence clearly speaks against projecting the Greek situation back to PIE. In turn, given that Doric has innovated with respect to Proto-Greek on this account, considerations of simplicity dictate that final consonant extrametricality should be treated as an inheritance into Greek, rather than an innovation of Proto-Greek.

6.4 Looking Forward: Paths to Modern Greek Prosody

Having now briefly looked backwards into the prosodic prehistory of Greek, a parallel glance into the later history of Greek prosody, beyond the 1st Millennium BCE is appropriate. Since much can be determined with relative precision about the prosody of Modern Greek, the combined knowledge of the prosodic systems of PIE, the 1st Millennium Greek dialects, and the present-day language makes possible a fairly detailed accounting of one possible direction in which a prosodic system may develop over the course of millennia. This section lays the groundwork for a more systematic description of the changes that ultimately give rise to the specific prosodic grammar and lexicon found in Modern Greek.

Already in the latter half of the 1st Millennium BCE, the dialect of Athens and the Koiné based largely upon this dialect, which was spoken throughout the Near East, underwent crucial segmental changes in that vowel system that would have major consequences for the prosodic grammar and lexicon: phonemic distinctions of vowel length were lost (cf. generally Horrocks 2010: 163–70). According to the analysis of the epigraphical and non-literary material from this period by Teodorsson (1974, 1977, 1978), the merger of short and long vowels may have occurred earliest in non-prestige varieties of Athens, perhaps already by the middle of the 4th c. BCE. On the other hand, more conservative varieties of Koiné were exported outside of Attica and Greece, where contrasts in vowel length may have persisted among Greek-speakers for another century. By c. 200 BCE, speakers of the Koiné and its local varieties seem to have universally merged short and long vowels in short vowels.

Before considering the implications for the greater prosodic system result from the loss of phonemic distinctions in vowel length, a brief introduction to the word prosody of Modern Greek is necessary. There exists general agreement that the “default” rule of stress assignment in Modern Greek is to place primary stress on the antepenultimate syllable (Malikouti-Drachman and Drachman 1999), which can be analyzed either as assignment to the rightmost syllabic trochee, given the extrametricality of final syllables, or alternatively, using the constraint typology in Gordon 2002, as a preference for the most prominent syllable to be aligned with the left edge of the word ($\text{ALIGN}(X_i, L)$), outranked by $\ast\text{EXTENDED LAPSE RIGHT}$, which penalizes a sequence of more than two stressless syllables at the right edge of the word.

Important to note is that, in contrast to Ancient Greek, Modern Greek has no quantity sensitivity: vowels do not contrast in length and coda consonants do not contribute to phonological syllable weight. The acoustic correlates of stress are comparable to Italian; see generally Revithiadou 1999: 85–6, with references. Revithiadou and Lengeris (2016), meanwhile, found that the perception of stress on *wug*-forms for Greek-speakers was dependent upon the apparent morphological category and lexical statistics: speakers’ expectations as to where stress is likely to occur appear to shape their perception of stress. In general, the proportion of tokens perceived with antepenultimate or penultimate stress corresponded to the proportion of types of that morphological category with that stress pattern. However, stress on the final syllable was perceived more rarely than lexical statistics would predict. One possible explanation for this behavior is that violations of NONFINALITY and FTBIN together make final stress comparatively disharmonic. Thus, when an underlying lexical accent ($\sim$ foot edge) is present, MAX-PROM ($\sim \text{ANCHOR-L/R}_f$) will still allow for final syllable stress, but, the default phonological constraints strongly disprefer it, just as in Italian (cf. 2.4.2.1) or Portuguese (cf. Garcia 2017b).

Unlike in the history of Latin and the Romance languages, where the loss of a phonemic distinction between long and short vowels partly only partly opacified the conditions for stress calculation in what was an otherwise transparent metrical stress system, the quantity sensitivity of the Attic-Ionic system created a greater number of analytical problems. Under the analysis advanced in Section 6.2.3, high tone and stress do not co-occur in a large proportion of the lexicon (i.e., in most recessively accented words). Abandoning quantity sensitivity thus potentially enables the reanalysis of recessively accented words with a light final syllable (e.g., $\acute{\alpha}\nu\theta\rho\omega\pi\omicron\varsigma$ or $\rho\omicron\mu\pi\acute{o}\lambda\upsilon\kappa\iota$) in such a way that high tone and primary stress would more straightforwardly align, if parsed as syllabic trochees with final syllable extrametricality, thus [$\acute{\text{á}}\text{n.t}^{\text{h}}\text{r}\acute{\text{o}}\text{:}$].(pos)].

Contrastive vowel length, of course, need not require that a stress assignment algorithm be quantity-sensitive (compare Vedic and Estonian), but it can provide internal analytical support for the acquisition of such a contrast. Conceivably, bias towards the relatively high ranking of the constraints preferring the association of high tone and foot heads ($\ast\text{HD/L}$, $\ast\text{NON-HD/H}$; see 2.4.1.2) could have promoted the abandonment of a quantity-sensitive stress system, including phonemic vowel length. Again, the Romance languages offer a point of contrast: while vowel length is not contrastive in Italian, Spanish, or Portuguese, coda consonants do contribute to syllable weight (as in Latin), and heavy penultimate syllables over-

whelmingly receive primary stress (as the Classical Latin stress rule would demand). If high tone and stress were always consistently aligned in Ancient Greek, explaining what motivated later Greek to abandon quantity-sensitivity altogether, rather than maintain vestiges thereof (as in Romance), becomes more challenging.

Thus, in broad terms any words of shapes like ἄνθρωπος [ántʰrɔ:pos] or πομπόλυκι [pom-póluki], with high pitch on the antepenult, lend themselves naturally towards reanalysis as bearing stress on the antepenult. Recessive disyllables, whether circumflex (κῆρυξ) or acute (φίλος) can likewise be parsed in a way compatible with an innovative antepenultimate stress grammar. However, longer words with final heavy syllables – whether closed by two consonants, like πολυπίδαξ [polupî:daks], or containing a long vowel, like ἀνθρώπου [antʰrɔ:po:] – do not fall out correctly. Precisely these cases are susceptible to synchronic analysis in Modern Greek as containing lexical accents: a stem may be marked as stress-preferring, or an inflectional ending as preaccenting, in order to yield exceptional penultimate stress.

For instance, the Attic-Ionic forms NOM.SG [ántʰrɔ:pos] and GEN.SG [antʰrɔ:po:] develop in MG into ['anθropos] and [an'θropu]: the position of the prominence remains unchanged, but because the stress assignment algorithm of Modern Greek would predict GEN.SG ^X['anθropu], with stress on the first syllable, the penultimate stress must be accounted for through a new lexical feature (in this case, a pre-accenting inflectional suffix /-u)/ for the GEN.SG). Compare analyses of this pattern in Revithiadou 1999: Ch. 3 and Apoussidou 2007: Ch. 6.

(124) 'man': Ancient vs. Modern Greek Nominal Stress Assignment

	Ancient		Modern	
	SG	PL	SG	PL
NOM	[ántʰrɔ:pos]	[ántʰrɔ:poi]	['anθrop-os]	['anθrop-i]
GEN	[antʰrɔ:po:]	[antʰrɔ:po:n]	[an'θrop-u]	[an'θrop-on]
ACC	[ántʰrɔ:pon]	[antʰrɔ:po:s]	['anθrop-o]	[an'θrop-us]
DAT	[antʰrɔ:po:i]	[antʰrɔ:pois]	–	–
VOC	[ántʰrɔ:pe]	[ántʰrɔ:poi]	['anθrop-e]	['anθrop-i]

Remarkable here is that the number of syllables is diachronically constant (each case form shows three syllables), and so is the position of the Ancient Greek the high tone relative to the Modern Greek stress. In Ancient Greek, the position of the high tone in this noun was certainly phonologically predictable, whatever the correct analysis may be; in Modern Greek, it is not. Instead, in Modern Greek, the stress differences between case forms may be attributed to differing lexical features for prosody: the forms with antepenultimate stress have an unaccented case ending (e.g., NOM.PL. /-i/), while the case forms with penultimate stress have pre-accenting case endings (e.g., GEN.PL. /-on/). Consequently, the role for lexical marking in prosodic entries of Ancient and Modern Greek is substantially different:

- In Ancient Greek, lexical prosodic marking (/H*/) is largely limited to derivational morphemes, monosyllabic nominal roots, and some nominal inflectional suffixes (perhaps essentially restricted to endings for monosyllabic stems).

- In Modern Greek, lexical stress is much more pervasive: polysyllabic stems may mark their final or penultimate syllables as stress preferring, derivational affixes may attract or reject stress, and inflectional suffixes of all inflectional classes all can bear contrastive prosodic information.

In this regard, compare the underlying representations of the words for ‘sea’ and ‘man’ in Modern Greek in (125) below. Besides unaccented stems like /θalas-/ and /anθrop-/, however, the language possesses accented stems, in which the primary stress may be associated with any vowel (e.g., *a*-stems /γondol-/ ‘gondola’, /sta(fið-/ ‘raisin’; *o*-stems /klivan-/ ‘kiln’, /fan(tar-/ ‘soldier’); in addition, some *o*-stems with original H* on the final vowel may be treated be post-accenting, forcing the primary stress away from the stem (e.g., /uran-/ ‘sky’). These more extensive, lexically-driven paradigmatic alternations in the position of stress in Modern Greek means that MAX-PROM (or similar) plays a greater role in in the phonological grammar of Modern Greek than in that of Ancient Greek, where HEADFAITH-H* comes into play principally in morphologically complex stems.

(125) Underlying Representations: Modern Greek *o*- and *a*-Stems

	-a		-os	
	Sg.	Pl.	Sg.	Pl.
NOM	/θalas-a/	/θalas-es/	/anθrop-os/	/anθrop-i/
GEN	/θalas-as/	/θalas-(on/	/anθrop-u/	/anθrop-on)/
ACC	/θalas-a/	/θalas-es/	/anθrop-o/	/anθrop-us/
VOC	/θalas-a/	/θalas-es/	/anθrop-e/	/anθrop-i/

In the lexicon, because the high tone of suffixes in Ancient Greek fell within the three-syllable window, the prosodic properties of derivational suffixes are generally maintained as such. For instance, unaccented derivational suffixes (e.g., abstract-forming /-si-/) are continued as unaccented, while derivational suffixes originally marked with H* (e.g., diminutive /-íko-/) and inflectional suffixes (e.g., GEN.PL of *a*-stems) are continued as accented. However, new accented and unaccentable stems have emerged – but these result largely from morphological reanalysis or loss of morphological structure, and loanwords. For instance, old oxytone *o*-stems (e.g., οὐρανός [uranós] from /uran(-os/) become unaccentable stems as a result of parsing the derivational suffix /-ó-/ as an inflectional ending. In the inflectional morphology of *o*-stems, new preaccenting inflectional suffixes emerged from historically heavy, unaccented inflectional suffixes that regularly conditioned high tone on the penultimate syllable, e.g., GEN.SG /-o:/ > /-u/ (cf. [an'θropu] above).

Although it seems remarkable, at first glance, that Modern Greek conserves the position of the Ancient Greek high tone as stress extremely faithfully, this outcome should perhaps not be terribly surprising. Instead, this development again appears to instantiate the principle “same output, different grammar (and lexicon)”. Indeed, this preservation of overt indications of prominence on the surface in the same position, through whatever means, is in fact what the straightforward interpretation of phonotactic learning with respect to metrical and lexical stress would predict; see the discussion under 3.3. PERTINACITY, in the sense of Lahiri 2015, comes through in the grammar in terms of the same foot shape (the trochee) and

the same alignment of feet and primary stress towards the right edge of the prosodic word. On the other hand, the loss of quantity distinctions means that the type of foot is simply a syllabic trochee, and that the final consonant extrametricality of Ancient Greek has been replaced by final syllable extrametricality. Note that moraic trochees and final consonant extrametricality are only possible if a distinction between heavy vs. light syllables is part of the phonology. In effect, the grammar has been minimally restructured so as to generate default antepenultimate stress wherever possible.

In short: the grammar and lexicon have both undergone relatively minimal changes in a way that essentially maintains the “same output”, once the syllabic position of the Ancient Greek high tone is taken to always indicate the position of primary stress. Such developments indicate that the cost of lexical restructuring is perhaps comparatively low, perhaps especially when other components of the grammar already required lexical marking of prosodic properties. From the perspective of the Greek learner in the latter centuries BC or early centuries AD, acquiring lexically specified prosodic information would necessarily have been part of the process, of constructing a grammar of word prosody, so to acquire lexical information that differs diachronically in such a way as to maintain the similar surface outputs might constitute the most straightforward type of reanalysis.

6.5 Prospects for the Continuing Study of (Ancient) Greek Prosody

This chapter has sought to provide a comprehensive overview and analysis of the word-prosodic system of Ancient Attic-Ionic Greek, and placed this well-studied system in relief against the word-prosodic systems of other Greek dialects of the 1st Millennium BCE (Lesbian and Doric) as well as the prosody of Modern Greek. 6.2 presented an overview of the apparent facts of Attic-Ionic and offered an extensive overview of some major attempts at the formal analysis of its word prosody, before constructing a novel analysis aiming for greater comprehensiveness; 6.3 dealt with differences in Lesbian and Doric in comparison to Attic-Ionic, while 6.4 outlined major changes in prosody between Ancient and Modern Greek. Here, I summarize the major claims advanced concerning the prosody of Attic-Ionic and its sister dialects, before discussing some potential directions for future study.

I advance the argument that Ancient Attic-Ionic prosody, and the similar prosody of Doric, is in many respects typologically comparable to Stockholm Swedish, characterized by obligatory and culminative word-level metrical structure that determines the position of stress, a default word-level intonational pitch accent (“prominence tone”), and a privative contrast between a single lexical tone, /H*/ and $\emptyset$, though a limited contrast between /H*/ and /H*L/ may have developed. Whether a falling tone /H*L/ must be attributed to any lexical representations depends on how successfully surface [H*L] sequences can be derived from post-lexical vowel contractions. The surface distinction on high-toned syllables between “acute” and “circumflex” is, on the basis of musical evidence, argued to constitute a distinction between a level High and a Falling (HL) pitch contour. In order to facilitate mappings between tone and stress, the syllable, rather than the mora, is treated as the tone-bearing unit.

The lexical tone, /H*/, is contrastive on monosyllabic stems, common on derivational suffixes, and is borne by some inflectional endings (in particular GEN and DAT endings of consonant-stems) as well. Special faithfulness to this lexical tone, formally implemented with the constraint HEADFAITH-H*, captures the tendency for derivational suffixes to determine the position of word-level High tone. Lexical /H*/ competes for realization with a default pitch accent inserted with the construction of each prosodic word, H+L*; the default pitch accent results in level High tone on the syllable preceding the stressed syllable, but Falling tone when associated entirely to the stressed syllable. The prosodic word is thus characterized by an near-obligatory fall in pitch, which may result in transformation of /H*/ into a Falling pitch, H*L, or the insertion of a Low tone following a lexical H*. The “grave” tone, for its part, is interpreted as the lowering of lexical /H*/ to L*, not distinct in realization from syllables not associated with any tone; the availability of default H+L* in every prosodic word requires that grave be understood as lowering of /H*/, rather than deletion, since deletion of an underlying association would make space for the association of H+L* available.

Like other metrical analyses of Ancient Greek prosody (Steriade 1988, Sauzet 1989, Golston 1990, Steriade 2014), I hold that stress is assigned at a window at the right edge of the word. The specific analysis advanced here builds mainly on that of Golston (1990), according to which primary stress is assigned to the rightmost moraic trochee. The stress window is limited to the final two syllables, a consequence of self-ganging of the constraint ALL-FT-RIGHT, with the effect that no foot structure may be built to the left of the penultimate syllable. The two essential differences from Golston’s analysis concern words of the type πολυπιδᾶξ (VV penult following by VCC final) and φίλος (V penult followed by VC final): the former receive primary stress on the penult rather than the final due to the relatively greater heaviness of VV syllables (MAIN → VV); the latter build degenerate feet on the final syllable to allow for the default H+L* pitch accent to be fully realized, without floating or crowding of tones.

Rightward shifts of High tone, both across derivationally related forms and in the paradigms of monosyllabic nominal stems are due to the presence of lexical /H*/ on derivational or inflectional suffixes. A relatively strict association between tone and stress, specifically that every stressed syllable be associated with tone ($\sigma \leftarrow T$) and that every starred tone (pitch accent) on the surface be associated with primary stress and vice-versa ($T^* \Leftrightarrow \sigma$), alongside the need to preserve underlying High tones (MAX-H) conspires to ensure that lexical /H*/, when present, and primary stress regularly align with the same syllable. Differences between polysyllabic and monosyllabic nominal stems in terms of High tone mobility suggest the availability of prosodically distinct allomorphs for endings, e.g. unaccented DAT.SG /-i/ versus accented /-í/.

While the analysis of stress and tone association in minimal prosodic words can only be improved around the margins, or reanalyzed from different theoretical perspectives, two areas call for more empirical and philological investigation. First, continuing analysis of the testimony from Ancient Greek music, which has yielded new insights concerning the nature of the acute in this chapter, may help to better establish aspects of Ancient Greek intonational phonology. In particular, to have a better grasp on what boundary tones asso-

ciated with which prosodic domains are present would be a boon. Second, clarification of the likely accentuation pattern(s) of multiple enclitic sequences, applying the approach of Probert (2016) to more data, might yield fruit as well.

Beyond Attic-Ionic, I have made two larger claims about the prosody of Lesbian and Doric. Concerning Lesbian, we confront a problem similar in nature to the development of categorical stress grammars for Classical Sanskrit and Proto-Germanic: the wholesale loss of lexically specified prosody. In the particular case of Lesbian, however, the loss of lexical specific prosody was not accompanied by the emergence of any new stress grammar; the same default stress role as in Attic-Ionic holds for Lesbian as well. On this point, the employ of more sophisticated simulations might be able to capture why Lesbian could have lost lexical accents, whereas Modern Greek reanalyzed the Ancient high tone and developed novel lexical accents. With regard to Doric, I have argued that its superficial differences from Attic-Ionic prosody are attributable to a difference in just one parameter: final consonant extrametricality, which applies in Attic-Ionic, but not in Doric.

One particularly ripe domain for detailed further study is the transition between the Ancient and Modern Greek word-prosodic systems. On the one hand, a more detailed and systematic investigation of correspondences between the Ancient Greek high tone and Modern Greek stress in stems, derivational affixes, and inflectional affixes is needed to substantiate the claim that high tone is faithfully continued as stress, and to more precisely determine what lexical restructurings have occurred. On the other hand, a clearer picture of Greek prosody after the Classical period but prior to the Modern period is needed. Did the Ancient Greek prosodic system more or less directly evolve into the Modern system, with only comparatively minor changes having occurred during the past 2000 years, or have other significant changes occurred over that period? Unfortunately, the nature of the available data from the Roman and Byzantine periods, with its tendency towards employing the Classical Attic as a model, renders unobscured investigation all the more difficult. Horrocks (2010), for instance, has virtually nothing to say about word-prosody in the long period intervening between Ancient and Modern Greek.

In sum, the work carried out in this chapter has sought to substantiate, motivate, and formalize many existing claims concerning the prosody of Ancient Greek. While the larger project remains incomplete, and details of the analyses proposed here can doubtless be improved, I hope to provide a point of reference for subsequent investigations along similar lines.

CHAPTER 7

Germanic: From Lexical Accent System to Fixed-Stress System (Again)

7.1 Introduction: A Simple Hard Problem

The central focus of this chapter lies with the development of the Proto-Germanic (PGmc.) word-level prosodic system out of the word-level prosodic system reconstructable for Proto-Indo-European (PIE). To merely describe this development, in comparing the prosodic phonological grammars of PGmc. and PIE, is seems simple: Proto-Germanic generally assigned stress to the leftmost syllable of a minimal prosodic word (cf. Prokosch 1939: 118–20, Ringe 2006: 152), whereas the system of word-level prominence assigned in PIE depended upon the interplay of lexical accents, morphological structure, and phonological preferences.¹ To briefly review: the standard view of Indo-European prosodic history is that agreement between Attic-Ionic Greek and Vedic Sanskrit high tones generally reflects the surface position of prosodic prominence in Proto-Indo-European; given the unrestrictedness of high tone in Vedic Sanskrit and the free occurrence of high tone in Attic-Ionic within the bounds set by the LAW OF LIMITATION (cf. (2) under 6.1), word-initial stress in Germanic is to be considered an innovation, in terms of the *surface* position for some lexemes, but more importantly in terms of the *grammar* of stress assignment. Compare the cognate forms in (1), where Vedic and Greek agree in showing the prosodic prominence on a non-initial syllable, the forms reconstructable for (late) Proto-Germanic instead exhibit stress on the initial syllable.

(1) Accent in Vedic and Greek vs. Germanic

Vedic Sanskrit	Attic-Ionic Greek/Hittite	(Late) Proto-Germanic	GLOSS
[pa:dós]	[podós]	*[ˈfo:tiz]	‘foot:GEN.SG’
[pitá:]	[paté:r]	*[ˈfaðe:r]	‘father:NOM.SG’

To account for how and why such a development took place in Proto-Germanic – but not in all Indo-European daughter languages – constitutes a much more difficult problem. Given that grammars must be induced by learners data, to posit that the brains of a new generation of Proto-Germanic speakers simply “flipped a switch” would be absurd; indeed, all things being equal, one would expect for the acquired phonological grammars to remain diachronically stable as long as the structure of the primary linguistic data remains constant

¹ For this view of PIE as a LEXICAL ACCENT SYSTEM, as partly continued in Greek and Vedic, as well as Proto-Balto-Slavic and Hittite, see Kiparsky and Halle 1977, Steriade 1988, Probert 2006b, Kiparsky 2010a, Sandell 2015a: Ch. 7, Yates 2016, and Yates 2017.

(cf. the discussion of prosodic “pertinacity” in Lahiri et al. 1999 and Fikkert et al. 2006). Thus, the truly substantive question at hand is fundamentally similar to the one posed in 5 with respect to the innovative metrical stress pattern of Classical Sanskrit vis-à-vis Vedic Sanskrit: how and why did the structure of the primary linguistic data change between PIE and PGmc., such that a far-reaching prosodic reanalysis could take place?

Clear-eyed – and probably correct – statements on how to characterize the prosodic changes between PIE and PGmc. can be found in some existing scholarship: both Hutton (1995: 7) and Halle (1997: 275; 310) see the emergence of left-edge stress in Proto-Germanic as an expected consequence of the loss of lexical accents inherited from the PIE system. Under the view that the PIE assigned primary prosodic prominence to the leftmost syllable in a phonological string in the absence of any lexical accents (i.e., the BAP of Kiparsky and Halle 1977; cf. (8) in Chapter 4), the “loss” of any and all underlying lexical accents from a word form ought to, by default, lead to the assignment of left-edge prominence any daughter languages in which such a principle is essentially preserved.² Cases of such behavior at the level of individual lexemes or morphological classes have been observed in Yates 2015a (inherited nasal-infix presents in Hittite), Lundquist 2015a (Vedic *-ti-* abstracts) Sandell 2016 (Vedic *śakti-* ‘power, ability’ > *śákti-*), and Probert 2006b (nouns in [-ro-], [-lo-], and [-to-] in Greek), though in the last case with “default” application of the Attic-Ionic recessive accent rather than left-edge prominence.

In a chapter specifically devoted to an account of the Germanic “stress shift”, Hutton (1995: 200–9) pursues two lines of possible explanation. One account (pp. 202–4) would be a “complete and simultaneous elimination of all lexical accents,” which essentially amounts to a sudden and “catastrophic” (in Hutton’s words) reanalysis of the prosodic data as most compatible with fixed left-edge stress; on how, why, and whether sufficient and necessary conditions for such a reanalysis were present in the Proto-Germanic learning data, Hutton does not elaborate. Another account (pp. 204–5) would be that the lexical accents of individual lexemes were progressively lost to the point that the grammar was most compatible with fixed left-edge stress and no lexical accents. Although Hutton identifies two classes of forms that may have promoted the loss of lexical accents (“morphemes with both [surface] accented and unaccented allomorphs” and “primary cyclic morphemes”, which “were less likely to retain their accents unless they were the only stem formant present or followed by non-cyclic morphemes”), but does not pursue any systematic analysis of the Germanic lexicon. While Hutton’s ideas indeed constitute possible and plausible bases for the defining prosodic change of Proto-Germanic, it remains to be convincingly demonstrated, either through close engagement with the data or through simulation, what probably occurred. Furthermore, given the role that frequency and productivity likely play in the acquisition of morphological structure, which in turn can impact prosody in a lexical accent system (cf.

²An immediate complicating factor is the apparent tendency for most morphologically complex (nominal) stems in Vedic Sanskrit and Attic-Ionic Greek to show stress at the right edge of the stem. Under the analyses presented at 4.3 and 6.2.3, this behavior results from lexical properties of derivational suffixes. Under the analysis of Kiparsky 2010a, on the other hand, a rule (the OXYTONE RULE of (20) in Chapter 4) supposedly creates this typical stem-edge prominence. If the OXYTONE RULE reflects a markedness preference, then it and the BAP are in conflict as to the assignment of a “default” primary stress.

Probert 2006b: *passim*, Sandell 2015a: 192–214), this factor should also be taken into account in a systemic reckoning of Proto-Germanic’s prosodic history.³

The fundamental question with which this chapter is ultimately concerned is thus: how and why did the prosodic grammar of Proto-Germanic develop so divergently from Vedic, Greek, and many other Indo-European languages, which better attest aspects of the PIE lexical accent system? Is the path taken taken by Germanic (and other languages) a statistical accident, or does it represent a virtually certain result of some chain of (language-internal or -external) developments?

The relevance of this question extends beyond Germanic, given that a variety of other Indo-European languages exhibit similar developments. Halle (1997: 298) correctly points out that a development from lexical accent system to fixed word-initial stress perhaps applies to Italic (cf. Meiser 1998: 53), Old Irish and other Celtic languages, some Slavic languages (Czech and Slovak in particular), and Latvian. Thus insofar as the questions treated in this chapter with respect to Germanic can be satisfactorily answered, the basis of an account for a substantial prosodic change that has affected a number of Indo-European daughters becomes available. In antiquity, other “western” IE branches often also reconstructed with consistent stress on the leftmost syllable of the prosodic word: cf. Meiser 1998: 53 on Proto-Italic and Thurneysen 1946: 27 on Celtic. Indeed, even Greek in the 1st Millennium BCE partly attests to the possibility of change from prosodic systems with lexical accent to purely algorithmic stress assignment: In all Ancient Greek dialects, lexical accents have been lost entirely from virtually all finite verbs, which are regularly subject to the rule of recessive accent. In Lesbian Greek, lexical accents have been lost and a fixed-stress pattern (following the same rule as the Attic-Ionic recessive accent) has developed (cf. Probert 2003: 159–60, Probert 2006b: 72–3, 6.3.1). Thus, conclusions reached about Germanic in particular may have implications for the prosodic prehistory of other Indo-European languages as well.

Before attempting to look into the development of the Proto-Germanic prosodic system in detail, the first objective must be to establish precisely how that system functioned: what word-level prosodic structures did Proto-Germanic construct, and how was primary stress (and any possibly any secondary stress) assigned? These questions are pursued in the immediately following section 7.2. With a more precise formalization of word-level prosodic structure and stress assignment for Proto-Germanic in hand, I consider in 7.3 whether any smaller, individual phonological or morphological changes may have occurred in Proto-Germanic that could have impacted the distribution of primary stress; discussion there will be centered around the testimony and interpretation of so-called “Verner’s Law Doublets”. Finally, in 7.4, I again employ iterated agent-based models to test the hypothesis that the prosody of Proto-Germanic was substantively affected by contact with (a) language(s) whose L1 speakers regularly produced initial left-edge stress.

³Since Proto-Germanic itself is, of course, not directly attested, token frequency data is necessarily unavailable, though corpus investigations of the oldest Germanic languages might allow for some estimation of relative token frequencies. Type frequencies, on the other hand, can reasonably be assessed on the basis of the reconstructable Proto-Germanic lexicon.

7.2 Stress Assignment in Proto-Germanic

Approaches to the assignment of word-level prosodic prominence (here, stress) in Proto-Germanic fall essentially into two camps:⁴ the view that stress was wholly phonologically calculated, systematically assigned to the leftmost syllable of a word, and the view that stress was morphologically determined, regularly assigned to the root syllable of a word.⁵ When stated in such an overly simplistic fashion, both views encounter a some measure of difficulty, though by and large, both views, even when formulated simplistically, correctly predict the position of Proto-Germanic stress. Distinguishing between these two views is, at first glance, not straightforward, since, in the vast majority of Proto-Germanic word-forms, the leftmost syllable and the root syllable of a word were coextensive with one another. Some such examples, which can easily be multiplied many times over, are given in (2). One can apply either of the two simplistic stress assignment algorithms stated in (2) to the URs given in (2) in order to generate the corresponding surface forms.

- (2) Two Simple Algorithms for Proto-Germanic Stress Assignment
 - a. **Phonological:** Identify the leftmost syllable of a prosodic word (ω) and assign primary stress to it.⁶
 - b. **Morphological:** Identify the root element $\sqrt{X}$ contained within a prosodic word (ω) and assign primary stress to it.
- (3) Coextensivity of Leftmost and Root Syllable in PGmc.
 - a. $/\sqrt{\text{ward}}\text{-i:}\text{-}\theta\text{i}/ \rightarrow [\text{'wardi:}\theta\text{i}]$ 'destroy.PRES.3G' (cf. Ringe 2006: 166)
 - b. $/\sqrt{\text{lang}}\text{-ista-z}/ \rightarrow [\text{'lanɣistaz}]$ 'longest.NOM.SG' (cf. Ringe 2006: 284)
 - c. $/\sqrt{\text{geb}}\text{-o:}\text{-miz}/ \rightarrow [\text{'ge}\beta\text{o:miz}]$ 'gift.INST.PL' (cf. Ringe 2006: 279)

However, counterexamples to both of the algorithms in (2) can readily be mustered, in particular where prefixes are concerned. Prefixed nominal forms of the oldest Germanic languages show main stress of the initial syllable, not on the root, whereas prefixed verbal formations regularly show stress on the root, not the leftmost syllable; see the examples in (4)–(5).⁷ The first task in this chapter is therefore to establish whether and how the algorithms under (2) can best be extended and formalized in order to account for such counterexamples. This lays the groundwork for a systematic formalized description of necessary prosodic changes between PIE and Proto-Germanic in 7.3 and an explanatory account

⁴The word-level prosodic prominence of later Proto-Germanic is universally regarded as having exhibited stress in the narrowest sense. The brief review under 7.2.1 makes clear that Proto-Germanic prosodic prominence clearly fulfills the typological criteria for canonical stress laid out in Hyman 2009.

⁵Occasionally, scholars seem to conflate root and initial syllable with respect to Proto-Germanic, whether accidentally or deliberately (in order to avoid a full reckoning with the question); cf. the statement of Hutton (1995: 7): "It is a well known fact that primary stress in the old Germanic languages fell on the root or initial syllable of words."

⁶In terms of Simplified Bracketed Grid theory, such a system can be characterized as LLL on both Line 0 and Line 1, with Head L on Line 1; cf. the analysis of main stress in Koya in Halle and Isardi 1995: 406–9.

⁷Note especially the allomorphs [anda-] vs. [and-] and [fōra-] vs. [fōr-] in Gothic, whose distribution can be accounted for by differences in stress; see further 7.2.2.2 below.

thereof under 7.4. After briefly reviewing the evidence in favor of characterizing the Proto-Germanic system of prosodic prominence as a stress system in 7.2.1, a more extensive survey of the evidence for the position of stress in the oldest Germanic languages will be carried out under 7.2.2, and finally, the two competing accounts of primary stress assignment for Proto-Germanic, root stress or left-edge stress, are assessed under 7.2.3 and 7.2.4.

(4) Germanic Prefixed Nominal Formations

- a. Goth. [ʼanda-wōrdi] ‘answer.N.ACC.SG’ < PGMc. *[andi-wurdi] (cf. Germ. [ʼantvɔxt])
- b. Goth. [ʼfōra-ʒaŋgam] ‘steward, manager.M.DAT.PL’ < *[ʼfraʒaŋgamaz] (cf. Germ. [ʼvɔxʒaŋg] ‘process’)
- c. Goth. [ʼus-lō:si:n] ‘release, liberation.F.ACC.SG’ < PGMc. *[ʼuz-lausi:ni] (cf. OHG [ʼurlo:si:] ‘redemption’, ON [ʼo:rlausn] ‘solution; answer’)

(5) Germanic Prefixed Verbal Formations (cf. Ringe 2006: 291)

- a. Goth. [and-ʼbindan] ‘untie’ < PGMc. *[andi-ʼbindanā] ‘untie’ (cf. Germ. [ɛntʼbɪndən])
- b. Goth. [fōr-ʼgangan] ‘pass by’ < PGmc. *[fra-ganganā] (cf. Engl. [fɔɪˈgəʊ])
- c. Goth. [fōr-ʼbīuðan]* ‘forbid; command’ < PGmc. *[fra-ʼβeuðanā] ‘forbid’ (cf. Germ. [fɛxˈbi:tən], Engl. [fɔɪˈbɪd])⁸
- d. Goth. [us-ʼdri:βan] ‘drive out’ < PGmc. *[uz-ʼdri:βanā]

7.2.1 Properties of Proto-Germanic Prosodic Prominence

Among the earliest well-attested Germanic languages (Gothic, Old Norse, Old English, Old High German, and to a lesser extent, Old Saxon), neither are orthographic means employed to indicate the position of primary prosodic prominence, nor are any statements from native-speaker grammarians on the nature of accent in those languages available. Reconstruction of the position and properties of word-level prosodic prominence for Proto-Germanic thus depends upon the analysis of segmental sound changes between Proto-Germanic and these earliest daughters, synchronic phonological processes, and possible interactions of stress with meter (alliterative verse [*Stabreim*] in OE, OHG, and OS; *dróttkvætt* in ON).⁹ This section will review some of the telling evidence for the phonetic realization of prosodic prominence and its description as “stress” in the older Germanic languages; evidence for the position of primary stress and analyses thereof are deferred to the following section.

Virtually all attempts at the description of prosody in Old Germanic languages arrive unproblematically to, or can be interpreted as compatible with, the following set of state-

⁸ Attested Gothic forms of *faurbiudan** are PRES.ACT.IND.3.SG *faurbiudip* and PRES.ACT.OPT.2.SG *faurbiudais*.

⁹ For a general overview of metrics in all older Germanic languages, the main reference work remains Sievers 1893; a more recent survey is von See 1967. For Old English and Old Norse, the literature is extensive; Russom (1998: Ch. 2–9) offers a comparative treatment of these two metrical traditions. Recent analyses and overview of Old English meter include Fulk 1992, Minkova 2003, Terasawa 2011, and Weiskott 2016; on Old Norse, one may consult Gade 1995, Poole 2005, Ross 2005, or Ólason 2006. Treatments of the much more limited Old High German and Old Saxon metrical corpora are correspondingly fewer, but here Bostock 1976: 302–26, Hoffmann 1981: 1–3, and Russom 1998: Ch. 10–11 will serve.

ments: 1) every word had at least and at most one most prominent syllable; 2) syllables can be described as either having or lacking prominence (usually, a distinction between syllables with primary, secondary, or no stress is observed); 3) the occurrence of syllables with and without prominence can be largely explained by reference to some kind of foot structure. Examples illustrating these properties for Old English are given in (5); parallel forms with the same properties can be adduced for Gothic, Old Norse, Old High German, and Old Saxon. Insofar as all of the oldest Germanic languages exhibit these properties, the fulfill the criteria for a “stress-accent” language laid out in Hyman 2009: 220 (cf. discussion in 1.3.3.2): prosodic words exhibit OBLIGATORINESS and CULMINATIVITY, prominence relations are PRIVATIVE, and the position of prominence is conditioned by METRICALITY (see further treatment in 7.2.2 below). To refer to the prosodic prominences occurring in older Germanic languages as “stress” is thus fully justified.

(6) Basics of Old English Word Prosody (cf. Fikkert et al. 2006: 127–8)

- a. OBLIGATORINESS and CULMINATIVITY: [ˈæðe.liŋges] ‘prince.M.GEN.SG’;
no prosodic words like ^X[æðe.liŋges] or ^X[ˈæðe.liŋges].
- b. PRIVATIVITY: [ˈæðe.liŋges] shows primary, secondary, and unstressed syllables;
no distinct pitch contours with equivalent prominence, e.g., ^X[æðe.liŋges].
- c. METRICALITY:¹⁰ [(ˈæ.ðe).(liŋ).ge(s)];
formation of moraic trochees and left-edge orientation (largely) determines the
distribution of primary and secondary stress.¹¹

As far as the acoustic correlates of stress in Old Germanic are concerned, the diachronic evidence of vowel reduction, syncope, and apocope evident to some extent in all the languages is indicative of a realization similar to Present-Day English: greater duration, greater intensity, and a frequent association with raised F₀ (likely due to the attraction of intonational pitch accents). Examples in (7)–(9) illustrate some such developments in OE, OHG, and ON; see generally Ringe and Taylor 2014: 10–16; 44–8 for more examples of similar changes. Hutton (1995: 168) thus characterizes the accent of the Old Germanic languages as “dynamic stress”, which was becoming “strongly centered”, and that the tendency for reduction of unstressed vowels reflects a change to “stress-timing” from “syllable-timing”. Since instances of vowel reduction and loss (when not motivated by considerations of metrical structure) are largely not reconstructable for Proto-Germanic itself,¹² one may posit that the emergence

¹⁰Metrical structure given here follows the analysis of Goering 2016a; see further under 7.2.2.1.

¹¹See further below on words with leftmost primary stress beginning with a sequence of a light plus a heavy syllable, which would necessitate allowing either degenerate feet, [(ˈL) H...] or trimoraic feet of the shape [(ˈL H) ...].

¹²One exception in this regard is the apocope of $\begin{bmatrix} \text{—high} \\ \text{—long} \end{bmatrix}$ vowels in word-final position, e.g., PIE *[pénkʷe] ‘five’ > PGmc. *[fɪmf], PIE *[apo] ‘away’ > PGmc. *[ab], PIE *[ŋsmé] ‘us’ > PGmc. *[ˈuns] (cf. Ringe 2006: 116–8; for the position of the accent in *[ŋsmé], compare Ved. *asmá-bhyam* ‘1.PL.-DAT’). That this apocope had nothing to do with changes in the position or phonetic realization of stress in Proto-Germanic is shown by the PGmc. form [ˈuns]: if Verner’s Law had applied prior to apocope in *[unsmé], the ultimate PGmc. form would have been ^X[ˈunz]; thus, apocope must have applied even to vowels under the prominence inherited from PIE,

of such “dynamic stress” in Proto-Germanic promoted reduction phenomena affecting vowels.¹³

(7) Vowel Reduction and Loss in OE

- a. PGmc. *['geβō] ‘gift:NOM.SG’ > PNwGmc. *['geβū] > *['geβu] > OE *giefu*
- b. PGmc. *[kuningaz] ‘headman:NOM.SG’ > PWGmc. **kuning* > OE *cyning* (with *[u] > [y] by *i*-Umlaut)

(8) Vowel Reduction and Loss in OHG

- a. PGmc. *[dayō] ‘day:INST.SG’ > PNwGmc. *[dayū] > *['dayu] > OHG *tagu*
- b. PGmc. *[kuningaz] ‘headman:NOM.SG’ > PWGmc. **kuning* > OHG *kuning*

(9) Vowel Reduction and Loss in ON

- a. PGmc. *['geβō] ‘gift:NOM.SG’ > PNwGmc. *['geβū] > *['geβu] > ON *gjǫf* (with *[e] > [ɔ] by *u*-Umlaut, and subsequent apocope of short [u])
- b. PGmc. *[dayaz] ‘day:NOM.SG’ > ON *dagr*

7.2.2 Determining the Position of Proto-Germanic Prosodic Prominence

7.2.2.1 Old English

Our investigations into the calculation of prosodic prominence in the earliest Germanic languages begin with Old English, because the data in this language has been most thoroughly evaluated, and many competing theoretical analyses have been developed. For a survey of treatments of stress in Old English developed between 1970 and 1995, see Hutton 1995: 210–231; see Minkova 2014: 289–94 for an overview of evidence from orthography and verse for the position of stress in Old English. Clear from all of the analyses discussed in Hutton 1995

and in the case at hand, conditioned a change in the position of the prominence, which is necessary in order to obtain the consonant [s] rather than [z] in ['uns]. The diachronic derivation in below illustrates this set of developments.

- (1) PIE *['nsmé] ‘us’ > PGmc. *['uns]

*['nsmé]	PIE
*['unsmé]	ŕ > uR
*['únsm]	Apocope + Accent Shift
*['úns]	Final Sonorant Loss
*['úns]	Verner’s Law
*['uns]	PGmc.

The only further common Germanic sound change affecting the number of syllables in a word was the syncope of unstressed *[ə], where all such *[ə] vowels continue a Proto-Indo-European epenthetic vowel in certain consonant clusters (word initial */CHC/ or word-medial */CHCC/, depending upon the exact segments involved; see generally Byrd 2015: Ch. 2–3.

¹³Although comparatively limited in scope when compared with such developments in West and North Germanic, Gothic too, exhibits loss of vowels in final syllables: short *[a] and *[i] > Ø/ __C₀]ω (e.g., PGmc. *['xlaibaz] > Goth. ['hle:φs] ‘bread.NOM.SG; PGmc. *['gastiz] > Goth. ['gasts]). Furthermore, if the view argued for in Rauch 1981 is correct, then short /a, i, u/ in Gothic may have been subject to centralization when unstressed; see further 7.2.2.2.

as well as in later treatments (e.g., Fikkert et al. 2006, Goering 2016a) is that the position of primary stress in Old English is essentially undisputed, and that the occurrence of both primary and secondary stresses may be diagnosed from the alignment of stresses in OE alliterative meter (mainly represented by *Beowulf*).¹⁴ In short, the following generalizations apply (cf. the short summary in Hogg 2011: §2.84–2.90):

- Primary stress falls on the initial syllable of the prosodic word if there are no prefixes or proclitics.
- Prefixed verbs and deverbal nouns built from them have stress on the root syllable (i.e., first syllable to the right of a single prefix).
- Prefixed nouns and adjectives have primary stress on the leftmost syllable, and denominal or deadjectival verbs maintain stress on leftmost syllable.
- Compounds assign primary stress to the first member, including so-called “compound verbs” (= “separable” or “phrasal” verbs, in which the prefix may occur in a position away from the verb with suffixes).
- The prefix *ge-* (< PGmc. */ga-/) could never bear stress, whether it occurred on a noun or a verb (cf. Bennett 1972: 109–10 and Bammesberger 1981 on the history of this prefix).¹⁵ The same probably also applies to *be-*, insofar as it is used in nominal derivation.
- The prefixes *and-* ‘against’, *ed-* ‘again’ can both be used in both nominal and verbal forms, and always bear primary stress (though the former also has an unstressed allomorph *an-/on-/un-* that occurs only on verbs).

Thus, as illustrated through Gothic forms and their corresponding Proto-Germanic reconstructions in examples (4)–(5) above, Old English continues the distinction in position

¹⁴As discussed in Getty 2002: 52–61, a line of thought going back to Sievers 1893, for which Kuhn (1933) argues for even more strongly, holds that finite verbs in Old English (and presumably in Proto-Germanic as well) were more weakly stressed or clitic-like in their prosody, in comparison to nouns, adjectives, and verbal nouns; even Hutton (1995: 233) still repeats the opinion that OE verbs could be unstressed. This view even led other scholars, such as Wackernagel (1896b) to suggest that the V2 placement of verbs in main clauses in Germanic languages was due to their prosodic deficiency (i.e., finite verbs were like clitics and thus subject to “Wackernagel’s Law”), though V2 phenomena are more readily analyzed syntactically than prosodically (cf., e.g., Holmberg 2015). The OE evidence for this view was based on the supposed avoidance of finite verbs to participate in alliteration, which was taken by Sievers, Wackernagel, and Kuhn as an indication of lack of primary stress. The more systematic statistical investigation of the question in Getty 2002: 61–9 shows clearly, however, that finite verbs happily participated in alliteration; the empirical basis for the possibility of prosodic deficiency among finite verbs in Germanic – with the exception of auxiliaries and modal verbs – is thus entirely lacking.

¹⁵Under Bammesberger’s account, PGmc. */ga-/ probably stems from instances of unstressed, word-internal PIE */kom-/ ‘with’ in nouns, and from */kom-/ in verbs, where unstressedness would be expected. The unstressed */ga-/ in those positions ultimately drove out the stressed variant */ham-/, possibly preserved in the adjective PGmc. */ham-duga-/ (> Goth. *handugs* ‘wise’, ON *hǫndugr* ‘able’).

“Weak”	“Strong” (Primary Stress)
<i>of-</i>	<i>æf-</i>
<i>on-</i>	<i>and-</i>
<i>be-</i>	<i>(bī-)</i> ^a
<i>a-</i>	<i>ǣ-/or-</i>
<i>in-</i>	<i>inn-</i>
<i>op-</i>	<i>ūp-</i>
<i>// wip-</i>	<i>wiper-</i>
<i>ymb-</i>	<i>ymb-</i>

^aMany handbooks offer the pair *be-* vs. *bi-*, but the supposedly strong allomorph is in fact often attested as unstressed. It is not clear that a distinct “strong” allomorph can be said to exist at all. There is, however, the separate preposition/particle *bī-*, which may sometimes appear compounded with nominal or verbal stems in ways that make it indistinguishable from the genuine prefix.

Table 7.1: Prefix Allomorphy in Old English

of stress between prefixed verbs and prefixed nouns.¹⁶

Most other common prefixes possessed both a “strong” and a “weak” allomorph, where the latter consistently occurs on verbal bases, and points to stress on the verbal root following the prefix. These allomorphs are shown in Table 7.1; cf. Cambell 1959: 30–1, Getty 2002: 84–5, Minkova 2008: 26, and Hogg 2011: 48–9. *ymb-* is notable in exhibiting no segmental allomorphy, though differences in stress can be diagnosed from the metrical evidence (see immediately below): contrast *ymbgànc* ‘circuit’ and *ymbgán* ‘to go around’. Compare also the similar pattern of prefix allomorphy in Old High German given at 7.2.2.3. For a detailed treatment of the metrical evidence concerning the stress of prefixes, see Kendall 1991; for a full analysis of stress phenomena relating to prefixes in terms of the prosodic hierarchy, see Minkova 2008 (discussed in more detail below). Note, however, that the “strong” allomorphs listed here, with the exception of *inn-* and *wiper-*, are either already unproductive in or become unproductive during the Old English period (so Dietz 2004: 608, Minkova 2008: 26). Furthermore, many of the “strong” allomorphs are attested without primary stress, and are gradually replaced during the Old English period by the “weak” allomorphs, which then bear primary stress when expected, namely, when the base is a noun (Minkova 2008: 27).

In addition, Old English possess a set of adverbial and prepositional particles, some of which are also related or segmentally identical to the prefixes. These particles, when occurring with a nominal or verbal stem, can exhibit the same differentiation of stress as prefixes: primary stress falls on the particle when the stem is nominal, but on the root syllable of a

¹⁶*Pace* Hutton (1995: 212), the privative prefix *un-* probably did not always bear primary stress, but instead exhibited largely the same distribution between nouns and verbs as other prefixes. In Old English, *un-* predominantly occurred with nouns and adjectives (as is the case for the cognate prefix in Modern German; English verbal *un-*, cognate with German *ent-*, Gothic *and-*, was originally distinct), and thus normally had primary stress; on the few verbs with which *un-* occurs, e.g., *unbindan* (replacing earlier *an-/on-*), it is unstressed (cf. Hogg 2011: §2.90.2.e). The occurrence of *un-* with primary stress in nominal forms preceding *ġe-*, as in OE *un-ġe-þyld* ‘impatience’ (cf. Germ. *Ungeduld* [‘ungədølt]), poses an interesting problem; see further below.

verb. In short, Old English seems to have possessed exactly five prefixes whose stress properties were indifferent to the lexical category of the base (unstressed *ge-*, *be-*, and *for-*); stressed *and-* and *ed-*), while all other prefixes ~ particles took primary stress depending upon the lexical category of its base.

One crucial diagnostic with regard to distinguishing primary and secondary stress comes from permitted and disallowed resolutions of light syllables in Old English metrics. In general, a light syllable under primary stress may be resolved with any following syllable, whether long or short, but a light syllable under secondary stress may only resolve with a following light syllable, but not a heavy syllable. This latter restriction is termed KALUZA'S LAW (Kaluza 1896, Fulk 1992: 153–68; 381–90, Hutton 1995: 233, Goering 2016a: 180–5).¹⁷ Example (10) immediately below (verses taken from Goering 2016a and Bermúdez-Otero 2018) illustrate the behavior of stress and resolution in *Beowulf*.¹⁸ In (10)a., short syllables under primary stress freely resolve with following syllables; in (10)b., the secondary stress on the syllable [fa] permits resolution with the following syllable [tu] only because the latter is light. KALUZA'S LAW thus seems to constitute *prima facie* evidence for the possibility of a prosodic grouping (Light–Heavy) under some restricted conditions.

(10) Metrics, Stress, and Kaluza's Law in *Beowulf*

- a. under heofenes haðor (Beowulf 414a)

×× ˘×× ˘˘

under heaven:GEN.SG vault:DAT.SG

'under heaven's vault'

- b. bānfatu bænan (Beowulf 1116a)

˘˘× ˘˘

bone-vessels:NOM.PL burn:PTCP.PST

'bone-vessels to be burned'

To some extent, OE metrical licenses also reflect a diachronically somewhat earlier stage of the language, especially as far as vowel shortenings are concerned: namely, certain lexemes, which are transmitted with a short vowel, behave metrically as though they contain a long vowel, which would historically have been the case. One common instance of this behavior is found with Class II weak verbs, which historically contained a suffix /-o:-/ (cf. Goth. *salbon* ['salbo:n] 'anoint' vs. OE *sealfan* ['sæəlvan]), but which is regularly shortened and falls together with /a/; nevertheless, this resulting short /a/ may bear stress, including, in some instances, the lift associated with primary stress. In this fashion, the distribution of

¹⁷Goering (2016a: 184, fn. 22) offers the following definition of KALUZA'S LAW for OE metrics: "When immediately following a heavy syllable, a short syllable bearing some degree of stress resolves with the following syllable if, and only if, the result is a precisely bimoraic unit" (cf. also Goering 2016b). Goering points to instances where a ˘ – sequence that is not line initial does not show resolution. Hence, the sequence /˘ –/ exhibits variability with respect to resolution: consistently resolved as ˘ – if word- and line-initial, consistently not resolved if not word initial, and not resolved line-medially if following a heavy, despite being word initial.

¹⁸× represents a syllable whose length is metrically irrelevant; ˘ is a short syllable with primary stress, and ˘ is a short syllable with secondary stress; ˘ and ˘ mark the same for long syllables. An undertie ˘ between two elements indicates resolution of two syllables into one metrical position.

stresses in OE alliterative poetry sometimes better reflects a Pre-Old English prosody than the synchronic prosody of the language.

Finally, certain derivational suffixes (e.g., *-dōm-*, *-fæst-*, *-hād-*) may bear secondary, or rarely, even primary stress; this behavior may be due to the origin of these suffixes as the second member of a compound. Indeed, prosodically speaking, such “suffixes” should perhaps still be analyzed as forming independent prosodic words – words like *dōm* and *fæst* simply constitute productive second compound members. Whether such derivational suffixes show signs of bearing stress (i.e., occur in a metrically strong position in verse) also seems to depend on two further factors: first, only when an inflectional suffix follows (i.e., the derivational suffix does not constitute the final syllable of the word) does it occur in a strong position; second, all of these suffixes constitute heavy syllables. This behavior probably indicates that heavy syllables in particular preferred stress (WSP), but stress on final syllables was avoided (NONFINALITY).

The interaction of meter and stress in Old English also allows for some insights into phrasal stress and perhaps thereby into the intonational phonology of the language. In particular, verbs, prepositions, adverbs, and pronouns can behave metrically as though they contain no primary or secondary stress. This fact probably stems from the relative prominence of these elements within recursive prosodic words or phonological (and intonational) phrases; it does not indicate that they categorically could not build prosodic words with stress – a fact which should be self-evident, given that the same words may be scanned as stressed under different metrical conditions. A detailed investigation into the relation between plausible higher-level prosodic structures (φ and ι) and stress in Old English metrics remains a desideratum; see Getty 2002: 136–155 for first steps.

In the following, I briefly summarize three analyses of Old English stress assignment, developed in Hutton 1995, Getty 2002, and (more cursorily) Goering 2016a. The lattermost is not a comprehensive account of stress assignment in OE, but its explicitness concerning the construction of foot structure is worth presenting here. Since the structures generable for Old English can, modulo some minor adjustments in the distribution of prefix allomorphs, essentially be applied to Proto-Germanic, understanding what analytical options exist for the OE stress system will facilitate the task of reconstructing and analyzing Proto-Germanic stress as well.

Hutton 1995 / 1998. The analysis of OE stress developed in Hutton 1995: 231–51 and reproduced in Hutton 1998 consists in a lexicon with both stress-attracting roots and stress attracting prefixes (lexically specified foot heads), plus the cyclic (Level 1, after each derivational affix) and non-cyclic (Level 2, once, after inflection) construction of trochaic feet. Thus, prefixed nominals, like *ándsaca* ‘adversary’, will systematically show left-edge stress because of the lexically specified foot head of the prefix / $\langle$ and-/; a verb with a prefix without a lexically specified foot head, like the corresponding weak allomorph seen in *onsacan* ‘to dispute’, will cede stress to the lexically specified foot head of the verbal root / $\langle$ sak-/. However, the deeper motivation behind the assumption of thoroughgoing lexical stress as part of the lexical entries of roots would appear to be the same motivation that underlies the non-standard foot

shape (resolved moraic trochee) posited by Dresher and Lahiri (1991): primary stress can fall on a light syllable immediately followed by a heavy syllable.

The conundrum that Hutton sees in dealing with stress assignment in Old English in a purely phonological fashion stems not so much from the contrast between nominal and verbal prefix stress, but from the simple fact that word-initial 'LH sequences may bear primary stress on the initial syllable, while secondary stress can only fall on 'H or 'LL sequences. The primary stress assignment, for Hutton, looks like the construction of a syllabic trochee, while the secondary stress assignment evidently follows from the construction of moraic trochees. Given that, without a system of violable constraints to permit the construction of a super-heavy foot of the shape ('L H) at the left edge of a prosodic word within a moraic trochee system, the parallel construction of syllabic and moraic trochees would indeed seem contradictory.

Hutton also exploits a combination of morphological and phonological information to generate secondary stress: secondary stress is assigned exclusively to heavy penultimate syllables that belong to derivational suffixes. The algorithm for stress assignment in OE posited by Hutton then functions as follows:

1. Lexicon: Foot heads are marked in the lexicon (roots, certain other morphemes).
2. Morphology: word formation processes
3. Phonology Level 1:
 - Quantity-insensitive trochaic feet are built by adjoining unfooted syllables to a foot head to their left.
 - *Pruning* at the end of the Level 1 cycle removes all but the most marginal weak foot branches. Thus, foot heads and the leftmost moraic trochee remain.
 - Pruning will keep left edge primary stress, but open the right edge for secondary stress.
4. Phonology Level 2:
 - Quantity-sensitive feet are made around the foot heads preserved from from Level 1.
 - Secondary stress is assigned to a heavy penult, provided that the final syllable is an inflectional and not a derivational morpheme, building a syllabic trochee out of the penult and final syllable.

In his derivations (p. 239) of *andsaca* 'adversary' and *onsacan* 'dispute', stress essentially falls out from the underlying lexical foot heads plus a left-edge preference. The root /s(ak-/ is taken to have a lexical accent, as is the nominal prefix /and-/ in contrast to the verbal prefix /on-/. Thus /'and-'sac-an/ first prunes the structure to a single leftmost foot

and the left edge of another foot, thus [(‘and)(saca)]; /on-‘sak-an/, meanwhile, simply takes a foot around the root [on(sa)kan]. In Level 2, branching trochees are filled out to create [(‘and)(saka)] and [on(‘sakan)]. Secondary stress emerges from underlying /(skand-lik-nes:-ə/ ‘shame:ACC/GEN/DAT.SG’, which comes out as [(‘skand).lik.(‘nes.sə)]

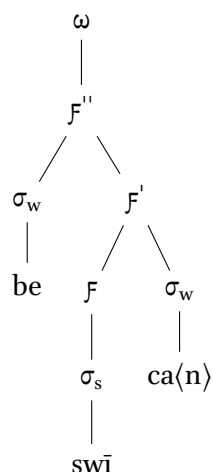
Empirically, Hutton’s predicted pattern of secondary stress is questionable, since secondary stress might also be permitted on light syllables forming moraic trochees (as Goering (2016a) allows for *bānfatu*, parsed as [(‘bān).(fa.tu)], at 10b. above), as well as on derivational suffixes ending in two consonants like agentive *-end*, even in absolute word-final position (see Goering 2016a: 179 and Russom 2001: 55–63 for more detail). Theoretically, Hutton’s machinery is intricate, and the set of rules and lexical representations results in many redundancies. In particular, the persistent lexical stress marking of roots, regardless of their lexical category, produces the right results, but seems superfluous outside of verbs, at least.

Getty 2002 As in Hutton’s analysis, the essential phenomena to be captured are the occurrence of primary stress on word-initial light syllables, even when followed by a heavy syllable, and the position of primary stress on the root syllable in prefixed verbs. The first problem is treated only implicitly, but one may infer from Getty’s analysis of the form *cyn-ing* ‘king:NOM.SG’, given the structure ([cy]_{strong} [nin(g)]_{weak})_{Ft} (Getty 2002: 110) and constraint ranking given for Getty’s treatment of Finnish stress (Getty 2002: 113), that word-initial stress in LX sequences follows from a ranking ALIGNLEFT(HEAD[ω],ω) ≫ FOOTMAX. In other words, the requirement that the left edge of the prosodic word be aligned with the strong member of a foot outranks the preference for a foot to contain maximally two morae.¹⁹

The basic claim and insight of Getty’s concerning unstressed prefixes is that OE feet may be recursively built in order to exhaustively parse all syllables into a foot without violating PARSE-σ, while generally requiring that the left edge of a root be aligned with the left edge of a foot through ALIGNLEFT(Root, Foot). The verb *beswīcan* ‘deceive:INF’ is thus analyzed as having the structure shown in (11). In this way, all syllables are ultimately incorporated into a foot, but because the minimal foot was built around [swi:] alone, the primary stress falls on that syllable. At the same time, Getty argues, in a fashion similar to Hutton, that prefixes are lexically specified as being strong or weak – in other words, semantically and phonologically parallel pairs of prefixes must possess lexically unstressed and lexically stressed allomorphs. In (11), the prefix /be-/ is presumed to be weak (unstressed), thereby allowing for stress to be assigned to the root syllable instead.

(11) Prosodic structure of *beswīcan* ‘deceive:INF’ after Getty 2002: 106

¹⁹See immediately and 7.2.4 below on deriving the foot structure (‘L H) without reference to an explicit size-restrictor constraint like FOOTMAX.



The analysis of these two phenomena encounters two major problems, however, one theoretical and one empirical. On the theoretical side, if foot recursivity is simply not an option provided by GEN, then words containing /H L/ sequences can simply accept an unparsed syllables (FTBIN $\gg$ PARSE- σ), and left edge stress in /L H/ can be generated by assuming that the need for alignment of a prosodic word with the left edge of a foot outranks WSP (given trochaic feet). Given that most current views on recursion of prosodic categories (Kabak and Revithiadou 2009b, Ito and Mester 2009, Ito and Mester 2013) see such recursion as a) in large part driven by the interface between phonology and syntactic structure, and b) probably therefore limited to prosodic categories at the level of the prosodic word and above, assuming recursion of feet below the level of the word, as Getty does, is not particularly attractive. Compare also the arguments of Golston (2021) against recursive feet, and see further 1.2.2 in Chapter 1 and 7.2.4 on the phenomenon of prosodic word recursion.

On the empirical side, the high ranking of the constraint ALIGNLEFT(Root, Foot) potentially encounters problems and redundancies in conjunction with the system of strong and weak prefixes. In prefixed nominal forms, since prefixes always bear primary stress (e.g., *wīpersaca* ['wīðer,saka] 'adversary', *ymbgang* ['ymbgang] 'circuit'), Getty must assume that only strong prefixes can combine with nouns, and that the preservation of their lexical stress outranks ALIGNLEFT(Root, Foot), while the corresponding weak forms of the prefixes occur exclusively with verbs. Given the phonological and semantic similarity of prefixes across nouns and verbs, to posit a single lexical entry in most cases would be more attractive. In the case of the rarer uniformly stressed (*and-* and *ed-*) or uniformly unstressed prefixes (*ge-*), only one lexical entry is needed, and the amount of prosodic information in the lexicon would be kept to a minimum.²⁰ However, given that the supposedly accented prefixes *and-*

²⁰ Another piece of data suggesting that lexically specified prosodic structure was rare or non-existent in the Old English lexicon comes from the treatment of loanwords from Old French: these are generally assigned left-edge stress (cf. Lahiri 2015: 233–7), with the exception of verbs that could be morphologically and prosodically parsed into a sequence of prefix + stem (e.g., *re-cōunt*). In light of the treatment of loanword prosody among languages whose native speakers are “stress-deaf” (cf. 3.3.4.3), the inability of Old English speakers to take over stress from Old French lexemes faithfully suggests that Old English prosody was algorithmically regular, predictable from phonological structure alone, with little or no lexical prosodic information. Compare also the

and *ut-*, like the negative prefix *un-*, never occur closer to a verbal root than an unstressed prefix, all three of these prefixes might be better analyzed as building independent prosodic words, thus participating in compound stress (cf. 7.2.4), where the primary stress of the leftmost member becomes the primary stress of the compound (cf. 33 in Chapter 1).

Summary: The Old English Moraic Trochee. The survey of data and analyses here may be summarized as follows: the assignment primary stress in Old English is most easily handled under the analysis of Goering 2016a. Under Goering’s analysis, moraic trochees are constructed from left to right, with trimoraic feet encompassing Light–Heavy sequences permitted only at the left edge of the word in order to satisfy the requirement that the left edge of the prosodic word be aligned with the left edge of a foot. Beyond the leftmost foot, all heavy syllables or sequences of two light syllables build a foot; (,L H) or (,H L) feet do not arise. A slight quandary arises concerning the footing of final [V(:)C], [VCC], and [L L] sequences, however, whose resolution goes beyond the present objectives. In short, metrical evidence suggests that words ending in [-VCC], such as *rīdend* ‘rider:NOM.SG’, bore secondary stress, but words ending in [-VC], such as **hēafudum* ‘heaven:DAT.PL’, did not, regardless of whether the penult was light or heavy. Furthermore, the contrast between **hēafudum*, which is subject to high vowel deletion (yielding attested *hēafdu*) and NOM/ACC.PL *hēafudu*, which does not undergo high vowel deletion (^X*hēafdu*) suggests that the latter had stress on the penult (footed as [(‘hēa).(fu.du)]), but the former did not (i.e., the penult was unfooted). A parse of **hēafudum* as [(‘hēa).fu.(,dum)] would suit the evidence of high vowel deletion, but not the meter, which instead suggests that the final syllable is unstressed (~ unfooted). Goering (2016a: 179–80) assumes that forms with final [-VC] sequences were footed, but subject to destressing, whereas forms with final [-VCC] were not (in virtue of the final consonant being treated as extrametrical in order to avoid a superheavy syllable, thus, e.g., [(‘rī).(den)(d)]). I tentatively assume then that **hēafudum*-type words were parsed as [(‘hēa).fu.(,dum)] and leave the difference in stress in meter with respect to VCC-final words aside.²¹

The tableaux in (12) express the crucial constraint rankings needed to derive the OE foot and stress patterns in cases where just a single minimal prosodic word is constructed. To summarize the ranking in prose: a minimally bimoraic trochaic foot must be constructed at the left edge of the word, where alignment of the left edge of the word is more important than satisfying WSP; otherwise, feet are exhaustively parsed from left to right, building a foot around every heavy syllable or sequence of two light syllables encountered; feet of the shape (H L) are categorically not permitted; primary stress is assigned to the leftmost foot.

(12) Ranking for Prosody of Old English $\omega_{[+min]}$:

LEFTMOST, TROCHEE, ALIGN- ω -L, RH-CONTOUR, FTBIN, *SUPERHEAVY- $\sigma \gg$ WSP, PARSE-C $\gg$ PARSE- σ , NONFINALITY- $\sigma \gg$ NONFINALITY-F, *CLASH, ALL-FT-L

similar discussion with respect to Latin loanwords into Old English in Minkova 2014: 289–90.

²¹Note also that assuming that high vowel deletion applies in the weak part of a (H L) foot does not lead to a viable solution either. If RH-CONTOUR-violating feet of the shape (H L) were permissible, then both *hēfudu* and **hēafudum* would be expected to parse the first two syllables into a single foot. In short, as long as *hēfudu* calls for the parse [(‘hēa).(fu.du)], with a foot allowed at the right edge of the word, and *rīdend* allows for stress on a final syllable, no parse of *hēafudum* except [(‘hēa).fu.(,dum)] can yield a coherent constraint ranking.

- a. /kyning-a/ → [('ky.niŋ).ga] 'king:GEN.PL'

/kyning-a/	LEFTMOST	TROCHEE	ALIGN-ω-L	RH-CONTOUR	FtBIN	WSP	PARSE-σ	NONFINALITY-ɸ	*CLASH	ALL-Ft-L
a. ☞ ('ky.niŋ).ga						*	*			
b. ('ky).niŋ.g					*!	*	**			
c. ky.('niŋ).ga			*!				**			*
d. ('ky).(niŋ.g)					*!			*	*	*
e. (ky).('niŋ.g)	*!			*!	*!			*	*	*
f. (ky.niŋ).ga		*!					*			

- b. /he:ʁfudu/ → [(he:ʁ).(vu.du)] 'heaven:NOM/ACC.SG'

/he:ʁfudu/	LEFTMOST	TROCHEE	ALIGN-ω-L	RH-CONTOUR	FtBIN	WSP	PARSE-σ	NONFINALITY-ɸ	*CLASH	ALL-Ft-L
a. ☞ (he:ʁ).(vu.du)								*	*	*
b. (he:ʁ).vu.du							*!*			
c. (he:ʁ.vu).du				*!			*			

- c. /ri:d-end/ → [(ri:).(den)(d)] 'rider:NOM.SG'

/ri:d-end/	LEFTMOST	RH-CONTOUR	FtBIN	*SUPERHEAVY	WSP	PARSE-C	PARSE-σ	NONFINALITY-σ	*CLASH	ALL-Ft-L
a. ☞ (ri:).(den)(d)						*		*	*	*
b. (ri:).dend				*!	*		*			
c. (ri:).(dend)				*!				*	*	*
d. (ri:).den(d)					*!	*	*			
e. (ri:d).('den)(d)	*!							*	*	*
f. (ri:de)(nd)		*!				**				

As we will see in the succeeding sections, primary stress assignment in the other Old Germanic languages can be handled through an essentially identical analysis. Furthermore, the reconstructability of Siever's Law for Proto-Germanic (see 7.2.2.2 below on the Gothic

evidence) makes valid the assumption of moraic trochees as the preferred foot form of Proto-Germanic (cf. Kiparsky 1998b, Bermúdez-Otero 2018), and thus, in the absence of strong evidence to the contrary, the perseverance of the moraic trochee as the preferred foot form in the early Germanic languages may be presumed (compare Lahiri et al. 1999 on inherited and historically perseverant prosodic parameters in Germanic languages).

As far as stress in prefixed sequences is concerned, essentially four facts must be made to fit with the evident basic initial syllable stress in what are presumably minimal prosodic words:²²

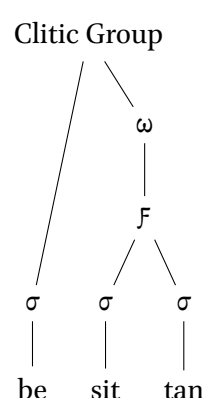
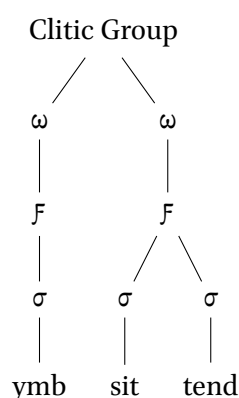
- Prefixes on nominal bases generally bear primary stress.
- Prefixes on verbal bases generally do not bear primary stress
- *ge-*, *be-*, and perhaps *for-* are persistently unstressed regardless of base.
- *and-* and *ed-* are persistently stressed regardless of base.

The analysis of Minkova (2008: 31–6) is fundamentally on the right track in attributing this complex of stress behaviors not to any lexical specification for stress attraction, as in the analysis of Hutton (1995) and Getty (2002) above, but instead to the prosodic structure constructed. Specifically, Minkova proposes that *all* prefixes of OE stand outside the prosodic word formed by the root and any suffixes; together, the prefix and stem form a clitic group, and in case the prefix can build a bimoraic foot on its own, is assumed to build its own minimal prosodic word. The structure posited for prefixed formations of Old English by Minkova is shown in (13).

- (13) Structure of Prefixed Formations by Prefix Size in Old English assumed in Minkova 2008: 33²³

a. Prefix = *f* in *ymb-sittend* ‘neighbor’

b. Prefix = *ȝ* in *besittan* ‘surround’



²²I ignore here inheritance of stress in derived nouns and derived verbs that preserves the primary stress of the base; this behavior is easily captured as a cyclic effect or through Output-Output faithfulness.

²³I reproduce here exactly the foot structures assumed by Minkova.

By keeping all prefixes outside the same ω -domain encompassing the stem, the consistent lack of stress on prefixes that clearly are not large enough to comprise a bimoraic foot (*be-*, *ge-*) can be captured: since they cannot make a foot, and will not build a foot together with a following heavy syllable (contrast the winner in (12)a.), they are ineligible for stress. However, Minkova smuggles a species of lexical specification into the analysis by allowing for verbs and nouns to apply a different relative ranking of two crucial constraints: $\text{ALIGN}(\omega, \text{L, Clitic Group, L})$, which prefers for the leftmost prosodic word in the Clitic Group to be the prosodic head of the Clitic Group, and ROOTMOST , which prefers for a morphological root to bear a primary stress. In nouns, the ranking $\text{ALIGN}(\omega, \text{L, Clitic Group, L}) \gg \text{ROOTMOST}$ holds, resulting in primary stress within the Clitic Group on the prefix (cf. 13a.); in verbs, this ranking is assumed to be reversed, with the consequence that primary stress within the Clitic Group will instead fall on the verbal root.

To refer to different lexical categories in constraints is licit, since morphosyntactic information must be visible to the phonology; to assume distinct rankings of the phonological constraints, by reference to the lexical categories is questionable, however (compare 2.4.2.2 on the use of lexically specific constraint rankings, which assume that higher-ranking copies of constraints are activated by certain morphemes). The same effect could be achieved by employing two versions of ROOTMOST , respectively ranked above and below $\text{ALIGN}(\omega, \text{L, Clitic Group, L})$: $\text{ROOTMOST}(\text{VERB})$ and $\text{ROOTMOST}(\text{NOUN})$. Finally, the persistently stressed prefixes are taken to simply have underlyingly nominal bases (e.g., *ándswarian* ‘answer:INF’ is taken to have initial stress in virtue of being a derivative from the noun *ándswaru* ‘answer’).²⁴ A more minor issue is the usage of the clitic group as the prosodic constituent encompassing the stem and prefixes, but a reformulation in terms of a non-minimal, recursive prosodic word after Ito and Mester (2009) could easily be applied. That is, the Clitic Group node in (13) could be replaced with a recursive prosodic word, $\bar{\omega}$, and each ω would be labeled as a $\omega_{[+min]}$. Old English prefixes would then generally constitute a species of compound of the structure $[\text{prefix}]_{\omega_{-min}} [\text{stem}]_{\omega_{-min}}]_{\bar{\omega}}$, while *be-* and *ge-* instead would be interpreted as affixal clitics of the structure $[\sigma [\text{stem}]_{\omega_{-min}}]_{\bar{\omega}}$.²⁵

A potentially fatal objection to Minkova’s account, however, is that it predicts that verbs that form a clitic group ($\sim \omega_{[-min]}$) with another element will exceptionlessly bear stress on the verbal root (except when the verb is denominal). Whether this is the case constitutes a fraught problem, since Old English certainly possessed adverbial particles, often identical to prepositions (e.g., *under-* ‘under’, *ofer-* ‘over’, *niper-* ‘down’, *wip-* ‘with’) that could appear as

²⁴This maneuver might be valid for Old English, and, insofar as there does not seem to be secure evidence for persistently stressed prefixes in Gothic, indicates that it is historically likely correct. In effect, there is no sense in referring “persistently stress prefixes” at all – though doing so is not uncommon in the literature on Old English stress. Getty (2002: 80) instead interprets *and-*, at least, as always building an independent minimal prosodic word that can be compounded with a stem, and thus generates primary stress on the first member (thus, e.g., *ándswàrian*) in accord with normal left-prominence in compounds.

²⁵In the case of productive nominal compounds in Old Germanic languages (e.g., OE *pēod-cýning-* ‘people-king’, Goth. *auga-dauro-* ‘eye-door = window’), I assume that these constitute a single non-minimal prosodic word, so that the morphological word built by the stems can be isomorphic to a prosodic word. If nominal compounds are instead interpreted as constituting a phonological phrase, the inflectional suffixes that scope over the entire stem would phonologically be adjoined to a ϕ , rather than a ω .

prefixes to verbs as well, e.g., *wipstándan* ‘withstand’. In some cases, like the just mentioned *wipstándan*, the adverbial particle is demonstrably unstressed (confirmed by both metrical evidence and stress in Modern English), but the same adverbial particles sometimes seem to act as full prosodic words that compound with stems and receive primary stress, and for several of the words of this class (*æfter*- ‘after’, *fore*- ‘before’, *nīper*- ‘down’, *wīper*- ‘against’), an unambiguously unstressed form is never attested (cf. Minkova 2008: 24). If it is the case that Old English possessed compound verbs, typologically comparable to the separable prefixed verbs of German or Dutch, then they would be expected to bear primary stress on the adverbial particle. That these adverbials can produce different outcomes with respect to stress, even regardless of “separability”, is shown by the behavior of certain German prefixes, such as *wider*- (cognate with OE *wīper*-), which cannot be separated from the verbal stem and which may bear either primary stress on the prefix (e.g., Germ. *widersagen* ‘contradict’) or on the verbal root (e.g., Germ. *widersprechen* ‘contradict’).²⁶ At least in origin, it is tempting to posit that these adverbial particles could build prosodic structures with verbal stems that could generate either primary stress on the adverbial particle or primary stress on the verbal root; as it stands, Minkova’s analysis predicts that only root stress on the verb should be generated, since all “prefixed” formations are taken to have a compound-like structure dominating two prosodic words. In short, if OE possessed no compound verbs of the type Germ. *widersagen*, Minkova’s analysis could stand in a modified form; if it did, a slightly different set of prosodic structures would need to be posited. See further Section 7.2.4 for such a solution, proposed to hold at the Proto-Germanic level.

7.2.2.2 Gothic

I next turn to stress in Gothic, since the early attestation (4th c. CE) of the language holds the potential for the preservation of more archaisms. Two relatively systematic treatments of stress assignment in Gothic are available: Bennett 1970 and Greiner 1994: 64–89; the latter further looks for evidence of stress through various phonological processes. Beyond these two treatments, shorter discussions of Gothic stress assignment are found in Voyles 1981: 42–5 and Streitberg 1920: 65; 160–3. To begin, I will summarize the views of Streitberg and Voyles, before more systematically looking at the Gothic evidence from the perspectives of Bennett and Greiner. For further exposition and analysis of many of the same facts, see Sandell and Goering 2020 and Goering and Sandell Accepted (2021)) as well, from which many examples are borrowed and whose thinking is essentially followed here.

Streitberg’s short paragraph dedicated to the accentuation of Gothic (Streitberg 1920: 65) makes three claims: 1) primary stress (“Wortkzent”) must have fallen on the first syllable, with the exception of prefixed verbs, where primary stress fell on the stem syllable (i.e., the verbal root); 2) compounds with monosyllabic first members, including cases with “mere particles”, have primary stress on the left most syllable (thus *gáburpai* [ˈgabʊrθe:]); 3) compounds with *un*- as a first member have two primary stresses, on *un*- and on the primary stress of the second member. The last statement, if taken at face value, is *a priori* hardly credible, and has neither direct evidence of any sort to support it in Gothic, nor support

²⁶Below, evidence for verbs with compound structures in Gothic will be discussed.

from comparison to the other Germanic languages. The first statement, meanwhile, essentially accords with the facts of Old English, while the second requires further investigation below.

The basic WORD-STRESS RULE posited for Gothic by Voyles (1981: 42), shown in (14), fundamentally accords with Streitberg's first statement.

(14) Gothic WORD-STRESS RULE of Voyles 1981: 42

[+vocalic] → [+stress] / ω [[−vocalic]]₀ __,
where __ is not contained in a verbal derivational prefix, enclitic particle, or reduplicative prefix.²⁷

In contrast to Streitberg, Voyles explicitly states that the prefix *ga-* does not bear stress, because it is to be classed as a “verbal derivational prefix”, even if it occurs on nouns as well.²⁸ This definition of “verbal prefix” then applies essentially to nearly all elements traditionally viewed as derivational prefixes in Gothic, with the exception of *un-* ‘not’, which is found exclusively on nouns and adjectives.²⁹ Voyles further allows the rule in (14) to apply cyclically, reassigning primary stress whenever a new environment for its application is created, with concomitant reduction of existing primary stress to secondary stress.³⁰ The derivation

²⁷Voyles' rule has here been partly rewritten, in that I have replaced his boundary symbol ## with ω [and placed the word-class conditions outside the rule itself. Although not explicitly stated, Voyles' ## boundary seems to mark either a) a morphological word or b) a minimal prosodic word or prosodic word at some level of recursion. He regularly places this symbol between any prefix and a root, and to the right of a word e.g., ##ga##baurgja## ‘fellow citizen’, ##un##barnahs## ‘childless’. For Voyles, the assumption of this boundary follows from the behavior of clear enclitics like *uh* ‘and’, which may only occur at the very end of a word, or after the first prefix of a word, as shown here:

- (1) Enclitic *uh* ‘and’
 - a. *habaidedun-uh* ‘have.PST.3PL-and’
 - b. *at-uh-gaf* ‘to-and-give.PST.3SG’

²⁸Whether *ga-* could bear primary stress or not is not entirely certain. The principal piece of evidence for primary stress on the first syllable of nouns with the prefix *ga-* comes from cases in which *ga-* is part of a denominal verb to a noun already prefixed with *ga-*, e.g., *ga-galeikon* ‘transform:INF’ to *galeiks* ‘similar’. Bammesberger (1981) defends the view that, although *ga-* represents an originally unstressed form of this prefix in verbs, when the stressed form **xam-* was replaced in nouns with *ga-*, then *ga-* received the primary stress, just like other nominal prefixes. This claim is at odds, however, with the metrical evidence for consistently unstressed *ge-* in Old English (see above). I see three possible interpretations: 1) the apparent stress [ga'gali:ko:n] ‘transform:INF’ is due to a *post-accenting* property of /ga-/, which results in primary stress on the syllable of the leftmost /ga-/, in which case *galeiks* ‘similar’ would also have peninitial primary stress; 2) Gothic has innovated in treating *ga-* on nouns like other nominal prefixes; 3) Old English has innovated in treating *ga-* as consistently unstressable.

²⁹One serious problem with Voyles' claim that the “verbal prefixes” can never bear primary stress, as pointed out by Greiner 1994: 68–9, is that nominals built with, e.g., /anda-/ (e.g., *andanems* ‘pleasant’) should have the same stress pattern as verbs like *andniman* ‘take, receive’, leaving the prefix allomorphy inexplicable (cf. further below). Voyles does not attempt to substantiate the claim that “verbal prefixes” on nouns and adjectives cannot bear primary stress.

³⁰Voyles further assumes that existing secondary stresses may be reduced to tertiary stresses, but, this as-

of *unanasiums** ‘invisible’ (cf. the attested adverb *unansiuniba* ‘invisibly’ at *Skeireins* 8.1) would thus proceed as in (15).

(15) Cyclic WORD-STRESS RULE Application

siuns	‘sight.NOM.SG’
‘siuns	WORD-STRESS RULE
ana-‘siuns	‘visible.NOM.SG
‘ana,siuns	WORD-STRESS RULE + stress reduction
un-‘ana,siuns	‘invisible.NOM.SG
‘un,ana,siuns	WORD-STRESS RULE + stress reduction

One clear issue with Voyles’ approach in distinguishing types of prefixes is the simple fact that many seemingly identical prefixes occur on both nouns and verbs, and yet probably are stressed differently, depending on whether the form is a noun or a verb (cf. (21)–(22) below).

Having now briefly reviewed two general views on Gothic stress assignment, which accord in the view that primary stress falls in the initial syllable in most cases, the following questions remain to be explored:

- what prosodic structures (i.e., feet and syllables) were constructed below the level of the ω ?
- what non-comparative evidence exists for a difference in primary stress between prefixed nouns/adjectives and prefixed verbs (as in Old English)?
- how did the reduplicant of reduplicated past tense forms (Class VIII strong verbs) behave in terms of stress?
- what prosodic structures were constructed involving compounding or cliticization above the $\omega_{[+min]}$?

Although the third issue is somewhat tangential, addressing these questions will help to further establish under what conditions prosodic words in Proto-Germanic may have exhibited primary stress other than on the leftmost syllable. In the following, I first present the main evidence for the evidence for the moraic trochee as the preferred foot structure of Gothic, before examining prefixed formations, reduplication, compounding, and cliticization in turn.

sumption either implies that Gothic did not admit of more than one secondary stress per word, which would be typologically unexpected, or that Gothic distinguished a theoretically infinite number of degrees of successively weaker stress, which seems theoretically undesirable. Thus, in the the final output of the derivation for *unanasiums** in Voyles 1981: 44, three degrees of stress are marked, where I mark one primary and two secondary stresses. Contrast the analysis of in Greiner 1994: 72, where only a distinction between primary, secondary, and unstressed syllables may be derived. With Greiner, Voyles’ cyclical process of stress demotion is empirically without basis.

Moraic Trochees in Gothic Just as in Old English, convincing language-internal evidence can be mustered to suggest that Gothic preferentially constructed moraic trochees, including WSP-violating feet of the shape ('L H) at the left edge of the word. In cases in which no prefixation or cliticization of any kind is involved, largely the same constraint ranking used for Old English in (12) above would suffice to generate Gothic stress patterns, though uncertainties concerning final segment extrametricality remain. Granting the construction of moraic trochees from right to left in Gothic produces parses in trisyllabic words like those shown in (16).

(16) Foot Construction in Trisyllabic Gothic Words: LEFTMOST, TROCHEE, FTBIN $\gg$ WSP³¹

- a. Heavy–Heavy: *stojandan* [('sto:).(jan).da(n)] 'judge:PTCP.NPST.ACC.SG.M'³²
- b. Heavy–Light: *airkniþa* [('ɛrk).(ni.θa)] 'sincerity:F.NOM.SG'
- c. Light–Light: *aqizi* [('a.k^wi).zi] 'axe:F.NOM.SG'
- d. Light–Heavy: *siponeis* [('si.po:).(ni:)(s)] 'disciple:M.NOM.SG'

The clearest evidence for the nature of the Gothic foot derives from the distribution of GEN.SG forms in *ja*-stems due to SIEVERS' LAW (Sievers 1878: 129–31): in short, when the last syllable of the root or stem preceding the suffix */-j-/* contains a short vowel and ends in a single consonant, as in (17)a., the suffix + ending sequence *-jis* [-jis] occurs, but when it ends in two consonants, as in (17)b., contains a long vowel followed by at least one consonant, as in (17)c., or is preceded by two light syllables, as in (17)d., the suffix + ending sequence *-eis* [-i:s] is found. The same pattern is observed in Class I weak verbs, also built with a suffix */-j-/* in the NPST.3SG, as illustrated in (18).

(17) NOM/GEN.SG in masculine *ja*-stems

- a. *har-jis* [('har).ji(s)] 'army'; *nip-jis* [('niθ).ji(s)] 'relative'
- b. *haird-eis* [('hɛr).(di:)(s)] 'shepherd'; *andeis* [('an).(di:)(s)] 'end'
- c. *lek-eis* [('le:).(ki:)(s)] 'doctor'; *sipon-eis* [('si.po:).(ni:)(s)] 'disciple'
- d. *ragin-eis* [('ra.ɣi).(ni:)(s)] 'counselor'

(18) NPST.3SG in Class I weak verbs³³

- a. *sat-jip* [('sat).ji(θ)] 'sets'; *sto-jip* 'judges' [('sto:).ji(θ)]
- b. *sand-eip* [('san).(di:)(θ)] 'sends'; *full-eip* [('ful).(li:)(θ)] 'fill'
- c. *fod-eip* [('fo:).(ði:)(θ)] 'feeds'; *brukeip* [('bru:).(ki:)(θ)]

³¹I tentatively assume, following Kiparsky 1998b, that consonants at the right edge of the word are extrametrical, with the consequence that final [VC] sequences are treated as light.

³²Whether successive sequences of heavy syllables, or a L L sequence after a heavy syllable, were always footed depends upon the relative rankings of WSP, PARSE-Syllable, ALL-Ft-L, and *CLASH. Whether stress clash was categorically avoided in Gothic or not is difficult to say. I assume here, without real consequence for the succeeding discussion, that Gothic satisfied WSP and PARSE-Syllable, in the face of resulting violations of ALL-Ft-L and *CLASH.

³³The same alternation of *-ji-* ~ *-ei-* is found in the NPST.2SG, ending in *-jis* or *-eis*, and the NPST.2PL, ending in *-jip* or *-eip*.

d. *riqiz-eip* [(‘ri.kʷi.).(zi:)<θ>] ‘becomes dark’; *mikileid* [(‘mi.ki.).(li:)<θ>] ‘magnify’

The members of both of these morphological classes may thus be divided into four prosodic structures based on the segments preceding the vowel [i] or glide [j] at the right edge of the stem, summarized in 19. In traditional terms, the shapes in a. constitute “light” stems, while b.–d. constitute “heavy” stems.

- (19) Prosodic shapes of stems preceding a suffix /-j-/³⁴
- a. /CVC-j-/ or /CV:-j-/;
 - b. /X₀CVCC-j-/;
 - c. /X₀CV:C-j-/;
 - d. /CVCVC-j/.

The basic distribution of the sequences [ji] ~ [i:] then appears straightforwardly straightforwardly explicable: if the sequence preceding the suffix /-j-/ can build a single bimoraic syllable, *-jis* / *-jip* occurs; in all other cases, *-eis* / *-eip* occurs. The behavior of forms like *ragineis* and *riqizeip* crucially shows that this pattern is motivated by foot structure: two light syllables – e.g., [(‘ra.ʏi).(ni:)<s>] – build a moraic trochee just as a single heavy syllable does – e.g., [(‘hər).(di:)<s>]. In contrast, a hypothetical ^x[(‘ha.ri:s] ‘army.NOM/GEN.SG’ would contain a light (monomoraic) syllable preceding the final syllable, whereas actually occurring [(‘har.jis] has a heavy (bimoraic) first syllable. Only words with an initial light–heavy sequence, like [(‘si.po:).(ni:)<s>] ‘disciple’ in (17)d. constitute an exception to foot bimoracity, but still fit the generalization that the first two syllables can build a licit – if dispreferred – foot shape.

Under the analysis of Kiparsky (1998a), this Siever’s Law pattern is thus driven by a preference for bimoraic feet, an avoidance of [Cj] onsets, and the preservation of all underlying segments. For instance, given GEN.SG /hird-j-is/ ‘shepherd’, the alternative X[(‘hər).dji<s>] is dispreferred because it contains a [Cj] onset, while X[(‘hərd).ji<s>] would contain a trimoraic syllable, and X[(‘hər).ji<s>] deletes an underlying /d/. The actually occurring form, [(‘hər).(di:)<s>] instead vocalizes underlying /j/. Given GEN.SG /har-j-is/ ‘army’, actually occurring [(‘har).ji<s>] is able to build a bimoraic foot at the left edge.³⁵

³⁴Whether this suffix is best represented as /-j-/, /-i-/, or a segment underspecified for the feature [+syllabic] (as assumed in Kiparsky 1998a) is not entirely clear. For concreteness, and because nothing crucial hinges on the choice here, I adopt the UR /-j-/ for the stem formant of *ja*-stems and Class I weak verbs.

³⁵The endings for the 2/3SG and 2PL forms of Class I weak verbs rigorously adhere to the distributions attributable to foot structure, and the masculine *ja*-stems attest no clear exceptions. Among neuter *ja*-stems, however, one often finds *-jis* where *-eis* would be expected: the two attested light stems, *frapi* ‘mind’ and *kuni* ‘race’, always have expected [-jis], while heavy stems predominantly have [-jis] (33/48 tokens, 15/23 types), less commonly [-i:s] (15/48 tokens, 8/23 types); see the data in Kim 2001: 103. Thus, expected [-i:s] is attested more rarely, but thankfully only on “heavy” bases (e.g., *andbahteis* ‘service:GEN.SG, where it would be historically expected. For accounts of these exceptions, see Schmierer 1977: 36–9, Vennemann 1985: 200–6, and Kim 2001.

Verbal and Nominal Prefixes As in Old English, good reasons exist to believe that prefix elements in Gothic exhibited different prosody depending upon the lexical category of the base. Perhaps the most unmistakable phonological evidence for a distinction between nouns or adjectives and verbs in terms of their stress patterns when prefixes were present comes from a process of vowel syncope that can be employed in order to unify the different allomorphs of a prefix that occurs on both nouns/adjectives and verbs. The allomorphs [anda-] and [and-], which still exhibit a perfect complementary distribution in Gothic, and [ϕɔra-] and [ϕɔr-], whose distribution remains largely complementary (cf. Miller 2019: 321), may be unified to /anda-/ ‘against, to’ and /ϕura-/ ‘before’ through a process of stress-conditioned apocope.³⁶ This process may be formulated restrictively as in (20), and used to derive the forms in (21) (contrast the forms in (22)).³⁷

(20) PRETONIC *a*-DELETION

$$\left[\begin{array}{c} a \\ -\text{stress} \end{array} \right] \rightarrow \emptyset / \left[\begin{array}{c} +\text{vocalic} \\ 2\text{stress} \end{array} \right] C_o [-\text{nasal}] \text{ — } C_o \left[\begin{array}{c} +\text{vocalic} \\ 1\text{stress} \end{array} \right]$$

An unstressed /a/ is deleted when immediately following a non-nasal consonant, standing between a vowel bearing secondary stress to its left and primary stress to its right.

(21) Verbal *and-* and *faur-*

- a. *andbeitan* [(and).(bi:).ta(n)] ‘blame:INF’
- b. *faurbiudan* [(ϕɔr).(bū).ða(n)] ‘forbid:INF’

(22) (35) Nominal *anda-* and *faura-*

- a. *andanahti* [(an).da.(nah).ti] ‘evening:NOM/ACC.SG’
- b. *fauradaurja* [(ϕɔ.ra).(ðɔr).ja] ‘street:ACC.PL’

This formulation of Gothic *a*-syncope will derive [and-] and [ϕɔr-] from /anda-/ and /ϕura-/ , under the hypothesis that the root syllable of a verb will receive primary stress and the leftmost vowel of the prefix a secondary stress. Including the condition [–nasal] preceding the /a/ can avoid application to /ana-/ ‘upon, over’.³⁸

The allomorphs [anda-] and [and-] may then be derived in parallel as in 23.

³⁶One nominal form, *andbahts* ‘servant’, would appear at first glance to disrupt the complementary distribution of nominal [anda-] and verbal [and-], but Bennett (1970: 469) plausibly argues that this form is a loanword from a Continental Celtic language, [‘ambaxtos], in which the first syllable became [and-] by folk etymology (cf. further Greiner 1994: 70).

³⁷When a prefix element is still available synchronically as a preposition, as is the case for *ana* ‘upon’ and *wipra* ‘against’, it is not subject to PRETONIC *a*-DELETION; see Greiner 1994: 78–9. That the preverb bore secondary stress on its first syllable is supported by the conditions for the application of high-vowel lowering (“Breaking”: /i, u/ → [ɛ, ɔ] / ___{r, h, ʌ}), which only occurs under stress. Consider the compound *fidurdags* [‘fiður.do:xs] ‘(of) four days’, where the /u/ in the second syllable is not subject to lowering; see further Greiner 1994: 112–8.

³⁸One possible source of underapplication comes from longer preverb sequences, in which preverbs such as *ga-* and *fra-* occur as the second preverb from the left, e.g., *mip-ga-timrjan* ‘build together’, *mip-fra-hinþan* ‘take prisoner with’ (but note that *fra-* as a second preverb occurs only in this verb, which is attested exclusively as a past participle, and so its status as a verb is somewhat questionable). Furthermore, the sequences *faur-bi-*

(23) Derivation of allomorphs [anda-] and [and-]³⁹

/[anda-ne:m-s] ‘pleasant’/	/anda-[nim-an]/ ‘receive, take:INF’	UR
'andane:ms	anda'niman	Primary Stress Assignment
'anda,ne:ms	,anda'niman	Secondary Stress Assignment
—	,and'niman	<i>α</i> -DELETION
'anda,ne:ms	,and'niman	SR

Thus, while Gothic does not exhibit extensive sets of prefix allomorphs as in Old English or Old High German, the limited process of *α*-DELETION in verbal forms strongly suggests that verbs regularly bore primary stress on the verbal root. If so, one means of enforcing primary stress on the root of a verb could be to require the left edge of stems belonging to the syntactic category of VERB to be aligned with the left edge of a minimal prosodic word (cf. the constraint LEX-TO- ω (L/R) in Ito and Mester 2009: 142, “every lexical word is right/left aligned with a prosodic word”), within which primary stress is determined.

Reduplication If indeed, Gothic requires for stems belonging to the syntactic category of VERB to have their left edge aligned with the left edge of a prosodic word, a resulting prediction is that the reduplicant should bear primary stress. Evidence in this regard comes from the behavior of some reduplicated stems with respect to the process of HIATUS LOWERING, which indicates that the root syllable in reduplicated forms was did not bear primary stress. The process of HIATUS LOWERING transforms underlying tense vowels /e:/ and /o:/ into lax vowels [ɛ:] and [ɔ:] when immediately preceding another vowel, as formulated in (24).

$$(24) \text{ HIATUS LOWERING } \left[\begin{array}{c} +\text{syllabic} \\ +\text{long} \\ +\text{stress} \end{array} \right] \rightarrow [-\text{tense}] / \text{ ___ } [+ \text{syllabic}]$$

Per Greiner (1994: 122–5) and Miller (2019: 43), the condition that HIATUS LOWERING applies only in [+stress] syllables is necessary in order to explain its failure to apply in the PST.3PL forms *lailoun* ['lelo:un] ‘reviled’ (not ^X[lɛ'lo:un]) and *waiwoun* ['wɛwo:un] ‘blew’ (not

and *faura-ga-* are both found. Assuming stress patterns [fɔrbi'gaggan] ‘precede’ and [fɔragɑ'ti:han] ‘predict’, [fɔrbi'gaggan] would show overapplication of syncope. If, on the other hand, the rule in (20) were adjusted to apply between to secondary stresses as well, and the stress patterns [fɔr,bi'gaggan] ‘precede’ and [fɔra,ga'ti:han], then [fɔra,ga'ti:han] would seem to suffer from underapplication; furthermore, since secondary stress on *ga-* would entail a degenerate foot, assuming as much is *a priori* unlikely. It is conceivable that the verb *faura-gateihan* and similar such verbs are in fact compounds, and thus stressed as [fɔragɑ'ti:han]. To ensure application in cases like *faurbigaggan*, (20 could be rewritten as follows, with an optional CV sequence following the target:

$$\left[\begin{array}{c} a \\ -\text{stress} \end{array} \right] \rightarrow \emptyset / \left[\begin{array}{c} +\text{vocalic} \\ 2\text{stress} \end{array} \right] C_0 [-\text{nasal}] \text{ ___ } (CV) C_0 \left[\begin{array}{c} +\text{vocalic} \\ 1\text{stress} \end{array} \right]$$

³⁹Square brackets here demarcate the domain of primary stress assignment. For the moment, I assume that primary stress is assumed to the leftmost syllable within the appropriate domain, and that secondary stress is assigned as predicted by the construction of moraic trochees.

^x[wɛ'wɔ:un]). Essentially, since final stress is improbable in such forms, and in forms in which HIATUS LOWERING clearly does apply, the affected vowel must be stressed (e.g., /se:an/ → ['sɛ:an] *saian* 'sow:INF'; cf. ['mana:se:θs] *mana-seþs* 'man-seed = mankind'), it follows that the vowel of the root syllable in *lailoun* and *waiwoun* is unstressed, leaving the reduplicant as the only possible position where primary stress could fall in such forms. The further implication is that reduplicants are indeed internal to the verbal stem, and prosodically distinct from the prefixes that are lexically affiliated with adverbials.

Compounding The reconstructable stress pattern of Gothic compounds – which by definition consist of two stems that can occur as independent morphological and prosodic words – indicates that each member of the compound built its own prosodic word, which was then merged under a larger prosodic structure, perhaps a recursively built prosodic word (see Ito and Mester 2007 on such balanced recursion in compounds) or perhaps a phonological phrase. The general prosodic structure of a compound with two members was as in (25):

(25) Prosodic Structure of Gothic Compounds

[[X]_{ω-min} [Y]_{ω-min}]_{ω-max/φ-min}

In compounds with more than two members, more non-minimal prosodic words or non-minimal phonological phrases would have been constructed in alignment with the morphological structure of the compound, but whether and exactly how that difference in prosodic structure would have been concretely manifested is less clear. In common two member compounds, primary stress of the compound fell on the syllable with primary stressed syllable of the leftmost member, and a secondary stress fell on the syllable of the second member that would be primary stress as a simplex. This fact is principally to be discerned from the occurrence of [ɛ] and [ɔ] resulting from BREAKING (high vowel lowering), which occurs only under primary or secondary stress, in the second member of compounds. See the data in (26)

(26) BREAKING in Compound Second Members

- a. *asilu-qairnus* ['asilu,kʷɛrnus] 'millstone (turned by a donkey):NOM.SG', lit. 'donkey-millstone'
- b. *auga-dauro* ['ɔ:ɣa,ðɔro:] 'window:NOM/ACC.SG', lit. 'eye-door'
- c. *gabaurpi-waurd* [ga'βɔrθi,wɔrd] 'genealogy:NOM/ACC.SG', lit. 'birth-word'
- d. *guda-faurts* ['guða,φɔrts] 'god-fearing: NOM.SG.M', lit. 'god-fear'

In derivational terms, one may conceive of the stress pattern that arises in compounds as the assignment of a stress to each ω_[+min], followed by either promotion of the leftmost stress to primary stress for the ω_{max}, or demotion of the primary stress in a ω_[+min] to secondary stress. The stress and prosodic structure of compounds holds further implications for the interpretation of prefixed verbs. Given that prefixed verbs such as *urreisan* [ur'ri:san] 'arise' or *andbindan* [and'bindan] 'release' do not exhibit primary stress on the prefix, such prefixes cannot be analyzed as building the same type of prosodic structure as compounds, since the

resulting prediction would be that stress in prefixed verbs would behave identically to stress in compounds: primary stress on the leftmost $\omega_{[+min]}$.

Enclitics This section concerns the behavior of two Gothic particles that consistently appear to behave as enclitics, in contrast to the apparently proclitic preverbs.⁴⁰ These elements include:

- the conjunction *-uh* ‘and’;
- and the question particle *-u*.

One of the most remarkable features of Gothic in comparison to the later-attested Germanic languages is these enclitic elements may intervene between verbal “prefixes” and the inflected verbal stem. These enclitics, *-uh* and *-u*, would thus seem to attach to the first prosodic constituent inside whatever prosodic constituent is built by the entire prosodic complex corresponding to some syntactic domain (on the syntax of these clitics, see Miller 2019: 511–19). If a verb is not preceded by any prefixes, the enclitic follows the stem and any inflectional suffixes, as shown in (27); I tentatively designate the stem + inflection as a $\omega_{[+min]}$ and treat the enclitic as an internal clitic (in the typology of Anderson 2005: 46), meaning that no prosodic word recursion applies, and the $\omega_{[+min]}$ is equivalent to a $\omega_{[+max]}$ in such cases.

(27) Enclitic Joined to a Simplex Prosodic Word

- magutsu* [[‘mayuts-u] $_{\omega-min}$] $_{\omega-max}$ ‘be.able :NPST.2DU-QUESTION’
- qīpiduh* [[‘kʷiθið-uh] $_{\omega-min}$] $_{\omega-max}$ ‘say:NPST.3SG-and’
- wileizu* [[‘wili:z-u] $_{\omega-min}$] $_{\omega-max}$ ‘want:NPST.2SG-QUESTION’

Conversely, if at least one “prefix” precedes the verbal stem + inflection complex, the enclitic attaches to the right of the first morphological element, as shown in (28). In this way, *-uh* or *-u* either attaches to or builds a domain for secondary stress with another linearly preverbal element. I continue to label the entire complex as a $\omega_{[+max]}$, without precisely designating the prosodic constituency of the preverbal complex. Thus, when a verb with preverb is clause-initial, the enclitic attaches after the preverb; schematically, the structure would be [[σ-u(h)]? ...[VERB] $_{\omega-min}$] $_{\omega-max}$

(28) Enclitic Joined to a Complex Prosodic Word

- gahmelida* [,ga-h-[‘meliða] $_{\omega-min}$] $_{\omega-max}$ ‘TELIC-and-write:PST.3SG’ ([-h-] here = /-uh)
- gauhvasehi* [,ga-u-hva-[‘semi] $_{\omega-min}$] $_{\omega-max}$ ‘TELIC-QUESTION-INDEF-see:PST.OPT.3SG’

⁴⁰I leave aside the sentence discourse particles such as *ibai*, which marks rhetorical questions, *an* ‘then’, *ip* ‘but’, *pan* ‘then’, *nu* ‘now’, or *auk* ‘for’, which sometimes appear to be enclitic, constituting part of a chain including a proclitic preverb and the enclitic *-uh* or *-u* (e.g., *diz-uh-pan-sat* ‘apart-and-then-sit.PST = and then seized’), but which can also host the pure enclitics in clause initial position (e.g., *pan-uh* ‘then-and’).

c. *uzuhiddja* [uz-uh-['id:ja]_{ω-min}]_{ω-max} 'out-and-go:PST.3SG'

Notice crucially here that, in examples (27)b. and (27)c., as well as (28)c., an underlying voiced fricative or stop (/b, d, g, z/) surfaces as voiced when followed by one of the enclitics *-uh* or *-u*. Typically, when one of these voiced segments occurs at the end of a prosodic word, in the case of inflectional endings, or, in the case of preverbs, if followed by another preverb or the verbal stem, it becomes voiceless, regardless of what the following segment is (cf. Miller 2019: 26–8 on this final devoicing of fricatives, where the voiced stops are first spirantized in the environment); see examples for fricative devoicing in 29. Thus, the preverb~preposition /uz/, for instance, frequently surfaces as [us], e.g., *us dauþaim* 'from the dead', *us-iddjan* 'go out:INF'. The question is then: what is the relevant prosodic boundary for fricative devoicing in Gothic? Evidently, *-uh* can go inside of a prosodic constituent in a fashion that blocks the application of fricative devoicing.

(29) FRICATIVE DEVOICING in Gothic

- a. /gab/ → ['gaɸ] *gaf* 'give:PST.1/3SG' (cf. *giban* ['giβan] 'give:INF')
- b. /go:d/ → [go:θ] *gop* 'good:ACC.SG' (cf. *goda* ['goða] 'good:ACC.PL')
- c. /mɛ:z/ → [mɛ:s] *mais* 'more' (cf. *maiz-uh* [mɛ:zuh] 'more-and')

Just as important to note is where enclitics may not attach. Specifically, *-uh* and *-u* may intervene between a preposition and a noun internal to a prepositional phrase, but there is no positive evidence that it these clitics could intervene between a prefix a nominal stem.⁴¹ Compare (28)c. with the following examples (where b. is inferred):

(30) (Non-)Attachment of Enclitics on Prepositions and Nouns

- a. *uz-uh himina* 'from-and heaven:DAT.SG
- b. *uzetin-u** 'manger:DAT.SG-Q' (^x*uz-u-etin* 'out-Q-eater')

Though the nature of the evidence does not allow for complete certainty, I cautiously infer from the absence of any evidence for intervention of clitics after a prefix on a nominal stem that no prosodic boundary was present between the prefix and the nominal stem. The further conclusion to draw is then that the lexically identical prefixes on nouns are likely incorporated into the prosodic word with the noun, i.e., *uzeta** 'manger' builds a $\omega_{[+min]}$: [('u.ze:).ta]_{ω-min}.

Summary on Gothic Stress and Prosodic Structure In light of the evidence reviewed above, the following relatively secure conclusions on Gothic stress and prosodic structure may be drawn:

⁴¹Note that *-uh* seems to be unable to occur on lexical nouns whatsoever. Rarer *-u* is attested as a clitic on pronouns and adjectives, though never on a lexical noun. See the complete data on the latter in Jones 1958.

- Primary stress is assigned to the leftmost foot (a moraic trochee) of a prosodic word. Primary stress on the first syllable of words with an initial light–heavy sequence (e.g., *liteinins* [(‘li.ti:).(,nins)] ‘supplication.ACC.PL’) may be accounted for by requiring that the left edge of a prosodic word be aligned with the left edge of a foot, in violation of the preference for maximally bimoraic feet (cf. 7.2.2.1 above on this behavior in Old English).
- In noun–noun compounds such as [[(‘wiðu)]_{ω-min} [(,wēr).(nan)(s)]_{ω-min}]_{ω-max} ‘orphan:ACC.PL’, each member of the compound builds a minimal prosodic word, and thus receives stress on its leftmost syllable, but the stress of the second member is “demoted” to a secondary stress. That is, primary stress in a non-minimal prosodic word falls on the leftmost $\omega_{[-max]}$, i.e., a prosodic constituent eligible to be the head of the $\omega_{[+max]}$.
- Nominal stems modified by a preposition allow the preposition to incorporate into their prosodic word when they together constitute a morphological word, and thus prosodically allow the resulting stems to behave identically to non-prefixed stems.
- Verbal stems with a prepositional prefix do not allow for the prefix to incorporate into the prosodic word. As shown by their susceptibility to the intervention of enclitics, prefixed verbs and prepositional phrases plausibly built identical prosodic structures.

As mentioned above, Gothic possesses some segmentally homophonous lexemes, such as *faura* ‘along; before’ with different syntactic categories of preposition and adverb. As the minimal stress pair (modulo stress-conditioned deletion of [a] in the latter) *fauragaggan** [‘fɔra.ganɡan] ‘go in front of’ versus *faurgaggan** [‘fɔr‘ganɡan] ‘pass by’ shows, the adverb could enter into a compound with a verb, ([‘fɔra.ganɡan]*), certainly constructing its own minimal prosodic word. The verb with preposition incorporation, on the other hand, seems to have necessarily built a distinct prosodic structure, so as to yield primary stress on the verbal stem and secondary stress on the prefix, as in ([‘fɔr‘ganɡan]*).

One means of accounting for primary stress on the root syllable of prefixed verbs could be through a requirement that morphs belonging to the syntactic category of VERB must have their left edge aligned with the left edge of a prosodic word, that prosodic words may be built recursively, and that a minimal prosodic word contains at least one stress, while a maximal prosodic word contains at most one primary stress (see Ito and Mester 2009). Under this account, verbs such as *andbindan* ‘untie:INF’ have the prosodic structure [[(,and)]_σ [(‘bin).da(n)]_{ω-min}]_{ω-max}: the preverb *and-* is incorporated into a recursively built prosodic word (as an affixal clitic), adjoined as a sister to the minimal prosodic word built around the verbal root *bindan* (see further 7.2.4 below). The structure of “prefixed” verbs like *ur-reisan* [ur‘ri:sa] ‘arise’ or *andbindan* [and‘bindan] ‘release’ is thus (schematically): [σ [VERB]_{ω-min}]_{ω-max}, where σ corresponds to the “prefix”. Since most of these prefixes either form a heavy syllable or are disyllabic, they may form a foot as well, resulting in secondary stress.

7.2.2.3 Old High German

The behavior of stress assignment in Old High German (specifically, the variety reflected in the text of *Isidor*) may be summarized very quickly, since its detailed analysis in Greiner 1994: Ch. 4 indicates that it was largely identical to the Gothic system. The one notable point of difference may be that, while verbal prefixes in Gothic seem to have carried secondary stress, the evidence in Isidorian Old High German indicates that such prefixes were unstressed in that language; regardless, both languages (and Old English generally as well) agree that the primary stress fell on the verbal root (or the reduplicant in Gothic, when present). Although some degree of variation in manuscripts occurs (see further Greiner 1994: 176–81), OHG shows a complementary distribution of phonologically and semantically related prefix allomorphs, where the verbal prefixes exhibit only the vowels *a* and *i* that are permissible in unstressed syllables (cf. Braune 1975: 72 on OHG prefixes).

- (31) Preverbs and Nominal Prefixes in OHG (after Greiner 1994: 182; cf. also Schmidt et al. 2013: 236)

Preverb	Nominal Prefix
<i>ar-</i>	<i>ur-</i>
<i>bi-</i>	<i>bi-</i>
<i>chi-</i>	<i>chi-</i>
<i>fir-</i>	<i>fora-</i>
<i>in-</i>	<i>ant-</i>
<i>zi-</i>	<i>zuo-</i>

At least in the case of *ar-* vs. *ur-* and *zi-* vs. *zuo-*, the preverb may be readily derived by vowel reduction in a pretonic syllable. For the pairs *fir-* vs. *fora-* and *in-* vs. *ant-*, additional phonological processes (*a*-DELETION, *t*-DELETION) would need to be assumed in order to posit unitary URs, and in the former case, semantic drift makes a single lexical entry perhaps less plausible as well. Regardless, the sum evidence indicates a reduced inventory of licit surface vowels in a presumably pretonic context.

As in Old English, the prefix *ga-* appears to have been systematically unstressed, whether in verbal or nominal formations; the prefix *za-* likewise shows an unexpected systematic unstressedness. Both prefixes consequently exhibit consistent vowel reduction to *gi-* and *zi-*, respectively. Beyond the behavior of prefixes, the persistent presence of primary stress on the leftmost syllable (again, except in prefixed verbs or the two just-mentioned prefixes) is evidenced by reduction of final syllables to [ə] (typically represented orthographically as ⟨e⟩), and the tendency for peninitial (thus post-tonic) short vowels in open syllables to assimilate in quality to the vowel of a neighboring (most commonly following) vowel.

7.2.2.4 Old Norse

In its broad outlines, the position of primary stress in Old Icelandic (Old West Norse; cf. Noreen 1923: §51) and Old Swedish (Old East Norse; Noreen 1904: §57) generally resembles the behavior found in Old English, Gothic, and Old High German as described above: pri-

mary stress fell on the leftmost syllable of a prosodic word, and in compounds, on the leftmost syllable of the first member, while prefixed verbs and their derivatives exhibited stress on the verbal root.⁴² However, certain derivational suffixes, and a not insignificant number of scattered monomorphemic lexemes and compounds show evidence for primary stress on the derivational suffix, or variation in accentuation. At least in the case of the derivational suffixes *-ing-* and *-ung-*, the data suggests that these suffixes attracted stress when the root syllable was light, but root-stressed variants could become heavy through gemination of a consonant.

- (32) Second syllable primary stress with suffix *-ing-* in Old Swedish (Noreen 1904: §57.I.B)
- a. [twi.'liŋ.ɡer] 'twin'
 - b. [pæ.'niŋ.ɡer] 'penny' as well as ['pæn.niŋ.ɡer]

Such data may suggest a tendency to avoid the WSP-violating ('L H) feet of the sort posited for Old English and Gothic above. Instead, to avoid a violation of WSP, primary stress was assigned to the leftmost heavy syllable, or alternatively, the initial syllable was made heavy. On the strength of the agreement between West Germanic and Gothic, however, such a treatment of /L H/ sequences in North Germanic should probably be regarded as a prosodic innovation vis-à-vis Proto-Germanic.

At the same time, analyses of the Old Icelandic *dróttkvætt* tend to operate with the same assumptions about stress generally made for Old English, and most crucially, it is assumed that a light syllable under primary stress could be resolved with a following syllable, regardless of its syllable weight (Gade 1995: 38–40). If it is correct to treat the principal alliterative meters of Old Icelandic and Old English as essentially identical in structure and behavior, then the word prosody of Old Icelandic must have been fairly similar in detail to the word prosody of Old English. At minimum, it could safely be assumed that all non-prefixed prosodic words bore primary stress on the initial syllable.

7.2.3 ROOTSTRESS is Unsatisfactory

In the immediately preceding sections, various analyses of stress assignment in older Germanic languages assess the role of the lexicon, morphology, and phonology in stress assignment differently. Concerning Old English, for instance, analyses like Hutton's (1995: 231–51) ascribe roots an underlying property of accentedness: they attract stress. Conversely, analyses like those in Minkova 2008 and Goering 2016a on Old English determination of primary and secondary stress purely via foot structure, without invoking any prespecified lexical prosody. The question pursued in this section is: is it necessary to outfit roots in Proto-Germanic with some sort of privileged status in order to correctly derive the assignment of primary stress?

⁴²Noreen (1904: §57.IA) characterizes the situation involving prefixed forms rather imprecisely, saying that the prefixes *be-*, *of-*, *æm-*, *for-*, *and-*, *mis-*, and *unt-* were always unstressed (regardless of the syntactic category of the root), while the data from prefixes that were sometimes stressed, sometimes unstressed (e.g., *til-*) indicates that the unstressed form occurred on verbs.

I propose here that the answer is “no”. As the data from the languages surveyed immediately above shows, the only class of lexemes that regularly and systematically exhibited primary stress on a non-initial syllable were prefixed verbs of the type represented by Gothic *andbindan* [and'bindan] ‘release’ (cf. Germ. *entbinden*). As will be argued in 7.2.4 immediately below, the stress pattern found in such prefixed verbs may be accounted for by analyzing these “prefixes” as a species of proclitic – they are prepositions and particles adjoined to a minimal prosodic word $\omega_{[+min]}$ that result in prosodic word recursion. That verbal prefixes in Germanic are not fully incorporated into the $\omega_{[+min]}$ is further unsurprising from a historical and comparative point of view: preverbs that alter the lexical semantics of a verb in Vedic Sanskrit and Homeric Greek may and not infrequently do construct apparently independent prosodic words, with other independent prosodic words intervening between them (so-called *tnesis*).

As argued with respect to the Old English data under 7.2.2.1 above, to assume that roots in general, regardless of their syntactic category, generally attracted stress, whether through lexical marking (as per Hutton 1995) or a high-ranking constraint that prefers root stress (as per Getty 2002, and also in a somewhat different form in Minkova 2008), can lead to the unwanted prediction that roots systematically attract stress. Outside of verbal formations, however, or nominal derivatives from prefixed verbs, such a prediction is evidently false. The consequence is, then, that lexically stressed versions of prefixes must be assumed to exist, which are categorically restricted to occurring with nouns and adjectives.

Although the idea of notion of some kind of highly ranked ROOTSTRESS as an overriding feature of stress grammars in some languages has been argued to hold for Russian, Japanese, and Cupeño by Alderete (2001a), these systems are open to alternative analyses that do not require any particular faithfulness to the lexical stress of roots (see Melvold 1989 and Revithiadou 1999 on Russian, and Yates 2017 on Cupeño); see the introduction to the notion of head faithfulness at 2.4.2.3. However, the particular cases treated by Alderete, under his analysis, exhibit *faithfulness* to the lexical accents of roots, not any tendency to attract stress to roots *per se*. Indeed, an analysis of the common Germanic stress system does not require any ROOTSTRESS (compare again the constraint ROOTMOST employed by Minkova (2008)) whatsoever. If verbal roots are generally assumed to be lexically accented, then the simple ranking MAX-Accent $\gg$ {ALIGN- ω -L, LEFTMOST} given any underlying distinction between unaccented preverbs and accented nominal prefixes, would work equally well. Alternatively, the preverbal prefixes could be lexically marked as *post-accenting* (e.g., /and(-/), with a left foot edge following the last vocalic peak of the prefix). Either way, some amount of lexical specification of prosodic properties, and a complete lexical separation of preverbs and nominal prefixes, despite overwhelming phonological and semantic similarity (especially at the Proto-Germanic level) would be necessary in order to account for the facts, if roots are presumed to somehow attract primary stress generally.

7.2.4 Salvaging Leftmost Stress

Based on the evidence in the earliest-attested Germanic languages, and in particular the extensive agreement in many details found between Old English and Gothic, a prosodic

grammar of minimal prosodic words that consistently assigns primary stress to the leftmost syllable of the word can be reconstructed. For the $\omega_{[+min]}$ in which the root is either coextensive with the leftmost syllable, or in which any prefixed material is incorporated into the $\omega_{[+min]}$, the stress grammar stated in (33), essentially identical to the basic stress grammar of Old English posited above, will suffice. This grammar constructs a bimoraic moraic trochee, or trochee of the shape ('L H), at the left edge of the prosodic word, and otherwise perfect moraic trochees.⁴³ Aspects of final syllable and consonant extrametricality remain rather uncertain, but seriously affect only secondary stress. In light of the evidence for shortening of long vowels in final syllables, and tendency of vowels in final closed syllables to weaken and centralize, I assume here that final heavy syllables were unstressed, driven by NONFINALITY- σ , though such stresslessness may have been variable (see the discussion concerning Old English final syllables above).

- (33) Grammar of Prosodic Structure in the $\omega_{[+min]}$ of Proto-Germanic:
 LEFTMOST, TROCHEE, ALIGN- ω -L, RH-CONTOUR, FTBIN $\gg$ PARSE-C, NONFINALITY- σ
 $\gg$ WSP $\gg$ PARSE- σ $\gg$ NONFINALITY- $\mathcal{F}$, *CLASH, ALL-Ft-L

The Proto-Germanic grammar for primary stress may be typologically compared and contrasted with the stress system of Nuu-chah-nulth (or “Nootka”; Wakashan, British Columbia). Nuu-chah-nulth constitutes a good example of a language with a prosodic system exhibiting left-edge-oriented moraic trochees, but in which the reverse ranking of ALIGN- ω -L relative to FTBIN and/or WSP occurs. Syllable weight in Nuu-chah-nulth is determined as in most older Indo-European languages, and, like in Proto-Germanic, primary stress is assigned to the initial syllable, except that it is instead assigned to the peninitial syllable when the leftmost syllable is light and the peninitial syllable is heavy. This behavior is expected when primary stress falls on the leftmost moraic trochee, but the construction of a degenerate or overlong foot to enforce the alignment of the left edge of the prosodic word with the left edge of the head foot is not permitted, thereby resulting in the “skipping” of the initial syllable in such cases.

- (34) Stress Assignment in Nuu-chah-nulth (cf. Wilson 1991: 249)
- a. Heavy–Heavy : ('ʔo:).(sim).fʰs'a
 - b. Heavy–Light : ('ʔin).k'i 'the fire'
 - c. Light–Light : ('ti.qʷi).lʔa 'now he sat down'
 - d. Light–Heavy : t'a.(ʰsi:).(ʔa.k'i) 'his door'
- Not ('t'a.si:).(ʔa.k'i), violating WSP, or ('t'a).(ʰsi:).(ʔa.k'i), violating FTBIN.

In the remainder of this section, I argue instead that only one additional and well-justified assumption is needed in order to account for counterexamples to the general phonological algorithm of that generates leftmost primary stress: prefixes on verbs were not part of

⁴³Nonetheless, there is some evidence within the individual Germanic languages for repairs of initial /L H/ sequences to [L L] or simply [H] (by deletion of the second vowel); various suffixes in Old and Middle High German present evidence for the shortening or deletion of vowels when [(L H)] would result, and Gothic and OHG sometimes show shortening of a heavy peninitial syllable in loanwords from Latin (e.g., Lat. *catēna* > OHG *ketīna* 'chain'). See generally Roethe 1919 and d'Alquen 1988: 20–3 on this evidence.

the same minimal prosodic word as the verbal stem ($\omega_{[+min]}$), but rather incorporated into a larger, recursively built prosodic word ($\omega_{[-min]}$). This assumption takes for granted that prosodic words (ω) in Proto-Germanic could be constructed recursively (as under the MIN-MAX model of prosody in Ito and Mester 2009, Ito and Mester 2012), and while OBLIGATORINESS was enforced at the level of the minimal prosodic word ($\omega_{[+min]}$), CULMINATIVITY was enforced in any larger prosodic words (see Sections 1.2.1 and 1.2.2).

With respect to prefixed verbs, the prefix is to be viewed as a proclitic, where the verb without prefix already constitutes a minimal prosodic word, and thus receives primary stress on its leftmost syllable. Since the “prefixes” are, at least at the Proto-Germanic level, and for the most part still relatively clearly in Gothic, function words, they are dispreferred as heads of non-minimal prosodic words. Thus, even if a “prefix” is able to build a bimoraic foot and receive stress, its stress will remain subordinate to the stress of the content word, the verb. Practically speaking when a prefix induces prosodic word recursion (forming a $\omega_{[-min]}$), the derivation remains faithful to the primary stress that has already been assigned at the $\omega_{[+min]}$ of the lexical verb.

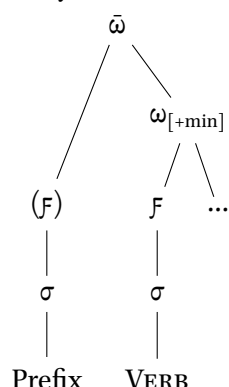
The fundamental problem then remains: what prosodic structure do the Germanic verbal “prefixes” construct? The answer hinges in part on the structure assumed for compounds as well, where the latter, consisting of at least two minimal prosodic words, have primary stress on the leftmost syllable of the entire compound. Taken together, two internally coherent analyses may be considered:

- **Analysis 1:** Proto-Germanic compounds exhibit balanced recursion, building a non-minimal prosodic word consisting of multiple minimal prosodic words. Prefixed verbs, on the other hand, have the structure of affixal clitics (in the sense of Anderson (2005: 46)), with a lower-level prosodic category (σ or $\mathfrak{f}$) adjoining to a $\omega_{[-min]}$.
- **Analysis 2:** Proto-Germanic compounds are phonological phrases, $\phi_{[+min]}$, built out of two or more prosodic words. Prefixed verbs, on the other hand, exhibit balanced recursion (in most cases), building a non-minimal prosodic word consisting of multiple minimal prosodic words.

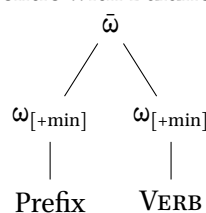
The prosodic structures shown in (35) show the two options considered for the prefixed verbs.

(35) Plausible Prosodic Structures for Proto-Germanic Prefixed Verbs

a. Analysis 1: Affixal Proclitic



b. Analysis 2: Prosodic Word Proclitic with Balanced Recursion



A choice between these two options is not straightforward to make. The difference in position of primary stress between compounds and prefixed verbs is easier to explain in the first case: because the preverbs do not project their own minimal prosodic word, their stress remains subordinate to the stress of the one minimal prosodic word. In the latter case, while the left-member stress of compounds may be understood as a phrasal stress rule, the stem stress of prefixed verbs would need to be attributed to some other factor, such as a preference for primary stress on content words rather than function words, or a rightmost orientation for stress in cases of balanced recursion. If Analysis 1 allows the stress facts to fall out more readily, some of the Gothic facts may speak in favor of Analysis 2: 1) the enclitics *-uh* and *-u* can certainly attach to a $\omega_{[+min]}$, but also to a verbal prefix; 2) fricative devoicing applies both at the right edge of a typical $\omega_{[+min]}$, but also at the right edge of a verbal prefix. Until the latter facts of Gothic and exactly which prosodic (or in the case of the clitics, syntactic) domains are relevant are better clarified, I will tentatively adopt Analysis 1 here, since it simplifies the treatment of the facts relating to the position of stress. Analysis 2 remains, however, a genuine possibility, and would fundamentally agree with the analysis proposed for Old English prefixed forms by Minkova (2008).

The first crucial distinction to derive is the consistent initial stress of Germanic nominal forms versus the peninitial stress of prefixed verbs. Gothic *['anda-wōrdi]* (cf. (4)a above) and *[and-'bindan]* (cf. (5)a above) can be derived through the structural differences shown in examples (38) and (39), where the noun has an internal clitic. What, however, should cause the verbal stem to build a prosodic word on its own, rather than incorporating the “prefix” (in terms of lexical category, normally a preposition)? I posit that the difference is attributable to different rankings of LEX-TO- ω constraints holding over verbs and nouns, respectively, relative to NORECURSION (see the lexical mapping constraints at (35) in Chapter 1, repeated here for convenience). Militating against the formation of prosodic words out of functional categories is the constraint in (37). The ranking VERB-TO- ω (Left), *FNC-TO- ω $\gg$ NORECURSION- ω $\gg$ NOUN-TO- ω (Left) will induce incorporation of functional material to the left of nominal stems, but allow for recursion of the ω in order to satisfy VERB-TO- ω (Left).

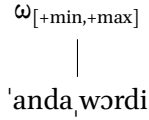
- (36) LEX-TO- ω (L, R): ALIGN(Lex, Left/Right, ω , Left/Right)

Every lexical word of the category LEX is left/right aligned with a prosodic word.

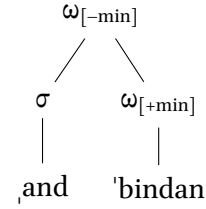
(37) *FNC-TO- ω

Function words do not project prosodic words.

(38) Goth. ['anda-₁wɔrdi] 'answer'



(39) Goth. [,and-'bindan] 'untie'^a



^aRecall from 7.2.2.2 above that the prefix /anda/ is subject to syncope in pretonic position, thus producing [and-] in verbs.

The tableaux in (41) show how the two lexical mapping constraints interact with NORECURSION to produce distinct prosodic structures and stress patterns. In order to ensure that any feet built around a prefix do not result in primary stress on the prefix, I assume undominated CULMINATIVITY at the level of the $\omega_{[-min]}$ (in contrast to Neapolitan Italian; cf. (37) in Chapter 1) and mark the $\omega_{[+min]}$ in cases of unbalanced recursion as the head of the $\omega_{[-min]}$ via the constraint in (40).

(40) HEADEDNESSINHERITANCE: In cases of recursion of a prosodic category X , the head of $X_{[-min]}$ must be another instance of X , with the result that the prosodic head of the subordinate X percolates upwards. Assign one violation * if the prosodic head of a $X_{[-min]}$ does not belong to the category X .

(41) Incorporation and Cliticization of Function Words in Gothic

a. *andawaurdi* 'answer' = [(₁'an).da.(₁wɔr).di]_{ω-min}

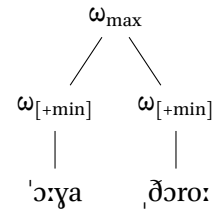
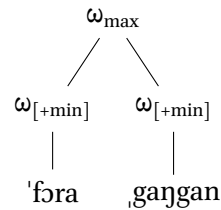
/[anda] _P -[wurdi] _N /		CULMINATIVITY HEADEDNESSINHERITANCE *FNC-TO-ω VERB-TO-ω=(Left) NORECURSION NOUN-TO-ω(Left)					
a.	 [(₁ 'an).da.(₁ wɔr).di] _{ω-min}						*
b.	[(₁ an).da.[(₁ 'wɔr).di] _{ω-min}] _ω					*!	
c.	[[(₁ 'an).da] _{ω-min} [(₁ 'wɔr).di] _{ω-min}] _ω	*!		*!			

b. *andbindan* 'untie' = [(₁and) [(₁'bin).dan]_{ω-min}]_ω

/[anda] _P -[bindan] _V /		CULMINATIVITY	HEADEDNESS	INHERITANCE	*FNC-TO- ω	VERB-TO- ω =(Left)	NO RECURSION	NOUN-TO- ω =(Left)
a.	[('and).(bin).dan] _{ω-min}				*!			
b.	[('an).d.(['bin).dan] _{ω-min}] _{$\bar{\omega}$}					*		
c.	[(['and)] _{ω-min} [(['wɔr).di] _{ω-min}] _{$\bar{\omega}$}	*!			*!			
d.	[('and) [(['bin).dan] _{ω-min}] _{$\bar{\omega}$}		*!			*		

In turn, stress assignment in compounds, where primary stress occurs on the primary stress of the leftmost member, can be accounted for by assuming that a compound consists of two minimal projections ($\omega_{[+min]}$) that combine to produce a maximal projection. Two Gothic compounds – a preposition + verb compound in (42) and a noun + noun compound in (43) – illustrate this behavior. In effect, the primary stress assigned to the second member as a minimal prosodic word then becomes a subordinate secondary stress.

- (42) Goth. /'fɔra/ + /'gangan/ → ['fɔra- (43) Goth. /'ɔ:ga/ + /'duro:/ → ['ɔ:ɣa,ðɔro:]
,gangan] 'go before' 'eye-door (= window)'



With this account in hand, crucial to observe, then, is that the the phonological component of Proto-Germanic lexical entries requires no sort of prosodic specification whatsoever (no underlying foot edges, diacritic marking, etc.); the position of primary stress is calculable based on the structure of prosodic words, their boundaries, and, to a limited extent, knowledge of syntactic categories. Further important is that the phonological grammar of Proto-Germanic not only assigned word-initial stress to phonological inputs without any prosodic prespecification, but that any such prespecification would be overwritten by the effects of TROCHEE, ALIGN-(ω , L, Ft, L), and LEFTMOST: a hypothetical input like /fo:t-íz/ (for exposition, assuming a diacritic lexical stress of the suffix /-íz/) would be realized as ['fo:t.iz], not ^X[fo:.'iz]. The inference is then that the ranking TROCHEE, ALIGN-(ω , L, Ft, L), and LEFTMOST $\gg$ MAX-PROMINENCE holds.⁴⁴

Evidence from Latin loanwords into Proto-Germanic – or, more likely, very early into its daughters, in the vast majority of instances – shows stress on the initial syllable where Latin would have assigned the stress to a heavy penult (cf. generally Ringe 2006: 296, Ringe 2017: 328–9): e.g., */ulbandu-z/ 'camel' (Goth. *ulbandus*, ON *úlfadi*) from Lat. *elephantus* (cf. Pronk-Tiethoff 2013: 147) or */rūmōn-ī/ 'Romans' from Lat. *Romānī*. Likewise, loanwords from Old French into Old English are generally assigned left-edge stress (Lahiri 2015: 233–7). This evidence indicates that either the phonological grammar of Proto-Germanic and its oldest daughters would systematically transform inputs with other stress patterns

⁴⁴Alternatively, if one specifies underlying foot edges, e.g., /fo:t-(iz/, the relevant constraint militating in favor of faithful realization of the foot edge and thus stress on the final syllable would ANCHOR-LEFT_{Foot} (cf. Özçelik 2014 and Yates 2019).

into outputs with left-edge stress, or that the speakers generally could not construct lexical entries with prespecified lexical prosody (cf. Section 3.3.4.3 on “stress deafness”).

The prosodic grammar for the $\omega_{[+min]}$ thus then be modified by adding undominated CULMINATIVITY, HEADEDNESSINHERITANCE, *FNC-TO- ω and VERB-TO- ω (Left) to the top stratum, NOUN-TO- ω (Left) and MAX-PROMINENCE to the bottom stratum, and NORECURSION to any of the intermediate strata. This grammar of metrical stress – and the absence of any contrastive lexical stress – in Proto-Germanic is thus generable through the constraint ranking in (44). This grammar fragment – the rankings of these constraints – represents the key target grammar that should have emerged at some point in the Proto-Germanic period. Under a traditional conception of the relative chronology of phonological developments in Proto-Germanic (assumed by, e.g., Schaffner 2001; cf. Ringe 2006: 152, Ringe 2017: 176), the systematic acquisition of this and in particular the relative rankings of LEFTMOST, WSP, FT-BIN, and MAX-PROM – are assumed to post-date the operation of VERNER’S LAW (Verner 1877; see further 7.3.3 below). But how and why should this grammar have come about, given the very different systems found in Vedic Sanskrit and Ancient Greek, in which lexical marking of prosodic structure plays a prominent role?

- (44) Grammar of Word-Level Prosodic Structure for Proto-Germanic:
 CULMINATIVITY, HEADEDNESSINHERITANCE, *FNC-TO- ω , VERB-TO- ω (Left), LEFTMOST,
 TROCHEE, ALIGN- ω -L, RH-CONTOUR, FTBIN $\gg$ PARSE-C, NONFINALITY- σ $\gg$ WSP, NORE-
 CURSION $\gg$ PARSE- σ $\gg$ NOUN-TO- ω (Left), NONFINALITY-f, *CLASH, ALL-FT-L, MAX-
 PROMINENCE

7.3 From PIE Prosody to Proto-Germanic Prosody: A Description

The survey of word prosody in the earliest attested Germanic languages has aimed to demonstrate that none of the languages considered (Old English and Gothic in detail, Old High German and Old Norse more superficially) present any convincing evidence for any lexical marking of prespecified properties or reference to the internal morphological constituency of stems (i.e., distinguishing between roots, derivation, and inflection). Rather, a standard array of prosodic markedness constraints, combined with knowledge of the prosodic structure involved in prefixed formations, is fundamentally sufficient to capture the data. While the exact prosodic structures constructed remain uncertain, the stress patterns of Old Germanic languages, reconstructable for Late Proto-Germanic, can be explained exclusively with reference to constraints on prosodic structure. This view contrasts sharply with the view of, e.g., Hutton (1995: 232), who holds that “stress assignment in pre- and post-shift [i.e., before and after the emergence of the tendency for word-initial stress] Germanic was morphological”.⁴⁵

For the remainder of the chapter, the foremost goal is to obtain a better understanding

⁴⁵That Hutton means for post-shift Proto-Germanic and its daughters to have possessed lexical accents is unambiguously stated: “It is clear that root stress was lexical and that certain affixes had lexical accents as part of their representation” (Hutton 1995: 235).

of how, and potentially why, the word-prosody of Late Proto-Germanic came to look fundamentally different from the word-prosody of its Proto-Indo-European ancestor. I first outline in 7.3.1 a tentative grammar of Proto-Indo-European (PIE) word-prosody, based on the testimony of Vedic Sanskrit (cf. Chapter 4), Ancient Greek (cf. Chapter 6), Hittite (Yates 2017), and to a lesser extent, Balto-Slavic. This reconstructed grammar lays the basis for a description of the major differences in word-prosody in terms of constraint rankings in 7.3.2. Two subsequent sections consider the testimony from so-called “Verner’s Law Variants” (7.3.3) and other various traces (7.3.4) that may be indicative of some innovations in the placement of stress during the Proto-Germanic period prior to the establishment of the fixed left-edge stress governed by the grammar in (44).

7.3.1 The Reconstruction of PIE Word Prosody

A metrical (in the sense of Hayes 1995) “fixed stress” system (in the sense of Revithiadou 1999: 12–13), like the one described above here for Late Proto-Germanic, governed exclusively by phonological markedness constraints assessing the well-formedness of output structures, stands in stark contrast to the LEXICAL ACCENT SYSTEM (or “morphological interface system”, in the terms of Revithiadou 1999: 14–24) reconstructable for PIE on the basis of the prosodic systems of Vedic Sanskrit, Ancient Greek, Hittite, and Balto-Slavic. As argued for PIE by Kiparsky and Halle (1977) and extensively illustrated on Vedic data in Kiparsky 2010a (see also Chapter 4), the assignment of primary stress was partly governed by the underlying prosodic properties of morphemes, by morphological structure, and a default stress rule. See further Steriade 1988, Probert 2006b and Sandell 2015a: 166–9; 190–2 on Ancient Greek, Yates 2017 on Hittite, and, on Vedic, Sandell 2015a: 163–6; 181–9, Yates 2019, and Chapters 4 and 6; an up-to-date synchronic analysis of the Proto-Balto-Slavic system is a desideratum (though some aspects may be gleaned from Kiparsky 2014 and Jasanoff 2017; see also Kushnir 2017 on some aspects of Lithuanian). While many details of how the lexical stress systems of these languages are best to be analyzed, exactly which properties of their grammars are reconstructable, and, in the case of individual morphemes, what the correct reconstruction of their lexically specified properties is, remains open for investigation and improvement, the general outlines of the PIE system seem reasonably clear (cf. the overviews of Yates 2016 and Yates 2017: Ch. 5).

Three properties are sufficiently robustly attested in the above-mentioned languages that they may be reconstructed for PIE without serious difficulty.

- Underlying Lexical Prosodic Specification: Morphemes (roots, derivational affixes, inflectional affixes) could certainly be stress-preferring (“accented”) or stress neutral (“unaccented”). “preaccenting” morphemes, which assigned stress to a preceding syllable, while largely absent from Greek and not (yet) known in Hittite, may be posited for PIE on the basis of Vedic and lexical correspondences (e.g., regular barytone stress among neuter *s*-stems). Evidence for the distinction between “strong” accented morphemes and “weak” accented morphemes at the PIE level, as posited for Vedic at 4.3.3, has not yet been carefully considered.

- “Leftmost Wins”: the leftmost lexical accent of a phonological string is realized as the primary word stress, or, in case no lexical accent is present, the leftmost syllable is assigned the primary stress (= the BASIC ACCENTUATION PRINCIPLE [BAP] of Kiparsky and Halle 1977 and Kiparsky 2010a; see also 4.2.1). The BAP is valid at least in monomorphemic stems of one or two syllables in Vedic and Hittite, which may be analyzed identically; a perhaps less restricted version appears valid for Balto-Slavic; BAP-like behaviors obtain in monosyllabic stems in Attic-Ionic Greek, but probably require a different analysis.⁴⁶
- Headedness Effects: certain “accented” or “pre-accenting” derivational affixes may impose their prosodic preference on the entire phonological string (e.g., past participle */-tó-/ > Vedic */-tá-/), when occurring as the outermost derivational suffix (i.e., the *head* of a morphologically complex stem); these affixes are “dominant” in the terminology of Kiparsky 2010a.⁴⁷ The special effect of the prosodic specification of a head in a morphologically complex stem recurs systematically in Vedic (4.2.4 and 4.3.3), Hittite (Yates 2017: 152–64), and Attic-Ionic (6.2.1.4 and 6.2.3.4); similar effects are well-established for the historical Balto-Slavic languages (e.g., Russian; cf. 2.4.2.3), but its relevance in Proto-Balto-Slavic is less clear (cf. Jasanoff 2017: 174–9).

The extent of the agreement between Vedic and Hittite, with supporting evidence from Attic-Ionic and the Balto-Slavic languages, permits the reconstruction of a phonological grammar fragment similar to that in (45). Undominated $\text{MAX-PROMINENCE}_{\text{Head}}$ (cf. Revithiadou 1999: § 1.1.3.3 and 2.4.2.3 on HEADFAITH) accounts for the “dominance” of outermost derivational suffixes; the ranking $\text{MAX-PROMINENCE} \gg \text{LEFTMOST}$ specifies that underlying prosodic specifications should be realized faithfully, while the markedness constraint LEFTMOST will drive the stress towards the left edge of the word when $\text{MAX-PROMINENCE}_{\text{Head}}$ is not relevant.⁴⁸ Compare also the final ranking given for Vedic at (88) under 4.4.

(45) Proto-Indo-European Stress Grammar

$\text{MAX-PROMINENCE}_{\text{Head}} \gg \text{MAX-PROMINENCE} \gg \text{MAINLEFT}$ ⁴⁹

⁴⁶Specifically, the synchronic contrast between acute and circumflex monosyllables in synchronic Attic-Ionic calls for some kind of lexical specification, while high tone may be mobile in the paradigms of monosyllabic stems. See the discussion and analysis at 6.2.1.3 and 6.2.3.5.

⁴⁷Note that the term “dominant” is employed somewhat differently in literature on Balto-Slavic, where also inflectional suffixes may be labeled “dominant”. Kiparsky and Halle (1977: 215–9) continue this usage in their treatment of Lithuanian, and in some descriptions, morphemes labeled “dominant” seem equivalent to what Kiparsky (2010a) would label as “accented”. Compare also Jasanoff 2017: 172–4.

⁴⁸I consider it likely that lexical prosodic specification in PIE is best regulated by underlying foot edges, as argued for Vedic Sanskrit in 4.3, but I operate here with MAX-PROMINENCE to simplify issues involving the behavior of “accented” and “preaccenting” affixes.

⁴⁹ MAINLEFT here stands in for other relatively high-ranked prosodic markedness or constraints whose interaction would regularly generate initial primary stress in a minimal prosodic word. The ranking $\text{TROCHEE}, \text{ALL-Ft-L} \gg \text{PARSE-}\sigma$ would be sufficient, in virtue of compelling the construction of just a single trochaic foot at the word’s left edge.

Both the Vedic and Hittite systems seem largely to admit of viable analyses as “unbounded” stress systems (cf. Hayes 1995: 296–99), consisting of a single foot encompassing all syllables in the word, or (more likely) just a single small foot marking the stressed syllable.⁵⁰ If stress in PIE should then be interpreted in terms of foot structure of some kind, a preference for stress on the left side of a binary foot (as in reconstructions like *[pód̥m̥] = [(pó.d̥m̥)]) constitutes *prima facie* evidence for trochaic structures. In the absence of evidence for secondary stress of any kind, the prosodic word may be assumed to have constructed only a single foot, which specified where the primary stress fell. A preference for aligning foot edges with the left edge of the prosodic word above a preference to parse syllables into feet limits foot construction and excludes secondary stress. Assuming structures like *[pe.(dí)], with degenerate feet, would entail a relatively low ranking of FTBIN.⁵¹

(46) Foot-Based Stress Grammar of PIE

TROCHEE, MAX-PROMINENCE_{Head} >> MAX-PROMINENCE >> LEFTMOST, ALIGN-(Ft, L, ω, L), ALL-FT-L >> PARSE-σ, FTBIN, WSP

Less certain in terms of their Indo-European background and nature are the OXYTONE RULE (defined in (47) after Kiparsky 2014; see also 4.2.2) and the extent of “genuine” (as opposed to “secondary”) intraparadigmatic stress mobility.

(47) OXYTONE RULE (OR)

$\sigma \rightarrow \acute{\sigma} / [...\sigma ___]_{\text{Stem}} \text{ Infl}$

“...a final accent is assigned to *all* inflected polysyllabic stems.”

The OR, whatever its nature (cf. Sandell 2015a: 181–9 for the proposal that it reflects the attraction of stress to a morphological head, and 4.3.3 for the suggestion that its apparent effects are reducible to various HEADFAITH effects), appears descriptively valid for Vedic nominal morphology (see discussion of problems at 4.2.3). On the other hand, the OR seems to play no obvious role in the prosodic grammars of Attic-Ionic or Hittite. Within Vedic, the OR or something similar is largely responsible for the fact that only transparently monomorphemic stems that are monosyllabic in their weak cases exhibit genuine stress mobility. Attic-Ionic similarly limits genuine mobility of the high tone to monosyllabic stems (as opposed to pseudo-mobility conditioned by the *Dreimorengesetz*), though probably for different reasons; the similar surface behavior, however, suggests a common surface restriction in their ancestor. Likewise, cases of intraparadigmatic stress mobility in Hittite are visible only when either the stem is monosyllabic on the surface, the stem has a heteroclitic suffix (in which case distinct accentual properties may be assumed), and in a handful of neuter

⁵⁰That Vedic had no sort of secondary stress seems uncontroversial. The evidence of “double plene” spelling in Hittite might potentially constitute evidence of secondary stress in some instances, but a thorough evaluation of the significance of double plene is wanting.

⁵¹Alternatively, if FTBIN dominates TROCHEE, which in turn dominates IAMB, then the foot structure in such cases would be *[pe.dí)]. Decisive evidence for the relative ranking of TROCHEE and FTBIN is not readily forthcoming.

nominative/accusative plural forms (e.g., *uttār* [ut:á:r] ‘word:NOM/ACC.PL, *witār* [witá:r] ‘water:NOM/ACC.PL’ vs. *uttar* [út:ar] and *wātar* [wá:tar] in the NOM/ACC.SG); otherwise, stress appears to remain on the same syllable / morpheme throughout an inflectional paradigm.⁵²

The situation is rather different in Balto-Slavic, where not only is there little trace of an OXYTONE RULE, but genuine intraparadigmatic mobility and indeed “long-distance” stress mobility is possible (see an example in (48) below, where stress moves between the initial and final syllable in trisyllabic forms).⁵³ While the historical developments that have led to the existence of relatively unrestricted edge-to-edge long-distance mobility in Baltic and Slavic remain debated (see proposals in Olander 2009, Kiparsky 2014, and Jasanoff 2017), the *communis opinio* is that such mobility is an innovation vis-à-vis Vedic, Greek, and Anatolian (in contrast to the view of Stang 1957). The important conclusion to draw for present purposes, then, is that PIE likely restricted stress mobility to Vedic- and Hittite-like patterns. The precise reasons – that is, constraints – that conditioned such restricted mobility are not entirely clear, but the OR in some form provides a prosodic system in which stress mobility would necessarily have been restricted. In the following, I will tentatively assume that the absence of stress mobility in polymorphemic paradigms is due to the properties of derivational suffixes: nearly all are accented (either “strongly” or “weakly”, as posited for Vedic at 4.3.3) or preaccenting.

(48) Lithuanian ‘apple tree’ (*i*-stem, Accent Pattern 3)

- a. ACC.SG *’óbelj*
- b. GEN.PL *obe’lū*

Furthermore, disagreements in terms of surface prosody between Vedic on the one hand versus Attic-Ionic and Hittite on the other provide some reason to believe that the latter languages have either “lost” the OR in some sense, or simplified the array of prosodic specification allowed in the lexicon via restructuring. The Vedic suffixes /-ti-/ (e.g., *matí-* ‘thought’) and /-vant-/ ~ /-mant-/ (e.g., *padvánt-* ‘having feet’) provide two possible examples. In Vedic, /-ti-/ regularly bears the surface stress in primary derivation, but cedes it to a following derivational suffix, including suffixes like /-vant-/ ~ /-mant-/ (e.g., *śruṣṭivántam* ‘obedient:ACC.SG’), which themselves cede stress to clearly accented stems (e.g., *devávant-* from *devá-*); /-ti-/ must therefore be unaccented (or “weakly” accented), and receives stress on the surface only via the OR (or HEADFAITH). The same properties may be attributed to /-vant-/ ~ /-mant-/.

⁵²See now Yates 2021 for an account of this stress mobility in Hittite (and perhaps PIE) as due to a preaccenting inflectional suffix NEUT.NOM/ACC.PL */-h₂/. In some of the same lexemes, e.g., ‘word’, the weak cases show spellings, e.g., GEN.SG *uddanāš* that could be interpreted as exhibiting “long distance” mobility, i.e., [ut:anás]; however, these are probably best interpreted as disyllabic, thus [ut:nás], with no “long-distance” mobility (cf. *sa-ak-na-aš* ‘shit.GEN.SG’, which can only be read as disyllabic).

⁵³Kiparsky (2014) argues that Balto-Slavic has both lost an inherited OR and eliminated lexical accents at the right edges of stems. At the same time, the DERIVATIONAL ACCENT RULE (DAR) proposed by Jasanoff (2017: 177) for Balto-Slavic strongly resembles the OR (though it is difficult for me to see that the DAR or similar is a necessary means of accounting for the facts that it is supposed to explain). The contrast between the mobile paradigm of lexemes like Lith. *sūny* ‘son’ and immobile *sūnikas* ‘son:DIM’ indicates that, at some historical level, the suffix /-nu-/ (which may be analyzed as unaccented or “weakly” accented) could receive an accent, but only surface as stress when not stem-final.

In Greek, the cognate suffixes *-σι-* and *-εις*, *-εντος* are, meanwhile, consistently recessive – they never attract the high tone.⁵⁴ In Hittite, the evidence from derivational adjectives in *-want-* best lends itself to the interpretation that the suffix is unaccented (see Maier 2013 for most of the data and Bondi 2020 for an assessment and analysis).

Given in particular that words built with PIE **/-ti-/* regularly exhibit zero grade of the root when phonotactically permissible (see Vine 2004), a plausible interpretation of this complex is that these suffixes were unaccented in PIE, but could receive the surface stress via the OR (or “weakly” accented, receiving stress only when attached to unaccented bases). This situation would then be preserved in Vedic, but, due to the loss of the OR in Greek and Anatolian (or loss of the distinction between “weak” accents and unaccented morphemes), the corresponding default stress assignment processes in these languages applies instead (“recessive” accentuation in Attic-Ionic, initial stress in Hittite).

In sum, the inclusion of the OR or a further distinction in degrees of “accentedness” (operationalized in 4.3.3 as a distinction between faithfulness to a left foot edge alone versus faithfulness to a left and right foot edge, e.g., Vedic */-(ti-)* vs. */-(ta-)*) means that the phonological default stress in the case of most derived words lies close to the right edge of the stem, usually corresponding to the penultimate or final syllable of the prosodic word. Undominated MAX-PROMINENCE_{Head} is essentially responsible for this effect, tending to attract stress to the same penultimate or final syllable, since morphological heads of words in PIE overwhelmingly map onto the rightmost (= outermost) derivational suffix. The preference for the stressed syllable to lie closer to the left edge of the word, however, means that other lexical accents not on the outermost derivational suffix may result in stress on or closer to the left edge.

The stress grammar in (46) seems, for the most part, to have been inherited into Indic and Anatolian; Greek presents additional complications, but also supports the ranking of HEADSTRESS above general prosodic markedness constraints. Anatolian and Greek, on the other hand, agree on the absence of effects of something like the OR or “weak” lexical accents. Although the OR could, in principle constitute an innovation of Indic, such an innovation would only be credible if it could be understood as the reranking of relevant prosodic constraints; discussion at 4.3.4.3 offers reasons to believe that no well-motivated ranking of prosodic markedness constraints alone can generate the effects of the OR in Vedic. The conclusion to be drawn is that the OR’s effects probably rest on a distinction between different types of lexically specified prosody: because Vedic derivational suffixes like */-ti-/* and */-vant-/* are but “weakly” accented, they receive stress only when they are both attached to an unaccented base and are the outermost derivational suffix.

The apparent “loss” of the OR in Greek and Anatolian could then perhaps be attributed to the loss of the distinction between “weakly” accented morphemes and unaccented morphemes. Since root-attaching “weakly” accented derivational suffixes will regularly cede stress in further derivation, they are frequently neutralized on the surface with genuinely

⁵⁴The data from *-εις*, *-εντος* might not, however, be fully probative, since the suffix really only occurs to vowel-final bases, especially *-ο-* and *-η-/̑-*, and in terms of synchronic productivity, the suffix is perhaps best analyzed as *-όεις*.

unaccented derivational suffixes. While interparadigmatic stress alternations like the one exemplified in (49) could, in principle, allow for a learner to posit a “weak” lexical accent, the surface neutralization could promote restructuring at the level of individual lexemes or even for entire classes of lexemes sharing a derivational suffix. While the same neutralization can occur even for derivational suffixes with “strong” accents (namely, when another “strongly” accented suffix is attached), the neutralization to unstressedness will be less frequent and persistent.

(49) Surface Neutralization of “Weak” Lexical Accent as Unstressed in Vedic

- a. /rs-(ti-/ ‘spear’ → *ṛṣtí-* (ACC.SG *ṛṣtím*, INST.PL *ṛṣtibhíḥ*)
- b. /rs-(ti-(mant-/ → *ṛṣtímant-* ‘having spears’ (ACC.SG *ṛṣtimántam*, INST.PL *ṛṣtimádbhiḥ*)

7.3.2 PIE and Late Proto-Germanic Word Prosody: What Differs?

From the constraint rankings laid out in (44) and (46), and the set of surface stress patterns that those grammars are capable of generating, one can see that the prosodic grammars of Late Proto-Germanic and PIE appear to coincide in some important respects, though their differences, especially as far as faithfulness constraints pertaining to underlying prosodic features of the lexicon are concerned, are more striking. The metrical stress constraints of PIE alone would regularly generate stress on the leftmost syllable, regardless of the syllable weight of the two initial syllables, just as the ranking in (44) does; this behavior follows from the ranking $\text{ALIGN-}\omega\text{-L} \gg \text{WSP}$ present in both grammars. The tableaux in (50), carrying over the respective rankings of metrical stress constraints alone from (44) and (46), show that, as far as the position of primary stress is concerned, the two grammars yield equivalent outputs. Under current assumptions, the only surface distinction in terms of the construction of metrical structure would be the presence of secondary stress in PGmc. due to inversion in the ranking of WSP, FTBIN and PARSE- σ with respect to ALL-FT-L.

(50) Metrical Stress Assignment in words of the shape /L H H L/ in PIE and Proto-Germanic⁵⁵

- a. PIE: [(‘L H) H L]

⁵⁵Throughout, I employ the cover labels L for a light syllable and H for a heavy syllable. In non-final syllables, L = [(C)V̆] while H = [(C)V:] or [(C)V(:)C₁]

/L H H L/	TROCHEE	ALIGN-(ω , L, Ft, L)	ALL-Ft-L	PARSE- σ	FtBIN	WSP
a.  ('L H) H L				**		**
b. ('L) H H L				***	*	*!*
c. L ('H) H L		*!	*!	***		
d. ('L H) (H L)			*!			
e. (L 'H) H L	*!			**	*	
f. (L H) ('H L)			*!*		**	

b. Proto-Germanic: [('L H) (H) L]

/L H H L/	TROCHEE	ALIGN-(ω , L, Ft, L)	LEFTMOST	FtBIN	RH-CONTOUR	WSP	PARSE- σ	ALL-Ft-L
a. ('L H) H L						*!	**	
b. ('L) H H L				*!		**	***	
c. L ('H) H L		*!	*!			*	***	*
d.  ('L H) (H) L							*	**
e. ('L H) (H L)					*!	*		**
f. (L 'H) H L	*!					*	**	
g. (L H) ('H L)			*!		*!	*		**

In terms of these metrical stress constraints, the contrast between the effects of ALL-Ft-L in PIE vs. ALIGN-(ω , L, Ft, L) in PGmc. may be characterized in terms of a change of a stress *preference* (a foot should ideally be close to the left edge) into a *requirement* (every minimal prosodic word must begin with a foot). The virtual “demotion” of ALL-Ft-L in PGmc. may also be viewed as a consequence of the fact that this constraint now bears less responsibility for outcomes in the language as a whole. Among the metrical constraints, TROCHEE and ALIGN- ω -L play the most crucial role in both grammars, but the undominated ranking of the latter in Proto-Germanic is responsible for the predictable occurrence of primary stress at the beginning of the word.

As alluded to above, the more crucial distinction involves the role of constraints that refer to underlying prosodic properties of morphemes or to morphological structure: the constraints MAX-PROMINENCE_{Head} MAX-PROMINENCE (let us call these the “lexical accent constraints”), which dominate constraints on metrical stress in PIE, are evidently wholly inactive in Proto-Germanic. In effect, these constraints appear to have been demoted below the crucial constraints that ensure that primary stress always surfaces on the leftmost syllable (LEFTMOST, ALIGN- ω -L, and TROCHEE). In any grammar in which these constraints dominates WSP, as well as all of the “lexical accent constraints”, fixed initial-syllable stress becomes essentially unavoidable. Other ranking permutations of the particular lexical accent

constraints with respect to the “initial stress constraints” are possible, and will result in different sorts of surface outputs. Including a cover constraint OXYTONE RULE to simulate the effects of (47) more likely attributable to intricate lexical accent interactions, various ranking permutations of MAX-PROMINENCE_{Head}, MAX-PROMINENCE, MAINLEFT (standing for undominated LEFTMOST, ALIGN-ω-L, and TROCHEE) and OXYTONE RULE will result in different patterns of surface outputs. Some of these possibilities are sketched in (51).

(51) Ranking permutations of MAINLEFT and lexical stress constraints

- a. PIE Ranking: MAX-PROMINENCE_{Head} >> OXYTONE RULE >> MAX-PROMINENCE >> MAINLEFT
Default left edge stress occurs only in morphologically simplex (monosyllabic?) stems, where no lexical accents are present in the input.
- b. MAX-PROMINENCE_{Head} >> OXYTONE RULE >> MAINLEFT >> MAX-PROMINENCE
Default left edge stress occurs on all morphologically simplex (monosyllabic?) stems; lexical accents of inflectional affixes are effectively suppressed.
- c. MAX-PROMINENCE_{Head} >> MAINLEFT >> OXYTONE RULE, MAX-PROMINENCE
Lexically accented morphological heads attract stress, but stress otherwise falls on the leftmost syllable.
- d. MAX-PROMINENCE_{Head} >> MAX-PROMINENCE >> MAINLEFT >> OXYTONE RULE
Inactivity of the OR allows lexically accented inflectional suffixes to attract stress, when no other lexical accents are present in a string.
- e. MAX-PROMINENCE >> MAINLEFT >> OXYTONE RULE, MAX-PROMINENCE_{Head}
The leftmost lexical accent receives the stress; morphological structure (and/or number of syllables) plays no role.
- f. OXYTONE RULE >> MAX-PROMINENCE >> MAINLEFT >> MAX-PROMINENCE_{Head}
No “dominance” effects of outermost lexical accents.
- g. MAX-PROMINENCE >> OXYTONE RULE >> MAINLEFT >> MAX-PROMINENCE_{Head}
Stress may be attracted to lexically accented final syllables in preference to the right edge of the stem, which occurs only in strings without lexical accents; no “dominance” effects.
- h. OXYTONE RULE >> MAINLEFT >> MAX-PROMINENCE_{Head}, MAX-PROMINENCE
Lexical accents play no role; stress is either assigned to the right edge of a disyllabic/polymorphemic stem, otherwise to the leftmost syllable.
- i. Post-Verner’s Law PGmc.: MAINLEFT >> MAX-PROMINENCE_{Head}, OXYTONE RULE, and MAX-PROMINENCE
Lexical accents and effects of morphological structure are suppressed in outputs (insofar as they exist).

All of the rankings b.-h. under (51) afford MAINLEFT a more decisive role in the grammar. Of particular interest and relevance would be to try to see whether evidence for any of the rankings b.-h. can potentially be inferred for some stage of PGmc.; Kiparsky (2014), for instance, proposes developments that could correspond to grammars d. or e. The same

applies to the relation of other lexical stress constraints to the metrical stress constraints; if, for instance, WSP came to dominate MAX-PROMINENCE, primary stress would be realized on heavy syllable in preference to a lexical accent on a light syllable (e.g., hypothetical PGMc. */fo:t-iz/ → [(‘fo:).ti(z)], if final consonants are assumed to be extrametrical). The fundamental question, though, remains the following: how and why did speakers of Proto-Germanic come to systematically arrive at a constraint ranking under which constraints interfacing with prosodic features of the lexicon or morphological structure played little or no role? As discussed at the outset of this chapter, the basic issue would seem to consist in the fact the clear evidence in the learning data inherited from PIE for lexical stress constraints appears to have been “ignored” or “lost”, in some sense.

Hutton (1995: 232), for instance, is helpfully explicit in his view of the sequence of prosodic developments leading to the prosodic system of Proto-Germanic was as follows:

1. Leveling of initial-syllable stress.
2. Leveling of root stress, with surface distinction between primary and secondary stress now permitted.

⇒ Stress fell either on a root or on a nominal prefix.

3. Construction of moraic trochees used to determine the assignment of secondary stress.

In this schematized account, what remains entirely unclear is the motivation behind any sort of “leveling” of stress on the part of learners. In other words, if the learning data did not transparently or predominantly point to a system of initial-syllable (or root) stress, how and why should such a system have been acquired in the first place?

Yet more explicit in its subdivision into individual “stress shifts” is the account presented in d’Alquen 1988: Ch. 1 (see esp. pp. 41–3). Alongside other developments affecting vowels and glides, D’Alquen posits four stages of prosodic changes, encompassing five distinct stress shifts (D’Alquen’s notations are freely modified here), which can, in many cases, be formulated in terms of new rankings with respect to MAX-PROMINENCE.

1. **Germanic 1:** Retraction of stress from a light open final syllable to the penult:

$[\sigma_o \sigma ' \check{V} \#] > [\sigma_o ' \sigma \check{V} \#]$
 (FTBIN $\gg$ MAX-PROM)

2. **Germanic 2:**

- Retraction of stress from a short vowel in a final syllable followed by a single consonant to the penult:

$[\sigma_o \sigma ' \check{V} C \#] > [\sigma_o ' \sigma \check{V} C \#]$
 (WSP $\gg$ NONFINALITY- σ $\gg$ MAX-PROM)

- Retraction of stress from a non-final light syllable to a preceding heavy syllable:

$[H 'L \sigma_1] > [H L \sigma_1]$

3. **Germanic 3:** Assignment of stress to all heavy syllables
(WSP $\gg$ MAX-PROM, ALL-FT-L)
4. **Germanic 4:** Assignment of (primary) stress to all roots
(ALIGN- ω -L, LEFTMOST $\gg$ WSP)

D'Alquen's schema has the virtue that reckons with exact phonological developments that can potentially feed one another by shifting the balance of evidence in favor of further innovative rankings. For instance, if stress is banned from final light syllables to satisfy FT-BIN, less evidence for stress on final syllables and feet at the right edge of the word, can in turn promote the acquisition of rankings that avoid stress on final syllables more generally. Indeed, Kuryłowicz (1968: 191–4), departing from the observation that prosodic words in Latin, Old Irish, and Germanic of the shape [C $\check{V}$] are illicit, proposes that stress was generally banned from short final syllables in the Italic, Celtic, and Germanic branches. From a theoretical perspective, a bimoraic minimal word requirement and the blocking of primary stress on a short final syllable (assuming a stress system with trochaic feet) can only be directly related insofar as one assumes that a) the prosodic word constructed but a single foot (i.e., no secondary stress) and b) the prosodic word obligatorily dominated a bimoraic foot. In effect, a higher ranking of FTBIN, granting undominated TROCHEE, would exclude stress from final syllables. Crucially, however, FTBIN would then need to dominate MAX-PROM, such that lexical accents could never compel the construction of degenerate feet (at the right edge of the word or anywhere). Furthermore, a ban on stress in final short syllables due to high-ranking FTBIN could conceivably high-rank NONFINALITY-F as well, penalizing the alignment of feet with the right edge of the word. In a system with moraic trochees, also sequences of the type [X ('L $\check{V}$)] would then be banned, resulting in stress on an antepenultimate syllable.

The series of developments posited by D'Alquen are in themselves credible, and consist essentially in the avoidance of final stress and attraction of stress to heavy syllables, both of which, in the context of a grammar that already preferred for feet to stand closer to the left edge of the word, would increase the total amount of stress falling on the initial syllable, particularly in disyllables and trisyllables. The direct evidence for reconstructing such a sequence of events is, however, rather thin: D'Alquen's (1989: 20–33) main source of evidence consists in accent marks in the OHG manuscripts of Notker, finding in them indications of preferences for certain syllable configurations to bear stress. While certainly interesting for the study of OHG prosody and relevant to the reconstruction of Late Proto-Germanic prosody, to read a historical chain of events into such evidence is methodological questionable.

Therefore, the objective of the following sections is to explore evidence that the primary prosodic prominence in Proto-Germanic may have gradually become rarer in non-initial syllables, either through independently justifiable prosodic innovations, or morphological innovations that would cause the primary prominence to default to the left edge of the prosodic word.

7.3.3 Verner's Law Variants: Evidence for Prosodic Change?

The question with which this section concerns itself is thus: did Proto-Germanic inherit high-ranking HEADSTRESS and/or HEADFAITH, and if so, is there evidence that these constraints may have been demoted into irrelevance prior to the total loss of lexical accents and generalization of left-edge stress?

One potentially relevant set of data comes from so-called “Verner’s Law variants” (VLV). Generally speaking, a “Verner’s Law variant” refers to a lexeme or morpheme of Germanic that – usually across different languages, but sometimes within the same language – exhibits different consonants due to the application or non-application of VERNER’S LAW (VL) to a voiceless fricative in Proto-Germanic (Verner 1877), stated in (52). When an original Proto-Germanic voiceless fricative */ ϕ , θ , s, x, x^w / was subject to VL, it became a voiced fricative, thus */ β , δ , z, γ , γ^w / (later phonologized to and partly merged with voiced stops, once left-edge stress was generalized).

For present purposes, VLVs are of interest precisely because VL attests to the position of the word-level prominence at the time that VL applied.⁵⁶ As the classic examples in (53) illustrate, the persistence of a voiceless fricative between two [+sonorant] segments indicates that the nearest syllable nucleus to the left of the fricative bore the prosodic prominence

$$(52) \text{ VERNER'S LAW: } \begin{bmatrix} -\text{sonorant} \\ +\text{continuant} \end{bmatrix} \rightarrow [+voice] / \begin{bmatrix} +\text{syllabic} \\ -\text{stress} \end{bmatrix} ([+\text{sonorant}]) \text{---} \left\{ \begin{array}{l} +\text{sonorant} \\]_{\omega} \end{array} \right\}$$

A fricative becomes voiced after a vowel or sonorant consonant preceding another sonorant or the end of the word, provided that the nearest syllable nucleus to the left of the fricative is unaccented.

(53) VERNER’S LAW at work

- a. PIE */ $p\acute{a}h_2tér$ / ‘father:NOM.SG’ > Pre-VL [fa θ é:r] > Post-VL [fa δ é:r] > Goth. *fadar* [‘fa δ ar] (cf. Ved. *pítā*)
- b. PIE */ $b^hráh_2tér$ / ‘brother:NOM.SG’ > Pre-VL [brá:θe:r] > Post-VL [brá:ðe:r] > Goth. *broþar* [‘bro:ðar] (cf. Ved. *bhrátā*)

Although the existence of VLVs has been evident since the work of Barber (1932), which remains the most detailed collection and organization of data concerning accentuation in Germanic nominal and adjectival stems, attempts to account for the data have only been offered during the past thirty years. Especially important is the treatment in Schaffner 2001, where all known cases of etyma exhibiting VLVs are systematically discussed. Bammesberger (1990) and Ringe (2006) outline the issue in general, and subscribe to the main line of explanation (forms of paradigmatic leveling) given in Schaffner 2001. Particularities of the Gothic material are discussed in Bernharðsson 2001 (esp. Ch. 3) and Casaretto 2004. Most recently,

⁵⁶The exact phonetic grounding for VERNER’S LAW has long been debated. One recent plausible account is offered by Johnsen (2011), who argues that preceding high tone allowed the vocal folds to maintain tension and not vibrate, whereas otherwise fricatives between sonorants became voiced as a natural co-articulatory effect.

partly novel accounts of the VLV data have been presented in Kiparsky 2010a / Kiparsky 2014 as well as Garrett 2011; these recent accounts will form the basis of discussion under 2.3.1–2.⁵⁷

VLVs come in two “flavors”: **Lexical Doublets** and **Stem-Class Doublets**. Lexical Doublets (see (54)) are single lexemes of Germanic that attest different voicing values for a PGmc. fricative in different daughter languages (typically only one form per language, but occasionally both fricatives in the same language). Stem-Class Doublets (see (55)), meanwhile, are different lexemes within the same stem class (e.g., fem. **ti*-stems) may or may not exhibit effects of Verner’s Law. For instance, the masculine **tu*-stems seen in (55) show a Proto-Germanic variant [-*þu*-] in the lexeme for ‘death’, but a variant [-*ðu*-] in the lexeme for ‘flood’. For convenience, in the following, word forms with prosodic prominence on the initial syllable will be labeled *barytone*, while forms with prominence on the last syllable of the stem or the word will be labeled *oxytone*. In general, Proto-Germanic forms with barytone accentuation (like ‘brother’ in 53b.) did not result in fricative voicing, whereas forms with oxytone accentuation (like ‘father’ in 53a.) did.

- (54) **Lexical Doublet: ‘debt’** (fem. **ti*-stem)
- a. Barytone: Pre-VL **[skúlpi-]* > Post-VL **[ˈskulpi-]* > OHG *sculd*⁵⁸
 - b. Oxytone: Pre-VL **[skulpí-]* > Post-VL **[ˈskulði-]* > OE *scyld*, ON *sculd*
- (55) **Stem-Class Doublets: **tu*-stems**
- a. Barytone: **[dáuþu-]* ‘death’ > **[ˈdauþu-]* > Goth. *daupus*, OE *dēap*, OHG *tōd*
 - b. Oxytone: **[flōþú-]* ‘flood’ > *[ˈflōðu-]* > OE *flōd*, OHG *fluot*, ON *flóð*

All investigations into and accounts of VLVs depart from the assumption that Verner’s Law, as a regular sound change, generated allomorphy. This assumption in turn means that the position of stress prior to the application of Verner’s Law is ultimately to be held responsible for the emergence of the doublets. Nonetheless, in specific cases, a doublet might conceivably have emerged as the result of productive derivation from a verbal stem containing a different consonant. That is, if a noun and verb containing the same root exhibited different stress patterns, use of the verbal allomorph in productive derivation could give the appearance of a VLV, as in (56) (cf. Schaffner 2001: 377–8 on feminine PGmc. **ō*-stems).

- (56) **Productive derivation >> Verner’s Law variant**
- a. PGmc. **[ˈpwaxilō-]* ‘washcloth’ (with **[x]* from the Class 6 strong verb **[pwaxe/a-]* / ‘wash’) versus older **[ˈpwayilō-]* < **[pwaxiló-]* (Schaffner 2001: 412)
 - b. PGmc. **[ˈrōβō-]* ‘number’ (with **[β]* from the Class 2 weak verb **[rōβōje/a-]* / ‘count’) versus older **[ˈrōφō-]* ‘number’ < **[róφō-]* (Schaffner 2001: 403)

⁵⁷The proposals advanced in Kiparsky 2010a / Kiparsky 2014 in fact go back to Kiparsky 1973: 845, where essentially the same core claim – that VLVs stem from PIE oxytone stems (i.e., stems with stress on the rightmost syllable of the stem preceding an inflectional suffix) through the development of Germanic edge-to-edge accentual mobility – is advanced.

⁵⁸The OHG [d] reflects the regular development of Proto-Germanic [θ]; Proto-Germanic [ð] would instead exhibit [t] in OHG (cf. ‘flood’ in (55)b. below).

The existence of lexical doublets is especially problematic. Comparison with the word-prosody of Vedic Sanskrit (see discussion above and in Chapter 4) suggests that nominals should have “columnar” stress, remaining on the same syllable of a stem throughout the paradigm. If stress in Early Proto-Germanic were determined as in Vedic, lexical doublets should not be expected to arise as a consequence of regular sound change alone. Especially troubling are lexical doublets involving derivational suffixes that, according to the present *communis opinio*, were had strictly immobile stress in PIE, such as **o*-stems; see one such doublet in (57). In theory, stem-class doublets are less problematic, since a lexically accented root could result in left-edge prominence where an unaccented root could allow prominence on the derivational suffix; however, given the relative rarity of accented roots in Vedic, that the Germanic stem-class doublets result from the different lexical prosodic properties of roots is *a priori* unlikely.

- (57) ‘blood’ (neut. *a*-stem)
 a. Goth. *bloþ*, *bloþis* < *[blóþa-]
 b. OHG *bluot*, OE *blōd* < *[blōþá-]

In the case of lexical doublets, the most frequently invoked explanation is the paradigmatic leveling of different stem allomorphs that could have resulted Post-VL in stems that had accentual mobility. That is, if an inflectional paradigm exhibited mobility of the accent between the initial syllable (barytone) and some non-initial syllable (oxytone), then a fricative following the leftmost [+syllabic] segment could have remained voiceless in some forms of the paradigm, but become voiced in others. Two major lines of account, connected with views on the accentuation of nominal paradigms in Proto-Indo-European, emerged during the 20th century. Under both accounts, VLVs largely reflect stress patterns in nominal accentuation inherited from PIE. On the one hand, Barber (1932) interpreted the existence of VLVs in almost all Germanic nominal classes as evidence that both Indo-European consonant stems (including **i*- and **u*-stems) and the “thematic” **o*- and **eh₂*-stems exhibited intraparadigmatic stress mobility (see also Stang 1957 and Halle 1997 for versions of this view). On the other, only the consonant stems of PIE showed intraparadigmatic stress mobility, whereas **o*- and *eh₂*-stems did not.

Thus, paradigmatic leveling of a stem allomorph is the principal, though not exclusive, explanation for VLVs given in Schaffner 2001. In Schaffner’s conception, the relevant type of mobility could be found in three types of paradigms: so-called “proterokinetic” paradigms with movement of the stress in PIE between the root and the derivational suffix, as shown in (58), so-called “amphikinetic” paradigms, with movement between the root and inflectional suffix, as shown in (59), and movement between the root and inflectional suffix in the NOM./ACC.PL of certain neuter nouns.⁵⁹ Stress on the root in such paradigms ultimately results in word-internal voiceless fricatives, while stress on a derivational or inflectional suffix results in a word-internal voiced fricative.

⁵⁹Schaffner and others sometimes view the relationship between singular and plural in neuter nouns as a derivational relationship through the formation of a “collective” stem (= plural). For simplicity, I will speak of inflectional categories singular and plural.

- (58) Proterokinetic Mobility⁶⁰
- a. Strong: $\acute{R}$ -D-I
 - b. Weak: R- $\acute{D}$ -I
- (59) Amphikinetic Mobility
- a. Strong: $\acute{R}$ -D-I
 - b. Weak: R-D- $\acute{I}$
- (60) Singular–Plural Mobility in Neuters
- a. SG. Pre-VL $*[x^w \acute{e}x^w l\ddot{a}]$ ‘wheel’ > Post-VL $*[x^w \acute{e}x^w l\ddot{a}]$ (> OE *hweohl*)
 - b. PL. Pre-VL $*[x^w ex^w l\acute{a}:]$ > Post-VL $*[x^w ey^w l\acute{a}:]$ (> OE *hewog(u)l*)

One substantial problem with Schaffner’s version (basically accepted in Ringe 2017: 301–3) of VLVs as resulting from paradigmatic accentual mobility is that the “proterokinetic” inflectional paradigms most likely never existed. Specifically, the empirical basis for proterokinetic accentual mobility is very weak to non-existent (Kümmel 2014, Lundquist 2015a, Lundquist 2017: Ch. 2). Thus, the lexemes belonging to stem classes often assigned a “proterokinetic” reconstruction (e.g., **ti*-stems) in handbooks (e.g., Fortson 2010) were more likely largely oxytone, as they largely are in Vedic (e.g., *matí*- ‘mind’, *śaktí*- ‘power, ability’ [later *śakti*-]).

As concerns mobility in “amphikinetic” paradigms, the former were comparatively rare in PIE (practically non-existent in Vedic), which leads Schaffner to posit that this pattern of was somehow analogically adopted by various lexemes (see Schaffner 2001: 517–35 on **n*-stems). Finally, neuter plural forms to **o*-stems cannot be assumed to have had stress mobility if the derivational suffix **/-o-/* otherwise conditioned fixed stress throughout the paradigm.⁶¹ Furthermore, according to the data in Barber 1932, VLVs are slightly more prevalent in masculine **o*-stems (25/280 = 9%) than in neuter **o*-stems (14/210 = 6.67%), even though there are is, under Schaffner’s account, no systematic explanation for VLVs in the masculines. If indeed the particular inflectional suffix of the neuter plural suffix were a potential conditioning factor for VLVs, a higher proportion of VLVs is to be expected for neuter nouns.

However, an alternative type of intraparadigmatic mobility, which I term “edge-to-edge” mobility, would be equally capable of generating VLVs. While such edge-to-edge mobility is typologically and theoretically unproblematic, it is unknown in Vedic and Ancient Greek on morphologically complex stems (i.e., outside of root nouns). On the other hand, such edge-to-edge mobility is frequently encountered in Balto-Slavic nominal inflection (cf. generally

⁶⁰R = root; D = derivational suffix; I = inflectional suffix.

⁶¹Yates (2021) allows for the neuter NOM/ACC.PL suffix **/-h₂/* to condition stress on the final syllable of a stem, but only if the stem was otherwise not lexically accented. Kim (2013), meanwhile, takes the inflectional ending of the NOM/ACC.PL to be dominant, i.e., able to erase other lexical accents on a stem. On the interpretation that the NOM/ACC.PL suffix is properly a derivational, not inflectional suffix (i.e. “collective”), HEADFAITH could be invoked to explain rightward attraction of stress.

Stang 1957).⁶² Given the fact that Germanic presents VLVs in stem classes that are uncontroversially immobile in Vedic and Greek, e.g., **o*-stems, as illustrated in (57), some mechanism seems to be required to generate accentual patterns in Germanic that did not exist in Vedic or Greek.

(61) Edge-to-Edge Mobility

- a. $/[\sigma\sigma_o]-\sigma/ \rightarrow [\sigma\sigma_o\sigma]$.

Stress surfaces on the leftmost syllable where neither stem nor inflectional suffix are lexically accented.

- b. $/[\sigma\sigma_o]-\acute{\sigma}/ \rightarrow [\sigma\sigma_o'\sigma]$.

Stress surfaces on the final syllable of a word with a monosyllabic lexically accented inflectional suffix, when the stem is not lexically accented.

The following sections, 7.3.3.1 and 7.3.3.2, will engage with the novel explanations for VLVs worked out in Kiparsky 2010a / Kiparsky 2014 and Garrett 2011, respectively. The ultimate objective is to determine the extent to which VLVs potentially present evidence for instances of left-edge prominence that are not to be found in Vedic or Greek (and thus presumably were not inherited into Germanic from PIE).

7.3.3.1 Verner's Law Variants after Kiparsky: Loss of the OXYTONE RULE

The essence of Kiparsky's view on VLVs is that they result from paradigmatic leveling on a lexeme-by-lexeme, language-by-language basis, after stem allomorphs in paradigms with edge-to-edge mobility were generated. Under Kiparsky's account, polysyllabic, originally barytone (initial-stressed), and polysyllabic, originally oxytone (originally stem-edge stressed) stems of PIE were subject to separate accentual developments (in both Balto-Slavic and Germanic): the former remained persistently barytone in Germanic, whereas the latter developed edge-to-edge mobility. Some inflectional forms of the word for 'debt' (cf. (54) above) would have thus developed as shown in (62): Germanic inherited from PIE a stem with oxytone accentuation, which came to have a paradigm with edge-to-edge mobility, thus giving rise to the alternation in voicing of the fricative $[\theta]$ ($<$ PIE $*[t]$).

(62) 'debt' via Edge-to-Edge Mobility

- a. ACC.SG: PIE $*/skel-tí-m/ \rightarrow *[sk_l\acute{t}ím] > \text{Pre-VL } *[skúl\theta in] > \text{Post-VL } *['skul\theta i]$
(cf. OHG *sculd*)

- b. DAT.SG: PIE $*/skel-ti-éi/ \rightarrow *[sk_l\acute{t}éjei] > \text{Pre-VL } *[skul\theta ijí:] > \text{Post-VL } ['skul\theta i:]$
(cf. OE *scyld*, ON *sculd*)

⁶²Detailed accounts of the historical developments of patterns of accentual mobility in Balto-Slavic are presented in Olander 2009 and Jasanoff 2017. The alternative explanation of Balto-Slavic mobility presented in Kiparsky 2014 is attractive in its simplicity, but open to some (but not all) of the same objections as Kiparsky's account of edge-to-edge mobility in Germanic discussed below, since both branches would have developed edge-to-edge mobility for the same reasons.

Recall here that Kiparsky (2010a) accounts for the systematic appearance of the surface accent in Vedic at the right edge of morphologically complex stems when the underlying string possesses no lexical accents farther to the left through the OXYTONE RULE (see (47) above), repeated in (63) for convenience. For instance, the Vedic compound *śítipád-* (NOM.PL *śítipádas*) ‘having white feet’ has the UR /śíti-pad-/ and receives a stem-edge accent due to the OXYTONE RULE. In contrast, stems having a lexical accent further to the left, e.g., /gó-mant-/ (ACC.SG *gómantam*, GEN.SG *gómatas*) allow that lexical accent to surface faithfully. The posited distinction in behavior between barytone and oxytone stems in Germanic would then reduce to the following generalizations: barytone stems preserve lexical prosodic information in the PIE lexicon: such barytone stems contained either a lexically accented root or a pre-accenting derivational suffix; oxytone stems become lexically unaccented stems, which permits edge-to-edge mobility in Germanic between initial syllable and lexically accented inflectional suffixes.

- (63) OXYTONE RULE
 $\sigma \rightarrow \acute{\sigma} / [\dots \sigma ___]_{\text{Stem}} \text{ Infl}$
 The rightmost syllable of a polysyllabic stem receives an accent.
- (64) Derivation of PIE *[skl̥téjei̯] (cf. (62)b. above)
- | | |
|----------------|----------------------|
| */skel-ti-éi̯/ | UR |
| skel-tei̯-éi̯ | <i>e</i> -EPENTHESIS |
| skel-téi̯-éi̯ | OXYTONE RULE |
| skl̥-téi̯-éi̯ | ZERO GRADE ABLAUT |
| skl̥-téi̯-ei̯ | BAP |
| *[skl̥téjei̯] | SR |

The claimed emergence of edge-to-edge mobility in Germanic raises two important questions. First, how and why did oxytone nouns lose their oxytonesis? Second, why would different lexemes analogically level different stem allomorphs within Proto-Germanic itself or individual daughter languages? The edge-to-edge hypothesis also makes the strong prediction that original barytone stems should only ever attest forms with voiceless fricatives (no stems with voiced fricatives, no VLVs), while originally oxytone stems might surface in the daughter languages with any variety of fricative, variable or not.

For the prediction that VLVs are associated with original oxytones, some weak statistical support is available. Based on the data in Barber 1932, there is a moderately positive correlation (Pearson's $r = 0.499$) between the proportion of stems with *Endbetonung* and the number of stems with *schwankender Betonung* within a stem class. If both *Endbetonung* and *schwankende Betonung* are attributable *only* to inherited oxytones, then a positive correlation between their frequencies is to be expected.⁶³

Under Kiparsky's account, the emergence of default left-edge prominence in unaccented stems, on the one hand, but faithful realization of the lexical accents of inflectional endings

⁶³For this analysis, I excluded the Barber's consonant stems and *s*-stems, since these categories exhibit no *Endbetonung*. I further excluded Barber's *u*-stems (except the masculine *u*-stems), neuter *n*-stems, and feminine *m*-stems, since these categories all contain less than 10 types each.

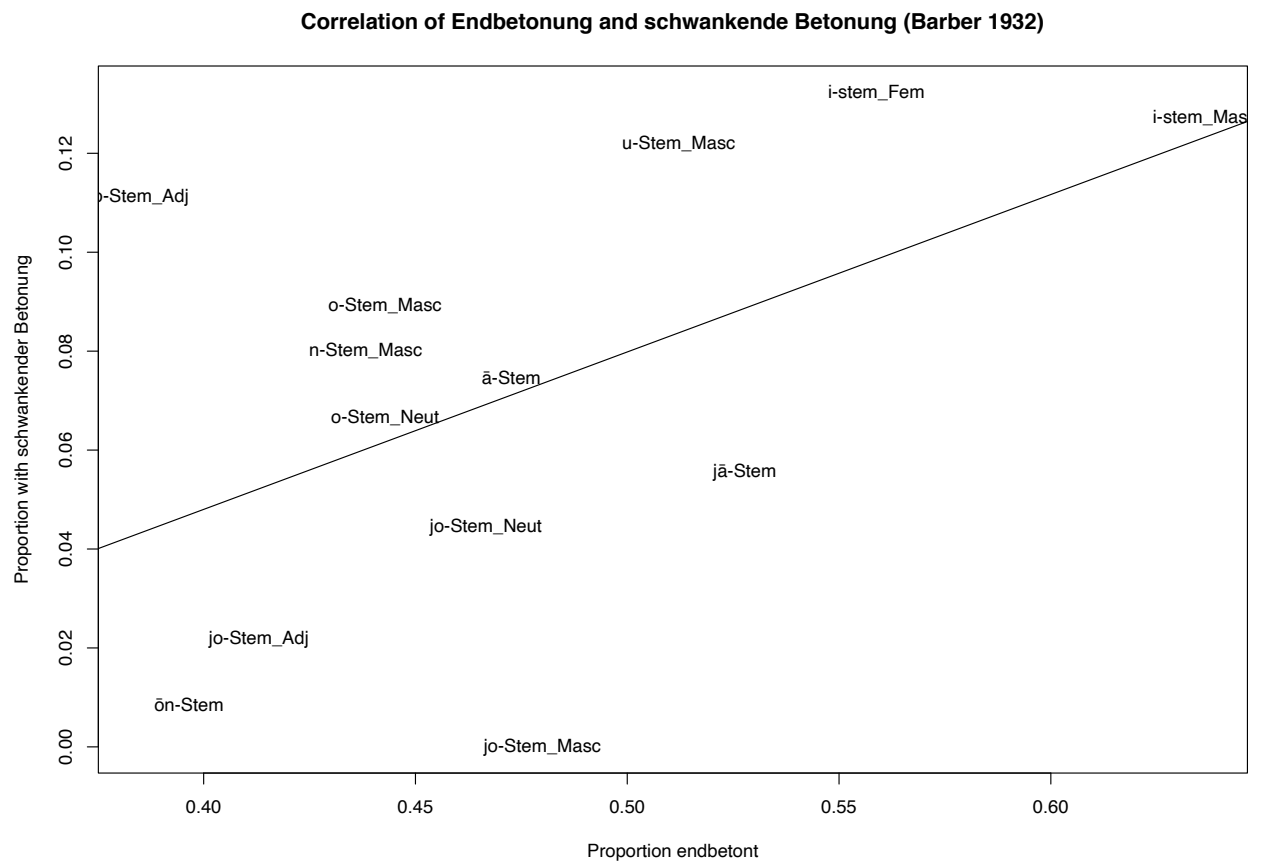


Figure 7.1: Correlation of *Endbetonung* and *schwankende Betonung* for Germanic stem classes in the data of Barber 1932

(such as DAT.SG */-éi/ > /-í:/) on the other, is due to a historical process of of PREDESINENTIAL DEACCENTUATION (PD).

(65) PREDESINENTIAL DEACCENTUATION (Kiparsky 2014: 29)

...σ̣]_{Stem} σ... → ... σ]_{Stem} σ...

Predesinential syllables are deaccented.

Although Kiparsky formalizes such a process and derives edge-to-edge mobility in Germanic and Balto-Slavic through its application, his more detailed explanation of how stem-edge accentuation in these languages came to be lost suggests that he envisions PREDESINENTIAL DEACCENTUATION as the result of a more complex morphophonological development. Kiparsky (2014: 29) says:

The distinction between inherently end-accented stems [oxytone stems that are oxytone because of a lexically accented derivational suffix] and inherently unaccented stems [i.e., oxytone stems that are oxytone because of Kiparsky's OXYTONE RULE] stems bears a minuscule functional load in the later stages of the proto-language [i.e., PIE]: most root nouns and verbs are movable, and in longer stems the distinction between oxytones and unaccented stems is neutralized by the Oxytone Rule. As a poorly entrenched opposition, it was liable to be lost by suppression of the marked member, in this case oxytone accent, whether lexically marked or assigned by (6) [the OXYTONE RULE]. The loss of this accentual opposition is a type of analogical change in the sense that it eliminates unmotivated complexity in the grammar, but as far as I can see is not expressible as proportional analogy. It is an instance of non-exemplar-based analogy (Kiparsky 2010), a structural simplification comparable, say, to the loss of the distinction between neuter and masculine gender in nouns in Baltic.

Whether Kiparsky's usage of the term "analogy" in the case at hand is appropriate or not is debatable, but the path of development on oxytone stems that Kiparsky envisions is comprehensible, and can be expressed more clearly. First, Kiparsky assumes here that, because both underlyingly accented derivational suffixes (e.g., PIE */-ó-/) and presumably unaccented derivational suffixes (e.g., PIE */-ti-/) were regularly realized on the surface as accented, due to the application of the OXYTONE RULE to the latter, all lexically accented suffixes were reanalyzed as unaccented. In other words, PIE */-ó-/ becomes PGmc. /-a-/, because the instances of surface *[-á-] could be attributed in many instances to the OXYTONE RULE. The lexicon of Proto-Germanic would have thus developed so as to lose lexically accented derivational suffixes; the possible combinations of morphology and lexical prosody would then be restricted to accented roots, pre-accenting derivational suffixes, accented inflectional endings, and unaccented counterparts. As a compositional system, the only accentual possibilities within an inflection paradigm would then be persistent root stress, persistent stress on the syllable immediately preceding a derivational suffix (usually the root), or persistent stress on the outermost derivational suffix, if root and all derivational suffixes were unaccented.

The subsequent – and crucial – innovation to the prosodic grammar of Proto-Germanic would then be the loss of the OXYTONE RULE from the phonological grammar. With this development, all stems that exhibited persistent stress on the outermost derivational suffix would subsequently become stems with edge-to-edge mobility, due to the application of default left-edge stress on fully unaccented strings, but stress on an inflectional suffix when lexically accented. See the data on reconstructable **o-*, **eh₂-*, and feminine **i-*stems in Kiparsky 2010a: 161–5.

With edge-to-edge mobility established prior to the operation of VL, allomorphy in the value of [voice] on fricatives is predicted to emerge. The remaining issue, in principle, is then to account for why which stem allomorphs have been leveled on which lexemes in which languages. Precisely the problem with undertaking this further explanatory step is that the individual Germanic languages exhibit scarcely any rhyme or reason on this account: individual languages seem to have leveled different stem allomorphs from lexeme to lexeme. For example, OHG *nōt* ‘need’ (< **[ˈnaʊði-]*) would have chosen the weak stem, whereas *sculd* (< **[ˈskulpi-]*) would have chosen the strong stem, even though both lexemes belong to the same stem class. Garrett (2011) therefore criticizes accounts of VLVs that rely substantially on later paradigmatic leveling, because they require what Garrett terms BIDIRECTIONAL LEVELING. Given the improved understanding of paradigmatic leveling afforded us by the work of Albright (Albright 2002b, Albright 2005, Albright 2008a, Albright 2010), it should indeed be surprising to find such a large-scale case in which the BASE of leveling is chosen at the level of individual lexemes, rather than at the level of the stem class or morphosyntactic category.

Besides this general issue with the asystematicity of paradigmatic leveling required by such accounts of VLVs, two further observations may be made concerning intraparadigmatic allomorphy in Germanic. First, as is well known, West and North Germanic verbal inflection maintained consonant alternations due to VL (*grammatischer Wechsel*) until a comparatively recent date (and some irregular patterns may still preserve lexicalized traces, e.g., Germ. NPST *zieh-* vs. PST *zog-*). Secondly, some inflectional classes in Germanic nouns happily preserve patterns of stem allomorphy, e.g., masculine *n*-stems in Gothic show three stem allomorphs, in *-a-*, *-an-* and *-in-*, which cannot be generated through productive phonological processes. These facts make the supposed leveling of VLVs just in nouns and adjectives suspicious: if VLVs were tolerated in verbs, and stem allomorphy of other types was acquisible, why should just stem allomorphy of the VLV-type have disappeared on nouns and adjectives alone?

Two further minor objections might be leveled concerning the behavior of root nouns and kinship nouns. Among Germanic root nouns in voiceless obstruents, Barber (1932) lists 12 stems in monosyllabic (root) stems in obstruents, which almost certainly mobile paradigms. With the exception of **/stup-/~/stuð-/* (OE *stupu-*, *studu-* ‘pillar’; cf. Schaffner 2001: 635–40), all attest only the voiceless fricative. This fact might indicate a general tendency to level the strong stem, and more specifically, the stem of the nominative singular, as illustrated with the development of the paradigm for ‘mouse’ in (66). One might perhaps have expected to encounter some instances in which the stem allomorph with the voiced fricative.

(66) ‘mouse’: PIE > Proto-Germanic > Old English

- a. NOM.SG PIE *[mús] > *['mu:s] > OE *mūs* (cf. Gk. μῦς [mû:s])
- b. GEN.PL PIE *[musóm] > ['muzō:] >> OE *mūsa* (cf. Gk. μῶν [muô:n])

If one takes the process of PREDESINENTIAL DEACCENTUATION under (65) literally, then the behavior of the well-attested kinship terms ‘mother’, ‘father’, and ‘brother’ is also perhaps somewhat surprising. Namely, the two oxytones ‘father’ (PIE *[pəh₂tér-] > Goth. *fadar*) and ‘mother’ (PIE *[mah₂tér-] > OE *mōdor*), should have been subject to PD, giving rise to stems with edge-mobility. It would have to be a (notable) coincidence that both of these nouns would have analogically leveled the stem allomorph with the voiced fricative, in all Germanic languages. Even if edge-to-edge mobility should be reconstructed for some of the Proto-Germanic lexicon, a means to maintain some inherited oxytones as oxytones would be desirable.

Perhaps the most serious objection of all, however, comes from the perspective of learnability, and reveals a certain internal contradiction in Kiparsky’s account. Consider again the two crucial prosodic innovations needed to transform an oxytone noun or adjective like PIE */h₁reud^h-ró-/ ‘red’ → */h₁reud^h-ró-/ (cf. Gk. ἐρυθρός) into a form with edge-to-edge mobility. First, the derivational suffix /-ó-/ must be reanalyzed as unaccented, so that PGmc. */raud-ro-/ → [raudró-] (by OXYTONE RULE); subsequently, the OXYTONE RULE must be lost from the grammar. The second step is precisely the problem: learners of Proto-Germanic would have necessarily acquired the OXYTONE RULE (or similar) in order to explain why stems like [raudró-] exhibited stem-edge stress. To simply fail to acquire the OXYTONE RULE would have substantial consequences on the surface outputs of stress in many lexemes, so as to generate forms that would be totally unfaithful to the learning data. Such unfaithfulness to the learning data precisely runs counter to the tendencies seen in the histories of the Romance languages and Greek, which tend to reproduce inherited prominence in the same position, if even a different grammar and lexicon is needed. If confronted with data like [raudró-], one expects by default for Proto-Germanic learners to have acquired a combination of lexicon and grammar that would permit the word prosody of the lexeme to be reproduced faithfully. The fact that Kiparsky’s account precisely requires such unfaithfulness is a considerable point against it.⁶⁴

Finally, the presumption that *all* inherited oxytone nominal stems would have developed edge-to-edge mobility confronts a difficult quantitative question: why are VLVs so comparatively infrequent? In any given stem class, based on Barber 1932 lexemes exhibiting VLVs constitute no more than about 13% of types (feminine *i*-stems exhibit this highest rate of *schwankender Betonung*). Since the number of stems attesting exclusively a voiced fricative are much more common (among the masculine *a*-stems listed in Barber 1932, they represent $\frac{124}{279} = 44.4\%$), one would have to assume that Proto-Germanic ultimately had a strong tendency to paradigmatically level the allomorph of the weak stem. To a certain extent, this observation could then counter the objection that paradigmatic leveling in Germanic nominal paradigms was asystematic; the generalization to draw would be that the weak stem,

⁶⁴Nevertheless, one must reckon with the fact that edge-to-edge stress mobility is an indisputable fact for the Balto-Slavic languages, while it is merely an artifact of one account of VLVs for Germanic.

with voiced fricative, was leveled in the majority of stems (among masculine *a*-stems, more than 80%).⁶⁵ The strong tendency to level the weak stem allomorph would, however, seem to contrast with the behavior of root nouns, where leveling of the NOM.SG allomorph seems to have predominated (cf. ‘mouse’ above, as well as */fo:t-/ ‘foot’). The selection of different stem allomorphs as a base between root nouns and derived stems might be motivated by their different morphological structures. The VLVs themselves still then present the problem of analogical aystematicity at the level of the individual languages.

Although Kiparsky (2010a: 159–66) convincingly demonstrates that an account of VLVs based on assumed edge-to-edge mobility makes better predictions for the attested Germanic data than the accounts offered in Schaffner 2001, the extent and depth of objections to which this “edge-to-edge theory” is susceptible should hinder its direct acceptance. In order to make the “edge-to-edge” theory fully palatable, better answers to at least the two core issues raised here must be addressed:

1. Why should the OXYTONE RULE (or its equivalent in a constraint-based formalism) have been lost from the grammar of Proto-Germanic, if it was essentially for learners to acquire it in order to explain a surface accents in a large number of types?
2. How and why was paradigmatic leveling of Verner’s Law-conditioned allomorphy so regular at the Proto-Germanic level (in favor of the weak stem, with voiced fricative), but so irregular in the individual Germanic languages (i.e., the clearly attested VLVs)?

7.3.3.2 Verner’s Law Variants after Garrett: Demorphologization in Germanic

Into this breach steps an entirely different account, forwarded by Garrett (2011). The essence of Garrett’s proposal is that the inherited distinction between oxytone and barytone stems is largely stable in Germanic, but that a limited number of oxytones may have become barytone via DEMORPHOLOGIZATION. The direction of prominence shift, oxytone $\Rightarrow$ barytone, is compatible with the data, and is potentially more restrained than the model of edge-to-edge mobility, limiting the occurrence of new barytone forms to just the forms for which VLVs are attested. However, if the prosodic change affects lexemes such that all inflectional forms of a word should become barytone, one might expect to find forms in Germanic with voiceless fricatives alongside forms with oxytone accentuation in Greek or Vedic (a good candidate in this regard is PGmc. *[láuxa-] ‘clearing’ (cf. ON *lō*, OE *lēah*, OHG *lōh*) < PIE *[loukó-] (cf. Ved. *loká-* ‘open space’), but no variance between the Germanic languages.

First, the operations of so-called DEMORPHOLOGIZATION must be clarified; subsequently, the problem of how DEMORPHOLOGIZATION might be leveraged or supplemented, in Garrett’s account, so as to permit the existence of VLVs, can be addressed. As formulated by Probert (2006b: 291), “demorphologization” amounts to a loss of morphological structure

⁶⁵Kiparsky nowhere explicitly observes that large-scale paradigmatic leveling in favor of weak stem allomorphs would have needed to occur in order to explain the large number of polymorphemic stems that attest exclusively to a voiced fricative, but this conclusion seems basically inescapable.

(i.e., a complex stem becomes simplex), with the consequence that the accentuation of the stem may undergo change.

When a word has undergone ‘demorphologization’ its accentuation can no longer be determined by the presence of an inherently accented suffix as the suffix is no longer treated synchronically as present. The word may retain its non-recessive accentuation but the necessary accentual property now becomes a characteristic of the whole — synchronically unanalyzed — stem. On the other hand, the word may lose its inherent accent altogether, in which case a recessive accent will be assigned by default.

In Probert’s (2006b) investigation of thematic oxytone derivational categories in Greek (e.g., -ρός-, -τός-), lexemes with barytone accent (e.g., βόθρος [bótʰros] ‘hole, trench’, ἄροτος [árotos] ‘field, crop’), the barytone accent is attributed to loss of morphological structure that allows for the default phonological algorithm – in Greek, the recessive accent (cf. 6.2.1.2) – to apply. That is, a lexeme like βόθρος [bótʰros] would originally have been derived as */botʰ-ró-/ → *[botʰró-] (with lexical high tone), but later reanalyzed as morphologically simplex /botʰro-/ → [bótʰro-]. Since internal the internal structure of a lexeme in the lexicon is of a speaker is not itself visible, simplex stems may nevertheless have lexical prosodic marking (e.g., perhaps νεκρός [nekrós] ‘corpse’, in Probert’s view, is /nekró-/). Thus, while stems showing recessive accentuation with an etymologically accented suffix would constitute *prima facie* evidence for demorphologized forms, other forms could still exhibit a lexical high tone while also being regarded by speakers as free of internal morphological structure. This issue – maintenance of a lexical accent as a property of the stem as a whole versus emergence of default word-level prominence – remains in need of explanation.⁶⁶

Regardless, the same general phenomenon of leftward stress retraction in cases where originally morphologically complex forms have likely become morphologically simplex is now also supported by further cases from Hittite (Yates 2015a) and Vedic (Lundquist 2015b, Sandell 2016). In particular, Yates (2015a) explains the clear word-initial stress of nasal-infix presents in Hittite (for which Vedic shows stress on the derivational suffix) by the same means: because this verbal derivational suffix */-né-/ became unproductive in Hittite, verbal stems built with it were reanalyzed as morphological simplexes, and the stem lost its lexical accent, resulting in default application of left-edge stress (on left-edge stress as the default in Hittite, cf. further Yates 2017: Ch. 4).

- (67) PIE */-né-/ results in forms with word-initial stress in Hittite (see Yates 2015a: 146)
a. PIE */h₂u-né-g-ti] → *[h₂unékti] > Hitt. /χo:nik-fsi/ [χó:nikf̥si] ‘batters’

⁶⁶The explanation on this account forwarded by Probert, namely, that stems having high token frequency should maintain a non-default lexical accent, while low token frequency stems should shift to word-level prominence assigned by the default algorithm, is unfortunately not really compatible with the Greek cases that she treats. As more attentive statistical analysis of Probert’s Greek cases of demorphologization shows, there is no genuinely secure correlation between token frequency and accentuation, at least for the frequency “bins” employed by Probert (Sandell 2015a: 192–206, Sandell 2015b); a careful reconsideration of the question is needed.

- b. PIE */ni-né-k-ti/ → [ninékti] > Hitt. /ni:nik-ṫsi/ [ní:nikṫsi] ‘mobilizes’

An account of VLVs in Germanic based on “demorphologization” must proceed from the assumption that left-edge prominence is the default result of a phonological algorithm applying to a string without lexical accents or morphological structure. Granting that the OXYTONE RULE either depended on polymorphemicity (as argued in Sandell 2015a: Ch. 7 and Sandell 2016), the demorphologization of a stem could allow for the determination of prominence by MAINLEFT. Thus, an original oxytone /dau-θu-/ → *[daʊθú-] ‘death’ might lose its morphological compositionality, giving */dauθu-/ with surface form *[dáʊθu-] (cf. (55)a. above), due to the ranking of MAINLEFT over other purely phonological constraints on primary stress. Conversely, a stem like *[flo:θú-] ‘flood’ might maintain its morphological compositionality (perhaps due to the availability of a primary verbal root */flo:-/ ‘flow’), and continues to be generated with stem-edge prominence until after the application of VL. In this fashion, to generate stem-class doublets, as exemplified under (55), is unproblematic. The situation in Germanic would be wholly akin to the situation of thematic stems in Greek per Probert (2006b).

Demorphologization with an accompanying oxytone-to-barytone prominence shift alone does not, however, readily predict the existence of lexical doublets of the sort exemplified under (54). Given a unitary Proto-Germanic language, all daughter languages ought to have inherited identical lexemes with identical fricatives. For precisely this reason, Garrett posits that Germanic was already substantially dialectally diversified prior to the “diffusion” of Verner’s Law across the Germanic dialect continuum. The existence of lexical doublets would then simply reflect the fact that different lexemes underwent demorphologization in different dialects. More specifically, given the almost complete absence of lexical doublets among the North Germanic languages, but many between Old English and Old High German, or between Gothic and the other languages. The implication is then that East, West, and North Germanic were already distinct, and that West Germanic itself was also dialectally distinct, prior to the diffusion of Verner’s Law. Nevertheless, Verner’s Law cannot be a late development of the Proto-Germanic period: all branches of the family have participated in a considerable number of sound changes that must post-date Verner’s Law (cf. the relative chronology of sound changes in Ringe 2006: 152, Ringe 2017: 176).

Example (68) illustrates the presumed sequence of developments on the basis of OE *ēarð* ‘plough-land’ and OHG *art* ‘ploughing’. Originally, the /-pi-/ stem received stem-edge prominence as a result of the OXYTONE RULE (or similar). Following sufficient dialect diversification, this stem was subject to semantic concretization and concomitant demorphologization in certain dialects (here, an direct ancestor of Old English); when demorphologized, a left-edge prominence could arise; following diffusion of VERNER’S LAW, dialectally distributed variants with different fricatives result.

(68) ‘ploughing’ and ‘plough-land’

- a. Original oxytone noun: /ar-pi-/ → *[arþí-] ‘ploughing’
- b. Dialect diversification: forerunners of OE and OHG become distinct

- c. Demorphologization (due to semantic concretization), barytonesis in the ancestor of OE:
/ar-pi-/² → *[árpi-] ‘plough-land’
- d. Diffusion of VERNER’S LAW, diffusion of initial stress:
*[árpi-] > *[arpi-] and *[arþi-] > [arði-].
- e. Attested forms: OE *ēarð* ‘plough-land’ vs. OHG *art* ‘ploughing’

The cost of Garrett’s overall approach – depending upon one’s views about the internal unity of proto-languages and the “spread” of sound changes – is itself considerable. Succinctly put: there would be virtually no unitary Proto-Germanic. At the same time, the supposedly early dialect differences would be almost exclusively lexical in nature, apparently limited to identifications through such sporadic instances of demorphologization with shift from oxytone to barytone prominence that give rise to VLVs.

The particular weaknesses of the Kiparsky’s and Garrett’s approaches to VLVs can, in light of the above discussion, be summarized as follows:

- Kiparsky: the mechanism and motivation for wholesale loss / demotion of the OXYTONE RULE is unexplained; presumed edge-to-edge mobility in all inherited oxytones risks substantial overgeneration.
- Garrett: Proto-Germanic – and even West Germanic – must have been dialectally distinct at a very early date; nevertheless, many sound changes that must post-date Verner’s Law, which is responsible for creating the segmental differences, were able to exceptionlessly and completely diffuse through the entirety of the Germanic dialect area, without leaving any leaving any dialect zones with preserved phonological archaisms (as far as is known).

Given these arguably considerable disadvantages, one would prefer an account of VLVs that neither risks overgeneration, nor create a portrait of Proto-Germanic that contains dialects recognizable almost exclusively from a handful of lexical differences in stress. An alternative may be to adopt demorphologization as a basic explanation for VLVs, but to presume that demorphologization in a lexeme allowed for edge-to-edge mobility. That is, demorphologized stems become unaccented stems, thereby allowing for left-edge prominence in inflectional forms with unaccented inflectional suffixes, but final prominence in forms with lexically accented inflectional suffixes. (69) schematically illustrates the sequence of developments with ‘debt’.⁶⁷

(69) ‘debt’ via demorphologization

1. Inherited Oxytone: PIE */skl-(ti-/ → *[skl̥tí-]
 (“weak” lexical accent produces oxytone stress)

⁶⁷It should further be noted that the plausible cases of “demorphologization” with concomitant stress changes at the lexical level thus far adduced for Greek, Hittite, and Vedic all share another property: in the “demorphologized” forms, stress becomes fixed on the leftmost syllable of the stem and does not admit of intraparametric stress mobility. In contrast, the Germanic scenario sketched here would precisely need to allow for stress mobility between stem and inflection.

2. Demorphologization in Proto-Germanic: /skulpi-/ →
ACC.SG [skúlpi-n] ~ DAT.SG [skulpi-jǐ]
3. VERNER'S LAW (+ other sound changes): ACC.SG ['skulpi] ~ DAT.SG ['skulðǐ]

Although this usage of demorphologization can avoid the potential overgeneration resulting from presumed edge-to-edge mobility in all original oxytones, it is nevertheless subject to the same concerns about bidirectional paradigmatic leveling. Without some means of seriously being able to predict how and why different lexemes of the same inflectional class should have leveled distinct stems, a more restricted recourse to edge-to-edge mobility via demorphologization is perhaps preferable to the Kiparskian account only in degree, but not in kind.

7.3.3.3 Summary: The Relevance of Verner's Law Variants for Proto-Germanic Prosody

However the Verner's Law Variants of Germanic are best to be accounted for, they do suggest that the number of tokens with left-edge prominence may have increased in number at some stage of Proto-Germanic, prior to the wholesale generalization of left-edge stress. Under the Kiparskian account, a huge number of word types – all word type that exhibit oxytone accentuation in Vedic – would have acquired left-edge prominence in their strong cases (nominative and accusative) forms. The “demorphologization” account favored by Garrett, on the other hand would certainly have affected fewer stems, but since all inflectional forms of the affected stems would be predicted to become barytone, the effect would still be considerable. However, because the “demorphologization” account explains the existence of VLVs as a result of a dialectally diversified Proto-Germanic, the total number of affected stems in any given dialect or dialect group would have been less than the total number of attested VLVs. Some number of stems might nonetheless have undergone demorphologization in all dialects or have been demorphologized prior to any dialect differentiation; a good candidate is PGmc. *[láuxa-] ‘clearing’ (cf. ON *lō*, OE *lēah*, OHG *lōh*) < PIE *[lou̯kó-] (cf. Ved. *loká-* ‘open space’).

Ultimately difficult to assess is the total quantitative extent to which the type and token frequency of left-edge accented forms was increased through the prosodic changes that led to the emergence of VLVs. Examination of the masculine *a*-stems in Barber 1932 reveals that fixed barytone and oxytone stems were about equally frequent (131 vs. 124, ~ 47% and 44%), while variant forms were considerably less common (24 stems, ~ 9%). Assuming that some amount of “demorphologization” in this stem class resulted in the complete change of an original oxytone to a barytone in Germanic as a whole, such that no inner-Germanic trace of the oxytone remains, while others became variant, the result must have been a change from a lexicon in which oxytone stems represented the majority pattern (~55-60% of primary stems), to a lexicon in which barytones and oxytones were of approximately equal frequency. Given that so many nominal and adjectival stems are attested exclusively with a voiced fricative, a large number of stems must still have exhibited either consistent stem-edge accent or edge-to-edge mobility when their stress universally shifted to the initial syllable of the word.

7.3.4 Other Possible Evidence for Changes in Prosody in Proto-Germanic

Hutton (1995) hypothesized that the gradual attrition of derivational suffixes that consistently attracted primary prosodic prominence (“dominant accented” suffixes, in the suffix typology of Kiparsky 2014), combined with loss of stressed allomorphs for derivational suffixes that could appear as either stressed or unstressed (“recessively accented”, or suffixes subject to the OXYTONE RULE), might result in a language in which left-edge prominence became the norm. In order to substantiate such a proposed development, the derivational morphology of Proto-Germanic, and especially its *productive* derivational morphology, requires some assessment.

One particularly tantalizing fact that points in the direction of reduced quantities of stem-edge stress is that fact that Germanic oxytone /-á-/ (cf. Vedic -á-, Gk. -ό-) may have been non-productive. In PIE, as well as in Greek and Vedic, nouns built with PIE */-ó-/ were fairly common. Ringe (2006: 292) notes that “few PIE agent nouns of the type Gk. τροχός [‘wheel’] survived in PGmc.” This fact could mean that the accented derivational suffix PIE */-ó-/ became unproductive and fell out of use in PGmc., such that only preaccenting /-‘a-/, generating initially stressed nouns, remained in use.

At the same time, at least two derivational suffixes with likely stem-edge prominence seemed to have generated many new types in Proto-Germanic: masculines in */-i-/ (e.g., *[k^widiz] ‘saying’; cf. OE *cwide*) and feminines in */-θi-/. Root ablaut in formations built with these suffixes – typically zero grade – suggests stem-edge prominence, and the Vedic analogues to these affixes support stem-edge prominence, too. Moreover, among both masculine and feminine *i*-stems listed in Barber 1932 (of which forms in */-θi-/ are the most common among the feminines), forms attesting to non-initial prominence are much more common than forms with initial prominence (72 types vs. 34 types, masculines and feminines taken together). Even newly created suffix complexes, such as masculine */-o:θu-/ (used to build action nouns to Class II weak verbs; described as “common and productive” by Ringe (2006: 293)) seem to have usually borne the prominence on a non-initial syllable, probably most often [-ó:θu-] when derived from a verb, but sometimes also [-o:θú-] (cf. OHG *arnōt* ‘harvest’ < *['arno:ðu-] < *[arno:θú-]).

Already these cases suffice to provide an indication that, at least preceding the operation (or diffusion) of Verner’s Law, non-initial prominence was hardly uncommon, and in many derivational categories, even predominant. Without evidence to the contrary, the default assumption should be that productive derivational morphology tended to maintain the same pattern of prominence assignment. While a more systematic investigation of evidence for the position of prominence in productive derivational morphology is nevertheless a desideratum, this short investigation suggests that productive derivational affixes that attracted non-initial prominence in Proto-Germanic may well have continued to attract non-initial prominence until left-edge stress became the only option.

7.3.5 Other Quantitative Data on the Stress Distributions of Proto-Germanic

7.3.5.1 Stressed and Unstressed Allomorphs in Inflectional Morphology

As mentioned just above, Hutton (1995: 204–5) hypothesized that some morphemes might be learned as lexically unaccented, in the event that they exhibited surface allomorphy between stressed and unstressed allomorphs. The acquisition of an unaccented morpheme might be particularly likely to impact morphemes that only rarely surfaced as stressed. To this end, the relative frequencies of the stressed and unstressed allomorphs of some lexically accented suffixes in the RV can be compared. I mark the suffixes here with left and right foot edges, as in 4.3, where a single left edge indicates a “weak” accent (stress-attracting only in the absence of any other foot-edges, or away from other “weak” accents when acting as the head of the word) and a left and right foot edge indicates a “strong” accent.

(70) Token Frequency of Stressed and Unstressed Suffix Allomorphs in the RV

Function	Stressed	Unstressed	SUM
INST.PL /-(bhis/	199	2705	2905
INST.SG /-(ā/	1020	2376	3396
GEN.PL /-(ām/	550	1485	2035
GEN.SG /-(ás/	205	714	919
LOC.PL /-(su/	140	326	466
LOC.SG /-(i/	209	444	653
DAT.SG /-(e/	684	2336	3020
Adjectival /-(vant-/	132	1241	1373
Adjectival /-(in)/	1349	21	1371

With the exception of the dominant, “strongly” accented suffix /-(in)/, the unstressed allomorph is more common in all instances. However, for only two of the suffixes examined here (INST.PL /-(bhis/ and adjectival /-(vant/) does the stressed allomorph occur in less than 10% of the tokens. If learners are presumed to generally be able to make the “correct” analytical inferences, given sufficient data, some affixes with especially rare stressed allomorphs might well be acquired as unaccented, but the relatively systematic disappearance of accented suffix in such a fashion appears implausible.

To test the effects of non-primary derivation on the creation of unstressed allomorphs of derivational suffixes that typically surface as stressed, I queried all instances of thematic nouns and adjectives built with thematic /-(a)/ in the RV (15199 tokens, 1466 types).⁶⁸ I then queried all lemmata containing an unaccented form of the same noun or adjective stem followed by at least one segment, to find instances in which a further accented derivational suffix or second member of a compound evidently overrides the accent of /-á/. Such cases are infrequent in comparison to the bases: only 2285 tokens among 709 types.⁶⁹ The Vedic

⁶⁸Pronouns whose stems terminate in -á were excluded, namely: *yá-*, *sá-*, *tá-*, *ká-*, *syá-*, *tyá-*, and *svá-*.

⁶⁹The true value is certainly somewhat lower, due to evident false positives in the query, where, e.g., *sahasrín-* or *sahas'ya-* was matched to *sahá-*. A yet lower incidence of derivatives to the bases would only strengthen the conclusion drawn here, however.

data would thus seem to indicate that failure to acquire a lexical accent – at least in the case of a productive derivational suffix – is unlikely, since the stressless allomorph is comparatively infrequent.

Insofar as the Vedic data considered here may be taken as representative of the lexical accent systems in operation in the prehistory of other Indo-European branches, the loss of lexical accents as a result of stress allomorphy does not appear to offer particularly promising prospects. The specific inflectional and derivational suffixes examined here are, of course, not always relevant to Germanic specifically, but the general conclusion likely holds.

7.3.5.2 Prosodic Shapes in the Germanic Lexicon

Obtaining a clear view of the frequency of 1) different stress patterns or 2) the frequency of words / stems with particular prosodic shapes for Germanic in particular poses a special challenge. The nature of the Gothic corpus, and its relatively late attestation relative to the period of Proto-Germanic relevant for the present study, means that the value of token frequencies in that corpus would at best be limited, if at all meaningful. Instead, one may try to estimate the *type* frequency of particular stress patterns based on the data in Barber 1932, and the type frequency of particular prosodic shapes in initial syllables of stems (e.g., /H L X/, /L L X/, etc.) based on reconstructed etyma (Kroonen 2013).

On the basis of Kroonen 2013, one can obtain a first approximation of the frequencies of particular stem shapes.⁷⁰ An electronic search indicates that Kroonen 2013 contains approximately 2550–2600 lemmata. The vast majority of the lemmata present disyllabic stems (~ 91%), with a much smaller number of trisyllabic (~ 7%), a few monosyllabic (~ 1%) and vanishingly few quadrisyllabic stems (< 1%).

Among the disyllabic and trisyllabic stems together, at least 1722 (~ 65%) lemmata contain an inevitably heavy first syllable (i.e., contain a long vowel or two consonants following the root vowel); the true number must be slightly higher.⁷¹ Among trisyllables, the incidence of heavy first syllables appears comparatively lower.⁷² Overall, the only relatively secure conclusion possible at the moment on the basis of Kroonen's data is that lexemes with heavy first syllables were typewise predominant in the Germanic lexicon.

(71) Distribution of Prosodic Shapes of Lemmata (Stems) in Kroonen 2013

⁷⁰The frequencies reported here were obtained by first converting a .pdf of Kroonen 2013 with OCR to a plain-text file, then manually eliminating the front-matter (before the first lemma) and end-matter (after the last lemma). This plain text file was then read into R (Version 3.6.2), broken on newline characters. With regular expression queries, the headwords of the lemmas could be extracted, as well as searches for particular segmental sequences in the headwords.

⁷¹Macrons were not preserved in the conversion from .pdf to plain-text, with the consequence that forms of the shape /C₀VCV/ have not been systematically accounted for; only 17 types of vowel-initial words with a long-vowel in the first syllable were added to the count by hand. Further diminishing the count of heavy first syllables are some instances in which a sequence of two consonants, e.g., *ht* in the .pdf was replaced by a whitespace in conversion to plain-text. A reasonable estimate would be perhaps 67–70% of types.

⁷²Again, the number of light syllables is likely an overestimate, meaning that the type frequency of heavy syllables is likely somewhat higher.

	1st Heavy	1st Light		
Disyllables	1643	667		
	LL	LH	HL	HH
Trisyllables	88	30	36	12

The data from Barber 1932, meanwhile, provides a potential portrait as to the position of stress in Proto-Germanic for many nominal and adjectival lexemes at the time of the operation of Verner's Law. As a reminder, Barber (1932) classified data mainly into three accentual categories based on the reconstructability of voiceless or voiced fricatives in given lexemes: stems with initial stress (*Anfangsbetonung*), stems stressed on the last vowel of the stem or derivational suffix (*Endbetonung*), and stems attesting to both initial and final stress (*schwankende Betonung*). If the data is taken at face value, and a lexeme like */furxō-/ 'furrow', attested only with a voiceless fricative, means a stem with fixed leftmost stress, while a lexeme like */lauyō-/ 'lye', attested only with a voiced fricative, means a stem with fixed stress on the stem-final syllable, then one can, at least within the nominal domain, consider the frequency of initial-stress vs. non-initial stress.⁷³ Besides the forms with *Endbetonung*, a smaller number of disyllabic obstruent-final stems (*t*-stems, *s*-stems) exhibit stress on the second syllable; for other stem classes (e.g., *n*-stems), one cannot clearly distinguish between stress on the suffix or ending (or suffix and ending are fused in one syllable, as is often the case for PGmc. **a*- and **ō*-stems), for which reason all are classed as *endbetont*, when the stress clearly was not on the initial syllable. The distribution of different stress positions in the totality of Barber's data is then given in (72).

(72) Distribution of Stress Patterns in Barber 1932

	Anfangsbetonung	2. Silbe	Endbetonung	schwankende Betonung
Type Frequency	732 (46.3%)	48 (3%)	682 (43.1%)	118 (7.5%)

While fixed stress on the leftmost syllable is the most common pattern, as is the case for both tokens and types in Vedic (cf. Tables 5.3 and 5.5), it is far from overwhelmingly predominant. The typewise distribution of stress in the Germanic lexicon at the time of the operation of Verner's Law can, therefore, hardly be considered to have possessed a particularly strong tendency towards left-edge stress.

If, however, all forms classified by Barber as having *Endbetonung* and *schwankende Betonung* are taken to have exhibited left-edge stress in at least some (namely, strong) case forms, as the Kiparskian hypothesis of edge-to-edge mobility in Germanic would presume, then virtually the entirety of the nominal lexicon (~ 97%) would have sometimes exhibited left-edge stress. If one assumes that strong and weak case forms were equally frequent among nouns ultimately classed as having *Endbetonung* or *schwankende Betonung*, and that all types were tokenwise equally well represented, then, one might reckon with about 716

⁷³However, under the view put forth by Kiparsky (2010a: 159–66), many originally oxytone lexemes across many stem classes would have, in the relevant stage of Proto-Germanic, exhibited paradigmatic alternation between word-initial and suffixal stress. The proportion of *tokens* with initial stress would then have been considerably higher, since all lexemes that show a voiced fricative must have been mobile, having only presumably paradigmatically leveled the stem allomorph that originally had suffixal stress.

tokens of stress on the leftmost syllable out of 1000 tokens. Left-edge stress at the token level would then appear predominant, while stress in any other position elsewhere would still constitute a significant minority.

From this approximative data on the type frequency of Proto-Germanic prosodic patterns, one can draw the inference that a stressed, heavy initial syllable was likely the most common single surface pattern. Apart from the open question of possible edge-to-edge mobility in Proto-Germanic inflectional paradigms, on the one hand, and innovations in verbal stem formation on the other, the distribution of prosodic patterns in surface word forms of Proto-Germanic may not have been radically different from the picture presented by Vedic. For the purposes of simulation in the following sections, I therefore take the frequencies of two–five syllable words Vedic (see the data in presented at 5.2.4) to constitute a basis for further inquiry.

7.4 Simulating the Development of Proto-Germanic Prosody: Language-External Factors

At this juncture, the difficulties posed by the “simple hard problem” at the outset of this chapter remain unfortunately all too difficult. Fundamentally, two problems stand in the way of a clear language-internal account of a transition from a prosodic system characterized by lexical accents to a prosodic system with regular left-edge stress.

First, as long as prosodic words with primary prosodic prominence other than on the leftmost syllable constituted a considerable proportion of the learning data (admitting that “considerable” in this context remains to be more precisely defined), the default expectation is that Proto-Germanic learners will have continued to acquire a grammar and lexicon that allowed for the unrestricted production of primary prosodic prominence on different positions in a prosodic word. This expectation is bolstered both by research into the learning of constraint rankings and URs by algorithm (see 3.2), and the fact that the histories of English, many Romance languages, and Greek speak in favor of an overriding faithfulness to surface prominence of lexemes: speakers can and do seem to assemble whatever combination of grammar and lexicon is necessary to reproduce the same primary stress as found in the learning data. From this point of view, the great diachronic puzzle is not surface stability, but rather, change (compare the discussion at 3.3)

Second, the data in Barber 1932 indicate that, at least at the time of the operation of Verner’s Law, many lexemes of PGmc. may have had neither fixed-initial nor mobile stress. If either the account of Garrett (2011) or the modified account of VLVs presented at the end of 7.3.3.2 is correct, then summing together the proportion of nominal stems with *Anfangs-betonung* or *schwankende Betonung*, only 53.8% of types would consistently or occasionally have borne initial stress. A large proportion of types (46.2%) would have presented consistently non-initial stress. Ultimately, for so many voiced fricatives to have survived in the nominal lexicon, the totality of Germanic material relating to Verner’s Law suggests that the change from a grammar and lexicon defined by lexical accents to a grammar and lexicon that preferentially generated left-edge stress was either relatively abrupt. In the case of an

abrupt transition, the sudden application of left-edge stress across the board would allow for massive restructuring of URs with surface voiced fricatives, before the constraints on stress position could interact with the constraints on fricative voicing to render the latter inert.

If a purely language-internal account of the emergence of left-edge stress in Germanic is difficult to understand and justify, given our assumptions about learnability theory and the Proto-Germanic lexicon, the search for an explanation leads in an unhappy, but all-too-often assumed direction: language contact. The view that language contact played a role in or was responsible for the dramatic change in the Proto-Germanic system has frequently been suggested or defended, though the fundamental problem of precisely *how* substrate should affect the phonological system of a superstrate language remains. The problem is obviously more acute for prehistory, where the substrate is necessarily uncertain.

Nevertheless, proponents of a language contact-based account of Proto-Germanic prosody have usually seen a likely suspect among the Finno-Ugric languages. A Finno-Ugric substrate as the basis for prosodic change in Germanic is, for instance, judged by Salmons (1992: 173) to be “perhaps more plausible than the alternatives, given the reconstructed Proto-Finno-Ugric accent and early and profound contacts by trade and as long-time neighbors.” Similarly, Garrett (2011) “The Baltic [viz., Latvian] and Germanic shifts [viz., to word-initial stress] were probably both influenced by language shift, perhaps in the case Germanic involving Finnic or Celtic (or other unknown) languages.” Contact with Finnic and Hungarian, respectively, have been treated as the basis for word-initial stress found in Latvian (Lehiste 1988: 67, Veenker 1967: 74) and Czech and Slovak (Veenker 1967: 197; on Uralic substrata in Baltic and Slavic languages more generally, cf. Thomason and Kaufman 1988: 238–51). Likewise, Andersen (2009: 13–16) points towards a tendency for a leftward “accent shift” in Slavic dialects that have Finnic (Karelian) or Hungarian substrata.

The line of inquiry pursued in this section thus departs from the following assumption: Proto-Germanic speakers had extensive contacts with some language(s) (likely belonging to the Finno-Ugric family), whose word-level prosody was defined by consistent word-initial primary stress (as in, e.g., Finnish). Given this premise, the problem then becomes a matter of understanding precisely how and why such contact could have lead to a new prosodic state of affairs in Proto-Germanic. Once the prerequisites for a prosodic change of the Proto-Germanic sort are better understood, then the likelihood of the Proto-Germanic stress shift under such conditions can better estimated or simulated.

Language contact might conceivably affect the structure of the learning data for the purposes of prosodic acquisition in one of two ways. One means would be through the adoption of loanwords (cf. the case of English as described in Fikkert et al. 2006 and Lahiri 2015: 231–43, or German as described by Speyer 2009); another would be through L2 speakers that fail to perfectly acquire the target prosodic grammar and lexicon, but whose errors become learning data for subsequent generations. Could either of these situations have resulted in proportionally greater amounts of left-edge stress in Proto-Germanic, so as to gradually “tip” acquisition in favor of fixed initial stress?

In the particular (imagined) contact situation at hand, a robust quantity of Finnic loanwords into Proto-Germanic is not present. Although a considerable proportion of the Proto-

Germanic lexicon has been claimed to have a non-Indo-European origin (Feist 1932)), because their source language is unknown, it is impossible to make verifiable claims about the original word prosody of such presumed non-IE loanwords. Furthermore, the reevaluation of Schuhmann (2014) of Feist's original claim suggests that more of the Germanic lexicon may in fact have a plausible Indo-European etymology than long assumed.

Conversely, if the contact language indeed had a stress grammar similar to that of Finnish, it is likely that the L₁ speakers of this language would have been “stress deaf” (Peperkamp 2004; see 3.3.4.3). In their L₂ acquisition of (Early) Proto-Germanic, such speakers would have had considerable difficulties in consistently reproducing the lexical stress contrasts of Proto-Germanic with much accuracy. Even if the L₁ speakers of the contact language were not strictly “stress-deaf”, if their L₁ stress grammar was at least left-edge oriented, some amount of error in the production of stress in L₂ forms in favor of a position at the left edge of the word would be expected (cf. 3.3.4.4). The consequence, in either case, would be an increase in the total amount of tokens in the ambient language data exhibiting word-initial stress.

7.4.1 Simulation Design

The simulations in this section concern themselves with the hypothesis of language contact sketched above: what is the response of learners to data in which the same lexemes are produced with systematically different stress patterns by different adult speakers, more specifically, when some adult speakers are liable to erroneously generate a percept with primary stress on an initial syllable?⁷⁴ In the case of a single learner exposed to data from a “mixed” population of L₁ speakers with a lexical accent grammar and L₂ speakers, whose L₁ exhibits only initial stress, the phonotactic grammar of the learner may turn out differently, depending upon the precise balance of L₁ and L₂ teachers. Assuming that a regular influx of L₂ speakers of Proto-Germanic entered the Proto-Germanic speech community over some generations, the scenario of prosodic acquisition and transmission might look roughly as in Figure 7.2. In this highly simplified portrait, each child has an L₁ Proto-Germanic-speaking parent and an L₂ Proto-Germanic-speaking parent, where the L₂-speaking parent is assumed to randomly produce incorrect stress patterns with some regularity (e.g., 20%) according to the stress rules of her L₁.

To simulate the potential loss of lexically contrastive stress in favor of a grammar with a fixed metrical stress pattern, I employ here the same fundamental simulation design used at 5.3, more specifically in simulation set 5 (5.3.3.1). For the initial state, the same frequency distribution of tokens for two–five syllable words as found in the *R̥gveda* is used; this choice is mainly practical in nature, given the absence of both a suitable reference corpus for Early Proto-Germanic. The single learning agent in each generation receives tokens from either the initial distribution (in generation 1) or the learning agent of the preceding generation. Each learning agent is equipped with a Maximum Entropy grammar for determining the

⁷⁴See the folder for Chapter 7 in the GitHub repository at <https://github.com/rpsandell/Sandell-LMU-Habilitati> for relevant code.

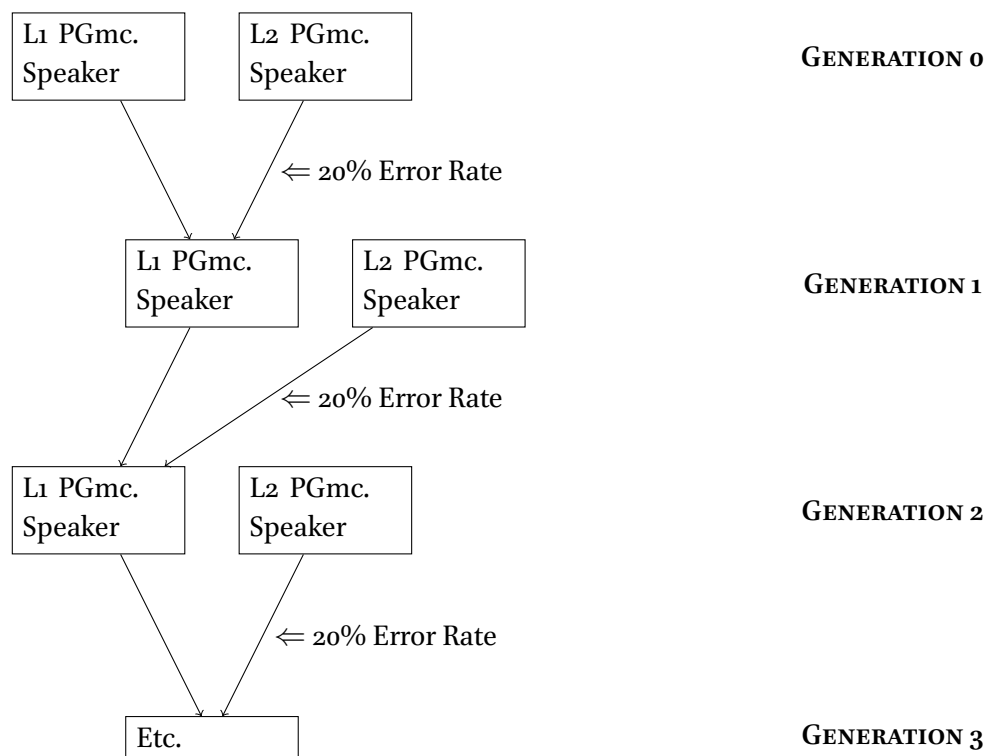


Figure 7.2: Transmission of Prosodic Data in Proto-Germanic Given L2 Speakers

probabilities of winning candidates, assessed by their violation profiles on 16 prosodic markedness constraints and a single prosodic faithfulness constraint, MAX-PROMINENCE; see again details at 5.3.

Although Figure 7.2 might suggest as much, the simulation conducted here does not, technically, employ a multi-agent model, since only a single agent in each generation is initialized. The effect of encountering tokens from an L2 speaker is simulated by simply randomly transforming the overt form of some proportion of sampled tokens into tokens with initial stress. For instance, if the presence of L2 speakers in the learning environment is assumed to result in 20% of the learning data being generated according to the L1 prosodic grammar of those L2 speakers (in this case, fixed initial stress), then 20% of all sampled tokens will be “rewritten” with initial stress. That, if the intended output of the teacher is, e.g., [H L H₁ VC], this output has a 20% chance of being transformed to [H₁ L H₁ VC] before being submitted to the learning agent. Note, though, that also intended outputs that would have initial stress in any case, e.g., [H₁ L H VC], are subject to this erroneous application of quantity-insensitive initial stress as well.

7.4.2 Simulation Results

To model the effect of language contact with stress-deaf speakers of an L1 having quantity-insensitive initial stress, the code for the simulation at 5.3.3.1 was modified to randomly change 20% of the overt forms presented to the learning agent in each generation to an overt form with stress on the leftmost syllable.⁷⁵ Given the overall results from 5.3 in general and 5.3.3.1 in particular, it is known that, if learning agents are allowed to learn from a sufficiently large quantity of data, and update their constraint weights quickly enough, then the faithfulness constraint MAX-PROM will always obtain a sufficiently high weight to ensure the faithful intergenerational transmission of lexically contrastive stress. Conversely, if the learning rate for MAX-PROM is low, or the amount of learning data limited, MAX-PROM will tend to decline in weight over successive generations of learning.

The crucial question to be considered here is whether, and to what extent, the addition of more tokens of initial stress impact the likelihood that a (quantity-insensitive) left-edge stress grammar will emerge and stabilize during the course of a simulation. Since the design employed here should have the effect of introducing at least some additional tokens with left-edge stress into the learner data at each generation, the learning agent in each generation ought, intuitively, have more opportunities to increase the weights of constraints like ALIGN- ω -L and TROCHEE that favor a (QI) Initial stress grammar.⁷⁶

Since a gradual “loss” of lexical stress in the limit is very likely, as shown at 5.3.3.1, in order

⁷⁵The error rate of 20% was chosen arbitrarily, based on the intuition that it would create a sufficient bias to affect the overall likelihood of outcomes without trivially guaranteeing the acquisition and transmission of quantity-insensitive initial stress.

⁷⁶Since no secondary stress is included in these simulations, the acquisition of a grammar like the one corresponding to the constraint ranking in (44) above in any given generation is relatively unlikely, though not impossible. Since the QI Initial grammar is simpler, and indistinguishable from the Proto-Germanic prosodic grammar, its emergence is *a priori* more likely.

to reduce run-time, the amount of learning data here was limited to 3000 tokens per generation. 65 simulations of 50 generations were carried out, with all markedness constraints set to an initial weight of 5 and MAX-PROM to a weight of 0; the markedness constraints were given a learning rate of 0.1, MAX-PROM a rate of 0.01. Table 7.2 reports the type of one of 16 fixed-stress patterns with the best performance in generation 50, as determined by the lowest Sum Squared Error (SSE) based on the predictions of that fixed-stress grammar in comparison to the predictions of the best-fit grammar in generation 50. The grammar with the lowest SSE obtained at any point during the simulation is also shown.

Grammar	<i>n</i> Best in Gen. 50	Mean SSE	<i>n</i> Best Overall	Mean SSE
Perfect Lexical	0	–	20 (30.8%)	55.94
QI Initial	25 (38.5%)	60.24	28 (43.1%)	5.65
QI Peninitial	2 (3.1%)	413.82	0	–
QI Penult	9 (13.8%)	225.82	4 (6.2%)	40.01
QI Final	0	–	1 (1.5%)	16.19
Left Iambs	1 (1.5%)	223.58	0	–
Right Trochees	10 (15.3%)	191.52	6 (9.2%)	22.47
DTS Left	3 (4.6%)	325.98	0	–
DTS Right FSM	2 (3.1%)	296.36	0	–
DTS Left 3 σ FSM	13 (20%)	134.33	6 (9.2%)	45.48

Table 7.2: Frequencies of Stress Grammars with Best Performance in Generation 50 and Overall (65 simulations, 50 generations per simulation)

As would be intuitively expected, the QI Initial pattern is the best option by all measures. For one, it is most often the pattern with the lowest SSE at the termination of a simulation, and its mean SSE is then significantly lower than all other options. Except in the simulations in which the QI Initial pattern stabilizes, achieving a very low SSE (below 1), the best performing grammar at any point is typically the Perfect Lexical pattern, in the first generation. That is, under the simulation conditions employed here, no other predictable, fixed-stress pattern stands a particularly good chance of stabilizing; only comparatively rarely do the QI Final or Right Trochee perform fairly well at any given point. Of course, as already established at 5.3, simple quantity-insensitive stress patterns are most likely to emerge out of the pool of variation present in the seed data, and the QI Initial pattern is particularly favored (mostly likely due to the slight predominance of initial stress and comparative infrequency of final stress in the RVic prominence data). Hence, it should ultimately come as no surprise that the QI initial pattern should arise so frequently under the conditions assumed here. A relatively typical instance in which the simulation can be said to have stably arrived to a QI initial grammar is shown in Figure 7.3, where the SSEs of the Perfect Lexical and DTS-L patterns are included for comparison.

That “adding” L2 speakers to the simulation indeed results in a greater likelihood that the QI Initial pattern, specifically, will appear as the best-performing grammar at the end of a simulation appears evident through comparison of the results in Table 7.2 with the results in Table 5.14, where the same initial starting weights were employed for markedness

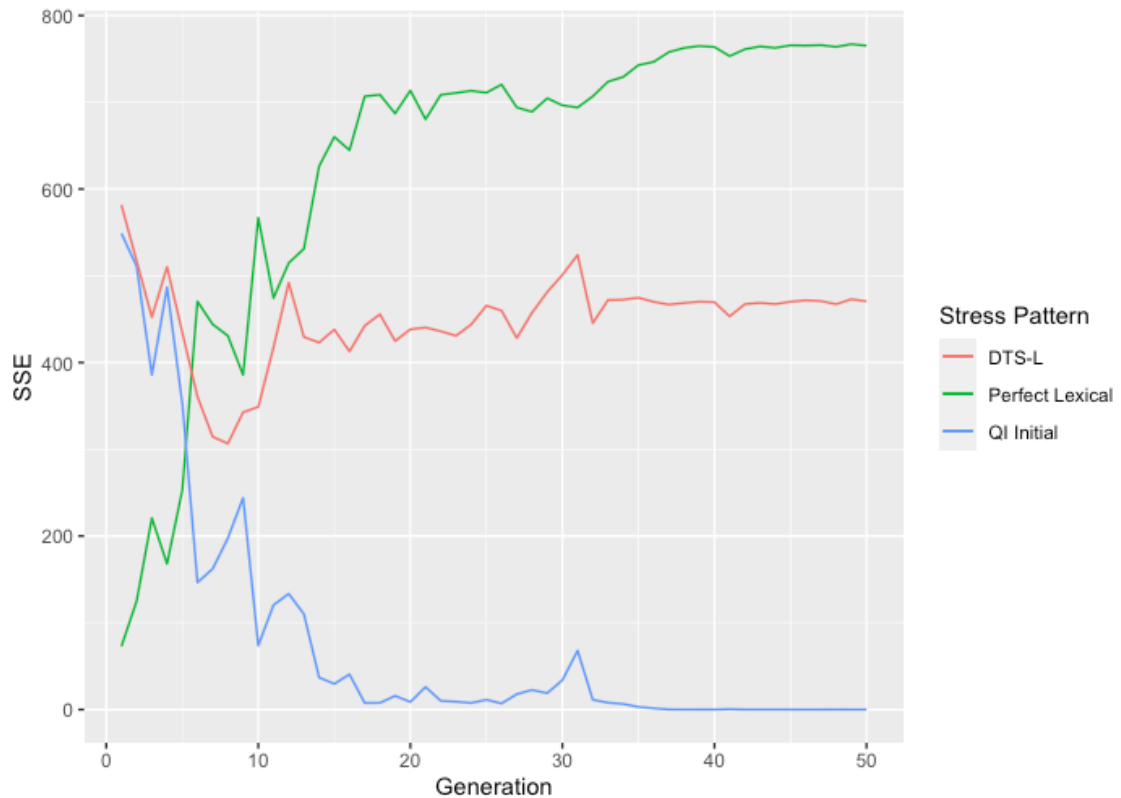


Figure 7.3: SSEs of QI Initial, DTS-L, and Perfect Lexical Patterns in Germanic Simulation 1.c

constraints. There, the QI Initial pattern had the lowest SSE in generation 50 in 22% of simulations, respectively; the difference in proportions is statistically significant ($\chi^2 = 4.4634$, $p = 0.034$). Simply put, by allowing for the possibility that the number of tokens with initial stress in the learning data is increased, the likelihood that learning of stress grammars over successive generations will converge on a pattern similar to the QI initial pattern increases as well.

Whether the particular assumptions adopted here concerning a specific error rate in stress production for “stress-deaf” L2 speakers are entirely accurate is difficult to say with certainty. Based on the data from experiments in Kijak 2009: 291–305, L1 speakers of French and Czech had error rate of $\sim 75\%$ and $\sim 65\%$ in primary stress placement, respectively, in a production task using actual lexemes of Polish (cf. 3.3.4.4). Given the relatively low amount of unpredictable stress in Polish as compared to what would be reconstructed for Proto-Germanic, one might suspect that error rates for “stress-deaf” L2 speakers could lie even higher. As a proof of concept, however, the highly oversimplified simulation conducted here adds plausibility to the hypothesis that language contact with speakers of language exhibiting fixed-initial stress may have substantially contributed to the likelihood that such a stress pattern (with quantity-sensitive secondary stress, however) would come to predominate during the prehistory of Germanic.

7.5 Conclusions and Further Implications

This chapter has been devoted to two questions:

- how is the grammar of stress assignment for Proto-Germanic to be reconstructed?
- what factors may have contributed to the development of a fixed-stress pattern out of a Vedic-like lexical accent system during the Proto-Germanic period.

The answer to the first question, at least as concerns the position of primary stress, is from a practical point of view, uncontroversial: primary stress stood on the leftmost syllable of all prosodic words, with the exception of prefixed verbs. Through closer consideration of the data from Old English and Gothic, in particular, following especially the earlier conclusions of Greiner (1994), Kiparsky 1998b, Minkova (2008), Goering (2016a), and Bermúdez-Otero (2018), I argued in favor of a grammar that constructed moraic trochees from left to right, but obligatorily placed the left edge of a foot at the left edge of the prosodic word. More importantly, this grammar of stress assignment need not make reference to any body of lexical exceptions: the apparent non-initial stress of prefixed verbs is attributable to a difference in prosodic structure, due to a requirement that the left edge of the lexical category VERB be aligned with the left edge of a (minimal) prosodic word. While the details of the exact prosodic structures constructed above the level of the $\omega_{[+min]}$ are open to further refinement, the important conclusion for present purposes is that the Proto-Germanic lexicon can then be regarded as entirely free of lexically prespecified prosody.

With the firm conclusion that Proto-Germanic underwent a total loss of lexically contrastive prosodic marking (in contrast to Vedic Sanskrit and Ancient Greek), the relevant issue becomes how the structure of the learning data made available to learners of Proto-Germanic might have changed. In this regard, the guiding question was: did Proto-Germanic undergo other phonological or morphological developments that would increase the overall proportion of left-edge primary stress in the data? In support of this investigation, Section 7.3.3 undertook a reevaluation of some hypotheses concerning the source of so-called “Verner’s Law Variants”, i.e., lexemes or derivational suffixes attested with multiple outcomes result from Verner’s Law. In particular, under the account of this phenomenon advanced by Kiparsky (2010a: 159–66), the existence of VLVs would constitute evidence for considerable “edge-to-edge” stress mobility in Germanic nominal inflection, and thus an increased number of tokens of initial stress. Kiparsky’s hypothesis is, however, open to serious doubt, even if no currently available alternative proposal is particularly convincing either.

Like attested changes in the prosodic grammar of English (compare Fikkert et al. 2006, Drescher 2013, or Lahiri 2015), the Germanic “stress shift” was thus not likely to have been a sudden, “catastrophic” event, but rather, a gradual evolution away from a grammar and lexicon that happily permitted non-initial stress, to a radically different system that permitted exclusively left-edge stress. Since prosodic change – specifically, change in the algorithm assigning stress – depends primarily changes in the learning data, the emergence of exclusively left-edge stress in Proto-Germanic necessarily required the gradual loss of lexemes with non-initial word stress.

Although some language-internal developments might increase the type and token frequency of word-initial prosodic prominence in Proto-Germanic, the available evidence from Verner's Law Variants and the prosodic properties of derivational suffixes hardly provides decisive evidence. If anything, the general picture available from such language internal data suggests that non-initial prominence was reasonably common in Proto-Germanic at the time that Verner's Law applied in Proto-Germanic. On the other hand, contact with a language or languages possessing fixed initial stress, especially if their native speakers became L2 speakers of Proto-Germanic, could have introduced substantial new token variation into Proto-Germanic stress patterns. Given that, in terms of the dynamics of individual lexemes, one variant is likely to "win" in the long run (compare the small simulation at 3.3.3), and the frequency distributions in Proto-Germanic the lexicon itself were already biased towards the initial syllable, the way to the acquisition of a fixed left-edge system then stands open.

At this point, the simulation carried out at 7.4 lends credence to the hypothesis that the prosody of Proto-Germanic was affected by prehistoric language contact, as has frequently been claimed (see e.g., Salmons 1992, with reference to earlier literature). If indeed "persistence" is diachronically the default modality of stress systems, including faithfulness to lexical stress, the new data or "noise" added to the data by a population of L2 speakers may offer one of the easiest paths towards a systemic prosodic change.

CHAPTER 8

Global Summary, Conclusions, and Future Directions

Having now conducted closer investigations into the synchronic and diachronic prosodies of Indic, Greek and Germanic with assistance from the toolkit developed in Part I, this closing chapter now answers the following questions:

- What has been learned about the prosody of Vedic Sanskrit, Classical Sanskrit, Ancient Greek, and the earliest Germanic languages (as well as their last common ancestor)?
- What larger implications do the results of the individual case studies hold for the study of prosody generally, stress assignment in particular, investigations into the prosody of (older) Indo-European languages specifically, and work on prosodic change?
- Which future research directions should or should not be pursued, in light the the results obtained and their implications?

8.1 Principal Results of Part II

Two major threads running through all of Part II, but emphasized especially in the chapters on Vedic Sanskrit and Ancient Greek, concern the typological characterization of their word-prosodic systems and the proper representations to assume in the analysis of lexically contrastive prominence. Taking a cue from Hyman (2009) (see the discussion under 1.3.3.3), I eschew the traditional characterization of Vedic Sanskrit and Ancient Greek as “pitch accent” languages or languages with “*musikalischem Akzent*” (Hirt 1895: 10), and instead attempt to situate these languages in terms of a property-oriented typology with relation to canonical stress and canonical tone languages. Both Vedic Sanskrit and Ancient Greek are argued to in fact essentially fulfill the criteria for canonical stress: word prominence is obligatory, culminative, privative, syllable-oriented, and the principles for determining its placement are amenable to description with metrical structure. What distinguishes these languages from canonical stress languages that make only an intonational use of tone (e.g., many varieties of present-day West Germanic languages) is the persistent availability of default “pitch accents” (in the intonational sense) at the level of minimal prosodic words. This property characterizes other languages with metrical stress, such as Egyptian Arabic (Hellmuth 2006, Hellmuth 2007) and most present-day varieties of Swedish (Riad 2014) and Norwegian (Kristoffersen 2000). Ancient Greek is argued to further share the property of possessing a privative contrastive in lexical tone, /H*/ vs. /Ø/, similar to Stockholm Swedish.

While the metricality of Ancient Greek word prosody, based on the evident limitation of High tone to a three-syllable window at the right edge of the word, is taken by many analysts for granted (e.g., Steriade 1988, Golston 1990), this feature is less obvious for Vedic Sanskrit, given the unboundedness of its word-level prominence. Building upon earlier analyses (Kiparsky 2010a, Kiparsky 2014, Sandell 2015a: Ch. 7, and Yates 2020), I show that it is possible to reproduce the empirical coverage of the Kiparskian analyses of Vedic accentuation in monostratal OT by referring to underlying foot edges (4.3) and the head of morphologically complex words. With the exception of certain problematic categories (privative compounds, active participles in *-ant-*), whose stress remains difficult to explain under any virtually any account (4.3.5), prospects for the analysis are good, and the typology of lexical entries predicted by the availability of both left and right foot edges correctly predicts all attested degrees of “strength” in Vedic lexical accents, when combined with effects of HEADFAITH (Revithiadou 1999). Instances of apparent violations of OBLIGATORINESS in Vedic word prosody are proposed to result from the competition of the default H*(+L) pitch accent with other L* pitch accents under specific morphological or syntactic conditions (4.1.3.1).

The role of special faithfulness to the prosody of morphological heads recurs in the treatment of Ancient Greek prosody as well. There, however, the phonology entity to be faithfully preserved is taken to be an autosegmentally associated tone, /H*/, rather than a foot edge. Under the assumption that Vedic prosody represents the historically conservative state, the Greek innovation in the representation of the “strongly stress-attracting unit” would constitute a type of optimization of underlying representations: because Greek requires underlying /H*/ to account for the contrast in monosyllables (level high tone *κῆρ* [ké:r] ‘fate’ versus *κῆρ* [kê:r] ‘heart’), this representation can be exploited by learners, together with constraints preferring the mutual cooccurrence of primary stress and starred tone, to restructure underlying representations. Thus, a Vedic stress-attracting suffix like past participle */(ta)/* is restructured as */-tó-/* in Greek, with /H*/ (< PIE stress-attracting **/-(to)-/*). The feature of head dominance established firmly for Vedic in Chapter 4 and reemphasized for Greek in Chapter 6 (compare similar observations in Steriade 1988 and Kiparsky 2014), when combined with the analysis of Yates (2017) of Hittite prosody, provide strong grounds for reconstructing head dominance, regulated by a constraint similar to Revithiadou’s HEADFAITH, for Proto-Indo-European (see discussion at 7.3).

Investigations into the prosody of Ancient Greek made two further novel discoveries, one synchronic and one diachronic. First, a statistical analysis of melismatic settings in a corpus of Ancient Greek music (see 6.2.1.6) firmly established that the so-called “acute” accent of Ancient Greek must be a level High tone (H), not a rising tone (LH), as often assumed.¹ This result substantially simplifies aspects of the synchronic analysis of how tone associates to stress in Ancient Greek. On the diachronic side, a new consideration of divergences between Attic-Ionic and Doric Greek with respect to the position of High tone could be explained by a difference in a single prosodic parameter, if Doric is assumed to otherwise be identical to Attic-Ionic in terms of its word prosody: Attic-Ionic exhibits extrametricality of word-

¹Thanks are due once again to Dieter Gunkel, who was responsible for building the electronic corpus of Ancient Greek music on the basis of Pöhlmann and West 2001, without which this analysis could not have been undertaken.

final consonants, while Doric does not. Insofar as attributing all differences between these dialects to a single parameter depends on the details of the synchronic phonological analysis of Attic-Ionic word prosody advanced at 6.2.3, Doric provides some further support for the claims made about Attic-Ionic.

Chapters 5 and 7, meanwhile, sought to better understand how the prosodic system of Vedic Sanskrit or the Vedic-like Proto-Indo-European could undergo developments that result in fixed metrical stress systems without contrastive lexical stress. To this end, the first natural step was to reconstruct the word-prosodics of Classical Sanskrit and Proto-Germanic, respectively. In the case of Classical Sanskrit, though the language itself is amply attested, little to nothing about its word prosody is directly attested, in contrast to the older Vedic.² Likely data upon which to build an analysis of a “Classical Sanskrit stress rule” had to be pieced together from observations on the practices of 19th century *paṇḍits* in India, variant forms in Classical Sanskrit, and historical developments in Middle Indic and Modern Indic languages. The available data point towards the existence of a quantity-sensitive, right-edge oriented trochaic system with a *four*-syllable window and final syllable extrametricality in words of three or more syllables; such a system is very similar to, though not identical with, the stress grammar likely to have existed in Old Latin (see 2.2.7.1 and 3.3.1.1). From the point of view of stress typology, the confidence with which this system can be reconstructed is of some importance, given the rarity of systems with such a large window (compare 3.3.5.1 and Staubs 2014 on the typology of stress windows in quantity-insensitive systems). A complete analysis of this “Classical Sanskrit” metrical stress grammar establishes two secure points in the prosodic history of Indic: the early Vedic system of c. 1200–1000 BCE, and the Classical Sanskrit system (of unclear date, but unlikely to be older than Pāṇini, and likely at least somewhat, if not considerably, younger).

The latter part of Chapter 5 was concerned, on the one hand, with establishing the learnability of the Classical Sanskrit stress rule, and on the other, determining whether any clear connections between it and the older Vedic word-prosodic system could be discovered. Answers to both questions were pursued primarily by means of computational simulations designed on the basis of the research into phonological learnability and acquisition, introduced in Chapter 3. At least for Maximum Entropy grammars equipped with the prosodic markedness constraints introduced at 2.3, the Classical Sanskrit pattern can indeed be learned using Robust Interpretive Parsing and SGA update rule, when the learner is presented with unambiguous data free of variation, and longer words of 4- and 5-syllables are presented just as frequently as short 2- and 3-syllable words. Relatively good results, obtaining accuracy above 91% in the worst case, were found for Stochastic OT and Noisy Harmonic Grammar as well. However, the complexity of the pattern is evident in the fact that relatively large amounts of learning data (20000 tokens or more for the MaxEnt SGA learner) were needed to obtain statistical performance indistinguishable from categoricity. Compare the result in Staubs 2014: 35–6 on quantity-insensitive stress systems, all of which (even the typologically very rare postpeninitial system) could be learned as near-categorical after exposure to

²Furthermore, as emphasized in Chapter 5, the word-prosodic dealt with there may not even properly constitute a development between Vedic and Classical Sanskrit directly, but rather a development between Vedic and some varieties of Middle Indic, whose stress rules were adopted for the pronunciation of Classical Sanskrit.

substantially less data (c. 6000 tokens).

The latter question – the relation between Vedic and Classical Sanskrit prosody – was investigated by first closely examining the statistical distribution of Vedic word prominence (the *udātta*) in the *R̥gveda*. In terms of token frequency, some statistically significant distributions indeed came to light: an underrepresentation of stress on final syllables, and, depending on word shape, an overrepresentation of stress on heavy syllables. However, the lexically and morphologically governed nature of Vedic stress means that no fixed-stress pattern correctly predicts the position of stress on more than c. 45% of all tokens in the *R̥gveda*.

Simulations employing single-agent iterated learning aimed to discover 1) what fixed stress patterns might emerge out of the Vedic stress distributions and 2) how a decline of lexically contrastive stress might come about. Given the complexities involved in simulating the acquisition of underlying representations beyond toy cases (like those in Apoussidou 2007: Ch. 6), the choice was made to focus on the phase of so-called phonotactic learning (see the end of 3.2.1.1). This choice constitutes a massive simplification of the overall learning problem, but allows the behaviors of one step posited for phonological acquisition to be tested in relative isolation.³ The effects of different parameters (learning rate, number of tokens presented per generation, weights of constraints in the initial state) were tested throughout. Simulations without the inclusion of a prosodic faithfulness constraint, MAX-PROMINENCE, were used to attack the first question, while simulations including this constraint (including different strategies for updating its weight) were intended to investigate conditions under which simple lexical stress might “die out”. In the main, results confirmed, in line with earlier work (Pater and Moreton 2012, Staubs 2014, Hughto 2020), that MaxEnt grammars exhibit a profound simplicity bias: grammars that require fewer constraint interactions are more likely to emerge and be stably transmitted. Nonetheless, the influence of the Vedic seed data was seen in the comparatively poor performance of quantity-insensitive final stress, and the tendency for categorical initial stress to emerge most often. Quantity-sensitive, right-edge oriented patterns with non-finality effects only seemed to emerge rarely. As concerns lexical stress, its behavior, given an initial MARKEDNESS $\gg$ FAITHFULNESS bias, was found to depend essentially on the amount of learning data and learning rate; if on or the other parameter is sufficiently small, then lexical stress (as reflected by the weight of MAX-PROM) tends to decline over generations of learning. Thus, impoverishment of learning data – especially if combined with a categoricity bias (Kirby and Sondregger 2015) in learning – might constitute one path for the elimination of lexical stress.

Ultimately, the simulations described at 5.3 demonstrate that the frequency distribution of stress in early Vedic is not alone a sufficient basis to account for the specific development of the Classical Sanskrit stress rule. While many aspects of the simulation design (MaxEnt grammar, interpretative parsing of overt forms) are fundamentally plausible, an enormous range of factors not considered (e.g., the possible effects of changes in morphological productivity on the frequencies of stress patterns) leaves much room open for further investigation.

³One could also imagine the results as pertinent to a hypothetical language with contrastive lexical stress, but no morphologically complex words, or in which perhaps only roots have lexically contrastive properties, and the addition of prefixes and suffixes determines the rest of a word’s prosodic shape.

For its part, Chapter 7 served to undergird the view that Proto-Germanic and its earliest-attested daughter languages possessed a purely fixed, metrical stress system. Apparent exceptions to a rule of initial primary stress can be explained as due to prosodic structures involving the cliticization of function words (prepositions, adverbials). Detailed engagement with the origin of so-called “Verner’s Law Variants” (Barber 1932) in Germanic also reconfirmed that this particular problem still awaits a satisfactory solution. Finally, the possibility that contact with “stress-deaf” speakers whose L1 applied fixed initial stress could be shown through agent-based simulations to increase the likelihood of acquiring just such a grammar.

8.2 Implications for the Study of (Indo-European) Prosody and Prosodic Change

The first and foremost methodological point to be reemphasized here is that the study of synchronic systems must precede the study of their diachrony, especially where reconstruction is concerned. While the traditional method of comparative reconstruction yields some usable results, for the study of change in phonological grammars, discussing language change is perilous without a fairly clear portrait of the synchronic grammars being compared. That the synchronic analyses may require adjustment or improvement should not lead historical linguists and Indo-Europeans to judge that detailed, formal work on synchrony is unlikely to be profitable. Much to the contrary, Indo-Europeanists, with detailed philological knowledge of the languages in question, must continue to adopt and apply tools of linguistic analysis with general purchase in order to help advance knowledge about the objects of investigation through communication with non-specialists (in the domain of phonology, see effective examples in Cooper 2015, Byrd 2015, DeLisi 2015, Yates 2017, or Zukoff 2017b).

The studies on Indic and Greek, and to a lesser extent examinations of Latin stress at 2.2.7.1 and 3.3.1.1, contribute to the work on the typology of prosodic systems and theoretical analysis of stress in three respects. First, the role of head dominance in Vedic and Greek word prosody, the reconstructability of this property for Proto-Indo-European (7.5), and apparent existence of head dominance in Slavic as well, allow for this property to be identified as a defining feature of Indo-European languages. The research program initiated by Revithiadou (1999) may continue to bear fruit. Second, the agreement of Classical Sanskrit and Old Latin on the existence and learnability of stress systems with a four-syllable window means that such systems should at least tentatively be considered in future work on stress typology. Finally, Attic-Ionic (and Doric) Greek appear to show that properties of stress attraction to High tone (in accord with the predictions made in de Lacy 2002) may coexist alongside the attraction of Low tone to stress. In addition, Attic-Ionic Greek should likely be added to the dossier of languages that observe a distinction between VV (long vowel or diphthong) and VC syllables for the purposes of stress assignment (Gordon 2006, Ryan 2019), and offers support for the relatively greater markedness of rising tone (Gordon 2001, Zhang 2004, Kubozono 2017).

Similarly, although the central focus has been the construction of grammars for the assignment of primary stress, or stress-to-tone association, the need to attend to aspects of

intonational phonology – insofar as possible – and the prosodic hierarchy above the level of the minimal prosodic word has shown itself to be indispensable. In this respect, the concern with intonation phonology in Vedic and Greek is preceded by Gunkel (2013, 2017) and Goldstein 2015. Closer attention to the interactions of stress, word-level pitch accent, and intonational tone in future work on the prosody of older Indo-European languages. Indeed, given the impact that boundary tones may have on word-level pitch accents diachronically (compare again the studies of Basque dialects in Hualde 2003a and Hualde 2022), hunting for traces of intonation in older Indo-European languages above the level of the word – or perhaps even attempting to reconstruct it based on the evidence of modern languages – is a desideratum.

As far as change in grammars of stress assignment is concerned, the puzzle of lexical stress loss (see 3.3.2) remains very much real. By seeking to put our knowledge of the stress grammars of Proto-Germanic and post-Vedic Sanskrit on somewhat firmer grounding, the studies in Chapter 5 and 7 have made evident that the wholesale disappearance of lexically contrastive stress in (some members of) these branches must be explained. While misanalysis at the phonological level may suffice as an explanation for change between two stress grammars that make overwhelmingly overlapping predictions (e.g., the Old Latin and Classical Latin systems), these instances of lexical stress loss are not easily shoehorned into such an account. At the same time, the sensitivity of phonological grammars to the frequency of patterns, how frequency shapes the trajectory of acquisition, and the ease with which particular stress patterns may be acquired, should certainly inform our priors on the likelihood of historical developments. From the point of view, that Proto-Germanic happened to develop fixed initial stress should not be a great surprise: (quantity-insensitive) initial stress is easily learned, and among virtually all fixed-stress patterns, probably exhibited the most accidental agreement with the lexical stress grammar of Proto-Indo-European. Conversely, the precise conditions that must have obtained to favor the acquisition of the Classical Sanskrit stress rule in some Indic languages appear all the more mysterious.

One final methodological point to be made is that grasping prosodic changes generally and changes in stress assignment specifically is not trivial. While phonological reanalysis always constitutes a serious possibility, it must be motivated as plausible, and kept free from unfounded speculation as much as possible. An example of over-reliance on reanalysis is found in Lahiri 2015: 225–6, where the author suggests that the development of the later Latin stress patterns could have resulted from the overlap between the predictions of a fixed initial stress grammar in often posited for Archaic Latin (cf. Nishimura 2014); beyond three-syllable words, the predictions of a fixed initial-stress grammar and the (Old or Classical) Latin stress rules become too divergent for phonological reanalysis to constitute a credible explanation. Similarly, Ramers (1999: 42–3) speculates, for instance, that initial syllables in Proto-Germanic all carried a secondary stress (i.e., the left edge of a trochaic foot was obligatorily aligned with the left edge of the word), which was subsequently reinterpreted as primary stress in all cases. Bringing overly facile explanations of these sorts to bear on difficult historical problems does not do much to aid the overall cause.

8.3 Future Questions and Continuing Research

Despite the significant headway made in the course of the present work, a substantial number of questions, remain open, but can now be formulated more precisely. A particularly promising focus of future research remains one of the central questions, stated at (51) in Chapter 3: what should cause perfectly learnable phonological grammars to be mislearned? Given the state of knowledge about phonological learnability generally, changes that impact the structure of the learning data in some fashion, whether language internal (loss of quantity contrast in vowels in Greek) or language external (substantial contact and bilingualism), remain fundamentally most plausible. While segmental developments with implications for word-level stress are usually readily recoverable through the application of traditional methods in historical linguistics, the influence of language-external factors, especially before the existence of a historical record, is much more difficult to firmly establish. On this point, the study of Ito and Feldman (2022) (discussed at 3.3.8) provides a helpful model: the authors implemented agent-based models to test the hypothesis that an early dialectal division in Korean was responsible for differences in the tone systems of modern languages. Similarly, an expansion of and improvement on the pilot simulation used in 7.4 would help to lend further credibility to, or potentially disprove, the hypothesis of language contact in the Proto-Germanic period.

Although the grammars constructed throughout Part II rest on centuries of philological work, not all issues surrounding the data itself are always entirely clear. Hence, many of the most potentially profitable immediate future directions of research are fundamentally philological in nature. With respect to Greek, a small degree of uncertainty persists about the correct accentuation of some classes of orthotonic words (the type $\kappa\acute{\eta}\rho\nu\varsigma$ [kê:ruks]; see note 7 in Chapter 6), and especially the accentuation of multiple enclitic sequences (see 6.2.1.5 and 6.2.3.7). Furthermore, I believe that the potential utility of the corpus of Ancient Greek music has not yet been fully exploited; the digitization of this corpus and the application of improved statistical methods to its testimony may yet yield new insights into the phonology of Ancient Greek intonation above the word level (see Devine and Stephens 1994 for earlier work along these lines). On the Indic side, more detailed work into the accentual behavior of specific morphological categories (e.g., privative compounds in *a(n)-*, *vṛddhi*-derivation) in Vedic is needed, as is a closer reassessment of the testimony of the *Śatapathabrāhmaṇa* from the perspective of a phonologist than could be undertaken at 5.1.2.

At various points throughout this work, though particularly at 5.1.1 and 7.3.3.2, the notion of “demorphologization” as a potential factor behind changes in the position of word prominence at the level of individual lexemes has been discussed. Unfortunately, no comprehensive treatment of the phenomenon, introduced by Probert (2006b), has been undertaken. If genuine, I suspect that this phenomenon may be akin to the “nativization” of loanwords in languages known to exhibit different strata of phonological processes for different clusters in the lexicon (cf. Ito and Mester 2003a on Japanese lexical strata and their phonology). That is, both phenomena would involve a transformation from “exceptional” to “regular” in the phonology. Sandell 2015a: 198–213 contains a preliminary examination of relationships between morphological productivity of derivational processes and stress in Vedic Sanskrit.

More corpus-based, quantitative investigations into modern languages with lexical accent (e.g., Russian, Japanese) that could reveal cases of potential “demorphologization” and stress shift would be desirable.

Several immediate expansions and improvements upon the agent-based simulations undertaken in Chapters 5 and 7 could readily be implemented. At minimum, the simulation at 7.4 should be redesigned as a multi-agent simulation, in which both L2 agents, already outfitted with initial-stress grammars could be made to acquire the variable, contrastive stress data alongside L1 learners, receiving data from both L1 speakers and L2 speakers. In this way, no artificial assumptions concerning an error rate as a further parameter need to be built in; however, a slower learning rate for the L2 speakers should be considered, and, under the assumption that such speakers are “stress deaf”, their capacity to increase the weight of prosodic faithfulness constraints should be assumed to be very limited. In addition, contrasting the behavior of the Stochastic Gradient Ascent learners with a batch Maximum Entropy learner would also be desirable, in order to explicitly test whether the non-convergent behavior of MaxEnt SGA is indeed desirable in the context of modeling language change. Implementing work on the acquisition of underlying representations to build a model that not only carries out phonotactic, but also morphophonological learning, would constitute a considerable advance, but must overcome some important theoretical and technical hurdles (see discussion under 3.2.2.2).

A major question for continuing work on change in stress grammars concerns the notion of “persistence”, specifically: is “persistence” (in the sense of Lahiri 2015) indeed the diachronic default with respect to stress patterns? That is, are learners generally faithful to the evidence for the position of primary stress in their learning data? And do fixed-stress patterns tend to be stably transmitted diachronically? A bird’s eye view of some language families suggests that the answer to both questions is generally “yes”. As to the first question, the remarkable diachronic persistence of the position of primary stress in most the Romance languages with respect to Latin, or in Modern Greek with respect to the ancient Ancient Greek acute or circumflex (see 6.4) may be interpreted as a bias towards constructing grammars that do not deviate from the learning data on the point; the research on L1 acquisition of stress patterns (see 3.3.4.1 and 3.3.4.2) also appears to accord with this conclusion. Outside the Indo-European family, the persistence of particular prosodic parameters across language families at substantial time depth is also of note: Finno-Ugric languages rarely exhibit stress patterns other than left-oriented trochees (see Vaysman 2009 and Gordon 2016 on partial exceptions), and right-edge trochees predominate across the Austronesian family (see van Zanten et al. 2011: 89–92).

Finally, in terms of the goal of better understanding the prosodic history of the Indo-European language family, the research here into Vedic, Greek, and Germanic should be extended to other branches. Especially Baltic and Slavic contain many further potential case studies for concerning the loss of contrastive lexical stress. Connecting the research on Vedic prosody with prosodic developments Indic languages, into the present-day languages, as well as Iranian, would advance the cause of constructing a diachronic typology of prosodic systems and prosodic changes.

These tasks may be viewed as the initial blueprint for a larger research program. The present work has constructed a theoretical scaffolding to support such a program in Part I, and added some of the first load-bearing pillars in Part II. In the years to come, seeing these plans executed will, hopefully, help to build up a larger edifice that can offer a comprehensive portrait of what forms prosodic change can take on, how it is actuated, and whether it is possible to predict the seemingly unpredictable paths that a change in motion might ultimately walk.

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